

A TEXTBOOK OF
FLUID MECHANICS

in

SI UNITS

A TEXTBOOK OF

FLUID MECHANICS

(“Fluid Mechanics and Hydraulic Machines”–Part-I)

**[For Engineering Students of Various Disciplines and Aspirants of Various
Competitive Examinations]**

in

SI UNITS

Er. R.K. RAJPUT

M.E. (Hons.), Gold Medallist; Grad. (Mech.Engg. & Elect. Engg.); M.I.E. (India);
M.S.E.S.I.; M.I.S.T.E.; C.E. (India)

Recipient of:

“Best Teacher (Academic) Award”

“Distinguished Author Award”

*“Jawahar Lal Nehru Memorial Gold Medal”
for an outstanding research paper
(Institution of Engineers–India)*

Principal (Formerly):

- *Thapar Polytechnic College;*
- *Punjab College of Information Technology,*

Patiala



S Chand And Company Limited
(ISO 9001 Certified Company)



S Chand And Company Limited

(ISO 9001 Certified Company)

Head Office: Block B-1, House No. D-1, Ground Floor, Mohan Co-operative Industrial Estate, New Delhi – 110 044 | Phone: 011-66672000

Registered Office: A-27, 2nd Floor, Mohan Co-operative Industrial Estate, New Delhi – 110 044
Phone: 011-49731800

WWW.schandpublishing.com; e-mail: info@schandpublishing.com

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Kolkata	: Ph: 23357458, 23353914; kolkata@schandpublishing.com
Lucknow	: Ph: 4003633; lucknow@schandpublishing.com
Mumbai	: Ph: 25000297; mumbai@schandpublishing.com
Nagpur	: Ph: 2250230; nagpur@schandpublishing.com
Patna	: Ph: 2260011; patna@schandpublishing.com
Ranchi	: Ph: 2361178; ranchi@schandpublishing.com
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Jurisdiction: All disputes with respect to this publication shall be subject to the jurisdiction of the Courts, Tribunals and Forums of New Delhi, India only.

First Edition 1998

Subsequent Editions and Reprints 2002, 2003, 2005, 2006, 2007, 2008 (Twice), 2009 (Twice); Fourth Edition 2010; Reprints 2010, 2011, 2012, Fully Revised Multicolour Edition 2013, Reprint 2014; Sixth Revised Edition 2014; Reprint 2015, 2016, 2017, 2019

LPSPE 2019

ISBN: 978-93-5283-720-5

Product Code: H3FME62FMEC10ENAF19L

PRINTED IN INDIA

By Vikas Publishing House Private Limited, Plot 20/4, Site-IV, Industrial Area Sahibabad, Ghaziabad – 201 010 and Published by S Chand And Company Limited, A-27, 2nd Floor, Mohan Co-operative Industrial Estate, New Delhi – 110 044

To my wife
Ramesh Rajput

PREFACE TO THE SEVENTH EDITION

I am pleased to present the **Seventh Edition (Multicolour)** of this book. The warm reception, which the previous editions and reprints of this book have enjoyed all over India and abroad has been a matter of satisfaction to me.

Besides revising the book thoroughly in general, **GATE and UPSC Examinations Questions with Answers/Solutions (Latest-Selected)** which are divided into:

A. Conventional Questions with Answers/Solutions.

B. Multiple-Choice Questions with Answers and Explanations
have been *updated* in particular by adding **latest questions**.

I'm thankful to the Management Team and the Editorial Department of S Chand And Company Limited, New Delhi for all help and support in the publication of this book.

Any constructive suggestions for the improvement of this book are most welcome.

Er. R.K. Rajput
(Author)

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PREFACE TO THE FIRST EDITION

This treatise on "*Fluid Mechanics*", contains comprehensive treatment of the subject matter in simple, lucid and direct language and envelopes a large number of solved problems properly graded, including typical examples from examination point of view.

The book comprises 16 chapters. All chapters of the book are saturated with much needed text supported by simple and self-explanatory figures and a large number of *Worked Examples* including *Typical Examples* (for competitive examinations). At the end of each chapter *Highlights*, *Objective Type Questions*, *Theoretical Questions* and *Unsolved Examples* have been added to make the book a comprehensive and a complete unit in all respects.

The book will prove to be a boon to the students preparing for engineering undergraduate, AMIE Section B (India) and competitive examinations.

The author's thanks are due to his wife Ramesh Rajput for extending all cooperation during preparation of the manuscript and proof reading.

In the end the author wishes to express his gratitude to Shri Ravindra Kumar Gupta, Director, S. Chand & Company Ltd., New Delhi, for taking a lot of pains in bringing out the book, with extremely good presentation, in a short span of time.

Although every care has been taken to make the book free of errors both in the text as well as in solved examples, yet the author shall feel obliged if errors present are brought to his notice. Constructive criticism of the book will be warmly received.

Er. R.K. Rajput
(Author)

NOMENCLATURE

a	Acceleration
A	Area
A_s	Area of suction pipe, surge tank
A_d	Area of delivery pipe
B	Width of wheel (turbine)
b	Width, bed width of rectangular or trapezoidal channel
c_p	Specific heat at constant pressure
CP	Centipoise
C_v	Specific heat at constant volume
C	Chezy's discharge coefficient
C	Celerity of a pressure wave
C_c	Coefficient of contraction
C_d	Discharge coefficient of weirs, orifice plates
C_D	Drag coefficient
C_D	Local drag coefficient
C_v	Coefficient of velocity
d	Diameter of orifice plate, pipe, particle
D	Diameter of pipe, wheel
d_d	Diameter of delivery pipe
d_s	Diameter of suction pipe
e	Linear strain
E	Young's modulus of elasticity of material
f	Darcy Weisbach friction coefficient, frequency
F	Force
F_B	Force exerted by boundary on the fluid
F_D	Drag force on the body
F_L	Lift force
F_r	Froude number
g	Gravitational acceleration
h	Piezometric head, specific enthalpy
h_d	Delivery head
h_f	Frictional loss of head
h_s	Suction head
H_g	Gross head
H	Total energy head, net head
h_{ad}	Acceleration head for delivery pipe
h_{as}	Acceleration head for suction pipe
I	Moment of inertia (of area), moment of inertia (of mass)
l_d	Length of delivery pipe
l_s	Length of suction pipe
l_d'	Length of delivery pipe between cylinder to air vessel
l_s'	Length of suction pipe between cylinder and air vessel

k	Roughness height
K	Conveyance
K	Head loss coefficient, bulk modulus of elasticity, blade friction coefficient
K_t	Vane thickness factor
K_u	Speed ratio
K_f	Flow ratio
m	Mass
M	Momentum, Mach number
n	ratio B/D
N	Manning's roughness coefficient, revolutions per minute
N_s	Specific speed
p, p_s	Pressure, stagnation pressure
P	Power, shaft power (turbine), Poise, force
q	Discharge per unit width, discharge per jet
Q	Discharge, heat
r	Distance from the centre
R	Radius of pipe, hydraulic radius, radius of pipe bend
R_o	Universal gas constant
Re	Reynolds number
S	Specific gravity, bed slope of channel
t	Thickness, time
T	Absolute temperature in Kelvins
T	Torque, water surface width
u	Instantaneous velocity at a point in X-direction
u_f	Shear friction velocity
U	Free stream velocity
V_d	Velocity of flow in delivery pipe
V	Velocity of flow in the cylinder
V_s	Velocity of flow in suction pipe
v	Instantaneous velocity at a point in Y-direction
v	Specific volume
v_c	Critical velocity
V_a	Velocity of approach
v	Time averaged velocity at a point in Y-direction
V_r	Relative velocity
V_f	Velocity of flow (in turbines and pumps)
V_w	Velocity of swirl (in turbines and pumps)
V	Volume
w	Weight density, Instantaneous velocity at a point in Z-direction
W	Weight of fluid, workdone
x	Distance in X-direction
y	Distance in Y-direction, depth of flow
y_c	Critical depth
x	Depth of centroid of area below water surface
Z	Number of buckets/vanes
z	elevation

Greek Notations

α	Energy correction factor, Mach angle, angle
β	Momentum correction factor, angle
γ	Ratio of specific heats
δ	Boundary layer thickness
δ'	Laminar sub-layer thickness
δ	Displacement thickness of boundary layer
Δs	Change in entropy
η	Efficiency, dimensionless distance (y/δ)
θ	Angle, momentum thickness of boundary layer
μ	Coefficient of dynamic viscosity
ν	Kinematic viscosity
ρ	Mass density of fluid
σ	Coefficient of surface tension, cavitation number (Thoma number)
τ	Shear stress
τ_0	Bottom shear stress
ϕ	Angle, velocity potential
ψ	Stream function
ω	Angular velocity
Γ	Circulation
Ω	Vorticity

Subscript 0	refer to any quantity at reference section
Subscripts 1, 2	refer to any quantity at section 1 or 2
Subscripts x, y, z	refer to any quantity in x, y, z direction
Subscripts m, p	refer to any quantity in model and prototype
Subscript r	refer to the ratio of any quantity in model to that in prototype.

CONTENTS

1. PROPERTIES OF FLUIDS	1–42
1.1. Introduction	1
1.2. Fluid	2
1.3. Liquids and their Properties	3
1.4. Density	3
1.4.1. Mass density	3
1.4.2. Weight density	3
1.4.3. Specific volume	3
1.5. Specific Gravity	3
1.6. Viscosity	4
1.6.1. Newton's law of viscosity	5
1.6.2. Types of fluids	5
1.6.3. Effect of temperature on viscosity	8
1.6.4. Effect of pressure on viscosity	8
1.7. Thermodynamic Properties	23
1.8. Surface Tension and Capillarity	25
1.8.1. Surface tension	25
1.8.1.1. Pressure inside a water droplet, soap bubble and a liquid jet	26
1.8.2. Capillarity	28
1.9. Compressibility and Bulk Modulus	34
1.10. Vapour Pressure	37
<i>Highlights</i>	39
<i>Objective Type Questions</i>	40
<i>Theoretical Questions</i>	41
<i>Unsolved Examples</i>	41
2. PRESSURE MEASUREMENT	43—96
2.1. Pressure of a Liquid	43
2.2. Pressure Head of a Liquid	43
2.3. Pascal's Law	45
2.4. Absolute and Gauge Pressures	48
2.5. Measurement of Pressure	53
2.5.1. Manometers	54
2.5.1.1. Simple manometers	54
2.5.1.2. Differential manometers	63
2.5.1.3. Advantages and limitations of manometers	81
2.5.2. Mechanical gauges	81
2.6. Pressure at a Point in Compressible Fluid	83

<i>Highlights</i>	91
<i>Objective Type Questions</i>	92
<i>Theoretical Questions</i>	93
<i>Unsolved Examples</i>	93
3. HYDROSTATIC FORCES ON SURFACES	97—159
3.1. Introduction	97
3.2. Total Pressure and Centre of Pressure	97
3.3. Horizontally Immersed Surface	97
3.4. Vertically Immersed Surface	98
3.5. Inclined Immersed Surface	116
3.6. Curved Immersed Surface	129
3.7. Dams	140
3.8. Possibilities of Dam Failure	142
3.9. Lock Gates	151
<i>Highlights</i>	155
<i>Objective Type Questions</i>	156
<i>Theoretical Questions</i>	157
<i>Unsolved Examples</i>	157
4. BUOYANCY AND FLOATATION	160—191
4.1. Buoyancy	160
4.2. Centre of Buoyancy	160
4.2. Types of Equilibrium of Floating Bodies	165
4.3.1. Stable equilibrium	165
4.3.2. Unstable equilibrium	165
4.3.3. Neutral equilibrium	165
4.4. Metacentre and Metacentric Height	165
4.5. Determination of Metacentric Height	166
4.5.1. Analytical method	166
4.5.2. Experimental method	167
4.6. Oscillation (Rolling of a Floating Body)	187
<i>Highlights</i>	189
<i>Objective Type Questions</i>	189
<i>Theoretical Questions</i>	190
<i>Unsolved Examples</i>	190
5. FLUID KINEMATICS	192—258
5.1. Introduction	192
5.2. Description of Fluid Motion	192
5.2.1. Lagrangian method	192
5.2.2. Eulerian method	193

5.3.	Types of Fluid Flow	195
5.3.1.	Steady and unsteady flows	195
5.3.2.	Uniform and non-uniform flows	196
5.3.3.	One, two and three dimensional flows	196
5.3.4.	Rotational and irrotational flows	197
5.3.5.	Laminar and turbulent flows	197
5.3.6.	Compressible and incompressible flows	197
5.4.	Types of Flow Lines	198
5.4.1.	Path line	198
5.4.2.	Stream line	198
5.4.3.	Stream tube	198
5.4.4.	Streak line	199
5.5.	Rate of Flow or Discharge	207
5.6.	Continuity Equation	207
5.7.	Continuity Equation in Cartesian Co-ordinates	209
5.8.	Equation of Continuity in Polar Coordinates	211
5.9.	Circulation and Vorticity	218
5.10.	Velocity Potential and Stream Function	227
5.10.1.	Velocity potential	227
5.10.2.	Stream function	228
5.10.3.	Relation between stream function and velocity potential	231
5.11.	Flow Nets	231
5.11.1.	Methods of drawing flow nets	231
5.11.2.	Uses and limitations of flow nets	232
	<i>Highlights</i>	253
	<i>Objective Type Questions</i>	255
	<i>Theoretical Questions</i>	257
	<i>Unsolved Examples</i>	257
6.	FLUID DYNAMICS	259—385
6.1.	Introduction	259
6.2.	Different Types of Heads (or Energies) of a Liquid in Motion	259
6.3.	Bernoulli's Equation	260
6.4.	Euler's Equation for Motion	262
6.5.	Bernoulli's Equation for Real Fluid	276
6.6.	Practical Applications of Bernoulli's Equation	291
6.6.1.	Venturimeter	291
6.6.1.1.	Horizontal venturimeters	292
6.6.1.2.	Vertical and inclined venturimeters	298
6.6.2.	Orificemeter	303
6.6.3.	Rotameter and elbow meter	308
6.6.3.1.	Rotameter	308
6.6.3.2.	Elbow meter	309

6.6.4. Pitot Tube	310
6.7. Free Liquid Jet	313
6.8. Impulse-Momentum Equation	320
6.9. Kinetic Energy and Momentum Correction Factors (Coriolis Co-efficients)	336
6.10. Moment of Momentum Equation	343
6.11. Vortex Motion	345
6.11.1. Forced vortex flow	345
6.11.2. Free vortex flow	346
6.11.3. Equation of motion for vortex flow	346
6.11.4. Equation of forced vortex flow	347
6.11.5. Rotation of liquid in a closed cylindrical vessel	354
6.11.6. Equation of free vortex flow	361
6.12. Liquids in Relative Equilibrium	364
6.12.1. Liquid in a container subjected to uniform acceleration in the horizontal direction	364
6.12.2. Liquid in a container subjected to uniform acceleration in the vertical direction	373
6.12.3. Liquid in container subjected to uniform acceleration along inclined plane	375
<i>Highlights</i>	376
<i>Objective Type Questions</i>	379
<i>Theoretical Questions</i>	381
<i>Unsolved Examples</i>	382
7. DIMENSIONAL AND MODEL ANALYSIS	386—456
DIMENSIONAL ANALYSIS	
7.1. Dimensional Analysis—Introduction	386
7.2. Dimensions	387
7.3. Dimensional Homogeneity	389
7.4. Methods of Dimensional Analysis	390
7.4.1. Rayleigh's method	390
7.4.2. Buckingham's π -method/theorem	394
7.4.3. Limitations of dimensional analysis	415
MODEL ANALYSIS	
7.5. Model Analysis—Introduction	415
7.6. Similitude	416
7.7. Forces Influencing Hydraulic Phenomena	417
7.8. Dimensionless Numbers and their Significance	418
7.8.1. Reynolds number (Re)	418
7.8.2. Froude's number (Fr)	419
7.8.3. Euler's number (Eu)	419
7.8.4. Weber number (We)	419
7.8.5. Mach number (M)	420

7.9. Model (or Similarity) Laws	420
7.10. Reynolds Model Law	420
7.11. Froude Model Law	434
7.12. Euler Model Law	445
7.13. Weber Model Law	446
7.14. Mach Model Law	447
7.15. Types of Models	449
7.15.1. Undistorted models	449
7.15.2. Distorted models	449
7.16. Scale Effect in Models	450
7.17. Limitations of Hydraulic Similitude	451
<i>Highlights</i>	451
<i>Objective Type Questions</i>	453
<i>Theoretical Questions</i>	454
<i>Unsolved Examples</i>	454
8. FLOW THROUGH ORIFICES AND MOUTHPIECES	457—507
8.1. Introduction	457
8.2. Classification of Orifices	457
8.3. Flow Through an Orifice	458
8.4. Hydraulic Co-efficients	458
8.4.1. Co-efficient of contraction (C_c)	458
8.4.2. Co-efficient of velocity (C_v)	459
8.4.3. Co-efficient of discharge	459
8.4.4. Co-efficient of resistance (C_r)	459
8.4. Experimental Determination of Hydraulic Co-efficients	460
8.5.1. Determination of co-efficient of velocity (C_v).	460
8.5.2. Determination of co-efficient of discharge (C_d)	461
8.5.3. Determination of co-efficient of contraction (C_c)	462
8.5.4. Loss of head in orifice flow	462
8.6. Discharge Through a Large Rectangular Orifice	470
8.7. Discharge Through Fully Submerged Orifice	472
8.8. Discharge Through Partially Submerged Orifice	473
8.9. Time Required for Emptying a Tank Through an Orifice at its Bottom	474
8.10. Time Required for Emptying a Hemispherical Tank	483
8.11. Time Required for Emptying a Circular Horizontal Tank	487
8.12. Classification of Mouthpieces	490
8.13. Discharge Through an External Mouthpiece	490
8.14. Discharge Through a Convergent-divergent Mouthpiece	493
8.15. Discharge Through an Internal Mouthpiece (or Re-entrant or Borda's Mouthpiece)	496
8.15.1. Mouthpiece running free	496
8.15.2. Mouthpiece running full	497

<i>Highlights</i>	503
<i>Objective Type Questions</i>	505
<i>Theoretical Questions</i>	506
<i>Unsolved Examples</i>	506

9. FLOW OVER NOTCHES AND WEIRS

508—533

9.1. Definitions	508
9.2. Types/Classification of Notches and Weirs	508
9-2-1. Types of notches	508
9-2-2. Types of weirs	509
9.3. Discharge Over a Rectangular Notch or Weir	509
9.4. Discharge Over a Triangular Notch or Weir	511
9.5. Discharge Over a Trapezoidal Notch or Weir	513
9.6. Discharge Over a Stepped Notch	514
9.7. Effect on Discharge Over a Notch or Weir due to Error in the Measurement of Head	516
9.8. Velocity of Approach	518
9.9. Empirical Formulae for Discharge Over Rectangular Weir	518
9.10. Cippoletti Weir or Notch	521
9.11. Discharge Over a Broad Crested Weir	522
9.12. Discharge Over a Narrow-crested Weir	523
9.13. Discharge Over an Ogee Weir	523
9.14. Discharge Over Submerged or Drowned Weir	523
9.15. Time Required to empty a Reservoir or a Tank with Rectangular and Triangular Weirs or Notches	526
<i>Highlights</i>	528
<i>Objective Type Questions</i>	530
<i>Theoretical Questions</i>	532
<i>Unsolved Examples</i>	533

10. LAMINAR FLOW

534—604

10.1. Introduction	534
10.2. Reynolds Experiment	535
10.3. Navier-Stokes Equations of Motion	537
10.4. Relationship between Shear Stress and Pressure Gradient	540
10.5. Flow of Viscous Fluid in Circular Pipes—Hagen Poiseuille Law	541
10.6. Flow of Viscous Fluid through an Annulus	567
10.7. Flow of Viscous Fluid Between Two Parallel Plates	570
10.7.1. One plate moving and other at rest—couette flow	570
10.7.2. Both plates at rest	572
10.7.3. Both plates moving in opposite directions	572
10.8. Laminar Flow through Porous Media	582

10.9.	Power Absorbed in Bearings	583
10.9.1.	Journal bearing	583
10.9.2.	Foot-step bearing	585
10.9.3.	Collar bearing	586
10.10.	Loss of Head due to Friction in Viscous flow	587
10.11.	Movement of Piston in Dashpot	589
10.12.	Measurement of Viscosity	591
10.12.1.	Rotating cylinder method	591
10.12.2.	Falling sphere method	594
10.12.3.	Capillary tube method	595
10.12.4.	Efflux Viscometers	597
	<i>Highlights</i>	597
	<i>Objective Type Questions</i>	601
	<i>Theoretical Questions</i>	602
	<i>Unsolved Examples</i>	602
11.	TURBULENT FLOW IN PIPES	605—637
11.1.	Introduction	605
11.2.	Loss of Head due to Friction in Pipe Flow—Darcy Equation	606
11.3.	Characteristics of Turbulent Flow	608
11.4.	Shear Stresses in Turbulent Flow	609
11.4.1.	Boussinesq's theory	609
11.4.2.	Reynolds theory	610
11.4.3.	Prandtl's mixing length theory	610
11.5.	Universal Velocity Distribution Equation	610
11.6.	Hydrodynamically Smooth and Rough Boundaries	612
11.6.1.	Velocity distribution for turbulent flow in smooth pipes	613
11.6.2.	Velocity distribution for turbulent flow in rough pipes	615
11.7.	Common Equation for Velocity Distribution for both Smooth and Rough Pipes	618
11.8.	Velocity Distribution for Turbulent Flow in Smooth Pipes by Power Law	620
11.9.	Resistance to Flow of Fluid in Smooth and Rough Pipes	621
	<i>Highlights</i>	633
	<i>Objective Type Questions</i>	635
	<i>Theoretical Questions</i>	636
	<i>Unsolved Examples</i>	637
12.	FLOW THROUGH PIPES	638—724
12.1.	Introduction	638
12.2.	Loss of Energy (or Head) in Pipes	638
12.3.	Major Energy Losses	638

12·3·1. Darcy-weisbach formula	639
12·3·2. Chezy's formula for loss of head due to friction	639
12.4. Minor Energy Losses	645
12·4·1. Loss of head due to sudden enlargement	645
12·4·2. Loss of head due to sudden contraction	652
12·4·3. Loss of head due to obstruction in pipe	656
12·4·4. Loss of head at the entrance to pipe	657
12·4·5. Loss of head at the exit of a pipe	657
12·4·6. Loss of head due to bend in the pipe	657
12·4·7. Loss of head in various Pipe Fittings	657
12.5. Hydraulic Gradient and Total Energy Lines	657
12.6. Pipes in Series or Compound Pipes	668
12.7. Equivalent Pipe	671
12.8. Pipes in Parallel	674
12.9. Syphon	699
12.10. Power Transmission through Pipes	703
12.11. Flow through Nozzle at the End of a Pipe	706
12·11·1. Power transmitted through the nozzle	707
12·11·2. Condition for transmission of maximum power through nozzle	707
12·11·3. Diameter of the nozzle for transmitting maximum power	708
12.12. Water Hammer in Pipes	711
12·12·1. Gradual closure of valve	711
12·12·2. Instantaneous closure of valve in rigid pipes	712
12·12·3. Instantaneous closure of valve in elastic pipes	713
12·12·4. Time required by pressure wave to travel from the valve to the tank and from tank to valve	714
<i>Highlights</i>	716
<i>Objective Type Questions</i>	719
<i>Theoretical Questions</i>	721
<i>Unsolved Examples</i>	721
13. BOUNDARY LAYER THEORY	725—784
13.1. Introduction	725
13.2. Boundary Layer Definitions and Characteristics	726
13.2.1. Boundary layer thickness (δ)	727
13.2.2. Displacement thickness (δ^*)	727
13.2.3. Momentum thickness (θ)	728
13.2.4. Energy thickness (δ_e)	729
13.3. Momentum Equation for Boundary Layer by Von Karman	739
13.4. Laminar Boundary Layer	742
13.5. Turbulent Boundary Layer	766
13.6. Total Drag due to Laminar and Turbulent Layers	769
13.7. Boundary Layer Separation and its Control	774

<i>Highlights</i>	778
<i>Objective Type Questions</i>	780
<i>Theoretical Questions</i>	782
<i>Unsolved Examples</i>	782

14. FLOW AROUND SUBMERGED BODIES—DRAG AND LIFT **785—824**

14.1. Introduction	785
14.2. Force Exerted by a Flowing Fluid on a Body	785
14.3. Expressions for Drag and Lift	786
14.4. Dimensional Analysis of Drag and Lift	788
14.5. Streamlined and Bluff Bodies	798
14.6. Drag on a Sphere	798
14.6.1. Terminal velocity of a body	799
14.6.2. Applications of stokes' law	800
14.7. Drag on a Cylinder	804
14.8. Circulation and Lift on a Circular Cylinder	804
14.8.1. Flow patterns and development of lift	804
14.8.2. Position of stagnation points	806
14.8.3. Pressure at any point on the cylinder surface	807
14.8.4. Expression for lift on cylinder (kutta- joukowski theorem)	807
14.8.5. Expression for lift coefficient for rotating cylinder	809
14.8.6. Magnus effect	810
14.9. Lift on an Airfoil	815
<i>Highlights</i>	818
<i>Objective Type Questions</i>	820
<i>Theoretical Questions</i>	822
<i>Unsolved Examples</i>	823

15. COMPRESSIBLE FLOW **825—879**

15.1. Introduction	825
15.2. Basic Thermodynamic Relations	825
15.2.1. The characteristics equation of state	825
15.2.2. Specific heats	826
15.2.3. Internal energy	826
15.2.4. Enthalpy	827
15.2.5. Energy, work and heat	827
15.3. Basic Thermodynamic Processes	827
15.4. Basic Equations of Compressible Fluid Flow	829
15.4.1. Continuity equation	829
15.4.2. Momentum equation	829
15.4.3. Bernoulli's or energy equation	829
15.5. Propagation of Disturbances in Fluid and Velocity of Sound	837

15.5.1. Derivation of sonic velocity (velocity of sound)	837
15.5.2. Sonic velocity in terms of bulk modulus	838
15.5.3. Sonic velocity for isothermal process	839
15.5.4. Sonic velocity for adiabatic process	839
15.6. Mach Number	840
15.7. Propagation of Disturbance in Compressible Fluid	841
15.8. Stagnation Properties	844
15.8.1. Expression for stagnation pressure (p_s) in compressible flow	844
15.8.2. Expression for stagnation density (ρ_s)	846
15.8.3. Expression for stagnation temperature (T_s)	847
15.9. Area-velocity Relationship and Effect of Variation of Area for Subsonic, Sonic and Supersonic Flows	850
15.10. Flow of Compressible Fluid Through a Convergent Nozzle	852
15.11. Variables of Flow in terms of Mach Number	857
15.12. Flow Through Laval Nozzle (Convergent-Divergent Nozzle)	860
15.13. Shock Waves	865
15.13.1. Normal shock wave	866
15.13.2. Oblique shock wave	868
15.13.3. Shock strength	868
15.14. Measurement of Compressible Flow	870
15.15. Flow of Compressible Fluid Through Venturimeter	870
<i>Highlights</i>	873
<i>Objective Type Questions</i>	876
<i>Theoretical Questions</i>	878
<i>Unsolved Examples</i>	878

16. FLOW IN OPEN CHANNELS

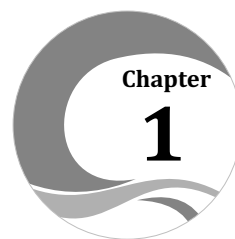
880—958

A. UNIFORM FLOW

16.1. Introduction	880
16.1.1. Definition of an open channel	880
16.1.2. Comparison between open channel and pipe flow	880
16.1.3. Types of channels	881
16.2. Types of Flow in Channels	881
16.2.1. Steady flow and unsteady flow	882
16.2.2. Uniform and non-uniform (or varied) flow	882
16.2.3. Laminar flow and turbulent flow	882
16.2.4. Subcritical flow, critical flow and supercritical flow	882
16.3. Definitions	883
16.4. Open Channel Formulae for Uniform Flow	884
16.4.1. Chezy's formula	884
16.5. Most Economical Section of a Channel	889

16.5.1. Most economical rectangular channel section	890
16.5.2. Most economical trapezoidal channel section	892
16.5.3. Most economical triangular channel section	908
16.5.4. Most economical circular channel section	910
16.6. Open Channel Section for Constant Velocity at all Depths of Flow	914
B. NON-UNIFORM FLOW	
16.7. Non-uniform Flow Through Open Channels	917
16.8. Specific Energy and Specific Energy Curve	917
16.9. Hydraulic Jump or Standing Wave	923
16.10. Gradually Varied Flow	938
16.10.1. Equation of gradually varied flow	938
16.10.2. Back water curve and afflux	940
16.11. Measurement of Flow of Irregular Channels	948
16.11.1. Area of flow	948
16.11.2. Mean velocity of flow	948
<i>Highlights</i>	951
<i>Objective Type Questions</i>	954
<i>Theoretical Questions</i>	956
<i>Unsolved Examples</i>	957
SHORT ANSWER QUESTIONS	959—967
GATE AND UPSC EXAMINATIONS' QUESTIONS WITH ANSWERS/SOLUTIONS (Latest-Selected)	968—1025
A. Conventional Questions with Answers/Solutions.	
B. Multiple-Choice Questions with Answers and <i>Explanations</i>.	
Index	i—iv

PART - I
FLUID MECHANICS



PROPERTIES OF FLUIDS

- 1.1. Introduction
- 1.2. Fluid
- 1.3. Liquids and their properties
- 1.4. Density-mass density-weight density-specific volume
- 1.5. Specific gravity
- 1.6. Viscosity-Newton's law of viscosity-types of fluids-effect of temperature on viscosity-effect of pressure on viscosity
- 1.7. Thermodynamic properties
- 1.8. Surface tension and capillarity
- 1.9. Compressibility and bulk modulus
- 1.10. Vapour pressure

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

1.1. INTRODUCTION

Hydraulics:

Hydraulics (this word has been derived from a Greek work 'Hudour' which means water) may be defined as follows :

“It is that branch of Engineering-science, which deals with water (at rest or in motion).”

or

“It is that branch of Engineering-science which is based on experimental observation of water flow.”

Fluid Mechanics:

Fluid mechanics may be defined as *that branch of Engineering-science which deals with the behaviour of fluid under the conditions of rest and motion.*

The fluid mechanics may be divided into three parts: *Statics, kinematics and dynamics.*

Statics. The study of incompressible fluids under static conditions is called *hydrostatics* and that dealing with the compressible static gases is termed as *aerostatics*.

Kinematics. It deals with the velocities, accelerations and the patterns of flow only. Forces or energy causing velocity and acceleration are *not* dealt under this heading.

Dynamics. It deals with the relations between velocities, accelerations of fluid *with the forces or energy causing them.*

Properties of Fluids-General Aspects:

The matter can be classified on the basis of the *spacing between the molecules* of the matter as follows:

1. Solid state, and
2. Fluid state,
 - (i) Liquid state, and
 - (ii) Gaseous state.

In *solids*, the molecules are very closely spaced whereas in *liquids* the spacing between the different molecules is relatively large and in *gases* the spacing between the molecules is still large. It means that inter-molecular cohesive forces are *large* in solids, *smaller* in liquids and *extremely small* in gases, and on account of this fact, solids possess compact and rigid form, liquid molecules can move freely within the liquid mass and the molecules of gases have greater freedom of movement so that the gases fill the container completely in which they are placed.

A *solid* can resist tensile, compressive and shear stresses upto a certain limit whereas a fluid has no tensile strength or very little of it and it can resist the compressive forces only when it is kept in a container. When a fluid is subjected to a shearing force it deforms continuously as long as the force is applied. The amount of shear stress in a fluid depends on the magnitude of the rate of deformation of the fluid element.

Liquids and *gases* exhibit different characteristics. The liquids under ordinary conditions are quite difficult to compress (and therefore they may for most purposes be regarded as incompressible) whereas gases can be compressed much readily under the action of external pressure (and when the external pressure is removed the gases tend to expand indefinitely).

1.2. FLUID

A fluid may be defined as follows:

“A fluid is a substance which is capable of flowing.”

or

“A fluid is a substance which deforms continuously when subjected to external shearing force.”

A fluid has the following *characteristics*:

1. It has no definite shape of its own, but conforms to the shape of the containing vessel.
2. Even a small amount of shear force exerted on a liquid/fluid will cause it to undergo a deformation which continues as long as the force continues to be applied.

A fluid may be *classified* as follows:

A. (i) *Liquid*, (ii) *Gas*, (iii) *Vapour*.

B. (i) *Ideal fluids* (ii) *Real fluids*.

Liquid

- A liquid is a fluid which possesses a *definite volume* (which varies only slightly with temperature and pressure).

- Liquids have bulk elastic modulus when under compression and will store up energy in the same manner as a solid. As the contraction of volume of a liquid under compression is extremely small, it is usually ignored and the *liquid is assumed to be incompressible*. A liquid will withstand a slight amount of tension due to molecular attraction between the particles which will cause an apparent shear resistance, between two adjacent layers. This phenomenon is known as **viscosity**.

- All known liquids vaporise at narrow pressures above zero, depending on the temperature.

Gas. It possesses *no definite volume* and is *compressible*.

Vapour. It is a gas whose temperature and pressure are such that it is very near the liquid state (e.g., steam).

Ideal fluids. An ideal fluid is one which has *no viscosity* and *surface tension* and is *incompressible*. In true sense no such fluid exists in nature. However fluids which have low viscosities such as water and air can be treated as ideal fluids under certain conditions. The assumption of ideal fluids helps in simplifying the mathematical analysis.

Real fluids. A real practical fluid is one which has viscosity, surface tension and compressibility in addition to the density. The real fluids are actually available in nature.

Continuum. A continuous and homogeneous medium is called **continuum**. From the continuum view point, the overall properties and behaviour of fluids can be studied without regard for its atomic and molecular structure.

1.3. LIQUIDS AND THEIR PROPERTIES

- Liquid can be easily distinguished from a solid or a gas.
- Solid has a definite shape.
- A liquid takes the shape of vessel into which it is poured.
- A gas completely fills the vessel which contains it.

The properties of water are of much importance because the subject of hydraulics is mainly concerned with it. Some important properties of water which will be considered are:

- | | | |
|------------------------|-------------------------|-----------------------|
| (i) Density, | (ii) Specific gravity, | (iii) Viscosity, |
| (iv) Vapour pressure, | (v) Cohesion, | (vi) Adhesion, |
| (vii) Surface tension, | (viii) Capillarity, and | (ix) Compressibility. |

1.4. DENSITY

1.4.1 Mass Density

The *density* (also known as *mass density* or *specific mass*) of a liquid may be defined as *the mass per unit volume* $\left(\frac{m}{V}\right)$ at a standard temperature and pressure. It is usually denoted by ρ (rho).

Its units are kg/m^3 , i.e., $\rho = \frac{m}{V}$... (1.1)

1.4.2 Weight Density

The weight density (also known as specific weight) is defined as the *weight per unit volume at the standard temperature and pressure*. It is usually denoted by w .

$$w = g \quad \dots (1.2)$$

For the purposes of all calculations, relating to Hydraulics and hydraulic machines, the specific weight of water is taken as follows:

- | | |
|------------------|--|
| In S.I. Units: | $w = 9.81 \text{ kN/m}^3$ (or $9.81 \times 10^{-6} \text{ N/mm}^3$) |
| In M.K.S. Units: | $w = 1000 \text{ kg}_f/\text{m}^3$ |

1.4.3 Specific volume

It is defined as *volume per unit mass of fluid*. It is denoted by v .

Mathematically, $v = \frac{V}{m} = \frac{1}{\rho}$... (1.3)

1.5. SPECIFIC GRAVITY

Specific gravity is the ratio of the specific weight of the liquid to the specific weight of a standard fluid. It is dimensionless and has no units. It is represented by S .

For liquids, the standard fluid is pure water at 4°C.

$$\therefore \text{Specific gravity} = \frac{\text{Specific weight of liquid}}{\text{Specific weight of pure water}} = \frac{w_{\text{liquid}}}{w_{\text{water}}}$$

Example 1.1. Calculate the specific weight, specific mass, specific volume and specific gravity of a liquid having a volume of 6 m³ and weight of 44 kN.

Solution: Volume of the liquid = 6 m³

Weight of the liquid = 44 kN

Specific weight, w :

$$w = \frac{\text{Weight of liquid}}{\text{Volume of liquid}} = \frac{44}{6} = 7.333 \text{ kN/m}^3 \text{ (Ans.)}$$

Specific mass or mass density, ρ :

$$\rho = \frac{w}{g} = \frac{7.333 \times 1000}{9.81} = 747.5 \text{ kg/m}^3 \text{ (Ans.)}$$

$$\text{Specific volume, } v = \frac{1}{\rho} = \frac{1}{747.5} = 0.00134 \text{ m}^3/\text{kg} \text{ (Ans.)}$$

Specific gravity, S :

$$S = \frac{w_{\text{liquid}}}{w_{\text{water}}} = \frac{7.333}{9.81} = 0.747 \text{ (Ans.)}$$

1.6. VISCOSITY

Viscosity may be defined as the *property of a fluid which determines its resistance to shearing stresses*. It is a measure of the internal fluid friction which causes resistance to flow. It is primarily *due to cohesion and molecular momentum exchange between fluid layers*, and as flow occurs, these effects appear as shearing stresses between the moving layers of fluid.

An *ideal fluid has no viscosity*. There is no fluid which can be classified as a perfectly ideal fluid. However, the fluids with very little viscosity are sometimes considered as ideal fluids.

Viscosity of fluids is due to *cohesion and interaction between particles*.

Refer to Fig 1.1. When two layers of fluid, at a distance ' dy ' apart, move one over the other at different velocities, say u and $u + du$, the viscosity together with relative velocity causes a shear stress acting between the fluid layers. The top layer causes a shear stress on the adjacent lower layer while the lower layer causes a shear stress on the adjacent top layer. This shear stress is proportional to the rate of change of velocity with respect to y . It is denoted by τ (called Tau).

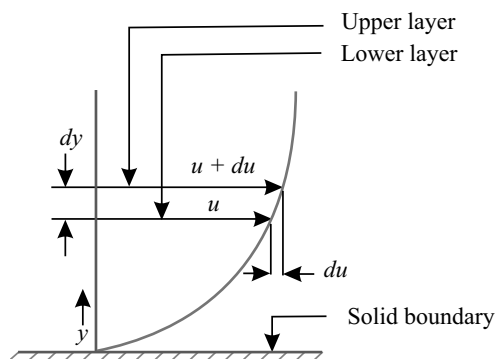


Fig. 1.1 Velocity variation near a solid boundary.

$$\text{Mathematically} \quad \tau \propto \frac{du}{dy}$$

$$\text{or} \quad \tau = \mu \cdot \frac{du}{dy} \quad \dots(1.4)$$

where, μ = Constant of proportionality and is known as *co-efficient of dynamic viscosity* or *only viscosity*.

$\frac{du}{dy}$ = Rate of shear stress or rate of shear deformation or velocity gradient.

From Fig. 1.1, we have
$$\mu = \frac{\tau}{\left[\frac{du}{dy}\right]} \quad \dots(1.5)$$

Thus viscosity may also be defined as the *shear stress required to produce unit rate of shear strain*.

Units of Viscosity:

In S.I. Units: N.s/m²

In M.K.S. Units: kg_f.sec/m²

$$\left[\because \mu = \frac{\text{force/area}}{(\text{length/time}) \times \frac{1}{\text{length}}} = \frac{\text{force/length}^2}{\frac{1}{\text{length}}} = \frac{\text{force} \times \text{time}}{(\text{length})^2} \right]$$

The unit of viscosity in C.G.S. is also called *poise* = $\frac{\text{dyne} - \text{sec}}{\text{cm}^2}$. One poise = $\frac{1}{10}$ N.s/m²

Note. The viscosity of water at 20°C is $\frac{1}{100}$ poise or one centipoise.

Kinematic Viscosity :

Kinematic viscosity is defined as the *ratio between the dynamic viscosity and density of fluid*. It is denoted by ν (called nu).

$$\text{Mathematically, } \nu = \frac{\text{Viscosity}}{\text{Density}} = \frac{\mu}{\rho} \quad \dots(1.6)$$

Units of kinematic viscosity:

In SI units: m²/s

In M.K.S. units: m²/sec.

In C.G.S. units the kinematic viscosity is also known as stoke (= cm²/sec.)

One stoke = 10⁻⁴ m²/s

Note: Centistoke means $\frac{1}{100}$ stoke.

1.6.1. Newton's Law of Viscosity

This law states that the *shear stress* (τ) on a fluid element layer is directly proportional to the *rate of shear strain*. The constant of proportionality is called the *co-efficient of viscosity*.

$$\text{Mathematically, } \tau = \mu \frac{du}{dy} \quad \dots(1.7)$$

The fluids which follow this law are known as *Newtonian fluids*.

1.6.2. Types of Fluids

The fluids may be of the following types:

Refer to Fig.1.2.

1. Newtonian fluids. These fluids follow Newton's viscosity equation (i.e. eqn. 1.7). For such fluids μ does not change with rate of deformation.

Examples. Water, kerosene, air etc.

2. Non-Newtonian fluids. Fluids which do not follow the linear relationship between shear stress and rate of deformation (given by eqn. 1.7) are termed as *Non-Newtonian fluids*. Such fluids are relatively uncommon.

Examples. Solutions or suspensions (slurries), mud flows, polymer solutions, blood etc. These fluids are generally complex mixtures and are studied under *rheology*, a science of deformation and flow.

3. Plastic fluids. In the case of a plastic substance which is non-Newtonian fluid an initial yield stress is to be exceeded to cause a continuous deformation. These substances are represented by straight line intersecting the vertical axis at the "yield stress" (Refer to Fig. 1.2).

An *ideal plastic* (or Bingham plastic) has a definite yield stress and a constant linear relation between shear stress and the rate of angular deformation. Examples: *Sewage sludge, drilling muds etc.*

A *thixotropic substance*, which is non-Newtonian fluid, has a non-linear relationship between the shear stress and the rate of angular deformation, beyond an initial yield stress. The *printer's ink* is an example of thixotropic substance.

4. Ideal fluid. An ideal fluid is one which is incompressible and has zero viscosity (or in other words shear stress is always zero regardless of the motion of the fluid). Thus an ideal fluid is represented by the horizontal axis ($\tau = 0$).

A *true elastic solid* may be represented by the vertical axis of the diagram.

Summary of relations between shear stress (τ) and rate of angular deformation for various types of fluids:

$$(i) \text{ Ideal fluids: } \tau = 0, \quad (ii) \text{ Newtonian fluids: } \tau = \mu \cdot \frac{du}{dy},$$

$$(iii) \text{ Ideal plastics: } \tau = \text{const.} + \mu \cdot \frac{du}{dy}, \quad (iv) \text{ Thixotropic fluids: } \tau = \text{const.} + \mu \left(\frac{du}{dy} \right)^n, \text{ and}$$

$$(v) \text{ Non-Newtonian fluids: } \tau = \left(\frac{du}{dy} \right)^n.$$

In case of non-Newtonian fluids, if n is less than unity then are called **pseudo-plastics** (e.g., *paper pulp, rubber suspension paints*) while fluids in which n is greater than unity are known as **dilatents**. (e.g., *Butter, printing ink*).

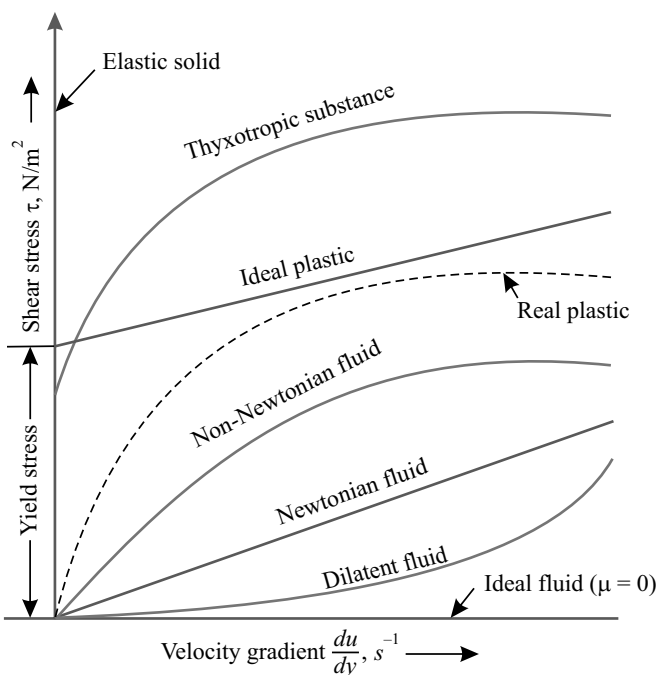


Fig. 1.2. Variation of shear stress with velocity gradient.

Ostwald-de-Waele Equation. It is an empirical solution to express steady-state shear stress as a function of velocity gradient, and is given as

$$\tau_{yx} = \alpha \left| \frac{du}{dy} \right|^{n-1} \frac{du}{dy}$$

If $n = 1$, this reduces to Newton's law of viscosity, with $\alpha = \mu$

Example 1.2. (a) What are the characteristics of an ideal fluid ?

(b) The general relation between shear stress and velocity gradient of a fluid can be written as

$$\tau = A \left(\frac{du}{dy} \right)^n + B$$

where A , B and n are constants that depend upon the type of fluid and conditions imposed on the flow. Comment on the value of these constants so that the fluid may behave as:

- (i) an ideal fluid,
 - (ii) a Newtonian fluid and
 - (iii) A non-Newtonian fluid.
- (c) Indicate whether the fluid with the following characteristics is a Newtonian or non-Newtonian.
- (i) $\tau = Ay + B$ and $u = C_1 + C_2y + C_3y^2$
 - (ii) $\tau = Ay^{n(n-1)}$ and $u = Cy^n$

Solution. (a) An ideal fluid has the following characteristics:

- No viscosity (i.e., $\mu = 0$)
- No surface tension.
- Incompressible (i.e., $\rho = \text{constant}$)

An ideal fluid can slip near a solid boundary and cannot sustain any shear force however small it may be.

(b) $\tau = A \left(\frac{du}{dy} \right)^n + B$

(i) An ideal fluid:

Since an ideal fluid has zero viscosity (i.e., shear stress is always zero regardless of the motion of the fluid), therefore.

$$A = B = 0$$

(ii) A Newtonian fluid:

Since a Newtonian fluid follows Newton's law of viscosity;

$$\tau = \mu \frac{du}{dy}, \text{ therefore:}$$

- $n = 1$ and $B = 0$
- The constant A takes the value of dynamic viscosity μ for the fluid.

Air, water, kerosene etc. behave as Newtonian fluids under normal working conditions.

(iii) A non-Newtonian fluid:

Depending on the value of power index n , the non-Newtonian fluids are classified as:

- If $n > 1$ and $B = 0$... **Dilatant fluids.**

Examples: Sugar solution, aqueous suspension and printing ink.

- If $n < 1$ and $B = 0$.. **Pseudo plastic fluids.**

Examples : *Blood, milk, liquid cement and clay.*

- If $n = 1$ and $B = \tau_0$ **Bingham fluid or ideal plastic.**

An ideal plastic fluid has a definite yield stress and a constant-linear relation between shear stress developed and rate of deformation:

$$i.e. \quad \tau = \tau_0 + \mu \frac{du}{dy}$$

Examples: *Sewage sludge, water suspension of clay and flyash, etc.*

$$(c) \quad (i) \quad \tau = Ay + B \text{ and } u = C_1 + C_2y + C_3y^2$$

$$\text{Now,} \quad \frac{du}{dy} = \frac{d}{dy} (C_1 + C_2y + C_3y^2) = C_2 + 2C_3y$$

$$\text{For Newtonian fluid,} \quad \tau = \mu \frac{du}{dy}$$

$$\therefore \quad \tau = \mu(C_2 + 2C_3y) = 2\mu C_3y + \mu C_2$$

which can be rewritten as

$$\tau = Ay + B \text{ where } A = 2\mu C_3 \text{ and } B = \mu C_2$$

Since this has the same form as the given shear stress, therefore the fluid characteristics correspond to that of an *ideal fluid*.

$$(ii) \quad \tau = Ay^{n(n-1)} \text{ and } u = Cy^n$$

$$\text{Now,} \quad \frac{du}{dy} = \frac{d}{dy} (Cy^n) = Cn(y)^{n-1}$$

$$\text{For a Newtonian fluid} \quad \tau = \mu \frac{du}{dy} = \mu Cn(y)^{n-1}$$

This expression does not conform to the value of shear stress and as such the fluid is *non-Newtonian* in character.

1.6.3. Effect of Temperature on Viscosity

Viscosity is effected by temperature. The viscosity of *liquids decreases* but that of *gases increases with increase in temperature*. This is due to the reason that in *liquids* the shear stress is due to the inter-molecular cohesion which *decreases* with increase of temperature. In *gases* the inter-molecular cohesion is negligible and the shear stress is due to exchange of momentum of the molecules, normal to the direction of motion. The molecular activity increases with rise in temperature and so does the viscosity of gas.

$$\text{For liquids:} \quad \mu_T = Ae^{\beta/T} \quad \dots(1.8)$$

$$\text{For gases:} \quad \mu_T = \frac{bT^{1/2}}{1 + a/T} \quad \dots(1.9)$$

where,

μ_T = Dynamic viscosity at absolute temperature T ,

A, β = Constants (for a given liquid), and

a, b = Constants (for a given gas).

1.6.4. Effect of Pressure on Viscosity

The viscosity under ordinary conditions is not appreciably affected by the changes in pressure. However, the viscosity of some oils has been found to increase with increase in pressure.

Example 1.3. *A plate 0.05 mm distant from a fixed plate moves at 1.2 m/s and requires a force of 2.2 N/m² to maintain this speed. Find the viscosity of the fluid between the plates.*

Solution: Velocity of the moving plate, $u = 1.2 \text{ m/s}$

Distance between the plates, $dy = 0.05 \text{ mm} = 0.05 \times 10^{-3} \text{ m}$

Force on the moving plate, $F = 2.2 \text{ N/m}^2$

Viscosity of the fluid, μ :

We know, $\tau = \mu \cdot \frac{du}{dy}$

where τ = shear stress or force per unit area = 2.2 N/m^2 ,

du = change of velocity

$= u - 0 = 1.2 \text{ m/s}$ and

dy = change of distance

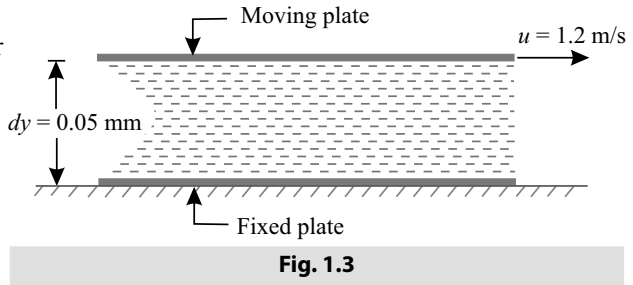
$= 0.05 \times 10^{-3} \text{ m}$.

$$\therefore 2.2 = \mu \times \frac{1.2}{0.05 \times 10^{-3}}$$

$$\text{or, } \mu = \frac{2.2 \times 0.05 \times 10^{-3}}{1.2} = 9.16 \times 10^{-5} \text{ N.s/m}^2$$

$$\left[\because 1 \text{ poise} = \frac{1}{10} \frac{\text{N.s}}{\text{m}^2} \right]$$

$$= 9.16 \times 10^{-4} \text{ poise (Ans.)}$$



Example 1.4. A plate having an area of 0.6 m^2 is sliding down the inclined plane at 30° to the horizontal with a velocity of 0.36 m/s . There is a cushion of fluid 1.8 mm thick between the plane and the plate. Find the viscosity of the fluid if the weight of the plate is 280 N .

Solution: Area of plate, $A = 0.6 \text{ m}^2$

Weight of plate, $W = 280 \text{ N}$

Velocity of plate, $u = 0.36 \text{ m/s}$

Thickness of film, $t = dy = 1.8 \text{ mm} = 1.8 \times 10^{-3} \text{ m}$

Viscosity of the fluid, μ :

Component of W along the plate $= W \sin \theta = 280 \sin 30^\circ = 140 \text{ N}$

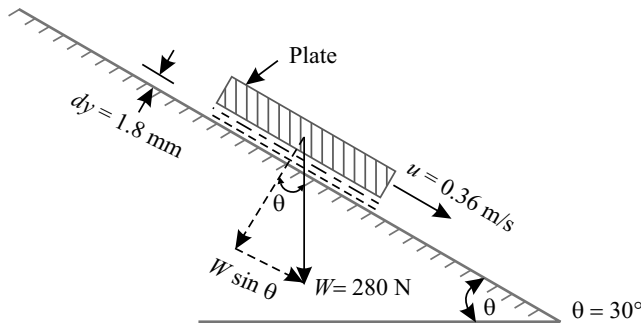


Fig. 1.4

$\therefore$ Shear force on the bottom surface of the plate, $F = 140 \text{ N}$ and shear stress,

$$\tau = \frac{F}{A} = \frac{140}{0.6} = 233.33 \text{ N/m}^2$$

We know,

$$\tau = \mu \cdot \frac{du}{dy}$$

Where,

$$du = \text{change of velocity} = u - 0 = 0.36 \text{ m/s}$$

$$dy = t = 1.8 \times 10^{-3} \text{ m}$$

$$\therefore 233.33 = \mu \times \frac{0.36}{1.8 \times 10^{-3}}$$

$$\text{or, } \mu = \frac{233.33 \times 1.8 \times 10^{-3}}{0.36} = 1.166 \text{ N.s/m}^2 = \mathbf{11.66 \text{ poise (Ans.)}}$$

Example 1.5. The space between two square flat parallel plates is filled with oil. Each side of the plate is 720 mm. The thickness of the oil film is 15 mm. The upper plate, which moves at 3 m/s requires a force of 120 N to maintain the speed. Determine:

(i) The dynamic viscosity of the oil;

(ii) The kinematic viscosity of oil if the specific gravity of oil is 0.95.

Solution. Each side of a square plate = 720 mm = 0.72 m

The thickness of the oil, $dy = 15 \text{ mm} = 0.015 \text{ m}$

Velocity of the upper plate = 3 m/s

$\therefore$ Change of velocity between plates, $du = 3 - 0 = 3 \text{ m/s}$

Force required on upper plate, $F = 120 \text{ N}$

$$\therefore \text{Shear stress, } \tau = \frac{\text{force}}{\text{area}} = \frac{120}{0.72 \times 0.72} = 231.5 \text{ N/m}^2$$

(i) **Dynamic viscosity, μ :**

We know that,

$$\tau = \mu \cdot \frac{du}{dy}$$

$$231.5 = \mu \cdot \frac{3}{0.015}$$

$$\therefore \mu = \frac{231.5 \times 0.015}{3} = \mathbf{1.16 \text{ N.s/m}^2 \text{ (Ans.)}}$$

(ii) **Kinematic viscosity, ν :**

Weight density of oil, $w = 0.95 \times 9.81 \text{ kN/m}^2 = 9.32 \text{ kN/m}^2 = \text{or } 9320 \text{ N/m}^3$

$$\text{Mass density of oil, } \rho = \frac{w}{g} = \frac{9320}{9.81} = 950$$

$$\text{Using the relation: } \nu = \frac{\mu}{\rho} = \frac{1.16}{950} = 0.00122 \text{ m}^2/\text{s}$$

$$\text{Hence } \nu = \mathbf{0.00122 \text{ m}^2/\text{s} \text{ (Ans.)}}$$

Example 1.6. The velocity distribution for flow over a plate is given by $u = 2y - y^2$ where u is the velocity in m/s at a distance y metres above the plate. Determine the velocity gradient and shear stress at the boundary and 1.5 m from it.

Take dynamic viscosity of fluid as 0.9 N.s/m^2 .

$$\text{Soluton. } u = 2y - y^2 \text{ ...(given)} \quad \therefore \frac{du}{dy} = 2 - 2y$$

(i) **Velocity gradient, $\frac{du}{dy}$:**

$$\text{At the boundary : At } y = 0, \left(\frac{du}{dy} \right)_{y=0} = \mathbf{2 \text{ s}^{-1} \text{ (Ans.)}}$$

At 0.15 m from the boundary:

$$\text{At } y = 0.15 \text{ m, } \left(\frac{du}{dy} \right)_{y=0.15} = 2 - 2 \times 0.15 = \mathbf{1.7 \text{ s}^{-1} \text{ (Ans.)}}$$

(ii) Shear stress, τ :

$$(\tau)_{y=0} = \mu \cdot \left(\frac{du}{dy} \right)_{y=0} = 0.9 \times 2 = \mathbf{1.8 \text{ N/m}^2 \text{ (Ans.)}}$$

$$\text{and, } (\tau)_{y=0.15} = \mu \left(\frac{du}{dy} \right)_{y=0.15} = 0.9 \times 1.7 = \mathbf{1.53 \text{ N/m}^2 \text{ (Ans.)}}$$

[Where $\mu = 0.9 \text{ N.s/m}^2 \dots$ (given)]

Example 1.7. A lubricating oil of viscosity μ undergoes steady shear between a fixed lower plate and an upper plate moving at speed V . The clearance between the plates is t . Show that a linear velocity profile results if the fluid does not slip at either plate.

Solution. For the given geometry and motion, the shear stress τ is constant throughout. From Newton's law of viscosity, we have

$$\frac{du}{dy} = \frac{\tau}{\mu} = \text{constant}$$

$$\text{or } u = ly + m$$

The constants l and m are evaluated from the no slip conditions at the upper and lower plates.

$$\text{At } y = 0, u = 0 \quad \therefore m = 0$$

$$\text{At } y = t, u = V$$

$$\therefore V = lt + 0 \text{ or } l = \frac{V}{t}$$

$\therefore$ The velocity profile between plates is then given by:

$$u = \frac{Vy}{t} \text{ and is linear as indicated in Fig 1.5 (Ans.)}$$

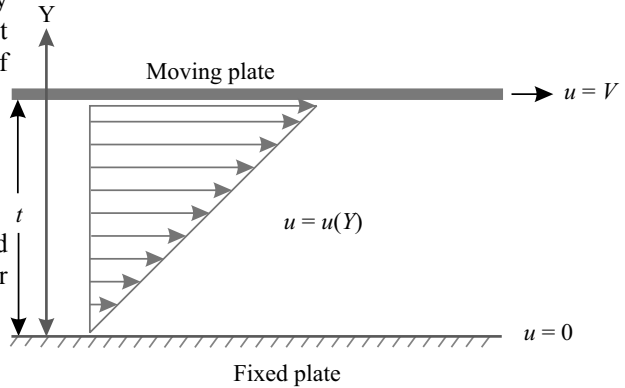


Fig. 1.5

Example 1.8. The velocity distribution of flow over a plate is parabolic with vertex 30 cm from the plate, where the velocity is 180 cm/s. If the viscosity of the fluid is 0.9 N.s/m^2 find the velocity gradients and shear stresses at distances of 0, 15 cm and 30 cm from the plate.

Solution. Distance of the vertex from the plate = 30 cm.

Velocity at vertex, $u = 180 \text{ cm/s}$

Viscosity of the fluid = 0.9 N.s/m^2

The equation of velocity profile, which is parabolic, is given by

$$u = ly^2 + my + n \quad \dots(1)$$

where l , m and n are constants. The values of these constants are found from the following boundary conditions:

$$(i) \text{ At } y = 0, u = 0,$$

$$(ii) \text{ At } y = 30 \text{ cm, } u = 180 \text{ cm/s and}$$

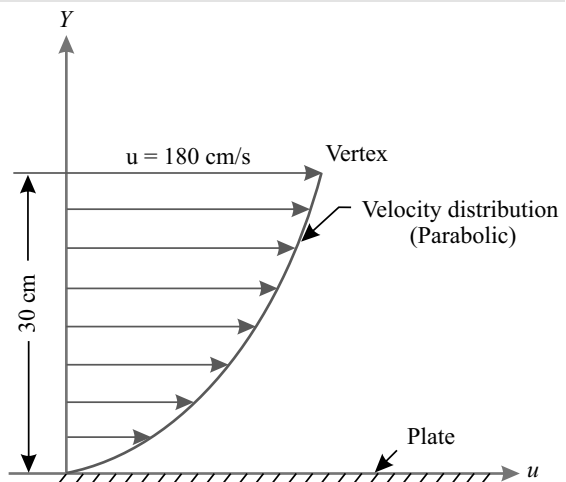


Fig. 1.6

(iii) At $y = 30$ cm, $\frac{du}{dy} = 0$.

Substituting boundary conditions (i) in eqn. (1), we get

$$0 = 0 + 0 + n \quad \therefore n = 0$$

Substituting boundary conditions (ii) in eqn. (1), we get

$$180 = l \times (30)^2 + m \times 30 \quad \text{or} \quad 180 = 900l + 30m \quad \dots(2)$$

Substituting boundary conditions (iii) in eqn. (1), we get

$$\frac{du}{dy} = 2ly + m \quad \therefore 0 = 2l \times 30 + m \quad \text{or} \quad 0 = 60l + m \quad \dots(3)$$

Solving eqns. (2) and (3), we have $l = -0.2$ and $m = 12$.

Substituting the values of l , m and n in eqn. (1), we get $u = -0.2y^2 + 12y$

Velocity gradients, $\frac{du}{dy}$:

$$\frac{du}{dy} = -0.2 \times 2y + 12 = -0.4y + 12$$

At $y = 0, \left(\frac{du}{dy}\right)_{y=0} = 12/\text{s (Ans.)}$

At $y = 15$ cm, $\left(\frac{du}{dy}\right)_{y=15} = -0.4 \times 15 + 12 = 6/\text{s (Ans.)}$

At $y = 30$ cm, $\left(\frac{du}{dy}\right)_{y=30} = -0.4 \times 30 + 12 = 0 \text{ (Ans.)}$

Shear stresses, τ :

We know, $\tau = \mu \frac{du}{dy}$

At $y = 0, (\tau)_{y=0} = \mu \cdot \left(\frac{du}{dy}\right)_{y=0} = 0.9 \times 12 = 10.8 \text{ N/m}^2 \text{ (Ans.)}$

At $y = 15, (\tau)_{y=15} = \mu \cdot \left(\frac{du}{dy}\right)_{y=15} = 0.9 \times 6 = 5.4 \text{ N/m}^2 \text{ (Ans.)}$

At $y = 30, (\tau)_{y=30} = \mu \cdot \left(\frac{du}{dy}\right)_{y=30} = 0.9 \times 0 = 0 \text{ (Ans.)}$

Example 1.9. A fluid has an absolute viscosity of 0.048 Pa-s and a specific gravity of 0.913. For flow of such a fluid over a solid flat surface, the velocity at a point 75 mm away from the surface is 1.125 m/s. Calculate the shear stresses at the solid boundary and also at points 25 mm, 50 mm and 75 mm away from the surface in normal direction, if the velocity distribution across the surface is (i) linear, (ii) parabolic with vertex at the point 75 mm away from the surface.

(UPTU)

Solution. (i) Linear velocity distribution:

If velocity distribution is linear, $\frac{du}{dy}$ is same at every point within the boundary layer and is

equal to $\frac{du}{dy} = \frac{1.125}{0.075}$ per s.

Shear stress for all the locations,

$$\tau = \mu \frac{du}{dy} = 0.048 \times \frac{1.125}{0.075} = 0.72 \text{ N/m (Ans.)}$$

(ii) Parabolic velocity distribution:

For parabolic velocity distribution, let the velocity profile be $u = ly^2 + my + n$ where the constants, l , m , and n are found from the boundary conditions.

At $y = 0$, $u = 0$, giving $n = 0$

At $y = 0.075 \text{ m}$, $u = 1.125 \text{ m/s}$, giving

$$1.125 = (0.075)^2 l + 0.075 m \quad \dots(i)$$

or $1.125 = 5.625 \times 10^{-3} l + 0.075 m$

At $y = 0.075 \text{ m}$, $\frac{du}{dy} = 0 = 2ly + m$

or $0 = 2l \times 0.075 + m$ or $m = -0.15 l \quad \dots(ii)$

Substituting (ii) in (i), we get

$$\begin{aligned} 1.125 &= 5.625 \times 10^{-3} l - 0.075 \times 0.15 l \\ &= l(5.625 \times 10^{-3} - 0.075 \times 0.15) = -0.005625 l \end{aligned}$$

$$\therefore l = -\frac{1.125}{0.005625} = -200$$

and from (ii), we have $m = 30$.

Hence the velocity distribution becomes $u = -200y^2 + 30y$, and $\frac{du}{dy} = 30 - 400y$

Hence the shear stresses at the required locations, y , are determined in the table below:

$y \text{ (m)}$	0	0.025	0.05	0.075
$\frac{du}{dy} \text{ (persecond)}$	30	20	10	0
Shere stress = $\mu \frac{du}{dy} \text{ N/m}^2$	1.44	0.96	0.48	0

(Ans.)

Example 1.10. A 400 mm diameter shaft is rotating at 200 r.p.m. in a bearing of length 120 mm. If the thickness of oil film is 1.5 mm and the dynamic viscosity of the oil is 0.7 N.s/m^2 , determine:

(i) Torque required to overcome friction in bearing;

(ii) Power utilised in overcoming viscous resistance.

Assume a linear velocity profile.

Solution. Diameter of the shaft, $d = 400 \text{ mm} = 0.4 \text{ m}$

Speed of the shaft, $N = 200 \text{ r.p.m.}$

Thickness of the oil film, $t = 1.5 \text{ mm} = 1.5 \times 10^{-3} \text{ m}$

Length of the bearing, $l = 120 \text{ mm} = 0.12 \text{ m}$

Viscosity, $\mu = 0.7 \text{ N.s/m}^2$

$$\text{Tangential velocity of the shaft, } u = \frac{\pi d N}{60} = \frac{\pi \times 0.4 \times 200}{60} = 4.19 \text{ m/s}$$

(i) Torque required to overcome friction, T :

We know, $\tau = \mu \cdot \frac{du}{dy}$

where du = change of velocity = $u - 0 = 4.19$ m/s

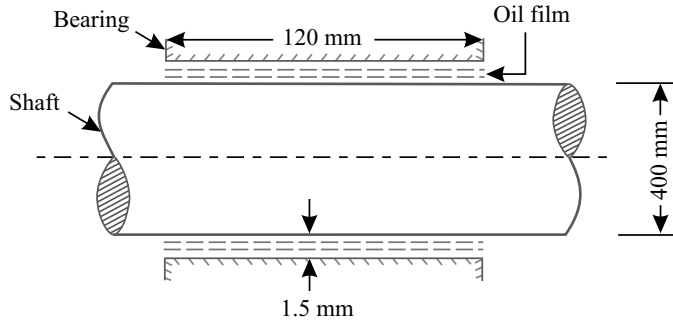


Fig. 1.7

$$dy = t = 1.5 \times 10^{-3} \text{ m}$$

$$\therefore \tau = 0.7 \times \frac{4.19}{1.5 \times 10^{-3}}$$

$$= 1955.3 \text{ N/m}^2.$$

$$\therefore \text{Shear force, } F = \text{shear stress} \times \text{area}$$

$$= \tau \cdot \pi dl$$

$$= 1955.3 \times \pi \times 0.4 \times 0.12$$

$$= 294.85 \text{ N}$$

$$\text{Hence, viscous torque} = F \times d/2 = 294.85 \times \frac{0.4}{2}$$

$$= 58.97 \text{ Nm (Ans.)}$$

(ii) Power utilised, P :

$$P = T \times \frac{2\pi N}{60} \text{ watts, where } T \text{ is in Nm}$$

$$P = 58.97 \times \frac{2\pi \times 200}{60} = 1235 \text{ W or } 1.235 \text{ kW (Ans.)}$$

Example 1.11. A 150 mm diameter shaft rotates at 1500 r.p.m. in a 200 mm long journal bearing with 150.5 mm internal diameter. The uniform annular space between the shaft and the bearing is filled with oil of dynamic viscosity 0.8 poise. Calculate the power dissipated as heat.

(Anna University)

Solution. Given: $d_{\text{shaft}} = 150$ mm; $d_{\text{bearing}} = 150.5$ mm; $l = 200$ mm = 0.2 m

$N = 1500$ r.p.m.; $\mu = 0.8$ poise = $0.8 \times 0.1 = 0.08$ Ns/m²

Power dissipated as heat:

$$\text{Radial thickness of the oil, } dy = \frac{(150.5 - 150)/2}{1000} \text{ m} = 0.00025 \text{ m}$$

$$\text{Tangential velocity of the shaft, } u = \frac{\pi d N}{60} = \frac{\pi \times (150 \times 10^{-3}) \times 1500}{60} = 11.78 \text{ m/s}$$

∴ Change of velocity,

$$du = u - 0 = 11.78 \text{ m/s}$$

Tangential stress in the oil layer,

$$\tau = \mu \cdot \frac{du}{dy}$$

$$\therefore \tau = 0.08 \times \frac{11.78}{0.00025} = 3769.6 \text{ N/m}^2$$

Power dissipated as heat = shear force $\times$ tangential velocity of this shaft

$$= [\tau \times (\pi dl)] \times u$$

$$= 769.6 \times \pi \times (150 \times 10^{-3}) \times 0.2 \times 11.78$$

$$= 4185 \text{ W or } \mathbf{4.185 \text{ kW (Ans.)}}$$

Example 1.12. A vertical cylinder of diameter 180 mm rotates concentrically inside another cylinder of diameter 181.2 mm. Both the cylinders are 300 mm high. The space between the cylinders is filled with a liquid whose viscosity is unknown. Determine the viscosity of the fluid if a torque of 20 Nm is required to rotate the inner cylinder at 120 r.p.m.

Solution. Given: Diameter of inner cylinder, $d = 180 \text{ mm} = 0.18 \text{ m}$

Diameter of outer cylinder, $D = 181.2 \text{ mm} = 0.1812 \text{ m}$

Length of each cylinder, $l = 300 \text{ mm} = 0.3 \text{ m}$

Speed of the inner cylinder, $N = 120 \text{ r.p.m.}$

Torque, $T = 20 \text{ Nm.}$

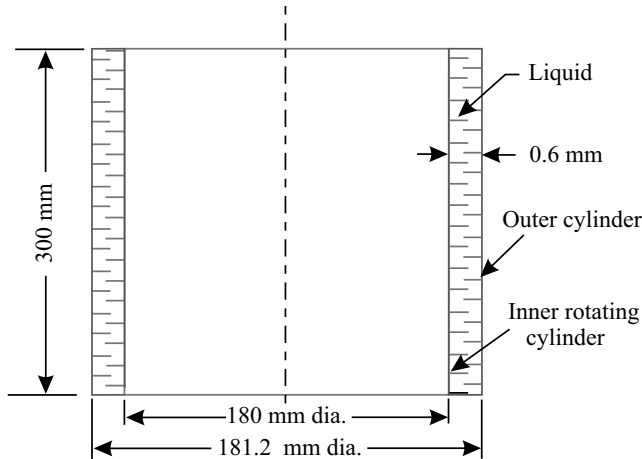


Fig. 1.8

Viscosity of the liquid, μ :

Tangential velocity of the inner cylinder

$$u = \frac{\pi d N}{60} = \frac{\pi \times 0.18 \times 120}{60} = 1.13 \text{ m/s}$$

Surface area of the inner cylinder,

$$\begin{aligned} A &= \pi dl = \pi \times 0.18 \times 0.3 \\ &= 0.1696 \text{ m}^2 \end{aligned}$$

Using the relation:

$$\tau = \mu \cdot \frac{du}{dy}$$

where,

$$\begin{aligned} du &= u - 0 = 1.13 - 0 \\ &= 1.13 \text{ m/s} \end{aligned}$$

and

$$dy = \frac{0.1812 - 0.180}{2} = 0.0006 \text{ m}$$

$$\tau = \mu \times \frac{1.13}{0.0006} = 1883.33\mu$$

$$\therefore \text{Shear force, } F = \tau \times A = 1883.33 \mu \times 0.1696 \text{ N}$$

$$\begin{aligned} \therefore \text{Torque, } T &= F \times F \times \frac{d}{2} \\ &= 1883.33 \mu \times 0.1696 \times \frac{0.18}{2} \end{aligned}$$

$$\text{or } 20 = 1883.33 \mu \times 0.1696 \times 0.09$$

$$\text{or } \mu = \frac{20}{1883.33 \times 0.1696 \times 0.09} = 0.696 \text{ N s/m}^2$$

$$\text{i.e., } \mu = \mathbf{6.96 \text{ poise (Ans.)}}$$

Example 1.13. A circular disc of diameter D is slowly rotated in a liquid of large viscosity (μ) at a small distance (h) from a fixed surface. Derive an expression of torque (T) necessary to maintain an angular velocity (ω). (M.U.)

Solution. The arrangement is shown in Fig. 1.9.

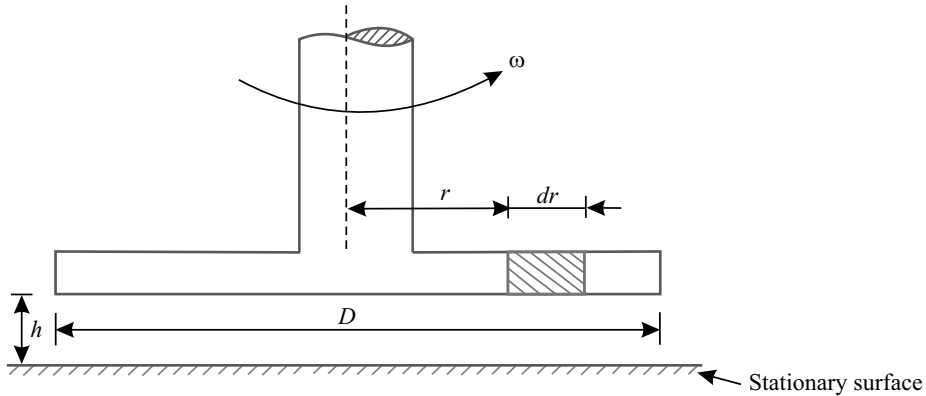


Fig. 1.9

Consider an elementary ring of disc at radius r and having a width dr . Linear velocity at this radius is ωr .

$$\text{Shear stress, } \tau = \mu \frac{du}{dy}$$

$$\begin{aligned} \text{Torque} &= \text{shear stress} \times \text{area} \times r \\ &= \tau \times 2\pi r \, dr \times r \\ &= \mu \frac{du}{dy} \times 2\pi r^2 \times dr \end{aligned}$$

Assuming the gap h to be small so that the velocity distribution may be assumed linear.

$$\frac{du}{dy} = \frac{\omega r}{h}$$

∴ Torque on the element

$$dT = \mu \frac{\omega r}{h} \times 2\pi r^2 \times dr = \frac{2\pi\mu\omega}{h} r^3 \times dr$$

∴ Total torque, $T = \int_0^{\pi/2} \frac{2\pi\mu\omega}{h} r^3 \times dr$

or
$$T = \frac{2\pi\mu\omega}{h} \left[\frac{r^4}{4} \right]_0^{D/2} = \frac{2\pi\mu\omega}{h} \cdot \frac{1}{4} \left(\frac{D}{2} \right)^4$$

or
$$T = \frac{\pi\mu\omega D^4}{32h}, \text{ which is the required expression. (Ans.)}$$

Example 1.14. A 120 mm disc rotates on a table separated by an oil film of 1.8 mm thickness. Find the viscosity of oil if the torque required to rotate the disc at 60 r.p.m is 3.6×10^{-4} Nm.

Assume the velocity gradient in the oil film to be linear.

Solution. Given: Diameter of the disc, $D = 120 \text{ mm} = 0.12 \text{ m}$

Thickness of oil film, $t = 1.8 \text{ mm} = 1.8 \times 10^{-3} \text{ m}$

Torque, $T = 3.6 \times 10^{-4} \text{ Nm}$

Speed of the disc, $N = 60 \text{ r.p.m.}$

∴ Angular speed of the disc, $\omega = \frac{2\pi N}{60} = \frac{2\pi \times 60}{60} = 2\pi \text{ rad/s}$

Viscosity, μ :

We know that when the velocity gradient is linear,

$$\frac{du}{dy} = \frac{u}{t}$$

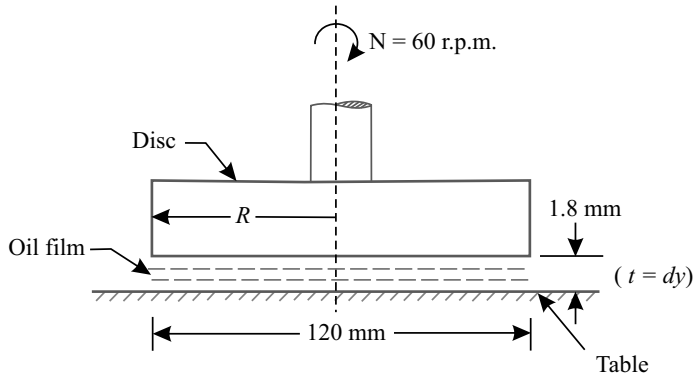


Fig. 1.10

Shearing stress, $\tau = \mu \cdot \frac{u}{t}$.

Shearing force = Shearing stress $\times$ Area

$$= \mu \cdot \frac{u}{t} \cdot 2\pi r dr \quad (\text{considering an element at radius } r \text{ and thickness } dr)$$

$$= \mu \cdot \frac{\omega r}{t} \cdot 2\pi r dr = \frac{2\pi\mu\omega r^2 \cdot dr}{t} \quad (\text{where } u = \omega r; \omega \text{ being the angular velocity})$$

$$\therefore \text{Viscous torque} = \text{Shearing force} \times r$$

$$= \frac{2\pi\mu\omega r^2 \cdot dr}{t} \cdot r = \frac{2\pi\mu\omega r^3 \cdot dr}{t}$$

$\therefore$ Total viscous torque,

$$T = \int_0^R \frac{2\pi\mu\omega r^3}{t} dr = \frac{2\pi\mu\omega}{t} \int_0^R r^3 dr = \frac{\pi\mu\omega R^4}{2t}$$

i.e.,
$$T = \frac{\pi\mu\omega R^4}{2t}$$

Substituting the values, we get:

$$3.6 \times 10^{-4} = \frac{\pi \times \mu \times 2\pi \times (0.12/2)^4}{2 \times 1.8 \times 10^{-3}}$$

or
$$\mu = \frac{3.6 \times 10^{-4} \times 2 \times 1.8 \times 10^{-3}}{\pi \times 2\pi \times (0.06)^4} = 0.00506 \text{ N.s/m}^2 = 0.0506 \text{ poise.}$$

Hence,
$$\mu = \mathbf{0.0506 \text{ poise (Ans.)}}$$

Example 1.15. A solid cone of maximum radius R and vertex angle 2θ is to rotate at angular velocity ω . An oil of viscosity μ and thickness t fills the gap between the cone and the housing. Derive an expression for the torque required and the rate of heat dissipation in the bearing.

Solution. Given: Maximum radius of the cone = R

Vertex angle = 2θ

Viscosity the oil = μ

Thickness of oil = t

Refer Fig. to 1.11.

Consider an elementary area dA at radius r of the cone.

$$dA = 2\pi r ds \times \frac{dr}{\sin \theta}$$

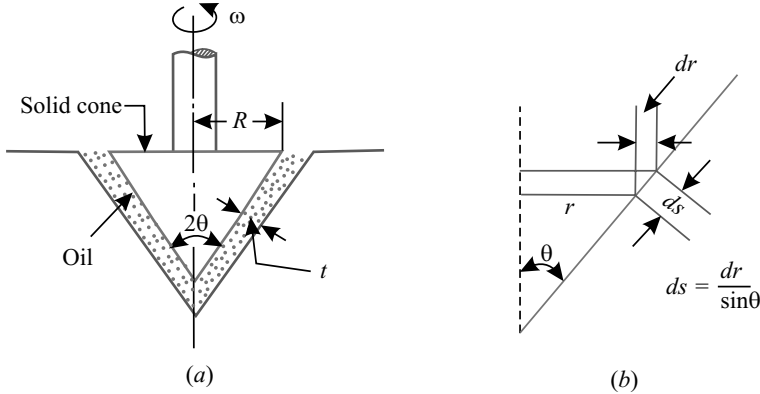


Fig. 1.11

$$\text{Shear stress } \tau = \mu \frac{du}{dy} = \mu \frac{u}{t}$$

Shear force = shear stress $\times$ area of the element

$$= \mu \frac{u}{t} \left(2\pi r \times \frac{dr}{\sin \theta} \right)$$

$$\text{Viscous torque on the element, } dT = \mu \frac{u}{t} \left(2\pi r \times \frac{dr}{\sin \theta} \right) \times r$$

Since the cone rotates with angular velocity ω rad/sec., the tangential velocity, $u = \omega r$

or,
$$dT = \mu \frac{\omega r}{t} \left(2\pi r \times \frac{dr}{\sin \theta} \right) \times r = \frac{2\pi\mu\omega}{t \sin \theta} r^3 dr$$

$\therefore$ Total torque, $T = \frac{2\pi\mu\omega}{t \sin \theta} \int_0^R r^3 dr$

i.e.,
$$T = \frac{2\pi\mu\omega}{t \sin \theta} \times \frac{R^4}{4} = \frac{\pi\mu\omega}{2t \sin \theta} R^4 \text{ (Ans.)}$$

Power utilised in overcoming the resistance (or rate of heat dissipation in the bearing),

$$P = T\omega = \left(\frac{\pi\mu\omega^2}{2t \sin \theta} R^4 \right) \text{ (Ans.)}$$

Example 1.16. Two large fixed parallel planes are 12 mm apart. The space between the surfaces is filled with oil of viscosity 0.972 N.s/m². A flat thin plate 0.25 m² area moves through the oil at a velocity of 0.3 m/s. Calculate the drag force:

- When the plate is equidistant from both the planes, and
- When the thin plate is at a distance of 4 mm from one of the plane surfaces.

Solution. Given: Distance between the fixed parallel planes = 12 mm = 0.012 m

Area of thin plate, $A = 0.25 \text{ m}^2$

Velocity of plate, $u = 0.3 \text{ m/s}$

Viscosity of oil = 0.972 N.s/m²

Drag force, F :

- When the plate is equidistant from both the planes:

Let, F_1 = Shear force on the upper side of the thin plate,

F_2 = Shear force on the lower side of the thin plate,

F = Total force required to drag the plate
($= F_1 + F_2$).

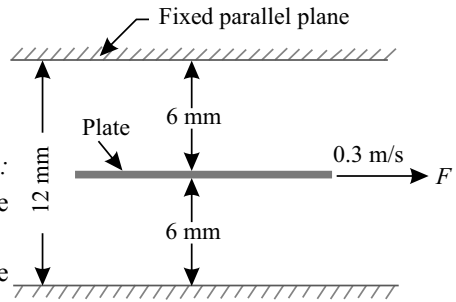


Fig. 1.12

The shear τ_1 , on the upper side of the thin plate is given by:

$$\tau_1 = \mu \cdot \left(\frac{du}{dy} \right)_1$$

where, $du = 0.3 \text{ m/s}$ (relative velocity between upper fixed plane and the plate), and $dy = 6 \text{ mm} = 0.006 \text{ m}$ (distance between the upper fixed plane and the plate)

(Thickness of the plate neglected).

$$\therefore \tau_1 = 0.972 \times \frac{0.3}{0.006} = 48.6 \text{ N/m}^2$$

$$\therefore \text{Shear force, } F_1 = \tau_1 \cdot A = 48.6 \times 0.25 = 12.15 \text{ N}$$

Similarly shear stress (τ_2) on the lower side of the thin plate is given by

$$\tau_2 = u \cdot \left(\frac{du}{dy} \right)_2 = 0.972 \times \frac{0.3}{0.06} = 48.6 \text{ N/m}^2$$

and $F_2 = \tau_2 \cdot A = 48.6 \times 0.25 = 12.15 \text{ N}$

$$\therefore F = F_1 + F_2 = 12.15 + 12.15 = \mathbf{24.30 \text{ N (Ans.)}}$$

- When the thin plate is at a distance of 40 mm from one of the plane surfaces: Refer to Fig. 1.13.

The shear force on the upper side of the thin plate,

$$F_1 = \tau_1 \cdot A = \mu \cdot \left(\frac{du}{dy} \right)_1 \times A$$

$$= 0.972 \times \frac{0.3}{0.008} \times 0.25 = 9.11 \text{ N}$$

The shear force on the lower side of the thin plate,

$$F_2 = \tau_2 \times A = \mu \cdot \left(\frac{du}{dy} \right)_2 \times A$$

$$= 0.972 \times \left(\frac{0.3}{0.004} \right) \times 0.25 = 18.22 \text{ N}$$

$$\therefore \text{Total force } F = F_1 + F_2 = 9.11 + 18.22 = \mathbf{27.33 \text{ N (Ans.)}}$$

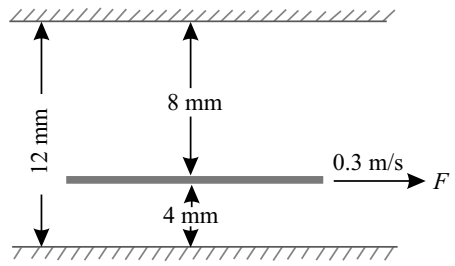


Fig. 1.13

Example 1.17. In the Fig. 1.14 is shown a central plate of area 6 m^2 being pulled with a force of 160 N . If the dynamic viscosities of the two oils are in the ratio of $1:3$ and the viscosity of top oil is 0.12 N.s/m^2 determine the velocity at which the central plate will move.

Solution: Area of the plate, $A = 6 \text{ m}^2$

Force applied to the plate, $F = 160 \text{ N}$

Viscosity of top oil, $\mu = 0.12 \text{ N.s/m}^2$

Velocity of the plate, u :

Let F_1 = Shear force in the upper side of thin (assumed) plate,

F_2 = Shear force on the lower side of the thin plate, and

F = Total force required to drag the plate
($= F_1 + F_2$)

Then, $F = F_1 + F_2 = \tau_1 \times A + \tau_2 \times A$

$$= \mu \left(\frac{\partial u}{\partial y} \right)_1 \times A + 3\mu \left(\frac{du}{dy} \right)_2 \times A$$

(where τ_1 and τ_2 are the shear stresses on the two sides of the plate)

$$160 = 0.12 \times \frac{u}{6 \times 10^{-3}} \times 6 + 3 \times 0.12 \times \frac{u}{6 \times 10^{-3}} \times 6$$

$$\text{or } 160 = 120u + 360u = 480u \quad \text{or } u = \frac{160}{480} = \mathbf{0.333 \text{ m/s (Ans.)}}$$

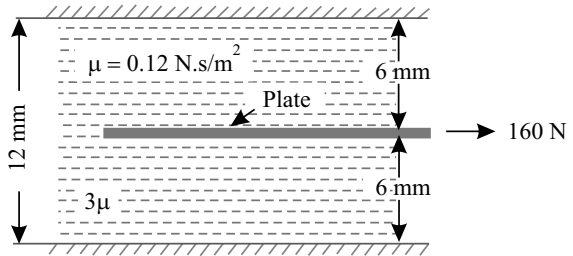


Fig. 1.14

Example 1.18. A metal plate $1.25 \text{ m} \times 1.25 \text{ m} \times 6 \text{ mm}$ thick and weighing 90 N is placed midway in the 24 mm gap between the two vertical plane surfaces as shown in the Fig. 1.15. The gap is filled with an oil of specific gravity 0.85 and dynamic viscosity 3.0 N.s/m^2 . Determine the force required to lift the plate with a constant velocity of 0.15 m/s .

Solution. Given: Dimensions of the plate = $1.25 \text{ m} \times 1.25 \text{ m} \times 6 \text{ mm}$

$\therefore$ Area of the plate, $A = 1.25 \times 1.25 = 1.5625 \text{ m}^2$

Thickness of the plate = 6 mm

$$\therefore t_1 = t_2 = \frac{24 - 6}{2} = 9 \text{ mm}$$

(Since the plate is situated midway in the gap)

Specific gravity of oil = 0.85

Dynamic viscosity of oil = 3 N.s/m²

Velocity of the plate = 0.15 m/s

Weight of the plate = 90 N

Force required to lift the plate:

Drag force (or viscous resistance) against the motion of the plate,

$$F = \tau_1 \cdot A + \tau_2 \cdot A$$

(where τ_1 and τ_2 are the shear stresses on two sides of the plate)

$$= \mu \cdot \left(\frac{du}{dy} \right)_1 \times A + \mu \left(\frac{du}{dy} \right)_2 \times A$$

$$= \mu \cdot \frac{u}{t_1} \times A + \mu \cdot \frac{u}{t_2} \times A$$

$$= \mu A u \cdot \left(\frac{1}{t_1} + \frac{1}{t_2} \right)$$

or
$$F = 3 \times 1.5625 \times 0.15 \left(\frac{1}{9 \times 10^{-3}} + \frac{1}{9 \times 10^{-3}} \right)$$

$$= 3 \times 1.5625 \times 0.15 \times \frac{2}{9 \times 10^{-3}} = 156.25 \text{ N}$$

Upward thrust or buoyant force on the plate = specific weight $\times$ volume of oil displaced

$$= 0.85 \times 9810 \times (1.25 \times 1.25 \times 0.006) = 78.17 \text{ N}$$

Effective weight of the plate = 90 – 78.17 = 11.83 N

$\therefore$ Total force required to lift the plate at velocity of 0.15 m/s = F + effective weight of the plate

$$= 156.25 + 11.83 = \mathbf{168.08 \text{ N (Ans.)}}$$

Example 1.19. A square metal plate 1.8 m side and 1.8 mm thick weighing 60 N is to be lifted through a vertical gap of 30 mm of infinite extent. The oil in the gap has a specific gravity of 0.95 and viscosity of 3 N.s/m². If the metal plate is to be lifted at a constant speed of 0.12 m/s, find the force and power required.

Solution. Area of metal plate, $A = 1.8 \times 1.8 = 3.24 \text{ m}^2$

$$\text{Thickness of the oil film, } t = dy = \frac{30 - 1.8}{2 \times 1000} = 0.0141$$

Speed of the metal plate, $u = 0.12 \text{ m/s}$.

Change of speed,

$$du = 0.12 - 0 = 0.12 \text{ m/s}$$

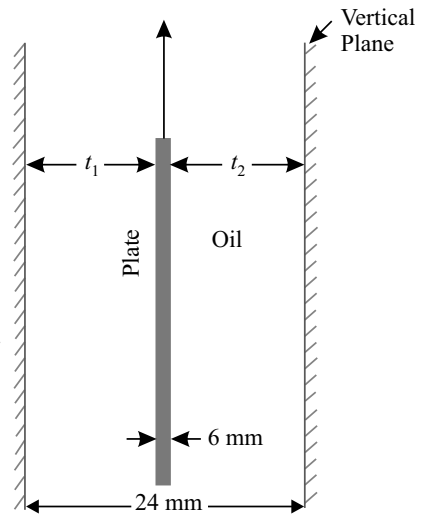


Fig. 1.15

Viscosity, $\mu = 3 \text{ N.s/m}^2$

We know, shear stress,

$$\tau = \mu \cdot \frac{du}{dy}$$

$$\therefore \tau = 3 \times \frac{0.12}{0.0141} = 25.53 \text{ N/m}^2$$

Force required, F :

$$F = W + 2(\tau \cdot A)$$

[where W = weight of the plate

$$= 60 \text{ N (given)}]$$

$$= 60 + 2 \times 25.53 \times 3.24 = 225.4 \text{ N}$$

Hence $F = 225.4 \text{ N (Ans.)}$

Power required, P :

$$P = F \times u = 225.4 \times 0.12 = 27.05 \text{ W}$$

Hence $P = 27.05 \text{ (Ans.)}$

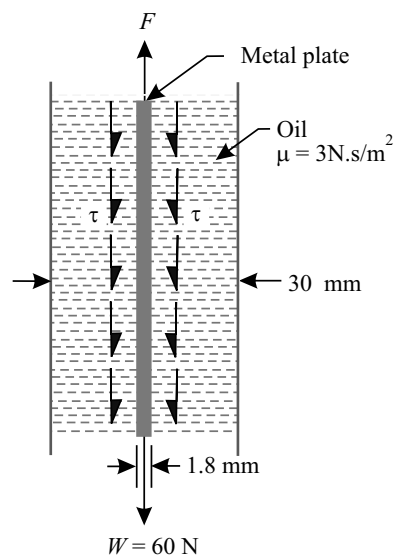


Fig. 1.16

Example 1.20. A thin plate of very large area is placed in a gap of height h with oils of viscosities μ' and μ'' on the two sides of the plate. The plate is pulled at a constant velocity V . Calculate the position of plate so that :

(i) The shear force on the two sides of the plate is equal

(ii) The force required to drag the plate is minimum.

Assume viscous flow and neglect all end effects.

Solution. Given : Height of the gap = h

Viscosities of oils = μ' and μ''

Velocity of the plate = V

Position of the plate, y :

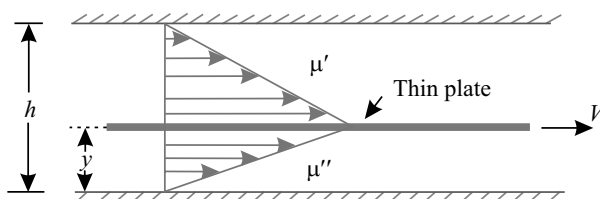


Fig. 1.17

Let y = The distance of the thin plate from one of the surfaces of the gap.

Force on the upper side of the plate,

$$F_{upper} = \mu \frac{du}{dy} = \mu' \times \frac{V}{(h-y)} A$$

Force on the lower side of the plate, $F_{lower} = \mu'' \times \frac{V}{y} A$

(i) Since the forces on the two sides of the plate are equal (given) we have,

$$i.e., F_{upper} = F_{lower}$$

$$\therefore \mu' \times \frac{V}{(h-y)} A = \mu'' \times \frac{V}{y} A$$

$$\text{or,} \quad \frac{\mu'}{h-y} = \frac{\mu''}{y} \quad \text{or} \quad \mu' y = \mu'' (h-y)$$

$$\therefore y = \frac{\mu'' h}{\mu' + \mu''} \quad (\text{Ans.})$$

(ii) Total drag force = sum of the forces on the upper and lower surfaces of the plate.

$$i.e., \quad F = F_{upper} + F_{lower}$$

$$\text{or,} \quad F = \mu' \times \frac{V}{h-y} \times A + \mu'' \times \frac{V}{y} A$$

For the drag force to be *minimum* $\frac{dF}{dy} = 0$

$$i.e., \quad \frac{d}{dy} \left[\mu' \times \frac{V}{h-y} \times A + \mu'' \times \frac{V}{y} A \right] = 0$$

$$\text{or,} \quad \frac{\mu' VA}{(h-y)^2} - \frac{\mu'' VA}{y^2} = 0$$

$$\text{or,} \quad \frac{\mu'}{\mu''} = \frac{(h-y)^2}{y^2} = \frac{h^2 + y^2 - 2hy}{y^2} = \frac{h^2}{y^2} + 1 - \frac{2h}{y}$$

$$\therefore \quad = \frac{h^2}{y^2} - \frac{2h}{y} + \left(1 + \frac{\mu'}{\mu''} \right)$$

$$\text{or,} \quad \frac{h}{y} = \frac{2 \pm \sqrt{4 - 4(1 - \mu'/\mu'')}}{2} = 1 \pm \sqrt{(\mu'/\mu'')}$$

Since, $\frac{h}{y}$ cannot be less than unity, therefore

$$\frac{h}{y} = 1 + \sqrt{\mu'/\mu''} \quad \text{or} \quad y = \frac{h}{1 + \sqrt{\mu'/\mu''}} \quad (\text{Ans.})$$

1.7. THERMODYNAMIC PROPERTIES

The thermodynamic properties need to be considered when a fluid is influenced by change of temperature. The following equation, known as the *characteristic equation of a state of a perfect gas*, is used for this purpose.

$$pV = mRT \quad \dots(1.10)$$

where, p = Absolute pressure, m = Mass of gas,
 V = Volume of m kg of gas, R = Characteristic gas constant, and
 T = Absolute temperature.

The characteristic equation in *another form*, can be derived by using *kilogram-mole as a unit*. The *kilogram-mole* is defined as a quantity of a gas equivalent to M kg of the gas, where M is the molecular weight of the gas (*i.e.*, since the molecular weight of oxygen is 32, then 1 kg mole of oxygen is equivalent to 32 kg of oxygen).

As per definition of the kilogram-mole, for m kg of a gas, we have:

$$m = nM \quad \dots(1.11)$$

where, n = No. of moles.

Note. Since the standard of mass is the kg, kilogram-mole will be written simply as mole.

Substituting for m from eqn. 1.11 in Eqn. 1.10 gives:

$$pV = nMRT \quad \text{or} \quad MR = \frac{pV}{nT}$$

According to Avogadro's hypothesis the volume of 1 mole of any gas is the same as the volume of 1 mole of any other gas, when the gases are at same temperature and pressure. Therefore, $\frac{V}{n}$ is the same for all gases at the same value of p and T . That is the quantity $\frac{pV}{nT}$ is a constant for all gases. This constant is called '*universal gas constant*', and is given the symbol, R_0 .

$$\text{i.e.,} \quad MR = R_0 = \frac{pV}{nT} \quad \text{or} \quad pV = nR_0T \quad \dots(1.12)$$

$$\text{Since,} \quad MR = R_0, \text{ then } R = \frac{R_0}{M} \quad \dots(1.13)$$

It has been found experimentally that the volume of 1 mole of any perfect gas at 1 bar and 0°C is approximately 22.71 m^3 . Therefore from eqn. 1.12,

$$R_0 = \frac{pV}{nT} = \frac{1 \times 10^5 \times 22.71}{1 \times 273.15} = 8314.3 \text{ Nm/mole K}$$

Using eqn. 1.13, the gas constant for any gas can be found when the molecular weight is known.

Example. For oxygen which has a molecular weight of 32, the gas constant

$$R = \frac{R_0}{M} = \frac{8314}{32} = 259.8 \text{ Nm/kg K.}$$

If the value of R is known, the specific weight of any gas can be computed at any temperature. The density can be changed by changing temperature or pressure.

(i) When the change in the state of the fluid system is affected at *constant pressure* the process is known as **isobaric** or **constant pressure process**.

$$\text{Here } \frac{V}{T} = \text{constant; (Charle's law) or } \frac{v}{T} = \text{constant or } \frac{v}{T} = \frac{1}{\rho T} = \text{constant} \quad \dots(1.14)$$

(ii) When the change in the state of the fluid system is affected at *constant temperature* the process is known as **isothermal process**.

$$\text{Here } pv^\gamma = \text{constant; (Boyle's Law) or } pv = \frac{p}{\rho} = \text{constant} \quad \dots(1.15)$$

(iii) When no heat is transferred to or from the fluid during the change in the state of fluid system, the process is called **adiabatic process**.

$$\text{Here, } pv^\gamma = \text{constant or } \frac{p}{\rho^\gamma} = \text{constant} \quad \dots(1.16)$$

$$\text{where} \quad \gamma = \frac{c_p}{c_v},$$

c_p = Specific heat of gas at constant pressure, and

c_v = Specific heat of gas at constant volume.

γ depends upon the molecular structure of the gas.

Note. For details regarding compression and expansion of gases please refer to chapter on "*Compressible flow*."

Example 1.21. The pressure and temperature of carbon-dioxide in a vessel are 600 kN/m^2 abs. and 30°C respectively. Find its mass density, specific weight and specific volume.

Solution. Given: Pressure of $\text{CO}_2 = 600 \text{ kN/m}^2$ abs.

Temperature of $\text{CO}_2 = 30 + 273 = 303 \text{ K}$

Molecular weight of $\text{CO}_2 = 12 + 2 \times 16 = 44$

Universal gas constant, $R_0 = 8314.3 \text{ Nm/mole K}$

$\therefore$ Characteristic gas constant, $R = \frac{R_0}{M} = \frac{8314.3}{44} = 189 \text{ Nm/kg K}$

(i) Mass density, ρ :

We know, $pV = mRT \quad \therefore \quad \frac{m}{V} = \frac{p}{RT}$

or, $\rho = \frac{p}{RT} = \frac{600 \times 10^3}{189 \times 313} = 10.14 \text{ kg/m}^3$

i.e., $\rho = 10.14 \text{ kg/m}^3$ (Ans.)

(ii) Specific weight, w :

$w = \rho g = 10.14 \times 9.81 = 99.47 \text{ N/m}^3$ (Ans.)

(iii) Specific volume v :

$v = \frac{1}{\rho} = \frac{1}{10.14} = 0.0986 \text{ m}^3/\text{kg}$ (Ans.)

1.8. SURFACE TENSION AND CAPILLARITY

1.8.1. Surface Tension

Cohesion. Cohesion means intermolecular attraction between *molecules of the same liquid*. It enables a liquid to resist small amount of tensile stresses. Cohesion is a tendency of the liquid to remain as one *assemblage of particles*. “Surface tension” is due to cohesion between particles at the free surface.

Adhesion. Adhesion means attraction between the molecules of a liquid and the molecules of a solid boundary surface in contact with the liquid. This property enables a liquid to stick to another body.

Capillary action is due to both cohesion and adhesion.

Surface tension is caused by the force of cohesion at the free surface. A liquid molecule in the interior of the liquid mass is surrounded by other molecules all around and is in equilibrium. At the free surface of the liquid, there are no liquid molecules above the surface to balance the force of the molecules below it. Consequently, as shown in Fig. 1.18, there is a net inward force on the molecule. The force is normal to the liquid surface. At the free surface a thin layer of molecules is formed. This is because of this film that a thin small needle can float on the free surface (the layer acts as a membrane).

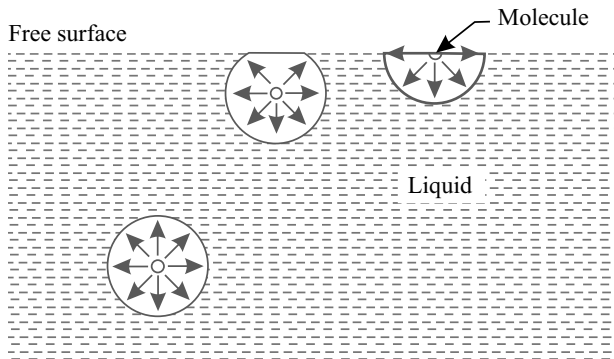


Fig. 1.18

Some important examples of phenomenon of surface tension are as follows:

- (i) Rain drops (A falling rain drop becomes spherical due to cohesion and surface tension).
- (ii) Rise of sap in a tree.
- (iii) Bird can drink water from ponds.
- (iv) Capillary rise and capillary siphoning.
- (v) Collection of dust particles on water surface.
- (vi) Break up of liquid jets.

Dimensional formula for surface tension:

The dimensional formula for surface tension is given by:

$$\left[\frac{E}{L} \right] \text{ or } \left[\frac{M}{T^2} \right]$$

It is usually expressed in N/m. The value of surface tension depends upon the following *factors*:

- (i) Nature of the liquid,
- (ii) Nature of the surrounding matter (*e.g.*, solid, liquid or gas), and
- (iii) Kinetic energy (and hence the temperature of the liquid molecules).

Note. As compared to pressure and gravitational forces surface tension forces are generally negligible but *become quite significant when there is a free surface and the boundary conditions are small as in the case of small scale models of hydraulic engineering structures.*

Surface tension of water and mercury when in contact with air:

Water-air	... 0.073 N/m at 20°C;	Water-air	... 0.058 N/m at 100°C;
Mercury-air	... 0.1 N/m length.		

1.8.1.1. Pressure Inside a Water Droplet, Soap Bubble and a Liquid Jet

Case I. Water droplet:

Let, p = Pressure inside the droplet above outside pressure (*i.e.*, $\Delta p = p - 0 = p$ above atmospheric pressure)

d = Diameter of the droplet and

σ = Surface tension of the liquid.

From free body diagram (Fig. 1.19 *d*), we have:

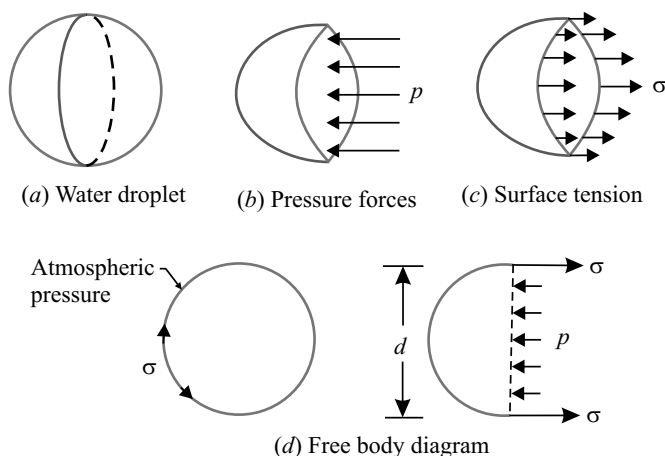


Fig. 1.19. Pressure inside a water droplet.

(i) Pressure force = $p \times \frac{\pi}{4} d^2$, and

(ii) Surface tension force acting around the circumference = $\sigma \times \pi d$.

Under equilibrium conditions these two forces will be equal and opposite,

$$\text{i.e.,} \quad p \times \frac{\pi}{4} d^2 = \sigma \times \pi d$$

$$\therefore \quad p = \frac{\sigma \times \pi d}{\frac{\pi}{4} d^2} = \frac{4\sigma}{d} \quad \dots(1.17)$$

Eqn. 1.17 shows that *with an increase in size of the droplet the pressure intensity decreases.*

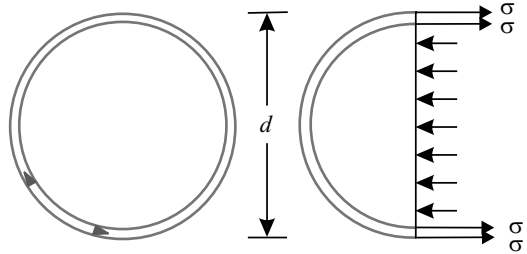
Case II. Soap (or hollow) bubble:

Soap bubbles have two surfaces on which surface tension σ acts.

From the free body diagram (Fig. 1.20), we have

$$p \times \frac{\pi}{4} d^2 = 2 \times (\sigma \times \pi d)$$

$$\therefore \quad p = \frac{2\sigma \times \pi d}{\frac{\pi}{4} d^2} = \frac{8\sigma}{d} \quad \dots(1.18)$$



Free body diagram

Fig. 1.20. Pressure inside a soap bubble.

Since the soap solution has a high value of surface tension σ , even with small pressure of blowing a soap bubble will tend to grow larger in diameter (hence formation of large soap bubbles).

Case III. A Liquid jet:

Let us consider a cylindrical liquid jet of diameter d and length l . Fig. 1.21 shows a semi-jet.

Pressure force = $p \times l \times d$

Surface tension force = $\sigma \times 2l$

Equating the two forces, we have:

$$p \times l \times d = \sigma \times 2l$$

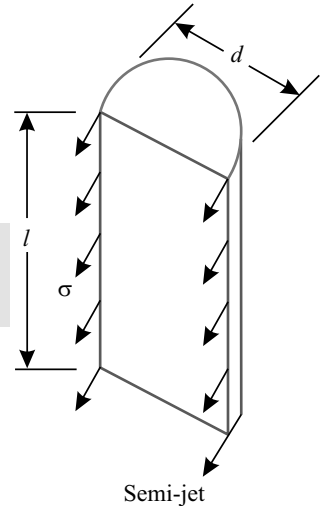
$$\therefore \quad p = \frac{\sigma \times 2l}{l \times d} = \frac{2\sigma}{d} \quad \dots(1.19)$$

Example 1.22. If the surface tension at air-water interface is 0.069 N/m, what is the pressure difference between inside and outside of an air bubble of diameter 0.009 mm?

Solution. Given: $\sigma = 0.069$ N/m; $d = 0.009$ mm

An air bubble has only one surface. Hence,

$$\begin{aligned} p &= \frac{4\sigma}{d} \\ &= \frac{4 \times 0.069}{0.009 \times 10^{-3}} = 30667 \text{ N/m}^2 \\ &= \mathbf{30.667 \text{ kN/m}^2 \text{ or kPa (Ans.)}} \end{aligned}$$



Semi-jet

Fig. 1.21. Forces on liquid jet.

Example 1.23. If the surface tension at the soap-air interface is 0.09 N/m, calculate the internal pressure in a soap bubble of 28 mm diameter.

Solution. Given: $\sigma = 0.09$ N/m; $d = 28$ mm.

In a soap bubble there are two interfaces. Hence,

$$p = \frac{8\sigma}{d} = \frac{8 \times 0.09}{28 \times 10^{-3}} \\ = 25.71 \text{ N/m}^2 \text{ (above atmospheric pressure) (Ans.)}$$

Example 1.24. In order to form a stream of bubbles, air is introduced through a nozzle into a tank of water at 20°C . If the process requires 3.0 mm diameter bubbles to be formed, by how much the air pressure at the nozzle must exceed that of the surrounding water?

What would be the absolute pressure inside the bubble if the surrounding water is at 100.3 kN/m^2 ?

Take surface tension of water at $20^\circ\text{C} = 0.0735 \text{ N/m}$.

Solution. Diameter of a bubble, $d = 3.0 \text{ mm} = 3 \times 10^{-3} \text{ m}$

Surface tension of water at 20°C , $\sigma = 0.0735 \text{ N/m}$

The excess pressure intensity of air over that of surrounding water, $\Delta p = p$.

We know,

$$p = \frac{4\sigma}{d} = \frac{4 \times 0.0735}{3 \times 10^{-3}} = 98 \text{ N/m}^2 \text{ (Ans.)}$$

Absolute pressure inside the bubble, p_{abs} :

$$p_{\text{abs}} = p + p_{\text{atm}} \\ = 98 \times 10^{-3} + 100.3 \\ = 0.098 + 100.3 = 100.398 \text{ kN/m}^2 \text{ (Ans.)}$$

Example 1.25. A soap bubble 62.5 mm diameter has an internal pressure in excess of the outside pressure of 20 N/m^2 . What is tension in the soap film?

Solution. Given: Diameter of the bubble, $d = 62.5 \text{ mm} = 62.5 \times 10^{-3} \text{ m}$;

Internal pressure in excess of the outside pressure, $p = 20 \text{ N/m}^2$.

Surface tension, σ :

Using the relation,

$$p = \frac{8\sigma}{d} \\ \text{i.e., } 20 = \frac{8\sigma}{62.5 \times 10^{-3}} \therefore \sigma = 20 \times \frac{62.5 \times 10^{-3}}{8} = 0.156 \text{ N/m (Ans.)}$$

Example 1.26. What do you mean by surface tension? If the pressure difference between the inside and outside of the air bubble of diameter 0.01 mm is 29.2 kPa , what will be the surface tension at air-water interface? (N.U.)

Solution. Surface tension is defined as the tensile force acting on the surface of a liquid in contact with a gas or on the surface between two immiscible liquids such that the contact surface behaves like a membrane under tension. The magnitude of this force per unit length of the free surface will have the same value as the surface energy per unit area. It is denoted by the letter σ and is expressed as N/m .

$$p \times \frac{\pi}{4} d^2 = \sigma (\pi d)$$

or

$$\sigma = p \times \frac{d}{4}$$

Substituting the values; $d = 0.01 \times 10^{-3} \text{ m}$; $p = 29.2 \times 10^3 \text{ Pa}$ (or N/m^2), we get

$$\sigma = 29.2 \times 10^3 \times \frac{0.01 \times 10^{-3}}{4} = 0.073 \text{ N/m (Ans.)}$$

1.8.2. Capillarity

Capillarity is a phenomenon by which a liquid (depending upon its specific gravity) rises into a thin glass tube above or below its general level. This phenomenon is due to the combined effect of cohesion and adhesion of liquid particles.

Fig. 1.22 shows the phenomenon of rising water in the tube of smaller diameters.

Let,

d = Diameter of the capillary tube,

θ = Angle of contact of the water surface,

σ = Surface tension force for unit length, and
 w = Weight density (ρg).

Now, upward surface tension force (lifting force) = weight of the water column in the tube (gravity force)

$$\pi d \cdot \sigma \cos \theta = \frac{\pi}{4} d^2 \times h \times w$$

$$\therefore h = \frac{4\sigma \cos \theta}{wd} \quad \dots(1.20)$$

For water and glass: $\theta \approx 0$.

Hence the capillary rise of water in the glass tube,

$$h = \frac{4\sigma}{wd} \quad \dots(1.21)$$

In case of mercury there is a capillary depression as shown in Fig. 1.23, and the angle of depression is $\theta \approx 140^\circ$. (It may be noted that here $\cos \theta = \cos 140^\circ = \cos (180 - 40^\circ) = -\cos 40^\circ$, therefore, h is *negative* indicating capillary depression).

Following points are worth noting:

(i) Smaller the diameter of the capillary tube, greater is the capillary rise or depression.

(ii) The measurement of liquid level in laboratory capillary (glass) tubes should not be smaller than 8 mm.

(iii) Capillary effects are negligible for tubes longer than 12 mm.

(iv) For *wetting liquid* (water): $\theta < \pi/2$. For water: $\theta = 0$ when pure water is in contact with clean glass. But θ becomes as high as 25° when water is slightly contaminated.

For *non-wetting liquid* (mercury): $\theta > \pi/2$.

(For mercury: θ varies between 130° to 150°)

Refer Fig. 1.24 which illustrates the liquid gas interface with a solid surface.

(v) The effects of surface tension are *negligible* in many flow problems *except those involving*.

- capillary rise;
- formation of drops and bubbles;
- the break up of liquid jets, and
- hydraulic model studies where the model or flow depth is small.

Capillary inversion. Due to surface tension the liquid passing out of an elliptical orifice tends to assume a circular or minimum perimeter cross-section. Here transformation of surface energy into

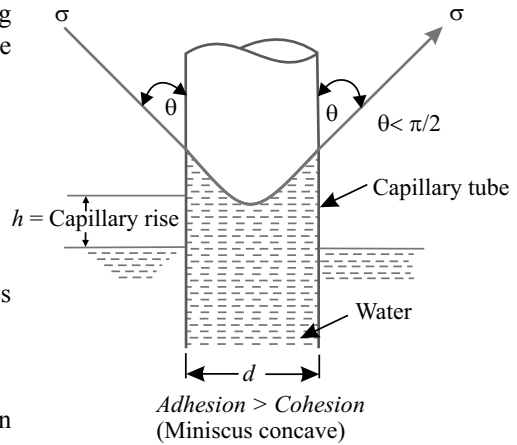


Fig. 1.22. Effect of capillarity.

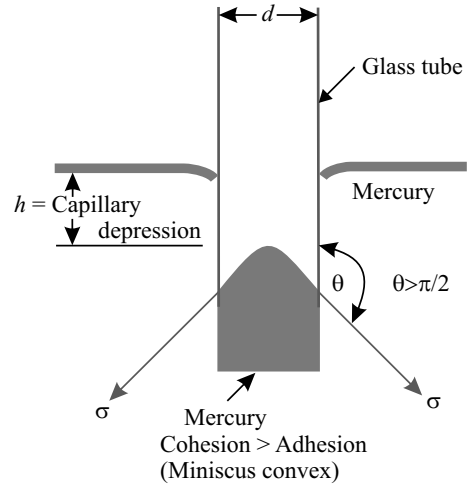


Fig. 1.23

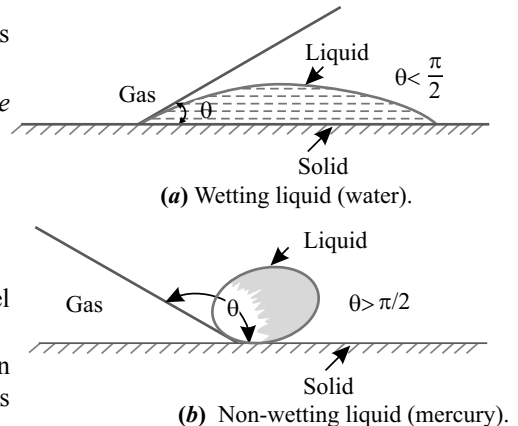


Fig. 1.24

kinetic energy takes place; the flow pattern varies as the Weber number changes and the motion continues giving rise to a series of standing waves. This phenomenon is known as *capillary inversion* of jet for orifices of *non-circular* cross-section. As shown in the Fig. 1.25 the jet issuing from a small elliptical orifice can be observed to undergo two inversion cycles in a given length.

The phenomenon of capillary inversion of jets is significant for industries involving the production and size control of liquid droplets like:

- paint,
- molten shot, and
- agricultural insecticides, *etc.*

Example 1.27. A clean tube of diameter 2.5 mm is immersed in a liquid with a coefficient of surface tension = 0.4 N/m. The angle of contact of the liquid with the glass can be assumed to be 135° . The density of the liquid = 13600 kg/m^3 .

What would be the level of the liquid in the tube relative to the free surface of the liquid inside the tube.

Solution. Given: $d = 2.5 \text{ mm}$; $\sigma = 4 \text{ N/m}$, $\theta = 135^\circ$; $\rho = 13600 \text{ kg/m}^3$

Level of the liquid in the tube, h :

The liquid in the tube rises (or falls) due to capillarity. The capillary rise (or fall),

$$\begin{aligned}
 h &= \frac{4\sigma \cos \theta}{wd} \quad \dots [\text{Eqn. (1.20)}] \\
 &= \frac{4 \times 0.4 \times \cos 135^\circ}{(9.81 \times 13600) \times 2.5 \times 10^{-3}} \quad (\because w = \rho g) \\
 &= -3.39 \times 10^{-3} \text{ m or } -3.39 \text{ mm}
 \end{aligned}$$

Negative sign indicates that there is a capillary **depression (fall) of 3.39 mm.** (Ans.)

Example 1.28. Assuming that the interstices in a clay are of size equal to one tenth the mean diameter of the grain, estimate the height to which water will rise in a clay soil of average grain diameter of 0.048 mm. Assume surface tension at air-water interface as 0.074 N/m.

Solution. Given: Diameter of the pores, $d = \frac{1}{10} \times 0.048 = 0.0048 \text{ mm}$; $\sigma = 0.074 \text{ N/m}$

Assuming

$$\theta = 0^\circ$$

$$h = \frac{4\sigma}{wd} = \frac{4 \times 0.074}{(9.81 \times 1000) \times 0.0048 \times 10^{-3}} = 6.286 \text{ m (Ans.)}$$

Example 1.29. Calculate the work done in blowing a soap bubble of diameter 100 mm. Assume the surface tension of soap solution = 0.038 N/m.

Solution. Given: $d = 100 \text{ mm or } 0.1 \text{ m}$; $\sigma = 0.038 \text{ N/m}$.

The soap bubble has two interfaces.

$\therefore$ Work done = Surface tension $\times$ total surface area

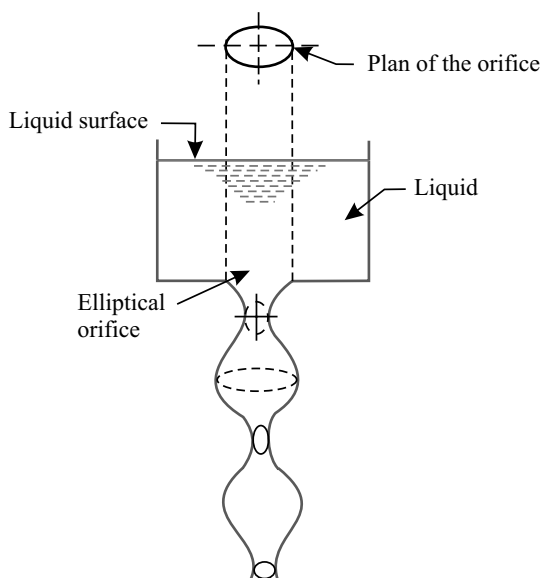


Fig. 1.25. Capillary inversion of a liquid jet.

$$= 0.038 \times 4\pi \times \left(\frac{0.1}{2}\right)^2 \times 2$$

$$= \mathbf{0.002388 \text{ Nm (Ans.)}}$$

Example 1.30. Determine the minimum size of glass tubing that can be used to measure water level, if the capillary rise in the tube is not to exceed 0.3 mm. Take surface tension of water in contact with air as 0.0735 N/m.

Solution. Given : Capillary rise, $h = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$

Surface tension, $\sigma = 0.0735 \text{ N/m}$

Specific weight of water, $w = 9810 \text{ N/m}^3$.

Size of glass tubing, d :

$$\text{Capillary rise, } h = \frac{4\sigma \cos \theta}{wd} = \frac{4\sigma}{wd}$$

(Assuming $\theta = 0$ for water)

$$0.3 \times 10^{-3} = \frac{4 \times 0.0735}{9810 \times d}$$

$$\therefore d = \frac{4 \times 0.0735}{0.3 \times 10^{-3} \times 9810} = 0.1 \text{ m} = \mathbf{100 \text{ mm (Ans.)}}$$

Example 1.31. A U-tube is made up of two capillaries of bores 1.2 mm and 2.4 mm respectively. The tube is held vertical and partially filled with liquid of surface tension 0.06 N/m and zero contact angle. If the estimated difference in the level of two menisci is 15 mm, determine the mass density of the liquid.

Solution. Given: Bores of the capillaries:

$$d_1 = 1.2 \text{ mm} = 0.0012 \text{ m}$$

$$d_2 = 2.4 \text{ mm} = 0.0024 \text{ m}$$

Difference of level, $h_1 - h_2 = 15 \text{ mm} = 0.015 \text{ m}$; Angle of contact, $\theta = 0$

Mass density of the liquid, ρ :

$$h_1 = \frac{4\sigma \cos \theta}{wd_1}, \quad \text{and} \quad h_2 = \frac{4\sigma \cos \theta}{wd_2}$$

[where $w (= \rho g)$ = weight density of the liquid]

$$\therefore h_1 - h_2 = \frac{4\sigma}{w} \left[\frac{1}{d_1} - \frac{1}{d_2} \right] \quad (\because \theta = 0)$$

$$0.015 = \frac{4 \times 0.06}{\rho \times 9.81} \left[\frac{1}{0.0012} - \frac{1}{0.0024} \right] = \frac{0.02446}{\rho} \times 416.67$$

$$\therefore \rho = \frac{0.02446 \times 416.67}{0.015} = \mathbf{679.45 \text{ kg/m}^3 \text{ (Ans.)}}$$

Example 1.32. Derive an expression for the capillary rise at a liquid having surface tension σ and contact angle θ between two vertical parallel plates at a distance W apart. If the plates are of glass, what will be the capillary rise of water having $\sigma = 0.073 \text{ N/m}$, $\theta = 0^\circ$? Take $W = 1 \text{ mm}$.

(Anna University)

Solution. Refer to Fig. 1.26. Consider two vertical parallel plates immersed in a liquid whose weight density is w .

Given : σ = Surface tension;

θ = Contact angle.

Let, h = Height of liquid between plates above general liquid surface.

Under a state of equilibrium, the weight of liquid of height h is balanced by the force at the surface of liquid between the plates.

Then weight of liquid of height h is balanced by the force between the plates

$$\begin{aligned} &= \text{volume of liquid of height } h \text{ between the plates} \times w \\ &= W \times L \times h \times w \end{aligned} \quad \dots(1)$$

where, L = length of plate, and w = weight density of the liquid.

Vertical component of surface tensile force

$$\begin{aligned} &= (\sigma \times \text{circumference}) \times \cos \theta \\ &= \sigma \times 2L \times \cos \theta \end{aligned} \quad \dots(2)$$

For equilibrium, eqns. (1) and (2) must balance.

$$\therefore W \times L \times h \times w = \sigma \times 2L \times \cos \theta \quad \dots(3)$$

or,
$$h = \frac{2\sigma \cos \theta}{W \times w}$$

Eqn. (3) is the expression for capillary rise. (Ans.)

When plates are of glass,

$$\theta = 0^\circ, \sigma = 0.073 \text{ N/m}$$

$$W = 1 \text{ mm} = 0.001 \text{ m}, w = 9810 \text{ N/m}^3$$

$$\begin{aligned} \text{Capillary rise of water, } h &= \frac{2\sigma \cos \theta}{W \times w} \\ &= \frac{2 \times 0.073 \times \cos 0^\circ}{0.001 \times 9810} = 0.0149 \text{ m or } 14.9 \text{ mm} \end{aligned}$$

Hence, capillary rise = **14.9 mm (Ans.)**

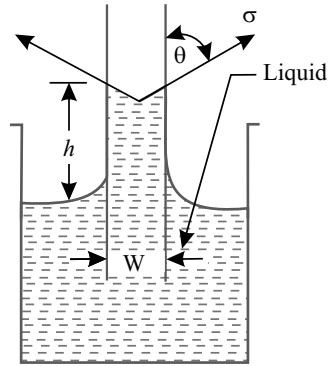


Fig. 1.26

Example 1.33. A single column U-tube manometer, made of glass tubing having a nominal inside diameter of 2.4 mm, has been used to measure pressure in a pipe or vessel containing air. If the limb opened to atmosphere is 10 percent oversize, find the error in mm of mercury in the measurement of air pressure due to surface tension effects. It is stated that mercury is the manometric fluid for which surface tension $\sigma = 0.52 \text{ N/m}$ and angle of contact $\alpha = 140^\circ$

Solution. Given: $d_1 = 2.4 \text{ mm}$; $d_2 = 2.4 \times 1.1 = 2.64 \text{ mm}$; $\sigma = 0.52 \text{ N/m}$; $\alpha = 140^\circ$.

Error in measurement due to surface tension effects:

The surface tension manifests the phenomenon of capillary action due to which rise or depression of manometric liquid in a tube is given by

$$h = \frac{4\sigma \cos \theta}{wd_1}$$

Now,
$$h_1 = \frac{4 \times 0.52 \times \cos 140^\circ}{(13.6 \times 9810) \times (2.4 \times 10^{-3})} = 4.97 \times 10^{-3} \text{ m}$$

(Negative sign indicates capillary depression)

$$h_2 = \frac{4 \times 0.52 \times \cos 140^\circ}{(13.6 \times 9810) \times (2.64 \times 10^{-3})} = -4.52 \times 10^{-3} \text{ m}$$

Hence, error in measurement due to surface tension effects

$$= (4.97 - 4.52) \times 10^{-3} = 0.45 \times 10^{-3} \text{ m} = \mathbf{0.45 \text{ mm (Ans.)}}$$

Example 1.34. Calculate the capillary effect in millimetres in a glass tube of 4 mm diameter, when immersed in (i) water and (ii) mercury. The temperature of the liquid is 20°C and the values of surface tension of water and mercury at 20°C in contact with air are 0.0735 N/m and 0.51 N/m respectively. The contact angle for water $\theta = 0^\circ$ and for mercury $\theta = 130^\circ$. Take specific weight of water at 20°C as equal to 9790 N/m³. [Engg. Services]

Solution. Given: Diameter of glass tube, $d = 4 \text{ mm} = 0.004 \text{ m}$

Surface tension at 20°C, σ :

$$\sigma_{\text{water}} = 0.0735 \text{ N/m}, \quad \sigma_{\text{mercury}} = 0.051 \text{ N/m}$$

Specific weight of water at 20°C = 9790 N/m³

The rise or depression h of a liquid in a capillary tube is given by

$$h = \frac{4\sigma \cos \theta}{w d}$$

where, σ = surface tension, θ = angle of contact, and w = specific weight.

(i) Capillary effect for water:

$$h = \frac{4 \times 0.0735 \times \cos 0^\circ}{9790 \times 0.004} \quad (\because \theta_{\text{water}} = 0^\circ \text{ ...given})$$

$$= 7.51 \times 10^{-3} \text{ m} = \mathbf{7.51 \text{ mm (rise) (Ans.)}}$$

(ii) Capillary effect for mercury

$$h = \frac{4 \times 0.051 \times \cos 130^\circ}{(13.6 \times 9790) \times 0.004} \quad (\because \theta_{\text{mercury}} = 130^\circ \text{ ... given})$$

$$\text{or,} \quad = -2.46 \times 10^{-3} \text{ m} = -2.46 \text{ mm}$$

$$\text{i.e.,} \quad h = \mathbf{2.46 \text{ mm (depression) (Ans.)}}$$

Example 1.35. In measuring the unit energy of a mineral oil (specific gravity = 0.85) by the bubble method, a tube having an internal diameter of 1.5 mm is immersed to depth of 12.5 mm in oil. Air is forced through the tube forming a bubble at the lower end. What magnitude of the unit surface energy will be indicated by a maximum bubble pressure intensity of 150 N/m². [Engg. Services]

Solution. Sp. gravity of oil = 0.85

Internal diameter of the tube,

$$d = 1.5 \text{ mm} = 0.0015 \text{ m}$$

Depth, $h = 12.5 \text{ mm} = 0.0125 \text{ m}$

Gauge pressure inside the bubble

$$p_i = 150 \text{ N/m}^2$$

Unit surface energy, σ :

Gauge pressure outside the bubble,

$$p_0 = wh = (0.85 \times 9810) \times 0.0125 = 104.23 \text{ N/m}^2$$

$\therefore$ Net pressure attributable to surface tension

$$p = p_i - p_0 = 150 - 104.23 = 45.77 \text{ N/m}^2$$

$$\text{Also,} \quad p_i - p_0 = \frac{4\sigma}{d}$$

Assuming diameter of bubble equal to that of the tube,

$$45.77 = \frac{4\sigma}{0.0015}$$

$$\therefore \quad \sigma = \frac{45.77 \times 0.0015}{4} = \mathbf{0.0172 \text{ N/m (Ans.)}}$$

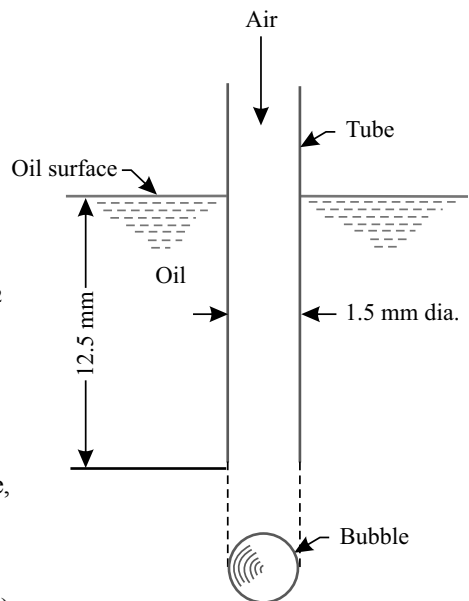


Fig. 1.27

Example 1.36. Two coaxial glass tubes forming an annulus with small gap are immersed in water in a trough. The inner and outer radii of the annulus are r_i and r_o respectively. What is the capillary rise if σ is the surface tension of water in contact with air? (PTU)

Solution. Refer to Fig. 1.28. If the angle of contact between the liquid and the curved tube surface is θ , the water in the annulus will continue to rise until the vertical component of the surface tension force which acts over the wetted length (outer curve of the inner tube and inner curve of the outer tube) equals the height of the water column, or

$T \cos \theta = \pi (r_o^2 - r_i^2) h \rho g$, where $T = \sigma \pi (r_o + r_i)$; substituting for T , we get

$$\sigma \pi (r_o + r_i) \cos \theta = \pi (r_o^2 - r_i^2) h \rho g$$

or,
$$h = \text{capillary rise} = \frac{\sigma \cos \theta}{(r_o - r_i) \rho g}$$

For pure water and clean glass $\theta \approx 0$ and $h = \frac{\sigma}{(r_o - r_i) \rho g}$

Under *actual conditions*, neither water is pure, nor glass is clean.

Gibson has obtained the value of θ as $25^\circ 32'$.

Thus,
$$h = \frac{\sigma \cos 25^\circ 32'}{(r_o - r_i) \rho g} = \frac{0.902 \sigma}{(r_o - r_i) \rho g} \quad (\text{Ans.})$$

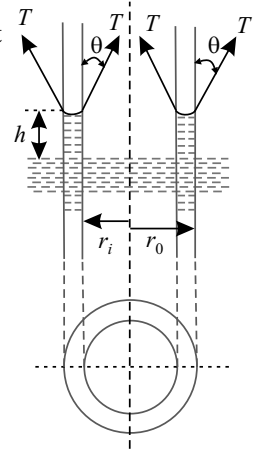


Fig. 1.28

1.9. COMPRESSIBILITY AND BULK MODULUS

The property by virtue of which fluids undergo a change in volume under the action of external pressure is known as **compressibility**. It decreases with the increases in pressure of fluid as the volume modulus increases with the increase of pressure.

The variation in volume of water, with variation of pressure, is so small that for all practical purposes it is neglected. Thus, the water is considered to be an incompressible liquid. However in case of water flowing through pipes when sudden or large changes in pressure (e.g. water hammer) take place, the compressibility cannot be neglected. The *compressibility* in Fluid Mechanics is considered mainly when the velocity of flow is high enough reaching 20 percent of speed of sound in the medium.

Elasticity of fluids is measured in terms of **bulk modulus of elasticity** (K) which is defined as the *ratio of compressive stress to volumetric strain*. Compressibility is the reciprocal of bulk modulus of elasticity.

Consider a cylinder fitted with a piston as shown in Fig. 1.29.

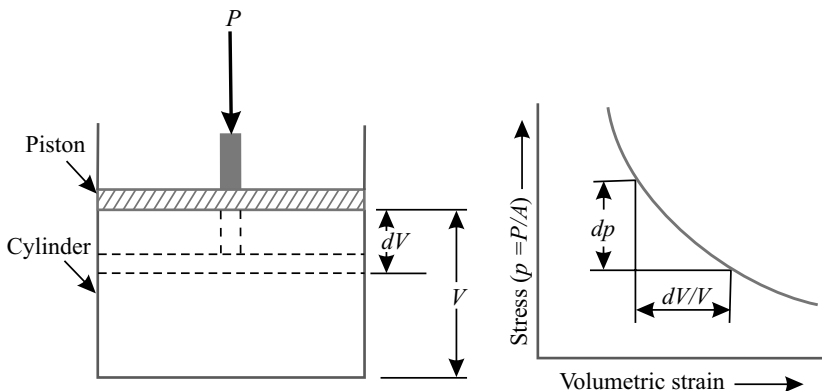


Fig. 1.29

Let,

V = Volume of gas enclosed in the cylinder, and

p = Pressure of gas when volume is V

= $\frac{P}{A}$, where A is the area of cross-section of the cylinder.

Let the pressure is increased to $p + dp$, the volume of gas decreases from V to $V - dV$.

Then increase in pressure = dp ; Decrease in volume = dV

$$\therefore \text{Volumetric strain} = -\frac{dV}{V}$$

(Negative sign indicates *decrease in volume with increase of pressure*)

$$\therefore \text{Bulk modulus, } K = \frac{dp(\text{increase of pressure})}{-dV/V(\text{volumetric strain})}$$

$$\text{i.e., } K = \frac{dp}{-dV/V}$$

$$\left(\text{Compressibility} = \frac{1}{K} \right) \quad \dots(1.22)$$

Steepening of the curve (Fig. 1.29) with increasing pressure shows that as fluids are compressed it becomes increasingly difficult to compress them further. In other words, the *value of K increases with increase of pressure*.

The following points are worth noting:

1. The bulk modulus of elasticity (K) of a fluid is not constant, but it increases with increase in pressure. This is so because when a fluid mass is compressed its molecules become close together and its resistance to further compression increases *i.e.*, K increases. (e.g. the value of K roughly doubles as the pressure is raised from 1 atmosphere to 3500 atmosphere).

2. The bulk modulus of elasticity (K) of the fluid is affected by the temperature of the fluid. In the case of *liquids* there is a *decrease* of K with *increase of temperature*. However, for gases since pressure and temperature are inter-related and as temperature increases, pressure also increases, an *increase in temperature* results in an *increase* in the value of K .

3. At NTP (normal temperature and pressure):

$$K_{\text{water}} = 2.07 \times 10^6 \text{ kN/m}^2,$$

$$K_{\text{air}} = 101.3 \text{ kN/m}^2$$

Example 1.37. When the pressure of liquid is increased from 3.5 MN/m^2 to 6.5 MN/m^2 its volume is found to decrease by 0.08 percent. What is the bulk modulus of elasticity of the liquid?

Solution. Initial pressure = 3.5 MN/m^2

Final pressure = 6.5 MN/m^2

$$\therefore \text{Increase in pressure, } dp = 6.5 - 3.5 = 3.0 \text{ MN/m}^2$$

$$\text{Decrease in volume} = 0.08 \text{ percent} \quad \therefore -\frac{dV}{V} = \frac{0.08}{100}$$

Bulk modulus (K) is given by:

$$K = \frac{dp}{-\frac{dV}{V}} = \frac{3 \times 10^6}{\frac{0.08}{100}} = 3.75 \times 10^9 \text{ N/m}^2 \text{ or } 3.75 \text{ GN/m}^2$$

Hence,

$$K = 3.75 \text{ GN/m}^2 \text{ (Ans.)}$$

Example 1.38. When a pressure of 20.7 MN/m^2 is applied to 100 litres of a liquid its volume decreases by 1 litre. Find the bulk modulus of the liquid and identify this liquid.

Solution. Net pressure applied, $dp = 20.7 \text{ MN/m}^2$

Decrease in volume, $dV = 1$ litre

$$\text{Initial volume, } V = 100 \text{ litres} \quad \therefore \quad -\frac{dV}{V} = \frac{1}{100}$$

Bulk modulus K:

$$K = -\frac{dp}{-dV/V} = \frac{20.7 \times 10^6}{1/100} = 20.7 \times 10^8 \text{ N/m}^2 = 2.07 \text{ GN/m}^2$$

i.e.,

$$K = \mathbf{2.07 \text{ GN/m}^2 \text{ (Ans.)}}$$

Evidently the liquid is **water (Ans.)**.

Example 1.39. Define compressibility of a fluid. Gas A at 125 kPa (abs.) is compressed isothermally and gas B at 100 kPa (abs.) is compressed isentropically ($\gamma = 1.4$). Which gas is more compressible? (N.U.)

Solution. Compressibility is the measure of relative change of volume (For density) when the fluid is subjected to a pressure change. It is the reciprocal of the bulks modulus of elasticity (K).

It is expressed mathetically as:

$$Z = \frac{1}{K} = \frac{-(dV/V)}{dp}$$

For an ideal gas, if the compression is isothermal, $Z = \frac{1}{p}$, and if the compression is isentropic, $Z = \frac{1}{\gamma p}$.

$$\text{For the given gas A,} \quad Z_A = \frac{1}{p} = \frac{1}{125} = 0.008 \text{ m}^2/\text{kN}$$

$$\text{For the gas B,} \quad Z_B = \frac{1}{\gamma p} = \frac{1}{1.4 \times 100} = 0.007143 \text{ m}^2/\text{kN}$$

Hence **gas A more compressible. (Ans.)**

Example 1.40. Find an expression for isothermal bulk modulus of elasticity for a gas which obeys Van der Waals' law of state according to the equation:

$$p = pRT \left[\frac{1}{1 - bp} - \frac{ap}{RT} \right]$$

where a , b are constants and p , ρ , R and T have their usual meanings.

(P.E.C.)

Solution. Bulk modulus of elasticity,

$$K = -\frac{dp}{\left(\frac{dV}{V}\right)} = -V \frac{dp}{dV},$$

where, V is volume $= -v \frac{dp}{dv}$, v is specific volume.

$$\text{Since,} \quad v = \frac{1}{\rho} \text{ or } \rho v = 1, \quad \frac{dv}{v} = -\frac{d\rho}{\rho}$$

$$\therefore \quad K = \rho \frac{dp}{d\rho}; \quad p = \rho RT \left[\frac{1}{1 - bp} - \frac{ap}{RT} \right] \quad \dots (\text{Given})$$

$$\therefore \quad \frac{dp}{d\rho} = RT \left[\frac{1}{1 - bp} - \frac{ap}{RT} \right] + \rho RT \left[\frac{b}{(1 - bp)^2} - \frac{a}{RT} \right] = \frac{RT}{1 - bp} - ap + \frac{bpRT}{(1 - bp)^2} - ap$$

$$= \frac{RT}{1 - b\rho} \left[1 + \frac{b\rho}{1 - b\rho} \right] - 2a\rho = \frac{RT}{(1 - b\rho)^2} - 2a\rho$$

and,

$$K = \rho \frac{dp}{d\rho} = \frac{\rho RT}{(1 - b\rho)^2} - 2a\rho^2 \quad \dots \text{Required expression (Ans.).}$$

1.10. VAPOUR PRESSURE

All liquids have a tendency to evaporate or vaporize (*i.e.*, to change from the liquid to the gaseous state). Molecules are continuously projected from the free surface to the atmosphere. These ejected molecules are in a gaseous state and exert their own partial vapour pressure on the liquid surface. This pressure is known as the vapour pressure of the liquid (p_v). If the surface above the liquid is confined, the partial vapour pressure exerted by the molecules increases till the rate at which the molecules re-enter the liquid is equal to the rate at which they leave the surface. When the equilibrium condition is reached, the vapour pressure is called saturation vapour pressure (p_{vs}).

The following points are worth noting:

1. If the pressure on the liquid surface is *lower than or equal to the saturation vapour pressure*, boiling takes place.
2. Vapour pressure increases with the rise in temperature.
3. Mercury has a *very low vapour pressure and hence*, it is an excellent fluid to be used in a barometer.

Table 1.1. Summary of Fluid Characteristics

Sr. No.	Characteristics	Symbol	Definition	Dimensions	Units
1.	Mass density	ρ	Mass per unit volume, $\frac{m}{V}$	ML^{-3}	kg/m^3
2.	Weight density (or specific weight)	w	Weight per unit volume, $\frac{w}{V}$	FL^{-3}	N/m^3
3.	Specific volume	v	Volume per unit mass $\frac{V}{m} = \frac{1}{\rho}$	L^3M^{-1}	m^3/kg
4.	Specific gravity	S	$\frac{\text{Specific weight of liquid}}{\text{Specific weight of pure water}}$ $= \frac{w_{\text{liquid}}}{w_{\text{water}}}$		
5.	Dynamic viscosity	μ	Newton's law: $\tau = \mu \cdot \frac{du}{dy}$	FTL^{-2}	$N.s/m^2$ poise, centipoise
6.	Kinematic viscosity	ν	$\nu = \frac{\mu}{\rho}$	L^2T^{-1}	m^2/s stoke, centistoke
7.	Bulk modulus	K	$K = -\frac{\Delta p}{dV/V}$	FL^{-2}	N/m^2
8.	Surface tension	σ	Force per unit length	FL^{-1}	N/m
9.	Vapour pressure	p	$p_v = \frac{F}{A}$	FL^{-2}	N/m^2

Table 1.2. Properties of Some Common Fluids at 20°C and Atmospheric Pressure

Fluid	Mass density $\rho(\text{kN/m}^3)$	Specific weight $w(\text{kN/m}^3)$	Dynamic viscosity μ		Kinematic viscosity ν		Modulus of elasticity $E(\text{N/m}^2)$	Surface tension in contact with air, σ (N/m)	Vapour pressure (N/m^2)
			Poise	kg/ms	Stoke	m^2/s			
Air	1.208	0.01185	1.85×10^{-4}	1.85×10^{-5}	1.53×10^{-1}	1.53×10^{-5}	—	—	—
Benzene	860	8.434	0.007	7.00×10^{-4}	8.14×10^{-3}	8.14×10^{-7}	1.0356×10^9	0.0255	1.000×10^4
Castor oil	960	9.414	9.800	9.80×10^{-1}	1.00×10^1	1.00×10^3	1.441×10^9	0.0392	—
Carbon tetrachloride	1594	15.632	0.010	1.00×10^{-3}	6.04×10^{-3}	6.04×10^{-7}	1.104×10^9	0.0265	1.275×10^4
Ethyl alcohol	789	7.737	0.012	1.20×10^{-3}	1.52×10^{-2}	1.52×10^{-6}	1.118×10^9	0.0216	5.786×10^3
Glycerine	1260	12.356	8.350	8.35×10^{-1}	6.63	6.63×10^{-4}	4.354×10^9	0.0637	1.373×10^{-2}
Kerosene	800	7.845	0.020	2.00×10^{-3}	2.50×10^{-2}	2.50×10^{-6}	—	0.0235	—
Mercury	13550	132.880	0.016	1.60×10^{-3}	1.18×10^{-3}	1.18×10^{-7}	2.431×10^{10}	0.510	1.726×10^{-1}

HIGHLIGHTS

1. *Hydraulics* is that branch of Engineering science, which deals with water at rest or in motion.
2. *Fluid mechanics* may be defined as that branch of Engineering science which deals with the behaviour of fluid under the conditions of rest and motion.
3. A *fluid* is substance which is capable of flowing.
4. *Mass density* is the mass per unit volume whereas weight density (or specific weight) is the weight per unit volume at the standard temperature and pressure.
5. *Specific gravity* is the ratio of the specific weight of the liquid to the specific weight of a standard fluid. It is dimensionless and has no units.
6. *Viscosity* is the property of a fluid which determines its resistance to shearing stresses.

Newton's law of viscosity states that the shear stress (τ) on a fluid element layer is directly proportional to the rate of shear strain. The constant of proportionality is called the co-efficient of viscosity.

Mathematically,
$$\tau = \mu \cdot \frac{du}{dy},$$

where μ = co-efficient of dynamic viscosity, and $\frac{du}{dy}$ = rate of shear deformation or velocity gradient.

Kinematic viscosity is the ratio between the dynamic viscosity and density of fluid. It is denoted by ν (nu).

i.e.,
$$\nu = \frac{\mu}{\rho}$$

7. *Cohesion and adhesion*:

Cohesion means intermolecular attraction between molecules of the same liquid.

Adhesion means attraction between molecules of a liquid and the molecules of a solid boundary surface in contact with the liquid.

8. *Surface tension* (σ) is caused by the force of cohesion at the free surface. It is usually expressed in N/m.

Pressure inside:

(a) Water droplet: $p = \frac{4\sigma}{d},$ (b) Soap bubble : $p = \frac{8\sigma}{d},$ and

(c) Liquid jet: $p = \frac{2\sigma}{d}$ (where d stands for diameter).

9. *Capillarity* is a phenomenon by which a liquid (depending upon its specific gravity) rises into a thin glass tube or below its general level.

$$h = \frac{4\sigma \cos \theta}{wd}$$

where,

h = Height of capillary rise,

d = Diameter of the capillary tube,

θ = Angle of contact of the water surface,

σ = Surface tension per unit length, and

w = Weight density (ρg).

10. *Compressibility* is the property by virtue of which fluid undergoes a change in volume under the action of external pressure. It is the *reciprocal* of bulk modulus of elasticity (K).

$$K = dp \text{ (increase of pressure)} / - \frac{dV}{V} \text{ (volumetric strain)}$$

$$\left(\text{compressibility} = \frac{1}{K} \right)$$

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer:

- The branch of Engineering-science, which deals with water at rest or in motion is called
 - hydraulics
 - fluid mechanic s
 - applied mechanics
 - kinematics.
- A solid can resist which of the following stresses?
 - Tensile
 - Compressive
 - Shear
 - All of the above.
- possesses no definite volume and is compressible.
 - Solid
 - Liquid
 - Gas
 - Vapour.
- A real practical fluid possesses which of the following?
 - Viscosity
 - Surface tension
 - Compressibility
 - density.
- The ratio of the specific weight of the liquid to the specific weight of a standard fluid is known as
 - specific volume
 - weight density
 - specific gravity
 - viscosity.
- The property of a fluid which determines its resistance to shearing stress is called
 - viscosity
 - surface tension
 - compressibility
 - none of the above.
- Newton's law of viscosity is given by the relation:
 - $\tau = \mu^2 \frac{du}{dy}$
 - $\tau = \sqrt{\mu} \frac{du}{dy}$
 - $\tau = \mu \cdot \frac{du}{dy}$
 - $\tau = (\mu)^{3/2} \frac{du}{dy}$
- Fluids which do not follow the linear relationship between shear stress and rate of deformation are termed as fluids.
 - Newtonian
 - Non-Newtonian
 - dilatant
 - ideal
- The printer's ink is an example of
 - Newtonian fluid
 - Non-Newtonian
 - Thixotropic substance
 - Elastic solid.
- The viscosity of liquids with increase in temperature.
 - decreases
 - increases
 - first decreases and then increases
 - first increases and then decreases.
- Surface tension is caused by the force of at the free surface.
 - cohesion
 - adhesion
 - both (a) and (b)
 - none of the above.
- Which of the following is an example of phenomenon of surface tension?
 - Rain drops
 - Rise of sap in a tree
 - Break up of liquid jets
 - All of the above.
- Surface tension is expressed in
 - N/m
 - N/m²
 - N²/m
 - N/m³.
- Pressure inside a water droplet is given by the relation
 - $p = \frac{4\sigma}{d}$
 - $p = \frac{3\sigma}{d}$
 - $p = \frac{8\sigma}{d}$
 - $p = \frac{16\sigma}{d}$
- is a phenomenon by which a liquid rises into a thin glass tube above or below its general level.
 - Surface tension
 - Capillarity
 - Cohesion
 - Adhesion.
- The capillary rise of water in the glass tube is given by
 - $h = \frac{2\sigma}{wd}$
 - $h = \frac{3\sigma}{wd}$
 - $h = \frac{4\sigma}{wd}$
 - $h = \frac{6\sigma}{wd}$

17. Elasticity of fluids is measured in terms of
 - (a) Young's modulus of elasticity
 - (b) shear modulus of elasticity
 - (c) bulk modulus of elasticity
 - (d) none of the above.
18. Compressibility is the reciprocal of
 - (a) bulk modulus of elasticity
 - (b) shear modulus of elasticity
 - (c) Young's modulus of elasticity
 - (d) any of the above.
19. Bulk modulus of elasticity is the ratio of
 - (a) tensile stress to tensile strain
 - (b) compressive stress to compressive strain
 - (c) compressive stress to volumetric strain
 - (d) none of the above.
20. The value of bulk modulus of elasticity with increase of pressure.
 - (a) increases
 - (b) decreases
 - (c) either of the above
 - (d) none of the above.

ANSWERS

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (d) | 3. (c) | 4. (e) | 5. (c) | 6. (a) |
| 7. (c) | 8. (b) | 9. (c) | 10. (a) | 11. (a) | 12. (d) |
| 13. (a) | 14. (a) | 15. (b) | 16. (c) | 17. (c) | 18. (a) |
| 19. (c) | 20. (a) | | | | |

THEORETICAL QUESTIONS

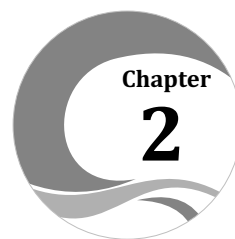
1. Define the following:
 - (i) Hydraulics
 - (ii) Fluid mechanics
 - (iii) Fluid
 - (iv) Aerostatics.
2. What is a fluid? How are fluids classified?
3. What is the difference between an ideal and a real fluid?
4. Name some important properties of liquids.
5. Explain briefly the following terms:
 - (i) Mass density
 - (ii) Weight density
 - (iii) Specific volume
 - (iv) Specific gravity.
6. What do you mean by the term 'Viscosity'?
7. State and explain the Newton's law of viscosity.
8. What is dynamic viscosity? What are its units?
9. What is kinematic viscosity? What are its units?
10. What is a Newtonian fluid?
11. Define the term vapour pressure. How does it vary with temperature?
12. What is the difference between cohesion and adhesion?
13. Explain briefly the following:
 - (i) Surface tension, and
 - (ii) Compressibility.
14. What is capillarity? Derive expression for height of capillary rise.

UNSOLVED EXAMPLES

1. Determine the mass density, specific volume and specific weight of a liquid whose specific gravity is 0.85.
[Ans. 850 kg/m^3 , $0.00118 \text{ m}^3/\text{kg}$, 8350 N/m^3]
2. A liquid has a specific gravity of 1.9 and kinematic viscosity of 6 stokes. What is its dynamic viscosity? [Ans. 11.38 poise]
3. The space between two parallel plates 5 mm apart is filled with crude oil. A force of 2 N is required to drag the upper plate at a constant velocity of 0.8 m/s. The lower plate is stationary. The area of the upper plate is 0.09 m^2 . Determine: (i) The dynamic viscosity, and (ii) the kinematic viscosity of the oil in stokes if the specific gravity of oil is 0.9. [Ans. (i) 1.39 poise, (ii) 1.52 stokes]
4. A plate has an area of 1 m^2 . It slides down an inclined plane, having angle of inclination 45° to the horizontal, with a velocity of 0.5 m/s. The thickness of oil film between the plane and the plate is 1 mm. Find the viscosity of the fluid if the weight of the plate is 70.72 N. [Ans. 1 poise]
5. The velocity distribution over a plate is given by

$$u = \frac{3}{2}y - \frac{1}{2}y^2$$
 where, u = velocity, m/s, and
 y = distance from the plate boundary, m.
 If the viscosity of the fluid is 8 poise find the

- shear stress at the plate boundary and at $y = 0.15$ m from the plate. [Ans. 1.20 N/m^2 , 1.08 N/m^2]
6. A flat plate weighing 0.45 kN has a surface area of 0.1 m^2 . It slides down an inclined plane at 30° to the horizontal, at a constant speed of 3 m/s . If the inclined plane is lubricated with an oil of viscosity 0.1 N.s/m^2 , find the thickness of the oil film. [Ans. 0.133 mm]
 7. A flat thin plate is dragged at a constant velocity of 4 m/s on the top of a 5 mm deep liquid layer of viscosity 20 centipoise. If the area of the plate is 1 m^2 find the drag force.
Assume variation of velocity in the liquid to be linear. [Ans. 16 N]
 8. A square metal plate 1.5 m side and 1.5 mm thick weighing 50 N is to be lifted through a vertical gap of 25 mm of infinite extent. The oil in the air gap has a specific gravity of 0.95 and viscosity of 2.5 N.s/m^2 . If the metal plate is to be lifted at a constant speed of 0.1 m/s find the force and power required. [Ans. 145.7 N , 14.57 W]
 9. Inside a 60 mm diameter cylinder a piston of 59 mm diameter rotates concentrically. Both the cylinder and piston are 80 mm long. If the space between the cylinder and piston is filled with oil of viscosity of 0.3 N.s/m^2 and a torque of 1.5 Nm is applied, find:
 - (i) The r.p.m. of the piston, and
 - (ii) The power required.
 [Ans. (i) 1850 r.p.m. (ii) 290.5 W]
 10. Two large fixed parallel planes are 240 mm apart. The space between the surfaces is filled with oil of viscosity 0.81 N.s/m^2 . A flat thin plate 0.5 m^2 area moves through the oil at a velocity of 0.6 m/s . Calculate the drag force (i) when the plate is equidistant from both the planes, and (ii) when the thin plate is at a distance of 80 mm from one of the plane surfaces.
[Ans. (i) 40.5 N (ii) 45.54 N]
 11. A cylinder of 100 mm diameter and 300 mm length rotates about a vertical axis inside a fixed cylindrical tube of 105 mm diameter and 300 mm length. If the space between the tube and the cylinder is filled with liquid of dynamic viscosity of 0.125 N.s/m^2 , determine the speed of rotation of the cylinder which will be obtained if an external torque of 1 Nm is applied to it.
[Ans. 81.03 r.p.m.]
 12. Determine the mass density, specific weight, and specific volume of CO_2 contained in a vessel at a pressure of 800 kN/m^2 and temperature 25°C .
[Ans. 14.2 kg/m^3 , 139.4 N/m^3 , $0.0703 \text{ m}^3/\text{kg}$]
 13. A soap bubble 50 mm diameter has an internal pressure in excess of the outside pressure of 25 N/m^2 . Calculate tension in the soap film.
[Ans. 0.156 N/m]
 14. Air is introduced through a nozzle into a tank of water (at 20°C) to form a stream of bubbles. If the process requires 2.5 mm diameter bubbles to be formed, by how much the air pressure at the nozzle must exceed that of surrounding water.
Take surface tension of water at $20^\circ\text{C} = 0.0735 \text{ N/m}$. [Ans. 117.4 N/m^2]
 15. Determine the minimum size of glass tubing that can be used to measure water level, if the capillary rise in the tube is not to exceed 0.25 mm . Take surface tension of water in contact with air as 0.0735 N/m . [Ans. 120 mm]
 16. A U-tube is made up of two capillaries of bore 1 mm and 2 mm respectively. The tube is held vertically and is partially filled with liquid of surface tension 0.05 N/m and zero contact angle. Calculate the mass density of the liquid if the estimated difference in the level of two menisci is 12.5 mm . [Ans. 816 kg/m^3]
 17. Determine the bulk modulus of elasticity of a liquid, if the pressure of the liquid is increased from 7 MN/m^2 to 13 MN/m^2 . The volume of liquid decreases by 0.15% . [Ans. 4 GN/m^2]
 18. A 20 mm wide gap between two vertical plane surfaces is filled with an oil of specific gravity 0.85 and dynamic viscosity 2.5 N.s/m^2 . A metal plate $1.25 \text{ m} \times 1.25 \text{ m} \times 2 \text{ mm}$ thick and weighing 30 N is placed midway in the gap. Determine the force required to lift the plate with a constant velocity of 0.18 m/s . [Ans. 160.2 N]



PRESSURE MEASUREMENT

- 2.1. Pressure of a liquid
 - 2.2. Pressure head of a liquid
 - 2.3. Pascal's law
 - 2.4. Absolute and gauge pressures.
 - 2.5. Measurement of pressure—Manometers—Mechanical gauges
- Highlights**
Objective Type Questions
Theoretical Questions
Unsolved Examples

2.1. PRESSURE OF A LIQUID

When a fluid is contained in a vessel, it exerts force at all points on the sides and bottom and top of the container. The *force per unit area* is called **pressure**.

If, $P =$ The force, and

$A =$ Area on which the force acts; then

$$\text{intensity of pressure, } p = \frac{P}{A} \quad \dots(2.1)$$

The pressure of a fluid on a surface will *always act normal to the surface*.

2.2. PRESSURE HEAD OF A LIQUID

A liquid is subjected to pressure due to its own weight, this pressure increases as the depth of the liquid increases.

Consider a vessel containing liquid, as shown in Fig. 2.1. The liquid will exert pressure on all sides and bottom of the vessel. Now, let cylinder be made to stand in the liquid, as shown in the figure.

Let, $h =$ Height of liquid in the cylinder,

$A =$ Area of the cylinder base,

$w =$ Specific weight of the liquid,

and, $p =$ Intensity of pressure.

Now, Total pressure on the base of the cylinder = Weight of liquid in the cylinder

$$\text{i.e., } p \cdot A = wAh$$

$$p = \frac{wAh}{A} = wh \quad \text{i.e., } p = wh \quad \dots(2.2)$$

As $p = wh$, the intensity of pressure in a liquid due to its depth will vary directly with depth.

As the pressure at any point in a liquid depends on height of the free surface above that point, it is sometimes convenient to express a liquid pressure by the *height of the free surface* which would cause the pressure, i.e.,

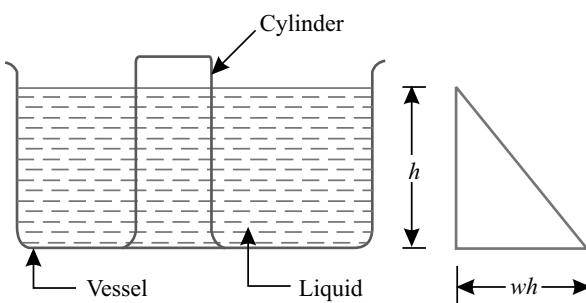


Fig. 2.1. Pressure head.

$$h = \frac{p}{w} \quad [\text{from eqn. (2.2)}]$$

The height of the free surface above any point is known as the *static head* at that point. In this case, static head is h .

Hence, the intensity of pressure of a liquid may be expressed in the following two ways:

1. As a force per unit area (*i.e.*, N/mm^2 , N/m^2), and
2. As an equivalent static head (*i.e.*, metres, mm or cm of liquid).

Alternatively:

Pressure variation in fluid at rest:

In order to determine the pressure at any point in a fluid at rest “*hydrostatic law*” is used; the law states as follows:

“The rate of increase of pressure in a vertically downward direction must be equal to the specific weight of the fluid at that point.”

The proof of the law is as follows.

Refer to Fig. 2.2

Let, p = Intensity of pressure on face LM,

ΔA = Cross-sectional area of the element,

Z = Distance of the fluid element from free surface, and

ΔZ = Height of the fluid element.

The forces acting on the element are:

(i) Pressure force on the face

$LM = p \times \Delta A$... (acting downward)

(ii) Pressure force on the face $ST = \left(p + \frac{\partial p}{\partial Z} \times \Delta Z \right) \times \Delta A$... (acting upward)

(iii) Weight of the fluid element = Weight density $\times$ volume
 $= w \times (\Delta A \times \Delta Z)$

(iv) Pressure forces on surfaces MT and LS are equal and opposite.

For equilibrium of the fluid element, we have:

$$p \times \Delta A - \left[p + \frac{\partial p}{\partial Z} \times \Delta Z \right] \times \Delta A + w \times (\Delta A \times \Delta Z) = 0$$

$$\text{or, } p \times \Delta A - p \times \Delta A - \frac{\partial p}{\partial Z} \times \Delta Z \times \Delta A + w \times \Delta A \times \Delta Z = 0$$

$$\text{or, } \frac{\partial p}{\partial Z} \Delta Z \times \Delta A + w \times \Delta A \times \Delta Z = 0$$

$$\text{or, } \frac{\partial p}{\partial Z} = w \text{ (cancelling } \Delta Z \times \Delta A \text{ from both the sides)}$$

$$\text{or, } \frac{\partial p}{\partial Z} = \rho \times g \quad (\because w = \rho \times g) \quad \dots(2.3)$$

Eqn. (2.3.) states that rate of increase of pressure in a vertical direction is equal to weight density of the fluid at that point. This is “**hydrostatic law**”.

On integrating the eqn. (2.3), we get:

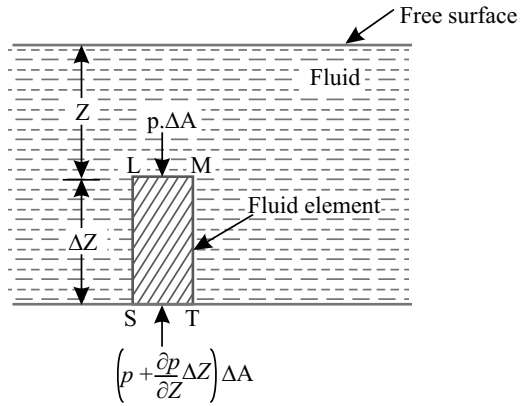


Fig. 2.2. Forces acting on a fluid element.

$$\int dp = \int \rho g \cdot dZ$$

$$\text{or,} \quad p = \rho g \cdot Z (= wZ) \quad \dots(2.4)$$

where, p is the pressure above atmospheric pressure.

From eqn. (2.4), we have:

$$Z = \frac{p}{\rho \cdot g} \left(= \frac{p}{w} \right) \quad \dots (2.5)$$

Here Z is known as *pressure head*.

Example 2.1. Find the pressure at a depth of 15 m below the free surface of water in a reservoir.

Solution. Depth of water, $h = 15$ m

Specific weight of water, $w = 9.81$ kN/m³

Pressure p :

We know that, $p = wh = 9.81 \times 15 = 147.15$ kN/m²

i.e., $p = 147.15$ kN/m² = **147.15 kPa (Ans.)**

Example 2.2. Find the height of water column corresponding to a pressure of 54 kN/m².

Solution. Intensity of pressure, $p = 54$ kN/m²

Specific weight of water, $w = 9.81$ kN/m³

Height of water column, h :

Using the relation: $p = wh$; $h = \frac{p}{w} = \frac{54}{9.81} = \mathbf{5.5 \text{ m (Ans.)}}$

2.3. PASCAL'S LAW

The Pascal's law states as follows :

“The intensity of pressure at any point in a liquid at rest, is the same in all directions”.

Proof. Let us consider a very small wedge shaped element LMN of a liquid, as shown in Fig. 2.3.

Let, p_x = Intensity of horizontal pressure on the element of liquid,

p_y = Intensity of vertical pressure on the element of liquid,

p_z = Intensity of pressure on the diagonal of the right angled triangular element,

α = Angle of the element of the liquid,

P_x = Total pressure on the vertical side LN of the liquid,

P_y = Total pressure on the horizontal side MN of the liquid, and

P_z = Total pressure on the diagonal LM of the liquid.

$$\text{Now,} \quad P_x = p_x \times LN \quad \dots(i)$$

$$\text{and,} \quad P_y = p_y \times MN \quad \dots(ii)$$

$$\text{and,} \quad P_z = p_z \times LM \quad \dots(iii)$$

As the element of the liquid is at rest, therefore the *sum of horizontal and vertical components of the liquid pressures must be equal to zero*.

Resolving the forces *horizontally*:

$$P_z \sin \alpha = P_x$$

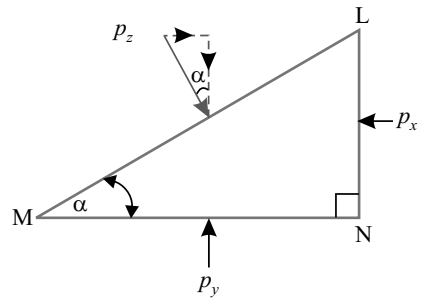


Fig. 2.3. Pressure on a fluid element at rest.

$$p_z \cdot LM \cdot \sin \alpha = p_x \cdot LN \quad (\because P_z = p_z \cdot LM)$$

But,

$$LM \cdot \sin \alpha = LN$$

... From Fig 2.3

$\therefore$

$$p_z = p_x$$

...(iv)

Resolving the forces vertically:

$$P_z \cdot \cos \alpha = P_y - W$$

(where, W = weight of the liquid element)

Since the element is very small, neglecting its weight, we have:

$$P_z \cos \alpha = P_y \quad \text{or} \quad p_z \cdot LM \cos \alpha = p_y \cdot MN$$

But,

$$LM \cos \alpha = MN$$

...From Fig 2.3

$\therefore$

$$p_z = p_y$$

...(v)

From (iv) and (v), we get: $p_x = p_y = p_z$

which is independent of α .

Hence, at any point in a fluid at rest the intensity of pressure is exerted equally in all directions, which is called **Pascal's law**.

Example 2.3. The diameters of ram and plunger of an hydraulic press are 200 mm and 30 mm respectively. Find the weight lifted by the hydraulic press when the force applied at the plunger is 400 N.

Solution. Diameter of the ram, $D = 200 \text{ mm} = 0.2 \text{ m}$

Diameter of the plunger, $d = 30 \text{ mm} = 0.03 \text{ m}$

Force on the plunger, $F = 400 \text{ N}$

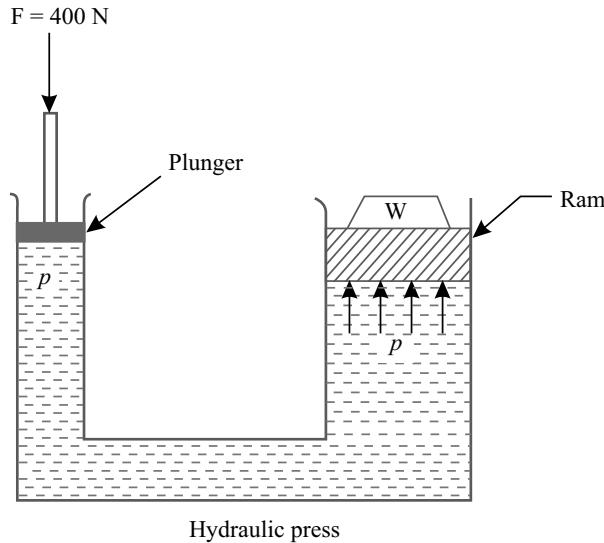


Fig. 2.4

Load lifted, W :

$$\text{Area of ram, } A = \frac{\pi}{4} D^2 = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$$

$$\text{Area of plunger, } a = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times 0.03^2 = 7.068 \times 10^{-4} \text{ m}^2$$

Intensity of pressure due to plunger,

$$p = \frac{F}{a} = \frac{400}{7.068 \times 10^{-4}} = 5.66 \times 10^5 \text{ N/m}^2$$

Since the intensity of pressure will be equally transmitted (due to Pascal's law), therefore the intensity of pressure at the ram is also

$$= p = 5.66 \times 10^5 \text{ N/m}^2$$

$$\text{But intensity of pressure at the ram} = \frac{\text{Weight}}{\text{Area of ram}} = \frac{W}{A} = \frac{W}{0.0314} \text{ N/m}^2$$

$$\therefore \frac{W}{0.0314} = 5.66 \times 10^5 \text{ or } W = 0.0314 \times 5.66 \times 10^5 \text{ N} = 17.77 \times 10^3 \text{ N or } \mathbf{17.77 \text{ kN (Ans.)}}$$

Example 2.4. For the hydraulic jack shown in Fig. 2.5 find the load lifted by the large piston when a force of 400 N is applied on the small piston. Assume the specific weight of the liquid in the jack is 9810 N/m³.

Solution. Diameter of small piston, $d = 30 \text{ mm} = 0.03 \text{ m}$

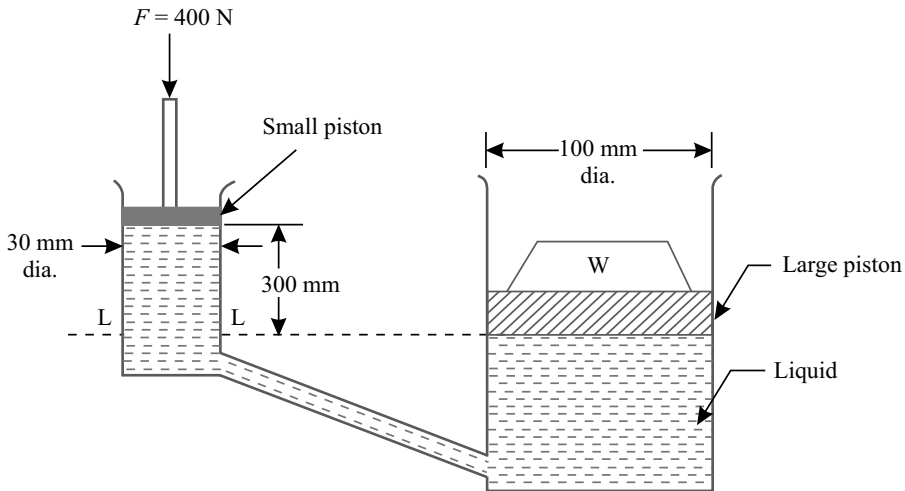


Fig. 2.5

$$\text{Area of small piston, } a = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times 0.03^2 = 7.068 \times 10^{-4} \text{ m}^2$$

$$\text{Diameter of the large piston, } D = 100 \text{ mm} = 0.1 \text{ m}$$

$$\text{Area of large piston, } A = \frac{\pi}{4} D^2 = \frac{\pi}{4} \times 0.1^2 = 7.854 \times 10^{-3} \text{ m}^2$$

$$\text{Force on small piston, } F = 400 \text{ N}$$

Load lifted, W :

$$\text{Pressure intensity on small piston, } p = \frac{F}{a} = \frac{400}{7.068 \times 10^{-4}} = 5.66 \times 10^5 \text{ N/m}^2$$

Pressure intensity at section LL,

$$\begin{aligned} p_{LL} &= \frac{F}{a} + \text{Pressure intensity due to height of 300 mm of liquid} \\ &= \frac{F}{a} + wh = 5.66 \times 10^5 + 9810 \times \frac{300}{1000} \\ &= 5.66 \times 10^5 + 2943 = 5.689 \times 10^5 \text{ N/m}^2 \end{aligned}$$

Pressure intensity transmitted to the large piston = $5.689 \times 10^5 \text{ N/m}^2$

$$\begin{aligned}\text{Force on the large piston} &= \text{Pressure intensity} \times \text{area of large piston} \\ &= 5.689 \times 10^5 \times 7.854 \times 10^{-3} = 4468 \text{ N}\end{aligned}$$

Hence, *load lifted by the large piston* = **4468 N (Ans.)**

2.4. ABSOLUTE AND GAUGE PRESSURES

Atmospheric pressure:

The atmospheric air exerts a normal pressure upon all surfaces with which it is in contact, and it is known as *atmospheric pressure*. The atmospheric pressure is also known as '*Barometric pressure*'.

The atmospheric pressure at sea level (above absolute zero) is called '*Standard atmospheric pressure*'.

Note. The local atmospheric pressure may be a little lower than these values if the place under question is higher than sea level, and higher values if the place is lower than sea level, due to the corresponding decrease or increase of the column of air standing, respectively.

Gauge pressure:

It is the pressure, measured with the help of pressure measuring instrument, in which the atmospheric pressure is taken as datum. The atmospheric pressure on the scale is marked as zero.

Gauges record pressure above or below the local atmospheric pressure, since they measure the difference in pressure of the liquid to which they are connected and that of surrounding air. If the pressure of the liquid is *below* the local atmospheric pressure, then the gauge is designated as '*vacuum gauge*' and the recorded value indicates the amount by which the pressure of the liquid is *below* local atmospheric pressure, i.e. *negative pressure*.

(*Vacuum pressure* is defined as the pressure *below the atmospheric pressure*).

Absolute pressure:

It is necessary to establish an absolute pressure scale which is independent of the changes in atmospheric pressure. A pressure of absolute zero can exist only in complete vacuum.

Any pressure measured above the absolute zero of pressure is termed as an 'absolute pressure'.

A schematic diagram showing the gauge pressure, vacuum pressure and the absolute pressure is given in Fig. 2.6.

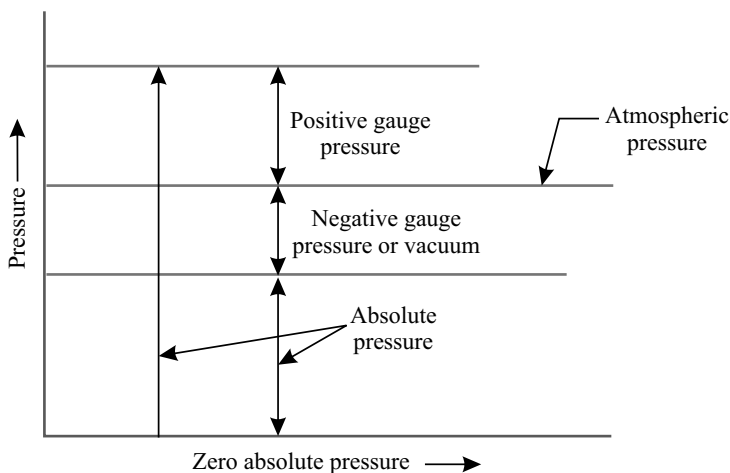


Fig. 2.6. Relationship between pressures.

Mathematically:

1. *Absolute pressure* = Atmospheric pressure + gauge pressure

$$\text{i.e., } p_{abs} = p_{atm} + p_{gauge}$$

2. *Vacuum pressure* = Atmospheric pressure – absolute pressure

Units for pressure:

The fundamental S.I. unit of pressure is newton per square metre (N/m^2). This is also known as *Pascal*.

Low pressures are often expressed in terms of mm of water or mm of mercury. This is an abbreviated way of saying that the pressure is such that will support a liquid column of stated height.

Note. When the local atmospheric pressure is not given in a problem, it is taken as 100 kN/m^2 or 10 m of water for simplicity of calculations.

Standard atmospheric pressure has the following equivalent values:

101.3 kN/m^2 or 101.3 kPa ; 10.3 m of water; 760 mm of mercury; 1013 mb (millibar) $\approx 1 \text{ bar}$
 $\approx 100 \text{ kPa} = 10^5 \text{ N/m}^2$.

Example 2.5. *Given that:*

Barometer reading = 740 mm of mercury;

Specific gravity of mercury = 13.6; Intensity of pressure = 40 kPa.

Express the intensity of pressure in S.I. units, both gauge and absolute.

Solution. Intensity of pressure, $p = 40 \text{ kPa}$

Gauge pressure:

$$(i) \ p = 40 \text{ kPa} = 40 \text{ kN/m}^2 = 0.4 \times 10^5 \text{ N/m}^2 = \mathbf{0.4 \text{ bar (Ans.)}}$$

$$(1 \text{ bar} = 10^5 \text{ N/m}^2)$$

$$(ii) \ h = \frac{p}{w} = \frac{0.4 \times 10^5}{9.81 \times 10^3} = \mathbf{4.077 \text{ m of water (Ans.)}}$$

$$(iii) \ h = \frac{p}{w} = \frac{0.4 \times 10^5}{9.81 \times 10^3 \times 13.6} = \mathbf{0.299 \text{ m of mercury (Ans.)}}$$

$$\left[\begin{array}{l} \text{Where, } w = \text{specific weight;} \\ \text{For water : } w = 9.81 \text{ kN/m}^3 \\ \text{For mercury : } w = 9.81 \times 13.6 \text{ kN/m}^3 \end{array} \right]$$

Absolute pressure:

Barometer reading (atmospheric pressure)

$$= 740 \text{ mm of mercury} = 740 \times 13.6 \text{ mm of water}$$

$$= \frac{740 \times 13.6}{1000} = 10.6 \text{ m of water}$$

$$\text{Absolute pressure } (p_{abs.}) = \text{Atmospheric pressure } (p_{atm.}) + \text{gauge pressure } (p_{gauge}).$$

$$\begin{aligned} \therefore p_{abs} &= 10.06 + 4.077 = \mathbf{14.137 \text{ m of water (Ans.)}} \\ &= 14.137 \times (9.81 \times 10^3) = \mathbf{1.38 \times 10^5 \text{ N/m}^2 \text{ (Ans.)}} \quad (p = wh) \\ &= \mathbf{1.38 \text{ bar (Ans.)}} \quad (1 \text{ bar} = 10^5 \text{ N/m}^2) \\ &= \frac{14.137}{13.6} = \mathbf{1.039 \text{ m of mercury. (Ans.)}} \end{aligned}$$

Example 2.6. *Calculate the pressure at a point 5 m below the free water surface in a liquid that has a variable density given by relation:*

$$\rho = (350 + Ay) \text{ kg/m}^3$$

where, $A = 8 \text{ kg/m}^4$ and y is the distance in metres measured from the free surface.

Solution. As per hydrostatic equation

$$dp = \rho \cdot g \cdot dy = g(350 + Ay)dy$$

Integrating both sides, we get:

$$\int dp = \int_0^5 g(350 + Ay) dy = g \int_0^5 (350 + 8y) dy$$

$$p = g \left[350y + 8 \times \frac{y^2}{2} \right]_0^5$$

$$= 9.81 \left(350 \times 5 + 8 \times \frac{5^2}{2} \right) = 18148 \text{ N/m}^2 \approx \mathbf{18.15 \text{ kN/m}^2}$$

(Ans.)

Example 2.7. On the suction side of a pump a gauge shows a negative pressure of 0.35 bar. Express this pressure in terms of:

- (i) Intensity of pressure, kPa,
- (ii) N/m^2 absolute,
- (iii) Metres of water gauge,
- (iv) Metres of oil (specific gravity 0.82) absolute, and
- (v) Centimetres of mercury gauge,

Take atmospheric pressure as 76 cm of Hg and relative density of mercury as 13.6.

Solution. Given: Reading of the vacuum gauge = 0.35 bar

(i) Intensity of pressure, kPa:

$$\begin{aligned} \text{Gauge reading} &= 0.35 \text{ bar} = 0.35 \times 10^5 \text{ N/m}^2 \\ &= 0.35 \times 10^5 \text{ Pa} = \mathbf{35 \text{ kPa (Ans.)}} \end{aligned}$$

(ii) N/m^2 absolute:

Atmospheric pressure, $p_{\text{atm.}} = 76 \text{ cm of Hg}$

$$= (13.6 \times 9810) \times \frac{76}{100} = 101396 \text{ N/m}^2$$

Absolute pressure = Atmospheric pressure – Vacuum pressure

$$\begin{aligned} p_{\text{abs.}} &= p_{\text{atm.}} - p_{\text{vac.}} \\ &= 101396 - 35000 = \mathbf{66396 \text{ N/m}^2 \text{ absolute (Ans.)}} \end{aligned}$$

(iii) Metres of water gauge:

$$p = \rho gh = wh$$

$$\therefore h_{\text{water}} (\text{gauge}) = \frac{p}{w} = \frac{0.35 \times 10^5}{9810} = \mathbf{3.567 \text{ m (gauge) (Ans.)}}$$

(iv) Metres of oil (sp. gr. = 0.82) absolute:

$$h_{\text{oil}} (\text{absolute}) = \frac{66396}{0.82 \times 9810} = \mathbf{8.254 \text{ m of water (absolute) (Ans.)}}$$

(v) Centimetres of mercury gauge:

$$h_{\text{mercury}} (\text{gauge}) = \frac{0.35 \times 10^5}{13.6 \times 9810} = 0.2623 \text{ m of mercury}$$

$$= \mathbf{26.236 \text{ cm of mercury (Ans.)}}$$

Example 2.8. The inlet to pump is 10.5 m above the bottom of sump from which it draws water through a suction pipe. If the pressure at the pump inlet is not to fall below 28 kN/m^2 absolute, work out the minimum depth of water in the tank.

Assume atmospheric pressure as 100 kPa .

Solution. Given: $p_{\text{atm.}} = 100 \text{ kPa} = 100 \text{ kN/m}^2$; $p_{\text{abs.}} = 28 \text{ kN/m}^2$.

Minimum depth of water in the tank:

Let, $p_{\text{vac.}} =$ The vacuum (suction) pressure at the pump inlet.

Then, $p_{\text{vac.}} = p_{\text{atm.}} - p_{\text{abs.}}$
 $= (100 - 28) = 72 \text{ kN/m}^2$ or 72000 N/m^2

Further, let h be the distance between the pump inlet and free water surface in the sump.

Invoking hydrostatic equation, we have:

$$p = wh$$

$$72000 = 9810 \times h$$

or,
$$h = \frac{72000}{9810} = 7.339 \text{ m}$$

$\therefore$ Minimum depth of water in the tank
 $= 10.5 - 7.339 = \mathbf{3.161 \text{ m (Ans.)}}$

Example 2.9. A cylindrical tank of cross-sectional area 600 mm^2 and 2.6 m height is filled with water upto a height of 1.5 m and remaining with oil of specific gravity 0.78 . The vessel is open to atmospheric pressure. Calculate:

- Intensity of pressure at the interface.
- Absolute and gauge pressures on the base of the tank in terms of water head, oil head and N/m^2 .
- The net force experienced by the base of the tank.

Assume atmospheric pressure as 1.0132 bar .

Solution. Given: Area of cross-section of the tank, $A = 600 \text{ mm}^2 = 600 \times 10^{-6}$; Sp.gr. of oil $= 0.78$; $p_{\text{atm.}} = 1.0132 \text{ bar}$.

(i) **Intensity of pressure at the interface:**

The pressure intensity at the interface between the oil and water is due to 1.1 m of oil and is given by:

$$p_{\text{interface}} = wh$$

$$= (0.78 \times 9810) \times 1.1$$

$$= \mathbf{8417 \text{ N/m}^2 \text{ (Ans.)}}$$

(ii) **Absolute and gauge pressures on the base of the tank:**

Pressure at the base of the tank

= Pressure at the interface (due to 1.1 m of oil) + pressure due to 1.5 m of water,

$$\text{i.e., } p_{\text{base (gauge)}} = 8417 + (9810 \times 1.5)$$

$$= \mathbf{23132 \text{ N/m}^2 \text{ (gauge) (Ans.)}}$$

$$= \frac{23132}{9810} = \mathbf{2.358 \text{ m of water (gauge) (Ans.)}}$$

$$= \frac{23132}{0.78 \times 9810} = \mathbf{3.023 \text{ m of oil (gauge) (Ans.)}}$$

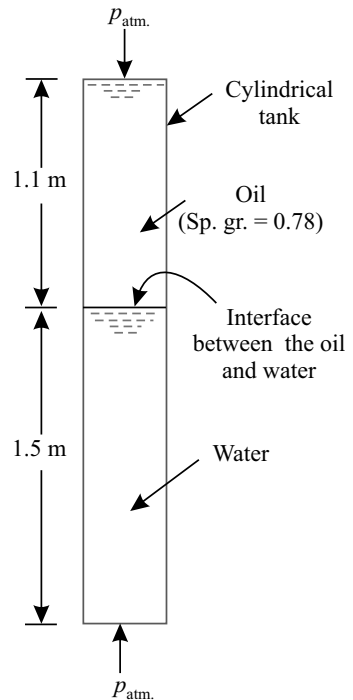


Fig. 2.7

$$\begin{aligned}
 \text{Atmospheric pressure, } p_{\text{atm.}} &= 1.0132 \text{ bar} \\
 &= 1.0132 \times 10^5 \text{ N/m}^2 \\
 &= \frac{1.0132 \times 10^5}{9810} = 10.328 \text{ m of water} \\
 &= \frac{1.0132 \times 10^5}{0.78 \times 9810} = 13.241 \text{ m of oil}
 \end{aligned}$$

Absolute pressure = Atmospheric pressure + gauge pressure

$$\begin{aligned}
 p_{\text{base}} (\text{absolute}) &= 10.328 + 2.358 = \mathbf{12.686 \text{ m of water (Ans.)}} \\
 &= 13.241 + 3.023 = \mathbf{16.264 \text{ m of oil (Ans.)}} \\
 &= 101320 + 23132 = \mathbf{124452 \text{ N/m}^2 (\text{Ans.})}
 \end{aligned}$$

(iii) The net force experienced by the base of the tank:

$$\begin{aligned}
 F (= P) &= p_{\text{base}}(\text{gauge}) \times \text{cross-sectional area} \\
 &= 23132 \times 600 \times 10^{-6} = \mathbf{13.879 \text{ N (Ans.)}}
 \end{aligned}$$

Example 2.10. (a) What is hydrostatic paradox?

(b) A cylinder of 0.25 m diameter and 1.2 m height is fixed centrally on the top of a large cylinder of 0.9 m diameter and 0.8 m height. Both the cylinders are filled with water. Calculate:

- (i) Total pressure at the bottom of the bigger cylinder, and
- (ii) Weight of total volume of water.

What is hydrostatic paradox between the two results and how this difference can be reconciled?

Solution. (a) Hydrostatic paradox:

Fig. 2.8 shows three vessels 1, 2 and 3 having the same area A at the bottom and each filled with a liquid upto the same height h .

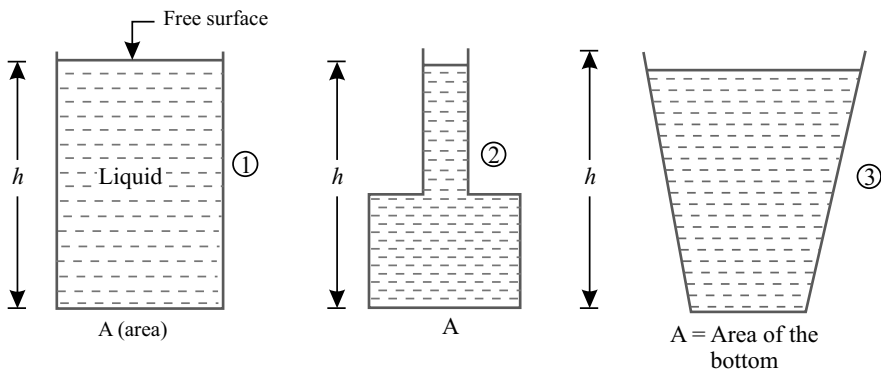


Fig. 2.8. Hydrostatic paradox.

According to the hydrostatic equation, $p = wh$; the intensity of pressure (p) depends only on the height of the column and not at all upon the size of the column. Thus, in all these vessels of different shapes and sizes, the same intensity of pressure would be exerted on the bottom of each of these vessels. Since each of the vessels has the same area A at the bottom, the pressure force $P = p \times A$ on the base of each vessel would be same. This is independent of the fact that the weight of liquid in each vessel is different. This situation is referred to as **hydrostatic paradox**.

(b) Area at the bottom:

$$A = \frac{\pi}{4} \times (0.9)^2 = 0.6362 \text{ m}^2$$

Intensity of pressure at the bottom

$$p = wh = 9810 \times (1.2 + 0.8) \\ = 19620 \text{ N/m}^2$$

Total pressure force at the bottom

$$P = p \times A = 19620 \times 0.6362 = 12482 \text{ N}$$

Weight of total volume of water contained in the cylinders,

$$W = w \times \text{volume of water} \\ = 9810 \left[\frac{\pi}{4} \times 0.9^2 \times 0.8 + \frac{\pi}{4} \times 0.25^2 \times 1.2 \right] \\ = 5571 \text{ N}$$

From the above calculations it may be observed that the *total pressure force at the bottom of the cylinder is greater than the weight of total volume of water (W) contained in the cylinders. This is hydrostatic paradox.*

The following is the *explanation of the hydrostatic paradox*: Refer to Fig. 2.9.

Total pressure force on the bottom of bigger tank = 12482 N (downward). A reaction at the roof of the lower tank is caused by the upward force which equals,

$$wAh = 9810 \times \frac{\pi}{4} (0.9^2 - 0.25^2) \times 1.2 = 6911 \text{ N (upward)}$$

The distance h corresponding to depth of water in the cylinder fixed centrally on the top of larger cylinder.

Net downward force exerted by water = $12482 - 6911 = 5571 \text{ N}$ and it equals the weight of water in the two cylinder.

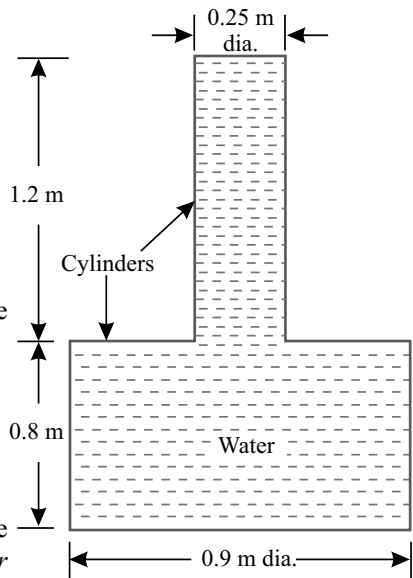


Fig. 2.9

2.5. MEASUREMENT OF PRESSURE

The pressure of a fluid may be measured by the following devices:

1. Manometers:

Manometers are defined as the devices used for measuring the pressure at a point in a fluid by balancing the column of fluid by the same or another column of liquid. These are classified as follows:

(a) Simple manometers:

- (i) Piezometer, (ii) U-tube manometer, and (iii) Single column manometer.

(b) Differential manometers.

2. Mechanical gauges:

These are the devices in which the pressure is measured by balancing the fluid column by spring (elastic element) or dead weight. Generally these gauges are used for measuring *high pressure* and where *high precision is not required*. Some commonly used mechanical gauges are:

- (i) Bourdon tube pressure gauge, (ii) Diaphragm pressure gauge,
(iii) Bellow pressure gauge, and (iv) Dead-weight pressure gauge.

2.5.1 Manometers

2.5.1.1. Simple manometers

A “simple manometer” is one which consists of a glass tube whose one end is connected to a point where pressure is to be measured and the other end remains open to atmosphere.

Common types of simple manometers are discussed below:

1. Piezometer:

A piezometer is the simplest form of manometer which can be used for measuring *moderate pressures* of liquids. It consists of a glass tube (Fig 2.10) inserted in the wall of a vessel or of a pipe, containing liquid whose pressure is to be measured. The tube extends vertically upward to such a height that liquid can freely rise in it without overflowing. The pressure at any point in the liquid is indicated by the height of the liquid in the tube above that point, which can be read on the scale attached to it. Thus if w is the specific weight of the liquid, then the pressure at point $A(p)$ is given by:

$$p = wh$$

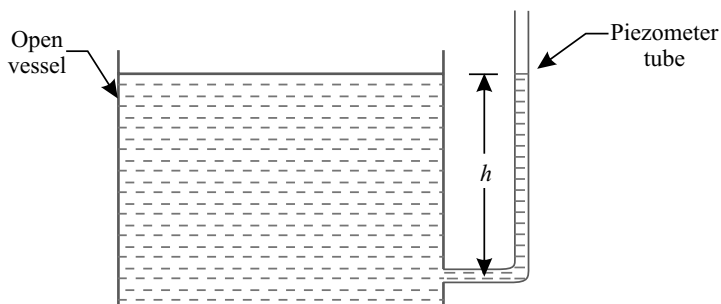


Fig. 2.10. (a) Piezometer tube fitted to open vessel.

Piezometers measure *gauge pressure only* (at the surface of the liquid), since the surface of the liquid in the tube is subjected to atmospheric pressure. A piezometer tube is *not suitable* for measuring *negative pressure*; as in such a case the air will enter in pipe through the tube.

2. U-tube manometer:

Piezometers cannot be employed when large pressures in the *lighter liquids* are to be measured, since this would require *very long tubes*, which cannot be handled conveniently. Furthermore gas pressures cannot be measured by the piezometers because a *gas forms no free atmospheric surface*. These limitations can be overcome by the use of U-tube manometers.

A U-tube manometer consists of a glass tube bent in U-shape, one end of which is connected to a point at which pressure is to be measured and other end remains open to the atmosphere as shown in Fig. 2.11.

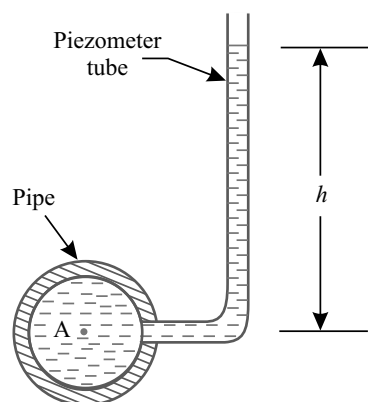


Fig. 2.10. (b) Piezometer tube fitted to a closed pipe.

(i) For positive pressure:

Refer to Fig. 2.11 (a).

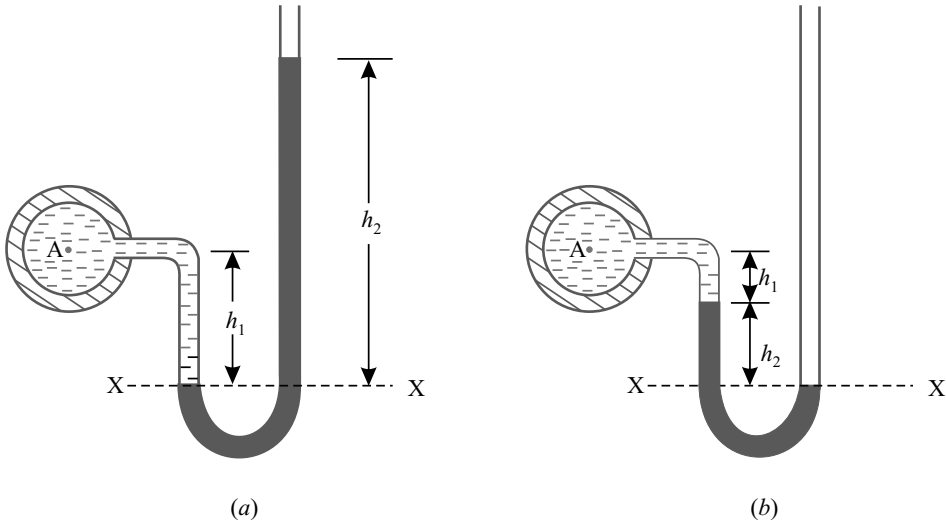


Fig. 2.11. U-tube manometer.

Let, A be the point at which pressure is to be measured. $X-X$ is the datum line as shown in Fig. 2.11 (a).

- Let, h_1 = Height of the light liquid in the left limb above the datum line,
 h_2 = Height of the heavy liquid in the right limb above the datum line,
 h = Pressure in pipe, expressed in terms of head,
 S_1 = Specific gravity of the light liquid, and
 S_2 = Specific gravity of the heavy liquid.

The pressures in the left limb and right limb above the datum line $X-X$ are equal (*as the pressures at two points at the same level in a continuous homogeneous liquid are equal*).

Pressure head above $X-X$ in the *left limb* = $h + h_1 S_1$

Pressure head above $X-X$ in the *right limb* = $h_2 S_2$

Equating these two pressures, we get:

$$h + h_1 S_1 = h_2 S_2 \quad \text{or} \quad h = h_2 S_2 - h_1 S_1 \quad \dots(2.6)$$

(ii) For negative pressure:

Refer to Fig. 2.11 (b).

Pressure head above $X-X$ in the *left limb* = $h + h_1 S_1 + h_2 S_2$

Pressure head above $X-X$ in the *right limb* = 0.

Equating these two pressures, we get:

$$h + h_1 S_1 + h_2 S_2 = 0 \quad \text{or} \quad h = -(h_1 S_1 + h_2 S_2) \quad \dots(2.7)$$

Example 2.11. In a pipeline water is flowing. A manometer is used to measure the pressure drop for flow through the pipe. The difference in level was found to be 20 cm. If the manometric fluid is CCl_4 , find the pressure drop in S.I units (density of $\text{CCl}_4 = 1.596 \text{ g/cm}^3$). If the manometric fluid is changed to mercury ($\rho = 13.6 \text{ gm/cm}^3$) what will be the difference in level?

(UPTU)

Solution. Given: $h_{\text{CCl}_4} = 20 \text{ cm} = 0.2 \text{ m}$; $\rho_{\text{CCl}_4} = 1.596 \text{ g/cm}^3$

$$= 1.596 \times 10^3 \text{ kg/m}^3$$

$$\rho_{Hg} = 13.6 \times 10^3 \text{ kg/m}^3$$

$$\begin{aligned} \text{Pressure drop, } \Delta p &= \rho_{CCl_4} g h_{CCl_4} \\ &= 1.596 \times 10^3 \times 9.81 \times 0.2 \text{ N/m}^2 \\ &= 3131.3 \text{ N/m}^2 \text{ or Pa} = \mathbf{3.131 \text{ kPa (Ans.)}} \end{aligned}$$

The difference in level with mercury,

$$\begin{aligned} h_{Hg} &= h_{CCl_4} \times \frac{\rho_{CCl_4}}{\rho_{Hg}} = 0.20 \times \frac{1.596 \times 10^3}{13.6 \times 10^3} \\ &= 0.02347 \text{ m or } \mathbf{2.347 \text{ cm (Ans.)}} \end{aligned}$$

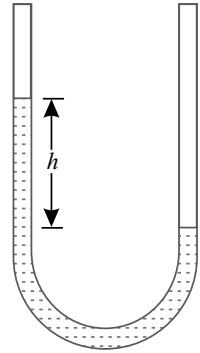


Fig. 2.12

Example 2.12. A U-tube manometer is used to measure the pressure of oil of specific gravity 0.85 flowing in a pipe line. Its left end is connected to the pipe and the right-limb is open to the atmosphere. The centre of the pipe is 100 mm below the level of mercury (specific gravity = 13.6) in the right limb. If the difference of mercury level in the two limbs is 160 mm, determine the absolute pressure of the oil in the pipe.

Solution. Specific gravity of oil, $S_1 = 0.85$

Specific gravity of mercury, $S_2 = 13.6$

Height of the oil in the left limb,

$$h_1 = 160 - 100 = 60 \text{ mm} = 0.06 \text{ m}$$

Difference of mercury level,

$$h_2 = 160 \text{ mm} = 0.16 \text{ m.}$$

Absolute pressure of oil:

Let, h_1 = Gauge pressure in the pipe in terms of head of water, and

p = Gauge pressure in terms of kN/m^2 .

Equating the pressure heads above the datum line $X-X$, we get:

$$h + h_1 S_1 = h_2 S_2$$

$$\text{or, } h + 0.06 \times 0.85 = 0.16 \times 13.6 = 2.125 \text{ m}$$

The pressure p is given by:

$$\begin{aligned} p &= wh \\ &= 9.81 \times 2.125 \text{ kN/m}^2 \\ &= 20.84 \text{ kPa} \quad (\because w = 9.81 \text{ kN/m}^3 \text{ in S.I. units}) \end{aligned}$$

Absolute pressure of oil in the tube,

$$\begin{aligned} P_{abs.} &= P_{atm.} + P_{gauge} \\ &= 100 + 20.84 = \mathbf{120.84 \text{ kPa (Ans.)}} \end{aligned}$$

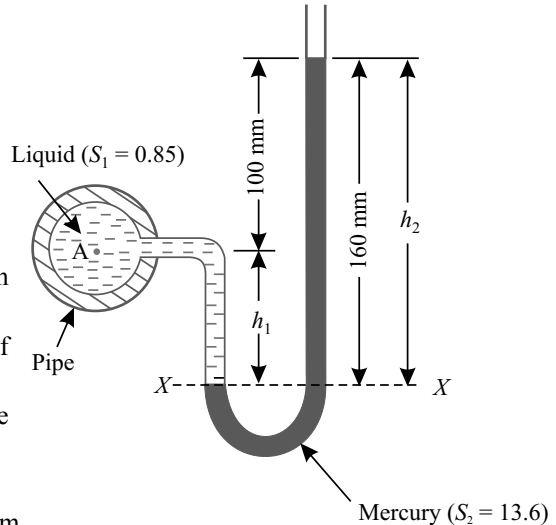


Fig. 2.13

Example 2.13. U-tube manometer containing mercury was used to find the negative pressure in the pipe, containing water. The right limb was open to the atmosphere. Find the vacuum pressure in the pipe, if the difference of mercury level in the two limbs was 100 mm and height of water in the left limb from the centre of the pipe was found to be 40 mm below.

Solution. Specific gravity of water, $S_1 = 1$

Specific gravity of mercury, $S_2 = 13.6$

Height of water in the left limb,

$$h_1 = 40 \text{ mm} = 0.04 \text{ m}$$

Height of mercury in the left limb,

$$h_2 = 100 \text{ mm} = 0.1 \text{ m}$$

Let, h = Pressure in the pipe in terms of head of water (*below* the atmosphere).

Equating the pressure heads above the datum line

$X-X$, we get:

$$h + h_1 S_1 + h_2 S_2 = 0$$

$$\begin{aligned} \text{or, } h &= -(h_1 S_1 + h_2 S_2) \\ &= -(0.04 \times 1 + 0.1 \times 13.6) \\ &= -1.4 \text{ m of water} \end{aligned}$$

Pressure p is given by:

$$\begin{aligned} p &= wh \\ &= 9.81 \times (-1.4) \text{ kN/m}^2 \\ &= -13.73 \text{ kPa} \\ &= \mathbf{13.73 \text{ kPa (vacuum) (Ans.)}} \end{aligned}$$

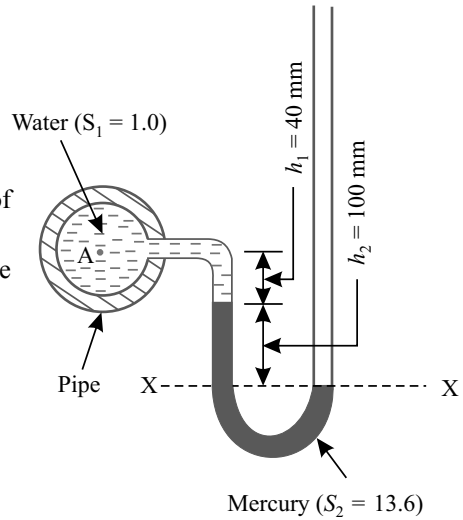


Fig. 2.14

Example 2.14. A simple U-tube manometer is installed across an orificemeter. The manometer is filled with mercury (sp. gravity = 13.6) and the liquid above the mercury is carbon tetrachloride (sp. gravity = 1.6). The manometer reads 200 mm. What is the pressure difference over the manometer in newtons per square metre.

Solution. Specific gravity of heavier liquid, $S_{hl} = 13.6$

Specific gravity of lighter liquid, $S_{ll} = 1.6$

Reading of the manometer, $y = 200 \text{ mm}$

Pressure difference over the manometer : p

Differential head,

$$\begin{aligned} h &= y \left[\frac{S_{hl}}{S_{ll}} - 1 \right] \\ 200 \left[\frac{13.6}{1.6} - 1 \right] &= 1500 \text{ mm of carbon tetrachloride} \end{aligned}$$

Pressure difference over manometer,

$$p = wh = (1.6 \times 9810) \times \left(\frac{1500}{1000} \right)$$

$$\text{or } p = \mathbf{23544 \text{ N/m}^2 \text{ (Ans.)}}$$

Example 2.15. In Fig. 2.15 is shown a conical vessel having its outlet at L to which U-tube manometer is connected. The reading of the manometer given in figure shows when the vessel is empty. Find the reading of the manometer when the vessel is completely filled with water.

Solution. When vessel is empty: (Refer to Fig. 2.15)

Let, h_1 = Height of water above $X-X$

Specific gravity of water, $S_1 = 1.0$

Specific gravity of mercury, $S_2 = 13.6$

Equating the pressure heads about the datum line $X-X$, we get:

$$h_1 S_1 = h_2 S_2 \quad \text{or} \quad h_1 \times 1.0 = 150 \times 13.6 \quad \text{or} \quad h_1 = 2040 \text{ mm}$$

When vessel is full of water:

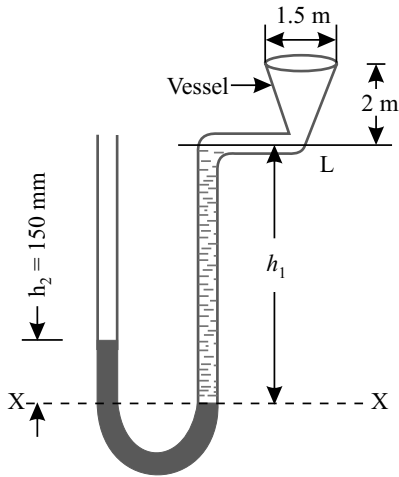


Fig. 2.15. Vessel is empty.

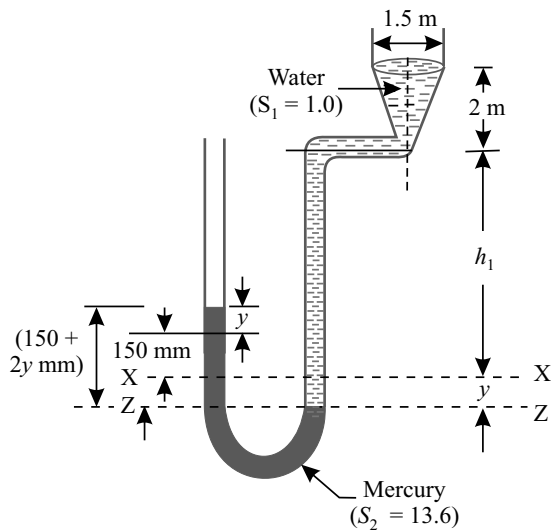


Fig. 2.16. Vessel is full of water.

Refer to Fig. 2.16. Consider the vessel to be completely filled with water. As a result of this let the mercury level go down by y mm in the right limb, and the mercury level go up by the same amount in the left limb. Now the datum line is $Z-Z$.

Equating the pressure heads above the datum line $Z-Z$, we get:

$$(150 + 2y) \times 13.6 = (h_1 + y + 2000) \times 1$$

$$\text{or,} \quad 150 \times 13.6 + 2y \times 13.6 = 2040 + y + 2000 \quad [\because h_1 = 2040 \text{ mm, calculated earlier}]$$

$$\text{or,} \quad 2040 + 27.2y = 4040 + y = 76.3 \text{ mm}$$

Thus the reading of the manometer when the vessel is completely filled with water

$$= (150 + 2y) = 150 + 2 \times 76.3 = 302.6 \text{ mm}$$

Hence, reading of the manometer 302.6 mm or **0.3026 m (Ans.)**

Example. 2.16. Fig. 2.17 shows a pressure gauge with the following particulars:

Cross-sectional area of each of the bulbs L and $M = 1200 \text{ mm}^2$;

Cross-sectional area of each vertical limb $= 30 \text{ mm}^2$;

Specific gravity of the liquid filled in bulb $M = 0.9$;

If the surface of separation is in the limb attached to M find the displacement of surface of separation when the pressure on the surface in M is greater than that in L by an amount equal to 20 mm of head of water.

Solution.

Cross-sectional area of each vertical limb, $a = 30 \text{ mm}^2$

Specific gravity of water, $S_1 = 1.0$

Specific gravity of the liquid, $S_2 = 0.9$.

Let,

$X-X$ = Initial level of separation,

h_L = Height of water above $X-X$, and

h_M = Height of liquid ($S_2 = 0.9$) above $X-X$.

Pressure head above $X-X$ in the left limb $= h_L$

Pressure head above $X-X$ in the right limb $= S_2 h_M = 0.9 h_M$

Equating the pressure heads above $X-X$, we get:

$$h_L = 0.9 h_M \quad \dots(i)$$

When the pressure on the surface in bulb M is increased by 20 mm of water, let the separation level fall by an amount equal to y . Then $Z-Z$ is the *new separation level*.

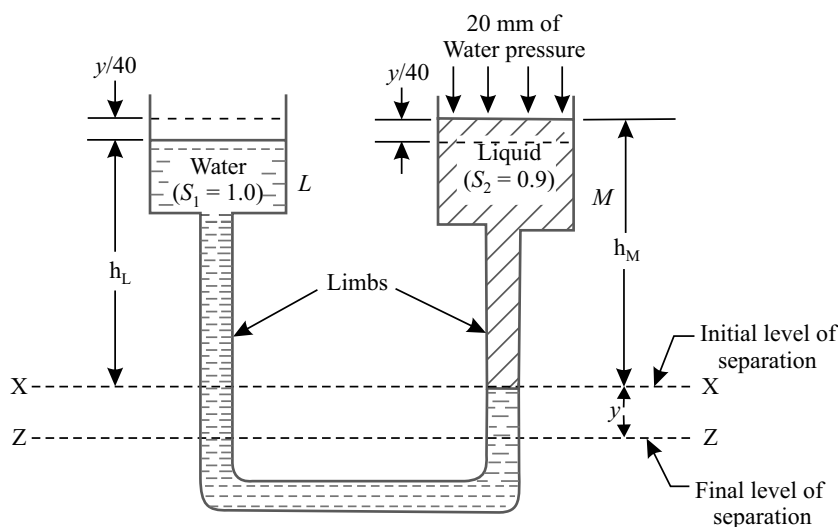


Fig. 2.17

Now, $A \times$ fall in separation level in bulb $M = a \times$ fall in separation level in the limb (y).

$$\text{Fall in separation level in bulb } M = \frac{a \times y}{A} = \frac{30 \times y}{1200} = \frac{y}{40}$$

Also, fall in separation level in bulb $M = \text{Rise in surface level of } L = \frac{y}{40}$

Considering pressure heads above $Z-Z$, we have:

$$\text{Pressure head in the left limb} = \left[\frac{y}{40} + h_L + y \right]$$

$$\text{Pressure head in the right limb} = \left(h_M + y - \frac{y}{40} \right) \times 0.9 + 20$$

Equating the pressure heads, we get:

$$\left[\frac{y}{40} + h_L + y \right] = \left[h_M + y - \frac{y}{40} \right] \times 0.9 + 20$$

$$\text{or,} \quad \frac{y}{40} + 0.9 h_M + y = 0.9 h_M + \frac{39y}{40} \times 0.9 + 20 \quad (\because h_L = 0.9 h_M)$$

$$\text{or,} \quad \frac{41y}{40} = \frac{39y}{40} \times 0.9 + 20 \quad \text{or} \quad \frac{41y}{40} - \frac{39y}{40} \times 0.9 = 20$$

$$\text{or,} \quad 1.025y - 0.877y = 20 \quad \text{or} \quad y = 135.1 \text{ mm}$$

Hence, *displacement of the surface of separation* = **135.1 mm (Ans.)**

3. Single column manometer (micro-manometer):

The U-tube manometer described above usually requires reading of fluid levels at two or more points since a change in pressure causes a rise of liquid in one limb of the manometer and a drop in the other. This difficulty is however overcome by using single column manometers. A single column

manometer is a modified form of a U-tube manometer in which a shallow reservoir having a large cross-sectional area (about 100 times) as compared to the area of the tube is connected to one limb of the manometer, as shown in Fig. 2.18. For any variation in pressure, the change in the liquid level in the reservoir will be so small that it may be neglected, and the pressure is indicated by the height of the liquid in the other limb. As such only one reading in the narrow limb of the manometer need be taken for all pressure measurements. The narrow limb may be vertical or inclined. Thus there are two types of single column manometer as given below:

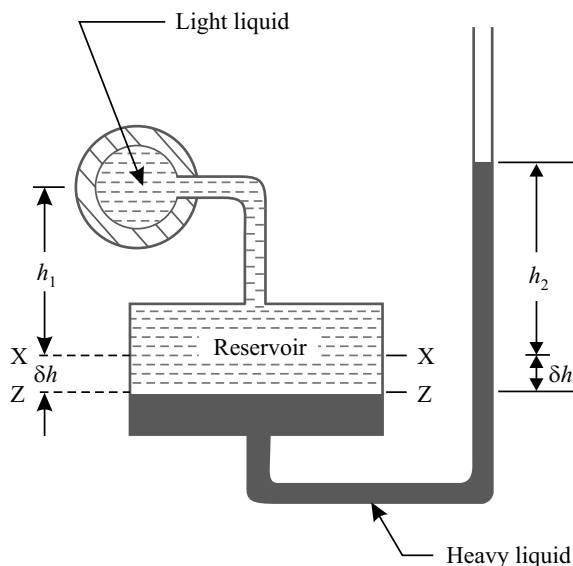


Fig. 2.18. Vertical single column manometer.

(a) Vertical single column manometer, and

(b) Inclined single column manometer.

(a) Vertical single column manometer:

Refer to Fig. 2.18

Let $X-X$ be the datum line in the reservoir when the single column manometer is not connected to the pipe. Now consider that the manometer is connected to a pipe containing light liquid under a very high pressure. The pressure in the pipe will force the light liquid to push the heavy liquid in the reservoir downwards. As the area of the reservoir is very large, the fall of the heavy liquid level will be very small. This downward movement of the heavy liquid, in the reservoir, will cause a considerable rise of the heavy liquid in the right limb.

Let,

h_1 = Height of the centre of the pipe above $X-X$,

h_2 = Rise of heavy liquid (after experiment) in the right limb,

δh = Fall of heavy liquid level in the reservoir,

h = Pressure in the pipe, expressed in terms of head of water,

A = Cross-sectional area of the reservoir,

a = Cross-sectional area of the tube (right limb),

S_1 = Specific gravity of light liquid in pipe, and

S_2 = Specific gravity of the heavy liquid.

We know that fall of heavy liquid in reservoir will cause a rise of heavy liquid level in the right limb.

Thus, $A \times \delta h = a \times h_2$ or $\delta h = \frac{a \times h_2}{A}$... (i)

Let us now consider pressure heads above the datum line $Z-Z$ as shown in Fig. 2.18.

Pressure head in the left limb $= h + (h_1 + \delta h)S_1$

Pressure head in the right limb $= (h_2 + \delta h)S_2$

Equating the pressure heads, we get:

$$h + (h_1 + \delta h)S_1 = (h_2 + \delta h)S_2 \text{ or } h = (h_2 + \delta h)S_2 - (h_1 + \delta h)S_1$$

$$= \delta h(S_2 - S_1) + h_2S_2 - h_1S_1$$

But, $\delta h = \frac{a \times h_2}{A}$... [Eqn. (i)]

$$h = \frac{a \times h_2}{A} (S_2 - S_1) + h_2S_2 - h_1S_1 \quad \dots (2.8)$$

When the area A is very large as compared to a , then the ratio $\frac{a}{A}$ becomes very small, and thus is neglected. Then the above equation becomes

$$h = h_2S_2 - h_1S_1 \quad \dots (2.9)$$

(b) Inclined single column manometer:

This type of manometer is useful for the measurement of *small pressures and is more sensitive than the vertical tube type*. Due to inclination the distance moved by the heavy liquid in the right limb is *more*.

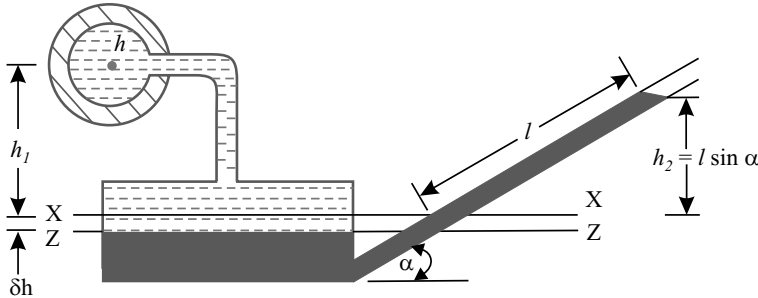


Fig. 2.19. Inclined single column manometer.

Let,

l = Length of the heavy liquid moved in right limb,

α = Inclination of right limb horizontal, and

h_2 = Vertical rise of liquid in right limb from $X-X = l \sin \alpha$.

Putting the value of h_2 in eqn. 2.9, we get:

$$h = l \sin \alpha \times S_2 - h_1S_1 \quad \dots (2.10)$$

Example. 2.17. Fig. 2.20 shows a single column manometer connected to a pipe containing liquid of specific gravity 0.8. The ratio of area of the reservoir to that of the limb is 100. Find the pressure in the pipe.

Take specific gravity of mercury as 13.6.

Solution. Specific gravity of liquid in the pipe, $S_1 = 0.8$.

Specific gravity of mercury, $S_2 = 13.6$

$$\frac{\text{Area of reservoir}}{\text{Area of right limb}} = \frac{A}{a} = 100$$

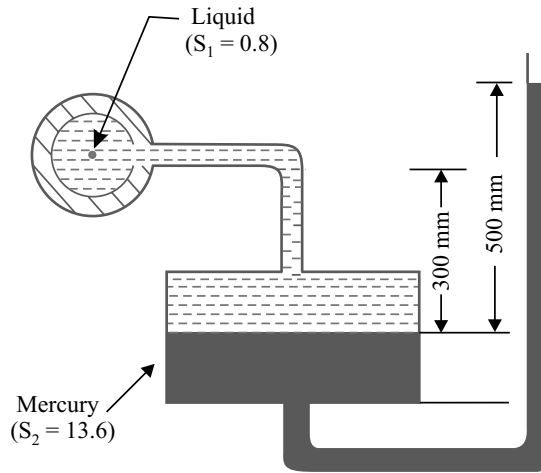


Fig. 2.20

Height of the liquid in the left limb,

$$h_1 = 300 \text{ mm}$$

Height of mercury in the right limb,

$$h_2 = 500 \text{ mm}$$

Let,

$$h = \text{Pressure head in the pipe.}$$

Using the relation:

$$h = \frac{a}{A} h_2 (S_2 - S_1) + h_2 S_2 - h_1 S_1$$

$$\text{or, } h = \frac{1}{100} \times 500 (13.6 - 0.8) + 500 \times 13.6 - 300 \times 0.8 \text{ mm of water}$$

$$= 6624 \text{ mm of water or } 6.624 \text{ m of water}$$

Pressure,

$$p = wh = 9.81 \times 6.624$$

$$= 64.98 \text{ kN/m}^2 \text{ or } 64.98 \text{ kPa}$$

i.e.,

$$p = \mathbf{64.98 \text{ kPa (Ans.)}}$$

Example 2.18. A manometer consists of an inclined glass tube which communicates with a metal cylinder standing upright; liquid fills the apparatus to a fixed zero mark on the tube when both the cylinder and the tube are open to atmosphere. The upper end of the cylinder is then connected to a gas supply at a pressure p and manometric liquid rises through a distance l in the tube. Establish the relation:

$$h = Sl \left[\sin \alpha + \left(\frac{d}{D} \right)^2 \right]$$

for the pressure head h of water column in terms of inclination α of the tube, specific gravity S of the liquid, and ratio of diameter d of the tube to the diameter D of the cylinder.

Also determine the value of $\left(\frac{D}{d} \right)$ so that the error due to disregarding the change in level in the cylinder will not exceed 0.1 percent when $\alpha = 25^\circ$.

Solution. Vertical rise in the tube $= l \sin \alpha$

$$\text{Fall of liquid level in the cylinder} = l \times \frac{a}{A} = l \times \left(\frac{d}{D} \right)^2$$

∴ Net change in level of manometric liquid due to the applied pressure

$$= l \sin \alpha + l \times \left(\frac{d}{D} \right)^2 = l \left[\sin \alpha + \left(\frac{d}{D} \right)^2 \right]$$

Now the increase in pressure,

$p = \text{Specific weight of manometric liquid} \times \text{net change in the level of manometric head}$

$$= w_m l \left[\sin \alpha + \left(\frac{d}{D} \right)^2 \right]$$

or, in terms of water column,

$$\frac{p}{w} = \frac{w_m}{w} \times l \left[\sin \alpha + \left(\frac{d}{D} \right)^2 \right]$$

or,

$$h = S \times l \left[\sin \alpha + \left(\frac{d}{D} \right)^2 \right] \quad \text{...Proved}$$

Change in the liquid level when variation of liquid in the cylinder is *considered*

$$= l \left[\sin \alpha + \left(\frac{d}{D} \right)^2 \right]$$

Change in the liquid level when variation of liquid in the cylinder is *disregarded* $= l \sin \alpha$

$$\therefore \frac{Sl \left[\sin \alpha + \left(\frac{d}{D} \right)^2 \right] - Sl \sin \alpha}{Sl \left[\sin \alpha + \left(\frac{d}{D} \right)^2 \right]} = 0.001 \quad \text{...(Given)}$$

$$\text{or,} \quad 1 - \frac{\sin \alpha}{\sin \alpha + \left(\frac{d}{D} \right)^2} = 0.001$$

$$\text{or,} \quad 1 - \frac{\sin 25^\circ}{\sin 25^\circ + \left(\frac{d}{D} \right)^2} = 0.001$$

$$\text{or,} \quad 1 - \frac{0.4226}{0.4226 + \left(\frac{d}{D} \right)^2} = 0.001$$

$$\text{or,} \quad \frac{0.4226}{0.4226 + \left(\frac{d}{D} \right)^2} = 0.999$$

$$\text{or,} \quad \left(\frac{d}{D} \right)^2 = \frac{0.4226}{0.999} - 0.4226 = 0.000423$$

$$\text{or,} \quad \frac{d}{D} = 0.02057$$

$$\text{or,} \quad \frac{D}{d} = 48.61 \text{ (Ans.)}$$

2.5.1.2. Differential Manometers

A differential manometer is used to measure the difference in pressures between two points in a pipe, or in two different pipes. In its simplest form a differential manometer consists of a

U-tube, containing a heavy liquid, whose two ends are connected to the points, whose difference of pressures is required to be found out. Following are the most commonly used types of differential manometers:

1. U-tube differential manometer.
2. Inverted U-tube differential manometer.

1. U-tube differential manometer:

A U-tube differential manometer is shown in Fig. 2.21.

Case I. Fig. 2.21 (a) shows a differential manometer whose two ends are connected with two different points A and B at the same level and containing same liquid.

- Let,
- h = Difference of mercury levels (heavy liquid) in the U-tube,
 - h_1 = Distance of the centre of A from the mercury level in the right limb,
 - $S_1 (= S_2)$ = Specific gravity of liquid at the two points A and B
 - S = Specific gravity of heavy liquid or mercury in the U-tube,
 - h_A = Pressure head at A, and
 - h_B = Pressure head at B.

We know that the pressures in the left limb and right limb, above the datum line, are equal.
Pressure head in the *left limb*

$$= h_A + (h_1 + h) S_1$$

Pressure head in the *right limb*

$$= h_B + h_1 \times S_1 + h \times S$$

$$h_A + (h_1 + h) S_1 = h_B + h_1 S_1 + h S$$

$$\text{or, } h_A - h_B = h_1 S_1 + h S - (h_1 + h) S_1$$

$$= h_1 S_1 + h S - h_1 S_1 - h S_1 = h (S - S_1)$$

i.e., Difference of pressure head,

$$h_A - h_B = h (S - S_1) \quad \dots(2.11)$$

Case II. Fig. 2.21 (b) shows a differential manometer whose two ends are connected to two different points A and B at different levels and containing different liquids.

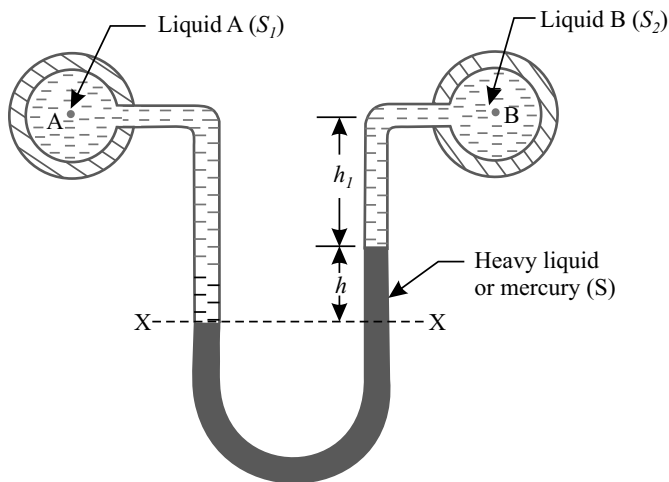


Fig. 2.21. (a) Two pipes at same level.

Let,

- h = Difference of mercury levels (heavy liquid) in the U-tube,
 h_1 = Distance of the centre of A , from the mercury level in the left limb,
 h_2 = Distance of the centre of B , from the mercury level in the right limb,
 S_1 = Specific gravity of liquid in pipe A ,
 S_2 = Specific gravity of liquid in pipe B ,
 S = Specific gravity of heavy liquid or mercury,
 h_A = Pressure head at A , and
 h_B = Pressure head at B .

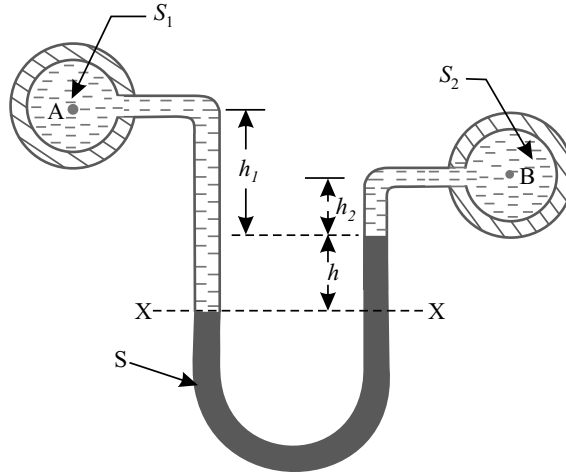


Fig. 2.21. (b) U-tube differential manometers.

Considering the pressure heads above the datum line $X-X$, we get:

$$\text{Pressure head in the left limb} = h_A + (h_1 + h) S_1$$

$$\text{Pressure head in the right limb} = h_B + h_2 \times S_2 + h \times S$$

Equating the above pressure heads, we get:

$$\begin{aligned}
 h_A + (h_1 + h) S_1 &= h_B + h_2 \times S_2 + h \times S \\
 (h_A - h_B) &= h_2 \times S_2 + h \times S - (h_1 + h) S_1 \\
 &= h_2 \times S_2 + h \times S - h_1 S_1 - h S_1 = h (S - S_1) + h_2 S_2 - h_1 S_1
 \end{aligned}$$

i.e., Difference of pressure heads at A and B ,

$$h_A - h_B = h (S - S_1) + h_2 S_2 - h_1 S_1 \quad \dots(2.12)$$

Example 2.19. A differential manometer connected at the two points A and B in a pipe containing an oil of specific gravity of 0.9 shows a difference in mercury levels as 150 mm. Find the difference in pressures at the two points.

Solution. Specific gravity of oil, $S_1 = 0.9$

Specific gravity of mercury, $S = 13.6$

Difference of mercury levels, $h = 150$ mm

Let, $h_A - h_B$ = Difference of pressures between A and B , in terms of head of water, and

$p_A - p_B$ = Difference of pressures between A and B .

Using the relation: $h_A - h_B = h (S - S_1)$

[Eqn. (2.11)]

$$= 150 (13.6 - 0.9) = 1905 \text{ mm} = \mathbf{1.905 \text{ m of water (Ans.)}}$$

Now, using the relation,

$$p_A - p_B = wh, \text{ we have, } p_A - p_B = 9.81 \times 1.905 = 18.68 \text{ kN/m}^2 = \mathbf{18.68 \text{ kPa (Ans.)}}$$

Example 2.20. Fig. 2.22 shows a U-tube differential manometer connecting two pressure pipes at A and B. The pipe A contains a liquid of specific gravity 1.6 under a pressure of 110 kN/m^2 . The pipe B contains oil of specific gravity 0.8 under a pressure of 200 kN/m^2 . Find the difference of pressure measured by mercury as fluid filling U-tube.

Solution. Specific gravity of liquid at A, $S_1 = 1.6$

Specific gravity of liquid at B, $S_2 = 0.8$

Pressure at A, $p_A = 110 \text{ kN/m}^2$

Pressure head at A,

$$h_A = \frac{p_A}{w} = \frac{110}{9.81} = 11.21 \text{ m of water}$$

Pressure at B, $p_B = 200 \text{ kN/m}^2$

Pressure head at B,

$$h_B = \frac{p_B}{w} = \frac{200}{9.81} = 20.38 \text{ m of water}$$

Taking X-X as the datum line:

Pressure head above X-X in the left limb

$$= h_A + (2.6 + 1.0) S_1 + h \times 13.6 \text{ m of water}$$

Pressure head above X-X in the right limb

$$= h_B + (1.0 + h) \times S_2 \text{ m of water}$$

Equating the above pressure heads, we get:

$$h_A + (2.6 + 1.0) S_1 + h \times 13.6 = h_B + (1.0 + h) S_2$$

$$11.21 + 5.76 + 13.6 h = 20.38 + (1.0 + h) \times 0.8$$

$$\text{or, } 16.97 + 13.6 h = 20.38 + 0.8 + 0.8 h \text{ or } 12.8 h = 4.21$$

$$\text{or, } h = 0.329 \text{ m or } \mathbf{329 \text{ mm (Ans.)}}$$

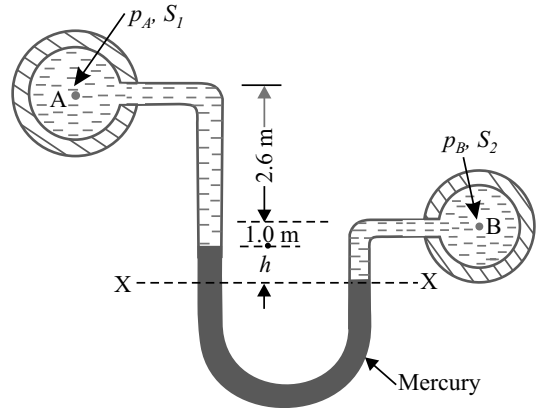


Fig. 2.22

Example 2.21. Fig. 2.23 shows a differential manometer connected at two points A and B. At A air pressure is 100 kN/m^2 . Find the absolute pressure at B.

Solution. Pressure of air at A,

$$p_A = 100 \text{ kN/m}^2$$

Pressure head at A,

$$h_A = \frac{100}{9.81} = 10.2 \text{ m}$$

Let the pressure at B is p_B .

Then, pressure head at B = $\frac{p_B}{w}$

Considering pressure heads above the datum line X-X, we have:

Pressure head in the left limb

$$= \frac{650}{1000} + h_A = 0.65 + 10.2 = 10.85 \text{ m}$$

Pressure head in the right limb

$$= h_B + \frac{250}{1000} \times 0.85 + \frac{150}{1000} \times 13.6$$

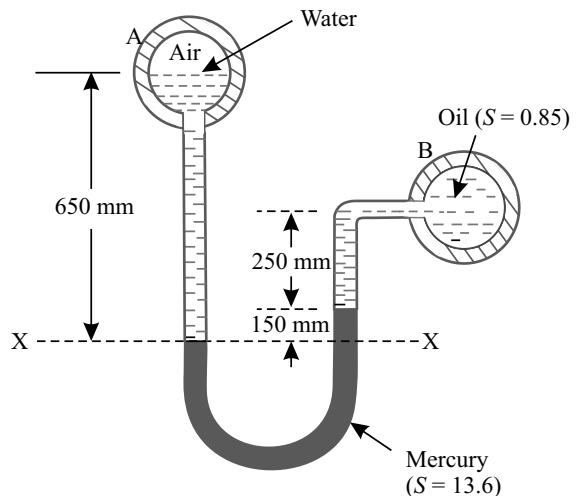


Fig. 2.23

$$= h_B + 0.212 + 2.04 = h_B + 2.25$$

Equating the above pressure heads, we get:

$$10.85 = h_B + 2.25 \quad \text{or} \quad h_B = 8.6 \text{ m}$$

But,

$$h_B = \frac{P_B}{w}$$

$$p_B = wh_B = 9.81 \times 8.6 = 84.36 \text{ kN/m}^2$$

or,

$$p_B = \mathbf{84.36 \text{ kPa (Ans.)}}$$

2. Inverted U-tube differential manometer:

This type of manometer is used for measuring difference of two pressures where *accuracy is the major consideration*.

Refer to Fig. 2.24. It consists of an inverted U-tube, containing *light liquid*, whose two ends are connected to the points, (*A* and *B*) whose difference of pressures is to be found out. Let the pressure at *A* is more than the pressure at *B*.

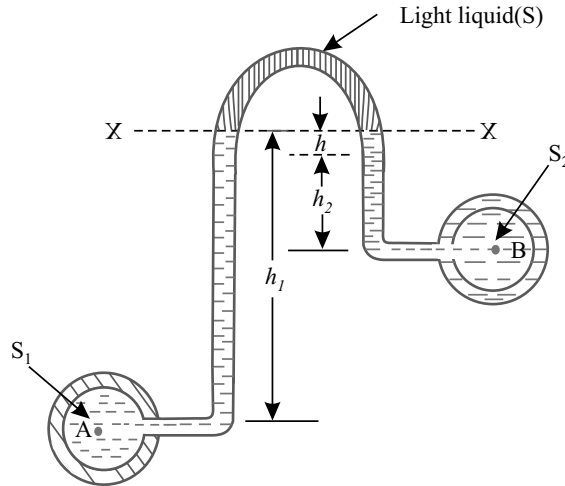


Fig. 2.24. Inverted differential manometer.

Let, h_1 = Height of liquid in the left limb below the datum line $X-X$,

h_2 = Height of liquid in the right limb below the datum line,

h = Difference of levels of the light liquid in the right and left limbs (also known as manometer reading),

S_1 = Specific gravity of the liquid in the left limb,

S_2 = Specific gravity of the liquid in the right limb,

S = Specific gravity of the light liquid,

h_A = Pressure head at *A*, and

h_B = Pressure head at *B*.

We know that pressure heads in the left limb and the right limb below the datum line $X-X$ are equal.

Pressure head in the *left limb* below $X-X = h_A - h_1 \times S_1$

Pressure head in the *right limb* below $X-X = h_B - h_2 \times S_2 - h \times S$

Equating the above heads, we get:

$$h_A - h_1 \times S_1 = h_B - h_2 \times S_2 - h \times S$$

$$h_A - h_B = h_1 \times S_1 - h_2 \times S_2 - h \times S$$

$$\text{i.e.,} \quad h_A - h_B = h_1 S_1 - h_2 S_2 - h S \quad \dots(2.13)$$

Example 2.22. Fig. 2.25 shows an inverted differential manometer having an oil of specific gravity 0.8 connected to two different pipes carrying water under pressure. Determine the pressure in the pipe B. The pressure in pipe A is 2.0 metres of water.

Solution. Height of water in the left limb,

$$h_1 = 300 \text{ mm}$$

Height of water in the right limb,

$$h_2 = 100 \text{ mm}$$

Height of light liquid in right limb,

$$h = 150 \text{ mm}$$

Pressure in pipe A, $h_A = 2.0 \text{ m of water}$

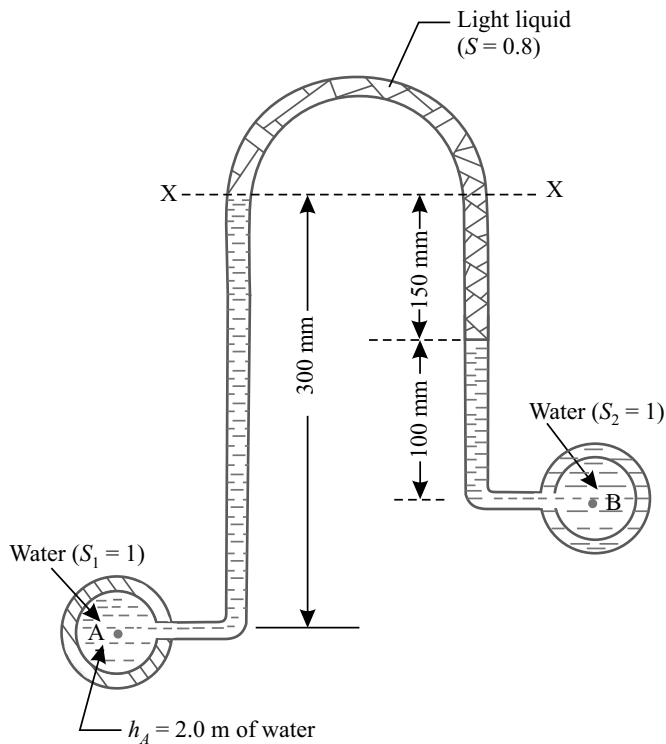


Fig. 2.25

Let, $S_1, S_2 = 1$ (sp. gr. of water)

We know that pressure heads in the left and right limbs below the datum line $X-X$ are equal.

Pressure head in the *left limb* below $X-X$

$$\begin{aligned} &= h_A - h_1 S_1 \\ &= 2.0 - \frac{300}{1000} \times 1 = 1.7 \text{ m} \end{aligned}$$

Pressure head in the *right limb* below $X-X$

$$\begin{aligned}
 &= h_B - h_2 S_2 - hS \\
 &= h_B - \frac{100}{1000} \times 1 - \frac{150}{1000} \times 0.8 \\
 &= h_B - 0.1 - 0.12 = h_B - 0.22
 \end{aligned}$$

Equating the two pressure heads, we get: $1.7 = h_B - 0.22$

or, $h_B = 1.92 \text{ m (Ans.)}$

Also, $p_B = wh_B = 9.81 \times 1.92 = 18.8 \text{ kN/m}^2$
 $= 18.8 \text{ kPa (Ans.)}$

Example 2.23. An inverted differential manometer is connected to two pipes A and B carrying water under pressure as shown in Fig. 2.26. The fluid in the manometer is oil of specific gravity 0.75. Determine the pressure difference between A and B.

Solution. Specific gravity of oil, $S = 0.75$

Specific gravity of water, $S_1, S_2 = 1$

Difference of oil in the two limbs = $(450 + 200) - 450 = 200 \text{ mm}$

We know that pressure heads on the left and right limbs below the datum line X-X are equal.

Pressure head in the left limb below X-X

$$= h_A - \frac{450}{1000} \times 1 = h_A - 0.45$$

Pressure head in the right limb below X-X

$$\begin{aligned}
 &= h_B - \frac{450}{1000} \times 1 - \frac{200}{1000} \times 0.75 \\
 &= h_B - 0.45 - 0.15 = h_B - 0.6
 \end{aligned}$$

Equating the two pressure heads, we get:

$$h_A - 0.45 = h_B - 0.6$$

$$h_B - h_A = 0.15 \text{ m (Ans.)}$$

$$\text{or, } \frac{p_B}{w} - \frac{p_A}{w} = 0.15 \quad \text{or } p_B - p_A$$

$$= w \times 0.15 = 9.81 \times 0.15 = 1.47 \text{ kN/m}^2 = 1.47 \text{ kPa (Ans.)}$$

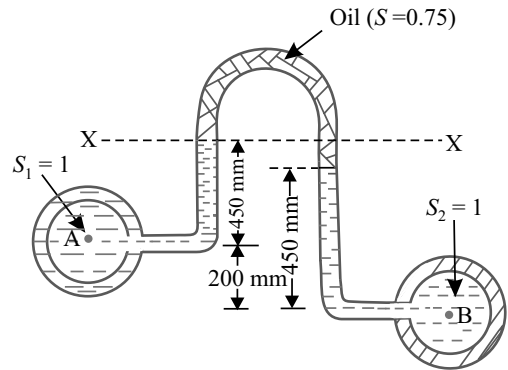


Fig. 2.26

Example 2.24. Describe, giving a sketch, a micromanometer. Explain how it could be used for measuring small pressure difference. (N.U.)

Solution: Micromanometer. It is shown in the Fig. 2.27 and is used for measuring small pressure differences. It utilizes two manometer liquids which are immiscible with each other and also with the fluid whose pressure difference is to be measured. The heavier liquid fills the lower part of the U-tube upto 0-0 and then the lighter liquid is added on both sides filling the tanks C and D upto the level X-X. The fluid (liquid or a gas) whose pressure difference is to be measured fills the space above X-X. When the pressure p_A is slightly greater than p_B , the liquid levels will be as shown in the figure. The volume of the liquid displaced in each tank is equal to the volume of liquid displaced in the U-tube. If a is the cross-sectional area of the U-tube, and A that of the tank, then

$$A \Delta Z = \frac{h}{2} a \quad \dots(i)$$

Let S_1 be the specific gravity of heavier manometric liquid, and S_2 be that of the lighter manometric liquid. An expression relating p_A and p_B may be obtained by equating pressures along L-L in the U-tube. If w is specific weight of water, then:

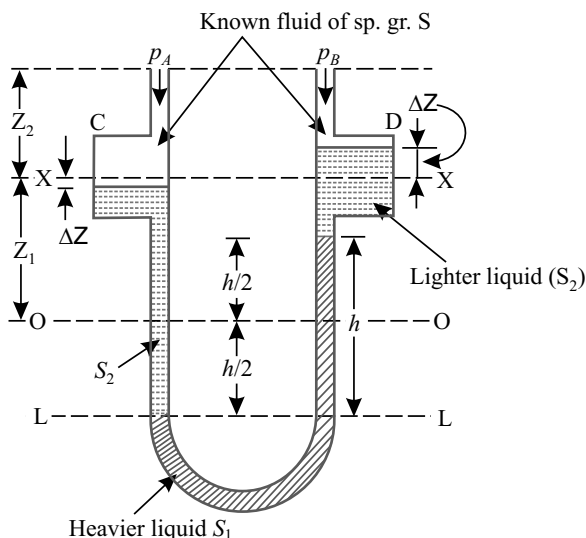


Fig. 2.27

$$\begin{aligned}
 \frac{p_A}{w} + (Z_2 + \Delta Z) S + \left(Z_1 - \Delta Z + \frac{h}{2} S_2 \right) \\
 &= \frac{p_B}{w} + (Z_2 + \Delta Z) S + \left(Z_1 + \Delta Z - \frac{h}{2} \right) S_2 + h S_1 \\
 \text{or,} \quad &= \frac{p_A}{w} + Z_2 S + \Delta Z S + Z_1 S_2 - \Delta Z S_2 + \frac{h}{2} S_2 \\
 &= \frac{p_B}{w} + Z_2 S - \Delta Z S + Z_1 S_2 + \Delta Z S_2 - \frac{h}{2} S_2 + h S_1
 \end{aligned}$$

$$\therefore \text{Pressure difference, } \frac{p_A - p_B}{w} = \Delta Z (S_2 - S + S_2 - S) + h \left(S_1 - \frac{S_2}{2} - \frac{S_2}{2} \right)$$

Substituting for $\Delta Z = \frac{ha}{2A}$ from (i) and simplifying, we get:

$$\begin{aligned}
 \text{or,} \quad \frac{p_A - p_B}{w} &= \frac{ha}{2A} [(2S_2 - 2S)] + h[S_1 - S_2] \\
 &= h \left[\frac{a}{A} (S_2 - S) + (S_1 - S_2) \right] = hK \text{ (Ans.)}
 \end{aligned}$$

The quantity K within the bracket is a *constant* for a given manometer and given manometric liquids of specific gravities S_1 , S_2 and known fluid of specific gravity S .

Example 2.25. Fig. 2.28. shows a fuel gauge, for a gasoline tank in car, which reads proportional to the bottom gauge. The tank is 30 cm deep and accidentally contains 1.8 cm of water in addition to the gasoline. Determine the height of air remaining at the top when the gauge erroneously reads full.

$$\begin{aligned}
 \text{Take:} \quad w_{\text{gasoline}} &= 6.65 \text{ kN/m}^3, \text{ and} \\
 w_{\text{air}} &= 0.0118 \text{ kN/m}^3.
 \end{aligned}$$

(Punjab University)

Solution. When the tank is full of gasoline,

$$p_{\text{gauge}} = wh = 6.65 \times \frac{30}{100} = 1.995 \text{ kN/m}^2$$

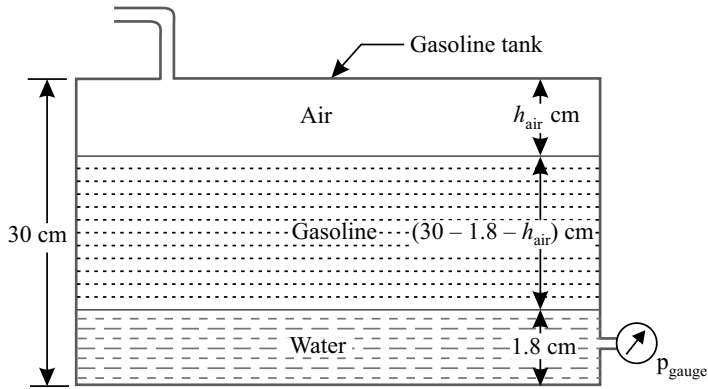


Fig. 2.28

The gauge would erroneously read 1.995 kN/m^2 even when $h \text{ cm}$ of air remains at the top; evidently when water is also accidentally present.

$\therefore$ Pressure due to $h \text{ cm}$ height of air + pressure due to $[30 - 1.8 - h_{\text{air}}] \text{ cm}$ height of gasoline + pressure due to 1.8 cm of water = 1.995

$$\text{or,} \quad 0.0118 \times \frac{h_{\text{air}}}{100} + 6.65 \times \frac{(30 - 1.8 - h_{\text{air}})}{100} + 9.81 \times \frac{1.8}{100} = 1.995$$

$$\text{or,} \quad 0.0118 h + 187.53 - 6.65 h_{\text{air}} + 17.658 = 199.5$$

$$\text{or,} \quad h_{\text{air}} = \frac{187.53 + 17.658 - 199.5}{6.638} = \mathbf{0.857 \text{ cm (Ans.)}}$$

Example 2.26. For the Fig. 2.29 determine the pressure difference between pipes A and B. Take $Z_1 = 0.45 \text{ m}$, $Z_2 = 0.225 \text{ m}$, $Z_3 = 0.675 \text{ m}$ and $Z_4 = 0.3 \text{ m}$.

Neglect pressure due to pressure of air column in the inclined tube.

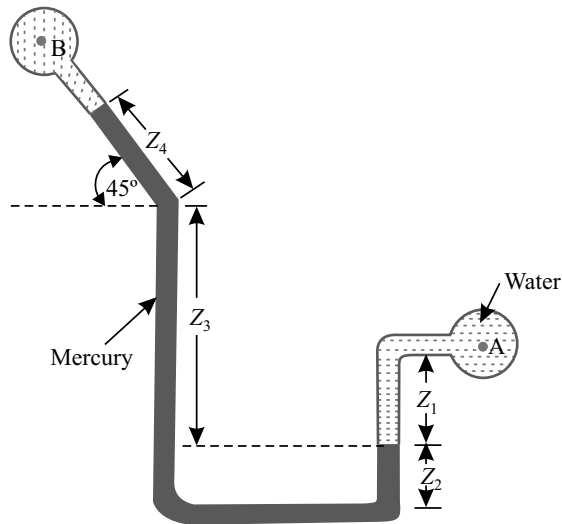


Fig. 2.29

Solution. Starting from point A, the governing manometric equation is:

$$p_A + w_w Z_1 - w_m (Z_3 + Z_4 \sin 45^\circ) = p_B$$

∴ Pressure difference,

$$\begin{aligned}
 p_A - p_B &= -w_m Z_1 + w_m (Z_3 + Z_4 \sin 45^\circ) \\
 &= -9.81 \times 0.45 + 13.6 \times 9.81 (0.675 + 0.3 \sin 45^\circ) \\
 &= -4.414 + 118.357 = \mathbf{113.943 \text{ kN/m}^2 \text{ (Ans.)}}
 \end{aligned}$$

Example 2.27. From the Fig. 2.30 determine the absolute pressure in pipe A that contains oil of specific gravity = 0.88. Take $Z_1 = 0.66 \text{ m}$, $Z_2 = 0.33 \text{ m}$, $Z_3 = 0.165 \text{ m}$ and $Z_4 = 0.11 \text{ m}$.

Assume an atmospheric pressure 105 kPa.

(Madras University)

Solution. Starting from F.W.S (free water surface) in tank (at atmospheric pressure), we get

$$p_{atm} + w_w Z_1 - w_w Z_2 - w_m Z_3 + w_o (Z_3 + Z_4) = p_A$$

$$105 + 9.81 \times 0.66 - 9.81 \times 0.33 - 13.6 \times 9.81 \times 0.165 + 0.88 \times 9.81 \times (0.165 + 0.11) = p_A$$

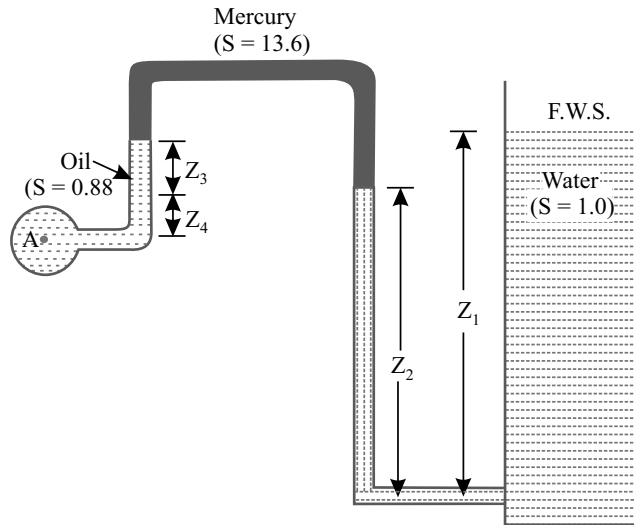


Fig. 2.30

or,

$$\begin{aligned}
 p_A &= 105 + 6.475 - 3.237 - 22.014 + 2.374 \\
 &= \mathbf{88.6 \text{ kN/m}^2 \text{ (absolute) (Ans.)}}
 \end{aligned}$$

Example 2.28. Find the pressure difference between L and M in Fig. 2.31.

Solution. $p_L - p_M$:

$$\begin{aligned}
 &\frac{p_L}{w} + h \times 1.5 - 0.15 \times 0.8 \\
 &\text{(at L)} \quad \text{(at N)} \quad \text{(at U = V)} \\
 &+ (0.15 + 0.2 - h) \times 1.5 = \frac{p_M}{w}
 \end{aligned}$$

$$\frac{p_L}{w} + 1.5 h - 0.12 + 0.525 - 1.5 h = \frac{p_M}{w}$$

or,

$$\frac{p_L - p_M}{w} = -0.405 \text{ m}$$

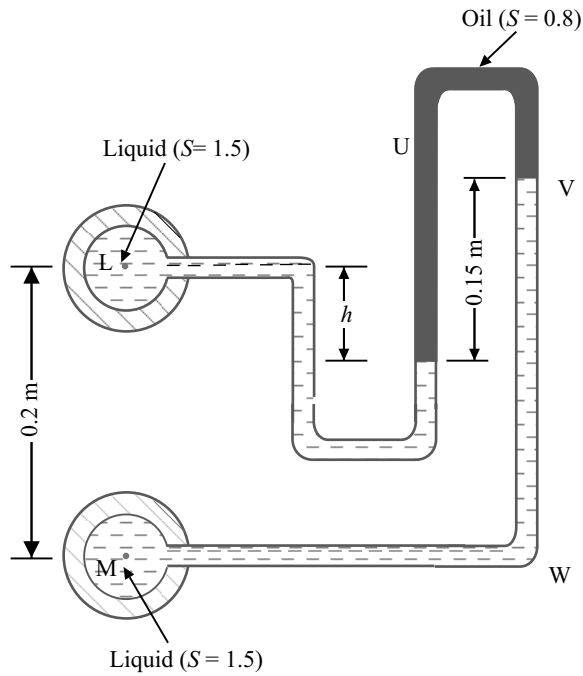


Fig. 2.31

Negative sign indicates $p_M > p_L$

$$\begin{aligned} \text{i.e.,} \quad p_M - p_L &= 0.405 \times 9.81 \\ &= \mathbf{3.97 \text{ kN/m}^2 \text{ (Ans.)}} \end{aligned}$$

Example 2.29. In the Fig. 2.32, if the local atmospheric pressure is 755 mm of mercury (sp. gravity = 13.6), calculate:

- Absolute pressure of air in the tank;
- Pressure gauge reading at L.

Solution. (i) Absolute pressure of air, $(p_{abs})_{air}$:

Starting from the open end, we have:

$$0 - (13.6 \times w) \times 0.6 = p_{air} \text{ (pressure of air)}$$

$$\text{i.e., } p_{air} = -13.6 \times 9.81 \times 0.6 = -80 \text{ kN/m}^2$$

$p_{atm.}$ = (atmospheric pressure)

$$= \frac{755}{1000} \times 13.6 \times 9.81 = 100.73 \text{ kN/m}^2$$

$$(p_{abs})_{air} = p_{air} + p_{atm.} = -80 + 100.73 = 20.73 \text{ kN/m}^2$$

Hence, $(p_{abs})_{air} = \mathbf{20.73 \text{ kN/m}^2 \text{ (Ans.)}}$

- Pressure gauge reading at L:

Pressure at $L = p_{abs.} \text{ (air)} + wh$

$$p_L = 20.73 + 9.81 \times 2 = 40.35 \text{ kN/m}^2 \text{ abs.}$$

Now, $40.35 = p_{gauge} + p_{atm.}$

$$\begin{aligned} p_{gauge(L)} &= 40.35 - p_{atm.} = 40.35 - 100.73 \\ &= -60.38 \text{ kN/m}^2 \end{aligned}$$

i.e., Vacuum pressure = 60.38 kN/m²

Hence, pressure gauge reading at L = $\mathbf{60.38 \text{ kN/m}^2 \text{ (vacuum) (Ans.)}}$

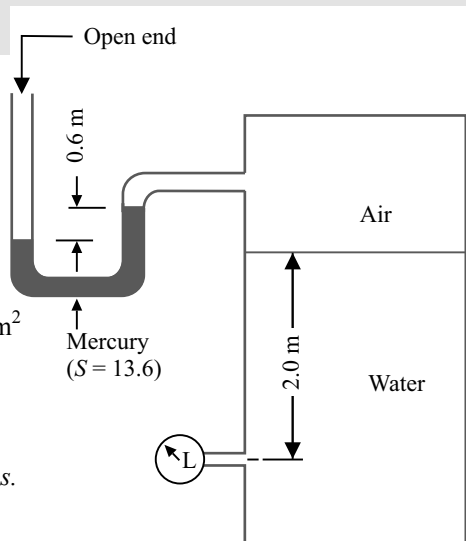


Fig. 2.32

Example 2.30. Find the gauge readings at L and M in Fig. 2.33 if the local atmospheric pressure is 755 mm of mercury.

Solution. Assuming the vapour pressure of mercury (Hg) and pressure due to short column of air (w_{air} is very low) to be negligible, we have:

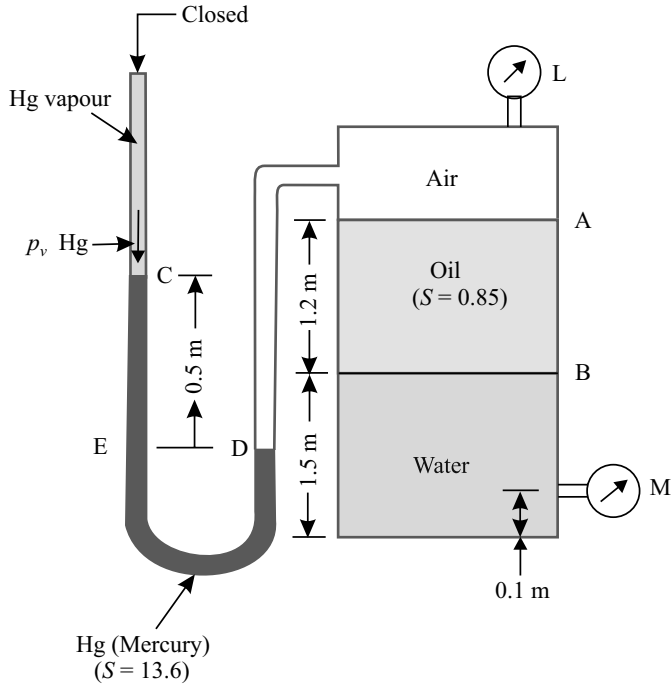


Fig. 2.33

(i) $(p_{gauge})_L$:

$$\begin{aligned} (p_v Hg \approx 0) + 0.5 \times 13.6 \\ \text{(at C)} \quad \quad \quad \text{(at D = E = A)} \\ = 6.8 \text{ m of water abs.} \\ \text{(at A)} \end{aligned}$$

$$p_{gauge} + p_{atm.} = p_{abs.}$$

$$\text{But,} \quad p_{atm.} = \frac{755}{1000} \times 13.6 = 10.27 \text{ m of water}$$

$$\therefore p_{gauge} + 10.27 = 6.8$$

$$\begin{aligned} \text{or,} \quad p_{gauge} &= -3.47 \text{ m of water} \\ &= -3.47 \times 9.81 \\ &= -34 \text{ kN/m}^2 \end{aligned}$$

Hence, gauge reading at $L = 34 \text{ kN/m}^2$ (vacuum) (Ans.)

(ii) $(p_{gauge})_M$:

$$6.8 + 1.2 \times 0.85 + (1.5 - 0.1) = 9.22 \text{ m of water abs.}$$

$$\text{(at A)} \quad \quad \text{(at B)} \quad \quad \text{(at M)}$$

$$p_{gauge} + p_{atm.} = p_{abs.}$$

$$p_{gauge} + 10.27 = 9.22$$

or, $p_{\text{gauge}} = -1.05 \text{ m of water}$
 $= -1.05 \times 9.81 = -10.3 \text{ kN/m}^2$

Hence, gauge reading at $M = 10.3 \text{ kN/m}^2$ (vacuum) (Ans.)

Example 2.31. For the Fig 2.34 determine specific gravity of gauge liquid B if the gauge pressure at A is -18 kN/m^2 .

Solution. Sp. gravity of liquid B:

$$\text{Pressure at } L = \text{pressure at } M$$

$$\text{i.e., } -18 + (1.5 \times 9.81 \times 0.6) = p_M$$

or, $p_M = -9.17 \text{ kN/m}^2$

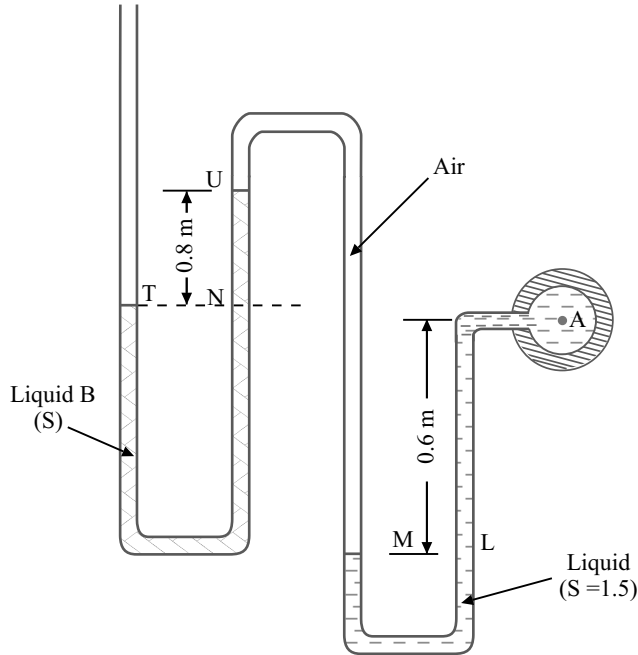


Fig. 2.34

Between points M and U , since there is an air column which can be neglected, therefore,

$$p_M = p_U (= -9.17 \text{ kN/m}^2)$$

Also, pressure at N = pressure at T .

But point T being at atmospheric pressure,

$$p_T = 0 = p_N$$

Thus, $p_N = p_U + S \times 9.81 \times 0.8 = 0$

or, $0 = -9.17 + 7.848 S$

$$S = 1.17 \text{ (Ans.)}$$

Example 2.32. (Compound manometer). In the Fig. 2.35 is shown a compound manometer. Find the gauge pressure at A if the manometric fluid is mercury and the fluid in the pipe and in the tubing which connects the two U-tubes is water.

Solution. Gauge pressure at A, p_A :

$$\text{Pressure at } B = \text{Pressure at } C$$

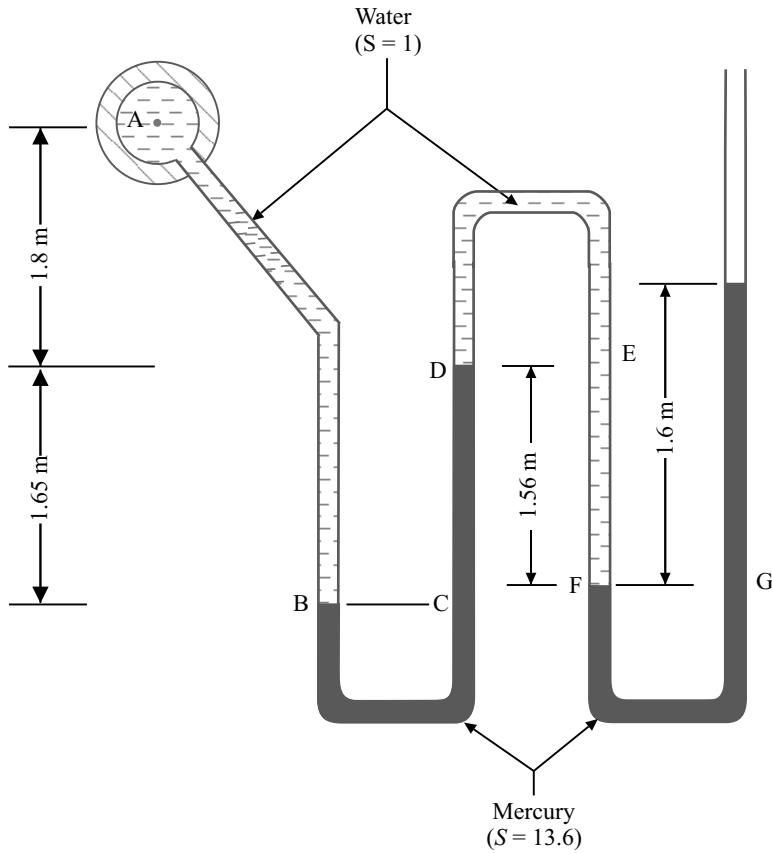


Fig. 2.35

$$\therefore \quad \frac{p_B}{w} = \frac{p_A}{w} + (1.8 + 1.65) = \frac{p_C}{w}$$

Further,
Pressure at D,

$$\begin{aligned} \frac{p_D}{w} &= \frac{p_C}{w} - 1.65 \times 13.6 \\ &= \frac{p_A}{w} + (1.8 + 1.65) - 1.65 \times 13.6 \\ &= \frac{p_A}{w} + 3.45 - 22.44 \end{aligned}$$

$$\text{or,} \quad \frac{p_D}{w} = \frac{p_A}{w} - 18.99 \quad \dots(1)$$

$$\text{Also,} \quad p_D = p_E \quad \text{and} \quad p_F = p_G$$

$$\text{But,} \quad \frac{p_F}{w} = \frac{p_E}{w} + 1.56$$

$$\text{and,} \quad \frac{p_G}{w} = 1.6 \times 13.6 = 21.76$$

$$\text{i.e.,} \quad \frac{p_E}{w} + 1.56 = 21.76 \quad (\because p_F = p_G)$$

$$\text{or,} \quad \frac{p_E}{w} = 20.2 \quad \text{or} \quad \frac{p_D}{w} = 20.2 \quad (\because p_D = p_E)$$

$$\text{Substituting the value of } \frac{p_D}{w} \text{ in eqn. (1), we get:} \quad 20.2 = \frac{p_A}{w} - 18.99$$

$$\text{or,} \quad \frac{p_A}{w} = 20.2 + 18.99 = 39.19 \text{ m of water.}$$

$$\text{i.e.,} \quad p_A = 9.81 \times 39.19 = \mathbf{384.4 \text{ kN/m}^2} \text{ (Ans.)}$$

Example 2.33. (Compound manometer). In the Fig. 2.36 is shown a compound manometer. Calculate pressure difference between the points A and B. Take $w_w = 10 \text{ kN/m}^3$ for water, $w_m = 136 \text{ kN/m}^3$ for mercury and $w_o = 8.5 \text{ kN/m}^3$ for oil. (Punjab University)

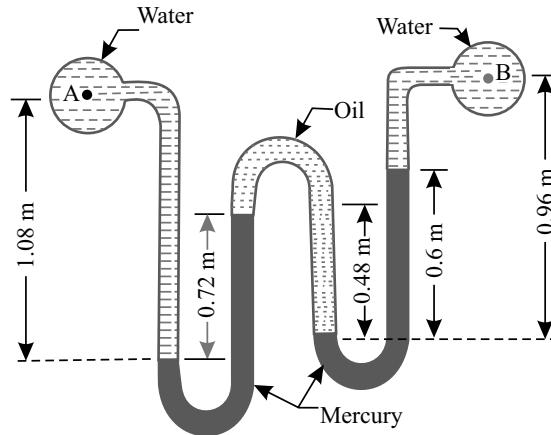


Fig. 2.36

Solution. Given: $w_w = 10 \text{ kN/m}^3$; $w_m = 136 \text{ kN/m}^3$; $w_o = 8.5 \text{ kN/m}^3$

$$p_A - p_B:$$

Starting from point A, the governing manometric equation is:

$$p_A + w_w \times 1.08 - w_m \times 0.72 + w_o \times 0.48 - w_m \times 0.6 - w_w (0.96 - 0.6) = p_B$$

$$\text{or, } p_A = 10 \times 1.08 - 136 \times 0.72 + 8.5 \times 0.48 - 136 \times 0.6 - 10 (0.96 - 0.6) = p_B$$

$$\text{or, } p_A + 10.8 - 97.92 + 4.08 - 81.6 - 3.6 = p_B$$

$$\text{or, } p_A - p_B = \mathbf{168.24 \text{ kN/m}^2} \text{ (Ans.)}$$

Example 2.34. A cylindrical bucket (empty) 450 mm in diameter and 750 mm long is forced with its open end first into water until its lower edge is 6 m below the surface. Determine the force required to maintain position, assuming the trapped air remains at constant temperature during the whole operation. Atmospheric pressure = 1.01 bar.

The wall thickness and weight of the bucket may be considered as negligible.

Solution. Diameter of the bucket, $d = 450 \text{ mm} = 0.45 \text{ m}$

Length of the bucket, $l = 750 \text{ mm} = 0.75 \text{ m}$

Atmospheric pressure, $p_{atm} = 1.01 \text{ bar}$.

Force required to maintain position, F: Refer to Fig 2.37.

Let, p_{air} = Absolute pressure of compressed air trapped in the cylindrical bucket, and

y = Depth of water raised in the bucket.

Then, since the temperature of air remains constant, therefore, as per isothermal condition, we have:

$$p_{atm.} \times \frac{\pi}{4} \times 0.45^2 \times 0.75 = p_{air} \times \frac{\pi}{4} \times 0.45^2 \times (0.75 - y)$$

($\because p_1 V_1 = p_2 V_2$ for isothermal process)

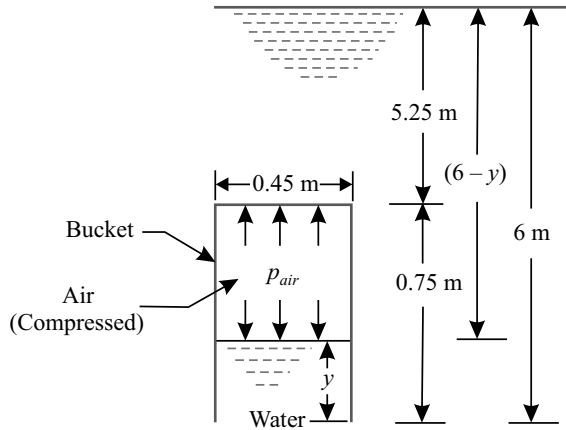


Fig. 2.37

$$\text{or, } p_{air} = \left(\frac{0.75}{0.75 - y} \right) p_{atm.} \quad \dots(i)$$

$$\text{Also, } p_{air} = p_{atm.} + wh = p_{atm.} + 9810 \times (6 - y) \quad \dots(ii)$$

($\because w = 9810 \text{ N/m}^3$)

From eqn. (i) and (ii), we have:

$$\left(\frac{0.75}{0.75 - y} \right) p_{atm.} = p_{atm.} + 9810 (6 - y)$$

$$\text{or, } \left(\frac{0.75}{0.75 - y} \right) \times 1.01 \times 10^5 = 1.01 \times 10^5 + 9810 (6 - y)$$

$$\text{or, } 1.01 \times 10^5 \left(\frac{0.75}{0.75 - y} - 1 \right) = 9810 (6 - y)$$

$$\text{or, } \frac{1.01 \times 10^5}{9810} \left(\frac{0.75 - 0.75 + y}{0.75 - y} \right) = 6 - y$$

$$\text{or, } 10.29 \times \left(\frac{y}{0.75 - y} \right) = 6 - y \quad \text{or} \quad \frac{10.29y}{0.75 - y} + y = 6$$

$$\text{or, } 10.29y + y(0.75 - y) = 6(0.75 - y)$$

$$\text{or, } 10.29y + 0.75y - y^2 = 4.5 - 6y \quad \text{or} \quad y^2 - 17.04y + 4.5 = 0$$

$$\text{or, } y = \frac{17.04 \pm \sqrt{17.04^2 - 4 \times 4.5}}{2} = \frac{17.04 \pm 16.5}{2} = 0.27 \text{ m}$$

(ignoring + ve sign, being not possible)

Substituting the value of y in (i), we get:

$$p_{air} = \left(\frac{0.75}{0.75 - 0.27} \right) \times 1.01 = 1.578 \text{ bar}$$

The force tending to move the bucket in *upward direction*,

$$\begin{aligned} P_1 &= p_{air} \times \frac{\pi}{4} \times 0.45^2 \\ &= (1.578 \times 10^5) \times \frac{\pi}{4} \times 0.45^2 \times 10^{-3} \text{ kN} = 25.097 \text{ kN} \end{aligned}$$

The force acting on the bucket in the *downward direction*,

$$P_2 = (1.01 \times 10^5 + 9810 \times 5.25) \times \frac{\pi}{4} \times 0.45^2 \times 10^{-3} \text{ kN} = 24.254 \text{ kN}$$

$\therefore$ The force required to maintain the bucket in position,

$$F = P_1 - P_2 = 25.097 - 24.254 = \mathbf{0.843 \text{ kN (Ans.)}}$$

Example 2.35. A glass tube of uniform bore is bent into the form of a square of sides l and filled with equal amounts of three invisible liquids of densities ρ_1 , ρ_2 and ρ_3 . It is known that $\rho_1 < \rho_2 < \rho_3$. If the tube arrangement is placed in a vertical plane (i.e. two sides vertical) and if one of the vertical sides is completely filled with the liquid of density ρ_2 :

(i) Show that $\frac{1}{3}(2\rho_3 + \rho_1) > \rho_2 > \frac{1}{3}(\rho_3 + 2\rho_1)$

(ii) If the relative densities of the first and third liquids are 1.0 and 1.22 respectively, find the range of the relative densities of the second liquid which makes the above arrangement possible.

Solution. Refer to Fig 2.38.

(i) To prove, $\frac{1}{3}(2\rho_3 + \rho_1) > \rho_2 > \frac{1}{3}(\rho_3 + 2\rho_1)$:

Let E , F and G be the interfaces, and

$$EA = x$$

Then, $DE = l - x$

Total length of the glass tube = $4l$

$\therefore$ Length of each liquid = $\frac{4}{3}l$

For liquid-1:

$$EG = \frac{4}{3}l$$

$$DG = \frac{4}{3}l - (l - x) = \frac{1}{3}l + x$$

For liquid-3:

$$GC = l - \left(\frac{1}{3}l + x \right) = \frac{2}{3}l - x$$

$$FB = l - \left(\frac{2}{3}l + x \right) = \frac{1}{3}l - x$$

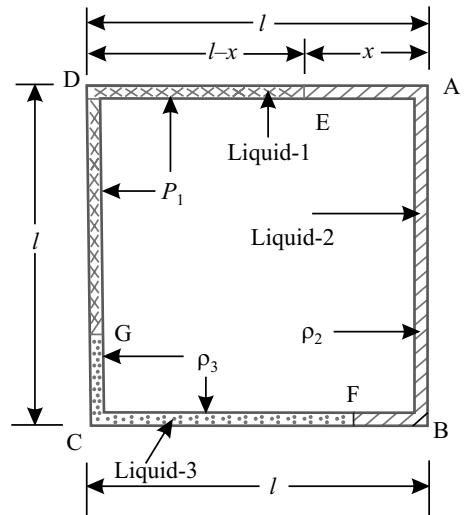


Fig. 2.38

$$\left(\because FC = \frac{4l}{3} - GC = \frac{4l}{3} - \left(\frac{2}{3}l - x \right) = \frac{2}{3}l + x \right)$$

$$\left[\text{Check: } FB + BA + AE = \left(\frac{1}{3}l - x \right) + l + x = \frac{4}{3}l \right]$$

The pressure balance at the interface F is given by:

Pressure of (column DG + column GC) = pressure of column AB

$$\rho_1 g \left(\frac{1}{3}l + x \right) + \rho_3 g \left(\frac{2}{3}l - x \right) = \rho_2 g l$$

$$\text{or,} \quad \rho_1 \left(\frac{1}{3}l + x \right) + \rho_3 \left(\frac{2}{3}l - x \right) = \rho_2 l$$

$$x(\rho_1 - \rho_3) = \frac{1}{3}l (-\rho_1 + 3\rho_2 - 2\rho_3)$$

$$\text{or,} \quad x = \frac{\frac{1}{3}l (-\rho_1 + 3\rho_2 - 2\rho_3)}{(\rho_1 - \rho_3)} = \frac{\frac{1}{3}l (2\rho_3 - 3\rho_2 + \rho_1)}{(\rho_3 - \rho_1)}$$

It is known that $x > 0$ and also $x < \frac{1}{3}l$

$$\text{Hence,} \quad 0 < x < \frac{l}{3}$$

$$\therefore \quad 0 < \frac{(2\rho_3 - 3\rho_2 + \rho_1)}{(\rho_3 - \rho_1)} < 1$$

Also, since $\rho_1 < \rho_2 < \rho_3$ the denominator $(\rho_3 - \rho_1)$ is positive.

Hence, numerator is:

$$0 < (2\rho_3 - 3\rho_2 + \rho_1) < 1$$

$$\text{or,} \quad \rho_1 + 2\rho_3 > 3\rho_2$$

$$\text{or,} \quad \rho_2 < \frac{1}{3}(2\rho_3 + \rho_1) \quad \dots(i)$$

$$\text{Also, since} \quad \frac{2\rho_3 - 3\rho_2 + \rho_1}{\rho_3 - \rho_1} < 1$$

$$\text{or,} \quad 2\rho_3 - 3\rho_2 + \rho_1 < (\rho_3 - \rho_1)$$

$$\text{or,} \quad 3\rho_2 > \rho_3 + 2\rho_1$$

$$\text{or,} \quad \rho_2 > \frac{1}{3}(\rho_3 + 2\rho_1) \quad \dots(ii)$$

Hence, from inequalities (i) and (ii), we have:

$$\frac{1}{3}(2\rho_3 + \rho_1) > \rho_2 > \frac{1}{3}(\rho_3 + 2\rho_1)$$

(ii) **Range of relative densities of the second liquid:**

$$\text{Given:} \quad \rho_1 = 1.0; \quad \rho_3 = 1.22$$

$$\begin{aligned} \text{Now,} \quad \rho_2 &> \frac{1}{3}(\rho_3 + 2\rho_1) \\ &> \frac{1}{3}(1.22 + 2 \times 1.0) > 1.0733 \end{aligned}$$

$$\begin{aligned} \text{Also,} \quad \rho_2 &> \frac{1}{3}(2\rho_3 + \rho_1) \\ &> \frac{1}{3}(2 \times 1.22 + 1.0) > 1.1467 \end{aligned}$$

$$\text{Hence,} \quad 1.0733 < \rho_2 < 1.1467 \text{ (Ans.)}$$

2.5.1.3. Advantages and Limitations of Manometers

Advantages:

1. Easy to fabricate and relatively inexpensive.
2. Good accuracy.
3. High sensitivity.
4. Require little maintenance.
5. Not affected by vibration.
6. Specially suitable for low pressure and low differential pressures.
7. It is easy to change the sensitivity by affecting a change in the quantity of manometric liquid in the manometer.

Limitations:

1. Usually bulky and large in size.
2. Being fragile, get broken easily.
3. Readings of the manometers are affected by changes in temperature, altitude and gravity.
4. A capillary effect is created due to surface tension of manometric fluid.
5. For better accuracy meniscus has to be measured by accurate means.

2.5.2. Mechanical Gauges

The manometers (discussed earlier) are suitable for *comparatively low pressures*. For high pressures they become unnecessarily larger even when they are filled with heavy liquids. Therefore, for measuring medium and high pressures we make use of *elastic pressure gauges*. They employ different forms of elastic systems such as tubes, diaphragms or bellows etc. to measure the pressure. The elastic deformation of these elements is used to show the effect of pressure. Since these elements are deformed within the elastic limit only, therefore these gauges are sometimes called *elastic gauges*. Sometimes they are also called *secondary instruments, which implies that they must be calibrated by comparison with primary instruments such as manometers etc.*

Some of the important types of these gauges are enumerated and discussed below:

1. Bourdon tube pressure gauge,
2. Diaphragm gauge, and
3. Vacuum gauge.

1. Bourdon tube pressure gauge:

Bourdon tube pressure gauge is used for measuring high as well as low pressures. A simple form of this gauge is shown in Fig. 2.39. In this case, the pressure element consists of a metal tube of approximately elliptical cross-section. This tube is bent in the form of a segment of a circle and responds to pressure changes. When one end of the tube which is attached to the gauge case, is connected to the source of pressure, the internal pressure causes the tube to expand, whereby circumferential stress *i.e.*, hoop tension is set up. The free end of the tube moves and is in turn connected by suitable levers to a rack, which engages with a small pinion mounted on the same spindle as the pointer. Thus the pressure applied to the tube causes the rack and pinion to move. The pressure is indicated by the pointer over a dial which can be graduated in a suitable scale.

The Bourdon tubes are generally made of *bronze or nickel steel*. The *former* is generally used for *low pressures* and the *latter* for *high pressures*.

Depending upon the purpose for which they are required Bourdon tube gauges are made in different forms, some of them are:

- (i) *Compound Bourdon tube*—used for measuring pressures both *above and below atmospheric pressure*.

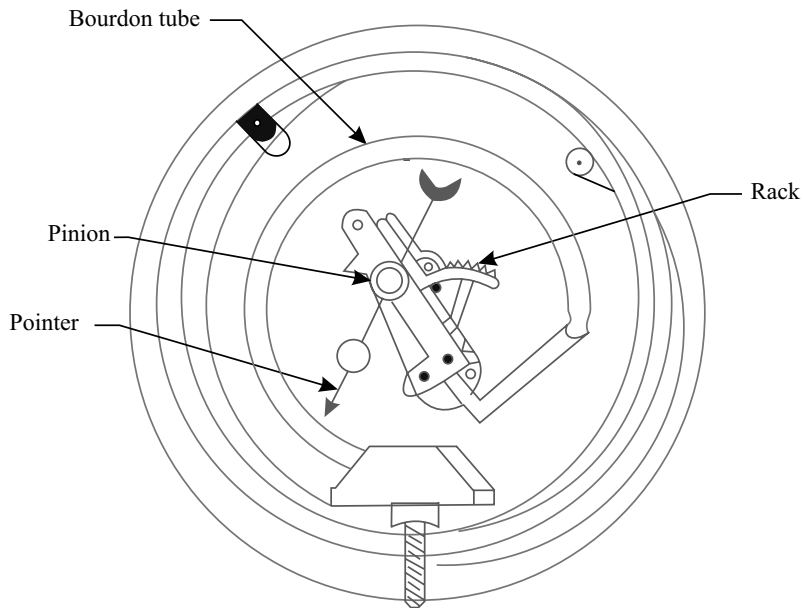


Fig. 2.39. Bourdon tube pressure gauge.

(ii) *Double Bourdon tube*—used where vibrations are encountered.

2. Diaphragm gauge:

This type of gauge employs a metallic disc or diaphragm instead of a bent tube. This disc or diaphragm is used for *actuating* the indicating device.

Refer to Fig. 2.40. When pressure is applied on the lower side of the diaphragm it is deflected upward. This movement of the diaphragm is transmitted to a rack and pinion. The latter is attached to the spindle of needle moving on a graduated dial. The dial can again be graduated in a suitable scale.

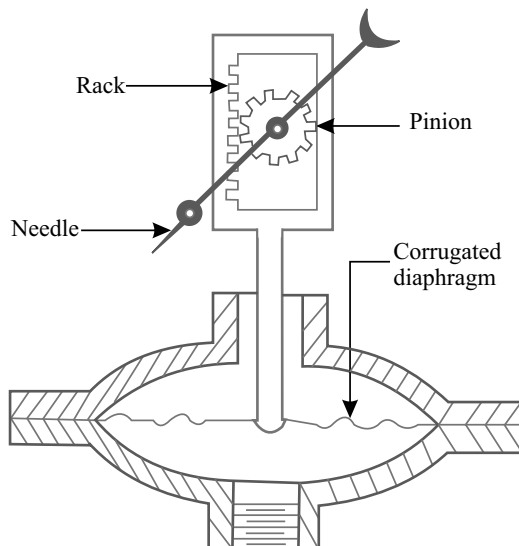


Fig. 2.40. Diaphragm gauge.

3. Vacuum gauge:

Bourdon gauges discussed earlier can be used to measure vacuum instead of pressure. Slight changes in the design are required in this purpose. Thus, in this case, the tube be *bent inward instead of outward* as in pressure gauges. *Vacuum gauges* are graduated in millimetres of mercury below atmospheric pressure. In such cases, therefore, absolute pressure in millimetres of mercury is the difference between barometer reading and vacuum gauge reading.

Vacuum gauges are used to measure the vacuum in the condensers, etc. If there is leakage, the vacuum will drop.

The pressure gauge installation requires the following considerations:

1. Flexible copper tubing and compression fittings are recommended for most installations.
2. The installation of a gauge cock and tee in the line close to the gauge is recommended because it permits the gauge to be removed for testing or replacement without having to shut down the system.
3. Pulsating pressures in the gauge line are not required.
4. The gauge and its connecting line is filled with an inert liquid and as such liquid seals are provided. Trapped air at any point of gauge lines may cause serious errors in pressure reading.

2.6. PRESSURE AT A POINT IN COMPRESSIBLE FLUID

In case of compressible fluids, the density (ρ) changes with the change of pressure and temperature. In the fields of meteorology, oceanography and aeronautics, we come across with problems involving atmospheric air where density, pressure and temperature change with the elevation. Therefore, for fluids having variable density eqn. (2.4) cannot be intergrated unless relation between ρ and p is known.

The 'equation of state' for gases is given as:

$$p = \rho RT \quad \dots(2.13)$$

or,
$$\rho = \frac{p}{RT}$$

We know that,
$$\frac{dp}{dZ} = w = \rho \times g \quad \text{(Eqn.2.4)}$$

$$= \frac{p}{RT} \times g$$

$$\therefore \frac{dp}{p} = \frac{g}{RT} \times dZ \quad \dots(2.14)$$

When Z is measured *vertically upward*, the above equation reduces to:

$$\frac{dp}{p} = \frac{-g}{RT} \times dZ \quad \dots(2.15)$$

Isothermal process:

In an isothermal process, temperature T remains constant, therefore, integrating eqn. (2.15) we get:

$$\int_{p_0}^p \frac{dp}{p} = - \int_{Z_0}^Z \frac{g}{RT} dZ = - \frac{g}{RT} \int_{Z_0}^Z dZ$$

or,
$$\ln\left(\frac{p}{p_0}\right) = \frac{g}{RT} (Z - Z_0)$$

where, p_0 is the pressure, and Z_0 is the height.

If the datum line is taken at Z_0 , then Z_0 becomes zero and p_0 becomes the pressure at datum line.

$$\ln\left(\frac{p}{p_0}\right) = -\frac{g}{RT} \cdot Z$$

or,
$$\frac{p}{p_0} = e^{(-gZ/RT)}$$

or, Pressure at a height Z is given by:

$$p = p_0 e^{(-gZ/RT)} \quad \dots(2.16)$$

Adiabatic process:

When the process follows an adiabatic law, the relation between pressure and density is given by:

$$\frac{p}{\rho^\gamma} = \text{constant} = C \quad \dots(i)$$

where, γ is the ratio of specific heats.

$$\therefore \rho^\gamma = \frac{p}{C}$$

or,
$$\rho = \left(\frac{p}{C}\right)^{1/\gamma} \quad \dots(ii)$$

Now eqn. (2.4) becomes:

$$\frac{dp}{dZ} = -\rho g = -\left(\frac{p}{C}\right)^{1/\gamma} \cdot g \quad \dots Z \text{ measured vertically up}$$

or,
$$\frac{dp}{\left(\frac{p}{C}\right)^{1/\gamma}} = -g \cdot dZ \quad \text{or} \quad C^{1/\gamma} \times \frac{dp}{(p)^{1/\gamma}} = -g \cdot dZ$$

Integrating the above equation, we get:

$$C^{1/\gamma} \left[\frac{p^{\left(-\frac{1}{\gamma}+1\right)}}{-\frac{1}{\gamma}+1} \right]_{p_0}^p = -g[Z]_{Z_0}^Z$$

or,
$$\left[\frac{C^{1/\gamma} p^{\left(-\frac{1}{\gamma}+1\right)}}{-\frac{1}{\gamma}+1} \right]_{p_0}^p = -g[Z]_{Z_0}^Z$$

From eqn. (i), we have, $C^{1/\gamma} = \left[\frac{p}{\rho^\gamma} \right]^{1/\gamma} = \frac{p^{1/\gamma}}{\rho}$ (C being constant, can be taken inside)

Substituting this value of $C^{1/\gamma}$ in the above equation, we get:

$$\left[\frac{p^{1/\gamma} p^{\left(-\frac{1}{\gamma}+1\right)}}{\rho^{\left(\frac{\gamma-1}{\gamma}\right)}} \right]_{p_0}^p = -g[Z-Z_0]$$

$$\begin{aligned} \text{or,} \quad \frac{\gamma}{\gamma-1} \left[\frac{p}{\rho} \right]_{p_0}^p &= -g(Z - Z_0) \\ &= \frac{\gamma}{\gamma-1} \left[\frac{p}{\rho} - \frac{p_0}{\rho_0} \right] = -g(Z - Z_0) \end{aligned}$$

If datum line is taken at Z_0 (where pressure, temperature and density are p_0 , T_0 , ρ_0), then $Z_0 = 0$.

$$\therefore \frac{\gamma}{\gamma-1} \left[\frac{p}{\rho} - \frac{p_0}{\rho_0} \right] = -gZ$$

$$\text{or,} \quad \left(\frac{p}{\rho} - \frac{p_0}{\rho_0} \right) = -gZ \left(\frac{\gamma-1}{\gamma} \right)$$

$$\text{or,} \quad \frac{p}{\rho} = \frac{p_0}{\rho_0} - gZ \left(\frac{\gamma-1}{\gamma} \right) = \frac{p_0}{\rho_0} \left[1 - \frac{\gamma-1}{\gamma} gZ \times \frac{\rho_0}{p_0} \right]$$

$$\text{or,} \quad \frac{p}{\rho} \times \frac{\rho_0}{p_0} = \left[1 - \frac{\gamma-1}{\gamma} gZ \times \frac{\rho_0}{p_0} \right] \quad \dots(iii)$$

$$\text{But from eqn. (i),} \quad \frac{p}{\rho^\gamma} = \frac{p_0}{\rho_0^\gamma} \quad \text{or} \quad \left(\frac{\rho_0}{\rho} \right)^\gamma = \frac{p_0}{p} \quad \text{or} \quad \frac{\rho_0}{\rho} = \left(\frac{p_0}{p} \right)^{1/\gamma}$$

Substituting the value of $\frac{\rho_0}{\rho}$ in eqn. (iii), we get:

$$\frac{p}{p_0} \times \left(\frac{p_0}{p} \right)^{1/\gamma} = \left[1 - \frac{\gamma-1}{\gamma} gZ \times \frac{\rho_0}{p_0} \right]$$

$$\text{or,} \quad \frac{p}{p_0} \times \left(\frac{p}{p_0} \right)^{-1/\gamma} = \left[1 - \frac{\gamma-1}{\gamma} gZ \times \frac{\rho_0}{p_0} \right]$$

$$\text{or,} \quad \left(\frac{p}{p_0} \right)^{1-\frac{1}{\gamma}} = \left[1 - \frac{\gamma-1}{\gamma} gZ \times \frac{\rho_0}{p_0} \right]$$

$$\text{or,} \quad \left(\frac{p}{p_0} \right)^{\frac{\gamma-1}{\gamma}} = \left[1 - \frac{\gamma-1}{\gamma} gZ \times \frac{\rho_0}{p_0} \right]$$

$$\text{or,} \quad \frac{p}{p_0} = \left[1 - \frac{\gamma-1}{\gamma} gZ \times \frac{\rho_0}{p_0} \right]^{\frac{\gamma}{\gamma-1}}$$

$\therefore$ Pressure (p) at a height Z from the ground level is given by:

$$p = p_0 \left[1 - \frac{\gamma-1}{\gamma} gZ \times \frac{\rho_0}{p_0} \right]^{\frac{\gamma}{\gamma-1}} \quad \dots(2.17)$$

where,

p_0 = Pressure at ground level (when $Z_0 = 0$), and

ρ_0 = Density of air at ground level.

Also, equation of state is:

$$\frac{p_0}{\rho_0} = RT_0 \quad \text{or} \quad \frac{\rho_0}{p_0} = \frac{1}{RT_0}$$

Substituting the value of $\frac{\rho_0}{p_0}$ in eqn. (2.17), we get:

$$p = p_0 \left[1 - \frac{\gamma-1}{\gamma} \times \frac{gZ}{RT_0} \right]^{\frac{\gamma}{\gamma-1}} \quad \dots(2.18)$$

- *Temperature at any point in compressible fluid in an adiabatic process is calculated as follows:*

Equation of state at ground level and at a height Z from the ground level is written as:

$$\frac{p_0}{\rho_0} = RT_0, \quad \text{and} \quad \frac{p}{\rho} = RT$$

Dividing these equations, we have:

$$\frac{p_0}{\rho_0} \times \frac{\rho}{p} = \frac{RT_0}{RT}$$

$$\text{or,} \quad \frac{T}{T_0} = \frac{p}{p_0} \times \frac{\rho_0}{\rho} \quad \dots(iv)$$

But from eqn. (2.18), $\frac{p}{p_0}$ is given by:

$$\frac{p}{p_0} = \left[1 - \frac{\gamma-1}{\gamma} \times \frac{gZ}{RT_0} \right]^{\frac{\gamma}{\gamma-1}}$$

Also for *adiabatic process* $\frac{p}{\rho^\gamma} = \frac{p_0}{\rho_0^\gamma}$ or $\left(\frac{\rho_0}{\rho} \right)^\gamma = \frac{p_0}{p}$

$$\text{or,} \quad \frac{\rho_0}{\rho} = \left(\frac{p_0}{p} \right)^{1/\gamma} = \left(\frac{p}{p_0} \right)^{-1/\gamma}$$

$$\text{or,} \quad \frac{\rho_0}{\rho} = \left[1 - \frac{\gamma-1}{\gamma} \times \frac{gZ}{RT_0} \right]^{\left(\frac{\gamma}{\gamma-1} \right) \times \left(-\frac{1}{\gamma} \right)}$$

$$\text{or,} \quad \frac{\rho_0}{\rho} = \left[1 - \frac{\gamma-1}{\gamma} \times \frac{gZ}{RT_0} \right]^{-\frac{1}{\gamma-1}}$$

Substituting the values of $\frac{p}{p_0}$ and $\frac{\rho_0}{\rho}$ in eqn. (iv), we get:

$$\begin{aligned} \frac{T}{T_0} &= \left[1 - \frac{\gamma-1}{\gamma} \times \frac{gZ}{RT_0} \right]^{\frac{\gamma}{\gamma-1}} \times \left[1 - \frac{\gamma-1}{\gamma} \times \frac{gZ}{RT_0} \right]^{\frac{1}{\gamma-1}} \\ &= \left[1 - \frac{\gamma-1}{\gamma} \times \frac{gZ}{RT_0} \right]^{\frac{\gamma}{\gamma-1} + \frac{1}{\gamma-1}} = \left[1 - \frac{\gamma-1}{\gamma} \times \frac{gZ}{RT_0} \right] \end{aligned}$$

$$\therefore T = T_0 \left[1 - \frac{\gamma-1}{\gamma} \times \frac{gZ}{RT_0} \right] \quad \dots(2.19)$$

Temperature lapse rate (L):

It is defined as the *rate at which the temperature changes with elevation*. It can be obtained by differentiating the eqn. w.r.t. Z as follows:

$$\frac{dT}{dZ} = \frac{d}{dZ} \left[T_0 \left\{ 1 - \frac{\gamma-1}{\gamma} \right\} \times \frac{gZ}{RT_0} \right]$$

where, T_0 , γ , g and R are constant.

$$\therefore \frac{dT}{dZ} = T_0 \left\{ -\frac{\gamma-1}{\gamma} \times \frac{g}{RT_0} \right\} = -\frac{g}{R} \left(\frac{\gamma-1}{\gamma} \right)$$

$$\text{Hence, temperature lapse-rate, } L = \frac{dT}{dZ} = -\frac{g}{R} \left(\frac{\gamma-1}{\gamma} \right) \quad \dots(2.20)$$

— In this eqn., if $\gamma = 1$ the process is an isothermal one which means $\frac{dT}{dZ} = 0$; it indicates that temperature does not *change* with height.

— If $\gamma > 1$, lapse-rate is negative which means that *temperature decreases with increase of height*.

Following points are worth noting:

— The value of γ in atmosphere varies with height.

— Up to an elevation of 11 km, above sea level, the temperature of air *decreases* at the rate of 0.0065°C/m . From 11 km to 32 km, the temperature remains constant but *rises* above 32 km of height.

Example 2.36. Derive an expression for the pressure ratio in the troposphere if the absolute temperature is assumed to vary according to the law $T = T_0 - \alpha (Z - Z_0)$, where, T_0 is the absolute temperature at sea level and α is the temperature gradient. (Nagpur University)

Solution. The variation in altitude is given by:

$$dp = -\rho g dZ = -w dZ$$

where,

ρ = Density of air, kg/m^3 , and

w = Specific weight, N/m^3 .

Since, $\rho = \frac{p}{RT}$, substituting in above relation, we get $dp = \left(-\frac{p}{RT} \right) g dZ$

$$\text{or,} \quad \frac{dp}{p} = -\frac{g}{RT} dZ \quad \dots(i)$$

$$\text{Since,} \quad T = T_0 - \alpha (Z - Z_0) \quad \dots(\text{Given})$$

$$\therefore dT = -\alpha dZ$$

Substituting for dZ in (i), we have:

$$\frac{dp}{p} = \frac{g}{\alpha R} \frac{dT}{T}$$

Integrating between (p_0, p) to (Z_0, Z) , where suffix 0 denotes sea level conditions, we get:

$$\ln \left(\frac{p}{p_0} \right) = \frac{g}{\alpha R} \ln \left(\frac{T}{T_0} \right) = \frac{g}{\alpha R} \ln \left[\frac{T_0 - \alpha (Z - Z_0)}{T_0} \right]$$

$$\text{or,} \quad \frac{p}{p_0} = \left[1 - \frac{\alpha (Z - Z_0)}{T_0} \right] \quad \dots\text{Required expression (Ans.)}$$

Example 2.37. The temperature of the earth's atmosphere drops about 5°C for every 1000 m of elevation above the earth's surface. If the air temperature at the ground level is 15°C and the pressure is 760 mm Hg, at what elevation is the pressure 380 mm Hg? Assume that air behaves as an ideal gas. (Roorkee University)

Solution: Given: $T_0 = 15 + 273 = 288 \text{ K}$; $p_0 = 760 \text{ mm Hg}$; $p = 380 \text{ mm Hg}$;

$$\frac{dT}{dZ} = -\frac{5}{1000}^{\circ}\text{C/m}$$

Elevation Z:

Temperature lapse-rate, $L = \frac{dT}{dZ} = -\frac{g}{R} \left(\frac{\gamma - 1}{\gamma} \right)$

$$\therefore L = -\frac{5}{1000} = -\frac{g}{R} \left(\frac{\gamma - 1}{\gamma} \right)$$

or, $\frac{5}{1000} = \frac{9.81}{287} \left(\frac{\gamma - 1}{\gamma} \right)$

$$\therefore \frac{\gamma - 1}{\gamma} = 0.1463$$

Using the relation: $p = p_0 \left[1 - \frac{\gamma - 1}{\gamma} \cdot \frac{gZ}{RT_0} \right]^{\frac{\gamma}{\gamma - 1}}$

or, $380 = 760 \left[1 - 0.1463 \times \frac{9.81 \times Z}{287 \times 288} \right]^{\frac{1}{0.1463}}$

or, $\left(\frac{380}{760} \right)^{0.1463} = 1 - 0.1463 \times \frac{9.81 Z}{287 \times 288} = 1 - 1.736 \times 10^{-5} Z$

or, $1.736 \times 10^{-5} Z = 1 - \left(\frac{380}{760} \right)^{0.1463} = 0.0964$

or, $Z = \frac{0.0964}{1.736 \times 10^{-5}} = 5553 \text{ m (Ans)}$

Example 2.38. The barometric pressure at sea level is 760 mm of mercury while that on a mountain top is 735 mm. If the density of air is assumed constant at 1.2 kg/m^3 , what is the elevation of the mountain top? (Punjab University)

Solution. Given: Pressure at sea level, $p_0 = 760 \text{ mm of Hg}$

$$= \frac{760}{1000} \times (13.6 \times 1000) \times 9.81 = 101396 \text{ N/m}^2$$

Pressure at mountain, $p = 735 \text{ mm of Hg}$

$$= \frac{735}{1000} \times (13.6 \times 1000) \times 9.81 = 98060 \text{ N/m}^2$$

Density of air, $\rho = 1.2 \text{ kg/m}^3$

Height of the mountain top from sea-level, h:

It is a known fact that as the elevation above the sea-level increases, the atmospheric pressure decreases. As the density is constant (given), hence the pressure at any height 'h' above the sea-level is given by the equation,

$$p = p_0 - \rho gh \quad \text{or} \quad h = \frac{p_0 - p}{\rho g} = \frac{(101396 - 98060)}{1.2 \times 9.81} = 283.4 \text{ m (Ans.)}$$

Example 2.39. Determine the pressure at a height of 800 m above sea level if the atmospheric pressure is $10.139 \times 10^4 \text{ N/m}^2$ and temperature is 15°C at sea level assuming:

(i) Air is incompressible;

(ii) Pressure variation follows isothermal law;

(iii) Pressure variation follows adiabatic law.

Take: Density of air at sea level = 1.285 kg/m^3

Neglect variation of g with altitude.

(PTU)

Solution. Given: Height above sea-level, $Z = 8000 \text{ m}$

Pressure at sea-level, $p_0 = 10.139 \times 10^4 \text{ N/m}^2$

Temperature at sea level, $t_0 = 15^\circ\text{C} \quad \therefore T_0 = 15 + 273 = 288 \text{ K}$

Density of air, $\rho = \rho_0 = 1.285 \text{ kg/m}^3$.

Pressure p :

(i) When air is incompressible:

$$\text{We know that,} \quad \frac{dp}{dz} = -\rho g; \quad \therefore \int_{p_0}^p dp = - \int_{Z_0}^Z \rho g dz$$

$$\text{or,} \quad p - p_0 = -\rho g (Z - Z_0)$$

$$\begin{aligned} \text{or,} \quad p &= p_0 - \rho g Z \quad (\because Z_0 = \text{datum line} = 0) \\ &= 10.139 \times 10^4 - 1.285 \times 9.81 \times 8000 = \mathbf{543.2 \text{ N/m}^2} \text{ (Ans.)} \end{aligned}$$

(ii) When pressure variation follows isothermal law :

$$\text{We know that,} \quad p = p_0 e^{(-gZ/RT)} \quad \dots(\text{Eqn. 2.16})$$

$$= p_0 e^{(-gZ\rho_0/p_0)} \left[\because \frac{p_0}{\rho_0} = RT \text{ or } \frac{\rho_0}{p_0} = \frac{1}{RT} \right]$$

$$= 10.139 \times 10^4 \times e^{(-9.81 \times 8000 \times 1.285/101390)}$$

$$= 10.139 \times 10^4 \times e^{-0.9946} = 10.139 \times 10^4 \times 0.3699$$

$$= 37504 \text{ N/m}^2 \text{ or } \mathbf{3.75 \text{ N/cm}^2} \text{ (Ans.)}$$

(iii) When pressure variation follows adiabatic law ($\gamma = 1.4$):

$$\text{Using the equation: } p = p_0 \left[1 - \frac{\gamma-1}{\gamma} gZ \frac{\rho_0}{p_0} \right]^{\gamma/\gamma-1} \quad \dots[\text{Eqn.2.17}]$$

Substituting the values, we get:

$$\begin{aligned} p &= 10.139 \times 10^4 \left[1 - \frac{1.4-1.0}{1.4} \times 9.81 \times 8000 \times \frac{1.285}{10.139 \times 10^4} \right]^{\left(\frac{1.4}{1.4-1} \right)} \\ &= 101390 (1 - 0.2842)^{(1.4/0.4)} = 101390 \times (0.7158)^{3.5} \\ &= 31460 \text{ N/m}^2 \text{ or } \mathbf{3.146 \text{ N/cm}^2} \text{ (Ans)} \end{aligned}$$

Example 40. Determine the pressure and density of air at a height of 4500 m from sea-level where pressure and temperature of the air are 101400 N/m^2 and 15°C respectively. Density of air at sea-level is equal to 1.285 kg/m^3 and the temperature lapse-rate is 0.0065°C/m .

Solution. Given: Height, $Z = 4500 \text{ m}$

Pressure at sea-level; $p_0 = 101400 \text{ N/m}^2$

Temperature at sea-level, $T_0 = t + 273 = 15 + 273 = 288 \text{ K}$

Temperature lapse-rate, $L = \frac{dT}{dZ} = 0.0065^\circ \text{C/m}$

Density of air at sea level, $\rho_0 = 1.285 \text{ kg/m}^3$

We know that,
$$L = \frac{dT}{dZ} = -\frac{g}{R} \left(\frac{\gamma - 1}{\gamma} \right) \quad \dots [\text{Eqn. (2.20)}]$$

where,
$$R = \frac{p_0}{\rho_0 T_0} = \frac{101400}{1.285 \times 288} = 274$$

Substituting the values in the above equation, we get:

$$-0.0065 = -\frac{9.81}{274} \left(\frac{\gamma - 1}{\gamma} \right)$$

$$\therefore \frac{\gamma - 1}{\gamma} = \frac{0.0065 \times 274}{9.81} = 0.1815$$

$$\therefore \gamma (1 - 0.1815) = 1 \quad \text{or} \quad \gamma = \frac{1}{1 - 0.1815} = 1.22$$

Pressure (p) and density (ρ) of air at a height of 4500 m:

We know that,
$$p = p_0 \left[1 - \frac{\gamma - 1}{\gamma} gZ \frac{\rho_0}{p_0} \right]^{(\gamma/\gamma - 1)} \quad \dots [\text{Eqn. (2.17)}]$$

Substituting the values, we get:

$$\begin{aligned} p &= 101400 \left[1 - \frac{1.22 - 1}{1.22} \times 9.81 \times 4500 \times \frac{1.285}{101400} \right]^{\frac{1.22}{0.22}} \\ &= 101400 (1 - 0.1)^{5.545} = 56534 \text{ N/m}^2 \text{ or } \mathbf{5.6534 \text{ N/cm}^2 \text{ (Ans.)}} \end{aligned}$$

Also,
$$\frac{p}{\rho} = RT$$

where, p , ρ and T are pressure, density and temperature respectively at a height of 4500 m.

Now the value of T is calculated from temperature lapse-rate as follows :

$$t \text{ at } 4500 \text{ m} = t_0 + \frac{dT}{dZ} \times 4500 = 15 - 0.0065 \times 4500 = -14.25^\circ \text{C}$$

$$\therefore T = 273 + (-14.25) = 258.75 \text{ K}$$

$$\therefore \text{Density of air at a height of 4500 m; } \rho = \frac{p}{RT} = \frac{56534}{274 \times 258.75} = \mathbf{0.797 \text{ kg/m}^3 \text{ (Ans.)}}$$

Example 2.41. Calculate the pressure round an aeroplane which is flying at an altitude of 4200 m. The temperature lapse-rate is 0.0065 K/m. The pressure, temperature and density of air at ground level are 101400 N/m², 15°C and 1.285 kg/m³ respectively.

Variation of g with altitude may be neglected.

Solution. Given: Height, $Z = 4200 \text{ m}$; Lapse-rate, $L = \frac{dT}{dZ} = -0.0065 \text{ K/m}$;

$$p_0 = 101400 \text{ N/m}^2; T_0 = 15 + 273 = 288 \text{ K}; \rho_0 = 1.285 \text{ kg/m}^3$$

Pressure round the aeroplane, p ;

Let us first calculate the value of power index γ as follows:

We know that,
$$L = \frac{dT}{dZ} = -\frac{g}{R} \left(\frac{\gamma - 1}{\gamma} \right) \quad \dots [\text{Eqn. (2.20)}]$$

where,
$$R = \frac{p_0}{\rho_0 T_0} = \frac{101400}{1.285 \times 288} = 274$$

Substituting the value, we have:

$$-0.0065 = -\frac{9.81}{274} \left(\frac{\gamma - 1}{\gamma} \right)$$

or,
$$\frac{\gamma - 1}{\gamma} = \frac{0.0065 \times 274}{9.81} = 0.1815 \quad \text{or} \quad \gamma = 1.22$$

Pressure round the aeroplane is given as:
$$p = p_0 \left[1 - \frac{\gamma - 1}{\gamma} gZ \frac{\rho_0}{p_0} \right]^{\frac{\gamma}{\gamma - 1}} \quad \dots [\text{Eqn. (2.17)}]$$

$$= 101400 \left[1 - \frac{1.22 - 1}{1.22} \times 9.81 \times 4200 \times \frac{1.285}{101400} \right]^{\frac{1.22}{0.22}}$$

$$= 101400 [1 - 0.094]^{5.545} = 58656 \text{ N/m}^2 = \mathbf{5.8656 \text{ N/cm}^2 \text{ (Ans.)}}$$

HIGHLIGHTS

- The force (P) per unit area (A) is called pressure (p); mathematically, $p = \frac{P}{A}$
- Pressure head of a liquid, $h = \frac{P}{w}$ ($\because p = wh$)
where, w is the specific weight of the liquid.
- Pascal's law* states as follows:
"The intensity of pressure at any point in a liquid at rest, is the same in all directions".
- The atmospheric pressure at sea level (above absolute zero) is called standard atmospheric pressure.
 - Absolute pressure = atmospheric pressure + gauge pressure

$$p_{abs.} = p_{atm.} + p_{gauge}$$
 - Vacuum pressure = Atmospheric pressure – absolute pressure (Vacuum pressure is defined as the pressure below the atmospheric pressure)
- Manometers* are defined as the devices used for measuring the pressure at a point in fluid by balancing the column of fluid by the same or another column of liquid.
- Mechanical gauges* are the devices in which the pressure is measured by balancing the fluid column by spring (elastic element) or dead weight. Some commonly used mechanical gauges are:
 - Bourdon tube pressure gauge,
 - Diaphragm pressure gauge,
 - Bellow pressure gauge, and
 - Dead-weight pressure gauge.
- The pressure at a height Z in a static compressible fluid (gas) undergoing isothermal compression $\left(\frac{p}{\rho} = \text{const.} \right)$,

$$p = p_0 e^{(-gZ/RT)}$$

where,
 p_0 = Absolute pressure at sea-level or at ground level,
 Z = Height from sea or ground level,
 R = Gas constant, and
 T = Absolute temperature.
- The pressure and temperature at a height Z in a static compressible fluid (gas) undergoing adiabatic compression ($p/\rho^\gamma = \text{const.}$):

$$p = p_0 \left[1 - \frac{\gamma - 1}{\gamma} gZ \frac{\rho_0}{p_0} \right]^{\frac{\gamma}{\gamma - 1}} = p_0 \left[1 - \frac{\gamma - 1}{\gamma} \frac{gZ}{RT_0} \right]^{\frac{\gamma}{\gamma - 1}}$$

and, temperature,
$$T = T_0 \left[1 - \frac{\gamma - 1}{\gamma} \frac{gZ}{RT_0} \right]$$

where, p_0 , T_0 are pressure and temperature at sea-level; $\gamma = 1.4$ for air.

9. The rate at which the temperature changes with elevation is known as *Temperature Lapse-Rate*. It is given by,

$$L = \frac{-g}{R} \left(\frac{\gamma - 1}{\gamma} \right)$$

If (i) $\gamma = 1$, temperature is zero; (ii) $\gamma > 1$, temperature decreases with the increase of height.

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer:

- The force per unit area is called
(a) pressure (b) strain
(c) surface tension (d) none of the above.
- The pressure of a liquid on a surface will always act to the surface.
(a) parallel (b) normal
(c) 45° (d) 60° .
- The pressure as the depth of the liquid increases.
(a) increases
(b) decreases
(c) remain unchanged
(d) none of the above.
- The intensity of pressure in a liquid due to its depth will vary with depth.
(a) directly
(b) indirectly
(c) either of the above
(d) none of the above.
- The height of the free surface above any point is known as
(a) static head
(b) intensity of pressure
(c) either of the above
(d) none of the above.
- "The intensity of pressure at any point in a liquid at rest is the same in all directions."
The above statement is known as
(a) Kirchhoff's law (b) Pascal's law
(c) either of the above (d) none of the above.
- Any pressure measured above the absolute zero of pressure is termed as
(a) atmospheric pressure
(b) gauge pressure
(c) either of the above
(d) none of the above.
- The fundamental S.I. unit of pressure is N/m^2 ; this is also known as
(a) Pascal (b) Stoke
(c) Poise (d) none of the above.
- The devices used for measuring the pressure at a point in a fluid by balancing the column fluid by the same or another column of liquid are known as
(a) mechanical gauges
(b) manometers
(c) either of the above
(d) none of the above.
- The simplest form of manometer which can be used for measuring moderate pressures of liquid is
(a) piezometer
(b) differential manometer
(c) U-tube manometer
(d) none of the above
- Piezometers measure pressure only.
(a) absolute (b) gauge
(c) atmospheric (d) any of the above.
- A piezometer tube is not suitable for measuring pressure.
(a) positive (b) negative
(c) atmospheric (d) none of the above.
- Inclined single column manometer is useful for the measurement of pressures.
(a) small (b) medium
(c) high (d) negative.
- Which of the following is used to measure the difference in pressures between two points in a pipe, or in two different pipes?
(a) Piezometer
(b) Single column manometer
(c) Differential manometer
(d) None of the above.

15. The manometers are suitable for comparatively pressures.
 (a) low (b) high
 (c) very high (d) none of the above.
16. A Bourdon tube pressure gauge is used for measuring pressures.
 (a) low (b) high
 (c) high as well as low (d) none of the above.
17. The Bourdon tubes are generally made of
 (a) copper (b) tin
 (c) mild steel
 (d) bronze or nickel steel.
18. Which of the following is a mechanical gauge?
 (a) Diaphragm gauge
 (b) Dead weight pressure gauge
 (c) Bourdon tube pressure gauge
 (d) All of the above.
19. Which of the following is an advantage of manometers?
 (a) Good accuracy (b) High sensitivity
 (c) Little maintenance (d) All of the above.
20. Which of the following is limitation of manometers?
 (a) Fragile
 (b) Bulky and large in size
 (c) Capillary effect is created due to surface tension of manometric fluid
 (d) All of the above.

ANSWERS

- | | | | | | |
|---------|----------|---------|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (a) | 4. (a) | 5. (a) | 6. (b) |
| 7. (c) | 8. (a) | 9. (b) | 10. (a) | 11. (b) | 12. (b) |
| 13. (a) | 14. (c) | 15. (a) | 16. (c) | 17. (d) | 18. (d) |
| 19. (d) | 20. (d). | | | | |

THEORETICAL QUESTIONS

1. Define the term 'pressure'.
2. State and prove 'Pascal's Law'.
3. Define the following:
 - (i) Atmospheric pressure,
 - (ii) Gauge pressure,
 - (iii) Vacuum pressure, and
 - (iv) Absolute pressure.
4. How is pressure measured?
5. What are manometers?
6. How are manometers classified?
7. Explain briefly the following:
 - (i) Piezometer
 - (ii) U-tube manometer.
8. What are differential manometers?
9. What are mechanical gauges? Name three important mechanical gauges.
10. Explain briefly the following mechanical gauges:
 - (i) Bourdon tube pressure gauge, and
 - (ii) Diaphragm gauge.
11. Derive an expression for the pressure at a height Z from sea-level for a static air when the compression of the air is assumed isothermal. The pressure and temperature at sea-level are p_0 and T_0 respectively.
12. Prove that the pressure and temperature for an adiabatic process at a height Z from sea-level for static air are:

$$p = p_0 \left[1 - \frac{\gamma - 1}{\gamma} \cdot \frac{gZ}{RT_0} \right]^{\gamma/(\gamma - 1)}, \text{ and}$$

$$T = T_0 \left[1 - \frac{\gamma - 1}{\gamma} \cdot \frac{gZ}{RT_0} \right] \text{ where } p_0 \text{ and } T_0 \text{ are the}$$

pressure and temperature at sea-level.
13. What do you understand by the term 'Temperature lapse-Rate'? Obtain an expression for the Temperature Lapse-Rate.

UNSOLVED EXAMPLES

1. If a mercury barometer reads 700 mm and a Bourdon gauge at a point in a flow system reads 500 kN/m^2 , what is the absolute pressure at the point?
 [Ans. $595 \text{ kN/m}^2 \text{ abs.}$]
2. Find the depth of a point below water surface in sea where the pressure intensity is 100.55 kN/m^2 . Specific gravity of sea water is 1.025.
 [Ans. 10 m]

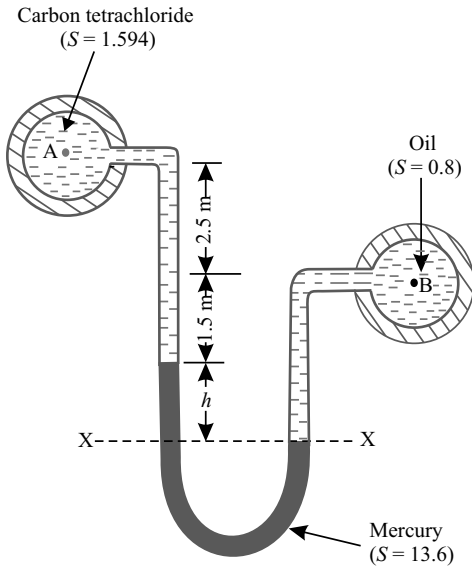


Fig. 2.41

3. Convert a pressure head of 100 m of water to
(i) kerosene of specific gravity 0.81, and
(ii) carbon tetrachloride of specific gravity 1.6.

[Ans. (i) 123.4 m, (ii) 62.5 m]

[Hint. $h_1 S_1 = h_2 S_2$]

4. As shown in Fig. 2.41, pipe A contains carbon tetrachloride of specific gravity 1.594 under a pressure of 103 kN/m^2 and pipe B contains oil of specific gravity 0.8. If the pressure in the pipe B is 171.6 kN/m^2 and the manometric fluid is mercury, find the difference h between the levels of mercury. [Ans. 142 mm]

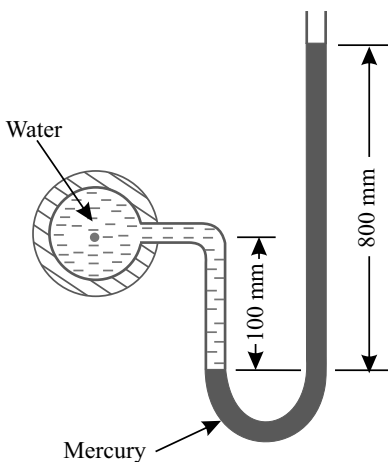


Fig. 2.42

5. The pressure of water in a pipeline was measured by means of a simple manometer containing

mercury. The reading of the manometer is shown in Fig. 2.42. Calculate the pressure of the oil, if the difference of mercury level be 0.5 m.

[Ans. 2.0 m; 19.62 kPa]

6. A U-tube containing mercury is used to measure the pressure of an oil of specific gravity 0.8 as shown in Fig. 2.43. Calculate the pressure of the oil, if the difference of mercury level be 0.5 m.

[Ans. 14 m]

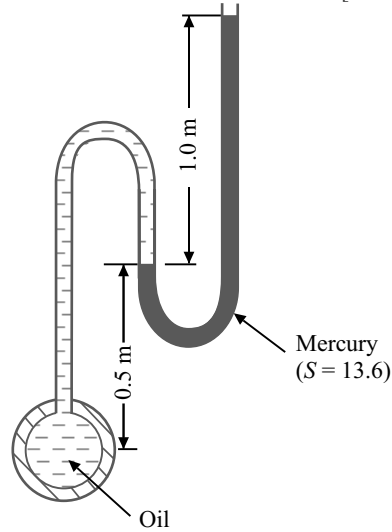
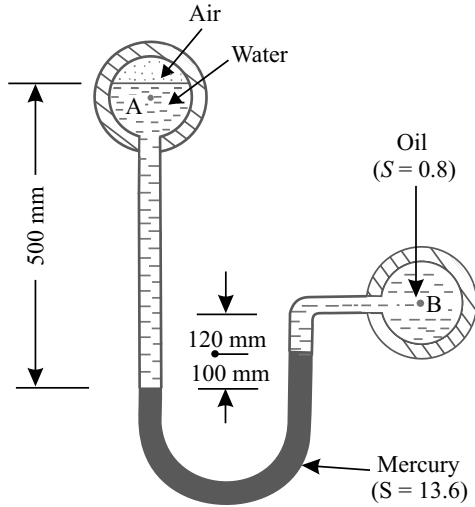


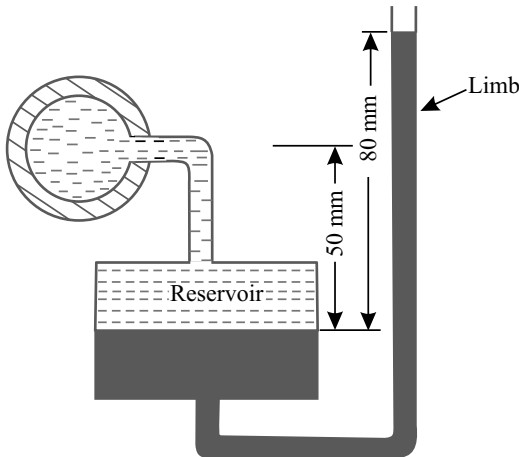
Fig. 2.43

7. A simple manometer (U-tube) containing mercury is connected to a pipe in which an oil of specific gravity 0.8 is flowing. The pressure in the pipe is vacuum. The other end of the manometer is open to atmosphere. Find the vacuum pressure in pipe, if the difference of mercury level in the two limbs is 200 mm and height of oil in the left-limb from the centre of the pipe is 150 mm below. [Ans. -278.6 kPa]
8. Fig. 2.44 shows a differential manometer connected at two points A and B. If at A air pressure is 78.5 kN/m^2 , find the absolute pressure at B. [Ans. 69.1 kN/m^2]
9. A single column vertical manometer is connected to a pipe containing oil of specific gravity 0.9. The area of the reservoir is 80 times the area of the manometer tube. The reservoir contains mercury of specific gravity 13.6. The level of mercury in the reservoir is at a height of 300 mm below the centre of the pipe and difference of mercury levels in the reservoir and right limb is 500 mm. Find the pressure in the pipe.

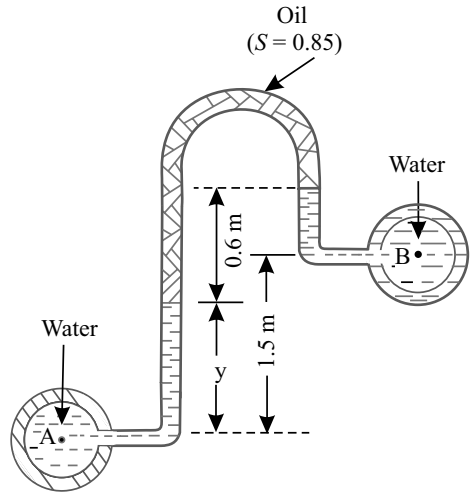
[Ans. 64.7 kN/m^2]

**Fig. 2.44**

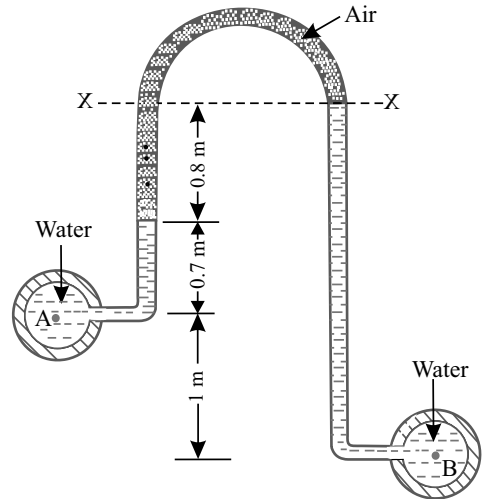
10. A micrometer, having a ratio of reservoir to limb areas as 40, was used to determine the pressure in a pipe containing water. Determine the pressure in the pipe for manometer reading shown in Fig. 2.45. [Ans. 688.8 mm]

**Fig. 2.45**

11. A U-tube mercury differential manometer is used to measure the difference of pressure between inlet throat of a venturimeter placed with its axis horizontal in a pipeline. Calculate the difference in pressure between inlet and throat when the manometer reading is 250 mm and water flows through the pipeline. [Ans. 3.15 m of water]
12. Calculate the pressure difference between two points A and B in Fig. 2.46. [Ans. 13.83 kN/m²]

**Fig. 2.46**

13. Find the difference in pressures between points A and B in Fig. 2.47. Neglect weight of air. [Ans. $p_B - p_A = 17.66 \text{ kN/m}^2$]

**Fig. 2.47**

14. An inverted differential manometer containing an oil of sp. gravity 0.9 is connected to find the difference of pressures at two points of a pipe containing water. If the manometer reading is 400 mm, find the difference of pressures. [Ans. 40 mm of water]
15. Fig. 2.48 shows a mercury manometer fitted to a venturimeter in which water is flowing. Determine the difference of pressures between the points A and B. [Ans. -18.94 kN/m^2]

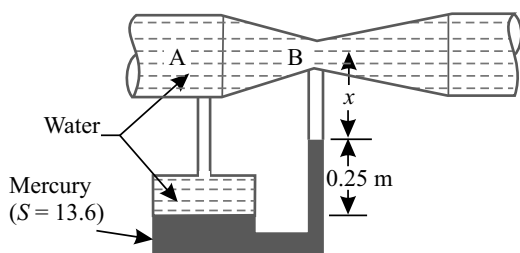


Fig. 2.48

16. Fig. 2.49 shows a glass funnel fitted to a U-type manometer. The manometer reading is 0.25 m when the funnel is empty. What is the manometer reading when the funnel is completely filled with water? [Ans. 0.4 m]

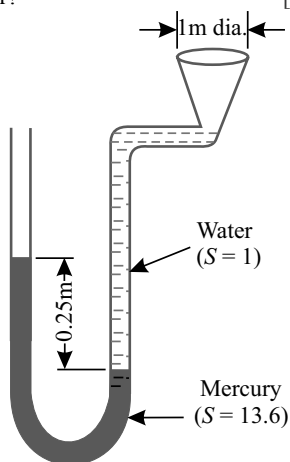


Fig. 2.49

17. Calculate the pressure at point A in Fig. 2.50. The slope of the tube is 4 horizontal to 1 vertical. The diameters of reservoir and tube are 50 mm and 5 mm respectively. The fluid in the pipe is air and that in the manometer is kerosene (sp. gr. = 0.8). [Ans. -0.494 kN/m^2]

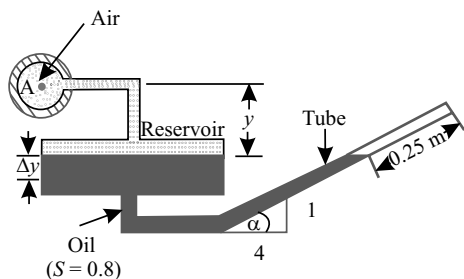


Fig. 2.50

18. Find the gauge readings at A and B in Fig. 2.51. [Ans. 17.39 kN/m^2 (vacuum), 12.61 kN/m^2]

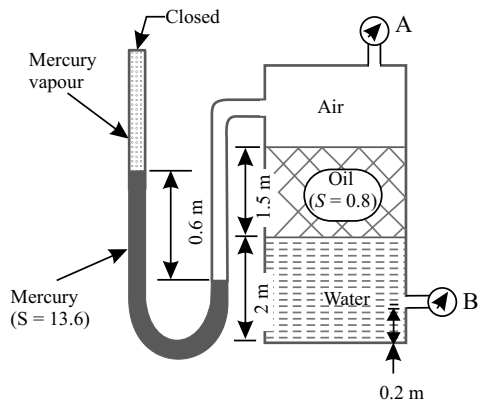


Fig. 2.51

19. Find the pressure between L and M in Fig. 2.52. [Ans. 8.32 kN/m^2]
20. Calculate the pressure at a height of 8000 m above sea-level if the atmospheric pressure is 101.3 kN/m^2 and temperature is 15°C at the sea-level assuming (i) air is incompressible, (ii) pressure variation follows adiabatic law, and (iii) pressure variation follows isothermal law. Take the density of air at the sea-level equal to 1.285 kg/m^3 . Neglect variation of g with altitude. [Ans. (i) 607.5 N/m^2 , (ii) 31.5 kN/m^2 , (iii) 37.45 kN/m^2]

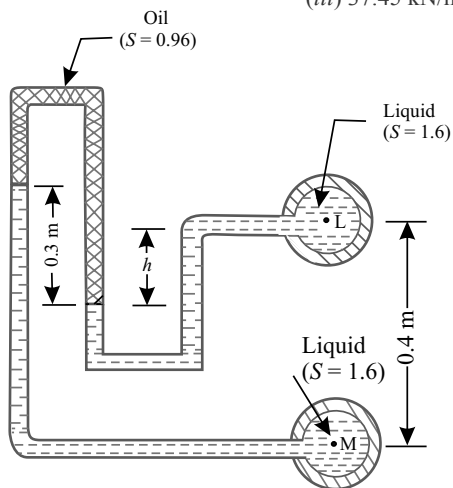
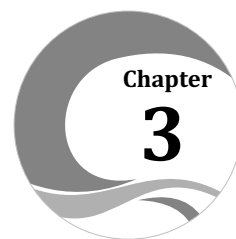
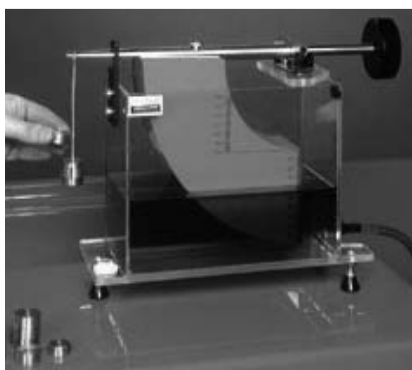


Fig. 2.52

21. The atmospheric pressure at the sea-level is 101.3 kN/m^2 and the temperature is 15°C . Calculate the pressure 8000 m above sea-level, assuming (i) air is incompressible, (ii) isothermal variation of pressure and density, and (iii) adiabatic variation of pressure and density. Assume density of air at sea-level as 1.285 kg/m^3 . Neglect variation of 'g' with altitude. [Ans. (i) 501.3 N/m^2 , (ii) 37.45 kN/m^2 , (iii) 31.5 kN/m^2]



HYDROSTATIC FORCES ON SURFACES

- 3.1. Introduction
- 3.2. Total pressure and centre of pressure
- 3.3. Horizontally immersed surface
- 3.4. Vertically immersed surface
- 3.5. Inclined immersed surface
- 3.6. Curved immersed surface
- 3.7. Dams.
- 3.8. Possibilities of dam failure

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

3.1. INTRODUCTION

In chapter 2, we have studied that a liquid, at rest, exerts some pressure on all sides of the container. The intensity of pressure (p) was related to specific weight w of the liquid and vertical depth h of the point by eqn. $p = wh$. In this chapter, we shall discuss the total pressure on a surface and its position. The term '*hydrostatics*' means the study of pressure, exerted by a liquid *at rest*. The direction of such a pressure is *always perpendicular to the surface, on which it acts*.

3.2. TOTAL PRESSURE AND CENTRE OF PRESSURE

Total pressure. It is defined as the *force exerted by static fluid on a surface (either plane or curved) when*

the fluid comes in contact with the surface. This force is always at right angle (or normal) to the surface.

Centre of pressure. It is defined as the *point of application of the total pressure on the surface.*

Now we shall discuss the total pressure exerted by a liquid on the immersed surface. The immersed surfaces may be:

- | | |
|------------------------------|----------------------------|
| 1. Horizontal plane surface; | 2. Vertical plane surface; |
| 3. Inclined plane surface; | 4. Curved surface. |

3.3. HORIZONTALLY IMMERSED SURFACE

Total Pressure (P):

Refer to Fig. 3.1. Consider a plane horizontal surface immersed in a liquid.

Let, A = Area of the immersed surface,

$\bar{x}$ = Depth of horizontal surface from the liquid,
and

w = Specific weight of the liquid.

The total pressure on the surface,

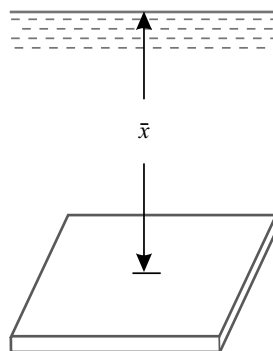


Fig. 3.1. Horizontally immersed surface.

$$\begin{aligned}
 P &= \text{Weight of the liquid above the immersed surface} \\
 &= \text{Specific weight of liquid} \times \text{volume of liquid} \\
 &= \text{Specific weight of liquid} \times \text{area of surface} \times \text{depth of liquid} \\
 &= wA\bar{x}
 \end{aligned}$$

3.4. VERTICALLY IMMERSED SURFACE

Consider a plane vertical surface of arbitrary shape immersed in a liquid as shown in Fig. 3.2.

Let, A = Total area of the surface,

G = Centre of the area of the surface,

$\bar{x}$ = Depth of centre of area,

OO = Free surface of liquid, and

$\bar{h}$ = Distance of centre of pressure from free surface of liquid.

(a) Total pressure (P):

Consider a thin horizontal strip of the surface of thickness dx and breadth b . Let the depth of the strip be x . Let the intensity of pressure on strip be p ; this may be taken as uniform as the strip is extremely small. Then,

$$p = wx$$

where,

w = specific weight of the liquid.

$$\text{Total pressure on the strip} = p \cdot b \cdot dx = wx \cdot b \cdot dx$$

$$\text{Total pressure on the whole area, } P = \int wx \cdot b \cdot dx = w \int b \cdot dx \cdot x$$

$$\text{But, } \int b \cdot dx \cdot x = \text{Moment of the surface area about the liquid level} = A\bar{x}$$

$$\therefore P = wA\bar{x} \quad \dots [\text{same as in Art. 3.3}]$$

or, the total pressure on a surface is equal to the *area multiplied by the intensity of pressure at the centre of area of the figure*.

The eqn., $P = wA\bar{x}$ holds good for all surfaces whether flat or curved.

(b) Centre of pressure ($\bar{h}$):

The intensity of pressure on an immersed surface is not uniform, but *increases with depth*. As the pressure is greater over the lower portion of the figure, therefore the resultant pressure, on any immersed surface will act at some point, below the centre of gravity of the immersed surface and towards the lower edge of the figure. *The point through which this resultant pressure acts is known as 'centre of pressure'* and is always expressed in terms of *depth* from the liquid surface.

Referring to Fig. 3.2, let C be the centre of pressure of the immersed figure. Then the resultant pressure P will act through the point.

Let, $\bar{h}$ = Depth of centre of pressure below free liquid surface, and

I_0 = Moment of inertia of the surface about OO .

Consider the horizontal strip of thickness dx . Total pressure on strip = $w \cdot x \cdot b \cdot dx$

$$\text{Moment of this pressure about free surface } OO = (w \cdot x \cdot b \cdot dx) x = w \cdot x^2 \cdot b \cdot dx$$

$$\text{Total moment of all such pressures for whole area, } M = \int w \cdot x^2 \cdot b \cdot dx = w \int x^2 \cdot b \cdot dx$$

$$\text{But, } \int x^2 \cdot b \cdot dx = I_0 = \text{Moment of inertia of the surface about the free surface } OO$$

(or second moment of area)

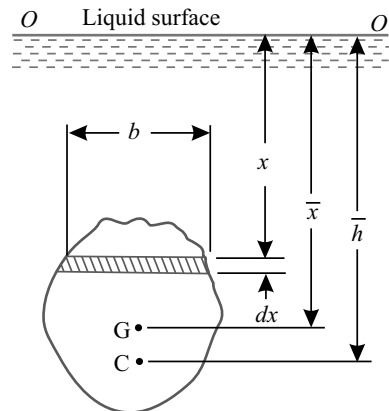


Fig. 3.2. Vertically immersed surface.

$$M = wI_0 \quad \dots(i)$$

The sum of the moments of the pressure is also equal to $P \times \bar{h}$... (ii)

Now equating eqns. (i) and (ii), we get:

$$P \times \bar{h} = wI_0$$

$$wA\bar{x} \times \bar{h} = wI_0 \quad (\because P = wA\bar{x})$$

$$\bar{h} = \frac{I_0}{A\bar{x}} \quad \dots(iii)$$

Also,

$$I = I_G + Ah^2 \quad \text{(Theorem of parallel axis)}$$

where,

I_G = Moment of inertia of the figure about horizontal axis through its centre of gravity, and

h = Distance between the free liquid surface and the centre of gravity of the figure ($\bar{x}$ in this case)

Thus rearranging equation (iii), we have

Table 3.1. The Centre of Gravity (G) and Moment of Inertia (I) of Some Important Geometrical Figures:

S.No.	Name of figure	C.G. from the base	Area	I about an axis passing through C.G. and parallel to the base	I about base
1.	Triangle Fig. 3.3	$x = \frac{h}{3}$	$\frac{bh}{2}$	$\frac{bh^3}{36}$	$\frac{bh^3}{12}$
2.	Rectangle Fig. 3.4	$x = \frac{d}{2}$	bd	$\frac{bd^3}{12}$	$\frac{bd^3}{3}$
3.	Circle Fig. 3.5	$x = \frac{d}{2}$	$\frac{\pi d^2}{4}$	$\frac{\pi d^4}{64}$	—
4.	Trapezium Fig. 3.6	$x = \left[\frac{2a+b}{a+b} \right] \frac{h}{3}$	$\left(\frac{a+b}{2} \right) h$	$\left(\frac{a^2 + 4ab + b^2}{3b(a+b)} \right) \times h^2$	—

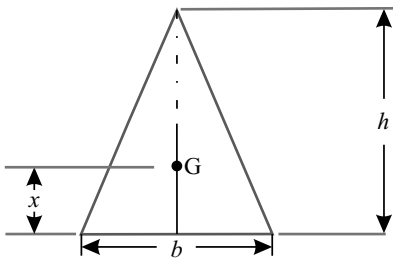


Fig. 3.3

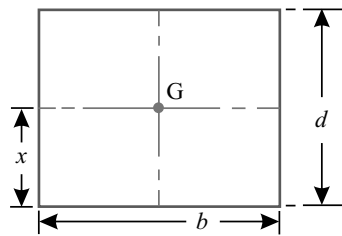


Fig. 3.4

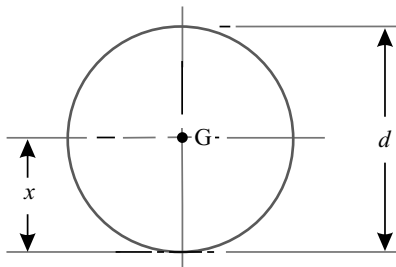


Fig. 3.5

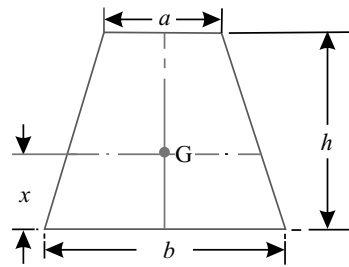


Fig. 3.6

$$\bar{h} = \frac{I_G + A\bar{x}^2}{A\bar{x}} = \frac{I_G}{A\bar{x}} + \bar{x}$$

Hence, centre of pressure,

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x} \quad \dots(3.2)$$

Example 3.1. Fig. 3.7 shows a circular plate of diameter 1.2 m placed vertically in water in such a way that the centre of the plate is 2.5 m below the free surface of water. Determine: (i) Total pressure on the plate. (ii) Position of centre of pressure.

Solution. Diameter of the plate, $d = 1.2$ m

Area,

$$A = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times 1.2^2 = 1.13 \text{ m}^2$$

$$\bar{x} = 2.5 \text{ m}$$

(i) **Total pressure, P:**

Using the relation:

$$\begin{aligned} P &= wA\bar{x} = 9.81 \times 1.13 \times 2.5 \\ &= 27.7 \text{ kN (Ans.)} \end{aligned}$$

(ii) **Position of centre of pressure, $\bar{h}$:**

Using the relation:

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x}$$

where,

$$I_G = \frac{\pi}{64} d^4 = \frac{\pi}{64} \times 1.2^4 = 0.1018 \text{ m}^4$$

$$\bar{h} = \frac{0.1018}{1.13 \times 2.5} + 2.5 = 2.536 \text{ m}$$

i.e.

$$\bar{h} = 2.536 \text{ m (Ans.)}$$

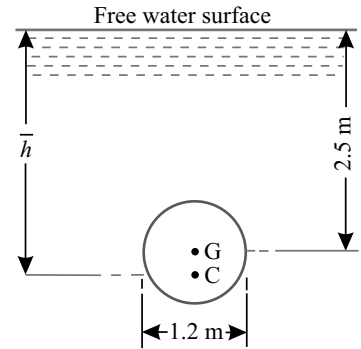


Fig. 3.7

Example 3.2. A rectangular plate 3 metres long and 1 metre wide is immersed vertically in water in such a way that its 3 metres side is parallel to the water surface and is 1 metre below it. Find: (i) Total pressure on the plate, and (ii) Position of centre of pressure.

Solution. Width of the plane surface, $b = 3$ m

Depth of the plane surface, $d = 1$ m

Area of the plane surface,

$$A = b \times d = 3 \times 1 = 3 \text{ m}^2$$

$$\bar{x} = 1 + \frac{1}{2} = 1.5 \text{ m}$$

(i) **Total pressure P:**

Using the relation:

$$\begin{aligned} P &= wA\bar{x} = 9.81 \times 3 \times 1.5 \\ &= 44.14 \text{ kN (Ans.)} \end{aligned}$$

(ii) **Centre of pressure, $\bar{h}$:**

Using the relation:

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x}$$

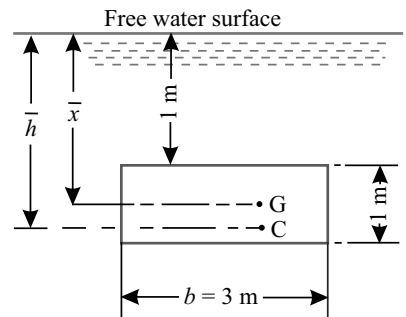


Fig. 3.8

But,
$$I_G = \frac{bd^3}{12} = \frac{3 \times 1^3}{12} = 0.25 \text{ m}^4$$

$$\therefore \bar{h} = \frac{0.25}{3 \times 1.5} + 1.5 = 1.556 \text{ m}$$

i.e.
$$\bar{h} = 1.556 \text{ m (Ans.)}$$

Example 3.3. An isosceles triangular plate of base 3 m and altitude 3 m is immersed vertically in an oil of specific gravity 0.8. The base of the plate coincides with the free surface of oil. Determine:

- (i) Total pressure on the plate; (ii) Centre of pressure.

Solution. Base of the plate, $b = 3 \text{ m}$

Height of the plate, $h = 3 \text{ m}$

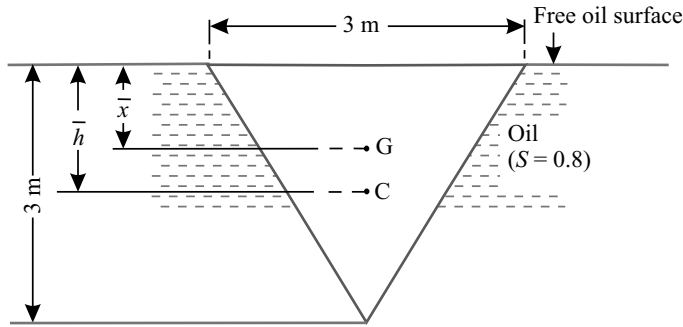


Fig. 3.9

Area,
$$A = \frac{b \times h}{2} = \frac{3 \times 3}{2} = 4.5 \text{ m}^2$$

Specific gravity of oil, $S = 0.8$

The distance of C.G. from the free surface of oil,

$$\bar{x} = \frac{1}{3}h = \frac{1}{3} \times 3 = 1 \text{ m}$$

- (i) Total pressure on the plate, P :

We know that,
$$P = wA\bar{x}$$

$$= (0.8 \times 9.81) \times 4.5 \times 1$$

$$P = 35.3 \text{ kN (Ans.)}$$

- (ii) Centre of pressure, $\bar{h}$:

Centre of pressure is given by the relation:

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x} = \frac{(bh^3/36)}{A\bar{x}} + \bar{x}$$

$$= \frac{(3 \times 3^3/36)}{4.5 \times 1} + 1$$

$$\bar{h} = 1.5 \text{ m (Ans.)}$$

Example 3.4. A circular opening, 2.5 m diameter, in a vertical side of tank is closed by a disc of 2.5 m diameter which can rotate about a horizontal diameter. Determine:

- (i) The force on the disc;
(ii) The torque required to maintain the disc in equilibrium in vertical position when the head of water above horizontal diameter is 3.5 m.

Solution. Diameter of the opening, $d = 2.5 \text{ m}$

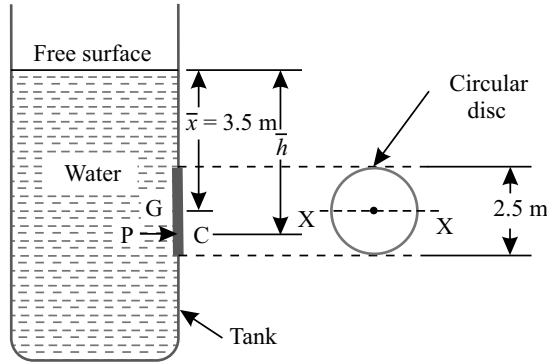


Fig. 3.10

∴ Area of the opening,

$$A = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times 2.5^2 = 4.91 \text{ m}^2$$

Depth of C.G.,

$$\bar{x} = 3.5 \text{ m}$$

(i) **Force on the disc, P:**

Using the relation:

$$\begin{aligned} P &= wA\bar{x} = 9.81 \times 4.91 \times 3.5 \\ &= \mathbf{168.6 \text{ kN (Ans.)}} \end{aligned}$$

(ii) **Torque required, T:**

In order to determine the torque (T) required to maintain the disc in equilibrium, let us first calculate the point of application of force acting on the disc, i.e. centre of pressure of the force P . The depth of centre of pressure ($\bar{h}$) is given by the relation:

$$\begin{aligned} \bar{h} &= \frac{I_G}{A\bar{x}} + \bar{x} = \frac{(\pi/64 \times d^4)}{(\pi/4 \times d^2) \bar{x}} + \bar{x} \quad \left[\because I_G = \frac{\pi}{64} \times d^4 \right] \\ &= \frac{(\pi/64 \times 2.5^4)}{(\pi/4 \times 2.5^2) \times 3.5} + 3.5 = 3.61 \text{ m} \end{aligned}$$

i.e., the force P is acting at a distance of 3.61 m from the free surface. Moment of this force about horizontal diameter $X-X$

$$\begin{aligned} &= P(\bar{h} - \bar{x}) = 168.6 (3.61 - 3.5) \\ &= \mathbf{18.55 \text{ kNm.}} \end{aligned} \quad (\text{anticlockwise})$$

Hence a torque (T) of **18.55 kNm** must be applied on the disc in the **clockwise direction** to maintain the disc in equilibrium position. (Ans.)

Example 3.5. A square aperture in the vertical side of a tank has one diagonal vertical and is completely covered by a plane plate hinged along one of the upper sides of the aperture. The diagonals of the aperture are 2.4 m long and the tank contains a liquid of specific gravity 1.2. The centre of aperture is 1.8 m below the free surface. Calculate:

(i) The thrust exerted on the plate by the liquid;

(ii) The position of its centre of pressure.

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Solution. Refer to Fig. 3.11

Diagonal of aperture, $PR = QS = 2.4 \text{ m}$

Area of square aperture, $A = \text{area of } \triangle PQR + \text{area of } \triangle PSR.$

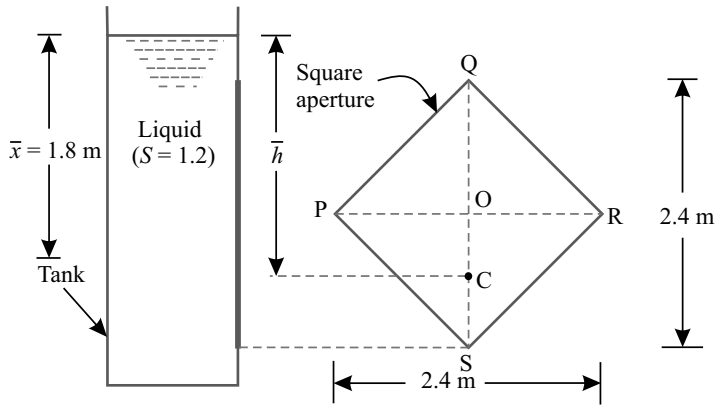


Fig. 3.11

$$\begin{aligned}
 &= \frac{1}{2} PR \times OQ + \frac{1}{2} PR \times OS \\
 &= \frac{1}{2} \times 2.4 \times \left(\frac{2.4}{2} \right) + \frac{1}{2} \times 2.4 \times \left(\frac{2.4}{2} \right) = 2.88 \text{ m}^2
 \end{aligned}$$

Depth of centre of aperture plate from free liquid surface, $\bar{x} = 1.8 \text{ m}$

(i) **Thrust exerted on the plate P :**

Pressure force or thrust on the plate,

$$P = wA\bar{x} = (1.2 \times 9.81) \times 2.88 \times 1.8 = \mathbf{61.026 \text{ kN (Ans.)}}$$

(ii) **The position of its centre of pressure, $\bar{h}$:**

Centre of pressure is given by the relation:

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x}$$

where,

I_G = M. O. I of $PQRS$ about diagonal PR .

= M.O.I. of $\triangle PQR$ + M.O.I. of $\triangle PSR$...about PR

$$= \frac{2.4 \times (1.2)^3}{12} + \frac{2.4 \times (1.2)^3}{12} = 0.6912 \text{ m}^4 \quad (\because OQ = OS)$$

[$\because$ The M.O.I. of a triangle about its base equals $\frac{\text{base} \times (\text{height})^3}{12}$]

$$\therefore \bar{h} = \frac{0.6912}{2.88 \times 1.8} + 1.8 = \mathbf{1.933 \text{ m (Ans.)}}$$

Example 3.6. A trapezoidal plate of parallel sides l and $2l$ and height h immersed vertically in water with its side of length l horizontal and topmost. The top edge is at a depth h below the water surface. Determine:

(i) The total force on one side of the plate.

(ii) The location of the centre of pressure.

Solution. Refer to Fig. 3.12, the trapezium can be considered to be made of:

(i) A rectangle: l (width) $\times$ h (height)

(ii) A triangle: l (base) $\times$ h (height)

(i) **Total force on one side of the plate P :**

Refer to Fig. 3.13.

For Rectangular part:

Pressure force P_1 on the rectangular part,

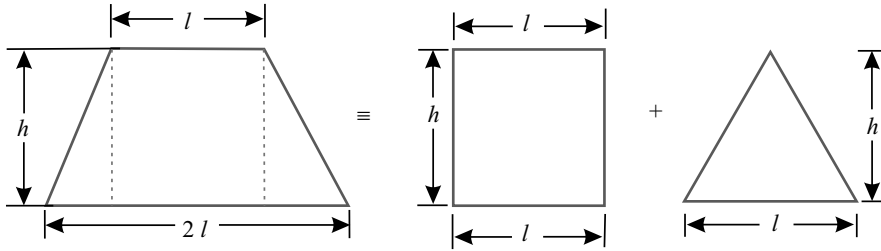


Fig. 3.12

$$P_1 = w(l \times h) \left(h + \frac{h}{2} \right) = \frac{3}{2} w l h^2$$

Centre of pressure of force P_1 ,

$$\begin{aligned} \bar{h}_1 &= \frac{I_G}{A\bar{x}} + \bar{x} = \frac{(l \times h^3 / 12)}{(l \times h) \times \left(h + \frac{h}{2} \right)} + \left(h + \frac{h}{2} \right) \\ &= \frac{h}{18} + \frac{3h}{2} = \frac{14}{9} h \end{aligned}$$

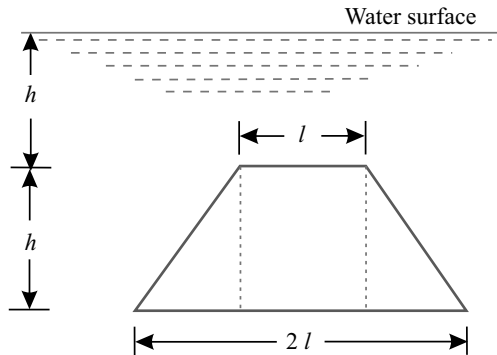


Fig. 3.13

For Triangular part:

Pressure force on the triangular part,

$$\begin{aligned} P_2 &= w \left(\frac{1}{2} \times l \times h \right) \times \left(h + \frac{2}{3} h \right) \\ &= \frac{5}{6} w l h^2 \end{aligned}$$

Centre of pressure of force P_2 ,

$$\begin{aligned} \bar{h}_2 &= \frac{I_G}{A\bar{x}} + \bar{x} = \frac{(l h^3 / 36)}{\left(\frac{1}{2} l \times h \right) \times \left(h + \frac{2}{3} h \right)} + \left(h + \frac{2}{3} h \right) \\ &= \frac{h}{30} + \frac{5h}{3} = \frac{51}{30} h \end{aligned}$$

$\therefore$ Total force/thrust, $P = P_1 + P_2$

$$= \frac{3}{2} w l h^2 + \frac{5}{6} w l h^2 = \frac{7}{3} w l h^2 \text{ (Ans.)}$$

(ii) The location of the centre of pressure, $\bar{h}$:

Centre of pressure of total force,

$$\begin{aligned}\bar{h} &= \frac{P_1 \bar{h}_1 + P_2 \bar{h}_2}{P} \\ &= \frac{\frac{3}{2} w l h^2 \times \frac{14}{8} h + \frac{5}{6} w l h^2 \times \frac{51}{30} h}{\frac{7}{3} w l h^2} = \frac{45}{28} h\end{aligned}$$

i.e.
$$\bar{h} = \frac{45}{28} h \text{ (Ans.)}$$

Example 3.7. A trapezoidal 2 m wide at the bottom and 1 m deep has side slopes 1: 1. Determine:

(i) Total pressure;

(ii) Centre of pressure on the vertical gate closing the channel when it is full of water.

Solution. Refer to Fig. 3.14

(i) Total Pressure, P :

For rectangle:

Area, $A_1 = 2 \times 1 = 2 \text{ m}^2$

$$\bar{x} = \frac{1}{2} = 0.5 \text{ m}$$

$$P_1 = w A \bar{x} = 9.81 \times 2 \times 0.5 = 9.81 \text{ kN}$$

This acts at a depth $\bar{h}_1$.

But,
$$\bar{h}_1 = \frac{I_G}{A \bar{x}} + \bar{x} = \frac{(2 \times 1^3 / 12)}{2 \times 0.5} + 0.5 = 0.6 \text{ m} \quad \dots \text{from the top}$$

For triangles:

Area, $A_2 = 2 \times \frac{1}{2} \times 1 \times 1 = 1 \text{ m}^2$ (there are two triangles); $\bar{x} = \frac{1}{3} \text{ m}$

$$P_2 = w A \bar{x} = 9.81 \times 1 \times \frac{1}{3} = 3.27 \text{ kN}$$

This acts at a depth of $\bar{h}_2$.

But,
$$\bar{h}_2 = \frac{I_G}{A \bar{x}} + \bar{x} = \frac{(2 \times 1^3 / 36)}{1 \times 1/3} + \frac{1}{3} = 0.5 \text{ m} \quad \dots \text{from the top.}$$

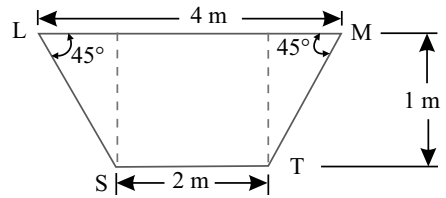
i.e.
$$\bar{h}_2 = 0.5 \text{ m}$$

Total pressure,

$$P = P_1 + P_2 = 9.81 + 3.27 = \mathbf{13.08 \text{ kN (Ans.)}}$$

(ii) Centre of pressure, $\bar{h}$:Taking moments about the top, we get: $P \times \bar{h} = P_1 \times \bar{h}_1 + P_2 \times \bar{h}_2$

or,
$$\bar{h} = \frac{P_1 \bar{h}_1 + P_2 \bar{h}_2}{P} = \frac{9.81 \times 0.66 + 3.27 \times 0.5}{13.08} = \mathbf{0.62 \text{ m (Ans.)}}$$

**Fig. 3.14**

Example 3.8. An isosceles triangle of base 3 metres and altitude 6 metres is immersed vertically in water, with its axis of symmetry horizontal, as shown in Fig. 3.15. If the head of water on it is 9 metres, determine:

(i) Total pressure on the plate, and (ii) The position of the centre of pressure.

Solution. Area of the triangle,

$$A = \frac{1}{2} \times 3 \times 6 = 9\text{m}^2$$

Depth of C.G. of the plate from the water surface,

$$\bar{x} = 9\text{m}$$

(i) **Total pressure, P :**

We know that,

$$P = wA\bar{x} = 9.81 \times 9 \times 9 \\ = 794.6 \text{ kN (Ans.)}$$

(ii) **Centre of pressure, $\bar{h}$:**

Using the relation:

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x}$$

But, I_G = moment of inertia of $\triangle ABD$ about AD + moment of inertia of $\triangle ACD$ about AD

$$= \frac{6 \times 1.5^3}{12} + \frac{6 \times 1.5^3}{12} \\ = 3.375 \text{ m}^4 \\ \bar{h} = \frac{3.375}{9 \times 9} + 9 = 9.04 \text{ m (Ans.)}$$

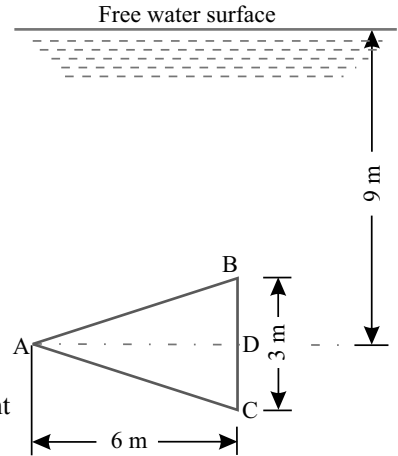


Fig. 3.15

Example 3.9. A circular lamina of radius R is kept immersed in a liquid such that its top most point A is on the free surface. Determine the depth and width of the horizontal chord BC so that the total thrust due to hydrostatic pressure on the triangle ABC is maximum. (UPTU)

Solution. Refer to Fig. 3.16.

The total thrust/pressure on the submerged triangle ABC is,

$$F = P = wA\bar{x} = w \times \left(\frac{1}{2} \times b \times h \right) \times \frac{2h}{3} = \frac{1}{3} w b h^2$$

But, $h = R + \sqrt{R^2 - b^2}$ (O is the centre of the circle)

$$\therefore F = \frac{1}{3} w b \left[R + \sqrt{R^2 - b^2} \right]^2$$

For F to be maximum, $\frac{dF}{db} = 0$

$$\text{i.e. } \frac{d}{db} \left[b(R + \sqrt{R^2 - b^2})^2 \right] = 0$$

$$b \times 2 \left(R + \sqrt{R^2 - b^2} \right) \times \frac{1}{2} (R^2 - b^2)^{-1/2} (-2b) + \left(R + \sqrt{R^2 - b^2} \right)^2 = 0$$

$$\frac{-2b^2}{\sqrt{R^2 - b^2}} + R + \sqrt{R^2 - b^2} = 0$$

$$\text{or, } -2b^2 + R\sqrt{R^2 - b^2} + R^2 - b^2 = 0$$

$$\text{or, } R\sqrt{R^2 - b^2} + R^2 - 3b^2 = 0$$

$$\text{or, } R\sqrt{R^2 - b^2} = 3b^2 - R^2$$

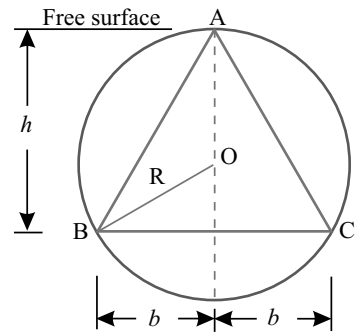


Fig. 3.16

Squaring both sides, we get:

$$R^2 (R^2 - b^2) = 9b^4 + R^4 - 6R^2b^2$$

$$R^4 - R^2b^2 = 9b^4 + R^4 - 6R^2b^2$$

$$9b^4 = 5R^2b^2$$

or

$$9b^2 = 5R^2$$

or

$$b = \sqrt{\frac{5}{9}} R = \frac{\sqrt{5}}{3} R$$

and

$$h = R + \sqrt{R^2 - \frac{5}{9}R^2} = \frac{5}{3} R$$

Hence, for maximum thrust, the depth and width of the chord are:

Depth, $h = \frac{5}{3} R$, and

width, $2b = \frac{2}{3}\sqrt{5} R$ (Ans.)

Example 3.10. Determine the total force and location of centre of pressure for plate *LMSUT* immersed vertically as shown in Fig. 3.17.

Solution. Area *LMST*, $A_1 = 2 \times 2 = 4 \text{ m}^2$

$$\text{Area } TSU, A_2 = \frac{1}{2} \times 2 \times 2 = 2 \text{ m}^2$$

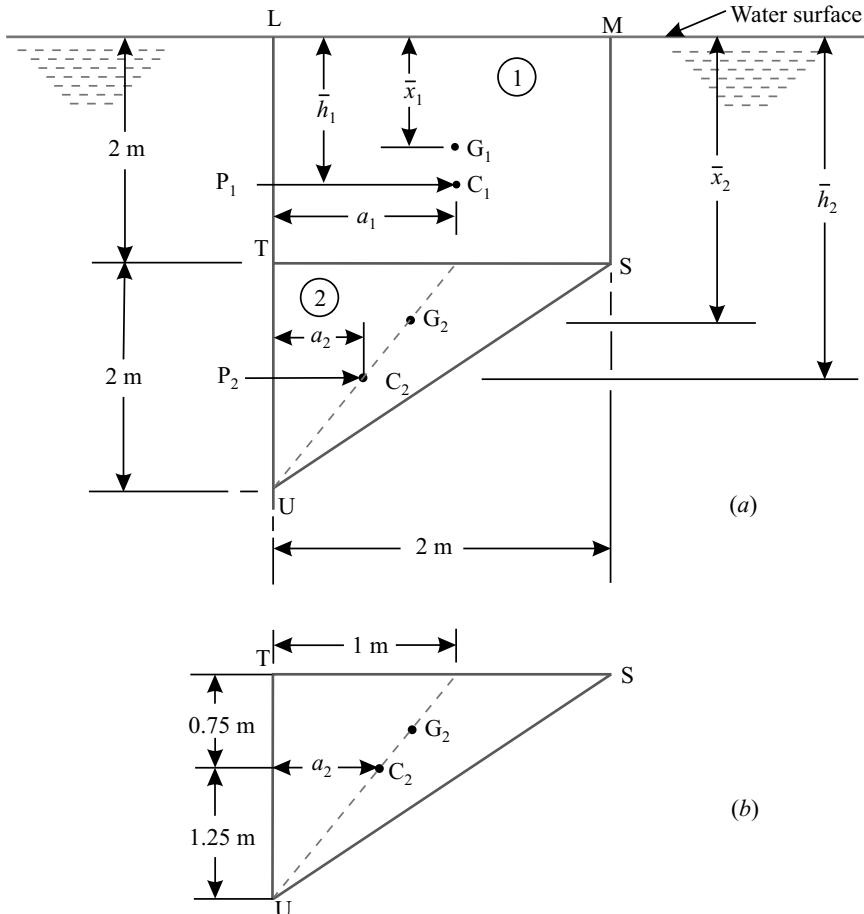


Fig. 3.17

Distance of centroid G_1 from water surface, $\bar{x}_1 = \frac{2}{2} = 1$ m

Distance of centroid G_2 from water surface, $\bar{x}_2 = 2 + \frac{2}{3} = 2.667$ m

Total pressure on area $LMST$, $P_1 = wA_1\bar{x}_1 = 9.81 \times 4 \times 1 = 39.24$ kN

Total pressure on area TSU , $P_2 = wA_2\bar{x}_2 = 9.81 \times 2 \times 2.667 = 52.33$ kN

Total pressure, $P = P_1 + P_2 = 39.24 + 52.33 = 91.57$ kN

Distance of centre of pressure (C_1) of area $LMST$ from free water surface,

$$\bar{h}_1 = \frac{I_{G_1}}{A_1\bar{x}_1} + \bar{x}_1 = \frac{\frac{2 \times 2^3}{12}}{2 \times 2 \times 1} + 1 = 1.333 \text{ m}$$

Distance of centre of pressure (C_2) of area TSU from the free water surface,

$$\bar{h}_2 = \frac{I_{G_2}}{A_2\bar{x}_2} + \bar{x}_2 = \frac{\frac{2 \times 2^3}{36}}{2 \times 2.667} + 2.667 = 2.75 \text{ m}$$

The depth ($\bar{h}$) at which the resultant force will act can be determined by taking moments of forces P_1 and P_2 about water surface.

$$\text{i.e.} \quad P_1 \times \bar{h}_1 + P_2 \times \bar{h}_2 = P \times \bar{h}$$

$$39.24 \times 1.333 + 52.33 \times 2.75 = 91.57 \times \bar{h}$$

$$\therefore \quad \bar{h} = \frac{39.24 \times 1.333 + 52.33 \times 2.75}{91.57}$$

= 2.14 m below the water surface (Ans.)

The horizontal location of centre of pressure can be obtained by taking moments of P_1 and P_2 about LTU . The force P_1 acts at 1 m from line LTU . The distance a_2 where force P_2 acts can be obtained as under:

$$\frac{1}{2} = \frac{a_2}{1.25} \quad [\text{from similarity of triangles (Fig. 3.17) (b)}]$$

$$\text{or,} \quad a_2 = 0.625 \text{ m}$$

$$\therefore \quad P_1 \cdot a_1 + P_2 \cdot a_2 = P\bar{a}$$

$$39.24 \times 1 + 52.33 \times 0.625 = 91.57 \times \bar{a}$$

$$\therefore \quad \bar{a} = \frac{39.24 \times 1 + 52.33 \times 0.625}{91.57} = 0.786 \text{ m}$$

Hence co-ordinates of centre of pressure are 2.14 m below water surface and 0.786 m from LTU . (Ans.)

Example 3.11. A sliding gate 3 m wide and 1.5 m high lies on a vertical plane and has a coefficient of friction of 0.2 between itself and guides. If the gate weighs 30 kN, find the vertical force required to raise the gate if its upper edge is at a depth of 9 m from free surface of water.

Solution. Width of the gate, $b = 3$ m

Depth/height of the gate,

$$d = 1.5 \text{ m}$$

$$\text{Area of the gate, } A = b \times d = 3 \times 1.5 = 4.5 \text{ m}^2$$

$$\text{Weight of the gate, } W = 30 \text{ kN}$$

$$\text{Co-efficient of friction, } \mu = 0.2$$

Vertical force required to raise the gate:

Depth of c.g. of the gate from water surface,

$$\bar{x} = 9 + \frac{1.5}{2} = 9.75 \text{ m}$$

Pressure force on the gate,

$$P = wA\bar{x} = 9.81 \times 4.5 \times 9.75 = 430.4 \text{ kN}$$

Force required to raise the gate

= Frictional force + weight of the gate

$$= \mu P + W$$

$$= 0.2 \times 430.4 + 30$$

$$= 116.08 \text{ kN (Ans.)}$$

Example 3.12. The hydrostatic water pressure acts only on one side and to a depth of 12 m from the top of a dock gate which is reinforced with three horizontal beams.

- Calculate the load taken by each beam.
- Locate the positions of beams in order that each carries an equal load.

Solution. Refer to Fig. 3.19. Consider an elementary strip of thickness dh at a depth h . Then for a unit width of the gate, we have:

Pressure/force on the element,

$$dP = w \times (dh \times 1) \times h = wh \, dh$$

Pressure on section 1,

$$P_1 = \int_0^{h_1} wh \, dh = \frac{w}{2} h_1^2$$

Pressure on section 2,

$$P_2 = \int_{h_1}^{h_2} wh \, dh = \frac{w}{2} (h_1^2 - h_2^2)$$

Pressure on section 3,

$$P_3 = \int_{h_2}^{h_3} wh \, dh = \frac{w}{2} (h_2^2 - h_3^2)$$

Total pressure on the gate,

$$\int_0^{h_3} wh \, dh = \frac{w}{2} h_3^2$$

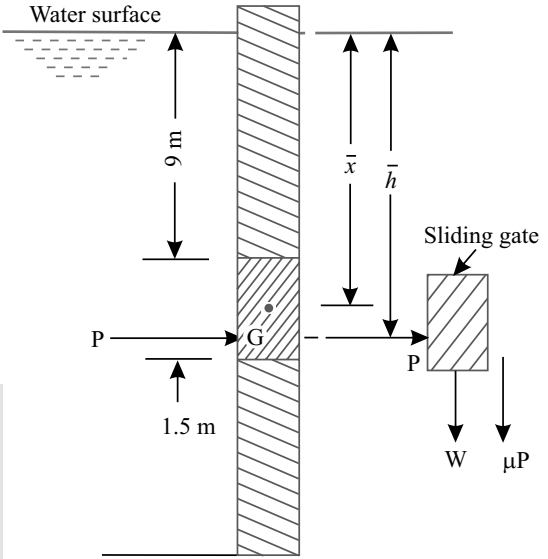
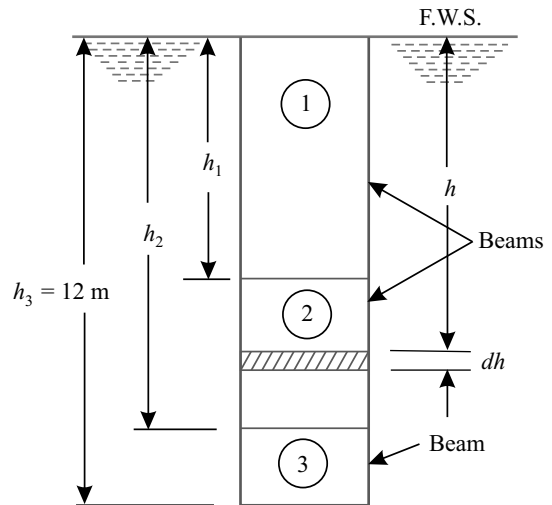
Load carried by each section is same and it equals $\frac{1}{3}$ rd of total pressure/force on the gate.

$$\text{Thus,} \quad \frac{w}{2} h_1^2 = \frac{w}{2} (h_2^2 - h_1^2) = \frac{w}{2} (h_3^2 - h_2^2) = \frac{1}{3} \times \frac{w}{2} h_3^2$$

$$\therefore \quad h_1^2 = h_2^2 - h_1^2 = 144 - h_1^2 = \frac{144}{3} = 48$$

Solving the above equations, we get:

$$h_1 = 6.93 \text{ m}; h_2 = 9.8 \text{ m}$$

**Fig. 3.18****Fig. 3.19**

(i) Load taken by each beam:

$$\text{Load taken by each beam} = \frac{w}{2} h_1^2 = \frac{9810}{2} \times 6.93^2 = 235562 \text{ (Ans.)}$$

(ii) Centres of pressure, $\bar{h}_1, \bar{h}_2, \bar{h}_3$:

$$\bar{h}_1 = \frac{2}{3} h_1 = \frac{2}{3} \times 6.93 = 4.62 \text{ m (Ans.)}$$

In order to obtain the centre of pressure for the section 2, taking moments of relevant forces about F.W.S., we get:

$$\frac{w}{2} (h_2^2 - h_1^2) \times \bar{h}_2 = \left(\frac{w}{2} h_2^2 \times \frac{2}{3} h_2 \right) - \left(\frac{w}{2} h_1^2 \times \frac{2}{3} h_1 \right)$$

$$\begin{aligned} \text{or, } \bar{h}_2 &= \frac{2}{3} \left[\frac{h_2^3 - h_1^3}{h_2^2 - h_1^2} \right] = \frac{2}{3} \left[\frac{(9.8)^3 - (6.93)^3}{(9.8)^2 - (6.93)^2} \right] \\ &= \frac{2}{3} \left[\frac{608.38}{48.015} \right] = 8.45 \text{ (Ans.)} \end{aligned}$$

Similarly for the bottom portion, the centre of pressure from the F.W.S.,

$$\begin{aligned} \bar{h}_3 &= \frac{2}{3} \left[\frac{h_3^3 - h_2^3}{h_3^2 - h_2^2} \right] = \frac{2}{3} \left[\frac{(12)^3 - (9.8)^3}{(12)^2 - (9.8)^2} \right] \\ &= \frac{2}{3} \left[\frac{786.81}{47.96} \right] = 10.94 \text{ m (Ans.)} \end{aligned}$$

Example 3.13. Fig. 3.20 shows a tank containing water and liquid (sp. gravity = 0.9) upto height 0.25 m and 0.5 m respectively. Calculate:

- Total pressure on the side of the tank;
- The position of centre of pressure from one side of the tank, which is 1.5 m wide. (U.P.S.C.)

Solution. Depth of water = 0.25 m
 Depth of liquid = 0.5 m
 Sp. gravity of liquid, $S = 0.9$
 Width of the tank = 1.5 m

(i) Total pressure on one side of the tank, P :

Total pressure (P) is calculated by drawing pressure diagram, which is shown in Fig. 3.21.

Intensity of pressure on top, $p_L = 0$

Intensity of pressure on T (or TS),

$$p_T = w_1 h_1 = (0.9 \times 9.81) \times 0.5 = 4.41 \text{ kN/m}^2$$

Intensity of pressure on the base (or MN),

$$\begin{aligned} p_M &= w_1 h_1 + w_2 h_2 = 4.41 + 9.81 \times 0.25 \\ &= 4.41 + 2.45 = 6.86 \text{ kN/m}^2 \end{aligned}$$

Now, force P_1 = area of the ΔLTS $\times$ width of the tank

$$= \frac{1}{2} \times LT \times TS \times 1.5$$

$$= \frac{1}{2} \times 0.5 \times 4.41 \times 1.5 = 1.65 \text{ kN}$$

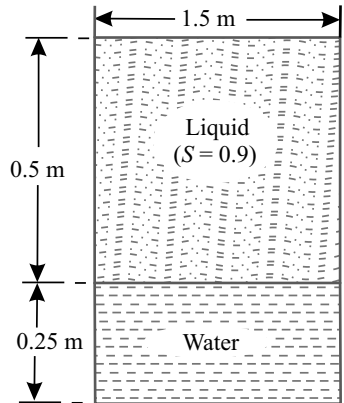


Fig. 3.20

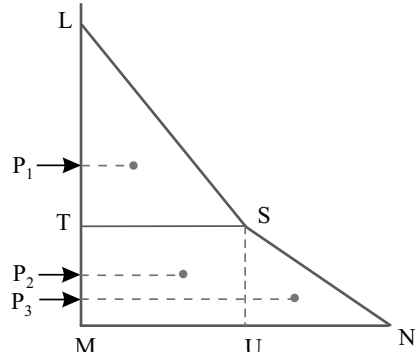


Fig. 3.21. Pressure diagram.

Force,

$$P_2 = \text{area of rectangle } MTSU \times \text{width of the tank} = MT \times TS \times 1.5 \\ = 0.25 \times 4.41 \times 1.5 = 1.65 \text{ kN}$$

$$P_3 = \text{area of } \triangle SUN \times \text{width of the tank} \\ = \frac{1}{2} \times SU \times UN \times 1.5 = \frac{1}{2} \times 0.25 \times 2.45 \times 1.5 = 0.46 \text{ kN}$$

Total pressure,

$$P = P_1 + P_2 + P_3 = 1.65 + 1.65 + 0.46 = \mathbf{3.76 \text{ kN (Ans.)}}$$

(ii) Centre of pressure, $\bar{h}$:

Taking moments of all the forces about L , we get:

$$P \times \bar{h} = P_1 \times \frac{2}{3} LT + P_2 \times \left(LT + \frac{1}{2} TM \right) + P_3 \times \left(LT + \frac{2}{3} MT \right) \\ 3.76 \times \bar{h} = 1.65 \times \frac{2}{3} \\ \times 0.5 + 1.65 \left(0.5 + \frac{1}{2} \times 0.25 \right) + 0.46 \left(0.5 + \frac{2}{3} \times 0.25 \right) \\ = 0.55 + 1.03 + 0.306 \\ \bar{h} = \mathbf{0.5016 \text{ m from the top (Ans.)}}$$

Example 3.14. An opening in a dam is covered by the use of a vertical sluice gate. The opening is 2 m wide and 1.2 m high. On the upstream of the gate the liquid of specific gravity 1.45 lies upto a height of 1.5 m above the top of the gate, whereas on the downstream side the water is available upto a height touching the top of the gate. Find:

- The resultant force acting on the gate and position of centre of pressure;
- The force acting horizontally at the top of the gate which is capable of opening it.

Assume that the gate is hinged at the bottom.

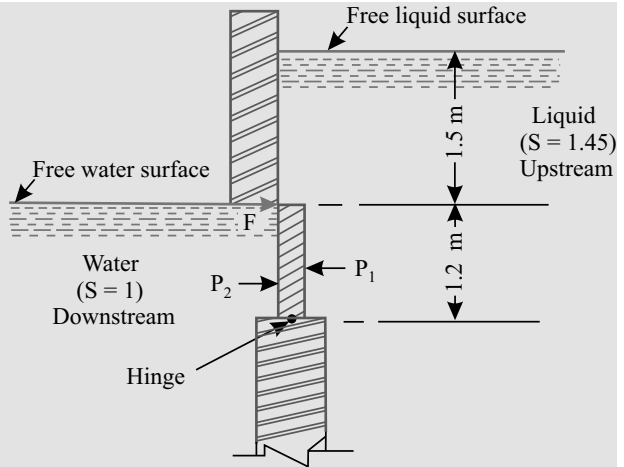


Fig. 3.22

(Rajasthan University)

Solution. Width of the gate, $b = 2 \text{ m}$

Depth of the gate, $d = 1.2 \text{ m}$

$$\text{Area, } A = b \times d = 2 \times 1.2 = 2.4 \text{ m}^2$$

Specific gravity of liquid = 1.45

Let,

P_1 = Force exerted by the liquid of sp. gravity 1.45 on the gate, and

P_2 = Force exerted by water on the gate.

(i) Resultant force, P :

Position of centre of pressure of resultant force:

We know that,

$$P_1 = wA\bar{x}_1$$

where,

$$w = 9.81 \times 1.45 = 14.22 \text{ kN/m}^3,$$

$$A = 2 \times 1.2 = 2.4 \text{ m}^2$$

Similarly,
where,

$$\bar{x}_1 = 1.5 + \frac{1.2}{2} = 2.1 \text{ m}$$

$$P_1 = 14.22 \times 2.4 \times 2.1 = 71.67 \text{ kN.}$$

$$P_2 = wA\bar{x}_2$$

$$w = 9.81 \text{ kN/m}^3.$$

$$A = 2.4 \text{ m}^2,$$

$$\bar{x}_2 = \frac{1.2}{2} = 0.6 \text{ m}$$

$$P_2 = 9.81 \times 2.4 \times 0.6 = 14.13 \text{ kN.}$$

Resultant force,

$$P = P_1 - P_2 = 71.67 - 14.13 \\ = 57.54 \text{ kN (Ans.)}$$

The force P_1 acts at a depth of $\bar{h}_1$ from free liquid surface, which is given by:

$$\bar{h}_1 = \frac{I_G}{A\bar{x}_1} + \bar{x}_1$$

where,

$$I_G = \frac{bd^3}{12} = \frac{2 \times 1.2^3}{12} = 0.288 \text{ m}^4$$

$$A = 2.4 \text{ m}^2, \bar{x} = 1.5 + \frac{1.2}{2} = 2.1 \text{ m}$$

$$\bar{h}_1 = \frac{0.288}{2.4 \times 2.1} + 2.1 = 2.157 \text{ m}$$

$$\therefore \text{Distance of } P_1 \text{ from the hinge} = (1.5 + 1.2) - \bar{h}_1 = 2.7 - 2.157 = 0.543 \text{ m}$$

Similarly the force P_2 acting at a depth of $\bar{h}_2$ from the liquid surface is given by:

$$\bar{h}_2 = \frac{I_G}{A\bar{x}_2} + \bar{x}_2$$

where,

$$I_G = 0.288 \text{ m}^4 \text{ (as above); } \bar{x}_2 = \frac{1.2}{2} = 0.6 \text{ m; } A = 2.4 \text{ m}^2$$

$$\therefore \bar{h}_2 = \frac{0.288}{2.4 \times 0.6} + 0.6 = 0.8 \text{ m}$$

$$\therefore \text{Distance of } P_2 \text{ from the hinge} = 1.2 - 0.8 = 0.4 \text{ m}$$

Now the resultant force will act at a distance given by:

$$\frac{71.67 \times 0.543 - 14.13 \times 0.4}{57.54} = 0.578 \text{ m above the hinge (Ans.)}$$

(ii) Force required to open the gate, F :

Taking moments of P_1 , P_2 and F about the hinge, we get:

$$F \times 1.2 + P_2 \times 0.4 = P_1 \times 0.543$$

$$\text{or, } F \times 1.2 + 14.13 \times 0.4 = 71.67 \times 0.543$$

$$\text{or, } F = \frac{71.67 \times 0.543 - 14.13 \times 0.4}{1.2} = 27.72 \text{ kN (Ans.)}$$

Example 3.15. For the system shown in Fig. 3.23 calculate the height H of the oil at which the rectangular hinged gate will just begin to rotate anticlockwise.

Solution.

Refer to Fig. 3.23.

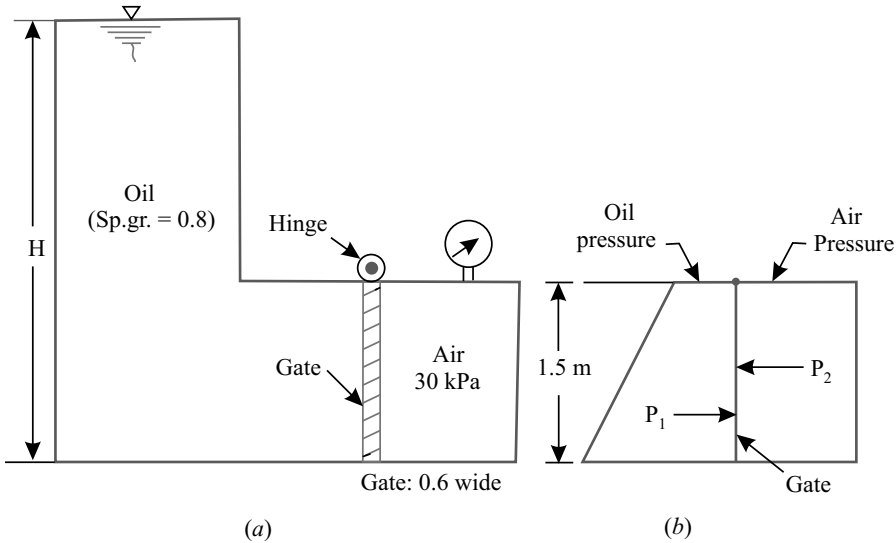
Height H of the oil:

- Force due to oil,

$$\begin{aligned}
 P_1 &= w_0 A \bar{x} \\
 &= (0.8 \times 9.81) \times (0.6 \times 1.5) \times \left[(H - 1.5) + \frac{1.5}{2} \right] \\
 &= 7.063 (H - 0.75) = 7.063 H - 5.297
 \end{aligned}$$

Centre of pressure of force P_1 ,

$$\bar{h}_1 = \frac{I_G}{A \bar{x}} + \bar{x} = \frac{0.6 \times (1.5)^3 / 12}{(0.6 \times 1.5) \times (H - 0.75)} + (H - 0.75)$$

**Fig. 3.23**

$$= \frac{0.1875}{(H - 0.75)} + (H - 0.75)$$

- Force due to air pressure,

$$\begin{aligned}
 P_2 &= p.A \\
 &= 30 \times (0.6 \times 1.5) = 27 \text{ kN}
 \end{aligned}$$

Centre of pressure of this force (below the oil surface),

$$\bar{h}_2 = (H - 0.75)$$

Taking moment about the hinge, we get:

$$\begin{aligned}
 P_1 \times [\bar{h}_2 - (H - 1.5)] &= P_2 \times [\bar{h}_2 - (H - 1.5)] \\
 (7.063 H - 5.297) \left\{ \frac{0.1875}{(H - 0.75)} + (H - 0.75) - (H - 1.5) \right\} \\
 &= 27 \times \{ (H - 0.75) - (H - 1.5) \} \\
 (7.063 H - 5.297) \left\{ \frac{0.1875}{(H - 0.75)} + 0.75 \right\} &= 27 \times 0.75
 \end{aligned}$$

On solving by trial and error, we get:

$$H = 4.324 \text{ m (Ans.)}$$

Example 3.16. A tank of 1m length and of cross-section shown in fig. 3.24 contains water. The tank is made of 4 mm steel plates.

- (i) What is the force on the bottom due to water?
 (ii) What are the longitudinal tensile stresses in the side walls AB if (a) the tank is suspended from the top and (b) it is supported at the bottom?

Solution.

Refer to Fig. 3.24

- (i) **Force on the bottom:**

Force on the bottom due to water,

$$\begin{aligned} P_{\text{bottom}} &= wA\bar{x} \\ &= 9.81 \times (0.6 \times 1.0) \times 0.75 \\ &= \mathbf{4.414 \text{ kN (Ans.)}} \end{aligned}$$

- (ii) **Longitudinal tensile stresses:**

Force on the surface AA,

$$\begin{aligned} P_{AA} &= 9.81 \times (0.3 \times 1.0) \times 0.45 \\ &= 1.324 \text{ kN} \end{aligned}$$

- (a) When suspended from the top the stress on the side walls,

$$\sigma = \frac{4.414}{(0.6 + 0.6 + 1.0 + 1.0) \times \frac{4}{1000}} = \mathbf{344.8 \text{ kN/m}^2 \text{ (Ans.)}}$$

- (b) When supported from bottom the stress on the side walls,

$$\sigma = \frac{1.324}{(0.6 + 0.6 + 1.0 + 1.0) \times \frac{4}{1000}} = \mathbf{103.4 \text{ kN/m}^2 \text{ (Ans.)}}$$

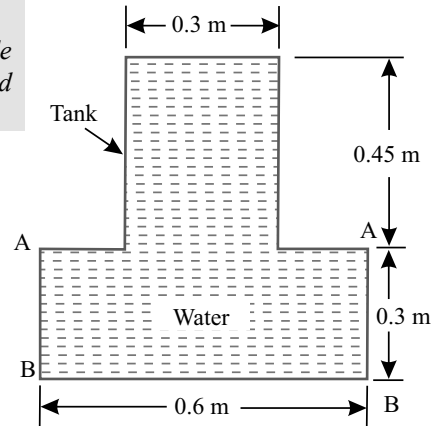


Fig. 3.24

Example 3.17. A vertical square $1.2 \text{ m} \times 1.2 \text{ m}$ is submerged in the water with upper edge 0.6 m below the water surface. Locate the horizontal line on the surface of the square such that the force on the upper portion equals the force on the lower portion.

Solution. Refer to Fig. 3.25. ABCD is the square plate submerged vertically in water with upper edge AB at a depth of 0.6 m below the free water surface (F.W.S.)

Let LM be the line such that force on ALMB equals the force on LDCM, and evidently the force on each portion equals half the total force on the entire plate ABCD.

Total pressure on the plate ABCD

$$\begin{aligned} &= wA\bar{x} \\ &= w \times (1.2 \times 1.2) \times \left(0.6 + \frac{1.2}{2}\right) \\ &= 1.728 w \end{aligned}$$

Total pressure on the position ALMB

$$= w \times (1.2 \times y) \times \left(0.6 + \frac{y}{2}\right) = 1.2 wy \left(0.6 + \frac{y}{2}\right)$$

Now, pressure force on ALMB = $\frac{1}{2} \times$ pressure force on ABCD

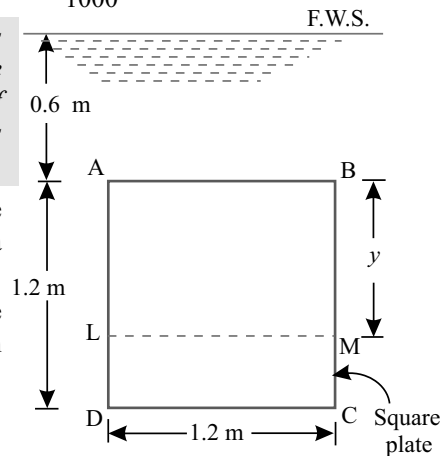


Fig. 3.25

$$1.2 wy \left(0.6 + \frac{y}{2} \right) = \frac{1}{2} \times 1.728 w$$

$$\text{or,} \quad 0.6y + \frac{y^2}{2} = 0.72$$

$$\text{or,} \quad y^2 + 1.2y - 1.44 = 0$$

$$\text{or,} \quad y = \frac{-1.2 \pm \sqrt{(1.2)^2 + 4 \times 1.44}}{2} = \frac{-1.2 \pm 2.683}{2}$$

$$= 0.7415 \text{ m or } -1.9415 \text{ m.}$$

$$\text{i.e.,} \quad y = \mathbf{0.7415 \text{ m (Ans.)}}$$

Example 3.18. A rectangular vertical door, 2.4 m (height) $\times$ 1.2 m (wide), is fastened by two hinges situated 18 cm below the top and 18 cm above the bottom on one vertical edge, and by one clamp at the centre of other vertical edge. The door is subjected to water pressure on one side and the depth of water above the top of door is 1.2 m. Calculate the reactions at the hinges and at the clamp.

Solution. Refer to Fig. 3.26.

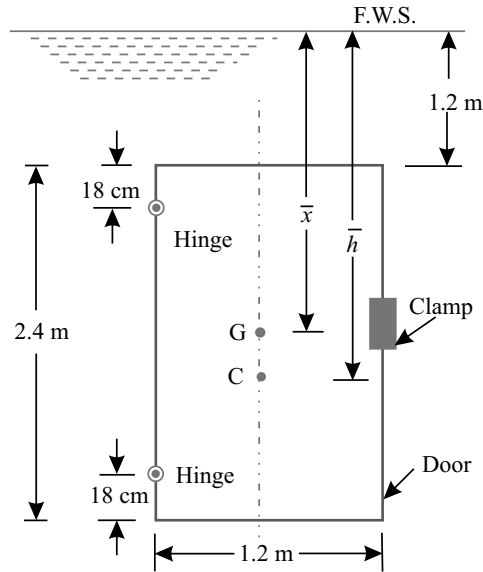


Fig. 3.26

$$\text{Depth of centroid of the door, } \bar{x} = 1.2 + \frac{2.4}{2} = 2.4 \text{ m}$$

$$\text{Area of the door, } A = 2.4 \times 1.2 = 2.88 \text{ m}^2$$

Total pressure on the door,

$$P = wA\bar{x} = 9.81 \times 2.88 \times 2.4 = 67.8 \text{ kN}$$

Depth of centre of pressure,

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x}$$

$$\frac{(1.2 \times 2.4^3 / 12)}{2.88 \times 2.4} + 2.4 = 2.6 \text{ m}$$

i.e., $(2.4 + 1.2) - 2.6 = 1 \text{ m}$ from the base

Let, $R_{th} = \text{Reaction at the top hinge,}$

$R_{bh} = \text{Reaction at the bottom hinge, and}$

$R_{cl} = \text{Reaction at the clamp.}$

The symmetry of the arrangement suggests that half of the total pressure force is taken by the two hinges and the other half by the clamp.

$$\therefore \text{Reaction at the clamp, } R_{cl} = \frac{67.8}{2} = 33.9 \text{ kN (Ans.)}$$

Taking moments of all forces about the horizontal axis through the bottom hinge, we get:

$$P \times (1 - 0.18) = R_{cl} \times \left(\frac{2.4}{2} - 0.18 \right) + R_{th} \times (2.4 - 0.18 - 0.18)$$

$$67.8 \times 0.82 = 33.9 \times 1.02 + R_{th} \times 2.04$$

$$\text{or, } R_{th} = 10.3 \text{ kN (Ans.)}$$

$$\therefore \text{Reaction at the bottom hinge, } R_{bh} = 33.9 - R_{th} = 33.9 - 10.3 = 23.6 \text{ kN (Ans.)}$$

3.5. INCLINED IMMERSED SURFACE

Refer to Fig. 3.27. Consider a plane inclined surface, immersed in a liquid.

Let, $A = \text{Area of the surface,}$

$\bar{x} = \text{Depth of centre of gravity of immersed surface from the free liquid surface,}$

$\theta = \text{Angle at which the immersed surface is inclined with the liquid surface, and}$

$w = \text{Specific weight of the liquid.}$

(a) Total pressure (P):

Consider a strip of thickness dx , width b at a distance l from O (A point, on the liquid surface, where the immersed surface will meet, if produced).

The intensity of pressure on the strip

$$= wl \sin \theta$$

$$\text{Area of the strip} = b \cdot dx$$

Pressure on the strip

$$= \text{Intensity of pressure} \times \text{area}$$

$$= wl \sin \theta \cdot b \cdot dx$$

Now total pressure on the surface,

$$P = \int wl \sin \theta \cdot b \cdot dx = w \sin \theta \int l \cdot b \cdot dx$$

$$\text{But, } \int l \cdot b \cdot dx = \text{moment of surface area about } OO$$

$$= \frac{A\bar{x}}{\sin \theta},$$

$$\therefore P = w \sin \theta \cdot \frac{A\bar{x}}{\sin \theta} = wA\bar{x} \text{ (same as in Arts. 3.3 and 3.4)}$$

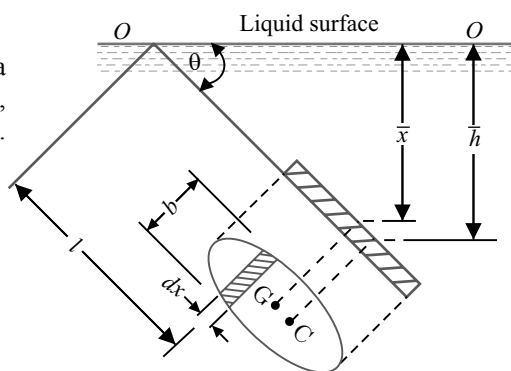


Fig. 3.27. Inclined immersed surface.

(b) Centre of pressure ($\bar{h}$):

Referring to Fig 3.27, let C be the centre of pressure of the inclined surface.

- Let,
- $\bar{h}$ = Depth of centre of pressure below free liquid surface,
 - I_G = Moment of inertia of the immersed surface about OO ,
 - $\bar{x}$ = Depth of centre of gravity of the surface from the liquid surface,
 - θ = Angle at which the immersed surface is inclined with the liquid surface, and
 - A = Area of the surface.

Consider a strip of thickness of dx , width b and at distance l from OO .

The intensity of pressure on the strip = $w \sin \theta$

Area of strip = $b \cdot dx$

$\therefore$ Pressure on the strip = Intensity of pressure $\times$ area = $wl \sin \theta \cdot b \cdot dx$

Moment of the pressure about OO = $(wl \sin \theta \cdot b \cdot dx) l = wl^2 \sin \theta \cdot b \cdot dx$

Now sum of moments of all such pressures about O ,

$$M = \int wl^2 \sin \theta \cdot b \cdot dx = w \sin \theta \int l^2 \cdot b \cdot dx$$

But, $\int l^2 \cdot b \cdot dx = I_0$ = moment of inertia of the surface about the point O (or the second moment of area)

$$M = w \sin \theta \cdot I_0 \quad \dots(i)$$

The sum of moments of all such pressures about O is also equal to $\frac{P\bar{h}}{\sin \theta}$ $\dots(ii)$

where, P is the total pressure on the surface.

Equating eqns. (i) and (ii), we get:

$$\begin{aligned} \frac{P\bar{h}}{\sin \theta} &= w \sin \theta \cdot I_0 \\ \frac{wA\bar{x}\bar{h}}{\sin \theta} &= w \sin \theta \cdot I_0 \quad (\because P = wA\bar{x}) \end{aligned}$$

$$\text{or,} \quad \bar{h} = \frac{I_0 \sin^2 \theta}{A\bar{x}} \quad \dots(iii)$$

Also, $I_0 = I_G + Ah^2$ $\dots$ Theorem of parallel axes.
 where, I_G = Moment of inertia of figure about horizontal axis through its centre of gravity, and

h = Distance between O and the centre of gravity of the figure = $l \left(= \frac{\bar{x}}{\sin \theta} \right)$ in this case.

Rearranging equation (iii), we have:

$$\begin{aligned} \bar{h} &= \frac{\sin^2 \theta}{A\bar{x}} (I_G + Ah^2) \\ &= \frac{\sin^2 \theta}{A\bar{x}} \left[I_G + A \left(\frac{\bar{x}}{\sin \theta} \right)^2 \right] = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x} \end{aligned}$$

$$\text{Hence, centre of pressure } \bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x} \quad \dots(3.3)$$

It will be noticed that if $\theta = 90^\circ$ eqn (3.3) becomes the same as equation (3.2).

Example 3.19. A 1 m wide and 1.5 m deep rectangular plane surface lies in water in such a way that its plane makes an angle of 30° with the free water surface. Determine the total pressure and position of centre of pressure when the upper edge is 0.75 m below the free water surface.

Solution. Width of the plane surface = 1 m
 Depth of the plane surface = 1.5 m
 Inclination, $\theta = 30^\circ$
 Distance of upper edge from free water surface = 0.75 m

(i) Total pressure, P :

Using the relation, $P = wA\bar{x}$

where, $w = 9.81 \text{ kN/m}^3$,

Area, $A = 1.5 \times 1 = 1.5 \text{ m}^2$,

$$\bar{x} = LU + UM = 0.75 + MN \sin 30^\circ$$

$$= 0.75 + \frac{1.5}{2} \times 0.5 = 1.125 \text{ m}$$

$$P = 9.81 \times 1.5 \times 1.125 \text{ m}$$

$$= \mathbf{16.55 \text{ kN (Ans.)}}$$

(ii) Centre of pressure, $\bar{h}$:

$$\text{Using the relation, } \bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x}$$

$$\text{where, } I_G = \frac{1 \times 1.5^3}{12} = 0.281 \text{ m}^4$$

$$\text{i.e., } \bar{h} = \frac{0.281 \times (0.5)^2}{1.5 \times 1.125} + 1.125 = \mathbf{1.166 \text{ m (Ans.)}}$$

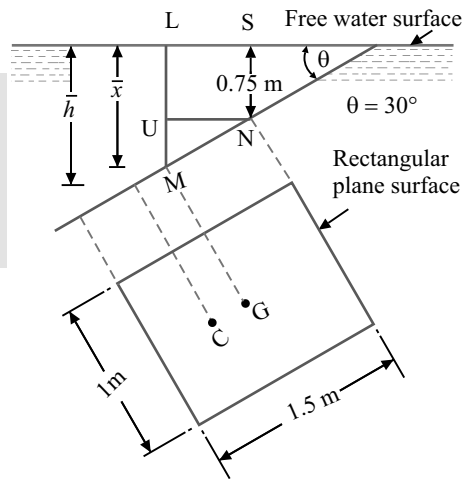


Fig. 3.28

Example 3.20. A circular plate 1.5 m diameter is submerged in water, with its greatest and least depths below the surface being 2 m and 0.75 m respectively. Determine:

(i) The total pressure on one face of the plate, and

(ii) The position of the centre of pressure.

Solution. Diameter of the plate, = 1.5 m

Area of the plate,

$$A = \frac{\pi}{4} \cdot d^2 = \frac{\pi}{4} \times 1.5^2 = 1.767 \text{ m}^2$$

Refer to Fig. 3.29

Distance, $SN = 0.75 \text{ m}$, $UM = 2 \text{ m}$

Distance of c.g. from free surface,

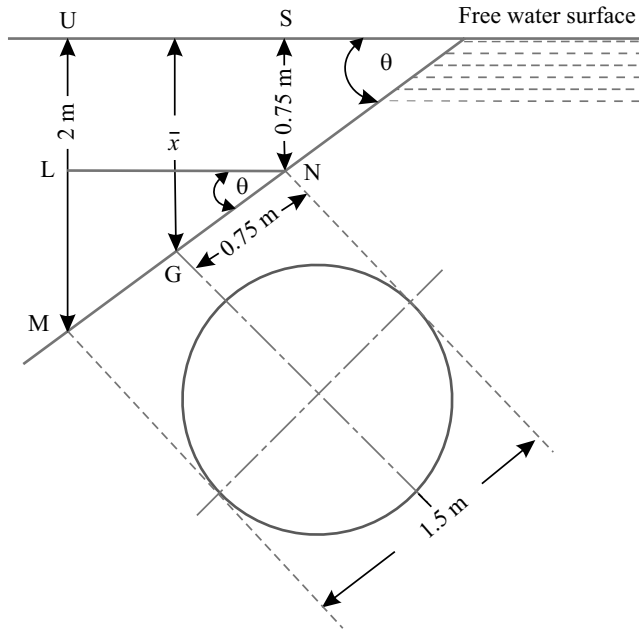


Fig. 3.29

$$\bar{x} = SN + GN \sin \theta$$

$$= 0.75 + 0.75 \sin \theta$$

But,

$$\sin \theta = \frac{LM}{MN} = \frac{UM - UL}{MN}$$

$$= \frac{2 - 0.75}{1.5} = 0.8333$$

$\therefore$

$$\bar{x} = 0.75 + 0.75 \times 0.8333 = 1.375 \text{ m}$$

(i) **Total pressure, P :**

We know that,

$$P = wA\bar{x} = 9.81 \times 1.767 \times 1.375$$

$$= \mathbf{23.83 \text{ kN (Ans.)}}$$

(ii) **Centre of pressure, $\bar{h}$:**

Using the relation,

$$\bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x}$$

$$= \frac{\pi / 64 \times 1.5^4 \times (0.8333)^2}{1.767 \times 1.375} + 1.375 = 1.446$$

i.e.,

$$\bar{h} = \mathbf{1.446 \text{ m (Ans.)}}$$

Example 3.21. An annular plate 2m external diameter and 1m internal diameter with its greatest and least depths below the surface being 1.5 m and 0.75 m respectively. Calculate the magnitude, direction and location of the force acting upon one side of the plate due to water pressure.

Solution. Refer to Fig. 3.30. From the geometry of the figure, we have:

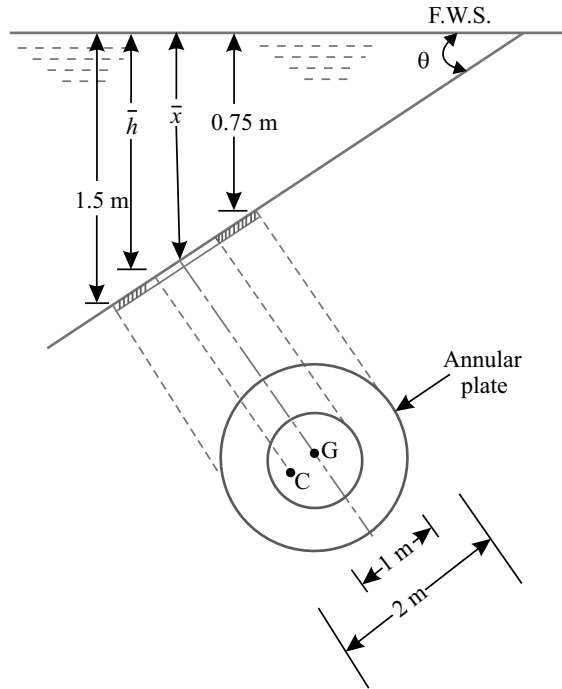


Fig. 3.30

$$\sin \theta = \frac{1.5 - 0.75}{2} = 0.375$$

$$\therefore \theta = \sin^{-1}(0.375) = 22^\circ$$

$$\text{Area of the plate, } A = \frac{\pi}{4} (2^2 - 1^2) = 2.356 \text{ m}^2$$

$$\text{Depth of centroid, } \bar{x} = \frac{1.5 + 0.75}{2} = 1.125 \text{ m}$$

Total pressure force,

$$P = wA\bar{x} = 9.81 \times 2.356 \times 1.125 = \mathbf{26 \text{ kN (Ans.)}}$$

This force acts perpendicular to the plate so it is acting in a direction which is $90^\circ - 22^\circ = \mathbf{68^\circ}$ to the vertical (Ans.)

Depth of centre of pressure,

$$\begin{aligned} \bar{h} &= \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x} \\ &= \frac{\frac{\pi}{64} (2^4 - 1^4) \times (\sin 22^\circ)^2}{\frac{\pi}{4} (2^2 - 1^2) \times 1.125} + 1.125 \\ &= \frac{0.1033}{2.651} + 1.125 = \mathbf{1.164 \text{ m (Ans.)}} \end{aligned}$$

Example 3.22. A triangular plate of 1 metre base and 1.5 metre altitude is immersed in water. The plane of the plate is inclined at 30° with free water surface and the base is parallel to and at a depth of 2 metres from water surface. Find the total pressure on the plate and the position of centre of pressure.

Solution. Refer to Fig. 3.31.

Area of the plate,

$$A = \frac{1}{2} \times 1 \times 1.5 = 0.75 \text{ m}^2$$

Inclination of the plate, $\theta = 30^\circ$

Total pressure on the plate, P :

The depth of c.g. of the plate from water surface,

$$\begin{aligned}\bar{x} &= 2 + \frac{1.5}{3} \sin 30^\circ \\ &= 2 + 0.5 \times 0.5 = 2.25 \text{ m}\end{aligned}$$

Using the relation,

$$\begin{aligned}P &= wA\bar{x} = 9.81 \times 0.75 \times 2.25 \\ &= \mathbf{16.55 \text{ kN (Ans.)}}\end{aligned}$$

Depth of centre of pressure, $\bar{h}$:

Moment of inertia of a triangular section about its c.g.,

$$I_G = \frac{1 \times 1.5^3}{36} = 0.09375 \text{ m}^4$$

Using the relation,

$$\begin{aligned}\bar{h} &= \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x} = \frac{0.09375 \sin^2 30^\circ}{0.75 \times 2.25} + 2.25 \\ &= \mathbf{2.264 \text{ m (Ans.)}}\end{aligned}$$

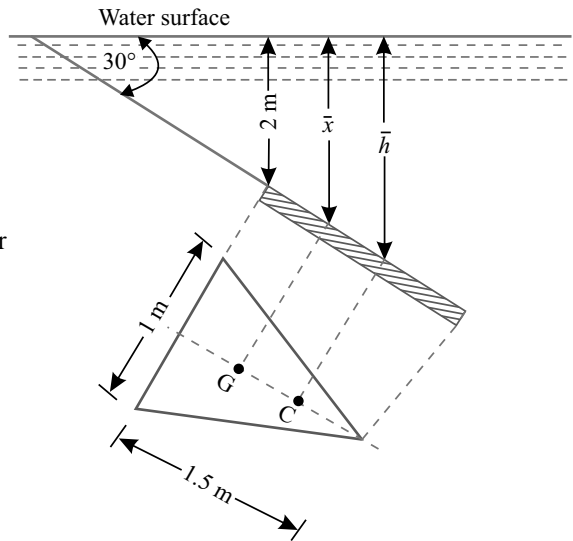


Fig. 3.31

Example 3.23. A trapezoidal plate measuring 1 m at the top edge and 1.5 m at the bottom edge is immersed in water with the plan making an angle of 30° to the free surface of water. The top and bottom edges lie at 0.5 m and 1 m respectively from the surface. Determine the hydrostatic force on the plate.

Solution.

Refer to Fig. 3.32, Given:

$$a = 1 \text{ m}; b = 1.5 \text{ m};$$

$$h = AB = \frac{1.0 - 0.5}{\sin 30^\circ} = 1 \text{ m}$$

Distance of centroid of a trapezium plate from its base,

$$\begin{aligned}h_G &= \frac{h}{3} \left(\frac{2a + b}{a + b} \right) \\ &= \frac{1}{3} \left(\frac{2 \times 1 + 1.5}{1 + 1.5} \right) = 0.467 \text{ m}\end{aligned}$$

Depth of centroid from the free water, surface,

$$\bar{x} = 1.0 - BC \sin 30^\circ = 1.0 - 0.467 \times \sin 30^\circ = 0.7665 \text{ m}$$

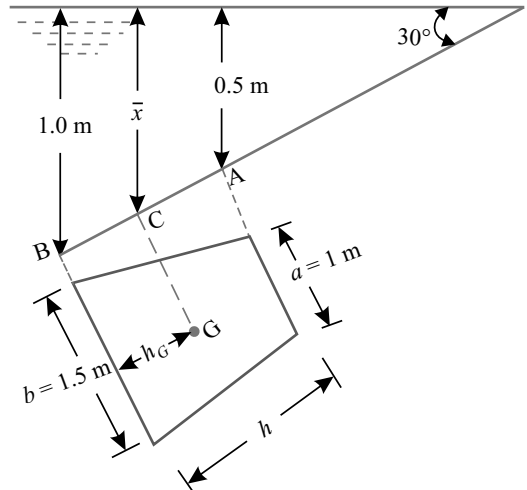


Fig. 3.32

Area of trapezium, $A = \left(\frac{a+b}{2} \right) h = \left(\frac{1+1.5}{2} \right) \times 1 = 1.25 \text{ m}^2$

$\therefore$ Hydrostatic force $P = wA\bar{x} = 9.81 \times 1.25 \times 0.7665 = 9.399 \text{ kN (Ans.)}$

Example 3.24. An inclined rectangular sluice gate AB 1.2 m by 5 m size as shown in Fig. 3.33 is installed to control the discharge of water. The end A is hinged. Determine the force normal to the gate applied at B to open it.

Solution. Size of the gate = 1.2 m $\times$ 5 m

Area of the gate = 1.2 $\times$ 5 = 6 m²

Refer to Fig. 3.33.

Depth of c.g. of the gate from free water surface,

$$\begin{aligned}\bar{x} &= 5 - BG \sin 45^\circ \\ &= 5 - 0.6 \times 0.707 = 4.576 \text{ m}\end{aligned}$$

The total pressure force (P) acting on the gate,

$$\begin{aligned}P &= wA\bar{x} \\ &= 9.81 \times 6 \times 4.576 = 269.3 \text{ kN}\end{aligned}$$

This force acts at a depth $\bar{h}$, given by the relation:

$$\bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x}$$

where, $I_G = M.O.I. \text{ of gate} = \frac{bd^3}{12} = \frac{5 \times 1.2^3}{12} = 0.72 \text{ m}^4, \theta = 45^\circ$

i.e., $\bar{h} = \frac{0.72 \times \sin^2 45^\circ}{6 \times 4.576} + 4.576 = 4.589 \text{ m}$

From Fig. 3.33, we have $\frac{\bar{h}}{OC} = \sin 45^\circ$

$$\text{Distance, } OC = \frac{\bar{h}}{\sin 45^\circ} = \frac{4.589}{0.707} = 6.49 \text{ m;}$$

$$\text{Distance, } OB = \frac{5}{\sin 45^\circ} = 7.072 \text{ m}$$

$\therefore$ Distance, $BC = OB - OC = 7.072 - 6.49 = 0.582 \text{ m}$

$$\text{Distance, } AC = AB - BC = 1.2 - 0.582 = 0.618 \text{ m}$$

Taking moments about the hinge A, we get:

$$F \times AB = P \times AC$$

or, $F = \frac{P \times AC}{AB} = \frac{269.3 \times 0.618}{1.2} = 138.69 \text{ kN (Ans.)}$

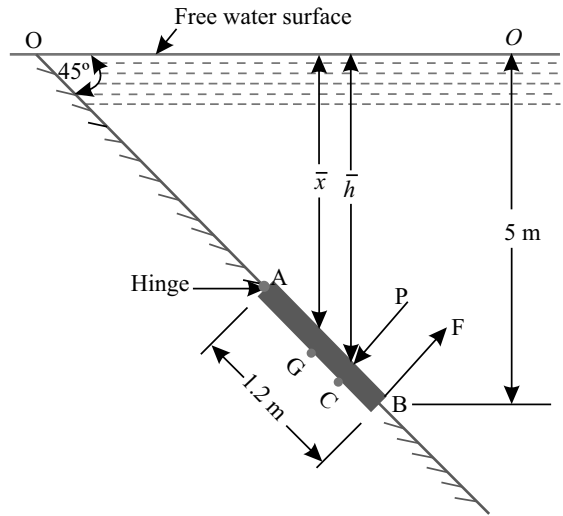


Fig. 3.33

Example 3.25. A 6 m $\times$ 2 m rectangular gate is hinged at the base and is inclined at an angle of 60° with the horizontal. The upper end of the gate is kept in position by a weight of 60 kN acting at angle of 90° as shown in Fig. 3.34. Neglecting the weight of the gate, find the level of water when the gate begins to fall.

Solution. Length of the gate, $l = 6 \text{ m}$
 Width of the gate, $b = 2 \text{ m}$
 Inclination, $\theta = 60^\circ$
 Weight, $W = 60 \text{ kN}$

Level of water when the gate begins to fall:

Refer to Fig. 3.34

Let, h = Height of free water surface from the bottom when the gate just begins to fall.

Then, length of gate in the shape of plate, submerged in water,

$$AD = \frac{AC}{\sin \theta} = \frac{h}{\sin 60^\circ} = \frac{h}{0.866}$$

$$= 1.1547 h$$

$\therefore$ Area of the gate immersed in water,

$$A = AD \times \text{width}$$

$$= 1.1547 h \times 2 = 2.309 h \text{ m}^2$$

Also depth of c.g. of the immersed area,

$$\bar{x} = \frac{h}{2} = 0.5 h$$

Total pressure on the gate,

$$P = wA\bar{x} = 9.81 \times 2.309 h \times 0.5 h$$

$$= 11.326 h^2 \text{ kN}$$

The centre of pressure of the immersed surface ($\bar{h}$) is given by:

$$\bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x}$$

where, I_G = moment of inertia of the immersed area

$$= \frac{b \times AD^3}{12} = \frac{2}{12} (1.1547 h)^3$$

$$= 0.2566 h^3$$

$$\bar{h} = \frac{0.2566 h^3 \times (\sin 60^\circ)^2}{2.309 h \times 0.5 h} + 0.5 h = 0.667 h \text{ metres.}$$

Distance of centre of pressure from the hinge (or pivot) along the length of the gate,

$$AE = \frac{h - \bar{h}}{\sin 60^\circ} = \frac{h - 0.667 h}{0.866} = 0.384 h$$

Taking moments about the hinge, we get:

$$P \times AE = 60 \times AB$$

$$11.326 h^2 \times 0.384 h = 60 \times 6$$

$$\text{or} \quad h^3 = \frac{60 \times 6}{11.326 \times 0.384} = 82.774$$

$$\text{or} \quad h = 4.36 \text{ m (Ans.)}$$

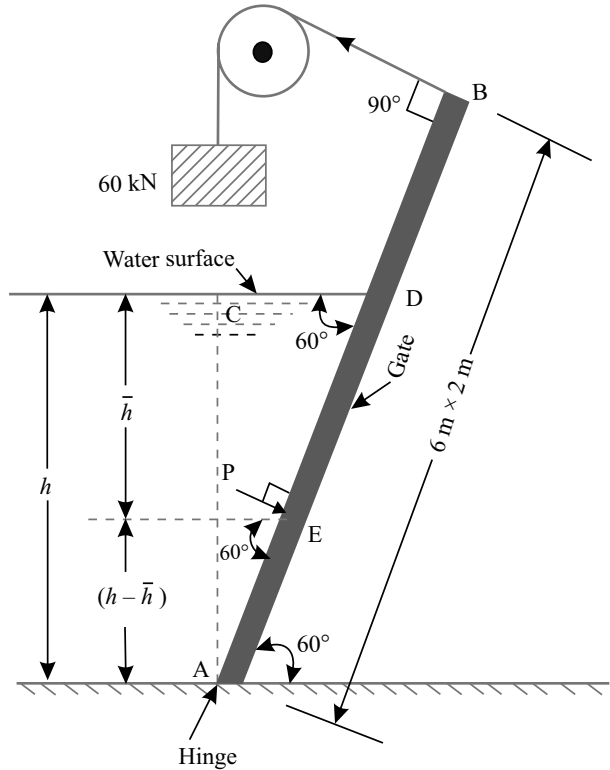


Fig. 3.34

Example 3.26. Fig. 3.35 shows a circular opening in the sloping wall of the reservoir closed by disc valve 0.9 m diameter. The disc is hinged at H and a balance weight W is just sufficient to hold the valve closed when the reservoir is empty. How much additional weight should be placed on the arm, 1.2 m from the hinge, in order that the valve shall remain closed until the water level is 0.72 m above the centre of the valve.

Solution. Dia. of the valve, $d = 0.9$ m

$$\text{Area of the valve, } A = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times 0.9^2 = 0.636 \text{ m}^2$$

Inclination, $\theta = 60^\circ$

Distance of c.g. of the valve from free water surface, $\bar{x} = 0.72$ m

Additional weight, W' :

Total pressure on the valve, $P = wA\bar{x} = 9.81 \times 0.636 \times 0.72 = 4.49$ kN

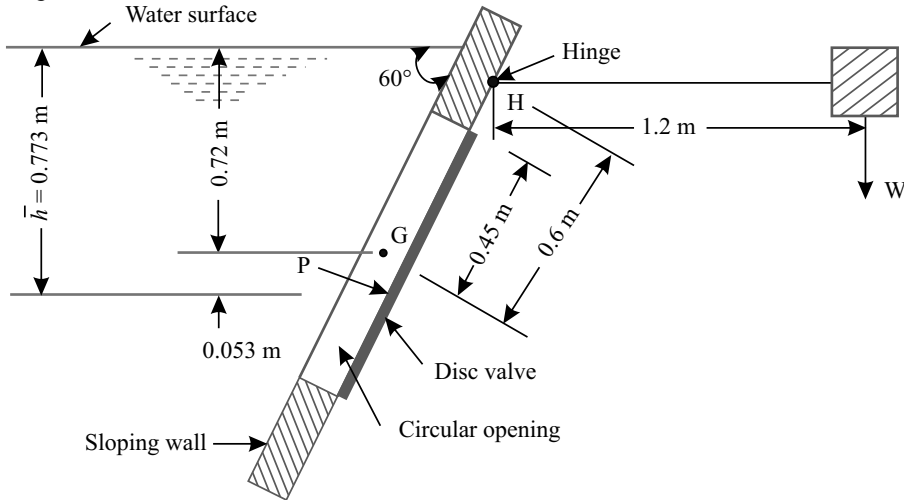


Fig. 3.35

Distance of centre of pressure ($\bar{h}$) is given by:

$$\bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x} = \frac{\pi/64 \times 0.9^4 \times (\sin 60^\circ)^2}{0.636 \times 0.72} + 0.72$$

$$= 0.773 \text{ m (from free water surface)}$$

$$= 0.053 \text{ m below the centroid } G.$$

Taking moments of all the forces about the hinge, we have:

$$P \left(\frac{0.053}{\sin 60^\circ} + 0.6 \right) = W' \times 1.2$$

$$\text{or, } 4.49 (0.0612 + 0.6) = W' \times 1.2 \text{ or } W' = 2.47 \text{ kN (Ans.)}$$

Example 3.27. Fig. 3.36 shows a gate supporting water. Taking the width of the gate as unity find: (i) Depth of water (h) so that the gate tips about the hinge; (ii) Reaction at the hinge.

Solution. Inclination of the gate, $\theta = 60^\circ$

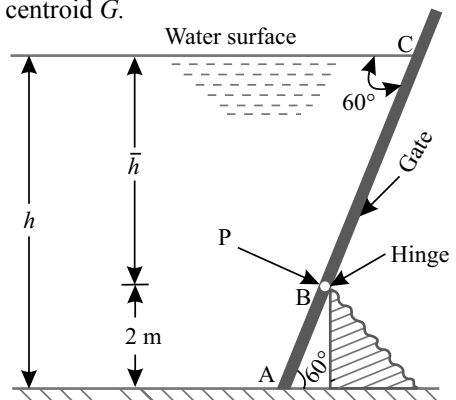


Fig. 3.36

(i) Depth of water, h :

As the depth of water increases, the total pressure P on the gate moves upwards, and just before tipping, P acts at the hinge.

$\therefore$ Depth of centre of pressure

$$\bar{h} = (h - 2) \text{ m}$$

But $\bar{h}$ is also given by:

$$\bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x}$$

Taking width of gate unity, we have:

$$\begin{aligned} \text{Area, } A &= AC \times 1 = \frac{h}{\sin 60^\circ} \times 1 \\ &= 1.1547 h \quad \left(\because AC = \frac{h}{\sin 60^\circ} \right) \end{aligned}$$

Distance of c.g. from water surface, $\bar{x} = \frac{h}{2} = 0.5 h$.

Moment of inertia, $I_G = \frac{1 \times AC^3}{12} = \frac{1}{12} \times (h / \sin 60^\circ)^3 = 0.1283 h^3$

$$\therefore \bar{h} = \frac{0.1283 h^3 \times (\sin 60^\circ)^2}{1.1547 h \times 0.5 h} + 0.5 h = 0.667 h$$

Equating the two values of h , we get: $h - 2 = 0.667 h$ or $0.333 h = 2$

or, $h = 6 \text{ m (Ans.)}$

Example 3.28. A gate supporting water takes the form of an inclined shield which swings around a hinged axis O (Fig. 3.37).

Determine: (i) The position x of the hinge at which a water level of $h = 5.1 \text{ m}$ on the left would cause the gate to tip over the hinge;

(ii) The magnitude of hydrostatic free on the gate just before it opens (tips about the hinge) automatically. Neglect frictional effects.

Solution. Refer to Fig. 3.37.

(i) The position x of the hinge:

The gate would tip about the hinge point O when the line of action of the resultant pressure force lies from O to B anywhere on the gate; the limiting condition being the situation when the resultant force passes through the hinged point O . The resultant also passes through the centroid of the pressure diagram, and the centroid lies at a distance $\frac{1}{3} \times AB$ from the bottom point A .

$$\therefore x = \frac{1}{3} \times AB$$

or, AB (length of the gate) $= 3x$

Depth of water, $h = 3x \times \sin 45^\circ$

i.e. $5.1 = 3x \times \sin 45^\circ$

$$\begin{aligned} \therefore x &= \frac{5.1}{3 \times \sin 45^\circ} \\ &= 2.4 \text{ m (Ans.)} \end{aligned}$$

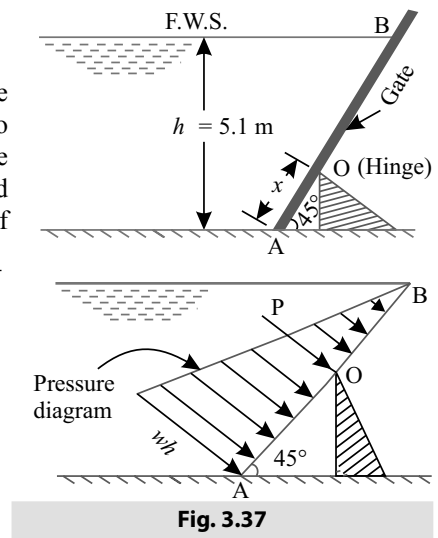


Fig. 3.37

(i) The magnitude of hydrostatic force P :

$$\begin{aligned}
 P &= \text{Area of pressure diagram} \times \text{width of gate} \\
 &= \left(\frac{1}{2} \times AB \times wh \right) \times 1 \quad \dots \text{considering unit width} \\
 &= \left[\frac{1}{2} \times (3 \times 2.4) \times 9.81 \times 5.1 \right] \times 1 \quad (\because AB = 3x) \\
 &= \mathbf{180.11 \text{ kN (Ans.)}}
 \end{aligned}$$

Example 3.29. A 3.6 m square gate provided in an oil tank is hinged at its top edge (Fig. 3.38). The tank contains gasoline (sp. gr = 0.7) upto a height of 1.8 m above the top edge of the plate. The space above the oil is subjected to a negative pressure of 8250 N/m^2 . Determine the necessary vertical pull to be applied at the lower edge to open the gate. (GATE)

Solution. Refer to Fig. 3.38.

Head of oil equivalent to negative pressure 8250 N/m^2 ,

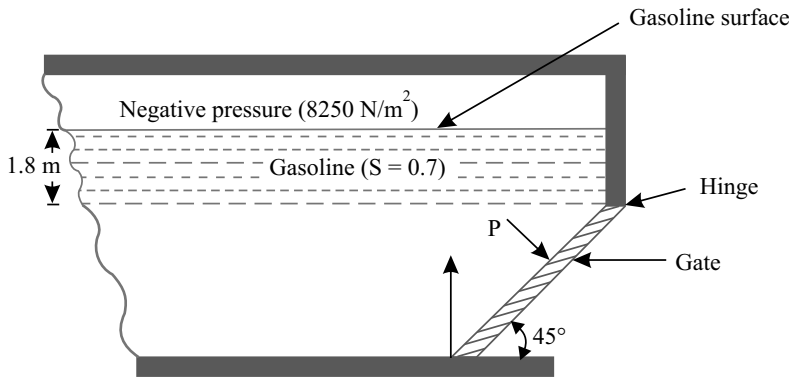


Fig. 3.38

$$h = \frac{p}{w} = \frac{8250}{0.7 \times 9810} = 1.2 \text{ m}$$

This negative pressure will reduce the oil head above the top edge of the gate from 1.8 m to 1.2 m (= 0.6 m) of oil. Calculations for the magnitude and location of the pressure force are thus to be made corresponding to 0.6 m of oil.

$$\text{Now,} \quad \bar{x} = 0.6 + \frac{3.6}{2} \sin 45^\circ = 1.873 \text{ m}$$

$$\text{Area, } A = 3.6 \times 3.6 = 12.96 \text{ m}^2$$

$$\text{Pressure, } P = wA\bar{x} = 0.7 \times 9810 \times 12.96 \times 1.873 = 166690 \text{ N}$$

$$\text{Centre of pressure, } \bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x}$$

$$\begin{aligned}
 &= \frac{\frac{1}{12} \times 3.6 \times (3.6)^3 \times (\sin 45^\circ)^2}{12.96 \times 1.873} + 1.873 = 2.16 \text{ m}
 \end{aligned}$$

$\therefore$ Vertical distance of centre of pressure below top edge of the gate

$$= 2.16 - 0.6 = 1.56 \text{ m}$$

Taking moments about the hinge, we get:

$$F \sin 45^\circ \times 3.6 = P \times \frac{1.56}{\sin 45^\circ}$$

Hence, vertical force,
$$F = \frac{P \times 1.56}{3.6 \times (\sin 45^\circ)^2} = \frac{166690 \times 1.56}{3.6 \times (\sin 45^\circ)^2} = 144465 \text{ N (Ans.)}$$

Example 3.30. There is an opening in a container shown in Fig. 3.39. Find the force F and the reaction at the hinge.

Solution. Gauge pressure = 23.5 kN/m^2

$$= \frac{23.5}{9.81 \times 0.8} \approx 3 \text{ m of oil} \quad \left(\because h = \frac{p}{w} \right)$$

The free liquid surface may be considered as 3 m above the hinge A (Fig. 3.40)

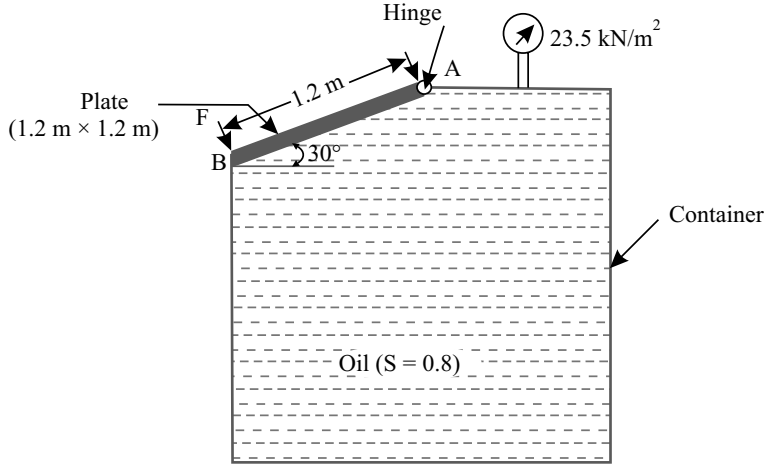


Fig. 3.39

Now, distance of centroid G of the plate from the oil surface,

$$\bar{x} = 3 + 0.6 \sin 30^\circ = 3.3 \text{ m}$$

Total pressure on the plate,

$$\begin{aligned} P &= wA\bar{x} \\ &= (9.81 \times 0.8) \times (1.2 \times 1.2) \times 3.3 \\ &= 37.29 \text{ kN} \end{aligned}$$

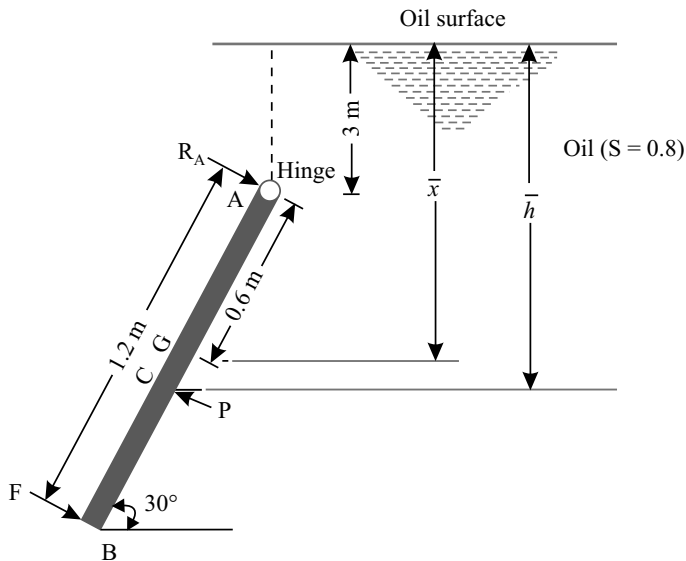


Fig. 3.40. Free body diagram.

Distance of centre of pressure ($\bar{h}$) is given by:

$$\bar{h} = \frac{I_G \sin^2 \theta}{A \bar{x}} + \bar{x} = \frac{1.2 \times 1.2^3}{12} \times (\sin 30^\circ)^2}{(1.2 \times 1.2) \times 3.3} + 3.3 = 3.309 \text{ m}$$

Taking moments about the hinge A, we get:

$$F \times 1.2 = P \times CA = P \times \left[\frac{(\bar{h} - 3)}{\sin 30^\circ} \right]$$

$$\text{or, } F \times 1.2 = 37.29 \times \frac{(3.309 - 3)}{\sin 30^\circ}$$

$$= 23.045$$

$$\text{or, } F = 19.2 \text{ kN}$$

Let, R_A = Reaction at the hinge,

$$\text{Then, } R_A + F = P$$

$$\text{or, } R_A = P - F$$

$$= 37.29 - 19.2$$

$$= \mathbf{18.09 \text{ kN (Ans.)}}$$

Example 3.31. Fig. 3.41. shows a rectangular sluice gate AB, 3 m wide and 4.5 m long hinged at A. It is kept closed by a weight fixed to the gate. The total weight of the gate and weight fixed to the gate is 515 kN. The centre of gravity of the weight and gate is at G. Find the height of the water h which will first cause the gate to open.

Solution. Width of gate, $b = 3 \text{ m}$, Length of gate; $l = 4.5 \text{ m}$

$$\text{Area, } A = 3 \times 4.5 = 13.5 \text{ m}^2$$

Weight of gate, $W = 515 \text{ kN}$; angle of inclination, $\theta = 45^\circ$

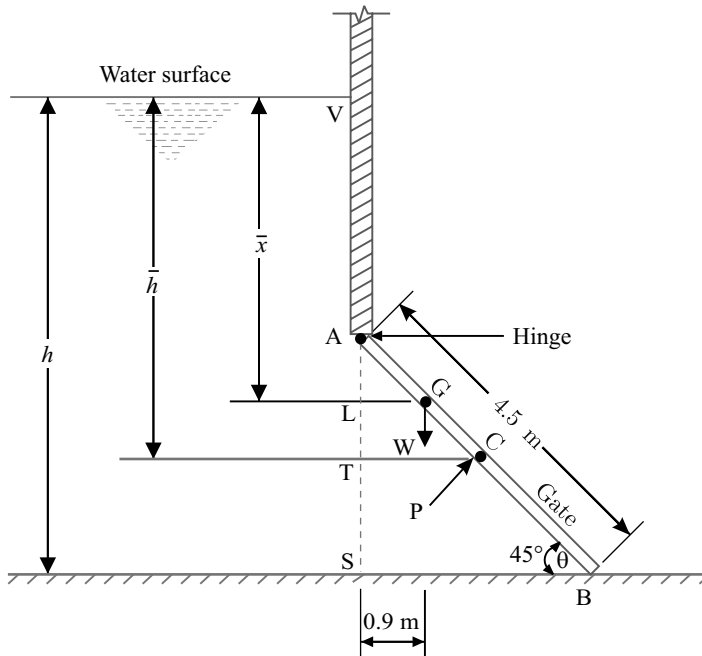


Fig. 3.41

Height of water, h :

$$\begin{aligned}\bar{x} &= h - LS = h - (AS - AL) = h - (AB \sin \theta - LG \tan \theta) \\ &= h - (4.5 \sin 45^\circ - 0.9 \tan 45^\circ) \\ &= h - (3.18 - 0.9) = (h - 2.28) \text{ m}\end{aligned}$$

The total pressure (P) is given by:

$$P = wA\bar{x} = 9.81 \times 13.5 \times (h - 2.28) = 132.43 (h - 2.28)$$

The total pressure is acting at centre of pressure at C as shown in the Fig. 3.41. The depth of C from the free surface is given by:

$$\bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x} = \frac{\frac{3 \times 4.5^3}{12} \times (\sin 45^\circ)^2}{13.5 \times (h - 2.28)} + (h - 2.28)$$

or,

$$\bar{h} = \frac{0.843}{(h - 2.28)} + (h - 2.28)$$

Now taking moments about hinge A , we get:

$$515 \times LG = P \times AC$$

or,

$$515 \times 0.9 = 132.43 (h - 2.28) \times \frac{AT}{\sin 45^\circ}$$

$$\therefore AT = \frac{515 \times 0.9 \times \sin 45^\circ}{132.43 (h - 2.28)} = \frac{2.47}{(h - 2.28)} \quad \dots(i)$$

But,

$$AT = \bar{h} - VA$$

or,

$$AT = \frac{0.843}{(h - 2.28)} + (h - 2.28) - VA \quad \dots(ii)$$

But,

$$\begin{aligned}VA &= VS - AS = h - 4.5 \sin 45^\circ \\ &= h - 3.18\end{aligned}$$

Substituting this value in (ii), we get:

$$\begin{aligned}AT &= \frac{0.843}{(h - 2.28)} + (h - 2.28) - (h - 3.18) \\ &= \frac{0.843}{(h - 2.28)} + 3.18 - 2.28\end{aligned}$$

or,

$$AT = \frac{0.843}{h - 2.28} + 0.9 \quad \dots(iii)$$

Equating the values of AT from (i) and (iii), we get:

$$\frac{2.47}{h - 2.28} = \frac{0.843}{h - 2.28} + 0.9$$

or,

$$2.47 = 0.843 + 0.9 (h - 2.28) = 0.843 + 0.9 h - 2.052$$

$$0.9 h = 2.47 - 0.843 + 2.052 = 3.679$$

$$h = \frac{3.679}{0.9} = 4.08 \text{ m (Ans.)}$$

3.6. CURVED IMMERSSED SURFACE

Consider a curved surface LM submerged in a static fluid as shown in Fig. 3.42. At any point on the curved surface, the pressure acts normal to the surface. Thus if dA is the area of a small element of the curved surface lying at a vertical depth of h from surface of the liquid, then the total pressure on the elemental area is,

$$dp = p \times dA = (wh) \times dA \quad \dots(3.4)$$

This force dP acts normal to the surface. Further integration of eqn. (3.4) would provide the total pressure on the curved surface and hence,

$$P = \int wh dA \quad \dots(3.5)$$

But, in case of curved surface the direction of the total pressures on the elementary areas are not in the same direction, but varies from point to point. Thus the integration of eqn. (3.5) for curved surface is impossible. The problem, however, can be solved by resolving the force P into horizontal and vertical components P_H and P_V . Then total force on the curved surface is,

$$P = \sqrt{P_H^2 + P_V^2} \quad \dots(3.6)$$

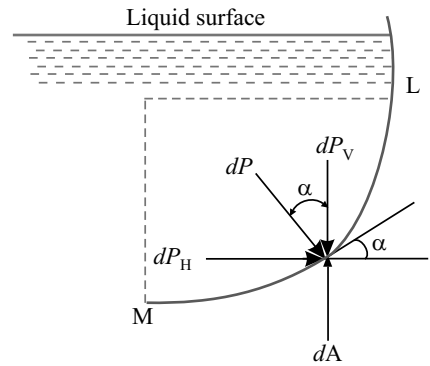


Fig. 3.42

The direction of the resultant force P with the horizontal is given by: $\tan \theta = \frac{P_V}{P_H}$

or, $\theta = \tan^{-1} \left(\frac{P_V}{P_H} \right) \quad \dots(3.7)$

Here,

P_H = Total pressure force on the projected area of the curved surface on vertical plane, and

P_V = Weight of the liquid supported by the curved surface upto free surface of liquid.

Example 3.32. The profile of a vessel is quadrant of a circle of radius R . Determine the horizontal and vertical components of the total pressure force, from the first principles.

Solution. Consider an elementary strip of radius R at depth h and subtending an angle as shown in Fig. 3.43.

Let the vessel has a unit depth perpendicular to the plane of paper. Then Area of the element,

$$dA = R d\alpha \times \text{unit depth} = R d\alpha$$

$$\text{Depth, } h = R \sin \alpha$$

$$\text{Intensity of pressure, } p = wh = w R \sin \alpha$$

$$\begin{aligned} \text{Pressure force, } dp &= p \times dA = w R \sin \alpha \times R d\alpha \\ &= w R^2 \sin \alpha d\alpha \end{aligned}$$

Vertical component of dP ,

$$dP_V = w R^2 \sin \alpha d\alpha \times \sin \alpha = w R^2 \sin^2 \alpha d\alpha$$

Horizontal component of dP ,

$$dP_H = w R^2 \sin \alpha d\alpha \times \cos \alpha = w R^2 \sin \alpha \cos \alpha d\alpha$$

$\therefore$ Total vertical pressure force,

$$\begin{aligned} P_V &= \int_0^{\pi/2} w R^2 \sin^2 \alpha d\alpha \\ &= \frac{w R^2}{2} \left[\int_0^{\pi/2} \left(\frac{1 - \cos 2\alpha}{2} \right) d\alpha \right] \end{aligned}$$

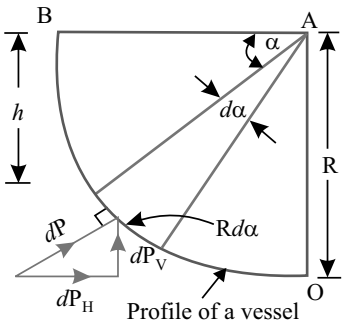


Fig. 3.43

$$= \frac{wR^2}{2} \left[\left| \alpha \right|_0^{\pi/2} - \left| \frac{\sin 2\alpha}{2} \right|_0^{\pi/2} \right] = \frac{wR^2\pi}{4}$$

$$= w \left(\frac{\pi R^2}{4} \times \text{unit length} \right)$$

$$= \text{specific weight} \times (\text{volume of liquid contained in curved surface}).$$

- Thus the vertical component of pressure force on a curved surface equals the weight of the volume liquid extending vertically from the curved surface to the free surface of liquid.

(Ans.)

Total horizontal pressure force,

$$\begin{aligned} P_H &= \int_0^{\pi/2} wR^2 \sin \alpha \cos \alpha d\alpha \\ &= \frac{wR^2}{2} \int_0^{\pi/2} 2 \sin \alpha \cos \alpha d\alpha \\ &= \frac{wR^2}{2} \int_0^{\pi/2} \sin 2\alpha d\alpha = \frac{wR^2}{2} \left[-\frac{\cos 2\alpha}{2} \right]_0^{\pi/2} = \frac{wR^2}{2} \\ &= w(R \times \text{unit length}) \times \frac{R}{2} \equiv wA\bar{x} \end{aligned}$$

- Thus the horizontal component of pressure force on a curved surface equals the force on projected area of curved surface on a vertical plane.

Example 3.33. A hemisphere projection of diameter 0.6 m exists on one of the vertical sides of a tank. If the tank contains water to an elevation of 1.5 m above the centre of the hemisphere, calculate the vertical and horizontal forces acting on the projection.

Solution. Refer to Fig. 3.44.

$$\begin{aligned} \text{Vertical force, } P_V &= P_{V_1} - P_{V_2} \\ &= \text{Weight volume of water MNST} - \text{weight of volume of water LNST} \\ &= \text{Weight of water contained by the hemisphere LNM} \\ &= w \times \frac{1}{2} \left(\frac{4}{3} \pi R^3 \right) \\ &= 9.81 \times \frac{1}{2} \times \frac{4}{3} \times \pi \times (0.3)^3 \\ &= 0.555 \text{ kN (Ans.)} \end{aligned}$$

Horizontal force, $P_H = wA\bar{x}$

$$= 9.81 \times \pi \times (0.3)^2 \times 1.5 = 4.16 \text{ kN (Ans.)}$$

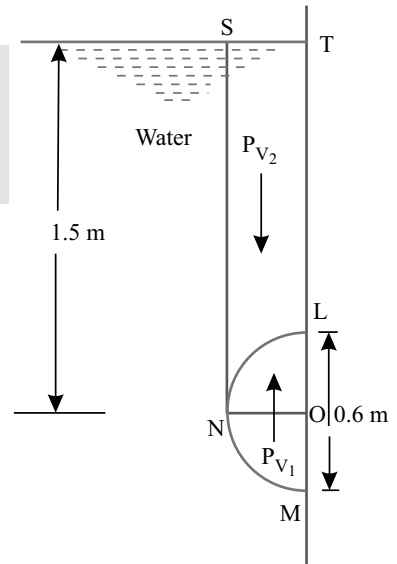


Fig. 3.44

Example 3.34. Fig. 3.45. shows a curved surface LM, which is in the form of a quadrant of a circle of radius 3 m, immersed in the water. If the width of the gate is unity, calculate the horizontal and vertical components of the total force acting on the curved surface.

Solution. Radius of the gate = 3 m

Width of the gate = 1 m

Refer to Fig. 3.45.

Distance $LO = OM = 3$ m

Horizontal component of total force, P_H :

Horizontal force (P_H) exerted by water on gate is given by,

P_H = Total pressure force on the projected area of curved surface LM on vertical plane

= Total pressure force on OM

(projected area of curved surface on vertical plane

$$= OM \times 1) = wA\bar{x}$$

But, $A = OM \times 1 = 3 \times 1 = 3 \text{ m}^2$ and $\bar{x} = 1 + \frac{3}{2} = 2.5$ m

$$P_H = 9.81 \times (3 \times 1) \times 2.5 = \mathbf{73.57 \text{ kN (Ans.)}}$$

The point of application of P_H is given by:

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x}$$

where, $I_G = \text{M.O.I. of } OM \text{ about its c.g.} = \frac{bd^3}{12} = \frac{1 \times 3^3}{12} = 2.25 \text{ m}^4$

$$\therefore \bar{h} = \frac{2.25}{(3 \times 1) \times 2.5} + 2.5 = \mathbf{2.8 \text{ m from water surface (Ans.)}}$$

Vertical component of total force, P_V :

Vertical force (P_V) exerted by water is given by:

P_V = Weight of water supported by LM upto free surface

= weight of portion $ULMOS$

= weight of $ULOS$ + weight of water in LOM

= w (volume of $ULOS$ + volume of LOM)

$$= 9.81 \left[UL \times LO + \frac{\pi \times (LO)^2}{4} \times 1 \right] = 9.81 \left[1 \times 3 + \frac{\pi \times 3^2}{4} \times 1 \right]$$

$$= 9.81 (3 + 7.068) \text{ kN} = \mathbf{98.77 \text{ kN (Ans.)}}$$

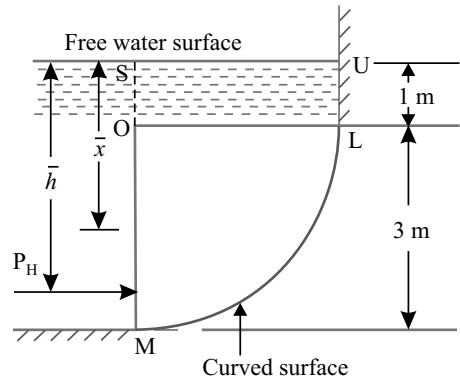


Fig. 3.45. Curved surface (gate).

Example 3.35. Fig. 3.46 shows a gate having a quadrant shape of radius of 1 m subjected to water pressure. Find the resultant force and its inclination with the horizontal. Take the length of the gate as 2 m.

Solution.

Radius of the gate, $r = 1$ m

Length of the gate = 2 m

Horizontal force, P_H :

P_H = Force on the projected area of the curved surface on vertical plane

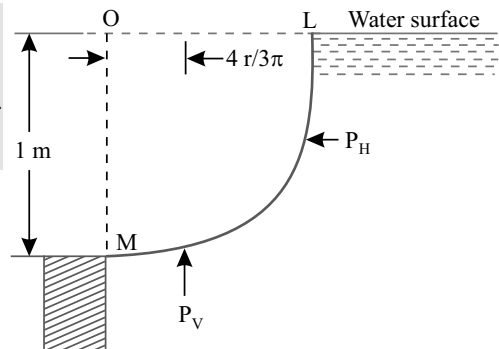


Fig. 3.46

where,

$$= \text{force on } MO = wA\bar{x}$$

$$w = 9.81 \text{ kN/m}^3,$$

$$A = \text{Area of } MO \text{ (projected area)} = 1 \times 2 = 2 \text{ m}^2$$

$$\bar{x} = \frac{1}{2} = 0.5 \text{ m}$$

$$P_H = 9.81 \times 2 \times 0.5 = 9.81 \text{ kN}$$

Vertical force, P_V :

$$P_V = \text{Weight of water (imagined) supported by } LM$$

$$= w \times \text{area of } LOM \times 2.0 = w \times \frac{\pi \times r^2}{4} \times 2$$

$$= 9.81 \times \frac{\pi}{4} \times 1^2 \times 2 = 15.4 \text{ kN}$$

(i) Resultant force P :

$$P = \sqrt{P_H^2 + P_V^2} = \sqrt{9.81^2 + 15.4^2} = 18.26 \text{ kN (Ans.)}$$

(ii) The angle made by the resultant force with the horizontal, θ :

We know that,

$$\tan \theta = \frac{P_V}{P_H} = \frac{15.4}{9.81} = 1.569 \quad \text{or} \quad \theta = 57.48^\circ \text{ (Ans.)}$$

Example 3.36. A liquid of specific gravity 0.9 is filled in a container, shown in Fig. 3.47, upto a depth of 2.4 m. Determine the magnitude and direction of hydrostatic pressure force per unit length of container exerted on its vertical face MN and curved corner NQ.

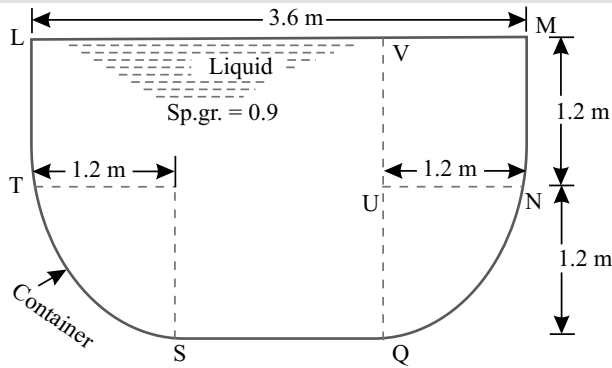


Fig. 3.47

Solution. Refer to Fig 3.47.

Vertical face MN:

$$\begin{aligned} P &= wA\bar{x} = w \times (MN \times \text{unit length}) \times \frac{MN}{2} \\ &= (9.81 \times 0.9) \times (1.2 \times 1) \times \frac{1.2}{2} = 6.357 \text{ kN (Ans.)} \end{aligned}$$

This force acts horizontally towards right and its point of application is given by:

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x} = \frac{1 \times \frac{1.2^3}{12}}{(1.2 \times 1) \times \frac{1.2}{2}} + \frac{1.2}{2} = 0.8 \text{ m (Ans.)}$$

Curved surface NQ:

Horizontal component of hydrostatic pressure force on the curved corner NQ ,

P_H = Specific weight $\times$ vertical projected area $\times$ depth of centre of vertical projection

$$= w \times (QU \times \text{unit length}) \times \left(VU + \frac{UQ}{2} \right)$$

$$= (9.81 \times 0.9) \times (1.2 \times 1) \times \left(1.2 + \frac{1.2}{2} \right) = 19.07 \text{ kN}$$

Vertical component of hydrostatic pressure force on the curved corner,

P_V = Weight of liquid contained in portion $MNQUV$

= Specific weight [Volume of liquid in portion $MNUV$ + volume of liquid in portion NQU]

$$= w [MN \times NU \times \text{unit length} + \frac{1}{4} \pi \times (NU)^2 \times \text{unit length}]$$

$$= (9.81 \times 0.9) [1.2 \times 1.2 \times 1 + \frac{1}{4} \times \pi \times (1.2)^2 \times 1] = 22.7 \text{ kN}$$

Resultant pressure force, $P = \sqrt{P_H^2 + P_V^2} = \sqrt{(19.07)^2 + (22.7)^2} = \mathbf{29.65 \text{ kN (Ans.)}$

The angle made by the resultant with the horizontal,

$$\theta = \tan^{-1} \left(\frac{P_V}{P_H} \right) = \tan^{-1} \left(\frac{22.7}{19.07} \right) = \mathbf{49.97^\circ \text{ (Ans.)}}$$

Example 3.37. A cylinder 2.2 m in diameter and 3.3 m long supported as shown in Fig. 3.48 retains water on one side. If the cylinder weighs 165 kN, calculate the vertical reaction at L and horizontal reaction at M.

Neglect the frictional effects.

Solution.

Radius of cylinder

$$= \frac{2.2}{2} = 1.1 \text{ m}$$

Length of cylinder = 3.3 m

Weight of cylinder = 165 kN

- The horizontal component of the resultant hydrostatic force acting on the gate is the horizontal force on the projected area of the curved surface on a vertical plane.

i.e. P_H = Hydrostatic pressure force on the curved area LSN projected on the vertical plane LON ,

$$= wA\bar{x}$$

$$= 9.81 \times (2.2 \times 3.3) \times \frac{2.2}{2} = 78.34 \text{ kN}$$

$\therefore$ Horizontal reaction at M = **78.34 kN (Ans.)**

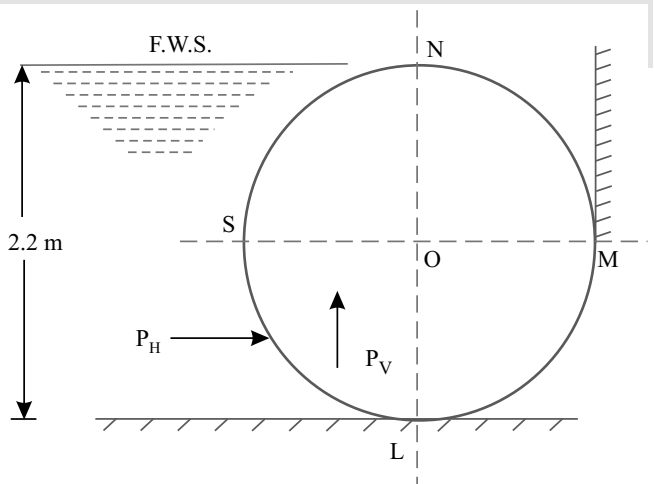


Fig. 3.48

-

Vertical force, $P_V =$ Weight of water displaced

= weight of volume equal to $LMU \times 3$.

Now,

$$\text{Area } LMU = \text{area } LOM - \text{area } MUO$$

$$= \pi R^2 \times \frac{60^\circ}{360^\circ} - \frac{1}{2} \times 2.6 \times 3 \cos 60^\circ$$

$$= \pi \times 3^2 \times 1/6 - \frac{1}{2} \times 2.6 \times 3 \times 0.5 = 4.712 - 1.95 = 2.762 \text{ m}^2$$

$$P_V = 2.762 \times 3 \times 9.81 = 81.28 \text{ kN};$$

$$P = \sqrt{P_H^2 + P_V^2} = \sqrt{99.47^2 + 81.28^2} = 128.45 \text{ kN}$$

Hence magnitude of resultant force = **128.45 kN (Ans.)**

Let,

θ = Inclination of P with horizontal.

$$\text{Then, } \tan \theta = \frac{P_V}{P_H} = \frac{81.28}{99.47} = 0.817 \text{ or } \theta = \mathbf{39.25^\circ \text{ (Ans.)}}$$

and P must pass through O .

As P_H acts at $(2.6 - 0.867) = 1.733 \text{ m}$ below water surface,

$$OT = \frac{ST}{\tan 39.25^\circ} = \frac{1.733}{0.817} = 2.12 \text{ m, and}$$

$$UT = OT - OU = 2.12 - 3 \cos 60^\circ = 0.62 \text{ m}$$

Hence point of application of P is 0.62 m to the left of MU and 1.733 m below water surface.

(Ans.)

Example 3.39. A cylinder having 3 m diameter and 1.5 m length is resting on the floor. On one side, water is filled upto half the depth while on the other side oil of relative density 0.8 filled upto the top (Fig 3.50). If the weight of the cylinder is 33.75 kN , determine the magnitudes of the horizontal and vertical components of the force which will keep the cylinder just touching the floor.

Solution. Given: Diameter of the cylinder, $d = 3 \text{ m}$; Length of the cylinder, $l = 1.5 \text{ m}$

Weight of the cylinder, $W = 30 \text{ kN}$; Relative density of the oil = 0.8

Specific weight of the oil, $w_{\text{oil}} = 9.81 \times 0.8 = 7.85 \text{ kN/m}^3$

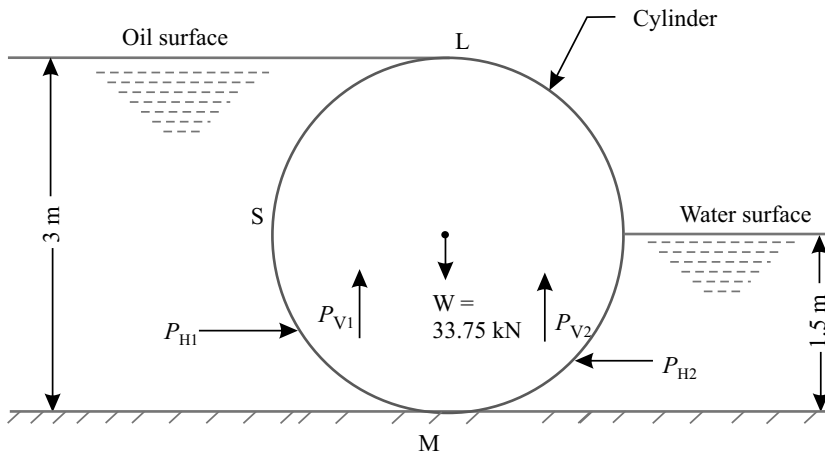


Fig. 3.50

Horizontal components:

$$\text{Horizontal force, } P_{H1} = 7.85 \times (3 \times 1.5) \times \frac{3}{2} = 52.98 \text{ kN}$$

This will act at $\frac{3}{3}$ or 1 m from bed.

$$\text{Horizontal force, } P_{H2} = 9.81 \times (1.5 \times 1.5) \times \frac{1.5}{2} = 16.55 \text{ kN}$$

This will act at $\frac{1.5}{3}$ or 0.5 m from bottom.

Hence, $(52.98 - 16.55) = \mathbf{36.43 \text{ kN}}$ force acting towards right is required to hold the cylinder stationary. (Ans.)

If it acts at a distance y , then taking moments about the bed, we get:

$$\begin{aligned} P_{H1} \times 1 - P_{H2} \times 0.5 &= (P_{H1} - P_{H2}) \times y \\ 52.98 - 16.55 \times 0.5 &= (52.98 - 16.55) \times h_1 \\ \therefore y &= \frac{52.98 - 16.55 \times 0.5}{52.98 - 16.55} = \frac{44.7}{36.43} = \mathbf{1.227 \text{ m (Ans.)}} \end{aligned}$$

Vertical components:

$$P_{V1} = 7.85 \times \frac{\pi \times 1.5^2}{2} \times 1.5 = 41.61 \text{ kN}$$

$$\text{It will act at } \frac{4 \times 1.5}{3\pi} = 0.636 \text{ m to left of LM;}$$

$$P_{V2} = 9.81 \times \frac{\pi \times 1.5^2}{4} \times 1.5 = 26 \text{ kN}$$

It will act at 0.636 m right of LM.

Since vertical forces must balance, therefore,

$$\text{External force required} = 41.61 + 26 - 33.75 = \mathbf{33.86 \text{ kN (Ans.)}}$$

This external force is required in *vertically downward* direction. To find out its line of action, taking moments about the vertical line along which P_{V2} acts, we get:

$$\begin{aligned} W \times 0.636 + 33.86 \times x &= P_{V1} \times (0.636 + 0.636) \\ 33.75 \times 0.636 + 33.86 x &= 41.61 \times 1.272 \\ x &= \mathbf{0.929 \text{ m (Ans.)}} \end{aligned}$$

Example 3.40. A tank is filled with water under pressure and the pressure gauge fitted at the top indicates a pressure of 18 kPa. The tank measures 3 m perpendicular to the plane of the paper, and the curved surface LM of the top is quarter of a circular cylinder of radius 2.4 m. Determine:

- Horizontal and vertical components of water pressure on the curved surface LM, and
- Magnitude and direction of the resultant force.

Solution. Refer to Fig. 3.51.

$$\text{Pressure indicated by pressure gauge, } p = 18 \text{ kPa} = 18 \times 10^3 \text{ N/m}^2$$

$\therefore$ The water head equivalent,

$$h = \frac{p}{w} = \frac{18 \times 10^3}{9810} = 1.835 \text{ m}$$

Hence the free water surface can be imagined to be 1.835 m above the top of the tank.

P_H (**Horizontal component**) = Hydrostatic pressure force on vertical projection MN or the curved surface LM

$$= wA\bar{x}$$

$$= 9.81 \times (2.4 \times 3) \times \left(1.835 + \frac{2.4}{2}\right)$$

$$= 214.37 \text{ kN} \rightarrow (\text{Ans.})$$

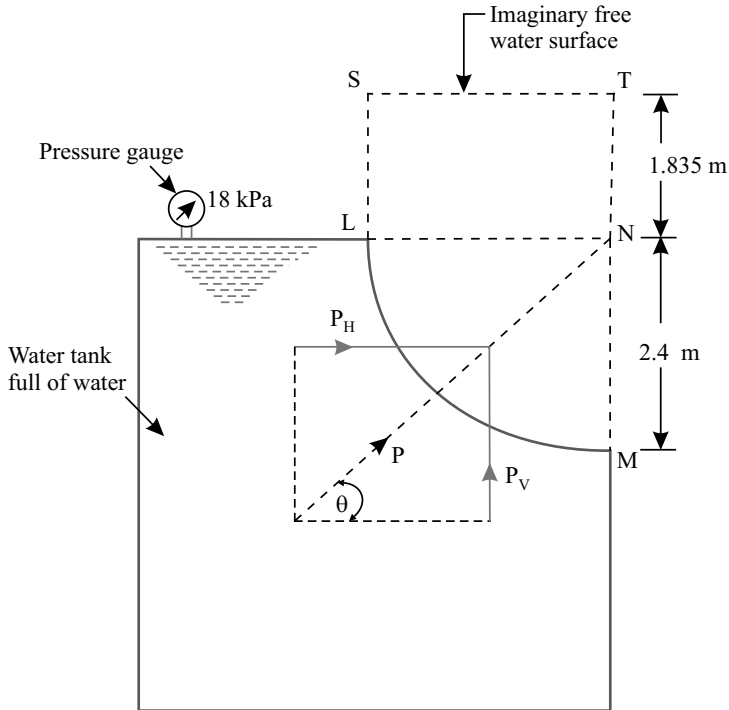


Fig. 3.51

P_V (Vertical component) = Weight of volume of water above LM upto imaginary water surface i.e., of volume $SLMNT$

$$= \left\{1.835 \times 2.4 + \frac{1}{4} \times \pi \times 2.4^2\right\} \times 3 \times 9.81 = 262.75 \text{ kN} \uparrow (\text{Ans.})$$

$$\text{The resultant force, } P = \sqrt{(P_H)^2 + (P_V)^2} = \sqrt{(214.37)^2 + (262.75)^2}$$

$$= 339.1 \text{ kN} (\text{Ans.})$$

The inclination of P with the horizontal,

$$\theta = \tan^{-1} \left(\frac{P_V}{P_H} \right) = \tan^{-1} \left(\frac{262.75}{214.37} \right) \approx 50.8^\circ (\text{Ans.})$$

Example 3.41. In the Fig. 3.52. is shown the cross-section of the tank full of water under pressure. The length of the tank is 3 m. An empty cylinder lies along the length of the tank on one of its corners as shown. Find the horizontal and vertical components of the force acting on the curved surface LMN of the cylinder.

Solution. Length of the tank = 3 m

Radius, $r = 1.5$ m

Pressure, $p = 30 \text{ kN/m}^2$

$$\text{Pressure head, } hp = \frac{p}{w} = \frac{30}{9.81} \approx 3 \text{ m}$$

Free water surface will be at a height of 3 from the top of the tank; equivalent free water surface is shown in Fig. 3.53.

(i) Horizontal component of force, P_H :

$$P_H = wA\bar{x}$$

where, w = Specific weight of water
(= 9.81 kN/m³)

$$A = \text{Area projected on vertical plane} \\ = 2.25 \times 3 = 6.75 \text{ m}^2$$

$$\bar{x} = 3 + \frac{2.25}{2} = 4.125 \text{ m}$$

$$P_H = 9.81 \times 6.75 \times 4.125 \\ = 273.15 \text{ kN (Ans.)}$$

(ii) Vertical component of force, P_V :

P_V = Weight of water enclosed or supported actually or imaginary by curved surface LMN

= Weight of water in the portion $NOTULMN$

= Weight of water in $NOTZMN$ – weight of water in $LUZM$.

But, weight of water in $NOTZMN$ = weight of water in NOM + weight of water in $OMZTO$

$$= w \left(\frac{\pi r^2}{4} + OM \times MZ \right) \times 3 \\ = 9.81 \left(\frac{\pi \times 1.5^2}{4} + 1.5 \times 3.75 \right) \times 3 \\ = 217.5 \text{ kN}$$

$$\text{Weight of water in } LUZM = w (\text{area of } LUZM) \times 3 \\ = 9.81 [\text{area of } LUZQ + LQMS - LSM] \times 3$$

$$\text{In } \triangle LSO, \sin \theta = \frac{LS}{OL} = \frac{0.75}{1.5} = 0.5, \therefore \theta = 30^\circ$$

$$MS = MO - SO = 1.5 - OL \cos \theta \\ = 1.5 - 1.5 \times \cos 30^\circ = 0.2 \text{ m}$$

$$\begin{aligned} \text{Area } LSM &= LMO - LSO \\ &= \pi r^2 \times \frac{30^\circ}{360^\circ} - \frac{1}{2} \times OS \times LS \\ &= \pi \times 1.5^2 \times \frac{1}{12} - \frac{1}{2} \times (1.5 \times \cos 30^\circ) \times (1.5 \sin 30^\circ) \\ &= 0.589 - 0.487 = 0.102 \text{ m} \end{aligned}$$

$\therefore$ Weight of water in $LUZM$

$$\begin{aligned} &= 9.81 [LQ \times ZQ + LQ \times QM - 0.102] \times 3 \\ &= 9.81 [0.2 \times 3 + 0.2 \times 1.5 \sin 30^\circ - 0.102] \times 3 \quad (\because LQ = MS) \\ &= 9.81 (0.6 + 0.15 - 0.102) \times 3 \\ &= 19.07 \text{ kN} \end{aligned}$$

$$P_V = 217.5 - 19.07 = 198.43 \text{ kN (Ans.)}$$

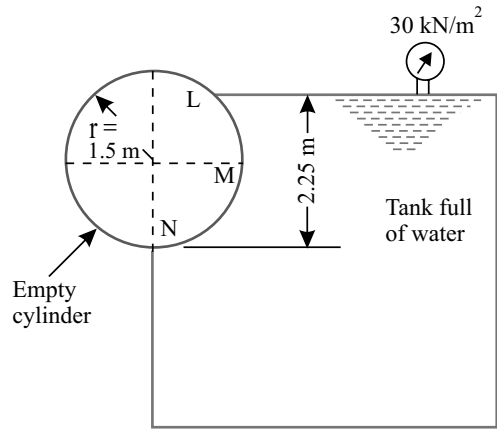


Fig. 3.52

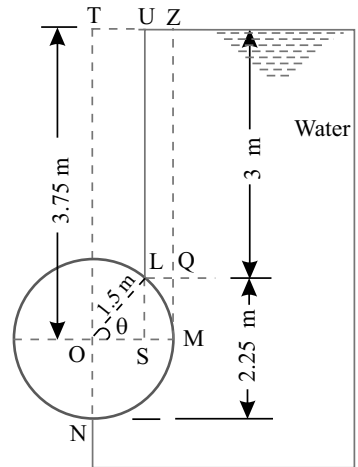


Fig. 3.53

Example 3.42. A cylindrical tank of 1.5 m diameter and height 0.75 m has a hemispherical dome. The tank contains oil of relative density 0.84 [Fig. 3.54]. The dome is joined to the cylinder portion by four equally spaced bolts. If the pressure gauge at a point M, 0.3 m, above the base of the tanks, reads 50 kPa determine the force on each bolt.

Solution. Equivalent of pressure p_L in terms of oil column,

$$p_L = w_o h_L$$

$$50 = (0.84 \times 9.81) h_L$$

$$\therefore h_L = \frac{50}{0.84 \times 9.81} = 6.07 \text{ m of oil}$$

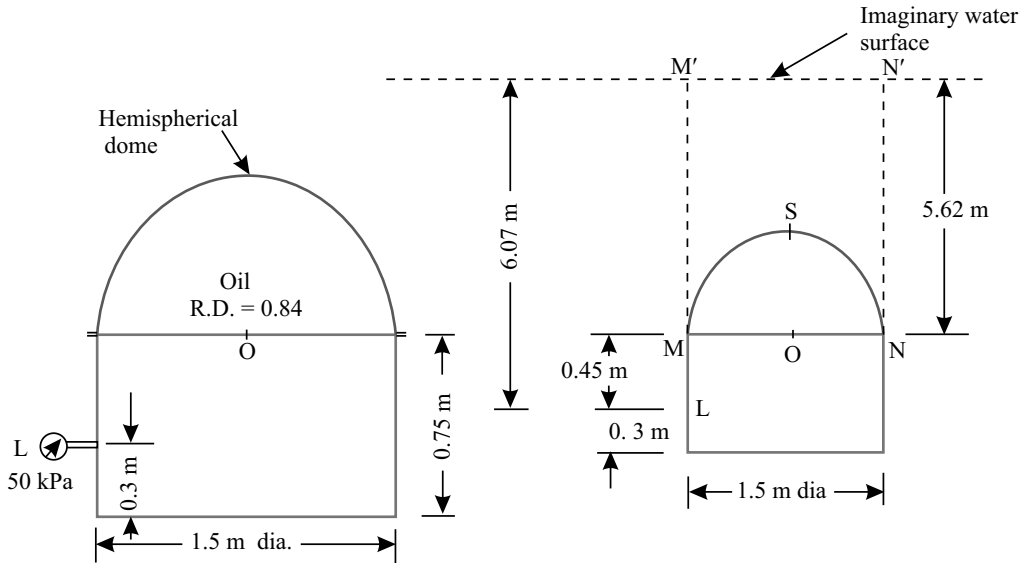


Fig. 3.54

Fig. 3.55

The imaginary oil surface at an elevation of $h_L = 6.07$ m is now considered (Fig. 3.55).

Above the base plane MN of the dome the elevation of the imaginary oil surface is

$$= 6.07 - 0.45 = 5.62 \text{ m}$$

By symmetry there is no horizontal force on the dome.

The vertical force P_V = Weight of oil above the dome surface upto the imaginary oil surface
= Weight of volume MSNN'M'

$$= 0.84 \times 9.81 \left[\left\{ \frac{\pi \times (1.5)^2}{4} \times 5.62 \right\} - \left\{ \frac{1}{2} \times \frac{4}{3} \pi (0.75)^3 \right\} \right]$$

$$= 8.24 (9.931 - 0.884) = 74.55 \text{ kN}$$

This force is shared by four bolts.

$$\therefore \text{ Tensile force on each bolt } = \frac{74.55}{4} = \mathbf{18.64 \text{ kN (Ans.)}}$$

3.7. DAMS

A **dam** is a massive structure, built up mostly with R.C.C. or stone or earth, across a river or a stream for the purpose of impounding or storing water. Its cross-section may be triangular, rectangular or trapezoidal. That side of the dam to which the water from the river or the stream

approaches is known *upstream* and the other, *downstream*. A dam which resists the water pressure by its own weight only, is termed as a *gravity dam* (viz Bhakra dam).

Fig 3.56 shows the trapezoidal cross-section of the dam with a vertical face and a straight slope or batter for the back.

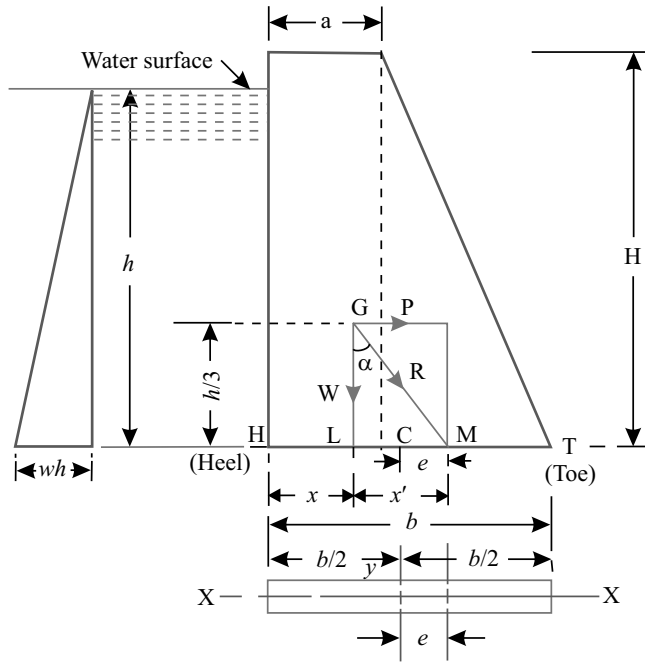


Fig. 3.56

Let,
 a = Top width of the dam,
 b = Base width of the dam,
 H = Height of the dam, and
 h = Height of water column.

Consider 1 m length of the dam.

Weight of masonry = Area $\times$ length $\times$ density of masonry

$$\therefore W = \left(\frac{a+b}{2} \right) \times H \times 1 \times \text{density of masonry} \quad \dots(3.8)$$

Let the c.g. of the section be at a distance x from the vertical face. Now dividing the trapezium into a rectangle and a triangle and taking moments about the vertical face, we get:

$$\begin{aligned} & a \times h \times \frac{a}{2} + \frac{1}{2} (b-a) \times h \left[a + \left(\frac{b-a}{3} \right) \right] \\ &= \left[a \times h + \left(\frac{b-a}{2} \right) \times h \right] \bar{x} \\ \therefore \bar{x} &= \frac{a \times h \times \frac{a}{2} + \frac{1}{2} (b-a) \times h \left[a + \frac{b-a}{3} \right]}{a \times h + \left(\frac{b-a}{2} \right) \times h} \end{aligned}$$

or,
$$\bar{x} = \frac{a^2 + ab + b^2}{3(a + b)} \quad \dots(3.9)$$

Total water pressure (P) = Area $\times$ average pressure

$$\therefore P = (l \times h) \times \frac{wh}{2} = \frac{wh^2}{2}$$

This pressure acts at $h/3$ from the base of dam. Let the resultant R of P and W cuts the base of the dam at the point M .

Then, from triangle GLM , we get:

$$\tan \alpha = \frac{LM}{GL} = \frac{P}{W}$$

i.e.,
$$\frac{x'}{h/3} = \frac{P}{W} \text{ or } x' = \frac{P}{W} \cdot h/3 \quad \dots(3.10)$$

The *eccentricity* of the resultant force, $e = (x + x') - b/2 \quad \dots(3.11)$

If e is +ve maximum stresses will develop towards the toe (T) and if it is -ve, maximum stresses will develop towards heel (H)

Stresses at the base:

Direct stress, $\sigma_d = \frac{\text{Weight of masonry}}{\text{Area at the base}} = \frac{W}{b \times 1} = \frac{W}{b}$ (compressive) $\dots(3.12)$

Bending stress, $\sigma_b = \pm \frac{My}{I} = \frac{(W.e) \times b/2}{1/12 \times l \times b^3}$

[Bending will take place about $Y-Y$ axis]

$$= \pm \frac{6 W.e}{b^2} \text{ (-ve sign stands for tensile stress here)} \quad \dots(3.13)$$

Maximum intensity of stress,

$$\begin{aligned} \sigma_{\max} &= \sigma_d + \sigma_b \\ &= +\frac{W}{b} + \frac{6We}{b^2} = +\frac{W}{b} \left(1 + \frac{6e}{b} \right) \text{ (Compressive)} \quad \dots(3.14) \end{aligned}$$

Minimum intensity of stress,

$$\sigma_{\min} = \sigma_d + (-\sigma_b) = \frac{W}{b} - \frac{6We}{b^2} = \frac{W}{b} \left(1 - \frac{6We}{b} \right) \text{ (Compressive)} \quad \dots(3.15)$$

It may be noted that $\sigma_{\min}$ may be tensile or compressive.

3.8. POSSIBILITIES OF DAM FAILURE

The following are the *possibilities of dam failure*:

- (i) Failure due to sliding along its base. (ii) Failure due to tension or compression.
- (iii) Failure due to shear at the weakest section. (iv) Failure due to overturning.
- (i) **Failure due to sliding along its base:**

The sliding of the dam is caused by the horizontal water pressure, P . The foundation offers frictional resistance which resists sliding. *The dam will be stable against sliding if the frictional resistance is more than the sliding (or driving) force P .*

Now, frictional force, $F = \mu W$

where, μ is the co-efficient of friction between two adjacent surfaces along which sliding is likely to take place.

Sliding or driving force = P

Factor of safety against sliding = $\frac{\mu W}{P}$ (This value *must be greater than unity*).

(ii) Failure due to tension or compression:

The dam will be stable if no tensile stress across the cross-section is produced. It means σ_d should be more than or equal to σ_b . In certain cases where tension cannot be avoided it should not increase more than 0.4 N/mm^2 .

$$\begin{aligned} \text{i.e.,} \quad \sigma_d &\geq \sigma_b \\ \frac{W}{b} &\geq \frac{6W.e}{b^2} \quad \text{or} \quad 1 \geq \frac{6 \times e}{b} \quad \text{or} \quad e \leq \frac{b}{6} \end{aligned} \quad \dots(3.16)$$

$$\begin{aligned} \text{Now, when} \quad e &= \frac{b}{6} \\ HM = x + x' &= \frac{b}{2} + e = \frac{b}{2} + \frac{b}{6} = \frac{2b}{3} \end{aligned} \quad \dots(3.17)$$

Therefore the resultant must always be in the *middle third* of the base.

(iii) Failure due to shear at the weakest section:

If, $A' =$ The least cross-sectional area of the dam at any section, and

$\sigma_{s(\max)} =$ Maximum safe shear stress of the dam material,

Then, resistance against shear = $\sigma_{s(\max)} \times A'$

Factor of safety against shear = $\frac{\sigma_{s(\max)} \times A'}{P'}$ (It must be *greater than unity*)

where, $P' =$ Total liquid pressure due to water column *above the section*.

(iv) Failure due to overturning:

Referring to Fig. 3.56 we find that water pressure tends to overturn the wall about the toe T whereas W tends to counteract the turning effect.

Taking moments of P and W about toe T , we get overturning moment = $P \times h/3$.

and resisting moment = $W(b - x)$

$$\begin{aligned} \text{Factor of safety against overturning} &= \frac{W(b - x)}{P \times (h/3)} \\ &= \frac{3W(b - x)}{Ph} \end{aligned} \quad \text{(It must be } \textit{greater than unity})$$

Example 3.43. A concrete dam of trapezoidal section having water on vertical face is 12 m high. The base of the dam is 8 metres wide and top 2 metres wide. Find the resultant thrust on the base per metre length of dam, and the point where it intersects the base. Take the specific weight of masonry as 240 kN/m^3 and water level coinciding with the top of the dam.

Solution. Refer to Fig. 3.57.

Top width = 2 m, base width = 8 m, height = 12 m

Consider 1 m length of dam.

Weight of masonry,

$$W = \left(\frac{8+2}{2} \right) \times 12 \times 1 \times 24 = 1440 \text{ kN}$$

Water pressure,

$$P = \frac{wh^2}{2} = \frac{9.81 \times 12^2}{2} = 706 \text{ kN}$$

$$\text{Resultant thrust, } R = \sqrt{P^2 + W^2}$$

$$= \sqrt{(706)^2 + (1440)^2} = \mathbf{1604 \text{ kN (Ans.)}}$$

The point, where the resultant thrust cuts the base:

Let x metres be the distance of c.g. from the vertical face. Dividing the trapezium into rectangle and triangle and taking moments about the vertical face, we have:

$$12 \times 2 \times \frac{2}{2} + \frac{1}{2} \times 12 \times 6 \left(2 + \frac{6}{3} \right)$$

$$= (12 \times 2 + \frac{1}{2} \times 12 \times 6) \times x$$

$$= 24 + 36 \times 4 = 60x$$

or,

$$x = \frac{24 + 36 \times 4}{60} = 2.8 \text{ m}$$

The value of ' x ' can also be found by using the relation:

$$x = \frac{a^2 + ab + b^2}{3(a+b)} = \frac{2^2 + 2 \times 8 + 8^2}{3(2+8)} = 2.8 \text{ m}$$

$$\text{From Fig. 3.57, } \tan \alpha = \frac{P}{W} = \frac{706}{1440}$$

$$\text{Also, } \tan \alpha = \frac{x'}{4} = \frac{706}{1440} \text{ or } x' = \frac{706 \times 4}{1440} = 1.96 \text{ m}$$

Now, the point where the resultant thrust cuts the base

$$= x + x' = 2.8 + 1.96 = \mathbf{4.76 \text{ m from H (heel) (Ans.)}}$$

Example 3.44. A masonry dam trapezoidal in cross-section is 4 m wide at the top, 8 m wide at the base and 10 m high. It retains water level with top against a vertical face. Obtain stress distribution at the base if specific gravity of masonry is 2.5.

Solution. Refer to Fig. 3.58.

Top width = 4 m; base width = 8 m; height = 10 m

Density of masonry = $2.5 \times 9.81 = 24.5 \text{ kN/m}^3$

Consider 1 m length of the dam.

Weight of masonry acting through c.g.,

$$W = \left(\frac{8+4}{2} \right) \times 10 \times 1 \times 24.5 = 1470 \text{ kN}$$

$$\text{Water pressure, } P = \frac{wh^2}{2} = \frac{9.81 \times 10^2}{2} = 490.5 \text{ kN}$$

Let x metres be the distance of c.g. from the vertical face. Dividing the trapezium into a rectangle and a triangle and taking moments about the vertical face, we have:

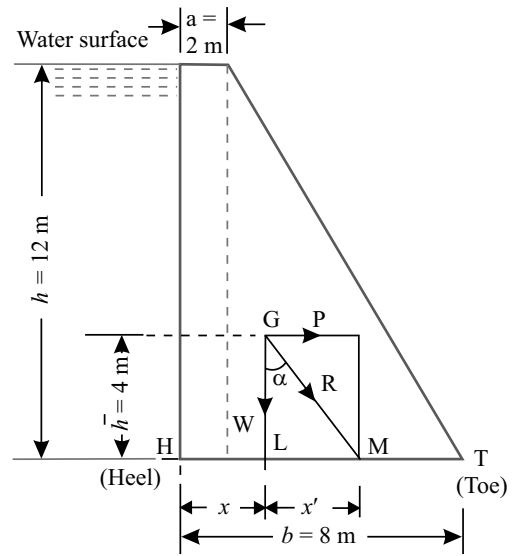


Fig. 3.57

$$10 \times 4 \times \frac{4}{2} + \frac{1}{2} \times 10 \times 4 \left(4 + \frac{4}{3} \right)$$

$$= \left(10 \times 4 + \frac{1}{2} \times 10 \times 4 \right) \times x$$

$$\text{or, } 80 + 20 \times 5.333 = 60x$$

$$x = \frac{80 + 20 \times 5.33}{60} = 3.11 \text{ m}$$

From Fig. 3.58.,

$$\tan \alpha = \frac{P}{W} = \frac{490.5}{1470}$$

$$\text{Also, } \tan \alpha = \frac{x'}{10/3}$$

$$\therefore \frac{x'}{10/3} = \frac{490.5}{1470} = \frac{1}{3}$$

$$\text{or, } x' = \frac{1}{3} \times \frac{10}{3} = \frac{10}{9} = 1.11 \text{ m}$$

Now, $x + x' = 3.11 + 1.11 = 4.22 \text{ m}$

which is well within two third base width.

Eccentricity of resultant thrust,

$$e = (x + x') - 4 = 4.22 - 4 = 0.22 \text{ m}$$

Bending stress,

$$\sigma_b = \frac{M}{I} y = \frac{(W.e).y}{I}$$

$$= \frac{1470 \times 0.22 \times 8/2}{\frac{1 \times 8^3}{12}} = \frac{1470 \times 0.22 \times 4 \times 12}{8^3} = \pm 30.3 \text{ kN/m}^2$$

Direct stress,

$$\sigma_d = \frac{\text{Weight of masonry}}{\text{Area of base}} = \frac{1470}{8 \times 1} = 183.7 \text{ kN/m}^2$$

$$\sigma_{\max} = \sigma_d + \sigma_b = 183.7 + 30.3 = 214 \text{ kN/m}^2 \text{ (comp.) (Ans.)}$$

$$\sigma_{\min} = \sigma_d - \sigma_b = 183.7 - 30.3 = 153.4 \text{ kN/m}^2 \text{ (comp.) (Ans.)}$$

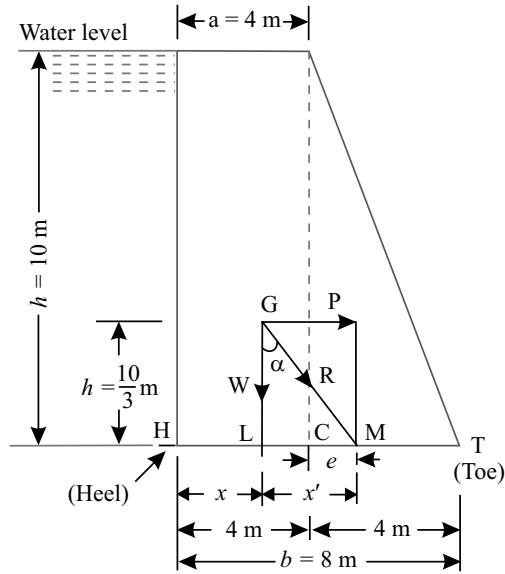


Fig. 3.58

Example 3.45. A masonry weir is of trapezoidal cross-section with a top width of 2m and of height 5m with upstream slope of 1 vertical in 0.1 horizontal and a downstream slope of 1 vertical in 0.75 horizontal. If the weir has water stored upto its crest on the upstream side and has a tail water of 2 m depth on the downstream, calculate, per unit length of weir:

- The resultant force on the base of the weir.
- The minimum and maximum stresses on the base of the weir.

Assume specific weight of masonry as 22 kN/m^3 and neglect uplift forces.

Solution. Refer to Fig. 3.59.

Consider 1 m length of the weir.

- Resultant force on the base:**

The forces acting on the weir are:

- (i) Weight of masonry, $W = W_1 + W_2 + W_3$
- (ii) Vertical force due to water on the upstream slope, P_{V1}
- (iii) Vertical force due to tail water on the downstream slope, P_{V2}
- (iv) Horizontal force due to upstream side, P_{H1}
- (v) Horizontal water force on the downstream side, P_{H2}

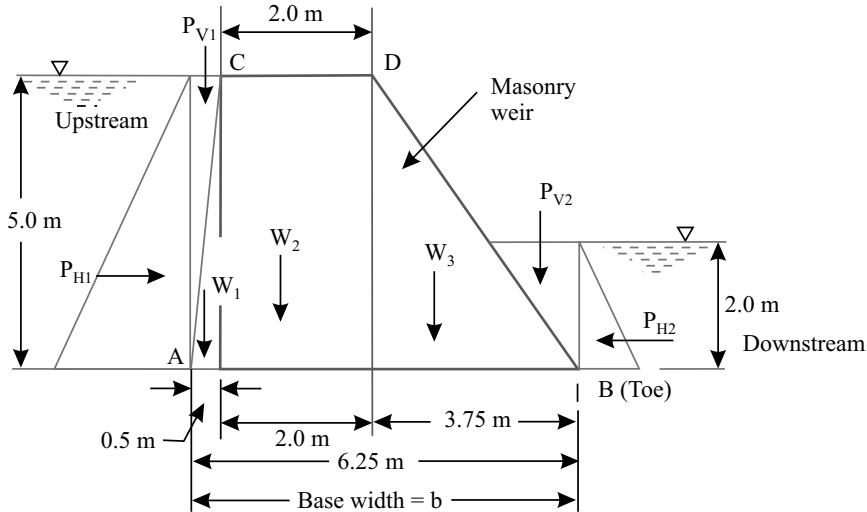


Fig. 3.59

The magnitudes of these forces, their distances from the toe of the weir (edge B) and the moments of these forces about B are tabulated in the table below:

Force	Description Magnitude, kN	Horiz. force	Magnitude, kN Vert. force	Lever arm about B (m)	Moment (Clockwise)	Moment (Anticlockwise)
W_1	$(0.5 \times 5) \times \frac{1}{2} \times 1 \times 22$		27.5	5.917		162.7
W_2	$(2.0 \times 5.0) \times 10 \times 22$		220.0	4.75		1045.0
W_3	$(3.75 \times 5) \times \frac{1}{2} \times 1 \times 22$		206.25	2.50		515.6
P_{V1}	$(0.5 \times 5) \times \frac{1}{2} \times 1 \times 9.81$		12.26	6.083		74.6
P_{V2}	$(1.5 \times 2.0) \times \frac{1}{2} \times 1 \times 9.81$		14.71	0.50		7.36
P_{H1}	$(5 \times 1) \times \frac{5}{2} \times 9.81$	122.62		1.667	204.4	
P_{H2}	$(2 \times 1) \times \frac{2}{2} \times 9.81$	-19.62		0.667		13.08
Sum		103	480.72		204.4	1818.34

Sum of vertical forces, $\Sigma V = 480.72 \text{ kN}$

Sum of horizontal forces, $\Sigma H = 103 \text{ kN}$

$$\begin{aligned}\text{Resultant, } R &= \sqrt{(\Sigma V)^2 + (\Sigma H)^2} \\ &= \sqrt{(480.72)^2 + (103)^2} = \mathbf{491.63 \text{ kN (Ans.)}}\end{aligned}$$

If θ is the *inclination* of the resultant to horizontal, then:

$$\tan \theta = \frac{\Sigma V}{\Sigma H} = \frac{480.72}{103} = 4.667$$

or, $\theta = \tan^{-1}(4.667) = \mathbf{77.9^\circ \text{ (Ans.)}}$

(ii) The minimum and maximum stresses; σ_1, σ_2 :

$$\Sigma M = 1818.34 - 204.4 = 1613.94 \text{ kNm}$$

x = Distance of point of action of the resultant from B

$$= \frac{\Sigma M}{\Sigma V} = \frac{1613.94}{480.72} = 3.357 \text{ m}$$

As b = base width = 6.25 m, the

$$\text{Eccentricity, } e = x - \frac{b}{2} = 3.357 - \frac{6.25}{2} = 0.232 \text{ m}$$

Maximum and minimum stresses,

$$\begin{aligned}\sigma_{1,2} &= \frac{\Sigma V}{b} \left(1 \pm \frac{6e}{b} \right) \\ &= \frac{480.72}{6.25} \left(1 \pm \frac{6 \times 0.232}{6.25} \right)\end{aligned}$$

$$\therefore \sigma_1 = \mathbf{94.04 \text{ kN/m}^2 \text{ (Ans.)}}$$

$$\sigma_2 = \mathbf{59.78 \text{ kN/m}^2 \text{ (Ans.)}}$$

Example 3.46. The curved face of a dam is shaped according to the relation $y = \frac{x^2}{12.25}$ as shown in the Fig. 3.60. If the width of the dam is unity and height of water retained by the dam is 12 m, determine the magnitude and direction of the resultant water pressure acting on the curved face of the dam.

Solution. Profile of the curved face of the dam:

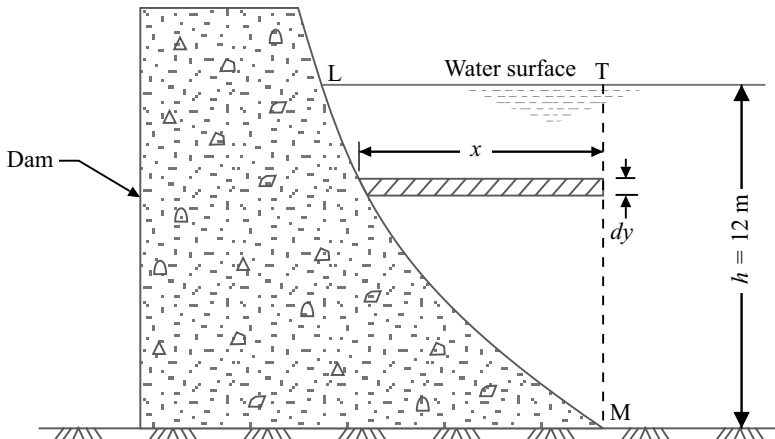


Fig. 3.60

$$y = \frac{x^2}{12.25}$$

or, $x^2 = 12.25 y$

or, $x = 3.5\sqrt{y}$

Height of water,

$$h = 12 \text{ m}$$

Width, $b = 1 \text{ m}$

Magnitude and direction of resultant water pressure:

Horizontal component, P_H = Pressure due to water on curved area projected on vertical plane

$$= \text{Pressure on area } MT = wA\bar{x}$$

where, $A = MT \times 1 = 12 \times 1 = 12 \text{ m}^2$

$$x = 12/2 = 6 \text{ m}$$

$\therefore P_H = 9.81 \times 12 \times 6 = 706.3 \text{ kN}$

Vertical component, P_V = Weight of water supported by the curve LM

= weight of water in portion LMT

$$= w \times (\text{Area of } LMT) \times \text{width of dam} = w \left[\int_0^{12} x \cdot dy \right] \times 1.0$$

where, $x \cdot dy$ = area of strip

$$= 9.81 \int_0^{12} 3.5 \sqrt{y} dy \quad (\because x = 3.5 \sqrt{y})$$

$$= 9.81 \times 3.5 \left[\frac{y^{3/2}}{3/2} \right]_0^{12} = 34.33 \times 2/3 [(12)^{3/2}] = 951.4 \text{ kN}$$

Resultant water pressure on the dam,

$$P = \sqrt{P_H^2 + P_V^2} = \sqrt{(706.3)^2 + (951.4)^2}$$

or, $P = 1184.9 \text{ kN (Ans.)}$

Direction of the resultant is given by:

$$\tan \alpha = \frac{P_V}{P_H} = \frac{951.4}{706.3} = 1.347$$

$\therefore \alpha = 53.4^\circ \text{ (Ans.)}$

Example 3.47. Fig. 3.61 shows a gate whose profile is given by $x = \sqrt{y}$. It holds water to a depth of 1 m behind it. If the width of the gate is 5 m, determine the moment M required to hold the gate in place.

Solution. Profile of the gate: $x = \sqrt{y}$

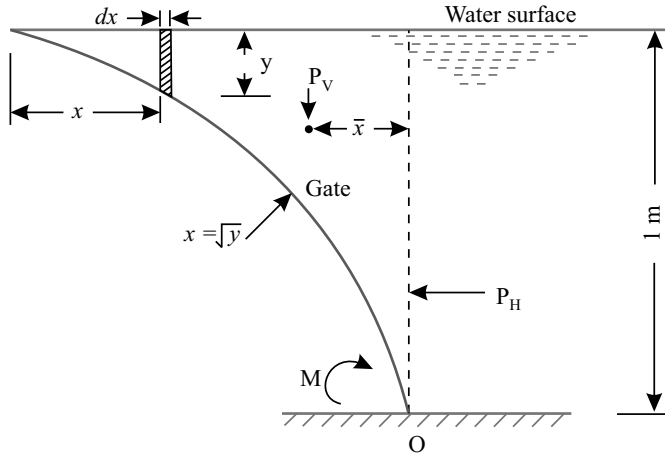


Fig. 3.61

$$\text{Horizontal force, } P_H = wA\bar{x} = 9.81 \times (1 \times 5) \times \frac{1}{2} = 24.25 \text{ kN}$$

It will act at $\frac{1}{3}$ m or 0.333 m from the bottom.

P_V = Weight of liquid above the gate

$$\begin{aligned} &= 5 \times w \int y \, dx \\ &= 5w \int_0^1 x^2 \, dx \quad \left[\because x = \sqrt{y} \right. \\ &\quad \left. \text{or } y = x^2 \right] \\ &= 5 \times 9.81 \left| \frac{x^3}{3} \right|_0^1 \\ &= 5 \times 9.81 \times \frac{1}{3} = 16.35 \text{ kN} \end{aligned}$$

The vertical line along which P_V will act is obtained by taking moments of elementary force $5wy \, dx$ about Y-axis and equating it to $P_V \times \bar{x}$.

$$\begin{aligned} \text{i.e.} \quad P_V \times \bar{x} &= \int_0^1 5wy(1-x) \, dx = \int_0^1 5wx^2(1-x) \, dx \\ &= 5w \left| \frac{x^3}{3} - \frac{x^4}{4} \right|_0^1 = 5w \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{5w}{12} \\ \therefore \quad \bar{x} &= \frac{5w}{12 \times P_V} = \frac{5 \times 9.81}{12 \times 16.35} = 0.25 \text{ m} \end{aligned}$$

Moment M:

Moment (M) of P_H and P_V about Z- axis passing through O is given by:

$$\begin{aligned} M &= P_H \times 0.333 + P_V \times 0.25 \\ &= 24.52 \times 0.333 + 16.35 \times 0.25 = 12.25 \text{ kNm} \end{aligned}$$

$$\text{i.e.} \quad M = \mathbf{12.25 \text{ kNm (Ans.)}}$$

Example 3.48. A dam has a parabolic shape $y = y_0 \left(\frac{x}{x_0} \right)^2$ as shown in Fig 3.62 having $x_0 = 6 \text{ m}$ and $y_0 = 9 \text{ m}$. The fluid is water with density $= 1000 \text{ kg/m}^3$. Compute the horizontal, vertical and the resultant thrust exerted by water per metre length of the dam.

Solution. Given: Width of the dam $= 1 \text{ m}$;

$$\text{Equation of the curve, } y = y_0 \left(\frac{x}{x_0} \right)^2; x_0 = 6\text{m}; y_0 = 9\text{m};$$

$$\text{Density of water, } \rho = 1000 \text{ kg/m}^3$$

Horizontal, vertical and resultant thrust exerted by water:

$$\begin{aligned} \text{Equation of the curve OL: } y &= y_0 \left(\frac{x}{x_0} \right)^2 \\ &= 9 \left(\frac{x}{6} \right)^2 = 9 \times \frac{x^2}{36} = \frac{x^2}{4} \end{aligned}$$

$$\text{or, } x^2 = 4y \quad \text{or} \quad x = 2\sqrt{y}$$

Horizontal thrust:

Horizontal thrust exerted by water,

F_x = Force exerted by water on vertical surface OM; the surface obtained by projecting the curved surface on vertical plane.

$$= \rho g A \bar{h}$$

$$= 1000 \times 9.81 \times (9 \times 1) \times \frac{9}{2} = 397305 \text{ N}$$

or,

$$397.3 \text{ kN (Ans.)}$$

Vertical thrust:

Vertical thrust exerted by water,

F_y = weight of water supported by curved surface OL upto free water surface

= weight of water in portion OLM $\times$ width of dam $\rho g \times \text{area of OLM} \times \text{width of dam}$

$$= 1000 \times 9.81 \times \left[\int_0^9 x \times dy \right] \times 1.0$$

$$= 1000 \times 9.81 \times \left[\int_0^9 2\sqrt{y} \times dy \right] \times 1.0$$

$$= 19620 \times \left[\frac{y^{3/2}}{3/2} \right]_0^9 = 19620 \times \frac{2}{3} \times (9)^{3/2}$$

$$= 353160 \text{ N or } 353.16 \text{ kN (Ans.)}$$

Resultant thrust:

Resultant thrust exerted by water,

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{(397.3)^2 + (353.16)^2} = 531.57 \text{ kN (Ans.)}$$

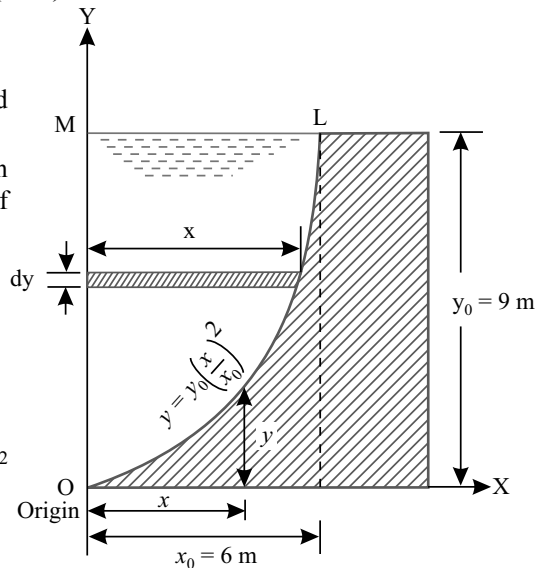


Fig. 3.62

Direction of the resultant is given by,

$$\tan \theta = \frac{F_y}{F_x} = \frac{353.16}{397.3} = 0.889$$

$$\therefore \theta = \tan^{-1}(0.889) = 41.63^\circ \text{ (Ans.)}$$

3.9. LOCK GATES

Lock gates are provided in navigation chambers to change the water level in a canal or river for navigation. There are two sets of gates G_1 and G_2 , one set on either side of the chamber. The working of the gates (Fig.3.63) is as follows:

Suppose the ship is at position 1 (on the left hand side of the chamber) and it is to be transferred to position 2 (on the right hand side). To do so the following *procedure* is adopted:

- (i) Open the sluice S_1 on the upstream gate G_1 and fill the chamber upto level $L-L$.
- (ii) Open the lock gate G_1 on the upstream and permit the ship to enter the chamber.
- (iii) Close the gate G_1 .
- (iv) Open the sluice S_2 and allow the water to fall to level MM .
- (v) Open the downstream gate G_2 and permit the ship to leave the chamber.
- (vi) Reverse the procedure in case the ship is to be transferred from position 2 to position 1.

Total pressure on the gates and reactions at top and bottom hinges:

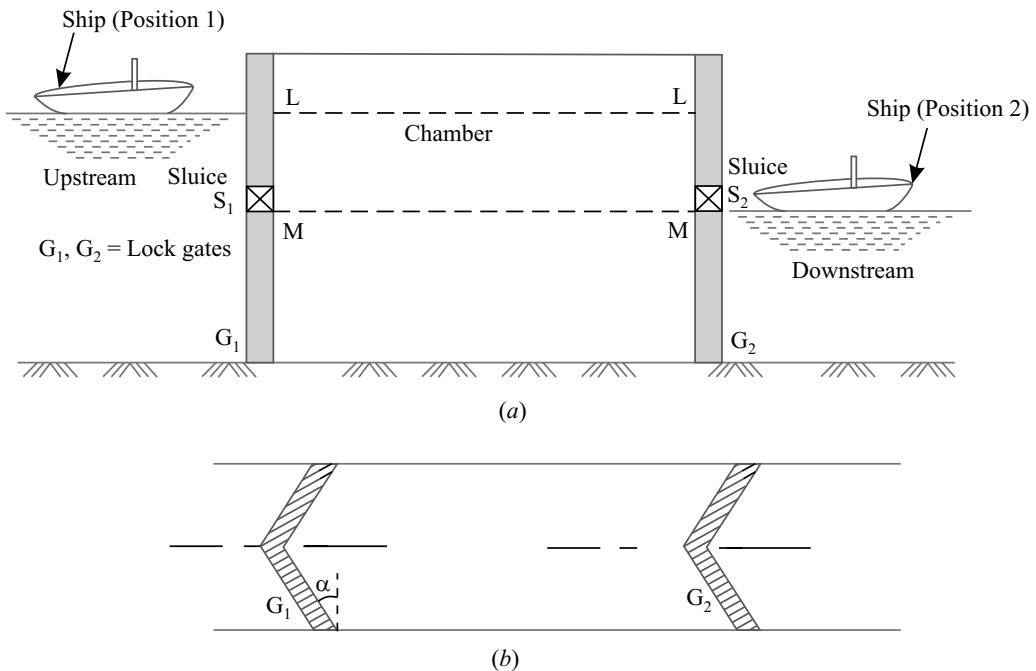


Fig. 3.63

Fig. 3.64 shows plan and elevation of a pair of lock gates. Let AB and BC be two lock gates, each carried on two hinges fixed on their top and bottom at both A and C . Due to action of water, the gates are tightly closed to one another at B .

Consider the gate AB :

Let,

P = Resultant force due to water acting at right angle to the gate,
 N = Reaction force supplied by gate BC to gate AB and acting perpendicular to the contact surfaces,

R = Resultant reaction of the top and bottom hinges (assumed to lie in the same horizontal plane in which P and N lie), and

α = Angle of inclination of gate to normal side of lock.

As the gate is in equilibrium under the forces, P , N and R , they will all intersect at one point. Let P and N intersect at D ; then R must pass through this point. Then triangle ABD will be isosceles, as $\angle DBA$ and $\angle DAB$ equal α .

Resolving the forces in a direction parallel to gate (AB),

$$R \cos \alpha = N \cos \alpha \quad \therefore R = N \quad \dots(3.18)$$

Resolving normal to the gate (AB), $P = R \sin \alpha + N \sin \alpha = (R + N) \sin \alpha = 2R \sin \alpha$

$$\therefore R = \frac{P}{2 \sin \alpha} \quad \dots(3.19)$$

(Also, inclination of R to centre line of gate = α)

Consider water pressure on the gate.

Let,

H_1 = Height of water to left of gate (*i.e.* upstream side),

H_2 = Height of water to right of gate (*i.e.* downstream side),

H = Height of top hinge from the bottom of gate,

P_1 = Total pressure of water to left of gate,

P_2 = Total pressure of water to right of gate,

R_t = Reaction of top hinge, and

R_b = Reaction of bottom hinge.

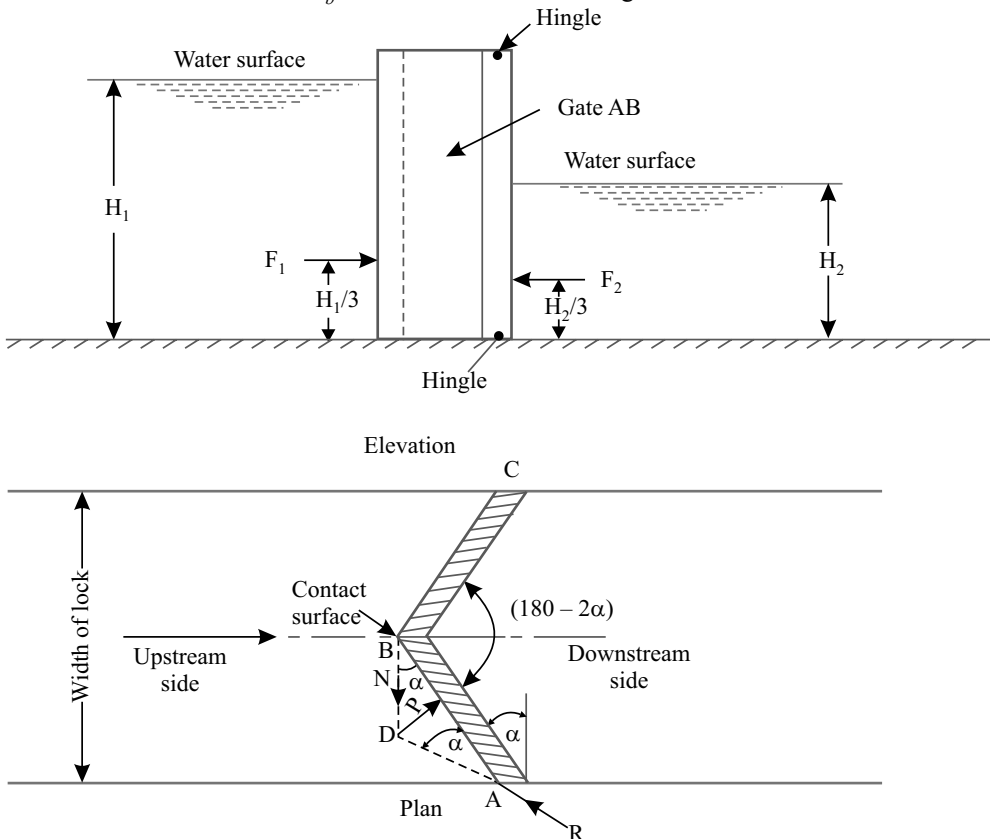


Fig. 3.64. Resultant pressure on lock gates.

Then, $R_t + R_b = R$

Also, $P_1 = \frac{wH_1}{2} \times \text{wetted area of the gate}$ (It will act at the centre of pressure

which is $\frac{H_1}{3}$ from the bottom).

and, $P_2 = \frac{wH_2}{2} \times \text{wetted area of the gate}$ (It will act at $\frac{H_2}{3}$ from the bottom).

Then, $P = P_1 - P_2$

It may be noted that only half the water pressure may be taken as acting on the hinge edge of the gate; the remaining half will be taken by the reaction of the gate BC .

Taking moments about the lower hinge, we have:

$$R_1 \sin \alpha \times H = \left(\frac{P_1}{2} \times \frac{H_1}{3} \right) - \left(\frac{P_2}{2} \times \frac{H_2}{3} \right) \quad \dots(i)$$

Resolving the forces *horizontally*, we get:

$$R_t \sin \alpha + R_b \sin \alpha = \frac{P_1}{2} - \frac{P_2}{2} \quad \dots(ii)$$

Then, from eqns. (i) and (ii), R_t and R_b may be found.

Example 3.49. Each gate of a lock is 6 m high and 5 m wide, supported on one side by two hinges, each 0.5m from the top and from bottom. The angle between the gates in closed position is 120° . If the water levels are 5m and 1.25 m on the upstream and downstream sides respectively, find:

- (i) The magnitude and position of the resultant water pressure on each gate,
- (ii) The magnitude of reaction between the gates, and
- (iii) The magnitudes of the reactions at the hinges.

Assume the reaction between the gates to be in the same horizontal plane as that of the resultant water pressure. [Jadavpur University]

Solution. Height of each gate = 6 m; width of each gate = 5 m

Height of water on upstream side, $H_1 = 5$ m

Height of water on downstream side, $H_2 = 1.25$ m

$$\text{Angle between the gates} = 120^\circ, \quad \therefore \alpha = \frac{180 - 120}{2} = 30^\circ$$

(i) **The magnitude and position of the resultant pressure:**

Upstream side: Wetted area of gate, $A_1 = 5 \times 5 = 25 \text{ m}^2$

$$\text{Total pressure on each gate, } P_1 = wA_1\bar{x}_1 = 9.81 \times 25 \times \frac{5}{2} = 613.12 \text{ kN}$$

$$\text{Position of centre of pressure } \bar{h}_1 = \frac{H_1}{3} \text{ from the bottom} = \frac{5}{3} \text{ m or } 1.667 \text{ m from the bottom}$$

Downstream side: Wetted area of gate, $A_2 = 5 \times 1.25 = 6.25 \text{ m}^2$

$$\text{Total pressure on each gate, } P_2 = wA_2\bar{x}_2 = 9.81 \times 6.25 \times \frac{1.25}{2} = 38.32 \text{ kN}$$

$$\text{Position of centre of pressure } \bar{h}_2 = \frac{1.25}{3} = 0.417 \text{ m from the bottom.}$$

Now, resultant water pressure on each gate,

$$P = P_1 - P_2 = 613.12 - 38.32 = \mathbf{574.8 \text{ kN (Ans.)}}$$

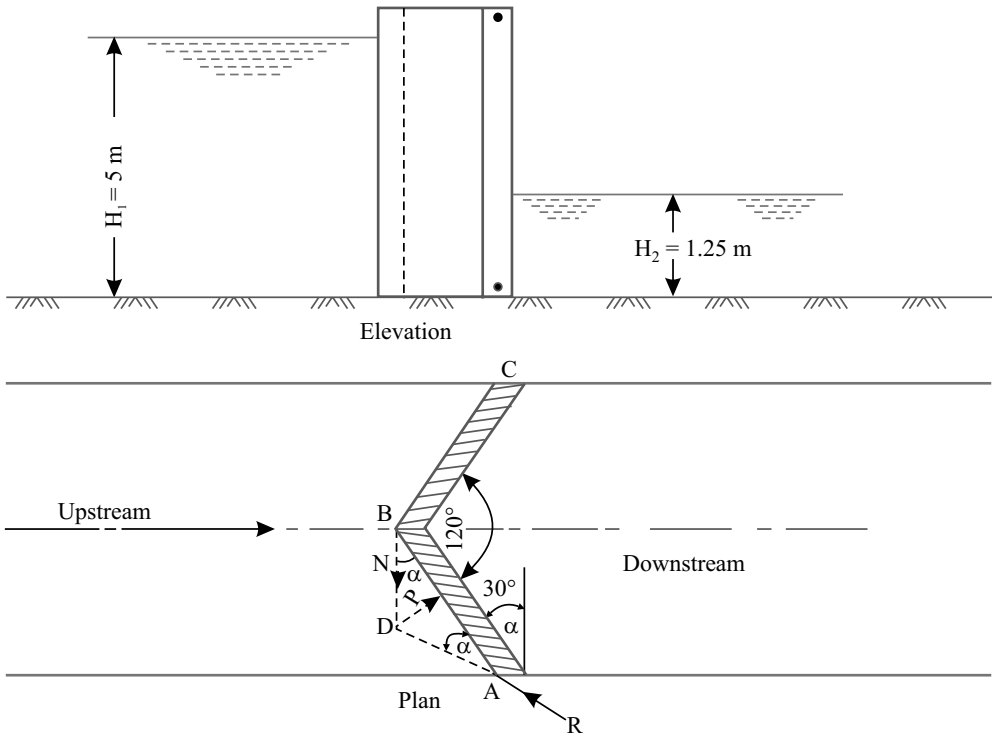


Fig. 3.65 (a). Elevation and plan of lock gates.

Let $\bar{h}$ is height of P from the bottom, then taking moments of P_1 , P_2 and P about the bottom, we get:

$$P \times \bar{h} = P_1 \times \bar{h}_1 - P_2 \times \bar{h}_2$$

or, $574.8 \times \bar{h} = 613.12 \times 1.667 - 38.32 \times 0.417 = 1006.09$

$\therefore \bar{h} = 1.75 \text{ m from the bottom (Ans.)}$

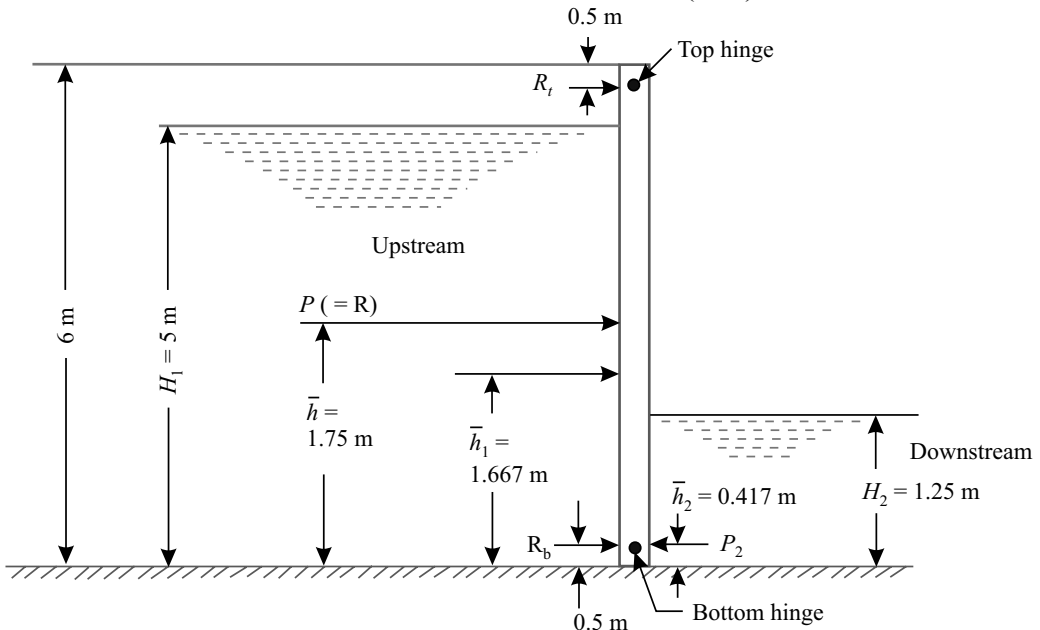


Fig. 3.65 (b). Resultant pressure and reactions at hinges of lock gates.

(ii) The magnitude of reaction between the gates, N:

Refer to Fig. 3.65 (a).

Resolving the forces at D in a direction parallel to gate (*i.e.* along AB), we have:

$$N \cos \alpha = R \cos \alpha \quad \text{or} \quad N = R$$

Resolving normally to gate (*i.e.* normal to AB), we have:

$$P = R \sin \alpha + N \sin \alpha = (R + N) \sin \alpha = 2N \sin \alpha$$

$$\therefore N (= R) = \frac{P}{2 \sin \alpha} = \frac{574.8}{2 \sin 30^\circ}$$

$$= 574.8 \text{ kN (Ans.)}$$

(iii) The magnitudes of the reactions at the hinges:

Refer to Fig. 3.65 (b).

Let,

R_t = Reaction at the top hinge, and

R_b = Reaction at the bottom hinge.

Then,

$$R_t + R_b = R = 574.8 \text{ kN}$$

Taking moments of hinge reactions R_t , R_b and $P (= R)$ about the bottom hinge, we have:

$$R_t (6 - 1) + R_b \times 0 = R \times (1.75 - 0.5)$$

$$5R_t = 574.8 \times (1.75 - 0.5)$$

$$\therefore R_t = 143.7 \text{ kN (Ans.)}$$

and,

$$R_b = 574.8 - 143.7 = 431.1 \text{ kN (Ans.)}$$

HIGHLIGHTS

1. The term hydrostatics means the study of pressure, exerted by a fluid at rest.
2. Total pressure (P) is the force exerted by a fluid on a surface (either plane or curved) when the fluid comes in contact with the surface.
For vertically immersed surface, $P = wA\bar{x}$
For inclined immersed surface, $P = wA\bar{x}$
where A = area of immersed surface, and $\bar{x}$ = depth of centre of gravity of immersed surface from the free liquid surface.
3. Centre of pressure ($\bar{h}$) is the point through which the resultant pressure acts and is always expressed in terms of depth from the liquid surface.

$$\text{For vertically immersed surface, } \bar{h} = \frac{I_G}{A\bar{x}} + \bar{x}$$

$$\text{For inclined immersed surface, } \bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x}$$

where I_G stands for moment of inertia of figure about horizontal axis through its centre of gravity.

4. The total force on a curved surface is given by:

$$P = \sqrt{P_H^2 + P_V^2}$$

where,

P_H = Horizontal force on curved surface

= Total pressure force on the projected area of the curved surface on the vertical plane = $wA\bar{x}$, and

P_V = Vertical force on submerged curved surface

= Weight of liquid actually or imaginary supported by curved surface.

The direction of the resultant force P with the horizontal is given by:

$$\tan \theta = \frac{P_V}{P_H} \quad \text{or} \quad \theta = \tan^{-1} \frac{P_V}{P_H}$$

5. Resultant force on a sluice gate $P = P_1 - P_2$

where,

P_1 = Pressure force on the upstream side of the sluice gate, and

P_2 = Pressure force on the downstream side of the sluice gate.

6. For a lock gate, the reaction between two gates is equal to the reaction at the hinge,

i.e., $N = R$

Also reaction between the two gates, $N = \frac{P}{2 \sin \alpha}$

where,

P = Resultant water pressure on the lock gate $= P_1 - P_2$, and

α = Inclination of the gate to normal side of lock.

OBJECTIVE TYPE QUESTION

Choose the correct Answer:

1. The intensity of pressure p is related to specific weight w of the liquid and vertical depth h of the point by the equation

- (a) $p = wh$ (b) $h = pw$
(c) $p = wh^2$ (d) $p = wh^3$

2. The point of application of the total pressure on the surface is

- (a) centroid of the surface
(b) centre of pressure
(c) either of the above
(d) none of the above.

3. If A is the area of the immersed surface, w is the specific weight of the liquid and $\bar{x}$ is the depth of horizontal surface from the liquid surface, then the total pressure P on the surface is given by

- (a) $p = wA^2\bar{x}$ (b) $p = w^2A\bar{x}$
(c) $p = wA\bar{x}$ (d) $p = wA\bar{x}^2$

4. Centre of pressure ($\bar{h}$) in case of an inclined immersed surface is given by

- (a) $\bar{h} = \frac{I_G \sin \theta}{A\bar{x}} + \bar{x}$ (b) $\bar{h} = \frac{I_G \sin \theta}{A^2\bar{x}} + \bar{x}$
(c) $\bar{h} = \frac{I_G^2 \sin \theta}{A\bar{x}} + \bar{x}$ (d) $\bar{h} = \frac{I_G \sin^2 \theta}{A\bar{x}} + \bar{x}$

5. The side of the dam to which the water from the river or the stream approaches is known as

(a) downstream (b) upstream

(c) either of the above (d) none of the above.

6. Which of the following is a possibility of dam failure?

- (a) Failure due to sliding along its base
(b) Failure due to tension or compression
(c) Failure due to shear at the weakest section
(d) Failure due to overturning
(e) All of the above.

7. Lock gates are provided to

- (a) change the water level in a canal or river for irrigation
(b) store water for irrigation purpose
(c) either of the above
(d) none of the above.

8. Total force on a curved surface is given by

- (a) $P = (P_H^2 + P_V^2)^{3/2}$ (b) $P = \sqrt{P_H^2 + P_V^2}$
(c) $P = (P_H^2 + P_V^2)^{5/2}$ (d) $P = P_H + P_V$

9. Resultant pressure on a sluice gate is given by

- (a) $P = P_1 - P_2$ (b) $P = P_1 + P_2$
(c) $P = \sqrt{P_1^2 + P_2^2}$ (d) $P = (P_1^2 + P_2^2)^{3/2}$

10. The term..... means the study of pressure exerted by a fluid at rest.

- (a) hydrostatics (b) fluid mechanics
(c) continuum (d) kinetics.

ANSWERS

- | | | | | | |
|--------|--------|--------|---------|--------|--------|
| 1. (a) | 2. (b) | 3. (c) | 4. (d) | 5. (b) | 6. (e) |
| 7. (a) | 8. (b) | 9. (a) | 10. (a) | | |

THEORETICAL QUESTIONS

- Define the following terms:
 - Total pressure, and
 - Centre of pressure.
- Derive expressions for total pressure and centre of pressure for a vertically immersed surface.
- Derive an expression for the depth of centre of pressure from free surface of liquid of an inclined plane surface submerged in the liquid.
- Derive an expression for the reaction between the gates as:

$$R = \frac{P}{2 \sin \alpha}$$
 where, P = Resultant force due to water acting at right angles to the gate, and
 α = Angle of inclination of gate to normal side of lock.

UNSOLVED EXAMPLES

- A rectangular plate $2\text{ m} \times 4\text{ m}$ is vertically immersed in water in such a way that 2 metres side is parallel to the water surface and 2.5 metres below it. Find the total pressure on the rectangular plate. Take $w = 9.81 \text{ kN/m}^3$.
 [Ans. 353.16 kN]
- A circular door of 0.5 m diameter closes on an opening in the vertical side of a bulk head, which retains water. The centre of the opening is at a depth of 2m from the water level. Determine the total pressure on the door. Take specific gravity of sea water as 1.03. [Ans. 3.968 kN]
- A circular plate of diameter 1.5 m is placed vertically in water in such a way that the centre of the plate is 3 m below the free surface of water. Determine:
 - Total pressure on the plate, and
 - Position of the centre of pressure
 [Ans. (i) 52 kN, (ii) 3.0468 m]
- A rectangular sluice gate is situated on the vertical wall of a lock. The vertical side of the sluice is 'd' metres in length and depth of centroid of the area is 'p' metres below the water surface. Prove that the depth of pressure is equal to $\left(p + \frac{d^2}{12p}\right)$.
- A circular opening, 3 m diameter, in a vertical side of a tank is closed by a disc of 3 m diameter which can rotate about a horizontal diameter. Calculate:
 - The force on the disc, and
 - The torque required to maintain the disc in equilibrium in the vertical position when the head of water above the horizontal diameter is 6 m.
 [Ans. 416 kN (ii) 39 kNm]
- An isosceles triangular plate of base 5 m and altitude 5 m is immersed vertically in an oil of specific gravity 0.8. The base of the plate is 1 m below the free water surface, determine:
 - The total pressure, and
 - The centre of pressure.
 [Ans. (i) 261.93 kN (ii) 3.19 m]
- Determine the total pressure and centre of pressure on an isosceles triangular plate of base 4 m and altitude 4 m when it is immersed vertically in an oil of specific gravity 0.9. The base of the plate coincides with the free oil surface.
 [Ans. 94.15 kN, 1.99 m]
- A tank contains water upto a height of 0.5 m above the base. An immiscible liquid of specific gravity 0.8 is filled on the top of water upto 1 m height. Calculate: (i) Total pressure on one side of the tank; (ii) The position of centre of pressure for one side of the tank, which is 2m wide.
 [Ans. 18.15 kN, 1.009 m from the top]

9. A circular plate of 1 m diameter is immersed in water in such a way that its plane makes an angle of 30° with the horizontal and its top edge is 1.25 m below the water surface. Find the total pressure on the plate and the point, where it acts.
[Ans. 11.56 kN, 1.51 m]
10. A triangular plate of 1 m base and 1.5 m altitude is immersed in water. The plane of the plate is inclined at 30° with water surface while the base is parallel to and at a depth of 2 m from the water surface. Calculate:
(i) The total pressure on the plate, and
(ii) The position of the centre of pressure
[Ans. 16.54 kN, (ii) 2.264 m]
11. A circular plate 3 metres in diameter is submerged in water in such a way that the greatest and least depths of the surface (below water surface) are 2 m and 1 m respectively, calculate:
(i) The total pressure on front face of the plate;
(ii) The position of centre of pressure.
[Ans. (i) 104 kN, (ii) 1.54 m]
12. A rectangular plane surface 1 m wide and 3 m deep lies in water in such a way that its plane makes an angle of 30° with the free water surface. Determine the total pressure and position of centre of pressure when the upper edge is 2 m below the free surface.
[Ans. 228.69 kN, 3.427 m from free surface]
13. Fig 3.66 shows a gate having a quadrant shape of radius of 3 m. Find the resultant force due to water per metre length of the gate. Also find the angle at which this resultant force will act.
[Ans. 82.2 kN, $57^\circ 31'$]

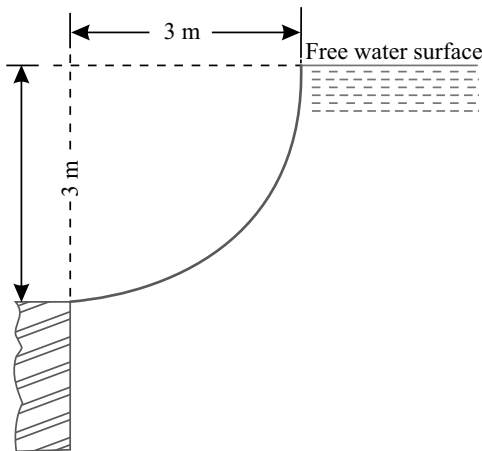
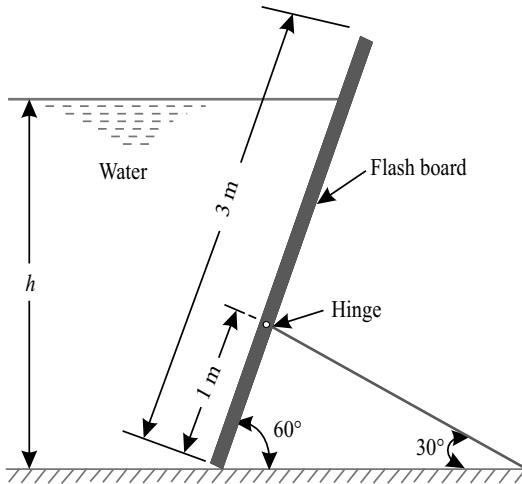


Fig. 3.66

14. A masonry dam 7 m high has a top width of 1.5 m and bottom width of 5 m. Maximum water level in the dam is 1.0 m below the top of the dam. Determine the maximum and minimum pressure intensities at the base when the dam is full. Take weight of water = 9.81 kN/m^3 and weight of masonry = 21.6 kN/m^3 .
[Ans. 117.5 kN/m^2 , 78.8 kN/m^2 both compressive]
15. The masonry dam of trapezoidal section has its upstream face vertical. The height is 10 m and top is 3 m wide. Find the minimum width of base if there is no tension at the base and water reaches the top of the dam. Take weight of water = 9.81 kN/m^3 and weight of masonry = 22 kN/m^3 . What is then maximum compressive stress at the base?
[Ans. $b = 6 \text{ m}$, 330 kN/m^2]
16. Each gate of a lock is 6 m high and is supported by two hinges placed at the top and bottom of the gate. When the gates are closed, they make an angle of 120° . The width of lock is 5 m. If the water levels are 4 m and 2 m on upstream and downstream sides respectively, find:
(i) Resultant water pressure on each gate, and
(ii) Reaction at the hinges.
[Ans. (i) 169.9 kN (ii) $R_t = 43.9 \text{ kN}$, $R_b = 126 \text{ kN}$]
[Hint : Width of each lock gate = $\frac{2.5}{\cos 30^\circ}$]
17. A gate which is 2 m wide and 1.2 m high lies in vertical plane and is hinged at the bottom. There is a liquid on upstream side of the gate which extends 1.5 m above the top of the gate and has a specific gravity of 1.45. On the downstream side of the gate there is water upto the top of the gate. Find:
(i) The resultant force acting on the gate,
(ii) The position of the centre of pressure, and
(iii) The least force acting horizontally on the top of the gate which is capable of opening it.
[IIT Kharagpur]
[Ans. (i) 57.48 kN, (ii) 2.123 m below the free surface of upstream, (iii) 27.66 kN]
18. Fig. 3.67 shows a flash board. Find the depth of water h and the compressive force in the strut per metre of the crest at the instant when the water is just ready to tip the flash board.
[IIT Kharagpur]
[Ans. 2.6 m, 38.26 kN]

**Fig. 3.67**

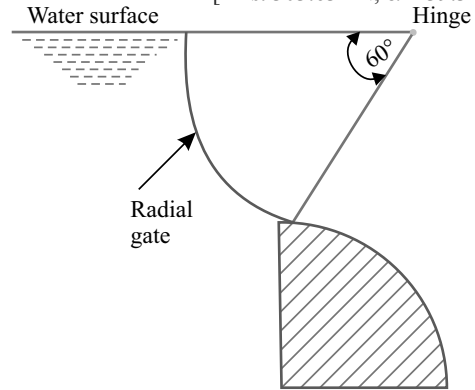
19. A rectangular door covering an opening $3\text{ m} \times 1.75$ high in a vertical wall, is hinged about its vertical edge by two points placed symmetrically 0.4 m from either end. The door is locked by clamp placed at the centre of other vertical

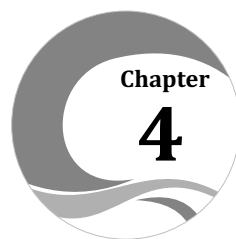
edge. Determine the reactions at the two hinges and the clamp, when the height of water is 1 m above the top edge of the opening.

[Ans. 48.28 kN ; 37.96 kN , 10.137 kN]

20. Determine the magnitude and direction of the resultant force acting on the radial gate shown in Fig. 3.68, if its length is 4 m .

[Ans. 303.63 kN , $\alpha = 39.32^\circ$]

**Fig. 3.68**



BUOYANCY AND FLOATATION

4.1. Buoyancy

4.2. Centre of buoyancy

4.3. Types of equilibrium of floating bodies

4.4. Metacentre and metacentric height

4.5. Determination of metacentric height—Analytical method—experimental method

4.6. Oscillation (rolling) of a floating body

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

4.1. BUOYANCY

Whenever a body is immersed wholly or partially in a fluid it is subjected to an upward force which tends to lift (or buoy) it up. This *tendency for an immersed body to be lifted up in the fluid, due to an upward force opposite to action of gravity is known as **buoyancy***. The force tending to lift up the body under such conditions is known as *buoyant force* or *force of buoyancy* or *upthrust*. The magnitude of the buoyant force can be determined by *Archimedes' principle* which states as follows:

“When a body is immersed in a fluid either wholly or partially, it is buoyed or lifted up by a force, which is equal to the weight of fluid displaced by the body.”

4.2. CENTRE OF BUOYANCY

The point of application of the force of buoyancy on the body is known as the **centre of buoyancy**. It is *always the centre of gravity of the volume of fluid displaced*.

Example 4.1. A wooden block of width 1.25 m, depth 0.75 m and length 3.0 m is floating in water. Specific weight of the wood is 6.4 kN/m^3 . Find:

(i) Volume of water displaced, and

(ii) Position of centre of buoyancy.

Solution. Width of the wooden block = 1.25 m

Depth of the wooden block = 0.75 m

Length of the wooden block = 3.0 m

Volume of the block = $1.25 \times 0.75 \times 3 = 2.812 \text{ m}^3$

Specific weight of wood, $w = 6.4 \text{ kN/m}^3$

Weight of the block = $6.4 \times 2.812 = 18 \text{ kN}$

(i) **Volume of water displaced:**

For equilibrium the weight of water displaced

= Weight of wooden block = 18 N

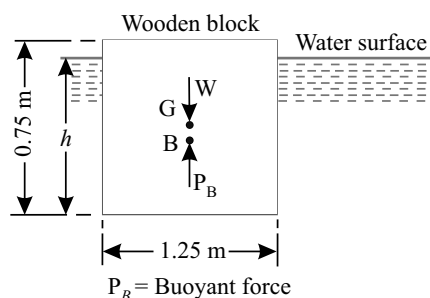


Fig. 4.1

Volume of water displaced

$$= \frac{\text{Weight of water displaced}}{\text{Weight density of water}}$$

$$= \frac{18}{9.81} = 1.835 \text{ m}^3 \text{ (Ans)} \quad (\because \text{Weight density of water} = 9.81 \text{ kN/m}^3)$$

(ii) Position of centre of buoyancy:

We know that,

Volume of wooden block in water = Volume of water displaced.

$$\text{or,} \quad 1.25 \times h \times 3.0 = 1.835$$

(where, h = depth of wooden block in water)

$$\therefore \quad h = \frac{1.835}{1.25 \times 3.0} = 0.489 \text{ m}$$

$$\text{Hence centre of buoyancy} = \frac{0.489}{2} = 0.244 \text{ from the base (Ans.)}$$

Example 4.2. A wooden block of specific gravity 0.7 and having a size of $2 \text{ m} \times 0.5 \text{ m} \times 0.25 \text{ m}$ is floating in water. Determine the volume of concrete of specific weight 25 kN/m^3 , that may be placed which will immerse (i) the block completely in water, and (ii) the block and concrete completely in water.

Solution. Size of the block = $2 \text{ m} \times 0.5 \text{ m} \times 0.25 \text{ m}$

$$\therefore \quad \text{Volume of the block} = 0.25 \text{ m}^3$$

Specific gravity of the block = 0.7

$$\text{Specific weight of the block} = 0.7 \times 9.81 = 6.867 \text{ kN/m}^3$$

$$\text{Weight of the block} = 6.867 \times 0.25 = 1.716 \text{ kN}$$

$$(\because \text{Specific weight of water} = 9.81 \text{ kN/m}^3)$$

Let, W_c = Weight of concrete required to be placed on the block, and

V_c = Volume of concrete required to be placed on the block.

$$\text{Total weight of the block} = W_c + 1.716 \text{ kN} \quad \dots(i)$$

(i) Immersion of the block only:

When the block is completely immersed, the volume of water displaced = 0.25 m^3

$\therefore$ Upward thrust at the time of complete immersion

$$= 0.25 \times 9.81 = 2.45 \text{ kN} \quad \dots(ii)$$

Now equating (i) and (ii), we get:

$$W_c + 1.716 = 2.45$$

$$\text{or} \quad W_c = 0.734 \text{ kN}$$

$$\text{Volume of concrete, } V_c = \frac{\text{Weight}}{\text{Sp. weight}} = \frac{0.734}{25} = 0.0294 \text{ m}^3 \text{ (Ans.)}$$

(ii) Immersion of block and concrete:

$$\text{Total weight of the block} = 25 V_c + 1.716 \quad \dots(i)$$

$$\text{Upward thrust} = (V_c + 0.25) \times 9.81 \quad \dots(ii)$$

Equating (i) and (ii), we get:

$$25 V_c + 1.716 = (V_c + 0.25) \times 9.81$$

$$\text{or,} \quad 25 V_c + 1.716 = 9.81 V_c + 2.45 \quad \text{or} \quad 15.19 V_c = 0.734$$

$$\text{or,} \quad V_c = 0.0483 \text{ m}^3 \text{ (Ans.)}$$

Example 4.3. Find the density of a metallic body which floats at the interface of mercury of specific gravity 13.6 and water such that 35 percent of its volume is submerged in mercury and 65 percent in water.

Solution. Let, V = Volume of the body, m^3 .

Then, volume of body submerged in mercury

$$= \frac{35}{100} \times V = 0.35 V \text{ m}^3$$

Volume of body submerged in water

$$= \frac{65}{100} \times V = 0.65 V \text{ m}^3$$

The body will be in equilibrium when,

Total buoyant (upward) force = weight of the body

But, Total buoyant force = Force of buoyancy due to water + force of buoyancy due to mercury
 = weight of water displaced by the body + weight of mercury displaced by the body
 = (weight density of water $\times$ volume of water displaced) + (weight density of mercury $\times$ volume of mercury displaced)
 = $9.81 \times 0.65 V$ (kN) + $13.6 \times 9.81 \times 0.35 V$ (kN)

and, Weight of the body = weight density $\times$ volume of the body
 = $w_{\text{body}} \times V$

(where, w_{body} = weight density of the metallic body)

For equilibrium, we have:

$$9.81 \times 0.65 V + 13.6 \times 9.81 \times 0.35 V = w_{\text{body}} \times V$$

$$\therefore w_{\text{body}} = 53.07 \text{ kN/m}^3 \text{ (Ans.)}$$

Example 4.4. A metallic cube 30 cm side and weighing 450 N is lowered into a tank containing a two-fluid layer of water and mercury. Determine the position of block at mercury-water interface when it has reached equilibrium.

(Anna University)

Solution. Refer to Fig. 4.3. The metallic cube sinks beneath the water surface and comes to rest at the water-mercury interface.

As per principle of floatation, we have weight of cubical block = buoyant force

= weight of water and mercury displaced by the block.

$$\begin{aligned} \text{Thus, } 450 &= 9810 (h_1 \times 0.3 \times 0.3) + 9810 \times 13.6 (h_2 \times 0.3 \times 0.3) \\ &= (h_1 + 13.6 h_2) \times (9810 \times 0.3 \times 0.3) \end{aligned}$$

$$\text{or, } (h_1 + 13.6 h_2) = \frac{450}{9810 \times 0.3 \times 0.3} = 0.509 \text{ m} \quad \dots(i)$$

$$\text{Also, } h_1 + h_2 = 0.3 \text{ m} \quad \dots(ii)$$

From (i) and (ii), we have the depth of cube below the water-mercury interface,

$$h_2 = \frac{(0.509 - 0.3)}{12.6} = 0.01658 \text{ m or } 16.58 \text{ mm (Ans.)}$$

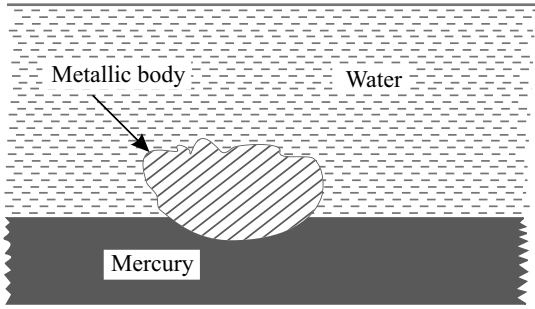


Fig. 4.2

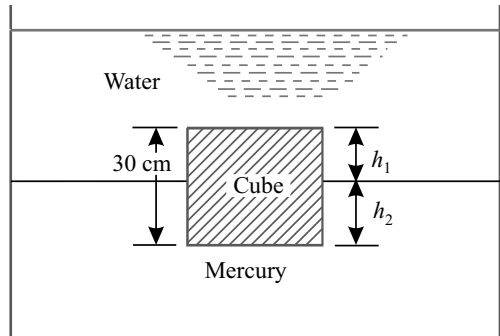


Fig. 4.3

Example 4.5. A 8 cm side cube weighing 4 N is immersed in a liquid of relative density 0.8 contained in a rectangular tank of cross-sectional area 12 cm $\times$ 12 cm. If the tank contained liquid to a height of 6.4 cm before the immersion, determine the levels of the bottom of the cube and the liquid surface.

Solution. Refer to Fig. 4.4.

Let, h_1 = Depth to which the bottom of the cube falls below original liquid surface (cm),

h_2 = Height of rise of liquid above the original liquid surface (cm), and

$(h_1 + h_2)$ = Depth and submergence of the cube (cm).

Now, Volume L = Volume M

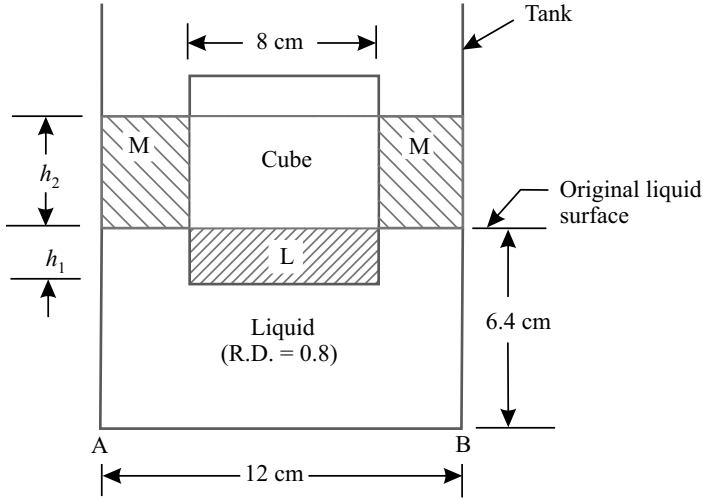


Fig. 4.4

$$8 \times 8 \times h_1 = (12^2 - 8^2) \times h_2$$

or, $h_1 = 1.25 h_2$

Weight of the cube, $W = 4\text{ N}$... (Given)

$$W = \text{Buoyant force} = \frac{(8 \times 8) \times (h_1 + h_2) \times 0.8 \times 9810}{10^6}$$

or, $4 = 0.5023 (h_1 + h_2)$

or, $4 = 0.5023 (1.25 h_2 + h_2) = 1.13 h_2$

$\therefore h_2 = 3.54 \text{ cm}$

and, $h_1 = 1.25 \times 3.54 = 4.425 \text{ cm}$

Level of bottom of cube above plane AB = $6.4 - h_1 = 6.4 - 4.425 = \mathbf{1.975 \text{ cm (Ans.)}}$

Level of the liquid surface above plane AB = $6.4 + h_2 = 6.4 + 3.54 = \mathbf{9.94 \text{ cm (Ans.)}}$

Example 4.6. A cube 50 cm side is inserted in a two-layer fluid with specific gravity 1.2 and 0.9 respectively. The upper and lower halves of the cube are composed of materials with specific gravity 0.6 and 1.4 respectively. What is the distance of the top of cube above interface? (UPSC)

Solution. Refer to Fig. 4.5.

$$\begin{aligned} \text{Weight of cube} &= [S_1 (= 0.6) \times 9.81 \times 0.5 \times 0.5 \times 0.25] + [S_2 (= 1.4) \times 9.81 \times 0.5 \times 0.5 \times 0.25] \\ &= 1.226 \text{ kN} \end{aligned}$$

Let, h = Height of top of the cube above the interface.

Then, Buoyant force = Weight of lighter and heavier liquids displaced by the block

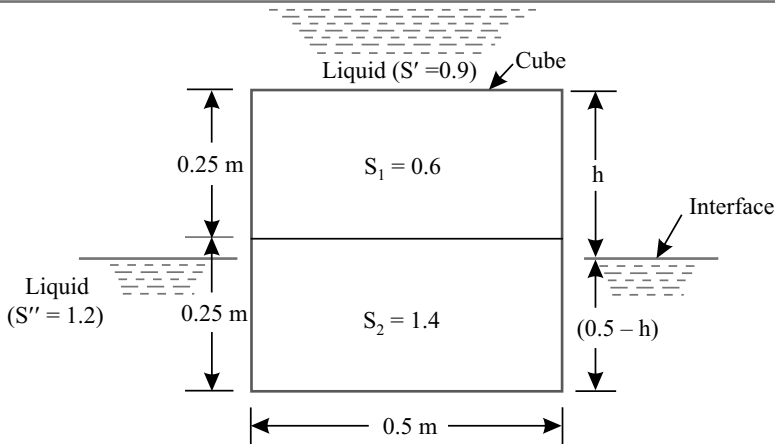


Fig. 4.5

$$= [S' (= 0.91) \times 9.81 \times 0.5 \times 0.5 \times h] + [S'' (= 1.2) \times 9.81 \times 0.5 \times 0.5 (0.5 - h)]$$

$$= 2.207 h + 1.471 - 2.943 h = -0.736 h + 1.471$$

As per principle of floatation, we have: Weight of block = Buoyant force

i.e. $1.226 = -0.736 h + 1.471$

$$\therefore h = \frac{(1.471 - 1.226)}{0.736} = 0.333 \text{ m or } 33.3 \text{ cm (Ans.)}$$

Example 4.7. A spherical object of 1.45 m diameter is completely immersed in a water reservoir and chained to the bottom. If the chain has a tension of 5.20 kN, find the weight of the object when it is taken out of the reservoir into the air:

Solution. Given: $d = 1.45 \text{ m}$; $T = 5.20 \text{ kN}$.

Weight of the object, W :

Buoyant force, $P_B = W$ (weight of the object) + T
(tension in the chain)

$$\therefore W = P_B - T$$

$$= \frac{4}{3} \pi \times \left(\frac{1.45}{2} \right)^3 \times 9.81 - 5.20$$

$$= 10.46 \text{ kN (Ans.)}$$

Example 4.8. A cylinder of mass 10 kg and area of cross-section 0.1 m^2 is tied down with string in a vessel containing two liquids as shown in figure 4.7. Calculate gauge pressure on the the cylinder bottom and the tension in the string. Density of water = 1000 kg/m^3 . Specific gravity of A = 0.8. Specific gravity of B (water) = 1.0 (GATE)

Solution. Given: Mass of cylinder, $m = 10 \text{ kg}$

$$\text{Area of cross-section} = 0.1 \text{ m}^2$$

$$\text{Density of water (liquid B)} = 1000 \text{ kg/m}^3$$

$$\text{Density of liquid A} = 0.8 \times 1000 = 800 \text{ kg/m}^3$$

Tension in string, T :

$$\text{Volume of liquid A displaced} = 0.1 \times 0.1$$

$$= 0.01 \text{ m}^3$$

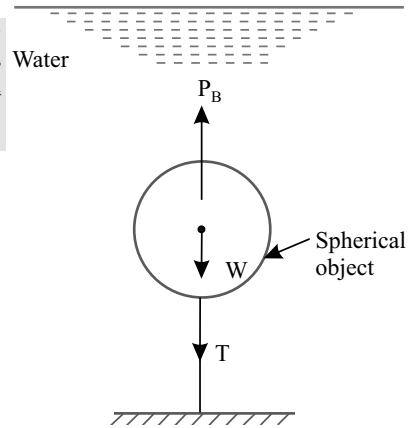


Fig. 4.6

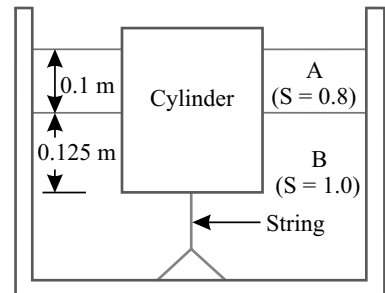


Fig. 4.7

- $\therefore$ Mass of liquid A displaced, $m_A = 0.01 \times 800 = 8 \text{ kg}$
 $\therefore$ Volume of liquid B displaced $= 0.1 \times 0.125 = 0.0125 \text{ m}^3$
 $\therefore$ Mass of liquid B displaced, $m_B = 0.0125 \times 1000 = 12.5 \text{ kg}$
 Total mass of liquid displaced $= m_A + m_B = 8 + 12.5 = 20.5 \text{ kg}$
 Upward thrust $= 20.5 \times 9.81 = 201.1 \text{ N}$
 Weight of cylinder $= mg = 10 \times 9.81 = 98.1 \text{ N}$
 Net upward thrust $= 201.1 - 98.1 = 103 \text{ N}$
 $\therefore$ Tension in the string, $T = 103 \text{ N (Ans.)}$

Pressure (gauge) on the cylinder bottom, p :

$$p = \frac{\text{Net upward thrust}}{\text{Area of cross-section}} = \frac{103}{0.1} = 1030 \text{ N/m}^2 \text{ (Ans.)}$$

4.2. TYPES OF EQUILIBRIUM OF FLOATING BODIES

The equilibrium of floating bodies is of the following types:

1. Stable equilibrium,
2. Unstable equilibrium, and
3. Neutral equilibrium.

4.3.1. Stable Equilibrium

When a body is given a small angular displacement (i.e. tilted slightly), by some external force, and then it returns back to its original position due to the internal forces (the weight and the upthrust), such an equilibrium is called **stable equilibrium**.

4.3.2. Unstable Equilibrium

If the body does not return to its original position from the slightly displaced angular position and heels farther away, when given a small angular displacement, such an equilibrium is called an **unstable equilibrium**.

4.3.3. Neutral Equilibrium

If a body, when given a small angular displacement, occupies a new position and remains at rest in this new position, it is said to possess a **neutral equilibrium**.

4.4. METACENTRE AND METACENTRIC HEIGHT

Metacentre :

Fig. 4.8 (a) shows body floating in a liquid in a state of equilibrium. When it is given a small angular displacement [see Fig. 4.8 (b)] it starts oscillating about some point (M). This point, about which the body starts oscillating, is called **metacentre**.

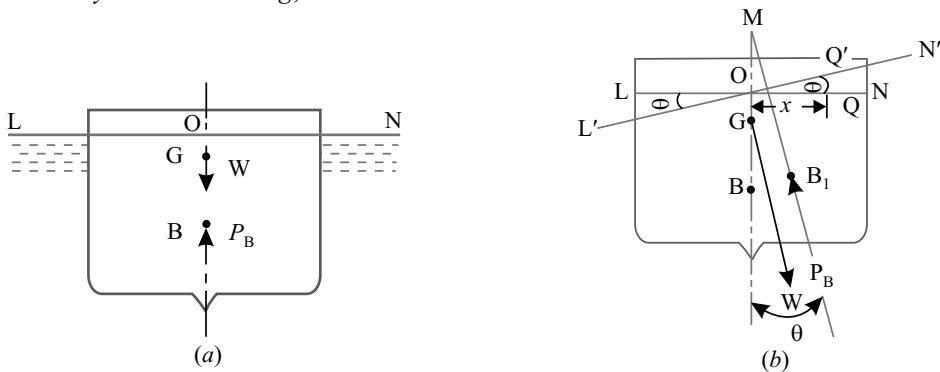


Fig. 4.8

The **metacentre** may also be defined as *a point of intersection of the axis of body passing through c.g. (G) and, original centre of buoyancy (B) and a vertical line passing through the centre of buoyancy (B_1) of the tilted position of the body.*

The position of metacentre, M remains practically constant for the small angle of tilt θ .

Metacentric height:

The distance between the centre of gravity of a floating body and the metacentre (i.e. distance GM as shown in Fig. 4.8 (b)) is called **metacentric height**.

- For stable equilibrium, the position of metacentre M remains *higher* than c.g. of the body, G .
- For unstable equilibrium, the position of metacentre M remains *lower* than G .
- For neutral equilibrium, the position of metacentre M *coincides* with G .

4.5 DETERMINATION OF METACENTRIC HEIGHT

The metacentric height may be determined by the following two methods:

1. Analytical method.
2. Experimental method.

4.5.1 Analytical Method

Refer to Fig. 4.8 (b). It shows the tilted position of the floating body, the line $L'ON'$ represents the water surface. The portion $N'ON$ of the body is submerged and the portion $L'OL$ is lifted because of tilting. As a result of this, the centre of buoyancy changes its position from B to B_1 . The intersection of axis of the body and the vertical line through B_1 locates the metacentre, M of the body.

To find the metacentric height GM consider an elementary cylindrical prism QQ' of portion $N'ON$ at a distance ' x ' from O . Let the area of this elementary prism be δA . The height of this elementary prism is given by $x.\theta$. The volume of this elementary prism is given by:

$$\delta V = x.\theta.\delta A. \quad \dots(i)$$

The upward force or buoyancy force acting at this prism (δP_B) is given by:

$$\delta P_B = w.\delta V = w.x.\theta.\delta A \quad \dots(ii)$$

(where, w = unit weight of liquid)

The moment of this buoyancy force about O

$$x.\delta P = w.\theta.x^2.\delta A \quad \dots(iii)$$

For the total portion $N'ON$, this moment is given by:

$$\int x.dP_B = \int w.\theta.x^2.dA = w\theta \int x^2 dA \quad \dots(iv)$$

$$\text{or,} \quad \int x.dP_B = w\theta.I$$

(where, I = moment of inertia of the sectional area at the water line about the axis through O)

$$\int x.dP_B \text{ gives the change in moment due to buoyancy.}$$

$$\text{Now,} \quad \int x.dP_B (= w\theta.I) = P_B \times BB_1$$

(where, P_B = total force of buoyancy)

$$\text{But,} \quad BB_1 = BM \times \theta \text{ and } P_B = W = w \times V$$

$$\therefore w.\theta.I = w.V.BM.\theta \quad \text{or} \quad BM = \frac{I}{V} \quad \dots(4.1)$$

Now metacentric height, $GM = BM \pm BG$

+ ve sign : when G is lower than B

– ve sign : when G is higher than B

4.5.2. Experimental Method

Refer to Fig. 4.9.

In this method, a known weight W_1 is shifted by a distance z across the axis of tilt. The change in moment due to this shift is $W_1 z$. Let the angle of tilt be θ . This angle of tilt may be measured experimentally by using a plumb bob. The change in moment due to this tilt is equal to $W.GG_1$ or $W.GM \tan \theta$.

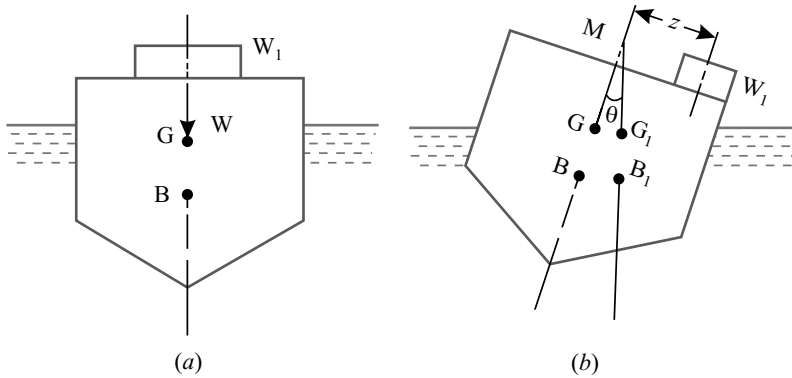


Fig. 4.9. Experimental method for determination of metacentric height.

$$\therefore W_1 z = W.Gm.\tan\theta \quad \text{or} \quad GM = \frac{W_1 \cdot z}{W \cdot \tan\theta} \quad \dots(4.2)$$

If, l = Length of plumb bob, and

d = Displacement of the plumb bob,

Then, $\tan \theta = \frac{d}{l}$

and, metacentric height is given by:

$$GM = \frac{W_1 \cdot z \cdot l}{W \cdot d} \quad \dots(4.3)$$

Example 4.9. A wooden block of specific gravity 0.75 floats in water. If the size of the block is $1 \text{ m} \times 0.5 \text{ m} \times 0.4 \text{ m}$, find its metacentric height.

Solution. Size (or dimensions) of the block = $1 \text{ m} \times 0.5 \text{ m} \times 0.4 \text{ m}$

Specific gravity of wood = 0.75

Specific weight $w = 0.75 \times 9.81 = 7.36 \text{ kN/m}^3$

Weight of wooden block = specific weight $\times$ volume
 $= 7.36 \times 1 \times 0.5 \times 0.4 = 1.472 \text{ kN}$

Let depth of immersion = h metres.

Weight of water displaced

= Specific weight of water $\times$ volume of the wood submerged in water

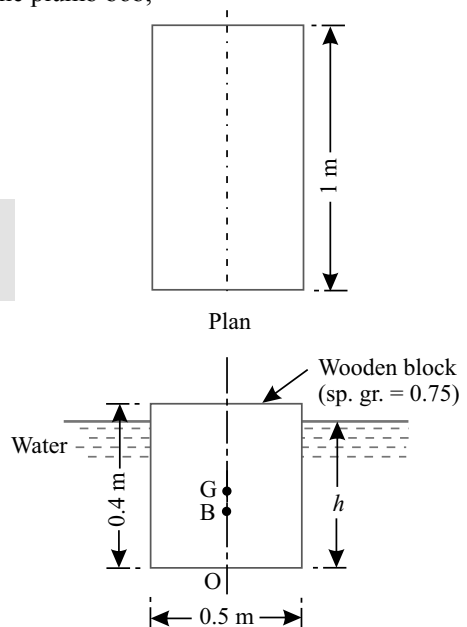


Fig. 4.10

$$= 9.81 \times 1 \times 0.5 \times h = 4.9 h \text{ kN}$$

Now, for equilibrium:

Weight of wooden block = Weight of water displaced *i.e.*, $1.472 = 4.9 h$

or,
$$h = \frac{1.472}{4.9} = 0.3 \text{ m}$$

$\therefore$ Distance of centre of buoyancy from bottom *i.e.*,

$$OB = \frac{h}{2} = \frac{0.3}{2} = 0.15 \text{ m}$$

and,
$$OG = \frac{0.4}{2} = 0.2 \text{ m}$$

$\therefore$
$$BG = OG - OB = 0.2 - 0.15 = 0.05 \text{ m}$$

Also,
$$BM = \frac{I}{V}$$

Where, I = Moment of inertia of a rectangular section

$$= \frac{1 \times 0.5^3}{12} = 0.014 \text{ m}^4$$

and, V = Volume of water displaced (or volume of wood in water)

$$= 1 \times 0.5 \times h = 1 \times 0.5 \times 0.3 = 0.15 \text{ m}^3$$

$$BM = \frac{I}{V} = \frac{0.0104}{0.15} = 0.069 \text{ m}$$

We know that the metacentric height,

$$GM = BM - BG \quad (\because G \text{ is above } B)$$

$$= 0.069 - 0.05 = \mathbf{0.019 \text{ m (Ans.)}}$$

Example 4.10. A solid cylinder 2 m in diameter and 2m high is floating in water with its axis vertical. If the specific gravity of the material of cylinder is 0.65 find its metacentric height. State also whether the equilibrium is stable or unstable.

Solution. Given: Diameter of cylinder, $d = 2 \text{ m}$; Height of cylinder, $h = 2 \text{ m}$; Specific gravity = 0.65

Depth of cylinder in water = Sp. gravity $\times h$

$$= 0.65 \times 2.0 = 1.3 \text{ m}$$

Distance of centre of buoyancy (B) from O,

$$OB = \frac{1.3}{2} = 0.65 \text{ m}$$

Distance of centre of gravity (G) from O,

$$OG = \frac{2.0}{2} = 1.0 \text{ m}$$

$$BG = OG - OB = 1.0 - 0.65 = 0.35 \text{ m}$$

Also,
$$BM = \frac{I}{V}$$

Where, I = Moment of inertia of the plan of the body about Y-Y

$$= \frac{\pi}{64} d^4 = \frac{\pi}{64} \times 2^4 = 0.785 \text{ m}^4$$

and, V = Volume of cylinder of water

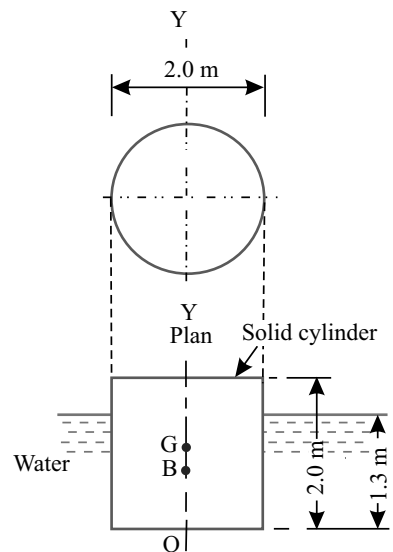


Fig. 4.11

$$= \frac{\pi}{4} d^2 \times \text{depth of cylinder in water}$$

$$= \frac{\pi}{4} \times 2^2 \times 1.3 = 4.084 \text{ m}^3$$

$$\therefore BM = \frac{I}{V} = \frac{0.785}{4.084} = 0.192 \text{ m}$$

We know that the metacentric height,

$$\begin{aligned} GM &= BM - BG = 0.192 - 0.35 \\ &= -0.158 \text{ m (Ans.)} \end{aligned}$$

–ve sign means that the metacentric (M) is *below* the centre of gravity (G). Thus the cylinder is in **unstable equilibrium**. (Ans.)

Example 4.11. A weight of 100 kN is moved through a distance of 8 metres across the deck of a pontoon of 7500 kN displacement floating in water. This makes a pendulum 2.5 metres long to move through 120 mm horizontally. Calculate the metacentric height of the pontoon.

Solution. Weight of the movable load, $W_1 = 100 \text{ kN}$

Distance through which load is moved, $z = 8 \text{ m}$

Weight of pontoon, $W = 7500 \text{ kN}$

Length of the plumb bob, $l = 2.5 \text{ m}$

Displacement of the plumb bob, $d = 120 \text{ mm} = 0.12 \text{ m}$

Let,

$GM =$ metacentric height of the pontoon.

Using the relation:

$$GM = \frac{W_1 \cdot z \cdot l}{W \cdot d} = \frac{100 \times 8 \times 2.5}{7500 \times 0.12} = 2.22 \text{ m (Ans.)}$$

Example 4.12. A body has the cylindrical upper portion of 4m diameter and 2.4 m deep. The lower portion, which is curved, displaces a volume of 800 litres of water and its centre of buoyancy is situated 2.6 m below the top of the cylinder. The centre of gravity of the whole body is 1.6 m below the top of the cylinder and the total displacement of water is 52 kN. Find the metacentric height of the body.

Solution. Given: Diameter of body, $d = 4 \text{ m}$

Depth of cylindrical portion = 2.4 m

Volume of curved portion = 800 litres = 0.8 m^3

Distance between centre of buoyancy of curved portion and top of body,

$$OB_1 = 2.6 \text{ m}$$

Distance between centre of gravity of the whole body and top of the cylinder,

$$OG = 1.6 \text{ m}$$

$$\text{Total volume of water displaced, } V = \frac{52}{9.81} = 5.3 \text{ m}^3$$

$$\begin{aligned} \text{Volume of water displaced by the cylindrical portion} \\ = 5.3 - 0.8 = 4.5 \text{ m}^3 \end{aligned}$$

If y is the distance between the water surface and the top of the body, then:

$$4.5 = \frac{\pi}{4} \times 4^2 \times (2.4 - y)$$

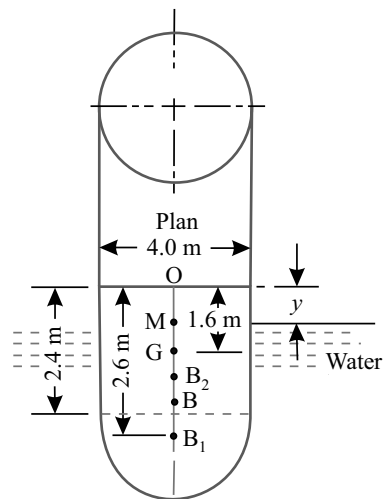


Fig. 4.12

$$y = 2.4 - \frac{4.5 \times 4}{\pi \times 4^2} = 2.04 \text{ m}$$

Distance of the centre of buoyancy of the cylindrical portion from the top of the body,

$$OB_2 = y + \left(\frac{2.4 - y}{2} \right) = 2.04 + \frac{2.4 - 2.04}{2} = 2.22 \text{ m}$$

If B be the centre of buoyancy of the whole body, then:

$$OB = \frac{(0.8 \times 2.6) + (4.5 \times 2.22)}{0.8 + 4.5} = 2.227 \text{ m}$$

Now,

$$BG = OB - OG = 2.227 - 1.6 = 0.677 \text{ m}$$

Now,

$$BM = \frac{I}{V}$$

where, I = Moment of inertia of the cylindrical portion (top portion) about its c.g.

$$= \frac{\pi}{64} \times 4^4 \text{ m}^4 = 12.566 \text{ m}^4$$

and,

$$V = 5.3 \text{ m}^3 \text{ (already calculated earlier)}$$

$$BM = \frac{12.566}{5.3} = 2.37$$

Metacentric height,

$$GM = BM - BG = 2.37 - 0.677 = \mathbf{1.693 \text{ m (Ans.)}}$$

Example 4.13. A solid cube of sides 0.5 m each is made of a material of relative density 0.5. The cube floats in a liquid of relative density 0.95 with two of its faces horizontal. Examine its stability.

(MDU, Haryana)

Solution. Given: Side of the cube = 0.5 m; Specific gravity of cube material = 0.5, Relative density of liquid = 0.95.

Depth of cube in liquid,

$$h = \frac{0.5 \times 0.5}{0.95} = 0.263 \text{ m}$$

Distance of centre of buoyancy (B) from O,

$$OB = \frac{0.263}{2} = 0.1315 \text{ m}$$

Distance of centre of gravity (G) from O,

$$OG = \frac{0.5}{2} = 0.25 \text{ m}$$

$$BG = OG - OB = 0.25 - 0.1315 \\ = 0.1185 \text{ m}$$

B lies below G.

$$BM = \frac{I}{V}$$

where, I = Moment of inertia of the plane of the body about YY

$$= \frac{1}{12} (0.5) (0.5)^3 = 0.005208 \text{ m}^4$$

V = Volume of liquid displaced

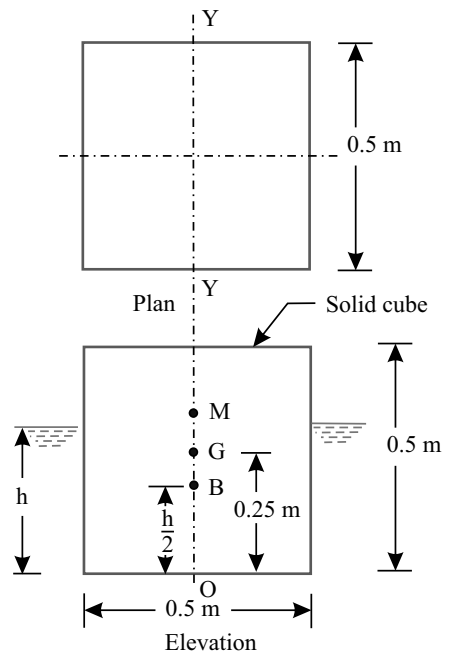


Fig. 4.13

$$= 0.5 \times 0.5 \times 0.263 = 0.06575 \text{ m}^3$$

$$\therefore BM = \frac{I}{V} = \frac{0.005208}{0.06575} = 0.0792 \text{ m}$$

$$\text{Metacentric height, } GM = BM - BG = 0.0792 - 0.1185 = -0.0393 \text{ m}$$

–ve sign means that the metacentre (M) is *below* the centre of gravity (G). Thus the cube will be **unstable**.

Example 4.14. A hollow cylinder closed at both ends has an outside diameter of 1.25 m, length 3.5 m and specific weight 75 kN/m^3 . If cylinder is to float just in stable equilibrium in sea water (specific weight 10 kN/m^3), find its minimum permissible thickness.

Solution. Given: $d = 1.25 \text{ m}$, $l = 3.5 \text{ m}$, $w_c = 75 \text{ kN/m}^3$; $w_w = 10 \text{ kN/m}^3$

Minimum permissible thickness, t :

Let, h = Depth of immersion, m .

Weight of sea water displaced

$$= \frac{\pi}{4} d^2 h \times w_w$$

$$= \frac{\pi}{4} \times 1.25^2 \times h \times 10 = 12.27 h \text{ kN}$$

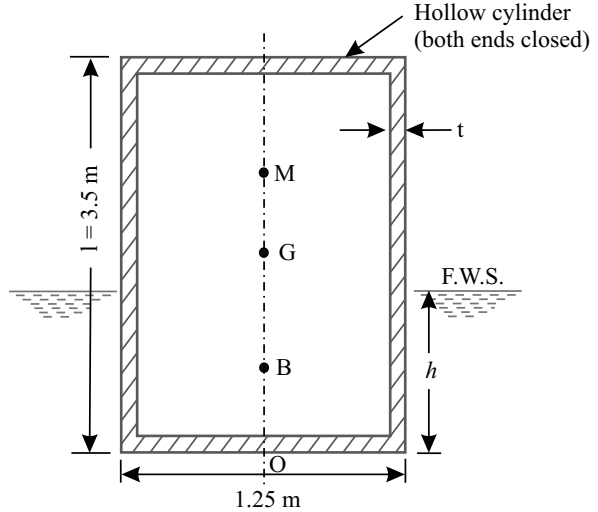


Fig. 4.14

Weight of cylinder = Volume of cylinder $\times$ Specific weight.

= (Volume of two end sections + volume of circular portion) $\times$ specific weight

$$= \left[2 \times \frac{\pi}{4} d^2 \times t + \frac{\pi}{4} \{ d^2 - (d - 2t)^2 \} l \right] \times w_c$$

$$= \left[2 \times \frac{\pi}{4} d^2 t + \pi d t l \right] \times w_c \quad (\text{assuming } t \ll l)$$

(Ignoring term involving t^2)

$$\left[2 \times \frac{\pi}{4} \times 1.25^2 \times t + \pi \times 1.25 \times t \times 3.5 \right] \times 75$$

$$= 1215 t \text{ kN}$$

Under equilibrium conditions:

Weight of cylinder = Weight of sea water displaced.

$$\text{i.e., } 12.27 h = 1215 t \quad \text{or} \quad h = 99 t$$

Volume of cylinder under water or volume of sea water displaced,

$$V = \frac{1215 t}{10} = 121.5 t$$

If M is the metacentre, then:

$$BM = \frac{I}{V} = \frac{\frac{\pi}{64} \times 1.25^4}{121.5 t} = \frac{0.00099}{t}$$

$$OB = \frac{h}{2} = \frac{99t}{2} = 49.5t$$

$$OG = \frac{3.5}{2} = 1.75 \text{ m}$$

$$BG = OG - OB = 1.75 - 49.5 t$$

For the cylinder to float just in stable equilibrium:

$$BG = BM \quad \text{i.e.,} \quad 1.75 - 49.5 t = \frac{0.00099}{t}$$

$$\text{or, } 49.5 t^2 - 1.75 t + 0.00099 = 0$$

$$\text{or, } t = \frac{1.75 \pm \sqrt{(1.75)^2 - 4 \times 49.5 \times 0.00099}}{2 \times 49.5} = \frac{1.75 \pm 1.69}{99}$$

$$= 0.0347 \text{ m} \quad \text{or} \quad 6.06 \times 10^{-4} \text{ m}$$

Hence, minimum permissible thickness = $6.06 \times 10^{-4} \text{ m}$ or **0.606 mm (Ans.)**

Example 4.15. A solid of 200 mm diameter and 800 mm length has its base 20 mm thick and of specific gravity 6. The remaining part of the cylinder is of specific gravity 0.6. State, if it can float vertically in water.

Solution. Given: Dia. of cylinder = 200 mm = 0.2 m

$$\text{Area of cylinder, } A = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$$

$$\text{Length of cylinder} = 800 \text{ mm} = 0.8 \text{ m}$$

$$\text{Thickness of base} = 20 \text{ mm} = 0.02 \text{ m}$$

$$\text{Sp. gr. of base} = 6, \text{ Sp. gr. of remaining portion} = 0.6$$

Distance between combined centre of gravity (G) and the bottom of the cylinder (O),

$$OG = \frac{\left[A \times 0.78 \times 0.6 \left(\frac{0.78}{2} + 0.02 \right) \right] + \left[A \times 0.02 \times 6 \times \frac{0.02}{2} \right]}{(A \times 0.78 \times 0.6) + (A \times 0.02 \times 6)}$$

(where, A = area of cylinder)

$$= \frac{0.1919 + 0.0012}{0.468 + 0.12} = 0.3284 \text{ m (or 328.4 mm)}$$

Combined sp. gr. of the cylinder

$$= \frac{(0.78 \times 0.6) + (0.02 \times 6)}{0.78 + 0.02} = 0.735$$

Depth of immersion of the cylinder

$$= 0.8 \times 0.735 = 0.588 \text{ m}$$

Distance of centre of buoyancy from the bottom of the cylinder,

$$OB = \frac{0.588}{2} = 0.294 \text{ (or 29.4 mm)}$$

$$BG = OG - OB = 0.3284 - 0.294$$

$$= 0.0344 \text{ m (or 34.4 mm)}$$

$$\text{Now, } BM = \frac{I}{V}$$

where, I = Moment of inertia of circular section

$$= \frac{\pi}{64} \times 0.2^4 = 2.5 \times 10^{-5} \pi \text{ m}^4$$

and, V = volume of water displaced

$$= \frac{\pi}{4} \times 0.2^2 \times 0.588 = 0.00588 \pi$$

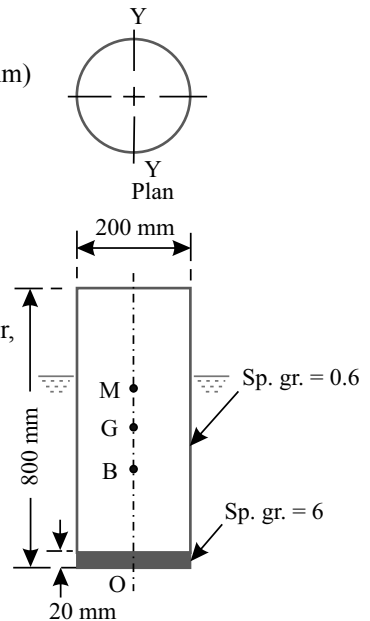


Fig. 4.15

$$\begin{aligned}\therefore BM &= \frac{2.5 \times 10^{-5} \pi}{0.00588 \pi} \\ &= 0.00425 \text{ m or } 4.25 \text{ mm}\end{aligned}$$

Now metacentric height,

$$GM = BM - BG = 4.25 - 34.4 \text{ mm} = -30.15 \text{ mm}$$

Negative sign means that the metacentre (M) is *below* the centre of gravity (G). Thus the cylinder is in *unstable equilibrium* and it **cannot float** vertically in water. (**Ans.**)

Example 4.16. An 80 mm diameter composite solid cylinder consists of an 80 mm diameter, 20 mm thick metallic plate having specific gravity 4.0 attached at the lower end of an 80 mm diameter wooden cylinder of specific gravity 0.8. Find the limits of the length of the wooden portion so that the composite cylinder can float in stable equilibrium in water (specific gravity 1.0) with its axis vertical. (MGU Kerala)

Solution. Refer to Fig. 4.16. Given: $d = 80 \text{ mm}$; $a = 20 \text{ mm}$; $S_1 = 4$; $S_2 = 0.8$

Limits of the length of the wooden portion:

The cylinder will float vertically in water if its metacentric height GM is +ve. To find the metacentric height, the locations of centre of gravity G and centre of buoyancy B of the combined cylinder is to be found.

The distance of the centre of gravity of the solid cylinder from O is given by:

$$\begin{aligned}OG &= \frac{\left[\frac{\pi}{4} d^2 \times a \times S_1 \right] \times \frac{a}{2} + \left[\frac{\pi}{4} d^2 \times b \times S_2 \right] \times \left[a + \frac{b}{2} \right]}{\left[\frac{\pi}{4} d^2 \times a \times S_1 + \frac{\pi}{4} d^2 \times b \times S_2 \right]} \\ &= \frac{\left(a \times S_1 \times \frac{a}{2} \right) + b \times S_2 \left(a + \frac{b}{2} \right)}{a \times S_1 + b \times S_2} \\ &= \frac{\left(20 \times 4 \times \frac{20}{2} \right) + b \times 0.8 \left(20 + \frac{b}{2} \right)}{20 \times 4 + b \times 0.8} \\ &= \left[\frac{800 + 0.8b \left(20 + \frac{b}{2} \right)}{80 + 0.8b} \right] = \left[\frac{1000 + 20b + \frac{b^2}{2}}{100 + b} \right] \dots (i)\end{aligned}$$

(Dividing numerator and denominator by 0.8 and simplifying)

Let, height of immersion of cylinder = $(h + a)$

Also, Weight of cylinder = Weight of water displaced

$$\begin{aligned}\text{or, } \frac{\pi}{4} d^2 \times a \times S_1 + \frac{\pi}{4} d^2 \times b \times S_2 &= \frac{\pi}{4} d^2 (h + a) \times S_{\text{water}} \\ a \times S_1 + b \times S_2 &= (h + a) \quad (\because S_{\text{water}} = 1) \\ \text{or, } 20 \times 4 + b \times 0.8 &= (h + a) \\ \text{i.e. } h + a &= 80 + 0.8b \dots (ii)\end{aligned}$$

$$\text{Now, } OB = \frac{h + a}{2} = \frac{80 + 0.8b}{2} = 40 + 0.4b \dots (iii)$$

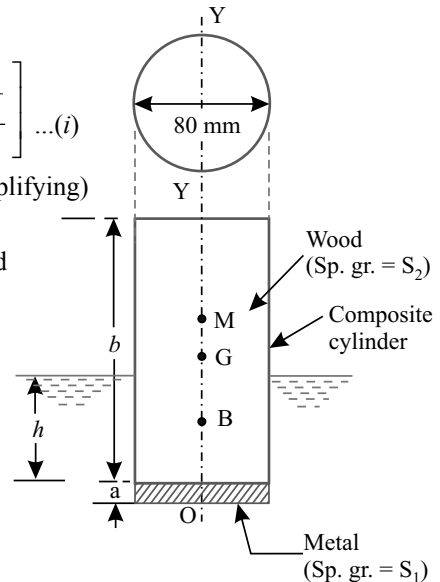


Fig. 4.16

$$\begin{aligned}
 BG &= OG - OB = \left[\frac{1000 + 20b + \frac{b^2}{2}}{100 + b} \right] \\
 &\quad - (40 + 0.4b) \quad \text{[From (i) and (iii)]} \\
 &= \frac{1}{(100 + b)} \left[\left(1000 + 20b + \frac{b^2}{2} \right) - (4000 + 80b + 0.4b^2) \right] \\
 &= \frac{1}{(100 + b)} [0.1b^2 - 60b - 3000] \quad \dots(iv)
 \end{aligned}$$

$$I = \text{Second moment of area of the section about } Y-Y = \frac{\pi d^4}{64}$$

$$V = \text{Volume of water displaced} = \frac{\pi}{4} d^2 (h + a)$$

$$\begin{aligned}
 BM &= \frac{I}{V} = \frac{\pi d^4 / 64}{\frac{\pi}{4} d^2 (h + a)} = \frac{\pi d^4}{64} \times \frac{4}{\pi d^2 (h + a)} = \frac{d^2}{16(h + a)} \\
 &= \frac{(80)^2}{16(80 + 0.8b)} \quad \text{using (ii) for } (h + a) \\
 &= \frac{400}{80 + 0.8b} = \frac{500}{100 + b} \quad \dots(v)
 \end{aligned}$$

$$\begin{aligned}
 GM &= BM - BG = \frac{500}{(100 + b)} - \frac{1}{(100 + b)} \\
 &\quad [0.1b^2 - 60b - 3000] \quad \text{[using (iv)] and (v)}
 \end{aligned}$$

$$\text{or,} \quad GM = \frac{3500 + 60b - 0.1b^2}{(100 + b)} \quad \dots(vi)$$

It should be +ve and in the limit = 0

$$\text{i.e.} \quad 0.1b^2 - 60b - 3500 = 0$$

$$\text{or,} \quad b^2 - 600b - 35000 = 0$$

$$\text{or,} \quad b = \frac{600 + \sqrt{(600)^2 + 4 \times 35000}}{2} \quad \text{(taking +ve value of } b)$$

$$\text{or,} \quad b = 653.55 \text{ mm. This is the upper limit for } b. \text{ (Ans.)}$$

The **lower limit** for b will be $b = h$, and from eqn. (ii), we have:

$$h + a = 80 + 0.8b$$

$$b + 20 = 80 + 0.8b$$

$$b = \mathbf{300 \text{ mm (Ans.)}}$$

It may be checked from eqn. (ii) that it gives a +ve value of GM.

Example 4.17. A hollow wooden cylinder of specific gravity 0.6 has an outer diameter of 600 mm and an inner diameter of 300 mm. It is required to float in oil of specific gravity 0.9. Calculate:

- The maximum length (height) of the cylinder so that it shall be stable when floating with its axis vertical;
- The depth to which it will sink.

Solution. Outer diameter of cylinder, $D = 600 \text{ mm} = 0.6 \text{ m}$

Inner diameter of cylinder, $d = 300 \text{ mm} = 0.3 \text{ m}$

Specific weight $= 0.6 \times 9.81 = 5.886 \text{ kN/m}^3$

$\therefore$ Weight of cylinder $=$ volume of cylinder $\times$ specific weight

$$= \frac{\pi}{4} (D^2 - d^2) \times l \times 5.886 = \frac{\pi}{4} (0.6^2 - 0.3^2) \times l \times 5.886$$

$$= 1.248 l \text{ kN}$$

(where, l = length/height of the cylinder)

This also represents the weight of oil displaced.

$$\therefore \text{Volume of oil displaced, } V = \frac{1.248l}{0.9 \times 9.81} = 0.1413 l$$

i.e. Volume of cylinder immersed in oil, $V = 0.1413 l$

Depth of immersion, $h = \frac{\text{volume of cylinder under oil}}{\text{cross-section area of cylinder}}$

$$= \frac{0.1413l}{\frac{\pi}{4}(0.6^2 - 0.3^2)} = 0.666 l$$

Height of centre of buoyancy (B) from O ,

$$\text{i.e. } OB = \frac{h}{2} = \frac{0.666l}{2} = 0.333 l$$

If M is the metacentre, then

$$BM = \frac{I}{V} = \frac{\frac{\pi}{64}(0.6^4 - 0.3^4)}{0.1413 l} = \frac{0.0422}{l}$$

$$OM = OB + BM = 0.333 l + \frac{0.0422}{l}$$

Distance of centre of gravity (G) from the point O ,

$$OG = \frac{l}{2} = 0.5 l$$

For stable equilibrium, M should be at a level greater than G , i.e. $OM > OG$

$$\text{or, } \left(0.333 l + \frac{0.0422}{l} \right) > 0.5 l$$

$$\text{or, } \frac{0.0422}{l} > 0.167 l \quad \text{or} \quad 0.0422 > 0.167 l^2$$

$$\text{or, } 0.167 l^2 < 0.0422 \quad \text{or} \quad l < \left(\frac{0.0422}{0.167} \right)^{1/2} < 0.503 \text{ m}$$

$$\therefore l_{\max} = 0.503 \text{ m (Ans.)}$$

$$\text{and, } h = 0.666 l = 0.666 \times 0.503 = 0.335 \text{ m (Ans.)}$$

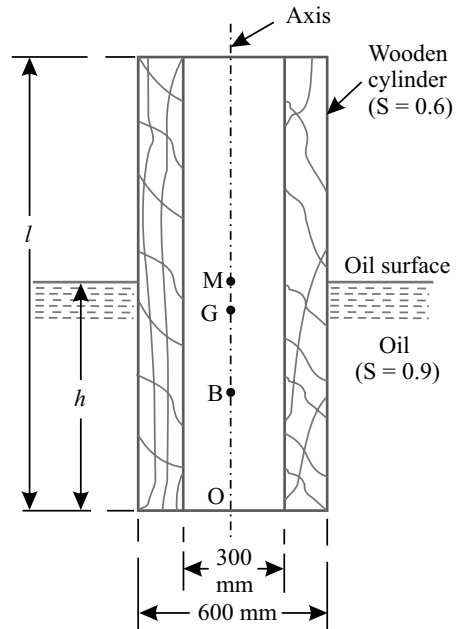


Fig. 4.17

Example 4.18. A rectangular pontoon 12 m long, 9 m wide and 3 m deep weighs 1380 kN and floats in sea water. The pontoon carries on its upper deck a boiler 6 m diameter and weighing 864 kN. The centre of gravity of each unit coincides with geometrical centre of the arrangement and lies in the same vertical line.

(i) What is the metacentric height?

(ii) Is the arrangement stable?

Take specific weight of sea water = 10 kN/m^3

Solution. Total weight of the arrangement, $W = 1380 + 864 = 2244 \text{ kN}$

This also represent the weight of water displaced.

Volume of sea water displaced,

$$\therefore V = \frac{\text{Weight of water displaced}}{\text{Specific weight of water}} = \frac{2244}{10} = 224.4 \text{ m}^3$$

i.e. Volume of the arrangement under water, $V = 224.4 \text{ m}^3$

Depth of immersion,

$$\begin{aligned} h &= \frac{\text{Volume of the arrangement under water}}{\text{Cross-sectional area of the pontoon}} \\ &= \frac{224.4}{9 \times 12} = 2.077 \text{ m} \end{aligned}$$

Distance of centre of buoyancy (B) from the base point O , $OB = \frac{2.077}{2} = 1.0385 \text{ m}$

Let, M be the metacentre.

$$\text{Then, } BM = \frac{I}{V} = \frac{\frac{1}{12} \times 12 \times 9^3}{224.4} = 3.248 \text{ m}$$

$$OM = OB + BM = 1.0385 + 3.248 = 4.286 \text{ m}$$

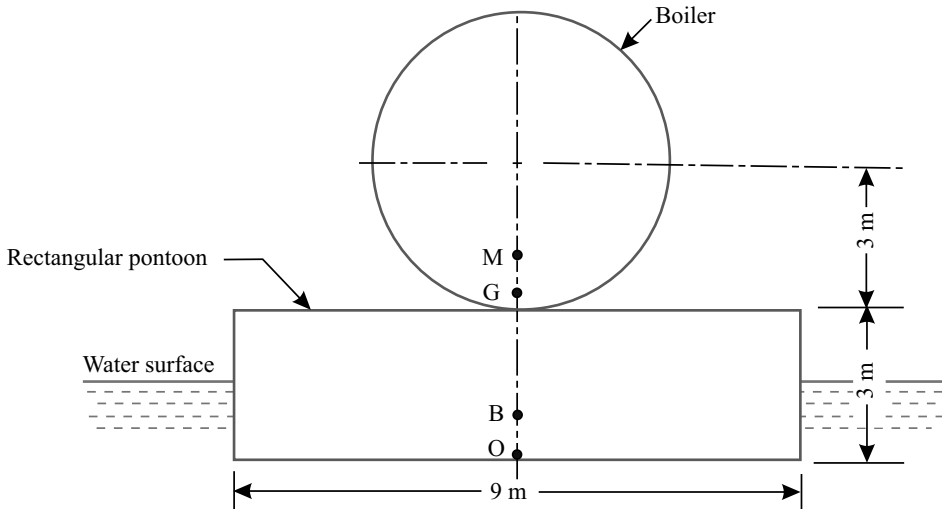


Fig. 4.18

To find the position of combined centre of gravity above the base point O , taking moments about O , we get:

$$1380 \times 1.5 + 864 \times 6 = 2244 \times OG$$

$$\therefore OG = \frac{1380 \times 1.5 + 864 \times 6}{2244} = 3.232 \text{ m}$$

(i) Metacentric height, GM : $GM = OM - OG = 4.286 - 3.232 = 1.054 \text{ m}$ (Ans.)

(ii) Stability of the arrangement:

Since $OM > OG$, M is at a higher level than G .

Hence the arrangement is **stable (Ans.)**

Example 4.19. A buoy having a diameter of 2.4 m and length 1.95 m is floating with its axis vertical in sea water (specific weight = 10 kN/m^3). Its weight is 16.5 kN and a load of 1.65 kN is placed centrally at its top. If the buoy is to remain in stable equilibrium, find the maximum permissible height of the centre of gravity of the load above the top of the buoy.

Solution. Given: Diameter of the buoy, $d = 2.4 \text{ m}$; Length of the buoy, $l = 1.95 \text{ m}$

Weight of the buoy, $W_{\text{buoy}} = 16.5 \text{ kN}$

Weight placed at the top of the buoy, $W = 1.65 \text{ kN}$

Specific weight of sea water = 10 kN/m^3

Total weight of the arrangement $W_t = W_{\text{buoy}} + W$
 $= 16.5 + 1.65 = 18.15 \text{ kN}$

This is also the weight of water displaced by the arrangement.

Volume of water displaced,

$$V = \frac{W_t}{\text{Sp. weight of water}}$$

$$= \frac{18.15}{10} = 1.815 \text{ m}^3$$

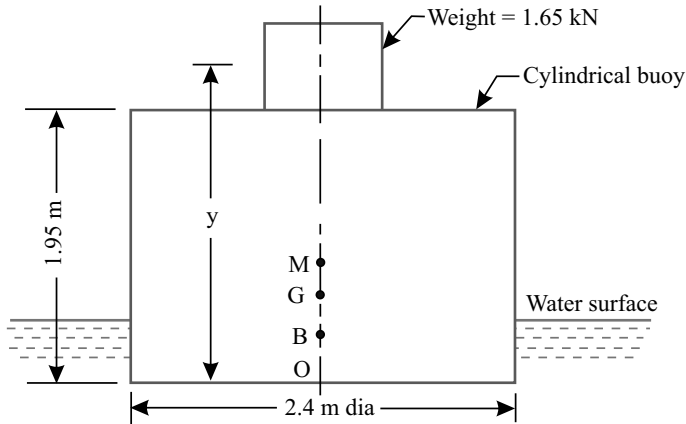


Fig. 4.19

Depth of immersion,

$$h = \frac{\text{Volume of water displaced}}{\text{Cross-sectional area of the buoy}}$$

$$= \frac{1.815}{(\pi/4) \times 2.4^2} = 0.4 \text{ m}$$

Height of centre of buoyancy (B) above base point O ,

$$OB = \frac{h}{2} = \frac{0.4}{2} = 0.2 \text{ m}$$

If M is the metacentre, then:

$$BM = \frac{I}{V} = \frac{(\pi/64) \times 2.4^4}{1.815}$$

$$OM = OB + BM = 0.2 + 0.897 = 1.097 \text{ m}$$

Let, y = Height of centre of gravity of the load above the base O .

To find the position of combined centre of gravity above the base point O , taking moments about O , we get:

$$16.5 \times \frac{1.95}{2} + 1.65 \times y = 18.15 \times OG$$

$$\text{or, } 16.087 + 1.65y = 18.15 \times OG$$

$$\text{or, } OG = \frac{16.087 + 1.65y}{18.15} = 0.886 + 0.091y$$

The equilibrium will be stable when $OM > OG$ i.e. $1.097 > (0.886 + 0.091y)$

$$\text{or, } 0.211 > 0.091y \quad \text{or} \quad 0.091y < 0.211 \quad \text{or} \quad y < 2.318 \text{ m}$$

But the height of buoy = 1.95 m

$\therefore$ The height of centre of gravity of the load above the buoy should *not be greater than* (2.318 – 1.95) or **0.368 m (Ans.)**

Example 4.20. A wooden cylinder (sp. gravity = 0.54) of diameter d and length l is required to float in oil (sp. gravity = 0.81). Find the l/d ratio for the cylinder to float with its longitudinal axis vertical in oil.

Solution. Given: Diameter of the cylinder = d ; Length of the cylinder = l ;

Sp. gravity, $S_1 = 0.54$; sp. gravity of oil, $S_2 = 0.81$.

$\frac{1}{d}$ ratio:

Let, h = Depth of cylinder immersed in oil.

Now, Weight of cylinder = Weight of oil displaced ... (principle of buoyancy)

$$\text{or, } \frac{\pi}{4} d^2 l \times S_1 = \frac{\pi}{4} d^2 h \times S_2$$

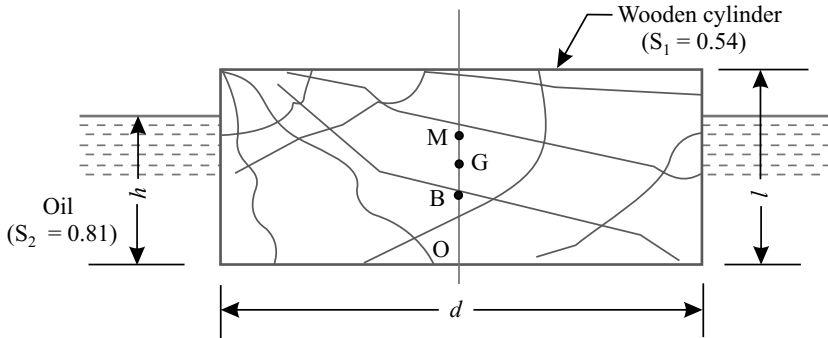


Fig. 4.20

$$l \times 0.54 = h \times 0.81 \quad \text{i.e., } h = \frac{0.54 l}{0.81} = \frac{2}{3} l$$

$\therefore$ The distance of centre of buoyancy B from O , $OB = \frac{h}{2} = \frac{1}{3} l$

The distance of centre of gravity G from base point O , $OG = l/2$

$\therefore$ If M is the metacentre, then:

$$BM = \frac{I}{V} = \frac{\frac{\pi}{64} d^4}{\frac{\pi}{4} d^2 h} = \frac{d^2}{16h} = \frac{d^2}{16 \times \frac{2}{3} l} = \frac{3d^2}{32l}$$

$$\therefore OM = OB + BM = \frac{1}{3}l + \frac{3d^2}{32l}$$

$$\text{For stable equilibrium: } OM > OG \text{ or } \left(\frac{1}{3}l + \frac{3d^2}{32l} \right) > \frac{l}{2}$$

$$\text{or, } \frac{3d^2}{32l} > \left(\frac{l}{2} - \frac{l}{3} \right) \text{ or } \frac{3d^2}{32l} > \frac{l}{6}$$

$$\text{or, } \frac{3d^2}{32l^2} > \frac{1}{6} \text{ or } \frac{18}{32} > \frac{l^2}{d^2}$$

$$\text{or, } \frac{l^2}{d^2} < \frac{18}{32} \text{ or } \frac{l}{d} < \left(\frac{18}{32} \right)^{1/2} \text{ or } (9/16)^{1/2} \text{ or } 3/4$$

$$\therefore \frac{l}{d} < 3/4 \text{ (Ans.)}$$

Example 4.21. A log of wood 0.9 m in diameter and 7.5 m long is floating in river water. If the specific gravity of log is 0.7, what is the depth of the wooden log in water?

Solution. Given: Diameter of the wooden log, $d = 0.9$ m;

Length of the log, $l = 7.5$ m

Specific gravity, $S = 0.7$

$$\text{Weight of the log} = (0.7 \times 9.81) \times \frac{\pi}{4} d^2 l$$

$$= 0.7 \times 9.81 \times \frac{\pi}{4} \times 0.9^2 \times 7.5 = 32.76 \text{ kN}$$

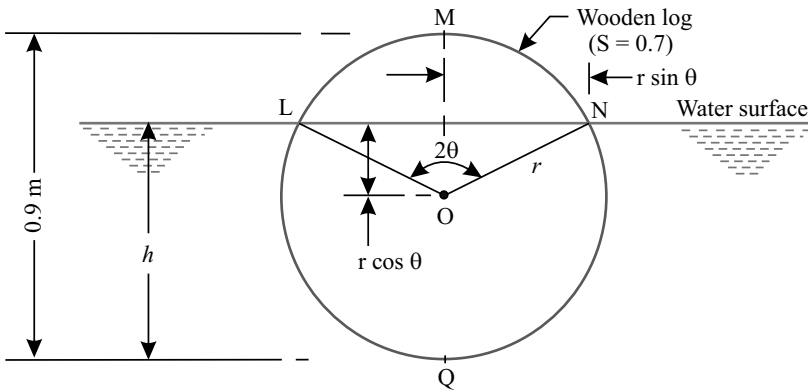


Fig. 4.21

This also represents the weight of water displaced.

$$\text{Volume of water displaced} = \frac{32.76}{9.81} = 3.34 \text{ m}^3$$

Let, h = Depth of immersion.

$$\therefore \text{Volume of log inside water} = \text{Volume of water displaced} = 3.34 \text{ m}^3$$

$$3.34 = \text{Area } LQNL \times 7.5$$

$$\text{Area, } LQNL = \frac{3.34}{7.5} = 0.4453 \text{ m}^2$$

$$\text{Also, Area, } LQNL = \text{Area } LQNL + \text{area } LON$$

$$= \pi r^2 \left[\frac{360 - 2\theta}{360} \right] + \frac{1}{2} \times 2r \sin \theta \times r \cos \theta$$

$$= \pi r^2 \left(1 - \frac{\theta}{180} \right) + r^2 \sin \theta \cos \theta$$

$$\therefore 0.4453 = \pi \times 0.45^2 \left(1 - \frac{\theta}{180} \right) + 0.45^2 \sin \theta \cos \theta$$

$$\text{or, } 0.4453 = 0.6362 - 0.003534 \theta + 0.2025 \sin \theta \cos \theta$$

$$\text{or, } 0.003534 \theta - 0.2025 \sin \theta \cos \theta = 0.6362 - 0.4453 = 0.1909$$

$$\text{or, } \theta - \frac{0.2025}{0.003534} \sin \theta \cos \theta = \frac{0.1909}{0.003534}$$

$$\text{or, } \theta - 57.3 \sin \theta \cos \theta = 54.02$$

$$\text{or, } \theta - 57.3 \sin \theta \cos \theta - 54.02 = 0$$

By hit and trial, we get, $\theta \approx 71.5^\circ$

$$\therefore \text{Depth of wooden log in water, } h \approx r + r \cos \theta$$

$$\approx 0.45 + 0.45 \cos 71.5^\circ \text{ or } h = \mathbf{0.593 \text{ m (Ans.)}}$$

Example 4.22. A float valve regulates the flow of oil of specific gravity 0.8 in a cistern. The spherical float is 150 mm in diameter. AOB is a weightless link carrying the float at one end, and a valve at the other end which closes the pipe through which oil flows into the cistern. The link is mounted in a frictionless hinge at O and angle AOB is 135° . The length of OA is 200 mm, and the distance between the centre of the float and hinge is 500 mm. When the flow is stopped AO will be vertical. The valve is to be pressed on to the seat with a force of 10 N to completely stop the flow of oil into the cistern. It was observed that the flow of oil is stopped when the free surface of oil in the cistern is 350 mm below the hinge. Determine the weight of the float. [UPSC Engg. Services]

Solution. Refer to Fig. 4.22.

Specific gravity of oil = 0.8

Diameter of the float, $d = 150 \text{ mm} = 0.15 \text{ m}$

$\angle AOB = 135^\circ$

Weight of the float, W :

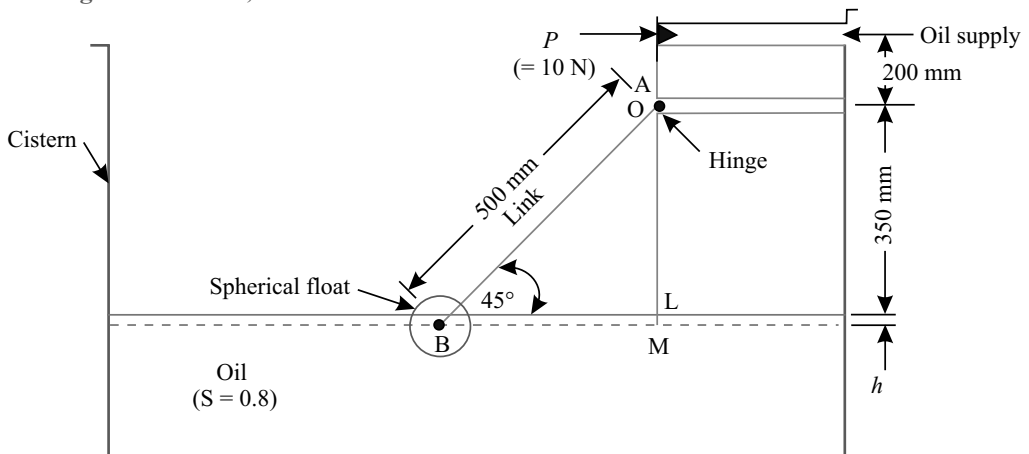


Fig. 4.22

When the oil flow is stopped, the level of oil is as shown in Fig. 4.22; the centre of float is below the level of oil by a depth h .

$$\begin{aligned}\text{In } \triangle OBM: \quad \sin 45^\circ &= \frac{OM}{OB} = \frac{OL + LM}{OB} = \frac{0.35 + h}{0.5} \\ 0.5 \sin 45^\circ &= 0.35 + h \quad \text{or} \quad h = 0.00355 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Volume of oil displaced} &= \frac{2}{3} \pi r^3 + \pi r^2 \times h \\ &= \frac{2}{3} \times \pi \left(\frac{0.15}{2} \right)^3 + \pi \times \left(\frac{0.15}{2} \right)^2 \times 0.00355 = 0.000946 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Buoyant force} &= \text{Weight of oil displaced} \\ &= \text{Volume of oil displaced} \times \text{sp. gravity of oil} \\ &= 0.000946 \times (0.8 \times 9.81) = 0.007424 \text{ kN or } 7.42 \text{ N}\end{aligned}$$

Since the buoyant force and the weight of the float passes through the same vertical line, therefore,

$$\text{Net force on float} = \text{Buoyant force} - \text{Weight of float} = 7.42 - W$$

Taking moments about the hinge O, we get:

$$P \times 0.2 = (7.42 - W) \times BM$$

$$\text{or,} \quad 10 \times 0.2 = (7.42 - W) \times 0.5 \cos 45^\circ$$

$$\therefore \quad W = 7.42 - \frac{10 \times 0.2}{0.5 \cos 45^\circ} = 1.76 \text{ N i.e. } W = \mathbf{1.76 \text{ N (Ans.)}}$$

Example 4.23. A cylindrical buoy is 2 m in diameter and 2.5 m long and weighs 22 kN. The specific weight of sea water is 10.25 kN/m^3 . Show that the buoy does not float with its axis vertical? What minimum pull should be applied to a chain attached to the centre of the base to keep the buoy vertical? [UPSC]

Solution. Given: Diameter of the buoy, $d = 2 \text{ m}$;

Length of the buoy, $l = 2.5 \text{ m}$;

Weight of the buoy, $W = 22 \text{ kN}$;

Specific weight of sea water = 10.25 kN/m^3 .

Part I: To show that the buoy does not float with its axis vertical:

$$V = \frac{\text{Weight of water displaced}}{\text{Specific weight of water}}$$

(Weight of the buoy = Weight of water displaced)

$$= \frac{22}{10.25} = 2.146 \text{ m}^3$$

i.e., Volume of buoy immersed in water

$$= 2.146 \text{ m}^3$$

Let, h = Depth of immersion.

$$\text{Then, } h = \frac{\text{Volume of buoy immersed in water}}{\text{Cross-sectional area of the buoy}}$$

$$= \frac{2.146}{(\pi/4) \times 2^2} = 0.683 \text{ m}$$

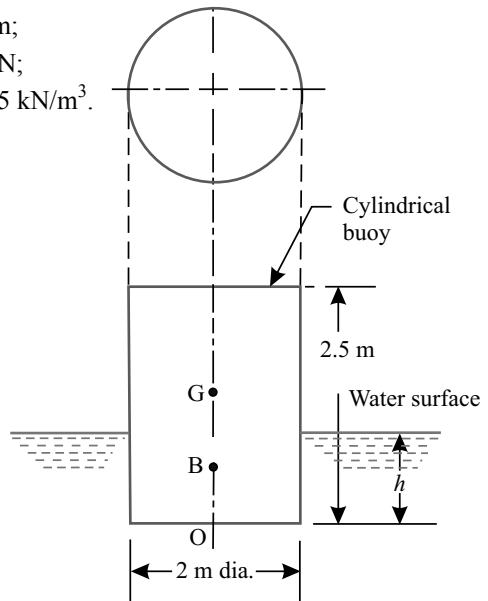


Fig. 6.23

Distance of centre of buoyancy (B) from the base point O,

$$OB = \frac{h}{2} = \frac{0.683}{2} = 0.342 \text{ m}$$

Let, M be the metacentre.

Then:
$$BM = \frac{I}{V} = \frac{(\pi/64) \times 2^4}{2.146} = 0.366 \text{ m}$$

$$OM = OB + BM = 0.342 + 0.366 = 0.708 \text{ m}$$

Distance of centre of gravity (G) from the base point O , $OG = \frac{2.5}{2} = 1.25 \text{ m}$

Since $OM < OG$, therefore, the buoy is **unstable** when floating with axis vertical. (Ans.)

Part II: Minimum pull required to keep the buoy vertical:

Let, T = Minimum pull (kN) which should be applied to chain attached to the centre of the base to keep the buoy vertical.

$$\text{Total downward force} = W + T = (22 + T)$$

$$\text{Displaced volume of water} = \left(\frac{22 + T}{10.25} \right) \text{ m}^3$$

$$\text{New depth of immersion, } h' = \frac{22 + T}{10.25 \times (\pi/4) \times 2^2} = \frac{22 + T}{32.2} \text{ m}$$

$$\therefore OB' = \frac{h'}{2} = \frac{22 + T}{2 \times 32.2} = \frac{22 + T}{64.4} \text{ m}$$

$$B'M' = \frac{I}{V} = \frac{(\pi/64) \times 2^4}{\frac{22 + T}{10.25}} = \frac{8.05}{22 + T}$$

To find new centre of gravity G' due to self weight acting at G and tension T in the chain, taking moments about point O , we get:

$$22 \times 1.25 = (22 + T) \times OG'$$

$$\therefore OG' = \frac{22 \times 1.25}{22 + T}$$

$$B'G' = OG' - OB' = \frac{22 \times 1.25}{22 + T} - \frac{22 + T}{64.4}$$

For stable equilibrium, M' must lie above G' , i.e.

$$B'M' > B'G'$$

$$\therefore \frac{8.05}{22 + T} > \left[\frac{22 \times 1.25}{22 + T} - \frac{22 + T}{64.4} \right]$$

$$\text{or, } \left[\frac{8.05}{22 + T} + \frac{22 + T}{64.4} \right] > \frac{22 \times 1.25}{22 + T}$$

$$\text{or, } \frac{8.05 \times 64.4 + (22 + T)^2}{(22 + T) \times 64.4} > \frac{22 \times 1.25}{22 + T}$$

$$\text{or, } 518.42 + (22 + T)^2 > 22 \times 1.25 \times 64.4$$

$$\text{or, } (22 + T)^2 > [22 \times 1.25 \times 64.4 - 518.42] \text{ or } > 1252.58$$

$$\text{or, } 22 + T > 35.39 \text{ or } T > 13.39 \text{ kN}$$

Hence, *minimum pull in the chain required to keep the buoy vertical* = **13.39 kN (Ans.)**

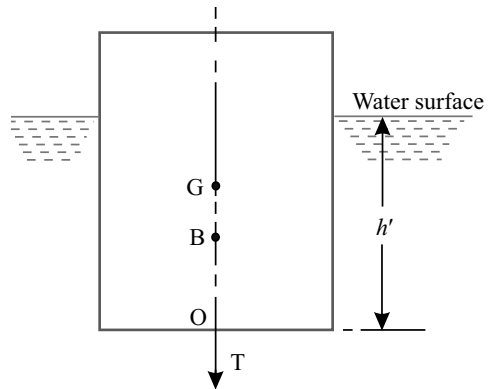


Fig. 4.24

Example 4.24. A solid cone floats in a liquid with its apex downwards. The specific gravity of the material of the cone and the liquid are 0.7 and 0.95 respectively. Determine the least apex angle of cone for stable equilibrium.

Solution. Specific weight of cone = $0.7 \times 9.81 = 6.87 \text{ kN/m}^2$

Specific weight of liquid = $0.95 \times 9.81 = 9.32 \text{ kN/m}^2$

Let,

H = Height of the cone,

h = Height of cone immersed in liquid, and

2α = Apex angle of the cone.

Weight of the cone = Volume $\times$ sp. weight

$$= \frac{1}{3} \pi R^2 H \times 6.87 = \frac{1}{3} \pi H^3 \tan^2 \alpha \times 6.87 \text{ kN}$$

$$\left[\because \tan \alpha = \frac{R}{H} \text{ i.e. } R = H \tan \alpha \quad \text{and} \quad r = h \tan \alpha \right]$$

$$\text{Weight of liquid displaced} = \frac{1}{3} \pi r^2 h \times 9.32 = \frac{1}{3} \pi h^3 \tan^2 \alpha \times 9.32 \text{ kN}$$

Now, Weight of cone = Weight of liquid displaced (since the cone is floating)

$$\text{i.e.,} \quad \frac{1}{3} \pi H^3 \tan^2 \alpha \times 6.87 = \frac{1}{3} \pi h^3 \tan^2 \alpha \times 9.32 \text{ kN}$$

$$\therefore h = H \left[\frac{6.87}{9.32} \right]^{1/3} = 0.9 H$$

$$\text{Distance of centre of buoyancy from the apex, } OB = \frac{3}{4} h = \frac{3}{4} \times 0.9 H = 0.675 H$$

$$\text{Distance of centre of gravity } G \text{ from the apex, } OG = \frac{3}{4} H = 0.75 H$$

For stable equilibrium, the metacentre (M) should be above G or may coincide with G .

$$\text{i.e. } BG \leq BM \text{ or } OG - OB \leq BM \quad \dots(i)$$

$$\text{Now, } BM = \frac{I}{V}$$

where, I = Moment of inertia of the circular section about the liquid level

$$= \frac{\pi r^4}{4} = \frac{\pi \times h^4 \tan^4 \alpha}{4}$$

and, V = Volume of liquid displaced

$$= \frac{1}{3} \pi h^3 \tan^2 \alpha$$

Substituting various values in (i), we get:

$$0.75 H - 0.675 H \leq \frac{\left(\frac{\pi h^4 \tan^4 \alpha}{4} \right)}{\left(\frac{1}{3} \pi h^3 \tan^2 \alpha \right)} \quad \text{or} \quad 0.75 H \leq \frac{h^4 \tan^4 \alpha}{4} \times \frac{3}{\pi h^3 \tan^2 \alpha}$$

$$\text{or,} \quad 0.075 H \leq 0.75 h \tan^2 \alpha$$

$$\text{or,} \quad 0.075 H \leq 0.75 \times (0.9 H) \tan^2 \alpha \quad (\because h = 0.9 H)$$

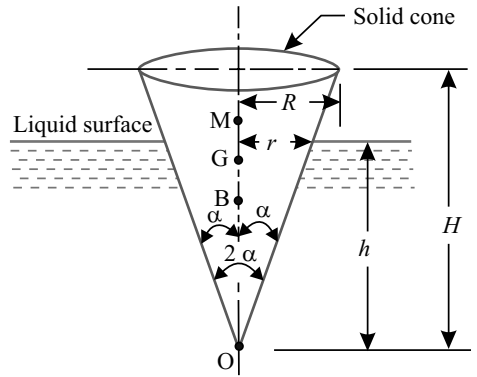


Fig. 4.25

or $\tan^2 \alpha \geq \frac{0.75}{0.75 \times 0.9} \geq 0.111$

or $\tan \alpha \geq 0.333$ or $\alpha \geq 18^\circ 24'$

$\therefore$ Least apex angle, $2\alpha = 36^\circ 48'$ (Ans.)

Example 4.25. A cone of specific gravity S , is floating in water with its apex downwards. It has a radius R and vertical height H . Show that for stable equilibrium of cone,

(i) $\sec^2 \alpha \propto \frac{H}{h}$ (ii) $H < \left[\frac{R^2 S^{1/3}}{1 - S^{1/3}} \right]^{1/2}$

where, h is the depth of immersion and α is the half apex angle.

Solution. Radius of the cone = R ; Height of the cone = H ;

Sp. gravity of the cone = S

Let, h = Depth of immersion, 2α = apex angle,
 r = Radius of cone at the water surface, O = apex of the cone,
 G = Centre of gravity of the cone,
 B = Centre of buoyancy, and
 M = Position of metacentre.

Then, $OG = 3/4 H$; $OB = 3/4 h$

(i) $\sec^2 \alpha \propto \frac{H}{h}$:

Now, $BM = \frac{I}{V}$

where, I = Moment of inertia

$$= \frac{\pi r^4}{4}, \text{ and}$$

V = Volume of water displaced

$$= \frac{1}{3} \pi r^2 h$$

$$BM = \frac{\pi r^4 / 4}{\frac{1}{3} \pi r^2 h} = \frac{3}{4} \times \frac{r^2}{h}$$

Substituting $r = h \tan \alpha$, we get:

$$\begin{aligned} BM &= \frac{3}{4} \times \frac{h^2 \tan^2 \alpha}{h} \\ &= \frac{3}{4} h \tan^2 \alpha \left(\because \frac{r}{h} = \tan \alpha \right) \\ OM = OB + BM &= \frac{3}{4} h + \frac{3}{4} h \tan^2 \alpha \\ &= \frac{3}{4} h (1 + \tan^2 \alpha) = \frac{3}{4} h \sec^2 \alpha \end{aligned}$$

For stable equilibrium:

$$BM > BG \quad \text{or} \quad OM > OG$$

$$= \frac{3}{4} h \sec^2 \alpha > \frac{3}{4} H \quad \text{or} \quad \sec^2 \alpha > \frac{H}{h}$$

...Proved (Ans.)

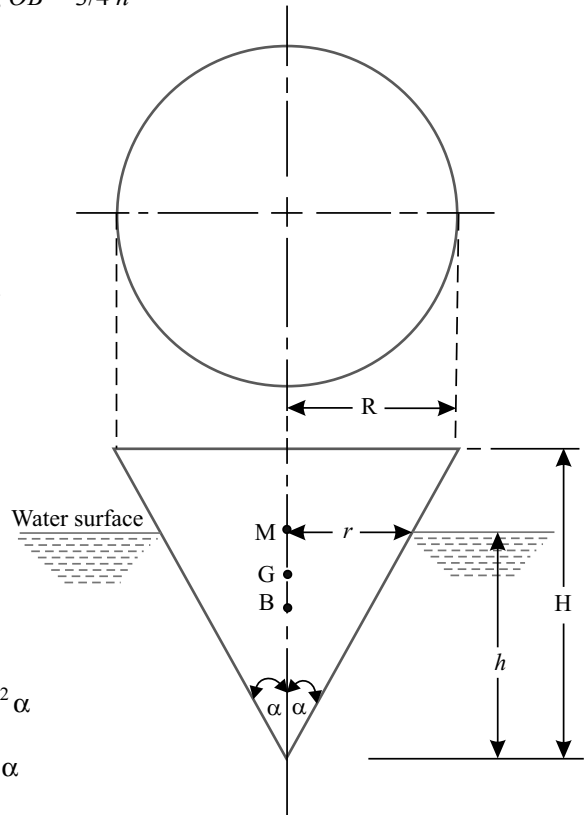


Fig. 4.26

$$(ii) \quad H < \left(\frac{R^2 S^{1/3}}{1 - S^{1/3}} \right)^{1/2} :$$

As per principle of floatation:

Weight of cone = weight of water displaced

$$\text{or,} \quad \frac{1}{3} \pi R^2 H \times w_c = \frac{1}{3} \pi r^2 h \times w$$

(where w_c and w are the specific weights of cone material and water respectively).

Substituting, $R = H \tan \alpha$, and $r = h \tan \alpha$, we get:

$$\frac{1}{3} \pi H^2 \tan^2 \alpha \times H \times w_c = \frac{1}{3} \pi h^2 \tan^2 \alpha \times h \times w$$

$$H^3 \tan^2 \alpha \times S = h^3 \tan^2 \alpha \quad \left(\because \frac{w_c}{w} = S \right)$$

$$\text{or,} \quad h^3 = H^3 S \text{ or } h = H.S^{1/3}$$

(where, S is the sp. gravity of cone material).

From the relations $\sec^2 \alpha > \frac{H}{h}$ and $h = H.S^{1/3}$ (derived above), we have $\sec^2 \alpha > \frac{1}{S^{1/3}}$

$$\text{or,} \quad (1 + \tan^2 \alpha) > \frac{1}{S^{1/3}} \quad \text{or} \quad \tan^2 \alpha > \frac{1}{S^{1/3}} - 1 \quad \text{or} \quad \tan^2 \alpha > \frac{1 - S^{1/3}}{S^{1/3}}$$

$$\text{Substituting, } \tan \alpha = \frac{R}{H}, \text{ we get } \frac{R^2}{H^2} > \frac{1 - S^{1/3}}{S^{1/3}}$$

$$\text{or,} \quad \frac{H^2}{R^2} < \frac{S^{1/3}}{1 - S^{1/3}} \quad \text{or} \quad H^2 < \frac{R^2 S^{1/3}}{1 - S^{1/3}}$$

$$\text{or,} \quad H < \left(\frac{R^2 S^{1/3}}{1 - S^{1/3}} \right)^{1/2}$$

...Proved (Ans.)

Example 4.26. A solid cone ($S = 0.8$) diameter 36 cm and height 30 cm floats with its vertex downwards in water as shown in fig. 4.27. Is this cone in stable equilibrium?

Solution. Given: $D = 36$ cm, $H = 30$ cm;
 $S = 0.8$

Let, θ = Semivertex angle,

$$\text{Then, } \tan \theta = \frac{18}{30} = 0.6$$

$$\text{or, } \theta = \tan^{-1}(0.6) = 30.96^\circ$$

Diameter of the cone at water surface,

$$d = 2y \tan \theta \quad \left(\because \frac{d/2}{h} = \tan \theta \right)$$

Weight of cone = Weight of water displaced

$$\frac{1}{3} \times \pi \left(\frac{D}{2} \right)^2 \times H \times (w \times S) = \frac{1}{3} \pi \left(\frac{d}{2} \right)^2 \times h \times w$$

$$\therefore D^2 HS = d^2 y$$

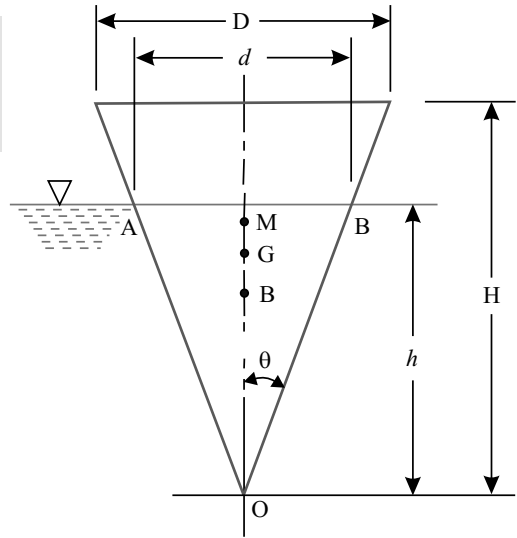


Fig. 4.27

$$= (2h \tan \theta)^2 \times h = 4h^3 \tan^2 \theta$$

$$= 4h^3 \left(\frac{D}{2H} \right)^2 \quad \left(\because \frac{D/2}{H} = \tan \theta \right)$$

$$\text{or,} \quad h^3 = H^3 S$$

$$\text{or,} \quad h = HS^{1/3}$$

$$= 30 \times (0.8)^{1/3} = 27.85 \text{ cm}$$

If B is the centre of buoyancy,

$$OB = \frac{3}{4}h = \frac{3}{4} \times 27.85 = 20.89 \text{ cm}$$

$$OG = \frac{3}{4}H = \frac{3}{4} \times 30 = 22.5 \text{ cm}$$

Now,

$$d = 2h \tan \theta = 2 \times 27.85 \times 0.6 = 33.42 \text{ cm}$$

$$BM = \frac{I}{V} = \frac{(\pi d^4)/64}{\frac{1}{3}\pi \left(\frac{d}{2}\right)^2 \times h} = \frac{3}{16} \left(\frac{d^2}{h} \right)$$

$$= \frac{3}{16} \times \frac{(33.42)^2}{27.85} = 7.52 \text{ cm}$$

$$OM = OB + BM$$

$$= 20.89 + 7.52 = 28.41 \text{ cm}$$

$$OG = 22.5 \text{ cm}$$

$$MG = OM - OG = 28.41 - 22.5 = 5.91 \text{ cm}$$

i.e., M is above G by 5.91 cm

Hence the cone is under **stable equilibrium**. (Ans.)

Example 4.27. A ship 63 m long and 9 m broad has a displacement of 16000 kN. When a weight of 200 kN is moved across the deck through a distance of 5.4 m, the ship is tilted through 5° . The second moment of area of the water line section about its fore-and-aft axis is 75 per cent of that of circumscribing rectangle, and centre of buoyancy is 2.1 m below the water line. Determine.

(i) The metacentric height, and (ii) The position of centre of gravity of ship.

Take specific weight of sea water = 10.25 kN/m^3

Solution. Length of the ship, $l = 63 \text{ m}$

Breadth of the ship, $b = 9 \text{ m}$

Displacement, $W = 16000 \text{ kN}$

Angle of tilt, $\theta = 5^\circ$

Movable weight, $W_1 = 200 \text{ kN}$

Distance moved by W_1 , $z = 5.4 \text{ m}$

(i) **Metacentric height, GM :**

We know,

$$GM = \frac{W_1 z}{W \tan \theta} \quad (\text{Eqn. 4.2})$$

$$= \frac{200 \times 5.4}{16000 \times \tan 5^\circ} = 0.77 \text{ m}$$

i.e. $GM = 0.77 \text{ m}$ (Ans.)

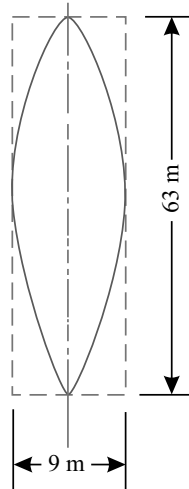


Fig. 4.28

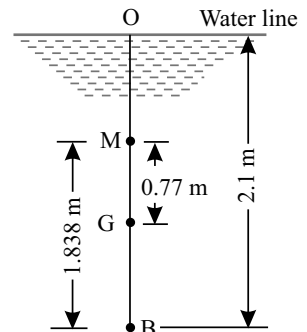


Fig. 4.29

(ii) The position of centre of gravity of the ship:

Distance between the metacentre M and the centre of buoyancy is given by: $BM = \frac{I}{V}$

where,

I = Second moment of area of the water line section

$$= 0.75 \times \left(\frac{63 \times 9^3}{12} \right) = 2870 \text{ m}^4$$

and,

V = Volume of water displaced by the vessel

$$= \frac{\text{Weight of the vessel}}{\text{Specific weight of vessel}} = \frac{16000}{10.25} = 1561 \text{ m}^3$$

$\therefore$

$$BM = \frac{2870}{1561} = 1.838 \text{ m}$$

Now,

$$OG = OM + MG = (OB - MB) + MG$$

$$= (2.1 - 1.838) + 0.77 = 1.032 \text{ m}$$

i.e.

$$OG = \mathbf{1.032 \text{ m (below the water line) (Ans.)}$$

4.6. OSCILLATION (ROLLING OF A FLOATING BODY)

It has been observed that whenever a body floating in a liquid is given a small angular displacement, it starts oscillating about its metacentre M (see Fig. 4.30) in the same manner as a pendulum oscillates about its point of suspension.

Let,

W = Weight of floating body,

θ = Angle (in radians) through which the body is depressed,

α = Angular acceleration of the body in rad/s^2 ,

T = Time of rolling (*i.e.* one complete oscillation) in seconds,

k = Radius of gyration about G , and

I = Moment of inertia of the body about its centre of gravity G

$$= \frac{W}{g} k^2$$

GM = Metacentric height of the body.

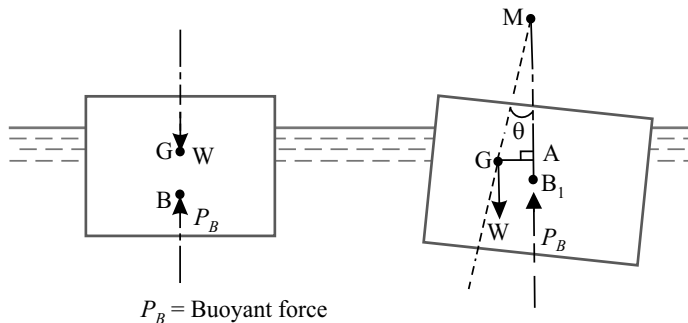


Fig. 4.30

When the force which has caused angular displacement is removed the only force acting on the body is due to the restoring couple due to the weight W of the body and the force of buoyancy P_B .

$$\therefore \text{Restoring couple} = W \times GA$$

$$= W \times GM \tan \theta$$

$$= W \cdot GM \cdot \theta$$

...(i)

[assuming θ to be small ($\tan \theta = \theta$)]

Angular acceleration of the body, $\alpha = -\frac{d^2\theta}{dt^2}$

– ve sign indicates that the force is acting in such a way that it tends to *decrease* the angle θ .

Also, Inertia torque = Moment of inertia $\times$ Angular acceleration

$$= I.\alpha = -\frac{W}{g}k^2 \times \frac{d^2\theta}{dt^2} \quad \dots(ii)$$

Equating (i) and (ii), we get:

$$W.GM.\theta = -\frac{W}{g}k^2 \times \frac{d^2\theta}{dt^2} \quad \text{or} \quad \frac{W}{g}k^2 \frac{d^2\theta}{dt^2} + W.GM.\theta = 0$$

Dividing both sides by W , we get:

$$\frac{k^2}{g} \times \frac{d^2\theta}{dt^2} + GM.\theta = 0$$

Again, dividing both sides by $\frac{k^2}{g}$, we get:

$$\frac{d^2\theta}{dt^2} + \frac{GM.g\theta}{k^2} = 0$$

The above equation is a differential equation of second degree, whose solution is:

$$Q = C_1 \sin \left[\sqrt{\frac{GM.g}{k^2}} \times t \right] + C_2 \cos \left[\sqrt{\frac{GM.g}{k^2}} \times t \right] \quad \dots(iii)$$

where, C_1 and C_2 are constants of integration.

The values of C_1 and C_2 are obtained from the following boundary conditions:

1. At $t = 0$, $\theta = 0$

$C_2 = 0$ [By substitution of $t = 0$, $\theta = 0$ in (iii)]

2. At $t = \frac{T}{2}$, $\theta = 0$

$$\therefore 0 = C_1 \sin \left[\sqrt{\frac{GM.g}{k^2}} \times \frac{T}{2} \right]$$

Since C_1 cannot be equal to zero, therefore:

$$\sin \left[\sqrt{\frac{GM.g}{k^2}} \times \frac{T}{2} \right] = 0 \quad \text{or} \quad \sqrt{\frac{GM.g}{k^2}} \times \frac{T}{2} = \pi \quad (\because \sin \pi = 0)$$

$$\text{or,} \quad T = 2\pi \sqrt{\frac{k^2}{GM.g}} \quad \dots(4.4)$$

Example 4.28. A ship of weight 32000 kN is floating in sea water. The centre of buoyancy is 1.6 meters below its centre of gravity. The moment of inertia of the ship area at the water level is 8320 m⁴. If the radius of gyration of the ship is 3.2 m, find its period of rolling.

Take sp. weight of sea water = 10.1 kN/m³

Solution. Given:

Weight of the ship, $W = 32000$ kN

Distance between centre of buoyancy and centre of gravity, $BG = 1.6$ m

Moment of inertia, $I = 8320$ m⁴

Radius of gyration, $k = 3.2$ m

Period of rolling of the ship, T :

Volume of sea water displaced,

$$V = \frac{\text{Weight}}{\text{Specific weight of sea water}} = \frac{32000}{10.1} = 3168.3 \text{ m}^3$$

$$\text{Using the relation, } BM = \frac{I}{V} = \frac{8320}{3168.3} = 2.626 \text{ m}$$

Also, the metacentric height, $GM = BM - BG = 2.626 - 1.6 = 1.026 \text{ m}$

$$\text{Now using the relation, } T = 2\pi \sqrt{\frac{k^2}{GM \cdot g}} = 2\pi \sqrt{\frac{3.2^2}{1.028 \times 9.81}} = \mathbf{6.33 \text{ s (Ans.)}}$$

HIGHLIGHTS

1. The tendency for an immersed body to be lifted up in the fluid due to an upward force opposite to action of gravity is known as *buoyancy*.
2. The floating bodies may have the following types of equilibrium:
 - (i) Stable equilibrium,
 - (ii) Unstable equilibrium, and
 - (iii) Neutral equilibrium.
3. The *metacentre* is defined as a point of intersection of the axis of body passing through *c.g.* (*G*) and original centre of buoyancy (*B*), and a vertical line passing through the centre of buoyancy (*B₁*) of the tilted position of the body.
4. The distance between the centre of gravity (*G*) of a floating body and the metacentre (*M*) is called *metacentric height*.
5. The metacentric height (*GM*) by experimental method is given by:

$$GM = \frac{W_1 \cdot z \cdot l}{W \cdot d} \left(= \frac{W_1 \cdot z}{W \tan \theta} \right)$$

where,

W_1 = Known weight,

z = Distance through which W_1 is shifted across the axis of the tilt,

l = Displacement of the plumb bob, and

θ = Angle of tilt $\left(\tan \theta = \frac{d}{l} \right)$.

6. Time of rolling,

$$T = 2\pi \sqrt{\frac{k^2}{GM \cdot g}}$$

where,

k = Radius of gyration about *c.g.* (*G*), and

GM = Metacentric height of the body.

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer.

1. The tendency for an immersed body to be lifted up in the fluid, due to an upward force opposite to the action of gravity is known as
 - (a) buoyancy
 - (b) centre of buoyancy
 - (c) buoyant force
 - (d) none of the above.
2. The magnitude of the buoyant force can be determined by
 - (a) Newton's second law of motion
 - (b) Archimedes' principle
 - (c) Principle of moments
 - (d) none of the above.
3. When a body is immersed in a fluid, partially or completely, the force of buoyancy is equal to

- (a) the weight of the body
 (b) the weight of the fluid displaced by the body
 (c) the weight of the volume of the fluid equal to the volume of body
 (d) none of the above.
4. The point of application of the force of buoyancy on the body is known as
 (a) centre of gravity
 (b) centre of buoyancy
 (c) metacentre
 (d) none of the above.
5. "When a body is immersed in a fluid either wholly or partially, it is buoyed or lifted up by a force which is equal to the weight of fluid displaced by the body".
 This principle was enunciated by
 (a) Archimedes (b) Newton
 (c) Pascal (d) Kirchhoff.
6. A floating body is in stable equilibrium when
 (a) the metacentre is below its centre of gravity
 (b) the metacentre is above its centre of gravity
 (c) the metacentric height is zero.
 (d) its centre of gravity is below the centre of buoyancy.
7. An ice-cube is floating in glass of water. As the cube melts the water level
 (a) remain constant (b) falls
 (c) rises (d) none of the above.
8. If the position of metacentre M remains lower than c.g. of the body, G, the body will remain in a state of
 (a) stable equilibrium
 (b) unstable equilibrium
 (c) neutral equilibrium
 (a) any of the above.
9. Metacentric height can be determined by
 (a) only analytical method
 (b) only experimental method
 (c) both (a) and (b)
 (d) none of the above.
10. If a body does not return to its original position from the slightly displaced angular position and heels farther away, when given a small angular displacement, such an equilibrium is called
 (a) stable equilibrium
 (b) unstable equilibrium
 (c) neutral equilibrium
 (d) any of the above.

ANSWERS

- | | | | | | |
|--------|--------|--------|----------|--------|--------|
| 1. (a) | 2. (b) | 3. (b) | 4. (b) | 5. (a) | 6. (b) |
| 7. (b) | 8. (b) | 9. (c) | 10. (b). | | |

THEORETICAL QUESTIONS

- What is buoyancy?
- What is centre of buoyancy?
- Explain briefly the following types of equilibrium of floating bodies:
 - Stable equilibrium,
 - Unstable equilibrium, and
 - Neutral equilibrium.
- Define and explain the following terms:
 - Metacentre, and
 - Metacentric height.
- Derive an expression for calculating time of rolling of a floating body.

UNSOLVED EXAMPLES

- A wooden block of width 2.5 m, depth 1.5 m and length 6 m is floating horizontally in water. If the specific gravity of block is 0.65 find:
 - The volume of water displaced, and
 - Position of centre of buoyancy.

[Ans. (i) 14.625 m^3 ; (ii) 0.4875 m from base]
- A wooden block $4 \text{ m} \times 1 \text{ m} \times 0.5 \text{ m}$ is floating in water. Its specific gravity is 0.76. Find the volume of concrete, of sp. gravity 2.5, that may be placed on the block which will immerse
 - the block completely in water, and
 - the block and concrete completely in water.

[Ans. (i) 0.2 m^3 ; (ii) 0.33 m^3]
- The following data relate to a pontoon floating in sea water:

Length = 5 m, width 3 m, height = 1.2 m
 The depth of immersion = 0.8 m
 Centre of gravity above the bottom of pontoon = 0.6 m

Sp. gravity of sea water = 1.025

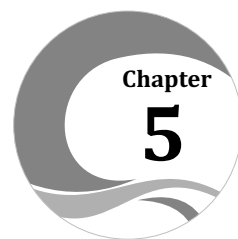
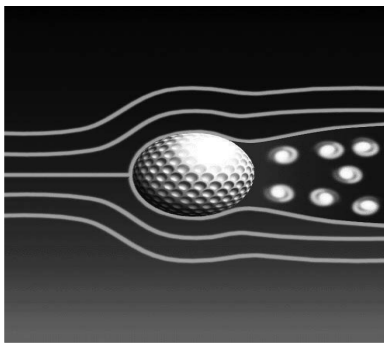
Determine the metacentric height.

[Ans. 0.7375 m]

4. A wooden block $1 \text{ m} \times 0.4 \text{ m} \times 0.3 \text{ m}$ is floating in water. If its specific gravity is 0.8 determine the metacentric height for tilt about its longitudinal axis. [Ans. 0.0255 m]
5. The following data relate to a body (consisting of cylindrical upper portion and curved lower portion) floating in water:
Diameter of cylindrical upper portion = 2 m
Depth of upper portion = 1.2 m
Volume of water displaced by the curved lower portion = 0.4 m^3
Centre of buoyancy of the lower portion below the top of the cylinder = 1.3 m
Centre of buoyancy of whole body below the top of cylinder = 0.8 m
Total displacement of water = 25.5 kN. Determine the metacentric height of the body. [Ans. GM = 0.182 m]
6. A wooden block of size $2 \text{ m} \times 1 \text{ m} \times 0.8 \text{ m}$ is floating in water. If its specific gravity is 0.7 find its metacentric height. [Ans. 0.0288 m]
7. A wooden cylinder of sp. gravity 0.6 and diameter 0.4 m is required to float in an oil of sp. gravity 0.8. Find the maximum length of the cylinder in order that it may float vertically in water. [Ans. 0.326 m]
8. A solid cylinder 4 m in diameter and 4 m high is floating in water with its axis vertical. If its specific gravity is 0.6, find the metacentric height. Also state whether the equilibrium is stable or unstable. [Ans. GM = -0.3833 m, unstable]
9. A weight 100 kN is moved through a distance of 9 metres across the deck of pontoon of 7500 kN displacement, floating in water. This makes a pendulum 2.7 metres long, move through 0.13 m horizontally. Calculate the metacentric height of the pontoon. [Ans. 2.5 m]
10. The following data correspond to a ship floating in a sea water:
Weight of the ship = 4000 tonnes
Centre of buoyancy below its c.g. (G) = 2 m
Moment of inertia of the ship area at the water level = 10400 m^4
Radius of gyration of the ship = 4 m
Sp. gravity of sea water = 1.03
Find the period of rolling of the ship. [Ans. 9.73 s]
11. Find the density of a metallic body which floats at the interface of mercury of specific gravity 13.6 and water such that 40 per cent of its volumes is submerged in mercury and 60 per cent in water. [Ans. 59.25 kN/m³]
12. A hollow wooden cylinder of specific gravity 0.56 has an outer diameter of 600 mm and an

inner diameter of 300 mm. It is required to float in oil of specific gravity 0.85. Calculate:

- (i) The maximum length/height of the cylinder so that it shall be stable when floating with its axis vertical.
- (ii) The depth to which it will sink. [Ans. (i) 0.5 m (ii) 0.3295 m]
13. A pontoon measuring 10 m (length) $\times$ 7.5 m (width) $\times$ 2.5 m (depth) weighing 800 kN floats in sea water. On its upper deck it carries a boiler 5 m in diameter and weighing 500 kN. The centre of gravity of each unit coincides with geometrical centre of the arrangement and lies in the same vertical line.
Calculate the metacentric height.
Take the specific weight of sea water = 10 kN/m^3 [Ans. 0.875 m]
14. A log of wood 0.6 m in diameter and 5 m long is floating in river water. If the specific gravity of log is 0.7 what is depth of the wooden log in water? [Ans. 0.395 m]
15. A wooden cylinder (specific gravity = 0.6) of circular cross-section is required to float in oil of specific gravity 0.8. If l and d are the length and diameter of the cylinder respectively, find the $\frac{l}{d}$ ratio for the cylinder to float with its longitudinal axis vertical in oil. [Ans. $\frac{l}{d} < 0.816$]
16. A solid cone (sp. gravity of material = 0.7) floats in water with its apex downward. Determine the least apex angle of cone for equilibrium. [Ans. $39^\circ 7'$]
17. A cylindrical buoy weighing 20 kN is floating in ocean. The buoy has a diameter of 2 m and height 2.5 m. Can the buoy float with its axis vertical? If now, a chain may be tied at the bottom of the buoy and anchored, what is the tension required in the anchor chain to make the buoy stable? Take sp. gravity of sea water = 1.025. [Ans. 12.9 kN]
18. A vessel has a length of 60 m, width 12 m and a displacement of 19620 kN. When a weight of 294.3 kN is rolled off transversely across the deck through a distance of 6.5 m, the vessel tilts through 5° . The second moment of area of the water line section about its fore-and-aft axis is 75 per cent of that of the circumscribing rectangle. The centre of buoyancy is 2.75 m below the water line. Find:
(i) The metacentric height;
(ii) The position of centre of gravity of the vessel.
Take specific weight of sea water = 10.104 kN/m^3
[Ans. (i) 1.1145 m (ii) 0.53 m below water surface]



FLUID KINEMATICS

- 5.1. Introduction
- 5.2. Description of fluid motion—
Langrangian method—
Eulerian method
- 5.3. Types of fluid flow—steady
and unsteady flows—
uniform and non-uniform
flows—one, two and three
dimensional flows—
rotational and irrotational
flows—laminar and turbulent
flows—compressible and
incompressible flows
- 5.4. Types of flow lines—path
line—stream line—stream
tube—streak line
- 5.5. Rate of flow or discharge
- 5.6. Continuity equation
- 5.7. Continuity equation in
cartesian co-ordinates
- 5.8. Equation of continuity in
polar co-ordinates
- 5.9. Circulation and vorticity
- 5.10. Velocity potential and
stream function—velocity
potential—stream
function—relation between
stream function and velocity
potential
- 5.11. Flow nets—methods of
drawing the flow nets—uses
and limitations of flow nets

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

5.1. INTRODUCTION

Fluid kinematics may be defined as follows:

Fluid kinematics is a branch of 'Fluid mechanics' which deals with the study of velocity and acceleration of the particles of fluids in motion and their distribution in space without considering any force or energy involved.

The motion of fluid can be described fully by an expression describing the location of a fluid particle in space at different times thus enabling determination of the magnitude and direction of velocity and acceleration in the flow field at any instant of time.

In the chapter we shall deal with the conception of fluid flow in general.

5.2. DESCRIPTION OF FLUID MOTION

The motion of fluid particles may be described by the following methods:

1. Langrangian method.
2. Eulerian method.

5.2.1. Langrangian Method

In this method, the observer *concentrates on the movement of a single particle*. The path taken by the particle and the changes in its velocity and acceleration are studied.

In the Cartesian system, the position of the fluid particle in space (x, y, z) at any time t from its position (a, b, c) at time $t = 0$ shall be given as:

$$\begin{aligned}x &= f_1(a, b, c, t) \\y &= f_2(a, b, c, t) \\z &= f_3(a, b, c, t) \quad \dots(5.1)\end{aligned}$$

The velocity and acceleration components (obtained by taking derivatives with respect to time) are given by:

$$\text{Velocity components:} \quad \left. \begin{aligned} u &= \frac{\partial x}{\partial t} \\ v &= \frac{\partial y}{\partial t} \\ w &= \frac{\partial z}{\partial t} \end{aligned} \right\} \quad \dots(5.2)$$

$$\text{Acceleration components :} \quad \left. \begin{aligned} a_x &= \frac{\partial^2 x}{\partial t^2} \\ a_y &= \frac{\partial^2 y}{\partial t^2} \\ a_z &= \frac{\partial^2 z}{\partial t^2} \end{aligned} \right\} \quad \dots(5.3)$$

At any point, the resultant velocity or acceleration shall be the *resultant* of three components of the respective quantity at that point.

$$\therefore \quad \text{Resultant velocity, } V = \sqrt{u^2 + v^2 + w^2} \quad \dots(5.4)$$

$$\text{Acceleration, } a = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad \dots(5.5)$$

Similarly, other quantities like pressure, density, etc. can be found.

This method entails the following *shortcomings*:

1. Cumbersome and complex.
2. The equations of motion are very difficult to solve and the motion is hard to understand.

5.2.2. Eulerian Method

In Eulerian method, the observer *concentrates on a point in the fluid system*. Velocity, acceleration and other characteristics of the fluid at that particular point are studied.

This method is almost *exclusively used* in fluid mechanics, especially because of its *mathematical simplicity*. In fluid mechanics, we are not concerned with the motion of each particle, but we study the general state of motion at various points in the fluid system.

The velocities at any point (x, y, z) can be written as:

$$\left. \begin{aligned} u &= f_1(x, y, z, t) \\ v &= f_2(x, y, z, t) \\ w &= f_3(x, y, z, t) \end{aligned} \right\} \quad \dots(5.6)$$

The components of acceleration of the fluid particle can be worked out by partial differentiation as follows:

$$\begin{aligned} du &= \frac{\partial u}{\partial x} \cdot dx + \frac{\partial u}{\partial y} \cdot dy + \frac{\partial u}{\partial z} \cdot dz + \frac{\partial u}{\partial t} \cdot dt \\ a_x &= \frac{du}{dt} = \left(\frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt} \right) + \frac{\partial u}{\partial t} \cdot \frac{dt}{dt} \end{aligned}$$

But,

$$\frac{dx}{dt} = u, \frac{dy}{dt} = v, \frac{dz}{dt} = w$$

$$\therefore \left. \begin{aligned} a_x &= \frac{du}{dt} = \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) + \frac{\partial u}{\partial t} \\ \text{Similarly, } a_y &= \frac{dv}{dt} = \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) + \frac{\partial v}{\partial t} \\ a_z &= \frac{dw}{dt} = \left(u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) + \frac{\partial w}{\partial t} \end{aligned} \right\} \quad \dots(5.7)$$

Now, resultant velocity: $V = \sqrt{u^2 + v^2 + w^2} \quad \dots(5.8)$

Resultant acceleration, $a = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad \dots(5.9)$

In vector notation:

Velocity vector: $V = ui + vj + wk \quad \dots(5.10)$

Acceleration vector: $a = \frac{dV}{dt} = \left(u \frac{\partial V}{\partial x} + v \frac{\partial V}{\partial y} + w \frac{\partial V}{\partial z} \right) + \frac{\partial V}{\partial t} \quad \dots(5.11)$

$$a = a_x i + a_y j + a_z k \quad \dots(5.12)$$

and, $|V| = \sqrt{u^2 + v^2 + w^2} \quad \dots(5.13)$

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad \dots(5.14)$$

Vectorially,

$$a = (V \cdot \nabla) V + \frac{\partial V}{\partial t} \quad \dots(5.15)$$

The velocity, in general, is a function of space(s) and time (t) *i.e.*

$$V = f(x, y, z, t)$$

or,

$$V = f(s, t)$$

and, The acceleration,

$$a = \frac{dV}{dt} = \frac{\partial V}{\partial s} \cdot \frac{ds}{dt} + \frac{\partial V}{\partial t}$$

$$\therefore a = V \frac{\partial V}{\partial s} + \frac{\partial V}{\partial t} \quad \dots(5.16)$$

Thus the acceleration consists of the two parts:

(i) $V \frac{\partial V}{\partial s}$: This part is due to change in position or movement and is called **convective acceleration**.

$$\therefore \text{Convective acceleration} = V \frac{\partial V}{\partial s} = \frac{1}{2} \frac{\partial (V)^2}{\partial s} \quad \dots(5.17)$$

= terms in the parenthesis of Eqn. (5.7)

(ii) $\frac{\partial V}{\partial t}$: This part is with respect to time at a given location and is called **local** (or *temporal*) **acceleration**.

$$\therefore \text{Local acceleration} = \frac{\partial V}{\partial t} \quad \dots(5.18)$$

$$= \frac{\partial u}{\partial t}, \frac{\partial v}{\partial t}, \frac{\partial w}{\partial t} \text{ in Eqn. (5.7)}$$

Tangential and normal acceleration: Refer to Fig. 5.1.

When the motion is curvilinear eqn. 5.16 gives the *tangential acceleration*. A particle moving in a curved path will always have a normal acceleration $a_n = \frac{V^2}{r}$ towards the centre of the curved path (r being the radius of the path), though its tangential acceleration (a_s) may be zero as in the case of uniform circular motion.

For motion along a curved path, in general,

$$\begin{aligned} a &= a_s + a_n \\ &= \left(V \frac{\partial V}{\partial s} + \frac{\partial V}{\partial t} + \frac{V^2}{r} \right) \end{aligned} \quad \dots(5.19)$$

5.3. TYPES OF FLUID FLOW

Fluids may be *classified* as follows:

1. Steady and unsteady flows
2. Uniform and non-uniform flows
3. One, two and three dimensional flows
4. Rotational and irrotational flows
5. Laminar and turbulent flows
6. Compressible and incompressible flows.

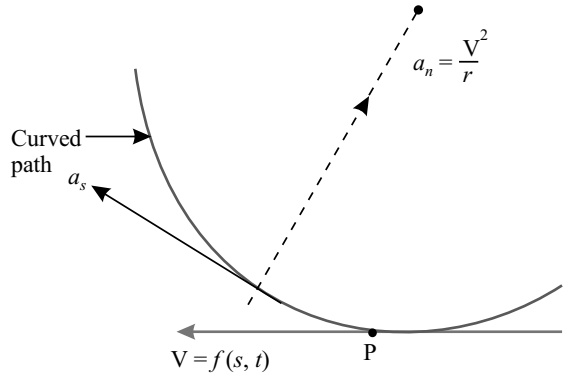


Fig. 5.1. Tangential and normal acceleration.

5.3.1. Steady and Unsteady Flows

Steady flow. The type of flow in which the fluid characteristics like velocity, pressure, density, etc. at a point *do not change* with time is called *steady flow*. Mathematically, we have:

$$\begin{aligned} \left(\frac{\partial u}{\partial t} \right)_{x_0, y_0, z_0} &= 0; \left(\frac{\partial v}{\partial t} \right)_{x_0, y_0, z_0} = 0; \left(\frac{\partial w}{\partial t} \right)_{x_0, y_0, z_0} = 0 \\ \left(\frac{\partial p}{\partial t} \right)_{x_0, y_0, z_0} &= 0; \left(\frac{\partial \rho}{\partial t} \right)_{x_0, y_0, z_0} = 0; \text{ and so on} \end{aligned}$$

where (x_0, y_0, z_0) is a fixed point in a fluid field where these variables are being measured *w.r.t.* time.

Example. Flow through a prismatic or non-prismatic conduit at a constant flow rate $Q \text{ m}^3/\text{s}$ is *steady*.

(A prismatic conduit has a constant size shape and has a velocity equation in the form $u = ax^2 + bx + c$, which is independent of time t).

Unsteady flow. It is that type of flow in which the velocity, pressure or density at a point *change* *w.r.t.* time. Mathematically, we have:

$$\begin{aligned} \left(\frac{\partial u}{\partial t} \right)_{x_0, y_0, z_0} &\neq 0; \left(\frac{\partial v}{\partial t} \right)_{x_0, y_0, z_0} \neq 0; \left(\frac{\partial w}{\partial t} \right)_{x_0, y_0, z_0} \neq 0 \\ \left(\frac{\partial p}{\partial t} \right)_{x_0, y_0, z_0} &\neq 0; \left(\frac{\partial \rho}{\partial t} \right)_{x_0, y_0, z_0} \neq 0; \text{ and so on} \end{aligned}$$

Example. The flow in a pipe whose valve is being opened or closed gradually (velocity equation is in the form $u = ax^2 + bxt$).

5.3.2. Uniform and Non-uniform Flows

Uniform flow. The type of flow, in which the velocity at any given time *does not change* with respect to space is called *uniform flow*. Mathematically, we have:

$$\left(\frac{\partial V}{\partial s} \right)_{t = \text{constant}} = 0$$

where,

∂V = Change in velocity, and

∂s = Displacement in any direction.

Example. Flow through a straight prismatic conduit (i.e. flow through a straight pipe of constant diameter).

Non-uniform flow. It is that type of flow in which the velocity at any given time *changes* with respect to space. Mathematically,

$$\left(\frac{\partial V}{\partial s} \right)_{t = \text{constant}} \neq 0$$

Example. (i) Flow through a non-prismatic conduit.

(ii) Flow around a uniform diameter pipe-bend or a canal bend.

5.3.3. One, Two and Three Dimensional Flows

One dimensional flow. It is that type of flow in which the flow parameter such as velocity is a function of time and one space co-ordinate only. Mathematically:

$$u = f(x), v = 0 \text{ and } w = 0$$

where u , v and w are velocity components in x , y and z directions respectively.

Example. Flow in a pipe where average flow parameters are considered for analysis.

Two dimensional flow. The flow in which the velocity is a function of time and two rectangular space coordinates is called *two dimensional flow*. Mathematically:

$$u = f_1(x, y)$$

$$v = f_2(x, y)$$

$$w = 0$$

Examples. (i) Flow between parallel plates of infinite extent.

(ii) Flow in the main stream of a wide river.

Three dimensional flow. It is that type of flow in which the velocity is a function of time and three mutually perpendicular directions. Mathematically:

$$u = f_1(x, y, z)$$

$$v = f_2(x, y, z)$$

$$w = f_3(x, y, z)$$

Examples. (i) Flow in a converging or diverging pipe or channel.

(ii) Flow in a prismatic open channel in which the width and the water depth are of the same order of magnitude.

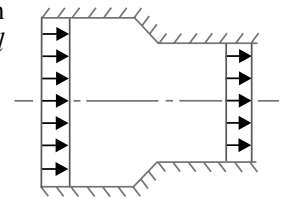


Fig. 5.2. One dimensional flow.

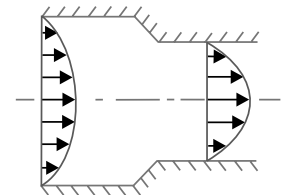


Fig. 5.3. Two dimensional flow.

5.3.4. Rotational and Irrotational Flows

Rotational flow. A flow is said to be *rotational* if the fluid particles while moving in the direction of flow *rotate* about their mass centres. *Flow near the solid boundaries is rotational.*

Example. Motion of liquid in a rotating tank.

Irrotational flow. A flow is said to be *irrotational* if the fluid particles while moving in the direction of flow *do not rotate* about their mass centres. Flow outside the boundary layer is generally considered irrotational.

Example. Flow above a drain hole of a stationary tank or a wash basin.

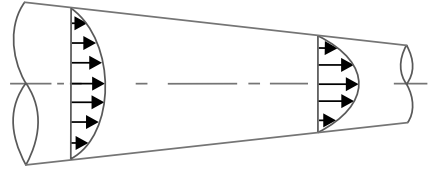


Fig. 5.4. Three dimensional flow.

Note. If the flow is irrotational as well as steady, it is known as **Potential flow**.

5.3.5. Laminar and Turbulent Flows

Laminar flow. A laminar flow is one in which *paths taken by the individual particles do not cross one another and move along well defined paths* (Fig. 5.5). This type of flow is also called *stream-line flow* or *viscous flow*.

- Examples.** (i) Flow through a capillary tube.
 (ii) Flow of blood in veins and arteries.
 (iii) Ground water flow.

Turbulent flow. A turbulent flow is that flow in which fluid *particles move in a zig zag way* (Fig. 5.6).

Example. High velocity flow in a conduit of large size. Nearly all fluid flow problems encountered in engineering practice have a turbulent character.

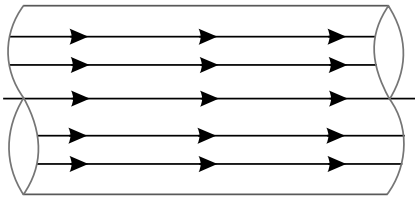


Fig. 5.5. Laminar flow.

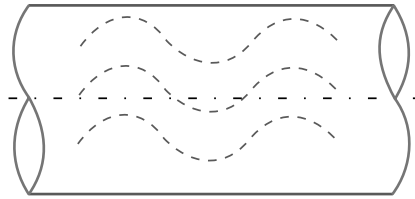


Fig. 5.6. Turbulent flow.

Laminar and turbulent flows are characterised on the basis of Reynolds number (refer to chapter 10).

- | | |
|---------------------------------------|--|
| For Reynolds number (Re) < 2000 | ... flow in pipes is <i>laminar</i> . |
| For Reynolds number (Re) > 4000 | ... flow in pipes is <i>turbulent</i> |
| For Re between 2000 and 4000 | ... flow in pipes <i>may be laminar or turbulent</i> . |

5.3.6. Compressible and Incompressible Flows

Compressible flow. It is that type of flow in which the *density (ρ) of the fluid changes* from point to point (or in other words *density is not constant for this flow*).

Mathematically: $\rho \neq \text{constant}$.

Example. Flow of gases through orifices, nozzles, gas turbines, etc.

Incompressible flow. It is that type of flow in which *density is constant for the fluid flow*. Liquids are generally considered flowing incompressibly.

Mathematically: $\rho = \text{constant}$.

Example. Subsonic aerodynamics.

5.4. TYPES OF FLOW LINES

Whenever a fluid is in motion, its innumerable particles move along certain lines depending upon the conditions of flow. Although flow lines are of several types, yet some important from subject point of view are discussed in the following subarticles.

5.4.1. Path line

A path line (Fig. 5.7) is the *path followed by a fluid particle in motion*. A path line shows the direction of particular particle as it moves ahead. In general, this is the curve in three-dimensional space. However, if the conditions are such that the flow is two-dimensional the curve becomes two-dimensional.

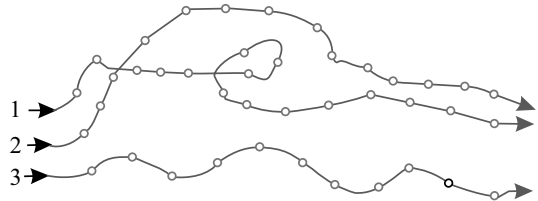


Fig. 5.7. Path lines.

5.4.2. Stream line

A **stream line** may be defined as an *imaginary line within the flow so that the tangent at any point on it indicates the velocity at that point*.

Equation of a stream line in a three-dimensional flow is given as:

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w} \quad \dots(5.20)$$

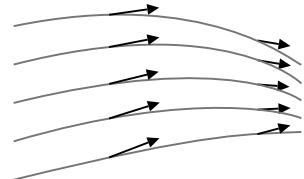


Fig. 5.8. Stream line.

Following *points* about streamlines are *worth noting*:

1. A streamline cannot intersect itself, nor two streamlines can cross.
2. There cannot be any movement of the fluid mass across the streamlines.
3. Streamline spacing varies inversely as the velocity; *converging of streamlines in any particular direction shows accelerated flow in that direction*.
4. Whereas a *path line* gives the path of *one particular particle* at successive instants of time, a *streamline* indicates the direction of a *number of particles* at the same instant.
5. The series of streamlines represent the flow pattern at an instant.

- In *steady flow*, the pattern of streamlines remains invariant with time. The path lines and streamlines will then be identical.

- In *unsteady flow*, the pattern of streamlines may or may not remain the same at the next instant.

5.4.3. Stream Tube

A **stream tube** is a *fluid mass bounded by a group of streamlines*. The contents of a stream tube are known as '*current filament*'.

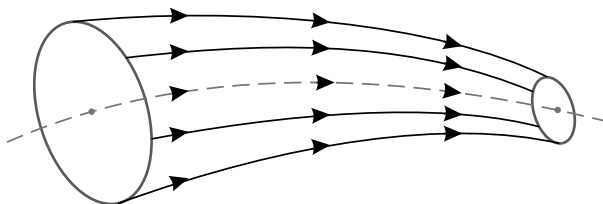


Fig. 5.9. Stream tube.

Examples of stream tube: Pipes and nozzles.

Following *points* about stream tube are *worth noting*:

1. The stream tube has finite dimensions.
2. As there is no flow perpendicular to stream lines, therefore, there is no flow across the surface (called *stream surface*) of the stream tube. The stream surface functions as if it were a solid wall.
3. The shape of a stream tube changes from one instant to another because of change in the position of streamlines.

5.4.4. Streak Line

The **streak line** is a curve which gives an instantaneous picture of the location of the fluid particles, which have passed through a given point.

Examples. (i) The path taken by smoke coming out of chimney (Fig. 5.10).

- (ii) In an experimental work to trace the motion of fluid particles, a coloured dye may be injected into the flowing fluid and the resulting coloured filament lines at a given location give the streak lines (Fig 5.11).

Note. In case of a *steady flow* there is no geometrical distinction between the streamlines, path lines and streak lines; they are coincident if they originate at the same point. For an *unsteady flow* (e.g. a person giving out whiff of smoke from a cigarette), the path, streak and stream lines are all different.

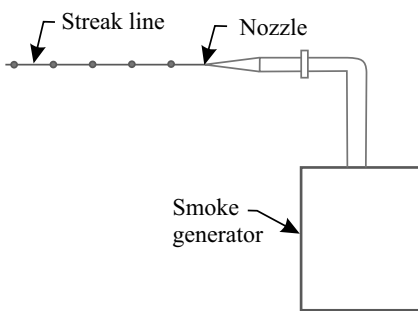


Fig. 5.10. Streak line of smoke issuing from a nozzle.

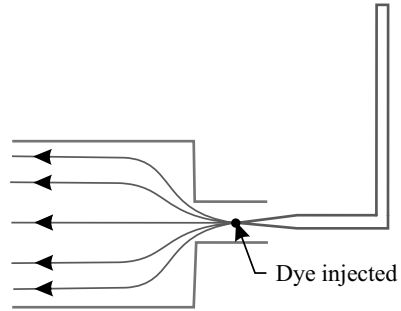


Fig. 5.11. Streak lines at $t = t_1$.

Example 5.1. In a fluid, the velocity field is given by

$$\mathbf{V} = (3x + 2y)\mathbf{i} + (2z + 3x^2)\mathbf{j} + (2t - 3z)\mathbf{k}$$

Determine:

- (i) The velocity components u, v, w at any point in the flow field;
- (ii) The speed at point $(1, 1, 1)$;
- (iii) The speed at time $t = 2s$ at point $(0, 0, 2)$.

Also classify the velocity field as steady, or unsteady, uniform or non-uniform and one, two or three dimensional.

Solution. Given: Velocity field, $\mathbf{V} = (3x + 2y)\mathbf{i} + (2z + 3x^2)\mathbf{j} + (2t - 3z)\mathbf{k}$

(i) **Velocity components:**

The velocity components are:

$$\mathbf{u} = 3x + 2y, \mathbf{v} = (2z + 3x^2), \mathbf{w} = (2t - 3z) \quad (\text{Ans.})$$

(ii) **Speed at point $(1, 1, 1)$, $\mathbf{V}_{(1,1,1)}$:**

Substituting $x = 1, y = 1, z = 1$ in the expressions for u, v and w , we have:

$$u = (3 + 2) = 5, v = (2 + 3) = 5, w = (2t - 3)$$

$$\therefore V^2 = u^2 + v^2 + w^2$$

$$\begin{aligned}
 &= 5^2 + 5^2 + (2t - 3)^2 \\
 &= 25 + 25 + 4t^2 - 12t + 9 \\
 &= 4t^2 - 12t + 59
 \end{aligned}$$

$$\therefore V_{(1,1,1)} = \sqrt{4t^2 - 12t + 59} \quad (\text{Ans.})$$

(iii) Speed at $t = 2s$ at point $(0, 0, 2)$:

Substituting $t = 2, x = 0, y = 0, z = 2$ in the expressions for u, v and w , we get:

$$u = 0, v = (2 \times 2 + 0) = 4, w = (2 \times 2 - 3 \times 2) = -2$$

$$\therefore V^2 = u^2 + v^2 + w^2 = 0 + 4^2 + (-2)^2 = 20$$

$$\text{or, } V_{(0,0,2)} = \sqrt{20} = 4.472 \text{ units } (\text{Ans.})$$

Velocity field, type:

(i) Since V at given (x, y, z) depends on t it is **unsteady flow**, (Ans.)

(ii) Since at given t velocity changes in the X direction it is **non-uniform flow**. (Ans.)

(iii) Since V depends on x, y, z it is three **dimensional flow**. (Ans.)

Example 5.2 Velocity for a two dimensional flow field is given by

$$V = (3 + 2xy + 4t^2) i + (xy^2 + 3t) j$$

Find the velocity and acceleration at a point $(1, 2)$ after 2 sec.

Solution. Given: Velocity field: $V = (3 + 2xy + 4t^2) i + (xy^2 + 3t) j$

Velocity at $(1, 2), V_{(1,2)}$:

Substituting $x = 1, y = 2, t = 2$ in the expression of velocity field, we get:

$$\begin{aligned}
 V &= (3 + 2 \times 1 \times 2 + 4 \times 2^2) i + (1 \times 2^2 + 3 \times 2) j \\
 &= (3 + 4 + 16) i + (4 + 6) j \\
 &= 23i + 10j
 \end{aligned}$$

$$\therefore V_{(1,2)} = \sqrt{23^2 + 10^2} = 25.08 \text{ units } (\text{Ans.})$$

Acceleration at point $(1, 2), a_{(1,2)}$:

$$\text{We know that: } a = \frac{dV}{dt} = \left(u \frac{\partial V}{\partial x} + v \frac{\partial V}{\partial y} \right) + \frac{\partial V}{\partial t}$$

$$\text{Also, } V = (3 + 2xy + 4t^2)i + (xy^2 + 3t)j \quad \dots (\text{Given})$$

$$\therefore \frac{\partial V}{\partial x} = 2yi + y^2 j,$$

$$\frac{\partial V}{\partial y} = 2xi + 2xyj, \text{ and}$$

$$\frac{\partial V}{\partial t} = 8ti + 3j$$

$$\begin{aligned}
 \therefore a &= (3 + 2xy + 4t^2) (2yi + y^2 j) + (xy^2 + 3t) (2xi + 2xyj) + (8ti + 3j) \\
 (\because u &= 3 + 2xy + 4t^2 \text{ and } v = xy^2 + 3t)
 \end{aligned}$$

Substituting the values, we get:

$$\begin{aligned}
 a &= (3 + 2 \times 1 \times 2 + 4 \times 2^2) (2 \times 2i + 2^2 j) + (1 \times 2^2 + 3 \times 2) \\
 &\quad (2 \times 1i + 2 \times 1 \times 2j) + (8 \times 2 \times i + 3j) \\
 &= (3 + 4 + 16) (4i + 4j) + (4 + 6) (2i + 4j) + (16i + 3j)
 \end{aligned}$$

$$\begin{aligned}
 &= 23(4i + 4j) + 10(2i + 4j) + (16i + 3j) \\
 &= 92i + 92j + 20i + 40j + 16i + 3j \\
 &= 128i + 135j
 \end{aligned}$$

$$\therefore a_{(1,2)} = \sqrt{128^2 + 135^2} = 186.03 \text{ units (Ans.)}$$

Example 5.3. Find the velocity and acceleration at a point (1, 2, 3) after 1 sec. for a three-dimensional flow given by $u = yz + t$, $v = xz - t$, $w = xy$ m/s.

Solution. Given: Three-dimensional flow field is given as:

$$u = yz + t, v = xz - t, w = xy \text{ m/s}$$

Velocity at a point 1, 2, 3 $V_{(1,2,3)}$:

Velocity at a point (1, 2, 3) after 1s is calculated as follows:

$$u = yz + t = 2 \times 3 + 1 = 7 \text{ m/s}, v = xz - t = 1 \times 3 - 1 = 2 \text{ m/s and}$$

$$w = xy = 1 \times 2 = 2 \text{ m/s.}$$

$$\begin{aligned}
 \therefore V_{(1,2,3)} &= 7i + 2j + 2k \\
 &= \sqrt{7^2 + 2^2 + 2^2} = 7.55 \text{ m/s}
 \end{aligned}$$

$$\text{Hence, } V_{(1,2,3)} = 7.55 \text{ m/s (Ans.)}$$

Acceleration, $a_{(1,2,3)}$:

Now,

$$V = (yz + t)i + (xz - t)j + xy k \text{ m/s}$$

Acceleration,

$$a = \frac{dV}{dt} = \left(u \frac{\partial V}{\partial x} + v \frac{\partial V}{\partial y} + w \frac{\partial V}{\partial z} \right) + \frac{\partial V}{\partial t}$$

$$a = (yz + t)(zi + yk) + (xz - t)(zi + xk) + xy(yi + xj) + (1i - 1j)$$

$$\begin{aligned}
 \therefore a_{(1,2,3)} &= 7(3j + 2k) + 2(3i + 1k) + 2(2i + 1j) + (1i - 1j) \\
 &= (21j + 14k) + (6i + 2k) + (4i + 2j) + (1i - 1j) \\
 &= (10i + 23j + 16k) + (1i - 1j)
 \end{aligned}$$

The convective acceleration components are: (10, 23, 16) m/s²

The local acceleration components are: (1, -1) m/s² along x and y directions.

The total acceleration of fluid particles at the points (1, 2, 3) is given by:

$$\begin{aligned}
 a_{(1,2,3)} &= \sqrt{(10 + 1)^2 + [23 + (-1)]^2 + 16^2} \\
 &= \sqrt{11^2 + 22^2 + 16^2} = 29.34 \text{ m/s}^2
 \end{aligned}$$

$$\text{Hence, } a_{(1,2,3)} = 29.34 \text{ m/s}^2 \text{ (Ans.)}$$

Example 5.4. The velocity along the centreline of a nozzle of length l is given by

$$V = 2t \left(1 - \frac{x}{2l} \right)^2$$

where V = velocity in m/s, t = time in seconds from commencement of flow, x = distance from inlet to nozzle. Calculate the local acceleration, convective acceleration and the total acceleration when $t = 6$ s, $x = 1$ m and $l = 1.6$ m.

$$\text{Solution. The velocity along the centreline of a nozzle, } V = 2t \left(1 - \frac{x}{2l} \right)^2 \quad \dots(\text{Given})$$

Local acceleration:

$$\text{Local acceleration} = \frac{\partial V}{\partial t} = 2 \left(1 - \frac{x}{2l} \right)^2$$

At $t = 6 \text{ s}$ and $x = 1 \text{ m}$,

$$\frac{\partial V}{\partial t} = 2 \left(1 - \frac{1}{2 \times 1.6} \right)^2 = 0.945 \text{ m/s}^2 \text{ (Ans.)}$$

Convective acceleration:

$$\begin{aligned} \text{Convective acceleration} &= V \frac{\partial V}{\partial x} \\ &= 2t \left(1 - \frac{x}{2l} \right)^2 \times 2t \times 2 \left(1 - \frac{x}{2l} \right) \left(-\frac{1}{2l} \right) \\ &= -\frac{4t^2}{l} \left(1 - \frac{x}{2l} \right)^3 \end{aligned}$$

At $t = 6 \text{ s}$ and $x = 1 \text{ m}$,

$$\begin{aligned} \text{Convective acceleration} &= -\frac{4 \times 6^2}{1.6} \left(1 - \frac{1}{2 \times 1.6} \right)^3 \\ &= -29.24 \text{ m/s}^2 \text{ (Ans.)} \end{aligned}$$

Total acceleration:

$$\begin{aligned} \text{Total acceleration} &= \text{Local acceleration} + \text{convective acceleration} \\ &= 0.945 + (-29.24) = -28.295 \text{ m/s}^2 \text{ (Ans.)} \end{aligned}$$

Example 5.5. A conical pipe diverges uniformly from 100 mm to 200 mm diameter over a length of 1 m. Determine the local and convective acceleration at the mid-section assuming

(i) Rate of flow is $0.12 \text{ m}^3/\text{s}$ and it remains constant;

(ii) Rate of flow varies uniformly from $0.12 \text{ m}^3/\text{s}$ to $0.24 \text{ m}^3/\text{s}$ in 5 sec., at $t = 2 \text{ sec}$.

Solution. Given:

Diameter at the inlet, $D_1 = 0.1 \text{ m}$.

Diameter at the outlet, $D_2 = 0.2 \text{ m}$

Length, $l = 1 \text{ m}$

Diameter at any distance x metres from the inlet,

$$\begin{aligned} D &= D_1 + \left(\frac{D_2 - D_1}{l} \right) \times x \\ &= 0.1 + \left(\frac{0.2 - 0.1}{1} \right) \times x \\ &= 0.1 + 0.1x = 0.1(1 + x) \end{aligned}$$

$\therefore$ Cross-sectional area,

$$\begin{aligned} A_x &= \frac{\pi}{4} \times D_x^2 = \frac{\pi}{4} \{0.1(1 + x)\}^2 \\ &= 0.00785 (1 + x)^2 \end{aligned}$$

$$\text{Velocity of flow, } u_x (= u) = \frac{Q}{A_x} = \frac{Q}{0.00785 (1 + x)^2}$$

$$\text{Velocity gradient, } \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left[\frac{Q}{0.00785 (1 + x)^2} \right] = \frac{-2Q}{0.00785 (1 + x)^3}$$

(i) Discharge $Q = 0.12 \text{ m}^3/\text{s} = \text{constant}$ (at any section)

$$\text{Acceleration} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x}$$

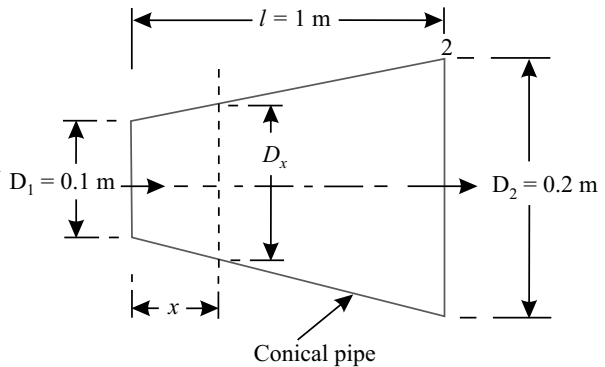


Fig. 5.12

(a) The local acceleration:

The local acceleration at *mid-section*

$$= \frac{\partial u}{\partial t} = 0, \text{ since the flow is steady (Ans.)}$$

(b) The convective acceleration:

The convective acceleration,

$$\begin{aligned} a_x &= u \frac{\partial u}{\partial x} = \frac{Q}{0.00785 (1+x)^2} \times \frac{-2Q}{0.00785 (1+x)^3} \\ &= \frac{-2Q^2}{(0.00785)^2 (1+x)^5} \end{aligned}$$

$\therefore$ The convective acceleration at *mid-section*,

$$\begin{aligned} (a_x)_{x=0.5\text{m}} &= \frac{-2 \times (0.12)^2}{(0.00785)^2 (1+0.5)^5} \\ &= -61.5 \text{ m/s}^2 \text{ (Ans.)} \end{aligned}$$

The $-ve$ sign indicates *decrease in velocity* along the direction of flow (this is so as the cross-sectional area is increasing)

(ii) Discharge Q varies w.r.t. time:

The discharge Q varies from $0.12 \text{ m}^3/\text{s}$ to $0.24 \text{ m}^3/\text{s}$ in 5 s .

At $t = 2 \text{ s}$, the discharge is,

$$Q = 0.12 + \frac{0.24 - 0.12}{5} \times 2 = 0.168 \text{ m}^3/\text{s}$$

(a) The local acceleration:

The local acceleration at *mid-section*,

$$\begin{aligned} &= \frac{\partial u}{\partial t} = \frac{\partial}{\partial t} \left[\frac{Q}{0.00785 (1+x)^2} \right] = \frac{1}{0.00785 (1+x)^2} \times \frac{\partial Q}{\partial t} \\ &= \frac{1}{0.00785 (1+0.5)^2} \times \left(\frac{0.168 - 0.12}{2} \right) \end{aligned}$$

[since discharge changes $0.12 \text{ m}^3/\text{s}$ to $0.168 \text{ m}^3/\text{s}$ in 2s]

$$= 1.36 \text{ m/s}^2 \text{ (Ans.)}$$

(b) The convective acceleration:

The convective acceleration at *mid-section*,

$$\begin{aligned} (a_x)_{x=0.5} &= \frac{-2Q^2}{(0.00785)^2 (1+x)^5} = \frac{-2 \times 0.168^2}{(0.00785)^2 (1+0.5)^5} \\ &= -120.6 \text{ m/s}^2 \text{ (Ans.)} \end{aligned}$$

Total acceleration along the main flow is,

$$\begin{aligned} (a)_{\text{total}} &= (a)_{\text{local}} + (a)_{\text{conv.}} \\ &= 1.36 - 120.6 = -119.24 \text{ m/s}^2 \text{ (Ans.)} \end{aligned}$$

Example 5.6. At entry to the pump intake the velocity is found to vary inversely as the square of radial distance from inlet to suction pipe. The velocity was found to be 0.6 m/s at a radial distance of 1.5 m . Calculate the acceleration of flow at radial distances of 0.5 m , 1.0 m and 1.5 m from the inlet. Consider the streamlines to be radial.

Solution. The distribution of velocity is given by the relation:

$$v = \frac{C}{r^2} \quad \dots(1)$$

(where, r = radial distance from the intake)

Since the streamlines are *radial*, normal acceleration is *zero*.

$$\text{Acceleration, } a = \frac{\partial v}{\partial t} + v \frac{\partial v}{\partial s}$$

But $\frac{\partial v}{\partial t} = 0$... the flow being steady (as the velocity is dependent only on the radial distance from intake).

$$\therefore a = v \cdot \frac{\partial v}{\partial s}$$

$$\text{Also, } v = \frac{C}{r^2} \quad \text{and} \quad \frac{\partial v}{\partial s} = \frac{\partial v}{\partial r}$$

(since, r is measured along the streamline)

$$\text{and, } \frac{\partial v}{\partial r} = -\frac{2C}{r^3}$$

$$\therefore a = v \frac{\partial v}{\partial s} = \frac{C}{r^2} \left(-\frac{2C}{r^3} \right) = -\frac{2C^2}{r^5} \quad \dots(2)$$

At $r = 1.5 \text{ m}$, $v = 0.6 \text{ m/s}$

Substituting these values in eqn. (1), we get:

$$0.6 = \frac{C}{1.5^2} \quad \text{or } C = 1.35 \text{ m}^3/\text{s}$$

Substituting, now, $C = 1.35$ in eqn. (2), we have:

$$a = -\frac{2 \times 1.35^2}{r^5} = -\frac{3.645}{r^5}$$

$$(i) \text{ Acceleration of flow at } r = 0.5 \text{ m} = -\frac{3.645}{(0.5)^5} = -116.64 \text{ m/s}^2 \text{ (Ans.)}$$

$$(ii) \text{ Acceleration of flow at } r = 1.0 \text{ m} = -\frac{3.645}{(1.0)^5} = -3.645 \text{ m/s}^2 \text{ (Ans.)}$$

$$(iii) \text{ Acceleration of flow at } r = 1.5 \text{ m} = -\frac{3.645}{(1.5)^5} = -0.48 \text{ m/s}^2 \text{ (Ans.)}$$

Example 5.7. For a three-dimensional flow the velocity distribution is given by $u = -x$, $v = 3 - y$ and $w = 3 - z$. What is the equation of a streamline passing through (1,2,2)?

Solution. Given: $u = -x$, $v = 3 - y$, $w = 3 - z$

...(velocity distribution)

Equation of a streamline passing through (1, 2, 2):

The streamlines are defined by:

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

Substituting for u, v and w , we get:

$$\begin{aligned} \frac{dx}{-x} &= \frac{dy}{3-y} = \frac{dz}{3-z} \\ (i) \quad (ii) \quad (iii) \end{aligned}$$

Considering the expressions (i) and (ii) and integrating, we get:

$$\begin{aligned} \int \frac{dx}{-x} &= \int \frac{dy}{(3-y)} \\ &= -\log_e x = -\log_e (3-y) + C_1 \end{aligned}$$

(where, C_1 = constant of integration).

Since the streamline passes through $x = 1, y = 2 \quad \therefore C_1 = 0$

$$\therefore (x)^{-1} = (3-y)^{-1} \quad \text{or} \quad x = (3-y) \quad \dots(1)$$

Considering the expressions (i) and (iii), and integrating, we get:

$$\int \frac{dx}{-x} = \int \frac{dz}{3-z}$$

$$\text{or,} \quad -\log_e x = -\log (3-z) + C_2$$

(where C_2 = constant of integration)

Since the streamline passes through $x = 1, z = 2 \quad \therefore C_2 = 0$

$$\begin{aligned} \therefore x^{-1} &= (3-z)^{-1} \\ \text{or,} \quad x &= (3-z) \quad \dots(2) \end{aligned}$$

From (1) and (2), the equation of the streamline passing through (1, 2, 2) is given as:

$$x = (3-y) = (3-z) \text{ (Ans.)}$$

Example 5.8. Obtain the equation to the streamlines for the velocity field given as:

$$V = 2x^3i - 6x^2yj$$

Solution. Given: Velocity field, $V = 2x^3i - 6x^2yj$

Here, $u = 2x^3, v = 6x^2y$

The streamlines in two dimensions are defined by:

$$\frac{dx}{u} = \frac{dy}{v}$$

$$\text{or,} \quad \frac{dy}{dx} = \frac{v}{u} = \frac{-6x^2y}{2x^3} = \frac{-3y}{x}$$

Separating the variables, we have:

$$\frac{dy}{y} = \frac{-3dx}{x}$$

Integrating, we get:

$$\ln(y) = -3\ln(x) + C_1$$

$$\text{or,} \quad \ln(y) + 3\ln(x) = C_1$$

$$\text{or,} \quad yx^3 = C \text{ (Ans.)}$$

Note. The streamlines in the first quadrant can be sketched by giving different values for the constant

$$C \left(y = \frac{C}{x^3} \right).$$

Example 5.9. For the following flows find the equation of the streamline passing through (2,2):

(i) $V = 3xi - 3yj$

(ii) $V = -y^2i - 6xj$

Solution. Equation of the streamline passing through (2, 2):

(i) $V = 3xi - 3yj$

$$u = 3x \text{ and } v = -3y$$

The equation of a streamline in two-dimensional flow is given as:

$$\frac{dx}{u} = \frac{dy}{v}$$

or,
$$\frac{dx}{3x} = -\frac{dy}{3y}$$

Integrating both sides, we get:

$$\int \frac{dx}{3x} = -\int \frac{dy}{3y}$$

$$\frac{1}{3} \ln(x) = -\frac{1}{3} \ln(y) + \frac{1}{3} \ln(C)$$

where, C is constant.

or, $\ln(xy) = \ln(C) \quad \text{or} \quad xy = C$

For the streamline passing through (2, 2),

$$C = 2 \times 2 = 4$$

Hence, the required streamline equation is: $xy = 4$ (Ans.)

(ii) $V = -y^2i - 6xj$

$$u = -y^2 \text{ and } v = -6x$$

$$-\frac{dx}{y^2} = -\frac{dy}{6x} \quad \text{or} \quad 6x \, dx = y^2 \, dy$$

$$\int 6x \, dx = \int y^2 \, dy$$

or,
$$\frac{6x^2}{2} = \frac{y^3}{3} + C$$

or,
$$3x^2 - \frac{y^3}{3} = C$$

Putting, $x = 2, y = 2$, we get:

$$3 \times (2)^2 - \frac{(2)^3}{3} = C \quad \text{or} \quad C = \frac{28}{3}$$

Hence the equations of the required streamline is:

$$3x^2 - \frac{y^3}{3} = \frac{28}{3}$$

$$9x^2 - y^3 = 28 \text{ (Ans.)}$$

Example 5.10. The velocity vector in a flow is given by:

$$V = 3xi + 4yj - 7zk$$

Determine the equation passing through a point $L(1, 2, 3)$.

Solution. The equation of a streamline is given as:

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

Here, $u = 3x$, $v = 4y$, and $w = -7z$

$$\therefore \frac{dx}{3x} = \frac{dy}{4y} = -\frac{dz}{7z}$$

Considering equations involving x and y , on integration we get:

$$\frac{1}{3} \ln(x) = \frac{1}{4} \ln(y) + \ln(C'_1) \text{ where } C'_1 = \text{a constant}$$

$$\text{or, } y = C_1 x^{4/3} \quad \dots(i)$$

where, C_1 is another constant:

Similarly, by considering equations with x and z and on integration, we have:

$$\frac{1}{3} \ln(x) = -\frac{1}{7} \ln(z) + \ln(C'_2), \text{ where } C'_2 = \text{a constant}$$

$$\text{or, } z = \frac{C_2}{x^{7/3}} \quad \dots(ii)$$

where, C_2 is another constant.

Inserting the coordinates of the point $L(1, 2, 3)$, we get:

$$\text{From eqn. (i)} \quad C_1 = \frac{y}{(x)^{4/3}} = \frac{2}{(1)^{4/3}} = 2$$

$$\text{From eqn. (ii)} \quad C_2 = zx^{7/3} = 3 \times (1)^{7/3} = 3$$

Hence, the streamline passing through L is given by:

$$y = 2x^{4/3} \text{ and } z = \frac{3}{x^{7/3}} \quad (\text{Ans.})$$

5.5. RATE OF FLOW OR DISCHARGE

Rate of flow (or discharge) is defined as the quantity of a liquid flowing per second through a section of pipe or a channel. It is generally denoted by Q . Let us consider a liquid flowing through a pipe.

Let, A = Area of cross-section of the pipe, and
 V = Average velocity of the liquid.

$$\therefore \text{Discharge, } Q = \text{Area} \times \text{average velocity i.e., } Q = A.V \quad \dots(5.21)$$

If area is in m^2 and velocity is in m/s , then the discharge,

$$Q = \text{m}^2 \times \text{m/s} = \text{m}^3/\text{s} = \text{cumecs.}$$

5.6. CONTINUITY EQUATION

The **continuity equation** is based on the principle of conservation of mass. It states as follows:

“If no fluid is added or removed from the pipe in any length then the mass passing across different sections shall be same.”

Consider two cross-sections of a pipe as shown in Fig 5.13

Let, A_1 = Area of the pipe at section 1–1,
 V_1 = Velocity of the fluid at section 1–1,

ρ_1 = Density of the fluid at section 1-1,

and A_2, V_2, ρ_2 are corresponding values at sections 2-2.

The total quantity of fluid passing through section 1-1 = $\rho_1 A_1 V_1$

and, the total quantity of fluid passing through section 2-2 = $\rho_2 A_2 V_2$

From the law of conservation of mass (theorem of continuity), we have:

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad \dots(5.22)$$

Eqn. (5.22) is applicable to the compressible as well as incompressible fluids and is called **Continuity Equation**. In case of *incompressible fluids*, $\rho_1 = \rho_2$ and the continuity eqn. (5.21) reduces to:

$$A_1 V_1 = A_2 V_2 \quad \dots(5.23)$$

Example 5.11. The diameters of a pipe at the sections 1-1 and 2-2 are 200 mm and 300 mm respectively. If the velocity of water flowing through the pipe at section 1-1 is 4 m/s, find:

- (i) Discharge through the pipe, and
- (ii) Velocity of water at section 2-2

Solution. Diameter of the pipe at section 1-1,

$$D_1 = 200 \text{ mm} = 0.2 \text{ m}$$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} D_1^2 = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$$

$$\text{Velocity, } V_1 = 4 \text{ m/s}$$

Diameter of the pipe at section 2-2,

$$D_2 = 300 \text{ mm}$$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} D_2^2 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$$

- (i) Discharge through the pipe, Q:

Using the relation,

$$Q = A_1 V_1, \text{ we have:}$$

$$Q = 0.0314 \times 4 = \mathbf{0.1256 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

- (ii) Velocity of water at section 2-2, V_2 :

Using the relation,

$$A_1 V_1 = A_2 V_2, \text{ we have:}$$

$$V_2 = \frac{A_1 V_1}{A_2} = \frac{0.0314 \times 4}{0.0707}$$

$$= \mathbf{1.77 \text{ m/s (Ans.)}}$$

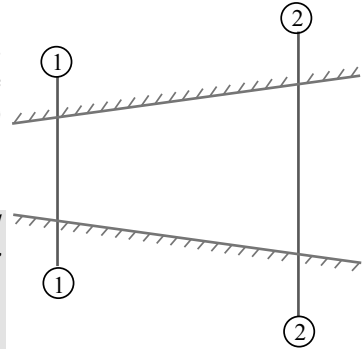


Fig. 5.13. Fluid flow through a pipe.

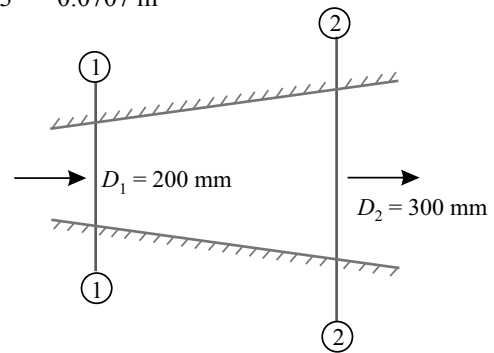


Fig. 5.14

Example 5.12. A pipe (1) 450 mm in diameter branches into two pipes (2 and 3) of diameters 300 mm and 200 mm respectively as shown in Fig. 5.15. If the average velocity in 450 mm diameter pipe is 3 m/s find:

- (i) Discharge through 450 mm diameter pipe;
- (ii) Velocity in 200 mm diameter pipe if the average velocity in 300 mm pipe is 2.5 m/s.

Solution. Diameter, $D_1 = 450 \text{ mm} = 0.45 \text{ m}$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.45^2 = 0.159 \text{ m}^2$$

Velocity, $V_1 = 3 \text{ m/s}$
 Diameter, $D_2 = 300 \text{ mm} = 0.3 \text{ m}$
 $\therefore$ Area, $A_2 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$
 Velocity, $V_2 = 2.5 \text{ m/s}$
 Diameter, $D_3 = 200 \text{ mm} = 0.2 \text{ m}$
 Area, $A_3 = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$

(i) Discharge through pipe (1) Q_1 :

Using the relation, $Q_1 = A_1 V_1 = 0.159 \times 3$
 $= 0.477 \text{ m}^3/\text{s}$ (Ans.)

(ii) Velocity in pipe of diameter 200 mm i.e. V_3 :

Let Q_1 , Q_2 and Q_3 be the discharge in pipes 1, 2 and 3 respectively.

Then, according to continuity equation:

$$Q_1 = Q_2 + Q_3 \quad \dots(i)$$

where, $Q_1 = 0.477 \text{ m}^3/\text{s}$ (calculated earlier)

and, $Q_2 = A_2 V_2 = 0.0707 \times 2.5 = 0.1767 \text{ m}^3/\text{s}$

$$\therefore 0.477 = 0.1767 + Q_3 \quad [\text{from eq. (i)}]$$

or, $Q_3 = 0.477 - 0.1767 \approx 0.3 \text{ m}^3/\text{s}$

But, $Q_3 = A_3 V_3$

$$\therefore V_3 = \frac{Q_3}{A_3} = \frac{0.3}{0.0314} = 9.55 \text{ m/s}$$

i.e. $V_3 = 9.55 \text{ m/s}$ (Ans.)

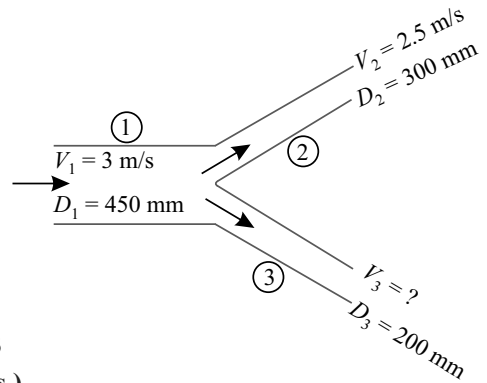


Fig. 5.15

5.7. CONTINUITY EQUATION IN CARTESIAN CO-ORDINATES

Consider a fluid element (control volume) – parallelepiped with sides dx , dy and dz as shown in Fig. 5.16.

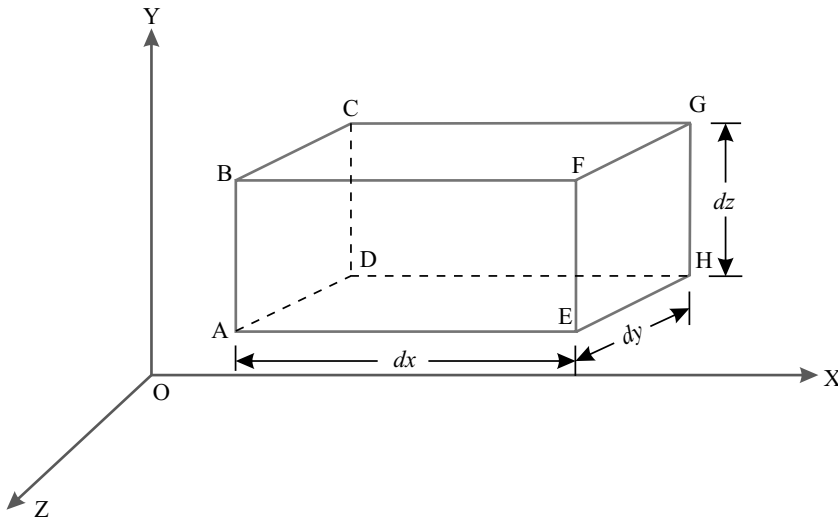


Fig. 5.16. Fluid element in three-dimensional flow.

Let, ρ = Mass density of the fluid at a particular instant;

u, v, w = Components of velocity of flow entering the three faces of the parallelopiped.

Rate of mass of fluid *entering* the face ABCD (*i.e.* fluid *influx*).

$$\begin{aligned} &= \rho \times \text{velocity in } X\text{-direction} \times \text{area of ABCD} \\ &= \rho u dy dz \end{aligned} \quad \dots(i)$$

Rate of mass of fluid *leaving* the face EFGH (*i.e.* fluid *efflux*).

$$= \rho u dy dz + \frac{\partial}{\partial x} (\rho u dy dz) dx \quad \dots(ii)$$

The gain in mass per unit time due to flow in the X-direction is given by the difference between the fluid influx and fluid efflux.

$\therefore$ Mass accumulated per unit time due to flow in X-direction

$$\begin{aligned} &= \rho u dy dz - \left[\rho u + \frac{\partial}{\partial x} (\rho u) dx \right] dy dz \\ &= - \frac{\partial}{\partial x} (\rho u) dx dy dz \end{aligned} \quad \dots(iii)$$

Similarly, the gain in fluid mass per unit time in the parallelopiped due to flow in Y and Z-directions

$$= - \frac{\partial}{\partial y} (\rho v) dx dy dz \quad (\text{in Y-direction}) \quad \dots(iv)$$

$$= - \frac{\partial}{\partial z} (\rho w) dx dy dz \quad (\text{in Z-direction}) \quad \dots(v)$$

The total (or net) gain in fluid mass per unit for fluid along three co-ordinate axes

$$= - \left[\frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) \right] dx dy dz \quad \dots(vi)$$

Rate of change of mass of the parallelopiped (control volume)

$$= \frac{\partial}{\partial t} (\rho dx dy dz) \quad \dots(vii)$$

Equations (vi) and (vii), we get:

$$- \left[\frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) \right] dx dy dz = \frac{\partial}{\partial t} (\rho dx dy dz)$$

Simplification and rearrangement of terms would reduce the above expression to:

$$\frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) + \frac{\partial \rho}{\partial t} = 0 \quad \dots(5.24)$$

This eqn. (5.24) is the general equation of continuity in three-dimensions and is applicable to *any type of flow* and for any fluid whether *compressible or incompressible*.

For *steady flow* $\left(\frac{\partial \rho}{\partial t} = 0 \right)$ *incompressible fluids* ($\rho = \text{constant}$) the equation reduces to:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \dots(5.25)$$

For *two dimensional flow*, eqn. (5.25) reduces to:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (\because w = 0)$$

For one dimensional flow, say in X -direction, eqn. (5.25) takes the form:

$$\frac{\partial u}{\partial x} = 0 \quad (\because v = 0, w = 0)$$

Integrating with respect to x , we get:

$$u = \text{constant} \quad \dots(5.26)$$

If the area of flow is a then the rate of flow is

$$Q = a.u = \text{constant for steady flow}$$

which is the same eqn. (5.23) and states that if area of flow a is constant the velocity of flow u will also be constant.

5.8. EQUATION OF CONTINUITY IN POLAR COORDINATES

Consider a fluid element LMST as shown in Fig. 5.17. The sides of the element has the following dimensions.

$$LT = MS = dr; LM = rd\theta \text{ and } ST = (r + dr)d\theta$$

Let, V_r = Component of the velocity in the radial direction, and
 V_θ = Component of the velocity in the tangential direction.

Further, let thickness of the element perpendicular to the plane of paper be *unity*. As the fluid flows through the element, changes will place in its velocity as well as in the density.

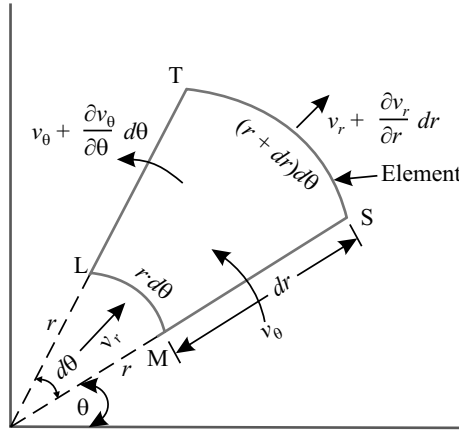


Fig. 5.17. Control volume for equation of continuity in polar coordinates.

Flow in radial direction:

Mass of fluid *entering the face LM* during time dt is given by:

$$\begin{aligned} \text{Fluid influx} &= \text{Density} \times (\text{velocity} \times \text{area}) \times \text{time} \\ &= \rho \times (v_r \times rd\theta) \times dt \end{aligned}$$

Mass of fluid *leaving the face ST* during the same time dt is given by:

$$\text{Fluid efflux} = \left[\rho v_r + \frac{\partial}{\partial r} (\rho v_r) dr \right] (r + dr) d\theta .dt$$

Mass accumulated in the element because of flow in radial direction

$$= \text{Fluid influx} - \text{fluid efflux}$$

$$\begin{aligned}
&= \rho \times (v_r \times r d\theta) \times dt - \left[\rho v_r + \frac{\partial}{\partial r} (\rho v_r) dr \right] (r + dr) d\theta dt \\
&= - \left[\rho v_r dr d\theta + \frac{\partial}{\partial r} (\rho v_r) dr \cdot d\theta \right] dt \\
&\quad [\text{Neglecting terms containing } (dr)^2]
\end{aligned}$$

Flow in tangential direction:

The mass accumulated due to flow in the tangential direction (by a similar treatment as discussed earlier).

$$= \left[\rho v_\theta dr - \left\{ \rho v_\theta + \frac{\partial}{\partial \theta} (\rho v_\theta) d\theta \right\} dr \right] dt = - \frac{\partial}{\partial \theta} (\rho v_\theta) dr d\theta dt$$

$\therefore$ Total gain in fluid mass

$$\begin{aligned}
&= - \left[\rho v_r dr d\theta + \frac{\partial}{\partial r} (\rho v_r) dr r d\theta + \frac{\partial}{\partial \theta} (\rho v_\theta) dr d\theta \right] dt \\
&= - \left[\rho v_r + \frac{\partial}{\partial r} (\rho v_r) r + \frac{\partial}{\partial \theta} (\rho v_\theta) \right] dr d\theta dt \quad \dots(i)
\end{aligned}$$

Also, the rate of change of fluid mass in the element LMST

$$\begin{aligned}
&= \frac{\partial}{\partial t} (\text{Density} \times \text{Volume}) dt \\
&= \frac{\partial}{\partial t} \left[\rho \times \frac{rd\theta + (r + dr)d\theta}{2} dr \right] dt \approx \frac{\partial}{\partial t} (\rho r d\theta dr) dt \quad \dots(ii)
\end{aligned}$$

As per law of conservation of mass:

The total gain in mass = The rate of change of fluid mass in the element LMST

$$\therefore - \left[\rho v_r + \frac{\partial}{\partial r} (\rho v_r) r + \frac{\partial}{\partial \theta} (\rho v_\theta) \right] dr d\theta dt = \frac{\partial}{\partial t} (\rho r d\theta dr dt)$$

$$\text{or, } \left[\rho v_r + \frac{\partial}{\partial r} (\rho v_r) r + \frac{\partial}{\partial \theta} (\rho v_\theta) \right] dr d\theta + \frac{\partial}{\partial t} (\rho r d\theta dr) = 0$$

For steady and compressible flow, $\frac{\partial}{\partial t} \rho r d\theta dr = 0$

$$\therefore \left[\rho v_r + \frac{\partial}{\partial r} (\rho v_r) r + \frac{\partial}{\partial \theta} (\rho v_\theta) \right] dr d\theta = 0 \quad \dots(5.27)$$

Further, for *incompressible flow*, $\rho = \text{constant}$.

$$\therefore v_r + \frac{\partial}{\partial r} (v_r) r + \frac{\partial}{\partial \theta} (v_\theta)$$

$$\text{or, } \frac{v_r}{r} + \frac{\partial v_r}{\partial r} + \frac{\partial v_\theta}{r \partial \theta} = 0 \quad \dots(5.28)$$

Example 5.13. Determine which of the velocity component sets given below satisfy the equation of continuity:

(i) $u = A \sin xy$

$$v = -A \sin xy$$

$$(ii) u = x + y$$

$$v = x - y$$

$$(iii) u = 2x^2 + 3y$$

$$v = -2xy + 3y^3 + 3zy$$

$$w = -\frac{3}{2}z^2 - 2xz - 6yz$$

Solution. (i) $u = A \sin xy$; $v = -A \sin xy$

$$\frac{\partial u}{\partial x} = Ay \cos xy; \frac{\partial v}{\partial y} = -Ax \cos xy$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = Ay \cos xy - Ax \cos xy \neq 0$$

i.e. Continuity equation is **not satisfied**. (Ans.)

$$(ii) u = x + y; v = x - y$$

$$\frac{\partial u}{\partial x} = 1; \frac{\partial v}{\partial y} = 1$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 1 + 1 = 2 \neq 0$$

i.e. Continuity equation is **satisfied** (Ans.)

$$(iii) u = 2x^2 + 3y; v = -2xy + 3y^3 + 3zy; w = -\frac{3}{2}z^2 - 2xz - 6yz$$

$$\frac{\partial u}{\partial x} = 4x; \frac{\partial v}{\partial y} = -2x + 9y^2 + 3z; \frac{\partial w}{\partial z} = -3z - 2x - 6y$$

$$\text{Hence, } \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 4x - 2x + 9y^2 + 3z - 3z - 2x - 6y \neq 0$$

i.e. Continuity equation is **not satisfied** (Ans.)

Example 5.14. Calculate the unknown velocity component in the following, so that the equation of continuity is satisfied.

$$(i) u = Ae^x$$

$$v = ?$$

$$(ii) u = A \ln \left(\frac{x}{l} \right)$$

$$v = ?$$

$$(iii) u = ?$$

$$v = Axy$$

Solution.

$$(i) u = Ae^x; v = ?$$

$$\frac{\partial u}{\partial x} = Ae^x = -\frac{\partial v}{\partial y}$$

$$v = \int -Ae^x dy = -Ae^x y + f(x) \quad (\text{Ans.})$$

$$(ii) u = -A \ln \left(\frac{x}{l} \right); v = ?$$

$$\frac{\partial u}{\partial x} = -\frac{A}{(x/l)} \times \frac{1}{l} = -\frac{A}{x} = -\frac{\partial v}{\partial y}$$

$$v = \int \frac{A}{x} dy = \frac{Ay}{x} + f(x) \text{ (Ans.)}$$

(iii) $u = ?$; $v = Axy$

$$\frac{\partial v}{\partial y} = Ax = -\frac{\partial u}{\partial x}$$

$$u = \int -Ax dx = -\frac{Ax^2}{2} + f(y) \text{ (Ans.)}$$

Example 5.15. In three-dimensional incompressible third flow, the velocity components in x and y -directions are:

$$u = x^2 + y^2z^3; \quad v = -(xy + yz + zx)$$

Use continuity equation to evaluate an expression for the velocity component w in the z -direction.

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Solution. The continuity equation for a steady, three-dimensional incompressible fluid flow is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \dots(i)$$

$$u = x^2 + y^2z^3; \quad v = -(xy + yz + zx)$$

$$\frac{\partial u}{\partial x} = 2x; \quad \frac{\partial v}{\partial y} = -(x + z)$$

Substituting these values in eqn. (i), we get:

$$2x - (x + z) + \frac{\partial w}{\partial z} = 0$$

$$\text{or,} \quad \frac{\partial w}{\partial z} = -x + z$$

Integrating w.r.t. z we have:

$$w = -xz + \frac{z^2}{2} + C$$

where C is a constant of integration which should be independent of z but may be function of x and/or y i.e. $C = f(x, y)$

$$\therefore \quad w = -x + \frac{z^2}{2} + f(x, y) \text{ (Ans.)}$$

Example 5.16. Given $u = \ln(y^2 + z^2)$ and $w = \ln(x^2 + y^2)$. What is the most general form of v so that the flow is possible for a steady three-dimensional incompressible flow?

Solution. $u = \ln(y^2 + z^2)$; $w = \ln(x^2 + y^2)$

$$\frac{\partial u}{\partial x} = 0; \quad \frac{\partial w}{\partial z} = 0$$

Substituting these values in continuity equation, we get:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\frac{\partial v}{\partial y} = 0$$

Upon integration w.r.t. z , we get:

$$v = f(x, z)$$

By symmetry, one of the values of velocity component could be

$$v = \ln(x^2 + z^2) \text{ (Ans.)}$$

Example 5.17. For an incompressible fluid the velocity components are: $u = x^3 - y^3 - z^2 x$, $v = y^3 - z^3$, $w = -3x^2 z - 3y^2 z + \frac{z^3}{3}$. Determine whether the continuity equation is satisfied.

Solution. Given: $u = x^3 - y^3 - z^2 x$, $v = y^3 - z^3$, $w = -3x^2 z - 3y^2 z + \frac{z^3}{3}$... velocity components

$$\text{Now, } \frac{\partial u}{\partial x} = 3x^2 - z^2$$

$$\frac{\partial v}{\partial y} = 3y^2$$

$$\frac{\partial w}{\partial z} = -3x^2 - 3y^2 + z^2$$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = (3x^2 - z^2) + 3y^2 + (-3x^2 - 3y^2 + z^2) = 0$$

Hence, the **continuity equation is satisfied.** (Ans.)

Example 5.18. In a three-dimensional incompressible flow, the velocity components in y and z directions are $v = ax^3 - by^2 + cz^2$; $w = bx^3 - cy^2 + az^2 x$. Determine the missing component of velocity distribution such that continuity equation is satisfied.

Solution. Given: $v = ax^3 - by^2 + cz^2$, and
 $w = bx^3 - cy^2 + az^2 x$

Missing component, u :

The continuity equation for an incompressible fluid flow is given as:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \dots(i)$$

From the given velocity components:

$$\frac{\partial v}{\partial y} = -2by; \quad \frac{\partial w}{\partial z} = 2azx$$

Substituting these values in eqn. (i), we get:

$$\frac{\partial u}{\partial x} - 2by + 2azx = 0$$

$$\text{or, } \frac{\partial u}{\partial x} = 2by - 2azx$$

Integrating w.r.t. x , we get:

$$u = 2byx - 2az \frac{x^2}{2} + C \text{ (Ans.)}$$

[where $C = f(y, z)$, the exact value will be known if the boundary conditions are known].

The constant of integration C is either a numerical constant or a function which is independent of x . If this constant is *omitted*, the velocity component may be expressed as:

$$u = 2byx - azx^2 \text{ (Ans.)}$$

Example 5.19. The velocity components in x and y directions are given as $u = 2xy^3/3 - x^2y$ and $v = xy^2 - 2yx^3/3$. Indicate whether the given velocity distribution is:

(i) A possible field of flow;

(ii) Not a possible field of flow.

[UPSC Exam.]

Solution. Given. $u = 2xy^3/3 - x^2y$, $v = xy^2 - 2yx^3/3$

...Velocity components

A possible flow field (two-dimensional) must satisfy the continuity equation.

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \dots(i)$$

$$\text{Now,} \quad \frac{\partial u}{\partial x} = \frac{2}{3}y^3 - 2xy, \quad \frac{\partial v}{\partial y} = 2xy - \frac{2}{3}x^3$$

Substituting these values in eqn. (i), we get:

$$\left(\frac{2}{3}y^3 - 2xy \right) + \left(2xy - \frac{2}{3}x^3 \right) = \frac{2}{3}(y^3 - x^3)$$

Since the continuity equation is *not* satisfied, the given velocity components, therefore, **do not represent a possible flow field. (Ans.)**

Example 5.20. In an incompressible flow, the velocity vector is given by:

$$V = (6xt + yz^2)i + (3t + xy^2)j + (xy - 2xyz - 6tz)k$$

(i) Verify whether the continuity equation is satisfied.

(ii) Determine the acceleration vector at point $L(2, 2, 2)$ at $t = 2.0$.

Solution. (i) $V = (6xt + yz^2)i + (3t + xy^2)j + (xy - 2xyz - 6tz)k \quad \dots(\text{Given})$
 $= ui + vj + wk$

$$u = 6xt + yz^2, \quad \frac{\partial u}{\partial x} = 6t$$

$$v = 3t + xy^2, \quad \frac{\partial v}{\partial y} = 2xy$$

$$w = xy - 2xyz - 6tz, \quad \frac{\partial w}{\partial z} = -2xy - 6t$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 6t + 2xy - 2xy - 6t = 0$$

Hence, the **continuity equation is satisfied (Ans.)**

(ii) Acceleration, $a = a_x i + a_y j + a_z k$

$$a_x = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}$$

$$= 6x + (6xt + yz^2)(6t) + (3t + xy^2)(z^2) + (xy - 2xyz - 6tz)(2yz)$$

At point $L(2, 2, 2)$ and at $t = 2$,

$$\begin{aligned} a_x &= 6 \times 2 + (6 \times 2 \times 2 + 2 \times 2^2)(6 \times 2) + (3 \times 2 + 2 \times 2^2)(2^2) \\ &\quad + (2 \times 2 - 2 \times 2 \times 2 \times 2 - 6 \times 2 \times 2)(2 \times 2 \times 2) \\ &= 12 + (32)(12) + (14)(4) + (-36)(8) = 164 \text{ units} \end{aligned}$$

$$a_y = \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z}$$

$$= 3 + (6xt + yz^2)(y^2) + (3t + xy^2)(2xy) + (xy - 2xyz - 6tz)(0)$$

At point $L(2, 2, 2)$ and at $t = 2$

$$\begin{aligned} a_y &= 3 + (6 \times 2 \times 2 + 2 \times 2^2) (2^2) + (3 \times 2 + 2 \times 2^2) (2 \times 2 \times 2) \\ &\quad + (2 \times 2 - 2 \times 2 \times 2 - 6 \times 2 \times 2) (0) \\ &= 3 + (32) (4) + (14) (8) = 243 \text{ units.} \end{aligned}$$

Similarly,

$$\begin{aligned} a_z &= \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \\ &= -6z + (6xt + yz^2) (y - 2yz) + (3t + xy^2) (x - 2xz) \\ &\quad + (xy - 2xyz - 6tz) (-2xy - 6t) \end{aligned}$$

At point $L(2, 2, 2)$ and at $t = 2$,

$$\begin{aligned} a_z &= -6 \times 2 + (6 \times 2 \times 2 + 2 \times 2^2) (2 - 2 \times 2 \times 2) + (3 \times 2 + 2 \times 2^2) \\ &\quad (2 - 2 \times 2 \times 2) + (2 \times 2 - 2 \times 2 \times 2 - 6 \times 2 \times 2) (-2 \times 2 \times 2 - 6 \times 2) \\ &= -12 + (32) (-6) + (14) (-6) + (-36) (-20) = 432 \text{ units.} \end{aligned}$$

Hence at $L(2, 2, 2)$ and at $t = 2$,

$$a = a_x i + a_y j + a_z k$$

or, $\mathbf{a} = 164 \mathbf{i} + 243 \mathbf{j} + 432 \mathbf{k} \text{ (Ans.)}$

Example 5.21. A two-dimensional incompressible flow in cylindrical polar coordinates is given by:

$$v_r = 2r \sin \theta \cos \theta; v_\theta = -2r \sin^2 \theta$$

Determine whether these velocity components represent a physically possible flow field.

Solution. The continuity equation for a steady, two-dimensional incompressible flow is

$$\frac{v_r}{r} + \frac{\partial v_r}{\partial r} + \frac{\partial v_\theta}{r \partial \theta} = 0 \quad \dots [\text{Eqn. (5.28)}]$$

From the given velocity components, we have:

$$\frac{\partial v_r}{\partial r} = \frac{\partial}{\partial r} (2r \sin \theta \cos \theta) = 2 \sin \theta \cos \theta$$

$$\frac{\partial v_\theta}{\partial \theta} = \frac{\partial}{\partial \theta} (-2r \sin^2 \theta) = -4r \sin \theta \cos \theta$$

Inserting these values in the above equation, we get:

$$\frac{2r \sin \theta \cos \theta}{r} + 2 \sin \theta \cos \theta - \frac{4r \sin \theta \cos \theta}{r} = 0$$

or $2 \sin \theta \cos \theta + 2 \sin \theta \cos \theta - 4 \sin \theta \cos \theta = 0$

i.e., $L. H. S. = 0$

Thus the continuity equation is satisfied and hence the **flow is physically possible. (Ans.)**

Example 5.22. The tangential component of velocity in a two-dimensional flow of incompressible fluid is

$$v_\theta = -\frac{C \sin \theta}{r^2}$$

where C is a constant.

(i) Using continuity equation, determine the expression for radial velocity v_r .

(ii) Find the magnitude and direction of resultant velocity.

Solution. Given: $v_\theta = -\frac{C \sin \theta}{r^2}$

(i) Expression for v_r :

The continuity equation for a two-dimensional, steady incompressible flow is

$$\frac{v_r}{r} + \frac{\partial v_r}{\partial r} + \frac{\partial v_\theta}{r \partial \theta} = 0 \quad [\text{Eqn. (5.28)}]$$

$$\text{or} \quad \frac{\partial}{\partial r}(rv_r) + \frac{\partial}{\partial \theta}(v_\theta) = 0 \quad \dots(i)$$

For the given velocity component:

$$\frac{\partial v_\theta}{\partial \theta} = \frac{\partial}{\partial \theta} \left(-\frac{C \sin \theta}{r^2} \right) = -\frac{C}{r^2} \cos \theta \quad \dots(ii)$$

From eqn. (i) and (ii), we have:

$$\frac{\partial}{\partial r}(rv_r) = \frac{C}{r^2} \cos \theta$$

Integrating both sides w.r.t. r we have:

$$rv_r = \int_0^r \frac{C}{r^2} \cos \theta dr = -\frac{C \cos \theta}{r}$$

$$\therefore \text{Radial component; } v_r = \frac{C \cos \theta}{r^2} \quad (\text{Ans.})$$

(ii) Resultant velocity:

$$\begin{aligned} \text{Resultant velocity} &= \sqrt{v_r^2 + v_\theta^2} \\ &= \sqrt{\left(-\frac{C \cos \theta}{r^2} \right)^2 + \left(-\frac{C \sin \theta}{r^2} \right)^2} \\ &= \frac{C}{r^2} \sqrt{(\cos^2 \theta + \sin^2 \theta)} = \frac{C}{r^2} \quad (\text{Ans.}) \end{aligned}$$

5.9. CIRCULATION AND VORTICITY

Let us consider a closed curve in a two-dimensional flow field shown in Fig. 5.18; the curve being cut by the stream lines. Let P be the point of intersection of the curve with one stream line, θ be the angle which the stream line makes with the curve. The component of velocity along the closed curve at the point of intersection is equal to $V \cos \theta$. **Circulation Γ** is defined mathematically as the line integral of the tangential velocity about a closed path (contour).

Thus,

$$\Gamma = \oint V \cos \theta ds$$

where, V = Velocity in the flow field at the element ds , and

θ = Angle between V and tangent to the path (in the positive anticlockwise direction along the path) at that point.

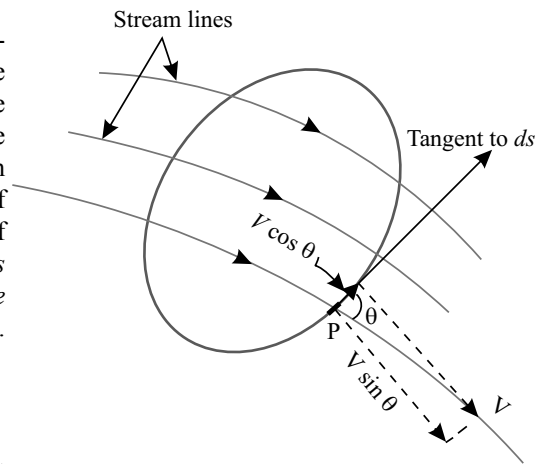


Fig. 5.18. Circulation in a two-dimensional flow.

Circulation around regular curves can be obtained by integration. Let us consider the circulation around an elementary box (fluid element ABCD) shown in Fig. 5.19.

Starting from *A* and proceeding anticlockwise, we have:

$$\begin{aligned} d\Gamma &= u\Delta x + \left(v + \frac{\partial v}{\partial x}\Delta x \right)\Delta y - \left(u + \frac{\partial u}{\partial y}\Delta y \right)\Delta x - v\Delta y \\ &= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)\Delta x.\Delta y \end{aligned}$$

The **vorticity** (Ω) is defined as the circulation per unit of enclosed area,

$$\Omega = \frac{\Gamma}{A}. \text{ Thus,}$$

$$\Omega = \frac{d\Gamma}{\Delta x.\Delta y} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \quad \dots(5.29)$$

If a flow possesses vorticity, it is rotational. **Rotation** ω (*omega*) is defined as one-half of the vorticity, or

$$\omega = \frac{1}{2} \left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right]$$

The flow is irrotational if rotation ω is zero.

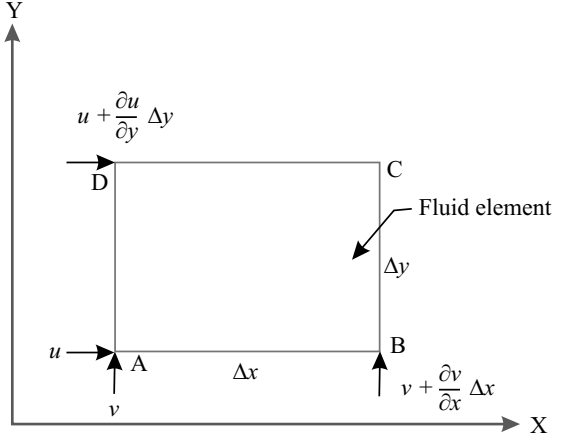


Fig. 5.19. Irrotational flow condition.

For a three-dimensional flow the rotation is possible about three axes. The expressions for rotation ω_z, ω_x and ω_y can be obtained in like manner:

$$\left. \begin{aligned} \omega_z &= \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \\ \omega_x &= \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \omega_y &= \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \end{aligned} \right\} \quad \dots(5.30)$$

In the vector notation, the above equation can be rewritten as:

$$\begin{aligned} \omega &= \frac{1}{2} [\omega_x i + \omega_y j + \omega_z k] \\ &= \frac{1}{2} (\Delta \times V) \end{aligned} \quad \dots(5.31)$$

The vector $(\Delta \times V)$ is the **curl** of velocity vector.

Vorticity in a fluid motion is taken numerically equal to *twice the value of rotation*.

$$\text{Vorticity, } \Omega = \text{curl } V = (\Delta \times V) \quad \dots(5.32)$$

Which may be expressed as:

$$\Omega = (\nabla \times V) \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

$$= \Omega_x i + \Omega_y j + \Omega_z k \quad \dots(5.33)$$

The vorticity components are separately given by:

$$\begin{aligned} \Omega_x &= 2\omega_x = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \Omega_y &= 2\omega_y = \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ \Omega_z &= 2\omega_z = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{aligned} \quad \dots(5.34)$$

The motion is described as **irrotational** when the components of rotation or vorticity are 'zero' throughout certain portion of the fluid.

When torque is applied to the fluid particle it will give rise to *rotation*; the torque is due to shear stress. Therefore, the rotation of fluid particle will always be associated with shear stress. As the shear stresses, in turn, depend upon the viscosity, the *rotational flow occurs where the viscosity effects are predominant*. However, in the cases where the viscosity effects are small, the flow is sometimes assumed to be irrotational. This simplifies analysis of problems of fluid flow.

Example 5.23. Given that

$$\begin{aligned} u &= -4ax(x^2 - 3y^2) \\ v &= 4ay(3x^2 - y^2) \end{aligned}$$

Examine whether these velocity components represent a physically possible two-dimensional flow; if so whether the flow is rotational or irrotational?

Solution. Given:

$$\begin{aligned} u &= -4ax(x^2 - 3y^2) \\ v &= 4ay(3x^2 - y^2) \end{aligned}$$

...Velocity components

A two-dimensional flow will be *continuous* if $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

$$\text{Now, } \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} [-4ax(x^2 - 3y^2)] = \frac{\partial}{\partial x} (-4ax^3 + 12axy^2) = -12ax^2 + 12ay^2$$

$$\text{and, } \frac{\partial v}{\partial y} = \frac{\partial}{\partial y} [4ay(3x^2 - y^2)] = \frac{\partial}{\partial y} [12ayx^2 - 4ay^3] = 12ax^2 - 12ay^2$$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = (-12ax^2 + 12ay^2) + (12ax^2 - 12ay^2) = 0$$

Hence the given velocity components **represent a physically possible two-dimensional flow.**

(Ans.)

The flow will be *irrotational* if,

$$\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

$$\text{Now, } \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} [-4ax(x^2 - 3y^2)] = \frac{\partial}{\partial y} (-4ax^3 + 12axy^2) = 24axy$$

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x} [4ay(3x^2 - y^2)] = \frac{\partial}{\partial x} [12ayx^2 - 4ay^3] = 24ayx$$

$$\therefore \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}, \text{ hence the flow is } \mathbf{irrotational. (Ans.)}$$

Example 5.24. Given that $u = xy$, $v = 2yz$. Examine whether these velocity components represent two or three-dimensional incompressible flow; if three-dimensional, determine the third component.

Solution. Given: $u = xy$, $v = 2yz$

... Velocity components

A two dimensional flow should satisfy the continuity equation,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

But,
$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(xy) = y$$

and,
$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y}(2yz) = 2z$$

$\therefore y + 2z \neq 0$

Hence, the flow is **not two-dimensional**.

For the flow to be *three-dimensional*, it should satisfy the continuity equation,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

or,
$$y + 2z + \frac{\partial w}{\partial z} = 0$$

or,
$$\frac{\partial w}{\partial z} = -(y + 2z)$$

or,
$$\begin{aligned} w &= [-(y + 2z)] dz + f(x, y, t) \\ &= -\left(yz + 2 \cdot \frac{z^2}{2}\right) + f(x, y, t) \\ &= -yz + z^2 + f(x, y, t) \end{aligned}$$

Hence, the third component,

$$w = -yz + z^2 + f(x, y, t) \text{ (Ans.)}$$

Example 5.25. For a two-dimensional flow, the velocity components are $u = x/(x^2 + y^2)$, $v = y/(x^2 + y^2)$. Determine: (i) The acceleration components a_x and a_y ; (ii) The rotation of w_z .

Solution. Given: $u = x/(x^2 + y^2)$, $v = y/(x^2 + y^2)$

... Velocity components

(i) The acceleration components, a_x and a_y :

We know that:

$$a_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \quad \dots(i)$$

$$a_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \quad \dots(ii)$$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left(\frac{x}{x^2 + y^2} \right) = \frac{(x^2 + y^2) \times 1 - x(2x)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left(\frac{x}{x^2 + y^2} \right) = \frac{\partial}{\partial y} [x(x^2 + y^2)^{-1}] = x \times [-(x^2 + y^2)^{-2} \times 2y] = -\frac{2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x} \left(\frac{y}{x^2 + y^2} \right) = \frac{\partial}{\partial x} [y(x^2 + y^2)^{-1}] = y \times -(x^2 + y^2)^{-2} \times 2x = -\frac{2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \left(\frac{y}{x^2 + y^2} \right) = \frac{(x^2 + y^2) \times 1 - y(2y)}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

Substituting these values in eqns. (i) and (ii), we get:

$$\begin{aligned} a_x &= \frac{x}{(x^2 + y^2)} \times \frac{(y^2 - x^2)}{(x^2 + y^2)^2} - \frac{y}{x^2 + y^2} \times \frac{2xy}{(x^2 + y^2)^2} \\ &= \frac{xy^2 - x^3}{(x^2 + y^2)^3} - \frac{2xy^2}{(x^2 + y^2)^3} = \frac{-x^3 - xy^2}{(x^2 + y^2)^3} = \frac{-x(x^2 + y^2)}{(x^2 + y^2)^3} = -\frac{x}{(x^2 + y^2)^2} \end{aligned}$$

Hence, $a_x = -\frac{x}{(x^2 + y^2)^2}$ (Ans.)

$$\begin{aligned} a_y &= \left(\frac{x}{x^2 + y^2} \right) \times \left[-\frac{2xy}{(x^2 + y^2)^2} \right] + \frac{y}{(x^2 + y^2)} \times \left(\frac{x^2 - y^2}{(x^2 + y^2)^2} \right) \\ &= -\frac{2x^2y}{(x^2 + y^2)^3} + \frac{y(x^2 - y^2)}{(x^2 + y^2)^3} = \frac{-2x^2y + x^2y - y^3}{(x^2 + y^2)^3} \\ &= \frac{-y(x^2 + y^2)}{(x^2 + y^2)^3} = -\frac{y}{(x^2 + y^2)^2} \end{aligned}$$

Hence, $a_y = -\frac{y}{(x^2 + y^2)^2}$ (Ans.)

(ii) The rotation of ω_z :

We know that:

$$\begin{aligned} \omega_z &= \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \\ &= \frac{1}{2} \left[-\frac{2xy}{(x^2 + y^2)^2} + \frac{2xy}{(x^2 + y^2)^2} \right] = 0 \end{aligned}$$

Hence the flow is irrotational. (Ans.)

Example 5.26. If the velocity field is given by $u = (16y - 8x)$, $v = (8y - 7x)$ find the circulation around the closed curve defined by $x = 4$, $y = 2$, $x = 8$, $y = 8$.

Solution. Given:

$$u = (16y - 8x), v = (8y - 7x)$$

...Velocity field

Refer to Fig 5.20.

$$\begin{aligned} \Gamma_{ABCD} &= \int_{ABCD} (u dx + v dy) \\ &= \int_{AB} (u dx + v dy) + \int_{BC} (u dx + v dy) + \int_{CD} (u dx + v dy) + \int_{DA} (u dx + v dy) \\ &= \int_4^8 (16y - 8x) dx + \int_2^8 (8y - 7x) dy + \int_8^4 (16y - 8x) dx + \int_8^2 (8y - 7x) dy \end{aligned}$$

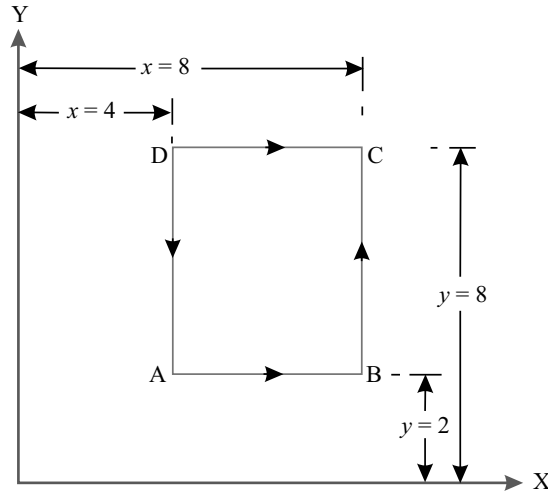


Fig. 5.20

$$= \underbrace{[16yx - 4x^2]_4^8}_{(i)} + \underbrace{[4y^2 - 7xy]_2^8}_{(ii)} + \underbrace{[16yx - 4y^2]_8^4}_{(iii)} + \underbrace{[4y^2 - 7xy]_8^2}_{(iv)}$$

In integral (i): $y = 2$

In integral (ii): $x = 8$

In integral (iii): $y = 8$

In integral (iv): $x = 4$

Substituting these values, we have:

$$\begin{aligned}\Gamma_{ABCD} &= [16 \times 2 \times 8 - 4 \times 8 \times 8 - 16 \times 2 \times 4 + 4 \times 4^2] \\ &\quad + [4 \times 8^2 - 2 \times 8 \times 8 - 4 \times 2^2 + 7 \times 8 \times 2] \\ &\quad + [16 \times 8 \times 4 - 4 \times 4^2 - 16 \times 8 \times 8 + 4 \times 8^2] \\ &\quad + [4 \times 2^2 - 7 \times 4 \times 2 - 4 \times 8^2 + 7 \times 4 \times 8] \\ &= [256 - 256 - 128 + 64] + [256 - 448 - 16 + 112] \\ &\quad + [512 - 64 - 1024 + 256] + [16 - 56 - 256 + 224] \\ &= -64 - 96 - 320 - 72 = -552\end{aligned}$$

$$\text{Area of the curve } ABCD = (8 - 4) \times (8 - 2) = 24$$

$$\therefore \text{Circulation per unit area} = -\frac{552}{24} = -23 \text{ (Ans.)}$$

Example 5.27. A fluid flow is given by

$$v_r = \left(1 - \frac{a}{r^2}\right) \cos \theta, \quad v_\theta = -\left(1 + \frac{a}{r^2}\right) \sin \theta$$

(i) Show that it represents a physically possible flow.

(ii) Determine whether the flow is rotational or irrotational.

Solution. Given: $v_r = \left(1 - \frac{a}{r^2}\right) \cos \theta, \quad v_\theta = -\left(1 + \frac{a}{r^2}\right) \sin \theta$

...Velocity components

(i) Is the flow physically possible?

The continuity equation for an incompressible fluid flow is given by:

$$\frac{v_r}{r} + \frac{\partial v_r}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} = 0 \quad \dots(1)$$

Now,
$$\frac{\partial}{\partial r} v_r = \frac{\partial}{\partial r} \left(1 - \frac{a}{r^2} \right) \cos \theta = \left[-a \cos \theta \left(-2 \right) \cdot \frac{1}{r^3} \right]$$

and,
$$\frac{\partial}{\partial \theta} v_\theta = \frac{\partial}{\partial \theta} \left[- \left(1 + \frac{a}{r^2} \right) \sin \theta \right] = \left[- \left(1 + \frac{a}{r^2} \right) \cos \theta \right]$$

Substituting these values in eqn. (1), we get:

$$\begin{aligned} & \frac{1}{r} \left(1 - \frac{a}{r^2} \right) \cos \theta + \left[-a \cos \theta + (-2) \cdot \frac{1}{r^2} \right] + \frac{1}{r} \left[- \left(1 + \frac{a}{r^2} \right) \cos \theta \right] \\ &= \frac{\cos \theta}{r} - \frac{a \cos \theta}{r^3} + \frac{2a \cos \theta}{r^2} - \frac{\cos \theta}{r} - \frac{a \cos \theta}{r^3} = 0 \end{aligned}$$

Since the continuity equation is satisfied, therefore, **the flow is physically possible.** (Ans.)

(ii) Flow—rotational or irrotational?

Let us check for rotationality.

Vorticity is given by:

$$\Omega = \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} - \frac{1}{r} \frac{\partial v_r}{\partial \theta} \quad \dots(2)$$

Now,
$$\frac{\partial v_\theta}{\partial r} = \frac{\partial}{\partial r} (v_\theta) = \frac{\partial}{\partial r} \left[- \left(1 + \frac{a}{r^2} \right) \sin \theta \right] = \left[-a \sin \theta \left(-2 \right) \cdot \frac{1}{r^3} \right]$$

and,
$$\frac{\partial v_r}{\partial \theta} = \frac{\partial}{\partial \theta} (v_r) = \frac{\partial}{\partial \theta} \left(1 - \frac{a}{r^2} \right) \cos \theta = \left[\left(1 - \frac{a}{r^2} \right) \times (-\sin \theta) \right]$$

Substituting these values in eqn. (2), we get:

$$\begin{aligned} \Omega &= \left[-a \sin \theta \left(-2 \right) \cdot \frac{1}{r^3} \right] + \frac{1}{r} \left[- \left(1 + \frac{a}{r^2} \right) \sin \theta \right] \\ &\quad - \frac{1}{r} \left[\left(1 - \frac{a}{r^2} \right) \times (-\sin \theta) \right] \\ &= \frac{2a \sin \theta}{r^3} - \frac{\sin \theta}{r} - \frac{a \sin \theta}{r^3} + \frac{\sin \theta}{r} - \frac{a \sin \theta}{r^3} = 0 \end{aligned}$$

Hence the **flow is irrotational.** (Ans.)

Example 5.28. The velocity components for a fluid flow are: $u = a + by - cz$, $v = d - bx - ez$, $w = f + cx - ey$ where a, b, c, d, e and f are arbitrary constants.

(i) Show that it is a possible case of fluid flow.

(ii) Is the fluid flow irrotational? If not, determine the vorticity and rotation.

[RGVP Bhopal]

Solution. Given: $u = a + by - cz$, $v = d - bx - ez$, $w = f + cx - ey$... Velocity components.

(i) **Possible case of fluid flow?**

Continuity equation is given as:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Now,
$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(a + by - cz) = 0,$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y}(d - bx - ez) = 0, \text{ and } \frac{\partial w}{\partial z} = \frac{\partial}{\partial z}(f + cx - ey) = 0$$

Since the equation of continuity is satisfied, therefore, the field is **possible case of fluid flow. (Ans.)**

(ii) Is the flow field irrotational?

For the flow to be irrotational, $\text{curl } V = 0$ i.e. $(\nabla \times V) = 0$

Now,
$$(\nabla \times V) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

Substituting velocity components, we have:

$$(\nabla \times V) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (a + by - cz) & (d - bx - ez) & (f + cx - ey) \end{vmatrix}$$

or,
$$\begin{aligned} (\nabla \times V) &= i \left[\frac{\partial}{\partial y}(f + cx - ey) - \frac{\partial}{\partial z}(d - bx - ez) \right] \\ &\quad + j \left[\frac{\partial}{\partial z}(a + by - cz) - \frac{\partial}{\partial x}(f + cx - ey) \right] \\ &\quad + k \left[\frac{\partial}{\partial x}(d - bx - ez) - \frac{\partial}{\partial y}(a + by - cz) \right] \\ &= i(-e + e) + j(-c - c) + k(-b - b) \end{aligned}$$

Since $(\nabla \times V) \neq 0$, the flow is **not irrotational (Ans.)**

Vorticity Ω :

Vorticity,
$$\begin{aligned} \Omega &= (\nabla \times V) = -2(cj + bk) \\ &= 2\sqrt{c^2 + b^2} \end{aligned}$$

Hence,
$$\Omega = 2\sqrt{c^2 + b^2} \text{ (Ans.)}$$

Rotation, ω

We know,
$$\begin{aligned} \omega &= \frac{\Omega}{2} \\ &= \frac{1}{2}[2\sqrt{c^2 + b^2}] = \sqrt{c^2 + b^2} \end{aligned}$$

Hence,
$$\text{Rotation, } \omega = \sqrt{c^2 + b^2} \text{ (Ans.)}$$

Example 5.29. Determine the components of rotation for the following velocity field pertaining to the flow of an incompressible fluid:

$$u = Cyz; v = Czx; w = Cxy, \text{ where } C = \text{constant.}$$

State whether the flow is rotational or irrotational.

Solution. Given: $u = Cyz; v = Czx; w = Cxy$

... Velocity field

The components of rotation are:

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) = \frac{1}{2} (Cx - Cx) = 0$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) = \frac{1}{2} (Cy - Cy) = 0$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} (Cz - Cz) = 0$$

Since each of the rotation components is zero, the given flow field represents **irrotational flow**. (Ans.)

Example 5.30. Determine the components of rotation about the various axes for the following flows:

(i) $u = y^2, v = -3x$

(ii) $u = 3xy, v = \frac{3}{2}x^2 - \frac{3}{2}y^2$

(iii) $u = xy^3z, v = -y^2z^2, w = yz^2 - \frac{y^3z^2}{2}$

Solution. The components of rotation about the various axes are:

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right)$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$$

(i) $u = y^2; v = -3x$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} (-3 - 2y) \text{ (Ans.)}$$

As the flow is two-dimensional in x - y plane, $\omega_x = \omega_y = 0$ (Ans.)

(ii) $u = 3xy, v = \frac{3}{2}x^2 - \frac{3}{2}y^2$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} (3x - 3x) = 0 \text{ (Ans.)}$$

As the flow is two-dimensional in the x - y plane, $\omega_x = \omega_y = 0$ (Ans.)

(iii) $u = xy^3z; v = -y^2z^2; w = yz^2 - \frac{y^3z^2}{2}$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} (0 - 3xy^2z) = -\frac{3}{2} xy^2z \text{ (Ans.)}$$

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) = \frac{1}{2} \left(z^2 - \frac{3y^2z^2}{2} + 2y^2z \right) \text{ (Ans.)}$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) = \frac{1}{2} (xy^3 - 0) = \frac{1}{2} xy^3 \text{ (Ans.)}$$

5.10. VELOCITY POTENTIAL AND STREAM FUNCTION

5.10.1. Velocity Potential

The **velocity potential** is defined as a scalar function of space and time such that its negative derivative with respect to any direction gives the fluid velocity in that direction. It is denoted by ϕ (phi). Thus mathematically the velocity potential is defined as:

$$\begin{aligned} \text{and,} \quad \phi &= f(x, y, z, t) && \dots \text{for unsteady flow,} \\ \phi &= f(x, y, z) && \dots \text{for steady flow;} \\ \text{such that:} \quad \left. \begin{aligned} u &= -\frac{\partial \phi}{\partial x} \\ v &= -\frac{\partial \phi}{\partial y} \\ w &= -\frac{\partial \phi}{\partial z} \end{aligned} \right\} && \dots (5.35) \end{aligned}$$

where, u , v and w are the components of velocity in the x , y and z directions respectively.

The *negative sign* signifies that ϕ decreases with an increase in the values of x , y and z . In other words it indicates that the flow is always in the direction of decreasing ϕ .

For an incompressible steady flow the continuity equation is given by:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

By substituting the values of u , v and w in terms of ϕ from eqn. 5.35, we get:

$$\begin{aligned} \frac{\partial}{\partial x} \left(-\frac{\partial \phi}{\partial x} \right) + \frac{\partial}{\partial y} \left(-\frac{\partial \phi}{\partial y} \right) + \frac{\partial}{\partial z} \left(-\frac{\partial \phi}{\partial z} \right) &= 0 \\ \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} &= 0 \end{aligned} \quad \dots (5.36)$$

This equation is known as **Laplace equation**.

Thus any function ϕ that satisfies the Laplace equation will correspond to some case of fluid flow.

The *rotational components* are given by [eqn. (5.30)]:

$$\begin{aligned} \omega_x &= \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \omega_y &= \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \end{aligned}$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

By substituting the values of u , v and w in term of ϕ from eqn. (5.35), we get:

$$\begin{aligned} \omega_x &= \frac{1}{2} \left[\frac{\partial}{\partial y} \left(-\frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(-\frac{\partial \phi}{\partial y} \right) \right] \\ &= \frac{1}{2} \left[-\frac{\partial^2 \phi}{\partial y \partial z} + \frac{\partial^2 \phi}{\partial z \partial y} \right] \\ \omega_y &= \frac{1}{2} \left[\frac{\partial}{\partial z} \left(-\frac{\partial \phi}{\partial x} \right) - \frac{\partial}{\partial x} \left(-\frac{\partial \phi}{\partial z} \right) \right] \\ &= \frac{1}{2} \left[-\frac{\partial^2 \phi}{\partial z \partial x} + \frac{\partial^2 \phi}{\partial x \partial z} \right] \\ \omega_z &= \frac{1}{2} \left[\frac{\partial}{\partial x} \left(-\frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(-\frac{\partial \phi}{\partial x} \right) \right] = \frac{1}{2} \left[-\frac{\partial^2 \phi}{\partial x \partial y} + \frac{\partial^2 \phi}{\partial y \partial x} \right] \end{aligned}$$

However, if ϕ is a continuous function then,

$$\frac{\partial^2 \phi}{\partial y \partial z} = \frac{\partial^2 \phi}{\partial z \partial y}; \frac{\partial^2 \phi}{\partial z \partial x} = \frac{\partial^2 \phi}{\partial x \partial z}; \text{ and } \frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$$

$$\therefore \omega_x = \omega_y = \omega_z = 0 \quad \text{i.e. the flow is irrotational.}$$

Thus if velocity potential (ϕ) satisfies the Laplace equation, it represents the possible steady, incompressible, irrotational flow. Often an irrotational flow is known as **potential flow**.

Equipotential line:

An **equipotential line** is one along which velocity potential ϕ is constant.

i.e. For equipotential line, $\phi = \text{constant}$.

$$\therefore d\phi = 0$$

But, $\phi = f(x, y)$ for steady flow.

$$\therefore d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy$$

$$\text{But, } \frac{\partial \phi}{\partial x} = -u \text{ and } \frac{\partial \phi}{\partial y} = -v$$

$$\therefore d\phi = -u dx - v dy = -(u dx + v dy)$$

For equipotential line, $d\phi = 0$

$$\text{or, } -(u dx + v dy) = 0$$

$$\text{or, } (u dx + v dy) = 0$$

$$\text{or, } \frac{dy}{dx} = -\frac{u}{v} \quad \dots(5.37)$$

where, $\frac{dy}{dx}$ = slope of equipotential line.

5.10.2. Stream Function

The **stream function** is defined as a function of space and time, such that its partial derivative with respect to any direction gives the velocity component at right angles to this direction. It is denoted by ψ (psi).

In case of two-dimensional flow, the stream function may be defined mathematically as

$$\begin{aligned} \psi &= f(x, y, t) && \dots \text{for } \textit{unsteady} \text{ flow, and} \\ \psi &= f(x, y) && \dots \text{for } \textit{steady} \text{ flow,} \end{aligned}$$

such that:

$$\left. \begin{aligned} u &= \frac{\partial \psi}{\partial y} \\ v &= -\frac{\partial \psi}{\partial x} \end{aligned} \right\} \dots (5.38)$$

For two-dimensional flow the continuity equation is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Substituting the values of u and v from eqn. (5.38), we get:

$$\begin{aligned} \frac{\partial}{\partial x} \left(\frac{\partial \psi}{\partial y} \right) + \frac{\partial}{\partial y} \left(-\frac{\partial \psi}{\partial x} \right) &= 0 \\ \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial x \partial y} &= 0 \end{aligned}$$

Hence, **existence of ψ means a possible case of fluid flow.**

— The flow may be ‘**rotational**’ or ‘**irrotational**’.

The rotational component ω_z is given by:

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

Substituting the values of u and v from eqn. (5.38), we get:

$$\omega_z = \frac{1}{2} \left[\frac{\partial}{\partial x} \left(-\frac{\partial \psi}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \psi}{\partial y} \right) \right]$$

$$\text{or,} \quad \omega_z = -\frac{1}{2} \left[\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \right] \dots (5.39)$$

*This equation is known as **Poisson’s equation**.* For an *irrotational flow*, since $\omega_z = 0$, eqn.(5.39) becomes:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0 \quad \text{i.e.,} \quad \Delta^2 \psi = 0$$

which is the **Laplace equation in ψ** .

In the *polar co-ordinates*:

$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}, \quad v_\theta = \frac{\partial \psi}{\partial r}$$

— Let $\psi(x, y)$ represent the stream line L (See Fig. 5.21). The $(\psi + d\psi)$ represents the adjacent stream line M . The velocity vector V perpendicular to the line AB has components u and v in the direction of X -axis and Y -axis respectively. From continuity consideration, we have:

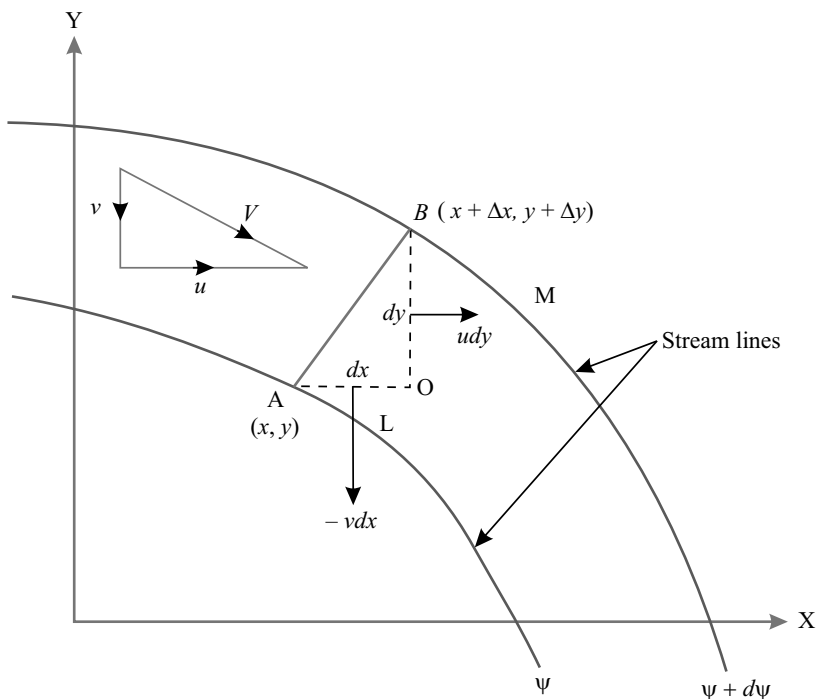


Fig. 5.21. Flow between two points and its relation to stream function.

Flow across, AB = Flow across AO + flow across OB

$$Vds = -vdx + udy$$

(The minus sign indicates that the velocity v is acting in the *downward* direction).

$$Vds = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = d\psi \quad \dots(5.40)$$

$$i.e. \quad dq = d\psi \quad \dots(5.41)$$

Obviously, the *stream function* can also be defined as the *flux or flow rate between two stream lines*. The units of ψ are m^3/s discharge per unit thickness of flow.

Properties of stream function

The properties of stream function are:

1. On any stream line, ψ is constant everywhere.

$$\left[\begin{array}{l} \psi = \text{constant, represents the family of stream lines.} \\ \psi = \text{constant, is a stream line equation.} \end{array} \right]$$

2. If the flow is *continuous*, the flow around any path in the fluid is *zero*.
3. The rate of change of ψ with distance in arbitrary direction is proportional to the component of velocity normal to that direction.
4. The algebraic sum of stream function for two incompressible flow patterns is the stream function for the flow resulting from the superimposition of these patterns.

$$i.e. \quad \frac{\partial \psi_1}{\partial s} + \frac{\partial \psi_2}{\partial s} = \frac{\partial (\psi_1 + \psi_2)}{\partial s} \quad \dots(5.42)$$

Cauchy Riemann equations:

From the above discussion of velocity potential function and stream function we arrive at the following conclusions:

- Potential function (ϕ) exists only for irrotational flow.
- Stream function (ψ) applies to both the rotational and irrotational flows (which are steady and incompressible).
- In case of irrotational flow, both the stream function and velocity function satisfy Laplace equation and as such they are interchangeable.

For irrotational incompressible flow, the following relationship between ϕ and ψ holds good:

$$\left. \begin{aligned} u &= -\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} \\ v &= -\frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x} \end{aligned} \right\} \quad \dots(5.43)$$

These equations, in hydrodynamics, are sometimes called “Cauchy Riemann equations”.

5.10.3. Relation between Stream Function and Velocity Potential

One of the properties of a stream function is that the *difference of its values at two points represent the flow across any line joining the points*. Thus if two lie on the same stream line, then, there being no flow across a stream line, the difference between the stream functions ψ_1 and $\psi_2 = 0$; this means the streamline is given by:

$$\psi = \text{constant.}$$

Similarly, $\phi = \text{constant}$, represents a case for which the velocity potential is same at every point, and hence it represents an *equipotential line*.

Let, two curves $\phi = \text{constant}$ and $\psi = \text{constant}$ intersect each other at any point. At the point of intersection the slopes are:

$$\text{For the curve } \phi = \text{constant: Slope} = \frac{\partial y}{\partial x} = \frac{\left(\frac{\partial \phi}{\partial x}\right)}{\left(\frac{\partial \phi}{\partial y}\right)} = \frac{-u}{-v} = \frac{u}{v}$$

$$\text{For the curve } \psi = \text{constant: Slope} = \frac{\partial y}{\partial x} = \frac{\left(\frac{\partial \psi}{\partial x}\right)}{\left(\frac{\partial \psi}{\partial y}\right)} = \frac{-v}{+u} = -\frac{v}{u}$$

Now, product of the slopes of these curves

$$= \frac{u}{v} \times -\frac{v}{u} = -1$$

It shows that these two sets of curves, viz *stream lines* and *equipotential lines* intersect each other **orthogonally** at all points of intersection.

5.11. FLOW NETS

A grid obtained by drawing a series of stream lines and equipotential lines is known as a **flow net**. The flow net provides a simple graphical technique for studying two-dimensional irrotational flows especially in the cases where mathematical relations for stream function and velocity function are either not available or are rather difficult and cumbersome to solve.

5.11.1. Methods of Drawing Flow Nets

The following *methods* are used for drawing flow nets:

1. Analytical method (or Mathematical analysis):

- Here, the equations corresponding to the curves ϕ and ψ are first obtained and the same are plotted to give the flow net pattern for the flow of fluid between the given boundary shape.
- This method can be applied to problems with simple and ideal boundary conditions.

2. Graphical method:

- A graphical method consists of drawing stream lines and equipotential lines such that they cut orthogonally and form curvilinear squares.
- This method consumes lot of time and requires lot of erasing to get the proper shape of a flow net.

3. Electrical analogy method:

- This method is a practical method of drawing a flow net for a particular set of boundaries.
- It is based on the fact that the flow of fluids and flow of electricity through a conductor are analogous. These two systems are similar in the respect that electric potential is analogous to the velocity potential, the electric current is analogous to the velocity of flow, and the homogeneous conductor is analogous to the homogeneous fluid.

4. Hydraulic models:

- Stream lines can be traced by injecting a dye in a seepage model or Heleshaw apparatus.
- Then, by drawing equipotential lines the flow net is completed.

Fig 5.22. shows some typical flow nets.

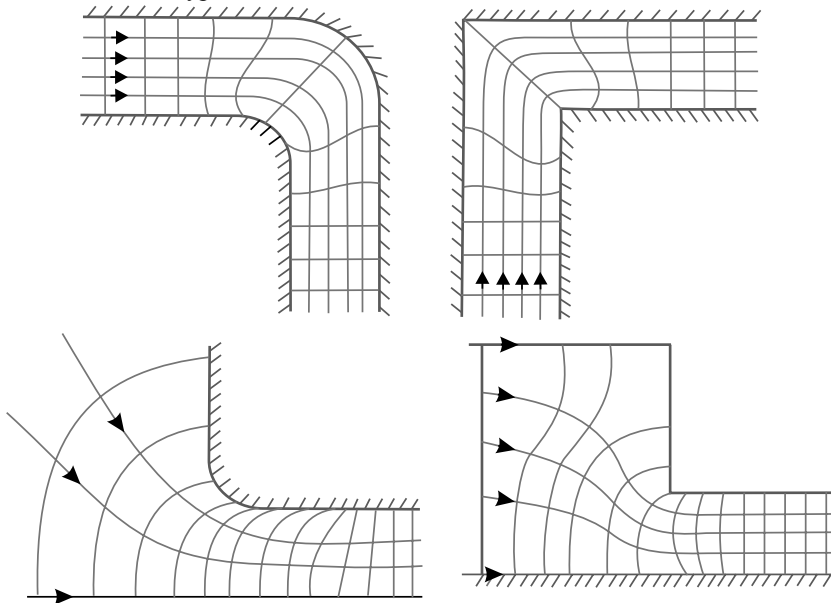


Fig. 5.22. Typical flow nets.

5.11.2. Uses and Limitations of Flow Nets

Use of flow nets:

The following are the *uses* of flow-net analysis:

1. To determine the stream lines and equipotential lines.
2. To determine quantity of seepage and upward lift pressure below hydraulic structure.

3. To determine the velocity and pressure distribution, for given boundaries of flow (provided the velocity distribution and pressure at any reference section are known).
4. To determine the design of the outlets for their streamlining.

Limitations of flow nets:

The following are the *limitations* of flow net:

1. The flow net analysis cannot be applied in the region close to the boundary where the effects of viscosity are predominant.
2. In case of a flow of a fluid past a solid body, while the flow net gives a fairly accurate picture of the flow pattern for the upstream part of the solid body, it can give little information concerning the flow conditions at the rear because of separation and eddies.

Example 5.31. *Verify whether the following functions are valid potential functions:*

(i) $\phi = A(x^2 - y^2)$ (ii) $\phi = A \cos x$

Solution. (i) $\phi = A(x^2 - y^2)$:

$$\frac{\partial \phi}{\partial x} = 2Ax; \quad \frac{\partial \phi}{\partial y} = -2Ay$$

$$\frac{\partial^2 \phi}{\partial x^2} = 2A; \quad \frac{\partial^2 \phi}{\partial y^2} = -2A$$

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 2A + (-2A) = 0$$

Hence, $\phi = A(x^2 - y^2)$ is a **valid potential function (Ans.)**

(ii) $\phi = A \cos x$:

$$\frac{\partial \phi}{\partial x} = -A \sin x; \quad \frac{\partial \phi}{\partial y} = 0$$

$$\frac{\partial^2 \phi}{\partial x^2} = -A \cos x; \quad \frac{\partial^2 \phi}{\partial y^2} = 0$$

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -A \cos x \neq 0$$

Hence, $\phi = A \cos x$ is **not a valid function (Ans.)**

Example 5.32. *Which of the following functions represent possible irrotational flow?*

(i) $\psi = A(x^2 - y^2)$ (ii) $\psi = xy$

(iii) $\phi = \left(r - \frac{2}{r}\right) \sin \theta$ (iv) $\phi = Ur \cos \theta + \frac{U}{r} \cos \theta$

Solution. For an irrotational fluid flow phenomenon ϕ as well ψ satisfy Laplace equation.

(i) $\psi = A(x^2 - y^2)$:

$$\frac{\partial \psi}{\partial x} = 2Ax; \quad \frac{\partial \psi}{\partial y} = -2Ay$$

$$\frac{\partial^2 \psi}{\partial x^2} = 2A; \quad \frac{\partial^2 \psi}{\partial y^2} = -2A$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 2A - 2A = 0$$

Hence, $\psi = A(x^2 - y^2)$ represents a **possible irrotational flow (Ans.)**

(ii) $\psi = xy$:

$$\frac{\partial \psi}{\partial x} = y; \quad \frac{\partial \psi}{\partial y} = x$$

$$\frac{\partial^2 \psi}{\partial x^2} = 0; \quad \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\therefore \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

Hence, $\psi = xy$ represents a **possible irrotational flow. (Ans.)**

$$(iii) \phi = \left(r - \frac{2}{r} \right) \sin \theta :$$

Laplace equation in radial coordinates (r, θ) is given as:

$$\frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0$$

$$\frac{\partial \phi}{\partial r} = \left(1 + \frac{2}{r^2} \right) \sin \theta$$

$$\frac{\partial^2 \phi}{\partial r^2} = -\frac{2}{r^3} \sin \theta$$

$$\frac{\partial \phi}{\partial \theta} = \left(r - \frac{2}{r} \right) \cos \theta$$

$$\frac{\partial^2 \phi}{\partial \theta^2} = -\left(r - \frac{2}{r} \right) \sin \theta = \left(\frac{2}{r} - r \right) \sin \theta$$

L.H.S. of Laplace equation is:

$$\begin{aligned} & \frac{1}{r} \times \left(1 + \frac{2}{r^2} \right) \sin \theta - \frac{2}{r^3} \sin \theta + \frac{1}{r^2} \left(\frac{2}{r} - r \right) \sin \theta \\ &= \sin \theta \left(\frac{1}{r} + \frac{2}{r^3} - \frac{2}{r^3} + \frac{2}{r^3} - \frac{1}{r} \right) \\ &= \sin \theta \left(\frac{2}{r^3} \right) \neq 0 \end{aligned}$$

Hence, the given function **doesnot represent any possible irrotational flow. (Ans.)**

$$(iv) \phi = Ur \cos \theta + \frac{U}{r} \cos \theta :$$

$$\frac{\partial \phi}{\partial r} = U \cos \theta - \frac{U}{r^2} \cos \theta$$

$$\frac{\partial^2 \phi}{\partial r^2} = \frac{2U}{r^3} \cos \theta$$

$$\frac{\partial \phi}{\partial \theta} = -Ur \sin \theta - \frac{U}{r} \sin \theta$$

$$\frac{\partial^2 \phi}{\partial \theta^2} = -Ur \cos \theta - \frac{U}{r} \cos \theta$$

The Laplace equation, in radial coordinates (r, θ) , is given by:

$$\frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0$$

Substituting for L.H.S. terms, we get:

$$\begin{aligned} &= \frac{1}{r} \left(U \cos \theta - \frac{U}{r^2} \cos \theta \right) + \frac{2U}{r^3} \cos \theta + \frac{1}{r^2} \left(-Ur \cos \theta - \frac{U}{r} \cos \theta \right) \\ &U \cos \theta \left(\frac{1}{r} - \frac{1}{r^3} + \frac{2}{r^3} - \frac{1}{r} - \frac{1}{r^3} \right) = 0 \end{aligned}$$

The Laplace equation is satisfied and hence the given function ϕ represents a possible irrotational flow. (Ans.)

Example 5.33. The velocity components in a fluid flow are given by:

$$u = 2xy; v = a^2 + x^2 - y^2$$

(i) Show that the flow is possible.

(ii) Derive the relative stream function.

Solution. Given: $u = 2xy; v = a^2 + x^2 - y^2$

...Velocity components

$$(i) \quad \frac{\partial u}{\partial x} = 2y; \quad \frac{\partial v}{\partial y} = -2y$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 2y + (-2y) = 0$$

The continuity equation for steady, incompressible flow is satisfied.

Hence, **flow is possible.** (Ans.)

(ii) The stream function ψ is related to u and v as:

$$u = \frac{\partial \psi}{\partial y} = 2xy$$

$$\text{or,} \quad \psi = \int 2xy \, dy = xy^2 + f(x) \quad \dots(i)$$

$$-\frac{\partial \psi}{\partial x} = -y^2 - f'(x) = v = a^2 + x^2 - y^2$$

$$\text{Hence,} \quad f(x) = -(a^2 + x^2)$$

$$\text{or,} \quad f(x) = -a^2x - \frac{x^3}{3} + \text{constant}$$

Inserting for $f(x)$ in eqn. (i) we get:

$$\psi = xy^2 - ax^2 - \frac{x^3}{3} + \text{constant}$$

$$\text{Thus,} \quad \text{the relative } \psi = xy^2 - a^2x - \frac{x^3}{3} + \text{constant (Ans.)}$$

Example 5.34. A stream function is given by:

$$\psi = 3x^2y + (3+t)y^2$$

Find the flow rates across the faces of the triangular prism having a thickness of 2.5 m in the Z- direction at the time instant $t = 3$ seconds.

Solution. Refer to Fig. 5.23.

$$\psi = 3x^2y + (3+t)y^2 \quad \dots \text{Given}$$

The coordinates of point M are: (0,1)

$$\therefore \psi_M = 0 + (3+3) \times 1^2 = 6$$

The coordinates of point L are: (1.5, 0)

$$\therefore \psi_L = 3 \times 1.5^2 \times 0 + (3+3) \times 0 = 0$$

The coordinates of point O are: (0,0)

$$\begin{aligned} \therefore \text{Flow rate across face MO} &= 2.5 (\psi_M - \psi_O) \\ &= 2.5 (6 - 0) = \mathbf{15 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

$$\begin{aligned} \text{Flow rate across face LO} &= 2.5 (\psi_L - \psi_O) \\ &= 2.5 (0 - 0) = 0 \end{aligned}$$

$$\begin{aligned} \therefore \text{Flow rate across face LM} &= 2.5 (\psi_M - \psi_L) \\ &= 2.5 (6 - 0) = \mathbf{15 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

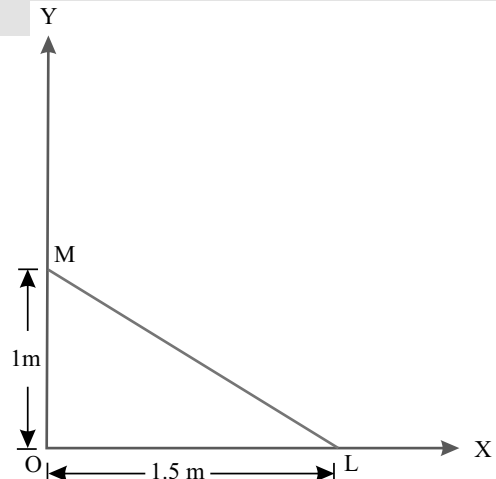


Fig. 5.23

Example 5.35. A flow is described by the stream function $\psi = 4xy$. Locate the point at which the velocity vector has a magnitude 7 units and makes an angle of 150° with X-axis.

Solution. Stream function, $\psi = 4xy$

...Given

The velocity components for the given flow field are:

$$u = \frac{\partial \psi}{\partial y} = \frac{\partial}{\partial y}(4xy) = 4x$$

$$v = -\frac{\partial \psi}{\partial x} = -\frac{\partial}{\partial x}(4xy) = -4y$$

$$V = \sqrt{u^2 + v^2} \quad \text{or} \quad 7 = \sqrt{(4x)^2 + (-4y)^2} = 4\sqrt{x^2 + y^2} \quad \dots(i)$$

$$\text{Also,} \quad \tan \theta = \frac{v}{u} \quad \text{or} \quad \tan 150^\circ = \frac{-4y}{4x} = -\frac{y}{x}$$

$$\text{or,} \quad -0.577 = -\frac{y}{x} \quad \text{or} \quad y = 0.577x$$

Substituting for y in eqn. (i), we get:

$$7 = 4\sqrt{x^2 + (0.577x)^2} = 4.62x$$

$$\therefore x = \frac{7}{4.62} = \mathbf{1.515 \text{ (Ans.)}}$$

Example. 5.36. Find a relevant stream function to each of the following sets of velocity components of steady, incompressible flow:

(i) $u = 2cx$; $v = -2cy$

(ii) $u = -cx/y$; $v = c \ln(xy)$

(iii) $u = x + y$; $v = x - y$

Solution. (i) $u = 2cx$; $v = -2cy$:

$$\frac{\partial u}{\partial x} = 2c; \frac{\partial v}{\partial x} = -2c$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 2i + (-2c) = 0. \text{ Hence the flow is possible and } \psi \text{ exists.}$$

$$u = \frac{\partial \psi}{\partial y} = 2cx$$

$$\psi = 2cxy + f(x)$$

$$-\frac{\partial \psi}{\partial x} = -2cy - f'(x) = v = -2cy$$

Hence, $f'(x) = 0$ and $f(x) = c_1 = a$ constant

$$\therefore \psi = 2cxy + c_1 \text{ (Ans.)}$$

(ii) $u = -cx/y$; $v = c \ln(xy)$:

$$\frac{\partial u}{\partial x} = -c/y; \frac{\partial v}{\partial y} = c/y$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = -c/y + c/y = 0$$

Hence, the flow is possible and ψ exists.

$$u = \frac{\partial \psi}{\partial y} = -cx/y$$

$$\psi = -cx \ln(y) + f(x)$$

$$-\frac{\partial \psi}{\partial x} = c \ln(y) - f'(x) = v = c \ln(xy) = c \ln(x) + c \ln(y)$$

or,

$$f'(x) = -c \ln(x)$$

$$f(x) = -\int c \ln(x).dx = -c(x \ln(x) - x) + c_2$$

where, c_2 = a constant.

Hence, the stream function representing this flow is:

$$\therefore \psi = -cx \ln(y) - c(x \ln(x) - x) + c_2$$

$$= -cx \ln(y) - cx \ln(x) + cx + c_2$$

or,

$$\psi = -cx \ln(xy) + cx + c_2 \text{ (Ans.)}$$

(iii) $u = x + y$; $v = x - y$:

$$\frac{\partial u}{\partial x} = 1; \frac{\partial v}{\partial y} = -1$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 1 + (-1) = 0. \text{ Hence the flow is possible and } \psi \text{ exists.}$$

$$u = \frac{\partial \psi}{\partial y} = x + y$$

$$\psi = xy + \frac{y^2}{2} + f(x)$$

$$-\frac{\partial \psi}{\partial x} = -y - f'(x) = v = x - y$$

or, $f'(x) = -x$

and, $f(x) = \int -x dx = -\frac{x^2}{2} + c$. where c = constant

$\therefore \psi = xy + \frac{y^2}{2} - \frac{x^2}{2} + c$

or, $\psi = \frac{1}{2}(y^2 - x^2) + xy + c$ (Ans.)

Example 5.37. For the following stream functions calculate velocity at a point (1, 2):

(i) $\psi = 3xy$

(ii) $y = 3x^2y - y^3$

Solution. (i) $\psi = 3xy$:

...(Given)

$$u = \frac{\partial \psi}{\partial y} = 3x$$

$$v = -\frac{\partial \psi}{\partial x} = -3y$$

At (1,2): $u = 3 \times 1 = 3$

$$v = -3 \times 2 = -6$$

$\therefore$

$$V = \sqrt{u^2 + v^2} = \sqrt{(3)^2 + (-6)^2} = \sqrt{45} \text{ units. (Ans.)}$$

(ii) $\psi = 3x^2y - y^3$:

...(Given)

$$u = \frac{\partial \psi}{\partial y} = 3x^2 - 3y^2$$

$$v = -\frac{\partial \psi}{\partial x} = -6xy$$

At (1,2): $u = 3 \times (1)^2 - 3 \times (2)^2 = -9$

$$v = -6 \times 1 \times 2 = -12$$

$$V = \sqrt{u^2 + v^2} = \sqrt{(-9)^2 + (-12)^2} = 15 \text{ (Ans.)}$$

Example 5.38. What is the irrotational velocity field associated with the potential $\phi = 3x^2 - 3x + 3y^2 + 16t^2 + 12zt$. Does the flow field satisfy the incompressible continuity equation?

[UPSC]

Solution. Given: $\phi = 3x^2 - 3x + 3y^2 + 16t^2 + 12zt$

The velocity field is represented by:

$$u = -\frac{\partial \phi}{\partial x}, \quad v = -\frac{\partial \phi}{\partial y}$$

$\therefore$

$$u = -\frac{\partial}{\partial x} (3x^2 - 3x + 3y^2 + 16t^2 + 12zt) = -6x + 3$$

and,
$$v = -\frac{\partial}{\partial y} (3x^2 - 3x + 3y^2 + 16t^2 + 12zt) = -6y$$

Also,
$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (-6x + 3) = -6$$

and,
$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} (-6y) = -6$$

The continuity equation for an incompressible fluid is:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Substituting the values, we get:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = -6 - 6 = -12$$

This shows that the given velocity field **does not satisfy the continuity equation.** (Ans.)

Example 5.39. The velocity potential function for a two-dimensional flow is $\phi = x(2y - 1)$. At a point $P(4, 5)$ determine:

(i) The velocity, and

(ii) The value of stream function.

[UPTU]

Solution. Given: $\phi = x(2y - 1)$...Velocity potential function.

(i) The velocity at a point $P(4, 5)$:

The velocity components in x and y directions are:

$$u = -\frac{\partial \phi}{\partial x} = -\frac{\partial}{\partial x} [x(2y - 1)] = -2y + 1$$

$$v = -\frac{\partial \phi}{\partial y} = -\frac{\partial}{\partial y} [x(2y - 1)] = -2x$$

$\therefore$ Resultant velocity,

$$V = \sqrt{u^2 + v^2} = \sqrt{(-2y + 1)^2 + (-2x)^2}$$

$\therefore$ At the point $P(4, 5)$ when $x = 4$ and $y = 5$, we have:

$$V = \sqrt{(-2 \times 5 + 1)^2 + (-2 \times 4)^2} = \sqrt{9^2 + 8^2} = 12.04 \text{ units (Ans.)}$$

(ii) The value of stream function at the point $(4, 5)$:

For stream function,

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

or,
$$d\psi = -v dx + u dy$$

or,
$$d\psi = +2x dx + (-2y + 1) dy$$

Integrating both sides, we get:

$$\psi = +2 \times \frac{x^2}{2} + \left(-2 \times \frac{y^2}{2} + y \right) + C$$

(where C = constant of integration)

For $\psi = 0$ at the origin, the constant $C = 0$

$$\therefore \psi = +x^2 - y^2 + y$$

At the point $P(4,5)$,

$$\psi = +(4)^2 - (5)^2 + 5 = -4 \text{ units (Ans.)}$$

Example 5.40. For a two-dimensional flow the velocity function is given by the expression, $\phi = x^2 - y^2$.

- (i) Determine velocity components in x and y directions.
- (ii) Show that the velocity components satisfy the conditions of flow continuity and irrotationality.
- (iii) Determine stream function and the flow rate between the stream lines $(2, 0)$ and $(2, 2)$.
- (iv) Show that the streamlines and potential lines intersect orthogonally at the point $(2, 2)$.

Solution. Given: $\phi = x^2 - y^2$... Velocity function.

(i) **Velocity components in x and y directions:**

The velocity components in x and y directions are:

$$u = -\frac{\partial \phi}{\partial x} = -\frac{\partial}{\partial x} (x^2 - y^2) = -2x \text{ (Ans.)}$$

$$v = -\frac{\partial \phi}{\partial y} = -\frac{\partial}{\partial y} (x^2 - y^2) = +2y \text{ (Ans.)}$$

(ii) **Continuity, irrotationality = ?**

From the velocity components, we have:

$$\frac{\partial u}{\partial x} = -2, \frac{\partial v}{\partial y} = +2$$

Conditions of flow continuity will be satisfied if:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Substituting the values, we get $-2 + 2 = 0$

Hence, the **velocity components satisfy the flow continuity conditions (Ans.)**

Now,

$$\begin{aligned} \nabla \times V &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -2x & +2y & 0 \end{vmatrix} \\ &= i \left[\frac{\partial}{\partial y} (0) - \frac{\partial}{\partial z} (+2y) \right] + j \left[\frac{\partial}{\partial z} (-2x) - \frac{\partial}{\partial x} (0) \right] \\ &\quad + \left[\frac{\partial}{\partial x} (+2y) - \frac{\partial}{\partial y} (-2x) \right] \end{aligned}$$

Since curl V is zero, hence the flow is **irrotational. ... (Proved)**

(iii) **Stream function and flow rate:**

The differential $d\psi$ for the stream function is (eqn. 5.40):

$$\begin{aligned} d\psi &= \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = -v dx + u dy \\ &= -(+2y) dx + (-2x) dy = -2d(xy) \end{aligned}$$

Integrating, we get:

$$\psi = -2xy + C \text{ (Ans.)}$$

(where, C = constant of integration)

Now, $\psi(2, 0) = 2 \times 2 \times 0 = 0$

and, $\psi(2, 2) = 2 \times 2 \times 2 = 8$

Hence, flow between the streamlines through (2, 0) and (2, 2)

$$= 8 - 0 = 8 \text{ m}^3/\text{s} \text{ (Ans.)}$$

(iv) Intersection of stream lines and potential lines orthogonally at point (2, 2) = ?

Slope of stream line,

$$\left(\frac{\partial y}{\partial x} \right)_{\psi = \text{const.}} = -\frac{v}{u} = -\frac{+2y}{-2x} = 1 \text{ at } (2, 2)$$

Slope of potential line,

$$\left(\frac{\partial y}{\partial x} \right)_{\phi = \text{const.}} = \frac{u}{v} = \left(\frac{-2x}{+2y} \right) = -1 \text{ at } (2, 2)$$

Thus, $\left(\frac{\partial y}{\partial x} \right)_{\psi = \text{const.}} \times \left(\frac{\partial y}{\partial x} \right)_{\phi = \text{const.}} = 1 \times (-1) = -1$

which shows that the **stream lines and the potential lines intersect orthogonally.(Proved)**

Example 5.41. A two-dimensional flow field is given by $\phi = 3xy$, determine:

(i) The stream function.

(ii) The velocity at L(2, 6) and M(6,6) and the pressure difference between the points L and M.

(iii) The discharge between the stream lines passing through the points L and M.

Solution. Given. $\phi = 3xy$...Flow field

(i) The stream function ψ :

We know that: $u = -\frac{\partial \phi}{\partial x} = -\frac{\partial}{\partial x}(3xy) = -3y$

$$v = -\frac{\partial \phi}{\partial y} = -\frac{\partial}{\partial y}(3xy) = -3x$$

Also, $u = \frac{\partial \psi}{\partial y} = -3y$, and $v = -\frac{\partial \psi}{\partial x} = -3x$

Again, $d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$ or $\partial \psi = 3x dx + (-3y) dy$

Integrating both sides, we get:

$$\begin{aligned} \psi &= \int 3x dx + \int (-3y) dy \\ &= 3 \times \frac{x^2}{2} - 3 \times \frac{y^2}{2} + C = \frac{3}{2}(x^2 - y^2) + C \end{aligned}$$

(where, C = constant of integration.)

For $\psi = 0$ at the origin, the constant $C = 0$

$\therefore \psi = \frac{3}{2}(x^2 - y^2) \text{ (Ans.)}$

(ii) Velocities at L and M:

At L(2, 6): $u = -3 \times 6 = -18$, $v = -3 \times 2 = -6$

$$\therefore \quad \mathbf{V}_L = \sqrt{u^2 + v^2} = \sqrt{(-18)^2 + (-6)^2} = \mathbf{18.97 \text{ units (Ans.)}}$$

At M(6, 6): $u = -3 \times 6 = -18$, $v = -3 \times 6 = -18$

$$\therefore \quad \mathbf{V}_M = \sqrt{u^2 + v^2} = \sqrt{(-18)^2 + (-18)^2} = \mathbf{25.45 \text{ units (Ans.)}}$$

Pressure difference between L and M:

For two-dimensional plane flow:

$$\frac{p_L}{w} + \frac{V_L^2}{2g} = \frac{p_M}{w} + \frac{V_M^2}{2g}$$

$$\therefore \quad \frac{p_L - p_M}{w} = \frac{1}{2g} (V_M^2 - V_L^2) = \frac{648 - 360}{2 \times 9.81} = \mathbf{14.68 \text{ units (Ans.)}}$$

(iii) The discharge between the streamlines, q :

$$\psi = \frac{3}{2} (x^2 - y^2)$$

$$\psi_{L(2,6)} = \frac{3}{2} (2^2 - 6^2) = -48 \text{ units.}$$

$$\psi_{M(2,6)} = \frac{3}{2} (6^2 - 6^2) = 0$$

$$\therefore \quad q = \psi_M - \psi_L = 0 - (-48) = \mathbf{48 \text{ units (Ans.)}}$$

Example 5.42. If $\phi = 3xy$, find x and y components of velocity at (1, 3) and (3, 3). Determine the discharge passing between stream lines passing through these points. [Roorkee University]

Solution. Given: $\phi = 3xy$... Velocity potential function.

The velocity components in terms of ϕ are given by:

$$u = -\frac{\partial \phi}{\partial x}, \quad v = -\frac{\partial \phi}{\partial y}$$

But, $\frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x} (3xy) = 3y$, and

$$\frac{\partial \phi}{\partial y} = \frac{\partial}{\partial y} (3xy) = 3x$$

$$\therefore \quad u = -3y \text{ and } v = -3x$$

Hence, the velocity components at (1, 3) and (3, 3) are:

$$\text{At (1, 3):} \quad \left. \begin{aligned} u &= -3 \times 3 = -9 \\ v &= -3 \times 1 = -3 \end{aligned} \right\} \text{(Ans)}$$

$$\text{At (3, 3):} \quad \left. \begin{aligned} u &= -3 \times 3 = -9 \\ v &= -3 \times 3 = -9 \end{aligned} \right\} \text{(Ans)}$$

Discharge between the streamlines:

The total derivative ψ may be written as:

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

But, $u = \frac{\partial \psi}{\partial y}$ and $v = -\frac{\partial \psi}{\partial x}$

$\therefore \partial \psi = -v dx + u dy$

or, $\partial \psi = 3x dx - 3y dy$

Integrating, we get: $\psi = \frac{3}{2}x^2 - \frac{3}{2}y^2 + C$

(where, C = constant of integration)

Discharge between the streamlines passing through (1, 3) and (3, 3)

$$= \psi_{(1,3)} - \psi_{(3,3)} = \frac{3}{2}(1 - 9) - \frac{3}{2}(9 - 9) = -12 \text{ units (Ans.)}$$

Example 5.43. The streamlines are represented by:

(a) $\psi = x^2 - y^2$ (b) $\psi = x^2 + y^2$

(i) Determine the velocity and its direction at (2, 2).

(ii) Sketch the streamlines and show the direction of flow in each case.

Solution. In a two-dimensional steady flow the velocity components in terms of ψ are given as:

$$u = \frac{\partial \psi}{\partial y} \text{ and } v = -\frac{\partial \psi}{\partial x}$$

Case (a) $\psi = x^2 - y^2$:

(i) Velocity and its direction at (2, 2):

$$\psi = x^2 - y^2 \quad \dots(1)$$

$$\therefore \frac{\partial \psi}{\partial y} = -2y \quad \text{and} \quad \frac{\partial \psi}{\partial x} = 2x$$

$$\therefore u = -2y \quad \text{and} \quad v = -2x$$

$$\therefore V = \sqrt{u^2 + v^2} = \sqrt{(-2y)^2 + (-2x)^2} = 2\sqrt{x^2 + y^2} \quad \dots(2)$$

$$\text{i.e.} \quad V = 2\sqrt{x^2 + y^2}$$

$$\therefore \text{Velocity at (2, 2)} = 2\sqrt{2^2 + 2^2} = 4\sqrt{2} \text{ units (Ans.)}$$

$$\text{Its direction has a slope, } \frac{\partial y}{\partial x} = -\frac{v}{u} = -\frac{-2x}{-2y} = +1 \quad \dots(3)$$

$\therefore$ Velocity vector is inclined at **45° to x-axis (Ans.)**

(ii) **Stream lines-sketch:**

The streamlines are lines of constant ψ , and for constant ψ , eqn. (1) represents hyperbola, which may be plotted for different values of ψ as shown in the table given below:

$$\left[\begin{array}{l} \psi = x^2 - y^2 \\ x = \pm \sqrt{y^2 + \psi} \end{array} \right]$$

	y	0	1	2	3
$\psi = 1$	$x = \pm\sqrt{y^2 + \psi}$	± 1	$\pm\sqrt{2}$	$\pm\sqrt{5}$	$\pm\sqrt{10}$
$\psi = 2$	$x = \pm\sqrt{y^2 + \psi}$	$\pm\sqrt{2}$	$\pm\sqrt{3}$	$\pm\sqrt{6}$	$\pm\sqrt{11}$
$\psi = 3$	$x = \pm\sqrt{y^2 + \psi}$	$\pm\sqrt{3}$	$\pm\sqrt{4}$	$\pm\sqrt{7}$	$\pm\sqrt{12}$

Fig. 5.24 shows the pattern of stream lines.

Case (b) $\psi = x^2 + y^2$:

(i) Velocity and its direction at (2, 2)

$$\psi = x^2 + y^2 \quad \dots(4)$$

$$\text{Now, } \frac{\partial \psi}{\partial y} = 2y \text{ and } \frac{\partial \psi}{\partial x} = 2x$$

$$\text{Hence, } u = 2y \text{ and } v = -2x$$

The resultant velocity,

$$\begin{aligned} V &= \sqrt{u^2 + v^2} = \sqrt{(2y)^2 + (-2x)^2} \\ &= 2\sqrt{x^2 + y^2} \end{aligned}$$

$$\therefore \text{Velocity at } (2, 2) = 2\sqrt{2^2 + 2^2} = 4\sqrt{2} \text{ units (Ans.)}$$

Its direction has a slope,

$$\frac{\partial y}{\partial x} = -\frac{v}{u} = -\frac{-2x}{2y} = \frac{x}{y} = \frac{2}{2} = 1$$

i.e., velocity makes an angle θ with the axis shown in Fig. 5.25, given by:

$$\tan \theta = 1 \text{ or } \theta = 45^\circ \text{ (Ans.)}$$

(ii) Stream lines-sketch:

For a streamlines $\psi = \text{constant}$, and for non-zero value of ψ , eqn. (4) represents *concentric circles* with centre at the origin (0, 0) and radius $\sqrt{\psi}$. In the 1st (i.e. positive) quadrant.

(x, y both positive), $u = 2y$ and $v = -2x$ is negative. Therefore, streamlines have clockwise direction as shown in Fig 5.26.

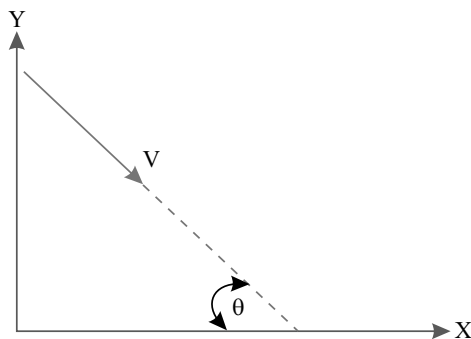


Fig. 5.25

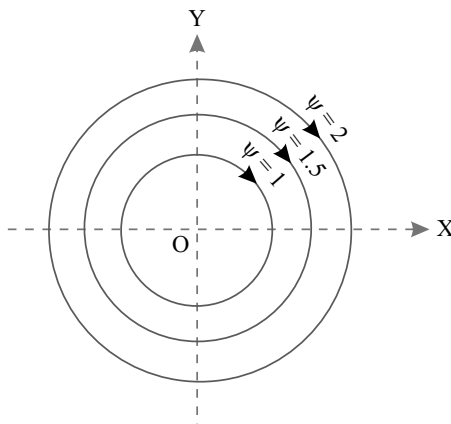


Fig. 5.26. Pattern of streamlines $\psi = x^2 + y^2$.

Example 5.44. If the expression for stream function is described by $\psi = x^3 - 3xy^2$, determine whether flow is rotational or irrotational. If the flow is irrotational, then indicate the correct value of the velocity potential.

(a) $\phi = y^3 - 3xy^2$

(b) $\phi = -3x^2y$

[UPSC]

Solution. Given: $\psi = x^3 - 3xy^2$...Stream function

A two-dimensional flow in $x-y$ plane will be irrotational if the vorticity vector in the z -direction is zero.

$$i.e., \quad \Omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \quad \dots(1)$$

We know, $u = \frac{\partial \psi}{\partial y} = \frac{\partial}{\partial y}(x^3 - 3xy^2) = -6xy$, and

$$v = -\frac{\partial \psi}{\partial x} = -\frac{\partial}{\partial x}(x^3 - 3xy^2) = -(3x^2 - 3y^2) = -3(x^2 - y^2)$$

$$\therefore \quad \frac{\partial u}{\partial y} = -6, \text{ and } \frac{\partial v}{\partial x} = -6x$$

Substituting these value in eqn. (1), we get:

$$\Omega_z = -6x - (-6x) = 0$$

Hence, the **flow is irrotational. (Ans.)**

For an irrotational flow Laplace equation in ϕ must be satisfied.

$$i.e., \quad \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

Let us check the validity for each expression for ϕ :

(a) $\phi = y^3 - 3xy^2$

$$\frac{\partial^2 \phi}{\partial x^2} = -6y \quad \text{and} \quad \frac{\partial^2 \phi}{\partial y^2} = 6y$$

$$\therefore \quad \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -6y + 6y = 0$$

(b) $\phi = -3x^2y$

$$\frac{\partial^2 \phi}{\partial x^2} = -6y \quad \text{and} \quad \frac{\partial^2 \phi}{\partial y^2} = 0$$

$$\therefore \quad \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} \neq 0$$

Hence, the correct value of $\phi = y^3 - 3xy^2$ (Ans.)

Example 5.45. In a two-dimensional incompressible flow, the fluid velocity components are given by $u = x - 4y$ and $v = -y - 4x$. Show that velocity potential exists and determine its form as well as stream function. [PTU]

Solution. Given: $u = x - 4y$ and $v = -y - 4x$...Velocity components.

The velocity potential will exist if flow is *irrotational*. Therefore, the vorticity component in the Z -direction must be zero.

Now,
$$\Omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Here,
$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x}(-y - 4x) = -4,$$

and,
$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(x - 4y) = -4$$

$$\therefore \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -4 - (-4) = 0$$

Since the *vorticity is zero*, the *flow is irrotational*; hence the **velocity potential exists. (Ans.)**

Total change in velocity potential,

$$\begin{aligned} d\phi &= \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy \\ &= -u dx - v dy = -(x - 4y)dx - (-y - 4x)dy \end{aligned}$$

or

$$d\phi = -x dx + 4y dx + y dy + 4x dy$$

Integrating, we get:

$$\begin{aligned} \phi &= -\frac{x^2}{2} + 4xy + \frac{y^2}{2} + 4xy + C \\ &= \frac{1}{2}(y^2 - x^2) + 8xy + C \end{aligned}$$

(where, C = constant of integration).

For $\phi = 0$ at the origin, the constant $C = 0$

$$\therefore \phi = \frac{1}{2}(y^2 - x^2) + 8xy \text{ (Ans.)}$$

For *stream function*,

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

We know,

$$u = \frac{\partial \psi}{\partial y} \quad \text{and} \quad v = -\frac{\partial \psi}{\partial x}$$

$$\therefore d\psi = -v dx + u dy = [-(-y - 4x)]dx + (x - 4y)dy$$

or

$$d\psi = (y + 4x)dx + (x - 4y)dy$$

Integrating, we get:

$$\begin{aligned} \psi &= xy + 4 \times \frac{x^2}{2} + xy - 4 \times \frac{y^2}{2} + C_1 \\ &= xy + 2x^2 + xy - 2y^2 + C_1 \\ &= 2(x^2 - y^2) + 2xy + C_1 \end{aligned}$$

(where C_1 = constant of integration)

For $\phi = 0$ at the origin, the constant $C_1 = 0$

$$\therefore \psi = 2(x^2 - y^2) + 2xy \text{ (Ans.)}$$

Example 5.46. In the two-dimensional incompressible flow field the velocity components are expressed as:

$$u = 2x - x^2y + \frac{y^3}{3}; v = xy^2 - 2y - \frac{x^3}{3}$$

- (i) Determine the velocity, and acceleration at point L ($x = 1$ m, $y = 3$ m).
 (ii) Is the flow possible? If so, obtain an expression for the stream function.
 (iii) What is the discharge between streamlines passing through (1, 3) and (2, 3)?
 (iv) Is the flow irrotational? If so, determine the corresponding velocity potential.
 (v) Show that each of the stream, and potential functions satisfy Laplace equation.

[MDU Haryana]

Solution. Given: $u = 2x - x^2y + \frac{y^3}{3}$; $v = xy^2 - 2y - \frac{x^3}{3}$...Velocity components

(i) **Velocity and acceleration at point L ($x = 1$ m, $y = 3$ m):**

$$u = 2x - x^2y + \frac{y^3}{3} = 2 \times 1 - 1^2 \times 3 + \frac{3^3}{3} = 8 \text{ m/s}$$

$$v = xy^2 - 2y - \frac{x^3}{3} = 1 \times 3^2 - 2 \times 3 - \frac{1^3}{3} = 2.67 \text{ m/s}$$

Resultant velocity, $V = \sqrt{u^2 + v^2} = \sqrt{(8)^2 + (2.67)^2} = \mathbf{8.43 \text{ m/s (Ans.)}}$

If the velocity is at an angle θ with X-axis, then:

$$\tan \theta = \frac{v}{u} = \frac{2.67}{8} = 0.3337 \quad \text{or} \quad \theta = 18.45^\circ$$

For a steady flow,

$$\begin{aligned} a_x &= u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \\ &= \left(2x - xy^2 + \frac{y^3}{3} \right) (2 - 2xy) + \left(xy^2 - xy - \frac{x^3}{3} \right) (y^2 - x^2) \end{aligned}$$

$\therefore$ Acceleration in the X-direction at $x = 1$ and $y = 3$,

$$a_x = 8(2 - 2 \times 1 \times 3) + 2.67(3^2 - 1^2) = -10.64 \text{ m/s}^2$$

Further,

$$\begin{aligned} a_y &= u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \\ &= \left(2x - xy^2 + \frac{y^3}{3} \right) (y^2 - x^2) + \left(xy^2 - xy - \frac{x^3}{3} \right) (2xy - 2) \\ &= 8(3^2 - 1^2) + 2.67(2 \times 1 \times 3 - 2) = 74.68 \text{ m/s}^2 \end{aligned}$$

Resultant acceleration, $a = \sqrt{a_x^2 + a_y^2} = \sqrt{(-10.64)^2 + (74.68)^2} = \mathbf{75.43 \text{ m/s}^2 \text{ (Ans.)}}$

(ii) **Is the flow physically possible:**

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = (2 - 2xy) + (2xy - 2) = 0$$

As the continuity equation is satisfied, hence the **flow is physically possible. (Ans.)**

Expression for stream function:

The differential $d\psi$ for stream function is,

$$\begin{aligned}
 d\psi &= \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy \\
 &= -v dx + u dy \\
 &= -\left(xy^2 - 2y - \frac{x^3}{3}\right) dx + \left(2x - x^2y + \frac{y^3}{3}\right) dy \\
 &= \left(-xy^2 + 2y + \frac{x^3}{3}\right) dx + \left(2x - x^2y + \frac{y^3}{3}\right) dy \\
 \text{or,} \quad d\psi &= \frac{x^3}{3} dx + \frac{y^3}{3} dy + 2d(xy) - d\left(\frac{x^2y^2}{2}\right)
 \end{aligned}$$

On integration, we get:

$$\psi = \frac{x^4}{12} + \frac{y^4}{12} + 2xy - \frac{x^2y^2}{2} \quad (\text{Ans.})$$

(iii) Discharge between stream lines passing through (1, 3) and (2, 3):

$$\begin{aligned}
 \psi_{(1,3)} &= \frac{1^4}{12} + \frac{3^4}{12} + 2 \times 1 \times 3 - \frac{1^2 \times 3^2}{2} = 8.33 \text{ m}^3/\text{s} \\
 \psi_{(2,3)} &= \frac{2^4}{12} + \frac{3^4}{12} + 2 \times 2 \times 3 - \frac{2^2 \times 3^2}{2} = 2.08 \text{ m}^3/\text{s}
 \end{aligned}$$

Hence, discharge between the stream lines

$$\psi_{(1,3)} - \psi_{(2,3)} = 8.33 - 2.08 = 6.25 \text{ m}^3/\text{s} \quad (\text{Ans.})$$

(iv) Is the flow irrotational?

$$\begin{aligned}
 \text{Rotation (angular velocity), } \omega_x &= \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \\
 &= \frac{1}{2} \left[(y^2 - x^2) - (y^2 - x^2) \right] = 0
 \end{aligned}$$

As the rotation is zero, the **flow is irrotational** and the potential function does exist. (Ans.)

Velocity potential:

$$\begin{aligned}
 d\phi &= \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy = u dx + v dy \\
 &= \left(2x - x^2y + \frac{y^3}{3}\right) dx + \left(xy^2 - 2y - \frac{x^3}{3}\right) dy \\
 &= 2x dx - 2y dy + \left(\frac{y^3}{3} dx + xy^2 dy\right) - \left(\frac{x^3}{3} dy + x^2y dx\right)
 \end{aligned}$$

$$\text{or,} \quad d\phi = 2x dx - 2y dy + \frac{1}{3} d(xy^3) - \frac{1}{3} d(x^3y)$$

On integration, we get:

$$\phi = x^2 - y^2 + \frac{xy^3}{3} - \frac{x^3y}{3} \quad (\text{Ans.})$$

(v) Check for Laplace equation:

We known that,

$$\psi = \frac{x^4}{12} + \frac{y^4}{12} + 2xy - \frac{x^2 y^2}{2}$$

$$\frac{\partial \psi}{\partial x} = \frac{x^3}{3} + 2y - xy^2; \quad \frac{\partial^2 \psi}{\partial x^2} = x^2 - y^2$$

$$\frac{\partial \psi}{\partial y} = \frac{y^3}{3} + 2x - yx^2; \quad \frac{\partial^2 \psi}{\partial y^2} = y^2 - x^2$$

$$\therefore \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = (x^2 - y^2) + (y^2 - x^2) = 0$$

Hence, stream function (ψ) satisfies Laplace equation. (Ans.)

Also,

$$\phi = x^2 - y^2 + \frac{xy^3}{3} - \frac{x^3 y}{3}$$

$$\frac{\partial \phi}{\partial x} = 2x + \frac{y^3}{3} - x^2 y; \quad \frac{\partial^2 \phi}{\partial x^2} = 2 - 2xy$$

$$\frac{\partial \phi}{\partial y} = -2y + xy^2 - \frac{x^3}{3}; \quad \frac{\partial^2 \phi}{\partial y^2} = -2 + 2xy$$

$$\therefore \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = (2 - 2xy) + (-2 + 2xy) = 0$$

Hence, potential function satisfies Laplace equation. (Ans.)

Example 5.47. In a two-dimensional flow the velocity components are $u = Cy$; $v = 0$ (where C is constant). Find the circulation about the circle $x^2 + y^2 - 2ay = 0$ situated in the flow (a is the radius of the circle).

Solution. Given: $u = Cy$; $v = 0$...Velocity components

Equation of the circle: $x^2 + y^2 - 2ay = 0$

(where, a = radius of the circle)

Circulation about the circle, Γ :

First method:

We know, $\Gamma = \Omega A$

where, Ω = Vorticity, and

A = Area.

Here, $\Omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 - \frac{\partial}{\partial y}(Cy) = -C$

Now, $A = \pi a^2$

$\therefore \Gamma = -C \times \pi a^2 = -C \pi a^2 \text{ m}^2/\text{s}$ (Ans.)

Second method:

Refer to Fig. 5.27.

We know, $\Gamma = \oint V_\theta \cdot ds$

(where, V_θ is the tangential velocity)

$$= \oint u \cdot dx = \oint C \cdot y \, dx$$

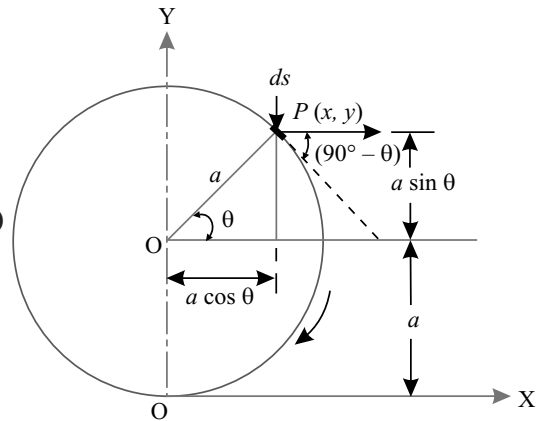


Fig. 5.27. Circulation about a circle.

From Fig. 5.27,

$$\begin{aligned}
 x &= a \cos \theta; y = a + a \sin \theta = a(1 + \sin \theta) \\
 \therefore dx &= -a \sin \theta d\theta \\
 \therefore \Gamma &= \int_0^{2\pi} Ca(1 + \sin \theta) \cdot (-a \sin \theta d\theta) \\
 &= -Ca^2 \int_0^{2\pi} (\sin \theta + \sin^2 \theta) d\theta \\
 &= -Ca^2 \int_0^{2\pi} \left(\sin \theta + \frac{1 - \cos 2\theta}{2} \right) d\theta \\
 &= -\frac{Ca^2}{2} \int_0^{2\pi} (2 \sin \theta + 1 - \cos 2\theta) d\theta \\
 &= -\frac{Ca^2}{2} \left[-2 \cos \theta + \theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi} \\
 &= -\frac{Ca^2}{2} [(-2 + 2\pi - 0) - (-2 + 0 - 0)] \\
 &= -\frac{Ca^2}{2} \times 2\pi = -C\pi a^2 \text{ m}^2/\text{s}
 \end{aligned}$$

i.e. $\Gamma = -C\pi a^2 \text{ m}^2/\text{s}$ (Ans.)

Example 5.48. The flow field of a fluid is given by $V = xyi + 2yzj - (yz + z^2)k$

- (i) Show that it represent a possible three-dimensional steady incompressible continuous flow.
(ii) Is this flow rotational or irrotational?

If rotational, determine at point A (2, 4, 6):

- (a) Angular velocity,
(b) Vorticity,
(c) Shear strains, and
(d) Dilatency.

Solution. Given:

$$V = xyi + 2yzj - (yz + z^2)k \quad \dots \text{Fluid flow field}$$

Here,

$$u = xy, v = 2yz, w = -(yz + z^2)$$

$$\frac{\partial u}{\partial x} = y, \frac{\partial v}{\partial y} = 2z, \frac{\partial w}{\partial z} = -(y + 2z)$$

(i) Is the flow physically possible?

Flow will be three-dimensional steady incompressible continuous flow, if continuity equation is satisfied,

$$i.e. \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \text{or} \quad y + 2z - (y + 2z) = 0$$

$$\text{Vectorially, } \nabla \cdot V = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(2yz) + \frac{\partial}{\partial z}[-(yz + z^2)] = y + 2z - (y + 2z) = 0$$

Hence, the given flow is a three-dimensional steady incompressible flow. (Ans.)

(ii) Flow-rotational or irrotational?

Now, $\frac{\partial u}{\partial y} = x, \frac{\partial v}{\partial x} = 0$

Since, $\frac{\partial u}{\partial y} \neq \frac{\partial v}{\partial x},$

therefore, the flow is **rotational**. (Ans.)

(a) Angular velocity, ω :

$$\begin{aligned} \text{We know, } \omega &= \frac{1}{2} (\Delta \times V) \begin{pmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & 2yz & -(yz + z^2) \end{pmatrix} \\ &= \frac{1}{2} \left[i \left\{ \frac{\partial}{\partial y} (-yz - z^2) - \frac{\partial}{\partial z} (2yz) \right\} + j \left\{ \frac{\partial}{\partial z} (xy) - \frac{\partial}{\partial x} \{ -(yz + z^2) \} \right\} \right. \\ &\quad \left. + k \left\{ \frac{\partial}{\partial x} (2yz) - \frac{\partial}{\partial y} (xy) \right\} \right] \\ &= \frac{1}{2} [i(-z - 2y) - j(0 - 0) + k(0 - x)] \end{aligned}$$

At A (2, 4, 6), $\omega = -\frac{1}{2}(14i + 2k)$ (Ans.)

(b) Vorticity, Ω :

$$\Omega = 2\omega = -(14i + 2k) \text{ (Ans.)}$$

(c) Shear strains:

$$\gamma_{xy} = \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) = \frac{1}{2} (0 + x) = \frac{1}{2} (0 + 2) = 1 \text{ (Ans.)}$$

$$\gamma_{yz} = \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) = \frac{1}{2} (2y - z) = \frac{1}{2} (2 \times 4 - 6) = 1 \text{ (Ans.)}$$

$$\gamma_{xz} = \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) = \frac{1}{2} (0 + 0) = 0 \text{ (Ans.)}$$

(d) Dilatency (linear strains):

$$e_x = \frac{\partial u}{\partial x} = y = 4 \text{ (Ans.)}$$

$$e_y = \frac{\partial v}{\partial y} = 2z = 2 \times 6 = 12 \text{ (Ans.)}$$

$$e_z = \frac{\partial w}{\partial z} = (y + 2z) = -(4 + 2 \times 6) = -16 \text{ (Ans.)}$$

Example 5.49. The stream function $\psi = 4xy$, in which ψ is in $\text{cm}^2/\text{second}$ and x and y are in metres, describe the incompressible flow between the boundaries shown below (Fig. 5.28).

Calculate:

(i) Velocity at B,

(ii) Convective acceleration at B, and

(iii) Flow rate per unit width across AB.

[UPSC]

Solution. Given: $\psi = 4xy$... Stream function
(where ψ is in cm^2/s and x and y in metres)

Co-ordinates of A are: (3, 0)

Co-ordinates of B are:

$$x = 3 \text{ m}, y = \frac{3}{x} = \frac{3}{3} = 1 \text{ m}$$

$$\left(\because xy = 3, \therefore y = \frac{3}{x} \right)$$

Hence, co-ordinates of B are: (3, 1)

(i) Velocity at B:

Velocity components are given by:

$$u = \frac{\partial \psi}{\partial y} = \frac{\partial}{\partial y}(4xy) = 4x, \text{ and}$$

$$v = -\frac{\partial \psi}{\partial x} = -\frac{\partial}{\partial x}(4xy) = -4y$$

At point B:

$$u = 4 \times 3 = 12 \text{ cm/s, and}$$

$$v = -4 \times 1 = -4 \text{ cm/s}$$

$$\therefore \text{Velocity at B, } V_B = \sqrt{u^2 + v^2} = \sqrt{12^2 + (-4)^2} = \mathbf{12.65 \text{ cm/s (Ans.)}}$$

(ii) Convective acceleration at B, a_B :

$$\text{Convective acceleration, } a = \sqrt{a_x^2 + a_y^2}$$

$$a_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}, \text{ and } a_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}$$

But,

$$u = 4x \text{ and } v = -4y$$

$$\therefore \frac{\partial u}{\partial x} = 4, \frac{\partial u}{\partial y} = 0; \frac{\partial v}{\partial x} = 0, \frac{\partial v}{\partial y} = -4$$

$$\therefore a_x = (4x)(4) + (-4y)(0) = 16x, \text{ and}$$

$$a_y = (4x)(0) + (-4y)(-4) = 16y$$

$$\text{Also, } a = \sqrt{a_x^2 + a_y^2} = \sqrt{(16x)^2 + (16y)^2}$$

$\therefore$ Convective acceleration at B (3,1) is,

$$a_B = \sqrt{(16 \times 3)^2 + (16 \times 1)^2} = 16\sqrt{3^2 + 1} = \mathbf{50.6 \text{ cm/s}^2 \text{ (Ans.)}}$$

(iii) Flow rate per unit width across AB, q_{AB} :

$$q_{AB} = \psi_B - \psi_A = \psi_{(3,1)} - \psi_{(3,0)}$$

$$= 4 \times 3 \times 1 - 4 \times 3 \times 0 = 12 \text{ cm}^3/\text{s/cm}$$

i.e.

$$\mathbf{q_{AB} = 12 \text{ cm}^3/\text{s/cm (Ans.)}}$$

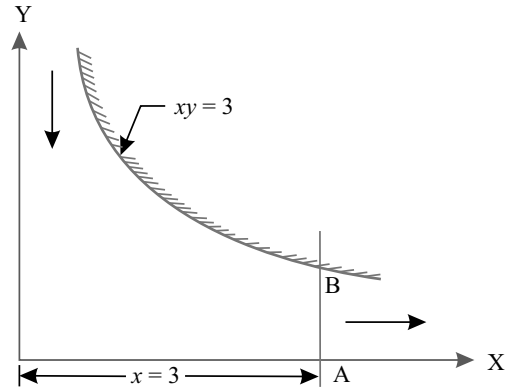


Fig. 5.28

HIGHLIGHTS

1. *Fluid kinematics* is a branch of fluid mechanics which deals with the study of velocity and acceleration of the particles of fluids in motion and their distribution in space without considering any forces or energy involved.
2. The motion of fluid particles may be described by the following methods:
 - (i) *Langrangian method*. In this method, the observer concentrates on the movement of a single particle. The path taken by the particle and the changes in its velocity and acceleration are studied.
 - (ii) *Eulerian method*. In Eulerian method, the observer concentrates on a point in the fluid system. Velocity, acceleration and other characteristics of the fluid at that particular point are studied.

The components of acceleration of the fluid particle are given by:

$$a_x = \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) + \frac{\partial u}{\partial t}$$

$$a_y = \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) + \frac{\partial v}{\partial t}$$

$$a_z = \left(u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) + \frac{\partial w}{\partial t}$$

Resultant velocity, $V = \sqrt{u^2 + v^2 + w^2}$

$$V = ui + vj + wk$$

... in vector notation.

Resultant acceleration, $a = \sqrt{a_x^2 + a_y^2 + a_z^2}$

$$a = a_x i + a_y j + a_z k$$

... in vector notation.

Also,
$$a = V \frac{\partial V}{\partial s} + \frac{\partial V}{\partial t}$$

where,
$$\frac{V \partial V}{\partial s}$$
 ... is called *convective acceleration*,

and,
$$\frac{\partial V}{\partial t}$$
 ... is called *local acceleration*.

3. Types of fluid flow:

- (i) Steady and unsteady flows
- (ii) Uniform and non-uniform flows
- (iii) One, two and three-dimensional flows
- (iv) Rotational and irrotational flows
- (v) Laminar and turbulent flows
- (vi) Compressible and incompressible flows.

4. Types of flow lines:

- (i) *Path line*. It is the path followed by a fluid particle in motion.
- (ii) *Stream line*. It is an imaginary line within the flow so that the tangent at any point on it indicates the velocity at that point.

(iii) *Stream tube*. It is a fluid mass bounded by a group of stream lines.

(iv) *Streak line*. It is a curve which gives an instantaneous picture of the location of the fluid particles which have passed through a given point.

5. The *continuity equation* based on the principle of conservation of mass is stated as follows:
“If no fluid is added or removed from the pipe in any length then the mass passing across different sections shall be same”.

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad \dots \text{in case of compressible fluids}$$

$$A_1 V_1 = A_2 V_2 \quad \dots \text{in case of incompressible fluids.}$$

The continuity equation in three dimensions in *cartesian co-ordinates* is given as:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \dots \text{for steady flow of incompressible fluid } (\rho = \text{constant})$$

The continuity equation in *polar co-ordinates* is given as:

$$\frac{1}{r}(\rho v_r) + \frac{\partial}{\partial r}(\rho v_r) + \frac{\partial}{r\partial\theta}(\rho v_\theta) = 0 \quad \dots \text{for compressible flow}$$

$$\frac{v_r}{r} + \frac{\partial v_r}{\partial r} + \frac{\partial v_\theta}{r\partial\theta} = 0 \quad \dots \text{for incompressible flow}$$

where,

v_r = Velocity component in radial direction, and

v_θ = Velocity component in tangential direction.

6. *Circulation* (Γ) is defined mathematically as the line integral of the tangential velocity about a closed path (contour).

$$\Gamma = \oint V \cos \theta \cdot ds$$

where,

V = Velocity in the flow field, and

θ = Angle between V and the tangent to the path (in the positive anticlockwise direction along the path) at that point.

Vorticity (Ω) is defined as the circulation per unit of enclosed area $\left(i.e. \Omega = \frac{\Gamma}{A} \right)$.

If a flow possesses vorticity, it is rotational. The flow is irrotational if rotation (ω) is zero.

The expression for rotation are:

$$w_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right); \quad w_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right); \quad w_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$$

7. *Velocity potential* (ϕ) is defined as a scalar function of space and time such that its negative derivative with respect to any direction gives the fluid velocity in that direction.

$$u = -\frac{\partial \phi}{\partial x}, \quad v = -\frac{\partial \phi}{\partial y}, \quad w = -\frac{\partial \phi}{\partial z}$$

Stream function (ψ) is defined as a scalar function of space and time, such that its partial derivative with respect to any direction gives the velocity components at right angles (in the counterclockwise direction) to this direction.

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

Existence of ψ means a possible case of flow.

$$\left. \begin{aligned} u &= -\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} \\ v &= -\frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x} \end{aligned} \right\} \dots \text{are known as Cauchy-Reimann equations.}$$

8. *Flow net.* A grid obtained by drawing a series of stream lines and equipotential lines is known as a flow net.

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer

- The motion of fluid particles may be described by which of the following methods?
 - Langrangian method
 - Eulerain method
 - Both (a) and (b)
 - None of the above.
- In which of the following methods, the observer concentrates on a point in the fluid system?
 - Langrangian method
 - Eulerian method
 - Any of the above
 - None of the above.
- Normal acceleration in fluid-flow situation exists only when
 - the flow is unsteady
 - the flow is two-dimensional
 - the streamlines are straight and parallel
 - the streamlines are curved.
- In a steady flow the velocity
 - does not change from place to place
 - at a given point does not change with time
 - may change its direction but the magnitude remains unchanged
 - none of the above.
- The flow in a pipe whose valve is being opened or closed gradually is an example of
 - steady flow
 - unsteady flow
 - rotational flow
 - compressible flow.
- The type of flow in which the velocity at any given time does not change with respect to space is called
 - steady flow
 - compressible flow
 - uniform flow
 - rotational flow.
- Flow in a pipe where average flow parameters are considered for analysis is an example of
 - incompressible flow
 - one-dimensional flow
 - two-dimensional flow
 - three-dimensional flow.
- The flow in a river during the period of heavy rainfall is
 - steady, non-uniform and three-dimensional
 - steady, uniform, two-dimensional
 - unsteady, uniform, three-dimensional
 - unsteady, non-uniform and three-dimensional.
- Flow between parallel plates of infinite extent is an example of
 - one-dimensional flow
 - two-dimensional flow
 - three-dimensional flow
 - compressible flow.
- If the flow is irrotational as well as steady it is known as
 - non-uniform flow
 - one-dimensional flow
 - potential flow
 - none of the above.
- High velocity flow in a conduit of large size is known as
 - laminar flow
 - turbulent flow
 - either of the above
 - none of the above.
- If the Reynolds number is less than 2000, the flow in a pipe is
 - laminar flow
 - turbulent flow
 - transition flow
 - none of the above.
- The path followed by fluid particle in motion is called a
 - streamline
 - path line
 - streak line
 - none of the above.
- A....is an imaginary line within the flow so that the tangent at any point on it indicates the velocity at that point.
 - streak line
 - stream line

- (c) path line (d) none of the above.
15. A stream line is one
 (a) in which stream function does not change
 (b) in which the flow cannot cross the bounding surface
 (c) which has a constant area throughout its length so that the velocity remains constant.
 (d) none of the above.
16. is a curve which gives an instantaneous picture of the location of the fluid particles which have passed through a given point.
 (a) Path line (b) Stream line
 (c) Streak line (d) None of the above.
17. In fluid mechanics, the continuity equation is a mathematical statement embodying the principle of
 (a) conservation of momentum
 (b) conservation of mass
 (c) conservation of energy
 (d) none of the above.
18. An irrotational flow is one in which
 (a) the stream lines of flow are curved and closely spaced
 (b) the fluid does not rotate as it moves along
 (c) the net rotation of fluid particles about their mass centres remains zero
 (d) none of the above.
19. In a fluid-flow the stream lines are lines
 (a) along which the vorticity is zero
 (b) along which the stream function $\psi = \text{constant}$
 (c) which are parallel to the equipotential lines
 (d) which exist in irrotational flow only.
20. is defined mathematically as the line integral of the tangential velocity about a closed path (contour).
 (a) circulation (b) vorticity
 (c) either of the above (d) none of the above.
21. The concept of stream function which is based on the principle of continuity is applicable to
 (a) irrotational flow only
 (b) two-dimensional flow only
 (c) three-dimensional flow
 (d) uniform flow only.
22. The motion is described as when the components of rotation or vorticity are zero throughout certain point of the fluid.
 (a) rotational (b) irrotational
 (c) either of the above (d) none of the above.
23. is defined as a scalar function of space and time such that its negative derivative with respect to any direction gives the fluid velocity in that direction.
 (a) Velocity potential function
 (b) Stream function
 (c) Circulation
 (d) Vorticity.
24. If velocity potential (ϕ) satisfies the Laplace equation, it represents the possible..... flow.
 (a) unsteady, compressible, rotational
 (b) steady, compressible, irrotational
 (c) unsteady, incompressible, rotational
 (d) steady, incompressible, irrotational.
25. A flownet is a graphical representation of stream lines and equipotential lines such that these lines
 (a) intersect each other orthogonally forming curvilinear squares
 (b) intersect each other at various different angles forming irregular-shaped nets
 (c) indicate the direction and magnitude of vector
 (d) none of the above.
26. The flow-net analysis can be used to determine
 (a) the stream lines and equipotential lines
 (b) quantity of seepage and upward lift pressure below hydraulic structures
 (c) the efficient boundary shapes, for which the flow does not separate.
 (d) the velocity and pressure distribution for given boundaries of flow (provided the velocity distribution and pressure at any reference section are known).
 (e) all of the above.

ANSWER

- | | | | | | |
|---------|----------|---------|---------|---------|---------|
| 1. (c) | 2. (d) | 3. (b) | 4. (b) | 5. (b) | 6. (c) |
| 7. (b) | 8. (d) | 9. (b) | 10. (c) | 11. (b) | 12. (a) |
| 13. (b) | 14. (b) | 15. (b) | 16. (c) | 17. (b) | 18. (c) |
| 19. (b) | 20. (a) | 21. (b) | 22. (b) | 23. (a) | 24. (d) |
| 25. (a) | 26. (e). | | | | |

THEORETICAL QUESTIONS

1. Differentiate between the Eulerian and Lagrangian methods of representing fluid flow.
2. Distinguish between pathlines, stream lines and streak lines.
3. Define convective and local accelerations.
4. Define tangential and normal accelerations.
5. How are fluid flows classified?
6. Define steady, non-steady, uniform and non-uniform flows.
7. Differentiate between the rotational and irrotational flows.
8. Sketch the velocity distribution for uniform irrotational flow.
9. How is the continuity equation based on the principle of conservation of mass stated?
10. Derive the continuity equation in cartesian co-ordinates.
11. How is 'circulation' defined?
12. What do you understand by vorticity?
13. Define and explain briefly the following:
(i) Velocity potential; (ii) Stream function.
14. In the analysis of two-dimensional irrotational flow what use can the velocity potential and stream function be put to?
15. To what type of flow is the concept of velocity potential and stream function applicable?
16. If stream function exists in a flow problem, does it imply that velocity potential also exists?
17. From the consideration of vorticity and rotation show that in case of ideal fluids the flow is irrotational.
18. Show that the stream lines and equipotential lines form a net of mutually perpendicular lines.
19. What is a 'flow-net'? Enumerate the methods of drawing flow nets.
20. Is the flow-net analysis applicable to rotational flow? If not, why?

UNSOLVED EXAMPLES

1. Given the velocity field:

$$V = (6 + 2xy + t^2)i - (xy^2 + 10t)j + 25k$$
 What is the acceleration of a particle at (3, 0, 2) at time $t = 1$? [IIT Bombay]
 [Ans. + 58.35 units]
2. The velocity along a stream line passing through the origin is given by:

$$V = 2\sqrt{x^2 + y^2}$$
 What is the velocity and acceleration at (4, 3)?
 [Ans. 10 m/s, 20 m/s²]
3. The velocity components in a steady flow are:
 $u = 2kx; v = 2ky; w = -4kz$.
 What is the equation of a stream line passing through the point (1, 0, 1)?
 [Ans. $y = 0, z = \frac{1}{x^2}$]
4. Determine whether the continuity equation is satisfied by the following velocity components for an incompressible fluid:
 $u = x^2y, v = 2xy - xy^2, w = x^2 - z^2$
 [Ans. Yes, the continuity equation is satisfied]
5. A conical pipe diverges uniformly from 0.1 m to 0.2 m diameter over a length of 1m. Determine the local and convective accelerations at the mid-section assuming (i) rate of flow is 0.1 m³/s and it remains constant, and (ii) rate of flow varies uniformly from 0.1 to 0.2 m³/s in 5 sec., at $t = 2$ sec.
 [Ans. (i) zero, - 42.76 m/s²
 (ii) 1.132 m/s², - 83.81 m/s²]
6. Determine the missing component of velocity distribution such that they satisfy continuity equation $u = 2x^2 + 2xy, v = 2yz^2 + 3z^2, w = ?$
 [Ans. $w = -4xz - 2yz - \frac{2}{3}z^3 + f(x, y, t)$]
7. The velocity components in a three-dimensional fluid flow are:
 $u = x^2 + y^2 z^3, v = -(xy + yz + zx)$
 Determine the missing component of velocity distribution such that continuity equation is satisfied.
 [Ans. $w = -xz + \frac{z^2}{2} + f(x, y, t)$]
8. The velocity components of fluid flow (incompressible) are:
 $u = x^2y, v = 2yz - xy^2, w = x^2 - z^2$
 Show that this flow is kinematically possible.
9. In a three-dimensional incompressible fluid flow, the velocity components are:

$$u = x^2 + z^2 + 5, v = y^2 + z^2 - 3$$

(i) Determine the third component of velocity.

(ii) Is the fluid flow irrotational?

[Ans. (i) $w = -2(x + y)z + f(x, y, t)$ (ii) No.]

10. The velocity potential function (ϕ) is given by:

$$\phi = -\frac{xy^3}{3} - x^2 + \frac{x^3y}{3} + y^2$$

Determine the velocity components in x and y directions and show that ϕ represents a possible case of flow.

$$\left[\text{Ans. } u = \frac{y^3}{3} + 2x - x^2y, v = xy^2 - \frac{x^3}{3} - 2y \right]$$

11. If $\phi = 2xy$ determine ψ .

$$[\text{Ans. } \psi = (y^2 - x^2) + C]$$

12. Does the velocity potential exist for two-dimensional incompressible flow prescribed by

$$u = x - 4y; v = -(y + 4x)?$$

If so determine its form as well as that of stream function.

$$\left[\begin{array}{l} \text{Ans. } \phi = \frac{x^2}{2} - \frac{y^2}{2} - 4xy + \text{constant} \\ \psi = 2x^2 - 2y^2 + xy + \text{constant} \end{array} \right]$$

13. A two-dimensional flow is described by the velocity components:

$$u = 5x^3 \text{ and } v = -15x^2y$$

Determine the stream function, velocity and acceleration at $P(x = 1 \text{ m}, y = 2 \text{ m})$.

$$[\text{Ans. } 10 \text{ m}^3/\text{s}, 30.41 \text{ m/s}, 167.70 \text{ m/s}^2]$$

14. If $u = ax$ and $v = ay$ and $w = -2az$ are the velocity components for a fluid flow in a particular case, check whether they satisfy the continuity equation. If they do, is the flow rotational or irrotational? Also obtain equation of stream line passing through the point $(2, 2, 4)$

$$[\text{Ans. Yes, irrotational, } x = y, xz^{1/2} = 4]$$

15. If the velocity field is given by $u = x^2 + 2xy$ and $v = (y^2 + 2xy)$, determine the circulation around

a closed curve defined by:

$$x = 1, x = 3, y = 1, y = 4. \quad [\text{Ans. } -86]$$

16. If the velocity field is given by

$$u = x^2 - y^2 + x \text{ and } v = -(2xy + y),$$

determine ϕ and ψ .

$$\left[\begin{array}{l} \text{Ans. } \phi = \frac{y^2}{2} - (2x - 1) - \frac{x^3}{3} + C, \\ \psi = -(2x + y) + C \end{array} \right]$$

17. In a two-dimensional flow field for an incompressible fluid the velocity components are:

$$u = \frac{y^3}{3} + 2x - x^2y$$

$$v = xy^2 - 2y - \frac{x^3}{3}$$

Find an expression for the stream function ψ .

$$\left[\text{Ans. } \psi = \frac{x^2y^2}{2} - 2xy - \frac{x^4}{12} - \frac{y^4}{12} \right]$$

18. From the law of conservation of mass, show that whether the flow field represented by

$$u = -3x + y^2 - \frac{1}{x} \text{ and } v = x^2 + 3y + y \log x$$

is a possible velocity field for two-dimensional incompressible fluid flow.

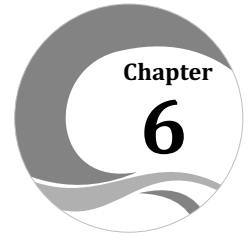
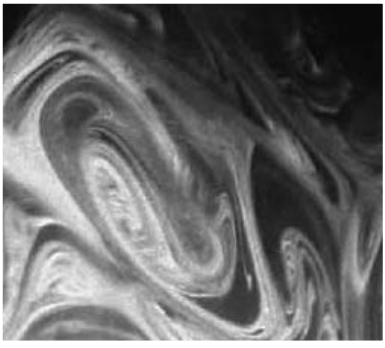
[IIT Delhi]

[Ans. Not possible]

19. If for two-dimensional flow the stream function is given by $\psi = 2xy$, calculate the velocity at the point $(3, 6)$. Show that velocity potential ϕ exists for this case and deduce it. Draw the stream lines corresponding to $\psi = 100$ and $\psi = 300$; also equipotential lines corresponding to $\phi = 100$ and $\phi = 300$. About six points on each line to represent its trend would suffice.

[UPSC]

$$[\text{Ans. } 2\sqrt{45}, \phi = y^2 + x^2].$$



FLUID DYNAMICS

- 6.1. Introduction
- 6.2. Different types of heads (or energies) of a liquid in motion
- 6.3. Bernoulli's equation
- 6.4. Euler's equation for motion
- 6.5. Bernoulli's equation for real fluids
- 6.6. Practical applications of Bernoulli's equation — Venturimeter-Orificemeter—Pitot tube
- 6.7. Free liquid jet
- 6.8. Impulse momentum equation
- 6.9. Kinetic energy and momentum correction factors (Coriolis co-efficients)
- 6.10. Moment of momentum equation
- 6.11. Vortex motion—Forced vortex flow-free vortex flow-equation of motion for vortex flow-equation of forced vortex-rotation of liquid in a closed cylindrical vessel

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

6.1. INTRODUCTION

When the fluids are at rest, the only fluid property of significance is the specific weight of the fluids. On the other hand, when a fluid is in motion various other fluid properties become significant, as such the nature of flow of a real fluid is complex. *The science which deals with the geometry of motion of fluids without reference to the forces causing the motion is known as “hydrokinematics”* (or simply *kinematics*). Thus, kinematics involves merely the description of the motion of fluids in terms of space-time relationship. *The science which deals with the action of the forces in producing or changing motion of fluids is known as “hydrokinetics”* (or simply *kinetics*). Thus, the study of fluids in motion involves the consideration of both the kinematics and kinetics. The dynamic equation of fluid motion is obtained by applying Newton's second law of motion to a fluid element considered as a free body. *The fluid is assumed to be incompressible and non-viscous.*

In fluid mechanics the basic equations are: (i) *Continuity equation*, (ii) *Energy equation*, and (iii) *Impulse-momentum equation*. In this chapter energy equation and impulse-momentum equations will be discussed (Continuity equation has already been discussed in Chapter 5).

6.2. DIFFERENT TYPES OF HEADS (OR ENERGIES) OF A LIQUID IN MOTION

There are three types of energies or heads of flowing liquids:

1. Potential head or potential energy:

This is due to configuration or position above some suitable datum line. It is denoted by z .

2. Velocity head or kinetic energy:

This is due to velocity of flowing liquid and is measured as $\frac{V^2}{2g}$ where, V is the velocity of flow and 'g' is the acceleration due to gravity ($g = 9.81$)

3. Pressure head or pressure energy:

This is due to the pressure of liquid and reckoned as $\frac{p}{w}$ where, p is the pressure, and w is the weight density of the liquid.

Total head/energy:

Total head of a liquid particle in motion is the sum of its potential head, kinetic head and pressure head. Mathematically,

$$\text{Total head, } H = z + \frac{V^2}{2g} + \frac{p}{w} \text{ m of liquid} \quad \dots[6.1 (a)]$$

$$\text{Total energy, } E = z + \frac{V^2}{2g} + \frac{p}{w} \text{ Nm/kg of liquid} \quad \dots[6.1 (b)]$$

Example 6.1. In a pipe of 90 mm diameter water is flowing with a mean velocity of 2 m/s and at a gauge pressure of 350 kN/m². Determine the total head, if the pipe is 8 metres above the datum line. Neglect friction.

Solution. Diameter of the pipe = 90 mm

Pressure, $p = 350 \text{ kN/m}^2$

Velocity of water, $V = 2 \text{ m/s}$

Datum head, $z = 8 \text{ m}$

Specific weight of water, $w = 9.81 \text{ kN/m}^3$

Total head of water, H:

$$\begin{aligned} H &= z + \frac{V^2}{2g} + \frac{p}{w} \\ &= 8 + \frac{2^2}{2 \times 9.81} + \frac{350}{9.81} = 43.88 \text{ m} \end{aligned}$$

$$H = 43.88 \text{ m (Ans.)}$$

6.3. BERNOULLI'S EQUATION

Bernoulli's equation states as follows:

"In an ideal incompressible fluid when the flow is steady and continuous, the sum of pressure energy, kinetic energy and potential (or datum) energy is constant along a stream line."

Mathematically,

$$\frac{p}{w} + \frac{V^2}{2g} + z = \text{constant}$$

where,

$\frac{p}{w}$ = Pressure energy,

$\frac{V^2}{2g}$ = Kinetic energy, and

z = Datum (or elevation) energy.

Proof:

Consider an ideal incompressible liquid through a non-uniform pipe as shown in Fig 6.1. Let us consider two sections LL and MM and assume that the pipe is running full and there is continuity of flow between the two sections;

Let, p_1 = Pressure at LL,
 V_1 = Velocity of liquid at LL,
 z_1 = Height of LL above the datum,
 A_1 = Area of pipe at LL, and
 p_2, V_2, z_2, A_2 = Corresponding values at MM.

Let the liquid between the two sections LL and MM move to $L' L'$ and $M' M'$ through very small lengths dl_1 and dl_2 as shown in Fig. 6.1. This movement of liquid between LL and MM is equivalent to the movement of the liquid between LL and $L' L'$ and MM and $M' M'$, the remaining liquid between $L' L'$ and MM being unaffected.

Let, W = Weight of liquid between LL and $L' L'$.

As the flow is continuous,

$$\therefore W = wA_1 \cdot dl_1 = wA_2 \cdot dl_2$$

$$\text{or, } A_1 \cdot dl_1 = \frac{W}{w} \quad \dots(i)$$

$$\text{Similarly, } A_2 \cdot dl_2 = \frac{W}{w} \quad \dots(ii)$$

$$\therefore A_1 \cdot dl_1 = A_2 \cdot dl_2$$

Work done by pressure at LL, in moving the liquid to $L' L'$

$$= \text{Force} \times \text{distance} = p_1 \cdot A_1 \cdot dl_1$$

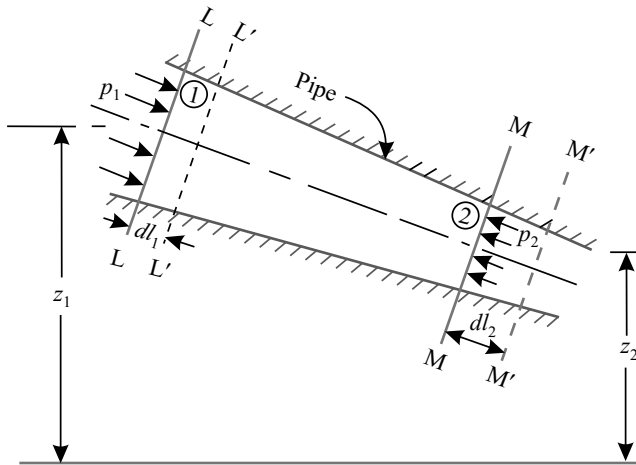


Fig. 6.1. Bernoulli's equation.

Similarly, work done by the pressure at MM in moving the liquid to $M' M'$ = $-p_2 \cdot A_2 \cdot dl_2$
 (– ve sign indicates that direction of p_2 is opposite to that of p_1)

$\therefore$ Total work done by the pressure

$$\begin{aligned} &= p_1 \cdot A_1 \cdot dl_1 - p_2 \cdot A_2 \cdot dl_2 \\ &= p_1 \cdot A_1 \cdot dl_1 - p_2 \cdot A_1 \cdot dl_1 \quad (\because A_1 \cdot dl_1 = A_2 \cdot dl_2) \\ &= A_1 \cdot dl_1 (p_1 - p_2) \end{aligned}$$

$$= \frac{W}{w} (p_1 - p_2) \quad \left(\because A_1 \cdot dl_1 = \frac{W}{w} \right)$$

$$\text{Loss of potential energy} = W(z_1 - z_2)$$

$$\text{Gain in kinetic energy} = W \left(\frac{V_2^2}{2g} - \frac{V_1^2}{2g} \right) = \frac{W}{2g} (V_2^2 - V_1^2)$$

Also, Loss of potential energy + work done by pressure = Gain in kinetic energy

$$\therefore W(z_1 - z_2) + \frac{W}{w} (p_1 - p_2) = \frac{W}{2g} (V_2^2 - V_1^2)$$

$$\text{or,} \quad (z_1 - z_2) + \left(\frac{p_1}{w} - \frac{p_2}{w} \right) = \left(\frac{V_2^2}{2g} - \frac{V_1^2}{2g} \right)$$

$$\text{or,} \quad \frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 \quad \dots(6.2)$$

which proves *Bernoulli's equation*.

Assumptions:

It may be mentioned that the following *assumptions* are made in the derivation of Bernoulli's equation:

1. The liquid is ideal and incompressible.
2. The flow is steady and continuous.
3. The flow is along the stream line, *i.e.*, it is one-dimensional.
4. The velocity is uniform over the section and is equal to the mean velocity.
5. The only forces acting on the fluid are the *gravity forces* and the *pressure forces*.

6.4. EULER'S EQUATION FOR MOTION

Consider steady flow of an ideal fluid along the stream tube. Separate out a small element of fluid of cross-sectional area dA and length ds from stream tube as a free body from the moving fluid.

Fig. 6.2 shows such a small element LM of fluid of cross-section area dA and length ds .

Let,

p = Pressure on the element at L,

$p + dp$ = Pressure on the element at M, and

V = Velocity of the fluid element.

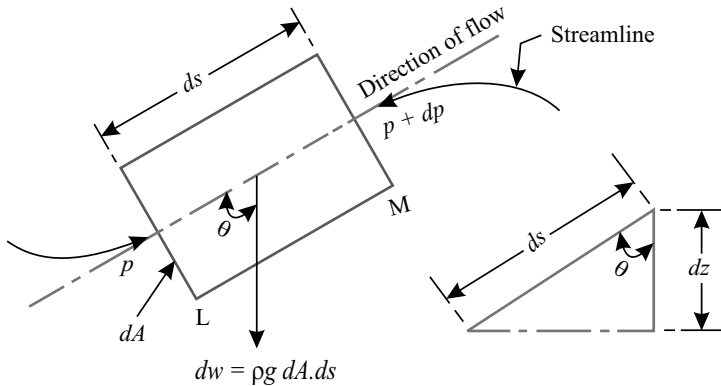


Fig. 6.2. Forces on a fluid element (Euler's equation).

The external forces tending to accelerate the fluid element in the direction of stream line are as follows:

1. Net pressure force in the direction of flow is,

$$p.dA - (p + dp).dA = -dp \cdot dA \quad \dots(i)$$

2. Component of the weight of the fluid element in the direction of flow is

$$\begin{aligned} &= -\rho.g.dA.ds.\cos\theta \\ &= -\rho.g.dA.ds\left(\frac{dz}{ds}\right) \quad \left(\because \cos\theta = \frac{dz}{ds}\right) \\ &= -\rho.g.dA.dz \quad \dots(ii) \end{aligned}$$

$$\text{Mass of the fluid element} = \rho.dA.ds \quad \dots(iii)$$

The acceleration of the fluid element

$$a = \frac{dV}{dt} = \frac{dV}{ds} \times \frac{ds}{dt} = V \cdot \frac{dV}{ds} \quad \dots(iv)$$

Now, according to Newton's second law of motion, Force = Mass $\times$ acceleration

$$\therefore -dp.dA - \rho.g.dA.dz = \rho.dA.ds \times V \cdot \frac{dV}{ds}$$

Dividing both sides by $\rho.dA$, we get:

$$\frac{-dp}{\rho} - g.dz = V.dV$$

$$\text{or,} \quad \frac{dp}{\rho} + V.dV + g.dz = 0 \quad \dots(6.3)$$

This is the required **Euler's equation** for motion, and is in the form of *differential equation*.

Integrating the above eqn., we get:

$$\frac{1}{\rho} \int dp + \int V.dV + \int g.dz = \text{constant}$$

$$\frac{p}{\rho} + \frac{V^2}{2} + gz = \text{constant}$$

Dividing by g , we get:

$$\frac{p}{\rho g} + \frac{V^2}{2g} + z = \text{constant}$$

$$\text{or,} \quad \frac{p}{w} + \frac{V^2}{2g} + z = \text{constant}$$

or, in other words,

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

which proves *Bernoulli's equation*.

Euler's equation in Cartesian coordinates:

Consider an infinitely small mass of fluid enclosed in an elementary parallelepiped of sides dx , dy and dz as shown in Fig. 6.3. The motion of the fluid element is influenced by the following forces:

(i) Normal forces due to pressure:

The intensities of hydrostatic pressure acting normal to each face of the parallelepiped are shown in Fig. 6.3.

The net pressure force in the X-direction

$$\begin{aligned}
 &= p \cdot dy \cdot dz - \left(p + \frac{\partial p}{\partial x} dx \right) dy \cdot dz \\
 &= -\frac{\partial p}{\partial x} dx \cdot dy \cdot dz
 \end{aligned}$$

(ii) Gravity or body force:

Let B be the body force *per unit mass of fluid* having components B_x , B_y and B_z in the X , Y and Z directions respectively.

Then, the body force acting on the parallelopiped in the direction of X -coordinate is $= B_x \cdot \rho \cdot dx \cdot dy \cdot dz$.

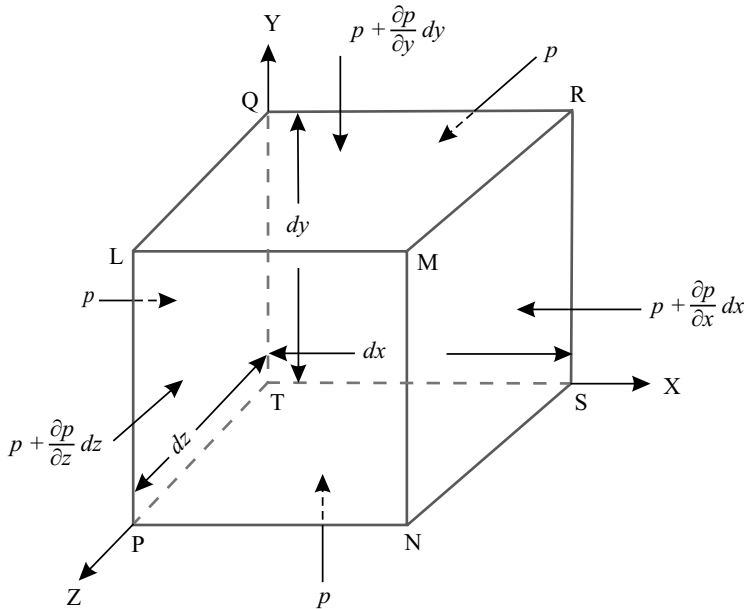


Fig. 6.3. Normal surface forces on a non-viscous fluid element.

(iii) Inertia forces:

The inertia force acting on the fluid mass, along the X -coordinate is given by,

$$\text{Mass} \times \text{acceleration} = \rho \cdot dx \cdot dy \cdot dz \cdot \frac{du}{dt}$$

As per Newton's second law of motion summation of forces acting in the fluid element in any direction equals the resulting inertia forces in that direction. Thus, along X -direction:

$$B_x \cdot \rho \cdot dx \cdot dy \cdot dz - \frac{\partial p}{\partial x} dx \cdot dy \cdot dz = \rho \cdot dx \cdot dy \cdot dz \cdot \frac{du}{dt}$$

Dividing both sides by $\rho \cdot dx \cdot dy \cdot dz$, we have:

$$B_x - \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{du}{dt} \quad \dots(i)$$

In this equation each term has dimensions of force per unit mass or acceleration. Obviously the total acceleration in a given direction is prescribed by the algebraic sum of the body force and the pressure gradient in that direction since the velocity components are functions of

position and time, *i.e.*, $u = f(x, y, z, t)$, therefore, the total derivative of velocity u in the X -direction can be written as:

$$du = \frac{\partial u}{\partial t} dt + \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz$$

or,
$$\frac{du}{dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$

Substituting, $\frac{dx}{dt} = u$, $\frac{dy}{dt} = v$ and $\frac{dz}{dt} = w$; we have:

$$\frac{du}{dt} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \quad \dots(ii)$$

Combining eqns. (i) and (ii), we get the force components as:

$$B_x - \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \quad \dots(iii)$$

Similarly,
$$B_y - \frac{1}{\rho} \frac{\partial p}{\partial y} = \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \quad \dots(iv)$$

and,
$$B_z - \frac{1}{\rho} \frac{\partial p}{\partial z} = \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \quad \dots(v)$$

For steady flow:
$$\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} = \frac{\partial w}{\partial t} = 0$$

Thus, the Euler's equation for a steady three-dimensional flow can be written as:

$$B_x - \frac{1}{\rho} \frac{\partial p}{\partial x} = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \quad \dots(vi)$$

$$B_y - \frac{1}{\rho} \frac{\partial p}{\partial y} = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \quad \dots(vii)$$

$$B_z - \frac{1}{\rho} \frac{\partial p}{\partial z} = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \quad \dots(viii)$$

In Euler's equation each term represents force per unit mass. Thus, if each equation is multiplied by the respective projections of the elementary displacement, the resulting equation would represent energy. Thus, in order to get total energy in the three-dimensional-steady-incompressible flow, the energy terms can be combined as follows:

$$B_x dx - \frac{1}{\rho} \frac{\partial p}{\partial x} dx = u \frac{\partial u}{\partial x} dx + v \frac{\partial u}{\partial y} dx + w \frac{\partial u}{\partial z} dx \quad \dots(ix)$$

$$B_y dy - \frac{1}{\rho} \frac{\partial p}{\partial y} dy = u \frac{\partial v}{\partial x} dy + v \frac{\partial v}{\partial y} dy + w \frac{\partial v}{\partial z} dy \quad \dots(x)$$

$$B_z dz - \frac{1}{\rho} \frac{\partial p}{\partial z} dz = u \frac{\partial w}{\partial x} dz + v \frac{\partial w}{\partial y} dz + w \frac{\partial w}{\partial z} dz \quad \dots(xi)$$

From the equation of a stream line in a three-dimensional flow, we have:

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

$$u dy = v dx; v dz = w dy; u dz = w dx$$

Substituting these values in eqns. (ix), (x) and (xi), we get:

$$B_x dx - \frac{1}{\rho} \frac{\partial p}{\partial x} dx = u \frac{\partial u}{\partial x} dx + u \frac{\partial u}{\partial y} dy + u \frac{\partial u}{\partial z} dz \quad \dots(xii)$$

$$B_y dy - \frac{1}{\rho} \frac{\partial p}{\partial y} dy = v \frac{\partial v}{\partial x} dx + v \frac{\partial v}{\partial y} dy + v \frac{\partial v}{\partial z} dz \quad \dots(xiii)$$

$$B_z dz - \frac{1}{\rho} \frac{\partial p}{\partial z} dz = w \frac{\partial w}{\partial x} dx + w \frac{\partial w}{\partial y} dy + w \frac{\partial w}{\partial z} dz \quad \dots(xiv)$$

Acceleration terms are of form $u \frac{\partial u}{\partial x}$ which can be replaced by $\frac{1}{2} \frac{\partial(u^2)}{\partial x}$. Thus,

$$B_x dx - \frac{1}{\rho} \frac{\partial p}{\partial x} dx = \frac{1}{2} \left[\frac{\partial}{\partial x}(u^2) dx + \frac{\partial}{\partial y}(u^2) dy + \frac{\partial}{\partial z}(u^2) dz \right] = \frac{1}{2} d(u^2) \quad \dots(xv)$$

Similarly, $B_y dy - \frac{1}{\rho} \frac{\partial p}{\partial y} dy = \frac{1}{2} d(v^2) \quad \dots(xvi)$

and, $B_z dz - \frac{1}{\rho} \frac{\partial p}{\partial z} dz = \frac{1}{2} d(w^2) \quad \dots(xvii)$

Adding eqns. (xv), (xvi) and (xvii), we get:

$$\begin{aligned} B_x dx + B_y dy + B_z dz - \frac{1}{\rho} \left(\frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz \right) \\ = \frac{1}{2} [d(u^2) + d(v^2) + d(w^2)] \end{aligned}$$

or, $B_x dx + B_y dy + B_z dz - \frac{1}{\rho} dp = \frac{1}{2} d(V^2) \quad \dots(xviii)$

where,

V = Total velocity vector.

When gravity is the only body force acting on the third element, then:

$$B_x = 0, B_z = 0 \text{ and } B_y = -g$$

$B_y = -g$ since the *gravitational force* acts in the *downward* direction which is negative 'with' respect to Y, which is positive upward. Inserting these values in (xviii), we get:

$$-g - \frac{1}{\rho} dp = \frac{1}{2} d(V^2)$$

or, $-g - \frac{1}{\rho} dp = V dV$

or, $\frac{dp}{\rho} + V dV + g = 0$ which is the same as Euler's equation (6.3).

Hydrostatic equation from Euler's equation:

If the fluid is at rest then the velocity terms in Euler's eqns, (vi), (vii) and (viii), vanish and we have:

$$B_x - \frac{1}{\rho} \frac{\partial p}{\partial x} = 0; \quad B_y - \frac{1}{\rho} \frac{\partial p}{\partial y} = 0; \quad B_z - \frac{1}{\rho} \frac{\partial p}{\partial z} = 0$$

Further, if gravity is the only body force, then:

$$B_x = 0; \quad B_y = -g; \quad B_z = 0$$

$\therefore \frac{1}{\rho} \frac{\partial p}{\partial x} = 0; \quad \frac{1}{\rho} \frac{\partial p}{\partial z} = 0 \quad \dots(xix)$

and, $-g - \frac{1}{\rho} \frac{\partial p}{\partial y} = 0 \quad \dots(xx)$

Eqn. (xix) signifies that fluid pressure p is independent of x and z . In that case $\frac{\partial p}{\partial y} = \frac{dp}{dy}$ and

$$-g - \frac{1}{\rho} \frac{dp}{dy} = 0$$

or, $dp = -\rho g dy$ or $dp = -w dy$

Integrating both sides, we get:

$$\int_1^2 dp = -w \int_1^2 dy$$

or, $(p_2 - p_1) = -w (y_2 - y_1)$

or, $dp = w dy$

which represents the hydrostatic equation. Thus, hydrostatic equation is merely a *corollary of Euler's equation*.

Example 6.2. A discharge through a 24 cm diameter horizontal pipe increases linearly from 30 to 120 litres/sec. of water in 4 seconds.

(i) What pressure gradient must exist to produce this acceleration?

(ii) What is the difference in pressure intensity that exists between two sections that lie 9 m apart?

Solution. The Euler's equation for one-dimensional flow along the pipe axis may be written as:

$$B_x - \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} \quad \dots(i)$$

As the pipe is of uniform cross-sectional area, the velocity remains constant along the flow direction and therefore,

$$\frac{\partial u}{\partial x} = 0$$

Further, since the pipe has been laid horizontally, therefore, the body forces per unit volume in the direction $X = 0$

Thus, the eqn. (i) reduces to:

$$-\frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{\partial u}{\partial t}$$

The change in velocity when the flow changes from 30 to 120 litres/sec

$$\partial u = (u_2 - u_1) = \frac{120 \times 10^{-3}}{\frac{\pi}{4} \times (0.24)^2} - \frac{30 \times 10^{-3}}{\frac{\pi}{4} \times (0.24)^2} = 1.989 \text{ m/s}$$

This change takes place in 4 sec,

$$\therefore \frac{\partial u}{\partial t} = \frac{1.989}{4} = 0.497 \text{ m/s}^2$$

(i) Pressure gradient, $\frac{\partial p}{\partial x}$:

$$\frac{\partial p}{\partial x} = -\rho \frac{\partial u}{\partial t} = -1000 \times 0.497 = -497 \text{ N/m}^2/\text{m (Ans.)}$$

(ii) Difference in pressure intensity between the sections:

Difference in pressure intensity between two sections that lie 9 m apart

$$= \frac{\partial p}{\partial x} \times 9 = -497 \times 9 = -4473 \text{ N/m}^2 \text{ (Ans.)}$$

Example 6.3. Brine of specific gravity 1.15 is draining from the bottom of a large open tank through an 80 mm pipe. The drain pipe ends at a point 10 m below the surface of the brine in the tank. Considering a stream line starting at the surface of the brine in the tank and passing through the centre of the drain line to the point of discharge and assuming the friction is negligible, calculate the velocity of flow along the stream line at the point of discharge from the pipe.

(UPTU)

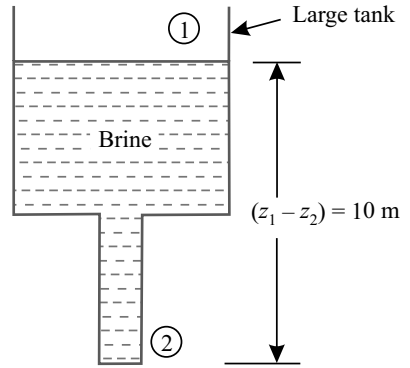


Fig. 6.4

Solution. Refer to Fig. 6.4.

Section 1– The surface of brine in the tank

Section 2 – The point of discharge.

Applying Bernoulli's equation between point 1 and 2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

Here,

$$p_1 = p_2 = p_{atm.} \text{ (atmospheric pressure),}$$

$$V_1 = 0 \quad \text{and} \quad (z_1 - z_2) = 10 \text{ m}$$

$\therefore$

$$V_2^2 = 2g(z_1 - z_2) = 2g \times 10 = 2 \times 9.81 \times 10 = 196.2$$

or,

$$V_2 \approx 14 \text{ m/s (Ans.)}$$

Example 6.4. A pipeline (Fig. 6.5) is 15 cm in diameter and it is at an elevation of 100 m at section A. At section B it is at an elevation of 107 m and has diameter of 30 cm. When a discharge of 50 litre/sec of water is passed through this pipeline, pressure at A is 35 kPa. The energy loss in pipe is 2 m of water. Calculate pressure at B if flow is from A to B. (Anna University)

Solution. Given:

$$D_A = 15 \text{ cm} = 0.15 \text{ m}; D_B = 30 \text{ cm} = 0.3 \text{ m};$$

$$p_A = 35 \text{ kPa}; Q = 50 \text{ litre/sec} = 0.05 \text{ m}^3/\text{s};$$

$$h_f = 2 \text{ m of water; Direction of flow: from A to B}$$

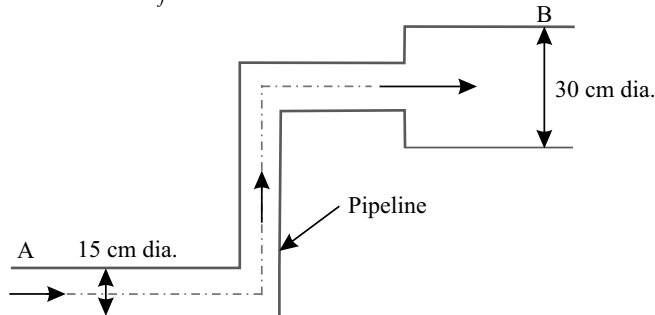


Fig. 6.5

Pressure at B, p_B :

$$V_A = \frac{Q}{\frac{\pi}{4} \times D_A^2} = \frac{0.05}{\frac{\pi}{4} \times 0.15^2} = 2.829 \text{ m/s}$$

$$V_B = \frac{Q}{\frac{\pi}{4} \times D_B^2} = \frac{0.05}{\frac{\pi}{4} \times (0.3)^2} = 0.707 \text{ m/s}$$

Applying Bernoulli's equation between sections A and B, we get:

$$\frac{P_A}{w} + \frac{V_A^2}{2g} + z_A = \frac{P_B}{w} + \frac{V_B^2}{2g} + z_B + h_f$$

$$\text{or,} \quad \frac{P_B}{w} = \frac{P_A}{w} + \left(\frac{V_A^2 - V_B^2}{2g} \right) + (z_A - z_B) - h_f$$

$$\begin{aligned} \text{or,} \quad p_B &= p_A + w \left[\left(\frac{V_A^2 - V_B^2}{2g} \right) + (z_A - z_B) - h_f \right] \\ &= 35 + \frac{(1000 \times 9.81)}{1000} \left[\left(\frac{2.829^2 - 0.707^2}{2 \times 9.81} \right) + (100 - 107) - 2 \right] \end{aligned}$$

$$35 + 9.81 (0.3824 - 7 - 2) = -49.54 \text{ kPa.}$$

i.e., $p_B = -49.54 \text{ kPa}$. This shows that the given pressure at A, 35 kPa is *gauge pressure* and hence there is *vacuum at point B*. (Ans.)

Example 6.5. An open circuit wind tunnel draws in air from the atmosphere through a well contoured nozzle. In the test section, where the flow is straight and nearly uniform, a static pressure tap is drilled into the tunnel wall. A manometer connected to the tap shows that the static pressure within the tunnel is 45 mm of water below atmosphere. Assume that air is incompressible and at 25°C , pressure is 100 kPa (absolute). Calculate the velocity in the wind tunnel section (Refer to Fig. 6.6). Density of water is 999 kg/m^3 and characteristic gas constant for air is 287 J/kg K . (GATE)

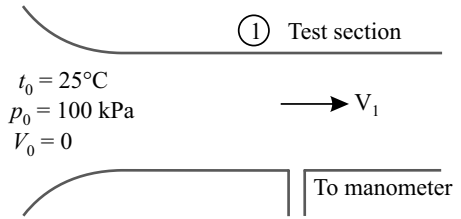


Fig. 6.6

Solution. Given: $T_0 = 25 + 273 = 298 \text{ K}$;

$$p_0 = 100 \text{ kPa (abs.)}; V_0 = 0;$$

Velocity in the wind tunnel section V_1 :

As per the problem, air is assumed as incompressible (*i.e.*, $\rho_0 = \rho_1 = \rho$). Velocity at test section can be found by using the equation:

$$\frac{p_0}{w} + \frac{V_0^2}{2g} + z_0 = \frac{p_1}{w} + \frac{V_1^2}{2g} + z_1$$

where,

$$z_0 = z_1; V_0 = 0, p_0 = 100 \text{ kPa (abs.)}$$

$$p_1 = 45 \text{ mm of water below atmosphere}$$

$$= 999 \times 9.81 \frac{45}{1000} \text{ Pa}$$

$$= 999 \times 9.81 \times \frac{45}{1000} \times 10^{-3} \text{ kPa} = 0.44 \text{ kPa}$$

$$= 999 \times 9.81 \frac{45}{1000} \times 10^{-3} \text{ kPa} = 0.44 \text{ kPa below atmosphere}$$

$$\begin{aligned} \therefore p_1(\text{absolute}) &= P_{\text{atm.}} (\text{in kPa}) - 0.44 \text{ kPa} \\ &= 100 - 0.44 = 99.56 \text{ kPa} \end{aligned}$$

$$\text{Also,} \quad pV = mRT = \rho RT \quad (\text{where, } \rho = \frac{m}{V})$$

$$\text{or,} \quad \rho = \frac{p}{RT} = \frac{100 \times 10^3}{287 \times 298} = 1.169 \text{ kg/m}^3$$

$$\therefore w = \rho g = 1.169 \times 9.81 = 11.468 \text{ N/m}^3$$

Substituting these values in (i), we get:

$$\frac{100 \times 10^3}{11.468} = \frac{99.56 \times 10^3}{11.468} + \frac{V_1^2}{2 \times 9.81}$$

$$8719.9 = 8681.5 + \frac{V_1^2}{2 \times 9.81}$$

$$\therefore V_1 = \sqrt{(8719.9 - 8681.5) \times 2 \times 9.81} = 27.45 \text{ m/s (Ans.)}$$

Example 6.6. Water flows in a circular pipe. At one section the diameter is 0.3 m, the static pressure is 260 kPa gauge, the velocity is 3 m/s and the elevation is 10 m above ground level. The elevation at a section downstream is 0 m, and the pipe diameter is 0.15 m. Find out the gauge pressure at the downstream section.

Frictional effects may be neglected. Assume density of water to be 999 kg/m^3 .

(RGPV, Bhopal)

Solution. Refer to Fig. 6.7. $D_1 = 0.3 \text{ m}$; $D_2 = 0.15 \text{ m}$; $z_1 = 0$; $z_2 = 10 \text{ m}$; $p_1 = 260 \text{ kPa}$, $V_1 = 3 \text{ m/s}$; $\rho = 999 \text{ kg/m}^3$.

From continuity equation, $A_1 V_1 = A_2 V_2$,

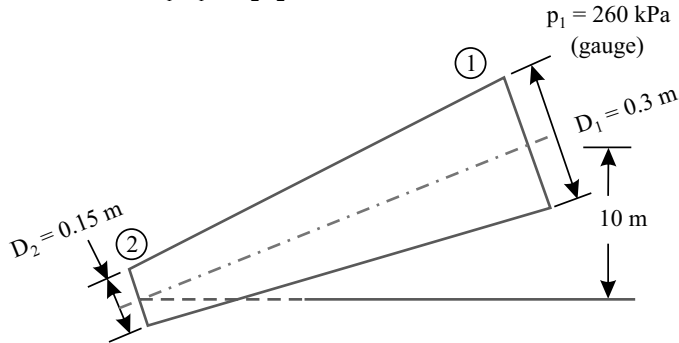


Fig. 6.7

$$\begin{aligned} V_2 &= \frac{A_1}{A_2} \times V_1 = \left(\frac{\frac{\pi}{4} D_1^2}{\frac{\pi}{4} D_2^2} \right) \times V_1 \\ &= \left(\frac{D_1}{D_2} \right)^2 \times V_1 = \left(\frac{0.3}{0.15} \right)^2 \times 3 = 12 \text{ m/s} \end{aligned}$$

Weight density of water, $w = \rho g = 999 \times 9.81 = 9800.19 \text{ N/m}^3$

From Bernoulli's equation between sections 1 and 2 (neglecting friction effects as given), we have:

$$\begin{aligned} \frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 &= \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 \\ \frac{260 \times 1000}{9800.19} + \frac{(3)^2}{2 \times 9.81} + 10 &= \frac{p_2}{9800.19} + \frac{(12)^2}{2 \times 9.81} + 0 \end{aligned}$$

$$= \frac{p_2}{9800.19} + \frac{(12)^2}{2 \times 9.81} + 0$$

$$26.53 + 0.459 + 10 = \frac{p_2}{9800.19} + 7.34$$

or,

$$p_2 = 290566 \text{ N/m}^2 = \mathbf{290.56 \text{ kPa (Ans.)}}$$

Example 6.7. The water is flowing through a tapering pipe having diameters 300 mm and 150 mm at sections 1 and 2 respectively. The discharge through the pipe is 40 litres/sec. The section 1 is 10 m above datum and section 2 is 6 m above datum. Find the intensity of pressure at section 2 if that at section 1 is 400 kN/m².

Solution. At Section 1:

$$\text{Diameter, } D_1 = 300 \text{ mm} = 0.3 \text{ m}$$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$$

$$\text{Pressure, } p_1 = 400 \text{ kN/m}^2$$

$$\text{Height of upper end above the datum, } z_1 = 10 \text{ m}$$

At Section 2:

$$\text{Diameter, } D_2 = 150 \text{ mm} = 0.15 \text{ m}$$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

$$\text{Height of lower end above the datum, } z_2 = 6 \text{ m}$$

Rate of flow (i.e., discharge),

$$\begin{aligned} Q &= 40 \text{ litres/sec} = \frac{40 \times 10^3}{10^6} \\ &= 0.04 \text{ m}^3/\text{s} \end{aligned}$$

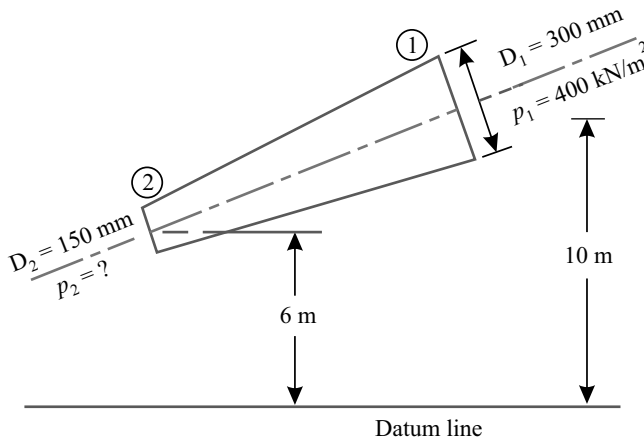


Fig. 6.8

Intensity of pressure at section 2, p_2 :

$$\text{Now, } Q = A_1 V_1 = A_2 V_2$$

$$\therefore V_1 = \frac{Q}{A_1} = \frac{0.04}{0.0707} = 0.566 \text{ m/s}$$

and,
$$V_2 = \frac{Q}{A_2} = \frac{0.04}{0.01767} = 2.264 \text{ m/s}$$

Applying Bernoulli's equation at sections 1 and 2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

and,
$$\begin{aligned} \frac{p_2}{w} &= \frac{p_1}{w} + \left(\frac{V_1^2}{2g} - \frac{V_2^2}{2g} \right) + (z_1 - z_2) \\ &= \frac{400}{9.81} + \frac{1}{2 \times 9.81} (0.566^2 - 2.264^2) + (10 - 6) \\ &\quad (\because w = 9.81 \text{ kN/m}^3) \\ &= 40.77 - 0.245 + 4 = 44.525 \text{ m} \\ \therefore p_2 &= 44.525 \times w = 44.525 \times 9.81 = \mathbf{436.8 \text{ kN/m}^2 \text{ (Ans.)}} \end{aligned}$$

Example 6.8. A pipe 200 m long slopes down at 1 in 100 and tapers from 600 mm diameter at the higher end to 300 mm diameter at the lower end, and carries 100 litres/sec of oil (sp. gravity 0.8). If the pressure gauge at the higher end reads 60 kN/m², determine:

- Velocities at the two ends;
- Pressure at the lower end.

Neglect all losses.

Solution. Length of the pipe, $l = 200 \text{ m}$; diameter of the pipe at the higher end, $D_1 = 600 \text{ mm} = 0.6 \text{ m}$,

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.6^2 = 0.283 \text{ m}^2$$

Diameter of the pipe at the lower end,

$$D_2 = 300 \text{ mm} = 0.3 \text{ m}$$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$$

Height of the higher end, above datum,

$$z_1 = \frac{1}{100} \times 200 = 2 \text{ m}$$

Height of the lower end, above datum $z_2 = 0$

Rate of oil flow, $Q = 100 \text{ litres/sec} = 0.1 \text{ m}^3/\text{s}$

Pressure at the higher end, $p_1 = 60 \text{ kN/m}^2$

(i) Velocities, V_1, V_2 :

$$\text{Now, } Q = A_1 V_1 = A_2 V_2$$

where, V_1 and V_2 are the velocities at the higher and lower ends respectively.

$$\therefore V_1 = \frac{Q}{A_1} = \frac{0.1}{0.283} = \mathbf{0.353 \text{ m/s (Ans.)}}$$

$$\text{and } V_2 = \frac{Q}{A_2} = \frac{0.1}{0.0707} = \mathbf{1.414 \text{ m/s (Ans.)}}$$

(ii) Pressure at the lower end p_2 :

Using Bernoulli's equation for both ends of pipe, we have:

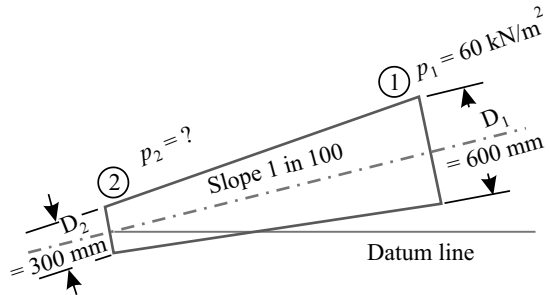


Fig. 6.9

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

$$\frac{60}{0.8 \times 9.81} + \frac{0.353^2}{2 \times 9.81} + 2 = \frac{p_2}{0.8 \times 9.81} + \frac{1.414^2}{2 \times 9.81} + 0$$

$$7.64 + 0.00635 + 2 = \frac{p_2}{0.8 \times 9.81} + 0.102$$

$$\therefore \frac{p_2}{0.8 \times 9.81} = 9.54 \text{ m}$$

or, $p_2 = 74.8 \text{ kN/m}^2 \text{ (Ans.)}$

Example 6.9. A 6 m long pipe is inclined at an angle of 20° with the horizontal. The smaller section of the pipe which is at lower level is of 100 mm diameter and the larger section of the pipe is of 300 mm diameter as shown in Fig. 6.10. If the pipe is uniformly tapering and the velocity of water at the smaller section is 1.8 m/s, determine the difference of pressures between the two sections.

Solution. Length of the pipe, $l = 6 \text{ m}$

Angle of inclination, $\theta = 20^\circ$

At Section 1:

Diameter, $D_1 = 100 \text{ mm} = 0.1 \text{ m}$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.1^2 = 0.00785 \text{ m}^2$$

Velocity, $V_1 = 1.8 \text{ m/s}$

Datum, $z_1 = 0$

At Section 2:

Diameter, $D_2 = 300 \text{ mm} = 0.3 \text{ m}$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$$

Datum, $z_2 = 0 + 6 \sin \theta = 6 \sin 20^\circ = 6 \times 0.342 = 2.05 \text{ m}$

Let, $p_1 = \text{Pressure at section 1 in kN/m}^2$, and

$p_2 = \text{Pressure at section 2 in kN/m}^2$.

Difference of pressures, $(p_1 - p_2)$:

From the equation of continuity, we know that:

$$A_1 V_1 = A_2 V_2$$

$$\therefore V_2 = \frac{A_1 V_1}{A_2} = \frac{0.00785 \times 1.8}{0.0707} = 0.2 \text{ m/s.}$$

Applying Bernoulli's equation to both sections of the pipe, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

or,
$$\left(\frac{p_1}{w} - \frac{p_2}{w} \right) = \left(\frac{V_2^2}{2g} - \frac{V_1^2}{2g} \right) + (z_2 - z_1)$$

$$= \frac{1}{2g} (V_2^2 - V_1^2) + (z_2 - z_1)$$

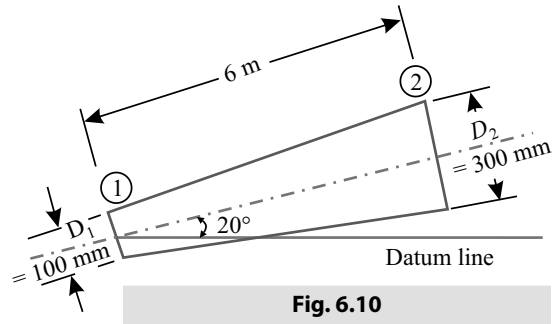


Fig. 6.10

$$= \frac{1}{2 \times 9.81} (0.2^2 - 1.8^2) + (2.05 - 0) = 1.88$$

$$\therefore (p_1 - p_2) = w \times 1.88 = 9.81 \times 1.88 = \mathbf{18.44 \text{ kN/m}^2 \text{ (Ans.)}}$$

$$[\because w \text{ (for water)} = 9.81 \text{ kN/m}^3]$$

Example 6.10. Water is flowing through a pipe having diameters 600 mm and 400 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 350 kN/m² and the pressure at the upper end is 100 kN/m². Determine the difference in datum head if the rate of flow through the pipe is 60 litres/sec.

Solution. At Section 1:

$$\text{Diameter, } D_1 = 600 \text{ mm} = 0.6 \text{ m}$$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.6^2 = 0.283 \text{ m}^2$$

$$\text{Pressure, } p_1 = 350 \text{ kN/m}^2$$

At Section 2:

$$\text{Diameter, } D_2 = 400 \text{ mm} = 0.4 \text{ m}$$

$$\text{Area, } A_2 = \frac{\pi}{4} \times 0.4^2 = 0.1257 \text{ m}^2$$

$$\text{Pressure, } p_2 = 100 \text{ kN/m}^2$$

Rate of flow,

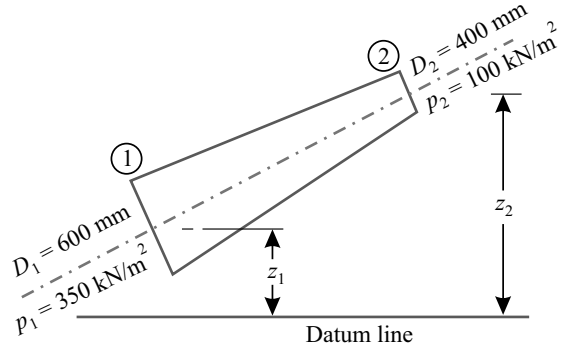


Fig. 6.11

$$Q = 60 \text{ litres/sec} = \frac{60}{1000} = 0.06 \text{ m}^3/\text{sec.}$$

$$\text{Now, } Q = A_1 V_1 = A_2 V_2$$

[where, V_1 and V_2 are the velocities at sections 1 and 2 respectively.]

$$\therefore V_1 = \frac{Q}{A_1} = \frac{0.06}{0.283} = 0.212 \text{ m/s}$$

$$\text{and, } V_2 = \frac{Q}{A_2} = \frac{0.06}{0.1257} = 0.477 \text{ m/s}$$

Applying Bernoulli's equation at sections 1 and 2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

$$\frac{350}{9.81} + \frac{0.212^2}{2 \times 9.81} + z_1 = \frac{100}{9.81} + \frac{0.477^2}{2 \times 9.81} + z_2$$

$$35.67 + 0.0023 + z_1 = 10.19 + 0.0116 + z_2$$

$$z_2 - z_1 = \mathbf{25.47 \text{ m (Ans.)}}$$

Example 6.11. Gasoline (sp. gr. 0.8) is flowing upwards a vertical pipeline which tapers from 300 mm to 150 mm diameter. A gasoline mercury differential manometer is connected between 300 mm and 150 mm pipe section to measure the rate of flow. The distance between the manometer tappings is 1 metre and gauge reading is 500 mm of mercury. Find:

(i) Differential gauge reading in terms of gasoline head;

(ii) Rate of flow.

Neglect friction and other losses between tappings.

[MGU, Kerala]

Solution. Sp. gravity of gasoline = 0.8

At Inlet:

$$\text{Diameter, } D_1 = 300 \text{ mm} = 0.3 \text{ m}$$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$$

At Outlet:

$$\text{Diameter, } D_2 = 150 \text{ mm} = 0.15 \text{ m}$$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

$$\text{Length of the pipe} = 1 \text{ m}$$

$$\text{Let datum of the pipe at inlet, } z_1 = 0$$

$$\therefore \text{Datum of the pipe at outlet, } z_2 = 0 + 1 = 1 \text{ m}$$

$$\text{Gauge reading, } h = 500 \text{ mm of mercury} = 0.5 \text{ m of mercury.}$$

(i) Differential gauge reading in terms of gasoline head:

$$\text{The gauge reading} = 0.5 \text{ m of mercury}$$

$$= \frac{13.6 - 0.8}{0.8} \times 0.5 \text{ of gasoline}$$

$$= \mathbf{8 \text{ m of gasoline (Ans.)}}$$

(ii) Rate of flow, Q:

Let,

V_1 = Velocity of gasoline at the inlet, and

V_2 = Velocity of gasoline at the outlet.

We know that, as per equation of continuity:

$$A_1 V_1 = A_2 V_2$$

$$\therefore V_2 = \frac{A_1 V_1}{A_2} = \frac{0.0707 \times V_1}{0.01767} = 4V_1$$

Now, using Bernoulli's equation for the inlet and outlet of the pipe, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

$$\left\{ \frac{p_1}{w} - \frac{p_2}{w} \right\} + \left\{ \frac{V_1^2}{2g} - \frac{V_2^2}{2g} \right\} + (z_1 - z_2) = 0$$

$$8 + \left[\frac{V_1^2}{2g} - \frac{(4V_1)^2}{2g} \right] + [0 - 1] = 0$$

$$\text{or, } 8 - \frac{15V_1^2}{2g} - 1 = 0$$

$$\text{or, } \frac{15V_1^2}{2g} = 7$$

$$\therefore V_1 = \left[\frac{7 \times 2 \times 9.81}{15} \right]^{1/2} = 3.026 \text{ m/s.}$$

$$\therefore \text{Rate of flow, } Q = A_1 V_1 = 0.0707 \times 3.026 = \mathbf{0.2139 \text{ m}^3/\text{s (Ans.)}}$$

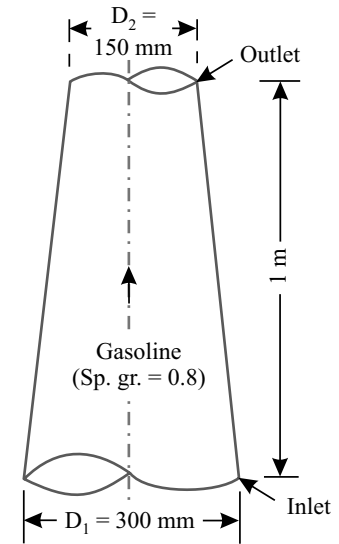


Fig. 6.12

Example 6.12. The suction pipe of a pump rises at a slope of 3 vertical in 4 along the pipe which is 12 cm in diameter. The pipe is 7.2 m long; its lower end being just below the water surface in the

reservoir. For design reasons, it is desirable that pressure at inlet to the pump shall fall to more than 75 kPa below atmospheric pressure. Neglecting friction, determine the maximum discharge that the pump may deliver. Take atmospheric pressure as 101.32 kPa. (Bangalore University)

Solution. Refer to Fig. 6.13. Given: $d = 12 \text{ cm} = 0.12 \text{ m}$; $l = 7.2 \text{ m}$; $p_{\text{atm.}} = 101.32 \text{ kPa} = 101.32 \text{ kN/m}^2$.

Applying Bernoulli's equation at point 1 (F.W.S.) and point 2 (suction point to pump), we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 \quad \dots(i)$$

Velocity V_1 on the free water surface (F.W.S.) = 0 (sump being very large)

$$p_1 = p_{\text{atm.}} = 101.32 \text{ kN/m}^2, \\ p_2 = 101.32 - 75 = 26.32 \text{ kN/m}^2$$

Taking point 1 as the datum head, we have:

$$z_1 = 0; z_2 = 7.2 \times \frac{3}{4} = 5.4 \text{ m}$$

Inserting the various values in eqn (i), we have:

$$\frac{101.32}{9.81} + 0 + 0 = \frac{26.32}{9.81} + \frac{V_2^2}{2g} + 5.4$$

$$\text{or,} \quad V_2 = 6.64 \text{ m/s}$$

$\therefore$ Discharge that the pump may deliver,

$$Q = A_2 \times V_2 = \frac{\pi}{4} \times (0.12)^2 \times 6.64 = 0.075 \text{ m}^3/\text{s} \text{ (Ans.)}$$

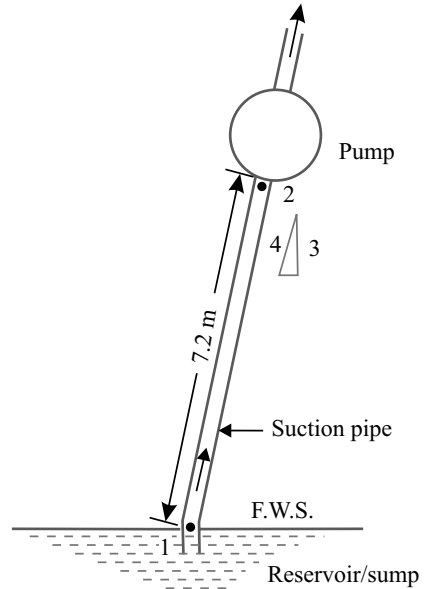


Fig. 6.13

6.5. BERNOULLI'S EQUATION FOR REAL FLUID

Bernoulli's equation earlier derived was based on the assumption that fluid is non-viscous and therefore frictionless. Practically, all fluids are real (and not ideal) and therefore are viscous as such there are always some losses in fluid flows. These losses have, therefore, to be taken into consideration in the application of Bernoulli's equation which gets modified (between sections 1 and 2) for *real fluids* as follows:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_L \quad \dots(6.4)$$

where, h_L = Loss of energy between sections 1 and 2.

Example 6.13. The following data relate to a conical tube of length 3.0 m fixed vertically with its smaller end upwards and carrying fluid in the downward direction.

The velocity of flow at the smaller end = 10 m/s.

The velocity of flow at the larger end = 4 m/s.

$$\text{The loss of head in the tube} = \frac{0.4 (V_1 - V_2)^2}{2g}$$

where, V_1 and V_2 are velocities at the smaller and larger ends respectively.

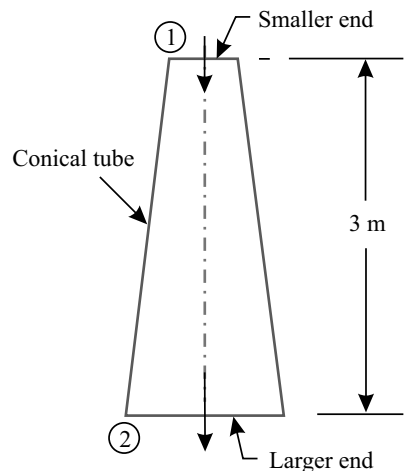


Fig. 6.14

Pressure head at the smaller end = 4 m of liquid.

Determine the pressure head at the larger end.

Solution. Length of tube, $l = 3.0$ m

Velocity, $V_1 = 10$ m/s.

Pressure head, $\frac{p_1}{w} = 4$ m of liquid.

Velocity, $V_2 = 4$ m/s.

$$\text{Loss of head, } h_L = \frac{0.4(V_1 - V_2)^2}{2g} = \frac{0.4(10 - 4)^2}{2 \times 9.81} = 0.73 \text{ m}$$

Pressure head at the larger end, $\frac{p_2}{w}$:

Applying Bernoulli's equation at sections (1) and (2), we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_L$$

Let the datum line passes through section (2).

Then, $z_2 = 0, z_1 = 3.0$ m

$$\therefore 4 + \frac{10^2}{2g} + 3.0 = \frac{p_2}{w} + \frac{4^2}{2g} + 0 + 0.73$$

$$\text{or, } (4 + 5.09 + 3.0) = \frac{p_2}{w} + 0.815 + 0.73$$

$$\text{or, } 12.09 = \frac{p_2}{w} + 1.54$$

$$\therefore \frac{p_2}{w} = 10.55 \text{ m of liquid (Ans.)}$$

Example 6.14. In a smooth inclined pipe of uniform diameter 250 mm, a pressure of 50 kPa was observed at section 1 which was at elevation 10 m. At another section 2 at elevation 12 m, the pressure was 20 kPa and the velocity was 1.25 m/s. Determine the direction of flow and the head loss between these two sections. The fluid in the pipe is water. The density of water at 20°C and 760 mm Hg is 998 kg/m³. (PTU)

Solution. Given:

$$D = 250 \text{ mm} = 0.25 \text{ m,}$$

$$p_1 = 50 \text{ kPa} = 50 \times 10^3 \text{ N/m}^2;$$

$$z_1 = 10 \text{ m; } z_2 = 12 \text{ m;}$$

$$p_2 = 20 \text{ kPa} = 20 \times 10^3 \text{ N/m}^2,$$

$$V_1 = V_2 = 1.25 \text{ m/s, } \rho = 998 \text{ kg/m}^3.$$

Refer to Fig. 6.15.

Loss of head h_L :

Total energy at section 1–1,

$$E_1 = \frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{50 \times 10^3}{998 \times 9.81} + \frac{1.25^2}{2 \times 9.81} + 10 = 15.187 \text{ m}$$

Total energy of section 2–2,

$$E_2 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 = \frac{20 \times 10^3}{998 \times 9.81} + \frac{1.25^2}{2 \times 9.81} + 12 = 14.122 \text{ m}$$

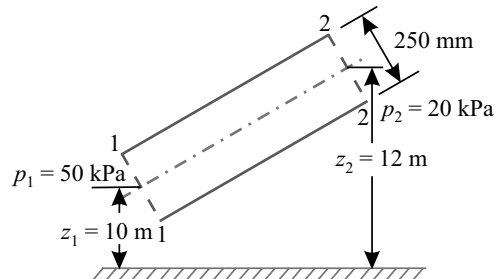


Fig. 6.15

∴ Loss of head, $h_L = E_1 - E_2 = 15.187 - 14.122 = \mathbf{1.065 \text{ m (Ans.)}}$

Direction of flow:

Since $E_1 > E_2$ direction of flow is from **section 1-1 to section 2-2. (Ans.)**

Example 6.15. A pipe line carrying oil (sp. gr. 0.8) changes in diameter from 300 mm at position 1 to 600 mm diameter at position 2 which is 5 metres at a higher level. If the pressures at positions 1 and 2 are 100 kN/m^2 and 60 kN/m^2 respectively and the discharge is 300 litres/sec., determine:

(i) Loss of head, and

(ii) Direction of flow.

Solution. Discharge, $Q = 300 \text{ litres/sec}$
 $= \frac{300}{1000} = 0.3 \text{ m}^3/\text{s}.$

Sp. gr. of oil = 0.8

∴ Weight of oil, $w = 0.8 \times 9.81 = 7.85 \text{ kN/m}^3$

At position '1':

Diameter of pipe, $D_1 = 300 \text{ mm} = 0.3 \text{ m}$

∴ Area of pipe, $A_1 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$

Pressure, $p_1 = 100 \text{ kN/m}^2$

If the datum line passes through section 1 (Fig. 6.16) then datum, $z_1 = 0$

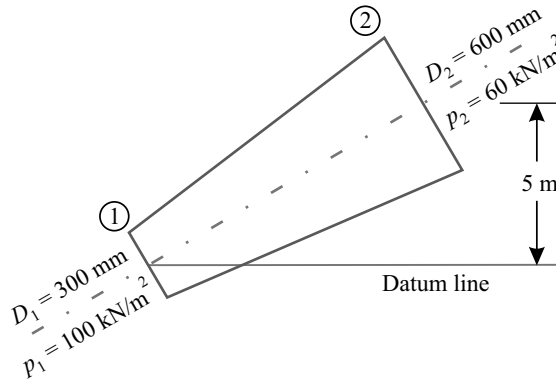


Fig. 6.16

Velocity, $V_1 = \frac{Q}{A_1} = \frac{0.3}{0.0707} = 4.24 \text{ m/s}$

At position '2':

Diameter of pipe, $D_2 = 600 \text{ mm} = 0.6 \text{ m}$

∴ Area of pipe, $A_2 = \frac{\pi}{4} \times 0.6^2 = 0.2888 \text{ m}^2$

Pressure, $p_2 = 60 \text{ kN/m}^2$

Datum, $z_2 = 5 \text{ m}$

Velocity, $V_2 = \frac{Q}{A_2} = \frac{0.3}{0.2828} = 1.06 \text{ m/s}.$

(i) Loss of head, h_L :

Total energy at position 1,

$$\begin{aligned}
 E_1 &= \frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 \\
 &= \frac{100}{7.85} + \frac{(4.24)^2}{2 \times 9.81} + 0 = 12.74 + 0.92 = 13.66 \text{ m}
 \end{aligned}$$

Total energy at position 2,

$$\begin{aligned}
 E_2 &= \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 \\
 &= \frac{60}{7.85} + \frac{(1.06)^2}{2 \times 9.81} + 5 = 7.64 + 0.06 = 7.7 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Loss of head, } h_L &= E_1 - E_2 \\
 &= 13.66 - 7.7 = 5.96 \text{ m}
 \end{aligned}$$

$$\text{i.e., } h_L = \mathbf{5.96 \text{ m (Ans.)}}$$

(ii) Direction of Flow:Since $E_1 > E_2$ therefore flow takes place **from 1 to 2**. (Ans.)

Example 6.16. A conical tube is fixed vertically with its smaller end upwards and it forms a part of pipeline. The velocity at the smaller end is 4.5 m/s and at the large end 1.5 m/s. Length of conical tube is 1.5 m. The pressure at the upper end is equivalent to a head of 10 m of water.

(i) Neglecting friction, determine the pressure at the lower end of the tube.

(ii) If head loss in the tube is $\frac{0.3(V_1 - V_2)^2}{2g}$, where V_1 is the velocity at the smaller end and V_2 is the velocity at the larger end, determine the pressure at the lower end (larger cross-section). (MDU, Haryana)

Solution. Given: $V_1 = 4.5 \text{ m/s}$; $V_2 = 1.5 \text{ m/s}$; $L = z_1 - z_2 = 1.5 \text{ m}$;

$$\frac{p_1}{w} = 10 \text{ m of water; } h_f = \frac{0.3(V_1 - V_2)^2}{2g}$$

Pressure at the lower end, p_2 :**(i) Neglecting friction:**

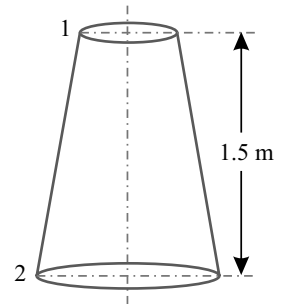
Applying Bernoulli's equation between points 1 and 2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

$$\begin{aligned}
 \text{or, } \frac{p_2}{w} &= \frac{p_1}{w} + \frac{1}{2g} (V_1^2 - V_2^2) + (z_1 - z_2) \\
 &= 10 + \frac{1}{2 \times 9.81} (4.5^2 - 1.5^2) + 1.5
 \end{aligned}$$

$$= 10 + 0.917 + 1.5 \approx 12.42 \text{ m of water}$$

$$\begin{aligned}
 \text{or, } p_2 &= 12.42 \times 9810 \text{ N/m}^2 = 12.42 \times 9810 \times 10^{-5} \text{ bar} \\
 &= \mathbf{1.218 \text{ bar (Ans.)}}
 \end{aligned}$$

**Fig. 6.17**

(ii) Considering loss of head (h_f) in the tube :

$$h_f = \frac{0.3(V_1 - V_2)^2}{2g}$$

Applying Bernoulli's equation between points 1 and 2, we have:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_f$$

$$\begin{aligned} \text{or,} \quad \frac{p_2}{w} &= \frac{p_1}{w} + \frac{1}{2g}(V_1^2 - V_2^2) + (z_1 - z_2) - h_f \\ &= 10 + \frac{4.5^2 - 1.5^2}{2 \times 9.81} + 1.5 - \frac{0.3(4.5 - 1.5)^2}{2 \times 9.81} \\ &= 10 + 0.917 + 1.5 - 0.138 = 12.279 \text{ m of water} \\ \text{or,} \quad p_2 &= 12.279 \times 9810 \times 10^{-5} \text{ bar} = \mathbf{1.204 \text{ bar (Ans.)}} \end{aligned}$$

Example 6.17. A drainage pump has tapered suction pipe. The pipe is running full of water. The pipe diameters at the inlet and at the upper end are 1 m and 0.5 m respectively. The free water surface is 2 m above the centre of the inlet and centre of upper end is 3 m above the top of free water surface. The pressure at the tip end of the pipe is 25 cm of mercury and it is known that loss of head by friction between top and the bottom section is one-tenth of the velocity head at the top section. Compute the discharge in litre/sec. Neglect loss of head at the entrance of the tapered pipe.

(UPTU)

Solution. Given: $D_1 = 1 \text{ m}$; $D_2 = 1.5 \text{ m}$;

$$p_1 = 76 \text{ cm of Hg} = \frac{76}{100} \times 13.6 = 10.336 \text{ of water};$$

$$p_2 = 25 \text{ cm of Hg} = \frac{25}{100} \times 13.6 = 3.4 \text{ m of water}; \quad h_f = \frac{1}{10} \frac{V_2^2}{2g}.$$

Discharge, Q:

Refer to Fig. 6.18. Applying continuity equation for the flow through pipe, we get:

$$A_1 V_1 = A_2 V_2$$

$$\frac{\pi}{4} D_1^2 V_1 = \frac{\pi}{4} D_2^2 V_2$$

$$\text{or,} \quad D_1^2 V_1 = D_2^2 V_2$$

$$\text{or,} \quad 1^2 \times V_1 = (0.5)^2 V_2$$

$$\text{or,} \quad V_2 = 4V_1$$

Now, applying Bernoulli's equation at 1-1 and 2-2,

we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_f$$

$$10.336 + \frac{V_1^2}{g} + 0 = 3.4 + \frac{16V_1^2}{2g} + 5 + \frac{1}{10} \times \frac{16V_1^2}{2g}$$

$$\frac{16V_1^2}{2g} + \frac{1.6V_1^2}{2g} - \frac{V_1^2}{2g} = 10.336 - 3.4 - 5 = 1.936$$

$$\text{or,} \quad 16.6 V_1^2 = 2 \times 9.81 \times 1.936 = 37.98$$

$$\therefore V_1 = 1.513 \text{ m/s}$$

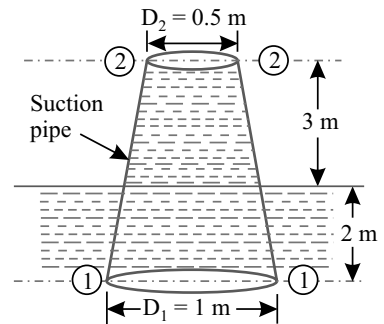


Fig. 6.18

$$\text{Discharge } Q = A_1 V_1 = \frac{\pi}{4} \times 1^2 \times 1.513 = 1.188 \text{ m}^3/\text{s} = \mathbf{1188 \text{ litres/sec. (Ans.)}}$$

Example 6.18. The closed tank of a fire engine is partly filled with water, the air space above being under pressure. A 6 cm bore connected to the tank discharges on the roof of a building 2.5 m above the level of water in the tank. The friction losses are 45 cm of water.

Determine the air pressure which must be maintained in the tank to deliver 20 litres/sec. on the roof. (Madras University)

Solution. Refer to Fig. 6.19 Given: Diameter of hose pipe $d = 6 \text{ cm} = 0.06 \text{ m}$; Friction, $h_f = 45 \text{ cm}$ or 0.45 m of water.

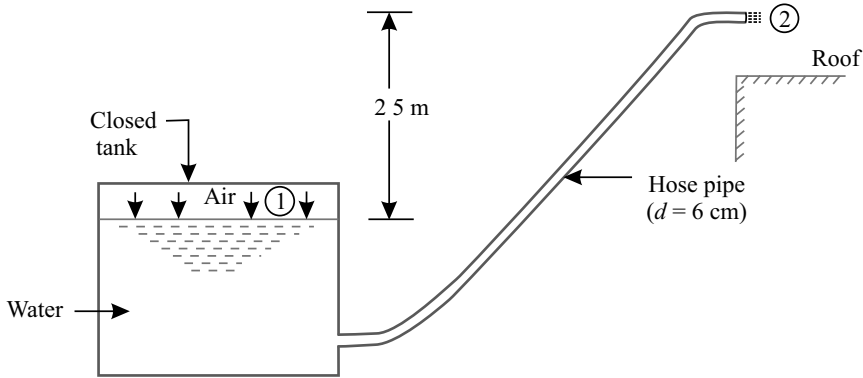


Fig. 6.19

$$\text{Discharge, } Q = 20 \text{ litres/sec. or } 0.02 \text{ m}^3/\text{s}.$$

$$\text{Velocity of water in the pipe, } V = \frac{Q}{A} = \frac{0.02}{\frac{\pi}{4} \times (0.06)^2} = 7.07 \text{ m/s}.$$

Applying Bernoulli's theorem at points 1 and 2 respectively, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_f$$

Here, $V_1 = 0$; $z_1 = 0$; $p_2 = 0$; $V_2 = 7.07 \text{ m/s}$; $z_2 = 2.5 \text{ m}$; $h_f = 0.45 \text{ m}$

Inserting the various values in the above equation, we get:

$$\frac{p_1}{w} + 0 + 0 = 0 + \frac{(7.07)^2}{2g} + 2.5 + 0.45$$

$$\text{or, } \frac{p_1}{9.81} = \frac{(7.07)^2}{2 \times 9.81} + 2.5 + 0.45$$

$$= 5.497 \text{ m of water (where } p_1 \text{ is in kN/m}^2\text{)}$$

$$\therefore p_1 = 9.81 \times 5.497 = \mathbf{53.93 \text{ kN/m}^2 \text{ (gauge) (Ans.)}}$$

Example 6.19. A siphon consisting of a pipe of 12 cm diameter is used to empty kerosene oil (Sp. gr. = 0.8) from the tank A. The siphon discharges to the atmosphere at an elevation of 1.2 m. The oil surface in the tank is at an elevation of 4.2 m. The centre line of the siphon pipe at its highest point C is at an elevation of 5.7 m. Determine:

- The discharge in the pipe.
- The pressure at point C.

The losses in the pipe may be assumed to be 0.45 m up to summit and 1.25 m from the summit to the outlet.

Solution. Consider points 1 and 2 at the surface of the oil in the tank *A* and at the outlet as shown in Fig. 6.20. The velocity V_1 can be assumed to be zero. Applying Bernoulli's equation at points 1 and 2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_{f(1-2)} \text{ (losses)}$$

$$0 + 0 + 4.2 = 0 + \frac{V_2^2}{2g} + 1.2 + (0.45 + 1.25)$$

or,

$$V_2 = 5.05 \text{ m/s}$$

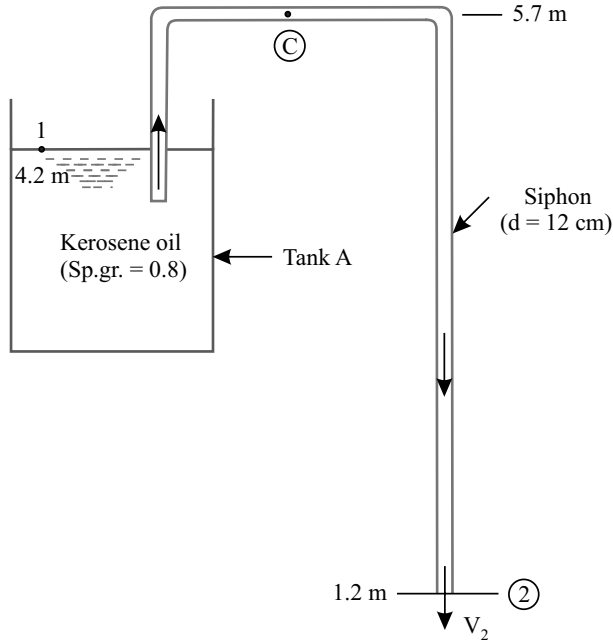


Fig. 6.20

(i) The discharge in the pipe, Q :

$$Q = A_2 V_2 = \frac{\pi}{4} \times (0.12)^2 \times 5.05 = \mathbf{0.057 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

(ii) The pressure at point C:

Applying Bernoulli's equation at points 1 and C, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_C}{w} + \frac{V_C^2}{2g} + z_C + h_{f(1-C)}$$

$$0 + 0 + 4.2 = \frac{p_C}{w} + \frac{(5.05)^2}{2 \times 9.81} + 5.7 + 0.45$$

or,
$$\frac{p_C}{w} = -3.25 \text{ m}$$

or,
$$p_C = (0.8 \times 9.81) \times (-3.25) \\ = -25.5 \text{ kN/m}^2 \text{ or } \mathbf{-25.5 \text{ kPa (gauge) (Ans.)}}$$

Example 6.20. The outlet at the bottom of a tank is so formed that velocity of water at point A (see Fig. 6.21) is 2.2 times the mean velocity within the outlet pipe. What is the greatest length of

pipe l which may be used without producing cavitation? Neglect all other losses.

Take atmospheric pressure = 96.24 kPa (abs.) and vapour pressure = 3.9 kPa (abs.)

Solution. Given: $V_A = 2.2 V_2$; $p_1 = p_2$
 $= p_{\text{atm.}} = 96.24 \text{ kPa} = 96.24 \text{ kN/m}^2$
 Vapour pressure, $p_A = 3.9 \text{ kPa}$
 $= 3.9 \text{ kN/m}^2 \text{ (abs.)}$

Applying Bernoulli's equation to points 1 and A, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_A}{w} + \frac{V_A^2}{2g} + z_A$$

$$\frac{96.24}{9.81} + 0 + 1.7 = \frac{3.9}{9.81} + \frac{V_A^2}{2 \times 9.81} + 0$$

$$\therefore V_A = 14.76 \text{ m/s}$$

$$V_2 = \frac{V_A}{2.2} = \frac{14.76}{2.2}$$

$$= 6.71 \text{ m/s}$$

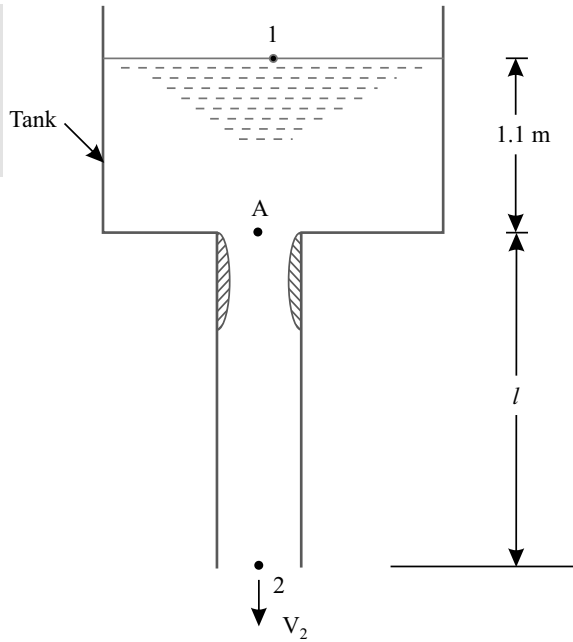


Fig. 6.21

Applying Bernoulli's equations to point 1 and 2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

$$\frac{96.24}{9.81} + 0 + (l + 1.1) = \frac{96.24}{9.81} + \frac{6.71^2}{2 \times 9.81} + 0$$

$$\therefore l = 1.195 \text{ m (Ans.)}$$

Example 6.21. A turbine has a supply line of diameter 45 cm and a tapering draft tube as shown in Fig. 6.22. When the flow in the pipe is $0.6 \text{ m}^3/\text{s}$ the pressure head at point L upstream of the turbine is 35 m and at a point M in the draft tube, where the diameter is 65 cm, the pressure head is -4.1 m . Point M is 2.2 m below the point L. Determine the power output of the turbine by assuming 92% efficiency.

Solution. $V_L = \frac{Q}{A_L} = \frac{0.6}{\frac{\pi}{4} \times (0.45)^2} = 3.77 \text{ m/s}$

$$V_M = \frac{Q}{A_M} = \frac{0.6}{\frac{\pi}{4} \times (0.65)^2} = 1.81 \text{ m/s}$$

Applying Bernoulli's equation to points L and M:

$$\frac{P_L}{w} + \frac{V_L^2}{2g} + z_L = \frac{P_M}{w} + \frac{V_M^2}{2g} + z_M + H_T$$

$$35 + \frac{(3.77)^2}{2 \times 9.81} + 2.2 = -4.1 + \frac{(1.81)^2}{2 \times 9.81} + 0 + H_T$$

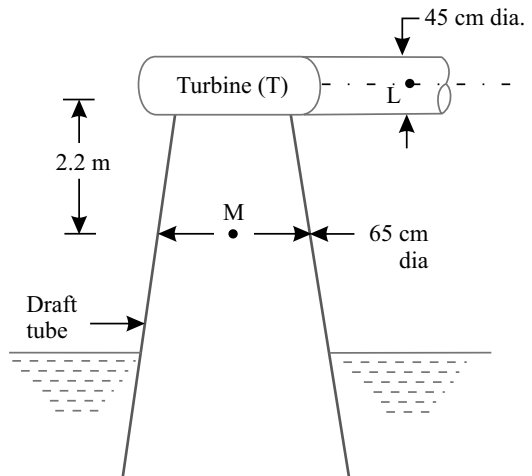


Fig. 6.22

$$\therefore H_T = 41.86 \text{ m}$$

Power output of the turbine,

$$P = wQH_T \times \eta$$

$$= 9.81 \times 0.6 \times 41.86 \times 0.92 = \mathbf{226.68 \text{ kW (Ans.)}}$$

Example 6.22. Fig 6.23 shows a pipe connecting a reservoir to a turbine which discharges water to the tail race through another pipe. The head loss between the reservoir and the turbine is 8 times the kinetic head in the pipe and that from the turbine to the tail race is 0.4 times the kinetic head in the pipe. The rate of flow is $1.2 \text{ m}^3/\text{s}$ and the pipe diameter in both cases is 1.1 m . Determine:

- (i) The pressure at the inlet and exit of the turbine.
(ii) The power generated by the turbine.

Solution. Given: $d = 1.1 \text{ m}$; $Q = 1.2 \text{ m}^3/\text{s}$; $h_{f(1-2)} = 8 \times \frac{V^2}{2g}$; $h_{f(3-4)} = 0.4 \times \frac{V^2}{2g}$

- (i) The pressure at the inlet and exit of the turbine; p_2, p_3 :

$$\text{Flow velocity in the pipe, } V = \frac{Q}{\frac{\pi d^2}{4}} = \frac{1.2}{\frac{\pi \times (1.1)^2}{4}} = 1.263 \text{ m/s}$$

Since the pipes before and after the turbine are of equal diameter, therefore,

$$V_2 = V_3 = 1.263 \text{ m/s}$$

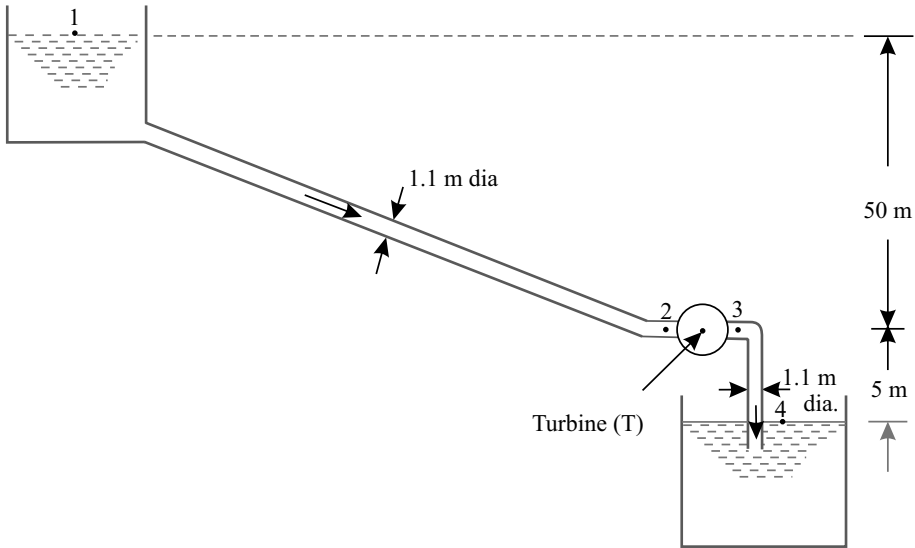


Fig. 6.23

Further, $V_1 = V_4 = 0$; $p_1 = p_4 = 0$ (atmospheric pressure)

Applying Bernoulli's equation between point 1 and 2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + 8 \times \frac{V_2^2}{2g}$$

$$0 + 0 + 50 = \frac{p_2}{w} + \frac{(1.263)^2}{2 \times 9.81} + 0 + 8 \times \frac{(1.263)^2}{2 \times 9.81}$$

or, $\frac{p_2}{w} = \mathbf{49.27 \text{ m of water} = 483.34 \text{ kN/m}^2 \text{ or kPa (Ans.)}$

Applying Bernoulli's equation between points 3 and 4, we have:

$$\frac{p_3}{w} + \frac{V_3^2}{2g} + z_3 = \frac{p_4}{w} + \frac{V_4^2}{2g} + z_4 + 0.4 \times \frac{V_3^2}{2g}$$

$$\frac{p_3}{w} + \frac{(1.263)^2}{2 \times 9.81} + 5 = 0 + 0 + 0 + 0.4 \times \frac{(1.263)^2}{2 \times 9.81}$$

or, $\frac{p_3}{w} = -5.049 \text{ m of water} = -49.53 \text{ kN/m}^2 \text{ or kPa (Ans.)}$

(ii) The power generated by the turbine, P:

Applying Bernoulli's equation between points 2 and 3, we get:

$$\frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 = \frac{p_3}{w} + \frac{V_3^2}{2g} + z_3 + H_T$$

where, H_T = Energy developed by the turbine per unit weight of liquid = Nm/N or m of liquid

$$49.27 = -5.049 + H_T \quad (\because V_2 = V_3 \text{ and } z_2 = z_3)$$

$$\therefore H_T = 54.32 \text{ m of water.}$$

Hence, power generated by the turbine,

$$P = wQH_T = 9.81 \times 1.2 \times 54.32 = 639.46 \text{ kW (Ans.)}$$

Example 6.23. In Fig. 6.24 is shown a turbine with inlet pipe and a draft tube. If the efficiency of turbine is 80 per cent and discharge is 1000 litres/s, find:

(i) The power developed by the turbine, and

(ii) The reading of the gauge G.

(Panjab University)

Solution. Efficiency of the turbine, $\eta = 80\%$

Discharge through the turbine $Q = 1000 \text{ litres/sec.} = 1 \text{ m}^3/\text{s}$

Diameter of the inlet pipe = 0.4 m

$\therefore$ Area of the inlet pipe,

$$A = (\pi/4) \times 0.4^2 = 0.1257 \text{ m}^2$$

$\therefore$ Velocity of water through the pipe,

$$V = \frac{Q}{A} = \frac{1}{0.1257} = 7.96 \text{ m/s}$$

(i) Power developed by the turbine, P:

Applying Bernoulli's equation at 1 and 2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + \text{losses} + H_T$$

where, H_T = Energy developed by the turbine per unit weight of liquid = Nm/N or m of liquid

$$\frac{350}{9.81} + \frac{7.96^2}{2 \times 9.81} + 5 = 0 + 0 + 0 + H_T \quad (\because V = V_1 = 7.96 \text{ m/s}) \quad (\text{neglecting losses})$$

$$\therefore H_T = 35.68 + 3.23 + 5 = 43.91 \text{ m} \quad (\text{where, } w = 9.81 \text{ kN/m}^3)$$

$$\text{Power developed, } P = wQH_T \times \eta \text{ kW} \quad (\text{where, } w = 9.81 \text{ kN/m}^3)$$

$$= 9.81 \times 1 \times 43.91 \times 0.8 = 344.6 \text{ kW}$$

i.e., $P = 344.6 \text{ kW (Ans.)}$

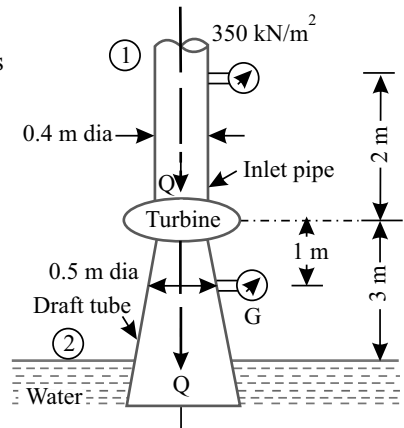


Fig. 6.24

(ii) Reading of the gauge G , p_G :

Applying Bernoulli's equation at 1 and G , we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_G}{w} + \frac{V_G^2}{2g} + z_G + H_T$$

$$\frac{350}{9.81} + \frac{7.96^2}{2 \times 9.81} + 3 = \frac{p_G}{w} + \frac{V_G^2}{2g} + 0 + 43.91 \quad \dots(i)$$

But,

$$V_G = \frac{Q}{A_G} = \frac{1}{(\pi/4) \times 0.5^2} = 5.09 \text{ m/s}$$

Substituting the value of V_G in (i) and rearranging, we get:

$$\frac{p_G}{w} = \frac{350}{9.81} + \frac{7.96^2}{2 \times 9.81} + 3 - \frac{5.09^2}{2 \times 9.81} - 43.91$$

$$= 35.68 + 3.23 + 3 - 1.32 - 43.91 = -3.32 \text{ m of water}$$

$\therefore p_G = 9.81 \times (-3.32) = -32.57 \text{ kN/m}^2 \text{ (Ans.)}$

Example 6.24. Fig. 6.25 shows a pump P pumping 72 litres/sec. of water from a tank.

- (i) What will be the pressures at points L and M when the pump delivers 12 kW of power to the flow? Assume the losses in the system to be negligible.
- (ii) What will be the pressure at M when the loss in the inlet up to the pump is negligible and between the pump and the point M , a loss equal to 1.8 times the velocity head at B takes place.

Solution. Given: $Q = 72 \text{ litres/sec.}$
 $= 0.072 \text{ m}^3/\text{s};$

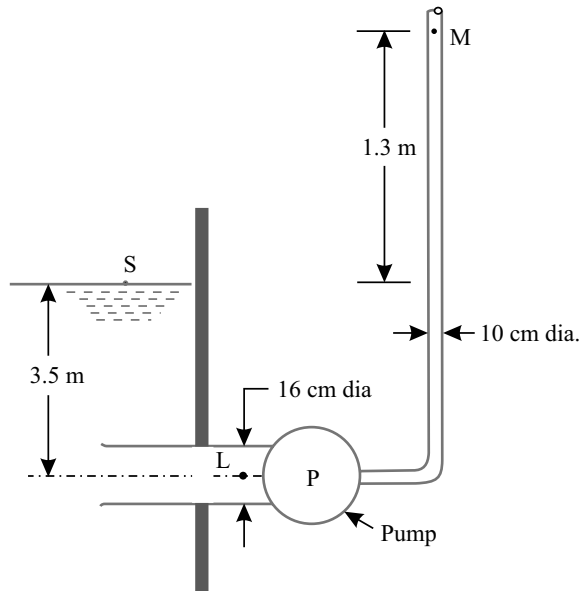


Fig. 6.25

Power delivered by the pump = 12 kW;

Also, $Q = 0.072 = A_L V_L = A_M V_M$

$$V_L = \frac{0.072}{A_L} = \frac{0.072}{\frac{\pi}{4} \times (0.16)^2} = 3.581 \text{ m/s}$$

$$V_M = \frac{0.072}{A_M} = \frac{0.072}{\frac{\pi}{4} \times (0.1)^2} = 9.167 \text{ m/s}$$

Power delivered by the pump P

$$= wQH_P = 12 \text{ kW}$$

or, $9.81 \times 0.072 \times H_P = 12$

$\therefore H_P = 16.99 \text{ m} = \text{Head delivered by the pump.}$

(i) Pressures at point L and M:

Applying Bernoulli's equation to points S and L, we have:

$$\frac{p_S}{w} + \frac{V_S^2}{2g} + z_S = \frac{p_L}{w} + \frac{V_L^2}{2g} + z_L$$

$$0 + 0 + 3.5 = \frac{p_L}{w} + \frac{(3.581)^2}{2 \times 9.81} + 0$$

or, $\frac{p_L}{w} = 2.846 \text{ m}$ or $p_L = 9.81 \times 2.846 = 27.92 \text{ kN/m}^2$ or **27.92 kPa (Ans.)**

By applying Bernoulli's equation between S and L with level at L as datum, we get:

$$\frac{p_S}{w} + \frac{V_S^2}{2g} + z_S + H_P = \frac{p_M}{w} + \frac{V_M^2}{2g} + z_M$$

$$0 + 0 + 3.5 + 16.99 = \frac{p_M}{w} + \frac{(9.167)^2}{2 \times 9.81} + (3.5 + 1.3)$$

or, $\frac{p_M}{w} = 11.407 \text{ m}$ or $p_M = 9.81 \times 11.407 = \mathbf{111.9 \text{ kN/m}^2 \text{ or kPa (Ans.)}$

(ii) Pressure at M when losses are considered:

By applying Bernoulli's equation between S and M with level at L as datum, *considering losses*, we have:

$$\frac{p_S}{w} + \frac{V_S^2}{2g} + z_S + H_P = \frac{p_M}{w} + \frac{V_M^2}{2g} + z_M + \text{losses} \left(\text{i.e. } 1.8 \times \frac{V_M^2}{2g} \right)$$

or, $0 + 0 + 3.5 + 16.99 = \frac{p_M}{w} + \frac{(9.167)^2}{2 \times 9.81} + (3.5 + 1.3) + 1.8 \times \frac{(9.167)^2}{2 \times 9.81}$

or, $3.5 + 16.99 = \frac{p_M}{w} + 16.79$

or, $\frac{p_M}{w} = 3.7 \text{ m}$ or $p_M = 9.81 \times 3.7 = \mathbf{36.3 \text{ kN/m}^2 \text{ or kPa (Ans.)}$

Example 6.25. Fig. 6.26 shows a pump drawing a solution (specific gravity = 1.8) from a storage tank through an 8 cm steel pipe in which the flow velocity is 0.9 m/s. The pump discharges through a 6 cm steel pipe to an overhead tank, the end of discharge is 12 m above the level of the solution in the feed tank. If the friction losses in the entire piping system are 5.5 m and pump efficiency is 65 per cent, determine:

(i) Power rating of the pump.

(ii) Pressure developed by the pump.

Solution. Given: $d_2 = 8 \text{ cm}$ or 0.08 m ; $d_3 = 6 \text{ cm}$ or 0.06 m ; $V_2 = 0.9 \text{ m/s}$, $\eta_{\text{pump}} = 65\%$

(i) Power rating of the pump:

From continuity equation, we have:

$$A_2 V_2 = A_3 V_3$$

$$\text{or, } V_3 (=V_4) = \frac{A_2 V_2}{A_3} = \frac{\frac{\pi}{4} \times (0.08)^2 \times 0.9}{\frac{\pi}{4} \times (0.06)^2} = 1.6 \text{ m/s}$$

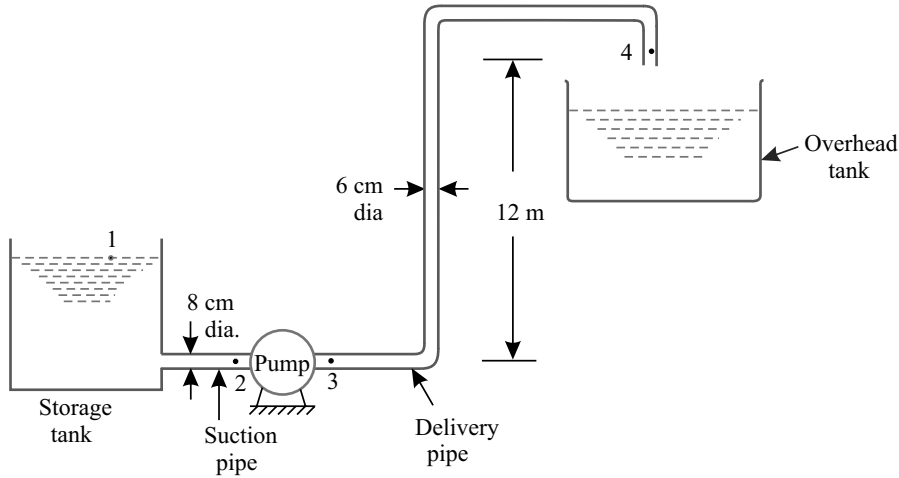


Fig. 6.26

Applying Bernoulli's equation between points 1 and 4, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 + H_p = \frac{p_4}{w} + \frac{V_4^2}{2g} + z_4 + \text{Losses}$$

(where, H_p = Energy added by the pump per unit weight of liquid in Nm/N or m of the liquid pumped)

$$0 + 0 + 0 + H_p = 0 + \frac{(1.6)^2}{2 \times 9.81} + 12 + 5.5$$

$$\text{or, } H_p = 17.63 \text{ m of liquid}$$

$$\therefore \text{Power rating of the pump} = \frac{wQH_p}{\eta_{\text{pump}}} = \frac{(9.81 \times 1.8) \times \left(\frac{\pi}{4} \times 0.08^2 \times 0.9 \right) \times 17.63}{0.65}$$

$$= \mathbf{2.167 \text{ kW (Ans.)}}$$

(ii) Pressure developed by the pump, ($p_3 - p_2$):

Applying Bernoulli's equation between points 2 and 3, we have:

$$\frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + H_p = \frac{p_3}{w} + \frac{V_3^2}{2g} + z_3$$

$$\left(\frac{p_3 - p_2}{w} \right) = \left(\frac{V_2^2 - V_3^2}{2g} \right) + H_p \quad \dots (\because z_2 = z_3)$$

$$= \frac{(0.9)^2 - (1.6)^2}{2 \times 9.81} + 17.63 = 17.54 \text{ m}$$

$$\text{or, } p_3 - p_2 = 17.54 \times (9.81 \times 1.8) = \mathbf{309.72 \text{ kN/m}^2 \text{ or kPa (Ans.)}}$$

Example 6.26. A pump is 2.2 m above the water level in the sump and has a pressure of -20 cm of mercury at the suction side. The suction pipe is of 20 cm diameter and the delivery pipe is short 25 cm diameter pipe ending in a nozzle of 8 cm diameter. If the nozzle is directed vertically upwards at an elevation of 4.2 m above the water sump level, determine:

- (i) The discharge.
 - (ii) The power input into the flow by the pump.
 - (iii) The elevation, above the water sump level, to which the jet would reach.
- Neglect all losses.

Solution. (i) The discharge, Q :

Applying Bernoulli's equation to points 1 and 2 (Fig 6.27), we get

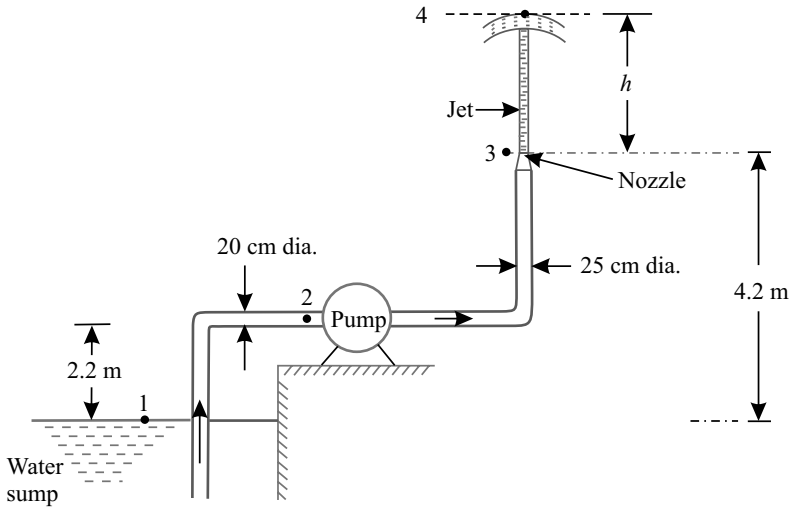


Fig. 6.27

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

$$0 + 0 + 0 = (-0.2 \times 13.6) + \frac{V_2^2}{2g} + 2.2$$

or, $V_2 = 3.194 \text{ m/s}$

$$\text{Discharge, } Q = \frac{\pi}{4} \times 0.21^2 \times 3.194 = 0.1 \text{ m}^3/\text{s (Ans.)}$$

(ii) The elevation, to which the jet will reach, h :

Also, $Q = A_2 V_2 = A_3 V_3$

$$\therefore \frac{\pi}{4} \times (0.2)^2 \times 3.194 = \frac{\pi}{4} \times (0.08)^2 \times V_3$$

or, $V_3 = 19.962 \text{ m/s}$

or, $\frac{V_3^2}{2g} = \frac{(19.962)^2}{2 \times 9.81} = 20.31 \text{ m}$

Hence, the height to which the jet will reach, $h = 20.31 \text{ m (Ans.)}$

(iii) The power input to the flow by the pump, P :

The elevation of point 4, the summit of the jet, is

$$= 4.2 + 20.31 = 24.51 \text{ m}$$

Applying Bernoulli's equation to points 1 and 3, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 + H_p = \frac{p_3}{w} + \frac{V_3^2}{2g} + z_3$$

$$0 + 0 + 0 + H_p = 0 + 20.31 + 4.2$$

$$\text{or, } H_p = 24.51 \text{ m}$$

Power delivered by the pump, $p = wQH_p$

$$= 9.81 \times 0.1 \times 24.51 = \mathbf{24.04 \text{ kW (Ans.)}}$$

Example 6.27. Fig 6.28 shows a pump employed for lifting water from a sump. If it is required to pump 60 litres/sec. of water through a 0.1 m diameter pipe from the sump to a point 10 m above, determine the power required. Also determine pressure intensities at L and M.

Assume an overall efficiency of 70 percent.

(Delhi University)

Solution. Quantity of water to be pumped, $Q = 60$ litres/sec.

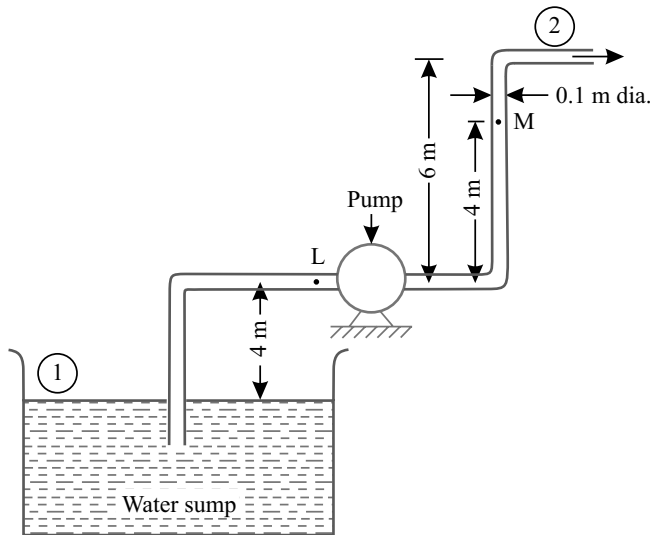


Fig. 6.28

$$= \frac{60}{1000} = 0.06 \text{ m}^3/\text{s}$$

$$\text{Dia of the pipe, } d = 0.1 \text{ m}$$

$$\therefore \text{Area of the pipe, } A = (\pi/4) \times 0.1^2 = 0.00785 \text{ m}^2$$

$$\text{Overall efficiency, } \eta_0 = 70\%$$

Power required, P :

As per continuity equation, $Q = AV$

[where, V = velocity of water in the pipe]

$$\therefore 0.06 = 0.00785 V$$

$$\text{or, } V = \frac{0.06}{0.00785} = 7.64 \text{ m/s}$$

Applying Bernoulli's equation at 1 and 2 points, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 + H_p = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

(where, H_p = Energy added by the pump per unit weight of liquid in Nm/N or m of the liquid pumped)

$$0 + 0 + 0 + H_p = 0 + \frac{(7.64)^2}{2 \times 9.81} + 10 \quad (\because V = V_2 = 7.64 \text{ m/s})$$

$$\therefore H_p = 12.97 \text{ m of water}$$

$\therefore$ Power required to run the pump,

$$P = \frac{wQH_p}{\eta_0} = \frac{9.81 \times 0.06 \times 12.97}{0.7} \text{ kW} \quad (\because w = 9.81 \text{ kN/m}^3)$$

$$\text{i.e., } P = 10.9 \text{ kW (Ans.)}$$

Pressure intensities at L and M:

Applying Bernoulli's equation at 1 and L, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_L}{w} + \frac{V_2^2}{2g} + z_2$$

$$0 + 0 + 0 = \frac{p_L}{w} + \frac{7.64^2}{2 \times 9.81} + 4$$

$$\therefore \frac{p_L}{w} = -\frac{7.64^2}{2 \times 9.81} - 4 = -6.97 \text{ m} \quad (\because V_L = V_2 = 7.64 \text{ m/s})$$

$$\text{or, } P_L = 9.81 \times (-6.97) = -68.4 \text{ kN/m}^2 \text{ (Ans.)}$$

Applying Bernoulli's equation at l and M, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 + H_p = \frac{p_M}{w} + \frac{V_M^2}{2g} + z_M$$

$$0 + 0 + 0 + 12.97 = \frac{p_M}{w} + \frac{7.64^2}{2 \times 9.81} + 8$$

$$\therefore \frac{p_M}{w} = 12.97 - \frac{7.64^2}{2 \times 9.81} - 8 = 12.97 - 2.97 - 8 = 2 \text{ m}$$

$$\text{or, } p_M = 9.81 \times 2 = 19.62 \text{ kN/m}^2 \text{ (Ans.)}$$

6.6. PRACTICAL APPLICATIONS OF BERNOULLI'S EQUATION

Although Bernoulli's equation is applicable in all problems of incompressible flow where there is involvement of energy considerations but here we shall discuss its applications in the following *measuring devices*:

1. Venturimeter
2. Orificemeter
3. Rotameter and elbow meter
4. Pitot tube.

6.6.1. Venturimeter

A venturimeter is one of the most important practical applications of Bernoulli's theorem. *It is an instrument used to measure the rate of discharge in a pipeline and is often fixed permanently at different sections of the pipeline to know the discharges there.*

A venturimeter has been named after the 18th century Italian engineer *Venturi*.

Types of venturimeters:

Venturimeters may be *classified* as follows:

1. Horizontal venturimeters.
2. Vertical venturimeters.
3. Inclined venturimeters.

6.6.1.1. Horizontal venturimeters

A venturimeter consists of the following *three* parts:

- (i) A short converging part,
- (ii) Throat, and
- (iii) Diverging part.

Expression for rate of flow:

Fig 6.29 shows a venturimeter fitted in horizontal pipe through which a fluid is flowing.

Let,

D_1 = Diameter at inlet or at section 1,

A_1 = Area at inlet $\left(= \frac{\pi}{4} D_1^2\right)$

p_1 = Pressure at section 1,

V_1 = Velocity of fluid at section 1,

and D_2, A_2, p_2 , and V_2 are the corresponding values at section 2.

Applying Bernoulli's equation at sections 1 and 2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 \quad \dots(i)$$

Here,

$$z_1 = z_2$$

... since the pipe is horizontal.

$$\therefore \frac{p_1}{w} + \frac{V_1^2}{2g} = \frac{p_2}{w} + \frac{V_2^2}{2g}$$

$$\text{or,} \quad \frac{p_1 - p_2}{w} = \frac{V_2^2}{2g} - \frac{V_1^2}{2g} \quad \dots(ii)$$

But, $\frac{p_1 - p_2}{w}$ = Difference of pressure heads at sections 1 and 2 and is equal to h .

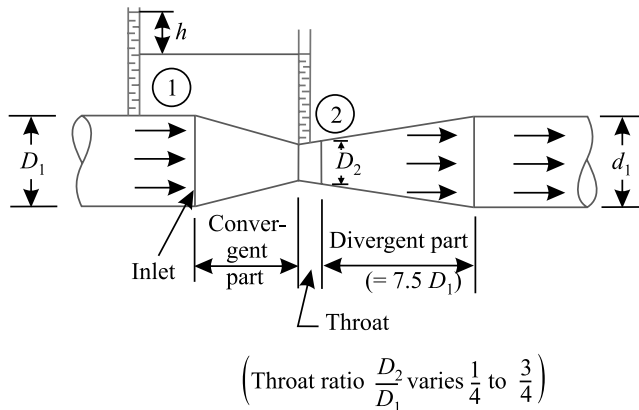


Fig. 6.29. Venturimeter.

i.e.,
$$\frac{p_1 - p_2}{w} = h$$

Substituting this value of $\frac{p_1 - p_2}{w}$ in eqn. (ii), we get:

$$h = \frac{V_2^2}{2g} - \frac{V_1^2}{2g} \quad \dots(iii)$$

Applying continuity equation at sections 1 and 2, we have:

$$A_1 V_1 = A_2 V_2 \quad \text{or} \quad V_1 = \frac{A_2 V_2}{A_1}$$

Substituting the value of V_1 in eqn. (iii), we get:

$$h = \frac{V_2^2}{2g} - \frac{\left(\frac{A_2 V_2}{A_1}\right)^2}{2g} = \frac{V_2^2}{2g} \left(1 - \frac{A_2^2}{A_1^2}\right)$$

or,
$$h = \frac{V_2^2}{2g} \left(\frac{A_1^2 - A_2^2}{A_1^2}\right) \quad \text{or} \quad V_2^2 = 2gh \left(\frac{A_1^2}{A_1^2 - A_2^2}\right)$$

or,
$$V_2 = \sqrt{2gh \left(\frac{A_1^2}{A_1^2 - A_2^2}\right)} = \frac{A_1}{\sqrt{A_1^2 - A_2^2}} \sqrt{2gh}$$

$\therefore$ Discharge, $Q = A_2 V_2 = A_2 \frac{A_1}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh}$

or,
$$Q = \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh} \quad \dots(6.5)$$

or,
$$Q = C \sqrt{h}$$

where, $C = \text{constant of venturimeter}$

$$= \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g}$$

Eqn. (6.5) gives the discharge under ideal conditions and is called *theoretical discharge*. Actual discharge (Q_{act}) which is less than the theoretical discharge ($Q_{th.}$) is given by:

$$Q_{act} = C_d \times \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh} \quad \dots(6.6)$$

where, $C_d = \text{Co-efficient of venturimeter}$ (or co-efficient of discharge) and its value is *less than unity* (varies between 0.96 and 0.98)

● Due to variation of C_d venturimeters are not suitable for very low velocities.

Value of 'h' by differential U-tube manometer:

Case. I. Differential manometer containing a liquid heavier than the liquid flowing through the pipe.

Let,

S_{hl} = Sp. gravity of heavier liquid,

S_p = Sp. gravity of liquid flowing through pipe, and

y = Difference of the heavier liquid column in U-tube.

$$\text{Then} \quad h = y \left[\frac{S_{hl}}{S_p} - 1 \right] \quad \dots(6.7)$$

Case. II. *Differential manometer containing a liquid lighter than the liquid flowing through the pipe.*

Let,
 S_{ll} = Sp. gravity of lighter liquid,
 S_p = Sp. gravity of liquid flowing through pipe, and
 y = Difference of lighter liquid column in U-tube.

$$\text{Then,} \quad h = y \left[1 - \frac{S_{ll}}{S_p} \right] \quad \dots(6.8)$$

Example 6.28. *A horizontal venturimeter with inlet diameter 200 mm and throat diameter 100 mm is used to measure the flow of water. The pressure at inlet is 0.18 N/mm^2 and the vacuum pressure at the throat is 280 mm of mercury. Find the rate of flow. The value of C_d may be taken as 0.98.*

Solution. Inlet diameter of venturimeter, $D_1 = 200 \text{ mm} = 0.2 \text{ m}$

$$\therefore \quad \text{Area of inlet, } A_1 = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$$

$$\text{Throat diameter, } D_2 = 100 \text{ mm} = 0.1 \text{ m}$$

$$\therefore \quad \text{Area of throat, } A_2 = \frac{\pi}{4} \times 0.1^2 = 0.00785 \text{ m}^2$$

$$\text{Pressure at inlet, } p_1 = 0.18 \text{ N/mm}^2 = 180 \text{ kN/m}^2$$

$$\therefore \quad \frac{p_1}{w} = \frac{180}{9.81} = 18.3 \text{ m}$$

Vacuum pressure at the throat,

$$\begin{aligned} \frac{p_2}{w} &= -280 \text{ mm of mercury} \\ &= -0.28 \text{ m of mercury} = -0.28 \times 13.6 = -3.8 \text{ m of water} \end{aligned}$$

Co-efficient of discharge, $C_d = 0.98$

$$\therefore \quad \text{Differential head, } h = \frac{p_1}{w} - \frac{p_2}{w} = 18.3 - (-3.8) = 22.1 \text{ m}$$

Rate of flow, Q:

Using the relation,

$$\begin{aligned} Q &= C_d \times \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh}, \text{ we have:} \\ &= 0.98 \times \frac{0.0314 \times 0.00785}{\sqrt{(0.0314)^2 - (0.00785)^2}} \times \sqrt{2 \times 9.81 \times 22.1} \\ &= \frac{0.000241}{0.0304} \times 20.82 \end{aligned}$$

$$\text{or} \quad Q = 0.165 \text{ m}^3/\text{s} \text{ (Ans.)}$$

Example 6.29. *A horizontal venturimeter with inlet diameter 200 mm and throat diameter 100 mm is employed to measure the flow of water. The reading of the differential manometer connected to the inlet is 180 mm of mercury. If the co-efficient of discharge is 0.98, determine the rate of flow.*

Solution. Inlet diameter of venturimeter, $D_1 = 200 \text{ mm} = 0.2 \text{ m}$

$$\therefore \text{Area at inlet, } A_1 = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$$

Throat diameter, $D_2 = 100 \text{ mm} = 0.1 \text{ m}$

$$\therefore \text{Area of throat, } A_2 = \frac{\pi}{4} \times 0.1^2 = 0.00785 \text{ m}^2$$

Reading of differential manometer, $y = 180 \text{ mm} (= 0.18 \text{ m})$ of mercury

Co-efficient of discharge, $C_d = 0.98$

Rate of flow, Q:

To find difference of pressure head (h) using the relation,

$$h = \left[\frac{S_{hl}}{S_p} - 1 \right], \text{ we have:}$$

where,

S_{hl} = Sp.gr. of mercury (heavy liquid) = 13.6, and

S_p = Sp. gr. of liquid through the pipe *i.e.*, water = 1

$$h = 0.18 \left[\frac{13.6}{1} - 1 \right] = 2.268 \text{ m}$$

To find Q, using the relation,

$$Q = C_d \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh}, \text{ we get:}$$

$$Q = 0.98 \times \frac{0.0314 \times 0.00785}{\sqrt{(0.0314)^2 - (0.00785)^2}} \times \sqrt{2 \times 9.81 \times 2.268}$$

or

$$Q = \frac{0.000241}{0.0304} \times 6.67 = \mathbf{0.0528 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

Example 6.30. A horizontal venturimeter with inlet and throat diameters 300 mm and 100 mm respectively is used to measure the flow of water. The pressure intensity at inlet is 130 kN/m^2 while the vacuum pressure head at the throat is 350 mm of mercury. Assuming that 3 per cent of head is lost in between the inlet and throat, find:

(i) The value of C_d (co-efficient of discharge) for the venturimeter, and

(ii) Rate of flow.

Solution. Inlet diameter of the venturimeter, $D_1 = 300 \text{ mm} = 0.3 \text{ m}$

$$\therefore \text{Area at inlet, } A_1 = \frac{\pi}{4} \times 0.3^2 = 0.07 \text{ m}^2$$

Throat diameter, $D_2 = 100 \text{ mm} = 0.1 \text{ m}$

$$\therefore \text{Area of throat, } A_2 = \frac{\pi}{4} \times 0.1^2 = 0.00785 \text{ m}^2$$

Pressure at inlet, $p_1 = 130 \text{ kN/m}^2$

$$\therefore \text{Pressure head, } \frac{p_1}{w} = \frac{130}{9.81} = 13.25 \text{ m}$$

Similarly, pressure head at throat,

$$\frac{p_2}{w} = -350 \text{ mm of mercury} = -0.35 \times 13.6 \text{ m of water} = -4.76 \text{ m}$$

(i) Co-efficient of discharge, C_d :

$$\text{Differential head, } h = \frac{p_1}{w} - \frac{p_2}{w} = 13.25 - (-4.76) = 18.01 \text{ m}$$

$$\text{Head lost, } h_f = 3\% \text{ of } h = \frac{3}{100} \times 18.01 = 0.54 \text{ m}$$

$$\therefore C_d = \sqrt{\frac{h - h_f}{h}} = \sqrt{\frac{18.01 - 0.54}{18.01}} = 0.985$$

(ii) Rate of flow, Q :

Using the relation,

$$Q = C_d \times \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh}, \text{ we have:}$$

$$\begin{aligned} Q &= 0.985 \times \frac{0.07 \times 0.00785}{\sqrt{0.07^2 - 0.00785^2}} \times \sqrt{2 \times 9.81 \times 18.01} \\ &= \frac{0.000541}{0.0956} \times 18.79 = \mathbf{0.146 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

Example 6.31. A venturimeter (throat diameter = 10.5 cm) is fitted to a water pipeline (internal diameter = 21.0 cm) in order to monitor flow rate. To improve accuracy of measurement, pressure difference across the venturimeter is measured with the help of an inclined tube manometer, the angle of inclination being 30° (Fig. 6.30). For a manometer reading of 9.5 cm of mercury, find the flow rate. Discharge co-efficient of venturimeter is 0.984. (GATE)

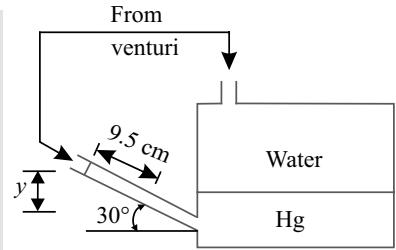


Fig. 6.30

Solution. Internal dia., $D_1 = 21.0 \text{ cm} = 0.21 \text{ m}$;

$$\text{Area of inlet, } A_1 = \frac{\pi}{4} D_1^2 = \frac{\pi}{4} \times (0.21)^2 = 0.0346 \text{ m}^2$$

$$\text{Throat dia, } D_2 = 10.5 \text{ cm} = 0.105 \text{ m}$$

$$\therefore \text{Area at throat, } A_2 = \frac{\pi}{4} \times D_2^2 = \frac{\pi}{4} \times (0.105)^2 = 0.00866 \text{ m}^2$$

Discharge co-efficient of venturimeter, $C_d = 0.984$

$$\text{Pressure head, } h = y \left[\frac{S_{Hg}}{S_{water}} - 1 \right] = (9.5 \sin 30^\circ) \left[\frac{13.6}{1} - 1 \right]$$

$$= 59.85 \text{ cm} = 0.5985 \text{ m}$$

Discharge (Q) through a venturimeter is given by:

$$\begin{aligned} Q &= C_d \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh} \\ &= 0.984 \times \frac{0.0346 \times 0.00866}{\sqrt{(0.0346)^2 - (0.00866)^2}} \times \sqrt{2 \times 9.81 \times 0.5985} \\ &= 0.984 \times 0.008945 \times 3.427 = \mathbf{0.0302 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

Example 6.32. Water at the rate of 30 litres/sec. is flowing through a 0.2 m. I.D. pipe. A venturimeter of throat diameter 0.1 m is fitted in the pipeline. A differential manometer in the pipeline has an indicator liquid M and the manometer reading is 1.16 m. What is the relative density of the manometer liquid M? Venturi co-efficient = 0.96; Density of water = 998 kg/m^3 .

(Anna University)

Solution. Given: $Q = 30 \text{ litres/sec.} = 30 \times 10^{-3} \text{ m}^3/\text{s} = 0.03 \text{ m}^3/\text{s} = D_1 = 0.2 \text{ m}; D_2 = 0.1 \text{ m}; C_d = 0.96; \rho_w = 998 \text{ kg/m}^3; y = 1.16 \text{ m}$

Assume venturimeter to be *horizontal*. The flow rate is given by,

$$Q = C_d \times \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh} \quad \dots(i)$$

Here,

$$A_1 = \frac{\pi}{4} \times D_1^2 = \frac{\pi}{4} \times 0.2^2 = 0.03141 \text{ m}^2, \text{ and}$$

$$A_2 = \frac{\pi}{4} \times D_2^2 = \frac{\pi}{4} \times 0.1^2 = 0.007854 \text{ m}^2$$

Substituting the various values in (i), we get:

$$0.03 = 0.96 \times \frac{0.03141 \times 0.007854}{\sqrt{0.03141^2 - 0.007854^2}} \times \sqrt{2 \times 9.81} \times \sqrt{h}$$

or,

$$0.03 = 0.96 \times 0.008112 \times 4.43 \times \sqrt{h}$$

or,

$$h = \left(\frac{0.03}{0.96 \times 0.008112 \times 4.43} \right)^2 = 0.756 \text{ m}$$

Also,

$$h = y \left(\frac{S_{hl}}{S_{ll}} - 1 \right) \quad [\text{Eqn. (6.7)}]$$

$$0.756 = 1.16 \left(\frac{S_{hl}}{0.998} - 1 \right)$$

$\therefore$

$$S_{hl} = \left(\frac{0.756}{1.16} + 1 \right) \times 0.998 = 1.648$$

Hence specific gravity/relative density of the manometer fluid $M = \mathbf{1.648 \text{ (Ans.)}}$

Example 6.33. A venturimeter is installed in a pipeline carrying water and is 30 cm in diameter. The throat diameter is 12.5 cm. The pressure in pipeline is 140 kN/m^2 , and the vacuum in the throat is 37.5 cm of mercury. Four percent of the differential head is lost between the gauges. Working from first principles find the flow rate in the pipeline in l/s assuming the venturimeter to be horizontal.

(PTU)

Solution. Refer to Fig. 6.29. Given: $D_1 = 30 \text{ cm} = 0.3 \text{ m}; D_2 = 12.5 \text{ cm} = 0.125 \text{ m}; p_1 = 140 \text{ kN/m}^2, p_2 = -37.5 \text{ cm of mercury}$

$$= -\frac{37.5 \times 13.6}{100} = -5.1 \text{ m of water;}$$

$$h_f = 4\% \text{ of differential head.}$$

Flow rate in pipeline, Q:

$$\frac{p_1}{\rho_w} = \frac{140 \times 10^3}{9810} = 14.27 \text{ m of water}$$

$$\frac{p_2}{\rho_w} = -5.1 \text{ m of water (Calculated above)}$$

$$h_f = 4\% \text{ of differential head}$$

$$= \frac{4}{100} \left(\frac{p_1}{\rho_w} - \frac{p_2}{\rho_w} \right) = \frac{4}{100} [(14.27 - (-5.1))] = 0.775 \text{ m of water.}$$

Applying Bernoulli's equation to the entrance (1) and throat (2) of the venturimeter, we get:

$$\frac{p_1}{\rho_w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho_w} + \frac{V_2^2}{2g} + z_2 + h_f$$

$$\text{or, } \frac{V_1^2 - V_2^2}{2g} = \left(\frac{p_2}{w} - \frac{p_1}{w} \right) + h_f \quad (\because z_1 = z_2)$$

$$\text{or, } \frac{V_1^2 - V_2^2}{2g} = -5.1 - 14.27 + 0.775 = -18.59$$

$$\frac{V_1^2}{2g} \left[1 - \left(\frac{V_2}{V_1} \right)^2 \right] = -18.59$$

$$\text{Also, } A_1 V_1 = A_2 V_2$$

$$\text{or, } \frac{V_2}{V_1} = \frac{A_1}{A_2} = \left(\frac{\frac{\pi}{4} D_1^2}{\frac{\pi}{4} D_2^2} \right) = \left(\frac{D_1}{D_2} \right)^2 = \left(\frac{0.3}{0.125} \right)^2 = 5.76$$

$$\text{or, } \frac{V_1^2}{2} [1 - (5.76)^2] = -18.59$$

$$\text{or, } \frac{V_1^2}{2} \times (-32.18) = 2 \times 9.81 \times (-18.59)$$

$$\text{or, } V_1 = \left(\frac{2 \times 9.81 \times 18.59}{32.18} \right)^{1/2} = 3.367 \text{ m/s}$$

$$\begin{aligned} \text{Hence, Discharge, } Q &= A_1 V_1 = \frac{\pi}{4} \times (0.3)^2 \times 3.367 \times 10^3 \text{ l/s} \\ &= \mathbf{238 \text{ l/s (Ans.)}} \end{aligned}$$

6.6.1.2. Vertical and inclined venturimeters

Vertical or inclined venturimeters are employed for measuring discharge on pipelines which are *not horizontal*. The same formula for discharge as used for horizontal venturimeter holds good in these cases as well.

$$\text{Here, } h = \left(\frac{p_1}{w} - \frac{p_2}{w} \right) + (z_1 - z_2)$$

[In horizontal venturimeters $z_1 - z_2 = 0$ as $z_1 = z_2$]

Vertical Venturimeters

Example 6.34. A 200 mm × 100 mm venturimeter is provided in a vertical pipe carrying water, flowing in the upward direction. A differential mercury manometer connected to the inlet and throat gives a reading of 220 mm. Find the rate of flow. Assume $C_d = 0.98$.

Solution. Diameter at the inlet, $D_1 = 200 \text{ mm} = 0.2 \text{ m}$

$$\therefore \text{Area of inlet, } A_1 = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$$

Diameter at the throat, $D_2 = 100 \text{ mm} = 0.1 \text{ m}$

$$\therefore \text{Area at the throat, } A_2 = \frac{\pi}{4} \times 0.1^2 = 0.00785 \text{ m}^2$$

Sp. gravity of heavy liquid (in the manometer), $S_{hl} = 13.6$

Sp. gravity of liquid flowing through pipe, $S_p = 1.0$

Co-efficient of discharge, $C_d = 0.98$

Reading of the differential manometer, $y = 220 \text{ mm} = 0.22 \text{ m}$

Rate of flow, Q:

Differential head,

$$h = \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) = y \left[\frac{S_{hl}}{S_p} - 1 \right]$$

$$= 0.22 \left(\frac{13.6}{1.0} - 1.0 \right) = 2.77 \text{ m}$$

Using the relation,

$$Q = C_d \cdot \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2gh}, \text{ we have}$$

$$Q = 0.98 \times \frac{0.0314 \times 0.00785}{\sqrt{0.0314^2 - 0.00785^2}} \times \sqrt{2 \times 9.81 \times 2.77}$$

$$= \frac{0.000241}{0.0304} \times 7.34 = \mathbf{0.0584 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

Example 6.35. A 300 mm × 150 mm venturimeter is provided in a vertical pipeline carrying oil of specific gravity 0.9, flow being upward. The difference in elevation of the throat section and entrance section of the venturimeter is 300 mm. The differential U-tube mercury manometer shows a gauge deflection of 250 mm. Calculate:

(i) The discharge of oil, and

(ii) The pressure difference between the entrance section and the throat section.

Take the co-efficient of meter as 0.98 and specific gravity of mercury as 13.6.

[UPTU]

Solution. Diameter at inlet, $D_1 = 300 \text{ mm} = 0.3 \text{ m}$

$$\therefore \text{Area of inlet, } A_1 = \frac{\pi}{4} \times 0.3^2 = 0.07 \text{ m}^2$$

Diameter at throat, $D_2 = 150 \text{ mm} = 0.15 \text{ m}$

$$\therefore \text{Area at throat, } A_2 = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

Specific gravity of heavy liquid (mercury)
in U-tube manometer, $S_{hl} = 13.6$

Specific gravity of liquid (oil) flowing
through pipe, $S_p = 0.9$

Reading of differential manometer,
 $y = 250 \text{ mm} = 0.25 \text{ m}$

The differential 'h' is given by:

$$h = \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right)$$

$$= y \left[\frac{S_{hl}}{S_p} - 1 \right] = 0.25 \left[\frac{13.6}{0.9} - 1 \right]$$

$$= 3.53 \text{ m of oil}$$

(i) **Discharge of oil, Q:**

Using the relation,

$$Q = C_d \times \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh}, \text{ we have:}$$

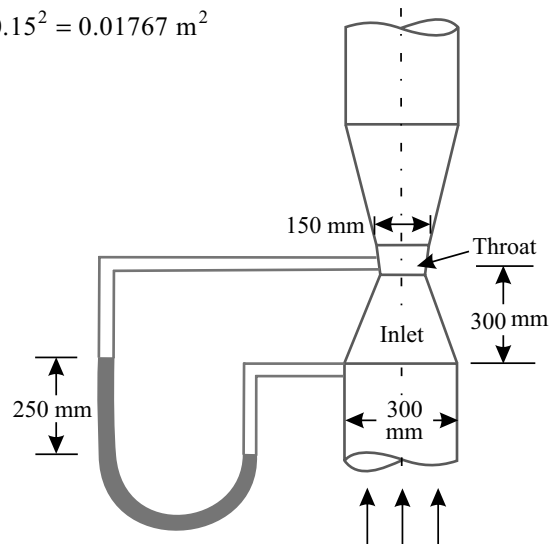


Fig. 6.31

$$Q = 0.98 \times \frac{0.07 \times 0.01767}{\sqrt{0.07^2 - 0.01767^2}} \times \sqrt{2 \times 9.81 \times 3.53}$$

$$= \frac{0.001212}{0.0677} \times 8.32 = \mathbf{0.1489 \text{ m}^3/\text{s. (Ans.)}$$

(ii) Pressure difference between entrance and throat sections, $p_1 - p_2$:

We know that,
$$h = \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) = 3.53$$

or,
$$\left(\frac{p_1}{w} - \frac{p_2}{w} \right) + (z_1 - z_2) = 3.53$$

But,
$$z_2 - z_1 = 300 \text{ mm or } 0.3 \text{ m} \quad \dots \text{ (Given)}$$

$$\therefore \left(\frac{p_1}{w} - \frac{p_2}{w} \right) - 0.3 = 3.53 \quad \text{or} \quad \frac{p_1 - p_2}{w} = 3.83$$

or,
$$p_1 - p_2 = (9.81 \times 0.9) \times 3.83 = \mathbf{33.8 \text{ kN/m}^2 \text{ (Ans.)}}$$

Example 6.36. A vertical venturimeter carries a liquid of relative density 0.8 and has inlet and throat diameters of 150 mm and 75 mm respectively. The pressure connection at the throat is 150 mm above that at the inlet. If the actual rate of flow is 40 litres/sec and the $C_d = 0.96$, calculate the pressure difference between inlet and throat in N/m^2 . (Anna University)

Solution. Given: Sp. gravity = 0.8, $D_1 = 150 \text{ mm} = 0.15 \text{ m}$; $D_2 = 75 \text{ mm} = 0.075 \text{ m}$; $z_2 - z_1 = 150 \text{ mm} = 0.15 \text{ m}$, $Q_{\text{act}} = 40 \text{ litres/sec.} = 0.04 \text{ m}^3/\text{s}$, $C_d = 0.96$.

Pressure difference ($p_1 - p_2$):

$$A_1 = \frac{\pi}{4} D_1^2 = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

$$A_2 = \frac{\pi}{4} D_2^2 = \frac{\pi}{4} \times (0.075)^2 = 0.00442 \text{ m}^2$$

$$Q_{\text{act}} = C_d \times \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh}, \text{ we get:}$$

$$0.04 = 0.96 \times \frac{0.01767 \times 0.00442}{\sqrt{0.01767^2 - 0.00442^2}} \times \sqrt{2 \times 9.81 \times \sqrt{h}}$$

or,
$$0.04 = 0.96 \times 0.004565 \times 4.429 \sqrt{h}$$

$$\therefore h = \left(\frac{0.04}{0.96 \times 0.004565 \times 4.429} \right)^2 = 4.247 \text{ m}$$

Also,
$$h = \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right)$$

or,
$$4.247 = \left(\frac{p_1}{w} - \frac{p_2}{w} \right) + (z_1 - z_2)$$

$$= \left(\frac{p_1 - p_2}{\rho g} \right) - 0.15 \quad (\because z_2 - z_1 = 0.15 \text{ m})$$

or,
$$(p_1 - p_2) = \rho g (4.247 + 0.15)$$

$$= (0.8 \times 1000 \times 9.81) (4.247 + 0.15) \text{ N/m}^2$$

$$= \mathbf{34.51 \text{ kN/m}^2 \text{ (Ans.)}}$$

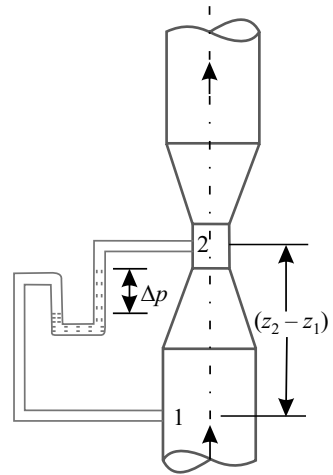


Fig. 6.32. Vertical venturimeter.

Inclined Venturimeters

Example 6.37. Determine the rate of flow of water through a pipe of 300 mm diameter placed in an inclined position where a venturimeter is inserted, having a throat diameter of 150 mm. The difference of pressure between the main and throat is measured by a liquid of sp. gravity 0.7 in an inverted U-tube which gives a reading of 260 mm. The loss of head between the main and throat is 0.3 times the kinetic head of the pipe.

Solution. Diameter at inlet,

$$D_1 = 300 \text{ mm} = 0.3 \text{ m}$$

∴ Area of inlet,

$$A_1 = \frac{\pi}{4} \times 0.3^2 = 0.07 \text{ m}^2$$

Throat diameter,

$$D_2 = 150 \text{ mm} = 0.15 \text{ m}$$

∴ Area at throat,

$$A_2 = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

Specific gravity of lighter liquid (U-tube), $S_{ll} = 0.7$

Specific gravity of liquid (water) flowing through pipe, $S_p = 1.0$

Reading of differential manometer,

$$y = 260 \text{ mm} = 0.26 \text{ m}$$

Difference of pressure head, h is given by:

$$\left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) = h$$

Also,

$$h = y \left(1 - \frac{S_{ll}}{S_p} \right) = 0.26 \left(1 - \frac{0.7}{1.0} \right) = 0.078 \text{ m of water}$$

Loss of head, $h_L = 0.3 \times \text{kinetic head of pipe}$

...(Given)

Now, applying Bernoulli's equation at sections '1' and '2', we get

$$\frac{p_1}{w} + z_1 + \frac{V_1^2}{2g} = \frac{p_2}{w} + z_2 + \frac{V_2^2}{2g} + h_L$$

$$\left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) + \frac{V_1^2}{2g} - \frac{V_2^2}{2g} = h_L$$

$$\text{But, } \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) = 0.078 \text{ m of water}$$

... (as above)

$$\text{and, } h_L = 0.3 \times \frac{V_1^2}{2g}$$

...(Given)

$$\therefore 0.078 + \frac{V_1^2}{2g} - \frac{V_2^2}{2g} = 0.3 \times \frac{V_1^2}{2g}$$

$$\text{or, } 0.078 + 0.7 \frac{V_1^2}{2g} - \frac{V_2^2}{2g} = 0$$

...(i)

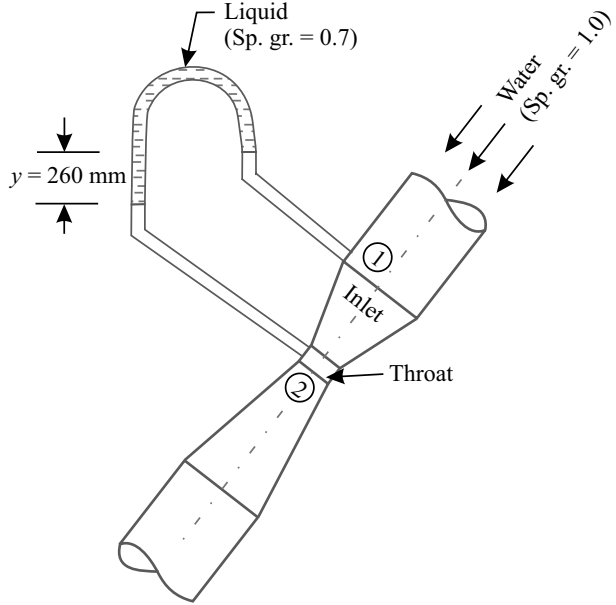


Fig. 6.33

Applying continuity equation at sections '1' and '2', we get:

$$A_1 V_1 = A_2 V_2$$

$$\therefore V_1 = \frac{A_2 V_2}{A_1} = \frac{\pi/4 \times 0.15^2}{\pi/4 \times 0.30^2} \times V_2 = \frac{V_2}{4}$$

Substituting this value of V_1 in eqn. (i), we get:

$$0.078 + 0.7 \times \frac{(V_2/4)^2}{2g} - \frac{V_2^2}{2g} = 0$$

$$\text{or, } 0.078 + \frac{V_2^2}{2g} \left(\frac{0.7}{16} - 1 \right) = 0$$

$$\text{or, } \frac{V_2^2}{2g} \times (-0.956) = -0.078$$

$$\text{or, } V_2^2 = \frac{0.078 \times 2 \times 9.81}{0.956} = 1.6 \text{ m}^2 \text{ or } V_2 = 1.26 \text{ m/s}$$

$$\therefore \text{Rate of flow, } Q = A_2 V_2 = 0.01767 \times 1.26 = \mathbf{0.0222 \text{ m}^3/\text{s. (Ans.)}}$$

Example 6.38. The following data relate to an inclined venturimeter:

Diameter of the pipeline = 400 mm

Inclination of the pipeline with the horizontal = 30°

Throat diameter = 200 mm

The distance between the mouth and throat of the meter = 600 mm

Sp. gravity of oil flowing through the pipeline = 0.7

Sp. gravity of heavy liquid (U-tube) = 13.6

Reading of the differential manometer = 50 mm

The co-efficient of the meter = 0.98

Determine the rate of flow in the pipeline.

(Delhi University)

Solution. Diameter at inlet, $D_1 = 400 \text{ mm} = 0.4 \text{ m}$

$$\therefore \text{Area of inlet, } A_1 = \frac{\pi}{4} \times 0.4^2 = 0.1257 \text{ m}^2$$

Throat diameter, $D_2 = 200 \text{ mm} = 0.2 \text{ m}$

$$\therefore \text{Area at throat, } A_2 = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$$

Reading of the differential manometer (U-tube),
 $y = 50 \text{ mm} = 0.05 \text{ m}$

Difference of pressure head h is given by:

$$h = y \left[\frac{S_{hl}}{S_p} - 1 \right]$$

where, S_{hl} = Sp. gravity of heavy liquid (i.e., mercury) in U-tube = 13.6, and

S_p = Sp. gravity of liquid (i.e., oil) flowing through the pipe = 0.7

$$\therefore h = 0.05 \left(\frac{13.6}{0.7} - 1 \right) = 0.92 \text{ m of oil}$$

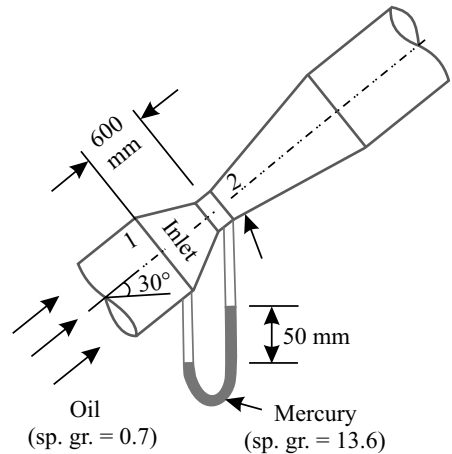


Fig. 6.34

Now, applying Bernoulli's equation at sections '1' and '2', we get:

$$\frac{p_1}{w} + z_1 + \frac{V_1^2}{2g} = \frac{p_2}{w} + z_2 + \frac{V_2^2}{2g} \quad \dots(i)$$

$$\text{or, } \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) + \frac{V_1^2}{2g} - \frac{V_2^2}{2g} = 0$$

$$\text{But, } \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) = h$$

$$\text{or, } \left(\frac{p_1}{w} - \frac{p_2}{w} \right) + (z_1 - z_2) = h$$

It may be noted that *differential gauge reading will include in itself the difference of pressure head and the difference of datum head.*

Thus, eqn. (i) reduces to:

$$h + \frac{V_1^2}{2g} - \frac{V_2^2}{2g} = 0 \quad \dots(ii)$$

Applying continuity equation at sections '1' and '2' we get:

$$A_1 V_1 = A_2 V_2$$

$$\text{or, } V_1 = \frac{A_2 V_2}{A_1} = \frac{(\pi/4) \times 0.2^2}{(\pi/4) \times 0.4^2} \times V_2 = \frac{V_2}{4}$$

Substituting the value of V_1 and h in eqn. (ii), we get:

$$0.92 + \frac{V_2^2}{16 \times 2g} - \frac{V_2^2}{2g} = 0$$

$$\text{or, } \frac{V_2^2}{2g} \left(1 - \frac{1}{16} \right) = 0.92 \quad \text{or} \quad V_2^2 \times \frac{15}{16} = 0.92$$

$$\text{or, } V_2^2 = \frac{0.92 \times 2 \times 9.81 \times 16}{15} = 19.25 \quad \text{or} \quad V_2 = 4.38 \text{ m/s}$$

$$\text{Rate of flow of oil, } Q = A_2 V_2 = 0.0314 \times 4.38 = \mathbf{0.1375 \text{ m}^3/\text{s}} \text{ (Ans.)}$$

6.6.2. Orificemeter

Orificemeter or orifice plate is a device (cheaper than a venturimeter) employed for measuring the discharge of fluid through a pipe. It also works on the same principle of a venturimeter.

It consists of a flat circular plate having a circular sharp edged hole (called orifice) concentric with the pipe. The diameter of the orifice may vary from 0.4 to 0.8 times the diameter of the pipe but its value is generally chosen as 0.5. A differential manometer is connected at section (1) which is at a distance of 1.5 to 2 times the pipe diameter upstream from the orifice plate, and at section (2) which is at a distance of about half the diameter of the orifice from the orifice plate on the downstream side.

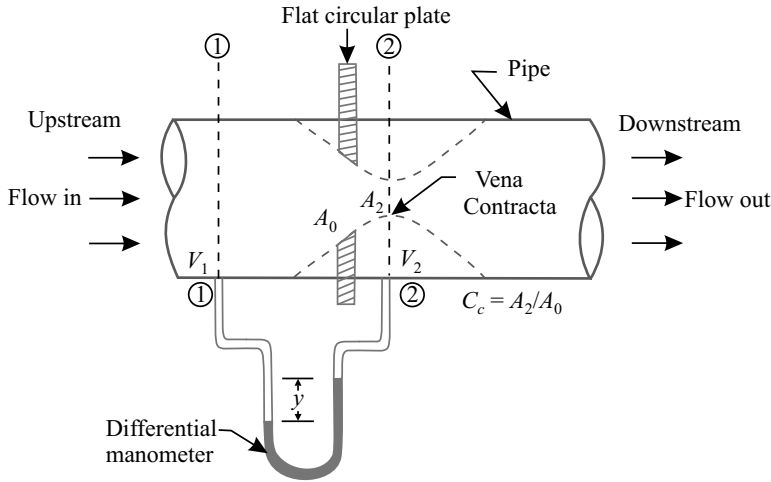
Let,

$$A_1 = \text{Area of pipe at section (1),}$$

$$V_1 = \text{Velocity at section (1),}$$

$$p_1 = \text{Pressure at section (1), and}$$

$$A_2, V_2 \text{ and } p_2 \text{ are corresponding values at section (2).}$$

**Fig. 6.35. Orificemeter**

Applying Bernoulli's equation at sections (1) and (2), we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

$$\text{or, } \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

$$\text{or, } h = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

$$\left[\because h = \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) = \text{differential head} \right]$$

$$\text{or, } \frac{V_2^2}{2g} = h + \frac{V_1^2}{2g} \quad \dots(i)$$

$$\text{or, } V_2 = \sqrt{2g \left(h + \frac{V_1^2}{2g} \right)} = \sqrt{2gh + V_1^2}$$

Now, section (2) is at *vena contracta* and A_2 represents the area at vena contracta. If A_0 is the area of orifice, then we have:

$$C_c = \frac{A_2}{A_0}$$

(where, C_c = co-efficient of contraction)

$$\therefore A_2 = A_0 C_c \quad \dots(ii)$$

Using continuity equation, we get:

$$A_1 V_1 = A_2 V_2 \quad \text{or} \quad V_1 = \frac{A_2 V_2}{A_1}$$

$$\text{or, } V_1 = \frac{A_0 C_c V_2}{A_1} \quad \dots(iii)$$

Substituting the value of V_1 in eqn. (i), we get:

$$V_2 = \sqrt{2gh + \frac{A_0^2 \cdot C_c^2 \cdot V_2^2}{A_1^2}}$$

$$\text{or,} \quad V_2^2 = 2gh + \left(\frac{A_0}{A_1}\right)^2 \cdot C_c^2 \cdot V_2^2$$

$$\text{or,} \quad V_2^2 \left[1 - \left(\frac{A_0}{A_1}\right)^2 C_c^2 \right] = 2gh$$

$$\therefore \quad V_2 = \frac{\sqrt{2gh}}{\sqrt{1 - (A_0/A_1)^2 C_c^2}}$$

$$\therefore \quad \text{The discharge, } Q = A_2 V_2 = A_0 \cdot C_c \cdot V_2 \quad [\because A_2 = A_0 \cdot C_c \dots \text{as above \{eqn. (ii)\}}]$$

$$= A_0 C_c \frac{\sqrt{2gh}}{\sqrt{1 - (A_0/A_1)^2 C_c^2}} \quad \dots(iv)$$

The above expression is simplified by using,

$$C_d = C_c \frac{\sqrt{1 - (A_0/A_1)^2}}{\sqrt{1 - (A_0/A_1)^2 C_c^2}}$$

(where, C_d = co-efficient of discharge)

$$C_c = C_d \frac{\sqrt{1 - (A_0/A_1)^2 C_c^2}}{\sqrt{1 - (A_0/A_1)^2}}$$

Substituting this value of C_c in eqn. (iv), we get:

$$\begin{aligned} Q &= A_0 \cdot C_d \frac{\sqrt{1 - (A_0/A_1)^2 C_c^2}}{\sqrt{1 - (A_0/A_1)^2}} \times \frac{\sqrt{2gh}}{\sqrt{1 - (A_0/A_1)^2 C_c^2}} \\ &= \frac{C_d \cdot A_0 \sqrt{2gh}}{\sqrt{1 - (A_0/A_1)^2}} = \frac{C_d \cdot A_0 \cdot A_1 \sqrt{2gh}}{\sqrt{A_1^2 - A_0^2}} \end{aligned}$$

$$\text{i.e.,} \quad Q = C_d \frac{A_0 \cdot A_1 \sqrt{2gh}}{\sqrt{A_1^2 - A_0^2}} \quad \dots(6.9)$$

It may be noted that C_d (co-efficient of discharge) of an orifice is *much smaller than that of a venturimeter*.

Difference between a venturimeter and an orificemeter:

A **venturimeter** is a device which is inserted into pipeline to measure incompressible fluid flow rates. It consists of a convergent section which reduces the diameter to between one-half to one-fourth of the pipe diameters. This is followed by a divergent section. The pressure difference between the position just before the venturi and at the throat of the venturi is measured by a differential manometer. The *working of the venturi is based on the Bernoulli's principle*, that is, *when the velocity head increases in an accelerated flow, there is a corresponding reduction in the piezometric head*.

The **orificemeter** is opening, usually round, located in the side wall of the tank or reservoir, for measuring the flow of a liquid. The main feature of the orificemeter is that *most of the potential energy of the liquid is converted into kinetic energy of the free jet issuing through the orifice*.

The main points of *difference* between a venturimeter and orificemeter are:

1. The venturimeter can be used for *measuring the flow rates of all incompressible flows*. (gases with low pressure variations, as well as liquids), whereas orificemeters are generally used for measuring the *flow rates of liquids*.
2. Venturimeter is installed *in pipeline only*, and the accelerated flow through the apparatus, is subsequently decelerated to the original velocity at the outlet of the venturimeter. The flow continues through the pipeline. In the orificemeter the entire potential energy of the fluid is converted to kinetic energy, and the jet *discharges freely into the open atmosphere*.
3. In venturimeter, the flow velocity is *measured by noting the pressure difference between the inlet and the throat of the venturimeter*, whereas in the orificemeter the discharge velocity is measured by using *Pitot tube* or by *trajectory method*.

Example 6.39. *The following data relate to an orificemeter:*

Diameter of the pipe = 240 mm

Diameter of the orifice = 120 mm

Sp. gravity of oil = 0.88

Reading of differential manometer = 400 mm of mercury

Co-efficient of discharge of the meter = 0.65.

Determine the rate of flow of oil.

Solution. Diameter of the pipe $D_1 = 240 \text{ mm} = 0.24 \text{ m}$

$$\therefore \text{Area of the pipe, } A_1 = \frac{\pi}{4} \times 0.24^2 = 0.0452 \text{ m}^2$$

Diameter of the orifice, $D_0 = 120 \text{ mm} = 0.12 \text{ m}$

$$\therefore \text{Area of the orifice, } A_0 = \frac{\pi}{4} \times 0.12^2 = 0.0113 \text{ m}^2$$

Co-efficient of discharge, $C_d = 0.65$

Sp. gravity of oil, $S_o = 0.88$

Reading of differential manometer, $y = 400 \text{ mm of mercury} = 0.4 \text{ m of mercury}$

$$\therefore \text{Differential head, } h = y \left[\frac{S_{hl}}{S_o} - 1 \right]$$

[where, S_{hl} = sp. gravity of heavier liquid = 13.6 (for mercury)]

$$= 0.4 \left[\frac{13.6}{0.88} - 1 \right] = 5.78 \text{ m of oil}$$

Discharge Q:

Using the relation, $Q = C_d \frac{A_0 \cdot A_1 \cdot \sqrt{2gh}}{\sqrt{A_1^2 - A_0^2}}$, we have:

$$\begin{aligned} Q &= 0.65 \times \frac{0.0113 \times 0.0452 \times \sqrt{2 \times 9.81 \times 5.78}}{\sqrt{(0.0452)^2 - (0.0113)^2}} \\ &= \frac{0.000353}{0.0437} = \mathbf{0.08 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

Example 6.40. *Water flows at the rate of $0.015 \text{ m}^3/\text{s}$ through a 100 mm diameter orifice used in a 200 mm pipe. What is the difference of pressure head between the upstream section and the vena contracta section? Take co-efficient of contraction $C_c = 0.60$ and $C_v = 1.0$. (Delhi University)*

Solution. Given: $Q = 0.015 \text{ m}^3/\text{s}$; $D_0 = 100 \text{ mm} = 0.1 \text{ m}$; $D_1 = 200 \text{ mm} = 0.2 \text{ m}$; $C_c = 0.60$; $C_v = 1.0$

Difference in pressure head h : Refer to Fig. 6.35.

$$A_1 = \frac{\pi}{4} D_1^2 = \frac{\pi}{4} \times 0.2^2 = 0.03142 \text{ m}^2$$

$$A_0 = \frac{\pi}{4} D_0^2 = \frac{\pi}{4} \times 0.1^2 = 0.007854 \text{ m}^2$$

$$C_d = C_c \times C_v = 0.60 \times 1.0 = 0.6$$

Using the relation: $Q = C_d \frac{A_0 A_1 \sqrt{2gh}}{\sqrt{A_1^2 - A_0^2}}$

or, $0.015 = 0.6 \times \frac{0.007854 \times 0.03142 \sqrt{2 \times 9.81 \times h}}{\sqrt{(0.03142)^2 - (0.007854)^2}} \quad \dots [\text{Eqn. (6.9)}]$

or, $0.015 = 0.6 \times \frac{0.001093 \sqrt{h}}{0.03042}$

or, $h = \left(\frac{0.015 \times 0.03042}{0.6 \times 0.001093} \right)^2 = \mathbf{0.484 \text{ m of water (Ans.)}$

Example 6.41. (a) Derive an expression for the volumetric flow rate of a fluid flowing through an orificemeter. Write down the advantages and disadvantages of using orificemeter over a venturimeter.

(b) Water is flowing through a pipeline of 50 cm ID at 30°C. An orifice is placed in the pipeline to measure the flow rate. Orifice diameter is 20 cm. If the manometer reads 30 cm of Hg, calculate the water flow rate and velocity of the fluid through the pipe.

$$\rho_{\text{water at } 30^\circ\text{C}} = 987 \text{ kg/m}^3$$

$$\rho_{\text{Hg}} = 13600 \text{ kg/m}^3$$

Orifice co-efficient. = 0.6

Sol. (a) Refer to Article 6.6.2.

Advantage of orificemeter over venturimeter is that its length is short and hence it can be used in a wide variety of application. Venturimeter has excessive length.

The **disadvantage** of orificemeter is that a sizeable pressure loss is increased because of the flow separation downstream of the plate. In a venturimeter the expanding section keeps boundary layer separation to a minimum, resulting in good pressure recovery across the meter.

(b) Given: $D_1 = 50 \text{ cm} = 0.5 \text{ m}$; $D_0 = 20 \text{ cm} = 0.2$; $y = 30 \text{ cm of Hg} = 0.3 \text{ m of Hg}$

$$\rho_{\text{water at } 30^\circ\text{C}} = 981 \text{ kg/m}^3, \rho_{\text{Hg}} = 13600 \text{ kg/m}^3;$$

$$C_0 = 0.6.$$

Water flow rate, Q :

$$A_1 = \frac{\pi}{4} D_1^2 = \frac{\pi}{4} \times 0.5^2 = 0.1963 \text{ m}^2$$

$$A_0 = \frac{\pi}{4} D_0^2 = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$$

$$h = y \left(\frac{\rho_{\text{Hg}}}{\rho_{\text{H}_2\text{O}}} - 1 \right) = 0.3 \left(\frac{13600}{987} - 1 \right) = 3.834 \text{ m}$$

Using the relation; $Q = C_d \frac{A_0 A_1 \sqrt{2gh}}{\sqrt{A_1^2 - A_0^2}}$, we get:

$$Q = 0.6 \times \frac{0.0314 \times 0.1963 \times \sqrt{2 \times 9.81 \times 3.834}}{\sqrt{(0.1963)^2 - (0.0314)^2}}$$

$$= 0.6 \times \frac{0.05346}{0.1938} = \mathbf{0.1655 \text{ m}^3/\text{s (Ans.)}$$

Velocity of water through pipe,

$$V_1 = \frac{Q}{A_1} = \frac{0.1655}{0.1963} = \mathbf{0.843 \text{ m/s (Ans.)}$$

6.6.3. Rotameter and Elbow meter

6.6.3.1. Rotameter. Refer to Fig. 6.36.

Construction: It consists of a tapered metering glass tube, inside of which is located a rotor or active element (float) of the meter. The tube is provided with inlet and outlet connections. The specific gravity of the float or bob material is higher than that of the fluid to be metered. On a part of the float spherical slots are cut which cause it (float) to rotate slowly about the axis of the tube and keep it centred. Owing to this spinning, accumulation of any sediment on the top or sides of float is checked. However, the stability of the bob may also be ensured by using a guide along which the float would slide.

Working : When the rate of flow increases the float rises in the tube and consequently there is an increase in the annular area between the float and the tube. Thus, the float rides higher or lower depending on the rate of flow.

The discharge through a rotameter is given by:

$$Q = C_d A_{ann} \cdot [2gV_f (\rho_f - \rho_f)/A_f \rho_f]^{1/2} \quad \dots (6.10)$$

where,

Q = Volume flow rate,

C_d = Co-efficient of discharge,

$A_{ann.}$ = Annular area between float and tube,

V_f = Volume of float,

ρ_f = Density of float material,

ρ_f = Density of fluid, and

A_f = Maximum cross-sectional area of the fluid.

As the flow area $A_{ann.}$ is a function of height of float in the tube, the flow rate scale can be engraved on the tube corresponding to a particular float.

Advantages :

1. Simpler in operation.
2. Handling and installation easy.
3. Wide variety of corrosive fluids can be handled.
4. Low cost, relatively.

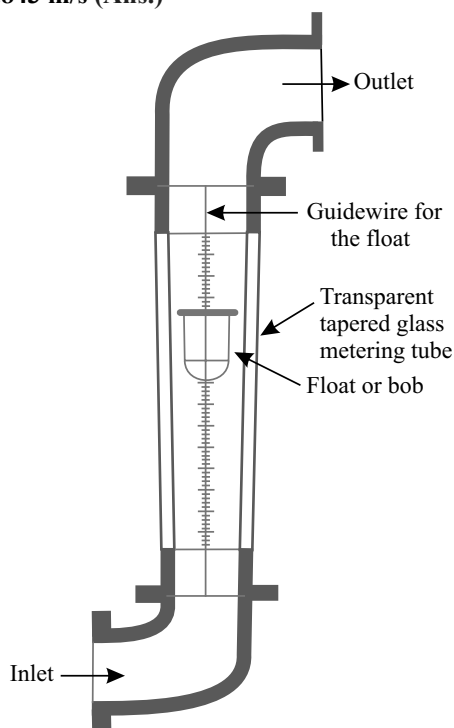


Fig. 6.36. Rotameter.

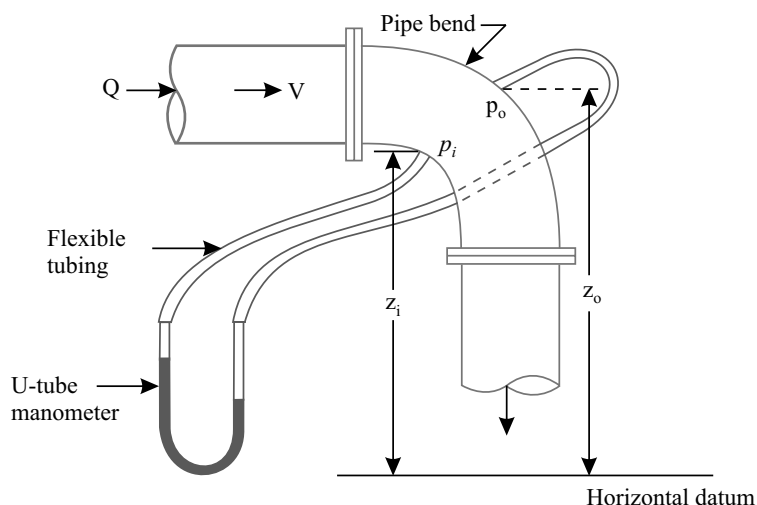
Limitations :

1. Mounted vertically, limited to small pipe sizes and capacities.
2. Less accurate, compared to venturimeter and orificemeter.

6.6.3.2. Elbow meter

When liquid flows around a pipe bend, there is an increase in pressure with radius, *i.e.* the pressure at the outer wall of the bend is more than that at the inner wall. This difference of pressure which exists between the outside and inside of the bend is used for the measurement of discharge in a pipeline.

As shown in Fig. 6.37. the pipe bend is provided with two pressure tappings, one each at the inner and outer walls of the bend. These tappings are connected to the limbs of U-tube manometer.

**Fig. 6.37.** Elbow meter.

As per literature, the following relation between velocity and pressure difference is available:

$$K \frac{V^2}{2g} = \left(\frac{p_o}{w} + z_o \right) - \left(\frac{p_i}{w} + z_i \right) \quad \dots(6.11)$$

where,

K = Constant (depends upon the shape and size of the bend),
ranges from 1.3 to 3.2, and

V = Velocity of flow.

Suffices 0 and i represent the conditions at the outer and inner walls of the pipe bend.

$$\text{or,} \quad V = \frac{1}{\sqrt{K}} \sqrt{2g} \sqrt{\left(\frac{p_o}{w} + z_o \right) - \left(\frac{p_i}{w} + z_i \right)} \quad \dots[6.11 (a)]$$

$$\therefore \quad \text{Discharge, } Q = AV = C_d A \sqrt{2g} \sqrt{\left(\frac{p_o}{w} + z_o \right) - \left(\frac{p_i}{w} + z_i \right)} \quad \dots(6.12)$$

where,

$$C_d = \frac{1}{\sqrt{k}} = \text{Co-efficient of discharge, and}$$

A = Cross-sectional area of the pipe.

(C_d varies between 0.56 and 0.88)

The following empirical relation has been suggested:

$$C_d = \sqrt{\frac{R_b}{D}},$$

where,

R_b = Radius of pipe bend, and

D = Diameter of the pipe.

- An elbowmeter can be conveniently used for the measurement of discharge in pipes which are fitted with elbows and bends.
- Its accuracy, with proper calibration, approaches that of a venturimeter or nozzle.

6.6.4. Pitot Tube

Pitot tube is one of the most accurate devices for velocity measurement. It works on the principle that if the velocity of flow at a point becomes zero, the pressure there is increased due to conversion of kinetic energy into pressure.

It consists of a glass tube in the form of a 90° bend of short length open at both its ends. It is placed in the flow with its bent leg directed upstream so that a *stagnation point* is created immediately in front of the opening (Fig. 6.38). The kinetic energy at this point gets converted into pressure energy causing the liquid to rise in the vertical limb, to a height equal to the stagnation pressure.

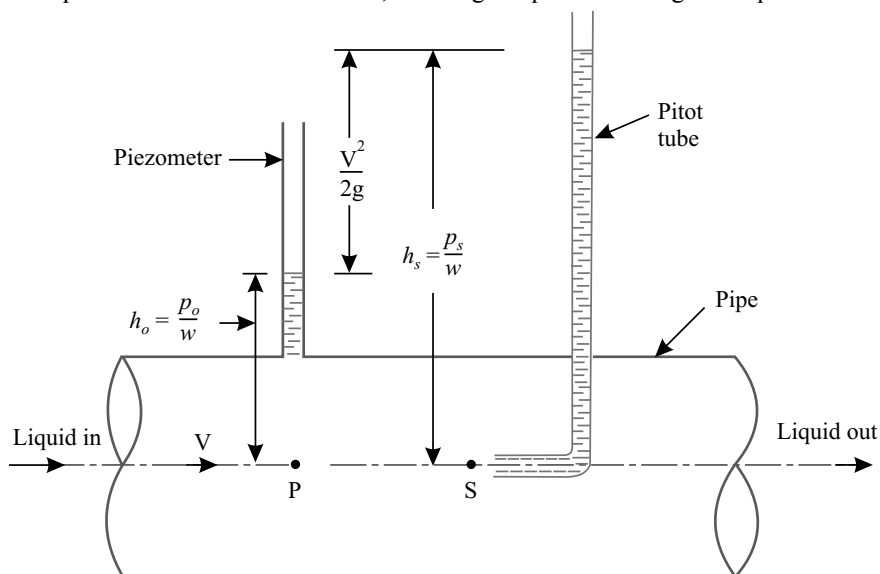


Fig. 6.38. Pitot tube.

Applying Bernoulli's equation between stagnation point (S) and point (P) in the undisturbed flow at the same horizontal plane, we get:

$$\frac{p_0}{w} + \frac{V^2}{2g} = \frac{p_s}{w} \quad \text{or} \quad h_0 + \frac{V^2}{2g} = h_s$$

$$\text{or,} \quad V = \sqrt{2g(h_s - h_0)} \quad \text{or} \quad \sqrt{2g \Delta h} \quad \dots(1)$$

where,

p_0 = Pressure at point 'P', i.e. static pressure,

V = Velocity at point 'P', i.e. free flow velocity,

p_s = Stagnation pressure at point 'S', and

Δh = Dynamic pressure

= Difference between stagnation pressure head (h_s) and static pressure head (h_0).

The height of liquid rise in the Pitot tube indicates the stagnation head. The static pressure head may be measured separately with a piezometer (Fig. 6.38).

Both the static pressure as well as stagnation pressure can be measured in a device known as **Pitot static tube**. (Fig. 6.39).

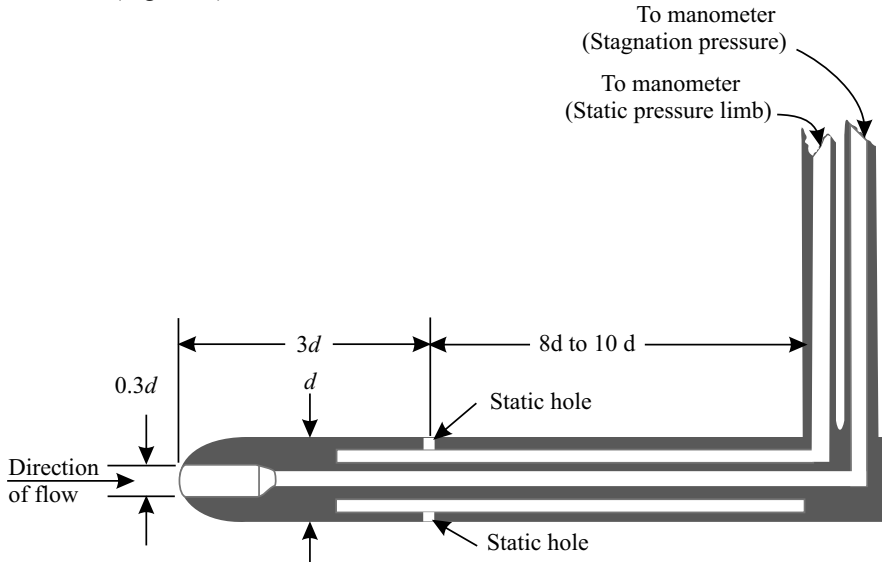


Fig. 6.39. Pitot static tube.

It consists of two concentric Pitot tubes with an annular space in between as shown in the figure. The outer tube has additional two or more holes drilled perpendicular to the direction of flow and thus the liquid level in it gives the static head, while the inner tube works as a normal Pitot tube. If a differential manometer is connected to the tubes of a Pitot static tube it will measure the dynamic pressure head.

If y is the manometric difference, then

$$\Delta h = y \left(\frac{S_m}{S} - 1 \right)$$

where,

S_m = Specific gravity of manometric liquid, and

S = Specific gravity of the liquid flowing through the pipe.

When a Pitot tube is placed in the fluid-stream the flow along its outer surface gets accelerated and causes the static pressure to decrease. Also the stem, which is perpendicular to the flow direction, tends to produce an excess pressure head. In order to take these effects into account eqn. (1) is modified to give the actual velocities as:

$$V = C \sqrt{2g\Delta h} \quad \dots(2)$$

where, C = A connective coefficient which takes into account the effect of stem and bent leg.

The most commonly used form of Pitot static tube known as the Prandle-Pitot-tube is so designed that the effect of stem and bent leg cancel each other, i.e., $C = 1$.

Example 6.42. A submarine fitted with a Pitot tube moves horizontally in sea. Its axis is 12 m below the surface of water. The Pitot tube fixed in front of the submarine and along its axis is connected to the two limbs of a U-tube containing mercury, the reading of which is found to be 200 mm. Find the speed of the submarine.

Take the specific gravity of sea water = 1.025 times fresh water.

Solution. Reading of the manometer, $y = 200 \text{ mm} = 0.2 \text{ m}$ of mercury

Sp. gravity of mercury, $S_{hl} = 13.6$

Sp. gravity of sea water, $S_l = 1.025$

To find the head, (h), using the relation: $h = y \left(\frac{S_{hl}}{S_l} - 1 \right)$, we have:

$$h = 0.2 \left(\frac{13.6}{1.025} - 1 \right) = 2.45$$

$\therefore$ Velocity of the submarine

$$V = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 2.45} = \mathbf{6.93 \text{ m/s or } 24.9 \text{ km/h (Ans.)}$$

Example 6.43. Petroleum oil (sp. gr. = 0.9 and viscosity = 13 cP) flows isothermally through a horizontal 5 cm pipe. A Pitot tube is inserted at the centre of a pipe and its leads are filled with the same oil and attached to a U-tube containing water. The reading on the manometer is 10 cm. Calculate the volumetric flow of oil in m^3/s . The co-efficient of Pitot tube is 0.98.

(Delhi University)

Solution. Given: Sp gr. of oil = 0.9; $\mu = 13 \text{ cP} = \frac{13}{100} \times 0.1 \text{ Ns/m}^2 = 0.013 \text{ Ns/m}^2$;

$y = 10 \text{ cm of Hg} = 0.1 \text{ m of Hg.}, D = 5 \text{ cm} = 0.05 \text{ m};$

Co-efficient of Pitot tube, $C_v = 0.98$

Volumetric flow of oil:

$$\text{Differential head, } h = y \left(\frac{S_{Hg}}{S_{Oil}} - 1 \right) = 0.1 \left(\frac{13.6}{0.9} - 1 \right) = 1.411$$

$\therefore$ Actual velocity of flow, $V = C_v \sqrt{2gh} = 0.98 \sqrt{2 \times 9.81 \times 1.411} = \mathbf{5.156 \text{ m/s}}$

$$\text{Volumetric flow of oil} = A \times V = \frac{\pi}{4} \times 0.05^2 \times 5.156 = \mathbf{0.01 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

Example 6.44. For the flow situation shown in Fig. 6.40 determine the ratio $\frac{h_1}{h_2}$ if the area

ratio $\frac{A_1}{A_2} = 1.8$.

Neglect losses due to friction.

Solution. Refer to Fig. 6.40.

For Pitot tube :

$$p_s + \rho_a g h_1 = p_2 + \rho_b g h_1 \quad \dots(i)$$

For piezometric tubes:

$$p_1 + \rho_a g h_2 = p_2 + \rho_b g h_2 \quad \dots(ii)$$

Subtracting (ii) from (i), we get:

$$(p_s - p_1) + \rho_a g (h_1 - h_2) = \rho_b g (h_1 - h_2)$$

$$(p_s - p_1) = (h_1 - h_2) (\rho_b - \rho_a) g$$

$$\frac{p_s - p_1}{\rho_a g} = (h_1 - h_2) \left(\frac{\rho_b}{\rho_a} - 1 \right) \quad \dots(iii)$$

.....Dividing by $\rho_a g$.

From piezometric tappings,

$$(p_1 - p_2) = h_2 g (\rho_b - \rho_a)$$

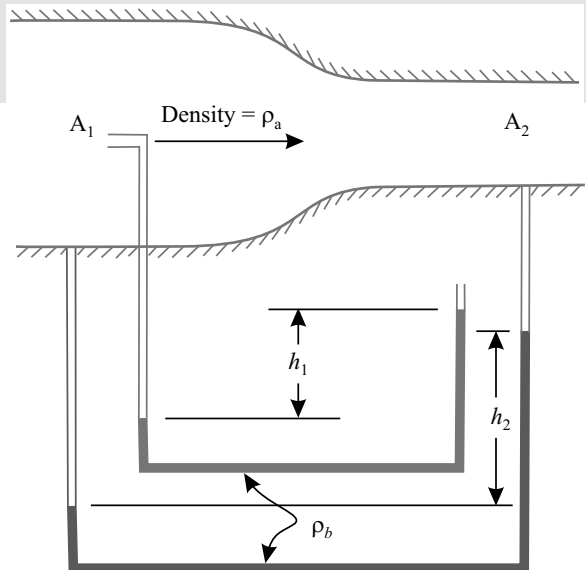


Fig. 6.40

$$\frac{p_1 - p_2}{\rho_a g} = h_2 \left(\frac{\rho_b}{\rho_a} - 1 \right) \quad \dots (iv)$$

...Dividing by $\rho_a g$

By Bernoulli's equation:

$$\frac{p_1}{\rho_a g} + \frac{V_1^2}{2g} = \frac{p_2}{\rho_a g} + \frac{V_2^2}{2g} \quad \dots (z_1 = z_2)$$

$$\frac{p_1 - p_2}{\rho_a g} = \frac{V_2^2 - V_1^2}{2g}$$

$$\text{or,} \quad \frac{p_1 - p_2}{\rho_a g} = \frac{V_1^2}{2g} \left[\left(\frac{V_2}{V_1} \right)^2 - 1 \right] = \frac{V_1^2}{2g} \left[\left(\frac{A_1}{A_2} \right)^2 - 1 \right] \quad \dots (v)$$

From (iii), (iv) and (v), we have:

$$\therefore h_2 \left(\frac{\rho_b}{\rho_a} - 1 \right) = (h_1 - h_2) \left(\frac{\rho_b}{\rho_a} - 1 \right) [(1.8)^2 - 1]$$

$$\text{or,} \quad \left(\frac{h_1}{h_2} - 1 \right) \times 2.24 = 1 \quad \left[\because \Delta h = \frac{p_s - p_1}{\rho_a g} = \frac{V_1^2}{2g} \right]$$

$$\text{or,} \quad \frac{h_1}{h_2} = 1.446 \text{ (Ans.)}$$

6.7. FREE LIQUID JET

Refer to Fig. 6.41. A jet of liquid issuing from the nozzle in atmosphere is called a *free liquid jet*. The parabolic path traversed by the liquid jet under the action of gravity is known as *trajectory*. Let the jet A make an angle θ with the horizontal direction. If U is the velocity of the water jet, then $U \cos \theta$ and $U \sin \theta$ are the horizontal and vertical components of this velocity respectively. Consider another point $P(x, y)$ on the centre line of the jet.

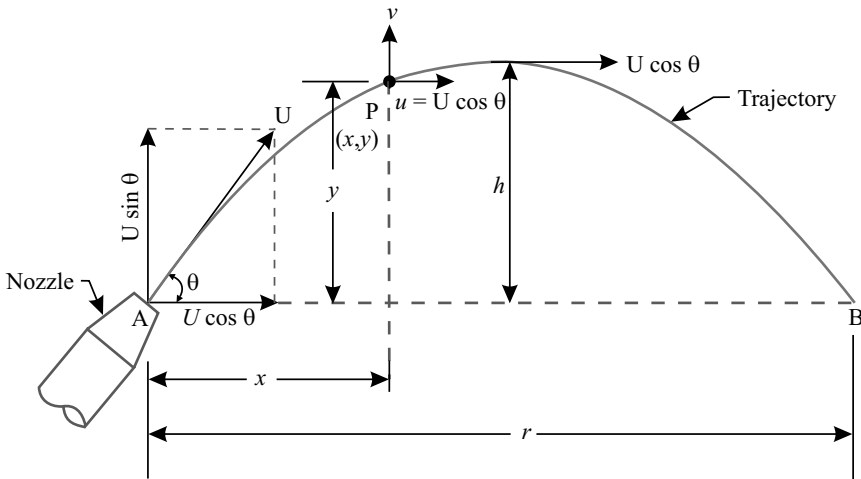


Fig. 6.41

Let,

u = Velocity of the jet at point P in X -direction,

v = Velocity of the jet at point P in Y -direction, and

t = Time taken by a liquid particle to reach from A to P .

Then, $x = u \times t = U \cos \theta \times t$ (where, $u = U \cos \theta$) ... (i)

$$y = U \sin \theta \times t - \frac{1}{2} g t^2 \quad \dots (ii)$$

(It may be noted that horizontal component of velocity U is $U \cos \theta$ which *remains constant* whereas the vertical component $U \sin \theta$ is *affected by gravity*.)

From eqn. (i) we have, $t = \frac{x}{U \cos \theta}$

Substituting the value of t in eqn. (ii), we get:

$$\begin{aligned} y &= U \sin \theta \times \frac{x}{U \cos \theta} - \frac{1}{2} g \times \frac{x^2}{U^2 \cos^2 \theta} \\ &= x \tan \theta - \frac{g x^2}{2 U^2 \cos^2 \theta} \\ y &= x \tan \theta - \frac{g x^2 \sec^2 \theta}{2 U^2} \quad \left(\because \frac{1}{\cos^2 \theta} = \sec^2 \theta \right) \quad \dots (6.13) \end{aligned}$$

This is the equation of a *parabola*.

(i) Maximum height attained by the jet, h :

Using the relation:

$$V_2^2 - V_1^2 = -2gh \quad (-ve \text{ sign is used as the particle is moving upward})$$

where,

V_1 = Initial vertical component = $U \sin \theta$, and

$V_2 = 0$ at the highest point.

$$\therefore 0 - (U \sin \theta)^2 = -2gh$$

or

$$h = \frac{U^2 \sin^2 \theta}{2g} \quad \dots (6.14)$$

(ii) Time of flight, T :

Time of flight is the time taken by the fluid particle in reaching from A to B (Fig. 6.41). From eqn. (ii), we have:

$$y = U \sin \theta \times t - \frac{1}{2} g t^2$$

When the particle reaches the point B , $y = 0$, $t = T$

Putting these values in the above equation, we get:

$$0 = U \sin \theta \times T - \frac{1}{2} g \times T^2$$

or,

$$T = \frac{2U \sin \theta}{g} \quad \dots (6.15)$$

$$\text{Time taken to reach the highest point, } T' = \frac{T}{2} = \frac{2U \sin \theta}{2g} = \frac{U \sin \theta}{g}$$

$$\text{i.e., } T' = \frac{U \sin \theta}{g} \quad \dots (6.16)$$

(iii) Horizontal range of the jet, r :

The range (r) of the jet is the total *horizontal distance travelled by the fluid particle*.

Then r ; (i.e., distance AB) = Velocity component in direction $\times$ time taken by the particle to reach from A to B

$$\begin{aligned}
 &= U \cos \theta \times T = U \cos \theta \times \frac{2U \sin \theta}{g} \\
 &= \frac{U^2 \times 2 \sin \theta \times \cos \theta}{g} = \frac{U^2 \sin 2\theta}{g}
 \end{aligned}$$

$$i.e., \quad r = \frac{U^2 \sin 2\theta}{g} \quad \dots(6.17)$$

The range will be maximum, when $\sin 2\theta = 1$

$$i.e., \quad 2\theta = 90^\circ \text{ or } \theta = 45^\circ$$

$$\text{Then maximum range, } r_{\max} = \frac{U^2 \sin(2 \times 45^\circ)}{g} = \frac{U^2}{g}$$

$$i.e., \quad r_{\max} = \frac{U^2}{g}$$

Example 6.45. A nozzle is situated at a distance of 1.2 m above the ground level and is inclined at 60° to the horizontal. The diameter of the nozzle is 40 mm and the jet of water from the nozzle strikes the ground at a horizontal distance of 5 m. Find the flow rate.

Solution. Distance of nozzle above the ground = 1.2 m

Angle of inclination, $\theta = 60^\circ$

Diameter of the nozzle,

$$d = 40 \text{ mm} = 0.04 \text{ m}$$

$\therefore$ Area of nozzle,

$$A = \frac{\pi}{4} \times 0.04^2 = 0.001256 \text{ m}^2$$

The horizontal distance, $x = 5 \text{ m}$

The co-ordinates of the point M , which on the centre line of the jet of water and is situated on the ground, with respect to L (origin) are: $x = 5 \text{ m}$, $y = -1.2 \text{ m}$ ($\because$ Point M is vertically down by 1.2 m)

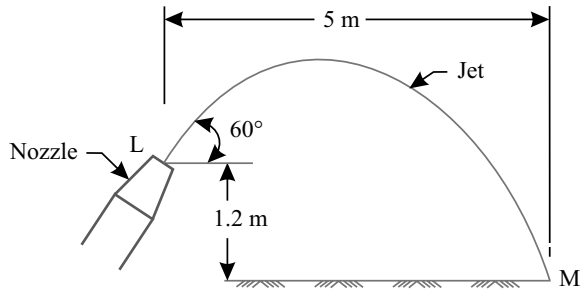


Fig. 6.42

The equation of the jet is given by :

$$y = x \tan \theta - \frac{gx^2}{2U^2 \cos^2 \theta} \quad \dots(i)$$

where,

U = Velocity of the jet.

Flow rate, Q :

The eqn. (i) can be written as:

$$\begin{aligned}
 y &= x \tan \theta - \frac{gx^2}{2U^2} \sec^2 \theta \\
 -1.2 &= 5 \tan 60^\circ - \frac{9.81 \times 5^2}{2U^2} (1 + \tan^2 60^\circ) \quad (\because \sec^2 \theta = 1 + \tan^2 \theta) \\
 -1.2 &= 5 \times 1.732 - \frac{122.62}{U^2} (1 + 3) \\
 -1.2 &= 8.66 - \frac{498.48}{U^2}
 \end{aligned}$$

or,

$$U^2 = \frac{498.48}{(8.66 + 1.2)} = 49.74 \quad \text{or} \quad U = 7.05 \text{ m/s}$$

Hence, flow rate, $Q = A \times U = 0.001256 \times 7.05 = \mathbf{0.00885 \text{ m}^3/\text{s}}$ (Ans.)

Example 6.46. It is required to place an orifice in the side of a tank at such an elevation that the jet will attain a maximum horizontal distance from the tank at the level of its base. What is the proper distance from the orifice to the free surface when the depth of liquid in the tank is maintained at 1.2 m?

Solution. Depth of liquid in the tank = 1.2 m

$$x = \sqrt{2gh} \times t \quad \dots(i)$$

$$\text{and, } y = -\frac{1}{2}gt^2 \quad \dots(ii)$$

Eliminating t , we get:

$$\begin{aligned} y &= -\frac{1}{2}g \times \left(\frac{x}{\sqrt{2gh}} \right)^2 \\ &= -\frac{1}{2} \times g \times \frac{x^2}{2gh} \end{aligned}$$

$$\text{or, } y = -\frac{x^2}{4h}$$

$$\text{Also, } 1.2 = h + y \text{ or } y = 1.2 - h$$

$$\therefore (1.2 - h) = -\frac{x^2}{4h}$$

$$\text{or, } x^2 = -4h(1.2 - h) = -4.8h + 4h^2$$

For horizontal distance x to be maximum $\frac{dx}{dh} = 0$

$$\therefore 2x \frac{dx}{dh} = -4.8 + 8h = 0 \text{ or } h = 0.6 \text{ m}$$

Thus, the orifice should be located at a distance of **0.6 m below the free surface.** (Ans.)

Example 6.47. Ten nozzles each 25 mm in diameter, all inclined at an angle of 45° with the horizontal are used in an ornamental fountain. The jet issuing from the nozzle falls into a basin at a point 1.5 m vertically beneath the nozzle and 4.5 m horizontally from it. The velocity co-efficient of nozzle is 0.97. Determine:

- (i) Pressure head at the nozzle, and
- (ii) Total discharge from the nozzles.

Solution. Diameter of each nozzle,

$$d = 25 \text{ mm} = 0.025 \text{ m}$$

Angle of inclination, $\theta = 45^\circ$

Velocity co-efficient of nozzle,

$$C_v = .97$$

(i) Pressure head at the nozzle, H :

Refer to Fig. 6.44.

Horizontal distance traversed,

$$x = 4.5 \text{ m}$$

Vertical distance traversed,

$$y = -1.5 \text{ m}$$

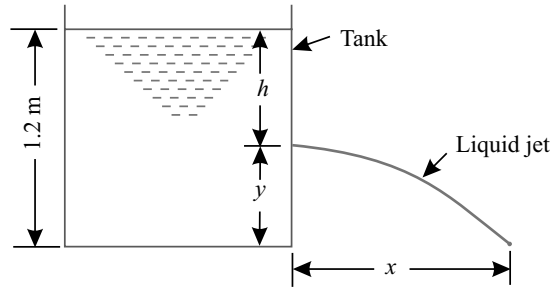


Fig. 6.43

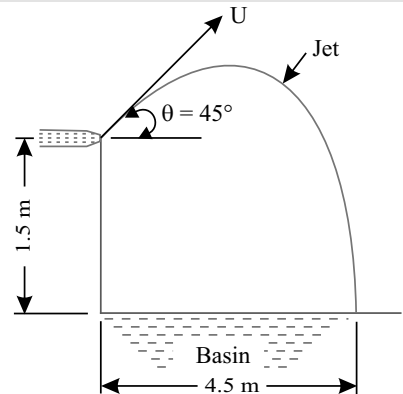


Fig. 6.44

For horizontal motion:

$$x = U \cos \theta \times t$$

or, $4.5 = U \cos 45^\circ \times t = 0.707 Ut$... (i)

For vertical motion:

$$y = U \sin \theta \cdot t - \frac{1}{2} gt^2$$

or, $-1.5 = 0.707 Ut - \frac{1}{2} gt^2$... (ii)

From (i) and (ii), we get:

$$-1.5 = 4.5 - \frac{1}{2} \times 9.81 \times t^2 \quad \text{or} \quad 4.905 t^2 = 6$$

or, $t = \left(\frac{6}{4.905} \right)^{1/2} = 1.106 \text{ s}$

From eqn. (i), we have:

$$U = \frac{4.5}{0.707 \times 1.106} = 5.75 \text{ m/s}$$

Also, $U = C_v \times \sqrt{2gh} \quad \text{or} \quad 5.75 = 0.97 \times \sqrt{2 \times 9.81 \times H}$

or, $5.93 = \sqrt{2 \times 9.81 \times H} \quad \text{or} \quad H = \frac{5.93^2}{2 \times 9.81} = 1.79 \text{ m}$

i.e., Pressure head at the nozzle = **1.79 m (Ans.)**

(ii) Total discharge from the nozzles:

Total discharge, $Q = (\pi/4 \times d^2 \times U) \times \text{number of nozzles}$
 $= \pi/4 \times 0.025^2 \times 5.75 \times 10 = 0.0282 \text{ m}^3/\text{s}$

i.e., Total discharge through nozzles = **0.0282 m³/s. (Ans.)**

Example 6.48. A fireman must reach a window 40 m above the ground with a water jet, issued from a nozzle 30 mm in diameter and discharging 30 kg/s. Assuming the nozzle height to be 2 m above the ground, determine the greatest horizontal distance from the building where the fireman can stand and still reach the jet into the window. (MDU, Haryana)

Solution Given: $D = 30 \text{ mm} = 0.03 \text{ m}$; $m = 30 \text{ kg/s}$

Refer to Fig. 6.45.

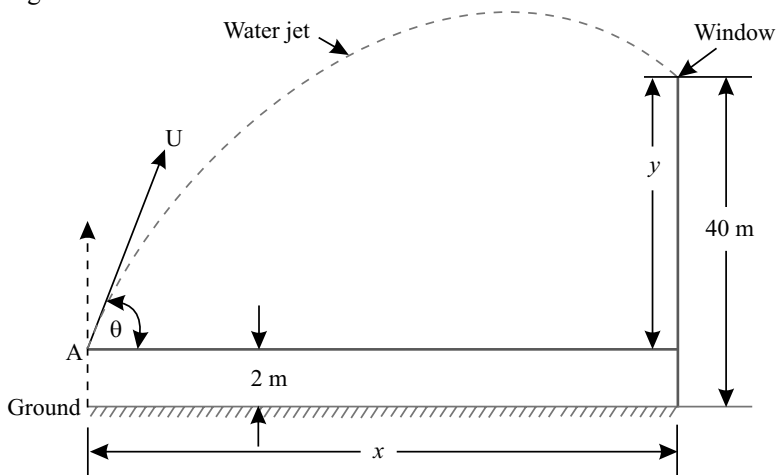


Fig. 6.45

Greatest horizontal distance, x :

Now, $m = \rho A U$ (where, U = velocity of water jet)

$$\therefore U = \frac{m}{\rho A} = \frac{30}{1000 \times \frac{\pi}{4} \times (0.03)^2} = 42.44 \text{ m/s}$$

Let, θ = Angle of inclination of the nozzle.

Then, $x = U \cos \theta \times t$ or $t = \frac{x}{U \cos \theta}$

$$\begin{aligned} y &= U \sin \theta \times t - \frac{1}{2} g t^2 \\ &= U \sin \theta \times \frac{x}{U \cos \theta} - \frac{1}{2} g \left(\frac{x}{U \cos \theta} \right)^2 \\ &= x \tan \theta - \frac{1}{2} g \frac{x^2}{U^2 \cos^2 \theta} = x \tan \theta - \frac{g x^2}{2 U^2} \sec^2 \theta \end{aligned}$$

$$\text{or, } x \tan \theta - \frac{g x^2}{2 U^2} \sec^2 \theta - y = 0 \quad \dots(i)$$

The maximum value of x is obtained by differentiating (i) w.r.t. θ , and putting $\frac{dx}{d\theta} = 0$

$$\therefore \left[x \sec^2 \theta + \tan \theta \times \frac{dx}{d\theta} \right] - \frac{g}{2 U^2} \left[x^2 \times 2 \sec \theta \times \sec \theta \tan \theta + \sec^2 \theta \times 2x \times \frac{dx}{d\theta} \right] = 0$$

Putting $\frac{dx}{d\theta} = 0$, we get:

$$x \sec^2 \theta - \frac{g}{2 U^2} (2 x^2 \sec^2 \theta \cdot \tan \theta) = 0$$

$$\text{or, } x \sec^2 \theta = \frac{g}{2 U^2} \times 2 x^2 \sec^2 \theta \cdot \tan \theta$$

$$\text{or, } x = \frac{U^2}{g \tan \theta}$$

$$\text{Also, } y = 40 - 2 = 38 \text{ m}$$

Substituting for x and y in eqn. (i), we get:

$$\begin{aligned} \frac{U^2}{g \tan \theta} \times \tan \theta - \frac{g}{2 U^2} \times \left(\frac{U^2}{g \tan \theta} \right) \sec^2 \theta - 38 &= 0 \\ \frac{U^2}{g} - \frac{U^2}{2 g \sin^2 \theta} - 38 &= 0 \end{aligned}$$

Substituting for $U = 42.44 \text{ m/s}$, we get:

$$\begin{aligned} \frac{(42.44)^2}{9.81} - \frac{(42.44)^2}{2 \times 9.81 \times \sin^2 \theta} - 38 &= 0 \\ 183.6 - \frac{91.8}{\sin^2 \theta} - 38 &= 0 \\ \sin^2 \theta &= \frac{91.8}{(183.6 - 38)} = 0.6305 \end{aligned}$$

or,

$$\sin \theta = 0.794 \quad \text{or} \quad \theta = \sin^{-1}(0.794) = 52.56^\circ$$

Hence,

$$x = \frac{U^2}{g \tan \theta} = \frac{(42.44)^2}{9.81 \times \tan(52.56^\circ)} = \mathbf{140.58 \text{ m (Ans.)}}$$

Example 6.49. The nozzle shown in Fig 6.46. has a jet diameter of 25 mm. The pressures on the water surface on the two sides of the arrangement are $p_1 = 170 \text{ kN/m}^2$ (gauge) and $p_2 = 300 \text{ mm of Hg}$. Determine:

(i) The discharge through the nozzle;

(ii) The maximum height of the free jet above the nozzle.

[IIT Delhi]

Solution. Diameter of the jet = 25 mm = 0.025 m

Pressure, $p_1 = 170 \text{ kN/m}^2$

Pressure, $p_2 = 300 \text{ mm or } 0.3 \text{ m of Hg}$

$= 0.3 \times 13.6 = 4.08 \text{ m of water.}$

(i) **Discharge through the nozzle, Q:**

Applying Bernoulli's equation to 1 (water surface) and 2 (the jet as it emerges from the nozzle), we get:

$$\frac{p_1}{\rho} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho} + \frac{V_2^2}{2g} + z_2$$

$$\frac{170}{9.81} + 0 + 3 = 4.08 + \frac{V_2^2}{2g} + 0$$

$$17.33 + 3 = 4.08 + \frac{V_2^2}{2g}$$

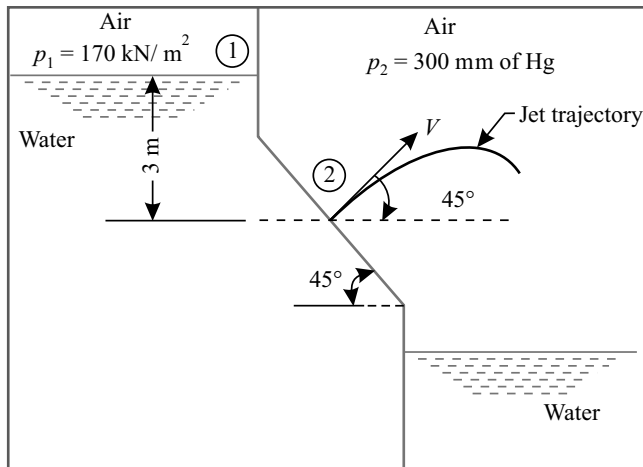


Fig. 6.46

$$\text{or,} \quad V_2^2 = [(17.33 + 3) - 4.08] \times 2g = 16.25 \times 2 \times 9.81$$

$$\therefore V_2 = 17.85 \text{ m/s} \quad (V_2 = V)$$

Hence,

$$Q = A \times V_2$$

$$= (\pi/4) \times 0.025^2 \times 17.85 = 0.00876 \text{ m}^3/\text{s} \text{ or } \mathbf{8.76 \text{ litres/sec. (Ans.)}}$$

(ii) **Maximum height of the free jet above the nozzle:**

$$\text{Vertical component of the jet velocity} = V \sin 45^\circ = 17.85 \sin 45^\circ = 12.62 \text{ m/s}$$

∴ Maximum height to which jet will rise,

$$h = \frac{12.62^2}{2 \times 9.81} = 8.12 \text{ m (Ans.)}$$

Example 6.50. A vertical jet of water 75 mm in diameter leaving the nozzle with a 9.2 m/s velocity strikes a horizontal and movable disc weighing 170 N (Fig.6.47). The jet is then deflected horizontally. Determine the vertical distance y above the nozzle tip at which the disc will be held in equilibrium. [Roorkee University]

Solution. Diameter of the jet, $d = 75 \text{ mm} = 0.075 \text{ m}$

Velocity of the jet at nozzle exit, $V = 9.2 \text{ m/s}$

Weight of the disc, $W = 170 \text{ N}$

Vertical Distance, y :

Let, v = Jet velocity at an elevation y .

Applying Bernoulli's theorem between the jet at nozzle exit and the jet at an elevation y , we get:

$$\frac{V^2}{2g} = \frac{v^2}{2g} + y$$

$$\text{or, } V^2 = v^2 + 2gy \quad \dots(i)$$

Also the momentum equation is written as

$$\frac{wQ}{g} \cdot v = 170$$

$$\text{where, } Q = (\pi/4) \times d^2 \times V = (\pi/4) \times 0.075^2 \times 9.2 \\ = 0.0406 \text{ m}^3/\text{s, and}$$

$$w = 9810 \text{ N/m}^3$$

$$\therefore \frac{9810 \times 0.0406 \times v}{9.81} = 170$$

$$\text{or, } v = \frac{170 \times 9.81}{9810 \times 0.0406} = 4.187 \text{ m/s}$$

Substituting this value of v in (i), we get:

$$9.2^2 = (4.187)^2 + 2 \times 9.81 \times y$$

$$\text{or, } 84.64 = 17.53 + 19.62 y$$

$$\text{or, } y = \frac{84.64 - 17.53}{19.62} = 3.43 \text{ m (Ans.)}$$

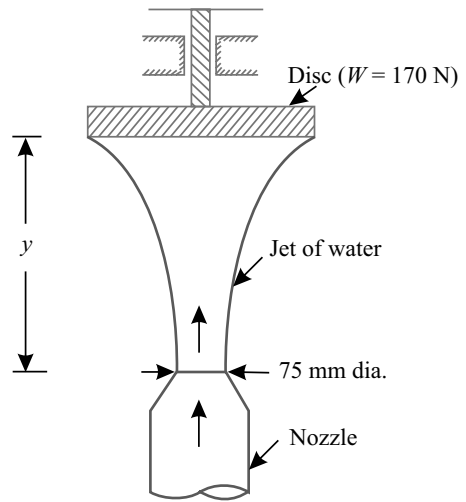


Fig. 6.47

6.8. IMPULSE-MOMENTUM EQUATION

The **impulse-momentum equation** is one of the basic tools (other being continuity and Bernoulli's equations) for the solution of flow problems. Its application leads to the solution of problems in fluid mechanics which cannot be solved by energy principles alone. Sometimes it is used in conjunction with the energy equation to obtain complete solution of engineering problems.

The momentum equation is based on the law of conservation of momentum or momentum principle which states as follows:

"The net force acting on a mass of fluid is equal to change in momentum of flow per unit time in that direction".

As per Newton's second law of motion,

$$F = ma$$

where,

m = Mass of fluid,

F = Force acting on the fluid, and

a = Acceleration (acting in the same direction as F).

But acceleration,

$$a = \frac{dv}{dt}$$

$\therefore$

$$F = m \cdot \frac{dv}{dt} = \frac{d(mv)}{dt} \quad \dots(6.18)$$

(‘ m ’ is taken inside the differential, being constant)

This equation is known as **momentum principle**. It can also be written as:

$$F \cdot dt = d(mv) \quad \dots(6.19)$$

This equation is known as **Impulse-momentum equation**. It may be stated as follows:

“The impulse of a force F acting on a fluid mass ‘ m ’ in a short interval of time dt is equal to the change of momentum $d(mv)$ in direction of force”.

The impulse-momentum equations are often called simply *momentum equations*.

Applications of impulse-momentum equation:

The impulse-momentum equation is used in the following types of problems:

1. To determine the resultant force acting on the boundary of flow passage by a stream of fluid as the stream changes its direction, magnitude or both. Problems of this type are:

(i) Pipe bends, (ii) Reducers, (iii) Moving vanes, (iv) Jet propulsion, etc.

2. To determine the characteristic of flow when there is an abrupt change of flow section. Problems of this type are:

(i) Sudden enlargement in a pipe, (ii) Hydraulic jump in a channel, etc.

Steady flow momentum equation:

The entire flow space may be considered to be made up of innumerable stream tubes. Let us consider one such stream tube lying in the X - Y plane (Fig 6.48) and having steady flow of fluid. Flow can be assumed to be uniform and normal to the inlet and outlet areas.

Let, V_1, ρ_1 = Average velocity and density (of fluid mass) respectively at the entrance, and

V_2, ρ_2 = Average velocity and density respectively at the exit.

Further let the mass of fluid in the region 1 2 3 4 shifts to new position 1' 2' 3' 4' due to the effect of external forces on the stream after a short interval. Due to gradual increase in the flow area in the direction of flow, velocity of fluid mass and hence the momentum is gradually reduced. Since the area 1' 2' 3' 4 is common to both the regions 1 2 3 4 and 1' 2' 3' 4', therefore, it will not experience any change in momentum. Obviously, then the changes in momentum of the fluid masses in the sections 1 2 2' 1' and 4 3 3' 4' will have to be considered.

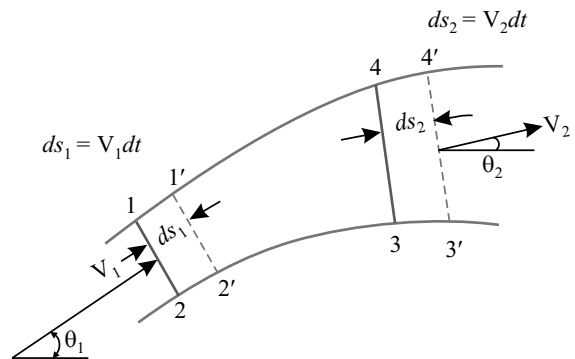


Fig. 6.48

According to the *principle of mass conservation*,

Fluid mass with the region 1 2 2' 1' = Fluid mass within the region 4 3 3' 4'

$$\rho_1 A_1 ds_1 = \rho_2 A_2 ds_2 \quad \dots(6.20)$$

$\therefore$ Momentum of fluid mass contained in the region 1 2 2' 1'

$$= (\rho_1 A_1 ds_1) V_1 = (\rho_1 A_1 V_1 \cdot dt) V_1$$

Momentum of fluid mass contained in the region 4 3 3' 4'

$$= (\rho_2 A_2 ds_2) V_2 = (\rho_2 A_2 V_2 \cdot dt) V_2$$

$$\therefore \text{Change in momentum} = (\rho_2 A_2 V_2 \cdot dt) V_2 - (\rho_1 A_1 V_1 \cdot dt) V_1$$

But, $\rho_1 = \rho_2 = \rho$...for steady incompressible flow

and, $A_1 V_1 = A_2 V_2 = Q$...from continuity considerations

$$\therefore \text{Change in momentum} = \rho Q (V_2 - V_1) dt$$

Using impulse-momentum principle, we have:

$$F dt = \rho Q (V_2 - V_1) dt \quad \dots(6.21)$$

$$\text{or,} \quad F = \frac{wQ}{g} (V_2 - V_1) \quad \dots(6.22)$$

The quantity $\frac{wQ}{g} = \rho Q$ is the mass flow per second and is called **mass flux**.

Resolving V_1 and V_2 along X -axis and Y -axis, we get:

Components along X -axis: $V_1 \cos \theta_1$ and $V_2 \cos \theta_2$

Components along Y -axis: $V_1 \sin \theta_1$ and $V_2 \sin \theta_2$

(where, θ_1 and θ_2 are the inclinations with the horizontal of the centre line of the pipe at 1-2 and 3-4).

$\therefore$ Components of force F along X -axis and Y -axis are:

$$\begin{aligned} F_x &= \frac{wQ}{g} (V_2 \cos \theta_2 - V_1 \cos \theta_1) \\ F_y &= \frac{wQ}{g} (V_2 \sin \theta_2 - V_1 \sin \theta_1) \end{aligned} \quad \dots(6.23)$$

Eqn. (6.23) represents the components of the force exerted by the *pipe bend on the fluid mass*. Usually, we are interested in the forces by the fluid on the pipe bend. Since action and reaction are equal and opposite (Newton's third law of motion), the fluid mass would exert the same force on the pipe bend but in *opposite direction* and as such the force components exerted by the fluid on the pipe bend are given as follows:

$$\left. \begin{aligned} F_x &= \frac{wQ}{g} (V_1 \cos \theta_1 - V_2 \cos \theta_2) \\ F_y &= \frac{wQ}{g} (V_1 \sin \theta_1 - V_2 \sin \theta_2) \end{aligned} \right\} \quad \dots(6.24)$$

Since the *dynamic* forces (eqn. 6.23) must be supplemented by the *static pressure forces* acting over the inlet and outlet sections, therefore, we have:

$$\left. \begin{aligned} F_x &= \frac{wQ}{g} (V_1 \cos \theta_1 - V_2 \cos \theta_2) + p_1 A_1 \cos \theta_1 - p_2 A_2 \cos \theta_2 \\ F_y &= \frac{wQ}{g} (V_1 \sin \theta_1 - V_2 \sin \theta_2) + p_1 A_1 \sin \theta_1 - p_2 A_2 \sin \theta_2 \end{aligned} \right\} \quad \dots(6.25)$$

The magnitude of the resultant force acting on the pipe bend,

$$F_R = \sqrt{F_x^2 + F_y^2} \quad \dots(6.26)$$

and, the direction of the resultant force with X -axis,

$$\theta = \tan^{-1} \left(\frac{F_y}{F_x} \right) \quad \dots[6.26 (a)]$$

Example 6.51. In a 45° bend a rectangular air duct of 1 m^2 cross-sectional area is gradually reduced to 0.5 m^2 area. Find the magnitude and direction of force required to hold the duct in position if the velocity of flow at 1 m^2 section is 10 m/s , and pressure is 30 kN/m^2 .

Take the specific weight of air as 0.0116 kN/m^3 .

[Anna University]

Solution. Refer to Fig. 6.49

Area at section, '1' = 1 m^2 ; Area at section '2' = 0.5 m^2

Velocity at section '1', $V_1 = 10 \text{ m/s}$

Pressure at section '1', $p_1 = 30 \text{ kN/m}^2$

Sp. weight of air, $w = 0.0116 \text{ kN/m}^3$

As per continuity equation,

$$A_1 V_1 = A_2 V_2$$

$$\therefore V_2 = \frac{A_1 V_1}{A_2} = \frac{1 \times 10}{0.5} = 20 \text{ m/s}$$

$$\text{Discharge, } Q = A_1 V_1 = 1 \times 10 = 10 \text{ m}^3/\text{s}$$

Applying Bernoulli's equation at sections '1' and '2', we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

But,

$$z_1 = z_2$$

$$\therefore \frac{p_1}{w} + \frac{V_1^2}{2g} = \frac{p_2}{w} + \frac{V_2^2}{2g}$$

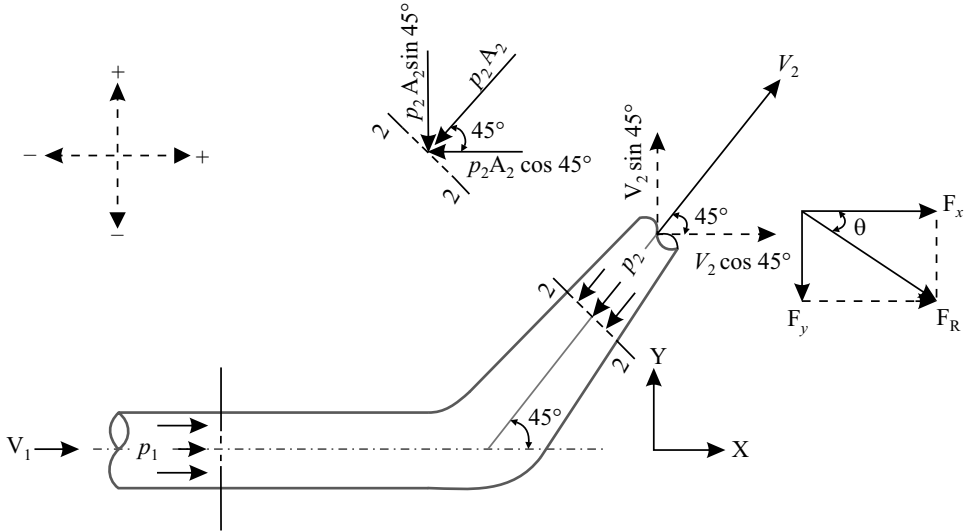


Fig. 6.49

$$\frac{30}{0.0116} + \frac{10^2}{2 \times 9.81} = \frac{p_2}{w} + \frac{20^2}{2 \times 9.81}$$

$$\text{or, } 2586 + 5.1 = \frac{p_2}{w} + 20.4$$

$$\text{or, } \frac{p_2}{w} = 2586 + 5.1 - 20.4 = 2570.7$$

or, $p_2 = 2570.7 \times 0.0116 = 29.82 \text{ kN/m}^2$

Magnitude and direction of force (resultant) F_R :

Force along X-axis:

$$F_x = \frac{wQ}{g} (V_{1x} - V_{2x}) + (p_1 A_1)_x + (p_2 A_2)_x$$

where, $V_{1x} = 10 \text{ m/s}$; $V_{2x} = V_2 \cos 45^\circ = 20 \times 0.707 = 14.14 \text{ m/s}$
 $(p_1 A_1)_x = p_1 A_1 = 30 \times 1 = 30 \text{ kN}$; $(p_2 A_2)_x = -p_2 A_2 \cos 45^\circ = -29.82 \times 0.5 \times 0.707 = -10.54 \text{ kN}$

$$\therefore F_x = \frac{0.0116}{9.81} \times 10 (10 - 14.14) + 30 - 10.54 = 19.41 \text{ kN } (\rightarrow)$$

Force along Y-axis:

$$F_y = \frac{wQ}{g} (V_{1y} - V_{2y}) + (p_1 A_1)_y + (p_2 A_2)_y$$

where, $V_{1y} = 0$; $V_{2y} = V_2 \sin 45^\circ = 20 \times 0.707 = 14.14 \text{ m/s}$
 $(p_1 A_1)_y = 0$; $(p_2 A_2)_y = -p_2 A_2 \sin 45^\circ = -29.82 \times 0.5 \times 0.707 = -10.54 \text{ kN}$

$$\therefore F_y = \frac{0.0116 \times 10}{9.81} (0 - 14.14) + 0 - 10.54 = -10.71 \text{ kN } (\downarrow)$$

$$\therefore \text{Resultant force, } F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{(19.41)^2 + (10.71)^2} = \mathbf{22.17 \text{ kN (Ans.)}}$$

The direction of F_R with X-axis is given as:

$$\tan \theta = \frac{F_y}{F_x} = \frac{10.71}{19.41} = 0.5518$$

$$\text{or, } \theta = \tan^{-1} 0.5518 = 28.88^\circ \text{ or } \mathbf{28^\circ 53' \text{ (Ans.)}}$$

Example 6.52. 250 litres/sec. of water is flowing in a pipe having a diameter of 300 mm. If the pipe is bent by 135° , find the magnitude and direction of the resultant force on the bend. The pressure of the water flowing is 400 kN/m^2 . Take specific weight of water as 9.81 kN/m^3 .

[Delhi University]

Solution. Diameter of the bend at inlet, $D_1 = 300 \text{ mm} = 0.3 \text{ m}$

Diameter of the bend at outlet, $D_2 = 300 \text{ mm} = 0.3 \text{ m}$

$$\therefore \text{Area, } A_1 = A_2 = (\pi/4) \times 0.3^2 = 0.07068 \text{ m}^2$$

$$\text{Discharge, } Q = 250 \text{ litres/sec.} = 0.25 \text{ m}^3/\text{s.}$$

$$\text{Pressure, } p_1 = p_2 = 400 \text{ kN/m}^2$$

$$\text{Velocity at section 1-1, } V_1 = \frac{Q}{A_1} = \frac{0.25}{0.07068} = 3.54 \text{ m/s}$$

$$\text{Velocity at section 2-2, } V_2 = V_1 = 3.54 \text{ m/s} \quad (\because A_1 = A_2)$$

Force along X-axis:

$$\begin{aligned} F_x &= \frac{wQ}{g} [V_1 - (-V_2 \cos 45^\circ)] + p_1 A_1 + p_2 A_2 \cos 45^\circ \\ &= \frac{9.81 \times 0.25}{9.81} [3.54 - (-3.54 \times 0.707)] \\ &\quad + (400 \times 0.07068) + (400 \times 0.07068 \times 0.707) \\ &= 0.25 \times (3.54 + 3.54 \times 0.707) + 28.27 + 19.98 \\ &= \mathbf{49.76 \text{ kN } (\rightarrow)} \end{aligned}$$

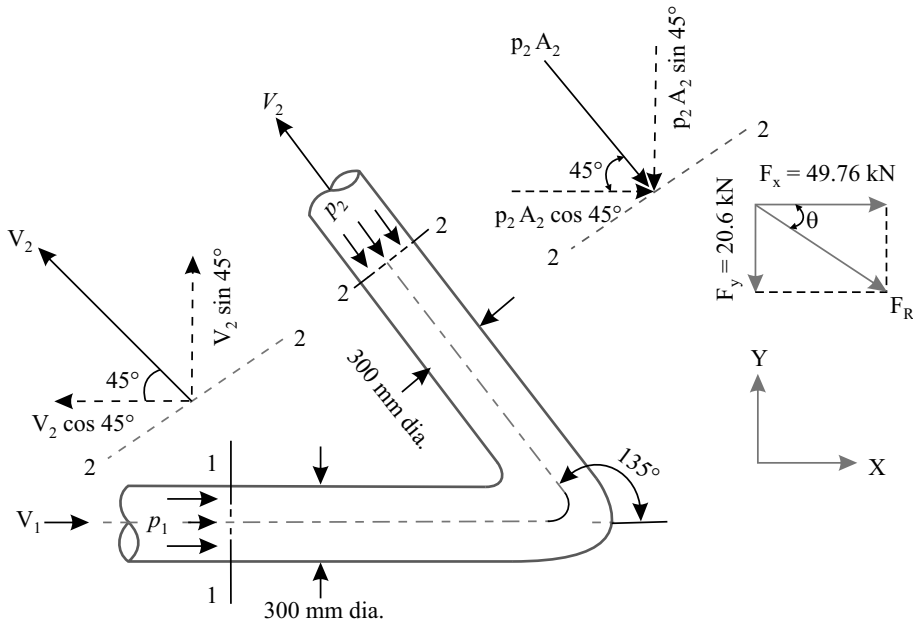


Fig. 6.50

Force along Y-axis:

$$\begin{aligned}
 F_y &= \frac{wQ}{g} [0 - V_2 \sin 45^\circ] - p_2 A_2 \sin 45^\circ \\
 &= \frac{9.81 \times 0.25}{9.81} (0 - 3.54 \times 0.707) - 400 \times 0.07068 \times 0.707 \\
 &= -0.625 - 19.98 = -20.6 \text{ kN} (\downarrow)
 \end{aligned}$$

The magnitude of the resultant force,

$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{49.76^2 + 20.6^2} = 53.85 \text{ kN (Ans.)}$$

The direction of F_R with X- axis is given as:

$$\tan \theta = \frac{F_y}{F_x} = \frac{20.6}{49.76} = 0.414$$

$$\therefore \theta = \tan^{-1} 0.414 = 22.5^\circ \text{ (Ans.)}$$

Example 6.53. 360 litres per second of water is flowing in a pipe. The pipe is bent by 120° . The pipe bend measures $360 \text{ mm} \times 240 \text{ mm}$ and volume of the bend is 0.14 m^3 . The pressure at the entrance is 73 kN/m^2 and the exit is 2.4 m above the entrance section.

Find the force exerted on the bend.

Solution. Discharge through the pipe, $Q = 360 \text{ litres/sec.} = 0.36 \text{ m}^3/\text{s}$

Volume of bend = 0.14 m^3

Diameter of the bend at 1-1, $D_1 = 360 \text{ mm} = 0.36 \text{ m}$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.36^2 = 0.1018 \text{ m}^2$$

Diameter of the bend at 2-2, $D_2 = 240 \text{ mm} = 0.24 \text{ m}$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.24^2 = 0.04524 \text{ m}^2$$

Velocity at section 1-1, $V_1 = \frac{Q}{A_1} = \frac{0.36}{0.1018} = 3.54 \text{ m/s}$

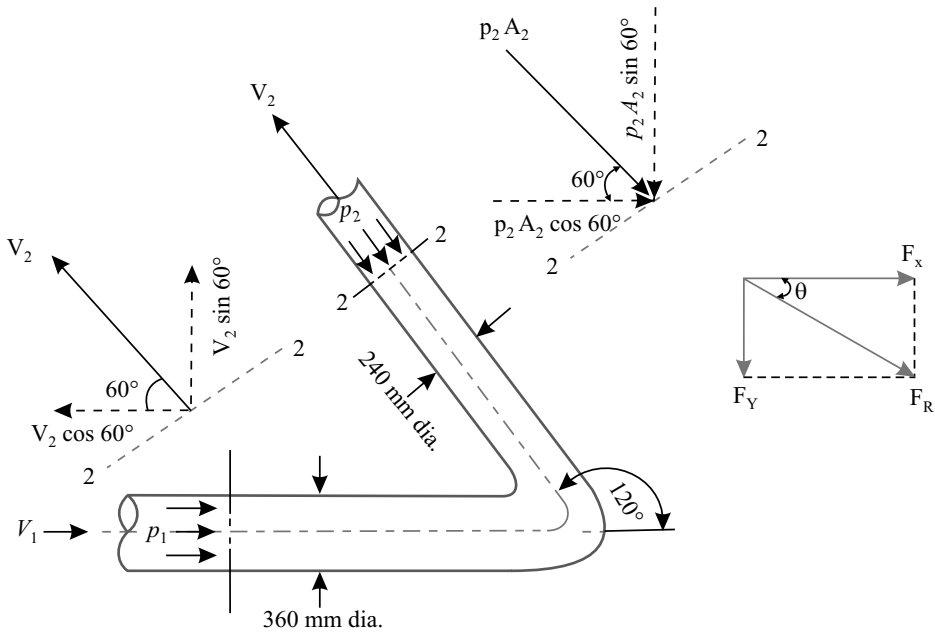


Fig. 6.51

Velocity at section 2-2, $V_2 = \frac{Q}{A_2} = \frac{0.36}{0.04524} = 7.96 \text{ m/s}$

Considering a horizontal line through the section 1-1 as datum for elevation head and applying Bernoulli's equation to the sections 1-1 and 2-2, we get:

$$\begin{aligned} \frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 &= \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 \\ \frac{72}{9.81} + \frac{3.54^2}{2 \times 9.81} + 0 &= \frac{p_2}{w} + \frac{7.96^2}{2 \times 9.81} + 2.4 \quad (\because p_1 = 72 \text{ kN/m}^2 \dots \text{Given}) \\ 7.34 + 0.64 &= \frac{p_2}{w} + 3.23 + 2.4 \end{aligned}$$

$$\therefore \frac{p_2}{w} = 2.35 \text{ or } p_2 = 2.35 \times 9.81 = 23.05 \text{ kN/m}^2$$

Force along the X-axis:

$$\begin{aligned} F_x &= \frac{wQ}{g} [V_1 - (-V_2 \cos 60^\circ)] + p_1 A_1 + p_2 A_2 \cos 60^\circ \\ &= \frac{9.81 \times 0.36}{9.81} [3.54 - (-7.96 \times 0.5)] + 72 \times 0.1018 + 23.05 \\ &\quad \times 0.04524 \times 0.5 \\ &= 0.36 (3.54 + 3.98) + 7.33 + 0.52 = 10.55 \text{ kN } (\rightarrow) \end{aligned}$$

Force along Y-axis:

$$F_y = \frac{wQ}{g} [0 - V_2 \sin 60^\circ] - p_2 A_2 \sin 60^\circ - \text{weight of water in the bend}$$

$$\begin{aligned}
 &= \frac{9.81 \times 0.36}{9.81} (0 - 7.96 \times 0.866) - 23.05 \times 0.04524 \times 0.866 - 0.14 \times 9.81 \\
 &= 0.36 (-6.89) - 0.9 - 1.37 = -4.75 \text{ kN } (\downarrow)
 \end{aligned}$$

Magnitude of the resultant force acting on the bend,

$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{10.55^2 + 4.75^2} = \mathbf{11.57 \text{ kN (Ans.)}}$$

Direction of the resultant force with the X-axis,

$$\tan \theta = \frac{F_y}{F_x} = \frac{4.75}{10.55} = 0.4502 \text{ or } \theta = \mathbf{24.24^\circ \text{ (Ans.)}}$$

Example 6.54. Fig. 6.52 shows a 90° reducer-bend through which water flows. The pressure at the inlet is 210 kN/m^2 (gauge) where the cross-sectional area is 0.01 m^2 . At the exit section, the area is 0.0025 m^2 and the velocity is 16 m/s . The pressure at the exit is atmospheric. Determine the magnitude and direction of the resultant force on the bend.

Solution. Area at section 1-1, $A_1 = 0.01 \text{ m}^2$

Area at section 2-2, $A_2 = 0.0025 \text{ m}^2$

Velocity at the exit, $V_2 = 16 \text{ m/s}$.

$$\text{Discharge, } Q = A_2 V_2 = 0.0025 \times 16 = 0.04 \text{ m}^3/\text{s}.$$

$$\therefore V_1 = \frac{Q}{A_1} = \frac{0.04}{0.01} = 4 \text{ m/s}$$

Assume the bend is horizontal and in XY plane.

Force along X-axis:

$$\begin{aligned}
 F_x &= \frac{wQ}{g} (V_1 - 0) + p_1 A_1 \\
 &= \frac{9.81 \times 0.04}{9.81} (4 - 0) + 210 \times 0.01 = 0.16 + 2.1 = 2.26 \text{ kN } (\rightarrow)
 \end{aligned}$$

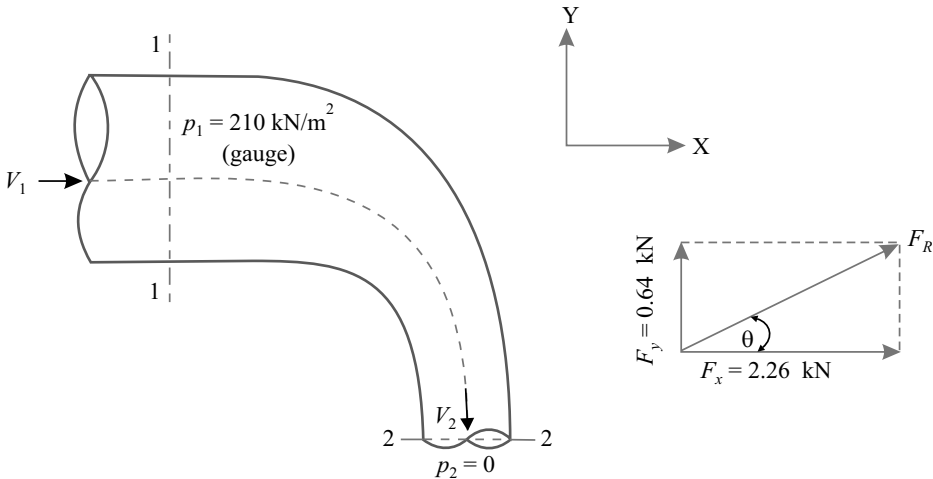


Fig. 6.52

Force along Y-axis:

$$F_y = \frac{wQ}{g} [0 - (-V_2)] + p_2 A_2$$

$$= \frac{9.81 \times 0.04}{9.81} (0 + 16) + 0 = 0.64 \text{ kN } (\uparrow)$$

Magnitude of the resultant force acting on the bend,

$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{2.26^2 + 0.64^2} = 2.35 \text{ kN (Ans.)}$$

Direction of the resultant force with the X-axis,

$$\tan \theta = \frac{F_y}{F_x} = \frac{0.64}{2.26} = 0.2832 \quad \therefore \theta = \tan^{-1} 0.2832 = 15.8^\circ$$

$$\therefore \theta = 15.8^\circ \text{ (Ans.)}$$

Example 6.55. Water enters a reducing pipe horizontally and comes out vertically in the downward direction. If the inlet velocity is 5 m/s and pressure is 80 kPa (gauge) and the diameters at the entrance and exit sections are 30 cm and 20 cm respectively, calculate the components of the reaction acting on the pipe. (RGPV, Bhopal)

Solution. Given: $D_1 = 30 \text{ cm} = 0.3 \text{ m}$; $D_2 = 20 \text{ cm} = 0.2 \text{ m}$; $V_1 = 5 \text{ m/s}$; $p_1 = 80 \text{ kPa} = 80 \text{ kN/m}^2$.

Components of the reaction acting on the pipe: Refer to Fig. 6.53.

From continuity equation, we have:

$$Q = A_1 V_1 = (\pi/4) \times 0.3^2 \times 5 = 0.3534 \text{ m}^3/\text{s}$$

$$A_1 V_1 = A_2 V_2 \text{ or } \frac{\pi}{4} \times 0.3^2 \times 5 = \frac{\pi}{4} \times 0.2^2 \times V_2$$

$$\therefore V_2 = 11.25 \text{ m/s}$$

Assume the bend is horizontal and in XY plane.

Applying Bernoulli's equation between sections (1) and (2), we have:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

$$\text{or, } \frac{p_2}{w} = \frac{p_1}{w} + \left(\frac{V_1^2 - V_2^2}{2g} \right)$$

$$(\because z_1 = z_2)$$

$$= \frac{80}{9.81} + \left(\frac{5^2 - 11.25^2}{2 \times 9.81} \right) = 2.978 \text{ m}$$

$$\text{or, } p_2 = 9.81 \times 2.978 = 29.22 \text{ kN/m}^2$$

Force along X-axis:

$$F_x = \frac{wQ}{g} (V_1 - 0) + p_1 A_1$$

$$= \frac{9.81 \times 0.3534}{9.81} (5 - 0) + 80 \times \frac{\pi}{4} \times 0.3^2 = 7.42 \text{ kN (Ans.)}$$

$$F_y = \frac{wQ}{g} [0 - (-V_2)] + p_2 A_2 = \frac{wQ V_2}{g} + p_2 A_2$$

$$= \frac{9.81 \times 0.3534 \times 11.25}{9.81} + 29.22 \times \frac{\pi}{4} \times 0.2^2 = 4.89 \text{ kN (Ans.)}$$

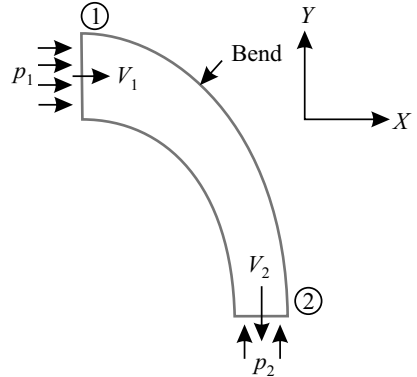


Fig. 6.53

Example 6.56. The angle of a reducing bend is 60° (that is deviation from initial direction to final direction). Its initial diameter is 300 mm and final diameter 150 mm and is fitted in a pipeline,

carrying a discharge of 360 litres/sec. The pressure at the commencement of the bend is 2.943 bar. The friction loss in the pipe bend may be assumed as 10 per cent of kinetic energy at exit of the bend. Determine the force exerted by the reducing bend. [UPSC Exams.]

Solution. Diameter at the inlet, $D_1 = 300 \text{ mm} = 0.3 \text{ m}$

$$\therefore \text{Area, } A_1 = (\pi/4) \times 0.3^2 = 0.07068 \text{ m}^2$$

Diameter at the outlet, $D_2 = 150 \text{ mm} = 0.15 \text{ m}$

$$\therefore \text{Area, } A_2 = (\pi/4) \times 0.15^2 = 0.01767 \text{ m}^2$$

Discharge through the bend, $Q = 3600 \text{ litres/sec.} = 0.36 \text{ m}^3/\text{s}$

Pressure at the inlet, $p_1 = 2.943 \text{ bar} = 294.3 \text{ kN/m}^2$

$$[\because 1 \text{ bar} = 10^5 \text{ N/m}^2 = 10^2 \text{ kN/m}^2]$$

$$\text{Velocity at the inlet, } V_1 = \frac{Q}{A_1} = \frac{0.36}{0.07068} = 5.09 \text{ m/s}$$

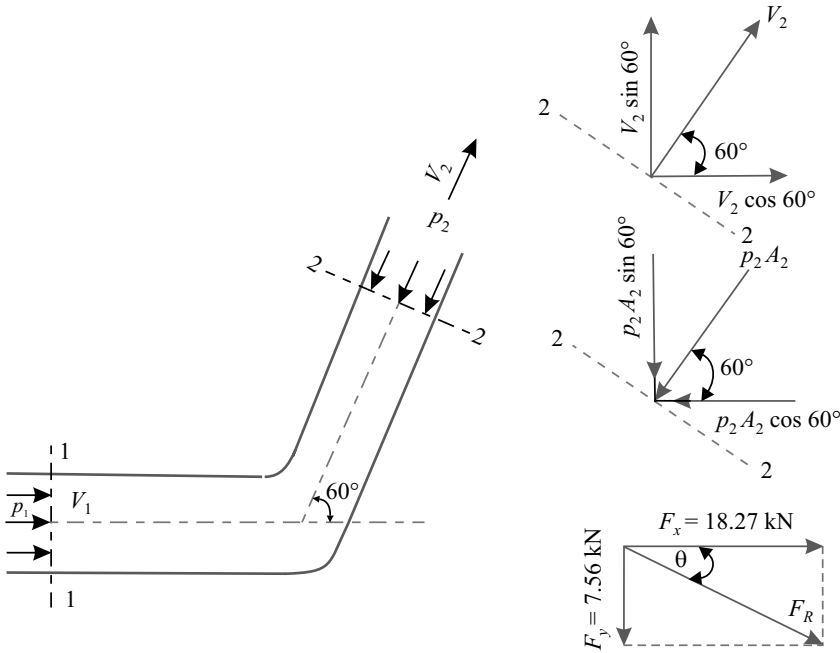


Fig. 6.54

$$\text{Velocity at the outlet, } V_2 = \frac{Q}{A_2} = \frac{0.36}{0.01767} = 20.37 \text{ m/s}$$

$$\text{Friction loss in the pipe} = 0.1 \times \frac{V_2^2}{2g}$$

...(Given)

Applying Bernoulli's equation at sections 1-1 and 2-2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + 0.1 \times \frac{V_2^2}{2g}$$

($\therefore z_1 = z_2$ since the bend lies in the horizontal plane)

$$\frac{294.3}{9.81} + \frac{5.09^2}{2 \times 9.81} = \frac{p_2}{w} + \frac{20.37^2}{2 \times 9.81} + 0.1 \times \frac{20.37^2}{2 \times 9.81}$$

$$30 + 1.32 = \frac{p_2}{w} + 21.15 + 2.11$$

$$\therefore \frac{p_2}{w} = 8.06 \quad \text{or} \quad p_2 = 9.81 \times 8.06 = 79.07 \text{ kN/m}^2$$

Force along X-axis:

$$\begin{aligned} F_x &= \frac{wQ}{g} [V_1 - V_2 \cos 60^\circ] + p_1 A_1 - p_2 A_2 \cos 60^\circ \\ &= \frac{9.81 \times 0.36}{9.81} (5.09 - 20.37 \times 0.5) + 294.3 \times 0.07068 \\ &\quad - 79.07 \times 0.01767 \times 0.5 \\ &= -1.8342 + 20.8 - 0.698 = 18.27 \text{ kN} (\rightarrow) \end{aligned}$$

Force along the Y-axis:

$$\begin{aligned} F_y &= \frac{wQ}{g} (0 - V_2 \sin 60^\circ) - p_2 A_2 \sin 60^\circ \\ &= \frac{9.81 \times 0.36}{9.81} (0 - 20.37 \sin 60^\circ) - 79.07 \times 0.01767 \times \sin 60^\circ \\ &= -6.35 - 1.21 = -7.56 \text{ kN or } 7.56 \text{ kN} (\downarrow) \end{aligned}$$

Magnitude of the resultant force acting on the bend,

$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{18.27^2 + 7.56^2} = \mathbf{19.77 \text{ kN}}$$

Direction of the resultant force with the X-axis,

$$\begin{aligned} \tan \theta &= \frac{F_y}{F_x} = \frac{7.56}{18.27} = 0.4138 \\ \theta &= \tan^{-1} 0.4138 = \mathbf{22.48^\circ} \end{aligned}$$

An equal and opposite force will be exerted by the reducing bend. (Ans.)

Example 6.57. A $0.4 \text{ m} \times 0.3 \text{ m}$, 90° vertical bend carries $0.5 \text{ m}^3/\text{s}$ oil of specific gravity 0.85 with a pressure of 118 kN/m^2 at inlet to the bend. The volume of the bend is 0.1 m^3 . Find the magnitude and direction of the force on the bend. Neglect friction and assume both inlet and outlet sections to be at same horizontal level. Also assume that water enters the bend at 45° to the horizontal. (PTU)

Solution. Given: $D_1 = 0.4 \text{ m}$, $\therefore A_1 = \frac{\pi}{4} \times 0.4^2 = 0.12566 \text{ m}^2$; $D_2 = 0.3 \text{ m}$;

$\therefore A_2 = \frac{\pi}{4} \times 0.3^2 = 0.07068 \text{ m}^2$; $Q = 0.5 \text{ m}^3/\text{s}$; $S_{oil} = 0.85$; $p_1 = 118 \text{ kN/m}^2$; Volume of bend = 0.1 m^3

Refer to Fig. 6.55.

$$V_1 = \frac{Q}{A_1} = \frac{0.5}{0.12566} = 3.98 \text{ m/s}$$

$$V_2 = \frac{Q}{A_2} = \frac{0.5}{0.07068} = 7.074 \text{ m/s}$$

Applying Bernoulli's equation between sections (1) and (2), we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + \text{losses}$$

Since, $z_1 = z_2$ and losses are negligible (Given),

$$\therefore \quad \frac{p_1}{w} + \frac{V_1^2}{2g} = \frac{p_2}{w} + \frac{V_2^2}{2g}$$

Substituting the values, we get:

$$\frac{118 \times 10^3}{(1000 \times 0.85 \times 9.81)} + \frac{(3.98)^2}{2 \times 9.81} = \frac{p_2}{(1000 \times 0.85 \times 9.81)} + \frac{(7.074)^2}{2 \times 9.81}$$

$$\text{or,} \quad \frac{118 \times 10^3}{850} + \frac{(3.98)^2}{2} = \frac{p_2}{850} + \frac{(7.074)^2}{2}$$

$$\text{or,} \quad p_2 = 850 \left[\frac{118 \times 10^3}{850} + \frac{(3.98)^2 - (7.074)^2}{2} \right] = 103464 \text{ N/m}^2$$

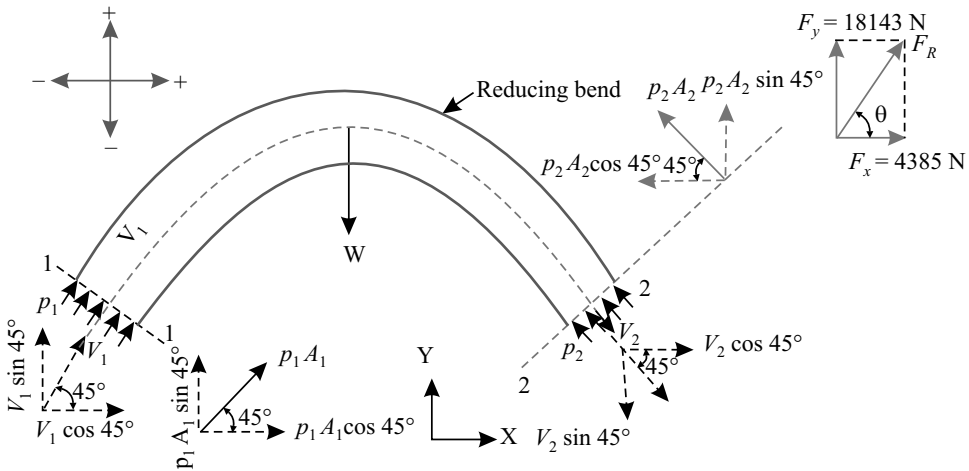


Fig. 6.55

Magnitude and direction of the force (resultant), F_R :

Force along X-axis:

$$F_x = \frac{wQ}{g} [V_{1x} - V_{2x}] + (p_1 A_1)_x + (p_2 A_2)_x$$

$$\text{where,} \quad V_{1x} = V_1 \cos 45^\circ = 3.98 \cos 45^\circ = 2.814 \text{ m/s,}$$

$$V_{2x} \cos 45^\circ = 7.074 \times \cos 45^\circ = 5.0 \text{ m/s}$$

$$(p_1 A_1)_x = p_1 A_1 \cos 45^\circ = 118 \times 10^3 \times 0.12566 \cos 45^\circ = 10484.89 \text{ N,}$$

$$(p_2 A_2)_x = -p_2 A_2 \cos 45^\circ = -103464 \times 0.07068 \cos 45^\circ = -5170.96 \text{ N}$$

$$\therefore F_x = \frac{(1000 \times 0.85 \times 9.81) 0.5}{9.81} [2.814 - 5.0] + 10484.89 + (-5170.96) = 4385 \text{ N (}\rightarrow\text{)}$$

Force along Y-axis:

$$F_y = \frac{wQ}{g} [V_{1y} - V_{2y}] + (p_1 A_1)_y + (p_2 A_2)_y - W$$

$$V_{1y} = V_1 \sin 45^\circ = 3.98 \sin 45^\circ = 2.814 \text{ m/s, } V_{2y} = -V_2 \sin 45^\circ = -7.074 \times \sin 45^\circ = -5.0 \text{ m/s}$$

$$(p_1 A_1)_y = p_1 A_1 \sin 45^\circ = 118 \times 10^3 \times 0.12566 \sin 45^\circ = 10484.89 \text{ N}$$

$$(p_2 A_2)_y = p_2 A_2 \sin 45^\circ = 103464 \times 0.07068 \times \sin 45^\circ = 5170.96 \text{ N}$$

$$W = 0.1(0.85 \times 1000) \times 9.81 = 833.85 \text{ N}$$

$$\therefore F_y = \frac{(1000 \times 0.85 \times 9.81)0.5}{9.81} [2.814 - (-5.0)] + 10484.89 + 5170.96 - 833.85 = 18143 \text{ N } (\uparrow)$$

$\therefore$ Resultant force on the bend,

$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{(4385)^2 + (18143)^2} = \mathbf{18665 \text{ (Ans.)}}$$

Inclination of F_x to the X -direction is,

$$\theta = \tan^{-1} \left(\frac{F_y}{F_x} \right) = \tan^{-1} \left(\frac{18143}{4385} \right) = \mathbf{76.4^\circ \text{ (Ans.)}}$$

Example 6.58. The following data refer to the Y-fitting shown in Fig. 6.56.

Reading of the pressure gauge at section 1-1 = 30 kN/m².

Discharge in at the section 1-1 = 15 litres/sec.

Discharge out from the section 3-3 = 5 litres/sec.

Assuming one-dimensional flow, neglecting elevation head and energy loss while making the energy balance, determine:

- (i) The pressures at the sections 2-2 and 3-3;
(ii) The force needed to hold the fitting in position.

(Roorkee University)

Solution. Refer to Fig. 6.56.

Given: $D_1 = 100 \text{ mm} = 0.1 \text{ m}$, $D_2 = 80 \text{ mm} = 0.08 \text{ m}$; $D_3 = 60 \text{ mm} = 0.06 \text{ m}$

$$\therefore \text{Area, } A_1 = (\pi/4) \times 0.1^2 = 0.007854 \text{ m}^2,$$

$$\text{Area, } A_2 = (\pi/4) \times 0.08^2 = 0.005026 \text{ m}^2, \text{ and}$$

$$\text{Area, } A_3 = (\pi/4) \times 0.06^2 = 0.002827 \text{ m}^2$$

Pressure at the section 1-1, $p_1 = 30 \text{ kN/m}^2$

$$\text{Now, } Q_1 = Q_2 + Q_3 \text{ or } 15 = Q_2 + 5$$

$$\therefore Q_2 = 10 \text{ litres/sec.}$$

$$\text{Velocity at the section 1-1, } V_1 = \frac{Q_1}{A_1} = \frac{15 \times 10^{-3}}{0.007854} = 1.91 \text{ m/s}$$

$$\text{Velocity at the section 2-2, } V_2 = \frac{Q_2}{A_2} = \frac{10 \times 10^{-3}}{0.005026} = 1.99 \text{ m/s}$$

$$\text{Velocity at the section 3-3, } V_3 = \frac{Q_3}{A_3} = \frac{5 \times 10^{-3}}{0.002827} = 1.77 \text{ m/s}$$

(i) Pressures at sections 2-2 and 3-3; p_2, p_3 :

Applying Bernoulli's equation between sections 1-1 and 2-2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

$$\frac{p_2}{w} - \frac{p_1}{w} = \frac{V_1^2}{2g} - \frac{V_2^2}{2g} \quad (\text{Neglecting elevation datum})$$

$$\begin{aligned} \text{or, } \frac{p_2 - p_1}{w} &= \frac{V_1^2 - V_2^2}{2g} \\ &= \frac{1.91^2 - 1.99^2}{2 \times 9.81} = -0.0159 \end{aligned}$$

$$\begin{aligned} \text{or, } p_2 - p_1 &= -w \times 0.0159 = -9.81 \times 0.0159 = -0.156 \text{ kN/m}^2 \\ &(\because w = 9.81 \text{ kN/m}^3 \text{ for water}) \end{aligned}$$

or,

$$p_2 = -0.156 + p_1 = -0.156 + 30 = 29.84 \text{ kN/m}^2 \text{ (Ans.)}$$

Similarly, for sections 1-1 and 3-3; we get:

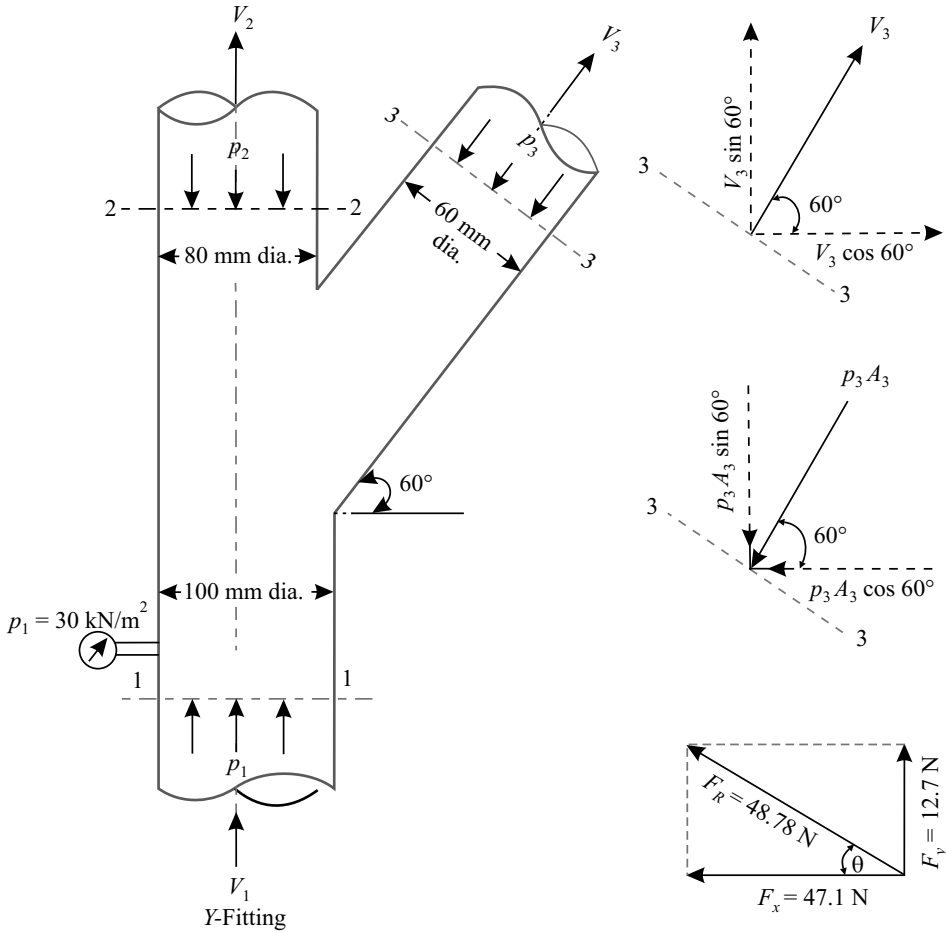


Fig. 6.56

$$\frac{p_3 - p_1}{w} = \frac{V_1^2 - V_3^2}{2g} = \frac{1.91^2 - 1.77^2}{2 \times 9.81} = 0.02626$$

or,

$$p_3 - p_1 = 9.81 \times 0.02626 = 0.26$$

$\therefore$

$$p_3 = 0.26 + p_1 = 0.26 + 30 = 30.26 \text{ kN/m}^2 \text{ (Ans.)}$$

(ii) Force needed to hold the fitting in position :

Force along X-axis:

$$F_x = \frac{wQ_3}{g} (0 - V_3 \cos 60^\circ) - p_3 A_3 \cos 60^\circ$$

(The velocities V_1 and V_2 and pressure p_1 and p_2 have no components in X-direction)

$$\begin{aligned} &= \frac{9.81 \times 5 \times 10^{-3}}{9.81} (0 - 1.77 \times 0.5) - 30.26 \times 0.002827 \times 0.5 \\ &= -0.004425 - 0.0427 = -0.00471 \text{ kN or } -47.1 \text{ N} \end{aligned}$$

or,

$$F_x = 47.1 \text{ N (}\leftarrow\text{)}$$

Force along Y-axis:

$$\begin{aligned}
 F_y &= \frac{wQ_1}{g} V_1 - \frac{wQ_2}{g} V_2 - \frac{wQ_3}{g} V_3 \sin 60^\circ + p_1 A_1 - p_2 A_2 - p_3 A_3 \sin 60^\circ \\
 &= \frac{w}{g} (Q_1 V_1 - Q_2 V_2 - Q_3 V_3 \sin 60^\circ) + p_1 A_1 - p_2 A_2 - p_3 A_3 \sin 60^\circ \\
 &= \frac{9.81}{9.81} (15 \times 10^{-3} \times 1.91 - 10 \times 10^{-3} \times 1.99 - 5 \times 10^{-3} \times 1.77 \\
 &\quad \times 0.866) + 30 \times 0.007854 - 29.84 \times 0.005026 - 30.26 \times 0.002827 \\
 &\quad \times 0.866 \\
 &= (0.02865 - 0.0199 - 0.007664) + 0.2356 - 0.1499 - 0.0741 \\
 &= 0.0127 \text{ kN or } 12.7 \text{ N } (\uparrow)
 \end{aligned}$$

The magnitude of resultant force acting on the fitting is

$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{47.1^2 + 12.7^2} = \mathbf{48.78 \text{ N (Ans.)}}$$

and the *direction* of the resultant force with X-axis (Fig. 6.56) is

$$\tan \theta = \frac{F_y}{F_x} = \frac{12.7}{47.1} = 0.2696$$

$$\therefore \theta = \mathbf{15.09^\circ \text{ (Ans.)}}$$

An equal and opposite force will be needed to hold the fitting in position. (Ans.)

Example 6.59. At inlet to a horizontal pipe of radius a , fitted at side of a vertical tank, the velocity distribution is uniform with magnitude V_0 . But at the outlet section, where the flow is fully developed, the velocity distribution is given by,

$$u = 2V_0 \left(1 - \frac{r^2}{a^2} \right)$$

where, u is the velocity at any radius r from the axis of the pipe. Determine the horizontal force required to hold the pipe in position. (UPSC)

Solution. For uniform velocity at inlet,

$$\text{Momentum} = \rho A V_0^2 = \rho \pi a^2 V_0^2 \quad (\text{where, } a = \text{radius of the pipe})$$

$$\text{At outlet, momentum} = \int \rho u^2 dA$$

$$\begin{aligned}
 &= \rho \int_0^a \left\{ 2V_0 \left(1 - \frac{r^2}{a^2} \right) \right\}^2 2\pi r \cdot dr \\
 &= 8\pi \rho \frac{V_0^2}{a^4} \int_0^a (a^2 - r^2)^2 r \cdot dr \\
 &= 8\pi \rho \frac{V_0^2}{a^4} \int_0^a (a^4 r - 2a^2 r^3 + r^5) dr \\
 &= 8\pi \rho \frac{V_0^2}{a^4} \left[a^4 \times \frac{r^2}{2} - 2a^2 \times \frac{r^4}{4} + \frac{r^6}{6} \right]_0^a \\
 &= 8\pi \rho \frac{V_0^2}{a^4} \left(\frac{a^6}{2} - \frac{a^6}{2} + \frac{a^6}{6} \right)
 \end{aligned}$$

$$= \frac{4}{3} \pi \rho V_0^2 a^2$$

- Assuming that the pressures at inlet and outlet are same, the force required is equal to change in momentum, or

Force, F = Momentum at outlet – Momentum at inlet

$$= \frac{4}{3} \pi \rho V_0^2 a^2 - \rho \pi a^2 V_0^2 = \pi \rho a^2 V_0^2 \left(\frac{4}{3} - 1 \right)$$

$$= \frac{1}{3} \pi \rho a^2 V_0^2 \text{ (Ans.)}$$

- If the pressures at inlet and outlet are different $(p_1 - p_0) = \Delta p \neq 0$, then the force required,

$$F = \frac{1}{3} \pi \rho a^2 V_0^2 + \Delta p \times \pi a^2 \text{ (Ans.)}$$

Example 6.60. Fig. 6.57 shows a rocket of circular cross-section with 2 m as its maximum diameter. The rocket is moving at 200 m/s and the jet of gases leaves at 1000 m/s relative to the rocket. The outside air pressure is 98.1 kN/m² and that of the jet is 93.1 kN/m². The density of outside air is 11.76 N/m³. The rocket breathes in air at the rate of 100 m³/s. Neglecting compressibility effects and assuming that the mass rate of flow of exhaust gases equals the mass flow rate of air breathed in, calculate:

- Thrust developed by the rocket.
- Energy supplied by the rocket to the air stream per unit weight of air flowing through the rocket.
- Energy supplied per second by the rocket to the air stream.
- Power developed by the rocket.

[IIT Delhi]

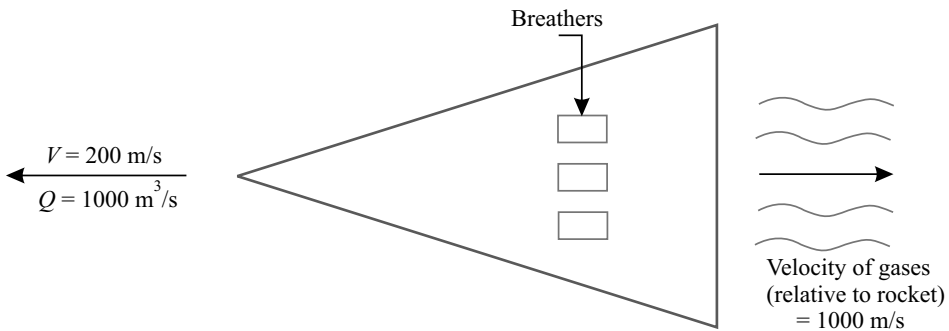


Fig. 6.57

Solution. Given: Maximum diameter of the rocket, $D = 2 \text{ m}$

Speed of the rocket, $V_1 = 200 \text{ m/s}$

Velocity of gases (relative to rocket), $V_2 = 1000 \text{ m/s}$

Outside air pressure, $p_1 = 98.1 \text{ kN/m}^2$

Jet pressure, $p_2 = 93.1 \text{ kN/m}^2$

Density of outside air, $w = 11.76 \text{ N/m}^3$

Rate of air breathed in by the rocket, $Q = 100 \text{ m}^3/\text{s}$

- Thrust developed by the rocket:**

Applying momentum equation to the control volume, we get:

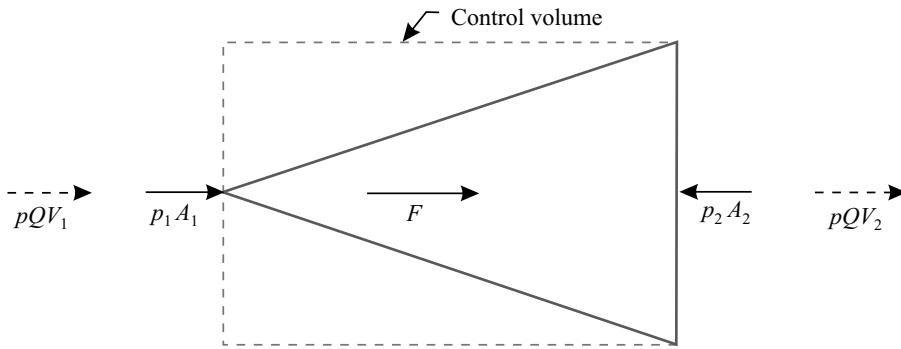


Fig. 6.58

$$F + p_1 A_1 - p_2 A_2 = \rho Q (V_2 - V_1)$$

or,

$$F = \rho Q (V_2 - V_1) + (p_2 - p_1) A_2$$

($\because A_1 = A_2$ and pressure p_1 is everywhere the same except in the jet)

or,

$$F = \frac{11.76}{9.81} \times 100 (1000 - 200) + (93.1 - 98.1) 10^3 \times \frac{\pi}{4} \times 2^2 \text{ N}$$

or,

$$F = 80.2 \times 10^3 \text{ N} = 80.2 \text{ kN}$$

Thrust on the rocket is equal and opposite to $F = 80.2 \text{ kN}$. (Ans.)

(ii) Energy supplied by the rocket:

Absolute velocity of the jet = $1000 - 200 = 800 \text{ m/s}$

Energy per unit weight of air at section (2)

$$= \frac{p}{w} + \frac{V^2}{2g} = \frac{93.1 \times 10^3}{11.76} + \frac{800^2}{2 \times 9.81}$$

$$= 7.92 \times 10^3 + 32.62 \times 10^3 = 40.54 \times 10^3 \text{ Nm/N of air}$$

Energy per unit weight of air at section (1)

$$= \frac{p}{w} = \frac{98.1 \times 10^3}{11.76} = 8.34 \times 10^3 \text{ Nm/N of air}$$

(Absolute velocity being zero)

$\therefore$ Energy supplied by the rocket per unit weight of air

$$= (40.54 - 8.34) \times 10^3 \text{ Nm} = 32.2 \text{ kNm/N (Ans.)}$$

(iii) Energy supplied per second by the rocket:

Energy supplied per second by the rocket

$$= wQ \times 32.2 \times 10^3 = 11.76 \times 100 \times 32.2 \times 10^3$$

$$= 37.86 \times 10^6 \text{ Nm (Ans.)}$$

(iv) Power developed by the rocket:

Power developed by the rocket = Energy supplied per second

$$= 37.86 \times 10^6 \text{ Nm/s} = 37.86 \times 10^3 \text{ kW (Ans.)}$$

6.9. KINETIC ENERGY AND MOMENTUM CORRECTION FACTORS (CORIOLIS CO-EFFICIENTS)

While deriving Bernoulli's equation, it is assumed that the velocity distribution across a single stream tube is uniform. But if there is an appreciable variation in the velocity distribution (on account of viscous and boundary resistance) correction factors α and β have to be applied to obtain the exact amount of kinetic energy or momentum available at a given cross-section.

Kinetic energy correction factor (α):

‘Kinetic energy correction factor’ is defined as the ratio of the kinetic energy of flow per second based on actual velocity across a section to the kinetic energy of flow per second based on average velocity across the same section. It is denoted by α .

Mathematically,
$$\alpha = \frac{\text{Kinetic energy per second based on actual velocity}}{\text{Kinetic energy per second based on average velocity}} \dots (6.27)$$

Refer to Fig. 6.59.

Let, $\bar{u}$ = Average velocity at the section LL,

u = Local or point or actual velocity,

dA = Elementary area, and

A = Area of cross-section.

For the velocity variation across the section LL of the stream tube the total K.E. for the entire section is given as:

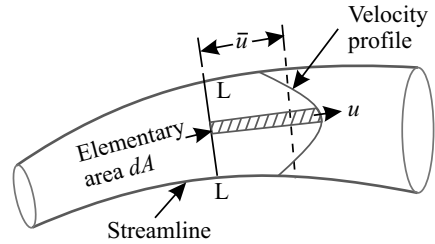


Fig. 6.59

$$K.E. = \frac{1}{2} m \bar{u}^2 = \frac{1}{2} (\rho A \bar{u}) \bar{u}^2 = \frac{1}{2} \rho A \bar{u}^3 \dots (i)$$

True K.E. for the entire cross-section

$$= \int \frac{1}{2} dm \cdot u^2 = \int \frac{1}{2} (\rho \cdot dA \cdot u) u^2 = \frac{\rho}{2} \int_A u^3 dA \dots (ii)$$

$$\alpha = \frac{\frac{\rho}{2} \int_A u^3 dA}{\frac{\rho}{2} A \bar{u}^3} = \frac{1}{A} \int \left(\frac{u}{\bar{u}} \right)^3 dA \dots (6.28)$$

$\alpha = 1$ for uniform velocity distribution and tends to become greater than 1 as the distribution of velocity becomes less and less uniform.

$\alpha = 1.02$ to 1.15 for turbulent flows.

$\alpha = 2$ for laminar flow.

It may be noted that in most of the fluid mechanics computations, α is taken as 1 without introducing much error, since the velocity is a small percentage of the total head.

Momentum correction factor (β):

‘Momentum correction factor’ is defined as the ratio of momentum of the flow per second based on actual velocity to the momentum of the flow per second based on average velocity across a section. It is denoted by β .

Mathematically,
$$\beta = \frac{\text{Momentum per second based on actual velocity}}{\text{Momentum per second based on average velocity}} \dots (6.29)$$

Refer to Fig. 6.59.

The momentum of fluid mass m is

$$= m \bar{u} = (\rho A \bar{u}) \bar{u} = \rho A \bar{u}^2 \dots (iii)$$

The true momentum at the section LL is given as:

$$\int_{LL} dm \cdot u = \int_{LL} (\rho dA \cdot u) u = \int_A \rho u^2 dA \dots (iv)$$

$$\beta = \frac{\int \rho u^2 dA}{\rho A \bar{u}^2} = \frac{1}{A} \int_A \left(\frac{u}{\bar{u}} \right)^2 dA \quad \dots(6.30)$$

$\beta = 1$ for uniform flow,

$\beta = 1.01$ to 1.07 for turbulent flow in pipes, and

$\beta = \frac{4}{3} = 1.33$ for laminar flow in pipes.

The value of β may be greater for open channel flow.

In most cases, β is taken as 1.

Note: Since majority of the flow situations are turbulent in character, the usual practice is to assign unit value to α and β .

Example 6.61. The velocity distribution for turbulent flow in pipe is given approximately by Prandtl's one-seventh power law.

$$u = U_m \left(\frac{y}{r_0} \right)^{1/7}$$

where u is the local velocity of flow at a distance y from the pipe wall, U_m is the maximum velocity at the centre line of the pipe and r_0 is the pipe radius. Find the following:

- (i) Average velocity,
- (ii) Kinetic energy correction factor, and
- (iii) Momentum correction factor.

[Delhi University]

Solution. (i) Average velocity:

Refer to Fig. 6.60. Consider an elementary area dA in the form of a ring at a radius $(r_0 - y)$ and of thickness dy , then,

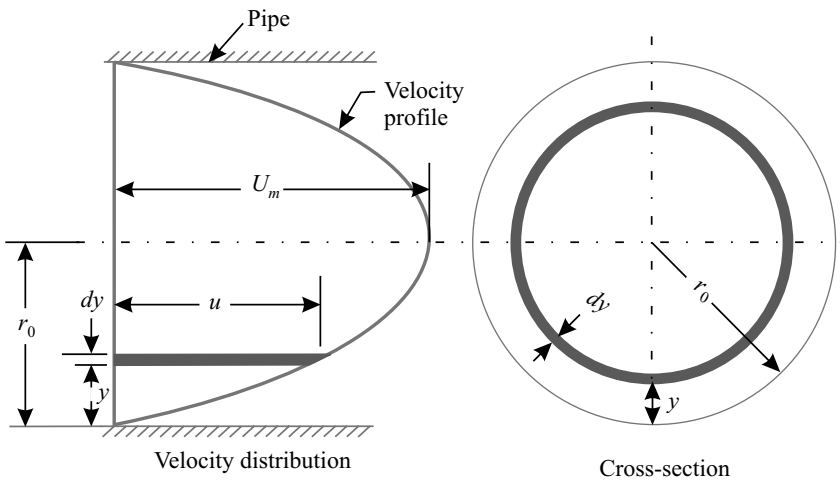


Fig. 6.60. Velocity distribution and cross-section of a circular pipe.

$$dA = 2\pi(r_0 - y) dy$$

Rate of fluid flowing through the ring

$$= dQ = \text{Area of ring element} \times \text{local velocity}$$

$$= 2\pi(r_0 - y) dy u$$

$$\begin{aligned}
 \therefore \quad \text{Total flow, } Q &= \int_0^{r_0} 2\pi u (r_0 - y) dy \\
 &= \int_0^{r_0} 2\pi U_m \left(\frac{y}{r_0} \right)^{1/7} (r_0 - y) dy \\
 &= \frac{2\pi U_m}{(r_0)^{1/7}} \int_0^{r_0} (y)^{1/7} (r_0 - y) dy \\
 &= \frac{2\pi U_m}{(r_0)^{1/7}} \int_0^{r_0} (r_0 y^{1/7} - y^{8/7}) dy \\
 &= \frac{2\pi U_m}{(r_0)^{1/7}} \left[\frac{7}{8} r_0 \cdot y^{8/7} - \frac{7}{15} y^{15/7} \right]_0^{r_0} \\
 &= \frac{2\pi U_m}{(r_0)^{1/7}} \left[\frac{7}{8} r_0 \cdot (r_0)^{8/7} - \frac{7}{15} (r_0)^{15/7} \right] \\
 &= \frac{2\pi U_m}{(r_0)^{1/7}} \left[\frac{7}{8} (r_0)^{15/7} - \frac{7}{15} (r_0)^{15/7} \right] \\
 &= \frac{2\pi U_m}{(r_0)^{1/7}} \times (r_0)^{15/7} \left[\frac{7}{8} - \frac{7}{15} \right] \\
 &= 2\pi U_m \left(\frac{49}{120} r_0^2 \right) \quad \dots(i)
 \end{aligned}$$

$$\text{If } \bar{u} \text{ is the average velocity, then } Q = A\bar{u} = \pi r_0^2 \bar{u} \quad \dots(ii)$$

From (i) and (ii), we get:

$$\begin{aligned}
 \pi r_0^2 \bar{u} &= 2\pi U_m \left(\frac{49}{120} r_0^2 \right) \\
 \therefore \quad \bar{u} &= \frac{2\pi U_m \left(\frac{49}{120} r_0^2 \right)}{\pi r_0^2} = \frac{49}{60} U_m
 \end{aligned}$$

$$\text{i.e.,} \quad \bar{u} = \frac{49}{60} U_m \text{ (Ans.)}$$

(ii) Kinetic energy correction factor, α :

$$\begin{aligned}
 \alpha &= \frac{1}{A\bar{u}^3} \int_0^{r_0} u^3 dA \quad \text{[Eqn. (6.28)]} \\
 &= \frac{1}{A\bar{u}^3} \int_0^{r_0} U_m^3 \left(\frac{y}{r_0} \right)^{3/7} 2\pi (r_0 - y) dy \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3 (r_0)^{3/7}} \int_0^{r_0} y^{3/7} (r_0 - y) dy
 \end{aligned}$$

$$\begin{aligned}
&= \frac{2\pi U_m^3}{A\bar{u}^3 (r_0)^{3/7}} \int_0^{r_0} (r_0 y^{3/7} - y^{10/7}) dy \\
&= \frac{2\pi U_m^3}{A\bar{u}^3 (r_0)^{3/7}} \left[r_0 \times \frac{7}{10} y^{10/7} - \frac{7}{17} y^{17/7} \right]_0^{r_0} \\
&= \frac{2\pi U_m^3}{A\bar{u}^3 (r_0)^{3/7}} \left[\frac{7}{10} (r_0)^{17/7} - \frac{7}{17} (r_0)^{17/7} \right] \\
&= \frac{2\pi U_m^3}{A\bar{u}^3 (r_0)^{3/7}} \left[\frac{49}{170} (r_0)^{17/7} \right]
\end{aligned}$$

Substituting the values, $A = \pi r_0^2$ and $\bar{u} = \frac{49}{60} U_m$, we get:

$$\alpha = \frac{2\pi U_m^3}{\pi r_0^2 \left(\frac{49}{60} U_m \right)^3 (r_0)^{3/7}} \left[\frac{49}{170} (r_0)^{17/7} \right] = \mathbf{1.06 \text{ (Ans.)}}$$

(iii) Momentum correction factor, β :

$$\beta = \frac{1}{A\bar{u}^2} \int u^2 dA \quad [\text{Eqn. (6. 29)}]$$

$$\begin{aligned}
&= \frac{1}{A\bar{u}^2} \int_0^{r_0} U_m^2 \left(\frac{y}{r_0} \right)^{2/y} 2\pi (r_0 - y) dy \\
&= \frac{2\pi U_m^2}{A\bar{u}^2 (r_0)^{2/7}} \int_0^{r_0} y^{2/7} (r_0 - y) dy \\
&= \frac{2\pi U_m^2}{A\bar{u}^2 (r_0)^{2/7}} \int_0^{r_0} (r_0 y^{2/7} - y^{9/7}) dy \\
&= \frac{2\pi U_m^2}{A\bar{u}^2 (r_0)^{2/7}} \left[r_0 \times \frac{7}{9} y^{9/7} - \frac{7}{16} y^{16/7} \right]_0^{r_0} \\
&= \frac{2\pi U_m^2}{A\bar{u}^2 (r_0)^{2/7}} \left[\frac{7}{9} (r_0)^{16/7} - \frac{7}{16} (r_0)^{16/7} \right] \\
&= \frac{2\pi U_m^2}{A\bar{u}^2 (r_0)^{2/7}} \left[\frac{49}{144} (r_0)^{16/7} \right]
\end{aligned}$$

Substituting the values, $A = \pi r_0^2$ and $\bar{u} = \frac{49}{60} U_m$, we get:

$$\beta = \frac{2\pi U_m^2}{\pi r_0^2 \left(\frac{49}{60} U_m \right)^2 (r_0)^{2/7}} \left[\frac{49}{144} (r_0)^{16/7} \right] = \mathbf{1.02 \text{ (Ans.)}}$$

Example 6.62. In a circular pipe the velocity profile is given as

$$u = U_m \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

where u is the velocity at any radius r , U_m is the velocity at the pipe axis, and R is the radius of the pipe. Find:

- (i) Average velocity,
- (ii) Energy correction factor, and
- (iii) Momentum correction factor.

[Anna University]

Solution. Refer to Fig. 6.61. Consider an elementary area dA in the form of a ring at a radius r and of thickness dr , then $dA = 2\pi r.dr$

Flow rate through the ring $= dQ = \text{Elemental area} \times \text{local velocity} = 2\pi r.dr.u$

$$\begin{aligned}
 \therefore \quad \text{Total flow } Q &= \int_0^R 2\pi r.u.dr \\
 &= \int_0^R 2\pi U_m \left(1 - \frac{r^2}{R^2}\right) r dr \\
 &= 2\pi U_m \int_0^R \left(r - \frac{r^3}{R^2}\right) dr = 2\pi U_m \left[\frac{r^2}{2} - \frac{r^4}{4R^2}\right]_0^R \\
 &= 2\pi U_m \left(\frac{R^2}{2} - \frac{R^2}{4}\right) = 2\pi U_m \left(\frac{R^2}{4}\right) \\
 \text{i.e.,} \quad Q &= 2\pi U_m \left(\frac{R^2}{4}\right) \quad \dots(i)
 \end{aligned}$$

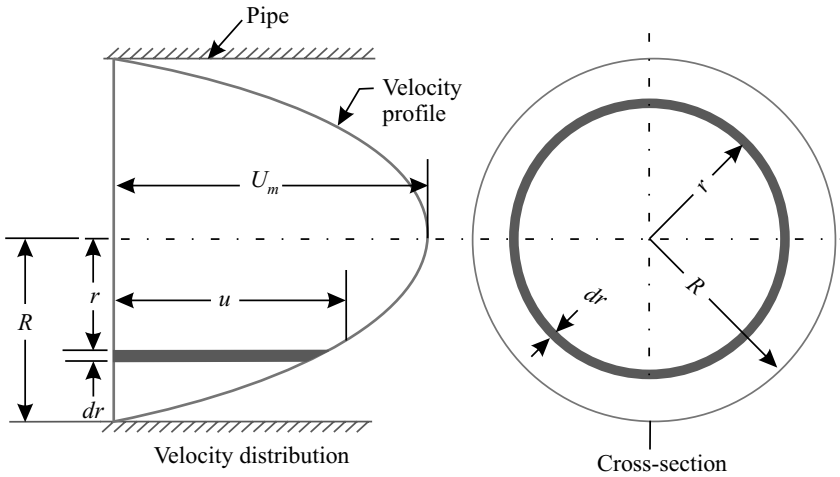


Fig. 6.61. Velocity distribution and cross-section of a pipe.

- (ii) Average velocity, $\bar{u}$:

If $\bar{u}$ is the average flow velocity, then:

$$Q = A\bar{u} = \pi R^2 \bar{u} \quad \dots(ii)$$

From (i) and (ii), we get:

$$\pi R^2 \bar{u} = 2\pi U_m \left(\frac{R^2}{4}\right)$$

$$\therefore \quad \bar{u} = \frac{2\pi U_m \left(\frac{R^2}{4}\right)}{\pi R^2} = \frac{U_m}{2} \quad (\text{Ans.})$$

(iii) Kinetic energy correction factor, α :

$$\begin{aligned}
 \alpha &= \frac{1}{A\bar{u}^3} \int_0^R u^3 dA && \dots[\text{Eqn. 6.28}] \\
 &= \frac{1}{A\bar{u}^3} \int_0^R U_m^3 \left[1 - \left(\frac{r}{R} \right)^2 \right]^3 2\pi r dr \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \int_0^R \left(1 - \frac{3r^2}{R^2} + \frac{3r^4}{R^4} - \frac{r^6}{R^6} \right) r dr \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \int_0^R \left(r - \frac{3r^3}{R^2} + \frac{3r^5}{R^4} - \frac{r^7}{R^6} \right) dr \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \left[\frac{r^2}{2} - \frac{3r^4}{4R^2} + \frac{3r^6}{6R^4} - \frac{r^8}{8R^6} \right]_0^R \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \left[\frac{R^2}{2} - \frac{3}{4}R^2 + \frac{R^2}{2} - \frac{R^2}{8} \right] \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \left(\frac{R^2}{8} \right)
 \end{aligned}$$

Substituting the values $A = \pi R^2$ and $\bar{u} = \frac{U_m}{2}$, we get:

$$\alpha = \frac{2\pi U_m^3}{\pi R^2 \times \left(\frac{U_m}{2} \right)^3} \times \left(\frac{R^2}{8} \right) = 2 \text{ (Ans.)}$$

(iv) Momentum correction factor, β :

$$\begin{aligned}
 \beta &= \frac{1}{A\bar{u}^2} \int u^2 dA && \dots[\text{Eqn. 6.29}] \\
 &= \frac{1}{A\bar{u}^2} \int_0^R U_m^2 \left[1 - \left(\frac{r}{R} \right)^2 \right]^2 2\pi r dr \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \int_0^R \left(1 - 2 \times \frac{r^2}{R^2} + \frac{r^4}{R^4} \right) r dr \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \int_0^R \left(r - 2 \times \frac{r^3}{R^2} + \frac{r^5}{R^4} \right) dr \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \left[\frac{r^2}{2} - 2 \times \frac{r^4}{4R^2} + \frac{1}{6} \times \frac{r^6}{R^4} \right]_0^R \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \left[\frac{R^2}{2} - \frac{R^2}{2} + \frac{R^2}{6} \right] \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \left(\frac{R^2}{6} \right)
 \end{aligned}$$

Substituting the values $A = \pi R^2$ and $\bar{u} = \frac{U_m}{2}$, we get:

$$\beta = \frac{2\pi U_m^2}{\pi R^2 \times \left(\frac{U_m}{2}\right)^2} \left(\frac{R^2}{6}\right) = 1.33 \text{ (Ans.)}$$

6.10. MOMENT OF MOMENTUM EQUATION

Moment of momentum equation is derived from *moment of momentum principle* which states as follows:

“The resulting torque acting on a rotating fluid is equal to the rate of change of moment of momentum”.

When the moment of momentum of flow leaving a control volume is different from that entering it, the result is a torque acting over the control volume.

Let,

- Q = Steady rate of flow of fluid,
- ρ = Density of fluid,
- V_1 = Velocity of fluid at section 1,
- r_1 = Radius of curvature at section 1, and
- V_2 and r_2 = Velocity and radius of curvature at section 2.

Momentum of fluid at section 1 = Mass $\times$ velocity = $\rho Q \times V_1$

$\therefore$ Moment of momentum per second of fluid at section 1 = $\rho Q \times V_1 \times r_1$

Similarly, moment of momentum per second of fluid at section 2 = $\rho Q \times V_2 \times r_2$

$\therefore$ Rate of change of moment of momentum = $\rho Q V_2 r_2 - \rho Q V_1 r_1 = \rho Q (V_2 r_2 - V_1 r_1)$

According to the moment of momentum principle,

Resultant torque = Rate of change of moment of momentum

$$T = \rho Q (V_2 r_2 - V_1 r_1) \quad \dots(6.31)$$

Eqn. (6.31) is known as *moment of momentum equation*. This equation is used:

- (i) To find torque exerted by water on sprinkler, and
- (ii) To analyse flow problems in turbines and centrifugal pumps.

Example 6.63. Fig. 6.62 shows an unsymmetrical sprinkler. It has a frictionless shaft and equal flow through each nozzle with a velocity of 8 m/s relative to the nozzle. Find the speed of rotation in r.p.m.

Solution. Refer to Fig. 6.62.

$r_A = 0.4 \text{ m}$, $r_B = 0.6 \text{ m}$

Velocity relative to the nozzle

$V_A (= V_B) = 8 \text{ m/s}$

Let, ω = Angular velocity of the sprinkler.

Absolute velocity, $V_1 = V_A + \omega r_A = 8 + \omega \times 0.4 = 8 + 0.4 \omega$

Absolute velocity, $V_2 = V_B - \omega r_B = 8 - \omega \times 0.6 = 8 - 0.6 \omega$

Speed of rotation, N (r.p.m.):

The moment of momentum of the fluid entering sprinkler is given zero and also there is no external torque applied on the sprinkler. Hence resultant torque is zero, i.e.

$$T = 0$$

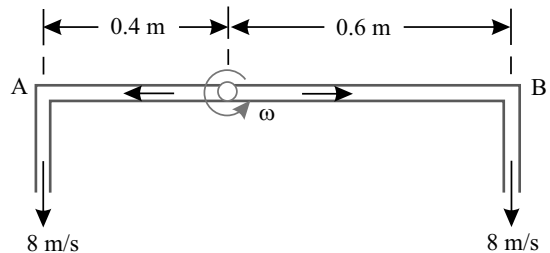


Fig. 6.62

$$\begin{aligned}
 \therefore \quad \rho Q (V_2 r_2 - V_1 r_1) &= 0 \\
 \text{or,} \quad V_2 r_2 - V_1 r_1 &= 0 \quad (\because \rho Q \neq 0) \\
 \text{or,} \quad (8 - 0.6 \omega) \times 0.6 &= (8 + 0.4 \omega) \times 0.4 \\
 \text{or,} \quad 4.8 - 0.36 \omega &= 3.2 + 0.16 \omega \\
 \text{or,} \quad 0.52 \omega &= 1.6 \\
 \text{or,} \quad \omega &= 3.077 \text{ rad/s} \\
 \text{But,} \quad \omega &= \frac{2\pi N}{60} \\
 \therefore \quad N &= \frac{60 \times \omega}{2\pi} = \frac{60 \times 3.077}{2\pi} = \mathbf{29.4 \text{ r.p.m. (Ans.)}}
 \end{aligned}$$

Example 6.64. A lawn sprinkler shown in Fig 6.63 has 12 mm diameter nozzle at the end of a rotating arm and discharges water with a velocity of 15 m/s. Determine:

- (i) Torque required to hold the rotating arm stationary, and
(ii) Constant speed of rotation of the arm, if free to rotate.

Solution. Diameter of each nozzle = 12 mm = 0.012 m

$\therefore$ Area of each nozzle

$$= \frac{\pi}{4} \times 0.012^2 = 0.000113 \text{ m}^2$$

Velocity of flow, $V_A (= V_B) = 15 \text{ m/s}$

$\therefore$ Discharge through each nozzle,

$$\begin{aligned}
 Q &= \text{Area} \times \text{velocity} \\
 &= 0.000113 \times 15 = 0.001695 \text{ m}^3/\text{s}
 \end{aligned}$$

- (i) Torque required to hold the rotating arm stationary:

Torque exerted by water coming through nozzle A on the sprinkler

$$= \rho Q V_A \times r_A = \frac{9810}{9.81} \times 0.001695 \times 15 \times 0.3 = 7.627 \text{ Nm}$$

Torque exerted by water coming through nozzle B on the sprinkler

$$= \rho Q V_B \times r_B = \frac{9810}{9.81} \times 0.001695 \times 15 \times 0.375 = 9.534 \text{ Nm}$$

$\therefore$ Total torque exerted by water on sprinkler

$$= 7.627 + 9.534 = 17.161 \text{ Nm}$$

$\therefore$ Torque required to hold the rotating arm stationary

$$\begin{aligned}
 &= \text{Torque exerted by water on sprinkler} \\
 &= \mathbf{17.161 \text{ Nm (Ans.)}}
 \end{aligned}$$

- (ii) Constant speed of rotation of the arm, if free to rotate, N (r.p.m.):

Let, ω = Angular speed of rotation of the sprinkler.

Then, absolute velocities of flow of water at the nozzles A and B are,

$$V_1 = 15 - 0.3 \omega$$

and

$$V_2 = 15 - 0.375 \omega$$

Torque exerted by water coming out at A, on sprinkler

$$\begin{aligned}
 &= \rho Q V_1 \times r_A = \frac{9810}{9.81} \times 0.001695 \times (15 - 0.3 \omega) \times 0.3 \\
 &= 0.5085 (15 - 0.3 \omega)
 \end{aligned}$$

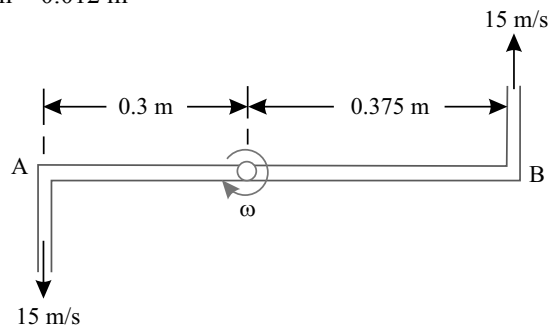


Fig. 6.63

Torque exerted by water, coming out at B , on sprinkler

$$\begin{aligned}
 &= PQV_2 \times r_B = \frac{9810}{9.81} \times 0.001695 \times (15 - 0.375 \omega) \times 0.375 \\
 &= 0.6356 (15 - 0.375 \omega)
 \end{aligned}$$

$\therefore$ Total torque exerted by water

$$= 0.5085 (15 - 0.3 \omega) + 0.6356 (15 - 0.375 \omega)$$

Since moment of momentum of the flow entering is zero and no external torque is applied on sprinkler, so the resultant torque on the sprinkler must be zero.

$$\therefore 0.5085 (15 - 0.3 \omega) + 0.6356 (15 - 0.375 \omega) = 0$$

$$7.627 - 0.1526 \omega + 9.534 - 0.238 \omega = 0$$

$$17.161 - 0.3906 \omega = 0$$

or,

$$\omega = \frac{17.161}{0.3906} = 43.93 \text{ rad/s}$$

Also,

$$\omega = \frac{2\pi N}{60} = 43.93$$

$\therefore$

$$N = \frac{60 \times 43.93}{2\pi} = 419.5 \text{ r.p.m. (Ans.)}$$

6.11. VORTEX MOTION

Vortex motion is defined as a motion in which the whole fluid mass rotates about an axis. A mass of fluid in rotation about a fixed axis is called **vortex**.

A vortex motion is characterised by a flow pattern wherein the stream lines are curved. When fluid flows between curved stream lines, centrifugal forces are set up and these are counter-balanced by the pressure force acting in the radial direction.

The vortex flow is of the following types:

1. Forced vortex flow, and
2. Free vortex flow.

6.11.1 Forced Vortex Flow

Forced vortex flow is one in which the fluid mass is made to rotate by means of some external agency. The external agency is generally the mechanical power which imparts a constant torque on the fluid mass. Then, in such a flow there is always expenditure of energy. The forced vortex motion is also called *flywheel vortex* or *rotational vortex*.

In this type of flow, the fluid mass rotates at a constant angular velocity ω . The tangential velocity of any fluid particle is given by:

$$v = \omega r \quad \dots(6.32)$$

(where, r = radius of the fluid particle from the axis of rotation)

$$\therefore \text{Angular velocity, } \omega = \frac{v}{r} = \text{constant} \dots[6.32 (a)]$$

Example:

1. Rotation of water through the runner of a turbine.
2. Rotation of liquid inside the impeller of a centrifugal pump.
3. Rotation of liquid in a vertical cylinder (Fig. 6.64).

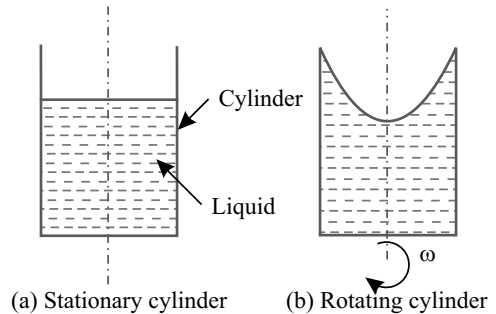


Fig. 6.64. Forced vortex flow.

6.11.2 Free Vortex Flow

Free vortex flow is one in which the fluid mass rotates without any external impressed contact force. The whole fluid mass rotates either due to fluid pressure itself or the gravity or due to rotation previously imparted. The free vortex motion is also called *potential vortex* or *irrotational vortex*.

Example:

1. Flow around a circular bend.
2. A whirlpool in a river.
3. Flow of liquid in a centrifugal pump casing after it has left the impeller.
4. Flow of water in a turbine casing before it enters the guide vanes.
5. Flow of liquid through a hole/outlet provided at the bottom of a shallow vessel (e.g., wash basin, bath tub, etc.)

In free vortex the relation between velocity and radius is obtained by putting the value of external torque equal to zero, or, the time rate change of angular momentum (i.e., moment of momentum) must be *zero*.

Let us consider a particle of mass m at a radius distance r from the axis of rotation, having a tangential velocity, v . Then:

$$\text{Moment of momentum} = (m \times v) \times r = mvr$$

$\therefore$ Time rate of change of momentum

$$= \frac{\partial}{\partial t}(mvr)$$

$$\text{But for the vortex, } = \frac{\partial}{\partial t}(mvr) = 0$$

Integrating, we get: $mvr = \text{constant}$

$$\text{Since } m \text{ is constant, } vr = \text{constant} = C \quad \dots(6.33)$$

where C is a constant and is known as *strength of vortex*.

$$\therefore v = \frac{C}{r} \quad \dots(6.34)$$

$$\text{or, } v \propto \frac{1}{r} \quad \dots(6.34 a)$$

i.e. tangential velocity is *inversely proportional to distance* r .

6.11.3. Equation of Motion for Vortex Flow.

Refer to Fig. 6.65. ABCD is fluid element rotating at a uniform velocity in a horizontal plane about an axis perpendicular to the plane of paper and passing through O.

Let, r = Radius of the element from O,

Δr = Radial thickness of the element,

ΔA = Area of cross-section of element, and

$\Delta \theta$ = Angle subtended by the element at O.

The various forces acting on the element are:

1. Centrifugal force, $\frac{mv^2}{r}$ acting away from the centre, O,
2. Pressure force $p\Delta A$ on the face AB, and

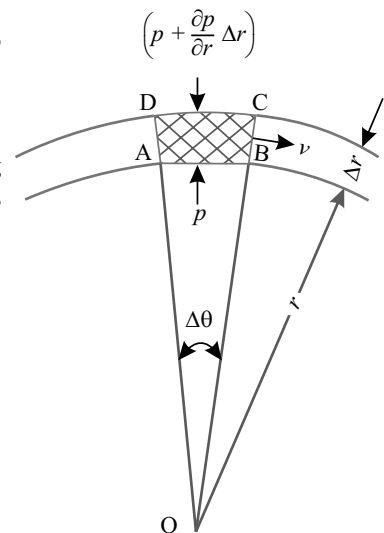


Fig. 6.65. Flow in a circular path.

3. Pressure force $\left(p + \frac{\partial p}{\partial r} \Delta r\right) \Delta A$ on the face CD .

Equating the forces in the radial direction, we get:

$$\left(p + \frac{\partial p}{\partial r} \Delta r\right) \Delta A - p \Delta A = \frac{mv^2}{r}$$

But, $m = \text{mass density} \times \text{volume} = \rho \times \Delta A \times \Delta r$

$$\therefore \left(p + \frac{\partial p}{\partial r} \Delta r\right) \Delta A - p \Delta A = \rho \Delta A \Delta r \frac{v^2}{r}$$

$$\text{or, } \rho \Delta A + \frac{\partial p}{\partial r} \Delta r \Delta A - p \Delta A = \rho \Delta A \Delta r \frac{v^2}{r}$$

$$\text{or, } \frac{\partial p}{\partial r} \Delta r \Delta A = \rho \Delta A \Delta r \frac{v^2}{r}$$

$$\text{or, } \frac{\partial p}{\partial r} = \frac{\rho v^2}{r} \quad \dots(6.35)$$

The expression $\frac{\partial p}{\partial r}$ is called *pressure gradient* in the radial direction.

(Since $\frac{\partial p}{\partial r}$ is +ve, therefore, pressure increases with the increase of radius r .)

In the vertical plane, the variation of pressure is given by the hydrostatic law, i.e.,

$$\frac{\partial p}{\partial z} = -\rho g \quad \dots(6.36)$$

As the pressure is a function of r and z , therefore total derivative of p ,

$$\partial p = \frac{\partial p}{\partial r} dr + \frac{\partial p}{\partial z} dz$$

Substituting the values of $\frac{\partial p}{\partial r}$ and $\frac{\partial p}{\partial z}$ from eqns. (6.35) and (6.36) respectively, we get:

$$dp = \frac{\rho v^2}{r} dr - \rho g dz \quad \dots(6.37)$$

Eqn. (6.37) gives the *variation of pressure of a rotating fluid in any plane*.

6.11.4. Equation of Forced Vortex Flow

In case of forced vortex flow,

$$v = \omega r \quad \dots[\text{Eqn. 6.32}]$$

(where, ω = constant angular velocity)

Putting the value of v in eqn. (6.37), we get:

$$dp = \frac{\rho \omega^2 r^2}{r} dr - \rho g dz$$

$$\text{or, } dp = \rho \omega^2 r dr - \rho g dz$$

Considering points 1 and 2 in the fluid having forced vortex flow (Fig. 6.66) and integrating the above eqn. for these points, we get:

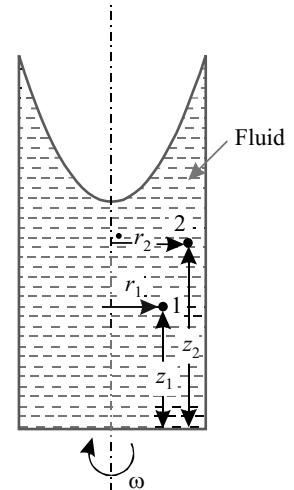


Fig. 6.66. Forced vortex flow.

$$\int_1^2 dp = \int_1^2 \rho \omega^2 r dr - \int_1^2 \rho g dz$$

$$\text{or,} \quad [p]_1^2 = \rho \omega^2 \left[\frac{r^2}{2} \right]_1^2 - \rho g [z]_1^2$$

$$\begin{aligned} \text{or,} \quad (p_2 - p_1) &= \frac{\rho \omega^2}{2} (r_2^2 - r_1^2) - \rho g (z_2 - z_1) \\ &= \frac{\rho}{2} (\omega^2 r_2^2 - \omega^2 r_1^2) - \rho g (z_2 - z_1) \\ &= \frac{\rho}{2} (v_2^2 - v_1^2) - \rho g (z_2 - z_1) \quad [\because v_1 = \omega_1 r_1 \text{ and } v_2 = \omega_2 r_2] \end{aligned}$$

— When the points 1 and 2 lie on the free surface of the liquid, then $p_1 = p_2$ and the above equation becomes:

$$0 = \frac{\rho}{2} (v_2^2 - v_1^2) - \rho g (z_2 - z_1)$$

$$\text{or,} \quad g(z_2 - z_1) = \left(\frac{v_2^2 - v_1^2}{2} \right)$$

$$\text{or,} \quad z_2 - z_1 = \frac{v_2^2 - v_1^2}{2g}$$

— When the point 1 lies on the axis of rotation, then:

$$v_1 = \omega r_1 = \omega \times 0 = 0; \text{ the above eqn. reduces to:}$$

$$z_2 - z_1 = \frac{v_2^2}{2g}$$

If, $z_2 - z_1 = z$ (say), then we have:

$$z = \frac{v_2^2}{2g} = \frac{\omega^2 r_2^2}{2g} \quad \dots(6.38)$$

Thus, z varies with square of r . Hence eqn. (6.38) is an equation of parabola which means that the free surface of the liquid is a paraboloid.

Example 6.65. Prove that in case of force vortex, the rise of liquid level at the ends is equal to the fall of liquid level at the axis of rotation.

Solution. Refer to Fig. 6.67.

Let, R = Radius of the cylinder, and OO = Initial liquid level when the cylinder is stationary.

Let the cylinder is rotated at constant angular velocity ω . The liquid will rise at the ends and will fall at the centre.

Let, y_r = Rise of liquid at the ends (from OO), and

y_f = Fall of liquid at the centre (from OO)

Now, initial height of liquid = $(h + y_f)$

$\therefore$ Volume of liquid in cylinder

$$= \pi R^2 (h + y_f) \quad \dots(i)$$

Volume of liquid = [Volume of cylinder up to level MM]

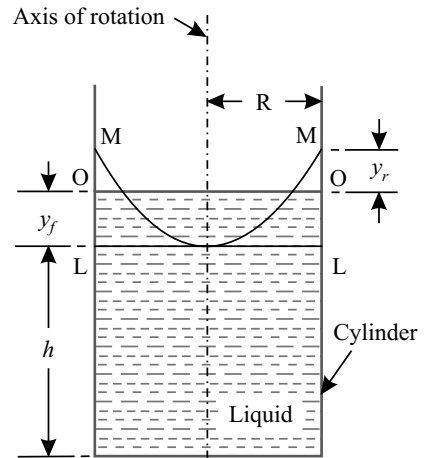


Fig. 6.67

$$\begin{aligned}
 & - [\text{volume of paraboloid}] \\
 & = (\pi R^2 \times \text{liquid height up to level } MM) \\
 & - \left(\frac{1}{2} \times \pi R^2 \times \text{height of paraboloid} \right) \\
 & = \pi R^2 \times (h + y_f + y_r) - \frac{\pi R^2}{2} (y_f + y_r) \\
 & = \pi R^2 h + \pi R^2 (y_f + y_r) - \frac{\pi R^2}{2} (y_f + y_r) \\
 & = \pi R^2 h + \frac{\pi R^2}{2} (y_f + y_r) \quad \dots(ii)
 \end{aligned}$$

Equating (i) and (ii), we get:

$$\pi R^2 (h + h_f) = \pi R^2 h + \frac{\pi R^2}{2} (y_f + y_r)$$

$$\pi R^2 h + \pi R^2 y_f = \pi R^2 h + \frac{\pi R^2}{2} y_f + \frac{\pi R^2}{2} y_r$$

$$\text{or,} \quad \pi R^2 y_f - \frac{\pi R^2}{2} y_f = \frac{\pi R^2}{2} y_r$$

$$\text{or,} \quad \frac{\pi R^2}{2} y_f = \frac{\pi R^2}{2} y_r$$

$$\text{or,} \quad y_f = y_r$$

i.e., Fall of liquid at centre = Rise of liquid at the ends**Proved.**

Example 6.66. A cylindrical tank 0.9 m in diameter and 2 m high open at top is filled with water to a depth of 1.5 m. It is rotated about its vertical axis at N r.p.m. Determine the value of N which will raise water level even with the brim. (GATE)

Solution. Refer to Fig. 6.68. Given: Radius, $R = \frac{0.9}{2} = 0.45$ m ; Length, = 2m; Initial height of water = 1.5 m.

Speed which will raise water level even with brim, N :

When the vessel is rotated, paraboloid is formed.

Volume of air before rotation = Volume of air after rotation

$$\pi R^2 \times 2 - \pi R^2 \times 1.5 = \frac{1}{2} \pi R^2 z$$

$$\text{or,} \quad z = 1.0 \text{ m}$$

Using the relation:

$$z = \frac{\omega^2 r^2}{2g}, \text{ we get:}$$

$$1.0 = \frac{\omega^2 R^2}{2 \times 9.81} \quad (\text{Here, } r = R)$$

$$\omega = \sqrt{\frac{1.0 \times 2 \times 9.81}{(0.45)^2}} = 9.843$$

$$\text{But,} \quad \omega = \frac{2\pi N}{60}$$

$$\therefore N = \frac{9.843 \times 60}{2\pi} = 93.99 \text{ r.p.m. (Ans.)}$$

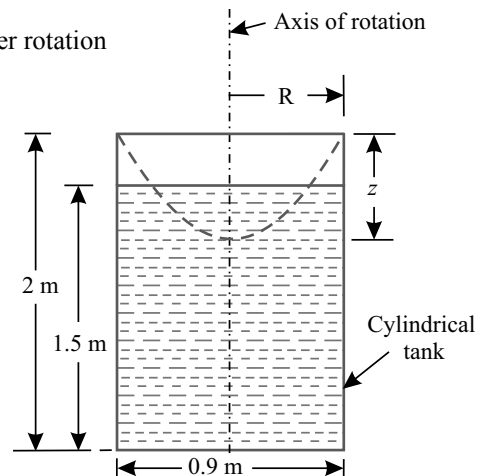


Fig. 6.68

Example. 6.67. Find the maximum speed of an open circular cylinder, having 180 mm diameter, 1200 mm length and containing water up to a height of 960 mm, at which it should be rotated about its vertical axis so that no water spills. (MU)

Solution. Given: Radius of the cylinder, $R = \frac{180}{2} = 90 \text{ mm} = 0.09 \text{ m}$

Length of the cylinder, $l = 1200 \text{ mm} = 1.2 \text{ m}$

Initial height of water, $h = 960 \text{ mm} = 0.96 \text{ m}$

Maximum speed of rotation, N :

Let, ω = Angular velocity of the cylinder when the water is about to spin.

We know, Rise of liquid at the ends = Fall of liquid at centre

But, Rise of liquid at the ends = Length of cylinder – initial height
 $= 1.2 - 0.96 = 0.24 \text{ m}$

$\therefore$ Fall of liquid at centre = 0.24 m

$\therefore$ Height of parabola = $0.24 + 0.24 = 0.48 \text{ m}$

$\therefore z = 0.48 \text{ m}$

Using the relation:

$$z = \frac{\omega^2 R^2}{2g}, \text{ we get}$$

$$0.48 = \frac{\omega^2 \times (0.09)^2}{2 \times 9.81}$$

$$\text{or, } \omega^2 = \frac{0.48 \times 2 \times 9.81}{(0.09)^2} = 1162.67 \quad \text{or } \omega = 34.09 \text{ rad/s}$$

$$\text{But, } \omega = \frac{2\pi N}{60} \quad \therefore 34.09 = \frac{2\pi N}{60}$$

$$\text{or, } N = \frac{34.09 \times 60}{2\pi} = 325.5 \text{ r.p.m (Ans.)}$$

Example 6.68. A 0.225m diameter cylinder is 1.5 m long and contains water up to a height of 1.05 m. Estimate the speed at which the cylinder may be rotated about its vertical axis so that the axial depth becomes zero.

Solution. Radius of the cylinder,

$$r = \frac{0.225}{2} = 0.1125 \text{ m}$$

Length of the cylinder, $l = 1.5 \text{ m}$

Initial height of water = 1.05 m

When axial depth is zero,

depth of paraboloid = 1.5 m

Speed of rotation N :

Using the relation:

$$z = \frac{\omega^2 R^2}{2g}, \text{ we get:}$$

$$1.5 = \frac{\omega^2 \times 0.1125^2}{2 \times 9.81}$$

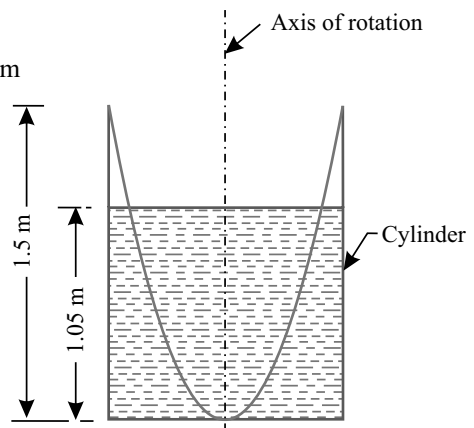


Fig. 6.69

$$\text{or,} \quad \omega^2 = \frac{1.5 \times 2 \times 9.81}{0.1125^2} = 2325.33$$

$$\text{or,} \quad \omega = 48.22 \text{ rad/s}$$

$$\text{But,} \quad \omega = \frac{2\pi N}{60}$$

$$\therefore \quad 48.22 = \frac{2\pi N}{60}$$

$$\text{or,} \quad N = \frac{48.22 \times 60}{2\pi} = \mathbf{460.46 \text{ r.p.m (Ans.)}}$$

Example 6.69. For example 6.68 find the difference in total pressure force due to rotation:

(i) At the bottom of cylinder, and

(ii) On the sides of the cylinder.

Solution. Given: Same as in example 6.68.

(i) **Difference in total pressure force at the bottom of the cylinder:**

Total pressure force at the bottom **before rotation**,

$$F_{\text{before rot.}} = wA\bar{h}$$

where, $w = 9810 \text{ N/m}^3$,

$$A = \text{Area of bottom} = \pi R^2 = \pi \times 0.1125^2 = 0.03976 \text{ m}^2$$

$$\bar{h} = 1.05 \text{ m.}$$

$$\therefore \quad F_{\text{before rot.}} = 9810 \times 0.03976 \times 1.05 = 409.55 \text{ N}$$

After rotation, the depth of water at the bottom is not constant and hence the pressure force due to the height of water *will not be constant*.

Consider an elementary ring of radius r and width dr as shown in Fig. 6.70. Let $z \left(= \frac{\omega^2 r^2}{2g} \right)$

be the height of water from the bottom of the tank up to free surface of water at a radius r .

Hydrostatic force on the ring at the bottom,

$$\begin{aligned} dF &= w \times \text{area of ring} \times z \\ &= 9810 \times 2\pi r \, dr \times \frac{\omega^2 r^2}{2g} \\ &= 9810 \times 2\pi r \times \frac{\omega^2}{2g} \times r^3 dr \\ &= 3141.6 \, \omega^2 r^3 dr \end{aligned}$$

Total pressure force at the bottom,

$$\begin{aligned} F_{\text{after rot.}} &= \int dF = \int_0^{0.1125} 3141.6 \, \omega^2 r^3 dr \\ &= 3141.6 \, \omega^2 \left[\frac{r^4}{4} \right]_0^{0.1125} \\ &[\because \omega = 48.22 \text{ rad/s, example 6.68}] \\ &= 3141.6 \times 48.22^2 \times \frac{0.1125^4}{4} \end{aligned}$$

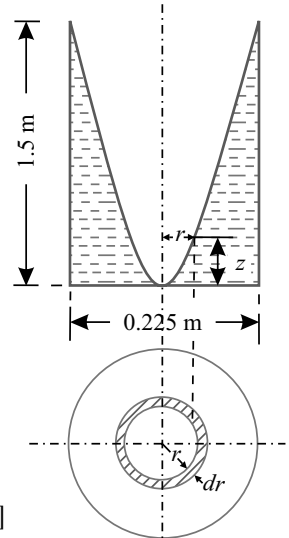


Fig. 6.70

$$= 292.52 \text{ N}$$

∴ Difference in pressure force at the bottom

$$\begin{aligned} &= F_{\text{before rot.}} - F_{\text{after rot.}} \\ &= 409.55 - 292.52 = \mathbf{117.03 \text{ N (Ans.)}} \end{aligned}$$

(ii) **Difference in total pressure force on the sides of the cylinder:**

Total pressure force on the sides of the cylinder **before rotation**,

$$F_{\text{before rot.}} = wA\bar{h}$$

where,

$$w = 9810 \text{ N/m}^3,$$

A = Surface area of the sides of the cylinder up to height of water

$$= \pi D \times \text{height of water}$$

$$= \pi \times 0.225 \times 1.05 = 0.7422 \text{ m}^2, \text{ and}$$

$\bar{h}$ = c.g. of the wetted area of the sides

$$= \frac{1}{2} \times 1.05 = 0.525 \text{ m.}$$

$$\therefore F_{\text{before rot.}} = 9810 \times 0.7422 \times 0.525 = 3822.5 \text{ N}$$

After rotation, the water is up to the top of the cylinder and force on the sides,

$$F_{\text{after rot.}} = w \times A \times \bar{h}$$

where,

$$w = 9810 \text{ N/m}^3$$

A = Wetted area of sides

$$= \pi D \times \text{height of water} = \pi \times 0.225 \times 1.5 = 1.06 \text{ m}^2, \text{ and}$$

$$\bar{h} = \frac{1}{2} \times \text{height of water} = \frac{1}{2} \times 1.5 = 0.75 \text{ m.}$$

$$\therefore F_{\text{after rot.}} = 9810 \times 1.06 \times 0.75 = 7798.95 \text{ N}$$

∴ Difference in pressure force on the sides

$$\begin{aligned} &= F_{\text{after rot.}} - F_{\text{before rot.}} \\ &= 7798.95 - 3822.5 = \mathbf{3976.45 \text{ N (Ans.)}} \end{aligned}$$

Example 6.70. An open cylindrical vessel 180 mm in diameter and 450 mm deep is filled with water up to the top. Estimate the volume of water left in the vessel when it is rotated about its vertical axis:

(i) With a speed of 200 r.p.m., and

(ii) With a speed of 400 r.p.m.

Solution. Radius of the vessel, $R = \frac{180}{2} = 90 \text{ mm} = 0.09 \text{ m}$

Initial height of water = 450 mm = 0.45 m

$$\therefore \text{Initial volume of water} = \pi \times 0.09^2 \times 0.45 = 0.01145 \text{ m}^3$$

(i) **Volume of water left at a speed of 200 r.p.m.:**

$$\text{Angular speed, } \omega = \frac{2\pi N}{60} = \frac{2\pi \times 200}{60} = 20.94 \text{ rad./s}$$

Height of parabola is given by:

$$z = \frac{\omega^2 R^2}{2g} = \frac{(20.94)^2 \times 0.09^2}{2 \times 9.81} = 0.181 \text{ m}$$

Since the vessel is initially full of water, water will be spilled when it is rotated.

Volume of water spilled = Volume of paraboloid.

$$\begin{aligned}\text{But, volume of paraboloid,} &= \frac{1}{2} (\text{Area of cross-section} \times \text{height of parabola}) \\ &= \frac{1}{2} \times \pi R^2 \times z = \frac{1}{2} \times \pi \times 0.09^2 \times 0.181 = 0.002303 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\therefore \text{Volume of water left} &= \text{Initial volume} - \text{volume of water spilled} \\ &= 0.01145 - 0.00203 = \mathbf{0.009147 \text{ m}^3 \text{ (Ans.)}}\end{aligned}$$

(ii) Volume of water left at a speed of 400 r.p.m.:

$$\text{Angular speed, } \omega = \frac{2\pi N}{60} = \frac{2\pi \times 400}{60} = 41.88 \text{ rad/s}$$

$$\text{Height of the parabola, } z = \frac{\omega^2 R^2}{2g} = \frac{(41.88)^2 \times 0.09^2}{2 \times 9.81} = 0.724 \text{ m}$$

Since the height of parabola is more than the height of the cylinder, therefore, the shape of the imaginary parabola will be as shown in Fig. 6.71

$$\begin{aligned}\text{Let, } r &= \text{Radius of the parabola at the bottom of the vessel} \\ &= 0.724 - 0.45 = 0.274 \text{ m}\end{aligned}$$

Volume of water left in the vessel

$$\begin{aligned}&= \text{Volume of water in the portions } LMN \text{ and } PQS \\ &= \text{Initial volume of water} - \text{volume of paraboloid } LOS + \text{volume of paraboloid } NOP\end{aligned}$$

Now, volume of paraboloid LOS

$$\begin{aligned}&= \frac{1}{2} (\pi R^2 \times \text{height of parabola}) \\ &= \frac{1}{2} \times \pi \times 0.092 \times 0.724 = 0.00921 \text{ m}^3\end{aligned}$$

For the imaginary parabola (NOP),

$$\omega = 41.88 \text{ r.p.m.}$$

$$z = 0.274 \text{ m}$$

$$r = \text{Radius at the bottom of the vessel}$$

Using the relation,

$$\begin{aligned}z &= \frac{\omega^2 r^2}{2g}, \text{ we get:} \\ 0.274 &= \frac{41.88^2 \times r^2}{2 \times 9.81} \\ \text{or, } r^2 &= \frac{0.274 \times 2 \times 9.81}{41.88^2} = 0.003065 \text{ m}^2\end{aligned}$$

$$\therefore r = 0.0554 \text{ m}$$

$\therefore$ Volume of paraboloid NOP

$$\begin{aligned}&= \frac{1}{2} (\text{area at the top of the imaginary parabola} \times \text{height of parabola}) \\ &= \frac{1}{2} \times \pi r^2 \times 0.274 \\ &= \frac{1}{2} \times \pi \times 0.0554^2 \times 0.274 = 0.00132 \text{ m}^3\end{aligned}$$

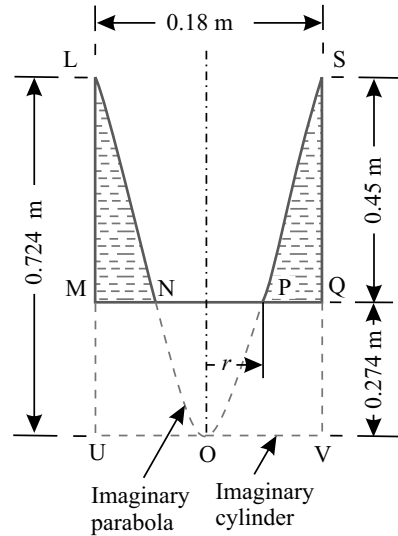


Fig. 6.71

∴ Volume of water left

$$= 0.01145 - 0.00921 + 0.00132$$

$$= \mathbf{0.00356 \text{ m}^3 \text{ (Ans.)}}$$

6.11.5. Rotation of Liquid in a Closed Cylindrical Vessel

When a cylindrical vessel sealed at the top and filled with some liquid is rotated about its vertical geometrical axis, the shape of paraboloid formed due to rotation of the vessel will be as shown in Fig 6.72 for different speeds of rotation.

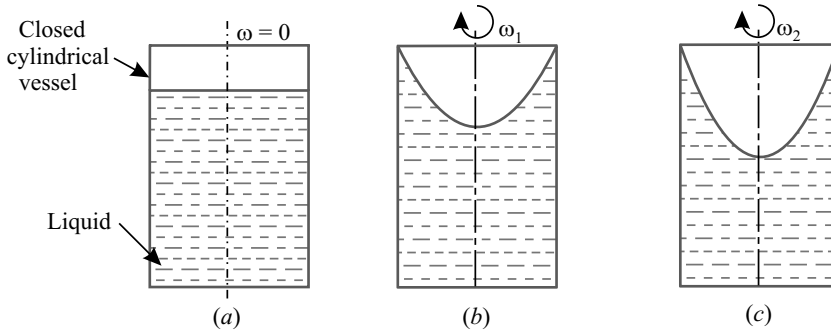


Fig. 6.72. Rotation of liquid in a closed cylindrical vessel.

- Fig. 6.72 (a) shows the cylindrical vessel when it is *stationary* (i.e., it is not rotated, $\omega = 0$)
- Fig. 6.72 (b) shows the shape of the paraboloid formed when the *speed of rotation* is ω_1 .
- Fig. 6.72 (c) shows the shape of the paraboloid formed when the *speed of rotation* is ω_2 ($\omega_2 > \omega_1$). In this case the following are unknown:

1. Radius of the parabola at the top of the vessel, and
2. Height of the parabola formed corresponding to the angular speed, ω_2 .

To solve these, two unknown equations are required:

(i) One equation is: $z = \frac{\omega^2 r^2}{2g}$...(i)

(ii) Second equation is from the fact that for closed Vessel:

Volume of air before rotation = Volume of air after rotation

Volume of air before rotation = Volume of closed vessel – volume of liquid in the vessel

Volume of air after rotation = Volume of paraboloid formed

$$= \frac{1}{2} \pi r^2 \times z.$$

Example 6.71. A cylindrical vessel closed at the top and bottom is 0.24 m in diameter, 1.44 m long and contains water up to height of 0.96 m.

- (i) Find the height of paraboloid formed, if it is rotated at 480 r.p.m about its vertical axis.
- (ii) Find the speed of rotation of the vessel, when axial depth of water is zero.

Solution. Radius of the vessel,

$$R = \frac{0.24}{2} = 0.12 \text{ m}$$

Length of the vessel, $L = 1.44 \text{ m}$

Initial height of water = 0.96 m

(i) Height of the paraboloid, z :Speed, $N = 480$ r.p.m.

$$\therefore \omega = \frac{2\pi N}{60} = \frac{2\pi \times 480}{60} = 50.26 \text{ rad/s}$$

When the vessel is rotated, paraboloid is formed (Fig. 6.73)

Let, r = Radius of paraboloid at the top of the vessel, and z = Height of the paraboloid.

As the vessel is closed one, therefore,

Volume of air before rotation = Volume of air after rotation

$$\text{or, } \pi R^2 L - \pi R^2 \times 0.96 = \frac{1}{2} \pi r^2 z$$

$$\text{or, } \pi R^2 (1.44 - 0.96) = \frac{1}{2} \pi r^2 z$$

$$\text{or, } r^2 z = 2 \times 0.12^2 (1.44 - 0.96) = 0.0138 \quad \dots(i)$$

Using the relation,

$$z = \frac{\omega^2 r^2}{2g}, \text{ we get:}$$

$$z = \frac{50.26^2 \times r^2}{2 \times 9.81} = 128.75 r^2$$

$$\therefore r^2 = \frac{z}{128.75}$$

Substituting this value of r^2 in (i), we get:

$$= \frac{z}{128.75} \times z = 0.0138$$

$$\therefore z^2 = 0.0138 \times 128.75 = 1.777$$

$$\text{or, } z = 1.333 \text{ m (Ans.)}$$

(ii) Speed of rotation, N :Let ω is the angular velocity, when axial depth is zero.

When axial depth is zero:

The height of paraboloid at the top = r

Using the relation,

$$z = \frac{\omega^2 r^2}{2g}, \text{ we get:}$$

$$1.44 = \frac{\omega^2 r^2}{2 \times 9.81}$$

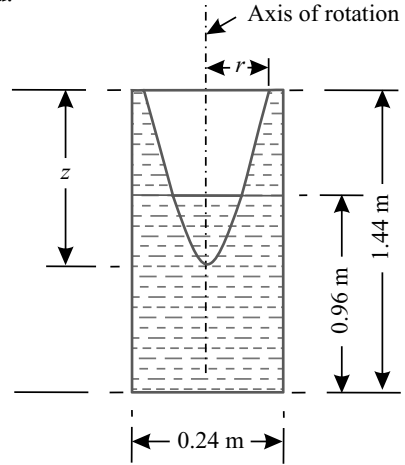
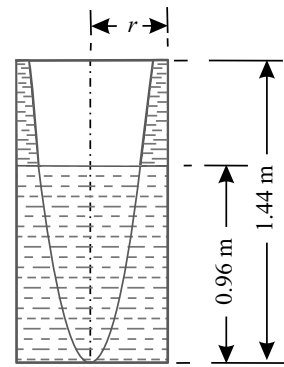
$$\therefore \omega^2 r^2 = 1.44 \times 2 \times 9.81 = 28.25 \quad \dots(i)$$

Volume of air before rotation = Volume of air after rotation

$$\therefore \pi R^2 (1.44 - 0.96) = \text{Volume of paraboloid}$$

$$= \frac{1}{2} \pi r^2 z = \frac{\pi r^2}{2} \times 1.44$$

$$\text{or, } \pi \times 0.12^2 \times 0.48 = 0.72 \pi r^2$$

**Fig. 6.73****Fig. 6.74**

$$\therefore r^2 = \frac{0.12^2 \times 0.48}{0.72} = 0.0096$$

Substituting the value r^2 in (i), we get:

$$\omega^2 \times 0.0096 = 28.25$$

$$\therefore \omega = \left(\frac{28.25}{0.0096} \right)^{1/2} = 54.25 \text{ rad/s}$$

But, $\omega = \frac{2\pi N}{60} \quad \therefore 54.25 = \frac{2\pi N}{60}$

or, $N = \frac{54.25 \times 60}{2\pi} = \mathbf{518 \text{ r.p.m. (Ans.)}}$

Example 6.72. A vessel, cylindrical in shape and closed at the top and bottom, is 0.24 m in diameter, 1.44 m long and contains water up to a height of 0.96 m. If it is rotated at 700 r.p.m. what is the area uncovered at the bottom of the tank?

Solution. Radius of vessel, $R = \frac{0.24}{2} = 0.12 \text{ m}$

Length of the vessel, $L = 1.44 \text{ m}$

Initial height of water = 0.96 m

Speed, $N = 700 \text{ r.p.m.}$

$\therefore$ Angular speed, $\omega = \frac{2\pi N}{60} = \frac{2\pi \times 700}{60} = 73.3 \text{ rad/s}$

Area uncovered at the bottom of the tank:

If the tank is not closed at the top and also is very long, then the height of parabola corresponding to $\omega = 73.3 \text{ rad/s}$ will be

$$= \frac{\omega^2 R^2}{2g} = \frac{73.3^2 \times 0.12^2}{2 \times 9.81} = 3.943 \text{ m}$$

From Fig. 6.75, $y_1 + 1.44 + y_2 = 3.943$

or, $y_1 + y_2 = 2.503 \text{ m}$

For the parabola GOH , we have:

$$(1.44 + y_1) = \frac{\omega^2 r_1^2}{2g} = \frac{73.3^2 \times r_1^2}{2 \times 9.81} = 273.85 r_1^2 \quad \dots(ii)$$

For the parabola IOJ , we have:

$$y_1 = \frac{\omega^2 r_2^2}{2g} = \frac{73.3^2 \times r_2^2}{2 \times 9.81} = 273.85 r_2^2 \quad \dots(iii)$$

Now, Volume of air before rotation

$$= \text{Volume of air after rotation}$$

Volume of air before rotation

$$= \pi R^2 (1.44 - 0.96) = \pi \times 0.12^2 \times 0.48$$

$$= 0.0217 \text{ m}^3 \quad \dots(iv)$$

Volume of air after rotation = Volume of paraboloid GOH – volume of paraboloid IOJ .

$$= \frac{1}{2} \times \pi r_1^2 \times (1.44 + y_1) - \frac{1}{2} \pi r_2^2 \times y_1 \quad \dots(v)$$

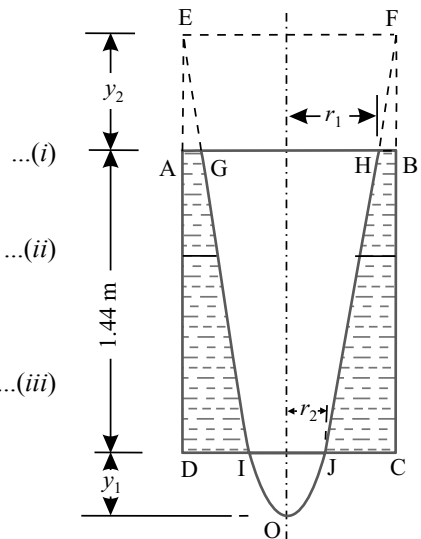


Fig. 6.75

Equating (iv) and (v), we get:

$$0.0217 = \frac{\pi}{2} [r_1^2 (1.44 + y_1) - r_2^2 y_1] \quad \dots(vi)$$

But from (ii), we have: $r_1^2 = \left(\frac{1.44 + y_1}{273.85} \right)$

Substituting this value of r_1^2 in (vi), we get:

$$0.0217 = \frac{\pi}{2} \left[\left(\frac{1.44 + y_1}{273.85} \right) (1.44 + y_1) - r_2^2 y_1 \right]$$

Now, substituting the value of y_1 from (iii) in the above eqn., we get:

$$0.0217 = \frac{\pi}{2} \left[\left(\frac{1.44 + 273.85 r_2^2}{273.85} \right) (1.44 + 273.85 r_2^2) - r_2^2 \times 273.85 r_2^2 \right]$$

$$\text{or, } \frac{0.0217 \times 2 \times 273.85}{\pi} = (1.44 + 273.85 r_2^2)^2 - (273.85)^2 r_2^4$$

$$\left[\text{Multiplying both sides by } 273.85 \times \frac{2}{\pi} \right]$$

$$\text{or, } 3.783 = (2.074 + 788.69 r_2^2 + 74994 r_2^4) - 74994 r_2^4$$

$$\text{or, } 788.69 r_2^2 = 1.709$$

$$\text{or, } r_2^2 = 0.002167 \text{ m}^2$$

$\therefore$ Area uncovered at the base

$$= \pi r_2^2 = \pi \times 0.002167 = \mathbf{0.0068 \text{ m}^2 \text{ (Ans.)}}$$

Example 6.73. A vessel, cylindrical in shape and closed at the top and bottom, is 0.45 m in diameter and 1.5 m long. It contains water up to a depth of 1.2 m. The air above the water surface is at a pressure of 90 kN/m². If the vessel is rotated at a speed of 300 r.p.m. about its vertical axis find the pressure head at the bottom of the vessel:

(i) At the centre, and (ii) At the edge.

Solution. Radius of the vessel,

$$R = \frac{0.45}{2} = 0.225 \text{ m}$$

Length of the vessel, $L = 1.5 \text{ m}$

Initial height of water = 1.2 m

Pressure of air above water, $p = 90 \text{ kN/m}^2$

$$\text{Head due to pressure, } h = \frac{p}{w} = \frac{90}{9.81} = 9.17 \text{ m}$$

($\because w$ for water = 9.81 kN/m³)

Speed of the vessel, $N = 300 \text{ r.p.m.}$

$\therefore$ Angular velocity,

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 300}{60} = 31.41 \text{ rad/s}$$

Let, y_1 = Height of paraboloid formed (assuming vessel to be very long and open at the top).

$$\text{Then, } y_1 = \frac{\omega^2 R^2}{2g} = \frac{31.41^2 \times 0.225^2}{2 \times 9.81} = 2.545 \text{ m} \quad \dots(i)$$

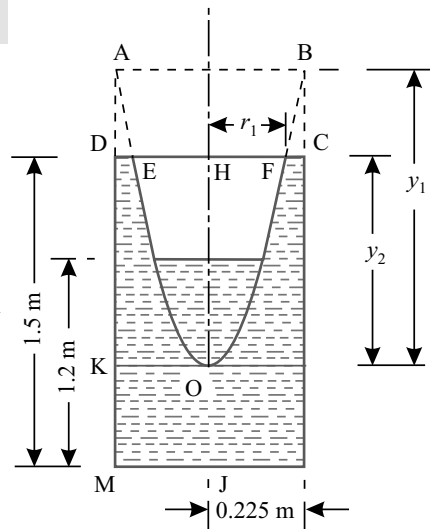


Fig. 6.76

Let, r_1 = Radius of actual parabola of height y_2 .

$$\text{Then, } y_2 = \frac{\omega^2 r_1^2}{2g} = \frac{31.41^2 \times r_1^2}{2 \times 9.81} = 50.28 r_1^2 \quad \dots(ii)$$

The volume of air before rotation

$$= \pi R^2 (1.5 - 1.2) = \pi \times 0.225^2 \times 0.3 = 0.0477 \text{ m}^3$$

Volume of air after rotation

$$= \frac{1}{2} \pi r_1^2 \times y_2$$

But, Volume of air before rotation = Volume of air after rotation

$$\therefore 0.0477 = \frac{1}{2} \pi r_1^2 y_2$$

$$\text{But from (ii), } y_2 = 50.28 r_1^2$$

$$\therefore 0.0477 = \frac{1}{2} \pi r_1^2 \times 50.28 r_1^2$$

$$\text{or, } r_1^4 = \frac{0.0477 \times 2}{\pi \times 50.28} = 6.039 \times 10^{-4} \text{ m}^4$$

$$\text{or, } r_1 = (6.039 \times 10^{-4})^{1/4} = 0.156 \text{ m}$$

Substituting the value of r_1 in (ii), we get:

$$y_2 = 50.28 r_1^2 = 50.28 \times (0.156)^2 = 1.224 \text{ m}$$

Pressure head at bottom of the vessel:

(i) *At the centre:*

The pressure head at the centre i.e., at J.

$$= \text{Pressure head due to air} + OJ$$

$$= 9.17 + (HJ - HO)$$

$$= 9.17 + (1.5 - 1.224)$$

$$= \mathbf{9.446 \text{ m of water (Ans.)}}$$

$$(\because OJ = HJ - HO)$$

$$\left\{ \begin{array}{l} HJ = 1.5 \text{ m} \\ HO = y_2 = 1.224 \text{ m} \end{array} \right\}$$

(ii) *At the edge:*

The pressure head at the edge M

$$= \text{Pressure head due to air} + \text{height of water above } M$$

$$= 9.17 + AM$$

$$= 9.17 + (AK + KM) = 9.17 + (y_1 + KM)$$

$$= 9.17 + (y_1 + OJ)$$

$$= 9.17 + 2.545 + 0.276$$

$$(\because y_1 = 2.545 \text{ m})$$

$$= \mathbf{11.99 \text{ m of water (Ans.)}}$$

$$\left[\begin{array}{l} OJ = HJ - HO \\ = 1.5 - 1.224 = 0.276 \text{ m} \end{array} \right]$$

Example 6.74. A vessel cylindrical in shape and closed at the top and bottom is of radius R and height H . The vessel is completely filled with water. If it is rotated about its vertical axis with a speed ω radians/sec., what is the total pressure force exerted by water on the top and bottom of the vessel?

Solution. Radius of the vessel = R

Height of the vessel = H

Angular speed = ω

As the vessel is closed and completely filled with water, and when it is rotated, the water will exert force on the complete top and bottom of the vessel.

Total pressure force exerted on the top of the vessel, F_{top} :

As the top of the vessel is in contact with water and is in horizontal plane, the pressure variation at any radius in horizontal plane is given as:

$$\begin{aligned}\frac{\partial p}{\partial r} &= \frac{\rho v^2}{r} & [\text{Eqn. (6.35)}] \\ &= \frac{\rho \omega^2 r^2}{r} = \rho \omega^2 r & [\because v = \omega r]\end{aligned}$$

Integrating both sides, we get:

$$\int dp = \int \rho \omega^2 r dr$$

$$\text{or, } p = \frac{\rho}{2} \omega^2 r^2$$

Refer to Fig. 6.77. Consider an elementary ring of radius r and width dr on the top of the vessel.

Area of the elementary ring = $2\pi r dr$

$$\begin{aligned}\text{Force on the elementary ring} &= \text{Intensity of pressure} \\ &\quad \times \text{area of ring} \\ &= p \times 2\pi r dr \\ &= \frac{\rho}{2} \omega^2 r^2 \times 2\pi r dr & \left(\because p = \frac{\rho}{2} \omega^2 r^2 \right)\end{aligned}$$

$\therefore$ Total force on the top of the vessel,

$$\begin{aligned}F_{\text{top}} &= \int_0^R \frac{\rho}{2} \omega^2 r^2 \times 2\pi r dr \\ &= \frac{\rho}{2} \omega^2 \times 2\pi \int_0^R r^3 dr \\ &= \frac{\rho}{2} \omega^2 \times 2\pi \left[\frac{r^4}{4} \right]_0^R = \frac{\rho}{2} \omega^2 \times 2\pi \times \frac{R^4}{4} \\ &= \frac{\rho}{4} \omega^2 \times \pi R^4\end{aligned}$$

$$\text{i.e., } F_{\text{top}} = \frac{\rho}{4} \omega^2 \pi R^4 \quad \dots(6.39)$$

Total pressure force exerted on the bottom of the vessel, F_{bottom} :

F_{bottom} = Total force on the top of the vessel + Weight of water in cylinder

$$= \frac{\rho}{4} \omega^2 \pi R^4 + \omega \times \pi R^2 \times H$$

$$\text{i.e., } F_{\text{bottom}} = \frac{\rho}{4} \omega \pi R^4 + \omega \pi R^2 H \quad \dots(6.40)$$

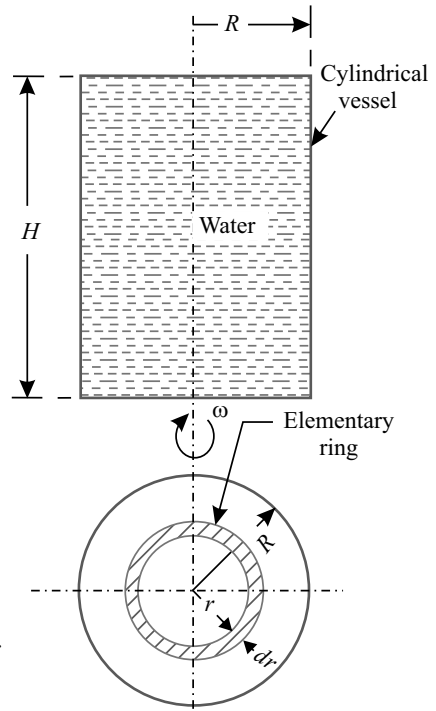


Fig. 6.77

Example 6.75. A vessel cylindrical in shape and closed at the top and bottom is 0.3 m in diameter and 0.225 m in height. The vessel is completely filled with water. If it is rotated about its vertical axis with a speed of 300 r.p.m., what is the total pressure force exerted by water on the top and bottom of vessel ?

Solution. Radius of the vessel, $R = \frac{0.3}{2} = 0.15 \text{ m}$

Height of the vessel, $H = 0.225 \text{ m}$.

Speed, $N = 300 \text{ r.p.m.}$

$\therefore$ Angular speed, $\omega = \frac{2\pi N}{60} = \frac{2\pi \times 300}{60} = 31.41 \text{ rad/sec}$

Total pressure force exerted by water on the top of the vessel, F_{top} :

Using the relation:
$$F_{\text{top}} = \frac{\rho}{4} \times \omega^2 \times \pi R^4 \quad [\text{Eqn. (6.39)}]$$

$$= \frac{w}{g \times 4} \omega^2 \pi R^4 \quad \left[\because \rho = \frac{w}{g}, w = 9.81 \text{ kN/m}^3 \right]$$

$$= \frac{9.81}{9.81 \times 4} \times (31.41)^2 \times \pi \times (0.15)^4 = 0.392 \text{ kN}$$

i.e., $F_{\text{top}} = 0.392 \text{ kN (Ans.)}$

Total pressure force exerted by water on the bottom of the vessel, F_{bottom} :

$$F_{\text{bottom}} = F_{\text{top}} + w \times \pi R^2 \times H$$

$$= 0.392 + 9.81 \times \pi \times 0.15^2 \times 0.225 = 0.548 \text{ kN}$$

$\therefore F_{\text{bottom}} = 0.548 \text{ kN (Ans.)}$

Example 6.76. A closed vertical cylinder 0.4 m in diameter and 0.4 m in height is completely filled with oil of specific gravity 0.80. If the cylinder is rotated about its vertical axis at 200 rpm, calculate the thrust of oil on top and bottom covers of the cylinder. (UPTU)

Solution. Given: $R = \frac{0.4}{2} = 0.2 \text{ m}; H = 0.4 \text{ m}; S = 0.8; N = 200 \text{ r.p.m.}$

F_{top} :

Using the relation:
$$F_{\text{top}} = \frac{\rho}{4} \omega^2 \times \pi R^4 \quad [\text{Eqn. (6.39)}]$$

$$= \frac{0.8 \times 1000}{4} \times \left(\frac{2\pi \times 200}{60} \right)^2 \times \pi \times (0.2)^4 = 440.98 \text{ N (Ans.)}$$

Again, using the relation:
$$F_{\text{bottom}} = F_{\text{top}} + w \times \pi R^2 \times H$$

$$= 440.98 + (0.8 \times 1000 \times 9.81) \times \pi \times 0.2^2 \times 0.4$$

$$= 835.46 \text{ N (Ans.)}$$

Example 6.77. A hollow sphere of radius R , completely filled with the liquid, is rotated about its vertical axis at an angular speed ω . Locate the circular line maximum pressure with respect to the centre of the sphere. (Delhi University)

Solution. The circular line of maximum pressure will be a horizontal circle on the internal surface of the circle aa . Let its location be at a distance h below the centre of the sphere. All points on the circle aa will be subjected to a centrifugal pressure $\frac{1}{2} \rho \omega^2 r^2$, and a hydrostatic pressure $\rho g (R + h)$.

The total pressure on any point,

$$\begin{aligned}
 p &= \frac{1}{2}\rho\omega^2 r^2 + \rho g(R + h) \\
 &= \frac{1}{2}\rho\omega^2 r^2 + \rho g \left\{ R + \sqrt{R^2 - r^2} \right\} \quad (\because h = \sqrt{R^2 - r^2})
 \end{aligned}$$

For p to be maximum:

$$\frac{dp}{dr} = 0 = \frac{1}{2}\rho\omega^2 \times 2r + \rho g \left\{ \frac{1}{2}(R - r^2)^{-1/2} \times (-2r) \right\}$$

$$\text{or, } \frac{\omega^2}{g} = \frac{1}{\sqrt{R^2 - r^2}} \quad \text{or} \quad (R^2 - r^2) = \left(\frac{g}{\omega^2} \right)^2$$

$$\text{or, } h^2 = R^2 - r^2 = \left(\frac{g}{\omega^2} \right)^2 \quad \text{or} \quad h = \frac{g}{\omega^2}$$

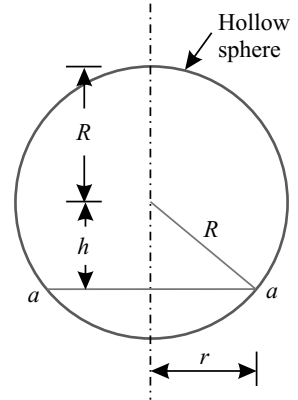


Fig. 6.78

Thus the circular line of maximum pressure is a horizontal circle, at a distance $h = \frac{g}{\omega^2}$ below the centre of the sphere. (Ans.)

6.11.6. Equation of Free Vortex Flow

In the case of free vortex flow, from eqn. (6.34), we have:

$$v = \frac{C}{r}$$

Substituting the value of v in eqn. (6.37), we get:

$$\begin{aligned}
 dp &= \frac{\rho v^2}{r} dr - \rho g dz \\
 &= \rho \times \frac{C^2}{r^2 \times r} dr - \rho g dz \\
 &= \frac{\rho C^2}{r^3} dr - \rho g dz
 \end{aligned}$$

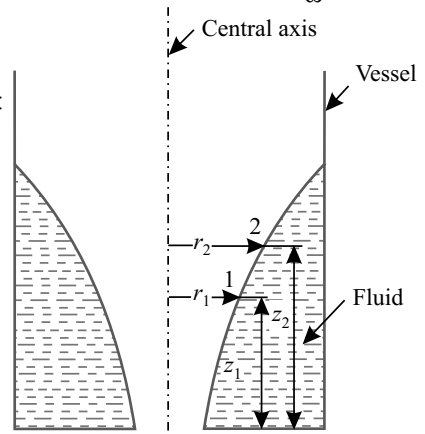


Fig. 6.79

Refer to Fig. 6.79. Consider two points 1 and 2 in the fluid having radii r_1 and r_2 respectively from the central axis, their heights being z_1 and z_2 from bottom of the vessel.

Integrating the above equation for the points 1 and 2, we get:

$$\int_1^2 dp = \int_1^2 \frac{\rho C^2}{r^3} dr - \int_1^2 \rho g dz$$

$$\begin{aligned}
 \text{or, } p_2 - p_1 &= \rho C^2 \int_1^2 \frac{dr}{r^3} - \rho g \int_1^2 dz \\
 &= \rho C^2 \left[\frac{r^{-3+1}}{-3+1} \right]_1^2 - \rho g (z_2 - z_1) \\
 &= \rho C^2 \left[-\frac{1}{2r^2} \right]_1^2 - \rho g (z_2 - z_1) \\
 &= -\frac{\rho C^2}{2} \left[\frac{1}{r_2^2} - \frac{1}{r_1^2} \right] - \rho g (z_2 - z_1)
 \end{aligned}$$

$$\begin{aligned}
 &= -\frac{\rho}{2}[v_2^2 - v_1^2] - \rho g(z_2 - z_1) & \left(\because v_2 = \frac{C}{r_2}, v_1 = \frac{C}{r_1} \right) \\
 &= \frac{\rho}{2}(v_1^2 - v_2^2) - \rho g(z_2 - z_1)
 \end{aligned}$$

Dividing both sides by ρg , we get:

$$\begin{aligned}
 \frac{p_2 - p_1}{\rho g} &= \frac{v_1^2 - v_2^2}{2g} - (z_2 - z_1) \\
 \text{or,} \quad \left(\frac{p_2}{\rho g} - \frac{p_1}{\rho g} \right) &= \left(\frac{v_1^2}{2g} - \frac{v_2^2}{2g} \right) + (z_1 - z_2) \\
 \text{or,} \quad \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 &= \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2 \quad \dots(6.41)
 \end{aligned}$$

Eqn. (6.41) is the Bernoulli's equation. Hence *Bernoulli's equation is applicable in the case of free vortex flow.*

Example 6.78. In a free cylindrical vortex flow, at a point in the fluid at a radius of 300 mm and a height of 150 mm, the velocity and pressure are 15 m/s and 120 kN/m² respectively. If the fluid is air having weight density of 0.012 kN/m³, find the pressure at a radius of 600 mm and at a height of 300 mm.

Solution. At point 1:

$$\begin{aligned}
 \text{Radius, } r_1 &= 300 \text{ mm} = 0.3 \text{ m} \\
 \text{Height, } z_1 &= 150 \text{ mm} = 0.15 \text{ m} \\
 \text{Velocity, } v_1 &= 15 \text{ m/s} \\
 \text{Pressure, } p_1 &= 120 \text{ kN/m}^2
 \end{aligned}$$

At point 2:

$$\begin{aligned}
 \text{Radius, } r_2 &= 600 \text{ mm} = 0.6 \text{ m} \\
 \text{Height, } z_2 &= 300 \text{ mm} = 0.3 \text{ m} \\
 \text{Density of air, } w &= 0.012 \text{ kN/m}^3
 \end{aligned}$$

Pressure, p_2 :

We know, for free vortex flow:

$$v \times r = \text{constant} \quad [\text{Eqn. (6.33)}]$$

$$\therefore v_1 r_1 = v_2 r_2$$

$$\text{or,} \quad v_2 = \frac{v_1 r_1}{r_2} = \frac{15 \times 0.3}{0.6} = 7.5 \text{ m/s}$$

Using the equation:

$$\begin{aligned}
 \frac{p_1}{w} + \frac{v_1^2}{2g} + z_1 &= \frac{p_2}{w} + \frac{v_2^2}{2g} + z_2 \\
 \text{or,} \quad \frac{120}{0.012} + \frac{15^2}{2 \times 9.81} + 0.15 &= \frac{p_2}{w} + \frac{7.5^2}{2 \times 9.81} + 0.3 \\
 \text{or,} \quad 10000 + 11.47 + 0.15 &= \frac{p_2}{w} + 2.86 + 0.3 \\
 \text{or,} \quad 10011.62 &= \frac{p_2}{w} + 3.16
 \end{aligned}$$

or, $\frac{p_2}{w} = 10008.46$

or, $p_2 = 0.012 \times 10008.46 = \mathbf{120.1 \text{ kN/m}^2 \text{ (Ans.)}}$

Example 6.79. Two stationary, horizontal flat plates with an external diameter of 400 mm are placed 10 mm apart. A vertical pipe 50 mm in diameter delivers $0.005 \text{ m}^3/\text{s}$ of water to the centre of the plates. The water is discharged to the periphery of the plates at atmospheric pressure of 98 kN/m^2 . Assuming radial flow and neglecting losses, determine the absolute pressure at the entrance of the flow. [UPSC Exams.]

Solution. Diameter of annular space, $d_i = 50 \text{ mm} = 0.05 \text{ m}$

External diameter of the plate, $d_0 = 400 \text{ mm} = 0.4 \text{ m}$

Distance between the plates, $t = 10 \text{ mm} = 0.01 \text{ m}$

Atmospheric pressure, $p_0 = 98 \text{ kN/m}^2$

Discharge, $Q = 0.005 \text{ m}^3/\text{s}$

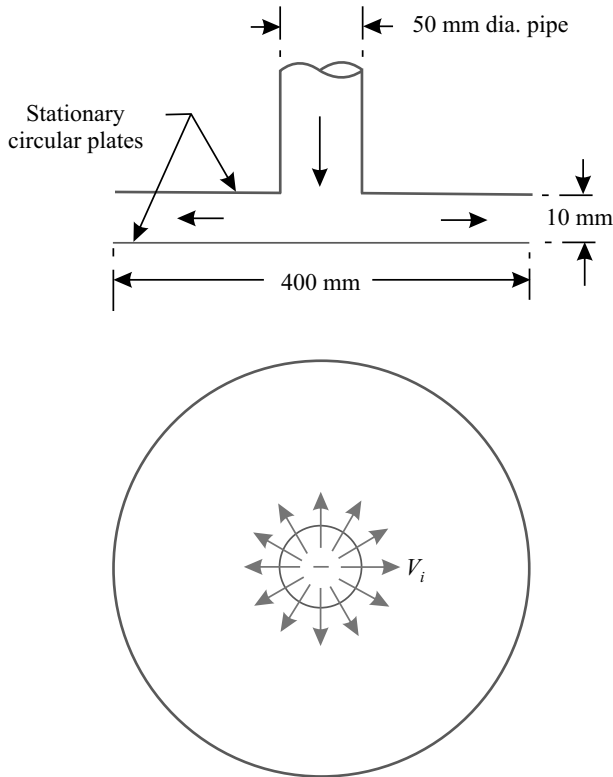


Fig. 6.80

Absolute pressure at the entrance of the flow, p_i :

Using the continuity equation, the velocity at the entrance to the annular space,

$$V_i = \frac{Q}{\pi d_i t} = \frac{0.005}{\pi \times 0.05 \times 0.01} = 3.18 \text{ m/s}$$

Velocity at the periphery of plates,

$$V_0 = \frac{Q}{\pi d_0 t} = \frac{0.005}{\pi \times 0.4 \times 0.01} = 0.398 \text{ m/s}$$

Applying the Bernoulli's equation between the inlet to and exit from plates, we get:

$$\frac{p_i}{w} + \frac{V_i^2}{2g} + z_i = \frac{p_0}{w} + \frac{V_0^2}{2g} + z_0$$

$$\frac{p_i}{w} + \frac{3.18^2}{2 \times 9.81} = \frac{98 \times 10^3}{9810} + \frac{0.398^2}{2 \times 9.81} \quad (\because z_i = z_0)$$

or, $\frac{p_i}{w} + 0.515 = 9.989 + 0.00807$

or, $\frac{p_i}{w} = 9.482$ or $p_i = 9810 \times 9.482 = 93018 \text{ N/m}^2$ or 93 kN/m^2

Hence, $p_i = 93 \text{ kN/m}^2$ (Ans.)

6.12. LIQUIDS IN RELATIVE EQUILIBRIUM

When a tank filled with a liquid is made to move with a constant acceleration, initially the fluid particles will move relative to each other and to the boundaries of the tank but, after a certain duration of time there will not be any relative movement between the fluid particles and boundaries of the container and the *whole fluid mass moves as a single unit* (A similar situation arises when the fluid mass is made to rotate with a uniform velocity). When such motion occurs, the fluids are said to be in “**relative equilibrium**”. Under such circumstances, since there is relative motion, the fluid is *not subjected to shearing forces*. Furthermore, the *fluid pressure acts normal to the surface in contact with it*.

Analysis of the fluid masses subjected to acceleration or deceleration can be made by using the principles of hydrostatics and giving due considerations to the effects of accelerating or decelerating forces.

6.12.1. Liquid in a Container Subjected to Uniform Acceleration in the Horizontal Direction

Consider a tank filled with liquid and being *accelerated horizontally to the right with uniform acceleration a_x* .

After slashing of the liquid particles for some time the motion of the liquid stabilizes and the liquid moves as a solid mass under the action of accelerating force. The final position of the liquid in the tank is as shown in Fig. 6.81, slope being upwards in the direction opposite to that of horizontal acceleration.

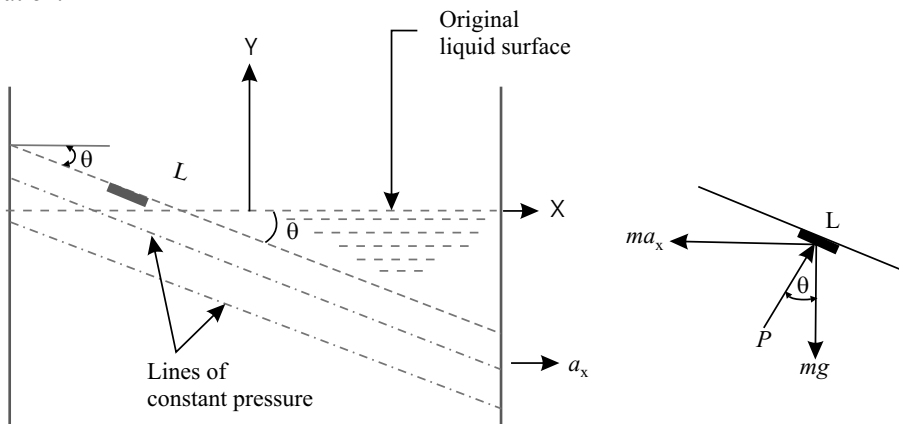


Fig. 6.81. Liquid under constant linear acceleration in horizontal direction.

Now, let us consider the equilibrium of a fluid particle L lying on the free surface. The pressure force P exerted by the surrounding fluid on particle L is normal to the free surface. The particle is subjected to the following forces:

- (i) Normal pressure force P ,
- (ii) Weight mg acting vertically downwards, and
- (iii) The accelerating force ma_x acting in horizontal direction.

Resolving horizontally and vertically respectively, we have:

$$P \sin \theta = ma_x \quad \dots(i)$$

$$P \cos \theta = mg \quad \dots(ii)$$

Dividing (i) by (ii), we get:

$$\tan \theta = \frac{a_x}{g} \quad \dots(6.42)$$

Since the term $\frac{a_x}{g}$ is constant at all points on the free liquid surface, hence $\tan \theta$ is constant and consequently the free surface is a straight line inclined at θ (downward) along the direction of acceleration (See Fig. 6.81).

Considering the equilibrium of a fluid element at depth h from the free surface we have:

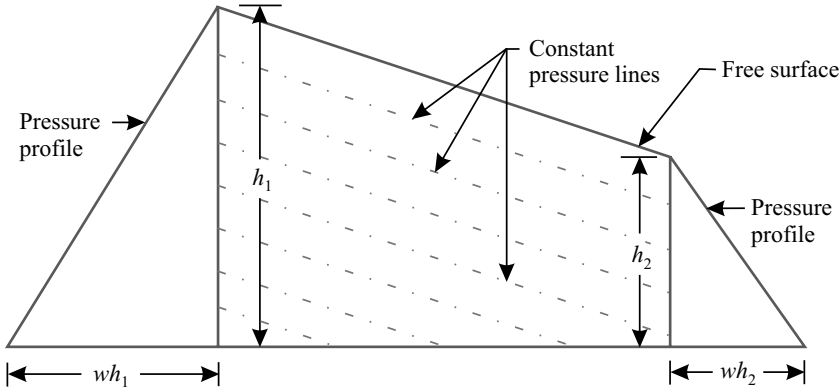


Fig. 6.82. Pressure distribution for horizontally accelerated fluid.

$$pdA = p_{\text{atm}}dA + whdA$$

where,

p_{atm} = Atmospheric pressure, and

dA = Cross-sectional area of an elementary prism.

or,

$$p = p_{\text{atm}} + wh; p = wh \text{ (gauge)} \quad \dots (6.43)$$

This means that *pressure at any point in a liquid subjected to constant horizontal acceleration equals the head above that point*. Thus, *lines of constant pressure will be parallel to the free liquid surface*. Fig. 6.82 shows the constant pressure lines and the variation in liquid pressure on rear and front of the tank. With the decrease in depth in the direction of acceleration, the pressure along the bottom of the tank also decreases.

- If the tank is completely filled with liquid and is closed at the top, the pressure builds up at the rear and is greater than that at the point (there being no preliminary adjustment in the surface elevation). The slope of the constant pressure lines is, however, still governed by the relation: $\tan \theta = \frac{a_x}{g}$.
- It may be noted that *so long as the container provides a continuous connection in the liquid mass, its shape does not matter*.

Note. The fuel tank of an aeroplane during take-off is an example of liquid in a container subjected to uniform acceleration in the horizontal direction.

Example 6.80. An open tank 6 m long, 2.4 m deep and 3.6 m wide contains oil of specific gravity 0.85 to a depth of 1.2 m. If the tank is accelerated along its length on a horizontal track at a constant acceleration 3.2 m/s^2 , determine:

- The new position of the oil surface.
- Pressures at the bottom of the tank at the front and rear edges.
- The amount of spill if the tank is given a horizontal acceleration of 4.8 m/s^2 instead of 3.2 m/s^2 .

Solution. Given: Tank dimensions: 6 m (length) $\times$ 3.6 m (width) $\times$ 2.4 m (depth),

Sp. gr. of oil = 0.85; $a_x = 3.2 \text{ m/s}^2$.

Refer to Fig. 6.83.

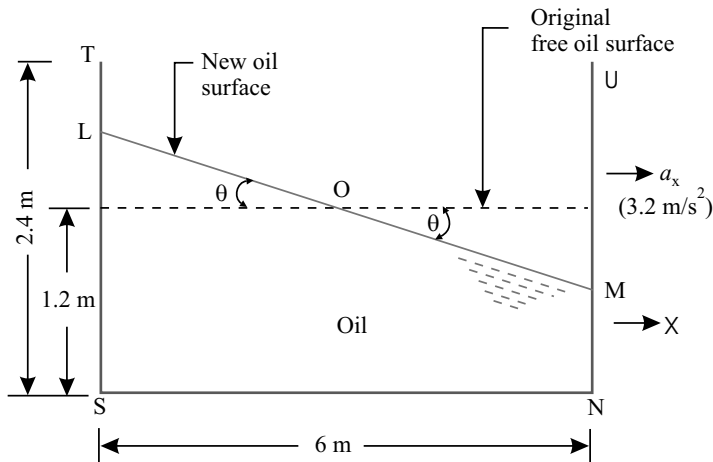


Fig. 6.83

- (i) New position of the oil surface, θ :

Inclination of the new oil surface (θ) is given by:

$$\tan \theta = \frac{a_x}{g} = \frac{3.2}{9.81} = 0.3262$$

$$\therefore \theta = \tan^{-1}(0.3262) = 18.07^\circ \text{ (Ans.)}$$

- (ii) Pressures at the bottom of the tank at the front and rear edges:

The depth of oil at the front edge N,

$$\begin{aligned} h_N &= 1.2 - \frac{6}{2} \times \tan \theta \\ &= 1.2 - 3 \times 0.3262 = 0.221 \text{ m} \end{aligned}$$

The depth of oil at the rear edges,

$$\begin{aligned} h_S &= 1.2 + \frac{6}{2} \times \tan \theta \\ &= 1.2 + 3 \times 0.3262 = 2.179 \text{ m} \end{aligned}$$

$$\therefore \text{Pressure at N, } p_N = wh_N = (9.81 \times 0.85) \times 0.221 = 1.843 \text{ kN/m}^2 \text{ (Ans.)}$$

$$\text{and, Pressure at S, } p_S = wh_S = (9.81 \times 0.85) \times 2.179 = 18.169 \text{ kN/m}^2 \text{ (Ans.)}$$

- (iii) The amount of spill with an acceleration of 4.8 m/s^2 :

Refer to Fig. [6.84 (i)].

Inclination of the oil surface,

$$\tan \theta' = \frac{a_x}{g} = \frac{4.8}{9.81} = 0.4893$$

$$(\theta' = \tan^{-1}(0.4893) = 26.07^\circ)$$

If there were no spill, the oil surface would swing about an axis at O. Piezometric head at N,

$$h'_S = 1.2 + \frac{6}{2} \tan \theta' = 1.2 + 3 \times 0.4893 = 2.668 \text{ m}$$

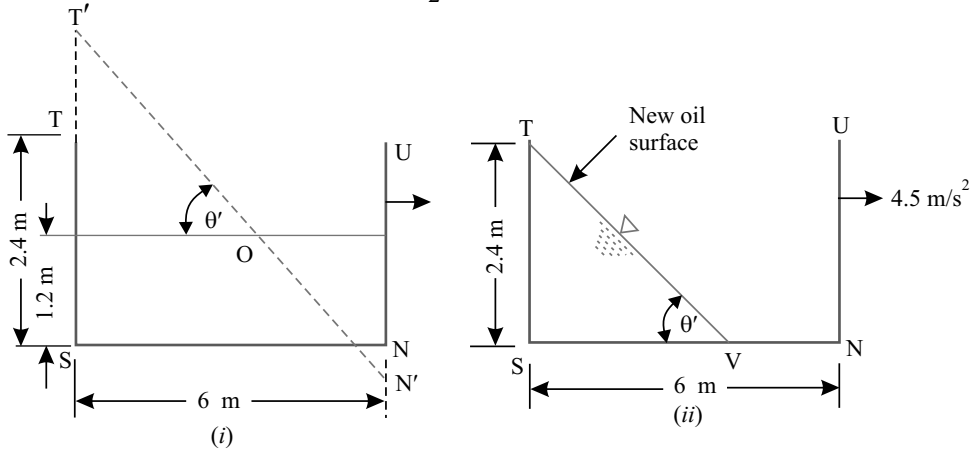


Fig. 6.84

Since this is larger than the depth of the tank there will be a spill of the oil. The new oil surface will have a depth of $h_S = \text{depth of tank} = 2.4 \text{ m}$ at S and a slope of θ' .

X-intercept of the surface at the bottom (SV) can be found from ΔTSV as follows.

$$\frac{TS}{SV} = \tan \theta'$$

$$\text{or, } SV = \frac{TS}{\tan \theta'} = \frac{2.4}{0.4893} = 4.9 \text{ m}$$

TV is the new oil surface [Fig. 6.84 (ii)].

Volume of oil = $\Delta TSV \times \text{width of tank}$

$$= \left(\frac{1}{2} \times 2.4 \times 4.9 \right) \times 3.6 = 21.17 \text{ m}^3$$

$$\text{Original volume of oil} = 6 \times 3.6 \times 1.2 = 25.92 \text{ m}^3$$

$$\therefore \text{Spill of oil} = 25.92 - 21.17 = \mathbf{4.75 \text{ m}^3} \text{ (Ans.)}$$

Example 6.81. A spherical tank of radius 1.5 m is half-filled with oil of specific gravity 0.9. If the tank is given a horizontal acceleration of 11 m/s^2 , calculate:

- (i) The inclination of the oil surface to the horizontal.
- (ii) Maximum pressure on the tank.

Solution. Given: Radius of the tank, $r = 1.5 \text{ m}$; Sp. gr. of oil = 0.9; $a_x = 11 \text{ m/s}^2$.

(i) The inclination of the oil surface to the horizontal θ :

$$\text{Refer to Fig. 6.85: } \tan \theta = \frac{a_x}{g} = \frac{11}{9.81} = 1.1213$$

$$\therefore \theta = \tan^{-1}(1.1213) = \mathbf{48.3^\circ} \text{ (Ans.)}$$

(LM is the original oil surface and RS is the new oil surface. The surface tilts around O).

(ii) **Maximum pressure on the tank:**

The maximum pressure acts on the boundary point where the *depth (measured normal to the free surface) is maximum*.

In this case maximum depth is $OT = r = 1.5 \text{ m}$

Hence,
$$\left(\frac{p}{w}\right)_{\max} = 1.5$$

or,
$$p_{\max} = w \times 1.5 = (9.81 \times 0.9) \times 1.5$$

$$= 13.24 \text{ kN/m}^2 \text{ (Ans.)}$$

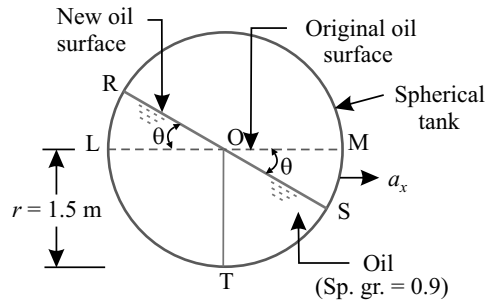


Fig. 6.85

Example 6.82. A closed tank 5 m long, 1.8 m wide and 1.6 m deep initially contains water to a depth of 1.1 m. The top has an opening in the front part to have air space at atmospheric pressure. If the tank is given acceleration at a constant value of 2.5 m/s^2 along its length, calculate the total pressure force on the top of the tank.

Solution. Given: Dimensions of the closed tank = 5 m $\times$ 1.8 m $\times$ 1.6 m; $a_x = 2.5 \text{ m/s}^2$

In the Fig. 6.86. EF is the original water surface.

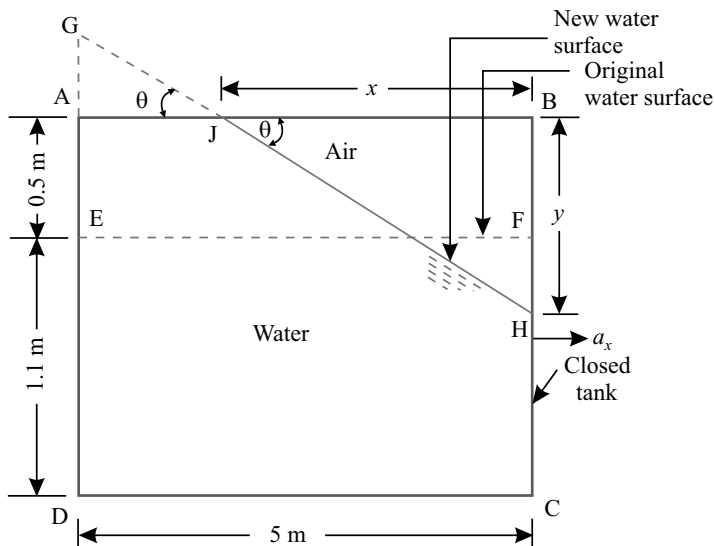


Fig. 6.86

After the acceleration of $a_x = 2.5 \text{ m/s}^2$, the water surface slope is

$$\tan \theta = \frac{a_x}{g} = \frac{2.5}{9.81} = 0.2548$$

or,
$$\theta = \tan^{-1}(0.2548) = 14.29^\circ$$

Since there is no spill of water, the air space will remain same as at start.

Air space volume,
$$V_{\text{air}} = 0.5 \times 5 \times 1.8 = 4.5 \text{ m}^3$$

Let JH be the new water surface at an inclination of θ to the horizontal.

If, JB = x and BH = y, b = breadth of the tank, then:

$$y = x \tan \theta$$

and,
$$V_{\text{air}} = \frac{1}{2} \times x \times y \times b = \frac{1}{2} \times x \times \tan \theta \times b = \frac{1}{2} x^2 b \tan \theta$$

or, $4.5 = \frac{1}{2}x^2 \times 1.8 \times 0.2548$

$\therefore x = 4.43 \text{ m}$, and $y = 4.43 \times 0.2548 = 1.13 \text{ m}$

Hence CH = Depth of water in the front = $1.6 - 1.13 = 0.47 \text{ m}$

$$AJ = 5 - x = 5 - 4.43 = 0.57 \text{ m}$$

$$AG = AJ \tan \theta = 0.57 \times 0.2548 = 0.145 \text{ m}$$

The pressure profile on the top is represented by the ΔAGJ extending over the width. Pressure force on the top,

$$\begin{aligned} P_{top} &= \left(\frac{1}{2} \times AG \times AJ \times \text{breadth} \right) \times w \\ &= \frac{1}{2} \times 0.145 \times 0.57 \times 1.8 \times 9.81 = \mathbf{0.73 \text{ kN (Ans.)}} \end{aligned}$$

The force acts vertically upwards at $\frac{AJ}{3} = \frac{0.57}{3} = 0.19 \text{ m}$ from A at the mid-width section.

Note: In this case free surface does not tilt at the mid-length. As there is no spill the volume of water and air volume are conserved.

Example 6.83. A closed tank 12 m long, 3.6 m high and 2.4 m wide contains oil of specific gravity 0.85 and is given a horizontal acceleration of 0.28 g to the right in the direction of 12 m side.

(i) Calculate: The pressure difference between (a) a point on the top rear edge and a point on the front edge, and (b) a point on the bottom front edge and a point on the top front edge.

(ii) Sketch the lines of equal pressure.

Solution. Given: Dimensions of the closed tank = 12 m $\times$ 2.4 m $\times$ 3.6 m;

Specific gravity of oil = 0.85; Horizontal acceleration, $a_x = 0.28 \text{ g}$

Refer to Fig. 6.87. At an acceleration of a_x let UV be the hydraulic gradient line. Its inclination is given by,

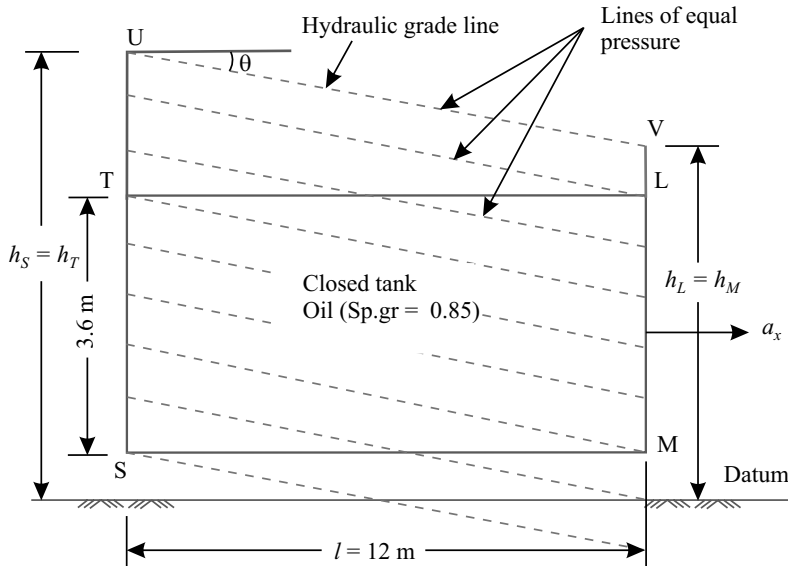


Fig. 6.87

$$\tan \theta = \frac{h_T - h_L}{l} = \frac{a_x}{g} = \frac{0.28g}{g} = 0.28$$

or,

$$\theta = \tan^{-1}(0.2) = 15.64^\circ$$

(a) $p_L - p_T$:

$$h_T - h_L = l \times \tan \theta = 12 \times 0.28 = 3.36 \text{ m}$$

But,

$$h_T - h_L = \left(\frac{p_T}{w} + z_T \right) - \left(\frac{p_L}{w} + z_L \right)$$

Here,

$$z_T = z_L$$

$\therefore$

$$p_T - p_L = w \times 3.36 = (9.81 \times 0.85) \times 3.36 = \mathbf{28.02 \text{ kN/m}^2 \text{ (Ans.)}}$$

(b) $p_M - p_L$:

Along ML the hydraulic gradient line is constant.

Hence,

$$h_M = h_L$$

$$\left(\frac{p_M}{w} + z_M \right) - \left(\frac{p_L}{w} + z_L \right) = 0$$

or,

$$\begin{aligned} p_M - p_L &= w(z_L - z_M) \\ &= (9.81 \times 0.85) \times 3.6 = \mathbf{30.02 \text{ kN/m}^2 \text{ (Ans.)}} \end{aligned}$$

- (ii) Since the pressure distribution is hydrostatic in any vertical direction and the hydraulic gradient line is inclined at θ to horizontal (line UV) the lines of equal pressure will be parallel to UV, as shown in Fig. 6.87.

Example 6.84. A tank LMNSTU shown in Fig. 6.88. is filled with water. A small opening at T keeps the pressure at T atmospheric.

- (i) Calculate the acceleration a_x required to cause onset of cavitation at L.

- (ii) What will be the pressure at that acceleration at points M, N, S and U?

Assume local atmospheric pressure head = 10 m of water and vapour pressure head = 0.48 m (abs.) of water.

Solution. (i) Acceleration a_x :

At the acceleration a_x the hydraulic gradient line will be inclined at θ , given by

$$\tan \theta = \frac{a_x}{g}$$

Since p_T = pressure at T = atmospheric pressure, the hydraulic gradient line will pass through T as shown in Fig. 6.89 by the line VTW.

Then above an arbitrary datum:

$$h_T = h_U, \text{ and}$$

$$h_L = h_M$$

$$\text{Also, } h_T - h_L = LU \tan \theta = 4.8 \tan \theta$$

At the onset of cavitation at L,

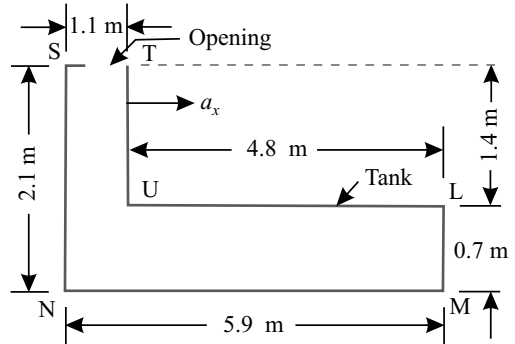


Fig. 6.88

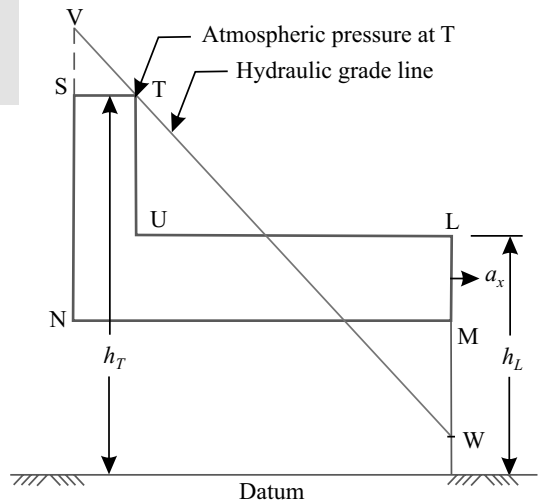


Fig. 6.89

$$p_L = p_v = \text{Vapour pressure.}$$

Considering the absolute pressures,

$$\begin{aligned} h_T - h_L &= \left(\frac{p_T}{w} + z_T \right) - \left(\frac{p_L}{w} + z_L \right) \\ &= \left(\frac{p_T}{w} - \frac{p_L}{w} \right) + (z_T - z_L) \\ &= (10 - 0.48) + (1.4) = 10.92 \text{ m} \end{aligned}$$

But,

$$h_T - h_L = 4.8 \tan \theta$$

Hence,

$$\tan \theta = \frac{a_x}{g} = \frac{h_T - h_L}{4.8} = \frac{10.92}{4.8} = 2.275$$

or

$$a_x = 9.81 \times 2.275 = \mathbf{22.32 \text{ m/s}^2} \text{ (Ans.)}$$

(ii) Pressures at point M, N, S and U:

Pressure at M, p_M :

$$\begin{aligned} h_M - h_L &= \left(\frac{p_M}{w} + z_M \right) - \left(\frac{p_L}{w} + z_L \right) = 0 \\ \frac{p_M}{w} &= \frac{p_T}{w} + (z_L - z_m) = 0.48 + 0.7 = 1.18 \text{ m} \\ \therefore p_M &= 9.81 \times 1.18 = \mathbf{11.576 \text{ kN/m}^2 \text{ (abs.)}} \text{ (Ans.)} \end{aligned}$$

Pressure at U, p_U :

$$\begin{aligned} h_T - h_U &= \left(\frac{p_S}{w} + z_T \right) - \left(\frac{p_U}{w} + z_u \right) = 0 \\ \frac{p_U}{w} &= \frac{p_T}{w} + (z_T - z_u) = 10 + 1.4 = 11.4 \text{ m} \\ \therefore p_U &= 9.81 \times 11.4 = \mathbf{111.834 \text{ kN/m}^2 \text{ (abs.)}} \text{ (Ans.)} \end{aligned}$$

Pressure at S, p_S :

$$\begin{aligned} h_S - h_T &= 1.1 \tan \theta = 1.1 \times 2.275 = 2.5 \text{ m} \\ \text{Also, } h_S - h_T &= \left(\frac{p_S}{w} + z_S \right) - \left(\frac{p_T}{w} + z_T \right) = 0 \\ \frac{p_S}{w} &= \frac{p_T}{w} + (z_T - z_S) + 2.5 \\ &= 10 + 0 + 2.5 = 12.5 \text{ m} \\ \therefore p_S &= 9.81 \times 12.5 = \mathbf{122.625 \text{ kN/m}^2 \text{ (Ans.)}} \end{aligned}$$

Pressure at N, p_N :

$$\begin{aligned} h_S - h_N &= \left(\frac{p_S}{w} + z_S \right) - \left(\frac{p_N}{w} + z_N \right) = 0 \\ \frac{p_N}{w} &= \frac{p_S}{w} + (z_S - z_N) \\ &= 12.5 + 2.1 = 14.6 \text{ m} \\ \therefore p_N &= 9.81 \times 14.6 = \mathbf{143.226 \text{ kN/m}^2 \text{ (abs.)}} \text{ (Ans.)} \end{aligned}$$

Example 6.85. A closed oil tanker 3.5 m long, 1.8 m wide and 2 m deep contains 1.6 m depth of oil of specific gravity 0.8. Calculate:

- (i) The acceleration which may be imparted to the tank in the direction of its length so that bottom front end of the tank is just exposed.

(ii) The net horizontal force acting on the tanker side and show that this equals the force necessary to accelerate the liquid mass in the tanker.

Take specific weight of water = 9.81 kN/m^3

(Roorkee University)

Solution. Given: Dimensions of the tank: 3.5 m (length) $\times$ 1.8 m (width) $\times$ 2 m (depth);
Depth of oil = 1.6 m ; Sp. gr. of oil = 0.8

(i) Acceleration a_x :

The following points are worth noting :

- Since the tank is closed, therefore, the liquid cannot spill from it under any acceleration imparted to it. The quantity of oil inside the tank remains the same.
- The oil surface which was initially horizontal (indicated by TU) assumes the profile MVS when the front bottom end M is just exposed.

Equating volumes of oil before and after the motion, we have:

Volume of rectangle LMUT = Volume of trapezium LMVS

$$3.5 \times 1.6 \times 1.8 = \frac{3.5 + (3.5 - x)}{2} \times 2 \times 1.8$$

$$\therefore x = 1.4 \text{ m}$$

$$\tan \theta = \frac{MN}{NV} = \frac{2}{1.4} = 1.428$$

$$\text{Also, } \tan \theta = \frac{a_x}{g} = 1.428$$

$$\therefore a_x = 9.81 \times 1.428 = \mathbf{14.01 \text{ m/s}^2} \text{ (Ans.)}$$

When the free surface MV is extended, it meets LS produced at W.

$$\frac{SW}{SV} = \tan \theta = 1.428$$

$$\therefore SW = (3.5 - 1.4) \times 1.428 = 3 \text{ m}$$

This represents an imaginary column of oil above S.

$$\text{Now, } p_S = w \times SW = (9.81 \times 0.8) \times 3 = 23.544 \text{ kN/m}^2$$

$$\text{and, } p_L = w \times LW = (9.81 \times 0.8) \times (3 + 2) = 39.24 \text{ kN/m}^2$$

$\therefore$ Pressure force on the trailing/rear face LS,

$$\begin{aligned} P_{LS} &= p_{\text{avg.}} \times \text{area} \\ &= \left(\frac{23.544 + 39.24}{2} \right) \times (2 \times 1.8) = \mathbf{113 \text{ kN}} \end{aligned}$$

$$\left[\text{Alternatively: } P_{LS} = wA\bar{x} = (9.81 \times 0.8) \times (2 \times 1.8) \times \left(3 + \frac{2}{2} \right) = 113 \text{ kN} \right]$$

The force needed to accelerate the liquid mass in the tank,

$$F = \text{Mass of oil} \times \text{uniform linear acceleration}$$

$$= (0.8 \times 1000) \times 3.5 \times 1.6 \times 1.8 \times 14.01 \times 10^{-3} \text{ kN} = \mathbf{113 \text{ kN (Ans.)}}$$

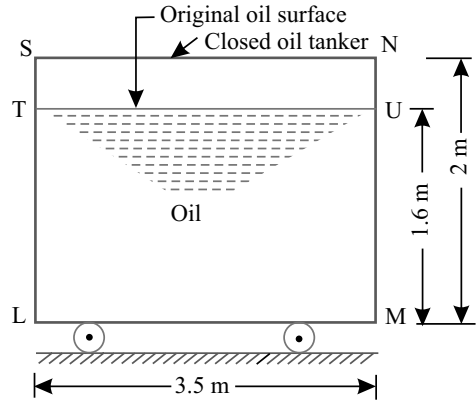


Fig. 6.90

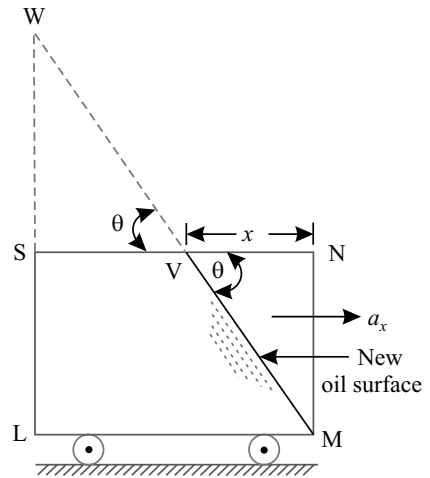


Fig. 6.91

Obviously the difference between the force on the two ends of the tanker is *equal* to the force necessary to accelerate the liquid mass in the tanker.

6.12.2. Liquid in a Container Subjected to Uniform Acceleration in the Vertical Direction

Consider a tank containing liquid and moving vertically *upwards* with uniform acceleration a_y (Fig 6.92). The liquid in the tank will have a free horizontal surface *but pressure intensity at any point in the liquid will be different* from what it would be when in a state of absolute rest:

Applying Newton's second law of motion, we have:

$$\Sigma F_y = m \times a_y$$

The force ΣF_y = Pressure force acting upwards – weight of prism acting downwards

$$= [p \text{ (intensity of pressure)} \times dA \text{ (area)}] - (w \times \text{volume of prism})$$

$$= p \times dA - w \times h \times dA$$

Also, $m = \frac{w}{g} \times \text{volume of elementary prism}$

$$= \frac{w}{g} \times (h \times dA)$$

$$\therefore p \times dA - w \times h \times dA = \frac{w}{g} (h \times dA) \times a_y$$

or,
$$p = wh \left(1 + \frac{a_y}{g} \right) \quad \dots(6.44)$$

This equation (6.44) reveals the following:

- The free liquid surface remains *horizontal*.
- The pressure variation in the vertical direction is *linear*.
- The pressure intensity at any point is *more* than the static pressure wh by an amount $wh \left(\frac{a_y}{g} \right)$ as shown in Fig. 6.93 (a).

If the liquid mass is uniformly accelerated vertically *downward* direction, a_y shall be negative and then eqn. (6.43) reduces to

$$p = wh \left(1 - \frac{a_y}{g} \right) \quad \dots(6.44)$$

i.e., Intensity of pressure at any point is *less* than static pressure wh by an amount $wh \left(\frac{a_y}{g} \right)$, as shown in (Fig. 6.93 (b)).

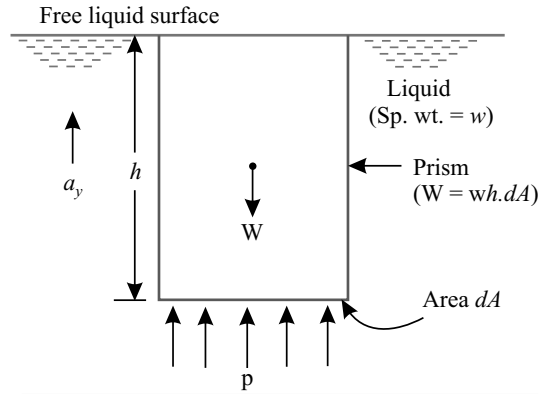


Fig. 6.92

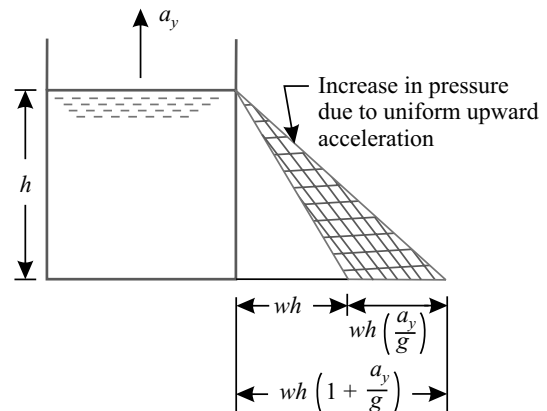


Fig. 6.93. (a)

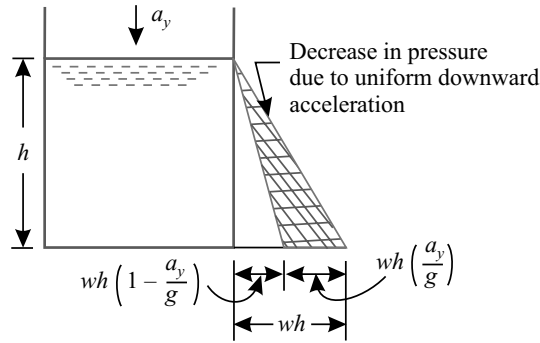


Fig. 6.93. (b)

If the tank is lowered vertically at the gravitational acceleration, then $a_y = g$ and eqn. (6.44) reduces to $p = 0$. Obviously the pressure is *uniform and equivalent to surrounding atmospheric pressure*, and no force acts either on the base or on the tank walls.

Example 6.86. An open cubical tank with each side 1.8 m contains oil of specific weight 8 kN/m^3 up to a depth of 1.8 m.

Calculate:

- (i) The force acting on side of the tank when it is being moved with an acceleration of $\frac{g}{3} \text{ m/s}^2$ in vertically upward and downward directions.
- (ii) The pressure at the bottom of the tank when acceleration rate is $g \text{ m/s}^2$ vertically downwards.

Solution: Given: Each side of cubical tank = 1.8 m; Sp. wt of oil, $w = 8 \text{ kN/m}^3$; a_y (upward and downward) = $\frac{g}{3} \text{ m/s}^2$; a_y (downward) = $g \text{ m/s}^2$.

(i) The force acting on the side of the tank:

(a) When a_y is vertically upward:

$$p = wh \left(1 + \frac{a_y}{g} \right) = 8 \times 1.8 \left(1 + \frac{g/3}{g} \right) = 8 \times 1.8 \left(1 + \frac{1}{3} \right) = 19.2 \text{ kN/m}^2$$

On a vertical side, the intensity of pressure varies linearly from zero at the top to 19.2 kN/m^2 at the bottom.

Force on the side of the tank = $p_{\text{avg}} \times \text{area}$

$$= \frac{19.2 + 0}{2} \times (1.8 \times 1.8) = \mathbf{31.1 \text{ kN (Ans.)}}$$

(b) When a_y is vertically downward:

$$p = wh \left(1 - \frac{a_y}{g} \right) = 8 \times 1.8 \left(1 - \frac{1}{3} \right) = 9.6 \text{ kN/m}^2$$

Force on the side of the tank = $p_{\text{avg.}} \times \text{area}$

$$= \frac{9.6 + 0}{2} \times (1.8 \times 1.8) = \mathbf{15.55 \text{ kN (Ans.)}}$$

(ii) The pressure at the bottom when $a_y = g$:

When the tank is lowered vertically at an acceleration, then:

$$p = wh \left(1 - \frac{a_y}{g} \right) = wh \left(1 - \frac{g}{g} \right) = 0$$

i.e., the liquid remains at the atmospheric pressure throughout and there is no force on the base or on the tank walls. (Ans.)

6.12.3. Liquid in Container Subjected to Uniform Acceleration Along Inclined Plane.

Fig. 6.94 shows a tank filled with a liquid being accelerated *up an inclined plane* with uniform acceleration a .

Horizontal component of acceleration, $a_x = a \cos \alpha$

Vertical component of acceleration, $a_y = a \sin \alpha$

A particle L of mass m lying on the liquid surface is *in equilibrium* under the action of following forces:

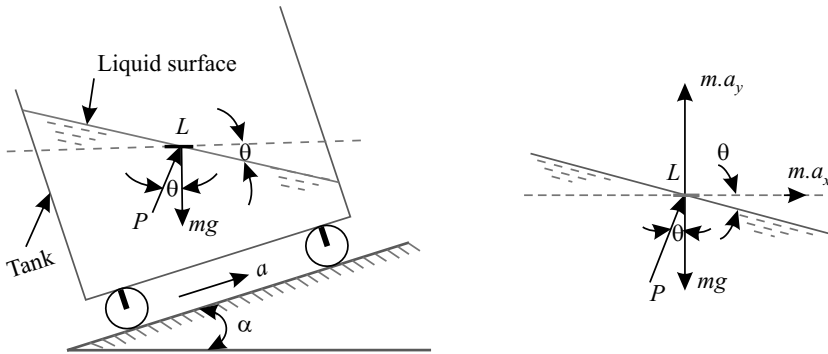


Fig. 6.94. Acceleration of fluid mass along an upward slope.

- (i) Weight mg acting vertically downward,
- (ii) Pressure force P acting *normal* to the surface of the fluid element, and
- (iii) Accelerating force, $(m.a)$ having component $m.a_x$ in the horizontal direction and a component $m.a_y$ in the vertical direction.

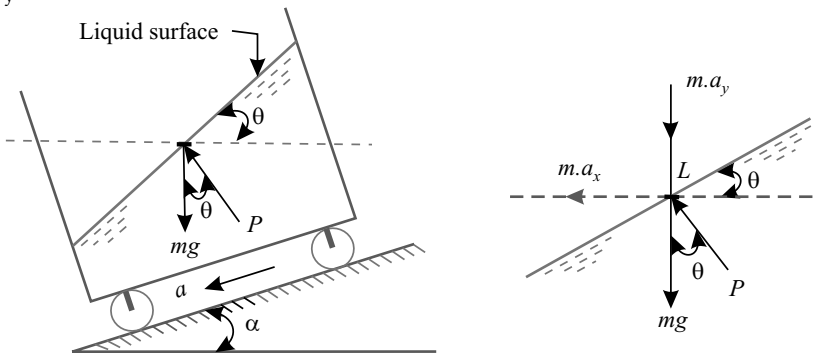


Fig. 6.95. Acceleration of fluid mass along a downward slope.

Resolving *horizontally* and *vertically* respectively, we get:

$$P \sin \theta = m.a_x \quad \dots(i)$$

$$P \cos \theta = m.a_y + mg \quad \dots(ii)$$

Dividing (i) by (ii), we get:

$$\tan \theta = \frac{a_x}{a_y + g} \quad \dots(6.45)$$

When the fluid mass is subjected to acceleration *down the slope*, we have:

$$P \sin \theta = m \cdot a_x \quad \dots(iii)$$

$$P \cos \theta = mg - m.a_y \quad \dots(iv)$$

Dividing (iii) by (iv), we get:

$$\tan \theta = \frac{a_x}{g - a_y} \quad \dots(6.46)$$

Example 6.87. An open rectangular tank 6 m long and 2.4 m wide is filled with water to a depth of 1.8 m. Find the slope of water surface when the tank moves with an acceleration of 2.5 m/s^2

(i) up a 30° inclined plane, and (ii) down a 30° inclined plane.

Solution. Given: Dimensions of the tank: 6 m (length) $\times$ 2.4 m (width);

Acceleration, $a = 2.5 \text{ m/s}^2$; Inclination, $\alpha = 30^\circ$.

Horizontal and vertical components of acceleration are:

$$a_x = a \cos \alpha = 2.5 \times \cos 30^\circ = 2.165 \text{ m/s}^2$$

$$a_y = a \sin \alpha = 2.5 \times \sin 30^\circ = 1.25 \text{ m/s}^2$$

Let, θ be the slope of the free liquid surface.

(i) When the tank moves with acceleration up the inclined plane:

$$\tan \theta = \frac{a_x}{a_y + g} = \frac{2.165}{1.25 + 9.81} = 0.196$$

$$\theta = \tan^{-1} (0.196) = 11.1^\circ \text{ (Ans.)}$$

(ii) When the tank moves with an acceleration down the inclined plane :

$$\tan \theta = \frac{a_x}{g - a_y} = \frac{2.165}{9.81 - 1.25} = 0.253$$

$$\theta = \tan^{-1} (0.253) = 14.2^\circ \text{ (Ans.)}$$

HIGHLIGHTS

1. The science which deals with the geometry of motion of fluids *without reference* to the forces causing the motion is known as *hydrokinematics*.
2. The science which *deals with* the action of forces in producing or changing motion of fluids is known as *hydrokinetics* (or simply kinetics).
3. Different types of heads are:
 - (i) Potential head (or potential energy)
 - (ii) Velocity head (or kinetic energy)
 - (iii) Pressure head (or pressure energy).
4. *Bernoulli's equation* states as follows:

"In an ideal, incompressible fluid when the flow is steady and continuous, then sum of pressure energy, potential (or datum) energy and kinetic energy is constant along a streamline." Mathematically,

$$\frac{p}{w} + \frac{V^2}{2g} + z = \text{constant}$$

where

$$\frac{p}{w} = \text{Pressure energy or head,}$$

$$\frac{V^2}{2g} = \text{Kinetic energy or head, and}$$

z = Datum (or elevation) energy or head.

5. *Euler's equation* for motion is given as:

$$\frac{dp}{\rho} + v \cdot dv + g \cdot dz = 0 \quad \dots \text{differential form}$$

6. *Bernoulli's equation for real fluid* is given as:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_L$$

where, h_L = loss of energy between sections 1 and 2.

7. *Practical applications of Bernoulli's equation* are:

- (i) Venturimeter; (ii) Orificemeter;
- (iii) Rotometer and elbow meter; (iv) Pitot tube.

In case of a *venturimeter*, the actual discharge (Q_{act}) is given as:

$$Q_{act} = C_d \times \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh}$$

where,

C_d = Co-efficient of discharge (varies between 0.96 and 0.98),

A_1 = Area at inlet,

A_2 = Area at outlet, and

h = Difference of pressure head at sections 1 and 2.

8. *Free liquid jet*:

A jet of liquid from the nozzle in atmosphere is called a *free liquid jet*. The parabolic path traversed by the liquid jet under the action of gravity is known as *trajectory*.

(i) Equation of the jet:

$$y = x \tan \theta - \frac{gx^2 \sec^2 \theta}{2U^2}$$

where,

x, y = Co-ordinates of any point on jet with respect to the nozzle,

U = Velocity of the jet of water issuing from the nozzle, and

θ = Inclination of the jet issuing from nozzle with horizontal.

(ii) Maximum height attained by the jet = $\frac{U^2 \sin^2 \theta}{2g}$

(iii) Time of flight, $T = \frac{2U \sin \theta}{g}$

(iv) Time taken to reach the highest point; $T' = \frac{U \sin \theta}{g}$

(v) Horizontal range of the jet, $r = \frac{U^2 \sin 2\theta}{g}$

(vi) Maximum range, $r_{\max} = \frac{U^2}{2g}$

(The range will be maximum when $\theta = 45^\circ$)

9. *Momentum principle* states as follows:

"The net force acting on a mass of fluid is equal to the change in momentum of flow per unit time in that direction".

10. *Impulse-momentum equation* $F \cdot dt = d(mv)$ may be stated as follows:

“The impulse of a force F acting on a fluid mass m in a short interval of time dt is equal to the change of momentum $d(mv)$ in the direction of force.”

11. *Kinetic energy correction factor* (α).

$$\alpha = \frac{\text{Kinetic energy per second based on actual velocity}}{\text{Kinetic energy per second based on average velocity}}$$

Momentum correction factor (β).

$$\beta = \frac{\text{Momentum per second based on actual velocity}}{\text{Momentum per second based on average velocity}}$$

The values of α and β may be calculated by using the following equations:

$$\alpha = \frac{1}{A} \int_A \left(\frac{u}{\bar{u}} \right)^2 dA \quad \text{and} \quad \beta = \frac{1}{A} \int_A \left(\frac{u}{\bar{u}} \right) dA$$

where,

A = Area of cross-section,

u = Local velocity,

$\bar{u}$ = Average velocity, and

dA = Elementary area.

12. *Moment of momentum equation.* This equation is derived from moment of momentum principle which states that the resulting torque acting on a rotating fluid is equal to the rate of change of momentum.

Mathematically, it is written as:

$$T = \rho Q (V_2 r_2 - V_1 r_1)$$

Vortex motion:

The pressure variation along the radial direction for vortex flow along a horizontal plane,

$$\frac{\partial p}{\partial r} = \frac{\rho v^2}{r}$$

and, pressure variation in the vertical plane,

$$\frac{\partial p}{\partial r} = -\rho g$$

Forced vortex flow:

Forced vortex flow is one in which the fluid mass is made to rotate by means of some external agency.

$$v = \omega \times r$$

$$z = \frac{v^2}{2g} = \frac{\omega^2 r^2}{2g} = \frac{\omega^2 R^2}{2g}$$

where,

z = Height of the paraboloid formed, and

ω = Angular velocity.

- For a forced vortex flow in an *open tank*:

Fall of liquid level at centre = Rise of liquid level at the ends

- In case of a *closed cylinder*:

Volume of air before rotation = volume of air after rotation.

- If a closed cylindrical vessel completely filled with water is rotated about its vertical axis, the total pressure force acting on the top and bottom are:

$$F_{\text{top}} = \frac{\rho}{4} \omega^2 \pi R^4$$

and,

$$F_{\text{bottom}} = F_{\text{top}} + \text{weight of water in cylinder} \\ = F_{\text{top}} + w \times \pi R^2 \times H$$

where,

ω = Angular velocity,

R = Radius of the vessel,

H = Height of the vessel, and

ρ = Density of fluid $\left(= \frac{w}{g} \right)$.

Free vortex flow:

When no external torque is required to rotate the fluid mass, type of flow is called free vortex flow. In case of free vortex flow:

$$v \times r = \text{constant} \quad \dots(i)$$

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2 \quad \dots(ii)$$

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer:

1. Velocity head is given by

- (a) $\frac{V}{g}$ (b) $\frac{V^2}{2g}$
(c) $\frac{V^3}{2g}$ (d) $\frac{V^2}{2g^2}$

2. Bernoulli's equation, mathematically is written as

- (a) $\frac{p}{w} + \frac{V}{2g} + z = \text{constant}$
(b) $\frac{p}{w^2} + \frac{V^2}{2g} + z = \text{constant}$
(c) $\frac{p}{w} + \frac{V^2}{2g} + z = \text{constant}$
(d) $\frac{p^2}{w} + \frac{V^2}{2g} + z = \text{constant}.$

3. Which of the following assumptions is made in the derivation of Bernoulli's equation?

- (a) The liquid is ideal and incompressible
(b) The flow is steady and continuous
(c) The flow is one-dimensional
(d) The velocity is uniform over the section and is equal to mean velocity
(e) All of the above.

4. Euler's equation (in differential form) is written as:

- (a) $\frac{dp}{\rho} + v^2 \cdot dv + g \cdot dz = 0$
(b) $\frac{dp}{\rho} + v \cdot dv + g \cdot dz = 0$
(c) $\frac{dp}{\rho} + v \cdot dv + g^2 \cdot dz = 0$
(d) $\frac{dp}{\rho^2} + v \cdot dv + g \cdot dz = 0.$

5. In which of the following measuring devices Bernoulli's equation is used:

- (a) Venturimeter (b) Orificemeter
(c) Pitot tube (d) All the above.

6. The co-efficient of discharge of an orificemeter isthat of a venturimeter.

- (a) equal to (b) much smaller than
(c) much more than (d) any of these.

7. Which of the following equations is known as momentum principle:

- (a) $F = \frac{d(m^2v)}{dt}$ (b) $F = \frac{dv}{dt}$
(c) $F = \frac{d(mv)}{dt}$ (d) $F = \frac{d(mv)}{dt^2}.$

8. The piezometric head is the summation of :
 - (a) velocity head and pressure head
 - (b) pressure head and elevation head
 - (c) velocity head and elevation head
 - (d) none of the above.
9. The total energy-line is always higher than the hydraulic gradient line, the vertical distance between the two representing:
 - (a) the pressure head
 - (b) the piezometric head
 - (c) the velocity head
 - (d) none of the above.
10. The total-energy-line in pipe flow is a graphical representation of the Bernoulli's equation and represents the sum of velocity head, pressure head and the elevation head above:
 - (a) the top of the pipeline
 - (b) the arbitrary horizontal datum
 - (c) the centre line of pipe
 - (d) the bottom of the pipe.
11. The total energy represented by the Bernoulli's equation $\left(\frac{p}{w} + \frac{V^2}{2g} + z \right)$ has the units:
 - (a) Nm/s
 - (b) Ns/m
 - (c) Nm/m
 - (d) Nm/N.
12. The Bernoulli's equation written in the conventional form $\frac{p}{w} + \frac{V^2}{2g} + z = \text{constant}$ represents total energy per unit of certain quantity. Identify this quantity from the choices given below:
 - (a) energy per unit mass
 - (b) energy per unit weight
 - (c) energy per unit volume
 - (d) energy per unit specific weight.
13. A venturimeter is used for measuring:
 - (a) pressure
 - (b) flow rate
 - (c) total energy
 - (d) piezometric head.
14. The co-efficient of discharge (C_d) of venturimeter lies within the limits:
 - (a) 0.95 to 0.99
 - (b) 0.8 to 0.85
 - (c) 0.7 to 0.8
 - (d) 0.6 to 0.7
15. A Pitot-tube is used for measuring:
 - (a) velocity of flow
 - (b) pressure of flow
 - (c) flow rate
 - (d) total energy.
16. When a Pitot-tube is put to use it must be ensured that its alignment is such that:
 - (a) the horizontal leg should be inclined at 45° in plan
 - (b) its horizontal leg is at right angles to the flow direction
 - (c) its opening faces upstream and the horizontal leg is perfectly aligned with the direction of flow
 - (d) none of the above.
17. The hydraulic gradient-line indicates the variation of which of the following:
 - (a) Velocity head in flow direction
 - (b) Piezometric head in the direction of flow
 - (c) Total energy of flow in the direction of flow
 - (d) None of the above.
18. The kinetic energy correction factor is expressed by:
 - (a) $\frac{1}{A} \int_A \left(\frac{u}{\bar{u}} \right) dA$
 - (b) $\frac{1}{A} \int_A \left(\frac{u}{\bar{u}} \right)^2 dA$
 - (c) $\frac{1}{A} \int_A \left(\frac{u}{\bar{u}} \right)^3 dA$
 - (d) $\frac{1}{A} \int_A \left(\frac{\bar{u}}{u} \right)^3 dA$
19. The momentum correction factor β is used to account for:
 - (a) change in direction of flow
 - (b) change in total energy
 - (c) non-uniform distribution of velocities at inlet and outlet sections
 - (d) change in mass rate of flow.
20. The change in moment of momentum of fluid due to flow along a curved path results in:
 - (a) a change in pressure
 - (b) torque
 - (c) a change in the total energy
 - (d) none of the above.
21. Which of the following is an example of free vortex flow?
 - (a) A whirlpool in a river
 - (b) Flow of liquid through a hole provided at the bottom of a container
 - (c) Flow of liquid around a circular bend in a pipe.
 - (d) All of the above.
22. In case of forced vortex, the rise of liquid level at the ends is the fall of liquid level at the axis of rotation.
 - (a) less than
 - (b) more than
 - (c) equal to
 - (d) none of the above.

23. In case of a closed cylindrical vessel sealed at the top and the bottom the volume of air before rotation is the volume of air after:
 (a) more than (b) less than
 (c) equal to (d) none of the above.
24. If a closed cylindrical vessel completely filled with water is rotated about its vertical axis, the total pressure force acting on the top is equal to:
 (a) $\frac{\rho}{4} \omega^2 \pi R$ (b) $\frac{\rho^2}{4} \omega \pi R^2$
 (c) $\frac{\rho}{4} \omega^2 \pi R^3$ (d) $\frac{\rho}{4} \omega^2 \pi R^4$.
25. For a free vortex flow the equation is:
 (a) $\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$
 (b) $\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$
 (c) $\frac{p_1^2}{\rho g} + \frac{V_1^2}{g} + z_1 = \frac{p_2^2}{\rho g} + \frac{V_2^2}{g} + z_2$
 (d) $\frac{p_1}{\rho g} + \frac{V_1^3}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^3}{2g} + z_2$.

ANSWERS

- | | | | | | |
|----------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (c) | 3. (e) | 4. (b) | 5. (d) | 6. (b) |
| 7. (c) | 8. (b) | 9. (c) | 10. (b) | 11. (d) | 12. (b) |
| 13. (b) | 14. (a) | 15. (a) | 16. (c) | 17. (b) | 18. (c) |
| 19. (c) | 20. (b) | 21. (d) | 22. (c) | 23. (c) | 24. (d) |
| 25. (b). | | | | | |

THEORETICAL QUESTIONS

- Explain briefly the following heads:
 (i) Potential head
 (ii) Velocity head
 (iii) Datum head.
- State and prove Bernoulli's equation.
- List the assumptions which are made while deriving Bernoulli's equation.
- Derive Euler's equation of motion.
- What are the limitations of the Bernoulli's equation?
- Describe an orificemeter and find an expression for measuring discharge of fluid through a pipe with this device.
- Why is co-efficient of discharge of an orifice-meter much smaller than that of venturimeter?
- What is a pitot tube? How is it used to measure velocity of flow at any point in a pipe or channel?
- What is a free jet of liquid? Derive an expression for the path travelled by free jet issuing from a nozzle?
- Prove that the equation of the free jet of liquid is given by the expression

$$y = x \tan \theta - \frac{gx^2}{2U^2} \sec^2 \theta$$
 where x, y = co-ordinates of a point on the jet,
 U = velocity of the jet, and
 θ = inclination of the jet with horizontal.
- What is an impulse-momentum equation?
- What is the moment of momentum equation?
- Define the terms: (i) Vortex flow, (ii) Forced vortex flow, and (iii) Free vortex flow.
- Differentiate between forced vortex flow and free vortex flow.
- Derive an expression for the depth of paraboloid formed by the surface of a liquid contained in a cylindrical tank which is rotated at a constant angular velocity ω about its vertical axis.
- Derive an expression for difference of pressure between two points in a free vortex flow.

UNSOLVED EXAMPLES

- The diameters of a tapering pipe at the sections 1-1 and 2-2 are 100 mm and 150 mm respectively. If the velocity of water flowing through the pipe at section 1-1 is 5 m/s, find:
 - Discharge through the pipe, and
 - Velocity of water at section 2-2.

[Ans. (i) 0.039 m³/s, (ii) 2.22 m/s]
- A pipe (1) 400 mm in diameter, conveying water, branches into two pipes (2 and 3 of diameters 300 mm and 200 mm respectively.
 - Find the discharge in pipe (1) if the average velocity of water in this pipe is 3 m/s.
 - Determine the velocity of water in 200 mm pipe if the average velocity in 300 mm diameter pipe is 2 m/s.

[Ans. (i) 0.377 m³/s, (ii) 7.5 m/s]
- The water is flowing through a pipe having diameters 200 mm and 100 mm at sections 1 and 2 respectively. The rate of flow through the pipe is 35 litres/s. The section 1 is 6 m above datum and section 2 is 4 m above datum. If the pressure at section 1 is 400 kN/m², find the intensity of pressure at section 2. [Ans. 410.5 kN/m²]
- A pipe 300 metres long has a slope of 1 in 100 and tapers from 1.0 m diameter at the higher end to 0.5 m at the lower end. Quantity of water flowing is 90 litre/s. If the pressure at higher end is 70 kN/m², find the pressure at the lower end. [Ans. 100 kN/m²]
- A pipe 5 m long is inclined at an angle of 15° with the horizontal. The diameters of pipe at smaller section (at lower level) and larger section are 80 mm and 240 mm respectively.
If the pipe is uniformly tapering and the velocity of water at the smaller section is 1 m/s, find the difference of pressures between the two sections. [Ans. 12.2 kN/m²]
- Water is flowing at the rate of 40 litres/s through a tapering pipe. The diameters at the bottom and upper ends are 300 mm and 200 mm respectively. If the intensities of pressure at the bottom and upper ends are 250 kN/m² and 100 kN/m² respectively, find the difference in datum head. [Ans. 13.7 m]
- A 2 m long conical tube is fixed vertically with its smaller end upwards. It carries liquid in downward direction. The flow velocities at the smaller and larger ends are 5 m/s and 2 m/s respectively. The pressure head at the smaller end is 2.5 m of liquid. If the loss of head in the tube

is $\frac{0.35(V_1 - V_2)^2}{2g}$ (V_1 and V_2 being the velocities

at the smaller and larger ends respectively) determine the pressure head at the larger end.

[Ans. 5.4 m of liquid]

- A pipeline carrying oil (sp. gr. = 0.87) changes in diameter from 200 mm diameter at position '1' to 500 mm diameter at position '2' which is 4 metres at a higher level. If the pressures at 1 and 2 are 100 kN/m² and 60 kN/m² respectively and the discharge is 0.2 m³/s, determine:
 - Loss of head, and
 - Direction of flow.

[Ans. (i) 2.6 m (ii) from 1 to 2]

- The following data relate to an orificemeter:

Diameter of the pipe = 300 mm
 Diameter of the orifice = 150 mm
 Reading of the differential manometer = 500 mm of mercury
 Sp. gravity of oil = 0.9
 Co-efficient of discharge of meter = 0.64
 Determine the rate of flow. [Ans. 0.137 m³/s]
- A venturimeter with 150 mm diameter at inlet and 100 mm at throat is laid with its axis horizontal and is used for measuring the flow of oil of sp. gr. 0.9. The oil mercury differential manometer shows a gauge difference of 200 mm. Calculate the discharge. Assume the co-efficient of meter as 0.98. [Ans. 0.06393 m³/s]
- A horizontal venturimeter with inlet and throat diameters 160 mm and 60 mm respectively is used to measure the flow of an oil of specific gravity 0.8. If the discharge of the oil is 0.05 m³/s, find the deflection of oil mercury gauge. Take venturimeter constant = 1. [Ans. 296 mm]
- A horizontal venturimeter 300 mm × 150 mm is used to measure the flow of oil of sp. gravity 0.8. The discharge of oil through venturimeter is 0.5 m³/s. Find the reading of oil-mercury differential manometer. Take venturimeter constant = 0.98. [Ans. 248.9 mm]
- A venturimeter with inlet and throat diameters 300 mm and 150 mm respectively is attached in a vertical pipe in which flow occurs from bottom to top. The distance between the point of entrance

and to the point of throat of the venturimeter is 750 mm. If the difference of mercury levels in the two limbs of differential gauge is 220 mm, find the discharge passing through the vertical pipe. Take co-efficient of discharge, $C_d = 0.98$.

[Ans. $0.146 \text{ m}^3/\text{s}$]

14. A venturimeter has its axis vertical, the inlet and throat diameters being 150 mm and 75 mm respectively. The throat is 225 mm above inlet and venturimeter constant = 0.96. Petrol of sp. gravity 0.78 flows up through the meter at a rate of $0.029 \text{ m}^3/\text{s}$. Find the pressure difference between inlet and throat. [Ans. 18.9 kN/m^2]
15. A venturimeter is used for measuring the flow of petrol in a pipeline inclined at 35° to horizontal. The sp. gravity of the petrol is 0.81 and throat area ratio is 4. If the difference in mercury levels in the gauge is 50 mm, calculate the flow in m^3/s if the pipe diameter is 300 mm. Take venturimeter constant = 0.975. [Ans. $0.07 \text{ m}^3/\text{s}$]
16. The following data relate to venturimeter fitted to an inclined pipe in which water is flowing.
Diameter of the pipe = 300 mm
Throat diameter = 150 mm
Sp. gravity of liquid used in U-tube manometer = 0.8
Reading of manometer = 400 mm
Loss of head between the inlet and throat = $0.3 \times$ kinetic head of the pipe.
Find the discharge. [Ans. $0.0226 \text{ m}^3/\text{s}$]
17. A vertical venturimeter of d/D ratio equal to 0.6 is fitted in a 100 mm diameter pipe. The throat is 200 mm above the inlet. The venturimeter constant = 0.92. Determine:
(i) Pressure difference as recorded by two gauges fitted at the inlet and throat.
(ii) Difference on a vertical differential mercury manometer (sp. gravity = 13.6) when a liquid of sp. gravity 0.8 flows through the meter at the rate of $0.05 \text{ m}^3/\text{s}$.
[Ans. (i) 130.47 kN/m^2 (ii) 1.026 m]
18. A 300 mm diameter 150° bend discharges $0.35 \text{ m}^3/\text{s}$ of water in the atmosphere. If the pressure of water entering the bend is 150 kN/m^2 (gauge), determine the force required to hold the bend in place. Assume the bend to be in horizontal plane.
[Ans. 10.86 kN , 4.57° with negative X-axis]
19. A 45° reducing bend is connected in pipeline, the diameters at the inlet and outlet of the bend being 400 mm and 200 mm respectively. Find

the force exerted by water on the bend if the intensity of pressure at inlet of the bend is 215.8 kN/m^2 . The rate of flow of water is $0.5 \text{ m}^3/\text{s}$.

[Ans. 22.7 kN ; $20^\circ 3.5'$]

20. A 300 mm diameter pipe carries water under a head of 20 metres with a velocity of 3.5 m/s . If the axis of the pipe turns through 45° , find the magnitude and direction of the resultant force at the bend. [Ans. 11.27 kN , $67^\circ 28'$]
21. A pipeline of 600 mm diameter, carrying oil (sp. gravity = 0.85) at the flow rate of 1800 litres/sec. has a 90° bend in the horizontal plane. The pressure at the entrance to the bend is 1.471 bar and the loss of head in the bend is 2 m of oil. Find the magnitude and direction of the force exerted by the oil on the bend and show the direction of the force on a sketch of the bend.

[UPSC Exams.]

[Ans. 69.357 kN , 42.3°]

22. A jet of water is coming out from a nozzle with a velocity of 20 m/s . The nozzle is situated at a distance of 20 m from a vertical wall 8 m high. Find the angle of projection of the nozzle to the horizontal so that the jet of water just clears the top of the wall. [Ans. $73^\circ 0.8'$ or $38^\circ 47'$]
23. A nozzle inclined at an angle of 45° to the horizontal is situated at a distance of 1 m above the ground level. If the diameter of the nozzle is 50 mm and the jet of water from the nozzle strikes the ground at a horizontal distance of 4 m, find the rate of flow of water. [Ans. $0.011 \text{ m}^3/\text{s}$]
24. A jet issuing from a 30 mm nozzle held at 0.6 m above the ground level at an angle of 30° to the horizontal strikes the ground 4 m away. Determine:
(i) The maximum height reached,
(ii) The range of the jet, and
(iii) The discharge.

[Ans. (i) 0.458 m above the nozzle tip

(ii) 3.18 m, $0.00425 \text{ m}^3/\text{s}$]

25. A 30 mm fire nozzle held at 1.5 m above ground discharging 10 lps has to reach a window in a wall 15 m away and 10 m above ground. At what angle or angles of the inclination to the horizontal, the nozzle is to be held? Neglect air resistance. [Ans. 32°]
26. In Fig 6.96 what is the power developed by the turbine assuming efficiency of 80%?

[Ans. 54 kW]

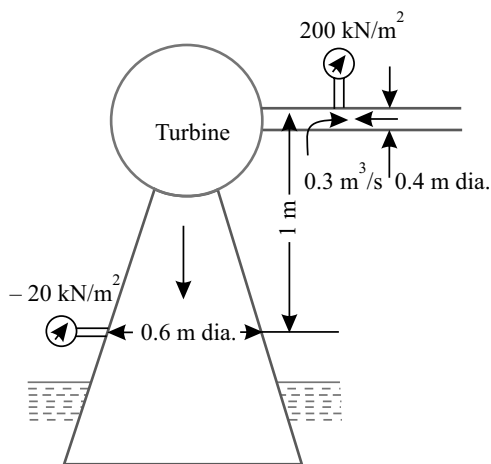


Fig. 6.96

27. The sprinkler shown in Fig 6.97 has nozzles of 5 mm diameter and carries a total discharge of 0.20 litres/sec. Determine:

- The angular speed of rotation of the sprinkler, and
- The torque required to hold the sprinkler stationary.

Assume no friction at the pivot.

[Ans. (i) 10.192 rad./s (ii) 0.05086 Nm]

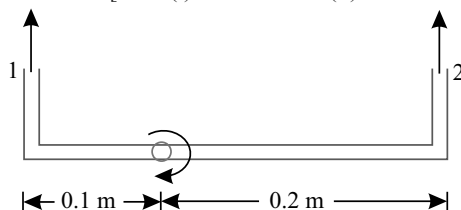


Fig. 6.97

28. Find the maximum speed of an open circular cylinder having 150 mm diameter, 1 m length and containing water up to a height of 800 mm at which it should be rotated about its vertical axis so that no water spills. [Ans. 356.67 r.p.m.]
29. A 150 mm diameter open circular cylinder is 1 m long and contains water up to a height of 0.7 m. Estimate the speed at which the cylinder may be rotated about its vertical axis so that the axial depth becomes zero. [Ans. 563.88 r.p.m.]
30. For the unsolved example 29, find the difference in total pressure force due to rotation:
- At the bottom of the cylinder, and
 - On the sides of the cylinder.
- [Ans. (i) 34.72 N (ii) 1178.4 N]
31. An open cylindrical vessel 120 mm in diameter and 300 mm deep is filled with water up to the

top. Estimate the volume of water left in the vessel when it is rotated about its vertical axis:

- With a speed of 300 r.p.m. and
- With a speed of 600 r.p.m.

[Ans. (i) 2369.37 cm³ (ii) 702.67 cm³]

32. A cylindrical vessel closed at the top is 0.2 m in diameter, 1.2 m long and contains water up to a height of 0.8 m.
- Find the height of the paraboloid formed, if it is rotated at 400 r.p.m. about its vertical axis.
 - Find the speed of rotation of the vessel when axial depth is zero.

[Ans. (i) 0.845 m (ii) 567.2 r.p.m.]

33. A vessel cylindrical in shape and closed at the top and bottom is 300 mm in diameter, 1 m long and contains water up to a depth of 0.8 m. The air above the water surface is at a pressure of 60 kN/m². If the vessel is rotated at a speed of 250 r.p.m. about its vertical axis find the pressure head at the bottom of the vessel
- At the centre, and
 - At the edge.

[Ans. (i) 6.439 m of water, (ii) 7.225 m of water.]

34. A vessel cylindrical in shape and closed at the top and bottom is 200 mm in diameter and 0.15 m in height. The vessel is completely filled with water. If it is rotated about its vertical axis with a speed of 200 r.p.m., what is the total pressure force exerted by water on the top and bottom of the vessel? [Ans. 0.0344 kN, 0.0806 kN]

35. An open tank 5m long, 2m deep and 3 m wide contains oil of relative density 0.9 to a depth of 0.9 m. If the tank is accelerated along its length on a horizontal track at a constant value of 3 m/s², determine:

- The new position of the oil surface.
- The pressures at the bottom of the tank at the front and rear edges.
- The amount of spill if the tank is given a horizontal acceleration of 4.5 m/s² instead of 3 m/s².

[Ans. (i) $\theta = 17^\circ$; (ii) 1.189 kN/m², 14.67 kN/m²; (iii) 0.42 m³]

36. A spherical tank of 1.2 m radius is half-filled with oil of relative density 0.8. If the tank is given a horizontal acceleration of 10 m/s², calculate:
- The inclination of the oil surface to horizontal.
 - The maximum pressure on the tank.

[Ans. (i) 45.55°; (ii) 9.4 kN/m²]

37. A closed tank 6 m long, 2 m wide and 1.8 m deep initially contains water to a depth of 1.2 m. The top has an opening in the front part to have air space at atmospheric pressure. If the tank is given a horizontal acceleration at a constant value of 2.4 m/s^2 along its length, calculate the total pressure force on the top of the tank.

[Ans. 0.792 kN]

38. An open cubical tank with each side 1.5 m contains oil of the specific weight 7.5 kN/m^3 . Calculate:
- (i) The force acting on side of the tank when it is being moved with an acceleration of

$\frac{g}{2} \text{ m/s}^2$ in vertically upward and downward directions.

- (ii) The pressure at bottom of the tank when the acceleration rate is $g \text{ m/s}^2$ vertically downwards.

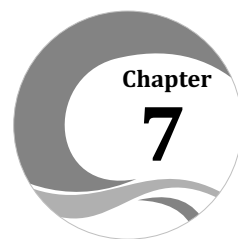
[Ans (i) 18.984 kN; 6.328 kN (ii) zero]

39. An open rectangular tank 5 m long $\times$ 2 m wide is filled with water to a depth of 1.5 m. Find the slope of water surface when tank moves with an acceleration of 3 m/s^2

(i) up a 30° inclined plane, and

(ii) down a 30° inclined plane.

[Ans. (i) 12.89° ; (ii) 17.37°]



DIMENSIONAL AND MODEL ANALYSIS

- 7.1. Dimensional analysis—Introduction.
- 7.2. Dimensions.
- 7.3. Dimensional homogeneity
- 7.4. Methods of dimensional analysis—Rayleigh's method—Buckingham's π —method/theorem—Limitations of dimensional analysis.
- 7.5. Model analysis—Introduction.
- 7.6. Similitude.
- 7.7. Forces influencing hydraulic phenomena.
- 7.8. Dimensionless numbers and their significance—Reynolds number (Re)—Froude number (Fr)—Euler number (Eu)—Weber number (We)—Mach number (M).
- 7.9. Model (or similarity) laws
- 7.10. Reynolds model law.
- 7.11. Froude model law.
- 7.12. Euler model law.
- 7.13. Weber model law.
- 7.14. Mach model law.
- 7.15. Types of models—undistorted models—distorted models.
- 7.16. Scale effect in models.
- 7.17. Limitations of hydraulic similitude.

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

DIMENSIONAL ANALYSIS

7.1. DIMENSIONAL ANALYSIS—INTRODUCTION

Dimensional analysis is a mathematical technique which makes use of the study of the dimensions for solving several engineering problems. Each physical phenomenon can be expressed by an equation giving relationship between different quantities, such quantities are dimensional and non-dimensional. Dimensional analysis helps in determining a systematic arrangement of the variables in the physical relationship, combining dimensional variables to form non-dimensional parameters. It is based on the *principle of dimensional homogeneity* and uses the dimensions of relevant variables affecting the phenomenon.

Dimensional analysis has become an important tool for analysing fluid flow problems. It is *especially useful in presenting experimental results in a concise form.*

Uses of dimensional analysis:

The uses of dimensional analysis may be summarised as follows:

1. To test the dimensional homogeneity of any equation of fluid motion.
2. To derive rational formulae for a flow phenomenon.
3. To derive equations expressed in terms of non-dimensional parameters to show the relative significance of each parameter.
4. To plan model tests and present experimental results in a systematic manner, thus making it possible to analyse the complex fluid flow phenomenon.

Advantages of dimensional analysis:

Dimensional analysis entails the following *advantages*:

1. It expresses the functional relationship between the variables in dimensionless terms.
2. In hydraulic model studies it reduces the number of variables involved in a physical phenomenon, generally by *three*.
3. By the proper selection of variables, the dimensionless parameters can be used to make certain logical deductions about the problem.
4. Design curves, by the use of dimensional analysis, can be developed from experimental data or direct solution of the problem.
5. It enables getting up a theoretical equation in a simplified dimensional form.
6. Dimensional analysis provides partial solutions to the problems that are too complex to be dealt with mathematically.
7. The conversion of units of quantities from one system to another is facilitated.

7.2. DIMENSIONS

The various physical quantities used in fluid phenomenon can be expressed in terms of **fundamental quantities** or *primary quantities*. The fundamental quantities are *mass, length, time* and *temperature*, designated by the letters, M, L, T, θ respectively. Temperature is specially useful in *compressible flow*. The quantities which are expressed in terms of the fundamental or primary quantities are called **derived or secondary quantities**, (e.g., velocity, area, acceleration etc.). The expression for a derived quantity in terms of the primary quantities is called the **dimension** of the physical quantity.

A quantity may either be expressed dimensionally in $M-L-T$ or $F-L-T$ system (some engineers prefer to use force instead of mass as fundamental quantity because the force is easy to measure). Table 7.1 gives the dimensions of various quantities used in both the systems.

Example. 7.1. Determine the dimensions of the following quantities:

- | | |
|-----------------|---------------------------|
| (i) Discharge, | (ii) Kinematic viscosity, |
| (ii) Force, and | (iv) Specific weight. |

Solution. (i) Discharge = Area $\times$ velocity

$$= L^2 \times \frac{L}{T} = \frac{L^3}{T} = L^3 T^{-1} \text{ (Ans.)}$$

(ii) Kinematic viscosity (v) $= \frac{\mu}{\rho}$

where μ is given by: $\tau = \mu \frac{du}{dy}$

$$\begin{aligned} \therefore \mu &= \frac{\tau}{\frac{du}{dy}} = \frac{\text{Shear stress}}{\frac{L}{T} \times \frac{I}{L}} = \frac{\text{Force/area}}{1/T} \\ &= \frac{\text{Mass} \times \text{acceleration}}{\text{Area} \times 1/T} = \frac{M \times \frac{L}{T^2}}{L^2 \times \frac{1}{T}} = \frac{ML}{L^2 T^2 \times \frac{1}{T}} \\ &= \frac{M}{LT} = ML^{-1} T^{-1} \text{ and } \rho = \frac{\text{Mass}}{\text{Volume}} = \frac{M}{L^3} = ML^{-3} \end{aligned}$$

$$\therefore \text{Kinematic viscosity } (\nu) = \frac{\mu}{\rho} = \frac{ML^{-1}T^{-1}}{ML^{-3}} = L^2T^{-1} \text{ (Ans.)}$$

(iii) Force = mass $\times$ acceleration

$$= M \times \frac{\text{length}}{\text{time}^2} = \frac{ML}{T^2} = MLT^{-2} \text{ (Ans.)}$$

$$(iv) \text{ Specific weight} = \frac{\text{Weight}}{\text{Volume}} = \frac{\text{Force}}{\text{Volume}} = \frac{MLT^{-2}}{L^3} = ML^{-2}T^{-2} \text{ (Ans.)}$$

Table 7.1. Quantities used in Fluid Mechanics and Heat Transfer and their Dimensions

S.No.	Quantity	Dimensions	
		M-L-T System	F-L-T System
	(a) Fundamental Quantities		
1.	Mass, M	M	FL^{-1}T^2
2.	Length, L	L	L
3.	Time, T	T	T
	(b) Geometric Quantities		
4.	Area, A	L^2	L^2
5.	Volume, $\forall$	L^3	L^3
6.	Moment of inertia	L^4	L^4
	(c) Kinematic Quantities		
7.	Linear velocity, u, V, U	LT^{-1}	LT^{-1}
8.	Angular velocity, ω ; rotational speed, N	T^{-1}	T^{-1}
9.	Acceleration, a	LT^{-2}	LT^{-2}
10.	Angular acceleration, α	T^{-2}	T^{-2}
11.	Discharge, Q	L^3T^{-1}	L^3T^{-1}
12.	Gravity, g	LT^{-2}	LT^{-2}
13.	Kinematic viscosity, ν	L^2T^{-1}	L^2T^{-1}
14.	Stream function, ψ , circulation, Γ	L^2T^{-1}	L^2T^{-1}
15.	Vorticity, Ω	T^{-1}	T^{-1}
	(d) Dynamic Quantities		
16.	Force, F	MLT^{-2}	F
17.	Density, ρ	ML^{-3}	FL^{-4}T^2
18.	Specific weight, w	$\text{ML}^{-2}\text{T}^{-2}$	FL^{-3}
19.	Dynamic viscosity, μ	$\text{ML}^{-1}\text{T}^{-1}$	FL^{-2}T
20.	Pressure, p ; shear stress, τ	$\text{ML}^{-1}\text{T}^{-2}$	FL^{-2}
21.	Modulus of elasticity, E, K	$\text{ML}^{-1}\text{T}^{-2}$	FL^{-2}
22.	Momentum	MLT^{-1}	FT
23.	Angular momentum or moment of momentum	ML^2T^{-1}	FLT
24.	Work, W ; energy, E	ML^2T^{-2}	FL
25.	Torque, T	ML^2T^{-2}	FL
26.	Power, P	ML^2T^{-3}	FLT^{-1}
	(e) Thermodynamic Quantities		
27.	Temperature	θ	θ
28.	Thermal conductivity	$\text{MLT}^{-3}\theta^{-1}$	$\text{FT}^{-1}\theta^{-1}$
29.	Enthalpy per unit mass	L^2T^{-2}	L^2T^{-2}

S.No.	Quantity	Dimensions	
		M-L-T System	F-L-T System
30.	Gas constant	$L^2 T^{-2} \theta^{-1}$	$L^2 T^{-2} \theta^{-1}$
31.	Entropy	$ML^2 T^{-2} \theta^{-1}$	$FL\theta^{-1}$
32.	Internal energy per unit mass	$L^2 T^{-2}$	$L^2 T^{-2}$
33.	Heat transfer	$ML^2 T^{-3}$	FLT^{-1}

7.3. DIMENSIONAL HOMOGENEITY

A physical equation is the relationship between two or more physical quantities. Any *correct equation* expressing a physical relationship between quantities *must be dimensionally homogeneous* (according to Fourier's principle of dimensional homogeneity) and *numerically equivalent*. *Dimensional homogeneity* states that *every term in an equation when reduced to fundamental dimensions must contain identical powers of each dimension*. A dimensionally homogeneous equation is applicable to all systems of units. In a dimensionally homogeneous equation, only quantities having the same dimensions can be added, subtracted or equated. Let us consider the equation:

$$p = wh$$

$$\text{Dimensions of L.H.S.} = ML^{-1}T^{-2}$$

$$\text{Dimensions of R.H.S.} = ML^{-2}T^{-2} \times L = ML^{-1}T^{-2}$$

$$\text{Dimensions of L.H.S.} = \text{Dimensions of R.H.S.}$$

$\therefore$ Equation $p = wh$ is dimensionally homogeneous; so it can be used in any system of units.

Applications of Dimensional Homogeneity:

The *principle of homogeneity* proves useful in the following ways:

1. It facilitates to determine the dimensions of a physical quantity.
2. It helps to check whether an equation of any physical phenomenon is dimensionally homogeneous or not.
3. It facilitates conversion of units from one system to another.
4. It provides a step towards dimensional analysis which is fruitfully employed to plan experiments and to present the results meaningfully.

Example 7.2. Determine the dimensions of E in the dimensionally homogeneous Einstein's equation

$$E = mc^2 \left[\frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} - 1 \right]$$

where v is the velocity and m is the mass.

Solution. Since the expression is dimensionally homogeneous, the term

$$\frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

should be dimensionless

$$\text{i.e.,} \quad [c] = [v] = \frac{L}{T}$$

$$\therefore [E] = m[c]^2 = M \left[\frac{L^2}{T^2} \right] = ML^2T^{-2}$$

i.e. E has the dimensions of **energy**. (Ans.)

7.4. METHODS OF DIMENSIONAL ANALYSIS

With the help of dimensional analysis the equation of a physical phenomenon can be developed in terms of dimensionless groups or parameters and thus reducing the number of variables. *The methods of dimensional analysis are based on the Fourier's principle of homogeneity.* The methods of dimensional analysis are:

1. Rayleigh's method
2. Buckingham's π -method
3. Bridgman's method
4. Matrix-tensor method
5. By visual inspection of the variables involved
6. Rearrangement of differential equations.

Here only first two methods will be dealt with.

7.4.1. Rayleigh's Method

This method gives a special form of relationship among the dimensionless groups, and has the *inherent drawback* that it *does not provide any information regarding the number of dimensionless groups to be obtained as a result of dimensional analysis.* Due to this reason this method has become *obsolete and is not favoured for use.*

Rayleigh's method is used for determining the expression for a variable which depends upon *maximum three or four variables only.* In case the number of independent variables becomes more than four, then it is very difficult to find the expression for the dependent variable.

In this method a functional relationship of some variables is expressed in the form of an exponential equation which must be dimensionally homogeneous. Thus if X is a variable which depends on $X_1, X_2, X_3, \dots, X_n$; the functional equation can be written as:

$$X = f(X_1, X_2, X_3, \dots, X_n) \quad \dots(7.1)$$

In the above equation X is a *dependent variable*, while $X_1, X_2, X_3, \dots, X_n$ are *independent variables*. A *dependent variable* is the one about which information is required while *independent variables* are those which govern the variation of dependent variable.

Eqn. (7.1) can also be written as:

$$X = C(X_1^a, X_2^b, X_3^c, \dots, X_n^n) \quad \dots(7.2)$$

where, C is a constant and $a, b, c, \dots$ are the arbitrary powers. The values of $a, b, c, \dots, n$ are obtained by comparing the powers of the fundamental dimensions on both sides. Thus the expression is obtained for dependent variable.

Example 7.3. Find an expression for the drag force on smooth sphere of diameter D , moving with a uniform velocity V in a fluid density ρ and dynamic viscosity μ . (PTU)

Solution. The force drag F is a function of

- | | |
|----------------------------------|--------------------------------|
| (i) Diameter D , | (ii) Velocity V , |
| (iii) Fluid density ρ , and | (iv) Dynamic viscosity μ . |

$$\text{Mathematically,} \quad F = f(D, V, \rho, \mu) \quad \text{or} \quad F = C(D^a \cdot V^b \cdot \rho^c \cdot \mu^d) \quad \dots(1)$$

where, C is a non-dimensional constant.

Using $M-L-T$ system the corresponding equation for dimensions is:

$$MLT^{-2} = [CL^a \cdot (LT^{-1})^b \cdot (ML^{-3})^c \cdot (ML^{-1}T^{-1})^d]$$

For dimensional homogeneity the exponents of each dimension on both sides of the equation must be identical. Thus:

$$\text{For M:} \quad 1 = c + d \quad \dots(i)$$

$$\text{For L:} \quad 1 = a + b - 3c - d \quad \dots(ii)$$

$$\text{For T:} \quad -2 = -b - d \quad \dots(iii)$$

There are *four unknowns* (a, b, c, d) but *equations are three* in number. Therefore, it is not possible to find the values of a, b, c and d . However, *three of them can be expressed in terms of fourth variable which is most important*. Here the role of viscosity is vital one and hence a, b, c are expressed in terms of d (i.e. power to viscosity)

$$\therefore \quad c = 1 - d \quad \dots \text{ from (i)}$$

$$b = 2 - d \quad \dots \text{ from (iii)}$$

Putting these values in (i), we get:

$$\begin{aligned} a &= 1 - b + 3c + d = 1 - 2 + d + 3(1 - d) + d \\ &= 1 - 2 + d + 3 - 3d + d = 2 - d \end{aligned}$$

Substituting these values of exponents in eqn. (1), we get:

$$\begin{aligned} F &= C[D^{2-d} \cdot V^{2-d} \cdot \rho^{1-d} \cdot \mu^d] \\ &= C[D^2 V^2 \rho(D^{-d} \cdot V^{-d} \cdot \rho^{-d} \cdot \mu^d)] = C \left[\rho D^2 V^2 \left(\frac{\mu}{\rho V D} \right)^d \right] \\ &= \rho D^2 V^2 \phi \left(\frac{\mu}{\rho V D} \right) \quad (\text{Ans.}) \end{aligned}$$

Example 7.4. The efficiency η of a fan depends on the density ρ , the dynamic viscosity μ of the fluid, the angular velocity ω , diameter D of the rotor and the discharge Q . Express η in terms of dimensionless parameters. [UPSC]

Solution. The efficiency η of a fan is a function of:

- (i) Density ρ , (ii) Viscosity μ ,
- (iii) Angular velocity ω , (iv) Diameter D , and
- (v) Discharge Q .

Mathematically, $\eta = f(\rho, \mu, \omega, D, Q)$

$$\text{or,} \quad \eta = C(\rho^a, \mu^b, \omega^c, D^d, Q^e) \quad \dots(1)$$

where C is a non-dimensional constant.

Using $M-L-T$ system, the corresponding equation for dimensions is:

$$M^0 L^0 T^0 = C[(ML^{-3})^a (ML^{-1}T^{-1})^b (T^{-1})^c (L)^d (L^3 T^{-1})^e]$$

For dimensional homogeneity the exponents of each dimension on both sides of the equation must be identical. Thus:

$$\text{For M:} \quad 0 = a + b$$

$$\text{For L:} \quad 0 = -3a - b + d + 3e$$

$$\text{For T:} \quad 0 = -b - c - e$$

There are *five variables* and we have only *three equations*. Experience has shown that recognized dimensionless groups appear if the exponents of D, ω and ρ are evaluated in terms of b and e (exponents of viscosity and discharge which are *more important*)

$$\therefore \quad a = -b; c = (b + e);$$

$$d = 3a + b - 3e = 3(-b) + b - 3e = -2b - 3e = -(2b + 3e)$$

Substituting these values of exponents in eqn. (1), we get:

$$\begin{aligned}\eta &= C(\rho^{-b} \cdot \mu^b \cdot \omega^{-(b+e)} \cdot D^{-(2b+3e)} \cdot Q^e) \\ &= C(\rho^{-b} \cdot \mu^b \cdot \omega^{-b} \cdot \omega^{-e} D^{-2b} \cdot D^{-3e} \cdot Q^e) \\ &= C \left[\left(\frac{\mu}{\rho \omega D^2} \right)^b \left(\frac{Q}{\omega D^3} \right)^e \right] \\ &= \phi \left[\left(\frac{\mu}{\rho \omega D^2} \right), \left(\frac{Q}{\omega D^3} \right) \right] \text{ (Ans.)}\end{aligned}$$

Example 7.5. The resistance force R of a supersonic plane during flight can be considered as dependent upon the length of the aircraft l , velocity V , air viscosity μ , air density ρ and bulk modulus of air K . Express the functional relationship between these variables and the resisting force.

[Anna University]

Solution. The resistance force R is a function of:

- (i) Length l , (ii) Velocity V ,
- (iii) Air viscosity μ , (iv) Air density ρ , and
- (v) Bulk modulus K .

Mathematically,

$$R = f(l, V, \mu, \rho, K)$$

or,

$$R = C(l^a \cdot V^b \cdot \mu^c \cdot \rho^d \cdot K^e) \quad \dots(1)$$

where, C is a non-dimensional constant.

Using M-L-T system, the corresponding equation for dimensions is:

$$MLT^{-2} = C[l^a \cdot (LT^{-1})^b \cdot (ML^{-1}T^{-1})^c \cdot (ML^{-3})^d \cdot (ML^{-1}T^{-2})^e]$$

For dimensional homogeneity the exponents of each dimension on both sides of the equation must be identical. Thus:

$$\text{For M:} \quad 1 = c + d + e$$

$$\text{For L:} \quad 1 = a + b - c - 3d - e$$

$$\text{For T:} \quad -2 = -b - c - 2e$$

There are five variables and we have only three equations. Expressing the three unknowns (a , b , d) in terms of c and e (exponents of viscosity and bulk modulus which are more important).

$$\begin{aligned}\therefore \quad d &= 1 - c - e \\ b &= 2 - c - 2e \\ a &= 1 - b + c + 3d + e \\ &= 1 - (2 - c - 2e) + c + 3(1 - c - e) + e \\ &= 1 - 2 + c + 2e + c + 3 - 3c - 3e + e = 2 - c\end{aligned}$$

Substituting these values of exponents in eqn. (1), we get:

$$\begin{aligned}R &= C(l^{2-c} \cdot V^{2-c-2e} \cdot \mu^c \cdot \rho^{1-c-e} \cdot K^e) \\ &= C l^2 \cdot V^2 \cdot \rho(l^{-c} \cdot V^{-c} \cdot \mu^c \cdot \rho^{-c}) \cdot V^{-2e} \cdot \rho^{-e} \cdot K^e \\ &= C l^2 V^2 \rho \left(\frac{\mu}{l V \rho} \right)^c \left(\frac{K}{V^2 \rho} \right)^e \\ &= l^2 V^2 \rho \phi \left[\left(\frac{\mu}{l V \rho} \right), \left(\frac{K}{V^2 \rho} \right) \right] \text{ (Ans.)}\end{aligned}$$

Example 7.6. A partially submerged body is towed in water. The resistance R to its motion depends on the density ρ , the viscosity μ of water, length l of the body, velocity V of the body and acceleration due to gravity g . Show that the resistance to motion can be expressed in the form

$$R = \rho l^2 V^2 \phi \left[\left(\frac{\mu}{\rho l V} \right), \left(\frac{lg}{V^2} \right) \right]$$

Solution. The resistance R is a function of:

- (i) Density ρ ,
- (ii) Viscosity μ ,
- (iii) Length l ,
- (iv) Velocity V , and
- (v) Acceleration due to gravity g .

Mathematically, $R = f(\rho, \mu, l, V, g)$

$$\text{or, } R = C (\rho^a \cdot \mu^b \cdot l^c \cdot V^d \cdot g^e) \quad \dots(1)$$

where, C is a non-dimensional constant.

Using M-L-T system, the corresponding equation for dimensions is:

$$MLT^{-2} = C[(ML^{-3})^a \cdot (ML^{-1}T^{-1})^b \cdot (L)^c \cdot (LT^{-1})^d \cdot (LT^{-2})^e]$$

For dimensional homogeneity the exponents of each dimension on both sides of the equation must be identical. Thus:

$$\text{For M: } 1 = a + b$$

$$\text{For L: } 1 = -3a - b + c + d + e$$

$$\text{For T: } -2 = -b - d - 2e$$

There are five variables and we have only three equations. To get the required result, we shall evaluate exponents of ρ, l, V (i.e. a, c, d) in terms of other unknowns (i.e. b, e)

$$\therefore a = 1 - b$$

$$d = 2 - b - 2e$$

$$\begin{aligned} c &= 1 + 3a + b - d - e = 1 + 3(1 - b) + b - (2 - b - 2e) - e \\ &= 1 + 3 - 3b + b - 2 + b + 2e - e = 2 - b + e \end{aligned}$$

Substituting these values of components in eqn. (1), we get:

$$\begin{aligned} R &= C[\rho^{1-b} \cdot \mu^b \cdot l^{2-b+e} \cdot V^{2-b-2e} \cdot g^e] \\ &= C[\rho l^2 V^2 (\rho^{-b} \cdot \mu^b \cdot l^{-b} \cdot V^{-b}) (l^e \cdot V^{-2e} \cdot g^e)] \\ &= C \left[\rho l^2 V^2 \left(\frac{\mu}{\rho l V} \right)^b \cdot \left(\frac{lg}{V^2} \right)^e \right] \\ &= \rho l^2 V^2 \phi \left[\left(\frac{\mu}{\rho l V} \right), \left(\frac{lg}{V^2} \right) \right] \quad \dots(\text{Proved.}) \end{aligned}$$

Example 7.7. The pressure drop Δp in a pipe of diameter D and length l depends on the density ρ and viscosity μ of fluid flowing, mean velocity V of flow and average height of protuberance t . Show that the pressure drop can be expressed in the form:

$$\Delta p = \rho V^2 \phi \left(\frac{l}{D}, \frac{\mu}{VD\rho}, \frac{t}{D} \right) \quad [\text{RGPV, Bhopal}]$$

Solution. The pressure drop Δp , is a function of : D, l, ρ, μ, V and t

Mathematically, $\Delta p = f(D, l, \rho, \mu, V, t)$

$$\text{or, } \Delta p = C(D^a \cdot l^b \cdot \rho^c \cdot \mu^d \cdot V^e \cdot t^f) \quad \dots(1)$$

where, C is a non-dimensional constant.

Using M-L-T system, the corresponding equation for dimensions is:

$$ML^{-1}T^{-2} = C [L^a \cdot L^b \cdot (ML^{-3})^c \cdot (ML^{-1}T^{-1})^d \cdot (LT^{-1})^e \cdot (L)^f]$$

For dimensional homogeneity the exponents of each dimension on both sides of the equation must be identical. Thus,

$$\text{For M:} \quad 1 = c + d$$

$$\text{For L:} \quad -1 = a + b - 3c - d + e + f$$

$$\text{For T:} \quad -2 = -d - e$$

There are *six variables* and we have only *three equations*. To get the required result, we shall evaluate exponents of D , ρ and V (i.e. a , c and e) in terms of the unknowns (i.e. b , d , f).

$$\begin{aligned} \therefore \quad c &= 1 - d; e = 2 - d \\ a &= -1 - b + 3c + d - e - f \\ &= -1 - b + 3(1 - d) + d - (2 - d) - f \\ &= -1 - b + 3 - 3d + d - 2 + d - f \\ &= -(b + d + f) \end{aligned}$$

Substituting these values of exponents in eqn. (1), we get:

$$\begin{aligned} \Delta p &= C (D^{-(b+d+f)} \cdot l^b \cdot \rho^{2-d} \cdot \mu^d \cdot V^{2-d} \cdot t^f) \\ &= C [\rho V^2 (D^{-b} \cdot l^b) (D^{-d}) \cdot \rho^{-d} \cdot \mu^d V^{-d} (D^{-f} \cdot t^f)] \\ &= C \left[\rho V^2 \left(\frac{l}{D} \right)^b \left(\frac{\mu}{VD\rho} \right)^d \left(\frac{t}{D} \right)^f \right] \\ &= \rho V^2 \phi \left(\frac{l}{D}, \frac{\mu}{VD\rho}, \frac{t}{D} \right) \quad \dots \text{Proved} \end{aligned}$$

From experiments it has been observed that Δp is a linear function of $\frac{t}{D}$.

$$\therefore \quad \Delta p = \rho V^2 \frac{l}{D} \phi \left(\frac{\mu}{VD\rho}, \frac{t}{D} \right)$$

$$\text{or,} \quad \frac{\Delta p}{w} = \frac{V^2}{g} \frac{l}{D} \phi \left(\frac{\mu}{VD\rho}, \frac{t}{D} \right) \quad \left(\because \rho = \frac{w}{g} \right)$$

This is usually written in the form,

$$h_f = \frac{4flV^2}{2gD} \quad (\text{Darcy-Weisbach formula})$$

where, f is the co-efficient of friction which depends upon the Reynolds number $\left(\frac{VD\rho}{\mu} \right)$ and

the surface finish of the pipe $\left(\frac{t}{D} \right)$.

7.4.2. Buckingham's π -Method/Theorem

When a large number of physical variables are involved Rayleigh's method of dimensional analysis becomes *increasingly* laborious and *cumbersome*. Buckingham's method is an *improvement* over Rayleigh's method. Buckingham designated the dimensionless group by the Greek capital letter π (Pi). It is therefore often called *Buckingham π -method*. The *advantage of this method over Rayleigh's method is that it lets us know, in advance, of the analysis, as to how many dimensionless groups are to be expected.*

Buckingham's π -theorem states as follows:

“If there are n variables (dependent and independent variables) in a dimensionally homogeneous equation and if these variables contain m fundamental dimensions (such as M , L , T , etc.) then the variables are arranged into $(n-m)$ dimensionless terms. These dimensionless terms are called π -terms.”

Mathematically, if any variable X_1 , depends on independent variables, $X_2, X_3, X_4, \dots, X_n$; the functional equation may be written as:

$$X_1 = f(X_2, X_3, X_4, \dots, X_n) \quad \dots(7.3)$$

Eqn. (7.3) can also be written as:

$$f_1(X_1, X_2, X_3, \dots, X_n) = 0 \quad \dots(7.4)$$

It is a dimensionally homogeneous equation and contains n variables. If there are m fundamental dimensions, then according to Buckingham's π -theorem, it [eqn. (7.4)] can be written in terms of number of π -terms (dimensionless groups) in which number of π -terms is equal to $(n-m)$. Hence, eqn. (7.4) becomes as:

$$f_1(\pi_1, \pi_2, \pi_3, \dots, \pi_{n-m}) = 0 \quad \dots(7.5)$$

Each dimensionless π -term is formed by combining m variables out of the total n variables with one of the remaining $(n-m)$ variables i.e. each π -term contains $(m+1)$ variables. These m variables which appear repeatedly in each of π -terms are consequently called **repeating variables** and are chosen from among the variables such that they together *involve all the fundamental dimensions and they themselves do not form a dimensionless parameter*. Let in the above case X_2, X_3 and X_4 are the repeating variables if the fundamental dimensions $m(M, L, T) = 3$, then each term is written as:

$$\left. \begin{aligned} \pi_1 &= X_2^{a_1} \cdot X_3^{b_1} \cdot X_4^{c_1} \cdot X_1 \\ \pi_2 &= (X_2^{a_2} \cdot X_3^{b_2} \cdot X_4^{c_2} \cdot X_5) \\ &\vdots \\ \pi_{n-m} &= (X_2^{a_{n-m}} \cdot X_3^{b_{n-m}} \cdot X_4^{c_{n-m}} \cdot X_n) \end{aligned} \right\} \quad \dots(7.6)$$

where $a_1, b_1, c_1; a_2, b_2, c_2$ etc. are the constants, which are determined by considering dimensional homogeneity. These values are substituted in eqn. (7.6) and values of $\pi_1, \pi_2, \pi_3, \dots, \pi_{n-m}$ are obtained. These values of π 's are substituted in eqn. (7.5). The final general equation for the phenomenon may then be obtained by expressing anyone of the π -terms as a function of the other as

$$\left. \begin{aligned} \pi_1 &= \phi(\pi_2, \pi_3, \pi_4, \dots, \pi_{n-m}) \\ \pi_2 &= \phi(\pi_1, \pi_3, \pi_4, \dots, \pi_{n-m}) \end{aligned} \right\} \quad \dots(7.7)$$

Selection of repeating variables:

The following points should be kept in view while selecting m repeating variables:

1. m repeating variables must contain *jointly* all the fundamental dimensions involved in the phenomenon. Usually the fundamental dimensions are M, L and T . However, if only two dimensions are involved, there will be 2 repeating variables and they must contain *together* the two dimensions involved.
2. The repeating variables *must not* form the non-dimensional parameters among themselves.
3. As far as possible, the dependent variable *should not* be selected as repeating variable.
4. *No two repeating variables should have the same dimensions.*
5. The repeating variables should be chosen in such a way that one variable contains **geometric property** (e.g. length, l ; diameter, d ; height, H etc.), other variable contains **flow property**

(e.g. velocity, V ; acceleration, a etc.) and third variable contains **fluid property** (e.g. mass density, ρ ; weight density, w dynamic viscosity, μ etc.).

The choice of repeating variables, in most of fluid mechanics problems, may be:

- (i) l, V, ρ
(ii) d, V, ρ
(iii) l, V, μ
(iv) d, V, μ .

The procedure for solving problems by Buckingham's π -theorem in the Example 7.8 below:

Example 7.8. The resistance R experienced by a partially submerged body depends upon the velocity V , length of the body l , viscosity of the fluid μ , density of the fluid ρ and gravitational acceleration g . Obtain a dimensionless expression for R . [UPTU]

Solution. Step 1. The resistance R is a function of:

- (i) Velocity V , (ii) Length l , (iii) Viscosity μ ,
(iv) Density ρ , and (v) Gravitational acceleration g .

$$\text{Mathematically,} \quad R = f(V, l, \mu, \rho, g) \quad \dots(i)$$
$$\text{or, } f_1(R, V, l, \mu, \rho, g) = 0 \quad \dots(ii)$$

$\therefore$ Total number of variables, $n = 6$

$$\left\{ \begin{array}{l} m \text{ is obtained by writing dimensions of each variable as:} \\ R = MLT^{-2}, V = LT^{-1}, \mu = ML^{-1}T^{-1}, \rho = ML^{-3}, g = LT^{-2}. \text{ Thus the} \\ \text{fundamental dimensions in the problem are } M, L, T \text{ and hence } m = 3 \end{array} \right\}$$

Number of dimensionless π -terms = $n - m = 6 - 3 = 3$

Thus three π -terms say π_1 , π_2 , and π_3 are formed.

The eqn. (ii) may be written as:

$$f_1(\pi_1, \pi_2, \pi_3) = 0 \quad \dots(iii)$$

Step 2. Selection of repeating variables: Out of six variables R, V, l, μ, ρ, g three variables (as $m = 3$) are to be selected as *repeating variables*. R is a dependent variable and should *not* be selected as a repeating variable. Out of the remaining five variables one variable should have *geometric property*, second should have *flow property* and third one should have *fluid property*; these requirements are met by selecting l, V and ρ as *repeating variables*. The repeating variables themselves should not form a dimensionless term and must contain *jointly all fundamental dimensions equal to m i.e. 3* here. Dimensions of l, V and ρ are L, LT^{-1}, ML^{-3} and hence the three fundamental dimensions exist in l, V and ρ and also *no dimensionless group is formed by them*.

Step 3. Each π -term ($= m + 1$ variables) is written as given in eqn. (7.6), i.e.,

$$\left. \begin{aligned} \pi_1 &= l^{a_1} . V^{b_1} . \rho^{c_1} . R \\ \pi_2 &= l^{a_2} . V^{b_2} . \rho^{c_3} . \mu \\ \pi_3 &= l^{a_3} . V^{b_3} . \rho^{c_3} . g \end{aligned} \right\} \dots (iv)$$

Step 4. Each π -term is solved by the *principle of dimensional homogeneity*, as follows:

π_1 -term:

$$\begin{aligned}\pi_1 &= l^{a_1} \cdot V^{b_1} \cdot \rho^{c_1} \cdot R \\ M^0 L^0 T^0 &= L^{a_1} \cdot (LT^{-1})^{b_1} \cdot (ML^{-3})^{c_1} \cdot (MLT^{-2})\end{aligned}$$

Equating the exponents of M , L and T respectively, we get:

For M: $0 = c_1 + 1$

For L : $0 = a_1 + b_1 - 3c_1 + 1$

For T: $0 = -b_1 - 2$

$$\therefore c_1 = -1; b_1 = -2$$

$$\text{and, } a_1 = -b_1 + 3c_1 - 1 = 2 - 3 - 1 = -2$$

Substituting the values of a_1 , b_1 , and c_1 in π_1 , we get:

$$\therefore \pi_1 = l^{-2} \cdot V^{-2} \cdot \rho^{-1} \cdot R = \frac{R}{l^2 V^2 \rho} \quad \dots(v)$$

π_2 -term:

$$\begin{aligned} \pi_2 &= l^{a_2} \cdot V^{b_2} \cdot \rho^{c_2} \cdot \mu \\ M^0 L^0 T^0 &= L^{a_2} \cdot (LT^{-1})^{b_2} \cdot (ML^{-3})^{c_2} \cdot (ML^{-1}T^{-1}) \end{aligned}$$

Equating the exponents of M , L and T respectively, we get:

$$\text{For M: } 0 = c_2 + 1$$

$$\text{For L: } 0 = a_2 + b_2 - 3c_2 - 1$$

$$\text{For T: } 0 = -b_2 - 1$$

$$\therefore c_2 = -1; b_2 = -1$$

$$\text{and, } a_2 = -b_2 + 3c_2 + 1 = 1 - 3 + 1 = -1$$

Substituting the values of a_2 , b_2 , and c_2 in π_2 , we get:

$$\therefore \pi_2 = l^{-1} \cdot V^{-1} \cdot \rho^{-1} \cdot \mu = \frac{\mu}{lV\rho}$$

π_3 -term:

$$\begin{aligned} \pi_3 &= l^{a_3} \cdot V^{b_3} \cdot \rho^{c_3} \cdot g \\ M^0 L^0 T^0 &= L^{a_3} \cdot (LT^{-1})^{b_3} \cdot (ML^{-3})^{c_3} \cdot (LT^{-2}) \end{aligned}$$

Equating the exponents of M , L and T respectively, we get:

$$\text{For M: } 0 = c_3$$

$$\text{For L: } 0 = a_3 + b_3 - 3c_3 + 1$$

$$\text{For T: } 0 = -b_3 - 2$$

$$\therefore c_3 = 0; b_3 = -2$$

$$\text{and, } a_3 = -b_3 + 3c_3 - 1 = 2 + 0 - 1 = 1$$

Substituting the values of a_3 , b_3 , and c_3 in π_3 , we get:

$$\therefore \pi_3 = l^1 \cdot V^{-2} \cdot \rho^0 \cdot g = \frac{lg}{V^2}$$

Step 5. Substitute the values of π_1 , π_2 , π_3 in eqn. (iii). The functional relationship becomes:

$$f_1\left(\frac{R}{l^2 V^2 \rho}, \frac{\mu}{lV\rho}, \frac{lg}{V^2}\right) = 0$$

$$\begin{aligned} \text{or, } \frac{R}{l^2 V^2 \rho} &= \phi\left(\frac{\mu}{lV\rho}, \frac{lg}{V^2}\right) \\ &= \phi\left(\frac{\rho lV}{\mu}, \frac{V}{\sqrt{lg}}\right) \end{aligned}$$

The above step has been made on the postulate that *reciprocal of pi-term and its square root is non-dimensional*.

$$R = l^2 V^2 \rho \phi\left(\frac{\rho lV}{\mu}, \frac{V}{\sqrt{lg}}\right) \quad (\text{Ans.})$$

The resistance R is thus a function of Reynolds number $\left(\frac{\rho V l}{\mu}\right)$ and Froude's number $\left(\frac{V}{\sqrt{lg}}\right)$.

Example 7.9. Using Buckingham's π -theorem, show that the velocity through a circular orifice is given by

$$V = \sqrt{2gH} \phi \left[\frac{D}{H}, \frac{\mu}{\rho V H} \right]$$

where,

H = Head causing flow,

D = Diameter of the orifice,

μ = Co-efficient of viscosity,

ρ = Mass density, and

g = Acceleration due to gravity.

[GATE]

Solution. V is a function of: H, D, μ, ρ and g

Mathematically, $V = f(H, D, \mu, \rho, g)$... (i)

or, $f_1(V, H, D, \mu, \rho, g) = 0$... (ii)

$\therefore$ Total number of variables, $n = 6$

Writing dimensions of each variable, we have:

$$V = LT^{-1}, H = L, D = L, \mu = ML^{-1}T^{-1}, \rho = ML^{-3}, g = LT^{-2}$$

Thus, number of fundamental dimensions, $m = 3$

$\therefore$ Number of π -terms = $n - m = 6 - 3 = 3$

Eqn. (ii) can be written as:

$$f_1(\pi_1, \pi_2, \pi_3) = 0 \quad \dots (iii)$$

Each π -term contains $(m + 1)$ variables, where $m = 3$ and is also equal to repeating variables.

Choosing H, g, ρ as repeating variables (V being a dependent variable should not be chosen as repeating variable), we get three π -terms as:

$$\pi_1 = H^{a_1} \cdot g^{b_1} \rho^{c_1} \cdot V$$

$$\pi_2 = H^{a_2} \cdot g^{b_2} \rho^{c_2} \cdot D$$

$$\pi_3 = H^{a_3} \cdot g^{b_3} \rho^{c_3} \cdot \mu$$

π_1 -term:

$$\pi_1 = H^{a_1} \cdot g^{b_1} \rho^{c_1} \cdot V$$

$$M^0 L^0 T^0 = L^{a_1} \cdot (LT^{-2})^{b_1} \cdot (ML^{-3})^{c_1} \cdot (LT^{-1})$$

Equating the exponents of M, L and T respectively, we get:

$$\text{For M:} \quad 0 = c_1$$

$$\text{For L:} \quad 0 = a_1 + b_1 - 3c_1 + 1$$

$$\text{For T:} \quad 0 = -2b_1 - 1$$

$$\therefore \quad c_1 = 0; b_1 = -\frac{1}{2}$$

$$\text{and,} \quad a_1 = -b_1 + 3c_1 - 1 = \frac{1}{2} + 0 - 1 = -\frac{1}{2}$$

Substituting the values of a_1, b_1 and c_1 in π_1 , we get:

$$\therefore \quad \pi_1 = H^{-1/2} \cdot g^{-\frac{1}{2}} \cdot \rho^0 \cdot V = \frac{V}{\sqrt{gh}}$$

π_2 -term:

$$\pi_2 = H^{a_2} \cdot g^{b_2} \cdot \rho^{c_2} \cdot D$$

$$M^0 L^0 T^0 = L^{a_2} \cdot (LT^{-2})^{b_2} \cdot (ML^{-3})^{c_2} \cdot L$$

Equating the exponents of M , L and T respectively, we get:

For M : $0 = c_2$

For L : $0 = a_2 + b_2 - 3c_2 + 1$

For T : $0 = -2b_2$

$\therefore c_2 = 0; b_2 = 0$

and, $a_2 = -b_2 + 3c_2 - 1 = -1$

Substituting the values of a_2 , b_2 , and c_2 in π_2 , we get:

$$\pi_2 = H^{-1} \cdot g^0 \cdot \rho^0 \cdot D = \frac{D}{H}$$

 π_3 -term:

$$\pi_3 = H^{a_3} \cdot g^{b_3} \cdot \rho^{c_3} \cdot \mu$$

$$M^0 L^0 T^0 = L^{a_3} \cdot (LT^{-2})^{b_3} \cdot (ML^{-3})^{c_3} \cdot ML^{-1} T^{-1}$$

Equating the exponents of M , L and T respectively, we get:

For M : $0 = c_3 + 1$

For L : $0 = a_3 + b_3 - 3c_3 - 1$

For T : $0 = -2b_3 - 1$

$\therefore c_3 = -1; b_3 = -\frac{1}{2}$

and, $a_3 = -b_3 + 3c_3 + 1 = \frac{1}{2} - 3 + 1 = -\frac{3}{2}$

Substituting the values of a_3 , b_3 , and c_3 in π_3 , we get:

$$\pi_3 = H^{-3/2} \cdot g^{-\frac{1}{2}} \cdot \rho^{-1} \cdot \mu = \frac{\mu}{H^{3/2} \rho \sqrt{g}}$$

$$= \frac{\mu}{H \rho \sqrt{gH}} = \frac{\mu V}{H \rho V \sqrt{gH}} \quad (\text{Multiply and divide by } V)$$

$$= \frac{\mu}{H \rho V} \cdot \pi_1 \quad \left(\because \frac{V}{\sqrt{gH}} = \pi_1 \right)$$

Substituting the values of π_1 , π_2 and π_3 in eqn. (iii), we get:

$$f_1 \left(\frac{V}{\sqrt{gH}}, \frac{D}{H}, \frac{\mu}{H \rho V} \cdot \pi_1 \right) = 0$$

or, $\frac{V}{\sqrt{gH}} = \phi \left[\frac{D}{H}, \frac{\mu}{H \rho V} \cdot \pi_1 \right]$

or, $V = \sqrt{2gH} \phi \left[\frac{D}{H}, \frac{\mu}{\rho V H} \right] \quad \dots \text{Proved.}$

(Multiplying or dividing by any constant does not change the character of π -terms).**Example 7.10.** Show that the lift F_L on airfoil can be expressed as

$$F_L = \rho V^2 d^2 \phi \left(\frac{\rho V d}{\mu}, \alpha \right)$$

where,

ρ = Mass density,

V = Velocity of flow,

d = A characteristic depth,

α = Angle of incidence, and

μ = Co-efficient of viscosity.

Solution. Lift F_L is a function of: ρ, V, d, μ, α

Mathematically, $F_L = f(\rho, V, d, \mu, \alpha)$... (i)

or, $f_1(F_L, \rho, V, d, \mu, \alpha)$... (ii)

$\therefore$ Total number of variables, $n = 6$

Writing dimensions of each variable, we have:

$$F_L = MLT^{-2}, \rho = ML^{-3}, V = LT^{-1}, d = L, \mu = ML^{-1}T^{-1}, \alpha = M^0L^0T^0$$

Thus, number of fundamental dimensions, $m = 3$

$\therefore$ Number of π -terms = $n - m = 6 - 3 = 3$

Eqn. (ii) can be written as: $f_1(\pi_1, \pi_2, \pi_3) = 0$... (iii)

Each π -term contains $(m + 1)$ variables, where $m = 3$ and is also equal to repeating variables.

Choosing d, V and ρ as repeating variables, we get these π -terms as:

$$\pi_1 = d^{a_1} \cdot V^{b_1} \cdot \rho^{c_1} \cdot F_L$$

$$\pi_2 = d^{a_2} \cdot V^{b_2} \cdot \rho^{c_2} \cdot \alpha$$

$$\pi_3 = d^{a_3} \cdot V^{b_3} \cdot \rho^{c_3} \cdot \mu$$

π_1 -term:

$$\pi_1 = d^{a_1} \cdot V^{b_1} \cdot \rho^{c_1} \cdot F_L$$

$$M^0L^0T^0 = L^{a_1} \cdot (LT^{-1})^{b_1} \cdot (ML^{-3})^{c_1} \cdot (MLT^{-2})$$

Equating the exponents of M, L and T respectively, we get:

For M: $0 = c_1 + 1$

For L: $0 = a_1 + b_1 - 3c_1 + 1$

For T: $0 = -b_1 - 2$

$\therefore c_1 = -1; b_1 = -2$

$\therefore a_1 = -b_1 + 3c_1 - 1 = 2 - 3 - 1 = -2$

Substituting the values of a_1, b_1 and c_1 in π_1 , we get:

$$\pi_1 = d^{-2} \cdot V^{-2} \cdot \rho^{-1} \cdot F_L = \frac{F_L}{\rho V^2 d^2}$$

π_2 -term:

$$\pi_2 = d^{a_2} \cdot V^{b_2} \cdot \rho^{c_2} \cdot \mu$$

$$M^0L^0T^0 = L^{a_2} \cdot (LT^{-1})^{b_2} \cdot (ML^{-3})^{c_2} \cdot (ML^{-1}T^{-1})$$

Equating the exponents of M, L and T respectively, we get:

For M: $0 = c_2 + 1$

For L: $0 = a_2 + b_2 - 3c_2 - 1$

For T: $0 = -b_2 - 1$

$\therefore c_2 = -1; b_2 = -1$

and, $a_2 = -b_2 + 3c_2 + 1 = 1 - 3 + 1 = -1$

Substituting the values of a_2, b_2 and c_2 in π_2 , we get:

$$\pi_2 = d^{-1} \cdot V^{-1} \cdot \rho^{-1} \cdot \mu = \frac{\mu}{\rho V d}$$

or,

$$\pi_2 = \frac{\rho V d}{\mu}$$

π_3 -term:

$$\pi_3 = d^{a_3} \cdot V^{b_3} \rho^{c_3} \cdot \alpha$$

$$M^0 L^0 T^0 = L^{a_3} \cdot (LT^{-1})^{b_3} \cdot (ML^{-3})^{c_3} \cdot (M^0 L^0 T^0)$$

Equating the exponents of M , L and T respectively, we get:

For M : $0 = c_3 + 0$

For L : $0 = a_3 + b_3 - 3c_3 + 0$

For T : $0 = -b_3 + 0$

$\therefore c_3 = 0; b_3 = 0$

and, $a_3 = -b_3 + 3c_3 = 0$

Substituting the values of a_3 , b_3 and c_3 in π_3 , we get:

$$\pi_3 = d^0 \cdot V^0 \cdot \rho^0 \cdot \alpha = \alpha$$

Substituting the values of π_1 , π_2 and π_3 in eqn. (iii), we get:

$$f_1 \left(\frac{F_L}{\rho V^2 d^2}, \frac{\rho V d}{\mu}, \alpha \right)$$

$$\frac{F_L}{\rho V^2 d^2} = \phi \left(\frac{\rho V d}{\mu}, \alpha \right)$$

or,

$$F_L = \rho V^2 d^2 \phi \left(\frac{\rho V d}{\mu}, \alpha \right)$$

...Proved.

Example 7.11. The pressure difference Δp in a pipe of diameter D and length l due to turbulent flow depends on the velocity V , viscosity μ , density ρ and roughness k . Using Buckingham's π -theorem, obtain an expression for Δp . [Delhi University]

Solution. The pressure difference Δp is a function of : D, l, V, μ, ρ, k

Mathematically, $\Delta p = f(D, l, V, \mu, \rho, k)$... (i)

or, $f_1(\Delta p, D, l, V, \mu, \rho, k) = 0$... (ii)

$\therefore$ Total number of variables, $n = 7$

Writing dimensions of each variable, we have:

$$\Delta p \text{ (dimensions of pressure)} = ML^{-1}T^{-2}, D = L, l = L, V = LT^{-1},$$

$$\mu = ML^{-1}T^{-1}, \rho = ML^{-3}, k = L$$

Thus, number of fundamental dimensions, $m = 3$

$\therefore$ Number of π -terms = $n - m = 7 - 3 = 4$

Eqn. (ii) can be written as:

$$f_1(\pi_1, \pi_2, \pi_3, \pi_4) = 0 \quad \dots (iii)$$

Each π -term contains $(m + 1)$ variables, where $m = 3$ and is also equal to repeating variables.

Choosing D, V and ρ as repeating variables, we get four π -terms as:

$$\pi_1 = D^{a_1} \cdot V^{b_1} \cdot \rho^{c_1} \cdot \Delta p$$

$$\pi_2 = D^{a_2} \cdot V^{b_2} \cdot \rho^{c_2} \cdot l$$

$$\pi_3 = D^{a_3} \cdot V^{b_3} \cdot \rho^{c_3} \cdot \mu$$

$$\pi_4 = D^{a_4} \cdot V^{b_4} \cdot \rho^{c_4} \cdot k$$

π_1 -term:

$$\pi_1 = D^{a_1} \cdot V^{b_1} \cdot \rho^{c_1} \cdot \Delta p$$

$$M^0 L^0 T^0 = L^{a_1} \cdot (LT^{-1})^{b_1} \cdot (ML^{-3})^{c_1} \cdot (ML^{-1}T^{-2})$$

Equating the exponents of M , L and T respectively, we get:

For M : $0 = c_1 + 1$

For L : $0 = a_1 + b_1 - 3c_1 - 1$

For T : $0 = -b_1 - 2$

$\therefore c_1 = -1; b_1 = -2$

and, $a_1 = -b_1 + 3c_1 + 1 = 2 - 3 + 1 = 0$

Substituting the values of a_1 , b_1 and c_1 in π_1 , we get:

$$\pi_1 = D^0 \cdot V^{-2} \cdot \rho^{-1} \cdot \Delta p = \frac{\Delta p}{\rho V^2}$$

π_2 -term:

$$\pi_2 = D^{a_2} \cdot V^{b_2} \cdot \rho^{c_2} \cdot l$$

$$M^0 L^0 T^0 = L^{a_2} \cdot (LT^{-1})^{b_2} \cdot (ML^{-3})^{c_2} \cdot L$$

Equating the exponents of M , L and T respectively, we get:

For M : $0 = c_2$

For L : $0 = a_2 + b_2 - 3c_2 + 1$

For T : $0 = -b_2$

$\therefore c_2 = 0; b_2 = 0$

and, $a_2 = -b_2 + 3c_2 - 1 = -1$

Substituting the values of a_2 , b_2 and c_2 in π_2 , we get:

$$\pi_2 = D^{-1} \cdot V^0 \cdot \rho^0 \cdot l = \frac{l}{D}$$

π_3 -term:

$$\pi_3 = D^{a_3} \cdot V^{b_3} \cdot \rho^{c_3} \cdot \mu$$

$$M^0 L^0 T^0 = L^{a_3} \cdot (LT^{-1})^{b_3} \cdot (ML^{-3})^{c_3} \cdot (ML^{-1} T^{-1})$$

Equating the exponents of M , L and T respectively, we get:

For M : $0 = c_3 + 1$

For L : $0 = a_3 + b_3 - 3c_3 - 1$

For T : $0 = -b_3 - 1$

$\therefore c_3 = -1; b_3 = -1$

and, $a_3 = -b_3 + 3c_3 + 1 = 1 - 3 + 1 = -1$

Substituting the values of a_3 , b_3 and c_3 in π_3 , we get:

$$\pi_3 = D^{-1} \cdot V^{-1} \cdot \rho^{-1} \cdot \mu = \frac{\mu}{DV\rho}$$

π_4 -term:

$$\pi_4 = D^{a_4} \cdot V^{b_4} \cdot \rho^{c_4} \cdot k$$

$$M^0 L^0 T^0 = L^{a_4} \cdot (LT^{-1})^{b_4} \cdot (ML^{-3})^{c_4} \cdot L$$

(Dimension of $k = L$)

Equating the exponents of M , L and T respectively, we get:

For M : $0 = c_4$

For L : $0 = a_4 + b_4 - 3c_4 + 1$

For T : $0 = -b_4$

$\therefore c_4 = 0; b_4 = 0$

and,

$$a_4 = -b_4 + 3c_4 - 1 = 0 + 0 - 1 = -1$$

Substituting the values of a_4 , b_4 and c_4 in π_4 , we get:

$$\pi_3 = D^{-1} \cdot V^0 \cdot \rho^0 \cdot k = \frac{k}{D}$$

Substituting the values of π_1 , π_2 , π_3 and π_4 in eqn. (iii), we get:

$$f_1\left(\frac{\Delta p}{\rho V^2}, \frac{l}{D}, \frac{\mu}{DV\rho}, \frac{k}{D}\right) = 0$$

or,

$$\frac{\Delta p}{\rho V^2} = \phi\left[\frac{l}{D}, \frac{\mu}{DV\rho}, \frac{k}{D}\right]$$

● **Expression for difference of pressure head (h_f):**

As observed from experiments, Δp is a linear function of $\frac{l}{D}$; therefore taking this out of function, we have:

$$\frac{\Delta p}{\rho V^2} = \frac{l}{D} \phi\left[\frac{\mu}{DV\rho}, \frac{k}{D}\right]$$

or,

$$\frac{\Delta p}{\rho} = V^2 \cdot \frac{l}{D} \phi\left[\frac{\mu}{DV\rho}, \frac{k}{D}\right]$$

Dividing both sides by g , we get:

$$\frac{\Delta p}{\rho g} = \frac{V^2}{g} \cdot \frac{l}{D} \phi\left[\frac{\mu}{DV\rho}, \frac{k}{D}\right]$$

Now $\phi\left[\frac{\mu}{DV\rho}, \frac{k}{D}\right]$ consists of following two terms:

(i) $\frac{\mu}{DV\rho}$ which is $\frac{1}{\text{Reynold number}}$ or $\frac{1}{Re}$

(ii) $\frac{k}{D}$...called roughness factor

$\therefore \phi\left[\frac{1}{Re}, \frac{k}{D}\right]$ is put equal to f

where, f = Co-efficient of friction (function of Reynold number and roughness factor)

$\therefore \frac{\Delta p}{\rho g} = \frac{4f}{2} \cdot \frac{V^2 l}{gD} \quad \left[\because f = \phi\left(\frac{\mu}{DV\rho}, \frac{k}{D}\right) \right]$

(Multiplying or dividing by any constant does not change the character of π -terms)

$\therefore \frac{\Delta p}{\rho g} = h_f = \frac{4f l V^2}{D \times 2g} \text{ (Ans.)}$

Example 7.12. The discharge Q of a centrifugal pump depends upon the mass density of fluid (ρ), the speed of the pump (N), the diameter of the impeller (D), the manometric head (H_m) and the viscosity of fluid (μ). Show that

$$Q = ND^3 \phi\left(\frac{gH}{N^2 D^2}, \frac{\mu}{\rho N D^2}\right)$$

Solution. The discharge Q is a function of: N, D, g, H, μ, ρ .

Mathematically, $Q = f(N, D, g, H, \mu, \rho)$... (i)

or, $f_1(Q, N, D, g, H, \mu, \rho) = 0$... (ii)

Total number of variables, $n = 7$

Writing dimensions of each variable, we have:

$$Q = L^3 T^{-1}, N = T^{-1}, D = L, g = LT^{-2}, \\ H = L, \mu = ML^{-1} T^{-1}, \rho = ML^{-3}$$

Thus, number of fundamental dimensions, $m = 3$

$\therefore$ Number of π -terms = $n - m = 7 - 3 = 4$

Eqn. (ii) can be written as:

$$f_1(\pi_1, \pi_2, \pi_3, \pi_4) = 0 \quad \dots (iii)$$

Each π -term contains $(m + 1)$ variables, where $m = 3$ and is also equal to repeating variables.

Choosing D, N, ρ as *repeating variables*, we get four π -terms as:

$$\pi_1 = D^{a_1} \cdot N^{b_1} \cdot \rho^{c_1} \cdot Q \\ \pi_2 = D^{a_2} \cdot N^{b_2} \cdot \rho^{c_2} \cdot g \\ \pi_3 = D^{a_3} \cdot N^{b_3} \cdot \rho^{c_3} \cdot H \\ \pi_4 = D^{a_4} \cdot N^{b_4} \cdot \rho^{c_4} \cdot \mu$$

π_1 -term:

$$\pi_1 = D^{a_1} \cdot N^{b_1} \cdot \rho^{c_1} \cdot Q \\ M^0 L^0 T^0 = L^{a_1} \cdot (T^{-1})^{b_1} \cdot (ML^{-3})^{c_1} (L^3 T^{-1})$$

Equating the exponents of M, L and T respectively, we get:

$$\text{For M:} \quad 0 = c_1$$

$$\text{For L:} \quad 0 = a_1 - 3c_1 + 3$$

$$\text{For T:} \quad 0 = -b_1 - 1$$

$$\therefore c_1 = 0; b_1 = -1$$

$$\text{and,} \quad a_1 = 3c_1 - 3 = 0 - 3 = -3$$

Substituting the values of a_1, b_1 and c_1 in π_1 , we get:

$$\pi_1 = D^{-3} \cdot N^{-1} \cdot \rho^0 \cdot Q = \frac{Q}{ND^3}$$

π_2 -term:

$$\pi_2 = D^{a_2} \cdot N^{b_2} \cdot \rho^{c_2} \cdot g \\ M^0 L^0 T^0 = L^{a_2} \cdot (T^{-1})^{b_2} \cdot (ML^{-3})^{c_2} \cdot (LT^{-2})$$

Equating the exponents of M, L and T respectively, we get:

$$\text{For M:} \quad 0 = c_2$$

$$\text{For L:} \quad 0 = a_2 - 3c_2 + 1$$

$$\text{For T:} \quad 0 = -b_2 - 2$$

$$\therefore c_2 = 0; b_2 = -2$$

$$\text{and,} \quad a_2 = 3c_2 - 1 = 0 - 1 = -1$$

Substituting the values of a_2, b_2 and c_2 in π_2 , we get:

$$\pi_2 = D^{-1} \cdot N^{-2} \cdot \rho^0 \cdot g = \frac{g}{N^2 D}$$

π_3 -term:

$$\pi_3 = D^{a_3} \cdot N^{b_3} \cdot \rho^{c_3} \cdot H$$

$$M^0 L^0 T^0 = L^{a_3} \cdot (T^{-1})^{b_3} \cdot (ML^{-3})^{c_3} \cdot L$$

Equating the exponents of M , L and T respectively, we get:

For M: $0 = c_3$

For L: $0 = a_3 - 3c_3 + 1$

For T: $0 = -b_3$

$\therefore c_3 = 0; b_3 = 0$

and, $a_3 = 3c_3 - 1 = 0 - 1 = -1$

Substituting the values of a_3 , b_3 and c_3 in π_3 , we get:

$$\pi_3 = D^{-1} \cdot N^0 \cdot \rho^0 \cdot H = \frac{H}{D}$$

π_4 -term:

$$\pi_4 = D^{a_4} \cdot N^{b_4} \cdot \rho^{c_4} \cdot \mu$$

$$M^0 L^0 T^0 = L^{a_4} \cdot (T^{-1})^{b_4} \cdot (ML^{-3})^{c_4} \cdot ML^{-1} T^{-1}$$

Equating the exponents of M , L and T respectively, we get:

For M: $0 = c_4 + 1$

For L: $0 = a_4 - 3c_4 - 1$

For T: $0 = -b_4 - 1$

$\therefore c_4 = -1; b_4 = -1$

and, $a_4 = 3c_4 + 1 = 3 \times (-1) + 1 = -2$

Substituting the values of a_4 , b_4 and c_4 in π_4 , we get:

$$\pi_4 = D^{-2} \cdot N^{-1} \cdot \rho^{-1} \cdot \mu = \frac{\mu}{\rho ND^2}$$

Substituting the values of π_1 , π_2 , π_3 and π_4 in eqn. (iii), we get:

$$f_1 \left(\frac{Q}{ND^3}, \frac{g}{N^2 D}, \frac{H}{D}, \frac{\mu}{\rho ND^2} \right) = 0$$

As the product of two π -terms is also dimensionless, the terms π_2 and π_3 can be replaced by $\frac{gH}{N^2 D^2}$. Thus:

$$f_1 \left(\frac{Q}{ND^3}, \frac{gH}{N^2 D^2}, \frac{\mu}{\rho ND^2} \right) = 0$$

or, $\frac{Q}{ND^3} = \phi \left(\frac{gH}{N^2 D^2}, \frac{\mu}{\rho ND^2} \right)$

or, $Q = ND^3 \phi \left(\frac{gH}{N^2 D^2}, \frac{\mu}{\rho ND^2} \right) \quad \dots \text{Proved.}$

Example 7.13. Derive on the basis of dimensional analysis suitable parameters to present the thrust developed by a propeller. Assume that the thrust P depends upon the angular velocity ω , speed of advance V , diameter D , dynamic viscosity μ , mass density ρ , elasticity of the fluid medium which can be denoted by the speed of sound in the medium C . [MDU, Haryana]

Solution. Thrust P is a function of : $\omega, V, D, \mu, \rho, C$

Mathematically, $P = f(\omega, V, D, \mu, \rho, C)$... (i)

or, $f_1(P, \omega, V, D, \mu, \rho, C) = 0$... (ii)

Total number of variables, $n = 7$

Writing dimensions of each variable, we get:

$$P = MLT^{-2}, \omega = T^{-1}, V = LT^{-1}, D = L, \\ \mu = ML^{-1}T^{-1}, \rho = ML^{-3}, C = LT^{-1}$$

Thus, number of fundamental dimensions, $m = 3$

$\therefore$ Number of π -terms = $n - m = 7 - 3 = 4$

Eqn. (ii) can be written as:

$$f_1(\pi_1, \pi_2, \pi_3, \pi_4) = 0 \quad \dots (iii)$$

Each π -term contains $(m + 1)$ variables, where $m = 3$ and also equal to repeating variables.

Choosing D, V and ρ as repeating variables, we get four π -terms as:

$$\pi_1 = D^{a_1} \cdot V^{b_1} \cdot \rho^{c_1} \cdot P$$

$$\pi_2 = D^{a_2} \cdot V^{b_2} \cdot \rho^{c_2} \cdot \omega$$

$$\pi_3 = D^{a_3} \cdot V^{b_3} \cdot \rho^{c_3} \cdot \mu$$

$$\pi_4 = D^{a_4} \cdot V^{b_4} \cdot \rho^{c_4} \cdot C$$

π_1 -term:

$$\pi_1 = D^{a_1} \cdot V^{b_1} \cdot \rho^{c_1} \cdot P$$

$$M^0 L^0 T^0 = L^{a_1} \cdot (LT^{-1})^{b_1} \cdot (ML^{-3})^{c_1} \cdot MLT^{-2}$$

Equating the exponents of M, L and T respectively, we get:

$$\text{For M:} \quad 0 = c_1 + 1$$

$$\text{For L:} \quad 0 = a_1 + b_1 - 3c_1 + 1$$

$$\text{For T:} \quad 0 = -b_1 - 2$$

$$\therefore \quad c_1 = -1; b_1 = -2$$

$$\text{and} \quad a_1 = -b_1 + 3c_1 - 1 = 2 - 3 - 1 = -2$$

Substituting the values of a_1, b_1 and c_1 in π_1 , we get:

$$\pi_1 = D^{-2} \cdot V^{-2} \cdot \rho^{-1} \cdot P = \frac{P}{D^2 V^2 \rho}$$

π_2 -term:

$$\pi_2 = D^{a_2} \cdot V^{b_2} \cdot \rho^{c_2} \cdot \omega$$

$$M^0 L^0 T^0 = L^{a_2} \cdot (LT^{-1})^{b_2} \cdot (ML^{-3})^{c_2} \cdot T^{-1}$$

Equating the exponents of M, L and T respectively, we get:

$$\text{For M:} \quad 0 = c_2$$

$$\text{For L:} \quad 0 = a_2 + b_2 - 3c_2$$

$$\text{For T:} \quad 0 = -b_2 - 1$$

$$\therefore \quad c_2 = 0; b_2 = -1$$

$$\text{and,} \quad a_2 = -b_2 + 3c_2 = 1 + 0 = 1$$

Substituting the values of a_2, b_2 and c_2 in π_2 , we get:

$$\pi_2 = D^1 \cdot V^{-1} \cdot \rho^0 \cdot \omega = \frac{D\omega}{V}$$

π_3 -term:

$$\pi_3 = D^{a_3} \cdot V^{b_3} \cdot \rho^{c_3} \cdot \mu$$

$$M^0 L^0 T^0 = L^{a_3} \cdot (LT^{-1})^{b_3} \cdot (ML^{-3})^{c_3} \cdot ML^{-1}T^{-1}$$

Equating the exponents of M , L and T respectively, we get:

For M: $0 = c_3 + 1$

For L: $0 = a_3 + b_3 - 3c_3 - 1$

For T: $0 = -b_3 - 1$

$\therefore c_3 = -1; b_3 = -1$

and, $a_3 = -b_3 + 3c_3 + 1 = 1 - 3 + 1 = -1$

Substituting the values of a_3 , b_3 and c_3 in π_3 , we get:

$$\pi_3 = D^{-1} \cdot V^{-1} \cdot \rho^{-1} \cdot \mu = \frac{\mu}{DV\rho}$$

π_4 -term:

$$\pi_4 = D^{a_4} \cdot V^{b_4} \cdot \rho^{c_4} \cdot C$$

$$M^0 L^0 T^0 = L^{a_4} \cdot (LT^{-1})^{b_4} \cdot (ML^{-3})^{c_4} \cdot (LT^{-1})$$

Equating the exponents of M , L and T respectively, we get:

For M: $0 = c_4$

For L: $0 = a_4 + b_4 - 3c_4 + 1$

For T: $0 = -b_4 - 1$

$\therefore c_4 = 0; b_4 = -1$

and, $a_4 = -b_4 + 3c_4 - 1 = 1 + 0 - 1 = 0$

Substituting the values of a_4 , b_4 and c_4 in π_4 , we get:

$$\pi_4 = D^0 \cdot V^{-1} \cdot \rho^0 \cdot C = \frac{C}{V}$$

Substituting the values of π_1 , π_2 , π_3 and π_4 in eqn. (iii), we get:

$$f_1\left(\frac{P}{D^2 V^2 \rho}, \frac{D\omega}{V}, \frac{\mu}{DV\rho}, \frac{C}{V}\right) = 0$$

or, $\frac{P}{D^2 V^2 \rho} = \phi\left(\frac{D\omega}{V}, \frac{\mu}{DV\rho}, \frac{C}{V}\right)$

or, $P = D^2 V^2 \rho \phi\left(\frac{D\omega}{V}, \frac{\mu}{DV\rho}, \frac{C}{V}\right)$ (Ans.)

Example 7.14. Using the method of dimensional analysis obtain an expression for the discharge Q over a rectangular weir. The discharge depends on the head H over the weir, acceleration due to gravity g , length of weir crest L , height of the weir crest over the channel bottom Z and the kinematic viscosity ν of the liquid.

Solution. The discharge Q over the weir is a function of : H, g, L, Z, ν

Mathematically, $Q = f(H, g, L, Z, \nu)$... (i)

or, $f_1(Q, H, g, L, Z, \nu)$... (ii)

Total number of variables, $n = 6$

Writing dimensions of each variable, we get:

$$Q = L^3 T^{-1}, H = L, g = LT^{-2}, L = L, Z = L, v = L^2 T^{-1}$$

Thus, number of fundamental dimensions, $m = 2$

$$\therefore \text{Number of } \pi\text{-terms} = n - m = 6 - 2 = 4$$

Eqn. (ii) can be written as:

$$f_1(\pi_1, \pi_2, \pi_3, \pi_4) = 0 \quad \dots(iii)$$

Each π -term contains $(m + 1)$ variables, where $m = 2$ and *also equal to repeating variables*. The head over the weir is the most important independent variable, and the other variable which affect the flow in open channel is the acceleration due to gravity g . These two variables H and g satisfy the requirement of repeating variables. Choosing H and g as repeating variables, we get four π -terms as:

$$\pi_1 = H^{a_1} \cdot g^{b_1} \cdot Q$$

$$\pi_2 = H^{a_2} \cdot g^{b_2} \cdot L$$

$$\pi_3 = H^{a_3} \cdot g^{b_3} \cdot Z$$

$$\pi_4 = H^{a_4} \cdot g^{b_4} \cdot v$$

π_1 -term:

$$\pi_1 = H^{a_1} \cdot g^{b_1} \cdot Q$$

$$M^0 L^0 T^0 = L^{a_1} \cdot (LT^{-2})^{b_1} \cdot (L^3 T^{-1})$$

Equating exponents of M , L and T respectively, we get:

$$\text{For M:} \quad 0 = 0$$

$$\text{For L:} \quad 0 = a_1 + b_1 + 3$$

$$\text{For T:} \quad 0 = -2b_1 - 1$$

$$\therefore \quad b_1 = -\frac{1}{2}$$

$$\text{and,} \quad a_1 = -b_1 - 3 = \frac{1}{2} - 3 = -5/2$$

Substituting the values of a_1 and b_1 in π_1 , we get:

$$\pi_1 = H^{-5/2} \cdot g^{-1/2} \cdot Q = \frac{Q}{g^{1/2} (H)^{5/2}}$$

π_2 -term:

$$\pi_2 = H^{a_2} \cdot g^{b_2} \cdot L$$

$$M^0 L^0 T^0 = L^{a_2} \cdot (LT^{-2})^{b_2} \cdot L$$

Equating exponents of M , L and T respectively, we get:

$$\text{For M:} \quad 0 = 0$$

$$\text{For L:} \quad 0 = a_2 + b_2 + 1$$

$$\text{For T:} \quad 0 = -2b_2$$

$$\therefore \quad b_2 = 0$$

$$\text{and,} \quad a_2 = -b_2 - 1 = -1$$

Substituting the values of a_2 and b_2 in π_2 , we get:

$$\pi_2 = H^{-1} \cdot g^0 \cdot L = \frac{L}{H}$$

π_3 -term:

$$\pi_3 = H^{a_3} \cdot g^{b_3} \cdot Z$$

$$M^0 L^0 T^0 = L^{a_3} \cdot (LT^{-2})^{b_3} \cdot L$$

Equating the exponents of M , L and T respectively, we get:

For M: $0 = 0$

For L: $0 = a_3 + b_3 + 1$

For T: $0 = -2b_3$

$\therefore b_3 = 0$

and, $a_3 = -b_3 - 1 = -1$

Substituting the values of a_3 and b_3 in π_3 , we get

$$\pi_3 = H^{-1} \cdot g^0 \cdot Z = \frac{Z}{H}$$

π_4 -term:

$$\pi_4 = H^{a_4} \cdot g^{b_4} \cdot v$$

$$M^0 L^0 T^0 = L^{a_4} \cdot (LT^{-2})^{b_4} \cdot (L^2 T^{-1})$$

Equating the exponents of M , L and T respectively, we get:

For M: $0 = 0$

For L: $0 = a_4 + b_4 + 2$

For D; $0 = -2b_4 - 1$

$$b_4 = -1/2 \text{ and } a_4 = -b_4 - 2 = +1/2 - 2 = -3/2$$

Substituting the values of a_4 and b_4 in π_4 , we get:

$$\pi_4 = H^{-3/2} \cdot g^{-1/2} \cdot v = \frac{v}{g^{1/2} H^{3/2}}$$

Substituting the values of π_1 , π_2 , π_3 and π_4 in eqn. (iii), we get:

$$f_1 \left(\frac{Q}{g^{1/2} H^{5/2}}, \frac{L}{H}, \frac{Z}{H}, \frac{v}{g^{1/2} H^{3/2}} \right) = 0$$

or,
$$\frac{Q}{g^{1/2} H^{5/2}} = \phi \left(\frac{v}{g^{1/2} H^{3/2}}, \frac{L}{H}, \frac{Z}{H} \right) \quad \dots(iv)$$

From the equation of continuity $Q = AV$ and, therefore, $\frac{Q}{AV}$ is dimensionless. In case of a rectangular weir, we may write:

$$A \propto LH \text{ and } V \propto g^{1/2} H^{1/2},$$

Hence,
$$\frac{Q}{LH g^{1/2} H^{3/2}} = \frac{Q}{Lg^{1/2} H^{3/2}} \text{ is dimensionless quantity.}$$

Thus the left hand side of eqn. (iv) may be transformed and written as $\frac{Q}{Lg^{1/2} H^{3/2}}$ and eqn. (iv)

may be expressed as:

$$\frac{Q}{Lg^{1/2} H^{3/2}} = \phi_1 \left(\frac{v}{g^{1/2} H^{3/2}}, \frac{L}{H}, \frac{Z}{H} \right) \quad \dots(v)$$

In eqn. (v), $\frac{v}{g^{1/2} H^{3/2}} = \frac{v}{H \sqrt{gH}}$ is the inverse of Reynolds number. Eqn. (v) may be written as:

$$Q = C \sqrt{g} L H^{3/2} \quad \dots(vi)$$

where,

$$C = \phi_2 \left(\frac{H \sqrt{gH}}{v}, \frac{H}{L}, \frac{H}{Z} \right)$$

and, since g is constant for a place, eqn. (vi) can be expressed as

$$Q = K L H^{3/2} \text{ (Ans.)}$$

Example 7.15. The resisting force F of a plane during flight can be considered as dependent upon the length of aircraft l , velocity v , air viscosity μ , air density ρ , and bulk modulus of air K . Express the functional relationship between these variables and the resisting force using dimensional analysis. Explain the physical meaning of the dimensionless groups.

[UPSC]

Solution. The resisting force F is a function of : l, v, μ, ρ, K

$$\text{Mathematically, } F = f(l, v, \mu, \rho, K) \quad \dots(i)$$

$$\text{or, } f_1(F, l, v, \mu, \rho, K) = 0 \quad \dots(ii)$$

Total number of variables, $n = 6$

Writing dimensions of each variable, we get:

$$F = MLT^{-2}, l = L, v = LT^{-1}, \mu = ML^{-1}T^{-1}, \rho = ML^{-3}, K = ML^{-1}T^{-2}$$

Thus, number of fundamental dimensions, $m = 3$

$$\therefore \text{Number of } \pi\text{-terms} = n - m = 6 - 3 = 3$$

Eqn. (ii) can be written as:

$$f_1(\pi_1, \pi_2, \pi_3) = 0 \quad \dots(iii)$$

Each π -term contains $(m + 1)$ variables, where $m = 3$ and also equal to repeating variables.

Choosing l, v and ρ as repeating variables, we get three terms as:

$$\pi_1 = l^{a_1} \cdot v^{b_1} \cdot \rho^{c_1} \cdot F$$

$$\pi_2 = l^{a_2} \cdot v^{b_2} \cdot \rho^{c_2} \cdot \mu$$

$$\pi_3 = l^{a_3} \cdot v^{b_3} \cdot \rho^{c_3} \cdot K$$

π_1 -term:

$$\pi_1 = l^{a_1} \cdot v^{b_1} \cdot \rho^{c_1} \cdot F$$

$$M^0 L^0 T^0 = L^{a_1} \cdot (LT^{-1})^{b_1} \cdot (ML^{-3})^{c_1} \cdot MLT^{-2}$$

Equating exponents of M, L and T respectively, we get:

$$\text{For M: } 0 = c_1 + 1$$

$$\text{For L: } 0 = a_1 + b_1 - 3c_1 + 1$$

$$\text{For T: } 0 = -b_1 - 2$$

$$\therefore c_1 = -1, b_1 = -2$$

$$\text{and, } a_1 = -b_1 + 3c_1 - 1 = 2 - 3 - 1 = -2$$

Substituting the values of a_1, b_1 and c_1 , in π_1 , we get:

$$\pi_1 = l^{-2} \cdot v^{-2} \cdot \rho^{-1} \cdot F = \frac{F}{l^2 v^2 \rho}$$

π_2 -term:

$$\pi_2 = l^{a_2} \cdot v^{b_2} \cdot \rho^{c_2} \cdot \mu$$

$$M^0 L^0 T^0 = L^{a_2} \cdot (LT^{-1})^{b_2} \cdot (ML^{-3})^{c_2} \cdot ML^{-1}T^{-1}$$

Equating exponents of M , L and T respectively, we get:

For M : $0 = c_2 + 1$

For L : $0 = a_2 + b_2 - 3c_2 - 1$

For T : $0 = -b_2 - 1$

$\therefore c_2 = -1; b_2 = -1$

and, $a_2 = -b_2 + 3c_2 + 1 = 1 - 3 + 1 = -1$

Substituting the values of a_2 , b_2 and c_2 in π_2 , we get:

$$\pi_2 = l^{-1} \cdot v^{-1} \cdot \rho^{-1} \cdot \mu = \frac{\mu}{lv\rho}$$

π_3 -term:

$$\pi_3 = l^{a_3} \cdot v^{b_3} \cdot \rho^{c_3} \cdot K$$

$$M^0 L^0 T^0 = L^{a_3} \cdot (LT^{-1})^{b_3} \cdot (ML^{-3})^{c_3} \cdot ML^{-1}T^{-2}$$

Equating exponents of M , L and T respectively, we get:

For M : $0 = c_3 + 1$

For L : $0 = a_3 + b_3 - 3c_3 - 1$

For T : $0 = -b_3 - 2$

$\therefore c_3 = -1; b_3 = -2$

and, $a_3 = -b_3 + 3c_3 + 1 = 2 - 3 + 1 = 0$

Substituting the values of a_3 , b_3 , and c_3 in π_3 , we get:

$$\pi_3 = l^0 \cdot v^{-2} \cdot \rho^{-1} \cdot K = \frac{K}{v^2 \rho}$$

Substituting the values of π_1 , π_2 and π_3 in eqn. (iii), we get the functional relationship as:

$$f_1\left(\frac{F}{l^2 v^2 \rho}, \frac{\mu}{lv\rho}, \frac{K}{v^2 \rho}\right) = 0$$

or, $\frac{F}{l^2 v^2 \rho} = \phi\left(\frac{\mu}{lv\rho}, \frac{K}{v^2 \rho}\right)$

or, $F = l^2 v^2 \rho \phi\left(\frac{\mu}{lv\rho}, \frac{K}{v^2 \rho}\right)$ (Ans.)

Physical meaning of dimensionless groups (π_1 , π_2 , π_3):

(i) $\pi_1 = \frac{F}{l^2 v^2 \rho}$: It is the ratio of F and dynamic force $l^2 v^2 \rho$. It indicates that the resisting force

experienced by an aircraft is dependent on the length and velocity. For any given aircraft, the resistance force will be proportional to the square of velocity.

(ii) $\pi_2 = \frac{\mu}{lv\rho}$: This dimensionless group is the reciprocal of Reynolds number and represents

the role of viscous action on the resistance.

- (iii) $\pi_3 = \frac{K}{\rho v^2}$: It takes into account the role of compressibility in influencing fluid resistance which an aircraft experiences.

Example 7.16. Show that the power P developed in a water turbine can be expressed as:

$$P = \rho N^3 D^5 \phi \left(\frac{D}{B}, \frac{\rho D^2 N}{\mu}, \frac{ND}{\sqrt{gH}} \right)$$

where,

ρ = Mass density of the liquid,

N = Speed in r.p.m.,

D = Diameter of the runner,

B = Width of the number, and

μ = Co-efficient of dynamic viscosity.

Under what conditions it can be used to determine the characteristic of a similar machine?

[UPTU]

Solution. The power P developed in a water turbine is a function of: ρ, N, D, B, μ, H, g

Mathematically, $P = f(\rho, N, D, B, \mu, H, g)$... (i)

or, $f_1(P, \rho, N, D, B, \mu, H, g) = 0$... (ii)

Total number of variables, $n = 8$

Writing dimensions of each variable, we get:

$$P = ML^2T^{-3}, \rho = ML^{-3}, N = T^{-1}, D = L, B = L, \mu = ML^{-1}T^{-1},$$

$$H = L, g = LT^{-2}$$

Thus, the number of fundamental dimensions, $m = 3$

$\therefore$ Number of π -terms = $n - m = 8 - 3 = 5$

Eqn. (ii) can be written as:

$$f_1(\pi_1, \pi_2, \pi_3, \pi_4, \pi_5) = 0 \quad \dots (iii)$$

Each π -term contains $(m + 1)$ variables, where $m = 3$ and also equal to repeating variables.

Choosing D, N and ρ as repeating variables, we get five π -terms as:

$$\pi_1 = D^{a_1} \cdot N^{b_1} \cdot \rho^{c_1} \cdot P$$

$$\pi_2 = D^{a_2} \cdot N^{b_2} \cdot \rho^{c_2} \cdot B$$

$$\pi_3 = D^{a_3} \cdot N^{b_3} \cdot \rho^{c_3} \cdot \mu$$

$$\pi_4 = D^{a_4} \cdot N^{b_4} \cdot \rho^{c_4} \cdot H$$

$$\pi_5 = D^{a_5} \cdot N^{b_5} \cdot \rho^{c_5} \cdot g$$

π_1 -term:

$$\pi_1 = D^{a_1} \cdot N^{b_1} \cdot \rho^{c_1} \cdot P$$

$$M^0 L^0 T^0 = L^{a_1} \cdot (T^{-1})^{b_1} \cdot (ML^{-3})^{c_1} \cdot ML^2 T^{-3}$$

Equating the exponents of M, L and T respectively, we get:

$$\text{For M:} \quad 0 = c_1 + 1$$

$$\text{For L:} \quad 0 = a_1 - 3c_1 + 2$$

$$\text{For T:} \quad 0 = -b_1 - 3$$

$$\therefore \quad c_1 = -1, b_1 = -3$$

$$\text{and,} \quad a_1 = 3c_1 - 2 = -3 - 2 = -5$$

Substituting the values of a_1 , b_1 and c_1 in π_1 , we get:

$$\pi_1 = D_1^{-5} \cdot N^{-3} \cdot \rho^{-1} \cdot P = \frac{P}{\rho N^3 D^5}$$

π_2 -term:

$$\begin{aligned}\pi_2 &= D^{a_2} \cdot N^{b_2} \cdot \rho^{c_2} \cdot B \\ M^0 L^0 T^0 &= L^{a_2} \cdot (T^{-1})^{b_2} \cdot (ML^{-3})^{c_2} \cdot L\end{aligned}$$

Equating the exponents of M , L and T respectively, we get:

$$\begin{aligned}\text{For M:} \quad & 0 = c_2 \\ \text{For L:} \quad & 0 = a_2 - 3c_2 + 1 \\ \text{For T:} \quad & 0 = -b_2 \\ \therefore \quad & c_2 = 0; b_2 = 0 \\ \text{and,} \quad & a_2 = 3c_2 - 1 = -1\end{aligned}$$

Substituting the values of a_2 , b_2 and c_2 in π_2 , we get:

$$\pi_2 = D^{-1} \cdot N^0 \cdot \rho^0 \cdot B = \frac{B}{D}$$

π_3 -term:

$$\begin{aligned}\pi_3 &= D^{a_3} \cdot N^{b_3} \cdot \rho^{c_3} \cdot \mu \\ M^0 L^0 T^0 &= L^{a_3} \cdot (T^{-1})^{b_3} \cdot (ML^{-3})^{c_3} \cdot (ML^{-1}T^{-1})\end{aligned}$$

Equating the exponents of M , L and T respectively, we get:

$$\begin{aligned}\text{For M:} \quad & 0 = c_3 + 1 \\ \text{For L:} \quad & 0 = a_3 - 3c_3 - 1 \\ \text{For T:} \quad & 0 = -b_3 - 1 \\ \therefore \quad & c_3 = -1; b_3 = -1 \\ \text{and,} \quad & a_3 = 3c_3 + 1 = -3 + 1 = -2\end{aligned}$$

Substituting the values of a_3 , b_3 and c_3 in π_3 , we get:

$$\pi_3 = D^{-2} \cdot N^{-1} \cdot \rho^{-1} \cdot \mu = \frac{\mu}{\rho D^2 N}$$

π_4 -term:

$$\begin{aligned}\pi_4 &= D^{a_4} \cdot N^{b_4} \cdot \rho^{c_4} \cdot H \\ M^0 L^0 T^0 &= L^{a_4} \cdot (T^{-1})^{b_4} \cdot (ML^{-3})^{c_4} \cdot L\end{aligned}$$

Equating the exponents of M , L and T respectively, we get:

$$\begin{aligned}\text{For M:} \quad & 0 = c_4 \\ \text{For L:} \quad & 0 = a_4 - 3c_4 + 1 \\ \text{For T:} \quad & 0 = -b_4 \\ \therefore \quad & c_4 = 0; b_4 = 0 \\ \text{and,} \quad & a_4 = 3c_4 - 1 = 0 - 1 = -1\end{aligned}$$

Substituting the values of a_4 , b_4 and c_4 in π_4 we get:

$$\pi_4 = D^{-1} \cdot N^0 \cdot \rho^0 \cdot H = \frac{H}{D}$$

π_5 -term:

$$\pi_5 = D^{a_5} \cdot N^{b_5} \cdot \rho^{c_5} \cdot g$$

$$M^0 L^0 T^0 = L^{a_5} \cdot (T^{-1})^{b_5} \cdot (ML^{-3})^{c_5} \cdot (LT^{-2})$$

Equating the exponents of M , L and T respectively, we get:

For M: $0 = c_5$
 For L: $0 = a_5 - 3c_5 + 1$
 For T: $0 = -b_5 - 2$
 $\therefore c_5 = 0; b_5 = -2$
 and, $a_5 = 3c_5 - 1 = 0 - 1 = -1$

Substituting the values of a_5 , b_5 and c_5 in π_5 we get:

$$\pi_5 = D^{-1} \cdot N^{-2} \cdot \rho^0 \cdot g = \frac{g}{DN^2}$$

Substituting the values of π_1 , π_2 , π_3 , π_4 and π_5 in eqn. (iii), we get:

$$f_1\left(\frac{P}{\rho N^3 D^5}, \frac{B}{D}, \frac{\mu}{\rho D^2 N}, \frac{H}{D}, \frac{g}{DN^2}\right) = 0$$

or,

$$f_1\left(\frac{P}{\rho N^3 D^5}, \frac{B}{D}, \frac{\mu}{\rho D^2 N}, \frac{gH}{D^2 N^2}\right) = 0$$

$$\left[\because \pi_4 \times \pi_5 \text{ (product of two terms)} = \frac{H}{D} \times \frac{g}{DN^2} = \frac{gH}{D^2 N^2} \text{ (non-dimensional)} \right]$$

or,

$$f_1\left(\frac{P}{\rho N^3 D^5}, \frac{B}{D}, \frac{\mu}{\rho D^2 N}, \frac{\sqrt{gH}}{ND}\right) = 0$$

$$\left(\because \text{Square root of } \frac{gH}{D^2 N^2} \text{ is also dimensionless} \right)$$

or,

$$f_1\left(\frac{P}{\rho N^3 D^5}, \frac{D}{B}, \frac{\rho D^2 N}{\mu}, \frac{ND}{\sqrt{gH}}\right) = 0$$

($\because$ Reciprocal of a π -term is non-dimensional)

or,

$$\frac{P}{\rho N^3 D^5} = \phi\left(\frac{D}{B}, \frac{\rho D^2 N}{\mu}, \frac{ND}{\sqrt{gH}}\right)$$

or,

$$P = \rho N^3 D^5 \phi\left(\frac{D}{B}, \frac{\rho D^2 N}{\mu}, \frac{ND}{\sqrt{gH}}\right) \text{ (Ans.)}$$

If the two water turbines are dynamically similar, then, all the dimensionless parameters must be the same in both the machines. In turbines, the viscous forces are less important as compared to gravity forces and, therefore, the parameter $\frac{\rho D^2 N}{\mu}$ has little significance, thus for complete

similarity between model and prototype of water turbines following conditions need to be satisfied:

(i) For geometric similarity:

$$\left(\frac{H}{D}\right)_p = \left(\frac{H}{D}\right)_m \text{ and } \left(\frac{D}{B}\right)_p = \left(\frac{D}{B}\right)_m$$

(ii) For kinematic similarity:

$$\left(\frac{ND}{\sqrt{gH}} \right)_p = \left(\frac{ND}{\sqrt{gH}} \right)_m$$

(iii) For dynamic similarity:

$$\left(\frac{P}{\rho N^3 D^5} \right)_p = \left(\frac{P}{\rho N^3 D^5} \right)_m$$

The subscripts p and m refer to the prototype and model respectively. (Ans.)

7.4.3. Limitations of Dimensional Analysis

Following are the *limitations* of dimensional analysis:

1. Dimensional analysis does not give any clue regarding the selection of variables. If the variables are wrongly taken, the resulting functional relationship is erroneous. It provides the information about the grouping of variables. In order to decide whether selected variables are pertinent or superfluous experiments have to be performed.
2. The complete information is not provided by dimensional analysis; it only indicates that there is some relationship between parameters. It does not give the values of co-efficients in the functional relationship. The values of co-efficients and hence the nature of functions can be obtained only from experiments or from mathematical analysis.

MODEL ANALYSIS

7.5. MODEL ANALYSIS—INTRODUCTION

In order to know about the performance of the hydraulic structures (e.g. dams, spillways etc.) or hydraulic machines (e.g. turbines, pumps etc.) before actually constructing or manufacturing them, their models are made and tested to get the required information. The **model** is the *small scale replica of the actual structure or machine*. The *actual structure or machine* is called **Prototype**. The models are not always smaller than the prototype, in some cases a model may be even larger or of the same size as prototype depending upon the need and purpose (e.g. the working of a wrist watch or a carburettor can be studied in a large scale model).

Advantages of model testing:

The following are the *advantages* of model analysis:

1. The model tests are quite *economical* and *convenient* (because the design, construction and operation of a model may be changed several times if necessary, without increasing much expenditure, till most suitable design is obtained).
2. With the use of models the performance of hydraulic structures/hydraulic machines can be predicted in advance.
3. While designing a particular portion of the structure if clear cut analytical and reliable method is not available then in such cases it becomes absolutely necessary to know about the safety and reliability of such parts which is possible by means of model testing.
4. Model testing can be used to detect and rectify the defects of an existing structure which is not functioning properly.

Applications of the model testing:

Following are the important fields where applications of the model testing is of great use:

1. Civil engineering structures such as *dams, spillways, weirs, canals* etc.

2. Flood control, investigation of silting, and scour in rivers, irrigation channels.
3. Turbines, pumps and compressors.
4. Design of harbours, ships and submarines.
5. Aeroplanes, rockets and missiles.
6. Tall buildings (to predict the wind loads on buildings, the stability characteristics of the buildings and airflow patterns in their vicinity).

7.6. SIMILITUDE

To find solutions to numerous complicated problems in hydraulic engineering and fluid mechanics model studies are usually conducted. In order that results obtained in the model studies represent the behaviour of prototype, the following *three similarities must be ensured* between the model and the prototype.

1. Geometric similarity,
2. Kinematic similarity, and
3. Dynamic similarity.

1. Geometric similarity:

For geometric similarity to exist between the model and the prototype, the ratios of corresponding lengths in the model and in the prototype must be same and the included angles between two corresponding sides must be the same. Models which are not geometrically similar are known as *geometrically distorted models*.

Let,
 L_m = Length of model,
 H_m = Height of model,
 D_m = Diameter of model,
 A_m = Area of model,
 V_m = Volume of model,

and, L_p, B_p, H_p, D_p, A_p and V_p = Corresponding values of the prototype.

Then, for *geometric similarity*, we must have the relation:

$$\frac{L_m}{L_p} = \frac{B_m}{B_p} = \frac{H_m}{H_p} = \frac{D_m}{D_p} = L_r \quad \dots(7.8)$$

where L_r is called the *scale ratio* or the *scale factor*.

$$\text{Similarly,} \quad A_r = \text{Area ratio} = \frac{A_m}{A_p} = L_r^2 \quad \dots(7.9)$$

$$\text{and,} \quad V_r = \text{Volume ratio} = \frac{V_m}{V_p} = L_r^3 \quad \dots(7.10)$$

2. Kinematic similarity:

Kinematic similarity is the *similarity of motion*. If at the corresponding points in the model and in the prototype, the velocity or acceleration ratios are same and velocity or acceleration vectors point in the *same direction*, the two flows are said to be *kinematically similar*.

Let,
 $(V_1)_m$ = Velocity of fluid at point 1 in the model,
 $(V_2)_m$ = Velocity of fluid at point 2 in the model,
 $(a_1)_m$ = Acceleration of fluid at point 1 in the model,
 $(a_2)_m$ = Acceleration of fluid at point 2 in the model,

and $(V_1)_p, (V_2)_p, (a_1)_p, (a_2)_p$ = Corresponding values at the corresponding points of fluid velocity and acceleration in the prototype.

Then, for kinematic similarity, we must have:

$$\frac{(V_1)_m}{(V_1)_p} = \frac{(V_2)_m}{(V_2)_p} = V_r \text{ velocity ratio} \quad \dots(7.11)$$

Similarly
$$\frac{(a_1)_m}{(a_1)_p} = \frac{(a_2)_m}{(a_2)_p} = a_r \text{ acceleration ratio} \quad \dots(7.12)$$

- The *directions* of the velocities in the model and prototype *should be same*.
- The *geometric similarity* is a *prerequisite* for kinematic similarity.

3. Dynamic similarity:

Dynamic similarity is the *similarity of forces*. The flows in the model and in prototype are dynamically similar if at all the corresponding points, identical types of forces are parallel and bear the same ratio. In dynamic similarity, the force polygons of the two flows can be superimposed by change in force scale.

Let, $(F_i)_m$ = Inertia force at a point in the model,
 $(F_v)_m$ = Viscous force at the point in the model,
 $(F_g)_m$ = Gravity force at the point in the model,

and, $(F_i)_p, (F_v)_p, (F_g)_p$ = Corresponding values of forces at the corresponding points in prototype.

Then for *dynamic similarity*, we have:

$$\frac{(F_i)_m}{(F_i)_p} = \frac{(F_v)_m}{(F_v)_p} = \frac{(F_g)_m}{(F_g)_p} \dots\dots = F_r \text{ (force ratio)} \quad \dots(7.13)$$

The directions of the corresponding forces at the corresponding points in the model and prototype should also be same.

7.7. FORCES INFLUENCING HYDRAULIC PHENOMENA

The forces which may affect/influence the flow characteristics of a problem are:

1. Inertia force (F_i):

- It always exists in the fluid flow problem (and hence it is customary to find out the force ratios with respect to inertia force).
- It is equal to the *product of mass and acceleration* of the flowing fluid and acts in the direction *opposite to the direction of acceleration*.

2. Viscous force (F_v):

- It is present in fluid flow problems where viscosity is to play an important role.
- It is equal to the *product of shear stress (τ) due to viscosity and surface area of the flow*.

3. Gravity force (F_g):

- It is present in case of open surface flow.
- It is equal to the *product of mass and acceleration due to gravity*.

4. Pressure force (F_p):

- This type of force is present in case of pipe flow.
- It is equal to the *product of pressure intensity and cross-sectional area of the flowing fluid*.

5. Surface tension force (F_s):

- It is equal to the *product of surface tension and length of surface of the flowing fluid*.

6. Elastic force (F_e):

It is equal to the *product of elastic stress and area of the flowing fluid*.

7.8. DIMENSIONLESS NUMBERS AND THEIR SIGNIFICANCE

The *dimensionless numbers* (also called non-dimensional parameters) are obtained by dividing the *inertia force* (which always exists when any mass in motion) by viscous force or gravity force or pressure force or surface tension force or elastic force. The important dimensionless numbers are :

1. Reynolds number
2. Froude's number
3. Euler's number
4. Weber's number
5. Mach's number.

7.8.1. Reynolds Number (Re)

It is defined as the *ratio of the inertia force to the viscous force*.

$$\text{Inertia force } (F_i) = \text{Mass} \times \text{acceleration}$$

$$= \rho \times \text{volume} \times \frac{\text{velocity}}{\text{time}}$$

$$= \rho \times \frac{\text{volume}}{\text{time}} \times \text{velocity}$$

$$= \rho \times AV \times V \quad \left[\begin{array}{l} \because \text{Volume per second} \\ = \text{area} \times \text{velocity} = AV \end{array} \right]$$

$$= \rho AV^2 \quad \dots(7.14)$$

$$\text{Viscous force } (F_v) = \text{Shear stress} \times \text{area} = \tau \times A$$

$$= \left(\mu \frac{du}{dy} \right) \times A$$

$$= \mu \frac{V}{L} \times A \quad \left(\because \frac{du}{dy} = \frac{V}{L} \right)$$

$$\therefore \text{Reynolds number, } Re = \frac{F_i}{F_v} = \frac{\rho AV^2}{\mu \times \frac{V}{L} \times A} = \frac{\rho VL}{\mu}$$

$$i.e. \quad Re = \frac{\rho VL}{\mu} = \frac{VL}{\mu/\rho} = \frac{VL}{\nu} \quad \left(\because \nu = \frac{\mu}{\rho} \right)$$

For pipe flow (where the linear dimension is taken as diameter d),

$$Re = \frac{Vd}{\nu} \quad \dots(7.15)$$

- Reynolds number *signifies* the relative predominance of the inertia to the viscous forces occurring in the flow systems.
- This number is taken as the criterion of dynamic similarity in the flow situations where the viscous forces predominate; **examples** being: (i) *Motion of submarine completely under water*, (ii) *Low velocity motion around automobiles and aeroplanes*, (iii) *Incompressible flow through pipes of smaller sizes*, and (iv) *Flow through low speed turbomachines*.

7.8.2. Froude's number (Fr)

It is defined as the square root of the ratio of the inertia force and the gravity force.

Mathematically, $Fr = \sqrt{\frac{F_i}{F_g}}$

where,

$$F_i = \rho AV^2 \quad (\text{Eqn. 7.14})$$

and,

$$\begin{aligned} F_g &= \text{Mass} \times \text{acceleration due to gravity} \\ &= \rho \times \text{volume} \times g \\ &= \rho L^3 g = \rho L^2 \cdot L \cdot g \\ &= \rho ALg \quad (\because L^2 = A = \text{area}) \end{aligned}$$

$$\therefore Fr = \sqrt{\frac{\rho AV^2}{\rho ALg}} = \frac{V}{\sqrt{Lg}} \quad \dots(7.16)$$

- Froude's number governs the dynamic similarity of the flow situations; where *gravitational force is most significant* and all other forces are comparatively negligible, **examples** being: (i) Flow over notches and weirs, (ii) Flow over the spillway of a dam, (iii) Flow through open channels, considering waves and jumps, and (iv) Motion of ship in rough and turbulent sea.

7.8.3. Euler's Number (Eu)

It is defined as the square root of the ratio of the inertia force to the pressure force.

Mathematically, $Eu = \sqrt{\frac{F_i}{F_p}}$

where,

$$F_i = \rho AV^2 \quad (\text{Eqn. 7.14})$$

and,

$$F_p = \text{Intensity of pressure} \times \text{area} = p \times A$$

$$\therefore Eu = \sqrt{\frac{\rho AV^2}{p \times A}} = \sqrt{\frac{V^2}{p/\rho}} = \frac{V}{\sqrt{p/\rho}} \quad \dots(7.17)$$

- The Euler number is important in the flow problems/situations in which a *pressure gradient exists*: **examples** being: (i) Discharge through orifices, mouthpieces and sluices, (ii) Pressure rise due to sudden closure of valves, (iii) Flow through pipes, and (iv) Water hammer created in penstocks.

7.8.4. Weber Number (We)

It is defined as the square root of the ratio of the inertia force to the surface tension force.

Mathematically, $We = \sqrt{\frac{F_i}{F_s}}$

where,

$$F_i = \rho AV^2 \quad [\text{Eqn. 7.14}]$$

and,

$$F_s = \text{Surface tension force} = \text{surface tension} \times \text{length}$$

$$\begin{aligned} \therefore We &= \sqrt{\frac{\rho AV^2}{\sigma L}} = \sqrt{\frac{\rho \times L^2 \times V^2}{\sigma L}} \quad (\because A = L^2) \\ &= \sqrt{\frac{\rho L \times V^2}{\sigma}} = \frac{V}{\sqrt{\sigma/\rho L}} \quad \dots(7.18) \end{aligned}$$

- This number assumes importance in the following flow situations: (i) Capillary movement of water in soils, (ii) Flow of blood in veins and arteries, and (iii) Liquid atomisation.

7.8.5. Mach Number (M)

It is defined as the square root of the ratio of the inertia force to the elastic force.

$$\text{Mathematically, } M = \sqrt{\frac{F_i}{F_e}} \quad (\text{Eqn. 7.14})$$

where,

$$F_i = \rho AV^2$$

and,

$$F_e = \text{Elastic force}$$

$$= \text{Elastic stress} \times \text{area}$$

$$= K \times A = K \times L^2$$

(where, K = elastic stress)

$$\therefore M = \sqrt{\frac{\rho AV^2}{KL^2}} = \sqrt{\frac{\rho L^2 V^2}{KL^2}} = \frac{V}{\sqrt{K/\rho}}$$

$$\text{But, } \sqrt{K/\rho} = C = \text{Velocity of sound in the fluid}$$

$$\therefore M = \frac{V}{C} \quad \dots(7.19)$$

- The Mach number is important in *compressible flow problems at high velocities, such as high velocity flow in pipes or motion of high-speed projectiles and missiles.*

7.9. MODEL (OR SIMILARITY) LAWS

To ensure *dynamic similarity* between the model and prototype it is necessary that the ratio of the corresponding forces acting at the corresponding points in the model and prototype be made equal. It implies that dimensionless numbers should be same for the model as well as the prototype; this condition is difficult to be satisfied for all the dimensionless numbers. Hence models are *designed on the basis of the force which is dominating in the flow situation. The laws on which the models are designed for dynamic similarity are called model or similarity laws; these are:*

1. Reynolds model law,
2. Froude model law,
3. Euler model law,
4. Weber model law, and
5. Mach model law.

7.10. REYNOLDS MODEL LAW

In flow situations where in addition to inertia, viscous force is the other predominant force, the similarity of flow in the model and its prototype can be established if *Reynolds number is same for both the systems*. This is known as Reynolds law and according to this law

$$\begin{aligned} (Re)_{\text{model}} &= (Re)_{\text{prototype}} \\ \frac{\rho_m V_m L_m}{\mu_m} &= \frac{\rho_p V_p L_p}{\mu_p} \quad \dots(7.20) \end{aligned}$$

where,

$$\rho_m = \text{Density of fluid in model,}$$

$$V_m = \text{Velocity of fluid in model,}$$

$$L_m = \text{Length or linear dimension of the model,}$$

$$\mu_m = \text{Viscosity of fluid in model,}$$

and ρ_p, V_p, L_p and μ_p are the corresponding values of density, velocity, linear dimension and viscosity of fluid in prototype.

$$\text{or,} \quad \frac{\rho_p}{\rho_m} \times \frac{V_p}{V_m} \times \frac{L_p}{L_m} \times \frac{1}{(\mu_p / \mu_m)} = 1$$

$$\text{or,} \quad \frac{\rho_r V_r L_r}{\mu_r} = 1 \quad \dots(7.21)$$

$$\left[\rho_r = \frac{\rho_p}{\rho_m}, V_r = \frac{V_p}{V_m}, L_r = \frac{L_p}{L_m} \right]$$

where, the various quantities with subscript r represent the corresponding *scale ratios*.

$$\text{Also,} \quad \text{Time scale ratio, } T_r = \frac{L_r}{V_r} \quad \left(\because V = \frac{L}{T} \text{ and } T = \frac{L}{V} \right)$$

$$\text{Acceleration scale ratio, } a_r = \frac{V_r}{T_r}$$

$$\begin{aligned} \text{Force scale ratio, } F_r &= (\text{mass} \times \text{acc.})_r \\ &= m_r \times a_r \\ &= \rho_r A_r V_r \times a_r = \rho_r L_r^2 V_r \times a_r \end{aligned}$$

$$\begin{aligned} \text{Discharge scale ratio, } Q_r &= (\rho A V)_r \\ &= \rho_r A_r V_r = \rho_r L_r^2 V_r \end{aligned}$$

- Following are some of the phenomena for which Reynolds model law can be a sufficient criterion for similarity of flow in the model and the prototype:

- (i) Motion of air planes,
- (ii) Flow of incompressible fluid in closed pipes,
- (iii) Motion of submarines completely under water, and
- (iv) Flow around structures and other bodies immersed completely under moving fluids.

Example 7.17. An oil of specific gravity 0.92 and viscosity 0.03 poise is to be transported at the rate of 2500 litres/sec. through a 1.2 m diameter pipe. Tests were conducted on a 12 cm diameter pipe using water at 20°C. If the viscosity water at 20°C is 0.01 poise, find:

(i) Velocity of flow in the model;

(ii) Rate of flow in the model.

Solution. Sp. gr. of oil, $S_p = 0.92$

$$\text{Viscosity of oil, } \mu_p = 0.03 \text{ poise} = 0.03 \times \frac{1}{10} \text{ Ns/m}^2 = 0.003 \text{ Ns/m}^2$$

$$\text{Rate of oil flow, } Q = 2500 \text{ litres/s} = 2.5 \text{ m}^3/\text{s}$$

$$\text{Diameter of prototype, } D_p = 1.2 \text{ m}$$

$$\text{Diameter of the model, } D_m = 12 \text{ cm} = 0.12 \text{ m}$$

$$\text{Viscosity of water at } 20^\circ\text{C, } \mu_m = 0.01 \text{ poise} = 0.01 \times \frac{1}{10} = 0.001 \text{ Ns/m}^2$$

(i) **Velocity of flow in the model, V_m :**

The dynamic similarity for pipe flow will be obtained if Reynolds number is *same* for both the model and prototype.

$$\therefore \quad \frac{\rho_m V_m D_m}{\mu_m} = \frac{\rho_p V_p D_p}{\mu_p} \quad \dots(\text{Eqn. 7.20})$$

(For pipe linear dimension is D)

$$\begin{aligned}
 \text{or, } \frac{V_m}{V_p} &= \frac{\rho_p}{\rho_m} \cdot \frac{D_p}{D_m} \cdot \frac{\mu_m}{\mu_p} \\
 &= \frac{(9.81 \times 0.92 / 9.81)}{(9.81 / 9.81)} \times \frac{1.2}{0.12} \times \frac{0.001}{0.003} = 3.067 \\
 \left[\because \rho_p &= \frac{w \times S_p}{g} = \frac{9.81 \times 0.92}{9.81} \text{ and } \rho_m = \frac{w}{g} = \frac{9.81}{9.81} \right]
 \end{aligned}$$

$$\text{But, } V_p = \frac{Q_p}{A_p} = \frac{2.5}{\frac{\pi}{4} \times 1.2^2} = 2.21 \text{ m/s}$$

$$\therefore V_m = 3.067 \times V_p = 3.067 \times 2.21 = \mathbf{6.78 \text{ m/s (Ans.)}}$$

(ii) Rate of flow in the model, Q_m :

$$\begin{aligned}
 Q_m &= A_m \times V_m = (\pi/4) \times D_m^2 \times V_m \\
 &= (\pi/4) \times 0.12^2 \times 6.78 = 0.07668 \text{ m}^3/\text{s} \text{ or } \mathbf{76.68 \text{ litres/s (Ans.)}}
 \end{aligned}$$

Example 7.18. A geometrically similar model of an air duct is built to $\frac{1}{25}$ scale and tested with water which is 50 times more viscous and 800 times denser than air. When tested under dynamically similar conditions, the pressure drop is 2 bar in the model. Find the corresponding pressure drop in the full scale prototype.
[Nagpur University]

Solution.

$$\begin{aligned}
 \text{Given: Scale ratio, } \frac{L_m}{L_p} &= \frac{1}{25} \\
 \frac{\mu_p}{\mu_m} &= \frac{1}{50}; \frac{\rho_p}{\rho_m} = \frac{1}{800}
 \end{aligned}$$

Pressure drop in the model = 2 bar

Pressure drop in the prototype:

For dynamic similarity between the prototype and its model, the Reynolds number for both of them should be equal.

$$\therefore \frac{\rho_p V_p L_p}{\mu_p} = \frac{\rho_m V_m L_m}{\mu_m}$$

$$\text{or, } \frac{V_p}{V_m} = \frac{\mu_p}{\mu_m} \times \frac{\rho_m}{\rho_p} \times \frac{L_m}{L_p}$$

Substituting the values, we get:

$$\frac{V_p}{V_m} = \frac{1}{50} \times 800 \times \frac{1}{50} = \frac{16}{25}$$

$$\text{The pressure, } p = \frac{F}{A} = \frac{\rho L^2 V^2}{L^2} = \rho V^2$$

$$\left[\because F = m \times a = \rho L^3 \times \frac{V}{T} = \rho L^2 \times \frac{L}{T} \times V \right. \\
 \left. = \rho L^2 V^2 \right]$$

$$\begin{aligned}\therefore \frac{\text{Pressure drop in prototype}}{\text{Pressure drop in model}} &= \frac{\rho_p}{\rho_m} \times \frac{V_p^2}{V_m^2} = \frac{1}{800} \times \left(\frac{16}{25}\right)^2 = 5.12 \times 10^{-4} \\ \therefore \text{Pressure drop in prototype} &= \text{Pressure drop in model} \times 5.12 \times 10^{-4} \\ &= 2 \times 5.12 \times 10^{-4} = 1.024 \times 10^{-3} \text{ bar} \\ &= 1.024 \times 10^{-3} \times 10^5 \text{ N/m}^2 = 102.4 \text{ N/m}^2 \\ &= \frac{102.4}{9810} \times 1000 \text{ mm} = \mathbf{10.44 \text{ mm (Ans.)}}\end{aligned}$$

Example 7.19. (a) The thrust T of a screw propeller is dependent upon the diameter D , speed of advance V , revolutions per second N , fluid density ρ and the co-efficient of viscosity μ . Experiments were performed with various models of propellers. What are the dimensionless groups to which the data should be plotted?

(b) The characteristics of a propeller of 4.8 m diameter and rotational speed 120 r.p.m. are examined by means of a geometrically similar model of 600 mm diameter. When the model is run at 480 r.p.m. by a torque of 30 Nm the thrust developed is 300 N and the speed of advance is 3 m/s. Determine the following for the full scale propeller:

- (i) Speed of advance, (ii) Thrust, and (iii) Torque.

Solution. (a) By means of dimensional analysis, it can be shown that the appropriate non-dimensional parameters are:

$$\frac{T}{\rho D^2 V^2} = \phi \left(\frac{VD\rho}{\mu}, \frac{DN}{V} \right)$$

Please solve this part (a) by using Rayleigh's method or Buckingham's π -theorem to obtain the above result.

The propeller thrust is thus governed by Reynolds number $\left(\frac{VD\rho}{\mu} \right)$ and the factor $\left(\frac{DN}{V} \right)$.

However, both the conditions *cannot be satisfied at the same time*. Usually the *effect of viscosity is neglected* and the factor $\left(\frac{DN}{V} \right)$ is *arranged to be the same* for the dynamic similarity between the model and prototype.

(b) **Given:** Diameter of propeller, (prototype) $D_p = 4.8 \text{ m}$

Speed, $N_p = 120 \text{ r.p.m.}$

Diameter of the model, $D_m = 600 \text{ mm} = 0.6 \text{ m}$

Rotational speed, $N_m = 480 \text{ r.p.m.}$

Torque of the model = 30 Nm

Thrust developed, $T_m = 300 \text{ N}$

Speed of advance, $V_m = 3 \text{ m/s.}$

(i) **Speed of advance for the propeller, V_p :**

$$\left(\frac{DN}{V} \right)_m = \left(\frac{DN}{V} \right)_p$$

$$\text{i.e.,} \quad \frac{D_m N_m}{V_m} = \frac{D_p N_p}{V_p}$$

$$\text{or,} \quad V_p = V_m \times \frac{D_p}{D_m} \times \frac{N_p}{N_m}$$

$$= 3 \times \frac{4.8}{0.6} \times \frac{120}{480} = 6 \text{ m/s (Ans.)}$$

(ii) Thrust of the propeller, T_p :

$$\left(\frac{T}{\rho D^2 V^2} \right)_m = \left(\frac{T}{\rho D^2 V^2} \right)_p$$

$$\text{i.e.} \quad \frac{T_m}{\rho_m D_m^2 V_m^2} = \frac{T_p}{\rho_p D_p^2 V_p^2}$$

$$\begin{aligned} \text{or,} \quad T_p &= T_m \times \frac{\rho_p}{\rho_m} \times \left(\frac{V_p}{V_m} \right)^2 \times \left(\frac{D_p}{D_m} \right)^2 \\ &= 300 \times 1 \times \left(\frac{6}{3} \right)^2 \times \left(\frac{4.8}{0.6} \right)^2 \quad \left[\because \rho_p = \rho_m, \text{ the fluid} \right. \\ &\quad \left. \text{medium being same} \right] \\ &= 76800 \text{ N (Ans.)} \end{aligned}$$

(iii) Torque of the propeller:

Efficiency of the model propeller

$$\begin{aligned} &= \frac{\text{Output}}{\text{Input}} = \frac{\text{Thrust} \times \text{speed of advance}}{\text{Torque} \times \text{angular speed}} \\ &= \frac{300 \times 3}{30 \times \left(\frac{2\pi \times 480}{60} \right)} = 0.597 \text{ or } 59.7\% \end{aligned}$$

Assuming that the prototype has the same efficiency as the model we have for the prototype,

$$0.597 = \frac{76800 \times 6}{\text{Torque} \times \left(\frac{2\pi \times 120}{60} \right)}$$

$$\begin{aligned} \therefore \text{Torque of the propeller} &= \frac{76800 \times 6 \times 60}{0.597 \times 2\pi \times 120} \\ &= 61422 \text{ Nm (Ans.)} \end{aligned}$$

Example 7.20. Resistance R to the motion of a completely submerged body is given by $R = \rho V^2 L^2 \phi \left(\frac{VL}{\nu} \right)$, where ρ and ν are density and kinematic viscosity of the fluid while L is the length of the body and V is the velocity of flow. If the resistance of a one-eighth scale airship model when tested in water at 12 m/s is 220 N, what will be the resistance in air of the airship at the corresponding speed? Kinematic viscosity of air is 13 times that of water and density of water is 810 times of air. [Delhi University]

$$\text{Solution. Given: Scale ratio} = \frac{L_m}{L_p} = \frac{1}{8}$$

$$\text{Velocity of model, } V_m = 12 \text{ m/s}$$

$$\text{Resistance of model, } R_m = 220 \text{ N}$$

The fluids for model and the prototype are *water* and *air* respectively.

$\therefore$ Kinematic viscosity of air = 13 $\times$ kinematic viscosity of water

$$\text{or,} \quad \nu_p = 13\nu_m$$

Density of water = $810 \times$ density of air

or, $\rho_m = 810 \rho_p$

Resistance of the airship in air, R_p :

The resistance, R , is given by:

$$R = \rho V^2 L^2 \phi \left(\frac{VL}{\nu} \right)$$

From the above equation, it is obvious that flow in the model will be dynamically similar if the Reynolds numbers are equal in both the systems. Thus, if

$$\left(\frac{VL}{\nu} \right)_m = \left(\frac{VL}{\nu} \right)_p \quad \dots(i)$$

$$\text{then,} \quad \left(\frac{R}{\rho V^2 L^2} \right)_m = \left(\frac{R}{\rho V^2 L^2} \right)_p \quad \dots(ii)$$

From eqn. (i), we have:

$$\frac{V_m L_m}{\nu_m} = \frac{V_p L_p}{\nu_p}$$

$$\text{or,} \quad V_p = V_m \cdot \frac{L_m}{L_p} \cdot \frac{\nu_p}{\nu_m} = 12 \times \frac{1}{8} \times 13 = 19.5 \text{ m/s}$$

At this prototype velocity, the resistance of the airship is obtained from eqn. (ii) as follows:

$$\frac{R_m}{\rho_m V_m^2 L_m^2} = \frac{R_p}{\rho_p V_p^2 L_p^2}$$

$$\begin{aligned} \text{or,} \quad R_p &= R_m \cdot \frac{L_p^2}{L_m^2} \cdot \frac{V_p^2}{V_m^2} \cdot \frac{\rho_p}{\rho_m} = 220 \times 8^2 \times \left(\frac{19.5}{12} \right)^2 \times \frac{1}{810} \\ &= \mathbf{45.9 \text{ N (Ans.)}} \end{aligned}$$

Example 7.21. (a) What are the various dimensionless groups in fluid mechanics? Under what circumstances is each of these groups important?

(b) The drag of a small submarine hull is desired when it is moving far below the free surface of water. A $\frac{1}{10}$ scale model is to be tested. What dimensionless group should be duplicated between the model and prototype and why? If the drag of the prototype at 1 knot is desired, at what speed should model be moved to give the drag to be expected by the prototype? Would this result still be true if the prototype were to be moved close to the surface? Explain. [Engg. Services]

Solution.

(a) The various dimensionless groups which often appear in fluid mechanics are given in the table (refer to page 426) along with their significance and fields of application.

$$(b) \quad \text{Scale ratio, } L_r = \frac{L_m}{L_p} = \frac{1}{10} \quad \dots(\text{Given})$$

When a submarine is moving far below the free water surface it corresponds to the flow situation where the body is *entirely submerged in an infinite mass of fluid at rest*; the dynamic similarity criterion is then prescribed by **Reynolds number** which must be duplicated for model and prototype. For dynamic similarity, equating the Reynolds number, we have:

$$\frac{\rho_m V_m L_m}{\mu_m} = \frac{\rho_p V_p L_p}{\mu_p}$$

or,
$$V_m = \frac{\rho_p}{\rho_m} \times \frac{\mu_m}{\mu_p} \times \frac{L_p}{L_m} \times V_p$$

But, $\rho_m = \rho_p$ and $\mu_m = \mu_p$ since water is the fluid both for model and prototype.

$\therefore V_m = 1 \times 1 \times 10 \times 1 = \mathbf{10 \text{ knots (Ans.)}}$

To evaluate the drag experienced by the prototype let us duplicate the drag co-efficient $F/\rho L^2 V^2$ for the model and the prototype as under:

$$\begin{aligned} F_p &= \frac{\rho_p}{\rho_m} \times \left(\frac{L_p}{L_m} \right)^2 \times \left(\frac{V_p}{V_m} \right)^2 \times F_m \\ &= 1 \times 10^2 \times \left(\frac{1}{10} \right)^2 \times F_m = F_m \end{aligned}$$

Table: Dimensionless Groups/Numbers

Sl. No.	Dimensionless number	Aspects			
		Symbol	Group of variables	Significance	Field of application
1.	<i>Reynolds number</i>	Re	$\frac{\rho V L}{\mu}$	$\frac{\text{Inertia force}}{\text{Viscous force}}$	Laminar viscous flow in confined passages (where <i>viscous effects are significant</i>)
2.	<i>Froude's number</i>	Fr	$\frac{V}{\sqrt{Lg}}$	$\sqrt{\frac{\text{Inertia force}}{\text{Gravity force}}}$	Free surface flows (where <i>gravity effects are important</i>)
3.	<i>Euler's number</i>	Eu	$\frac{V}{\sqrt{p/\rho}}$	$\sqrt{\frac{\text{Inertia force}}{\text{Pressure force}}}$	Conduit flow (where <i>pressure variations are significant</i>)
4.	<i>Weber's number</i>	We	$\frac{V}{\sqrt{\sigma/\rho L}}$	$\sqrt{\frac{\text{Inertia force}}{\text{Surface tension}}}$	Small surface waves, capillary and sheet flow (where <i>surface tension is important</i>)
5.	<i>Mach's number</i>	M	$\sqrt{\frac{V}{K/\rho}}$	$\sqrt{\frac{\text{Inertia force}}{\text{Elastic Force}}}$	High speed flow (where <i>compressibility effects are significant</i>).

Hence when a $\frac{1}{10}$ model moves at a speed of 10 knots, the drag experienced by it will be the same as that experienced by the prototype moving at a speed of 1 knot.

- When the submarine moves close to the surface, the *waves offer additional resistance*, therefore, the gravity effect must also be considered. The above result would therefore, *not be valid* then.

Example 7.22. A model of submarine is scaled down 1/20 of the prototype and is to be tested in a wind tunnel. The speed of the prototype at which we are to estimate the drag is 8 m/s. What should be the free-stream velocity of the air? What will be ratio of the drag between the model and the prototype? Explain why there would be no dynamic similarity if the submarine prototype is moved near the free surface.

$$v_{\text{sea water}} = 1.21 \times 10^{-2} \text{ cm}^2/\text{s}$$

$$\begin{aligned}v_{air} &= 1.64 \times 10^{-1} \text{ cm}^2/\text{s} \\ \rho_{sea\ water} &= 1027 \text{ kg/m}^3. \\ \rho_{air} &= 1.34 \text{ kg/m}^3.\end{aligned}$$

[IIT Delhi]

Solution. Given: Scale ratio = 1/20

Speed of the prototype, $V_p = 8 \text{ m/s}$

Free-stream velocity, V_m :

As the submarine is to overcome the *viscous resistance*, the similarity of *Reynolds number* is essential for dynamic similarity between the model and the prototype.

$$\left(\frac{VL}{\nu}\right)_p = \left(\frac{VL}{\nu}\right)_m$$

$$\begin{aligned}\therefore V_m &= V_p \times \frac{L_p}{L_m} \times \frac{\nu_m}{\nu_p} = 8 \times 20 \times \frac{1.64 \times 10^{-1}}{1.21 \times 10^{-2}} \\ &= \mathbf{2168.6 \text{ m/s. (Ans.)}}\end{aligned}$$

Ratio of drag force:

Also drag force, $F = \text{Mass} \times \text{acceleration}$

$$= \rho \times L^3 \times \frac{V}{T} = \rho L^2 \times \frac{L}{T} \times V = \rho L^2 V^2$$

$\therefore$ The ratio of drag force,

$$\begin{aligned}\frac{F_p}{F_m} &= \frac{\rho_p}{\rho_m} \times \left(\frac{L_p}{L_m}\right)^2 \times \left(\frac{V_p}{V_m}\right)^2 \\ &= \frac{1027}{1.34} \times (20)^2 \times \left(\frac{8}{2168.6}\right)^2 = \mathbf{4.17 \text{ (Ans.)}}\end{aligned}$$

- When the prototype moves closer to the surface, *gravity force will generate the surface waves and that too will contribute to the surface resistance* and hence there would be no dynamic similarity. The model, therefore, will have to be tested with the similarity of Reynolds and Froude's numbers.

Example 7.23. An orifice meter to carry water is calibrated with air in a geometrically similar model at 1/5 prototype scale. Dynamically similar flow will be obtained when the discharge ratio (air to water) is

(i) 0.4

(ii) 2.5

(iii) 62.5.

Assume the ratio of kinematic viscosity of air to water as 12.5.

[UPSC Exams; Fluid Mechanics]

Solution.

Given: Scale ratio, $\frac{L_m}{L_p} = \frac{1}{5}$

$$\frac{\nu_{air}}{\nu_{water}} = \frac{\nu_m}{\nu_p} = 12.5$$

Discharge ratio:

In case of an orifice which is to carry water and is calibrated with air, the dynamically similar flow will be obtained when the Reynolds numbers in model and prototype are equal, thus:

$$\frac{V_m L_m}{v_m} = \frac{V_p L_p}{v_p}$$

$$\text{or,} \quad \frac{V_m}{V_p} = \frac{v_m}{v_p} \times \frac{L_p}{L_m} = 12.5 \times 5 = 62.5$$

∴ Discharge ratio (air to water),

$$\frac{Q_m}{Q_p} = \frac{L_m^2}{L_p^2} \frac{V_m}{V_p} = \left(\frac{1}{5}\right)^2 \times 62.5 = \mathbf{2.5 \text{ (Ans.)}}$$

Example 7.24. A test was made on a pipe model 15 mm in diameter and 3 m long with water flowing through it at the corresponding speed for frictional resistance. The head loss was found by measurement to be 7 m of water. The prototype pipe is 300 mm in diameter and 240 m long through which air is flowing at 3.6 m/s. Density of water and air are 1000 kg/m³ and 1.22 kg/m³ respectively and co-efficients of viscosity of water and air are 0.01 and 1.8×10^{-4} poise respectively. Find:

(i) The corresponding speed of water in the model pipe for dynamic similarity;

(ii) Pressure drop in prototype pipe.

Neglect change of density of the air.

Solution. For model:

$$\text{Diameter of pipe, } D_m = 15 \text{ mm} = 0.015 \text{ m}$$

$$\text{Density of water, } \rho_m = 1000 \text{ kg/m}^3$$

$$\text{Dynamic viscosity of water, } \mu_m = 0.01 \text{ poise} = \frac{0.01}{10} = 0.001 \frac{\text{N.s}}{\text{m}^2}$$

$$\text{Length, } L_m = 3 \text{ m}$$

For prototype:

$$\text{Diameter of pipe, } D_p = 300 \text{ mm} = 0.3 \text{ m}$$

$$\text{Density of air, } \rho_p = 1.22 \text{ kg/m}^3$$

$$\text{Dynamic viscosity of air, } \mu_p = 1.8 \times 10^{-4} \text{ poise} = \frac{1.8 \times 10^{-4}}{10} = 1.8 \times 10^{-5} \frac{\text{N.s}}{\text{m}^2}$$

$$\text{Length, } L_p = 240 \text{ m}$$

$$\text{Velocity of air, } V_p = 3.6 \text{ m/s}$$

$$\text{Head lost in model} = 7 \text{ m}$$

$$\therefore \text{ Pressure drop in model, } (\Delta p)_m = 7 \times 9.81 = 68.67 \text{ kN/m}^2.$$

(i) Speed of water in the model pipe, V_m :

For dynamic similarity in the case of frictional resistance, the Reynolds number in the model and prototype must be same i.e.

$$\left(\frac{\rho V D}{\mu}\right)_m = \left(\frac{\rho V D}{\mu}\right)_p$$

$$\text{i.e.} \quad \frac{\rho_m V_m D_m}{\mu_m} = \frac{\rho_p V_p D_p}{\mu_p}$$

$$\text{or,} \quad V_m = V_p \times \frac{\rho_p}{\rho_m} \times \frac{D_p}{D_m} \times \frac{\mu_m}{\mu_p}$$

$$= 3.6 \times \frac{1.22}{1000} \times \frac{0.3}{0.015} \times \frac{0.001}{1.8 \times 10^{-5}}$$

$$= 4.88 \text{ m/s (Ans.)}$$

(ii) Pressure drop in prototype pipe, $(\Delta p)_p$:

We know that for frictional resistance,

$$\frac{R}{\rho L^2 V^2} = \phi \text{ (Reynolds number)}$$

where, R is the resistance and L^2 represents the characteristic area. In the case of pipe flow the characteristic area may be taken as the wetted area i.e., πDL where L is the length of pipe under consideration.

As the Reynolds number is the same in model and prototype, the above equation becomes:

$$\left(\frac{R}{\rho \pi DL V^2} \right)_m = \left(\frac{R}{\rho \pi DL V^2} \right)_p$$

Also,
$$R = \Delta p \times \frac{\pi D^2}{4}$$

where, Δp is the pressure drop.

$$\therefore \left(\frac{\Delta p \cdot D}{\rho L V^2} \right)_m = \left(\frac{\Delta p \cdot D}{\rho L V^2} \right)_p$$

or,
$$\frac{(\Delta p)_m \cdot D_m}{\rho_m \cdot L_m \cdot V_m^2} = \frac{(\Delta p)_p \cdot D_p}{\rho_p \cdot L_p \cdot V_p^2}$$

or,
$$(\Delta p)_p = (\Delta p)_m \cdot \frac{\rho_p}{\rho_m} \cdot \frac{D_m}{D_p} \cdot \frac{V_p^2}{V_m^2} \cdot \frac{L_p}{L_m}$$

$$= 68.67 \times \frac{1.22}{1000} \times \frac{0.015}{0.3} \times \frac{3.6^2}{4.88^2} \times \frac{240}{3}$$

$$= 0.182 \text{ kN/m}^2 \text{ (Ans.)}$$

Example 7.25. Water having a coefficient of kinematic viscosity of $1.12 \times 10^{-6} \text{ m}^2/\text{s}$ and a mass density of 1 Mg/m^3 flows at a mean speed of 1.75 m/s through a 75 mm diameter pipeline. What corresponding volumetric rate (measured at atmospheric pressure) of air flow through this pipeline would give rise to essentially similar dynamical flow conditions and why would this be so? Air may be assumed to have a coefficient of kinematic viscosity of $14.7 \times 10^{-6} \text{ m}^2/\text{s}$ and a mass density of 1.23 kg/m^3 . Determine for each fluid, the pressure drop which would occur in 10 m length of this pipeline. Take $f_1 = 0.010$ (Darcy's friction factor) for both fluids. (MU)

Solution. Given:

Water

$$\nu = 1.12 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\rho = 1 \text{ Mg/m}^3 = 1000 \text{ kg/m}^3$$

$$V = 1.75 \text{ m/s}$$

$$D = 75 \text{ mm} = 0.075 \text{ m}$$

$$L = 10 \text{ m}$$

$$f_1 = 0.010$$

Air

$$\nu = 14.7 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\rho = 1.23 \text{ kg/m}^3$$

$$V = ?$$

$$D = 75 \text{ mm} = 0.075 \text{ m}$$

$$L = 10 \text{ m}$$

$$f_1 = 0.010$$

For similar dynamical flow conditions, the ratio of corresponding forces acting at corresponding points in the model and prototype should be equal. The ratio of forces are dimensionless numbers. It means for dynamic similarity between model and prototype, dimensionless numbers should be same for model and prototype.

In the above question, we are dealing with the flow through pipe for which *Reynolds model law should hold good, which states that Reynolds number for the model must be equal to the Reynolds number for the prototype.*

$$\begin{aligned} (Re)_{water} &= (Re)_{air} \\ \left(\frac{\rho VD}{\mu} \right)_{water} &= \left(\frac{\rho VD}{\mu} \right)_{air} \\ \text{or,} \quad \left(\frac{VD}{\nu} \right)_{water} &= \left(\frac{VD}{\nu} \right)_{air} \\ V_{air} &= V_{water} \times \frac{D_{water}}{D_{air}} \times \frac{\nu_{air}}{\nu_{water}} \\ &= 1.75 \times \frac{0.075}{0.075} \times \frac{14.7 \times 10^{-6}}{1.12 \times 10^{-6}} = 22.97 \text{ m/s} \end{aligned}$$

$$\begin{aligned} \text{Volumetric flow rate of air} &= \text{Area} \times \text{velocity} \\ &= \frac{\pi}{4} D_{air}^2 \times V_{air} = \frac{\pi}{4} \times (0.075)^2 \times 22.97 \\ &= \mathbf{0.1015 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

For water:

$$Re = \frac{VD}{\nu} = \frac{1.75 \times 0.075}{1.12 \times 10^{-6}} = 117187.5$$

Since Reynolds number is greater than 4000, hence flow is *turbulent*, for which loss is given by:

$$\begin{aligned} h_f &= \frac{4fLV^2}{D \times 2g}, \text{ where } f = \text{co-efficient of friction.} \\ &= \frac{f_1 LV^2}{D \times 2g}, \text{ where } f_1 = \text{friction factor} \\ &= \frac{0.010 \times 10 \times 1.75^2}{0.075 \times 2 \times 9.81} = 0.2081 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Pressure drop} &= 0.2081 \text{ m water column} \\ &= 0.2081 \times 1000 \times 9.81 \text{ N/m}^2 = 2041.46 \text{ Pa} \approx \mathbf{2.04 \text{ kPa} \text{ (Ans.)}} \end{aligned}$$

For air:

$$Re = \frac{VD}{\nu} = \frac{22.97 \times 0.075}{14.7 \times 10^{-6}} = 117193.8$$

Since Reynolds number is greater than 4000, hence flow is turbulent for which loss of head is given by:

$$h_f = \frac{f_1 LV^2}{D \times 2g} = \frac{0.010 \times 10 \times (22.97)^2}{0.075 \times 2 \times 9.81} = 35.86 \text{ m}$$

$$\begin{aligned} \text{Pressure drop} &= 35.86 \text{ m of air column} \\ &= 35.86 \times 1.23 \times 9.81 \text{ N/m}^2 = 432.7 \text{ Pa} \approx \mathbf{0.433 \text{ kPa} \text{ (Ans.)}} \end{aligned}$$

Example 7.26. A model of submarine is scaled down to $\frac{1}{20}$ of the prototype and is to be tested in a wind tunnel where free stream pressure is 2.0 MPa absolute and temperature is 50°C . The speed of the prototype is 7.72 m/s. Determine the free stream velocity of air and the ratio of the drags between model and prototype. Assume kinematic viscosity of sea water as $1.4 \times 10^{-6} \text{ m}^2/\text{s}$ and viscosity of air as 0.0184 cP. (UPTU)

Solution. Given: $L_r = \frac{L_m}{L_p} = \frac{1}{20}$; $p = 2.0 \text{ MPa} = 2 \times 10^6 \text{ N/m}^2$; $t = 50^\circ\text{C}$
 $V_p = 7.72 \text{ m/s}$; $\nu_p = 1.4 \times 10^{-6} \text{ m}^2/\text{s}$; $\mu_{\text{air}} = 0.0184 \times 10^{-3} \text{ N s/m}^2$.
 V_m ; $\frac{(F_D)_m}{(F_D)_p}$:

$$\rho_{\text{air}} = \frac{p}{RT} = \frac{2 \times 10^6}{287 \times (50 + 273)} = 21.57 \text{ kg/m}^3$$

$$\nu_m = \frac{\mu_{\text{air}}}{\rho_{\text{air}}} = \frac{0.0184 \times 10^{-3}}{21.57} = 8.53 \times 10^{-7} \text{ m}^2/\text{s}$$

Since the submarine has to overcome the viscous resistance, there has to be dynamic similarity between the model and the prototype; which implies equality of Reynolds number.

i.e. $(Re)_p = (Re)_m$

or, $\frac{V_p L_p}{\nu_p} = \frac{V_m L_m}{\nu_m}$

or, $V_m = \frac{V_p L_p \nu_m}{\nu_p L_m} = \frac{7.72 \times 20 \times 8.53 \times 10^{-7}}{1.4 \times 10^{-6}} = 94.07 \text{ m/s}$

Hence, free stream velocity of air = **94.07 m/s (Ans.)**

Now, $\frac{(F_D)_m}{(F_D)_p} = \frac{\rho_m}{\rho_p} \times \left(\frac{L_m}{L_p}\right)^2 \times \left(\frac{V_m}{V_p}\right)^2 = \frac{\rho_m}{\rho_p} \times \left(\frac{L_m}{L_p} \frac{V_m}{V_p}\right)^2$

Assuming density of sea water as 1025 kg/m^3 ,

Ratio of drag forces,

$$\frac{(F_D)_m}{(F_D)_p} = \frac{21.57}{1025} \left[\frac{1}{20} \times \frac{94.07}{7.72} \right]^2 = \mathbf{0.00781 \text{ (Ans.)}}$$

Example 7.27. A torpedo shaped object, 900 mm diameter is to move in air at 60 m/s and its drag is to be estimated from tests in water on a half scale model. Determine the necessary speed of the model and the drag of the full scale object if that of the model is 1140 N. Given properties: air viscosity = $1.86 \times 10^{-5} \text{ N s/m}^2$, water viscosity = $1.01 \times 10^{-3} \text{ N s/m}^2$, air density = 1.2 kg/m^3 , water density = 1000 kg/m^3 . (Anna University)

Solution. Given: Torpedo shaped object (Prototype): $V_p = 60 \text{ m/s}$

$$\mu_p = \mu_{\text{air}} = 1.86 \times 10^{-5} \text{ N s/m}^2;$$

$$\rho_p = \rho_{\text{air}} = 1.2 \text{ kg/m}^3$$

Model half size tested in water: $\mu_m = \mu_{\text{water}} = 1.01 \times 10^{-3} \text{ N s/m}^2$;

$$\rho_m = \rho_{\text{water}} = 1000 \text{ kg/m}^3; (F_D)_m = 1140 \text{ N}$$

Also,
$$\frac{L_p}{L_m} = \frac{1}{(1/2)} = 2.$$

Speed of the model, V_m :

Now, $(Re)_{model} = (Re)_{prototype}$

$$\frac{\rho_m V_m L_m}{\mu_m} = \frac{\rho_p V_p L_p}{\mu_p}$$

$$\therefore V_m = \frac{\mu_m \rho_p V_p L_p}{\mu_p \rho_m L_m} = \frac{1.01 \times 10^{-3} \times 1.2 \times 60}{1.86 \times 10^{-5} \times 1000} \times 2 = 7.819 \text{ m/s (Ans.)}$$

Drag of the full scale object, $(F_D)_p$:

$$(C_D)_p = (C_D)_m;$$

$$\frac{A_p}{A_m} = \left(\frac{L_p}{L_m} \right)^2$$

$$\frac{(F_D)_p}{(F_D)_m} = \frac{(C_D)_p \times \frac{1}{2} \rho_p A_p V_p^2}{(C_D)_m \times \frac{1}{2} \rho_m A_m V_m^2} = \frac{(\rho_p L_p^2 V_p^2)}{(\rho_m L_m^2 V_m^2)}$$

$$(F_D)_p = (F_D)_m \times \left[\frac{\rho_p L_p^2 V_p^2}{\rho_m L_m^2 V_m^2} \right]$$

$$= 1140 \times \frac{1.2}{1000} \times (2)^2 \times \frac{(60)^2}{(7.819)^2}$$

$$= 322.2 \text{ N (Ans.)}$$

Example 7.28. A geometrically similar model of an air duct is built to $\frac{1}{25}$ scale and tested with water which is 50 times more viscous and 800 times denser than air. When tested under dynamically similar conditions, the pressure drop is 2 bar in model. Find corresponding pressure drop in prototype and express in water column. (MGU, Kerala)

Solution. Given: $\frac{L_p}{L_m} = 25; \frac{\mu_p}{\mu_m} = \frac{1}{50}; \frac{\rho_p}{\rho_m} = \frac{1}{800}; (\Delta p)_m = 2 \text{ bar}$

$(\Delta p)_p$:

For dynamic similarity, the Reynolds number must be equal in model and prototype.

$$\therefore \frac{\rho_m V_m L_m}{\mu_m} = \frac{\rho_p V_p L_p}{\mu_p}$$

or,
$$\frac{V_p}{V_m} = \frac{\mu_p}{\mu_m} \times \frac{\rho_m}{\rho_p} \times \frac{L_m}{L_p}$$

$$= \frac{1}{50} \times 800 \times \frac{1}{25} = \frac{16}{25}$$

$$\text{Pressure} = \frac{F}{A} = \frac{\rho L^2 V^2}{L^2} = \rho V^2$$

$$\begin{aligned}\frac{(\Delta p)_p}{(\Delta p)_m} &= \frac{\rho_p}{\rho_m} \left(\frac{V_p}{V_m} \right)^2 \\ &= \frac{1}{800} \times \left(\frac{16}{25} \right)^2 = 0.000512\end{aligned}$$

$$\begin{aligned}\therefore (\Delta)_p &= 2 \times 0.000512 \text{ bar} \\ &= 2 \times 0.000512 \times 10^5 \text{ N/m}^2 = \rho_w g \frac{h_w}{1000}\end{aligned}$$

(where, h_w = pressure drop in mm of water column)

$$\therefore h_w = \frac{2 \times 0.000512 \times 10^5 \times 1000}{1000 \times 9.81} = \mathbf{10.438 \text{ mm (Ans.)}}$$

$$(\because \rho_w = 1000 \text{ kg/m}^3)$$

Example 7.29. A ship 300 m long moves in sea-water, whose density is 1030 kg/m^3 . A 1 : 100 model of this ship is to be tested in a wind tunnel. The velocity of air in the wind tunnel around the model is 30 m/s and the resistance of the model is 60 N. Determine the velocity of ship in sea-water and also the resistance of the ship in sea-water. The density of air is given as 1.24 kg/m^3 . Take the kinematic viscosity of sea-water and air as 0.012 stokes and 0.018 stokes respectively.

(Delhi University, 2000)

Solution. Given:

Prototype	Model
$L_p = 300 \text{ m}$	$L_m = \frac{1}{100} \times 300 = 3 \text{ m}$
$\rho_p = 1030 \text{ kg/m}^3$	$\rho_m = 1.24 \text{ kg/m}^3$
$\nu_p = 0.012 \times 10^{-4} \text{ m}^2/\text{s}$	$\nu_m = 0.018 \times 10^{-4} \text{ m}^2/\text{s} \quad (\because 1 \text{ stoke} = 10^{-4} \text{ m}^2/\text{s})$
	$V_m = 30 \text{ m/s}$
	$R_m = 60 \text{ N.}$

V_p, R_p :

For dynamic similarity between the prototype and its model, the Reynolds number for both of them should be equal.

$$\therefore \frac{V_p \times L_p}{\nu_p} = \frac{V_m \times L_m}{\nu_m}$$

$$\begin{aligned}\text{or, } V_p &= \frac{\nu_p}{\nu_m} \times \frac{L_m}{L_p} \times V_m \\ &= \frac{0.012 \times 10^{-4}}{0.018 \times 10^{-4}} \times \frac{3}{300} \times 30 = 0.2 \text{ m/s}\end{aligned}$$

Resistance = Mass $\times$ acceleration

$$= \rho L^3 \times \frac{V}{t} = \rho L^2 \times \frac{V}{1} \times \frac{L}{t} = \rho L^2 V^2$$

$$\text{Then, } \frac{R_p}{R_m} = \frac{F_p}{F_m} = \frac{(\rho L^2 V^2)_p}{(\rho L^2 V^2)_m}$$

$$\begin{aligned}
 &= \frac{\rho_p}{\rho_m} \times \left(\frac{L_p}{L_m} \right)^2 \times \left(\frac{V_p}{V_m} \right)^2 \\
 &= \frac{1030}{1.24} \times \left(\frac{300}{3} \right)^2 \times \left(\frac{0.2}{30} \right)^2 = 369.17
 \end{aligned}$$

$$\therefore R_p = R_m \times 369.17 = 60 \times 369.17 = \mathbf{22150.2 \text{ N (Ans.)}}$$

7.11. FROUDE MODEL LAW

When the gravitational force can be considered to be the only predominant force which controls the motion in addition to the inertia force, the similarity of the flow in any two such systems can be established if the Froude's number for both the systems is the same. This is known as **Froude Model Law**. Some of the phenomena for which the Froude model law can be sufficient criterion for dynamic similarity to be established in the model and the prototype are:

- (i) Free surface flows such as flow over spillways, sluices etc.;
- (ii) Flow of jet from an orifice or nozzle;
- (iii) Where waves are likely to be formed on the surface;
- (iv) Where fluids of different mass densities flow over one another.

Let,

V_m = Velocity of fluid in model,

L_m = Length (or linear dimension) of the model,

g_m = Acceleration due to gravity (at a place where model is tested),

and V_p , L_p and g_p are the corresponding values of the velocity, length and acceleration due to gravity for the prototype.

Then according to Froude model law:

$$\begin{aligned}
 (Fr)_m &= (Fr)_p \\
 \text{or, } \frac{V_m}{\sqrt{g_m L_m}} &= \frac{V_p}{\sqrt{g_p L_p}} \quad \dots(7.22)
 \end{aligned}$$

As the value of g at the site of model testing will be practically the same as at the site of the proposed prototype, therefore, $g_m = g_p$ and the eqn. (7.22) becomes:

$$\frac{V_m}{\sqrt{L_m}} = \frac{V_p}{\sqrt{L_p}} \quad \dots(7.23)$$

$$\text{or, } \frac{V_p}{V_m} = \sqrt{\frac{L_p}{L_m}} = \sqrt{L_r} \quad \left(\because \frac{L_p}{L_m} = L_r \right)$$

where

L_r = Scale ratio for length.

$$\frac{V_p}{V_m} = V_r = \text{Scale ratio for velocity}$$

$$\therefore \frac{V_p}{V_m} = V_r = \sqrt{L_r} \quad \dots(7.24)$$

(i) Time scale ratio, T_r :

$$\text{We know, } T_m = \frac{L_m}{V_m} \text{ and } T_p = \frac{L_p}{V_p}$$

$$\begin{aligned}\therefore T_r &= \frac{T_p}{T_m} = \frac{L_p / V_p}{L_m / V_m} = \frac{L_p}{L_m} \times \frac{V_m}{V_p} \\ &= L_r \times \frac{1}{\sqrt{L_r}} = \sqrt{L_r}\end{aligned}$$

$$\text{i.e. } T_r = \sqrt{L_r} \quad \dots(7.25)$$

(ii) Acceleration scale ratio, a_r :

$$\text{We know, } a = \frac{V}{t}$$

$$\therefore a_r = \frac{a_p}{a_m} = \frac{V_p / T_p}{V_m / T_m} = \frac{V_p}{V_m} \times \frac{T_m}{T_p} = \sqrt{L_r} \times \frac{1}{\sqrt{L_r}} = 1$$

$$\text{i.e. } a_r = 1 \quad \dots(7.26)$$

(iii) Discharge scale ratio, Q_r :

$$\text{The discharge, } Q = A.V = L^2 \times \frac{L}{T} = \frac{L^3}{T}$$

$$\begin{aligned}\therefore \text{Discharge ratio, } Q_r &= \frac{Q_p}{Q_m} = \frac{(L^3 / T)_p}{(L^3 / T)_m} = \left(\frac{L_p}{L_m} \right)^3 \times \left(\frac{T_m}{T_p} \right) \\ &= L_r^3 \times \frac{1}{\sqrt{L_r}} = L_r^{2.5} \quad \dots(7.27)\end{aligned}$$

(iv) Force scale ratio, F_r :

The force may be expressed from the Newton's second law,

$$F = ma = \rho L^3 \times \frac{V}{T} = \rho L^2 \times \frac{L}{T} \times V = \rho L^2 V^2$$

$$\therefore \text{Force ratio, } F_r = \frac{F_p}{F_m} = \frac{\rho_p L_p^2 V_p^2}{\rho_m L_m^2 V_m^2} = \rho_r L_r^2 V_r^2$$

$$\text{If the same fluid is used in the model, then } \rho_r \left(= \frac{\rho_p}{\rho_m} \right) = 1$$

$$\begin{aligned}\therefore F_r &= L_r^2 \times (\sqrt{L_r})^2 = L_r^3 \\ &\quad \left(\because V_r = \sqrt{L_r} \right) \quad \dots(7.28)\end{aligned}$$

(v) Pressure intensity scale ratio, p_r :

$$\text{We know, } p = \frac{F}{A} = \frac{\rho L^2 V^2}{L^2} = \rho V^2$$

$$\therefore \text{Pressure ratio, } p_r = \frac{p_p}{p_m} = \frac{\rho_p V_p^2}{\rho_m V_m^2}$$

For the same fluid

$$\rho_p = \rho_m$$

$$\therefore p_r = \frac{V_p^2}{V_m^2} = (\sqrt{L_r})^2 = L_r \quad \dots(7.29)$$

(vi) *Energy or work done scale ratio E_r* :

$$\text{Energy} = \text{Force} \times \text{distance}$$

$$\therefore E = F \times L$$

$$\text{and, the energy ratio, } E_r = \frac{E_p}{E_m} = \frac{F_p}{F_m} \times \frac{L_p}{L_m} = F_r \cdot L_r$$

$$\text{or, } E_r = L_r^3 \cdot L_r = L_r^4 \quad \dots(7.30)$$

(vii) *The momentum or impulse scale ratio M_r* :

The momentum or impulse

$$= mV = \rho L^3 V$$

$$\begin{aligned} \text{and, the momentum ratio} &= \frac{M_p}{M_m} = \frac{\rho_p L_p^3 V_p}{\rho_m L_m^3 V_m} = \rho_r \cdot L_r^3 \cdot V_r \\ &= L_r^3 \sqrt{L_r} = L_r^{7/2} \end{aligned} \quad \dots(7.31)$$

$$(\because \rho_r = 1)$$

(viii) *Torque scale ratio, T_r^** :

The torque is given by the product of force and its perpendicular distance from the centre of rotation,

$$T^* = F \cdot L$$

$$\begin{aligned} \text{and the torque ratio, } T_r^* &= \frac{T_p^*}{T_m^*} = \frac{F_p}{F_m} \times \frac{L_p}{L_m} = F_r \cdot L_r \\ &= L_r^3 \cdot L_r = L_r^4 \end{aligned} \quad \dots(7.32)$$

(ix) *Power scale ratio, P_r* :

The power being the time rate of doing work, is given by: $P = \frac{F \times L}{T}$

$$\begin{aligned} \text{and, the power ratio, } P_r &= \frac{P_p}{P_m} = \frac{(F_p \times L_p) / T_p}{(F_m \times L_m) / T_m} = \frac{F_p}{F_m} \times \frac{L_p}{L_m} \times \frac{T_m}{T_p} \\ &= F_r \cdot L_r \cdot \frac{1}{T_r} = L_r^3 \cdot L_r \cdot \frac{1}{\sqrt{L_r}} = L_r^{3.5} \\ \text{i.e. } P_r &= L_r^{3.5} \end{aligned} \quad \dots(7.33)$$

Example 7.30. (a) *With Froude's number as the criterion of dynamic similarity for a certain flow situation, work out the scale factors for velocity, time, discharge, acceleration, force, work and power in terms of the scale factor for length.*

(b) *A geometrically similar model of spillway is to be laid to a scale of 1 in 50. Calculate the velocity ratio, discharge ratio and acceleration ratio.* [MDU, Haryana]

Solution. (a) Refer to Article 7.11.

$$(b) \quad \text{Scale ratio, } L_r = \frac{L_p}{L_m} = 50$$

$$\text{Velocity ratio [Eqn. (7.24)], } V_r = \frac{V_p}{V_m} = \sqrt{L_r} = \sqrt{50} = 7.07 \text{ (Ans.)}$$

Discharge ratio [Eqn. (7.27)], $Q_r = \frac{Q_p}{Q_m} = (L_r)^{2.5} = (50)^{2.5} = \mathbf{17677.6 \text{ (Ans.)}}$

Acceleration ratio, $a_r = \frac{a_p}{a_m} = \mathbf{1 \text{ (Ans.)}}$

Example 7.31. In the model test of a spillway the discharge and velocity of flow over the model were $2.5 \text{ m}^3/\text{s}$ and 1.5 m/s respectively. Calculate the velocity and discharge over the prototype which is 36 times the model size. [PTU]

Solution. Discharge over the model, $Q_m = 2.5 \text{ m}^3/\text{s}$

Velocity of flow over the model, $V_m = 1.5 \text{ m/s}$

Scale ratio (linear), $L_r = 36$

Velocity over the prototype, V_p :

Using Froude model law [Eqn. (7.24)], we have

$$\frac{V_p}{V_m} = \sqrt{L_r} = \sqrt{36} = 6$$

$\therefore V_p = 1.5 \times 6 = \mathbf{9 \text{ m/s (Ans.)}}$

Discharge over the prototype, Q_p :

Again, using Froude model law [Eqn. (7.27)], we have

$$\frac{Q_p}{Q_m} = (L_r)^{2.5} = (36)^{2.5} = 7776$$

$\therefore Q_p = 2.5 \times 7776 = \mathbf{19440 \text{ m}^3/\text{s (Ans.)}}$

Example 7.32. In a geometrically similar model of spillway the discharge per metre length is $0.2 \text{ m}^3/\text{s}$. If the scale of the model is $\frac{1}{36}$, find the discharge per metre run of the prototype.

Solution. Discharge per metre length for model = $0.2 \text{ m}^3/\text{s}$

Scale ratio (linear), $L_r = 36$

Discharge per metre run for the prototype, Q_p :

According to Froude law:

$$\frac{Q_p}{Q_m} = L_r^{2.5} \quad [\text{Eqn. (7.27)}]$$

Discharge ratio per metre length is given as:

$$\begin{aligned} \frac{q_p}{q_m} &= \frac{Q_p/L_p}{Q_m/L_m} = \frac{Q_p}{Q_m} \times \frac{L_m}{L_p} \\ &= L_r^{2.5} \times \frac{1}{L_r} = L_r^{1.5} \end{aligned}$$

$\therefore \frac{q_p}{q_m} = (36)^{1.5} = 216$

and, $q_p = 0.2 \times 216 = \mathbf{43.2 \text{ m}^3/\text{s (Ans.)}}$

Example 7.33. The force required to tow a 1: 30 scale model of a motor boat in a lake at a speed of 2 m/s is 0.5 N . Assuming that the viscous resistance due to water and air is negligible in comparison with the wave resistance, calculate the corresponding speed of the prototype for

dynamically similar conditions. What would be the force required to propel the prototype at that velocity in the same lake? [GATE Exam.]

Solution. Linear scale ratio, $L_r = \frac{L_p}{L_m} = 30$

Speed of the model, $V_m = 2 \text{ m/s}$

Force required to tow the model, $F_m = 0.5 \text{ N}$

Speed of the prototype, V_p :

Since the wave resistance is the dominant force in comparison with the viscous resistance, therefore, the dynamic similarity will be attained when the Froude numbers in model and prototype are equal.

$$\therefore \left(\frac{V}{\sqrt{Lg}} \right)_m = \left(\frac{V}{\sqrt{Lg}} \right)_p$$

$$\text{or, } \frac{V_m}{\sqrt{L_m g_m}} = \frac{V_p}{\sqrt{L_p g_p}}$$

$$\text{or, } \frac{V_p}{V_m} = \sqrt{\frac{L_p}{L_m}} = \sqrt{30} \quad (\because g_m = g_p)$$

$$\therefore V_p = 2 \times \sqrt{30} = \mathbf{10.95 \text{ m/s (Ans.)}}$$

Force required to propel the prototype, F_p :

$$\text{We know, } F_r = \frac{F_p}{F_m} = \frac{\rho_p L_p^2 V_p^2}{\rho_m L_m^2 V_m^2} \quad (\because F = \rho L^2 V^2)$$

$$\begin{aligned} \text{or, } \frac{F_p}{F_m} &= \left(\frac{L_p}{L_m} \right)^2 \times \left(\frac{V_p}{V_m} \right)^2 \quad [\because \rho_m = \rho_p, \text{ fluid being same}] \\ &= (30)^2 \times (\sqrt{30})^2 = 30^3 \end{aligned}$$

$$\therefore F_p = F_m \times 30^3 = 0.5 \times 30^3 = \mathbf{13500 \text{ N (Ans.)}}$$

Example 7.34. A spillway model is to be built to a geometrically similar scale of $\frac{1}{50}$ across a flume of 600 mm width. The prototype is 15 metres high and the maximum head on it is expected to be 1.5 metres.

- (i) What height of model and what head on model should be used?
- (ii) If flow over the model for a particular head is 12 litres/ second, what flow per metre length of prototype is expected?
- (iii) If the negative pressure in the model is 200 mm, what is the negative pressure in prototype? Is it practicable? [PTU]

Solution. Linear scale ratio, $L_r = \frac{L_p}{L_m} = 50$

Width of model, $B_m = 600 \text{ mm} = 0.6 \text{ m}$

Flow over model, $Q_m = 12 \text{ litres/sec.}$

Pressure in model, $h_m = -200 \text{ mm of water} = -0.2 \text{ m of water}$

Height of prototype, $H_p = 15 \text{ m}$

Head on prototype, $h_p = 1.5 \text{ m}$

(i) Height of model, H_m :

Head on model, h_m :

$$\text{Linear scale ratio, } L_r = \frac{H_p}{H_m} = \frac{h_p}{h_m} = 50$$

$$\therefore \text{Height of model, } H_m = \frac{H_p}{50} = \frac{15}{50} = \mathbf{0.3 \text{ m (Ans.)}}$$

$$\text{Head on model, } h_m = \frac{h_p}{50} = \frac{1.5}{50} = \mathbf{0.03 \text{ m (Ans.)}}$$

(ii) Discharge per metre length of prototype, q_p :

$$\text{Width of prototype, } B_p = B_m \times L_r = 0.6 \times 50 = 30 \text{ m}$$

$$\text{Now, discharge ratio [Eqn. 7.27], } \frac{Q_p}{Q_m} = L_r^{2.5} = (50)^{2.5} = 17677.66$$

$$\therefore Q_p = Q_m \times 17677.66 = 12 \times 17677.66 = 212132 \text{ litres/sec.}$$

Discharge per metre length of prototype

$$= \frac{Q_p}{L_p} = \frac{Q_p}{B_p} = \frac{212132}{30} = \mathbf{7071 \text{ litres/sec. (Ans.)}}$$

(iii) Negative pressure head in prototype:

Negative pressure head in prototype,

$$h_p = h_m \times L_r = -0.2 \times 50 = \mathbf{-10 \text{ m (Ans.)}}$$

Since the cavitation limits the maximum negative pressure head to 7.5 m, therefore, a negative head of 10 m is **not practicable. (Ans.)**

Example 7.35. The performance of a spillway of an irrigation project is to be studied by means of a model constructed to a scale of 1 : 9, neglecting the viscous and surface tension effects, determine:

(i) Rate of flow in model for a prototype discharge of 1200 m³/s;

(ii) The dissipation of energy in the prototype hydraulic jump, if the jump in the model dissipates 0.25 kW.

Solution. Linear scale ratio, $L_r = \frac{L_p}{L_m} = 9$

$$\text{Rate of flow in the prototype, } Q_p = 1200 \text{ m}^3/\text{s}$$

$$\text{Dissipation of energy in the model, } P_m = 0.250 \text{ kW}$$

(i) Rate of flow in the model, Q_m :

For dynamic similarity between the model and its prototype the Froude's numbers must be equal.

$$\therefore \left(\frac{V}{\sqrt{Lg}} \right)_p = \left(\frac{V}{\sqrt{Lg}} \right)_m$$

$$\therefore \frac{V_p}{V_m} = \sqrt{\frac{L_p}{L_m}} \quad (\because g_p = g_m)$$

$$\text{The discharge ratio, } Q_r = \frac{Q_p}{Q_m} = \frac{L_p^2 \times V_p}{L_m^2 \times V_m} = \left(\frac{L_p}{L_m} \right)^2 \times \frac{V_p}{V_m}$$

$$= (L_r)^2 \times \sqrt{L_r} = (L_r)^{5/2}$$

$$\therefore Q_m = \frac{Q_p}{(L_r)^{5/2}} = \frac{1200}{(9)^{5/2}} = 4.938 \text{ m}^3/\text{s} \text{ (Ans.)}$$

(ii) The dissipation of energy in the prototype hydraulic jump, P_p :

Power = wQH (where, H = head of fluid)

$$\therefore \text{Power ratio, } P_r = \frac{P_p}{P_m} = \frac{w_p Q_p H_p}{w_m Q_m H_m} = (L_r)^{5/2} \times L_r = (L_r)^{7/2} \quad (\because w_p = w_m)$$

$\therefore$ Power dissipated in the prototype,

$$P_p = P_m \times (L_r)^{7/2} = 0.25 \times (9)^{7/2} = 546.75 \text{ kW (Ans.)}$$

Example 7.36. The characteristics of the spillway are to be studied by means of a geometrically similar model constructed to the scale ratio of 1 : 10.

- (i) If the maximum rate of flow in the prototype is 28.3 cusecs, what will be the corresponding flow in model?
- (ii) If the measured velocity in the model at a point on the spillway is 2.4 m/s, what will be the corresponding velocity in prototype?
- (iii) If the hydraulic jump at the foot of the model is 50 mm high, what will be the height of jump in prototype?
- (iv) If the energy dissipated per second in the model is 3.5 Nm, what energy is dissipated per second in the prototype? (Anna University)

Solution. Given: Scale ratio, $\frac{L_p}{L_m} = 10$; $Q_p = 28.3 \text{ m}^3/\text{s}$; $V_m = 2.4 \text{ m/s}$;

$$H_m = 50 \text{ mm}; E_m = 3.5 \text{ Nm.}$$

(i) Flow in the model, Q_m :

$$\frac{Q_p}{Q_m} = (L_r)^{2.5} = (10)^{2.5} = 316.22$$

$$\therefore Q_m = \frac{Q_p}{316.22} = \frac{28.3}{316.22} = 0.0895 \text{ m}^3/\text{s} \text{ (Ans.)}$$

(ii) Velocity of flow in prototype, V_p :

$$\frac{V_p}{V_m} = \sqrt{L_r} = \sqrt{10} = 3.162$$

$$\therefore V_p = V_m \times 3.162 = 2.4 \times 3.162 = 7.589 \text{ m/s (Ans.)}$$

(iii) Height of jump in prototype, H_p :

$$\frac{H_p}{H_m} = L_r = 10$$

$$\therefore H_p = H_m \times 10 = 50 \times 10 = 500 \text{ mm (Ans.)}$$

(iv) Energy dissipated per second in the prototype, E_p :

$$\frac{E_p}{E_m} = (L_r)^{3.5} = (10)^{3.5} = 3162.28$$

$$\therefore E_p = E_m \times 3162.28 = 3.5 \times 3162.28 = 11067.98 \text{ Nm (Ans.)}$$

Example 7.37. A 1 : 64 model is constructed of an open channel in concrete which has Manning's $N = 0.014$. Find the value of N for the model. (Delhi University)

Solution. Given: $L_r = 64; N_p = 0.014$.

Value of N of the model, N_m :

The Manning's formula is given by :

$$V = \frac{1}{N} (R)^{2/3} (S)^{1/2}$$

where,

R = Hydraulic radius (or hydraulic mean depth) in metres;

S = Slope of the bed of the channel.

Now for the model, the Manning's formula becomes:

$$V_m = \frac{1}{N_m} (R_m)^{2/3} (S_m)^{1/2} \quad \dots(i)$$

and, for the prototype, the Manning's formula is written as:

$$V_p = \frac{1}{N_p} (R_p)^{2/3} (S_p)^{1/2} \quad \dots(ii)$$

Dividing (ii) by (i), we get:

$$\frac{V_p}{V_m} = \frac{\frac{1}{N_p} (R_p)^{2/3} (S_p)^{1/2}}{\frac{1}{N_m} (R_m)^{2/3} (S_m)^{1/2}} = \frac{N_m}{N_p} \times \left(\frac{R_p}{R_m} \right)^{2/3} \left(\frac{S_p}{S_m} \right)^{1/2} \quad \dots(iii)$$

For dynamic similarity, Froude model law is used.

We know that: $\frac{V_p}{V_m} = \sqrt{L_r} \quad \dots[\text{Eqn. (17.24)}]$

or, $\frac{V_p}{V_m} = \sqrt{64} = 8$

Also, $\frac{R_p}{R_m} = L_r$ and $\frac{S_p}{S_m} = 1$ as S_p and S_m are dimensionless.

Substituting the values in (iii), we get:

$$8 = \frac{N_m}{N_p} \times (L_r)^{2/3} \times 1 = \frac{N_m}{0.014} \times (64)^{2/3}$$

$$\therefore N_m = \frac{8 \times 0.014}{(64)^{2/3}} = \mathbf{0.007 \text{ (Ans.)}}$$

Example 7.38. A 7.2 m high and 15 m long spillway discharges 94 m³/s discharge under a head of 2.03. If 1: 9 scale model of this spillway is to be constructed, determine model dimensions, head over spillway model and the model discharge. If model experiences a force of 7500 N, determine force on the prototype. (Panjab University)

Solution. Given: Height, $h_p = 7.2$ m; $L_p = 15$ m; $Q_p = 94$ m³/s, head $H_p = 2.0$ m;
 $L_r = 9$; $F_m = 7500$ N

(i) Model dimensions (h_m, L_m):

$$\frac{h_p}{h_m} = \frac{L_p}{L_m} = L_r = 9$$

$$\therefore h_m = \frac{h_p}{9} = \frac{7.2}{9} = \mathbf{0.8 \text{ m (Ans.)}}$$

and,
$$L_m = \frac{L_p}{9} = \frac{15}{9} = 1.67 \text{ m (Ans.)}$$

(ii) Head over the model, H_m :

$$\frac{H_p}{H_m} = L_r = 9$$

$$\therefore H_m = \frac{H_p}{9} = \frac{2.0}{9} = 0.222 \text{ m (Ans.)}$$

(iii) Discharge through the model, Q_m :

$$\frac{Q_p}{Q_m} = (L_r)^{2.5} = (9)^{2.5} = 243$$

$$\therefore Q_m = \frac{Q_p}{243} = \frac{94}{243} = 0.387 \text{ m}^3/\text{s (Ans.)}$$

(iv) Force on the prototype F_p :

$$\frac{F_p}{F_m} = (L_r)^3 = (9)^3 = 729$$

$$\therefore F_p = F_m \times 729 = 7500 \times 729 = 5467500 \text{ N (Ans.)}$$

Example 7.39. A model of rectangular pier 1.5 m wide and 4.5 m long in the river is built to a scale of 1: 25. The average depth of water in the river is 3 m. The model was tested in a laboratory, where the velocity of flow was maintained constant at 0.6 m/s. It was observed that the force acting on the model was 3.6 N and the height of the standing wave was 30 mm. Determine the following for the prototype:

- (i) The corresponding speed,
- (ii) The force acting,
- (iii) The height of the standing wave at nose, and
- (iv) The co-efficient of drag resistance.

[Roorkee University]

Solution. Linear scale ratio, $L_r = \frac{L_p}{L_m} = 25$

Velocity of flow in the model, $V_m = 0.6 \text{ m/s}$

Force acting on the model, $F_m = 3.6 \text{ N}$

Height of standing wave in the model, $H_m = 30 \text{ mm}$

(i) The corresponding speed in the prototype V_p :

As the flow in a river is a *free surface flow affected by gravity*, the dynamic similarity between the model and its prototype will be achieved by equating the Froude's number.

$$\therefore \frac{V_p}{\sqrt{L_p g_p}} = \frac{V_m}{\sqrt{L_m g_m}}$$

$$\text{or, } \frac{V_p}{V_m} = \sqrt{\frac{L_p}{L_m}} = \sqrt{25} = 5 \quad (\because g_p = g_m)$$

$$\therefore V_p = V_m \times 5 = 0.6 \times 5 = 3 \text{ m/s (Ans.)}$$

(ii) The force acting on the prototype, F_p :

$$\text{Force} = \text{Mass} \times \text{acceleration} = \rho L^3 \times \frac{V}{T} = \rho L^3 \times \frac{V}{(L/V)}$$

$$= \rho L^3 \times \frac{V^2}{L} = \rho L^2 V^2$$

$$\text{Force ratio, } F_r = \frac{F_p}{F_m} = \frac{\rho_p}{\rho_m} \times \frac{L_p^2}{L_m^2} \times \frac{V_p^2}{V_m^2} = \rho_r \cdot L_r^2 \cdot V_r^2$$

$$= L_r^2 \times (\sqrt{L_r})^2 = L_r^3$$

$$\therefore F_p = F_m \times L_r^3 = 3.6 \times (25)^3 = \mathbf{56250 \text{ N (Ans.)}}$$

[$\because \rho_r = 1$, fluid being same in model and prototype]

(iii) The height of the standing wave in the prototype, H_p :

$$\frac{H_p}{H_m} = L_r = 25$$

$$\therefore H_p = H_m \times 25 = 30 \times 25 = \mathbf{750 \text{ mm (Ans.)}}$$

(iv) The co-efficient of drag resistance:

The co-efficient of drag resistance is defined by:

$$F = C_D \cdot \rho A \frac{V^2}{2}$$

$$\therefore C_D = \frac{F}{\frac{1}{2} \times \rho A V^2}$$

$$\text{or, } (C_D)_p = \frac{F_p}{\frac{1}{2} \times \rho_p A_p V_p^2}$$

where,

F_p = Force acting on the prototype (= 56250 N),

ρ_p = Density of water (= 1000 kg/m³),

A_p = Width of the pier $\times$ depth of water in the river
= $1.5 \times 3 = 4.5 \text{ m}^2$, and

V_p = Velocity of flow in the prototype (= 3 m/s).

$$\therefore (C_D)_p = \frac{56250}{\frac{1}{2} \times 1000 \times 4.5 \times 3^2} = \mathbf{2.777 \text{ (Ans.)}}$$

The drag co-efficient will be *same* for model and prototype.

Example 7.40. A 1 : 40 model of an ocean tanker is dragged through fresh water at 2 m/s with a total measured drag of 117.7 N. The skin (frictional) drag co-efficeint 'f' for model and prototype are 0.3 and 0.02 respectively in the equation $R_f = f AV^2$. The wetted surface area of the model is 25 m². Taking the densities for the prototype and the model as 1030 kg/m³ and 1000 kgN/m³ respectively, determine:

(i) The total drag on the prototype;

(ii) Power required to drive the prototype.

Solution. Given:

Linear scale ratio, $L_r = 40$

Velocity of model, $V_m = 2$ m/s

Total drag of model, $R_m = 117.7$ N

Wetted area of model, $A_m = 25$ m²

Co-efficient of friction for model, $f_m = 0.3$

Co-efficient of friction for prototype, $f_p = 0.02$

(i) Total drag on the prototype, R_p :

Frictional drag on model, $(R_f)_m = f_m A_m V_m^2 = 0.3 \times 25 \times 2^2 = 30$ N

$\therefore$ Wave drag on model, $(R_w)_m = R_m - (R_f)_m = 117.7 - 30 = 87.7$ N

The wave drag for the prototype can be evaluated by satisfying the following condition for dynamic similarity:

$$\left(\frac{R_w}{\rho L^2 V^2} \right)_p = \left(\frac{R_w}{\rho L^2 V^2} \right)_m$$

$$\therefore (R_w)_p = \frac{\rho_p}{\rho_m} \times \left(\frac{L_p}{L_m} \right)^2 \times \frac{V_p^2}{V_m^2} \times (R_w)_m$$

$$\text{But,} \quad \left(\frac{V_p}{V_m} \right)^2 = \frac{L_p}{L_m}$$

$$\left[\begin{array}{l} \therefore \frac{V_p}{\sqrt{L_p g_p}} = \frac{V_m}{\sqrt{L_m g_m}} \quad \dots \text{Froude's number} \\ \text{or } \frac{V_p}{V_m} = \sqrt{\frac{L_p}{L_m}} \text{ or } \left(\frac{V_p}{V_m} \right)^2 = \frac{L_p}{L_m} \quad (\because g_p = g_m) \end{array} \right]$$

$$\therefore (R_w)_p = \frac{\rho_p}{\rho_m} \times \left(\frac{L_p}{L_m} \right)^3 \times (R_w)_m$$

$$= \frac{1030}{1000} \times (40)^3 \times 87.7 \quad \left[\because \frac{L_p}{L_m} = L_r = 40 \right]$$

$$= 5781184 \text{ N}$$

The frictional drag on the prototype is given by:

$$(R_f)_p = f_p A_p V_p^2 \quad \dots (i)$$

where, V_p is velocity of prototype and is obtained from Froude model law as given below:

$$\frac{V_m}{\sqrt{L_m g_m}} = \frac{V_p}{\sqrt{L_p g_p}} \text{ or } \frac{V_m}{\sqrt{L_m}} = \frac{V_p}{\sqrt{L_p}} \quad (\because g_m = g_p)$$

$$\therefore V_p = V_m \times \sqrt{\frac{L_p}{L_m}} = 2 \times \sqrt{40} = 12.65 \text{ m/s}$$

$$\text{Also,} \quad \frac{A_p}{A_m} = L_r^2 = 40^2 = 1600$$

$$\therefore A_p = A_m \times 1600 = 25 \times 1600 = 40000 \text{ m}^2$$

Substituting these values in (i), we get:

$$(R_f)_p = 0.02 \times 40000 \times (12.65)^2 = 128018 \text{ N}$$

Total drag on the prototype,

$$\begin{aligned} R_p &= (R_w)_m + (R_f)_p = 5781184 + 128018 \\ &= 5909202 \text{ N or } \mathbf{5909.2 \text{ kN (Ans.)}} \end{aligned}$$

(ii) Power required to drive the prototype, P:

Power required = Total drag on prototype $\times$ velocity of prototype

$$\begin{aligned} \text{i.e. } P &= (R_f)_p \times V_p \\ &= 5909.2 \times 12.65 \text{ kW} \\ &= \mathbf{74751.4 \text{ kW (Ans.)}} \end{aligned}$$

7.12. EULER MODEL LAW

In a fluid system where *pressure forces alone are the controlling forces in addition to the inertia force*, the dynamic similarity is obtained by equating the Euler number for both the model and its prototype. This is known as **Euler model law**. According to this law:

$$(Eu)_{\text{model}} = (Eu)_{\text{prototype}} \quad \dots(7.34)$$

If, V_m = Velocity of fluid in model,
 p_m = Pressure of fluid in model,
 ρ_m = Density of fluid in model,

and V_p, p_p, ρ_p are the corresponding values in prototype, then by substituting these values in eqn. (7.34), we get:

$$\frac{V_m}{\sqrt{p_m/\rho_m}} = \frac{V_p}{\sqrt{p_p/\rho_p}} \quad \dots(7.35)$$

when, $\rho_m = \rho_p$ (i.e. same fluid in model and prototype) the above equation becomes:

$$\frac{V_m}{\sqrt{p_m}} = \frac{V_p}{\sqrt{p_p}} \quad \dots(7.36)$$

This law is applied in the following flow problems:

- (i) Enclosed fluid system where the turbulence is fully developed so that viscous forces are negligible and also the forces of gravity and surface tension are entirely absent;
- (ii) Where the phenomenon of cavitation occurs.

Example 7.41. In an aeroplane model of size $\frac{1}{10}$ of its prototype the pressure drop is 7.5 kN/m^2 . The model is tested in water. Find the corresponding pressure drop in the prototype.
 Take: Density of air = 1.24 kg/m^3 ; Density of water = 1000 kg/m^3 ;
 Viscosity of air = 0.00018 poise ; Viscosity of water = 0.01 poise .

Solution. Linear scale ratio, $L_r = 40$; Pressure drop in model, $(\Delta p)_m = 7.5 \text{ kN/m}^2$;
 Density of water, $\rho_m = 1000 \text{ kg/m}^3$; Viscosity of water, $\mu_m = 0.01 \text{ poise}$;
 Density of air, $\rho_p = 1.24 \text{ kg/m}^3$; Viscosity of air, $\mu_p = 0.00018 \text{ poise}$.

Pressure drop in the prototype $(\Delta P)_p$:

Since in the problem pressure and viscous forces are involved, therefore, for dynamic similarity between the model and prototype, Euler's number and Reynolds number should be considered.

Making Reynolds number equal, we get:

$$\left(\frac{\rho VL}{\mu} \right)_m = \left(\frac{\rho VL}{\mu} \right)_p$$

$$\text{or,} \quad \frac{\rho_m V_m L_m}{\mu_m} = \frac{\rho_p V_p L_p}{\mu_p}$$

$$\text{or,} \quad \frac{V_m}{V_p} = \frac{\rho_p}{\rho_m} \times \frac{L_p}{L_m} \times \frac{\mu_m}{\mu_p}$$

Substituting the values, we have:

$$\frac{V_m}{V_p} = \frac{1.24}{1000} \times 40 \times \frac{0.01}{0.00018} = 2.755$$

Now making Euler's number equal, we get:

$$\left(\frac{V}{\sqrt{\Delta p / \rho}} \right)_m = \left(\frac{V}{\sqrt{\Delta p / \rho}} \right)_p$$

$$\text{or,} \quad \frac{V_m}{\sqrt{(\Delta p)_m / \rho_m}} = \frac{V_p}{\sqrt{(\Delta p)_p / \rho_p}}$$

$$\text{or,} \quad \frac{V_m}{V_p} = \frac{\sqrt{(\Delta p)_m / \rho_m}}{\sqrt{(\Delta p)_p / \rho_p}} = \sqrt{\frac{(\Delta p)_m}{(\Delta p)_p}} \times \sqrt{\frac{\rho_p}{\rho_m}}$$

$$\text{or,} \quad \sqrt{\frac{(\Delta p)_m}{(\Delta p)_p}} = \frac{V_m}{V_p} \times \sqrt{\frac{\rho_m}{\rho_p}} = 2.755 \times \sqrt{\frac{1000}{1.24}} = 78.24$$

$$\therefore \quad \frac{(\Delta p)_m}{(\Delta p)_p} = (78.24)^2 = 6121.5$$

$$\text{or,} \quad (\Delta p)_p = \frac{(\Delta p)_m}{6121.5} = \frac{7.5 \times 1000}{6121.5} \text{ N/m}^2 = 1.225 \text{ N/m}^2$$

Hence, pressure drop in the prototype = **1.225 N/m² (Ans.)**

7.13. WEBER MODEL LAW

In a fluid system where *surface tension effects predominate in addition to inertia force*, the dynamic similarity is obtained by equating the Weber number for the model and its prototype, which is known as **Weber model law**. According to this law:

$$(We)_{\text{model}} = (We)_{\text{prototype}}$$

$$\text{where, } We \text{ (i.e. Weber number)} = \frac{V}{\sqrt{\sigma / \rho L}}$$

$$\begin{aligned} \text{If,} \quad & V_m = \text{Velocity of fluid in model,} \\ & \sigma_m = \text{Surface tension force in model,} \\ & \rho_m = \text{Density of fluid in model,} \\ & L_m = \text{Length of surface in model,} \end{aligned}$$

and $V_p, \sigma_p, \rho_p, L_p$ are the corresponding values of fluid in the prototype, then according to Weber model law, we have:

$$\frac{V_m}{\sqrt{\sigma_m / \rho_m L_m}} = \frac{V_p}{\sqrt{\sigma_p / \rho_p L_p}} \quad \dots(7.37)$$

Weber model law is applied in the following flow situations:

- (i) Flow over weirs involving very low heads;
- (ii) Very thin sheet of liquid flowing over a surface;
- (iii) Capillary waves in channels;
- (iv) Capillary rise in narrow passages;
- (v) Capillary movement of water in soil.

7.14. MACH MODEL LAW

When in any fluid system *only the forces resulting from elastic compression are significant in addition to inertial force*, then the dynamic similarity between the model and its prototype may be achieved by equating the Mach numbers, which is known as **Mach model law**. According to this law:

$$(M)_{\text{model}} = (M)_{\text{prototype}}$$

where, $M = \text{Mach number} = \frac{V}{\sqrt{K / \rho}}$

or,
$$\frac{V_m}{\sqrt{K_m / \rho_m}} = \frac{V_p}{\sqrt{K_p / \rho_p}} \quad \dots(7.38)$$

where, $V_m =$ Velocity of fluid in model,

$K_m =$ Elastic stress for model,

$\rho_m =$ Density of fluid in model,

and V_p, K_p, ρ_p are the corresponding values for prototype.

The similitude based on Mach model law finds application in the following:

- (i) Aerodynamic testing;
- (ii) Phenomena involving velocities exceeding the speed of sound;
- (iii) Hydraulic model testing for the cases of unsteady flow, especially water hammer problems;
and
- (iv) Under-water testing of torpedoes.

Example 7.42. (Model testing of ships). A 1: 20 model of a naval ship having a submerged area of 5 m^2 and length 8 metres has a total drag of 20 N when towed through water at a velocity of 1.5 m/s. Calculate the total drag on the prototype when moving at the corresponding speed. Use the relation $F_f = \frac{1}{2} C_f \rho A V^2$ for calculating the skin resistance. The value of C_f is given by, $C_f = 0.0735 / (Re)^{1.5}$.

Take kinematic viscosity of water (or sea water) as 0.01 stoke and the specific weight of water (or sea water) as 9810 N/m^3 .

Solution. Linear scale ratio, $L_r = 20$

Submerged area of the model, $A_m = 5 \text{ m}^2$

Length of the model, $L_m = 8 \text{ m}$

Total drag of model, $R_m = 20 \text{ N}$

Velocity of model, $V_m = 1.5 \text{ m/s}$

Let A_p, L_p, R_p, V_p be the corresponding values for the prototype.

Kinematic viscosity of sea water,

$$\nu_m = \nu_p = 0.01 \text{ stoke} = 0.01 \text{ cm}^2/\text{s} = 0.01 \times 10^{-4} \text{ m}^2/\text{s}$$

Total drag on the prototype:

(i) Analysis of model:

$$\begin{aligned} \text{Reynolds number, } Re &= \left(\frac{VL}{\nu} \right)_m = \frac{V_m L_m}{\nu_m} \\ &= \frac{1.5 \times 8}{0.01 \times 10^{-4}} = 12 \times 10^6 \end{aligned}$$

$$\text{Also, } C_f = \frac{0.0735}{(Re)^{1/5}} \quad \dots(\text{Given})$$

$$\therefore C_{fm} = \frac{0.0735}{(12 \times 10^6)^{1/5}} = 0.00282$$

Frictional or skin resistance of the model is given by:

$$F_f = \frac{1}{2} C_f \rho A V^2 \quad \dots(\text{Given})$$

$$\begin{aligned} \therefore (F_f)_m &= (R_f)_m = \frac{1}{2} C_{fm} \rho_m A_m V_m^2 = \frac{1}{2} \times 0.00282 \times \left(\frac{9810}{9.81} \right) \times 5 \times 1.5^2 \\ &= 15.86 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Also, } R_m &= (R_w)_m + (R_f)_m \\ &\left[\begin{array}{l} \text{where, } (R_w)_m = \text{Wave resistance experienced by the model, and} \\ (R_f)_m = \text{Frictional or skin resistance experienced by the model} \end{array} \right] \end{aligned}$$

$$\therefore 20 = (R_w)_m + 15.86$$

$$\text{or } (R_w)_m = 20 - 15.86 = 4.14 \text{ N}$$

(ii) Analysis of prototype:

Since resistance to wave formation exists, condition will be dynamically similar if the *Froude's number* are equal.

$$\text{i.e., } \frac{V_p}{\sqrt{L_p g_p}} = \frac{V_m}{\sqrt{L_m g_m}}$$

$$\text{or, } V_p = V_m \times \sqrt{\frac{L_p}{L_m}} = 1.5 \times \sqrt{20} = 6.708 \text{ m/s} \quad (\because g_p = g_m)$$

$$\text{Reynolds number, } Re = \left(\frac{VL}{\nu} \right)_p = \frac{V_p L_p}{\nu_p}$$

$$= \frac{6.708 \times 160}{0.01 \times 10^{-4}} = 10.73 \times 10^8$$

$$\left[\begin{array}{l} \because \frac{L_p}{L_m} = 20 \\ \therefore L_p = 8 \times 20 = 160 \text{ m} \end{array} \right]$$

$$\therefore C_{fp} = \frac{0.0735}{(10.73 \times 10^8)^{1/5}} = 0.001148$$

∴ Frictional or skin resistance of the prototype is given by:

$$(F_f)_p = (R_f)_p = \frac{1}{2} C_{fp} \rho_p A_p V_p^2 = \frac{1}{2} \times 0.001148 \times \frac{9810}{9.81} \times 2000 \times 6.708^2$$

$$= 51657 \text{ N} \quad \left[\because A_p = A_m \times L_r^2 = 5 \times 20^2 \right]$$

$$= 2000 \text{ m}^2$$

Wave resistance R_w for the prototype can be evaluated by satisfying the following condition for dynamic similarity:

$$\left(\frac{R_w}{\rho L^2 V^2} \right)_p = \left(\frac{R_w}{\rho L^2 V^2} \right)_m$$

$$\therefore (R_w)_p = \frac{\rho_p}{\rho_m} \times \left(\frac{L_p}{L_m} \right)^2 \times \frac{V_p^2}{V_m^2} \times (R_w)_m$$

But, $\rho_p = \rho_m$... (Given)

$$\text{and, } \left(\frac{V_p}{V_m} \right)^2 = \frac{L_p}{L_m} \quad \dots [\text{From the equivalent of Froude's number}]$$

$$\therefore (R_w)_p = 1 \times \left(\frac{L_p}{L_m} \right)^3 \times (R_w)_m = (20)^3 \times 4.14 = 33120 \text{ N}$$

Hence, total drag on the prototype,

$$R_p = (R_w)_p + (R_f)_p = 33120 + 51657 = \mathbf{84777 \text{ N (Ans.)}}$$

7.15. TYPES OF MODELS

The hydraulic models, in general, are *classified* into the following two broad categories:

1. Undistorted models;
2. Distorted models.

7.15.1. Undistorted models

An **undistorted model** is one which is geometrically similar to its prototype. The conditions of similitude are *completely satisfied* for such models, hence the results obtained from the model tests are easily used to predict the performance of prototype body. In such models the design and construction of the model and the interpretation of the model results are *simpler*.

7.15.2. Distorted models

A **distorted model** is one which is not geometrically similar to its prototype. In such a model *different scale ratios* for the linear dimensions are adopted. For example in case of a wide and shallow river it is not possible to obtain the same horizontal and vertical scale ratios, however, if these ratios are taken to be same then because of the small depth of flow the vertical dimensions of the model will become too less in comparison to its horizontal length. Thus in distorted models the *plan form is geometrically similar to that of prototype but the cross-section is distorted*.

A distorted model may have the following *distortions*:

- (i) **Geometrical distortion.** The geometric distortion can be either of dimensions or that of *configuration*.
 - The *distortion of dimensions* results due to adoption of different scales for vertical and horizontal dimensions. This type of distortion is frequently adopted in *river models* where a *different scale ratio for depth is adopted*.

- When the general configuration of the model does not bear a resemblance with it prototype, it becomes a *distortion of configuration*.
- (ii) *Material distortion*. It involves the use of different materials for the model and prototype.
- (iii) *Distortion of hydraulic quantities*. This type of distortion occurs due to certain uncontrollable hydraulic quantities such as velocity, discharge etc.

Typical examples for which distorted models are required to be prepared are:

- (i) Rivers,
- (ii) Dams across very wide rivers,
- (iii) Harbours etc.

Reasons for adopting distorted models:

The distorted models are adopted for:

- Maintaining accuracy in vertical dimensions;
- Maintaining turbulent flow;
- Accommodating the available facilities (such as money, water supply, space etc.);
- Obtaining suitable roughness condition;
- Obtaining suitable bed material and its adequate movement.

Merits and Demerits of Distorted Models:

Merits:

1. Due to increase in the depth of fluid or height of waves accurate measurements are made possible.
2. The surface tension can be reduced to minimum.
3. Model size can be sufficiently reduced, thereby *its operation is simplified* and also the *cost is lowered considerably*.
4. Sufficient tractive force can be developed to move the bed material of the model.
5. The Reynolds number of flow in a model can be increased that will yield better results.

Demerits:

1. The pressure and velocity distributions are not truly reproduced.
2. A model wave may differ in type and possibly in action from that of the prototype.
3. Slopes of river beds, earth cuts and dikes cannot be truly reproduced.
4. It is difficult to extrapolate and interpolate results obtained from distorted models.
5. The observer experiences an unfavourable psychological effect.

Note. Although the distorted models entail a number of demerits, yet by incorporating judicious allowances in the interpretation of the results obtained from such models, useful information can be obtained (which is otherwise not possible).

7.16. SCALE EFFECT IN MODELS

By model testing it is *not possible* to predict the *exact behaviour of the prototype*. The behaviour of the prototype as predicted by two models with different scale ratios is generally *not the same*. Such a *discrepancy or difference in the prediction of behaviour of the prototype is termed as “scale effect”*. The magnitude of the scale effect is affected by the type of the problem and the scale ratio used for the performance of experiments on models. The scale effect can be *positive* and *negative* and when applied to the results accordingly, the corrected results then hold good for prototype.

Since it is impossible to have complete similitude satisfying all the requirements, therefore, the discrepancy due to scale effect creeps in. During investigation of models only two or three forces which are *predominant are considered* and the effect of the rest of the forces which are not significant is *neglected*. *These forces which are not so important cause small but varying effect on the model depending upon the scale of the model*, due to which scale effect creeps in. Sometimes the *imperfect simulation* in different models causes the discrepancy due to scale effect.

In ship models both viscous and gravity forces have to be considered, however it is *not possible* to satisfy Reynolds and Froude's numbers *simultaneously*. Usually the models are tested satisfying only Froude's law, then the results so obtained are corrected by applying the scale effect due to viscosity.

In the models of weirs and orifices with very small scale ratio the scale effect is due to surface tension forces. The surface tension forces which are insignificant in prototype become quite important in small scale models with head less than 15 mm.

Scale effect can be known by testing a number of models using different scale ratios, and the exact behaviour of the prototype can then be predicted.

7.17. LIMITATIONS OF HYDRAULIC SIMILITUDE

Model investigation, although very important and valuable, may not provide ready solution to all problems. It has the following **limitations**:

1. The model results, in general, are qualitative but not quantitative.
2. As compared to the cost of analytical work, models are usually expensive.
3. Transferring results to the prototype requires some judgment (the scale effect should be allowed for).
4. The selection of size of a model is a matter of experience.

HIGHLIGHTS

1. *Dimensional analysis* is a mathematical technique which makes use of the study of the dimensions for solving several engineering problems.
2. *Dimensional homogeneity* states that every term in an equation when reduced to fundamental dimensions must contain identical powers of each dimension.
3. Dimensional analysis is generally performed by two methods namely Rayleigh's method and Buckingham's π -theorem.
4. *Rayleigh's method* of dimensional analysis is useful when the number of variables is small. In this method, the equations are expressed in exponential forms. The dimensionless parameters are obtained by first evaluating the exponents so that equation is dimensionally homogeneous, and then by grouping together the variables with like powers to form dimensionless parameters.
5. *Buckingham's π -theorem* states as follows:
 "If there are n variables (dependent and independent variables) in a dimensionally homogeneous equation and if these variables contain m fundamental dimensions (such as M , L , T etc.), the variables are arranged into $(n - m)$ dimensionless terms. These dimensionless terms are called π -terms".
6. *Model analysis* is an experimental method of finding solutions of complex flow problems.

7. A model is a small scale replica of the actual machine or structure. For the model to yield useful information about the characteristics of the prototype, the model must have *geometric*, *kinematic*, and *dynamic similarities* with the prototype.

For *geometric similarity*, the ratio of all linear dimensions of the model and of the prototype should be equal.

Kinematic similarity means the similarity of motion between model and prototype.

Dynamic similarity means the similarity of forces between the model and prototype.

8. *Reynolds number* is defined as the ratio of inertia force and viscous force of a flowing fluid. It is given by

$$Re = \frac{\rho V L}{\mu} = \frac{V L}{\nu} = \frac{V \times d}{\nu} \text{ for pipe flow}$$

where,

V = velocity of flow,

d = diameter of the pipe, and

ν = kinematic viscosity of the fluid.

9. *Froude number* is the ratio of the square root-of inertia and gravity force and is given by,

$$Fe = \sqrt{\frac{F_i}{F_g}} = \frac{V}{\sqrt{Lg}}$$

10. *Euler number* is the ratio of the square root of inertia force and pressure force and is given by,

$$Eu = \sqrt{\frac{F_i}{F_p}} = \frac{V}{\sqrt{p/\rho}}$$

11. *Weber number* is the ratio of the square root of inertia force to the surface tension force and is given by,

$$We = \sqrt{\frac{F_i}{F_s}} = \sqrt{\frac{V}{\sigma/\rho L}}$$

12. *Mach number* is the ratio of the square root of inertia force and elastic force and is given by,

$$M = \sqrt{\frac{F_i}{F_e}} = \frac{V}{\sqrt{K/\rho}} = \frac{V}{C}$$

13. The laws on which the models are designed for dynamic similarity are called *model* or *similarity laws*. The model laws are:

- | | |
|-------------------------|---------------------------|
| (i) Reynolds model law, | (ii) Froude model law, |
| (iii) Euler model law, | (iv) Weber model law, and |
| (v) Mach model law. | |

14. The drag experienced by a ship model (a partially submerged body) is obtained by *Froude's method*.

15. The hydraulic models are classified as:

- | | |
|-----------------------------|------------------------|
| (i) Undistorted models, and | (ii) Distorted models. |
|-----------------------------|------------------------|

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer:

- Dimensional analysis is used to
 - test the dimensional homogeneity of any equation of fluid motion
 - derive rational formulae for a flow phenomenon
 - derive equations expressed in terms of non-dimensional parameters
 - all of the above.
- Which of the following is an advantage of dimensional analysis?
 - It expresses the functional relationship between the variables in dimensionless terms
 - In hydraulic model studies it reduces the number of variables involved in a physical phenomenon, generally by three
 - By the use of dimensional analysis design curves can be developed from experimental data or direct solution of the problem
 - all of the above.
- A dimensionally homogeneous equation is applicable to
 - C.G.S. system only
 - F.P.S. system only
 - M.K.S. and SI systems
 - all systems of units.
- In which of the following methods of dimensional analysis, a functional relationship of some variables is expressed in the form of an exponential equation, which must be dimensionally homogeneous?
 - Buckingham's π -method
 - Rayleigh's method
 - Bridgman's method
 - Matrix-tensor method.
- In dimensional analysis the Buckingham's π -theorem is widely used and expresses the resulting equation in terms of
 - the repeating variables
 - geometric, kinematic and dynamic variables
 - $(n - m)$ dimensionless parameters
 - n dimensionless parameters.
- To apply Buckingham's π -theorem, m repeating variables are selected from amongst the n variables influencing the phenomenon. The repeating variables are selected such that they
 - belong to kinematic and dynamic category of variables
 - must always contain the dependent variables
 - in combination contain each of the m fundamental dimensions involved in the problem.
 - none of the above.
- In order that results obtained in model studies correctly represent the behaviour of the prototype, which of the following similarities must be ensured between the model and the prototype?
 - Geometric similarity
 - Kinematic similarity
 - Dynamic similarity
 - All of the above.
- Dynamic similarity between the model and prototype is the
 - similarity of motion
 - similarity of lengths
 - similarity of forces
 - None of the above.
- is equal to the product of shear stress due to viscosity and surface area of flow.
 - Viscous force
 - Inertia force
 - Pressure force
 - Gravity force.
- Kinematic similarity between model and prototype is the
 - similarity of discharge
 - similarity of shape
 - similarity of streamline pattern
 - none of the above.
- is the ratio of the inertia force to the viscous force.
 - Froude's number
 - Weber's number
 - Reynolds number
 - Mach's number.
- is the square root of the ratio of the inertia force to the pressure force.
 - Reynolds number
 - Mach's number
 - Euler's number
 - Froude's number.
- Euler number is important in which of the following flow situations?
 - Discharge through orifices, mouthpieces and sluices
 - Pressure rise due to sudden closure of valves
 - Flow through pipes
 - All of the above.

14. Mach's number is defined as the square root of the ratio of the
 - (a) inertia force to the pressure force
 - (b) inertia force to the surface tension force
 - (c) inertia force to the elastic force
 - (d) none of the above.
15. is important in compressible fluid flow problems at high velocities, such as high velocity flow in pipes or motion of high-speed projectiles and missiles.
 - (a) Euler's number
 - (b) Mach's number
 - (c) Reynolds number
 - (d) Froude's number.
16. Distorted models are required to be prepared for which of the following?
 - (a) Rivers
 - (b) Dams across very wide rivers
 - (c) Harbours
 - (d) All of the above.
17. The scale effect in models can be
 - (a) positive only
 - (b) negative only
 - (c) both positive and negative
 - (d) none of the above.

ANSWERS

- | | | | | | |
|---------|---------|---------|---------|----------|---------|
| 1. (d) | 2. (d) | 3. (d) | 4. (b) | 5. (c) | 6. (c) |
| 7. (d) | 8. (c) | 9. (a) | 10. (c) | 11. (c) | 12. (c) |
| 13. (d) | 14. (c) | 15. (b) | 16. (d) | 17. (c). | |

THEORETICAL QUESTIONS

1. What is dimensional analysis?
2. What are the uses of dimensional analysis?
3. What do you mean by fundamental units and derived units? Give examples.
4. Explain the term dimensional homogeneity.
5. Enumerate the applications of dimensional homogeneity.
6. What are the various methods of dimensional analysis to obtain a functional relationship between various parameters influencing a physical phenomenon.
7. Describe Rayleigh's method for dimensional analysis.
8. Describe Buckingham's method or π -theorem to formulate a dimensionally homogeneous equation between the various physical quantities effecting a certain phenomenon.
9. What are repeating variables? How are these selected by dimensional analysis?
10. What is model analysis?
11. What are the advantages of model testing?
12. What are applications of model testing?
13. What is meant by geometric, kinematic and dynamic similarities? Are these similarities truly attainable? If not why?
14. What are dimensionless numbers?
15. Define the following dimensionless numbers and state their significance for fluid flow problems.
 - (i) Reynolds number,
 - (ii) Froude's number, and
 - (iii) Mach's number.
16. Enumerate different laws on which models are designed for dynamic similarity. Where are they used?
17. How are hydraulic models classified?
18. What are distorted models? What are the reasons of constructing such models for rivers?
19. What are the merits and demerits of distorted models?
20. Explain scale effect in model testing. How is it found?
21. What are the limitations of hydraulic similitude?

UNSOLVED EXAMPLES

1. Show that the resistance R to the motion of a sphere of diameter D moving with a uniform velocity V through a real fluid of density ρ and viscosity μ is given by:

$$F = \rho D^2 V^2 f\left(\frac{\mu}{VD\rho}\right) \quad [\text{BHU}]$$

2. By dimensional analysis, show that the power P developed by a hydraulic turbine is given by:

$$P = \rho N^3 D^5 f\left(\frac{N^2 D^2}{gH}\right)$$

where, ρ = Mass density of liquid,
 N = Rotational speed,
 D = Diameter of runner,
 H = Working head, and
 g = Acceleration due to gravity.

3. The resistance R , to the motion of a completely submerged body depends upon the length of the body L , velocity of flow V , mass density of fluid ρ and kinematic viscosity of fluid ν . By dimensional analysis prove that

$$R = \rho V^2 L^2 \phi\left(\frac{VL}{\nu}\right)$$

4. Prove that velocity through an orifice can be expressed as

$$V = \sqrt{2gH} \phi\left[\frac{D}{H}, \frac{\mu}{\rho VH}, \frac{\sigma}{\rho V^2 H}\right]$$

where, H = Head causing flow,
 D = Diameter of orifice,
 μ = Co-efficient of viscosity,
 ρ = Mass density, and
 σ = Surface tension.

5. Prove that the shear stress (τ) in a fluid flowing through a pipe can be expressed by the equation:

$$\tau = \rho V^2 \phi\left[\frac{\mu}{\rho DV}, \frac{k}{D}\right]$$

where, D = Diameter of pipe,
 ρ = Mass density,
 V = Velocity,
 μ = Viscosity, and
 k = Height of roughness projection.

6. The pressure difference Δp in a pipe of diameter D and length L due to viscous flow depends on the velocity V , viscosity μ and density ρ . Using Buckingham's theorem, obtain an expression for Δp .

[Hint: Choose D , V and μ as repeating variables, μ has been chosen repeating variable (instead of ρ) since the flow is viscous.]

$$\left[\text{Ans. } \Delta p = \frac{\mu V}{D} \times \frac{L}{D} \phi\left(\frac{\rho DV}{\mu}\right) \right]$$

7. The frictional torque T of a disc of diameter D rotating at speed N in a fluid of viscosity μ and density ρ in a turbulent flow is given by:

$$T = D^5 N^2 \rho \phi\left(\frac{\mu}{D^2 N \rho}\right)$$

Prove this by the method of dimensions.

[UPSC]

8. An oil of sp. gr. 0.9 and viscosity 0.03 poise is to be transported at the rate of 3000 litres/sec. through a 1.5 m diameter pipe. Tests were conducted on a 15 cm diameter pipe using water at 20°C. If the viscosity of water at 20°C is 0.01 poise, find:
 (i) Velocity of flow in the model;
 (ii) Rate of flow in the model.

[Ans. (i) 5.09 m/s, (ii) 80.9 litres/s].

9. A model of a submarine of scale 1/40 is tested in a wind tunnel. Find the speed of air in wind tunnel if the speed of submarine in sea water is 15 m/s. Also find the ratio of the resistance between the model and its prototype. Take the values of kinematic viscosities for sea water and air as 0.012 stokes and 0.016 stokes respectively. The weight densities of sea water and air are given as 10.1 kN/m³ and 0.0122 kN/m³ respectively. [Ans. 800 m/s, $\frac{F_m}{F_p} = 0.0021$]

10. In 1 : 30 model of a spillway, the velocity and discharge are 1.5 m/sec. and 2.0 m³/sec. Find the corresponding velocity and discharge in the prototype. [Ans. 8.2 m/s, 9859 m³/s]
 11. A 120 m long surface vessel is to be tested by a 3 m long model. If the vessel travels at 10 m/s, at what speed must model be towed for dynamic similarity between model and prototype? If the drag of the model is 9.37 N, what prototype drag is to be expected?

[Ans. 1.58 m/s, 596.4 kN]

12. In an open channel water is flowing at a depth of 1.5 m with a velocity of 7.5 m/s. At a particular location, a hydraulic jump is formed and the depth increases to 2.2 m. Another channel is built where a similar jump is formed. If the flow depth in the new dynamically similar channel is 6 m, estimate the flow velocity and the height of jump. [GATE]

[Ans. 15 m/s, 0.8 m]

13. In order to predict the pressure drop in a large air duct a model is constructed with linear dimension $\frac{1}{10}$ th that of the prototype, and water is

used as the test fluid. If water is 1000 times denser than air and has 100 times the viscosity of air, determine the pressure drop in the prototype for the conditions corresponding to a pressure drop of 70 kPa in the model.

[Ans. 0.07 kPa]

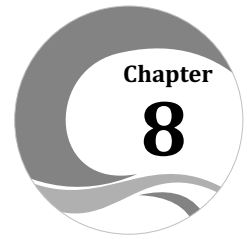
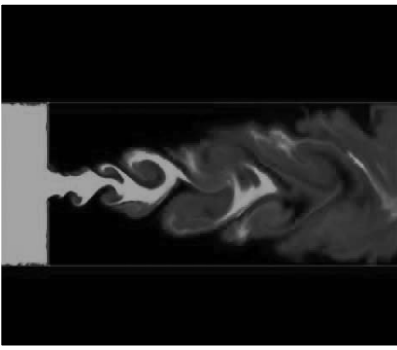
14. If model prototype ratio is 1:75, show that the ratio of discharges per unit width of spillway is given by $\left(\frac{1}{75}\right)^{3/2}$. [AMIE]
15. In an aeroplane model of size $\frac{1}{50}$ of its prototype the pressure drop is 4 bar. The model is tested in water. Find the corresponding pressure drop in the prototype. Take specific height of air = 0.0124 kN/m^3 . The viscosity of water is 0.01 poise while the viscosity of air is 0.00018 poise. [Ans. 0.00042 bar]
16. An air duct is to be modelled to a scale of 1:20 and tested with water which is 50 times viscous and 800 times more dense than air. When tested under dynamically similar conditions, the pressure drop between two sections in the model is 235 kPa. What is the corresponding pressure drop in the prototype? [Ans. 189.3 Pa]
17. Explain what is meant by dynamic similarity between a flow system and its model.

If the dynamic behaviour of the flow system for an overflow spillway is governed by the Froude law of similarity, what would be the discharge Q_p when the (model) scale ratio is L_r ? Show that the same result would be obtained if the weir formula is used instead and it is assumed that the prototype and the model spillways have the same co-efficient of discharge.

[UPSC Civil Services (IAS) Exam.]

18. A model of rectangular pier 1.2 m wide and 4 m long in the river is built to a scale of 1:20. The average depth of water in the river is 3 m. The model was tested in a laboratory, where the velocity of flow was maintained constant at 0.6 m/s. It was observed that the force acting on the model was 4 N and the height of the standing wave was 35 mm. Make calculations of speed, force, height of standing wave and the co-efficient of drag resistance for the prototype. Assume that the flow in the model and prototype are insensitive to changes in Reynolds number.

[Ans. 2.68 m/s, 32 kN, 700 mm, 2.428]



FLOW THROUGH ORIFICES AND MOUTHPIECES

- 8.1. Introduction
- 8.2. Classification of orifices.
- 8.3. Flow through an orifice.
- 8.4. Hydraulic co-efficient of resistance
- 8.5. Experimental determination of co-efficients—determination of co-efficient of velocity—determination of co-efficient of discharge
- 8.6. Discharge through a large rectangular orifice.
- 8.7. Discharge through fully submerged orifice
- 8.8. Discharge through partially submerged orifice
- 8.9. Time required for emptying a tank through an orifice at its bottom
- 8.10. Time required for emptying a hemispherical tank
- 8.11. Time required for emptying a circular horizontal tank
- 8.12. Classification of mouthpieces
- 8.13. Discharge through an external mouthpiece
- 8.14. Discharge through a convergent-divergent mouthpiece
- 8.15. Discharge through an internal mouthpiece (or Re-entrant or Borda's mouthpiece) mouthpiece running free-mouthpiece running full

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

8.1. INTRODUCTION

An **orifice** is an opening in the wall or base of a vessel through which the fluid flows. The top edge of the orifice is always below the free surface (If the free surface is below the top edge of the orifice, becomes a weir)

A **mouthpiece** is an attachment in the form of a small tube or pipe fixed to the orifice (the length of pipe extension is usually 2 to 3 times the orifice diameter) and is used to increase the amount of discharge.

- Orifices as well as mouthpieces are used to measure the discharge.

8.2. CLASSIFICATION OF ORIFICES

The orifices are *classified* as follows

1. According to size:

- (i) Small orifice (ii) Large orifice.

An orifice is termed *small* when its dimensions are small compared to the head causing flow. The velocity does not vary appreciably from top to the bottom edge of the orifice and is assumed to be uniform.

The orifice is *large* if the dimensions are comparable with the head causing flow. The variation in the velocity from the top to the bottom edge is considerable.

2. According to shape

- (i) Circular orifice (ii) Rectangular orifice
- (iii) Square orifice (iv) Triangular orifice.

3. Shape of upstream edge

- (i) Sharp-edged orifice
- (ii) Bell-mouthed orifice.

4. According to discharge conditions

- (i) Free discharge orifices

- (ii) Drowned or submerged orifices
 - (a) Fully submerged
 - (b) Partially submerged.

Note. An orifice or a mouthpiece is said to be discharging *free* when it discharges into *atmosphere*. It is said to be *submerged* when it discharges into *another liquid*.

8.3. FLOW THROUGH AN ORIFICE

Fig. 8.1 shows a small circular orifice with sharp edge in the side wall of a tank discharging free into the atmosphere. Let the orifice be at a depth H below the free surface. As the fluid flows through the orifice, it contracts and attains a parallel form (*i.e.*, stream lines become parallel) at a distance $\frac{d}{2}$ from the plane of the orifice. The point at which the stream lines first become parallel is termed as **vena-contracta** (the cross-sectional area of the jet at the vena contracta is less than that of orifice). Beyond this section, the jet *diverges* and is attracted in the downward direction by gravity.

Considering points 1 and 2 as shown in Fig. 8.1 and applying Bernoulli's theorem, we have:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2$$

But,

$$\begin{aligned} p_1 &= p_2 = p_a \\ z_1 &= z_2 + H \end{aligned}$$

(p_a = atmospheric pressure)

Further, if the cross-sectional area of the tank is very large, the liquid at point 1 is practically standstill and hence $V_1 = 0$

Thus,
$$\frac{V_2^2}{2g} = H$$

or,
$$V_2 = \sqrt{2gH} \quad \dots (8.1)$$

Equation (8.1) is known as **Torricelli's theorem**.

Note. In the problems of orifices it is convenient to work in terms of *gauge pressures*.

8.4. HYDRAULIC CO-EFFICIENTS

The hydraulic co-efficients (or orifice co-efficients) are enumerated and discussed below :

1. Co-efficient of contraction, C_c
2. Co-efficient of velocity, C_v
3. Co-efficient of discharge, C_d
4. Co-efficient of resistance, C_r

8.4.1. Co-efficient of Contraction (C_c)

The ratio of the area of the jet at vena-contracta to the area of the orifice is known as **Co-efficient of contraction**. It is denoted by C_c .

Let,

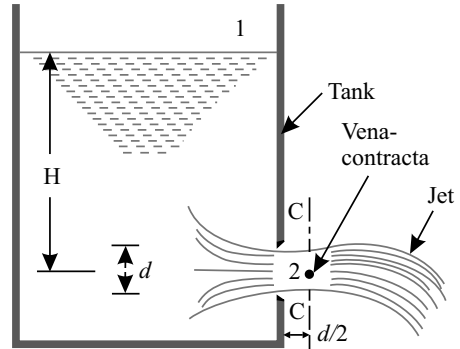
$$\begin{aligned} a_c &= \text{Area of jet at vena contracta, and} \\ a &= \text{Area of orifice.} \end{aligned}$$


Fig. 8.1. Orifice discharging free.

Then,
$$C_c = \frac{a_c}{a} \quad \dots (8.2)$$

The value of C_c varies slightly with the available head of the liquid, size and shape of the orifice; in practice it varies from 0.613 to 0.69 but the average value is taken as 0.64.

8.4.2. Co-efficient of Velocity (C_v)

The ratio of actual velocity (V) of the jet at vena-contracta to the theoretical velocity (V_{th}) is known as **Co-efficient of velocity**. It is denoted by C_v and mathematically, C_v is given as:

$$C_v = \frac{\text{Actual velocity of jet at vena contracta } (V)}{\text{Theoretical velocity } (V_{th})}$$

i.e.,
$$C_v = \frac{V}{\sqrt{2gH}} \quad \dots (8.3)$$

$$\left[\begin{array}{l} \text{where, } V = \text{Actual velocity, and} \\ H = \text{Head under which the fluid flows out of the orifice} \end{array} \right]$$

The value of C_v varies from 0.95 to 0.99, depending upon the shape of orifice and the head of liquid under which the flow takes place. For sharp-edged orifices the value of C_v is taken as 0.98.

8.4.3. Co-efficient of Discharge

The ratio of actual discharge (Q) through an orifice to the theoretical discharge, (Q_{th}) is known as **Co-efficient of discharge**. It is denoted by C_d .

Mathematically,
$$C_d = \frac{\text{Actual discharge } (Q)}{\text{Theoretical discharge } (Q_{th})}$$

$$= \frac{\text{Actual area} \times \text{actual velocity}}{\text{Theoretical area} \times \text{theoretical velocity}}$$

$$= \frac{\text{Actual area}}{\text{Theoretical area}} \times \frac{\text{actual velocity}}{\text{theoretical velocity}}$$

$\therefore C_d = C_c \times C_v \quad \dots (8.4)$

The value of C_d varies from 0.62 to 0.65 depending upon size and the shape of the orifice and the head of liquid under which the flow takes place.

8.4.4. Co-efficient of Resistance (C_r)

The ratio of loss of head (or loss of kinetic energy) in the orifice to the head of water (actual kinetic energy) available at the exit of the orifice is known as **Co-efficient of resistance**. It is denoted by C_r .

Mathematically,
$$C_r = \frac{\text{Loss of head in the orifice}}{\text{Head of water}}$$

The loss of head in the orifice takes place, because the walls of the orifice offer some resistance to the liquid, as it comes out. While solving numerical problems C_r is generally neglected.

Example 8.1. An orifice 50mm in diameter is discharging water under a head of 10 metres. If $C_d = 0.6$ and $C_v = 0.97$, find :

- (i) Actual discharge, and
- (ii) Actual velocity of the jet at vena contracta.

Solution. Diameter of the orifice, $d = 50 \text{ mm} = 0.05 \text{ m}$

$$\therefore \text{Area of the orifice, } a = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times (0.05)^2 = 0.001963 \text{ m}^2$$

$$\text{Head, } H = 10\text{m}; \quad C_d = 0.6; \quad C_v = 0.97$$

(i) **Actual discharge**

$$C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}} = 0.6 \quad \dots(\text{Given})$$

But theoretical discharge = Area of orifice $\times$ theoretical velocity

$$= a \times \sqrt{2gH}$$

$$= 0.001963 \times \sqrt{2 \times 9.81 \times 10}$$

$$= 0.02749 \text{ m}^3/\text{s}$$

$$\therefore \text{Actual discharge} = 0.6 \times 0.02749 = \mathbf{0.01649 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

(ii) **Actual velocity**

$$\text{We know that, } C_v = \frac{\text{actual velocity}}{\text{theoretical velocity}}$$

$$\therefore \text{Actual velocity} = C_v \times \text{theoretical velocity} \\ = 0.97 \times \sqrt{2gH} = 0.97 \times \sqrt{2 \times 9.81 \times 10} = \mathbf{13.58 \text{ m/s (Ans.)}}$$

Example 8.2. The head of water over the centre of an orifice of diameter 30 mm is 1.5m. The actual discharge through the orifice is 2.55 litres/sec. Find the co-efficient of discharge.

Solution. Diameter of the orifice, $d = 30 \text{ mm} = 0.03 \text{ m}$

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.03^2 = 0.0007068 \text{ m}^2$$

$$\text{Head, } H = 1.5 \text{ m}$$

Co-efficient of discharge, C_d

Actual discharge through the orifice, $Q = 2.55 \text{ litres/sec.} = 0.00255 \text{ m}^3/\text{s}$

Theoretical discharge, $Q_{th} = \text{Area of orifice} \times \text{theoretical velocity.}$

$$\text{But theoretical velocity, } V_{th} = \sqrt{2gH} = \sqrt{2 \times 9.81 \times 1.5} = 5.425 \text{ m/s}$$

$$\therefore Q_{th} = a \times V_{th} \\ = 0.0007068 \times 5.425 = 0.004166 \text{ m}^3/\text{s}$$

$$\text{Hence, } C_d = \frac{Q}{Q_{th}} = \frac{0.00255}{0.004166} = \mathbf{0.612 \text{ (Ans.)}}$$

8.4. EXPERIMENTAL DETERMINATION OF HYDRAULIC CO-EFFICIENTS

8.5.1. Determination of Co-efficient of Velocity (C_v).

Fig. 8.2 shows a tank containing water at a constant level, maintained by a constant supply. Let the water flow out of the tank through an orifice, fitted in one side of the tank. Let the section C–C represents the point of vena contracta. Consider a particle of water in the jet at P.

Let, x = Horizontal distance travelled by the particle in time ' t ',

y = Vertical distance between C–C and P,

V = Actual velocity of the jet at vena-contracta, and

H = Constant water head.

Then, horizontal distance, $x = V \times t$... (i)

and, vertical distance, $y = \frac{1}{2} g t^2$... (ii)

From eqn. (i), $t = \frac{x}{V}$

Substituting this value of 't' in eqn. (ii), we get:

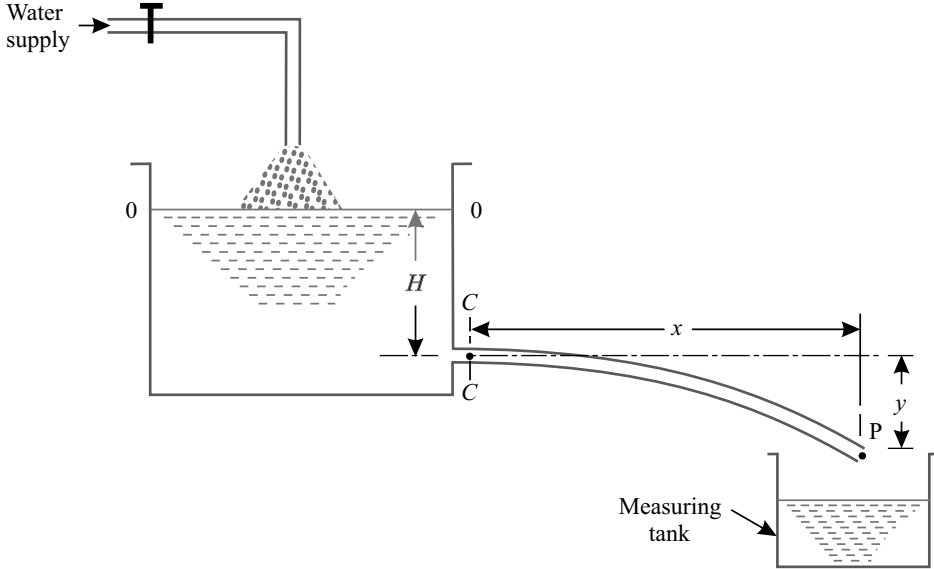


Fig. 8.2. Experiment for hydraulic co-efficients.

$$y = \frac{1}{2} g \times \left(\frac{x}{V} \right)^2 = \frac{gx^2}{2V^2}$$

$$\therefore V^2 = \frac{gx^2}{2y} \text{ or } V = \sqrt{\frac{gx^2}{2y}}$$

But, theoretical velocity, $V_{th} = \sqrt{2gH}$

$\therefore$ Co-efficient of velocity,

$$C_v = \frac{V}{V_{th}} = \frac{\sqrt{\frac{gx^2}{2y}}}{\sqrt{2gH}} = \sqrt{\frac{x^2}{4yH}}$$

$$i.e. \quad C_v = \frac{x}{\sqrt{4yH}} \quad \dots (8.5)$$

8.5.2. Determination of Co-efficient of Discharge (C_d)

The water flowing through the orifice under the constant head H is collected in a measuring tank for a known time ' t '. The rise of water level in the measuring tank is noted down. Then actual discharge through the orifice,

$$Q = \frac{\text{Area of measuring tank} \times \text{rise of water level in the measuring tank}}{\text{Time (t)}}$$

$$\text{Theoretical discharge, } Q_{th} = \text{Area of orifice} \times \sqrt{2gH}$$

$$\text{Hence, } C_d = \frac{Q}{Q_{th}} = \frac{Q}{a \times \sqrt{2gH}} \quad \dots(8.6)$$

8.5.3. Determination of Co-efficient of Contraction (C_c)

The co-efficient of contraction (C_c) can be found from the following relation:

$$C_d = C_c \times C_v$$

$$\therefore C_c = \frac{C_d}{C_v} \quad \dots(8.7)$$

8.5.4. Loss of head in Orifice Flow

The loss of head through an orifice can be determined by applying the Bernoulli's equation between points O and C (Fig. 8.2).

$$\frac{p_0}{w} + \frac{V_0^2}{2g} + z_0 = \frac{p_C}{w} + \frac{V_C^2}{2g} + z_C + \text{losses}$$

Substituting the proper values, we get:

$$0 + 0 + H = 0 + \frac{V^2}{2g} + 0 + h_f$$

Where, V is the actual flow velocity through the orifice.

$$\therefore h_f = H - \frac{V^2}{2g} = H \left(1 - \frac{V^2}{2gH} \right) \quad \dots[8.8(a)]$$

$$= H(1 - C_v^2)$$

$$\text{or } h_f = \frac{V^2}{2g} \left(\frac{2gH}{V^2} - 1 \right) = \frac{V^2}{2g} \left(\frac{1}{C_v^2} - 1 \right) \quad \dots[8.8(b)]$$

Example 8.3. A vertical sharp-edged orifice 120 mm in diameter is discharging water at the rate of 98.2 litres/sec. under a constant head of 10 metres. A point on the jet, measured from the vena contracta of the jet has co-ordinates 4.5 metres horizontal and 0.54 metre vertical. Find the following for the orifice.

- (i) Co-efficient of velocity, (ii) Co-efficient of discharge, and
(iii) Co-efficient of contraction.

Solution. Diameter of orifice, $d = 120 \text{ mm} = 0.12 \text{ m}$

$$\therefore \text{Area of orifice, } a = \frac{\pi}{4} \times 0.12^2 = 0.01131 \text{ m}^2$$

$$\text{Discharge, } Q = 98.2 \text{ litres/sec.} = \frac{98.2}{1000} = 0.0982 \text{ m}^3/\text{s}$$

$$\text{Head, } H = 10 \text{ m}$$

Horizontal distance of a point on the jet from vena contracta, $x = 4.5 \text{ m}$

$$\text{Vertical distance, } y = 0.54 \text{ m}$$

$$\text{Now theoretical velocity, } V_{th} = \sqrt{2gH} = \sqrt{2 \times 9.81 \times 10} = 14 \text{ m/s}$$

Theoretical discharge,

$$Q_{th} = \text{Area of orifice } (a) \times V_{th} \\ = 0.01131 \times 14 = 0.1583 \text{ m}^3/\text{s}$$

(i) Co-efficient of velocity, C_v :

$$C_v = \frac{x}{\sqrt{4yH}} \\ = \frac{4.5}{\sqrt{4 \times 0.54 \times 10}} = \mathbf{0.968 \text{ (Ans.)}}$$

(ii) Co-efficient of discharge, C_d :

$$C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}} \\ = \frac{0.0982}{0.1583} = \mathbf{0.62 \text{ (Ans.)}}$$

(iii) Co-efficient of contraction, C_c :

$$= \frac{C_d}{C_v} = \frac{0.62}{0.968} = \mathbf{0.64 \text{ (Ans.)}}$$

Example 8.4. A large tank has a sharp edged circular orifice of 930 mm^2 area at a depth of 3 m below constant water level. The jet issues horizontally and in a horizontal distance of 2.4 m , it falls by 0.53 m , the measured discharge is 4.3 lit/s . Derermine coefficients of velocity, contraction and discharge for the orifice.

Solution. Given : Area of the orifice, $a = 930 \text{ mm}^2$; $H = 3 \text{ m}$; $x = 2.4 \text{ m}$; $y = 0.53 \text{ m}$;

$Q = 4.3 \text{ litres/sec.} = 0.0043 \text{ m}^3/\text{s}$

C_v , C_c and C_d :

$$\text{Theoretical velocity, } V = \sqrt{2gH} = \sqrt{2 \times 9.81 \times 3} \\ = 7.67 \text{ m/s}$$

$$\text{Theoretical discharge} = a \times V_{th} \\ = 930 \times 10^{-6} \times 7.67 \\ = 0.00713 \text{ m}^3/\text{s}$$

Co-efficient of velocity,

$$C_v = \frac{x}{\sqrt{4yH}} = \frac{2.4}{\sqrt{4 \times 0.53 \times 3}} \\ = \mathbf{0.952 \text{ (Ans.)}}$$

Co-efficient of discharge,

$$C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}} = \frac{0.0043}{0.00713} \\ = \mathbf{0.603 \text{ (Ans.)}}$$

Co-efficient of contraction,

$$C_c = \frac{C_d}{C_v} = \frac{0.603}{0.952} = \mathbf{0.633 \text{ (Ans.)}}$$

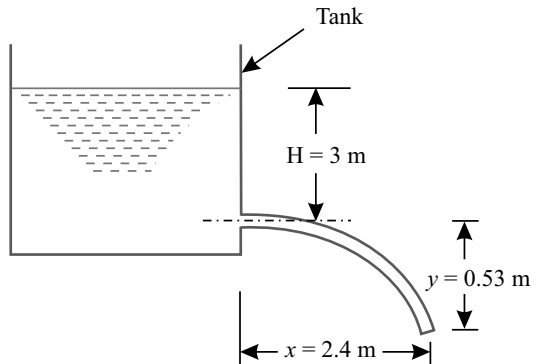


Fig. 8.3

Example 8.5. The head of water over an orifice of diameter 100 mm is 12 m. The water coming out from the orifice is collected in a rectangular tank 2 m × 0.9 m. The rise of water level in this tank is 1.2 m in 30 seconds. Find the co-efficient of discharge.

Solution. Head of water, $H = 12$ m

Diameter of orifice, $d = 100$ mm = 0.1 m

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.1^2 = 0.00785 \text{ m}^2$$

Area of the measuring tank, $A = 2 \times 0.9 = 1.8 \text{ m}^2$

Rise of water level (in $t = 30$ s), $h = 1.2$ m

Co-efficient of discharge, C_d

$$\begin{aligned} \text{Theoretical velocity, } V_{th} &= \sqrt{2gH} \\ &= \sqrt{2 \times 9.81 \times 12} = 15.34 \text{ m/s} \end{aligned}$$

$$\begin{aligned} \therefore \text{Theoretical discharge, } Q_{th} &= a \times V_{th} \\ &= 0.00785 \times 15.34 = 0.1204 \text{ m}^3/\text{s} \end{aligned}$$

$$\text{Actual discharge, } Q = \frac{A \times h}{t} = \frac{1.8 \times 1.2}{30} = 0.072 \text{ m}^3/\text{s}$$

$$\therefore \text{Co-efficient of discharge, } C_d = \frac{Q}{Q_{th}} = \frac{0.072}{0.1204} \approx \mathbf{0.6 \text{ (Ans.)}}$$

Example 8.6. A tank has two identical orifices in one of its vertical sides. The upper orifice is 1.5 m below the water surface and the lower one is 3 m below the water surface as shown in Fig. 8.4. Find the point, at which the two jets will intersect, if the co-efficient of velocity is 0.92 for both the orifices.

Solution. Height of water above the upper orifice, $H_1 = 1.5$ m

Height of water above the lower orifice, $H_2 = 3$ m

Co-efficient of velocity, $C_v = 0.92$

Let, x = Horizontal co-ordinate of the point of intersection A,

y_1 = Vertical co-ordinate of the point of intersection A from the orifice 1, and

y_2 = Vertical co-ordinate of the point of intersection A from the orifice 2.

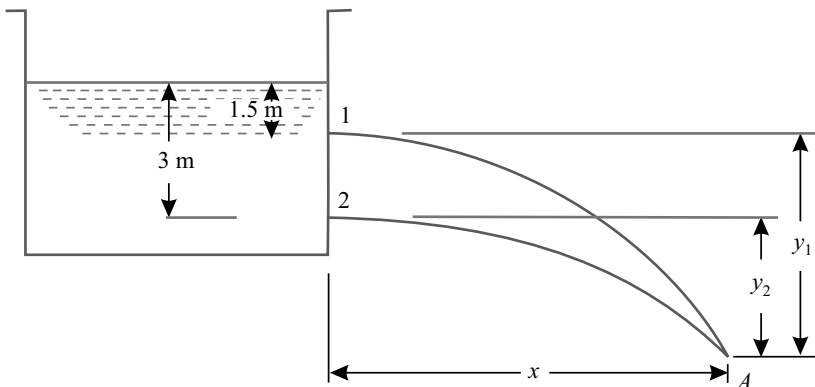


Fig. 8.4

Using the relation,

$$C_v = \sqrt{\frac{x^2}{4yH}}, \text{ with usual notations, we have:}$$

$$C_{v1} = \sqrt{\frac{x^2}{4y_1 \times 1.5}} \quad \dots(i)$$

and,

$$C_{v2} = \sqrt{\frac{x^2}{4y_2 \times 3.0}} \quad \dots(ii)$$

Since the two orifice are identical, therefore equating (i) and (ii), we get:

$$\sqrt{\frac{x^2}{4y_1 \times 1.5}} = \sqrt{\frac{x^2}{4y_2 \times 3.0}}$$

$$\therefore y_1 = 2y_2 \quad \dots(iii)$$

From the geometry of the tank, we know that,

$$y_1 = y_2 + (3 - 1.5) = y_2 + 1.5 \quad \dots(iv)$$

Solving eqns. (iii) and (iv), we get:

$$y_2 = 1.5 \text{ m and } y_1 = 3 \text{ m}$$

Again, using the relation,

$$C_v = \sqrt{\frac{x^2}{4y_1 \times H_1}}, \text{ with usual notations, we get:}$$

$$0.92 = \sqrt{\frac{x^2}{4 \times 3 \times 1.5}} = \sqrt{\frac{x^2}{18}} = \frac{x}{\sqrt{18}}$$

$$\therefore x = 0.92 \times \sqrt{18} = \mathbf{3.9 \text{ m (Ans.)}}$$

Example 8.7. Explain briefly how the co-efficient of velocity of a jet issuing through an orifice can be experimentally determined.

Find an expression for head loss in an orifice flow in terms of co-efficient of velocity and jet velocity.

The head lost in flow through a 50 mm diameter orifice under a certain head is 160 mm of water and the velocity of water in the jet is 7.0 m/s. If the co-efficient of discharge be 0.61, determine:

(i) Head on the orifice causing flow;

(ii) The co-efficient of velocity;

(iii) The diameter of the jet.

[UPSC]

Solution. Diameter of the orifice, $d = 50 \text{ mm} = 0.05 \text{ m}$

Head of water lost in flow, $h_f = 160 \text{ mm} = 0.16 \text{ m}$

Velocity of water in the jet, $V = 7.0 \text{ m/s}$

Co-efficient of discharge, $C_d = 0.61$

(i) Head on the orifice causing flow, H :

Bernoulli's equation between the reservoir surface and vena-contracta (see Fig. 8.1), yields:

$$H = \frac{V^2}{2g} + h_f = \frac{7^2}{2 \times 9.81} + 0.16$$

$$= 2.66 \text{ m (Ans.)}$$

(ii) The Co-efficient of velocity, C_v

$$C_v = \frac{V}{\sqrt{2gH}} = \frac{7}{\sqrt{2 \times 9.81 \times 2.66}}$$

$$= 0.97 \text{ (Ans.)}$$

(iii) The Diameter of the jet, d_j :

$$C_d = C_v \times C_c$$

$$\therefore C_c = \frac{C_d}{C_v} = \frac{0.61}{0.97} = 0.63$$

$$\text{But, } C_c = \frac{(\pi/4) \times d_j^2}{(\pi/4) \times d^2} = \frac{d_j^2}{d^2}$$

$$\therefore 0.63 = \frac{d_j^2}{(0.05)^2}$$

$$\text{or, } d_j^2 = 0.63 \times (0.05)^2 = 0.001575$$

$$\text{or, } d_j = 0.0397 \text{ m or } 39.7 \text{ mm (Ans.)}$$

Example 8.8. A 3 m high tank standing on the ground is kept full of water. There is a small orifice in its vertical side with its centre at depth h metres below the free surface of liquid in the tank. Find the value of h so that the liquid strikes the ground at the maximum distance from the tank. Assuming $C_v = 0.97$, calculate the maximum value of the horizontal distance.

Solution. Height of tank/water, $H = 3 \text{ m}$

Co-efficient of velocity, $C_v = 0.97$

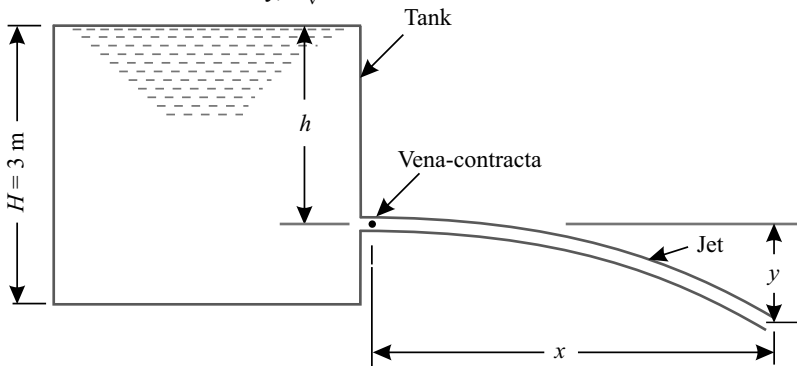


Fig. 8.5

Value of h :

From kinematics of flow,

$$x = Vt \quad \dots(i) \qquad y = \frac{1}{2}gt^2 \quad \dots(ii)$$

where, x, y = Co-ordinates, and

t = The time taken for the liquid particle to travel from the orifice to the ground.

Eliminating ' t ' from (i) and (ii), we get:

$$y = \frac{1}{2}g \times \left(\frac{x}{V}\right)^2$$

or,
$$x^2 = \frac{2V^2 y}{g}$$

Substituting, $V = C_v \sqrt{2gh}$ and $y = (H - h)$, we get:

$$x^2 = \frac{2 \times C_v^2 \times 2gh (H - h)}{g}$$

$$\therefore x = 2 C_v \sqrt{h (H - h)} \quad \dots(iii)$$

Horizontal distance x of the jet trajectory would be maximum when $h (H - h)$ is maximum or when,

$$\frac{d}{dh} [h (H - h)] = 0 \text{ or } \frac{d}{dh} (hH - h^2) = 0$$

or, $H - 2h = 0$

or,
$$h = \frac{H}{2} = \frac{3}{2} = 1.5 \text{ m (Ans.)}$$

Maximum value of horizontal distance, $(x)_{\max}$:

Substituting, $h = 1.5 \text{ m}$ and $C_v = 0.97$ in (iii), we get:

$$\begin{aligned} (x)_{\max} &= 2 \times 0.97 \sqrt{1.5 (3 - 1.5)} \\ &= 2.91 \text{ m (Ans)} \end{aligned}$$

Example 8.9. A tank containing water is provided with a sharp edged circular orifice of 7.5 mm diameter. The height of water in the tank is 1.44 m above the orifice. The jet strikes a wall 1.5 m away and 0.42 m vertically below the centre line of the contracted section of the jet. The actual discharge through the orifice is measured to be 35 litres in 4 minutes. Determine:

- (i) The orifice co-efficients; (ii) The power loss at the orifice.

Solution. Diameter of the orifice, $d = 7.5 \text{ mm} = 0.0075 \text{ m}$

$$\therefore \text{Area of the orifice, } a = \frac{\pi}{4} \times d^2 = \frac{\pi}{4} \times 0.0075^2 = 0.0000442 \text{ m}^2$$

Height of water in the tank above the orifice, $H = 1.44 \text{ m}$

Horizontal distance of a point on the jet from vena-contracta, $x = 1.5 \text{ m}$

Vertical distance, $y = 0.42 \text{ m}$

The actual discharge measured in 4 minutes = 35 litres

$$\text{i.e., } Q = \frac{35}{1000} \times \frac{1}{4 \times 60} = 1.46 \times 10^{-4} \text{ m}^3/\text{s}$$

(i) Orifice co-efficients

$$Q = C_d \cdot a \cdot \sqrt{2gH}$$

or, $1.46 \times 10^{-4} = C_d \times 0.0000442 \times \sqrt{2 \times 9.81 \times 1.44}$

$$\therefore C_d = \frac{1.46 \times 10^{-4}}{0.0000442 \times \sqrt{2 \times 9.81 \times 1.44}} = 0.62 \text{ (Ans.)}$$

From the measurement of jet co-ordinates,

$$\text{Co-efficient of velocity, } C_v = \frac{x}{\sqrt{4yH}} = \frac{1.5}{\sqrt{4 \times 0.42 \times 1.44}} = \mathbf{0.964 \text{ (Ans.)}}$$

The co-efficient of contraction follows from the relation,

$$C_d = C_c \times C_v$$

$$\therefore C_v = \frac{C_d}{C_v} = \frac{0.62}{0.964} = \mathbf{0.64 \text{ (Ans.)}}$$

(ii) The power loss at the orifice:

$$\begin{aligned} \text{Loss of head, } h_f &= H(1 - C_v^2) & [\text{Eqn. 8.8 (a)}] \\ &= 1.44(1 - 0.964^2) = 0.102 \text{ m} \end{aligned}$$

The *loss co-efficient* or the *co-efficient of resistance* prescribes the ratio of loss of head in the orifice to the total head available.

$$\text{Loss co-efficient} = \frac{h_f}{H} = \frac{0.102}{1.44} = \mathbf{0.071 \text{ (Ans.)}}$$

$$\text{Power loss} = wQh_f = 9810 \times 1.46 \times 10^{-4} \times 0.102 = \mathbf{0.146 \text{ W (Ans.)}}$$

Example 8.10. A 100 mm diameter orifice discharge 36 litres per second of water under a constant head of 2.6 m. A flat plate held normal to the jet just downstream from the orifice requires a force of 240 N to resist the impact of the jet. Determine the hydraulic co-efficients.

Solution. Dia. of the orifice, $d = 100 \text{ mm} = 0.1 \text{ m}$

$\therefore$ Area of the orifice,

$$A = \frac{\pi}{4}d^2 = \frac{\pi}{4} \times 0.1^2 = 0.00785 \text{ m}^2$$

Discharge through the orifice,

$$Q = 36 \text{ litres/s} = \frac{36}{1000} = 0.036 \text{ m}^3/\text{s}$$

Constant head of water above the orifice,

$$H = 2.6 \text{ m}$$

Force required to resist the jet, $F = 240 \text{ N}$

Hydraulic co-efficients:

From momentum equation, the force (F) required to hold the plate in position is

$$F = \frac{wQ}{g} V$$

where, $w = \text{Sp. weight of water } (= 9810 \text{ N/m}^3), \text{ and}$

$V = \text{Velocity at the vena-contracta.}$

Substituting the values in the above equation, we get:

$$240 = \frac{9810 \times 0.036}{9.81} \times V$$

$$\therefore V = \frac{240 \times 9.81}{9810 \times 0.036} = 6.66 \text{ m/s}$$

$$\text{Theoretical velocity, } V_{th} = \sqrt{2gH} = \sqrt{2 \times 9.81 \times 2.6} = 7.14 \text{ m/s}$$

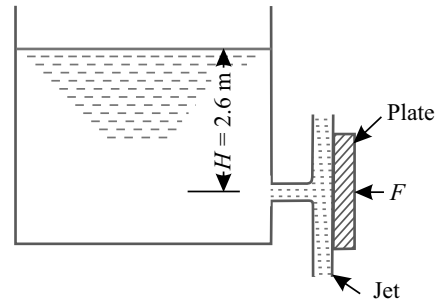


Fig. 8.6

$$\therefore \text{Co-efficient of velocity, } C_v = \frac{V}{V_{th}} = \frac{6.66}{7.14} = \mathbf{0.932 \text{ (Ans.)}}$$

$$\begin{aligned} \text{Theoretical discharge, } Q_{th} &= a \times \sqrt{2gH} = 0.00785 \times \sqrt{2 \times 9.81 \times 2.6} \\ &= 0.056 \text{ m}^3/\text{s} \end{aligned}$$

$$\therefore \text{Co-efficient of discharge, } C_d = \frac{Q}{Q_{th}} = \frac{0.036}{0.056} = \mathbf{0.643 \text{ (Ans.)}}$$

$$\text{Co-efficient of contraction, } C_c = \frac{C_d}{C_v} = \frac{0.643}{0.932} = \mathbf{0.69 \text{ (Ans.)}}$$

Example 8.11. A tank shown in Fig. 8.7 has a nozzle of exit diameter D_1 at a depth H_1 below the free surface. At the side opposite to that of nozzle 1, another nozzle is proposed at a depth $\frac{H_1}{2}$. What should be diameter D_2 in terms of D_1 so that the net horizontal force on the tank is zero? (UPTU)

Solution. Let, V = Velocity, m/s

m = Mass discharge, kg/s, and

F = Reactive force due to rate of change of momentum of the issuing jet.

At jet 1:

$$V_1 = \sqrt{2gH_1}$$

$$m_1 = \rho A_1 V_1 = \rho \times \frac{\pi}{4} D_1^2 \times \sqrt{2gH_1}$$

$$\begin{aligned} \text{and, } F_1 &= m_1 V_1 = \rho \times \frac{\pi}{4} D_1^2 \sqrt{2gH_1} \times \sqrt{2gH_1} \\ &= \rho \frac{\pi}{4} D_1^2 \times 2gH_1 \end{aligned}$$

At jet 2:

$$V_2 = \sqrt{2g \left(\frac{H_1}{2} \right)} = \sqrt{gH_1}; m_2 = \rho A_2 V_2 = \rho \times \frac{\pi}{4} D_2^2 \times \sqrt{gH_1}$$

$$\text{and, } F_2 = m_2 V_2 = \rho \times \frac{\pi}{4} D_2^2 \times \sqrt{gH_1} \times \sqrt{gH_1} = \rho \times \frac{\pi}{4} D_2^2 \times gH_1$$

For the net horizontal force to be zero, F_1 should be numerically equal to F_2 , since both of them act in opposite directions.

$$\therefore \rho \frac{\pi}{4} D_1^2 \cdot 2gH_1 = \rho \times \frac{\pi}{4} D_2^2 gH_1$$

$$\text{or, } 2D_1^2 = D_2^2$$

$$\text{or, } D^2 = D_1 \sqrt{2} = \mathbf{1.414D_1 \text{ (Ans.)}}$$

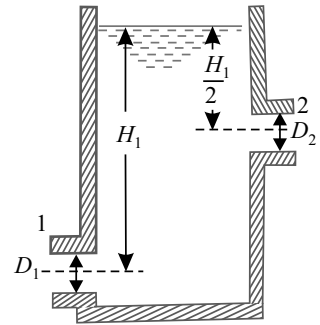


Fig. 8.7

Example 8.12. A closed tank, having an orifice of diameter 20 mm at the bottom of the tank, is partially filled with water upto a height of 2.5 m. The air is pumped into the upper part of the tank. Determine the pressure required for a discharge of 5 litres per second through the orifice, Take discharge co-efficient, $C_d = 0.6$ for the orifice.

Solution. Height of water above orifice, $H = 2.5$ m

Dia. of the orifice, $d = 20$ mm = 0.02 m

$$\therefore \text{Area of the orifice, } A = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times (0.02)^2 = 0.000314 \text{ m}^2$$

Discharge through the orifice, $Q = 5$ litres/sec.

$$= \frac{5}{1000} = 0.005 \text{ m}^3/\text{s}$$

Co-efficient of discharge, $C_d = 0.6$

Pressure required

Let p is the intensity of pressure required above water surface in kN/m^2 .

$$\text{Then, pressure head of air} = \frac{P}{\rho_w} = \frac{P}{981} = 0.102p \text{ metres of water.}$$

If V is the velocity at outlet of orifice, then:

$$V = \sqrt{2g \left(H + \frac{P}{\rho_w} \right)} = \sqrt{2 \times 9.81 (2.5 + 0.102p)}$$

$$\therefore \text{Discharge, } Q = C_d \times a \times V$$

$$\text{or} \quad 0.005 = 0.6 \times 0.000314 \times \sqrt{2 \times 9.81 (2.5 + 0.102p)}$$

$$\sqrt{2 \times 9.81 (2.5 + 0.102p)} = \frac{0.005}{0.6 \times 0.000314} = 26.5$$

Squaring both sides, we get:

$$\text{or, } 2 \times 9.81 (2.5 + 0.102p) = (26.5)^2 = 702.2$$

$$\text{or, } 2.5 + 0.102p = \frac{702.2}{2 \times 9.81} = 35.79$$

$$\therefore p = \frac{35.79 - 2.5}{0.102} = 326.4 \text{ kN/m}^2 \text{ (Ans.)}$$

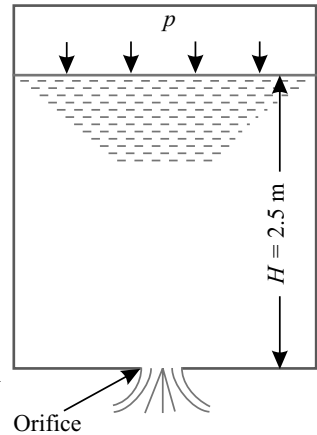


Fig. 8.8

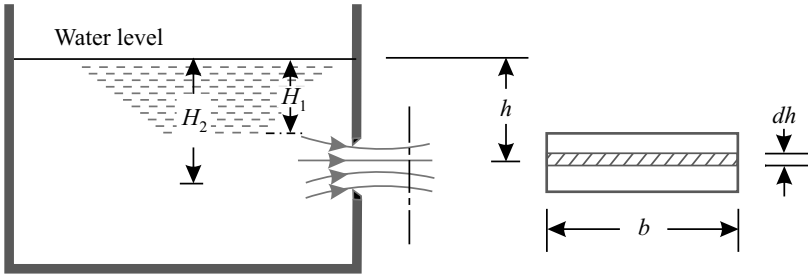
8.6. DISCHARGE THROUGH A LARGE RECTANGULAR ORIFICE

When the available head of a liquid is less than 5 times the height of the orifice, the orifice is called a **large orifice**. In case of a small orifice, the velocity is considered to be constant in the entire cross-section and the discharge can be calculated by the formula $Q = C_d \times a \times \sqrt{2gH}$. But in case of a large orifice, the velocity of a liquid, flowing through the orifice, varies with the available head of the liquid and hence Q cannot be calculated as mentioned above (i.e. $Q = C_d \times a \times \sqrt{2gH}$).

Consider a large rectangular orifice in one side of the tank discharging water freely into the atmosphere, as shown in Fig. 8.9.

- Let,
- H_1 = Height of liquid above the top of the orifice,
 - H_2 = Height of liquid above the bottom of the orifice,
 - b = Breadth of the orifice, and
 - C_d = Co-efficient of discharge.

Consider an elementary horizontal strip of depth ' dh ' at depth of ' h ' below the water level as shown in Fig. 8.9.

**Fig. 8.9.** Large rectangular orifice.

∴ Area of the strip = $b \cdot dh$

Theoretical velocity of water through the strip = $\sqrt{2gh}$

∴ Discharge through the strip,

$$\begin{aligned} dQ &= C_d \times \text{area of strip} \times \text{velocity} \\ &= C_d \times b \times dh \times \sqrt{2gh} \\ &= C_d \cdot b \cdot dh \sqrt{2gh} \end{aligned}$$

Total discharge through the whole orifice may be found out by integrating the above equation between the limits H_1 and H_2 .

$$\begin{aligned} \therefore Q &= \int_{H_1}^{H_2} C_d \cdot b \cdot dh \sqrt{2gh} \\ &= C_d \cdot b \times \sqrt{2g} \int_{H_1}^{H_2} \sqrt{h} \\ &= C_d \cdot b \sqrt{2g} \left[\frac{h^{3/2}}{3/2} \right]_{H_1}^{H_2} \\ &= \frac{2}{3} C_d \cdot b \sqrt{2g} (H_2^{3/2} - H_1^{3/2}) \quad \dots(8.9) \end{aligned}$$

Example 8.13. Find the discharge through a rectangular orifice 3.0 m wide and 2.0 m deep fitted to a water tank. The water level in the tank is 4.0 m above the top edge of the orifice. Take $C_d = 0.62$.

Solution. Width of the orifice, $b = 3.0$ m

Depth of the orifice, $d = 2.0$ m

Height of water above the top of the orifice, $H_1 = 4.0$ m

∴ Height of the water above the bottom of the orifice, $H_2 = 4 + d = 4 + 2 = 6$ m

Co-efficient of discharge, $C_d = 0.62$

Discharge through the orifice, Q :

Using the relation:

$$\begin{aligned} Q &= \frac{2}{3} C_d \cdot b \sqrt{2g} (H_2^{3/2} - H_1^{3/2}) \quad \text{with usual notations} \\ &= \frac{2}{3} \times 0.62 \times 3.0 \times \sqrt{2 \times 9.81} (6^{3/2} - 4^{3/2}) = 36.78 \text{ m}^3/\text{s} \end{aligned}$$

i.e. **$Q = 36.78 \text{ m}^3/\text{s}$ (Ans.)**

Example 8.14. A rectangular orifice 0.6 m wide and 0.8 m deep is discharging water from a vessel. The top edge of the orifice is 0.4 m below the water surface in the vessel. Find:

- (i) The discharge through the orifice if $C_d = 0.62$;
(ii) The percentage error if the orifice is treated as a small orifice.

Solution. Width of orifice, $b = 0.6$ m

Depth of orifice, $d = 0.8$ m

$$H_1 = 0.4 \text{ m}$$

$$H_2 = H_1 + d = 0.4 + 0.8 = 1.2 \text{ m}$$

Co-efficient of discharge, $C_d = 0.62$

(i) Discharge through the orifice, Q :

$$\begin{aligned} Q &= \frac{2}{3} C_d \times b \times \sqrt{2g} (H_2^{3/2} - H_1^{3/2}) \\ &= \frac{2}{3} \times 0.62 \times 0.6 \times \sqrt{2 \times 9.81} [1.2^{3/2} - 0.4^{3/2}] \text{ m}^3/\text{s} \\ &= 1.098 (1.314 - 0.253) = \mathbf{1.165 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

(ii) The percentage error if the orifice is treated as a small orifice:

Discharge for a small orifice,

$$Q' = C_d \times a \times \sqrt{2gh}$$

where, $h = H_1 + \frac{d}{2} = 0.4 + \frac{0.8}{2} = 0.8 \text{ m},$

and, $a = b \times d = 0.6 \times 0.8 = 0.48 \text{ m}^2$

$\therefore Q' = 0.62 \times 0.48 \times \sqrt{2 \times 9.81 \times 0.8} = 1.179 \text{ m}^3/\text{s}$

$$\% \text{ error} = \frac{Q' - Q}{Q} = \frac{1.179 - 1.165}{1.165} = 0.012 \text{ or } \mathbf{1.2 \% \text{ (Ans.)}}$$

8.7. DISCHARGE THROUGH FULLY SUBMERGED ORIFICE

If an orifice has its whole of the outlet side submerged under liquid so that it discharges a jet of liquid into the liquid of the same kind then it is known as **fully submerged** (or **drowned**) **orifice**.

Consider a fully submerged orifice as shown in Fig. 8.10.

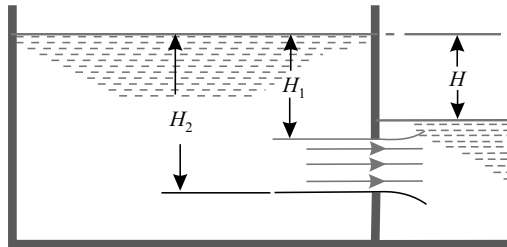


Fig. 8.10. Fully submerged orifice.

- Let, H_1 = Height of water (on the upstream side) above the top of the orifice,
 H_2 = Height of water (on the upstream side) above the bottom of the orifice,
 H = Difference between the two water levels on either side of the orifice,
 b = Width of orifice, and
 C_d = Co-efficient of discharge.

$\therefore$ Area of the orifice $= b (H_2 - H_1)$

We know that theoretical velocity of water through the orifice = $\sqrt{2gH}$

$\therefore$ Actual velocity of water = $C_v \sqrt{2gH}$

Since in this case co-efficient of contraction is 1, therefore, taking C_d equal to C_v , we find that the actual velocity of water = $C_d \times \sqrt{2gH}$

$\therefore$ Discharge through the orifice,

$$\begin{aligned} Q &= \text{Area of orifice} \times \text{actual velocity} \\ &= b(H_2 - H_1) \times C_d \sqrt{2gH} \\ &= C_d \cdot b(H_2 - H_1) \times \sqrt{2gH} \end{aligned} \quad \dots(8.10)$$

Sometimes, depth of submerged orifice (d) is given instead of H_1 and H_2 . In such cases, the discharge,

$$Q = C_d \cdot b \cdot d \sqrt{2gH} \quad \dots(8.11)$$

Example 8.15. Find the discharge through a totally drowned orifice 1.5 m wide and 1 m deep, if the difference of water levels on both the sides of the orifice be 2.5 m. Take $C_d = 0.62$.

Solution. Width of the orifice, $b = 1.5$ m

Difference of water levels, $H = 2.5$ m

Depth of the orifice, $d = 1$ m

Co-efficient of discharge, $C_d = 0.62$

Discharge, Q:

Using the relation,

$$\begin{aligned} Q &= C_d \cdot b \cdot d \sqrt{2gH} \\ &= 0.62 \times 1.5 \times 1 \times \sqrt{2 \times 9.81 \times 2.5} = 6.513 \text{ m}^3/\text{s} \end{aligned}$$

i.e.,

$$Q = 6.513 \text{ m}^3/\text{s} \text{ (Ans.)}$$

8.8. DISCHARGE THROUGH PARTIALLY SUBMERGED ORIFICE

If the outlet side of an orifice is only partly submerged (or drowned) under liquid then it is known as **partially submerged (or drowned) orifice** (Fig. 8.11). The upper portion behaves as an orifice discharging free, while the lower portion behaves as a submerged orifice. The total discharge is determined by computing separately the discharges through the free and the submerged portions and then adding together the two discharges thus computed.

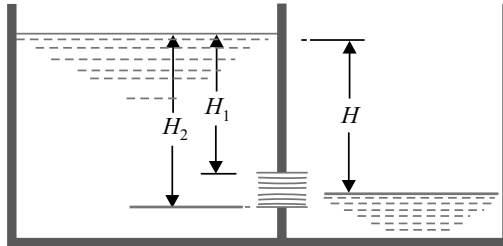


Fig. 8.11. Partially submerged orifice.

Discharge through the submerged portion,

$$Q_1 = C_d \cdot b \cdot (H_2 - H) \times \sqrt{2gH} \quad \dots(\text{As in Art. 8.7})$$

and, the discharge through the free portion,

$$Q_2 = \frac{2}{3} C_d \cdot b \cdot \sqrt{2g} (H^{3/2} - H_1^{3/2}) \quad \dots(\text{As in Art. 8.6})$$

Total discharge

$$Q = Q_1 + Q_2 \\ = C_d \cdot b \cdot (H_2 - H) \times \sqrt{2gH} + \frac{2}{3} \cdot C_d \cdot b \cdot \sqrt{2g} (H^{3/2} - H_1^{3/2}) \quad \dots(8.12)$$

Example 8.16. A rectangular orifice 1.5 m wide and 1.2 m deep is fitted in one side of a large tank. The water level on one side of the orifice is 2 m above the top edge of the orifice, while on the other side of the orifice, the water level is 0.4 m below its top edge. Calculate the discharge through the orifice if $C_d = 0.62$.

Solution. Width of orifice, $b = 1.5$ m

Depth of orifice, $d = 1.2$ m

Height of water level above the top of the orifice,

$$H_1 = 2 \text{ m}$$

Height of water level above the bottom of the orifice,

$$H_2 = 2 + 1.2 = 3.2 \text{ m}$$

Difference of water levels on both the sides,

$$H = 2.4 \text{ m}$$

Co-efficient of discharge, $C_d = 0.62$

Since the orifice is partially submerged, let us treat the upper portion as a free orifice and the lower portion as a submerged orifice.

Let, Q_1 = Discharge through the submerged portion, and

Q_2 = Discharge through the free portion.

Using the relation:

$$Q_1 = C_d \cdot b \cdot (H_2 - H) \times \sqrt{2gH} \quad \dots\text{with usual notations}$$

$$= 0.62 \times 1.5 (3.2 - 2.4) \times \sqrt{2 \times 9.81 \times 2.4} = 5.1 \text{ m}^3/\text{s}$$

Now, using the relation:

$$Q_2 = \frac{2}{3} C_d \cdot b \cdot \sqrt{2g} (H^{3/2} - H_1^{3/2}) \quad \dots\text{with usual notations}$$

$$= \frac{2}{3} \times 0.62 \times 1.5 \times \sqrt{2 \times 9.81} (2.4^{3/2} - 2^{3/2})$$

$$= 2.44 \text{ m}^3/\text{s}$$

The total discharge,

$$Q = Q_1 + Q_2 = 5.1 + 2.44 = 7.54 \text{ m}^3/\text{s}$$

$$\text{i.e. } Q = 7.54 \text{ m}^3/\text{s (Ans.)}$$

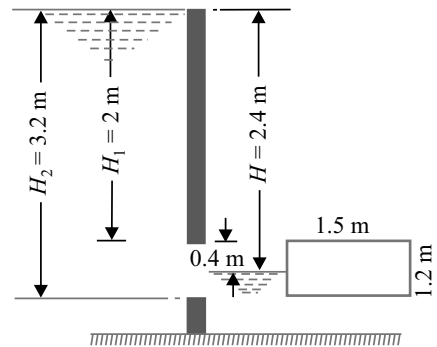


Fig. 8.12

8.9. TIME REQUIRED FOR EMPTYING A TANK THROUGH AN ORIFICE AT ITS BOTTOM

Consider a tank, of uniform cross-sectional area, containing some liquid, and having an orifice at its bottom as shown in Fig. 8.13.

Let, A = Cross-sectional area of the tank,
 a = Area of the orifice,
 H_1 = Initial height of liquid,
 H_2 = Final height of liquid
 T = Time in seconds, required to
 bring the level from H_1 to H_2

Let at some instant the height of the liquid be h above the orifice and let the liquid surface fall by an amount dh after a small interval for time dt .

Then, volume of the liquid that has passed the tank in time dt ,

$$dq = -A \cdot dh \quad \dots(i)$$

(- ve sign of dh is taken because the value of h decreases when the discharge increases). Also, theoretical velocity through the orifice, $v = \sqrt{2gh}$

$\therefore$ Discharge through the orifice in a small interval of time dt ,

dq = Co-efficient of discharge $\times$ area $\times$ theoretical velocity $\times$ time.

$$= C_d \cdot a \cdot \sqrt{2gh} \cdot dt \quad \dots(ii)$$

Equating (i) and (ii), we get:

$$-A \cdot dh = C_d \cdot a \cdot \sqrt{2gh} \cdot dt$$

$$\therefore dt = \frac{-A \cdot dh}{C_d \cdot a \cdot \sqrt{2gh}} = \frac{-A (h^{-1/2}) dh}{C_d \cdot a \sqrt{2g}}$$

Time taken (T) to lower the level from H_1 to H_2 is calculated by integrating the above equation between the limits H_1 and H_2 .

$$\begin{aligned} \text{i.e.} \quad T &= \int_{H_1}^{H_2} \frac{A (h^{-1/2}) dh}{C_d \cdot a \cdot \sqrt{2g}} = \frac{-A}{C_d \cdot a \sqrt{2g}} \int_{H_1}^{H_2} h^{-1/2} dh \\ &= \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \left[\frac{h^{1/2}}{1/2} \right]_{H_1}^{H_2} = \frac{-2A}{C_d \cdot a \sqrt{2g}} [h^{1/2}]_{H_1}^{H_2} \\ &= \frac{-2A}{C_d \cdot a \cdot \sqrt{2g}} - [\sqrt{H_2} - \sqrt{H_1}] = \frac{2A (\sqrt{H_1} - \sqrt{H_2})}{C_d \cdot a \sqrt{2g}} \quad \dots(8.13) \end{aligned}$$

If the tank is to be emptied completely, then $H_2 = 0$

$$\text{and,} \quad T = \frac{2A \sqrt{H_1}}{C_d \cdot a \sqrt{2g}} \quad \dots(8.14)$$

Example 8.17. A circular tank of diameter 3 m contains water upto a height of 4m. The tank is provided with an orifice of diameter 0.4 m at the bottom. Find the time taken by water,

(i) to fall from 4 m to 2 m, and

(ii) for completely emptying the tank.

Take $C_d = 0.6$

Solution. Dia. of the tank, $D = 3\text{ m}$

$$\therefore \text{Area, } a = (\pi/4) \times 3^2 = 7.068 \text{ m}^2$$

Dia. of the orifice, $d = 0.4 \text{ m}$

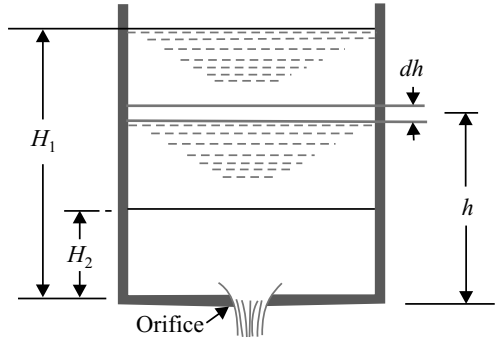


Fig. 8.13 Tank with an orifice at its bottom.

$$\therefore \text{Area, } a = (\pi/4) \times 0.4^2 = 0.1257 \text{ m}^2$$

$$\text{Initial height of water, } H_1 = 4 \text{ m}$$

$$\text{Final height of water, (i) } H_2 = 2 \text{ m (ii) } H_2 = 0$$

Case I. When $H_2 = 2 \text{ m}$

Using the relation,

$$T = \frac{2A (\sqrt{H_1} - \sqrt{H_2})}{C_d \cdot a \cdot \sqrt{2g}} \quad \dots \text{with usual notations}$$

$$= \frac{2 \times 7.068 (\sqrt{4} - \sqrt{2})}{0.6 \times 0.1257 \times \sqrt{2 \times 9.81}} = \frac{8.28}{0.334} = 24.8 \text{ s}$$

$$\text{i.e. } T = 24.8 \text{ s (Ans.)}$$

Case II. When $H_2 = 0$.

$$T = \frac{2A \sqrt{H_1}}{C_d \cdot a \sqrt{2g}} = \frac{2 \times 7.068 \times \sqrt{4}}{0.6 \times 0.1257 \sqrt{2 \times 9.81}} = \frac{28.27}{0.334} = 84.6 \text{ s}$$

$$\text{i.e. } T = 84.6 \text{ s (Ans.)}$$

Example 8.18. A swimming pool 12 m long and 7 m wide holds water to a depth of 2 m. If the water is discharged through an opening of area 0.2 m^2 at the bottom of the pool, find the time required to empty the tank. Take co-efficient of discharge for the opening as 0.6. (UPSC)

Solution. Area of the swimming pool, $A = 12 \times 7 = 84 \text{ m}^2$

$$\text{Area of the orifice (opening) } = 0.2 \text{ m}^2$$

$$\text{Co-efficient of discharge, } C_d = 0.6$$

$$\text{Initial height of water, } H_1 = 2 \text{ m}$$

Time required to empty the tank, T :

$$\text{Using the relation: } T = \frac{2A \sqrt{H_1}}{C_d \cdot a \sqrt{2g}} \quad [\text{Eqn. (8.14)}]$$

$$\text{Substituting the values, we get: } T = \frac{2 \times 84 \times \sqrt{2}}{0.6 \times 0.2 \sqrt{2 \times 9.81}} = 446.98 \text{ s (Ans.)}$$

Example 8.19. A 1 m diameter circular tank contains water upto a height of 4 m. At the bottom of tank an orifice of 40 mm is provided. Find the height of water above the orifice after 1.5 minutes. Take co-efficient of discharge for the orifice $C_d = 0.6$.

Solution. Dia. of tank, $D = 1 \text{ m}$

$$\therefore \text{Area, } A = \frac{\pi}{4} D^2 = \frac{\pi}{4} \times 1^2 = 0.785 \text{ m}^2$$

$$\text{Dia. of orifice, } d = 40 \text{ mm} = 0.04 \text{ m}$$

$$\therefore \text{Area, } a = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times 0.04^2 = 0.001257 \text{ m}^2$$

$$\text{Initial height of water, } H_1 = 4 \text{ m}$$

$$\text{Time, } T = 1.5 \text{ min.} = 1.5 \times 60 = 90 \text{ s}$$

Height of water above the orifice after 1.5 minutes:

Let H_2 be height of water above the orifice after 1.5 minutes.

Using the relation:
$$T = \frac{2A (\sqrt{H_1} - \sqrt{H_2})}{C_d \cdot a \sqrt{2g}} \quad (\text{Eqn.8.13})$$

Substituting the values, we get:

$$\begin{aligned} 90 &= \frac{2 \times 0.785 \times (\sqrt{4} - \sqrt{H_2})}{0.6 \times 0.001257 \times \sqrt{2 \times 9.81}} \\ &= 469.9 (2 - \sqrt{H_2}) \end{aligned}$$

or,
$$\sqrt{H_2} = 2 - \frac{90}{469.9} = 1.808$$

$\therefore H_2 = 3.269\text{m (Ans.)}$

Example 8.20. Fig. 8.14 shows a rectangular tank having the compartments 1 and 2, communicating by an orifice 100 mm square, its centre being 1 m above the bottom of the tank. The horizontal cross-sections of the compartments 1 and 2 are 12 m^2 and 24 m^2 respectively. At a certain instant the water stands 4 m deep in 1 and 2 m deep in 2. How soon thereafter will surface reach a common level?

Take the efficient of discharge, $C_d = 0.6$

Solution. Size of the orifice = 100 mm (or 0.1 m) square

$\therefore$ Area, $a = 0.1 \times 0.1 = 0.01\text{ m}^2$

Area of compartment 1, $A_1 = 12\text{ m}^2$

Area of compartment 2, $A_2 = 24\text{ m}^2$

Discharge co-efficient, $C_d = 0.6$

Time taken by the surface to reach a common level, T:

Let, y = Depth of water in '1' at any instant, and

dy = Change in depth during any interval of time dT .

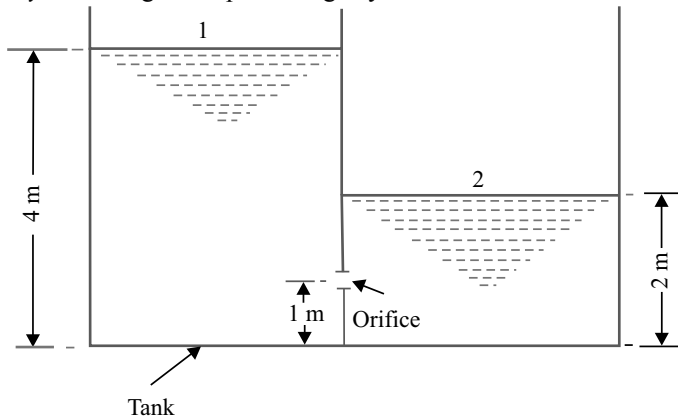


Fig. 8.14

Then, rise in 2's level during the same time will be $\frac{12}{24} \times dy$ and net change in head will be,

$$dh = dy - \left(-\frac{12}{24} \times dy\right) = \frac{3}{2} dy \text{ or } dy = \frac{2}{3} dh$$

Discharge through orifice in time dT

$$= Q \cdot dT = C_d \cdot a \cdot \sqrt{2gh} \cdot dT \quad \dots(i)$$

Also the decrease in volume of water in time dT

$$= 12 \times dy = 12 \times \frac{2}{3} dh = 8 dh \quad \dots(ii)$$

Equating (i) and (ii), we get:

$$-8 dh = C_d \cdot a \cdot \sqrt{2gh} \cdot dT$$

(Negative sign is introduced as with the increase in time, the height of water in '1' decreases)

$$\therefore dT = \frac{-8}{C_d \cdot a \cdot \sqrt{2g}} \cdot \frac{dh}{\sqrt{h}} = \frac{-8}{0.6 \times 0.01 \times \sqrt{2 \times 9.81}} \cdot \frac{dh}{\sqrt{h}} = -301 \frac{dh}{\sqrt{h}}$$

Integrating both the sides, we get:

$$\begin{aligned} \int_0^T dT &= -301 \int_2^0 \frac{dh}{\sqrt{h}} = 301 \int_0^2 h^{-1/2} dh \\ T &= 301 \left[\frac{h^{-1/2+1}}{-1/2+1} \right]_0^2 = [301 \times 2\sqrt{h}]_0^2 \\ &= 301 \times 2 \times \sqrt{2} = \mathbf{851.35 \text{ s (Ans.)}} \end{aligned}$$

Example 8.21. A vessel has compartments A and B communicating by an orifice 150 cm^2 , its centre being 1 m above the bottom of the vessel. The cross section of A is 10 m^2 and that of B is 20 m^2 . At a certain time the water stands 4 m in A and 2 m in B. How soon thereafter water will attain common level? Assume $C_d = 0.62$. (MU)

Solution. Refer to Fig. 8.15.

Let, y = Depth of water in compartment A at any instant,

and, dy = Change in depth during any interval of time dT .

Then, the rise in the level of water in compartment B during the time dT will be $\frac{10}{20} \times dy$ and the net change in head will be,

$$dh = dy - \left(-\frac{10}{20} dy \right) = \frac{3}{2} dy \text{ (or } dy = \frac{2}{3} dh)$$

Discharge through the orifice in time dT

$$= Q \cdot dT = C_d \cdot a \cdot \sqrt{2gh} \cdot dT \quad \dots(i)$$

(where, a = area of orifice)

Also decrease in volume of water in compartment A in time dT

$$= 10 \times dy = 10 \times \frac{2}{3} dh = \frac{20}{3} dh$$

Equating (i) and (ii), we get:

$$-\frac{20}{3} dh = C_d \cdot a \cdot \sqrt{2gh} \cdot dT$$

(Negative sign is introduced as with increase in time, the height of water in compartment A decreases)

$$\therefore dT = \frac{-6.667}{C_d \cdot a \cdot \sqrt{2g}} \times \frac{dh}{\sqrt{h}}$$

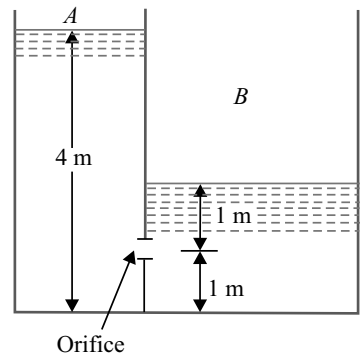


Fig. 8.15

$$= \frac{-6.667}{0.62 \times (150 \times 10^{-4}) \times \sqrt{2 \times 9.81}} \times \frac{dh}{\sqrt{h}} = -161.84 \times \frac{dh}{\sqrt{h}}$$

Integrating both sides, we get:

$$\int_0^T dT = -161.84 \int_2^0 \frac{dh}{\sqrt{h}} = 161.84 \int_0^2 h^{-1/2} dh$$

$$\therefore T = 161.84 \left[\frac{(h)^{\frac{-1}{2}+1}}{-\frac{1}{2}+1} \right]_0^2 = 161.84 [2\sqrt{h}]_0^2 = 457.7 \text{ s (Ans.)}$$

Example 8.22. A cylindrical tank 3 m in diameter and 6 m high has an orifice 150 mm in diameter at the bottom centre of the tank. A constant discharge of 85 litres of water per second is fed into the tank. At the same time water is being discharged through the orifice. Determine the time taken to lower the water surface level in the tank from 5 m to 2.5 m above the centre of orifice. Take $C_d = 0.72$. The top of the tank is open to the atmosphere. [UPSC Exams.]

Solution. Dia. of the tank, $D = 3 \text{ m}$

$$\therefore \text{Area, } A = \frac{\pi}{4} \times 3^2 = 7.068 \text{ m}^2$$

Dia. of the orifice, $d = 150 \text{ mm} = 0.15 \text{ m}$

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

Constant discharge of water fed into the tank, $q = 85 \text{ litres/s} = 0.085 \text{ m}^3/\text{s}$

Initial height of water, $H_1 = 5 \text{ m}$

Final height of water, $H_2 = 2.5 \text{ m}$

Time taken to lower the water surface level:

Let, $Q =$ Outflow i.e. discharge through the orifice, and

$dh =$ The fall in water surface in the tank in time dT under the conditions of simultaneous inflow and outflow.

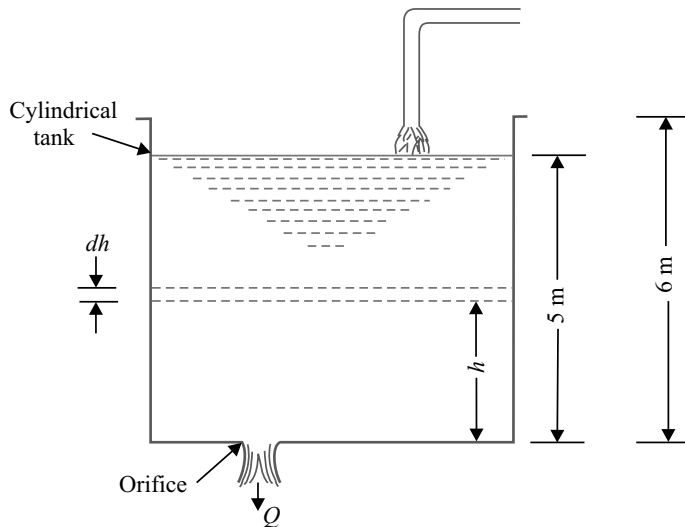


Fig. 8.16

From the principle of continuity, we have:

$$-A.dh = (Q - q) dT = (C_d.a.\sqrt{2gh} - q) dT$$

or,
$$dT = - \frac{A.dh}{(C_d.a.\sqrt{2gh} - q)}$$

Let,
$$Q = K\sqrt{h}$$

where,
$$K = C_d.a.\sqrt{2g}$$

and,
$$(K\sqrt{h} - q) = x,$$

then,
$$dx = \frac{Kdh}{2\sqrt{h}} \left(\text{or } dh = \frac{2\sqrt{h} \times dx}{K} \right)$$

Time taken to lower the water surface from a height, H_1 to H_2

$$T = - \int_{H_1}^{H_2} \frac{A.dh}{(K\sqrt{h} - q)} = - \int_{H_1}^{H_2} \frac{A \times 2\sqrt{h}.dx}{K(K\sqrt{h} - q)} \quad (\text{substituting the value of } dh)$$

Multiplying the numerator and denominator by K , we get:

$$\begin{aligned} &= - \int_{H_1}^{H_2} \frac{A.2\sqrt{h}.dx.K}{K^2(K\sqrt{h} - q)} = - \frac{2A}{K^2} \int (x + q) \frac{dx}{x} \quad (\because K\sqrt{h} = x + q) \\ &= - \frac{2A}{K^2} \int \left(1 + \frac{q}{x} \right) dx = - \frac{2A}{K^2} [x + q \ln(x)] \\ &= - \frac{2A}{K^2} [(K\sqrt{h} - q) + q \ln(K\sqrt{h} - q)]_{H_1}^{H_2} \\ &= \frac{2A}{K^2} [(K\sqrt{h} - q) + q \ln(K\sqrt{h} - q)]_{H_2}^{H_1} \\ &= \frac{2A}{K^2} \left[K(\sqrt{H_1} - \sqrt{H_2}) + q \ln \frac{(K\sqrt{H_1} - q)}{(K\sqrt{H_2} - q)} \right] \end{aligned}$$

Substituting the given data, we get $K = C_d.a.\sqrt{2g} = 0.72 \times 0.01767 \times \sqrt{2 \times 9.81} = 0.0564$

$$\begin{aligned} &= \frac{2 \times 7.068}{(0.0564)^2} \left[0.0564 (\sqrt{5} - \sqrt{2.5}) + 0.085 \ln \left\{ \frac{(0.0564\sqrt{5} - 0.085)}{(0.0564\sqrt{2.5} - 0.085)} \right\} \right] \\ &= 4443.94 \left[0.0369 + 0.085 \ln \left(\frac{0.0411}{0.00417} \right) \right] \\ &= \mathbf{1028.3 \text{ s (Ans.)}} \end{aligned}$$

Example 8.23. A swimming pool 30 m long and 10 m wide has vertical sides and bottom at a slope. The depths of water at the shallow and deep sides are 2 m and 5 m respectively. Two outlets, each of 0.4 m diameter, have been provided at each of the deep and shallow ends. Calculate the time taken to empty the pool if both the outlets are kept open. Take $C_d = 0.6$ for each opening.

Solution. Length of swimming pool = 30 m

Width of the pool = 10 m

$$\therefore \text{Area of the pool } A = 30 \times 10 = 300 \text{ m}^2$$

$$\text{Diameter of each outlet, } d = 0.4 \text{ m}$$

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.4^2 = 0.1256 \text{ m}^2$$

Discharge co-efficient for each opening, $C_d = 0.6$

Time taken to empty the pool, T:

Let, T_1 = Time required to bring the water level from LM to NP , and

T_2 = Time taken to empty the triangular element NPS .

Then, $T = T_1 + T_2$

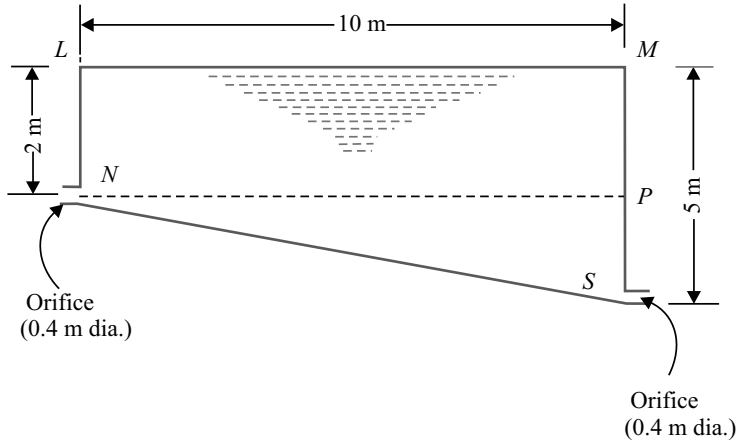


Fig. 8.17

(i) Time T_1 :

Refer to Fig. 8.17. The section NP corresponds to centre of the top orifice/opening. Let us consider an instant when the height of water above the centre of the top orifice is h metres. At that instant, the height of water above the bottom orifice equals $(h + 3)$ metres. If during a small time interval dT the water level falls by dh , then:

Volume of water leaving the pool in time dT

= Discharge through the two orifices in time dT

$$\text{i.e.,} \quad -A \cdot dh = C_d \cdot a \cdot \sqrt{2g(h+3)} dT + C_d \cdot a \cdot \sqrt{2gh} dT$$

or,

$$dT = \frac{-A dh}{C_d \cdot a \cdot \sqrt{2gh} (\sqrt{h+3} + \sqrt{h})}$$

$$\int_0^{T_1} dT = \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \int_{h=2}^{h=0} \frac{dh}{\sqrt{(h+3)} + \sqrt{h}}$$

$$T_1 = \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \int_{h=2}^{h=0} \frac{[\sqrt{(h+3)} - \sqrt{h}]}{[\sqrt{(h+3)} + \sqrt{h}][\sqrt{(h+3)} - \sqrt{h}]} \cdot dh$$

$$= \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \int_{h=2}^{h=0} \frac{[\sqrt{(h+3)} - \sqrt{h}]}{3} \cdot dh$$

$$\begin{aligned}
 &= \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \times \frac{1}{3} \left[\frac{(h+3)^{3/2}}{3/2} - \frac{h^{3/2}}{3/2} \right]_2^0 \\
 &= \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \times \frac{1}{3} \times \frac{2}{3} \left[(h+3)^{3/2} - h^{3/2} \right]_2^0
 \end{aligned}$$

Substituting the values, we get:

$$\begin{aligned}
 &= \frac{-300}{0.6 \times 0.1256 \times \sqrt{2 \times 9.81}} \times \frac{2}{9} \left[(3)^{3/2} - \{ (2+3)^{3/2} - 2^{3/2} \} \right] \\
 &= -898.73 \times \frac{2}{9} (5.196 - 8.352) = 630.3 \text{ s}
 \end{aligned}$$

(ii) Time T_2 :

Within the triangular element NPS , let h be the height of water above the bottom orifice at any instant; and let the water falls by dh during a small time interval dT . At that instant:

$$\text{Width of water surface} = \frac{h}{3} \times 10$$

$$\text{Area of water surface} = \left(\frac{h}{3} \times 10 \right) \times 30 = 100 h$$

Now, Volume of water leaving the pool in time dT

= Discharge through the bottom orifice in time dT

$$-A \cdot dh = C_d \cdot a \cdot \sqrt{2gh} \cdot dT$$

or

$$dT = \frac{-A \cdot dh}{C_d \cdot a \cdot \sqrt{2gh}} = \frac{-100h \cdot dh}{C_d \cdot a \cdot \sqrt{2gh}}$$

$$= \frac{-100}{C_d \cdot a \cdot \sqrt{2g}} h^{1/2} \cdot dh$$

$$\therefore \int_0^{T_2} dT = \frac{-100}{C_d \cdot a \cdot \sqrt{2g}} \int_{h=3}^{h=0} h^{1/2} \cdot dh$$

Substituting values, we get:

$$\begin{aligned}
 T_2 &= \frac{-100}{0.6 \times 0.1256 \times \sqrt{2 \times 9.81}} \left[\frac{2}{3} \times h^{3/2} \right]_3^0 \\
 &= -299.57 \left[\frac{2}{3} \times (0 - 3^{3/2}) \right] = 1037.7 \text{ s}
 \end{aligned}$$

$\therefore$ Total time, $T = T_1 + T_2 = 630.3 + 1037.7 = 1668 \text{ s (Ans.)}$

Example 8.24. Fig 8.18 shows a truncated cone having vertex angle $\theta = 60^\circ$. How long does it take to draw the liquid surface from $y = 3 \text{ m}$ to $y = 1 \text{ m}$? Take $C_d = 0.85$ [IIT Bombay]

Solution. Dia. of the orifice, $d = 0.1 \text{ m}$

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.1^2 = 0.007854 \text{ m}^2$$

Vertex angle, $\theta = 60^\circ$

Discharge co-efficient, $C_d = 0.85$

Time required, T:

From the geometry of the truncated cone,

$$\frac{0.5}{x} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\text{or, } x = 0.5 \times \sqrt{3} = 0.866 \text{ m}$$

Let r = Radius of truncated cone at a distance $(x + y)$ from the vertex.

$$\text{Then, } \frac{r}{x + y} = \tan 30^\circ$$

$$\begin{aligned} \text{or } r &= (0.866 + y) \tan 30^\circ \\ &= \frac{(0.866 + y)}{\sqrt{3}} \end{aligned}$$

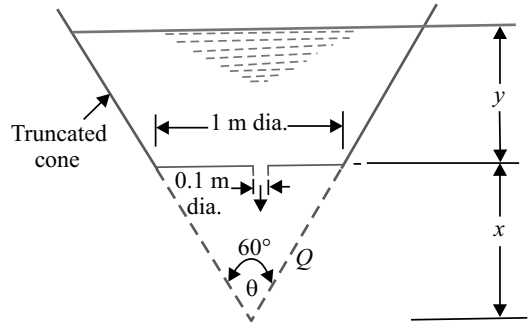


Fig. 8.18

Further, let the liquid surface falls a distance dy in time dT . Then from the continuity principle, we have:

$$\pi r^2 (-dy) = Q \cdot dT = C_d \cdot a \cdot \sqrt{2gy} \cdot dT$$

$$\text{or, } -\pi (0.866 + y)^2 \times \frac{1}{2} dy = 0.85 \times 0.007854 \times \sqrt{2 \times 9.81 \times y} \times dT$$

$$\text{or, } -1.047 (0.866 + y)^2 dy = 0.0296 \sqrt{y} \times dT$$

$$\text{or, } dT = \frac{1.047 (0.866 + y)^2 dy}{0.0296 \sqrt{y}} = -35.37 \times \frac{(0.866 + y)^2}{\sqrt{y}} \cdot dy$$

The time required to empty the cone from $y = 3$ m to 1 m ,

$$\begin{aligned} T &= 35.37 \int_3^1 \frac{(0.866 + y)^2}{\sqrt{y}} \cdot dy = -35.37 \int_3^1 \frac{(0.75 + y^2 + 1.732y)}{\sqrt{y}} \cdot dy \\ &= 35.37 \int_1^3 \left(\frac{0.75}{\sqrt{y}} + y^{3/2} + 1.732y^{1/2} \right) dy \\ &= 35.37 \left[0.75 \times \frac{y^{-1/2+1}}{-1/2+1} + \frac{y^{3/2+1}}{3/2+1} + 1.732 \times \frac{y^{1/2+1}}{1/2+1} \right]_1^3 \\ &= 35.37 \left[1.5y^{1/2} + \frac{2}{5}y^{5/2} + 1.732 \times 2/3 y^{3/2} \right]_1^3 \\ &= 35.37 \left[1.5 (3^{1/2} - 1) + \frac{2}{5} (3^{5/2} - 1) + 1.155 (3^{3/2} - 1) \right] \\ &= 35.37 (1.098 + 5.835 + 4.846) = \mathbf{416.6 \text{ s (Ans.)}} \end{aligned}$$

8.10. TIME REQUIRED FOR EMPTYING A HEMISPHERICAL TANK

Consider a hemispherical tank containing some liquid and fitted with an orifice at its bottom as shown in the Fig. 8.19.

Let, R = Radius of the tank,

- a = Area of the orifice,
 H_1 = Initial height of the liquid,
 H_2 = Final height of the liquid, and
 T = Time in seconds for the liquid to fall from height H_1 to H_2 .

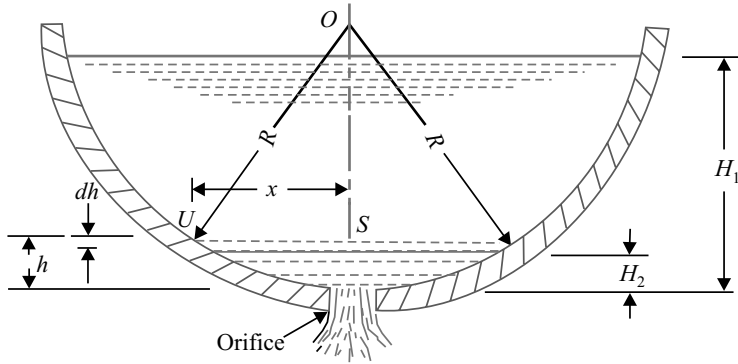


Fig. 8.19. Hemispherical tank.

Let at any instant of time, the height of liquid over the orifice is h and x be the radius of the liquid surface.

Then, area of liquid surface, $A = \pi x^2$

Theoretical velocity of liquid $= \sqrt{2gh}$

Let the height of liquid decrease by dh in a small interval of time dT . Then,

Volume of liquid leaving the tank in time dT

$$= A \cdot dh = \pi x^2 \times dh \quad \dots(i)$$

Also, volume of liquid flowing through the orifice in time dT

$$\begin{aligned}
 &= C_d \times \text{area of orifice} \times \text{velocity} \times dT \\
 &= C_d \cdot a \cdot \sqrt{2gh} \times dT \quad \dots(ii)
 \end{aligned}$$

Equating (i) and (ii), we get:

$$\pi x^2 (-dh) = C_d \cdot a \sqrt{2gh} \times dT$$

The negative sign accounts for the *decrease in head* on the orifice with *increase in time interval*.

$$dT = \frac{-\pi x^2 \cdot dh}{C_d \cdot a \cdot \sqrt{2gh}} \quad \dots(iii)$$

From Fig. 8.19, we have:

$$OU = R \text{ and } OS = (R - h)$$

$$\begin{aligned}
 \therefore x &= US = \sqrt{OU^2 - OS^2} = \sqrt{R^2 - (R - h)^2} \\
 &= \sqrt{R^2 - R^2 - h^2 + 2Rh} = \sqrt{2Rh - h^2}
 \end{aligned}$$

or $x^2 = (2Rh - h^2)$

Substituting this value of x^2 in eqn. (iii), we get:

$$dT = \frac{-\pi (2Rh - h^2) dh}{C_d \cdot a \cdot \sqrt{2gh}}$$

$$\begin{aligned}
 &= \frac{-\pi}{C_d \cdot a \cdot \sqrt{2g}} = (2Rh - h^2) h^{-1/2} dh \\
 &= \frac{-\pi}{C_d \cdot a \cdot \sqrt{2g}} (2Rh^{1/2} - h^{3/2}) dh
 \end{aligned}$$

The total time T required to bring the liquid level from H_1 to H_2 is obtained by integrating the above equation between the limits H_1 to H_2 .

$$\int_0^T dT = \int_{H_1}^{H_2} \frac{-\pi}{C_d \cdot a \cdot \sqrt{2g}} (2Rh^{1/2} - h^{3/2}) dh$$

$$T = \frac{-\pi}{C_d \cdot a \cdot \sqrt{2g}} \int_{H_1}^{H_2} (2Rh^{1/2} - h^{3/2}) dh$$

$$= \frac{-\pi}{C_d \cdot a \cdot \sqrt{2g}} \left[2R \times \frac{h^{1/2+1}}{\frac{1}{2}+1} - \frac{h^{3/2+1}}{\frac{3}{2}+1} \right]_{H_1}^{H_2}$$

$$= \frac{-\pi}{C_d \cdot a \cdot \sqrt{2g}} \left[\frac{2}{3} \times 2R h^{3/2} - \frac{2}{5} h^{5/2} \right]_{H_1}^{H_2}$$

$$= \frac{-\pi}{C_d \cdot a \cdot \sqrt{2g}} \left[\frac{4}{3} R (H_2^{3/2} - H_1^{3/2}) - \frac{2}{5} (H_2^{5/2} - H_1^{5/2}) \right]$$

$$\text{or} \quad T = \frac{\pi}{C_d \cdot a \cdot \sqrt{2g}} \left[\frac{4}{3} R (H_1^{3/2} - H_2^{3/2}) - \frac{2}{5} (H_1^{5/2} - H_2^{5/2}) \right] \quad \dots(8.15)$$

For emptying the tank completely, $H_2 = 0$ and hence,

$$T = \frac{\pi}{C_d \cdot a \cdot \sqrt{2g}} \left[\frac{4}{3} R H_1^{3/2} - \frac{2}{5} H_1^{5/2} \right] \quad \dots(8.16)$$

Example 8.25. A hemispherical tank of 2 m radius is provided with an orifice of 40 mm at its bottom. It contains water upto a height of 1.8 m. Find the time required by water

- (i) to fall from 1.8 m to 1.2 m, and
(ii) for completely emptying the tank.

Take $C_d = 0.62$.

Solution.

Radius of tank, $R = 2$ m

Diameter of the orifice, $d = 40$ mm = 0.04 m

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.04^2 = 1.2566 \times 10^{-3} \text{ m}^2$$

Initial height of water, $H_1 = 1.8$ m

Co-efficient of discharge for the orifice, $C_d = 0.62$

(i) Time required by water to fall from 1.8 m to 1.2 m:

In this case, $H_1 = 1.8$ m and $H_2 = 1.2$ m

Time T is given by :

$$T = \frac{\pi}{C_d \cdot a \cdot \sqrt{2g}} \left[\frac{4}{3} R (H_1^{3/2} - H_2^{3/2}) - \frac{2}{5} (H_1^{5/2} - H_2^{5/2}) \right] \quad \dots(\text{Eqn. 8.15})$$

$$\begin{aligned}
 &= \frac{\pi}{0.62 \times 1.2566 \times 10^{-3} \times \sqrt{2 \times 9.81}} \left[\frac{4}{3} \times 2(1.8^{3/2} - 1.2^{3/2}) - \frac{2}{5}(1.8^{5/2} - 1.2^{5/2}) \right] \\
 &= 910.36 [2.934 - 1.107] = 1663 \text{ seconds} \\
 &= \mathbf{27 \text{ min. } 43 \text{ sec. (Ans.)}
 \end{aligned}$$

(ii) Time required by water for completely emptying the tank:

Here $H_1 = 1.8 \text{ m}$ and $H_2 = 0$

$$\begin{aligned}
 \text{Time } T \text{ is given by: } T &= \frac{\pi}{C_d \cdot a \sqrt{2g}} \left(\frac{4}{3} R H_1^{3/2} - \frac{2}{5} H_1^{5/2} \right) \quad \dots \text{Eqn. (8.16)} \\
 &= \frac{\pi}{0.62 \times 1.2566 \times 10^{-3} \times \sqrt{2 \times 9.81}} \left(\frac{4}{3} \times 2 \times 1.8^{3/2} - \frac{2}{5} \times 1.8^{5/2} \right) \\
 &= 910.36 (6.44 - 1.738) = 4280.5 \text{ seconds} \\
 &= \mathbf{71 \text{ min. } 20.5 \text{ sec. (Ans.)}
 \end{aligned}$$

Example 8.26. A tank has an upper cylindrical portion of 1.25 m radius and 3 m height with hemispherical base. The tank is provided with an orifice of 150 mm diameter at its bottom. Find the time required to empty it if it is initially full of water.

Take $C_d = 0.62$ for the orifice.

Solution. Radius of the tank, $R = 1.25 \text{ m}$

$$\therefore \text{Area, } A = \pi R^2 = \pi \times 1.25^2 = 4.908 \text{ m}^2$$

$$\text{Dia. of orifice, } d = 150 \text{ mm} = 0.15 \text{ m}$$

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

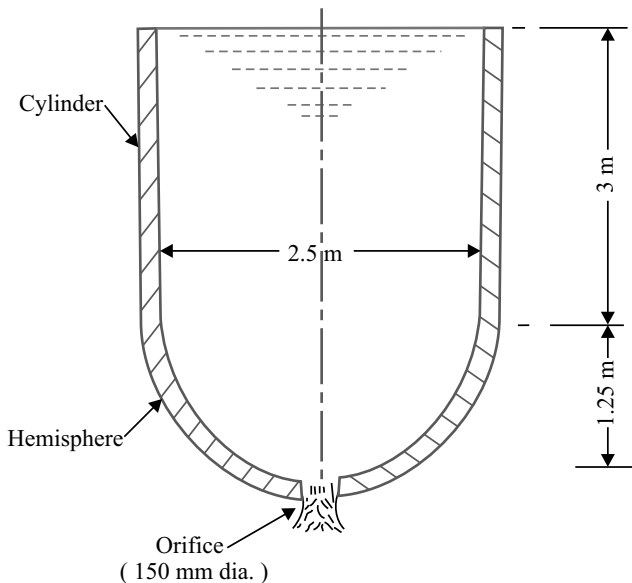


Fig. 8.20

The tank consists of two portions, cylindrical and hemispherical (Fig. 8.20).

Let, T_1 = Time required to empty the upper cylindrical portion, and

T_2 = Time required to empty the hemispherical portion.

Then, total time, = $T_1 + T_2$.

For cylindrical portion :

$$H_1 = 3 + 1.25 = 4.25 \text{ m}, H_2 = 3 \text{ m}$$

$$\begin{aligned} \text{Using the relation: } T_1 &= \frac{2A(\sqrt{H_1} - \sqrt{H_2})}{C_d \cdot a \cdot \sqrt{2g}} \quad \dots(\text{Eqn. 8.13}) \\ &= \frac{2 \times 4.908 (\sqrt{4.25} - \sqrt{1.25})}{0.62 \times 0.01767 \times \sqrt{2 \times 9.81}} \\ &= \frac{9.816 (2.061 - 1.118)}{0.0485} = 190.85 \text{ s} \end{aligned}$$

For hemispherical portion:

$$H_1 = 1.25 \text{ m}, H_2 = 0$$

Using the relation:

$$T_2 = \frac{\pi}{C_d \cdot a \cdot \sqrt{2g}} \left[\frac{4}{3} R H_1^{3/2} - \frac{2}{5} H_1^{5/2} \right] \quad \dots(\text{Eqn. 8.16})$$

Substituting the values, we get:

$$\begin{aligned} T_2 &= \frac{\pi}{0.62 \times 0.01767 \times \sqrt{2 \times 9.81}} \left[\frac{4}{3} \times 1.25 \times 1.25^{3/2} - \frac{2}{5} \times 1.25^{5/2} \right] \\ &= 64.74 (2.329 - 0.698) = 105.6 \text{ s} \end{aligned}$$

∴ Total time taken to empty the tank,

$$T = T_1 + T_2 = 190.85 + 105.6 = \mathbf{296.45 \text{ s (Ans.)}}$$

8.11. TIME REQUIRED FOR EMPTYING A CIRCULAR HORIZONTAL TANK

Consider a circular horizontal tank having an orifice at its bottom and containing some liquid.

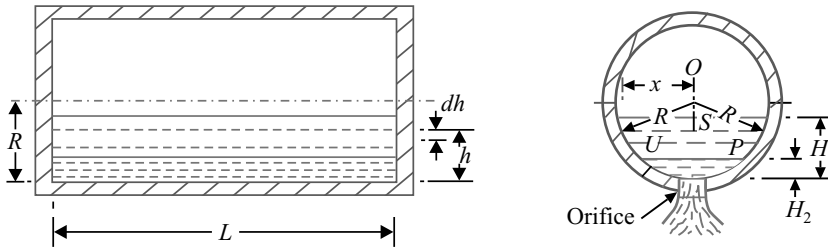


Fig. 8.21

Let,

R = Radius of the tank,

L = Length of the tank,

H_1 = Initial height of the liquid,

H_2 = Final height of the liquid, and

T = Time in seconds for the liquid to fall from height H_1 to H_2 .

Let at any time, the height of liquid over the orifice is h and it decreases dh in a small interval of time dT . Further, let x be the radius of liquid surface at this instant. Then,

$$\begin{aligned} \text{Volume of liquid leaving the tank in time } dT &= A \cdot dh \quad \dots(i) \\ &= A \cdot dh = UP \times L \times dh = 2xL \cdot dh \quad (\because UP = 2x) \end{aligned}$$

(where, A = surface area)

Velocity of liquid through the orifice = $\sqrt{2gh}$

Volume of liquid flowing through the orifice in time dT

$$= C_d \times a \times \sqrt{2gh} \times dT \quad \dots(ii)$$

Volume of liquid leaving the tank equals the volume of liquid flowing through the orifice.

$$\text{i.e.} \quad -2x L \cdot dh = C_d \times a \times \sqrt{2gh} \times dT$$

The negative sign accounts for the decrease in head on the orifice with increase in time interval.

$$\therefore dT = \frac{-2x L \cdot dh}{C_d \cdot a \cdot \sqrt{2gh}} \quad \dots(iii)$$

From Fig. 8.21, we have:

$$OU = R \text{ and } OS = (R - h)$$

$$\therefore x = US = \sqrt{OU^2 - OS^2} = \sqrt{R^2 - (R - h)^2} = \sqrt{2Rh - h^2}$$

Substituting this value of x in eqn. (iii), we get:

$$dT = \frac{-2\sqrt{(2Rh - h^2)} \times L \times dh}{C_d \cdot a \cdot \sqrt{2gh}} = \frac{-2L\sqrt{(2R - h)} \cdot dh}{C_d \cdot a \cdot \sqrt{2g}} \quad (\text{Taking } \sqrt{h} \text{ common})$$

The total time T required to bring the liquid level from height H_1 to H_2 can be found out by integrating the above equation within the limits H_1 and H_2 .

$$\therefore \int_0^T dT = \int_{H_1}^{H_2} \frac{-2L\sqrt{(2R - h)} \cdot dh}{C_d \cdot a \cdot \sqrt{2g}}$$

$$\begin{aligned} \text{or,} \quad T &= \frac{-2L}{C_d \cdot a \cdot \sqrt{2g}} \int_{H_1}^{H_2} (2R - h)^{1/2} dh \\ &= \frac{-2L}{C_d \cdot a \cdot \sqrt{2g}} \times \frac{2}{3} \times [(2R - h)^{3/2} \times (-1)]_{H_1}^{H_2} \end{aligned}$$

$$\text{or,} \quad T = \frac{4L}{3C_d \cdot a \cdot \sqrt{2g}} [(2R - H_2)^{3/2} - (2R - H_1)^{3/2}] \quad \dots(8.17)$$

For emptying the tank completely, $H_2 = 0$ and hence,

$$T = \frac{4L}{3C_d \cdot a \cdot \sqrt{2g}} [(2R)^{3/2} - (2R - H_1)^{3/2}] \quad \dots(8.18)$$

Example 8.27. A horizontal boiler drum 6 m long and 3 m in diameter is provided with an orifice 100 mm in diameter at its bottom. It contains water upto a height of 2.4 m. Calculate the time taken to empty the drum. Take discharge co-efficient, $C_d = 0.6$.

Solution. Length of the drum, $L = 6$ m

Diameter of the drum, $D = 3.0$ m

$\therefore$ Radius, $R = 1.5$ m

Dia. of the orifice, $d = 100$ mm = 0.1 m

$\therefore$ Area, $a = \frac{\pi}{4} \times 0.1^2 = 7.854 \times 10^{-3} \text{ m}^2$

Initial height of water, $H_1 = 2.4$ m

Final height of water, $H_2 = 0$

Time taken to empty the drum, T :

$$\begin{aligned} \text{Using the relation: } T &= \frac{4L}{3C_d \cdot a \cdot \sqrt{2g}} \left[(2R)^{3/2} - (2R - H_1)^{3/2} \right] \quad (\text{Eqn.8.18}) \\ &= \frac{4 \times 6}{3 \times 0.6 \times 7.854 \times 10^{-3} \times \sqrt{2 \times 9.81}} \left[(2 \times 1.5)^{3/2} - (2 \times 1.5 - 2.4)^{3/2} \right] \\ &= 383.26 (5.196 - 0.464) = 1813.6 \text{ s} \\ &= \mathbf{30 \text{ min. } 13.6 \text{ sec. (Ans.)}} \end{aligned}$$

Example 8.28. (Tank with two orifices). A 600 mm diameter cylindrical tank (vertical) contains water to a depth of 2.5 m. There are two orifices in the tank, one at the bottom and the other in one of the vertical sides at a height of 1.5 m above the bottom. if the area of each orifice is 12 cm^2 and discharge co-efficient 0.6, calculate the time required to empty the tank.

Solution. Dia, of the tank, $D = 600 \text{ mm} = 0.6 \text{ m}$

$$\therefore \text{Area, } A = \frac{\pi}{4} \times 0.6^2 = 0.2827 \text{ m}^2$$

Area of each orifice,

$$a = 12 \text{ cm}^2 = 0.0012 \text{ m}^2$$

Discharge co-efficient for each orifice, $C_d = 0.6$.

Time required to empty the tank, T :

Let, T_1 = Time required to bring the water level from LM to NP , and

T_2 = Time required to empty the portion below the section NP .

Total time, $T = T_1 + T_2$

For time T_1 :

Let us consider an instant when the height of water above the center of upper orifice is h metres. Then the height of water above the bottom, at that instant, equals $(h + 1.5)$ metres. If the water level falls by dh during a small time interval dT , then:

Volume of the water leaving the tank in time dT = Discharge through the two orifices in time dT .

$$-A \cdot dh = C_d \cdot a \cdot \sqrt{2g(h + 1.5)} \cdot dT + C_d \cdot a \cdot \sqrt{2gh} \cdot dT$$

$$\therefore dT = \frac{-A \cdot dh}{C_d \cdot a \sqrt{2g} [\sqrt{(h + 1.5)} + \sqrt{h}]}$$

$$\therefore \int_0^{T_1} dT = \frac{-A}{C_d \cdot a \sqrt{2g}} \int_{h=1}^0 \frac{dh}{\sqrt{(h + 1.5)} + \sqrt{h}}$$

$$T_1 = \frac{-A}{C_d \cdot a \sqrt{2g}} \int_{h=1}^{h=0} \frac{[\sqrt{(h + 1.5)} - \sqrt{h}]}{[\sqrt{(h + 1.5)} + \sqrt{h}][\sqrt{(h + 1.5)} - \sqrt{h}]} \cdot dh$$

$$\begin{aligned} T_1 &= \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \int_{h=1}^{h=0} \frac{\sqrt{(h + 1.5)} - \sqrt{h}}{1.5} \cdot dh \\ &= \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \times \frac{1}{1.5} \left[\frac{(h + 1.5)^{3/2}}{3/2} - \frac{h^{3/2}}{3/2} \right]_1^0 \end{aligned}$$

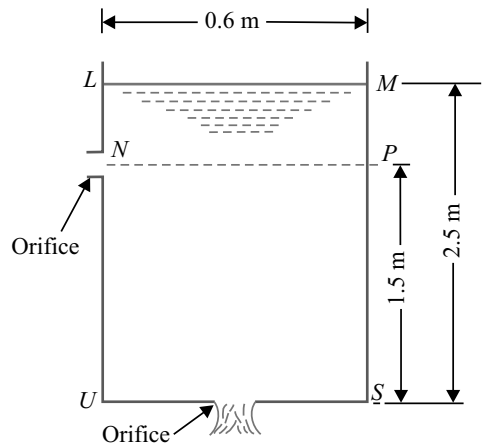


Fig. 8.22

Substituting the values, we get:

$$\begin{aligned}
 T_1 &= \frac{-0.2827}{0.6 \times 0.0012 \times \sqrt{2 \times 9.81}} \times \frac{1}{1.5} \times \frac{2}{3} [(h + 1.5)^{3/2} - h^{3/2}]_1^0 \\
 &= -39.39 [(1.5)^{3/2} - \{(2.5)^{3/2} - (1)^{3/2}\}] \\
 &= -39.39 (1.837 - 3.953 + 1) = 43.96 \text{ s}
 \end{aligned}$$

For time T_2 :

Below the section NP , let h be the height of water above the bottom orifice at any instant, and let the water be leaving in a small time interval dT , then:

Volume of water leaving the tank is time dT

= Discharge through the bottom orifice in time dT

$$-A \cdot dh = C_d \cdot a \cdot \sqrt{2gh} \cdot dT$$

$$\therefore dT = \frac{-A \cdot dh}{C_d \cdot a \cdot \sqrt{2gh}}$$

$$\text{or, } \int_0^{T_2} dT = \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \int_{h=1.5}^0 h^{-1/2} dh$$

$$\begin{aligned}
 \text{or, } T_2 &= \frac{-A}{C_d \cdot a \cdot \sqrt{2g}} \left[\frac{h^{1/2}}{1/2} \right]_{1.5}^0 \\
 &= \frac{-2A}{C_d \cdot a \cdot \sqrt{2g}} [h^{1/2}]_{1.5}^0
 \end{aligned}$$

$$\text{Substituting the values, we get: } T_2 = \frac{-2 \times 0.2827}{0.6 \times 0.0012 \times \sqrt{2 \times 9.81}} (0 - 1.5^{1/2})$$

$$= 177.28 \times 1.5^{1/2} = 217.12 \text{ s}$$

$$\text{Total time, } T = T_1 + T_2 = 43.96 + 217.12 = \mathbf{261.08 \text{ s (Ans.)}}$$

8.12. CLASSIFICATION OF MOUTHPIECES

The mouthpieces may be classified as follows :

1. *According to the position of the mouthpiece :*

- (i) Internal mouthpiece. (ii) External mouthpiece.

2. *According to the shape of the mouthpiece:*

- (i) Cylindrical mouthpiece. (ii) Convergent mouthpiece.
(iii) Convergent - divergent mouthpiece.

3. *According to nature of discharge:*

- (i) Mouthpiece running full. (ii) Mouthpiece running free.

Note : A mouthpiece is said to be *running free* if the jet of liquid after contraction does not touch the sides of the mouthpiece. But if the jet after contraction expands and fills the whole mouthpiece it is known as *running full*.

8.13. DISCHARGE THROUGH AN EXTERNAL MOUTHPIECE

A **mouthpiece** is a small tube (two or three times its diameter in length) attached to an orifice. An external mouthpiece is attached to the vessel such that it projects outside. Fig.8.23 shows a tank to which is attached an external cylindrical mouthpiece.

Let, a_1 = Area of mouthpiece at outlet,
 v_1 = Velocity of liquid at outlet,
 a_c = Area of flow at vena-contracta,
 v_c = Velocity of liquid at C-C section,
 H = Height of liquid above the centre of the mouthpiece, and
 C_c = Co-efficient of contraction.

Applying continuity equation at C-C and 1-1, we get:

$$a_c v_c = a_1 v_1$$

$$\therefore v_c = \frac{a_1 v_1}{a_c} = \frac{v_1}{a_c/a} = \frac{v_1}{C_c}$$

(where $a_c/a = C_c$ = co-efficient of contraction)

Taking $C_c = 0.62$, we get: $v_c = \frac{v_1}{0.62}$

From section C-C the jet of liquid suddenly enlarges at section 1-1; the loss of head due to sudden enlargement is given by:

$$\begin{aligned} h_L &= \frac{(v_c - v_1)^2}{2g} \\ &= \frac{\left(\frac{v_1}{0.62} - v_1\right)^2}{2g} \quad \left(\because v_c = \frac{v_1}{0.62}\right) \\ &= \frac{v_1^2}{2g} \left(\frac{1}{0.62} - 1\right)^2 \\ &= \frac{0.375 v_1^2}{2g} \end{aligned}$$

(Please refer to Art. 12.4.1 for loss of head due to sudden enlargement)

Applying Bernoulli's equation to point A and 1-1, we get:

$$\frac{P_A}{w} + \frac{v_A^2}{2g} + z_A = \frac{p_1}{w} + \frac{v_1^2}{2g} + z_1 + h_L$$

But $z_A = z_1$, $\frac{p_1}{w}$ = atmospheric pressure = 0, and v_A is negligible.

$$\therefore H + 0 = 0 + \frac{v_1^2}{2g} + \frac{0.375 v_1^2}{2g}$$

$$\text{or, } H = \frac{1.375 v_1^2}{2g} \quad \text{or } v_1 = \sqrt{\frac{2gH}{1.375}} = 0.855 \sqrt{2gH}$$

Theoretical velocity of liquid at outlet, $v_{th} = \sqrt{2gH}$

$\therefore$ Co-efficient of velocity for mouthpiece,

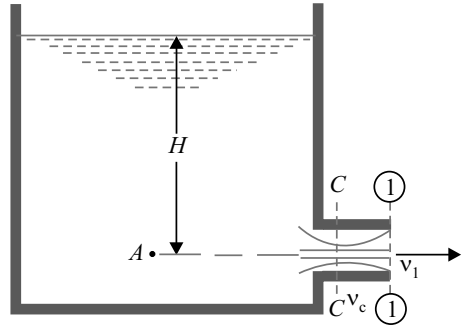


Fig. 8.23. External cylindrical mouthpiece.

$$C_v = \frac{\text{Actual velocity}}{\text{Theoretical velocity}} = \frac{0.855 \sqrt{2gH}}{\sqrt{2gH}} = 0.855$$

For a mouthpiece, since the area of jet of liquid at outlet is equal to the area of mouthpiece at outlet, therefore, $C_c = 1$.

$$\text{Hence } C_d = C_c \times C_v = 1 \times 0.855 = 0.855$$

Thus the value of C_d for mouthpiece is more than the value of C_d for orifice, and so *discharge through mouthpiece will be more.*

Note: In actual practice $C_v = C_d \approx 0.82$.

Example 8.29. Find the discharge from a 80 mm diameter external mouthpiece, fitted to a side of a large vessel, if the head over the mouthpiece is 6 m.

Solution. Dia. of the mouthpiece = 80 mm = 0.08 m

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.08^2 = 0.005026 \text{ m}^2$$

Head over the mouthpiece, $H = 6 \text{ m}$

$$C_d \text{ for the mouthpiece} = 0.855$$

$$\begin{aligned} \therefore \text{Discharge, } Q &= C_d \times \text{area} \times \text{velocity} \\ &= C_d \times a \times \sqrt{2gH} \\ &= 0.855 \times 0.005026 \times \sqrt{2 \times 9.81 \times 6} = \mathbf{0.0466 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

Example 8.30. An external cylindrical mouthpiece of diameter 120 mm is discharging water under a constant head of 6 m. If C_c for vena-contracta = 0.62, $C_d = 0.86$ and atmospheric pressure head = 10.3 m of water, find :

- (i) Discharge through the mouthpiece, and
- (ii) Absolute pressure head of water at vena-contracta.

Solution. Diameter of mouthpiece, $d = 120 \text{ mm} = 0.12 \text{ m}$

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.12^2 = 0.0113 \text{ m}^2$$

Head, $H = 6 \text{ m}$

$$C_c \text{ for vena-contracta} = 0.62$$

$$\text{Discharge co-efficient, } C_d = 0.86$$

$$\text{Atmospheric pressure head, } H_a = 10.3 \text{ m}$$

(i) Discharge through mouthpiece, Q :

$$\begin{aligned} \text{Discharge, } Q &= C_d \cdot a \cdot \sqrt{2gH} = 0.86 \times 0.0113 \times \sqrt{2 \times 9.81 \times 6} \\ &= \mathbf{0.1054 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

(ii) Absolute pressure head at vena-contracta, H_c :

Refer to Fig. 8-24. Applying Bernoulli's equation at A and C-C, we get:

$$\frac{p_A}{w} + \frac{v_A^2}{2g} + z_A = \frac{p_c}{w} + \frac{v_c^2}{2g} + z_c$$

$$\text{But, } \frac{p_A}{w} = H_a + H, \quad v_A = 0 \text{ and } z_A = z_c$$

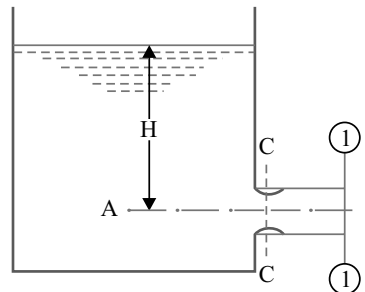


Fig. 8.24

$$\therefore (H_a + H) + 0 = \frac{p_c}{w} + \frac{v_c^2}{2g} = H_c + \frac{v_c^2}{2g}$$

$$\therefore H_c = H_a + H - \frac{v_c^2}{2g}$$

$$\text{But, } v_c = \frac{v_1}{0.62}$$

$$\therefore H_c = H_a + H - \left(\frac{v_1}{0.62} \right)^2 \times \frac{1}{2g} = H_a + H - \frac{v_1^2}{2g} \times \frac{1}{(0.62)^2}$$

$$\text{But, } H = 1.375 \frac{v_1^2}{2g} \text{ or } \frac{v_1^2}{2g} = \frac{H}{1.375} = 0.7272H$$

$$\begin{aligned} \therefore H_c &= H_a + H - 0.7272 H \times \frac{1}{(0.62)^2} \\ &= H_a + H - 1.89 H = H_a - 0.89 H \\ &= 10.3 - 0.89 \times 6 = \mathbf{4.96 \text{ m (absolute) (Ans.)}} \end{aligned}$$

8.14. DISCHARGE THROUGH A CONVERGENT-DIVERGENT MOUTHPIECE

Fig. 8.25. shows a convergent-divergent mouthpiece (which converges upto vena-contracta and then diverges). In this mouthpiece since there is no sudden enlargement of the jet, therefore, the loss of energy due to sudden enlargement is eliminated. For this mouthpiece, $C_d = 1$.

Let, H = Head of liquid over the mouthpiece,

H_a = Atmospheric pressure head, and

H_c = Absolute pressure head at vena-contracta.

Applying Bernoulli's equation at the free water surface and section C-C, we get:

$$\frac{p}{w} + \frac{v^2}{2g} + z = \frac{p_c}{w} + \frac{v_c^2}{2g} + z_c$$

Assuming that datum passes through the centre of the mouthpiece, we have:

$$\frac{p}{w} = H_a, v = 0, \frac{p_c}{w} = H, z_c = 0$$

$$\therefore H_a + 0 + H = H_c + \frac{v_c^2}{2g} + 0 \quad \dots(i)$$

$$\text{or, } \frac{v_c^2}{2g} = H_a + H - H_c \quad \dots(ii)$$

$$\text{or, } v_c = \sqrt{2g (H_a + H - H_c)}$$

Now applying Bernoulli's equation at sections C-C and 1-1, we get:

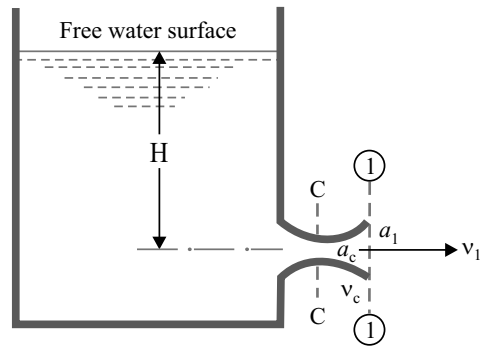


Fig. 8.25. Convergent-divergent mouthpiece.

$$\frac{p_c}{w} + \frac{v_c^2}{2g} + z_c = \frac{p_1}{w} + \frac{v_1^2}{2g} + z_1$$

But, $z_c = z_1$ and $\frac{p_1}{w} = H_a$

$$\therefore H_c + \frac{v_c^2}{2g} = H_a + \frac{v_1^2}{2g}$$

Also from eqn. (i), we have:

$$H_c + \frac{v_c^2}{2g} = H_a + H$$

$$\therefore H_a + \frac{v_1^2}{2g} = H_a + H$$

$$\therefore v_1 = \sqrt{2gH}$$

By continuity equation, we have: $a_c v_c = a_1 v_1$

or,
$$\frac{a_1}{a_c} = \frac{v_c}{v_1} = \frac{\sqrt{2g(H_a + H - H_c)}}{\sqrt{2gH}} = \sqrt{\frac{H_a}{H} + 1 - \frac{H_c}{H}}$$

i.e.
$$\frac{a_1}{a_c} = \sqrt{1 + \frac{H_a - H_c}{H}} \quad \dots(8.19)$$

The discharge, $Q = a_c \times \sqrt{2gH} \quad \dots(8.20)$

Example 8.31. A convergent-divergent mouthpiece having throat diameter 40 mm is discharging water under a constant head of 4 meters. Determine the maximum outlet diameter to avoid separation of the flow, if the maximum vacuum pressure is 8.5 meters of water. Find the discharge also.

Solution. Diameter of throat, $d_c = 40 \text{ mm} = 0.04 \text{ m}$

$$\therefore \text{Area, } a_c = \frac{\pi}{4} \times 0.04^2 = 0.001257 \text{ m}^2$$

Constant head, $H = 4 \text{ m}$

Maximum vacuum pressure head, $H_a - H_c = 8.5 \text{ m}$

Maximum diameter at outlet, d_1 :

Using the relation:

$$\frac{a_1}{a_c} = \sqrt{1 + \frac{H_a - H_c}{H}} \quad \text{with usual notations, we have:}$$

$$\frac{\frac{\pi}{4} \times d_1^2}{\frac{\pi}{4} \times d_c^2} = \sqrt{1 + \frac{8.5}{4}} = 1.767 \quad \text{or} \quad \frac{d_1}{d_c} = 1.329$$

or, $d_1 = 1.329 \times d_c = 1.329 \times 40 \approx \mathbf{53 \text{ mm (Ans.)}}$

Discharge, Q :

Using the relation,

$$Q = a_c \times \sqrt{2gH} \quad \text{with usual notations, we get:}$$

$$Q = 0.001257 \times \sqrt{2 \times 9.81 \times 4} = 0.01113 \text{ m}^3/\text{s (Ans.)}$$

Example 8.32. The diameters of the throat and exit of a convergent-divergent mouthpiece are 40 mm and 80 mm respectively. It is fitted to the vertical side of a tank, containing water. If the maximum vacuum pressure is 7.5 m of water find the maximum head of water for steady flow. Take atmospheric pressure = 10.3 m of water.

Solution. Dia. of throat, $d_c = 40 \text{ mm} = 0.04 \text{ m}$

Dia. of exit, $d_1 = 80 \text{ mm} = 0.08 \text{ m}$

Atmospheric pressure head, $H_a = 10.3 \text{ m of water}$

The maximum vacuum pressure will be at *throat only*, therefore, pressure head at throat = 7.5 m

$$\begin{aligned} \text{or,} \quad H_c &= H_a - 7.5 \text{ (absolute)} \\ &= 10.3 - 7.5 = 2.8 \text{ m (absolute)} \end{aligned}$$

Maximum head of water:

Let the maximum head of water over mouthpiece = H metres of water

Using the relation:

$$\frac{a_1}{a_c} = \sqrt{1 + \frac{H_a - H_c}{H}} \quad \dots(\text{Eqn.8.19})$$

$$\frac{\frac{\pi}{4} d_1^2}{\frac{\pi}{4} d_c^2} = \sqrt{1 + \frac{10.3 - 2.8}{H}} \quad \text{or} \quad \frac{0.08^2}{0.04^2} = \sqrt{1 + \frac{7.5}{H}}$$

$$\text{or,} \quad 4 = \sqrt{1 + \frac{7.5}{H}} \quad \text{or} \quad 16 = 1 + \frac{7.5}{H}$$

$$\text{or,} \quad \mathbf{H = 0.5 \text{ m of water (Ans.)}}$$

Example 8.33. A convergent-divergent mouthpiece is fitted to the side of a tank. It is discharging 5.5 litres/sec. of water under a constant head of 2.0 m. If the head lost in the divergent portion is $\frac{1}{10}$ th of the kinetic head at outlet and the separation pressure is 2.5 m, find the throat and exit diameters. Take atmospheric pressure = 10.3 m of water.

Solution. Discharge through the mouthpiece, $Q = 5.5 \text{ litres/sec.} = 0.0055 \text{ m}^3/\text{s}$

Constant head, $H = 2.0 \text{ m}$

Head lost in divergent portion = $0.1 \times \text{kinetic head at outlet}$

H_c or $H_{sep} = 2.5 \text{ m (abs.)}$

Atmospheric pressure head, $H_a = 10.3 \text{ m. of water}$

Diameter at throat; d_c :

Refer to Fig. 8.25. Applying Bernoulli's equation to the free water surface and throat section, we get:

$$\frac{p}{w} + \frac{v^2}{2g} + z = \frac{p_c}{w} + \frac{v_c^2}{2g} + z_c$$

Assuming that datum passes through the centre of the mouthpiece, we have:

$$H_a + 0 + H = H_c + \frac{v_c^2}{2g}$$

$$\therefore \frac{v^2}{2g} = H_a + H - H_c = 10.3 + 2.0 - 2.5 = 9.8 \text{ m of water}$$

$$\therefore v_c = \sqrt{2 \times 9.81 \times 9.8} = 13.866 \text{ m/s}$$

$$\text{Now, } Q = a_c \times v_c$$

$$\text{or, } 0.0055 = \frac{\pi}{4} d_c^2 \times 13.866$$

$$\text{or, } d_c = \left(\frac{0.0055 \times 4}{\pi \times 13.866} \right)^{1/2} = 0.0225 \text{ m} = \mathbf{22.5 \text{ mm (Ans.)}}$$

Diameter at outlet, d_1 :

Refer to Fig. 8.25. Applying Bernoulli's equation to the free water surface and outlet of mouthpiece, we get:

$$\frac{p}{w} + \frac{v^2}{2g} + z = \frac{p_1}{w} + \frac{v_1^2}{2g} + z_1 + h_L$$

$$H_a + 0 + H = H_a + \frac{v_1^2}{2g} + 0 + 0.1 \times \frac{v_1^2}{2g} \quad \left(\because \frac{p_1}{w} = H_a \right)$$

$$\therefore H = \frac{v_1^2}{2g} + 0.1 \times \frac{v_1^2}{2g} = 1.1 \frac{v_1^2}{2g}$$

$$\text{or, } v_1^2 = \frac{2gH}{1.1} = \frac{2 \times 9.81 \times 2}{1.1} = 35.673$$

$$\therefore v_1 = 5.97 \text{ m/s}$$

$$\text{Now, } Q = a_1 v_1$$

$$\text{or, } 0.0055 = \frac{\pi}{4} d_1^2 \times v_1 = \frac{\pi}{4} d_1^2 \times 5.97$$

$$\therefore d_1 = \left(\frac{0.0055 \times 4}{\pi \times 5.97} \right)^{1/2} = 0.03425 \text{ m} = \mathbf{34.25 \text{ mm (Ans.)}}$$

8.15. DISCHARGE THROUGH AN INTERNAL MOUTHPIECE (OR RE-ENTRANT OR BORDA'S MOUTHPIECE)

An **internal mouthpiece** is short cylindrical tube attached to an orifice in such a way that it (tube) projects inwardly to a tank. If the length of the tube is equal to diameter, the jet of liquid comes out from mouthpiece without touching the sides of the tube (Fig. 8.26); the mouthpiece is known as **running free**. But if the length of the tube is about 3 times its diameter, the jet comes out with its diameter equal to the diameter of mouthpiece at the outlet (Fig. 8.27); the mouthpiece is said to be **running full**.

8.15.1. Mouthpiece Running Free

Consider a mouthpiece *running free* as shown in Fig. 8.26.

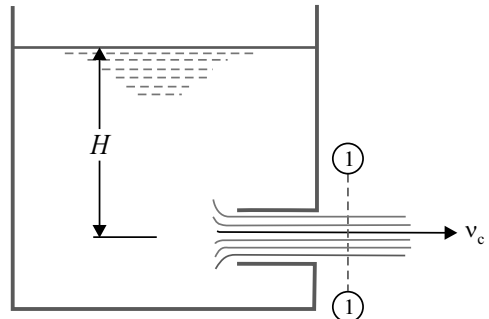


Fig. 8.26. Mouthpiece running free.

Let, H = Height of the liquid above the mouthpiece,
 a = Area of orifice or mouthpiece,
 a_c = Area of contracted jet, and
 v_c = Velocity through mouthpiece.

$\therefore$ Pressure of the liquid on the mouthpiece, $p = wH$

and, force acting on the mouthpiece

$$\begin{aligned} &= \text{Pressure} \times \text{area} \\ &= wH \times a \end{aligned} \quad \dots(i)$$

$$\text{Mass of liquid flowing per second} = \frac{wa_c v_c}{g}$$

Momentum of flowing liquid/sec.

$$\begin{aligned} &= \text{Mass} \times \text{velocity} = \frac{wa_c v_c \times v_c}{g} \\ &= \frac{wa_c v_c^2}{g} \end{aligned} \quad \dots(ii)$$

Since the water is initially at rest, therefore initial momentum = 0

$$\therefore \quad \text{Change of momentum} = \frac{wa_c v_c^2}{g}$$

As per Newton's second law of motion, the force is equal to the rate of change of momentum. Therefore equating (i) and (ii), we get:

$$wH \times a = \frac{wa_c v_c^2}{g}$$

$$H \times a = \frac{a_c v_c^2}{g}$$

$$\frac{v_c^2}{2g} \times a = \frac{a_c v_c^2}{g} \quad \left(\because H = \frac{v_c^2}{2g} \right)$$

$$\therefore \quad a = 2a_c \text{ or } \frac{a_c}{a} = \frac{1}{2} = 0.5$$

$$\therefore \quad \text{Co-efficient of contraction, } C_c = \frac{a_c}{a} = 0.5$$

Since there is no loss of head, co-efficient of velocity, $C_v = 1.0$

$$\therefore \quad \text{Co-efficient of discharge, } C_d = C_c \times C_v = 0.5 \times 1 = 0.5$$

$$\begin{aligned} \therefore \quad \text{Discharge, } Q &= C_d \times a \times \sqrt{2gH} \\ &= 0.5 \times a \times \sqrt{2gH} \end{aligned} \quad \dots(8.21)$$

8.15.2 Mouthpiece Running Full

Consider a mouthpiece *running full* as shown in Fig. 8.27.

Let, a_c = Area at vena-contracta,
 a = Area of orifice or mouthpiece,
 v_c = Velocity of the liquid at C-C (vena-contracta),

v_1 = Velocity of the liquid at 1-1 (or outlet), and

H = Height of liquid above the mouthpiece.

Since the liquid is flowing continuously, therefore from the continuity equation, we have:

$$a_c v_c = a_1 v_1 \quad (a_1 = a)$$

$$v_c = \frac{a_1 v}{a_c} \quad \dots(i)$$

We know that the co-efficient of contraction for an internal mouthpiece is 0.5. Substituting this value of

$$C_c \left(= \frac{a_c}{a_1} \right) = 0.5 \text{ in (i), we get:}$$

$$v_c = 2v_1 \quad \dots(ii)$$

The jet of liquid after passing through C-C, suddenly enlarges at section 1-1. Therefore, there will be loss of head due to sudden enlargement,

$$\begin{aligned} h_L &= \frac{(v_c - v_1)^2}{2g} = \frac{(2v_1 - v_1)^2}{2g} \quad (\because v_c = 2v_1) \\ &= \frac{v_1^2}{2g} \end{aligned}$$

Applying Bernoulli's equation to free water surface in tank and section 1-1 (or outlet), we get:

$$\frac{p}{w} + \frac{v^2}{2g} + z = \frac{p_1}{w} + \frac{v_1^2}{2g} + z_1 + h_L$$

Assuming datum line passing through the centre line of mouthpiece

$$0 + 0 + H = 0 + \frac{v_1^2}{2g} + 0 + \frac{v_1^2}{2g}$$

$$\therefore H = \frac{v_1^2}{2g} + \frac{v_1^2}{2g} = \frac{v_1^2}{g} \quad \text{or} \quad v_1 = \sqrt{gH}$$

Here v_1 is the actual velocity as losses have been taken into account.

But theoretical velocity,

$$v_{th} = \sqrt{2gH}$$

$$\therefore \text{Co-efficient of velocity, } C_v = \frac{v_1}{v_{th}} = \frac{\sqrt{gH}}{\sqrt{2gH}} = \frac{1}{\sqrt{2}}$$

As the area of the jet at outlet is equal to the area of the mouthpiece, hence co-efficient of contraction = 1

$$\therefore C_d = C_c \times C_v = 1 \times \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} = 0.707$$

$$\therefore \text{Discharge, } Q = C_d \times a \times \sqrt{2gH} = 0.707 \times a \times \sqrt{2gH} \quad \dots(8.22)$$

Example 8.34. An internal mouthpiece of 100 mm diameter is discharging water under a constant head of 5 m. Find the discharge through mouthpiece, when :

(i) The mouthpiece is running free, and

(ii) The mouthpiece is running full.

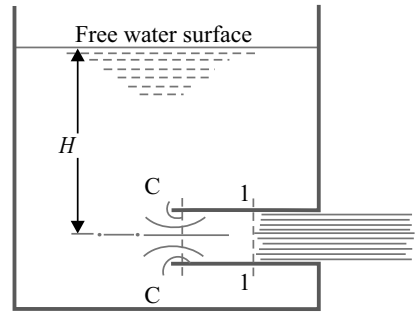


Fig. 8.27. Mouthpiece running full.

Solution. Dia. of mouthpiece, $d = 100 \text{ mm} = 0.1 \text{ m}$

$$\therefore \text{Area, } a = \frac{\pi}{4} \times 0.1^2 = 0.00785 \text{ m}^2$$

Constant head, $H = 5 \text{ m}$

Discharge, Q:

(i) When mouthpiece is running free:

Using the relation:

$$Q = 0.5 \times a \times \sqrt{2gH} \quad \dots[\text{Eqn. (8.21)}]$$

$$= 0.5 \times 0.00785 \times \sqrt{2 \times 9.81 \times 5}$$

$$= \mathbf{0.0388 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

(ii) When mouthpiece is running full:

Using the relation:

$$Q = 0.707 \times a \times \sqrt{2gH} \quad \dots[\text{Eqn. (8.22)}]$$

$$= 0.707 \times 0.00785 \times \sqrt{2 \times 9.81 \times 5}$$

$$= \mathbf{0.0549 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

Example 8.35. An external mouthpiece converges from inlet up to the vena-contracta to the shape of the jet and then diverges gradually. The diameter at vena-contracta is 20 mm and the head over the centre of the mouthpiece is 1.44 m. The head loss in the contraction may be taken as 1% and that in the divergent portion 5% of the total energy head before the inlet. What is the maximum discharge that can be drawn through the outlet and what should be the corresponding diameter at the outlet?

Assume that the pressure in the system may be permitted to fall upto 8 m below atmosphere, the liquid conveyed being water. [UPSC Exam, Fluid Mech.]

Solution. The dia. of mouthpiece at vena-contracta,

$$d_c = 20 \text{ mm} = 0.02 \text{ m}$$

$$\therefore \text{Area, } a_c = \frac{\pi}{4} \times 0.02^2 = 3.141 \times 10^{-4} \text{ m}^2$$

Head over the centre of the mouthpiece,
 $H = 1.44 \text{ m}$

Head loss in the contraction

$$= \frac{1}{100} \times H = \frac{1}{100} \times 1.44 = 0.0144 \text{ m}$$

Head loss in the divergent portion

$$= \frac{5}{100} H = \frac{5}{100} \times 1.44 = 0.072 \text{ m}$$

Head at the vena-contracta, $H_c = -8 \text{ m}$

Maximum discharge, Q:

Applying Bernoulli's equation at the free water surface and section C-C, we get:

$$\frac{p}{w} + \frac{v^2}{2g} + z = \frac{p_c}{w} + \frac{v_c^2}{2g} + z_c + \text{head loss in the contraction}$$

Assuming that datum passes through the centre of the mouthpiece, we have:

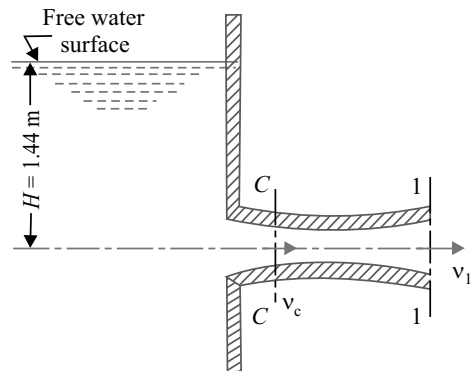


Fig. 8.28

$$0 + 0 + 1.44 = -8.0 + \frac{v_c^2}{2g} + 0 + 0.0144$$

$$\text{or, } \frac{v_c^2}{2g} = 1.44 + 8 - 0.0144 = 9.426$$

$$\therefore v_c = (2 \times 9.81 \times 9.426)^{1/2} = 13.6 \text{ m/s.}$$

$$\begin{aligned} \text{Maximum discharge, } Q &= a_c \cdot v_c \\ &= 3.141 \times 10^{-4} \times 13.6 = 4.272 \times 10^{-3} \text{ m}^3/\text{s} \text{ or } \mathbf{4.272 \text{ litres/s (Ans.)}} \end{aligned}$$

Diameter at the outlet, d_1 :

Now applying Bernoulli's equation at the free water surface and the exit end of the mouthpiece, we get:

$$\frac{p}{w} + \frac{v^2}{2g} + z = \frac{p_1}{w} + \frac{v_1^2}{2g} + z_1 + \text{total head loss in the contraction and divergent portion}$$

$$\text{or, } 0 + 0 + 1.44 = 0 + \frac{v_1^2}{2g} + 0 + (0.0144 + 0.072)$$

$$\therefore \frac{v_1^2}{2g} = 1.44 - (0.0144 + 0.072) = 1.354$$

$$\text{or } v_1 = (2 \times 9.81 \times 1.354)^{1/2} = 5.15 \text{ m/s}$$

$$\text{Also, Discharge } Q = a_1 \cdot v_1 \text{ or } 4.27 \times 10^{-3} = \frac{\pi}{4} \times d_1^2 \times 5.15$$

$$\therefore d_1 = \left(\frac{4.272 \times 10^{-3} \times 4}{\pi \times 5.15} \right)^{1/2} = 0.0325 \text{ m} = \mathbf{32.5 \text{ mm (Ans.)}}$$

Example 8.36. A streamlined nozzle of diameter d is supplied at constant head, the magnitude of the head being larger compared to d . The nozzle discharges directly into the atmosphere and is so shaped that the issuing jet is parallel at the nozzle exit. To increase the flow rate a shroud of the diameter D is firmly secured to the nozzle as shown in Fig. 8-29. The jet expands to fill the shroud and the shroud is long enough to ensure that the flow leaving is steady and parallel. Determine:

- (i) The diameter of the shroud so that the flow rate is maximised, and
- (ii) The percentage increase in discharge.

Neglect shear stresses at the walls of the shroud.

[UPSC Fluid Mech. and FluidM/C.]

Solution. $H > d$ (given).

(i) Diameter of the shroud, D

Applying Bernoulli's equation between section (1) and (2), we get:

$$\frac{p_1}{w} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{v_2^2}{2g} + z_2 + \text{loss due to sudden expansion}$$

$$\frac{p_1}{w} + \frac{v_1^2}{2g} = 0 + \frac{v_2^2}{2g} + \frac{(v_1 - v_2)^2}{2g} \quad (\because z_2 = z_2) \dots(i)$$

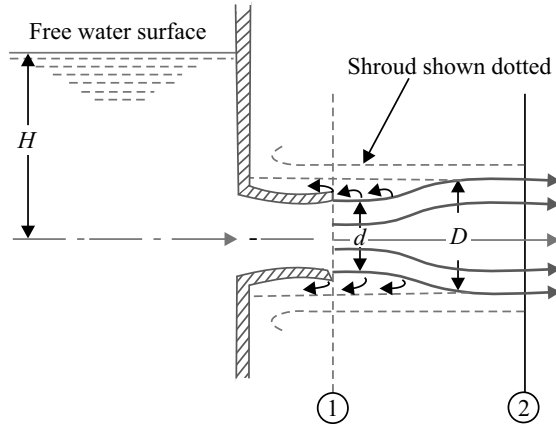


Fig. 8.29

Now, applying Bernoulli's equation between free water surface and section (1), we get:

$$\frac{p}{w} + \frac{v^2}{2g} + z = \frac{p_1}{w} + \frac{v_1^2}{2g} + z_1$$

$$0 + 0 + H = \frac{p_1}{w} + \frac{v_1^2}{2g} + 0$$

$$\text{or,} \quad H = \frac{p_1}{w} + \frac{v_1^2}{2g} \quad \dots(ii)$$

From continuity equation, we have:

$$Q = a_1 v_1 = a_2 v_2$$

$$\text{or,} \quad Q = \frac{\pi}{4} \times d^2 \times v_1 = \frac{\pi}{4} \times D^2 \times v_2 \quad \dots(iii)$$

Eliminating p_1 between eqns. (i) and (ii) and using eqn. (iii), we get:

$$\begin{aligned} H &= \frac{v_2^2}{2g} + \frac{(v_1 - v_2)^2}{2g} \\ &= \frac{1}{2g} \left(\frac{4Q}{\pi D^2} \right) + \frac{1}{2g} \left(\frac{4Q}{\pi d^2} - \frac{4Q}{\pi D^2} \right)^2 \\ &= \frac{8Q^2}{\pi^2 g} \left[\frac{1}{D^4} + \left(\frac{1}{d^2} - \frac{1}{D^2} \right)^2 \right] \\ &= \frac{8Q^2}{\pi^2 g} \left[\frac{1}{D^4} + \frac{1}{d^4} + \frac{1}{D^4} - \frac{2}{D^2 d^2} \right] \\ &= \frac{8Q^2}{\pi^2 g} \left[\frac{1}{D^4} + \frac{1}{d^4} - \frac{2}{D^2 d^2} \right] = \frac{8Q^2}{\pi^2 g} \left[\frac{2d^4 + D^4 - 2D^2 d^2}{D^4 d^4} \right] \\ &= \frac{8Q^2}{\pi^2 g} \left[\frac{d^4 + d^4 + D^4 - 2D^2 d^2}{D^4 d^4} \right] = \frac{8Q^2}{\pi^2 g} \left[\frac{d^4 + (D^2 - d^2)^2}{D^4 d^4} \right] \end{aligned}$$

$$\text{or, } Q^2 = \frac{\pi^2 gH}{8} \left[\frac{D^4 d^4}{d^4 + (D^2 - d^2)^2} \right]$$

$$\therefore Q = \frac{\sqrt{\pi^2 gH}}{8} \times \frac{D^2 d^2}{\sqrt{d^4 + (D^2 - d^2)^2}} \quad \dots(iv)$$

For maximum Discharge, $\frac{dQ}{dD} = 0$

$$\text{or, } \frac{d}{dD} \left[\sqrt{\frac{\pi^2 gH}{8}} \times \frac{D^2 d^2}{\sqrt{d^4 + (D^2 - d^2)^2}} \right] = 0$$

$$\text{or, } \frac{1}{\sqrt{d^4 + (D^2 - d^2)^2}} \times (2Dd^2) + D^2 d^2 \left[-\frac{1}{2} \{d^4 + (D^2 - d^2)^2\}^{-3/2} \right] 2(D^2 - d^2) \cdot 2D = 0$$

$$\text{or, } \frac{1}{\sqrt{d^4 + (D^2 - d^2)^2}} \times 2d^2 - 2D^2 d^2 \left[\frac{1}{\{d^4 + (D^2 - d^2)^2\}^{3/2}} \right] \times (D^2 - d^2) = 0$$

$$\text{or, } 2d^2 - 2D^2 d^2 (D^2 - d^2) \times \frac{1}{[d^4 + (D^2 - d^2)^2]} = 0$$

$$\text{or, } 2d^2[d^4 + (D^2 - d^2)^2] - 2D^2 d^2 (D^2 - d^2) = 0$$

$$\text{or, } d^4 + (D^2 - d^2)^2 - D^2 (D^2 - d^2) = 0$$

$$\text{or, } d^4 + (D^2 - d^2)^2 = D^2 (D^2 - d^2)$$

$$\text{or, } d^4 + D^4 + d^4 - 2D^2 d^2 = D^4 - D^2 d^2$$

$$\text{or, } 2d^4 = D^2 d^2$$

$$\text{or, } D = \sqrt{2}d \quad (\text{Ans.})$$

(ii) Percentage increase in discharge:

The maximum discharge,

$$Q_{\max} = \sqrt{\frac{\pi^2 gH}{8}} \times \frac{(\sqrt{2}d)^2 d^2}{\sqrt{d^4 + (2d^2 - d^2)^2}} \quad [\text{Substituting } D = \sqrt{2}d \text{ in eqn. (iv)}]$$

$$= \frac{\pi d^2}{2\sqrt{2}} \sqrt{2gH}$$

$\therefore$ Percentage increase in discharge

$$\begin{aligned} &= \frac{Q_{\max} - Q}{Q} \times 100 = \frac{\frac{\pi d^2}{2\sqrt{2}} \sqrt{2gH} - \frac{\pi d^2}{4} \sqrt{2gH}}{\frac{\pi d^2}{4} \sqrt{2gH}} \times 100 \\ &= (\sqrt{2} - 1) \times 100 = 41.42\% \quad (\text{Ans.}) \end{aligned}$$

HIGHLIGHTS

1. An *orifice* is an opening in the wall or base of a vessel through which the fluid flows. The top edge of the orifice is always *below* the free surface.
2. A *mouthpiece* is an attachment in the form of a small tube or pipe fixed to the orifice (the length of pipe extension is usually 2 to 3 times the orifice diameter) and is used to increase the amount of discharge.
3. Theoretical velocity of jet of water from orifice is given as: $V = \sqrt{2gH}$
4. There are three important hydraulic co-efficients namely

$$(i) \quad \text{Co-efficient of contraction, } C_c = \frac{\text{Area of jet at vena-contracta}}{\text{Area of orifice}}$$

$$(ii) \quad \text{Co-efficient of velocity, } C_v = \frac{\text{Actual velocity at vena-contracta}}{\text{Theoretical velocity}} \\ = \frac{x}{\sqrt{4yH}} \cdot$$

(where, x and y are the co-ordinates of any point of jet of water from vena-contracta)

$$(iii) \quad \text{Co-efficient of discharge, } C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}} = C_c \times C_v$$

5. Discharge through a *large rectangular orifice*,

$$Q = \frac{2}{3} C_d \times b \times \sqrt{2g} [H_2^{3/2} - H_1^{3/2}]$$

where,

C_d = Discharge co-efficient for the orifice,

b = Width of orifice,

H_1 = Height of liquid above top edge of orifice, and

H_2 = Height of liquid above bottom edge of orifice.

(A large orifice is one, where the head of liquid above the centre of orifice is *less* than 5 times the depth of orifice)

6. Discharge through *fully submerged orifice*,

$$Q = C_d \times b \times (H_2 - H_1) \times \sqrt{2gH}$$

where,

C_d = Discharge co-efficient for the orifice,

b = Width of orifice,

H_2 = Height of liquid above bottom edge of orifice on upstream side

H_1 = Height of liquid above top edge of orifice on upstream side, and

H = Difference of liquid levels on both sides of the orifice.

7. Discharge through *partially submerged orifice*,

$$Q = C_d \cdot b \cdot (H_2 - H) \sqrt{2gH} + \frac{2}{3} C_d \cdot b \cdot \sqrt{2g} (H^{3/2} - H_1^{3/2})$$

where C_d , b , H , H_1 , H_2 have usual meanings.

8. Time of emptying a tank through an orifice at its bottom,

$$T = \frac{2A (\sqrt{H_1} - \sqrt{H_2})}{C_d . a . \sqrt{2g}}$$

where,

A = Area of tank,

a = Area of orifice,

H_1 = Initial height of liquid in tank,

H_2 = Final height of liquid in tank, and

C_d = Co-efficient of orifice.

If the tank is to be completely emptied, then:

$$T = \frac{2A \sqrt{H}}{C_d . a . \sqrt{2g}}$$

9. Time of emptying a hemispherical tank by an orifice fitted at its bottom,

$$T = \frac{\pi}{C_d . a . \sqrt{2g}} \left[\frac{4}{3} R (H_1^{3/2} - H_2^{3/2}) - \frac{2}{5} (H_1^{5/2} - H_2^{5/2}) \right]$$

and for completely emptying the tank,

$$T = \frac{\pi}{C_d . a . \sqrt{2g}} \left[\frac{4}{3} R H_1^{3/2} - \frac{2}{5} H_1^{5/2} \right]$$

where,

a = Area of orifice,

R = Radius of the hemispherical tank,

H_1 = Initial height of liquid,

H_2 = Final height of liquid, and

C_d = Co-efficient of discharge.

10. Time of emptying a circular horizontal tank by an orifice at the bottom of the tank,

$$T = \frac{4L}{3 C_d . a . \sqrt{2g}} \left[(2R - H_2^{3/2}) - (2R - H_1)^{3/2} \right]$$

and for completely emptying the tank,

$$T = \frac{4L}{3 C_d . a . \sqrt{2g}} \left[(2R)^{3/2} - (2R - H_1)^{3/2} \right]$$

where, L = length of horizontal tank.

11. Co-efficient of discharge C_d

(i) External mouthpiece = 0.855

(ii) Internal mouthpiece; running full, = 0.707
running free = 0.50

(iii) Convergent - divergent mouthpiece = 1.0

12. Absolute pressure head, for an external mouthpiece, at vena-contracta,

$$H_c = H_a - 0.89 H$$

where,

H_a = Atmospheric pressure head = 10.3 m of water, and

H = Head of liquid above the mouthpiece.

13. In case of a convergent-divergent mouthpiece, the ratio of areas at outlet and at vena-contracta is given by:

$$\frac{a_1}{a_c} = \sqrt{1 + \frac{H_a - H_c}{H}}$$

where,

a_1 = Area of mouthpiece at outlet,

a_c = Area of mouthpiece at vena-contracta,

H_a = Atmospheric pressure head,

H_c = Absolute pressure head at vena-contracta, and

H = Height of liquid above mouthpiece.

OBJECTIVE TYPE QUESTIONS

Fill in the Blanks/Choose the Correct Answer

- An is an opening in the wall or base of a vessel through which the fluid flows.
- The top edge of the orifice is always below/above the free surface.
- A mouthpiece is used to decrease /increase the amount of discharge.
- An orifice is said to be submerged/discharging free when it discharges into another liquid.
- In orifice it is convenient to work in terms of gauge/absolute pressures.
- Co-efficient of contraction (C_c) is equal to
 - a_c/a
 - a/a_c
 - $a \times a_c$
 - $\sqrt{a_c/a}$
- For sharp-edged orifices the value of C_v is taken as,
 - 0.82
 - 0.84
 - 0.9
 - 0.98.
- The value of C_d varies from,
 - 0.2 to 0.3
 - 0.3 to 0.4
 - 0.4 to 0.5
 - 0.6 to 0.65.
- The discharge through a large rectangular orifice is given by
 - $\frac{1}{3} C_d \cdot b \cdot \sqrt{2g} (\sqrt{H_2} - \sqrt{H_1})$
 - $\frac{2}{3} C_d \cdot b \cdot 2g (\sqrt{H_2} - \sqrt{H_1})$
 - $\frac{2}{3} C_d \cdot b \cdot \sqrt{2g} (H_2^{3/2} - H_1^{3/2})$
 - $\frac{2}{3} C_d \cdot \sqrt{b} \cdot 2g (\sqrt{H_2} - \sqrt{H_1})$
- The discharge through a fully submerged orifice is given by,
 - $C_d \cdot b \cdot (H_2 - H_1) \times 2gH$
 - $C_d \cdot \sqrt{b} \cdot (H_2 - H_1) \times \sqrt{2gH}$
 - $C_d \cdot b^2 (H_2 - H_1) \times \sqrt{2gH}$
 - $C_d \cdot b \cdot (H_2 - H_1) \times \sqrt{2gH}$
- Time of emptying a tank through an orifice at its bottom is given by,
 - $\frac{2A \sqrt{H_1}}{C_d \cdot a^2 \cdot 2g}$
 - $\frac{2A \sqrt{H_1}}{C_d \cdot a \cdot \sqrt{2g}}$
 - $\frac{A \sqrt{H_1}}{C_d \cdot a \sqrt{2g}}$
 - $\frac{4A \sqrt{H_1}}{C_d \cdot a \sqrt{2g}}$
- Discharge through an internal mouthpiece running free is given by
 - $0.5 \times a \times \sqrt{2gH}$
 - $0.4 \times a^2 \times \sqrt{2gH}$
 - $0.707 \times a \times \sqrt{2gH}$
 - $0.3 \times a^2 \times \sqrt{2gH}$

ANSWERS

- | | | | | | |
|------------|----------|-------------|--------------|----------|----------|
| 1. Orifice | 2. Below | 3. Increase | 4. Submerged | 5. Gauge | 6. (a) |
| 7. (d) | 8. (d) | 9. (c) | 10. (d) | 11. (b) | 12. (a). |

THEORETICAL QUESTIONS

- What is an orifice? How are the orifices classified?
- What is a mouthpiece?
- What is Torricelli's theorem?
- Name and explain briefly the hydraulic co-efficients.
- Derive the expression $C_d = C_c \times C_v$.
- How is vena contracta defined?
- How are hydraulic co-efficients determined experimentally?
- What is the difference between a small and a large orifice?
- Obtain an expression for discharge through a large orifice.
- State the difference between a wholly submerged orifice and a partially submerged orifice.
- Derive an expression for discharge through fully submerged orifice.
- Obtain an expression for time of emptying a tank through an orifice at its bottom.
- Derive an expression for time of emptying a circular hemispherical tank.
- Obtain an expression for time of emptying a circular horizontal tank.
- How are mouthpieces classified?
- Obtain an expression for absolute pressure head at vena contracta for an external mouthpiece.
- Derive an expression for the ratio of diameters at outlet and at vena-contracta for a convergent-divergent mouthpiece in terms of absolute pressure head at vena-contracta, head of liquid above mouthpiece and atmospheric pressure head.

UNSOLVED EXAMPLES

ORIFICES

- An orifice 60 mm in diameter is discharging water under a head of 9 metres. If $C_d = 0.6$ and $C_v = 0.9$ find :
(i) Actual discharge, and
(ii) Actual velocity of the jet at vena-contracta.
[Ans. (i) 0.02254 m³/s, (ii) 11.26 m/s]
- The head of water over the centre of an orifice of diameter 20 mm is 1 m. The actual discharge through the orifice is 0.85 litres/s. Find the co-efficient of discharge. [Ans. 0.61]
- A jet of water issues from a circular orifice of 25 mm diameter, under a constant head of 1 metre. It falls 35 mm vertically down and strikes the ground at a distance of 350 mm from the center of the vena contracta. If the discharge through the jet is 1.35 litres/s find:
(i) Co-efficient of discharge;
(ii) Co-efficient of velocity;
(iii) Co-efficient of contraction.
[Ans. (i) 0.625, (ii) 0.935, (iii) 0.668]
- The water is coming out of an orifice of diameter 100 mm under a head of 10 m. It is collected in a circular tank of diameter 1.5 m. The rise of water level in this tank is 1.0 m. in 25 seconds. Also the co-ordinates of a point on the jet, measured from venacontracta are 4.3 m horizontal and 0.5 m vertical. Find the hydraulic co-efficients (i.e. co-efficient of discharge, co-efficient of velocity and co-efficient of contraction).
[Ans. 0.643; 0.96; 0.669]
- The head of water over an orifice of diameter 100 mm is 10 m. The water coming out from orifice is collected in circular tank of diameter 1.5 m. The rise of water level in this tank is 1.0 m in 25 seconds. Find the co-efficient of discharge. [Ans. 0.643]
- An orifice, 60 mm in diameter, is discharging water under a head of 9 metres. Calculate the discharge and actual velocity of the jet at vena contracta, if $C_d = 0.6$ and $C_v = 0.9$.
[Ans. 0.02254 m³/s; 11.96 m/s]
- A tank has two identical orifices in one of its vertical sides. The upper orifice is 2 metres below the water surface. and the lower one is 4 metres below the water surface. If the value of C_v for each orifice is 0.9, find the point of intersection of the two jets. [Ans. 5.1m]
- A closed tank has water to a height of 0.9 m, above 15 mm diameter sharp-edged orifice provided at the bottom of the tank. To what pressure air must be pumped into the tank above water if the discharge is to be 1.5 litres/sec. ?
Take discharge co-efficient, $C_d = 0.62$ for the orifice. [Ans. 91.25 kN/m²]
- A 100 mm diameter orifice discharges 45 litres/sec. of water under a head of 2.75 metres. A flat plate held normal to the jet just downstream from the vena contracta requires a force of 310 N to resist the impact of jet. Find C_c, C_v, C_d .
[Ans. 0.84, 0.927, 0.78]
- A rectangular orifice 1.5 m wide and 1.0 m deep is discharging water from a tank. If the water level in the tank is 3.0 m above the top edge

of the orifice, find the discharge through the orifice. Take $C_d = 0.6$ [Ans. $7.45 \text{ m}^3/\text{s}$]

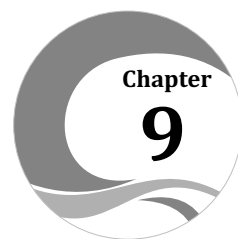
11. A submerged orifice 1 meter wide has height of water from the bottom and top of the orifice as 2.25 metres respectively. Find the discharge through the orifice, if the difference of water levels on both the sides of the orifice be 375 mm. Take $C_d = 0.62$ [Ans. $0.42 \text{ m}^3/\text{s}$]
12. An orifice, in one side of a large tank, is rectangular in shape 2 m broad and 1 m deep. The water level on one side of the orifice is 4 m above its top edge. The water level on the other side of the orifice is 0.5 m below its top edge. Calculate the discharge through the orifice per second, if $C_d = 0.625$ [Ans. $11.58 \text{ m}^3/\text{s}$]
13. A circular tank, of diameter 4 m contains water upto a height of 5 m. The tank is provided with an orifice of diameter 0.5 m at the bottom. Find the time taken by water,
 - (i) to fall from 5 m to 2 m, and
 - (ii) for completely emptying the tank.
 Take $C_d = 0.6$ [Ans. (i) 39.6 s, (ii) 107.7 s]
14. A 1.25 m diameter circular tank contains water upto a height of 5 m. At the bottom of the tank an orifice of 50 mm is provided. Find the height of water above the orifice after 1.5 minutes. Take $C_d = 0.62$. [Ans. 4.154 m]
15. A hemispherical cistern of 6 m radius is full of water. It is fitted with a 75 mm diameter sharp edged orifice at the bottom. Calculate the time

required to lower the level in the cistern by 2 meters. Assume co-efficient of discharge for the orifice as 0.6. [Ans. 2 h 18 min 42 s]

16. A tank has an upper cylindrical portion of 5 m diameter and 4 m high with hemispherical base. Find the time required to empty it through an orifice of 200 mm diameter at its bottom, if the tank is initially full of water. Take $C_d = 0.6$ for the orifice. [Ans. 1078 s]
17. A horizontal boiler 5 m long and 3 m in diameter is half full of water. It is provided with a 100 mm diameter orifice at its bottom. Calculate the time taken to empty the drum.
Take $C_d = 0.6$ [Ans. 17 min 50 sec.]
18. A 30 m long and 10 m wide swimming pool has vertical sides and bottom at a slope. The depth of water at the shallow and deep sides are 2 m and 5 m respectively. Two outlets, each 0.3 m diameter, have been provided at each of the deep and shallow ends. Calculate the time taken to empty the pool if both the outlets are kept open. Take $C_d = 0.62$ for each opening. [Ans. 2871 s]
19. A 750 mm diameter vertical cylindrical tank contains water to a depth of 2.5 m. There are two orifices in the tank; one at the bottom and the other in one of the vertical sides at a height of 1.5 m above the bottom. If the area of each orifice is 15 cm^2 and discharge co-efficient 0.6, calculate the time required to empty the tank. [Ans. 326.3 s]

MOUTHPIECES

20. Find the discharge from a 100 mm diameter external mouthpiece, fitted to a side of a vessel, if the head over the mouthpiece is 4 m. [Ans. $0.595 \text{ m}^3/\text{s}$]
21. An external cylindrical mouthpiece of diameter 150 mm is discharging water under a constant head of 6 m. If C_c for vena-contracta = 0.62, $C_d = 0.855$ and atmospheric pressure head = 10.3 m of water, find:
 - (i) Discharge through the mouthpiece, and
 - (ii) Absolute pressure head of water at vena contracta.
 [Ans. (i) $0.1639 \text{ m}^3/\text{s}$, (ii) 496 m (abs.)]
22. A convergent-divergent mouthpiece having throat diameter of 40 mm is discharging water under a constant head of 2 m. Determine the maximum outlet diameter for maximum discharge. Find the maximum discharge also. Take $H_a = 10.3$ m of water and $H_{\text{sep}} = 2.5$ m absolute. [Ans. 59.5 mm; $0.00787 \text{ m}^3/\text{s}$]
23. An internal mouthpiece of 80 mm diameter is discharging under a constant head of 8 m. Find the discharge through the mouthpiece, when
 - (i) the mouthpiece is running free and (ii) the mouthpiece is running full.
 [Ans. (i) $0.02226 \text{ m}^3/\text{s}$, (ii) $0.03147 \text{ m}^3/\text{s}$]
24. A convergent-divergent mouthpiece is fitted into the vertical side of a tank containing water. Assuming that there are no losses in the convergent part of the mouthpiece, and that the losses in the divergent part are equivalent to 0.2 times the velocity head at exit and that the maximum absolute pressure head at the throat is 2.44 m of water for a barometric pressure of 760 mm of mercury, determine the throat and exit diameters of the mouthpiece when the discharge is 4.25 lit./sec. for a head of 1.52 metres. [UPSC Exams.] [Ans. 20 mm, 33 mm]



FLOW OVER NOTCHES AND WEIRS

- 9.1. Definitions
- 9.2. Types of notches and weirs
- 9.3. Discharge over a rectangular notch or weir
- 9.4. Discharge over a triangular notch or weir
- 9.5. Discharge over a trapezoidal notch or weir
- 9.6. Discharge over a stepped notch.
- 9.7. Effect on discharge over a notch or weir due to error in the measurement of head
- 9.8. Velocity of approach
- 9.9. Empirical formulae for discharge over rectangular weir
- 9.10. Cippoletti weir or notch
- 9.11. Discharge over a broad-crested weir
- 9.12. Discharge over a narrow-crested weir
- 9.13. Discharge over an Ogee weir
- 9.14. Discharge over submerged or drowned weir
- 9.15. Time required to empty a reservoir or a tank with rectangular and triangular weirs or notches

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

9.1. DEFINITIONS

Notch. A **notch** may be defined as an opening provided in the side of a tank or vessel such that the liquid surface in the tank is below the top edge of the opening. A notch may be regarded as an orifice with the water surface below its upper edge. It is generally made of metallic plate. It is used for measuring the rate of flow of a liquid through a small channel or a tank.

Weir. A **weir** may be defined as any regular obstruction in an open stream over which the flow takes place. It is made of masonry or concrete. The conditions of flow in the case of a weir are practically the same as those of a rectangular notch. That is why, a notch is sometimes called as a weir and vice versa.

Weirs may be used for measuring the rate of flow of water in rivers or streams.

— *Nappe or vein.* The sheet of water flowing through a notch or over a weir is known as the *nappe* or *vein*.

— *Sill or crest.* The top of the weir over which the water flows is known as the *sill* or *crest*.

Note. The main difference between a notch and a weir is that the *notch* is of small size, but the *weir* is of a bigger one. Moreover a notch is usually made in a *plate*, whereas a weir is usually made of *masonry or concrete*.

9.2. TYPES/CLASSIFICATION OF NOTCHES AND WEIRS

9.2.1 Types of Notches

There are several types of notches, depending upon their shapes. However, the following are important from subject point of view:

1. Rectangular notch,
2. Triangular notch,
3. Trapezoidal notch, and
4. Stepped notch.

9.2.2 Types of Weirs

There are several types of weirs depending upon their shapes, nature of discharge, width of crest or nature of crest. However, the following are important from subject point of view:

1. *According to shape:*
 - (i) Rectangular weir, and
 - (ii) Cippolletti weir.
2. *According to nature of discharge:*
 - (i) Ordinary weir, and
 - (ii) Submerged or drowned weir.
3. *According to the width of crest:*
 - (i) Narrow-crested weir, and
 - (ii) Broad-crested weir.
4. *According to the nature of crest:*
 - (i) Sharp-crested weir, and
 - (ii) Ogee weir.

9.3. DISCHARGE OVER A RECTANGULAR NOTCH OR WEIR

Consider a rectangular notch or weir provided in a channel carrying water as shown in Fig. 9.1.

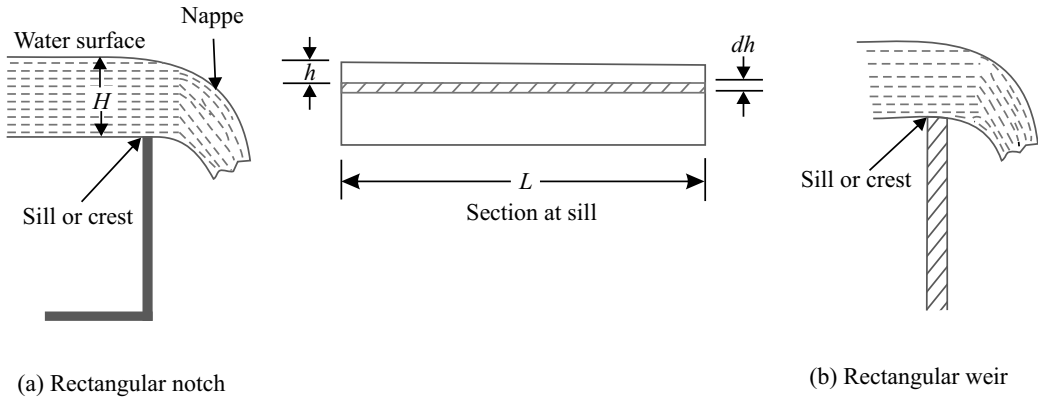


Fig. 9.1. Rectangular notch and weir.

Let, H = Height of water above sill of the notch,
 L = Length of notch or weir, and
 C_d = Co-efficient of discharge.

Let us consider a horizontal strip of water of thickness dh at a depth h from the water level as shown in Fig. 9.1.

$$\text{Area of strip} = L \times dh$$

$$\begin{aligned} \text{Theoretical velocity of water flowing through strip} \\ = \sqrt{2gh} \end{aligned}$$

The discharge through the strip,

$$\begin{aligned} dQ &= C_d \times \text{area of strip} \times \text{theoretical velocity} \\ &= C_d \times L \times dh \times \sqrt{2gh} \end{aligned} \quad \dots(i)$$

The total discharge, over the whole notch, may be found out by integrating the above equation within the limits 0 and H .

$$\begin{aligned}
 \therefore Q &= \int_0^H C_d \times L \times \sqrt{2gh} \times dh \\
 &= C_d \times L \times \sqrt{2g} \int_0^H (h)^{1/2} dh \\
 &= C_d \times L \times \sqrt{2g} \left[\frac{h^{1/2+1}}{\frac{1}{2}+1} \right]_0^H \\
 &= C_d \times L \times \sqrt{2g} \left[\frac{h^{3/2}}{3/2} \right]_0^H \\
 &= \frac{2}{3} \times C_d \times L \times \sqrt{2g} (H)^{3/2} \\
 \text{i.e. } Q &= \frac{2}{3} C_d \cdot L \sqrt{2g} (H)^{3/2} \quad \dots(9.1)
 \end{aligned}$$

Note. The expression for discharge over a rectangular notch or weir is same.

Example. 9.1. A rectangular notch 2.0 m wide has a constant head of 500 mm. Find the discharge over the notch, if co-efficient of discharge for the notch is 0.62.

Solution. Length of the notch, $L = 2.0$ m

Head over notch, $H = 500 \text{ mm} = 0.5$ m

Co-efficient of discharge, $C_d = 0.62$

Discharge, Q:

Using the relation,

$$\begin{aligned}
 Q &= \frac{2}{3} C_d \cdot L \sqrt{2g} (H)^{3/2} \\
 &= \frac{2}{3} \times 0.62 \times 2.0 \times \sqrt{2 \times 9.81} \times (0.5)^{3/2} \\
 &= \mathbf{1.294 \text{ m}^3/\text{s} \text{ (Ans.)}}
 \end{aligned}$$

Example. 9.2. A rectangular notch has a discharge of $0.24 \text{ m}^3/\text{s}$, when head of water is 800 mm. Find the length of the notch. Assume $C_d = 0.6$.

Solution. Discharge, $Q = 0.24 \text{ m}^3/\text{s}$

Head over notch, $H = 800 \text{ mm} = 0.8$ m

Co-efficient of discharge, $C_d = 0.6$

Length of the notch, L:

Using the relation :

$$\begin{aligned}
 Q &= \frac{2}{3} C_d \cdot L \times \sqrt{2g} (H)^{3/2} \\
 0.24 &= \frac{2}{3} \times 0.6 \times L \times \sqrt{2 \times 9.81} (0.8)^{3/2} = 1.267 L \\
 \therefore L &= \frac{0.24}{1.267} = 0.189 \text{ m or } 189 \text{ mm} \\
 \text{i.e. } L &= \mathbf{189 \text{ mm (Ans.)}}
 \end{aligned}$$

9.4. DISCHARGE OVER A TRIANGULAR NOTCH OR WEIR

Refer to Fig. 9-2. A triangular notch is also called a V -notch.

Let, H = Head of water above the apex of the notch,

θ = Angle of the notch, and

C_d = Co-efficient of discharge.

Consider a horizontal strip of water of thickness dh , and at a depth h from the water surface as shown in Fig. 9-2.

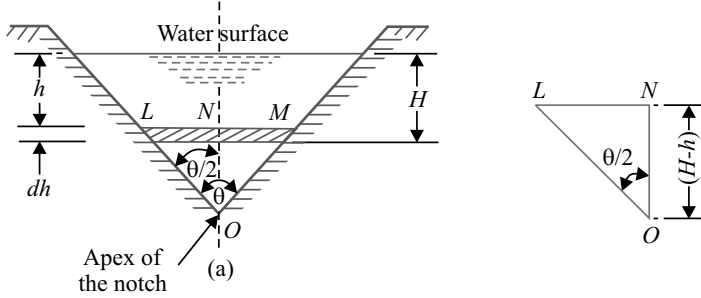


Fig. 9.2. The triangular notch.

From Fig. 9-2 (b), we have:

$$\tan \frac{\theta}{2} = \frac{LN}{ON} = \frac{LN}{H-h}$$

$$\therefore LN = (H-h) \tan \frac{\theta}{2}$$

$$\text{Width of strip} = LM = 2LN = 2(H-h) \tan \frac{\theta}{2}$$

$$\therefore \text{Area of the strip} = 2(H-h) \tan \frac{\theta}{2} \times dh$$

We know that theoretical velocity of water through the strip

$$= \sqrt{2gh}$$

$\therefore$ Discharge through the strip,

$$\begin{aligned} dQ &= C_d \times \text{area of strip} \times \text{theoretical velocity} \\ &= C_d \times 2(H-h) \tan \frac{\theta}{2} \times dh \times \sqrt{2gh} \end{aligned}$$

The total discharge, over the whole notch, may be found out by integrating the above equation, within the limits 0 and H .

$$\begin{aligned} \therefore Q &= \int_0^H C_d \times 2(H-h) \tan \frac{\theta}{2} \times \sqrt{2gh} \cdot dh \\ &= 2 C_d \sqrt{2g} \tan \frac{\theta}{2} \int_0^H (H-h) \sqrt{h} \cdot dh \\ &= 2 C_d \sqrt{2g} \tan \frac{\theta}{2} \int_0^H [Hh^{1/2} - h^{3/2}] dh \\ &= 2 C_d \sqrt{2g} \tan \frac{\theta}{2} \left[\frac{H \cdot h^{3/2}}{3/2} - \frac{h^{5/2}}{5/2} \right]_0^H \end{aligned}$$

$$\begin{aligned}
 &= 2C_d \sqrt{2g} \tan \frac{\theta}{2} \left[\frac{2}{3} H \cdot H^{3/2} - \frac{2}{5} H^{5/2} \right] \\
 &= 2 C_d \sqrt{2g} \tan \frac{\theta}{2} \left[\frac{2}{3} H^{5/2} - \frac{2}{5} H^{5/2} \right] \\
 &= 2 C_d \sqrt{2g} \tan \frac{\theta}{2} \left[\frac{4}{15} H^{5/2} \right] \\
 &= \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} H^{5/2} \quad \dots(9.2)
 \end{aligned}$$

For a *right angled V-notch*, if $C_d = 0.6$,

$$\left(\theta = 90^\circ, \therefore \tan \frac{\theta}{2} = 1 \right)$$

$$\begin{aligned}
 \text{Then,} \quad Q &= \frac{8}{15} \times 0.6 \sqrt{2 \times 9.81} \times 1 \times H^{5/2} \\
 &= 1.417 H^{5/2} \quad \dots(9.3)
 \end{aligned}$$

Advantages of a triangular notch over a rectangular notch:

A triangular notch claims the following *advantages* over a rectangular notch:

1. For a right angled *V-notch* or weir the expression for the computation of discharge is *very simple*.
2. For low discharges a triangular notch gives *more accurate results* than a rectangular notch.
3. In a given triangular notch, only one reading *i.e.*, head (H) is required to be taken for the measurement of discharge.
4. Ventilation of a triangular notch is not necessary.
5. The same triangular notch can measure a wide range of flows accurately.

Example 9.3. Find the discharge over a triangular notch of angle 60° when the head over the triangular notch is 0.2 m. Assume $C_d = 0.6$.

Solution. Angle of notch, $\theta = 60^\circ$

Depth of water, $H = 0.2$ m

Co-efficient of discharge, $C_d = 0.6$

Discharge, Q :

Using the relation :

$$\begin{aligned}
 Q &= \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} \times H^{5/2} \\
 &= \frac{8}{15} \times 0.6 \times \sqrt{2 \times 9.81} \times \tan \frac{60^\circ}{2} \times (0.2)^{5/2} \\
 &= \frac{8}{15} \times 0.6 \times 4.429 \times 0.577 \times 0.01788 \\
 &= \mathbf{0.01462 \text{ m}^3/\text{s} \text{ (Ans.)}}
 \end{aligned}$$

Example 9.4. During an experiment in a laboratory, 0.05 m^3 of water flowing over a right-angled notch was collected in one minute. If the head of the sill is 50 mm calculate the co-efficient of discharge of the notch.

Solution. Discharge, $Q = 0.05 \text{ m}^3/\text{min} = 0.000833 \text{ m}^3/\text{s}$

Angle of notch, $\theta = 90^\circ$

Head of the sill, $H = 50 \text{ mm} = 0.05 \text{ m}$

Co-efficient of discharge, C_d :

Using the relation:

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} \times H^{5/2}$$

$$\begin{aligned} 0.000833 &= \frac{8}{15} \times C_d \times \sqrt{2 \times 9.81} \times \tan \left(\frac{90^\circ}{2} \right) \times (0.05)^{5/2} \\ &= \frac{8}{15} \times C_d \times 4.429 \times 1 \times 0.000559 = 0.00132 C_d \end{aligned}$$

$$\therefore C_d = \frac{0.000833}{0.00132} = \mathbf{0.63 \text{ (Ans.)}}$$

Example 9.5. A rectangular channel 1.5 m wide has a discharge of $0.2 \text{ m}^3/\text{s}$, which is measured by a right-angled V-notch-weir. Find the position of the apex of the notch from the bed of the channel if the maximum depth of water is not to exceed 1 m. Assume $C_d = 0.62$.

Solution. Width of the rectangular channel, $L = 1.5 \text{ m}$

Discharge, $Q = 0.2 \text{ m}^3/\text{s}$

Depth of water in the channel = 1.0 m

Co-efficient of discharge, $C_d = 0.62$

Angle of the notch, $\theta = 90^\circ$

Position of the apex of the notch :

Using the relation :

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} \times H^{5/2}$$

$$\begin{aligned} 0.2 &= \frac{8}{15} \times 0.62 \times \sqrt{2 \times 9.81} \times \tan \left(\frac{90^\circ}{2} \right) \times H^{5/2} \\ &= 1.465 H^{5/2} \end{aligned}$$

$$\therefore H = \left(\frac{0.2}{1.465} \right)^{2/5} = 0.45 \text{ m}$$

Position of apex of the notch from the bed of channel

= Depth of water in channel – height of water over V-notch

= $1 - 0.45 = \mathbf{0.55 \text{ m (Ans.)}}$

9.5. DISCHARGE OVER A TRAPEZOIDAL NOTCH OR WEIR

Fig. 9-3 shows a trapezoidal notch or weir which is a combination of a rectangular and a triangular notch or weir. As such the discharge over such a notch or weir will be the *sum of the discharges over the rectangular and triangular notches or weirs.*

Let, H = Height of water over the notch,

L = Length of the rectangular portion (or crest) of the notch.,

C_{d1} = Co-efficient of discharge for the rectangular portion, and

C_{d2} = Co-efficient of discharge for the triangular portion.

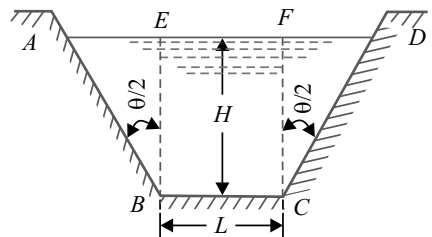


Fig. 9.3 The trapezoidal notch.

The discharge through the rectangular portion $BCFE$ is given by (Eqn. 9.1),

$$Q_1 = \frac{2}{3} C_{d1} L \sqrt{2g} H^{3/2}$$

The discharge through two triangular notches ABE and FCD is equal to the discharge through a single triangular notch of angle θ and is given by [Eqn. 9.2],

$$Q_2 = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} \times H^{5/2}$$

$\therefore$ Discharge through trapezoidal notch or weir $ABCD$,

$$\begin{aligned} Q &= Q_1 + Q_2 \\ &= \frac{2}{3} C_{d1} L \sqrt{2g} H^{3/2} + \frac{8}{15} C_{d2} \sqrt{2g} \tan \frac{\theta}{2} \times H^{5/2} \quad \dots(9.4) \end{aligned}$$

Example 9.6. Find the discharge through a trapezoidal notch which is 1.2 m wide at the top and 0.50 m at the bottom and is 0.4 m in height. The head of water on the notch is 0.3 m. Assume C_d for rectangular portion = 0.62, while for triangular portion = 0.60.

Solution. Top width = 1.2 m
Base width, L = 0.5 m
Head of water, H = 0.3 m
For rectangular portion, C_{d1} = 0.62
For triangular portion, C_{d2} = 0.60

Discharge, Q :

From $\triangle MNB$, we have:

$$\begin{aligned} \tan \frac{\theta}{2} &= \frac{MN}{NB} = \frac{(MS - NP) / 2}{NB} \\ &= \frac{(1.2 - 0.5) / 2}{0.4} \\ &= 0.875 \end{aligned}$$

Discharge through the trapezoidal notch is given by,

$$\begin{aligned} Q &= \frac{2}{3} C_{d1} \cdot L \sqrt{2g} H^{3/2} + \frac{8}{15} C_{d2} \sqrt{2g} \tan \frac{\theta}{2} \times H^{5/2} \\ &= \frac{2}{3} \times 0.62 \times 0.5 \times \sqrt{2 \times 9.81} \times 0.3^{3/2} + \frac{8}{15} \times 0.6 \times \sqrt{2 \times 9.81} \times 0.875 \times (0.3)^{5/2} \\ &= 0.1504 + 0.0611 = \mathbf{0.2115 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

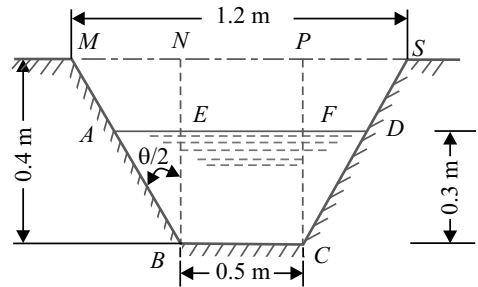


Fig. 9.4

9.6. DISCHARGE OVER A STEPPED NOTCH

A **stepped notch** is a combination of rectangular notches as shown in Fig. 9-5. The discharge through a stepped notch is equal to the sum of the discharges through the different rectangular notches.

Consider a stepped notch as shown in Fig. 9-5.

Let,

- H_1 = Height of water above sill of notch 1,
- L_1 = Length of notch 1,
- H_2, L_2 = Corresponding values for notch 2,
- H_3, L_3 = Corresponding values for notch 3, and
- C_d = Co-efficient of discharge for all notches.

The discharge over the notch 1,

$$Q_1 = \frac{2}{3} C_d \cdot L_1 \sqrt{2g} H_1^{3/2}$$

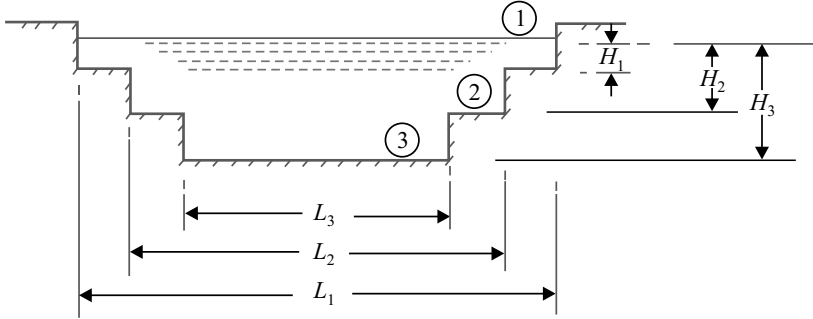


Fig. 9.5. The stepped notch.

Similarly, discharge over the notch 2,

$$Q_2 = \frac{2}{3} C_d \cdot L_2 \sqrt{2g} [H_2^{3/2} - H_1^{3/2}]$$

and, discharge over the notch 3,

$$Q_3 = \frac{2}{3} C_d \cdot L_3 \sqrt{2g} [H_3^{3/2} - H_2^{3/2}]$$

∴ Total discharge, $Q = Q_1 + Q_2 + Q_3$

Example 9.7. Find the discharge over a stepped rectangular notch, as shown in Fig. 9-6. Take co-efficient of discharge for all the portions as 0.62.

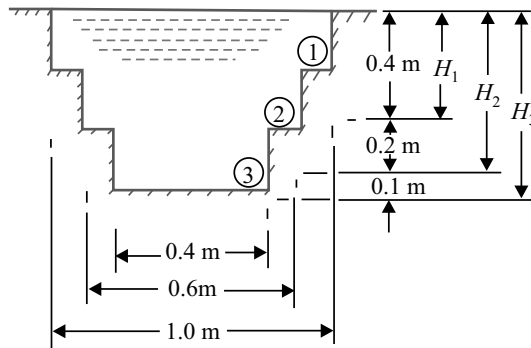


Fig. 9.6

Solution. Co-efficient of discharge, $C_d = 0.62$

Let, Q_1 = Discharge over the top portion,

Q_2 = Discharge over the middle portion, and

Q_3 = Discharge over the bottom portion.

The total discharge over the notch,

$$Q = Q_1 + Q_2 + Q_3$$

$$= \frac{2}{3} C_d \cdot L_1 \sqrt{2g} H_1^{3/2} + \frac{2}{3} C_d \cdot L_2 \sqrt{2g} [H_2^{3/2} - H_1^{3/2}] + \frac{2}{3} C_d \cdot L_3 \sqrt{2g} [H_3^{3/2} - H_2^{3/2}]$$

$$\begin{aligned}
&= \frac{2}{3} \times 0.62 \times 1.0 \times \sqrt{2 \times 9.81} \times (0.4)^{3/2} + \frac{2}{3} \times 0.62 \times 0.6 \times \sqrt{2 \times 9.81} [0.6^{3/2} - 0.4^{3/2}] \\
&\quad + \frac{2}{3} \times 0.62 \times 0.4 \times \sqrt{2 \times 9.81} [0.7^{3/2} - 0.6^{3/2}] \\
&= 0.463 + 0.232 + 0.0885 = \mathbf{0.783 \text{ m}^3/\text{s} \text{ (Ans.)}}
\end{aligned}$$

9.7. EFFECT ON DISCHARGE OVER A NOTCH OR WEIR DUE TO ERROR IN THE MEASUREMENT OF HEAD

The discharge over a rectangular notch or weir is proportional to $H^{3/2}$ and over a triangular notch or weir is proportional to $H^{5/2}$, where H is the height of liquid surface above the sill of the notch or weir. As such the accurate measurement of head H is quite essential in order to obtain an accurate value of the discharge over the notch or weir. However, if an error is introduced in the measurement of the head it will affect the computed discharge. The following cases of error in measurement of head will be considered:

(i) For rectangular notch or weir.

(ii) For triangular notch or weir.

(i) **Rectangular Notch or Weir :**

The discharge for a rectangular notch or weir is given by (Eqn. 9-1),

$$\begin{aligned}
Q &= \frac{2}{3} C_d \cdot L \sqrt{2g} H^{3/2} \\
&= KH^{3/2} \quad \dots(i)
\end{aligned}$$

$$(\text{where } K = \frac{2}{3} C_d \cdot L \sqrt{2g})$$

Differentiating the above equation, we get:

$$dQ = K \times 3/2 \times H^{1/2} dH \quad \dots(ii)$$

Dividing (ii) by (i), we get:

$$\frac{dQ}{Q} = \frac{K \times 3/2 \times H^{1/2} dH}{KH^{3/2}} = \frac{3}{2} \frac{dH}{H} \quad \dots(9.5)$$

Eqn. (9-5) shows that an error of 1% in measuring H will produce 1.5 % error in discharge over a rectangular notch or weir.

(ii) **Triangular Notch or Weir :**

The discharge over a triangular notch or weir is given by (Eqn. 9-2),

$$\begin{aligned}
Q &= \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} \times H^{5/2} \quad \dots(iii) \\
&= K \times H^{5/2}
\end{aligned}$$

$$\left(\text{where, } K = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} \right)$$

Differentiating Eqn. (iii), we get:

$$dQ = K \times 5/2 \times H^{3/2} dH \quad \dots(iv)$$

Dividing (iv) by (iii), we get:

$$\frac{dQ}{Q} = \frac{K \times 5/2 \times H^{3/2} dH}{K \times H^{5/2}} = \frac{5}{2} \frac{dH}{H} \quad \dots(9.6)$$

Eqn. (9-6) shows that an error of 1% in measuring H will produce 2.5% error in discharge over a triangular notch or weir.

Example 9.8. A rectangular notch 0.5 m long is used for measuring a discharge of $0.04 \text{ m}^3/\text{s}$. An error of 2 mm was made in measuring the head over the notch. Calculate the percentage error in the discharge. Take $C_d = 0.6$.

Solution. Length of notch, $L = 0.5 \text{ m}$

Discharge, $Q = 0.04 \text{ m}^3/\text{s}$

Error in head, $dH = 2 \text{ mm} = 0.002 \text{ m}$

Let, H = Height of water over rectangular notch.

Error in discharge, $\frac{dQ}{Q}$:

The discharge through a rectangular notch is given by,

$$Q = \frac{2}{3} C_d \cdot L \sqrt{2g} H^{3/2}$$

$$0.04 = \frac{2}{3} \times 0.6 \times 0.5 \times \sqrt{2 \times 9.81} \times H^{3/2}$$

$$= 0.886 H^{3/2}$$

$$\therefore H = \left(\frac{0.04}{0.886} \right)^{2/3} = 0.126 \text{ m}$$

Using the relation:

$$\frac{dQ}{Q} = \frac{3}{2} \times \frac{dH}{H}$$

$$= \frac{3}{2} \times \frac{0.002}{0.126} = 0.0238 \quad \text{or} \quad 2.38 \% \text{ (Ans.)}$$

Example 9.9. A discharge of $0.06 \text{ m}^3/\text{s}$ was measured over a right-angled notch. While measuring the head over the notch, an error of 1.5 mm was made. Determine the percentage error in the discharge, if the co-efficient of discharge for the notch is 0.6.

Solution. Discharge, $Q = 0.06 \text{ m}^3/\text{s}$

Angle of notch, $\theta = 90^\circ$

Error in measurement of head, $dH = 1.5 \text{ mm} = 0.0015 \text{ m}$.

Co-efficient of discharge, $C_d = 0.6$

Let, H = Height of water, above the apex of the notch.

Error in discharge, $\frac{dQ}{Q}$:

Using the relation:

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} H^{5/2}$$

$$0.06 = \frac{8}{15} \times 0.6 \times \sqrt{2 \times 9.81} \times \tan \left(\frac{90^\circ}{2} \right) \times H^{5/2}$$

$$= 1.417 H^{5/2}$$

$$\therefore H = \left(\frac{0.06}{1.417} \right)^{2/5} = 0.282 \text{ m}$$

Now, using the relation:

$$\frac{dQ}{Q} = \frac{5}{2} \times \frac{dH}{H} = \frac{5}{2} \times \frac{0.0015}{0.282}$$

$$= 0.0133 \text{ or } 1.33 \% \text{ (Ans.)}$$

9.8. VELOCITY OF APPROACH

The velocity with which the water approaches or reaches the weir or notch before it flows over it is known as 'velocity of approach'. Thus if V_a is the velocity of approach, then an additional head $H_a \left(= \frac{V_a^2}{2g} \right)$ due to the velocity of approach, is acting on water flowing over the notch or weir. Then

initial and final height of water over the notch or weir will be $(H + H_a)$ and H_a respectively.

The velocity of approach (V_a) is determined by finding the discharge over the weir or notch neglecting velocity of approach,

Let, Q = Discharge over weir or notch, and

A = Cross-sectional area of channel on the upstream side of the weir or notch.

Then the velocity of approach,

$$V_a = \frac{Q}{A}$$

This velocity of approach is used to find an additional head $\left(H_a = \frac{V_a^2}{2g} \right)$. Again the discharge is

calculated and above process is repeated for more accurate discharge.

Discharge over a rectangular weir, with velocity of approach

$$= \frac{2}{3} C_d \cdot L \sqrt{2g} \left[(H_1 + H_a)^{3/2} - (H_a)^{3/2} \right] \quad \dots(9.7)$$

Example 9.10. Find the discharge over a rectangular weir of length 80 m. The head of water over the weir is 1.2 m. The velocity of approach is given as 1.5 m/s. Take $C_d = 0.6$.

Solution. Length of weir, $L = 80$ m

Head of water, $H = 1.2$ m

Velocity of approach, $V_a = 1.5$ m/s

Co-efficient of discharge, $C_d = 0.6$

Discharge, Q:

The head due to velocity of approach,

$$H_a = \frac{V_a^2}{2g} = \frac{1.5^2}{2 \times 9.81} = 0.1146 \text{ m}$$

Using the relation:

$$\begin{aligned} Q &= \frac{2}{3} C_d \cdot L \sqrt{2g} \left[(H + H_a)^{3/2} - H_a^{3/2} \right] \\ &= \frac{2}{3} \times 0.6 \times 80 \times \sqrt{2 \times 9.81} \left[(1.2 + 0.1146)^{3/2} - (0.1146)^{3/2} \right] \\ &= 141.74 (1.507 - 0.0388) = \mathbf{208.1 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

9.9. EMPIRICAL FORMULAE FOR DISCHARGE OVER RECTANGULAR WEIR

The discharge over a rectangular weir,

$$Q = \frac{2}{3} C_d \cdot L \sqrt{2g} H^{3/2} \quad \dots \text{without velocity of approach} \dots(i)$$

$$= \frac{2}{3} C_d \cdot L \sqrt{2g} \left[(H + H_a)^{3/2} - H_a^{3/2} \right] \quad \dots \text{with velocity of approach} \dots(ii)$$

The equations (i) and (ii) are applicable to the weir or notch for which the *crest/sill length is equal to the width of the channel*; this type of weir is called **Suppressed weir**.

When the weir is not suppressed, the effect of end contractions is considered.

1. Francis's Formula:

On the basis of experimental analysis Francis established that:

- The end contraction decreases the effective length of the crest of weir and hence decreases the discharge.
- Each end contraction reduces the crest length by $0.1 H$, where H is the head over the weir.

For a rectangular weir there are *two end contractions only* and hence *effective length*

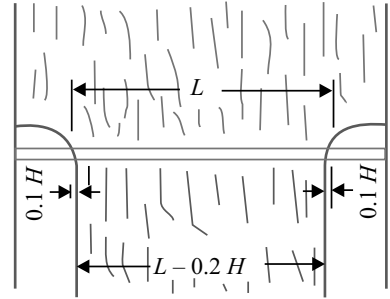


Fig. 9.7

$$= L - 0.1 \times 2 \times H = L - 0.2 H$$

and discharge,

$$Q = \frac{2}{3} \times C_d \times (L - 0.2 H) \times \sqrt{2g} H^{3/2} \quad \dots(9.8)$$

If there are n end contractions, we may write the empirical formula proposed by Francis as:

$$Q = \frac{2}{3} \times C_d \times (L - 0.1 nH) \times \sqrt{2g} H^{3/2} \quad \dots 9.8 (a)$$

When $C_d = 0.623$ and $g = 9.81 \text{ m/s}^2$, then:

$$\begin{aligned} Q &= \frac{2}{3} \times 0.623 \times (L - 0.2H) \times \sqrt{2 \times 9.81} H^{3/2} \\ &= 1.84 (L - 0.2 H) H^{3/2} \end{aligned} \quad \dots(9.9)$$

— When end contractions are *suppressed*, we have:

$$Q = 1.84 L H^{3/2} \quad \dots(9.10)$$

(When end contractions are *suppressed*, the value of n is taken as zero.)

When velocity of approach is considered, we have:

$$Q = 1.84 L \left[(H + H_a)^{3/2} - H_a^{3/2} \right] \quad \dots(9.11)$$

2. Bazin's Formula:

Bazin's formula for the discharge (Q) over a rectangular weir is given as follows:

$$\left. \begin{aligned} Q &= m \times L \times \sqrt{2g} \times H^{3/2} \\ m &= \frac{2}{3} \times C_d = 0.405 + \frac{0.003}{H} \end{aligned} \right\} \quad \dots(9.12)$$

where,

H = Height of water over the weir.

When velocity of approach is considered, we have:

where,

$$\left. \begin{aligned} Q &= m_1 \times L \times \sqrt{2g} \times (H + H_a)^{3/2} \\ m_1 &= 0.405 + \frac{0.003}{(H + H_a)} \end{aligned} \right\} \quad \dots(9.13)$$

Example 9.11. The head of water over a rectangular weir is 500 mm. If the length of the crest of the weir with end contractions suppressed is 1.4 m, find the discharge using the following formulae:

- (i) Francis's formula, and
- (ii) Bazin's formula.

Solution. Head of water over the weir, $H = 500 \text{ mm} = 0.5 \text{ m}$

Length of the weir, $L = 1.4 \text{ m}$

(i) **Discharge by Francis's formula, Q :**

Using Francis's formula,

$$Q = 1.84 LH^{3/2} = 1.84 \times 1.4 \times (0.5)^{3/2} \\ = 0.91 \text{ m}^3/\text{s. (Ans.)}$$

(ii) **Discharge by Bazin's formula, Q :**

Using Bazin's formula,

$$Q = m \times L \times \sqrt{2g} H^{3/2}$$

where,
$$m = 0.405 + \frac{0.003}{H} = 0.405 + \frac{0.003}{0.50} = 0.411$$

$\therefore$
$$Q = 0.411 \times 1.4 \times \sqrt{2 \times 9.81} \times (0.5)^{3/2} \\ = 0.901 \text{ m}^3/\text{s (Ans.)}$$

Example 9.12. A 30 metres long weir is divided into 10 equal bays by vertical posts, each 0.6 m wide. Using Francis's formula, calculate the discharge over the weir under an effective head of 1 metre.

Solution. Length of the weir = 30 m

Number of bays = 10

$\therefore$ Number of vertical posts = 10 – 1 = 9

Width of each post = 0.6 m

$\therefore$ Effective length, $L = 30 - 9 \times 0.6 = 24.6 \text{ m}$

Number of end contractions, $n = 2 \times 10 = 20$

(one bay has two end contractions)

Head of water, $H = 1 \text{ m}$

Discharge, Q :

Using Francis's formula,

$$Q = 1.84 (L - 0.1n H) H^{3/2} \\ = 1.84 (24.6 - 0.1 \times 20 \times 1) \times (1)^{3/2} \\ = 41.58 \text{ m}^3/\text{s (Ans.)}$$

Example 9.13. A 40 metres long weir is divided into 12 equal bays by vertical posts, each 0.6 m wide. Using Francis's formula, calculate the discharge over the weir if the head over the crest is 1.20 m and velocity of approach is 2 m/s.

Solution. Length of the weir = 40 m

Number of bays = 12

$\therefore$ Number of vertical post = 12 – 1 = 11

Width of each post = 0.6 m

$\therefore$ Effective length, $L = 40 - 11 \times 0.6 = 33.4 \text{ m}$

Number of end contractions, $n = 2 \times 12 = 24$

Head on weir, $H = 1.2 \text{ m}$

Velocity of approach, $V_a = 2 \text{ m/s}$

$\therefore$ Head due to V_a , $H_a = \frac{V_a^2}{2g} = \frac{2^2}{2 \times 9.81} = 0.2038 \text{ m}$

Discharge (Q) by Francis's formula with end contractions and velocity of approach is given by:

$$\begin{aligned}
 Q &= 1.84 [L - 0.1 n(H + H_a)] [(H + H_a)^{3/2} - H_a^{3/2}] \\
 &= 1.84 [33.4 - 0.1 \times 24 \times (1.2 + 0.2038)] [(1.2 + 0.2038)^{3/2} - (0.2038)^{3/2}] \\
 &= 1.84 [33.4 - 3.369] [1.663 - 0.092] \\
 &= 86.8 \text{ m}^3/\text{s} \text{ (Ans.)}
 \end{aligned}$$

9.10. CIPPOLETTI WEIR OR NOTCH

The Cippoletti weir is *trapezoidal weir*, having side slopes of 1 horizontal to 4 vertical as shown in Fig. 9-8. By providing slope on the sides, an increase in discharge through the triangular portions (AED and FBC) is obtained; without this slope the weir would be a rectangular one, and due to end contraction, the discharge would decrease. Thus the advantage of this weir is that the *factor of end contraction is not required* (while using Francis's formula).

Let us split the trapezoidal weir into the following:

- (i) Rectangular weir, and
- (ii) Triangular notch

The discharge over a rectangular weir (with two end contractions),

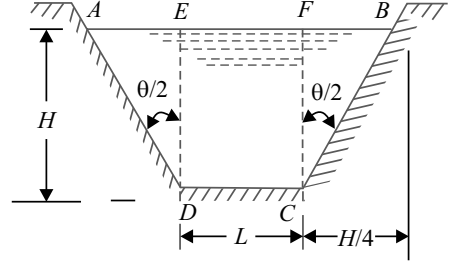


Fig. 9.8 Cippoletti weir.

$$Q_1 = \frac{2}{3} \times C_d \times (L - 0.2 H) \sqrt{2g} H^{3/2} \quad \dots(i)$$

and discharge over the triangular notch,

$$Q_2 = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} \times (H)^{5/2} \quad \dots(ii)$$

$\therefore$ Total discharge,

$$\begin{aligned}
 Q &= Q_1 + Q_2 \\
 &= \frac{2}{3} C_d (L - 0.2 H) \sqrt{2g} H^{3/2} + \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} \times H^{5/2} \quad \dots(9.14)
 \end{aligned}$$

To avoid the factor of end contraction, Cippoletti gave the formula for discharge,

$$Q = \frac{2}{3} C_d L \sqrt{2g} \times H^{3/2} \quad \dots(9.15)$$

Equating the eqns. (9.14) and (9.15), we get:

$$\frac{2}{3} C_d L \sqrt{2g} H^{3/2} = \frac{2}{3} C_d (L - 0.2 H) \sqrt{2g} H^{3/2} + \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} \times H^{5/2}$$

Dividing both sides by $\frac{2}{3} C_d \sqrt{2g} H^{3/2}$, we have:

$$L = L - 0.2 H + \frac{4}{5} \tan \frac{\theta}{2} \times H$$

$$\text{or, } \frac{4}{5} \tan \frac{\theta}{2} \times H = 0.2 H$$

$$\therefore \tan \frac{\theta}{2} = 0.2 \times \frac{5}{4} = \frac{1}{4}$$

The above relation indicates that in a trapezoidal weir having side slopes 1 horizontal to 4 vertical the factor of end contraction is *not* required for discharge, while using Francis's formula.

Example 9.14. Find the discharge over a Cippoletti weir of length 1.8 m when the head over the weir is 1.2 m. Take $C_d = 0.62$

Solution. Length of weir, $L = 1.8$ m

Head of water, $H = 1.2$ m

Co-efficient of discharge, $C_d = 0.62$

Discharge over the weir, Q :

Using the relation:

$$\begin{aligned} Q &= \frac{2}{3} C_d L \sqrt{2g} H^{3/2} \\ &= \frac{2}{3} \times 0.62 \times 1.8 \times \sqrt{2 \times 9.81} \times (1.2)^{3/2} \\ &= 4.33 \text{ m}^3/\text{s (Ans.)} \end{aligned}$$

9.11. DISCHARGE OVER A BROAD CRESTED WEIR

Fig. 9.9 shows a broad-crested weir. Let 1 and 2 be the upstream and downstream ends of the weir respectively.

Let,

H = Head of water in the upstream side of the weir,

h = Head of water on the downstream side of the weir,

v = Velocity of the water on the downstream side of the weir,

L = Length of the weir, and

C_d = Co-efficient of discharge.

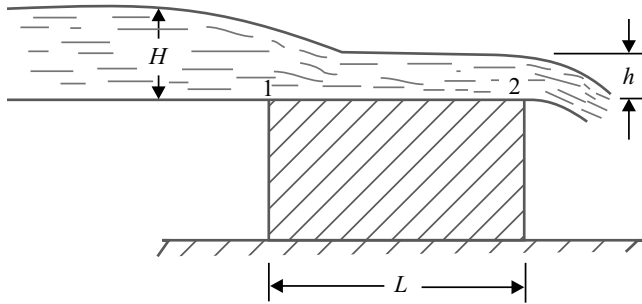


Fig. 9.9. Broad-crested weir.

Applying Bernoulli's equation at 1 and 2, we get:

$$0 + 0 + H = 0 + \frac{v^2}{2g} + h$$

$$\therefore \frac{v^2}{2g} = H - h$$

$$\text{or, } v = \sqrt{2g(H - h)}$$

$\therefore$ The discharge over weir, $Q = C_d \times \text{area of flow} \times \text{velocity}$

$$= C_d \times L \times h \times v$$

$$= C_d \times L \times h \times \sqrt{2g(H - h)}$$

$$= C_d \times L \times \sqrt{2g} \sqrt{Hh^2 - h^3}$$

...(9.16)

The discharge will be *maximum*, if $(Hh^2 - h^3)$ is maximum.

$$\text{or, } \frac{d}{dh} (Hh^2 - h^3) = 0$$

$$\text{or, } 2hH - 3h^2 = 0$$

$$\text{or, } 2H = 3h$$

$$\therefore h = \frac{2}{3} H$$

Substituting the value of h in eqn. (9.16), we get:

$$\begin{aligned} Q_{\max} &= C_d \times L \times \sqrt{2g} \sqrt{H \times \left(\frac{2}{3}H\right)^2 - \left(\frac{2}{3}H\right)^3} \\ &= C_d \times L \times \sqrt{2g} \sqrt{\frac{4}{9}H^3 - \frac{9}{27}H^3} \\ &= C_d \times L \times \sqrt{2g} \sqrt{\frac{4}{27}H^3} \\ &= C_d \times L \times \sqrt{2g} \times \frac{2}{3}H \sqrt{\frac{H}{3}} \\ &= \frac{2}{3\sqrt{3}} C_d \times L \times \sqrt{2g} \times H^{3/2} \\ &= 0.3849 \times C_d \times L \times \sqrt{2 \times 9.81} \times H^{3/2} \\ &= 1.705 \times C_d \times L \times H^{3/2} \end{aligned} \quad \dots(9.17)$$

9.12. DISCHARGE OVER A NARROW-CRESTED WEIR

In case of a narrow-crested weir, $2L < H$. This weir is similar to a rectangular weir or notch and hence, Q is given by:

$$Q = \frac{2}{3} \times C_d \times L \times \sqrt{2g} \times H^{3/2} \quad \dots(9.18)$$

9.13. DISCHARGE OVER AN OGEE WEIR

In the Fig. 9-10 is shown an Ogee weir, in which the crest of the weir rises upto maximum height of $1.115 H$ and then falls as shown (where, H = height of water above inlet of the weir). The discharge over an Ogee weir is the same as that of a rectangular weir and is given by:

$$Q = \frac{2}{3} \times C_d \times L \times \sqrt{2g} \times H^{3/2} \quad \dots(9.18)$$

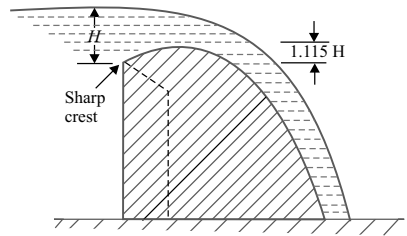
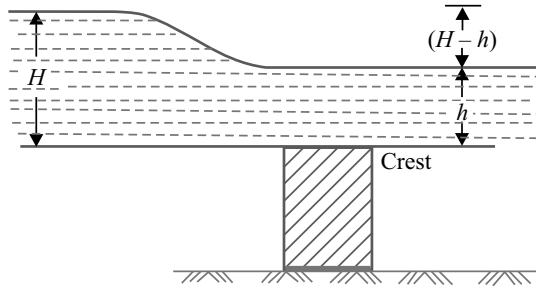


Fig. 9.10. An Ogee weir.

9.14. DISCHARGE OVER SUBMERGED OR DROWNED WEIR

A weir is said to be **submerged** or **drowned weir** if the water level on its downstream side is above its crest. Such a weir is shown in Fig. 9-11. The total discharge over the weir is obtained by dividing the weir into two parts. The portion between upstream and downstream water surfaces may be treated as **free weir** and portion between downstream water surface and crest as a **drowned weir**.

**Fig. 9.11.** Submerged weir.

Let, H = Height of water on the upstream side of the weir, and

h = Height of water on the downstream side of the weir.

Then, Q_1 = Discharge over upper portion

$$= \frac{2}{3} \cdot C_{d1} \cdot L \cdot \sqrt{2g} (H - h)^{3/2}$$

and, Q_2 = Discharge through drowned portion

$$= C_{d2} \times \text{area of flow} \times \text{velocity of flow}$$

$$= C_{d2} \cdot L \cdot h \cdot \sqrt{2g (H - h)}$$

where, C_{d1} and C_{d2} are the respective discharge co-efficients.

$$\begin{aligned} \therefore \text{Total discharge, } Q &= Q_1 + Q_2 \\ &= \frac{2}{3} \cdot C_{d1} \cdot L \cdot \sqrt{2g} (H - h)^{3/2} + C_{d2} \cdot L \cdot h \cdot \sqrt{2g (H - h)} \quad \dots(9.19) \end{aligned}$$

Example 9.15. A 45 m long broad-crested weir has 0.5 m of water above its crest. Find the maximum discharge over the weir. Take $C_d = 0.62$. Neglect velocity of approach.

Solution. Length of the weir, $L = 45$ m

Head of water, $H = 0.5$ m

$$C_d = 0.62$$

Maximum discharge, $Q_{\max}$:

$$\begin{aligned} Q_{\max} &= 1.705 C_d \times L \times H^{3/2} \\ &= 1.705 \times 0.62 \times 45 \times (0.5)^{3/2} \\ &= \mathbf{16.82 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

Example 9.16. Find the discharge over an narrow-crested weir 6 m long and having a head of 0.4 m of water. Take $C_d = 0.62$.

Solution. Length of weir, $L = 6$ m

Head of water, $H = 0.4$ m

Discharge co-efficient, $C_d = 0.62$

Discharge, Q :

$$\begin{aligned} Q &= \frac{2}{3} \times C_d \times L \times \sqrt{2g} \times H^{3/2} \quad \dots[\text{Eqn. 9.18}] \\ &= \frac{2}{3} \times 0.62 \times 6 \times \sqrt{2 \times 9.81} \times 0.4^{3/2} = \mathbf{2.779 \text{ m}^3/\text{s} \text{ (Ans.)}} \end{aligned}$$

Example 9.17. In a submerged weir of 2.5 m length the heights of water on the upstream and downstream sides are 0.2 m and 0.1 m respectively. Find the discharge over the weir if discharge co-efficients for free and drowned portions are 0.62 and 0.8 respectively.

Solution. Length of weir, $L = 25$ m

Height of water on upstream side, $H = 0.2$ m

Height of water on downstream side, $h = 0.1$ m

$$C_{d1} = 0.62$$

$$C_{d2} = 0.8$$

Discharge over the weir, Q :

$$\text{Total discharge, } Q = Q_1 \text{ (discharge through free portion)} + Q_2 \text{ (discharge through the drowned portion)}$$

$$= \frac{2}{3} C_{d1} \times L \times \sqrt{2g} (H - h)^{3/2} + C_{d2} \times L \times h \times \sqrt{2g (H - h)} \dots [\text{Eqn. (9.19)}]$$

$$= \frac{2}{3} \times 0.62 \times 2.5 \times \sqrt{2 \times 9.81} (0.2 - 0.1)^{3/2} + 0.8 \times 2.5 \times 0.1 \times \sqrt{2 \times 9.81 (0.2 - 0.1)}$$

$$= 0.1447 + 0.2801 = \mathbf{0.4248 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

Example 9.18. Water flows over a rectangular sharp-crested weir 1 m long, the head over the sill of the weir being 0.66 m. The approach channel is 1.4 m wide and depth of flow in the channel is 1.2 m. Starting from first principles, determine the rate of discharge over the weir. Consider also the velocity of approach and the effect of end contractions. Take the co-efficient of discharge for the weir as 0.6. [UPSC Exams.]

Solution. Length of weir, $L = 1$ m

Head of water, $H = 0.66$ m

Co-efficient of discharge, $C_d = 0.6$

Rate of discharge, Q :

(i) *Neglecting velocity of approach:*

The discharge over a rectangular sharp-crested contracted weir is given by Eqn. (9.8),

$$Q = \frac{2}{3} C_d (L - 0.1 n H) \sqrt{2g} H^{3/2}$$

(where, n = no. of end contractions)

$$= \frac{2}{3} \times 0.6 (1 - 0.1 \times 2 \times 0.66) \times \sqrt{2 \times 9.81} \times (0.66)^{3/2}$$

$$= 0.3472 \times 4.429 \times 0.536 = \mathbf{0.824 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

(ii) *Taking velocity of approach into consideration:*

$$\text{Velocity of approach, } V_a = \frac{Q}{A} = \frac{0.824}{1.4 \times 1.2} = 0.49 \text{ m/s}$$

$$\text{Head due to } V_a, h_a = \frac{V_a^2}{2g} = \frac{0.49^2}{2 \times 9.81} = 0.0122 \text{ m}$$

$$H + h_a = 0.66 + 0.0122 = 0.6722 \text{ m}$$

Discharge considering the velocity of approach,

$$Q = \frac{2}{3} C_d [L - 0.1 n (H + h_a)] \times \sqrt{2g} \times [(H + h_a)^{3/2} - (h_a)^{3/2}]$$

$$\begin{aligned}
&= \frac{2}{3} \times 0.6 (1 - 0.1 \times 2 \times 0.6722) \times \sqrt{2 \times 9.81} \left[(0.6722)^{3/2} - (0.0122)^{3/2} \right] \\
&= 0.3462 \times 4.429 (0.551 - 0.001347) \\
&= \mathbf{0.8428 \text{ m}^3/\text{s} \text{ (Ans.)}}
\end{aligned}$$

9.15. TIME REQUIRED TO EMPTY A RESERVOIR OR A TANK WITH RECTANGULAR AND TRIANGULAR WEIRS OR NOTCHES

(a) Rectangular weir or notch :

Consider a reservoir or a tank provided with a rectangular weir or notch in one of its sides.

Let, A = Uniform cross-sectional area of the tank,

L = Length of crest of the weir or notch,

H_1 = Initial height of liquid above the crest of notch,

H_2 = Final height of liquid above the crest of notch,

C_d = Co-efficient of discharge, and

T = Time required in seconds to lower the height of liquid from H_1 to H_2 .

Further, let h = The height of liquid surface above the crest of weir or notch at any instant, and

dh = The fall of liquid surface in a small time dT .

$$\begin{aligned}
\text{Then,} \quad -A \cdot dh &= Q \times dT \\
&= \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} \cdot h^{3/2} dT
\end{aligned}$$

(Negative sign indicates that as T increases, h decreases.)

$$\text{or,} \quad dT = \frac{-A \, dh}{\frac{2}{3} C_d \cdot L \cdot \sqrt{2g} \cdot h^{3/2}}$$

To obtain total time T , the above eqn. is integrated between the limits H_1 to H_2 .

$$\therefore \int_0^T dT = \int_{H_1}^{H_2} \frac{-A \, dh}{\frac{2}{3} C_d \cdot L \cdot \sqrt{2g} \cdot h^{3/2}}$$

$$\begin{aligned}
\text{or,} \quad T &= \frac{-A}{\frac{2}{3} C_d \cdot L \cdot \sqrt{2g}} \int_{H_1}^{H_2} h^{-3/2} \, dh \\
&= \frac{-3A}{2 C_d \cdot L \cdot \sqrt{2g}} \left[\frac{h^{-3/2+1}}{(-3/2)+1} \right]_{H_1}^{H_2} \\
&= \frac{-3A}{2 C_d \cdot L \cdot \sqrt{2g}} \left(-\frac{2}{1} \right) \left[\frac{1}{\sqrt{h}} \right]_{H_1}^{H_2} \\
&= \frac{3A}{C_d \cdot L \cdot \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right] \quad \dots(9.20)
\end{aligned}$$

(b) Triangular weir or notch:

Consider a reservoir or a tank provided with a triangular weir or notch in one of its sides.

Let, A = Uniform cross-sectional area of the tank,
 θ = Angle of the notch,
 H_1 = Initial height of liquid above the apex of notch,
 H_2 = Final height of liquid above the apex of notch, and
 C_d = Co-efficient of discharge.

Further, let, h = The height of liquid surface above the crest of weir or notch at any instant;
 and,

dh = The fall of liquid surface in a small time dT .

Then,
$$-A \cdot dh = Q \times dT$$

$$= \frac{8}{15} C_d \times \tan \frac{\theta}{2} \times \sqrt{2g} \times h^{5/2} dT$$

or,
$$dT = \frac{-A \cdot dh}{\frac{8}{15} C_d \times \tan \frac{\theta}{2} \times \sqrt{2g} \times h^{5/2}}$$

To obtain total time T , the above eqn. is integrated between the limits H_1 to H_2 .

$$\therefore \int_0^T dT = \int_{H_1}^{H_2} \frac{-A \cdot dh}{\frac{8}{15} C_d \times \tan \frac{\theta}{2} \times \sqrt{2g} \times h^{5/2}}$$

or,
$$T = \frac{-A}{\frac{8}{15} \times C_d \times \tan \frac{\theta}{2} \times \sqrt{2g}} \int_{H_1}^{H_2} h^{-5/2} dh$$

$$= - \frac{15A}{8 \times C_d \times \tan \frac{\theta}{2} \times \sqrt{2g}} \left[\frac{h^{-3/2}}{-3/2} \right]_{H_1}^{H_2}$$

$$= \frac{-15A}{8 \times C_d \times \tan \frac{\theta}{2} \times \sqrt{2g}} (-2/3) \left[\frac{1}{h^{3/2}} \right]_{H_1}^{H_2}$$

$$= \frac{5A}{4 \times C_d \times \tan \frac{\theta}{2} \times \sqrt{2g}} \left[\frac{1}{H_2^{3/2}} - \frac{1}{H_1^{3/2}} \right] \quad \dots(9.21)$$

Example 9.19. Find the time required to lower the water level from 3 m to 2 m in a reservoir of dimensions 70 m $\times$ 70 m, by

(i) a rectangular notch of length 1.2 m;

(ii) a right angled V-notch.

Take $C_d = 0.62$

Solution. Dimensions of the reservoir = 70 m $\times$ 70 m

$\therefore$ Area, $A = 70 \times 70 = 4900 \text{ m}^2$

Length of rectangular notch, $L = 1.2 \text{ m}$

Angle of notch, $\theta = 90^\circ$

Co-efficient of discharge, $C_d = 0.62$

Initial height of water, $H_1 = 3 \text{ m}$

Final height of water, $H_2 = 2 \text{ m}$

Time required, T:

(i) *Rectangular notch:*

$$T = \frac{3A}{C_d \cdot L \cdot \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right] \quad \dots[\text{Eqn. 9-20}]$$

Substituting the values, we get:

$$\begin{aligned} T &= \frac{3 \times 4900}{0.62 \times 1.2 \times \sqrt{2 \times 9.81}} \left[\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}} \right] \\ &= 4460.62 (0.707 - 0.577) = \mathbf{579.88 \text{ s (Ans.)}} \end{aligned}$$

(ii) *A right angled V-notch:*

$$\begin{aligned} T &= \frac{5A}{4 \times C_d \times \tan \frac{\theta}{2} \times \sqrt{2g}} \left[\frac{1}{H_2^{3/2}} - \frac{1}{H_1^{3/2}} \right] \quad \dots[\text{Eqn. 9-21}] \\ &= \frac{5 \times 4900}{4 \times 0.62 \times \tan \frac{90^\circ}{2} \times \sqrt{2 \times 9.81}} \left[\frac{1}{2^{3/2}} - \frac{1}{3^{3/2}} \right] \\ &= 2230.3 (0.3535 - 0.1924) = \mathbf{359.3 \text{ s (Ans.)}} \end{aligned}$$

HIGHLIGHTS

1. A *notch* may be defined as an opening provided in the side of a tank or vessel such that the liquid surface in the tank is *below the top edge of the opening*.
2. A *weir* may be defined as any regular obstruction is an open stream over which the flow takes place.
3. Discharge through a rectangular notch or weir is given by,

$$Q = \frac{2}{3} C_d \times L \times \sqrt{2g} \times H^{3/2}$$

where,

C_d = Co-efficient of discharge,

L = Length of notch or weir, and

H = Head of water over the notch or weir.

4. Discharge through a triangular notch or weir is given by,

$$Q = \frac{8}{15} \times C_d \times \sqrt{2g} \times \tan \frac{\theta}{2} \times H^{5/2}$$

where,

θ = angle of the notch.

5. Discharge through a trapezoidal notch or weir is given by,

$$Q = \frac{2}{3} C_{d1} \times L \times \sqrt{2g} \times H^{3/2} + \frac{8}{5} C_{d2} \sqrt{2g} \times \tan \frac{\theta}{2} \times H^{5/2}$$

where,

C_{d1} = Co-efficient of discharge for rectangular notch,

C_{d2} = Co-efficient of discharge for triangular notch, and

$\frac{\theta}{2}$ = Slope of the side of trapezoidal notch.

6. The error in discharge due to the error in the measurement of head over a rectangular or a triangular notch or weir is given by

$$\frac{dQ}{Q} = \frac{3}{2} \times \frac{dH}{H} \quad \dots \text{ for a rectangular notch or weir}$$

$$= \frac{5}{2} \times \frac{dH}{H} \quad \dots \text{for a triangular notch or weir}$$

where, Q = Discharge through rectangular or triangular notch or weir, and
 H = Head over the notch or weir.

7. The velocity with which the water approaches or reaches the weir or notch before it flows over it is known as '*Velocity of approach*'. It is denoted by V_a and is given by:

$$V_a = \frac{\text{Discharge over the notch or weir}}{\text{Cross-sectional area of channel}}$$

8. The head due to velocity of approach is given by,

$$H_a = \frac{V_a^2}{2g}$$

9. The discharge over a rectangular weir,

$$Q = \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} \cdot H^{3/2} \quad \dots \text{without velocity of approach}$$

$$= \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} [(H + H_a^{3/2}) - (H_a)^{3/2}] \quad \dots \text{with velocity of approach}$$

10. *Francis's formula* for discharge over a rectangular weir is given by:

$$Q = \frac{2}{3} C_d \times (L - 0.1 nH) \sqrt{2g} H^{3/2} \quad \dots \text{for } n \text{ end contractions}$$

$$= 1.84 (L - 0.2 H) H^{3/2} \quad \dots \text{for two end contractions}$$

$$= 1.84 L H^{3/2} \quad \dots \text{if end contractions are suppressed}$$

$$= 1.84 L [(H + H_a)^{3/2} - H_a^{3/2}] \quad \dots \text{if velocity of approach is considered}$$

(when $C_d = 0.623$, $g = 9.81 \text{ m/s}^2$)

11. *Bazin's formula* for discharge over a rectangular weir,

$$Q = m L \sqrt{2g} H^{3/2} \quad \dots \text{without velocity of approach}$$

$$= m L \sqrt{2g} [(H + H_a)^{3/2}] \quad \dots \text{with velocity of approach}$$

where $m = \frac{2}{3} C_d = 0.405 + \frac{0.003}{H} \quad \dots \text{without velocity of approach}$

$$= 0.405 + \frac{0.003}{(H + H_a)} \quad \dots \text{with velocity of approach}$$

12. The *Cippoletti weir* is a trapezoidal weir, having side slopes of 1 horizontal to 4 vertical. The discharge through the weir is given by,

$$Q = \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} H^{3/2} \quad \dots \text{without velocity of approach}$$

$$= \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} [(H + H_a)^{3/2} - H_a^{3/2}] \quad \dots \text{with velocity of approach.}$$

13. Discharge over a *broad-crested weir* is given by,

$$Q = C_d \cdot L \cdot \sqrt{2g} \sqrt{Hh^2 - h^3}$$

where, L = Length of the weir,
 H = Height of water, above crest, and
 h = Head of water on the downstream of the weir.

The condition for maximum discharge is

$$h = \frac{2}{3} H, \text{ and maximum discharge is given by,}$$

$$Q_{\max} = 1.705 C_d \cdot L \cdot H^{3/2}$$

14. Discharge over an *Ogee weir* is given by,

$$Q = \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} \times H^{3/2}$$

15. Discharge over *submerged or drowned weir* is given by,

$$\begin{aligned} Q &= \text{Discharge over upper portion} + \text{discharge through drowned portion} \\ &= \frac{2}{3} C_{d1} \cdot L \cdot \sqrt{2g} (H - h)^{3/2} + C_{d2} \cdot L \cdot h \cdot \sqrt{2g} (H - h) \end{aligned}$$

16. The time required to *empty a reservoir or a tank* by a rectangular or a triangular notch is given by,

$$T = \frac{3A}{C_d \cdot L \cdot \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right] \quad \dots \text{by a rectangular notch}$$

$$= \frac{5A}{4 C_d \times \tan \frac{\theta}{2} \times \sqrt{2g}} \left[\frac{1}{H_2^{3/2}} - \frac{1}{H_1^{3/2}} \right] \quad \dots \text{by a triangular notch}$$

where,

A = Cross-sectional area of a tank or a reservoir,

H_1 = Initial height of liquid above the apex of notch, and

H_2 = Final height of liquid above the apex of notch.

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer:

- An opening provided in the side of a tank or vessel such that the liquid surface in the tank is below the top edge of the opening is known as
(a) weir (b) notch
(c) orifice (d) none of the above.
- A notch is generally made of
(a) masonry or concrete (b) metallic plate
(c) plastic plate (d) any of these.
- Any regular obstruction in an open stream over which the flow takes place is known as
(a) notch (b) orifice
(c) weir (d) any of these.
- Which of the following may be used for measuring the rate of flow of water in rivers or streams?
(a) Notches (b) Orifices
(c) Weir (d) Any of these.
- Discharge over a rectangular notch or weir is given by

$$(a) \frac{1}{2} C_d \cdot L \cdot \sqrt{2g} (H)^{3/2}$$

$$(b) \frac{3}{4} C_d \cdot L \cdot \sqrt{2g} (H)^{3/2}$$

$$(c) C_d \cdot L \cdot \sqrt{2g} (H)^{5/2}$$

$$(d) \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} (H)^{3/2}.$$

where, C_d = Co-efficient of discharge,

L = Length of notch or weir, and

H = Height of water above sill of the notch.

6. Discharge, (Q) over a triangular notch or weir is given by

$$(a) \frac{2}{3} C_d \cdot \sqrt{2g} \tan \frac{\theta}{2} \cdot H^{3/2}$$

$$(b) C_d \cdot \sqrt{2g} \tan \frac{\theta}{2} \cdot H^{5/2}$$

$$(c) \frac{1}{2} C_d \cdot \sqrt{2g} \tan \theta H^{5/2}$$

$$(d) \frac{8}{15} C_d \cdot \sqrt{2g} \tan \frac{\theta}{2} \cdot H^{5/2}$$

where, θ is the angle of notch.

7. Discharge over a trapezoidal notch or weir is given by

$$(a) C_{d1} \cdot L \cdot \sqrt{2g} H^{3/2} + \frac{8}{15} C_{d2} \cdot \sqrt{2g} \tan \frac{\theta}{2} \cdot H^{3/2}$$

$$(b) \frac{2}{3} C_{d1} \cdot L \cdot \sqrt{2g} H^{3/2} + C_{d2} \cdot \sqrt{2g} \tan \theta H^{5/2}$$

$$(c) \frac{2}{3} C_{d1} \cdot L \cdot \sqrt{2g} H^{3/2} + \frac{8}{15} C_{d2} \sqrt{2g} \tan \frac{\theta}{2} \cdot H^{5/2}$$

$$(d) \frac{1}{2} C_{d1} \cdot L \cdot \sqrt{2g} H^{3/2} + \frac{8}{15} C_{d2} \cdot \sqrt{2g} \tan \frac{\theta}{2} \cdot H^{5/2}$$

Where C_{d1} = Co-efficient of discharge for the rectangular portion, and

C_{d2} = Co-efficient of discharge for the triangular portion.

8. An error of 1% in measuring H will produce ... error in discharge over a rectangular notch or weir.

- (a) 1% (b) 1.5%
(c) 2% (d) 2.5%.

9. An error of 1% in measuring H will produce ... error in discharge over a triangular notch or weir

- (a) 1% (b) 1.5%
(c) 2% (d) 2.5%.

10. The error in discharge due to the error in the measurement of head over a rectangular notch or weir is given by

$$(a) \frac{dQ}{Q} = \frac{1}{2} \frac{dH}{H} \quad (b) \frac{dQ}{Q} = \frac{3}{2} \frac{dH}{H}$$

$$(c) \frac{dQ}{Q} = \frac{3}{4} \frac{dH}{H} \quad (d) \text{ none of these.}$$

11. The error in discharge due to the error in the measurement of head over a triangular notch or weir is given by

$$(a) \frac{dQ}{Q} = \frac{1}{2} \frac{dH}{H} \quad (b) \frac{dQ}{Q} = \frac{dH}{H}$$

$$(c) \frac{dQ}{Q} = \frac{3}{2} \frac{dH}{H} \quad (d) \frac{dQ}{Q} = \frac{5}{2} \frac{dH}{H}.$$

12. The discharge over a rectangular weir, considering velocity of approach, is given by

$$(a) Q = \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} H^{3/2}$$

$$(b) Q = \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} \left[(H + H_a)^{3/2} - (H_a)^{3/2} \right]$$

$$(c) Q = \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} \left[(H + H_a)^{1/2} - (H_a)^{1/2} \right]$$

- (d) none of these.

where H_a is the additional head due to velocity of approach.

13. Francis's formula for a rectangular weir with n end contractions is given by

$$(a) Q = \frac{2}{3} C_d \cdot (L - 0.1 nH) \sqrt{2g} \cdot H^{3/2}$$

$$(b) Q = \frac{3}{2} C_d \cdot (L - 0.1 nH) \sqrt{2g} \cdot H^{5/2}$$

$$(c) Q = C_d \cdot (L - 0.1 nH) \sqrt{2g} \cdot H^{3/2}$$

- (d) none of these.

14. Discharge over a broad-crested weir is given by

$$(a) Q = C_d \cdot L \cdot \sqrt{2g} \sqrt{H - h}$$

$$(b) Q = \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} \sqrt{H - h}$$

$$(c) Q = \frac{3}{4} C_d \cdot L \cdot \sqrt{2g} \sqrt{H - h}$$

$$(d) Q = C_d \cdot L \cdot \sqrt{2g} \sqrt{Hh^2 - h^3}$$

where, H = Head of water on the upstream side of the weir,

h = Head of water on the downstream side of the weir,

L = Length of the weir, and

C_d = Co-efficient of discharge.

15. Maximum discharge over a broad-crested weir is given by.

$$(a) Q = C_d \cdot L \cdot H^{3/2}$$

$$(b) Q = 0.5 C_d \cdot L \cdot H^{5/2}$$

$$(c) Q = 1.705 C_d \cdot L \cdot H^{3/2}$$

$$(d) Q = 1.705 C_d \cdot L \cdot H^{5/2}$$

16. The time required to empty a reservoir or a tank by a rectangular notch is given by

$$(a) T = \frac{A}{C_d \cdot L \cdot \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right]$$

$$(b) T = \frac{2A}{C_d \cdot L \cdot \sqrt{2g}} \left[\frac{1}{H_2^{3/2}} - \frac{1}{H_1^{3/2}} \right]$$

$$(c) T = \frac{3A}{C_d \cdot L \cdot \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right]$$

- (d) none of these.

17. The time required to empty a reservoir or a tank by a triangular notch is given by

$$(a) T = \frac{A}{4C_d \cdot \tan \frac{\theta}{2} \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right]$$

$$(b) T = \frac{3A}{4C_d \cdot \tan \theta \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right]$$

$$(c) T = \frac{5A}{4C_d \cdot \tan \frac{\theta}{2} \sqrt{2g}} \left[\frac{1}{H_2^{3/2}} - \frac{1}{H_1^{3/2}} \right]$$

- (d) none of these.

ANSWERS

- | | | | | | |
|---------|---------|---------|---------|----------|---------|
| 1. (b) | 2. (b) | 3. (c) | 4. (c) | 5. (d) | 6. (d) |
| 7. (c) | 8. (b) | 9. (d) | 10. (b) | 11. (d) | 12. (b) |
| 13. (a) | 14. (d) | 15. (c) | 16. (c) | 17. (c). | |

THEORETICAL QUESTIONS

- Define the following terms:
 - Notch,
 - Weir,
 - Nappe or vein
 - Sill or crest.
- What is the main difference between a notch and a weir ?
- How are notches and weirs classified ?
- Derive an expression for the discharge over a rectangular notch or weir in terms of head of water over the crest of the notch or weir.
- Find an expression for the discharge over a triangular notch or weir in terms of head of water over the crest of the notch or weir.
- Prove that the error in discharge due to error in the measurement of head over a triangular notch or weir is given by

$$\frac{dQ}{Q} = \frac{5}{2} \frac{dH}{H}$$

where, Q = Discharge through the triangular notch, and

H = Head over the triangular notch.

- What are the advantages of a triangular notch over a rectangular notch ?
- What is 'velocity of approach' ?
- Derive an expression for the discharge over a rectangular weir with velocity of approach.
- What do you understand by 'end contraction' of a weir ?

- What is the effect of end contraction on the discharge through a weir ?
- Find an expression for the discharge over a Cippoletti weir ?
- Derive an expression for the maximum discharge over a broad-crested weir.
- Write short notes on the following:
 - Narrow-crested weir
 - Ogee weir.
 - Submerged or drowned weir.
- Derive an expression for the time required to empty a tank with a rectangular notch.
- Prove that the time (T) required to empty a tank with a triangular notch is given by

$$T = \frac{5A}{4C_d \cdot \tan \frac{\theta}{2} \sqrt{2g}} \left[\frac{1}{H_2^{3/2}} - \frac{1}{H_1^{3/2}} \right]$$

Where, A = Uniform cross-sectional area of the tank,

θ = Angle of notch,

H_1 = Initial height of liquid above the apex of notch,

H_2 = Final height of liquid above the apex of notch, and

C_d = Co-efficient of discharge.

UNSOLVED EXAMPLES

- Find the discharge of water flowing over a rectangular notch of 2.5 m length when the constant head over the notch is 400 mm. Take $C_d = 0.62$.
[Ans. $1.16 \text{ m}^3/\text{s}$]
- The head of water over a rectangular notch is 900 mm. The discharge is $0.3 \text{ m}^3/\text{s}$. Find the length of the notch when $C_d = 0.62$. [Ans. 192 mm]
- Find the discharge over a triangular notch of angle 60° when the head over the V-notch is 0.3 m. Take $C_d = 0.6$. [Ans. $0.04 \text{ m}^3/\text{s}$]
- A rectangular channel 2.0 m wide has a discharge of $0.25 \text{ m}^3/\text{s}$, which is measured by a right-angled V-notch. Find the position of the apex of the notch from the bed of the channel if the maximum depth of water is not to exceed 1.3 m. Assume $C_d = 0.62$. [Ans. 0.807 m]
- Find the discharge through a trapezoidal notch which is 1 m wide at the top and 0.4 m at the bottom and is 0.3 m in height. The head of water on the notch is 0.2 m. Assume C_d for rectangular portion = 0.62 while for triangular portion = 0.6. [Ans. $0.09084 \text{ m}^3/\text{s}$]
- Find the discharge over a stepped rectangular notch, as shown in Fig. 9.12. Take co-efficient of discharge for all the portions as 0.64.
[Ans. $0.712 \text{ m}^3/\text{s}$]
- A right angled V-notch is used for measuring a discharge of $0.03 \text{ m}^3/\text{s}$. An error of 1.5 mm was made while measuring the head over the notch. Calculate the percentage error in the discharge. Assume $C_d = 0.62$. [Ans. 1.77%]
- A discharge of $0.15 \text{ m}^3/\text{s}$ was measured over a V-notch under a constant head of 1 metre. Determine the percentage error, in the measurement of discharge, if an error of 10 mm has taken place while measuring the head of water. [Ans. 2.5%]
- Find the discharge over a rectangular weir of length 100 m. The head of water over the weir is 1.5 m. The velocity of approach is given as 0.5 m/s. Assume $C_d = 0.6$. [Ans. $329.35 \text{ m}^3/\text{s}$]
- The head of water over a rectangular weir is 400 mm. The length of the crest of the weir with end contraction suppressed is 1.5 m. Find the discharge using the following formulae:
(i) Francis's formula, and
(ii) Bazin's formula
[Ans. (i) $0.6982 \text{ m}^3/\text{s}$, (ii) $0.6932 \text{ m}^3/\text{s}$]
- A weir 36 metres long is divided into 12 equal bays by vertical posts, each 0.6 m wide. Determine the discharge over the weir if the head over the crest is 1.20 m and velocity of approach is 1.2 m/s. Use Francis's formula. [Ans. $75.246 \text{ m}^3/\text{s}$]
- Water is flowing over a Cippoletti weir 4 metres long under a head of 1 metre. Calculate the discharge, if the co-efficient of discharge for the weir is 0.6. [Ans. $7.086 \text{ m}^3/\text{s}$]
- A 40 m long broad-crested weir has 0.4 m height of water above its crest. Find the maximum discharge. Take $C_d = 0.6$. Neglect velocity of approach. [Ans. $10.35 \text{ m}^3/\text{s}$]
- In a submerged weir of 3 m length the heights of water on the upstream and downstream sides are 0.2 m and 0.1 m respectively. Find the discharge over the weir if the discharge co-efficients for free and drowned portions are 0.6 and 0.8 respectively. [Ans. $0.504 \text{ m}^3/\text{s}$]

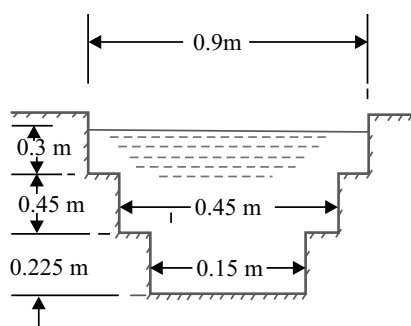


Fig. 9.12

- A rectangular notch 400 mm long is used for measuring a discharge of $0.03 \text{ m}^3/\text{s}$. An error of 1.5 mm was made, while measuring the head over the notch. Calculate the percentage error in the discharge. Assume $C_d = 0.6$.
[Ans. 1.85%]



Laminar Flow

LAMINAR FLOW

10.1. Introduction

10.2. Reynolds experiment

10.3. Navier-Stokes equations of motion

10.4. Relationship between shear stress and pressure gradient

10.5. Flow of viscous fluid in circular pipes—Hagen Poiseuille Law

10.6. Flow of viscous fluid through an annulus

10.7. Flow of viscous fluid between two parallel plates—One plate moving and other at rest—Couette flow—Both plates at rest—Both plates moving in opposite direction

10.8. Laminar flow through porous media

10.9. Power absorbed in bearings—Journal bearing—Foot-step bearing—Collar bearing

10.10. Loss of head due to friction in viscous flow

10.11. Movement of piston in dashpot

10.12. Measurement of viscosity—Rotating cylinder method—Falling sphere method—Capillary tube method—Efflux viscometers

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

10.1. INTRODUCTION

So far, in the preceding chapters, primarily the flow of an ideal fluid has been discussed. In the case of Newtonian fluid, the flows can be classified as (i) laminar (or viscous), and (ii) turbulent, depending on characteristic Reynolds number $\frac{\rho V l}{\mu}$, where l is the characteristic length.

Examples of laminar/viscous flow:

- (i) Flow past tiny bodies.
- (ii) Underground flow.
- (iii) Movement of blood in the arteries of a human body.
- (iv) Flow of oil in measuring instruments.
- (v) Rise of water in plants through their roots etc.

Characteristics of laminar flow:

- (i) 'No slip' at the boundary.
- (ii) Due to viscosity, there is a shear between fluid layers, which is given by $\tau = \mu \cdot \frac{du}{dy}$ for flow in X -direction.
- (iii) The flow is rotational.
- (iv) Due to viscous shear, there is continuous *dissipation of energy* and for maintaining the flow energy must be supplied externally.
- (v) Loss of energy is proportional to *first power of velocity* and *first power of viscosity*.
- (vi) No mixing between different fluid layers (except by molecular motion, which is very small).
- (vii) The flow remains laminar as long as $\frac{\rho V l}{\mu}$ is less than critical value of Reynolds number.

10.2. REYNOLDS EXPERIMENT

Osborne Reynolds in 1883, with the help of a simple experiment (see Fig. 10.1), demonstrated the existence of the following two types of flows:

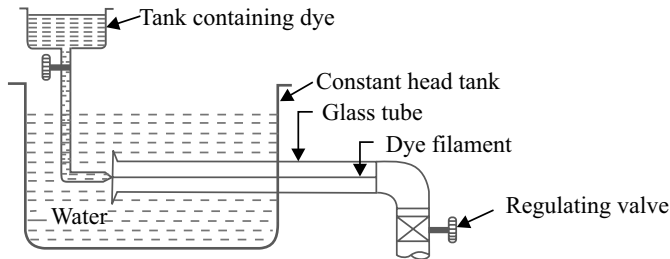


Fig.10.1. Reynolds apparatus.

1. Laminar flow (Reynolds number, $Re < 2000$)
2. Turbulent flow (Reynolds number, $Re > 4000$)

(Re between 2000 and 4000 indicates *transition from laminar to turbulent flow*)

Reynolds experiment:

Apparatus:

Refer to Fig. 10.1. Reynolds experiment apparatus consisted essentially of the following:

1. A constant head tank filled with water,
2. A small tank containing dye (sp. weight of dye same as that of water),
3. A horizontal glass tube provided with a bell mouth entrance, and
4. A regulating valve.

Procedure followed:

The water was made to flow from the tank through the glass tube into the atmosphere and the velocity of flow was varied by adjusting valve. The liquid dye was introduced into the flow at the bell mouth through a small tube as shown in Fig. 10.1.

Observations made:

1. When the *velocity* of flow was *low*, the dye remained in the form of a *straight and stable filament* passing through the glass tube so steadily that it scarcely seemed to be in motion. This was a case of **laminar flow** as shown in Fig. 10.2 (a).
2. With the increase of velocity a critical state was reached at which the dye filament showed irregularities and began to waver (see Fig. 10.2 b). This shows that the flow is no longer a laminar one. This was a **transitional state**.
3. With further increase in velocity of flow the fluctuations in the filament of dye became more intense and ultimately the dye diffused over the entire cross-section of the tube, due to the intermingling of the particles of the flowing fluid. This was the case of a **turbulent flow** as shown in Fig. 10.2 (c).

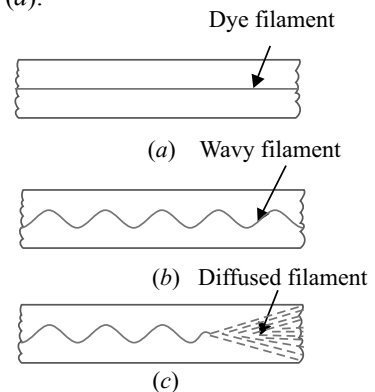


Fig. 10.2. Appearance of dye filament in (a) laminar flow, (b) transition, and (c) turbulent flow.

On the basis of his experiment Reynolds discovered that:

- (i) In case of **laminar flow**: The loss of pressure head $\propto$ velocity.
- (ii) In case of **turbulent flow**: The loss of head is approximately $\propto V^2$
[More exactly the loss of head $\propto V^n$ where n varies from 1.75 to 2.0]

Fig. 10.3 shows the apparatus used by Reynolds for estimating the loss of head in a pipe by measuring the pressure difference over a known length of the pipe.

- (i) The velocity of water in the pipe was determined by measuring the volume of water (Q) collected in the tank over a known period of time ($V = \frac{Q}{A}$, where A is the area of cross-section of the pipe.)

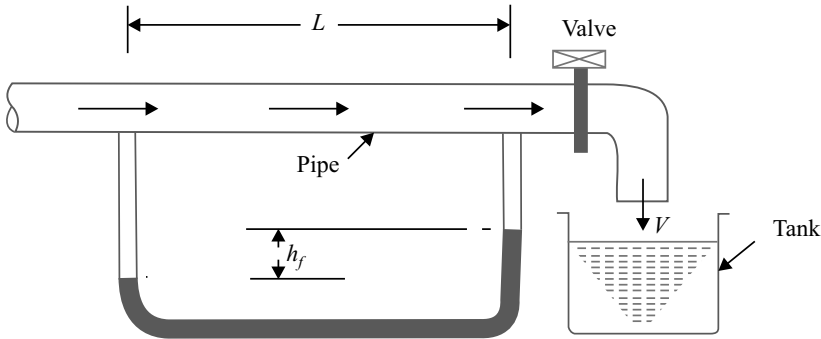


Fig. 10.3. Loss of head in a pipe.

- (ii) The velocity of flow (V) was changed and corresponding values of h_f (loss of head) were obtained.
- (iii) A graph was plotted between V (velocity of flow) and h_f (loss of head). Such a graph is shown in Fig. 10.4. It may be seen from the graph that:
 - (a) At *low velocities* the curve is a straight line, indicating that the h_f (loss of head) is *directly proportional to velocity*—the flow is **laminar** (or viscous),
 - (b) At *higher velocities* the curve is parabolic; in this range $h_f \propto V^n$, where the value of n lies between 1.75 to 2.0 — the flow is **turbulent**.
 - (c) In the intermediate region, there is a transition zone. This is shown by *dotted line*.

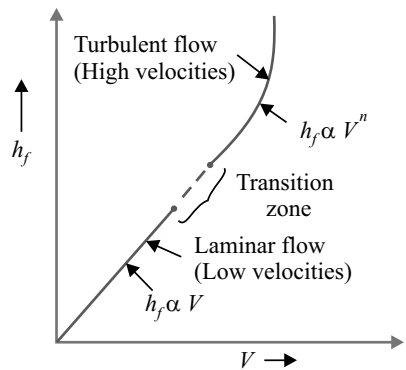


Fig. 10.4

Reynolds number :

Reynolds from his experiments found that the nature of flow in a *closed conduit* depends upon the following factors:

- (i) Diameter of the pipe (D),
- (ii) Density of the liquid (ρ),
- (iii) Viscosity of the liquid (μ), and
- (iv) Velocity of flow (V).

By combining the above variables Reynolds determined a non-dimensional quantity equal to $\frac{\rho V D}{\mu}$ which is known as **Reynolds number** (Re).

i.e. Reynolds number $Re = \frac{\rho V D}{\mu}$

(In general case D is replaced by L , known as characteristic length and we have, $Re = \frac{\rho VL}{\mu}$)

It may also be expressed as:

$$Re = \frac{VD}{\nu}$$

where, $\nu = \text{Kinematic viscosity} \left(= \frac{\mu}{\rho} \right)$

when, $Re < 2000$... the flow is *laminar* (or viscous)
 $Re > 4000$... the flow is *turbulent*.
 Re between 2000 and 4000 ... the flow is *unpredictable*.

Critical Reynolds number :

- All experiments agree that a lower limit of critical value of $(Re)_{cr}$ exists (though there appears to be no definite upper limit of the critical value of $(Re)_{cr}$ which characterises full attainment of turbulence) and its value is approximately 2000 (for circular pipe). This lower critical Reynolds number is of greater engineering importance as it defines the *limit below which all turbulence, no matter how severe, entering the flow from any source will eventually be damped out by viscous action*.
- It has been observed that the upper limit of critical Reynolds number $(Re)_{cr}$ depends upon the following factors:
 - (i) Initial turbulence in the flow (approach),
 - (ii) Shape of the pipe entrance, and
 - (iii) Roughness of pipe.

Reynolds found the upper limit of $(Re)_{cr}$ to lie between $12000 < (Re)_{cr} < 14000$; these values are of little practical interest and we may consider the upper limit of $(Re)_{cr}$ to be defined by $2700 < (Re)_{cr} < 4000$.

— For demarcating the regimes of laminar and turbulent flows, the concept of critical Reynolds number proves quite useful.

The *lower* critical Reynolds number for some important cases are as under:

- (i) $(Re)_{cr} = 1$... for sphere
- (ii) $(Re)_{cr} = 50$... for open channels
- (iii) $(Re)_{cr} = 1000$... for parallel plates.

10.3. NAVIER-STOKES EQUATIONS OF MOTION

Refer to Fig. 10.5.

Consider an infinitely small mass of fluid enclosed in an elementary parallelepiped of sides dx , dy and dz . The motion of the fluid element is influenced by the following forces :

- (i) *Normal forces due to pressure :*

The net pressure force in the X -direction

$$= p \cdot dy \cdot dz - \left[p + \frac{\partial p}{\partial x} dx \right] dy \cdot dz = - \frac{\partial p}{\partial x} dx \cdot dy \cdot dz$$

- (ii) *Gravity or body force :*

Let B be the body force *per unit mass of fluid* having components B_x , B_y and B_z in the X , Y and Z directions respectively.

Then, the body force acting on the parallelepiped in the direction of X -coordinate

$$= B_x \cdot \rho \cdot dx \cdot dy \cdot dz.$$

(iii) *Inertia forces :*

The inertia force acting on the fluid mass, along the X -co-ordinate is given by,

Mass $\times$ acceleration

$$= \rho \cdot dx \cdot dy \cdot dz \cdot \frac{du}{dt}$$

(iv) *Shear forces:*

Let S_x, S_y, S_z be the components of shear force per unit mass set up by viscous effects in X, Y and Z direction respectively. Then, the shear force acting on the parallelopiped in the direction of X -co-ordinate is $= S_x \cdot \rho \cdot dx \cdot dy \cdot dz$

As per Newton's second law of motion summation of forces acting in the fluid element in any direction equals the resulting inertia forces in that direction. Thus along X -direction:

$$-\frac{\partial p}{\partial x} dx \cdot dy \cdot dz + B_x \cdot \rho \cdot dx \cdot dy \cdot dz + S_x \cdot \rho \cdot dx \cdot dy \cdot dz = \rho \cdot dx \cdot dy \cdot dz \frac{du}{dt}$$

$$B_x - \frac{1}{\rho} \cdot \frac{\partial p}{\partial x} = \frac{du}{dt} - S_x$$

Similarly,

$$B_y - \frac{1}{\rho} \cdot \frac{\partial p}{\partial y} = \frac{dv}{dt} - S_y,$$

and,

$$B_z - \frac{1}{\rho} \cdot \frac{\partial p}{\partial z} = \frac{dw}{dt} - S_z \quad \dots(10.1)$$

Let us now find the **values of S_x, S_y and S_z** :

— The resistance (shear) force acting on the face AEHD $= -\mu \frac{\partial u}{\partial x} (dy \cdot dz)$

(Resistance force due to viscosity $= \tau \times \text{area}$)

The resistance force acting on the face BFGC

$$= \mu \frac{\partial}{\partial x} \left(u + \frac{\partial u}{\partial x} \cdot dx \right) (dy \cdot dz) = \mu \left(\frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} dx \right) (dy \cdot dz)$$

The resistance forces acting on the opposite faces have opposite signs. Further, *both of these forces are directed apposite to the respective opposite forces.*

The resultant force along the X -axis

= The algebraic sum of forces acting on the faces AEHD and BFGC

$$= -\mu \frac{\partial u}{\partial x} (dy \cdot dz) + \mu \left(\frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} dx \right) (dy \cdot dz)$$

$$= \mu \frac{\partial^2 u}{\partial x^2} dx \cdot dy \cdot dz \quad \dots(i)$$

— Similarly the X -component of the resistance force acting on the faces EFGH and ABCD respectively are :

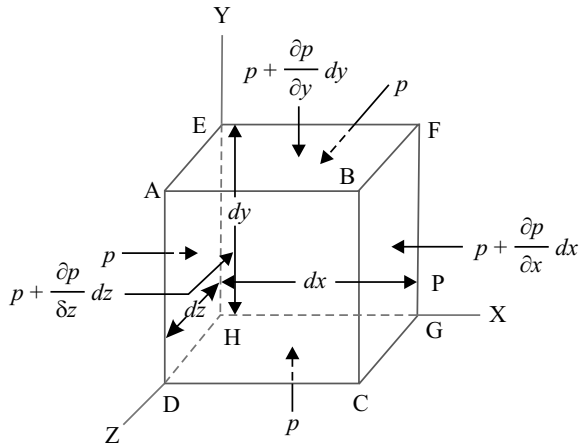


Fig. 10.5

$$- \mu \frac{\partial u}{\partial z} (dx \cdot dy) \quad \text{and} \quad \mu \left(\frac{\partial u}{\partial z} + \frac{\partial^2 u}{\partial z^2} \cdot dz \right) (dx \cdot dy)$$

The resultant force along the X -axis

$$= \text{The sum of the above components} = \mu \frac{\partial^2 u}{\partial z^2} \cdot dx \cdot dy \cdot dz \dots (ii)$$

— Similarly, the X -components of the resistance force acting on the faces DHGC and AEFB

$$= \mu \frac{\partial^2 u}{\partial y^2} dx \cdot dy \cdot dz \dots (iii)$$

The total viscous force, parallel to X -axis, on all the six faces of the parallelepiped is given by the sum of the quantities (i), (ii) and (iii) and is

$$\begin{aligned} &= \mu \cdot \frac{\partial^2 u}{\partial x^2} \cdot dx \cdot dy \cdot dz + \mu \cdot \frac{\partial^2 u}{\partial y^2} \cdot dx \cdot dy \cdot dz + \mu \cdot \frac{\partial^2 u}{\partial z^2} \cdot dx \cdot dy \cdot dz \\ &= \mu \cdot \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx \cdot dy \cdot dz \end{aligned}$$

The resistance (shear) per unit mass is obtained by dividing the above quantity by $\rho \, dx \cdot dy \cdot dz$.

$$\therefore S_x = \frac{\mu}{\rho} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right),$$

Similarly $S_y = \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right), \text{ and}$

$$S_z = \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right).$$

Putting these values of S_x , S_y and S_z in eqn. (10.1), we get:

$$\begin{aligned} B_x - \frac{1}{\rho} \cdot \frac{\partial p}{\partial x} &= \frac{du}{dt} - \nu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right] \\ B_y - \frac{1}{\rho} \cdot \frac{\partial p}{\partial y} &= \frac{dv}{dt} - \nu \left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right] \\ B_z - \frac{1}{\rho} \cdot \frac{\partial p}{\partial z} &= \frac{dw}{dt} - \nu \left[\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right] \end{aligned} \dots (10.2)$$

*These equations are called **Navier-Stokes equations** and are fundamental to general analysis of a viscous flow.*

- Navier-Stokes equations, in vector form, may be written as :

$$\frac{DV}{Dt} = B - \frac{1}{\rho} \text{grad. } p + \nu \nabla^2 V$$

Where ∇^2 denotes the Laplace operator,

$$i.e. \quad \nabla^2 = \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right]$$

- The number of unknown, for an incompressible flow is four viz. u , v , w and p . The Navier-Stokes equation plus incompressible continuity equation $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$ are the sufficient conditions for the determination of the flow characteristics.
- Since Navier-Stokes equation is a second order non-linear differential equation, therefore, its general solution has not been yet found out (the non-linearity arises from the convective terms in $\frac{DV}{Dt}$). Thus, *the solutions are available for flow situations where the fluid characteristics such as viscosity and density are constant and boundary configuration is simple.*

Some of the important applications of Navier-Stokes equations are:

1. Laminar flow in circular pipes.
2. Laminar flow between concentric rotating cylinders.
3. Laminar uni-directional flow between stationary parallel plates.
4. Laminar uni-directional flow between parallel plates having relative motion.

10.4. RELATIONSHIP BETWEEN SHEAR STRESS AND PRESSURE GRADIENT

Let us consider a fluid element having the form of an elementary parallelepiped shown in Fig. 10.6 (a). The velocity distribution is shown in Fig. 10.6 (b); the velocity distribution is non-uniform due to relative motion between different layers of fluid. The motion of the fluid element will be resisted by shearing or frictional forces which must be overcome by maintaining a pressure gradient in the direction of flow. Let us assume that the pressure is uniformly distributed at both the ends of the body.

Let, τ = Shear stress on the lower face $ABCD$ of the element, then

$$\tau + \frac{\partial \tau}{\partial y} \delta y = \text{Shear stress on the upper face } A'B'C'D' \text{ of the element.}$$

For two-dimensional steady flow there will be no shear stresses on the vertical faces $ABB'A'$ and $CDD'C'$. Thus the only forces acting on the element in the direction of flow (X -axis) will be the pressure and shear forces.

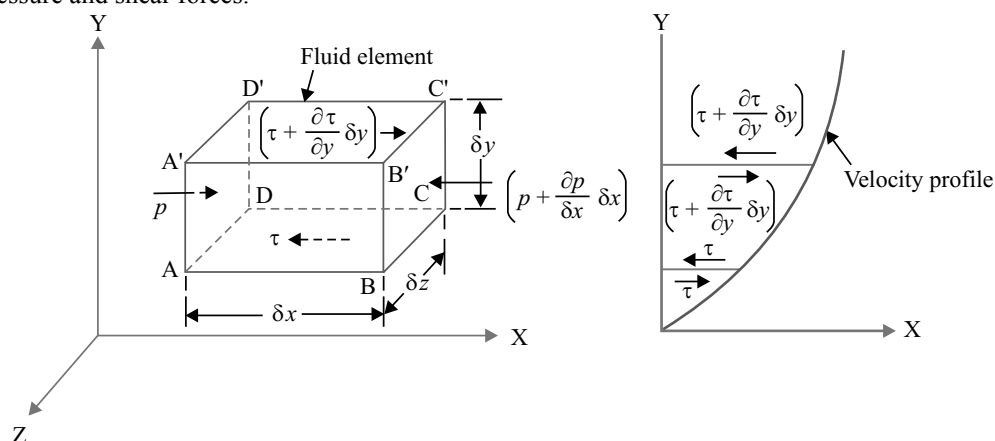


Fig. 10.6. Forces on a fluid element in laminar flow.

Net shearing force on the element

$$\begin{aligned}
 &= \left(\tau + \frac{\partial \tau}{\partial y} \delta y \right) \delta x \cdot \delta z - \tau \cdot \delta x \cdot \delta z \\
 &= \frac{\partial \tau}{\partial y} \delta x \cdot \delta y \cdot \delta z \quad \dots(i)
 \end{aligned}$$

Net pressure force on the element

$$\begin{aligned}
 &= p \cdot \delta y \cdot \delta z - \left(p + \frac{\partial p}{\partial x} \cdot \delta x \right) \delta y \cdot \delta z \\
 &= -\frac{\partial p}{\partial x} \delta x \cdot \delta y \cdot \delta z \quad \dots(ii)
 \end{aligned}$$

For the flow to be steady and uniform, there being no acceleration, the sum of the forces must be zero.

From (i) and (ii), we have:

$$\begin{aligned}
 \frac{\partial \tau}{\partial y} \cdot \delta x \cdot \delta y \cdot \delta z - \frac{\partial p}{\partial x} \cdot \delta x \cdot \delta y \cdot \delta z &= 0 \\
 \text{or,} \quad \frac{\partial \tau}{\partial y} &= \frac{\partial p}{\partial x} \quad \dots(10.3)
 \end{aligned}$$

This equation (10.3) indicates that the *pressure gradient in the direction of flow is equal to the shear gradient in the direction normal to the direction of flow*. This equation holds good for all types of flow and all types of boundary geometry.

10.5. FLOW OF VISCOUS FLUID IN CIRCULAR PIPES—HAGEN POISEUILLE LAW

Hagen–Poiseuille theory is based on the following *assumptions*:

1. The fluid follows Newton's law of viscosity.
2. There is no slip of fluid particles at the boundary (*i.e.* the fluid particles adjacent to the pipe will have zero velocity).

Fig. 10.7 shows a horizontal circular pipe of radius R , having laminar flow of fluid through it. Consider a small concentric cylinder (fluid element) of radius r and length dx as a free body.

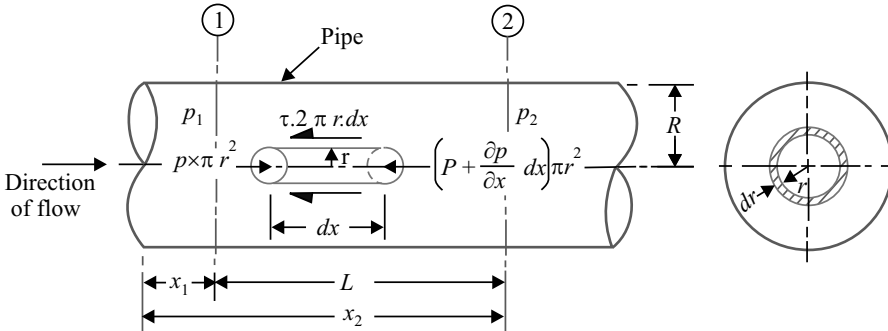


Fig. 10.7. Viscous/laminar flow through a circular pipe.

If τ is the shear stress, the shear force F is given by:

$$F = \tau \times 2\pi r \times dx$$

Let p be the intensity of pressure at left end and the intensity of pressure at the right end be $\left(p + \frac{\partial p}{\partial x} \cdot dx \right)$.

Thus the forces acting on the fluid element are:

1. The shear force, $\tau \times 2\pi r \times dx$ on the surface of fluid element.
2. The pressure force, $p \times \pi r^2$ on the left end.
3. The pressure force, $\left(p + \frac{\partial p}{\partial x} \cdot dx \right) \pi r^2$ on the right end.

For steady flow, the net force on the cylinder must be zero.

$$\therefore \left[p \times \pi r^2 - \left(p + \frac{\partial p}{\partial x} \cdot dx \right) \pi r^2 \right] - \tau \times 2\pi r \times dx = 0$$

$$\text{or,} \quad -\frac{\partial p}{\partial x} \cdot dx \times \pi r^2 - \tau \times 2\pi r \times dx = 0$$

$$\text{or,} \quad \tau = -\frac{\partial p}{\partial x} \cdot \frac{r}{2} \quad \dots(10.4)$$

— Eqn. (10.4) shows that flow will occur only if *pressure gradient exists in the direction of flow*.

The *negative sign* shows that *pressure decreases in the direction of flow*.

— Eqn. (10.2) indicates that the *shear stress* varies linearly across the section (see Fig. 10.8). Its value is zero at the centre of pipe ($r = 0$) and maximum at the pipe wall given by:

$$\tau_0 = -\frac{\partial p}{\partial x} \left(\frac{R}{2} \right) \quad \dots[10.4 (a)]$$

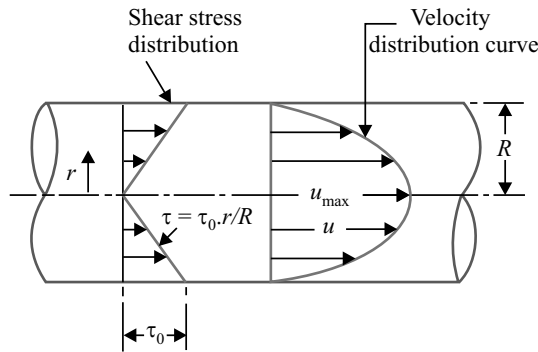


Fig. 10.8. Shear stress and velocity distribution across a section.

From Newton's law of viscosity,

$$\tau = \mu \cdot \frac{du}{dy} \quad \dots(i)$$

In this equation, the distance y is measured from the boundary. The radial distance r is related to distance y by the relation:

$$y = R - r \quad \text{or} \quad dy = -dr$$

The eqn. (i) becomes

$$\tau = -\mu \frac{du}{dr} \quad \dots(10.5)$$

Comparing two values of τ from eqns. 10.2 and 10.3, we have:

$$-\mu \frac{du}{dr} = -\frac{\partial p}{\partial x} \cdot \frac{r}{2}$$

$$\text{or,} \quad du = \frac{1}{2\mu} \left(\frac{\partial p}{\partial x} \right) r \cdot dr$$

Integrating the above equation w.r.t. ' r ', we get:

$$u = \frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} r^2 + C \quad \dots(10.6)$$

Where C is the constant of integration and its value is obtained from the boundary condition :

At,

$$r = R, \quad u = 0$$

∴

$$0 = \frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} R^2 + C \quad \text{or} \quad C = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} \cdot R^2$$

Substituting this value of C in eqn. (10.4), we get:

$$u = \frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} r^2 - \frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} R^2$$

or,

$$u = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} (R^2 - r^2) \quad \dots(10.7)$$

Eqn. (10.7) shows that the velocity distribution curve is a *parabola* (see Fig. 10.8). The maximum velocity occurs at the centre and is given by,

$$u_{\max} = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} \cdot R^2 \quad \dots(10.8)$$

From eqns. (10.7) and (10.8), we have:

$$u = u_{\max} \left[1 - \left(\frac{r}{R} \right)^2 \right] \quad \dots(10.9)$$

Eqn. (10.9) is the *most commonly used equation for the velocity distribution for laminar flow through pipes*. This equation can be used to calculate the *discharge* as follows:

The discharge through an elementary ring of thickness dr at radial distances r is given by:

$$\begin{aligned} dQ &= u \times 2\pi r \times dr \\ &= u_{\max} \left[1 - \left(\frac{r}{R} \right)^2 \right] 2\pi r \cdot dr \end{aligned}$$

$$\begin{aligned} \text{Total discharge, } Q &= \int dQ \\ &= \int_0^R u_{\max} \left[1 - \left(\frac{r}{R} \right)^2 \right] 2\pi r \cdot dr \\ &= 2\pi u_{\max} \int_0^R \left(r - \frac{r^3}{R^2} \right) dr \\ &= 2\pi u_{\max} \left[\frac{r^2}{2} - \frac{r^4}{4R^2} \right]_0^R = 2\pi u_{\max} \left[\frac{R^2}{2} - \frac{R^4}{4R^2} \right] \\ &= \frac{\pi}{2} u_{\max} R^2 \end{aligned}$$

$$\text{Average velocity of flow, } \bar{u} = \frac{Q}{A} = \frac{\frac{\pi}{2} u_{\max} R^2}{\pi R^2} = \frac{u_{\max}}{2} \quad \dots(10.10)$$

Eqn. (10.10) shows that the *average velocity is one-half the maximum velocity*.

Substituting the value of $u_{\max}$ from eqn. (10.8), we have:

$$\bar{u} = -\frac{1}{8\mu} \cdot \frac{\partial p}{\partial x} \cdot R^2$$

or,

$$-\partial p = \frac{8\mu \bar{u}}{R^2} \cdot \partial x$$

The pressure difference between two sections 1 and 2 at distance x_1 and x_2 (see Fig. 10.7). is given by

$$-\int_{p_1}^{p_2} \partial p = \frac{8\mu\bar{u}}{R^2} \int_{x_1}^{x_2} \partial x$$

$$\text{or,} \quad (p_1 - p_2) = \frac{8\mu\bar{u}}{R^2} (x_2 - x_1) = \frac{8\mu\bar{u}L}{R^2}$$

$$\text{or,} \quad (p_1 - p_2) = \frac{32\mu\bar{u}L}{D^2} \quad \dots(10.11)$$

where, D is the diameter of the pipe, and L is the length.

Eqn. (10.11) is known as the **Hagen-Poiseuille equation**.

Example 10.1. An oil of viscosity 9 poise and specific gravity 0.9 is flowing through a horizontal pipe of 60 mm diameter. If the pressure drop in 100 m length of the pipe is 1800 kN/m², determine:

- (i) The rate of flow of oil;
- (ii) The centre-line velocity;
- (iii) The total frictional drag over 100 m length;
- (iv) The power required to maintain the flow;
- (v) The velocity gradient at the pipe wall;
- (vi) The velocity and shear stress at 8 mm from the wall.

Solution. Viscosity of the oil, $\mu = 9 \text{ poise} = \frac{1}{10} \times 9 = 0.9 \text{ Ns/m}^2$

Sp. gr. of the oil = 0.9

Diameter of the pipe, $D = 60 \text{ mm} = 0.06 \text{ m}$

$\therefore$ Area of the pipe, $A = \frac{\pi}{4} \times 0.06^2 = 0.002827 \text{ m}^2$

Pressure drop in 100 m length of the pipe, $\Delta p = 1800 \text{ kN/m}^2$

(i) **The rate of flow, Q :**

$$(p_1 - p_2) = \Delta p = \frac{32\mu\bar{u}L}{D^2} \quad (\text{where } \bar{u} = \text{average velocity})$$

$$1800 \times 10^3 = \frac{32 \times 0.9 \times \bar{u} \times 100}{(0.06)^2}$$

$$\text{or,} \quad \bar{u} = \frac{1800 \times 10^3 \times (0.06)^2}{32 \times 0.9 \times 100} = 2.25 \text{ m/s}$$

$$\text{Reynolds number, } Re = \frac{\rho V D}{\mu} = \frac{0.9 \times 1000 \times 2.25 \times 0.06}{0.9} = 135$$

As Re is less than 2000, the flow is *laminar*.

$$\begin{aligned} \text{Rate of flow, } Q &= A \cdot \bar{u} = 0.002827 \times 2.25 \\ &= \mathbf{0.00636 \text{ m}^3/\text{s} \text{ or } 6.36 \text{ lit./s. (Ans.)} \end{aligned}$$

(ii) **The centre-line velocity, $u_{\max}$:**

$$u_{\max} = 2\bar{u} = 2 \times 2.25 = \mathbf{4.5 \text{ m/s (Ans.)}}$$

(iii) The total frictional drag over 100 m length, F_D :

Wall shear stress,

$$\tau_0 = -\frac{\partial p}{\partial x} \cdot \frac{R}{2} \quad \dots[\text{Eqn. 10.2 (a)}]$$

Now,
$$-\frac{\partial p}{\partial x} = -\frac{p_2 - p_1}{x_2 - x_1} = \frac{p_1 - p_2}{L} = \frac{\Delta p}{L} = \frac{1800 \times 10^3}{100} = 18000$$

$$\therefore \tau_0 = 18000 \times \frac{0.06/2}{2} = 270 \text{ N/m}^2$$

$\therefore$ Frictional drag for 100 m length,

$$F_D = \tau_0 \times \pi DL = 270 \times \pi \times 0.06 \times 100 = 5089 \text{ N or } \mathbf{5.089 \text{ kN (Ans.)}}$$

(iv) The power required to maintain the flow, P :

$$\begin{aligned} P &= F_D \times \bar{u} \\ &= 5.089 \times 2.25 = \mathbf{11.45 \text{ kW (Ans.)}} \end{aligned}$$

[Alternatively, $P = Q \cdot \Delta p = 0.00636 \times 1800 = 11.45 \text{ kW}$]

(v) The velocity gradient at the pipe wall, $\left(\frac{du}{dy}\right)_{y=0}$:

$$\tau_0 = \mu \cdot \left(\frac{\partial u}{\partial y}\right)_{y=0}$$

or,
$$\left(\frac{\partial u}{\partial y}\right)_{y=0} = \frac{\tau_0}{\mu} = \frac{270}{0.9} = \mathbf{300 \text{ s}^{-1} \text{ (Ans.)}}$$

(vi) The velocity and shear stress at 8 mm from the wall :

$$u = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} (R^2 - r^2) \quad \dots[\text{Eqn. (10.5)}]$$

Here, $y = 8 \text{ mm} = 0.008 \text{ m}$

But, $y = R - r$

$$\therefore 0.008 = 0.03 - r \quad \text{or} \quad r = 0.022 \text{ m}$$

$$\begin{aligned} \therefore u_{(8 \text{ mm})} &= -\frac{1}{4 \times 0.9} \times \frac{1800 \times 10^3}{100} (0.03^2 - 0.022^2) \\ &= \mathbf{2.08 \text{ m/s (Ans.)}} \end{aligned}$$

Also,
$$\frac{\tau}{r} = \frac{\tau_0}{R} \quad (\text{Refer to Fig. 10.9})$$

$$\begin{aligned} \therefore \tau_{(8 \text{ mm})} &= r \times \frac{\tau_0}{R} = 0.022 \times \frac{270}{0.03} \\ &= \mathbf{198 \text{ kN/m}^2 \text{ (Ans.)}} \end{aligned}$$

Example 10.2. Oil of absolute viscosity 1.5 poise and density 848.3 kg/m³ flows through a 30 cm I.D. pipe. If the head loss in 3000 m length of pipe is 20 m, assuming a laminar flow, determine (i) the velocity, (ii) Reynolds number and (iii) friction factor (Fanning's). (UPTU)

Solution. Given : $\mu = 1.5 \text{ poise} = 1.5 \times \frac{1}{10} = 0.15 \text{ N-s/m}^2$, $\rho = 848.3 \text{ kg/m}^3$; $D = 30 \text{ cm} = 0.3 \text{ m}$; $h_f = 20 \text{ m}$; $L = 3000 \text{ m}$; Flow-laminar.

(i) The velocity:

We know that,
$$\Delta p = \frac{32\mu\bar{u}L}{D^2} \quad (\text{where, } \bar{u} = \text{average velocity})$$

$$\text{or,} \quad \Delta p = \rho g h_f = \frac{32\mu \bar{u} L}{D^2}$$

$$\text{or,} \quad \bar{u} = \frac{\rho g h_f D^2}{32 \mu L} = \frac{848.3 \times 9.81 \times 20 \times 0.3^2}{32 \times 0.15 \times 3000} = \mathbf{1.04 \text{ m/s (Ans.)}}$$

(ii) Reynolds number; Re :

$$Re = \frac{\rho \bar{u} D}{\mu} = \frac{848.3 \times 1.04 \times 0.3}{0.15} = \mathbf{1764.5 \text{ (Ans.)}}$$

(iii) Friction coefficient, f :

$$f = \frac{16}{Re} = \frac{16}{1764.5} = \mathbf{0.00907 \text{ (Ans.)}} \quad [\text{Eqn. (10.11)}]$$

Example 10.3. A crude oil of viscosity 0.9 poise and relative density 0.9 is flowing through a horizontal circular pipe of diameter 120 mm and length 12 m. Calculate the difference of pressure at the two ends of the pipe, if 785 N of the oil is collected in a tank in 25 seconds.

Solution. Viscosity of the crude oil, $\mu = 0.9 \text{ poise} = 0.09 \text{ Ns/m}^2$

Relative density = 0.9

$\therefore$ Weight density = $0.9 \times 9810 = 8829 \text{ N/m}^3$

Diameter of the pipe, $D = 120 \text{ mm} = 0.12 \text{ m}$

Length of the pipe, $L = 12 \text{ m}$

Weight of the oil collected in 25 s = 785 N

Difference of pressure, $(p_1 - p_2)$:

The difference of pressure for viscous or laminar flow is given by

$$(p_1 - p_2) = \frac{32\mu \bar{u} L}{D^2}$$

$$\text{Now, weight of oil collected/sec.} = \frac{785}{25} = 31.4 \text{ N/s} = w \times Q$$

$$\therefore Q = \frac{31.4}{8829} = 0.00355 \text{ m}^3/\text{s}$$

$$\text{Average velocity, } \bar{u} = \frac{Q}{\text{Area}} = \frac{0.00355}{(\pi/4)D^2} = \frac{0.00355}{(\pi/4) \times 0.12^2} = 0.314 \text{ m/s}$$

$$\text{Reynolds number, } Re = \frac{\rho V D}{\mu} = \frac{(0.9 \times 1000) \times 0.314 \times 0.12}{0.09} = 376.8$$

Since $Re < 2000$, therefore, the flow is *laminar/viscous*.

Substituting the values in eqn. (i), we get

$$(p_1 - p_2) = \frac{32 \times 0.09 \times 0.314 \times 12}{(0.12)^2} = \mathbf{753.6 \text{ N/m}^2 \text{ (Ans.)}}$$

Example 10.4. A liquid with a specific gravity 2.8 and a viscosity 0.8 poise flows through a smooth pipe of unknown diameter, resulting in a pressure drop of 800 N/m^2 in 2 km length of the pipe. What is the pipe diameter if the mass flow rate is 2500 kg/h. (NU)

Solution. Given: Sp. gravity = 2.8, $\mu = 0.8 \times 0.1 = 0.08 \text{ Ns/m}^2$; $\Delta p = 800 \text{ N/m}^2$;

$$L = 2 \text{ km} = 2000 \text{ m}; m = 2500 \text{ kg/h} = \frac{2500}{3600} = 0.6944 \text{ kg/s}$$

Pipe diameter, D :

$$m = \rho A \bar{u} \quad \text{or} \quad \bar{u} = \frac{m}{\rho A}$$

$$\text{or,} \quad \bar{u} = \frac{0.6944}{(2.8 \times 1000) \times \frac{\pi}{4} D^2} = \frac{3.158 \times 10^{-4}}{D^2}$$

Assuming flow to be laminar, we have:

$$\Delta p = \frac{32\mu\bar{u}L}{D^2} \quad (\text{where, } \bar{u} = \text{average velocity}) \quad \dots[\text{Eqn. 10.11}]$$

$$800 = \frac{32 \times 0.08 \times 3.158 \times 10^{-4} \times 2000}{D^2 \times D^2}$$

$$\text{or,} \quad D = \left(\frac{32 \times 0.08 \times 3.158 \times 10^{-4} \times 2000}{800} \right)^{1/4} = \mathbf{0.212 \text{ m (Ans.)}}$$

Check for laminar flow,

$$Re = \frac{\rho\bar{u}D}{\mu} = \frac{(2.8 \times 1000) \times 3.158 \times 10^{-4} \times D}{D^2 \times 0.08} = \frac{2800 \times 3.158 \times 10^{-4}}{0.212 \times 0.08} = 52, \text{ which confirms that}$$

flow is laminar ($Re < 2000$).

Example 10.5. A fluid of viscosity 8 poise and specific gravity 1.2 is flowing through a circular pipe of diameter 100 mm. The maximum shear stress at the pipe wall is 210 N/m². Find:

- (i) The pressure gradient,
- (ii) The average velocity, and
- (iii) Reynolds number of flow.

Solution. Viscosity of fluid, $\mu = 8 \text{ poise} = 0.8 \text{ Ns/m}^2$

Specific gravity = 1.2

$\therefore$ Mass density, $\rho = 1.2 \times 1000 = 1200 \text{ kg/m}^3$

Diameter of the pipe, $D = 100 \text{ mm} = 0.1 \text{ m}$

Maximum shear stress, $\tau_0 = 210 \text{ N/m}^2$

(i) The pressure gradient, $\frac{\partial p}{\partial x}$:

$$\text{We know,} \quad \tau_0 = -\frac{\partial p}{\partial x} \cdot \frac{R}{2}$$

$$\text{or,} \quad 210 = -\frac{\partial p}{\partial x} \times \frac{(0.1/2)}{2}$$

$$\therefore \quad \frac{\partial p}{\partial x} = -\frac{210 \times 4}{0.1} = \mathbf{-8400 \text{ N/m}^2 \text{ per m (Ans.)}}$$

(ii) The average velocity, $\bar{u}$:

$$\text{We know,} \quad \bar{u} = \frac{1}{2} u_{\max}$$

$$= \frac{1}{2} \left[-\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} \cdot R^2 \right] \quad \dots[\text{Eqn. (10.8)}]$$

$$= \frac{1}{2} \left[-\frac{1}{4 \times 0.8} \times (-8400) \times (0.1/2)^2 \right]$$

$$= \mathbf{3.28 \text{ m/s (Ans.)}}$$

(iii) Reynolds number, Re :

$$Re = \frac{\rho V D}{\mu} = \frac{1200 \times 3.28 \times 0.1}{0.8} = 492 \text{ (Ans.)}$$

Example 10.6. A fluid of density 1200 kg/m^3 and viscosity 0.5 poise is flowing at a rate of $5 \text{ m}^3/\text{min}$ in a circular pipe of cross-section of 1 m^2 . Is the flow laminar or turbulent? Can you predict the maximum velocity of the fluid in the pipe?

Solution. Given: $\rho = 1200 \text{ kg/m}^3$; $\mu = 0.5 \text{ poise} = 0.5 \times \frac{1}{10} = 0.05 \text{ Ns/m}^2$

$$A = 1 \text{ m}^2; \quad Q = 5 \text{ m}^3/\text{min}$$

$$Q = AV = 1 \times \frac{5}{60} \text{ m}^3/\text{s}$$

$$V_{av} = \bar{u} = \frac{Q}{A} = \frac{5/60}{1} = \frac{5}{60} \text{ m/s}$$

$$A = 1 = \frac{\pi}{4} D^2 \quad \text{or} \quad D = 1.128 \text{ m}$$

$$\text{Reynolds number, } Re = \frac{\rho \bar{u} D}{\mu} = \frac{1200 \times (5/60) \times 1.128}{0.05} = 2256$$

Since $Re < 2300$, therefore the flow is **laminar**. (Ans.)

The velocity profile is parabolic, hence

$$u_{\max} = 2\bar{u} = 2 \times \frac{5}{60} = 0.1667 \text{ m/s (Ans.)}$$

Example 10.7. A lubricating oil of viscosity 1 poise and specific gravity 0.9 is pumped through a 30 mm diameter pipe. If the pressure drop per metre length of pipe is 20 kN/m^2 , determine:

- (i) The mass flow rate in kg/min ,
- (ii) The shear stress at the pipe wall,
- (iii) The Reynolds number of flow, and
- (iv) The power required per 50 m length of the pipe to maintain the flow.

Solution. Viscosity of oil, $\mu = 1 \text{ poise} = 0.1 \text{ Ns/m}^2$

Sp. gr. of oil, $= 0.9$

$\therefore$ Weight density, $w = 0.9 \times 9810 = 8829 \text{ N/m}^3$

Diameter of pipe, $D = 30 \text{ mm} = 0.03 \text{ m}$

$$\text{Area, } A = \frac{\pi}{4} \times 0.03^2 = 7.068 \times 10^{-4} \text{ m}^2$$

Pressure drop per metre length of pipe, $(p_1 - p_2) = 20 \text{ kN/m}^2$

(i) **Mass flow rate:**

Pressure drop for laminar flow through a pipeline is given by,

$$(p_1 - p_2) = \frac{32\mu\bar{u}L}{D^2}$$

$$\therefore 20 \times 10^3 = \frac{32 \times 0.01 \times \bar{u} \times 1}{(0.03)^2}$$

(where, $\bar{u}$ = average velocity of flow)

$$\therefore, \bar{u} = \frac{20 \times 10^3 \times (0.03)^2}{32 \times 0.1 \times 1} = 5.625 \text{ m/s}$$

$$\begin{aligned}\text{Flow rate, } Q &= A \times \bar{u} = 7.068 \times 10^{-4} \times 5.625 = 0.003975 \text{ m}^3/\text{s} \\ \therefore \text{Mass flow rate} &= (0.9 \times 1000) \times 0.003975 \times 60 \\ &= \mathbf{214.65 \text{ kg/min. (Ans.)}}\end{aligned}$$

(ii) Shear stress at the wall, τ_0 :

$$\tau_0 = -\frac{\partial p}{\partial x} \cdot \frac{R}{2} = 20 \times 10^3 \times \frac{(0.03/2)}{2} = \mathbf{150 \text{ N/m}^2 \text{ (Ans.)}}$$

(iii) Reynolds number of flow, Re :

$$Re = \frac{\rho V D}{\mu} = \frac{(0.9 \times 1000) \times 5.625 \times 0.03}{0.1} = \mathbf{1518.7 \text{ (Ans.)}}$$

(where, $V = \bar{u} = 5.625 \text{ m/s}$)

This is less than 2000 and hence the flow is *laminar*.

(iv) Power required, P :

$$\text{Loss of head, } h_f = \frac{p_1 - p_2}{w} = \frac{20 \times 10^3}{8829} = 2.265 \text{ m of oil}$$

$$\text{Power reqd. per metre} = w Q h_f = 8829 \times 0.003975 \times 2.265 \text{ W} = 79.49 \text{ W}$$

For 50 m length, power required,

$$P = 79.49 \times 50 = 3974.5 \text{ W or } \mathbf{3.974 \text{ kW (Ans.)}}$$

Example 10.8. In a pipe of 300 mm diameter the maximum velocity of flow is found to be 2 m/s. If the flow in the pipe is laminar, find:

(i) The average velocity and the radius at which it occurs, and

(ii) The velocity at 50 mm from the wall of the pipe.

Solution. Diameter of the pipe, $D = 300 \text{ mm} = 0.3 \text{ m}$

Maximum velocity, $u_{\max} = 2 \text{ m/s}$

(i) Average velocity, $\bar{u}$:

$$\text{We know, } \bar{u} = \frac{1}{2} u_{\max}$$

$$\therefore \bar{u} = \frac{2}{2} = \mathbf{1 \text{ m/s (Ans.)}}$$

Radius at which $\bar{u}$ occurs :

The velocity, u at any radius r is given by

$$u = -\frac{1}{4\mu} \frac{\partial p}{\partial x} (R^2 - r^2) \quad \dots[\text{Eqn. (10.7)}]$$

$$= -\frac{1}{4\mu} \frac{\partial p}{\partial x} R^2 \left[1 - \frac{r^2}{R^2} \right]$$

$$\text{Also, } u_{\max} = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} \cdot R^2 \quad \dots[\text{Eqn. (10.8)}]$$

$$\therefore u = u_{\max} \left[1 - \left(\frac{r}{R} \right)^2 \right] \quad \dots(i)$$

Here, $u = \bar{u} = 1 \text{ m/s}$

$$\therefore 1 = 2 \left[1 - \left(\frac{r}{0.15} \right)^2 \right] \quad \left(\because R = \frac{0.3}{2} = 0.15 \text{ m} \right)$$

$$\begin{aligned} \text{or,} \quad \frac{1}{2} &= 1 - \left(\frac{r}{0.15} \right)^2 \\ \text{or,} \quad \left(\frac{r}{0.15} \right)^2 &= 0.5 \\ \text{or,} \quad \frac{r}{0.15} &= \sqrt{0.5} = 0.707 \\ \therefore \quad r &= 0.106 \text{ m or } \mathbf{106 \text{ mm (Ans.)}} \end{aligned}$$

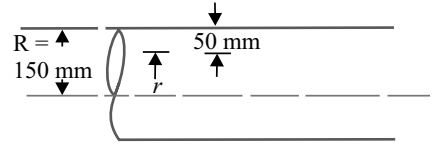


Fig. 10.9

(ii) Velocity at 50 mm from the wall :

$$r = 150 - 50 = 100 \text{ mm} = 0.1 \text{ m}$$

$\therefore$ Velocity at a radius 0.1 m or 50 mm from the pipe wall is given by [eqn. (i)]:

$$\begin{aligned} u &= u_{\max} \left[1 - \left(\frac{r}{R} \right)^2 \right] \\ &= 2 \left[1 - \left(\frac{0.1}{0.15} \right)^2 \right] = \mathbf{1.11 \text{ m/s (Ans.)}} \end{aligned}$$

Example 10.9. An oil of viscosity 0.15 Ns/m^2 and specific gravity 0.9 is flowing through a circular pipe of diameter 30 mm and of length 3 m at $\frac{1}{10}$ th of critical velocity for which Reynolds number is 2450 . Find:

- The velocity of flow through the pipe,
- The head in metres of oil across the pipe length required to maintain the flow, and
- The power required to overcome viscous resistance to flow of oil.

Solution. Viscosity of the oil, $\mu = 0.15 \text{ Ns/m}^2$

Specific gravity = 0.9

$\therefore$ Mass density, $\rho = 0.9 \times 1000 = 900 \text{ kg/m}^3$

Diameter of the pipe, $D = 30 \text{ mm} = 0.03 \text{ m}$

Length of the pipe, $L = 3 \text{ m}$

Velocity of flow, $\bar{u} = \frac{1}{10} \times \text{critical velocity (at Reynolds number 2450)}$

(i) Velocity of flow, $\bar{u}$:

We know, $(Re)_{cr} = \frac{\rho V_{cr} D}{\mu}$

(where, V_{cr} = critical velocity)

$$\text{or,} \quad 2450 = \frac{900 \times V_{cr} \times 0.03}{0.15}$$

$$\text{or,} \quad V_{cr} = \frac{2450 \times 0.15}{900 \times 0.03} = 13.61 \text{ m/s}$$

$\therefore$ Velocity of flow through the pipe,

$$\bar{u} = \frac{1}{10} \times 13.61 = \mathbf{1.361 \text{ m/s (Ans.)}}$$

(ii) Head required to maintain the flow:

For laminar flow through a pipeline,

$$p_1 - p_2 = \frac{32\mu\bar{u}L}{D^2}$$

or, Loss of head, $h_f = \frac{p_1 - p_2}{w} = \frac{32\mu\bar{u}L}{wD^2} = \frac{32\mu\bar{u}L}{(\rho g)D^2}$

$$\therefore h_f = \frac{32 \times 0.15 \times 1.361 \times 3}{900 \times 9.81 \times 0.03^2} = \mathbf{2.466 \text{ m (Ans.)}}$$

(iii) Power required, P :

The power required to overcome viscous resistance to flow of oil,

$$P = w Q h_f = (900 \times 9.81) \times \left(\frac{\pi}{4} \times 0.03^2 \times 1.361 \right) \times 2.466$$

$$= \mathbf{20.9 \text{ W (Ans.)}}$$

Example 10.10. An oil ($\mu = 20$ cP, $\rho = 1200$ kg/m³) flows through a 2.5 cm I.D. pipe 250 m long.

(i) What is the maximum flow in m³/s that will ensure laminar flow ?

(ii) What would be the pressure drop for this flow? **(Bombay University)**

Solution. Given: $\mu = 20$ c.P. $= 20 \times 10^{-2}$ poise $= 20 \times 10^{-2} \times \frac{1}{10}$ Ns/m² $= 0.02$ Ns/m²;
 $\rho = 1200$ kg/m³; $D_1 = 2.5$ cm $= 0.025$ m; $L = 250$ m.

Flow will be *laminar flow* if Reynolds number is less than 2000.

Now, $Re = \frac{\rho V D}{\mu}$

or, $2000 = \frac{1200 \times V \times 0.025}{0.02}$ or $V = \frac{2000 \times 0.02}{1200 \times 0.025} = 1.33$ m/s

(i) Maximum flow that will ensure laminar flow:

$$\text{Discharge} = A \times V = \frac{\pi}{4} \times (0.025)^2 \times 1.33 = \mathbf{6.528 \times 10^{-4} \text{ m}^3/\text{s (Ans.)}}$$

(ii) Pressure drop :

$$\text{Coefficient of friction, } f = \frac{16}{Re} = \frac{16}{2000} = 0.008$$

$$\text{Head lost due to friction, } h_f = \frac{4fLV^2}{D \times 2g}$$

$$= \frac{4 \times 0.008 \times 250 \times 1.33^2}{0.025 \times 2 \times 9.81} = 28.85 \text{ m}$$

$$\text{Pressure drop for the flow} = wh_f = (\rho g) \times h_f$$

$$= 1200 \times 9.81 \times 28.85 \text{ N/m}^2$$

$$= 339622.2 \text{ N/m}^2 = \mathbf{3.396 \text{ bar (Ans.)}} \quad (\because 1 \text{ bar} = 10^5 \text{ N/m}^2).$$

Example 10.11. Crude oil of $\mu = 1.5$ poise and relative density 0.9 flows through a 20 mm diameter vertical pipe. The pressure gauges fixed 20 m apart read 600 kN/m² and 200 kN/m², as shown in Fig. 10.10. Find the direction and rate of flow through the pipe. **[PTU]**

Solution. Dynamic viscosity, $\mu = 1.5$ poise $= 0.15$ Ns/m²

Relative density = 0.9

$\therefore$ Weight density of oil, $w = 0.9 \times 9.81 = 8.829$ kN/m³

Diameter of the pipe, $D = 20$ mm $= 0.02$ m

Length, $L = 20$ m

Pressure at A, $p_A = 600 \text{ kN/m}^2$

Pressure at B, $p_B = 200 \text{ kN/m}^2$

Direction of flow:

Since the pipe is of uniform diameter, *velocity head would be same* and as such flow direction will be indicated by the values of *piezometric head* at sections A and B. Taking the level at A as datum, we have:

Piezometric head at A

$$= \frac{p_A}{w} + z_A = \frac{600}{8.829} + 0 = 67.96 \text{ m}$$

Piezometric head at B

$$= \frac{p_B}{w} + z_B = \frac{200}{8.829} + 20 = 42.65 \text{ m}$$

Since piezometric head at A is greater than that at B, hence, flow takes place **from A to B** (i.e. upwards) (Ans.)

Rate of flow:

Loss of piezometric head = $h_f = 67.96 - 42.65 = 25.31 \text{ m}$

But the loss of pressure head for viscous flow through circular pipe is given by the Hagen-Poiseuille relation,

$$h_f = \frac{32 \mu \bar{u} L}{w D^2}$$

$$\text{or, } 25.31 = \frac{32 \times 0.15 \times \bar{u} \times 20}{(8.829 \times 10^3) \times (0.02)^2}$$

$$\text{or, } \bar{u} = \frac{25.31 \times (8.829 \times 10^3) \times (0.02)^2}{32 \times 0.15 \times 20} = 0.931 \text{ m/s}$$

$$\text{Reynolds number, } Re = \frac{\rho V D}{\mu} = \frac{(0.9 \times 1000) \times 0.931 \times 0.02}{0.15} = 111.72 \quad (\because V = \bar{u})$$

As Reynolds number is less than 2000, the flow is *laminar*.

$$\begin{aligned} \therefore \text{Flow rate, } Q &= \text{Average velocity} \times \text{area} \\ &= \bar{u} \times \pi/4 \times D^2 = 0.931 \times \pi/4 \times 0.02^2 \\ &= 2.925 \times 10^{-4} \text{ m}^3/\text{s} \quad \text{or} \quad 0.2925 \text{ litres/sec. (Ans.)} \end{aligned}$$

Example 10.12. Oil of specific gravity 0.82 is pumped through a horizontal pipeline 150 mm in diameter and 3 km long at the rate of $0.015 \text{ m}^3/\text{s}$. The pump has an efficiency of 68% and requires 7.5 kW to pump the oil.

(i) What is the dynamic viscosity of the oil?

(ii) Is the flow laminar?

[Panjab University]

Solution.

Sp. gr. of the oil = 0.82

Diameter of pipe, $D = 150 \text{ mm} = 0.15 \text{ m}$

$$\therefore \text{Area, } A = (\pi/4) \times 0.15^2 = 0.01767 \text{ m}^2$$

Length of pipe, $L = 3 \text{ km} = 3000 \text{ m}$

Discharge, $Q = 0.015 \text{ m}^3/\text{s}$

Efficiency of pump, $\eta = 68\%$

Power required to pump the oil, $P = 7.5 \text{ kW}$

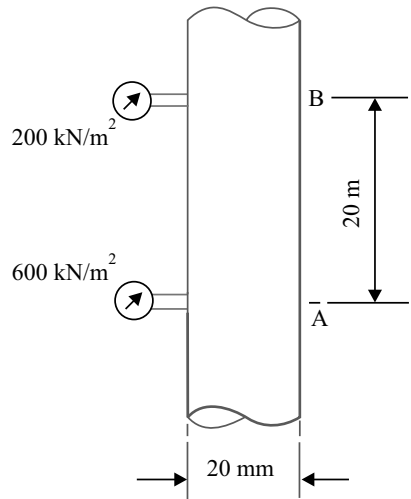


Fig. 10.10

(i) Dynamic viscosity of oil, μ :

$$\text{Average velocity, } \bar{u} = \frac{Q}{A} = \frac{0.015}{0.01767} = 0.849 \text{ m/s}$$

If h_f represents the loss of head, then:

$$w Q h_f = \eta \times P$$

$$\text{or, } h_f = \frac{\eta \times P}{w Q} = \frac{0.68 \times (7.5 \times 10^3)}{(0.82 \times 9810) \times 0.015} = 42.26 \text{ m}$$

Again, for viscous/laminar flow through a pipeline,

$$(p_1 - p_2) = \frac{32 \mu \bar{u} L}{D^2}$$

$$\text{or, Loss of head, } h_f = \left(\frac{p_1 - p_2}{w} \right) = \frac{32 \mu \bar{u} L}{w D^2}$$

$$\text{or, } 42.26 = \frac{32 \times \mu \times 0.849 \times 3000}{(0.82 \times 9810) \times 0.15^2}$$

$$\text{or, } \mu = \frac{42.26 \times (0.82 \times 9810) \times 0.15^2}{32 \times 0.849 \times 3000} = \mathbf{0.0938 \text{ Nsm/m}^2 \text{ (Ans.)}}$$

(ii) Is the flow laminar?

$$\text{Reynolds number, } Re = \frac{\rho V D}{\mu} = \frac{(0.82 \times 1000) \times 0.849 \times 0.15}{0.0938} \quad (\because V = \bar{u})$$

$$\text{or, } Re = 1113.3$$

This is less than 2000 and hence the flow is **laminar (Ans.)**

Example 10.13. A pipe 60 mm diameter and 450 m long slopes upwards at 1 in 50. An oil of viscosity 0.9 Ns/m^2 and specific gravity 0.9 is required to be pumped at the rate of 5 litres/sec.

(i) Is the flow laminar?

(ii) What pressure difference is required to attain this condition?

(iii) What is the power of the pump required assuming an overall efficient of 65%?

(iv) What is the centre-line velocity and the velocity gradient at pipe wall? (MU)

Solution. Diameter of the pipe, $D = 60 \text{ mm} = 0.06 \text{ m}$

$$\therefore \text{Area of the pipe} = \frac{\pi}{4} \times 0.06^2 = 0.00283 \text{ m}^2$$

Length of the pipe, $L = 450 \text{ m}$

Slope = 1 in 50

Viscosity of oil, $\mu = 0.9 \text{ Ns/m}^2$

Weight density, $w = 0.9 \times 9810 = 8829 \text{ N/m}^3$

Discharge, $Q = 5 \text{ litres/sec.}$

$$= 0.005 \text{ m}^3/\text{s.}$$

Overall efficiency, $\eta_0 = 65\%$

(i) Is the flow laminar?

$$\text{Average velocity, } \bar{u} = \frac{Q}{A} = \frac{0.005}{0.00283} = 1.767 \text{ m/s}$$

$$\therefore \text{Reynolds number, } Re = \frac{\rho V D}{\mu} = \frac{(0.9 \times 1000) \times 1.767 \times 0.06}{0.9} = 106$$

Since $Re < 2000$, therefore, **flow is laminar. (Ans.)**

(ii) Pressure difference required:

Applying Bernoulli's equation between section (1) and (2), we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_f$$

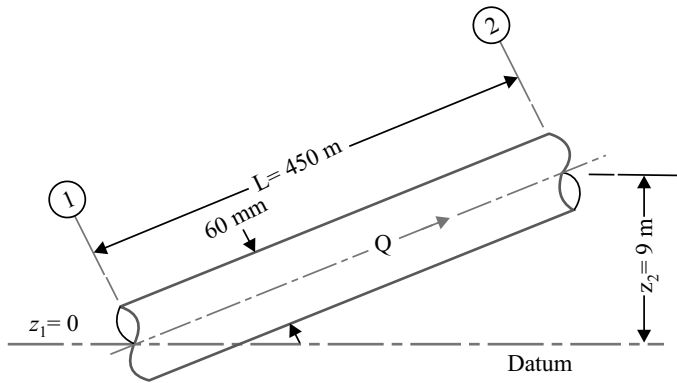
$$\text{or, } \frac{p_1}{w} + \frac{V_1^2}{2g} + 0 = \frac{p_2}{w} + \frac{V_2^2}{2g} + \frac{1}{50} \times 450 + \frac{32\mu\bar{u}L}{D^2 \times w}$$

$$\text{or, } \left(\frac{p_1 - p_2}{w} \right) = 9 + \frac{32\mu\bar{u}L}{D^2 \times w} \quad (\because V_1 = V_2 \text{ and } z_1 = 0)$$

$$\begin{aligned} \text{or, } p_1 - p_2 &= 9w + \frac{32\mu\bar{u}L}{D^2} \\ &= 9 \times 8829 + \frac{32 \times 0.9 \times 1.767 \times 450}{0.06^2} \\ &= 79461 + 6.36 \times 10^6 = 6.44 \times 10^6 \text{ N/m}^2 \text{ or } \mathbf{6.44 \text{ MN/m}^2} \text{ (Ans.)} \end{aligned}$$

(iii) Power of the pump:

$$P = Q(p_1 - p_2) = 0.005 \times 6.44 \times 10^3 = 32.2 \text{ kW}$$

**Fig. 10.11**

$$\text{Power of the pump} = \frac{32.2}{\eta_0} = \frac{32.2}{0.65} = 49.54 \text{ kW (Ans.)}$$

(iv) Centre-line velocity, $u_{\max}$:

$$u_{\max} = 2\bar{u} = 2 \times 1.767 = \mathbf{3.534 \text{ m/s (Ans.)}}$$

Velocity gradient at the pipe wall:

$$\begin{aligned} \tau_0 &= -\frac{\partial p}{\partial x} \cdot \frac{R}{2} \\ &= \frac{6.44 \times 10^6}{450} \times \frac{0.03}{2} = 214.67 \text{ N/m}^2 \end{aligned}$$

But,

$$\tau_0 = \mu \cdot \left(\frac{\partial u}{\partial y} \right)_{y=0}$$

$$\text{or, } \left(\frac{\partial u}{\partial y} \right)_{y=0} = \frac{\tau_0}{\mu} = \frac{214.67}{0.9} = \mathbf{238.5 \text{ s}^{-1} \text{ (Ans.)}}$$

Example 10.14. A total of 12 litres per second of oil is pumped through two pipes in parallel, one 12 cm in diameter and the other 10 cm in diameter, both pipes being 1000 metres long. The specific gravity of the oil is 0.97 and the kinematic viscosity $9 \text{ cm}^2 \text{ per second}$. Calculate the flow rate through each pipe and the power of the pump. [UPSC Exam, Fluid Machines]

Solution. Total discharge of oil through the two pipes,

$$Q = 12 \text{ litres/sec.} \\ = 0.012 \text{ m}^3/\text{s}$$

Diameter of pipe 1,

$$D_1 = 12 \text{ cm} = 0.12 \text{ m}$$

Diameter of pipe 2,

$$D_2 = 10 \text{ cm} = 0.1 \text{ m}$$

$$L_1 = L_2 = L = 1000 \text{ m}$$

(where, L_1 and L_2 are the lengths of the pipes 1 and 2 respectively)

Specific gravity of the oil = 0.97

Kinematic viscosity of the oil,

$$\nu = 9 \text{ cm}^2/\text{s} \\ = 9 \times 10^{-4} \text{ m}^2/\text{s}.$$

Let, Q_1 = Discharge in pipe 1, and

Q_2 = Discharge in pipe 2.

Since the pipes are in parallel, the total discharge Q is distributed in both the pipes. From the principle of continuity, we have

$$Q = Q_1 + Q_2$$

$$\text{or, } 0.012 = Q_1 + Q_2 \quad \dots(i)$$

Flow rate through each pipe, Q_1 and Q_2 :

Assuming laminar flow in both the pipes, the pressure difference between sections 'A' and 'B' is the same in both the pipes,

$$\Delta p = \frac{32\mu V_1 L}{D_1^2} = \frac{32\mu V_2 L}{D_2^2} \quad \dots(ii)$$

(where, V_1 and V_2 are the velocities of flow in the pipes 1 and 2 respectively).

From the continuity equation, we have:

$$V_1 = \frac{Q_1}{A_1} = \frac{Q_1}{\frac{\pi}{4} D_1^2} = \frac{4Q_1}{\pi D_1^2}$$

$$\text{and, } V_2 = \frac{Q_2}{A_2} = \frac{Q_2}{\frac{\pi}{4} D_2^2} = \frac{4Q_2}{\pi D_2^2}$$

From eqn. (ii), we have:

$$\frac{V_1}{D_1^2} = \frac{V_2}{D_2^2}$$

$$\therefore \frac{4Q_1}{\pi D_1^4} = \frac{4Q_2}{\pi D_2^4}$$

$$\text{From which, } \frac{Q_1}{Q_2} = \left(\frac{D_1}{D_2} \right)^4 \quad \text{or} \quad Q_2 = Q_1 \left(\frac{D_2}{D_1} \right)^4$$

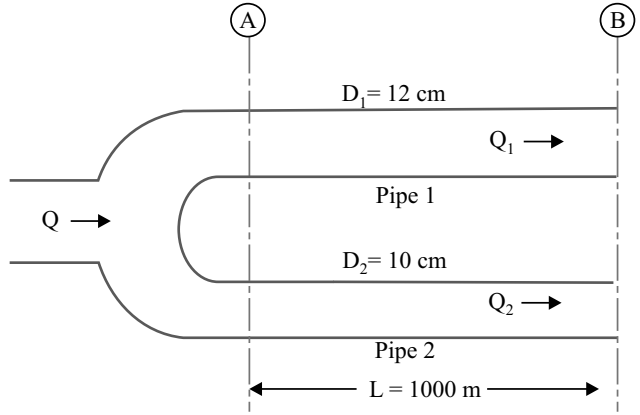


Fig. 10.12

Substituting the value of Q_2 in eqn. (i), we get:

$$\begin{aligned} 0.012 &= Q_1 + Q_1 \left(\frac{D_2}{D_1} \right)^4 = Q_1 \left[1 + \left(\frac{D_2}{D_1} \right)^4 \right] \\ &= Q_1 \left[1 + \left(\frac{0.1}{0.12} \right)^4 \right] = 1.482 Q_1 \end{aligned}$$

$$\therefore Q_1 = 0.008097 \text{ m}^3/\text{s} \text{ or } \mathbf{8.097 \text{ litres/sec. (Ans.)}}$$

$$\text{and, } Q_2 = 12 - 8.097 = \mathbf{3.903 \text{ litres/sec. (Ans.)}}$$

Power of the pump, P:

$$\text{Velocity of flow in pipe 2, } V_2 = \frac{4Q_2}{\pi D_2^2} = \frac{4 \times 0.003903}{\pi \times (0.1)^2} = 0.497 \text{ m/s}$$

$$\text{Velocity of flow in pipe 1, } V_1 = V_2 \left(\frac{D_2}{D_1} \right)^2 = 0.497 \times \left(\frac{0.1}{0.12} \right)^2 = 0.345 \text{ m/s}$$

Pressure drop in 1000 m length of pipe 2,

$$\Delta p = \frac{32\mu V_2 L}{D_2^2}$$

$$\begin{aligned} \text{Loss of head, } h_f &= \frac{\Delta p}{w} = \frac{32 \left(\frac{\mu}{\rho} \right) V_2 L}{g D_2^2} \\ &= \frac{32 \times 9 \times 10^{-4} \times 0.497 \times 1000}{9.81 \times (0.1)^2} = 145.9 \text{ m of oil} \end{aligned}$$

Power lost in viscous friction,

$$P = w Q h_f = (0.97 \times 9.81) \times 0.012 \times 145.9 = 16.66 \text{ kW}$$

Assuming the overall efficiency of the pump as 62%, the power of pump

$$= \frac{16.66}{0.62} = \mathbf{26.87 \text{ kW (Ans.)}}$$

$$\text{Reynolds number, } Re_2 = \frac{V_2 D_2}{\nu} = \frac{0.497 \times 0.1}{9 \times 10^{-4}} = 55.2$$

$$Re_1 = \frac{V_1 D_1}{\nu} = \frac{0.345 \times 0.12}{9 \times 10^{-4}} = 46$$

Since the Reynolds numbers in both the pipes are less than 2000, the assumption of laminar flow in both the pipes is *valid*.

Example 10.15. It is required to pump glycerine at the rate of 25 litres/sec. from a sump and deliver it freely at a point 120 m away and 8 m above the level of sump through a 150 mm pipe, Fig. 10.13.

(i) What is the power of the pump required assuming an overall efficiency of 65% ?

(ii) What would be the rate of rise of temperature due to viscous dissipation if the pipe is completely insulated ?

Sp. gr. of glycerine = 1.26; viscosity = 15 poise, specific heat = 248 J/N °C;

K.E. correction factor, $\alpha = 2$.

Solution. Rate of flow of glycerine, $Q = 25 \text{ litres/sec.} = 0.025 \text{ m}^3/\text{s}$

Diameter of the pipe, $D = 150 \text{ mm} = 0.15 \text{ m}$

$$\therefore \text{Area, } A = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

Overall efficiency, $\eta_0 = 65 \%$

Specific gravity of glycerine = 1.26

$$\therefore \text{Weight density, } w = 1.26 \times 9810 = 12361 \text{ N/m}^3$$

Viscosity, $\mu = 15 \text{ poise} = 1.5 \text{ Ns/m}^2$

Specific heat = 248 J/N °C

K.E. Correction factor, $\alpha = 2$.

(i) **Power of the pump required, P:**

$$\text{Velocity of flow, } V = \frac{Q}{A} = \frac{0.025}{0.01767} = 1.415 \text{ m/s}$$

$$\text{Reynolds number, } Re = \frac{\rho V D}{\mu} = \frac{(1.26 \times 1000) \times 1.415 \times 0.15}{1.5} = 178.3$$

Since the Reynolds number is less than 2000, the flow is *laminar*.

Applying Bernoulli's equation at sump (1) and free delivery point (2), we get:

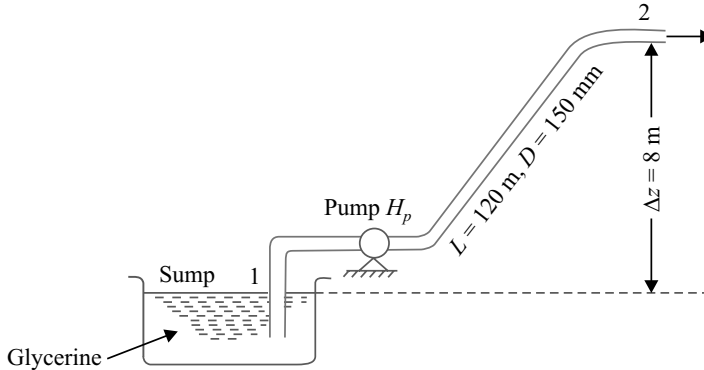


Fig. 10.13

$$\frac{p_1}{w} + \alpha \frac{V_1^2}{2g} + z_1 + H_p = \frac{p_2}{w} + \alpha \frac{V_2^2}{2g} + z_2 + h_f$$

$$0 + 0 + 0 + H_p = 0 + 2 \times \frac{1.415^2}{2 \times 9.81} + 8 + \frac{32\mu V_2 L}{w D^2}$$

$$\text{or, } H_p = \frac{2 \times 1.415^2}{2 \times 9.81} + 8 + \frac{32 \times 1.5 \times 1.415 \times 120}{12361 \times 0.15^2} = 37.51 \text{ m}$$

Power of the pump required,

$$P = \frac{w Q H_p}{\eta_0} = \frac{12361 \times 0.025 \times 37.51}{0.65 \times 1000} \text{ kW} = \mathbf{17.83 \text{ kW (Ans.)}}$$

(ii) **Rate of rise of temperature:**

Dissipation of energy per N per second

= (Energy on the discharge side of the pump – energy at the point of delivery) per N per second

$$= h_f \times \frac{V}{L}$$

(since h_f is energy lost per unit weight (N) of the fluid in a length L)

$$\begin{aligned}
 &= \frac{32 \mu VL}{wD^2} \times \frac{V \text{ N.m}}{L \text{ N.s}} \text{ or } \frac{\text{J}}{\text{s.N}} \\
 &= \frac{32 \times 1.5 \times 1.415^2}{12361 \times 0.15^2} = 0.345 \frac{\text{J}}{\text{s.N}}
 \end{aligned}$$

$\therefore$ Rate of rise of temperature

$$= \frac{0.345}{248} \times 60 \times 60 = 5^\circ\text{C/h (Ans.)}$$

Example 10.16. A horizontal circular tube of radius “ a ” has a fixed co-axial cylindrical core of radius b . τ_a and τ_b are the shear stresses along the tube and core surfaces when a viscous liquid is flowing through the annulus. The flow is laminar and the rate of variation of pressure along the length of the passage is $\left(-\frac{\partial p}{\partial l}\right)$. Show that :

$$\alpha \tau_a - b \cdot \tau_b = \frac{1}{2} (a^2 - b^2) \left(\frac{\partial p}{\partial l} \right)$$

[UPSC Exams., Fluid Mech. & Fluid Machines]

Solution. Radius of the circular tube = a

Radius of the co-axial cylindrical core = b

Shear stress along the tube = τ_a

Shear stress along the core surfaces = τ_b

Rate of variation of pressure = $-\frac{\partial p}{\partial l}$

We know, $\frac{\partial p}{\partial x} + \frac{\partial \tau}{\partial r} + \frac{\tau}{r} = 0$...Eqn. [10.12 (a)]

Since p is dependent on x and τ on r , the above equation can be written as:

$$\frac{\partial p}{\partial x} + \frac{1}{r} \cdot \frac{\partial}{\partial r} (\tau \cdot r) = 0$$

Integrating, w.r.t. r , we get:

$$\frac{r^2}{2} \cdot \frac{\partial p}{\partial x} + \tau \cdot r = C$$

(where, C = constant of integration)

At, $r = a, \tau = \tau_a$

At, $r = b, \tau = \tau_b$

Also, $\frac{\partial p}{\partial x} = -\frac{\partial p}{\partial l}$

Substituting the above values, we get:

$$\frac{a^2}{2} \left(-\frac{\partial p}{\partial l} \right) + a \cdot \tau_a = C \quad \dots(i)$$

$$\text{and, } \frac{b^2}{2} \left(-\frac{\partial p}{\partial l} \right) + b \cdot \tau_b = C \quad \dots(ii)$$

Subtracting (ii) from (i), we get:

$$\frac{1}{2} (a^2 - b^2) \left(-\frac{\partial p}{\partial l} \right) + a \cdot \tau_a - b \cdot \tau_b = 0$$

$$\text{or,} \quad a \cdot \tau_a - b \cdot \tau_b = \frac{1}{2} (a^2 - b^2) \left(\frac{\partial p}{\partial l} \right) \quad \dots(\text{Proved})$$

Example 10.17. Crude oil is pumped through a 150 mm diameter smooth pipe which is subjected to seasonal changes in temperature. At the maximum temperature of 38°C , when the kinematic viscosity is 0.28 stokes, a power input of 2.3 kW per 300 m is required to maintain a flow of 30 litres/sec. What power input would be required to maintain the same rate of flow at the minimum temperature of 0°C if the viscosity of the oil is then 10 times great ?

Assume a specific gravity of 0.9 at both temperatures.

[Roorkee University]

Solution. Diameter of the pipe, $D = 150 \text{ mm} = 0.15 \text{ m}$

Kinematic viscosity of the oil, $\nu = 0.28 \text{ stokes} = 0.28 \times 10^{-4} \text{ m}^2/\text{s}$

Length of pipe considered, $L = 300 \text{ m}$

Specific gravity = 0.9

Rate of flow, $Q = 30 \text{ litres/sec.} = 0.03 \text{ m}^3/\text{s}$

Specific gravity of the oil = 0.9

Power input required, P:

$$\text{Reynolds number at } 38^\circ\text{C}, (Re)_{38^\circ\text{C}} = \frac{VD}{\nu}$$

$$\left(\text{where velocity, } V = \frac{Q}{A} = \frac{0.03}{(\pi/4) \times 0.15^2} = 1.697 \text{ m/s} \right)$$

$$\text{or,} \quad (Re)_{38^\circ\text{C}} = \frac{1.697 \times 0.15}{0.28 \times 10^{-4}} = 9091$$

$$\therefore \text{ Reynolds number at } 0^\circ\text{C}, (Re)_{0^\circ\text{C}} = \frac{1}{10} \times 9091 = 909.1$$

It is thus obvious that the flow at 38°C is turbulent, while at 0°C , it is *laminar* (since $Re < 2000$).

Pressure drop in laminar flow is given by,

$$p_1 - p_2 = \frac{32\mu\bar{u}L}{D^2} = \frac{32 \times (\rho\nu \times 10) \bar{u}L}{D^2} \quad [\because (\mu)_{0^\circ\text{C}} = 10 \times (\mu)_{38^\circ\text{C}}]$$

Here, $\bar{u} = V = 1.697 \text{ m/s}$, $L = 300 \text{ m}$, $D = 0.15 \text{ m}$

$\rho = 0.9 \times 1000 = 900 \text{ kg/m}^3$, and $\nu = 0.28 \times 10^{-4}$

Substituting the values, we get:

$$\begin{aligned} p_1 - p_2 &= \frac{32 \times (800 \times 0.28 \times 10^{-4} \times 10) \times 1.697 \times 300}{(0.15)^2} \\ &= 182461 \text{ N/m}^2 \text{ or } 182.46 \text{ kN/m}^2 \end{aligned}$$

Power required to maintain flow

$$= Q(p_1 - p_2) = 0.03 \times 182.46 = \mathbf{5.474 \text{ kW (Ans.)}}$$

Example 10.18. A pipe of diameter 100 mm and length 1000 m is used to pump oil of viscosity 0.85 Ns/m^2 and specific gravity 0.92 at the rate of $1.2 \text{ m}^3/\text{min}$. The first 300 m of pipe is laid along the ground sloping upwards 10° to the horizontal and the remaining pipe is laid on the ground sloping upwards at 15° to the horizontal.

(i) State whether the flow is laminar or turbulent ?

(ii) Determine the pressure to be developed by the pump and the power of the driving motor if the pump efficiency is 65%.

Assume suitable data for friction factor, f , if required

[UPSC Exams., Fluid Mechanics and Hydraulic Machines]

Solution. Diameter of the pipe, $D = 100 \text{ mm} = 0.1 \text{ m}$

Length of the pipe, $L = 1000 \text{ m}$

Viscosity of the oil, $\mu = 0.85 \text{ Ns/m}^2$

Specific gravity = 0.92

Discharge, $Q = 1.2 \text{ m}^3/\text{min.}$

Pump efficiency, $\eta_p = 65\%$

(i) Flow—laminar or turbulent?

Reynolds number of flow,

$$Re = \frac{\rho V D}{\mu}$$

where,

$$V = \frac{Q}{\frac{\pi}{4} \times D^2} = \frac{(1.2 / 60)}{\frac{\pi}{4} \times 0.1^2} = 2.546 \text{ m/s}$$

$$\rho = 0.92 \times 1000 = 920 \text{ kg/m}^3$$

$$Re = \frac{920 \times 2.546 \times 0.1}{0.85} = 275.6$$

Since $Re < 2000$, therefore, the flow is **laminar. (Ans.)**

(ii) Pressure to be developed by the pump, p :

Height of the end point B of the pipeline above the pump centre P

$$= 300 \sin 10^\circ + 700 \sin 15^\circ = 233.26 \text{ m.}$$

The friction factor, for laminar flow, is given by:

$$f = \frac{64}{Re} = \frac{64}{275.6} = 0.2322$$

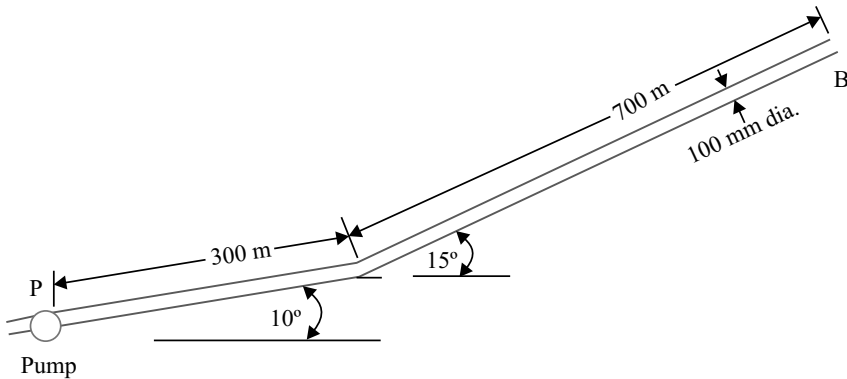


Fig. 10.14

Head lost in friction in 1000 m length of pipeline,

$$h_f = \frac{f L V^2}{D \times 2g} = \frac{0.2322 \times 1000 \times 2.546^2}{0.1 \times 2 \times 9.81} = 767.15 \text{ m}$$

$$\left[\begin{aligned} \text{Alternatively, } h_f &= \frac{32 \mu V L}{w D^2} = \frac{32 \times 0.85 \times 2.546 \times 1000}{0.92 \times 9810 \times 0.1^2} \\ &= 767.3 \text{ m} \end{aligned} \right]$$

Assuming pressure in the pipeline at B as atmospheric, taking horizontal through the pump centre as the datum and applying Bernoulli's equation between the pump outlet and the end of the pipeline, we get:

$$\begin{aligned}\frac{p}{w} + \frac{V^2}{2g} + 0 &= 0 + \frac{V^2}{2g} + 233.26 + h_f \\ &= 0 + \frac{V^2}{2g} + 233.26 + 767.15 = \frac{V^2}{2g} + 1000.41\end{aligned}$$

(where, p = pressure just at the pump outlet)

or, $\frac{p}{w} = 1000.41$ or $p = 0.92 \times 9.81 \times 1000.41 = 9028.9 \text{ kN/m}^2$ (Ans.)

Power of the driving motor,

$$\begin{aligned}P &= \frac{wQh}{\eta_p} = \frac{w \times Q \times p}{w \times \eta_p} = \frac{Q \times p}{0.65} = \frac{(1.2/60) \times 9028.9}{0.65} \\ &= 277.8 \text{ kW (Ans.)}\end{aligned}$$

Example 10.19. Derive a relation for the torque required to rotate the cone at a constant angular velocity ω for the conical thrust bearing shown in Fig. 10.15. For one such bearing, 120 W gets dissipated when a shaft with maximum cone radius 100 mm turns with 620 r.p.m. over a uniform fluid layer of thickness 1.2 mm. If semi-angle for the conical bearing is 30° , find the dynamic viscosity of the fluid. (Anna university)

Solution. Refer to Fig. 10.15.

Consider an elementary ring of bearing surface of radius r and thickness dh at a distance h from the cone apex.

Then, $r = h \tan \alpha$, and

bearing surface of the elementary ring, $dA = 2 \pi r (dh \sec \alpha) = 2 \pi h \tan \alpha \cdot dh \sec \alpha$.

Shear stress (viscous),

$$\tau = \mu \frac{du}{dy} = \mu \frac{V}{t}$$

But, V (linear velocity) $= \omega r$
 $= \omega \cdot h \tan \alpha$

$$\therefore \tau = \frac{\mu \omega h \tan \alpha}{t}$$

(where, t is the thickness of the oil film)

Tangential resistance on the ring,

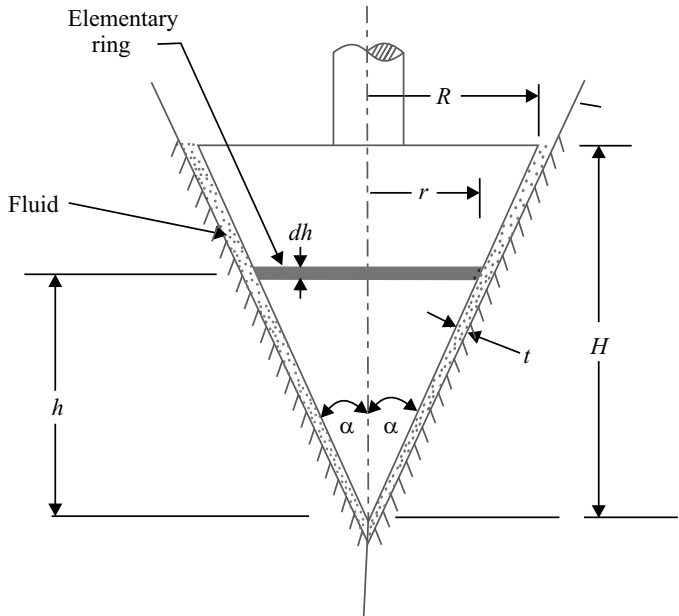


Fig. 10.15. Conical thrust bearing.

$$\begin{aligned}dF &= \text{Shear stress} \times \text{area of the ring} \\ &= \frac{\mu \omega h \tan \alpha}{t} \times 2 \pi h \cdot \tan \alpha \cdot dh \cdot \sec \alpha \\ &= 2 \pi \mu \tan^2 \alpha \cdot \sec \alpha \cdot \frac{\omega}{t} h^2 \cdot dh\end{aligned}$$

and, $\text{torque} = dF \times r = dF \cdot h \tan \alpha$

$$\therefore dT = 2 \pi \mu \tan^3 \alpha \cdot \sec \alpha \cdot \frac{\omega}{t} h^3 dh$$

$$\text{Torque, } T = 2 \pi \mu \tan^3 \alpha \cdot \sec \alpha \cdot \frac{\omega}{t} \int_0^H h^3 dh$$

$$= 2 \pi \mu \tan^3 \alpha \cdot \sec \alpha \cdot \frac{\omega}{t} \times \frac{H^4}{4}$$

$$\text{or, } T = \frac{\pi \mu \omega}{2t} H^4 \tan^3 \alpha \cdot \sec \alpha \quad \dots(i)$$

Given : Power dissipated, $P = 120 \text{ W}$

Maximum cone radius, $R = 100 \text{ mm} = 0.1 \text{ m}$

Speed, $N = 620 \text{ r.p.m.}$

$$\therefore \text{Angular velocity} = \frac{2\pi N}{60} = \frac{2\pi \times 620}{60} = 64.93 \text{ rad./s}$$

Thickness of fluid layer, $t = 1.2 \text{ mm} = 0.0012 \text{ m}$

Semi-angle for conical bearing, $\alpha = 30^\circ$

Dynamic viscosity, μ :

$$\text{Power, } P = T \cdot \omega$$

$$\text{where, } T = \frac{\pi \mu \omega}{2t} H^4 \cdot \tan^3 \alpha \cdot \sec \alpha$$

$$\therefore P = \frac{\pi \mu \omega^2}{2t} H^4 \cdot \tan^3 \alpha \cdot \sec \alpha$$

$$120 = \frac{\pi \times \mu \times (64.93)^2 \times (0.1732)^4 \times (\tan 30^\circ)^3 \times \sec 30^\circ}{2 \times 0.0012}$$

$$\left(\text{where, } H = \frac{R}{\tan \alpha} = \frac{0.1}{\tan 30^\circ} = 0.1732 \text{ m} \right)$$

$$\text{or, } \mu = \frac{120 \times 2 \times 0.0012}{\pi \times (64.93)^2 \times (0.1732)^4 \times (\tan 30^\circ)^3 \times \sec 30^\circ}$$

$$= \frac{0.288}{\pi \times (64.93)^2 \times (0.1732)^4 \times 0.1924 \times 1.1547}$$

$$\text{or, } \mu = \mathbf{0.1087 \text{ Ns/m}^2 \text{ (Ans.)}}$$

Example 10.20. The velocity distribution in a pipe is given by,

$$\frac{u}{u_{\max}} = 1 - \left(\frac{r}{R} \right)^n$$

where, $u_{\max}$ is the maximum velocity at the centre of the pipe, u is the velocity at a distance r from the centre and R is the pipe radius. Obtain an expression for mean velocity in terms of $u_{\max}$ and n . (PTU)

Solution. The velocity distribution in a pipe is:

$$\frac{u}{u_{\max}} = 1 - \left(\frac{r}{R} \right)^n \quad \dots (\text{given})$$

where,

u = Velocity at a distance r from the centre,

$u_{\max}$ = Maximum velocity at the centre of pipe, and
 R = Pipe radius.

Now discharge Q passing through any cross-section of the circular pipe can be obtained by integrating a small discharge passing through an elementary ring of the thickness dr at a distance, r :

$$dQ = u \times (2\pi r) dr = u_{\max} \left[1 - \left(\frac{r}{R} \right)^n \right] 2\pi r \cdot dr$$

Integrating, we get:

$$\begin{aligned} Q &= \int_0^R u_{\max} \left[1 - \left(\frac{r}{R} \right)^n \right] 2\pi r \cdot dr \\ &= 2\pi u_{\max} \int_0^R \left[r - \frac{(r)^{n+1}}{R^n} \right] dr \end{aligned}$$

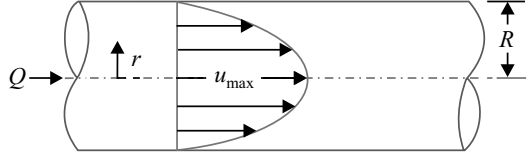


Fig. 10.16

$$\begin{aligned} &= 2\pi u_{\max} \left[\frac{r^2}{2} - \frac{(r)^{n+2}}{(n+2) R^n} \right]_0^R \\ &= 2\pi u_{\max} \left[\frac{R^2}{2} - \frac{(R)^{n+2}}{(n+2) R^n} \right] \\ &= 2\pi u_{\max} \left[\frac{R^2}{2} - \frac{R^2}{n+2} \right] \\ &= \pi R^2 \times u_{\max} \left[\frac{n}{n+2} \right] \\ &= A \times u_{\max} \left[\frac{n}{n+2} \right] \end{aligned}$$

[where, $A (= \pi R^2)$ is cross-sectional area of the pipe]

$$\therefore \text{Mean velocity, } \bar{u} = \frac{Q}{A} = u_{\max} \left(\frac{n}{n+2} \right) \text{ (Ans.)}$$

Example 10.21. (a) The radial velocity profile in a pipe is given by $u = u_{\max} \left(1 - \frac{r}{R} \right)^n$ where u is the velocity at a radial distance r , $u_{\max}$ is the maximum velocity and R is the radius of the pipe. Derive an equation for the average velocity in the pipe.

(b) For incompressible fluid in laminar flow prove that $f = \frac{16}{Re}$, where f is the Fanning friction coefficient and Re is the Reynolds number. (MDU, Haryana)

Solution. (a) $u = u_{\max} \left(1 - \frac{r}{R} \right)^n$... Given

$$\begin{aligned} \text{Average velocity, } \bar{u} &= \frac{1}{\pi R^2} \int_0^R u \times 2\pi r \, dr = \frac{2}{R^2} \int_0^R u r \, dr \\ &= \frac{2}{R^2} \int_0^R u_{\max} \left(1 - \frac{r}{R} \right)^n r \, dr \\ &= \frac{2u_{\max}}{R^2} \int_0^R r \left(1 - \frac{r}{R} \right)^n dr \end{aligned}$$

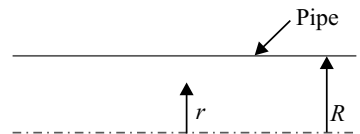


Fig. 10.17

The integral is evaluated first.

$$I = \int r \left(1 - \frac{r}{R}\right)^n dr$$

Put, $\left(1 - \frac{r}{R}\right)^n = Z$

$$I = r \int Z dr - \int \frac{dr}{dr} \left(\int Z dr \right) dr = r \int Z dr - \int \left(\int Z dr \right) dr$$

Now, $\int Z dr = \int \left(1 - \frac{r}{R}\right)^n dr$

Let, $1 - \frac{r}{R} = t$, then $\frac{-dr}{R} = dt$ or $dr = -Rdt$

Hence, $\int Z dr = \int (-Rt^n) dt = \frac{-Rt^{n+1}}{n+1}$

$$\therefore I = \frac{-rRt^{n+1}}{n+1} - \int \left[\left(\frac{-Rt^{n+1}}{n+1} \right) (-Rdt) \right]$$

$$= -\frac{R}{n+1} \left[rt^{n+1} + \int Rt^{n+1} dt \right]$$

$$= -\frac{R}{n+1} \left[r \left(1 - \frac{r}{R}\right)^{n+1} + \frac{R}{n+2} \left(1 - \frac{r}{R}\right)^{n+2} \right]$$

$$\therefore \bar{u} = \frac{2u_{\max}}{R^2} \left(-\frac{R}{n+1} \right) \left[r \left(1 - \frac{r}{R}\right)^{n+1} + \frac{R}{n+2} \left(1 - \frac{r}{R}\right)^{n+2} \right]_0^R$$

$$= \frac{2u_{\max}}{R(n+1)} (-1) \left[-\frac{R}{n+2} \right]$$

or, $\bar{u} = \frac{2u_{\max}}{(n+1)(n+2)} \text{ (Ans.)}$

(b) Refer to article 10.10.

Example 10.22. Show that the momentum correction factor and energy correction factor for laminar flow through a circular pipe are $\frac{4}{3}$ and 2 respectively.

Solution. (i) Momentum correction factor, β :

In a circular pipe, for laminar flow, the velocity distribution at any radius r is given by eqn. (10.7)

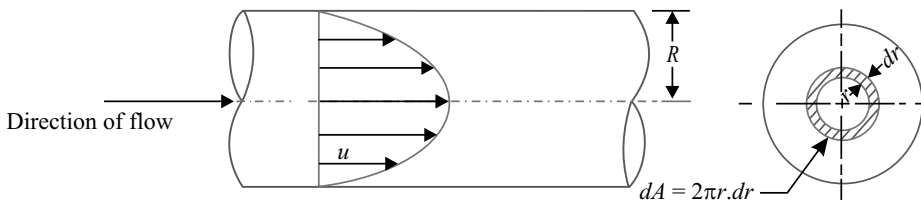


Fig. 10.18

$$u = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} (R^2 - r^2) \quad \dots(i)$$

Consider an elementary area dA in the form of a ring (Fig. 10.18) at a radius r and of width dr then

$$dA = 2\pi r \cdot dr$$

Discharge through the ring, $dQ = u \times 2\pi r \cdot dr$

Momentum of the fluid through the ring per second

$$\begin{aligned} &= \text{Mass of fluid} \times \text{velocity of flow} \\ &= (\rho \cdot dQ) \times u \\ &= (\rho \times u \times 2\pi r \cdot dr) \times u \\ &= 2\pi \rho u^2 r \cdot dr \end{aligned}$$

Total actual momentum of the fluid per second across the section

$$= \int_0^R 2\pi \rho u^2 r \cdot dr$$

Substituting the value of u from (i), we have:

Actual momentum of the fluid per second

$$\begin{aligned} &= 2\pi \rho \int_0^R \left[-\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} (R^2 - r^2) \right]^2 r \cdot dr \\ &= 2\pi \rho \left[\frac{1}{4\mu} \left(\frac{\partial p}{\partial x} \right) \right]^2 \int_0^R (R^2 - r^2)^2 r \cdot dr \\ &= \frac{2\pi \rho}{16\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 \int_0^R (R^4 + r^4 - 2R^2 r^2) r \cdot dr \\ &= \frac{\pi \rho}{8\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 \int_0^R (R^4 r + r^5 - 2R^2 r^3) dr \\ &= \frac{\pi \rho}{8\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 \left[\frac{R^4 r^2}{2} + \frac{r^6}{6} - \frac{2R^2 r^4}{4} \right]_0^R \\ &= \frac{\pi \rho}{8\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 \left[\frac{R^6}{2} + \frac{R^6}{6} - \frac{2R^6}{4} \right] \\ &= \frac{\pi \rho}{8\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 \left(\frac{6R^6 + 2R^6 - 6R^6}{12} \right) \\ &= \frac{\pi \rho}{48\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 R^6 \quad \dots(ii) \end{aligned}$$

Momentum of the fluid per second based on *average velocity*

$$\begin{aligned} &= \text{Mass of fluid/sec.} \times \text{average velocity} \\ &= \rho A \bar{u} \times \bar{u} = \rho A \bar{u}^2 \end{aligned}$$

where,

$$A = \text{Area of cross-section} = \pi R^2$$

$$\bar{u} = \text{Average velocity} = \frac{u_{\max}}{2}$$

$$\begin{aligned}
 &= \frac{1}{2} \left[-\frac{1}{4\mu} \left(\frac{\partial p}{\partial x} \right) \right] R^2 \quad \left[\because u_{\max} = -\frac{1}{4\mu} \cdot \left(\frac{\partial p}{\partial x} \right) R^2 \right] \\
 &= -\frac{1}{8\mu} \left(\frac{\partial p}{\partial x} \right) R^2
 \end{aligned}$$

$\therefore$ Momentum of the fluid per second based on average velocity

$$\begin{aligned}
 &= \rho \times \pi R^2 \times \left[-\frac{1}{8\mu} \left(\frac{\partial p}{\partial x} \right) R^2 \right]^2 \\
 &= \rho \pi \times \frac{1}{64\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 R^6 \quad \dots(iii)
 \end{aligned}$$

$\therefore$ $\beta = \frac{\text{Momentum/sec. based on actual velocity}}{\text{Momentum/sec. based on average velocity}}$

$$\begin{aligned}
 &= \frac{\frac{\pi \rho}{48\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 R^6}{\frac{\pi \rho}{64\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 R^6} = \frac{4}{3} \text{ (Ans.)}
 \end{aligned}$$

(ii) Energy correction factor, α :

K.E. of the fluid flowing through the elementary ring of radius r and thickness dr per second

$$\begin{aligned}
 &= \frac{1}{2} \times \text{mass} \times (\text{velocity})^2 \\
 &= \frac{1}{2} \times (\rho \cdot dQ) \times u^2 \\
 &= \frac{1}{2} \rho (u \times 2\pi r \times dr) \times u^2 \\
 &= \frac{1}{2} \rho \times 2\pi r u^3 dr
 \end{aligned}$$

Total actual K.E. of flow per second

$$\begin{aligned}
 &= \int_0^R \rho \pi r u^3 dr \\
 &= \int_0^R \rho \pi r \left[-\frac{1}{4\mu} \left(\frac{\partial p}{\partial x} \right) (R^2 - r^2) \right]^3 dr \\
 &= \rho \pi \left[-\frac{1}{4\mu} \left(\frac{\partial p}{\partial x} \right) \right]^3 \int_0^R (R^2 - r^2)^3 dr \\
 &= -\rho \pi \times \frac{1}{64\mu^3} \left(\frac{\partial p}{\partial x} \right)^3 \int_0^R (R^6 - r^6 - 3R^4 r^2 + 3R^2 r^4) r dr \\
 &= -\frac{\rho \pi}{64\mu^3} \left(\frac{\partial p}{\partial x} \right)^3 \left[\frac{R^6 r^2}{2} - \frac{r^8}{8} - \frac{3R^4 r^4}{4} + \frac{3R^2 r^6}{6} \right]_0^R \\
 &= -\frac{\rho \pi}{64\mu^3} \left(\frac{\partial p}{\partial x} \right)^3 \left[\frac{R^8}{2} - \frac{R^8}{8} - \frac{3R^8}{4} + \frac{3R^8}{6} \right]
 \end{aligned}$$

$$= \frac{\rho\pi}{512\mu^3} \left(\frac{\partial p}{\partial x} \right)^3 R^8 \quad \dots(iv)$$

K.E. of the flow based on average velocity

$$\begin{aligned} &= \frac{1}{2} \times \text{mass} \times (\text{average velocity})^2 \\ &= \frac{1}{2} \times \rho A \bar{u} \times \bar{u}^2 = \frac{1}{2} \rho A \bar{u}^3 \\ &= \frac{1}{2} \rho \times \pi R^2 \times \left[-\frac{1}{8\mu} \cdot \left(\frac{\partial p}{\partial x} \right) R^2 \right]^3 \\ &\quad \left[\because A = \pi R^2 \text{ and } \bar{u} = -\frac{1}{8\mu} \left(\frac{\partial p}{\partial x} \right) R^2 \right] \end{aligned}$$

$$\begin{aligned} &= -\frac{1}{2} \times \rho \times \pi R^2 \times \frac{1}{512\mu^3} \left(\frac{\partial p}{\partial x} \right)^3 \times R^6 \\ &= -\frac{\rho\pi}{1024} \cdot \left(\frac{\partial p}{\partial x} \right)^3 \times R^8 \quad \dots(v) \end{aligned}$$

$\therefore$

$$\begin{aligned} \alpha &= \frac{\text{K.E./sec. based on actual velocity}}{\text{K.E./sec. based on average velocity}} \\ &= \frac{-\frac{\rho\pi}{512\mu^3} \left(\frac{\partial p}{\partial x} \right)^3 \times R^8}{-\frac{\rho\pi}{1024} \left(\frac{\partial p}{\partial x} \right)^3 \times R^8} = 2.0 \text{ (Ans.)} \end{aligned}$$

10.6. FLOW OF VISCOUS FLUID THROUGH AN ANNULUS

Let us consider an annulus (horizontal) of outer radius R_1 and inner radius R_2 through which steady laminar flow of an incompressible fluid is taking place. A fluid element having a shape of small concentric cylindrical sleeve of length dx and thickness dr considered at a radial distance r is chosen as a free body. The forces acting on the fluid element as shown in Fig. 10.19, in the direction of flow, are:

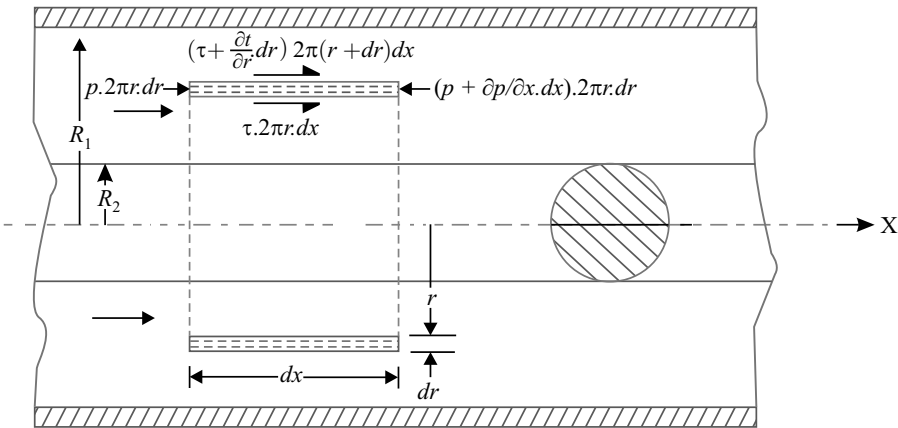


Fig. 10.19. Laminar flow through an annulus.

1. Normal pressure forces over the end areas,

$$p \cdot 2\pi r \cdot dr; \text{ and, } \left(p + \frac{\partial p}{\partial x} \cdot dx \right) 2\pi r \cdot dr$$

2. Shear forces over the inner and outer curved surfaces,

$$\tau \cdot 2\pi r \cdot dx, \text{ and, } \left(\tau + \frac{\partial \tau}{\partial r} \cdot dr \right) 2\pi (r + dr) \cdot dx$$

Since the flow is steady and uniform, the summation of the forces on the free body in the direction of flow must be zero.

$$\therefore p \cdot 2\pi r \cdot dr - \left[\left(p + \frac{\partial p}{\partial x} \cdot dx \right) 2\pi r \cdot dr \right] + \tau \cdot 2\pi r \cdot dx - \left[\left(\tau + \frac{\partial \tau}{\partial r} \cdot dr \right) 2\pi (r + dr) dx \right] = 0$$

Simplifying, we get:

$$\left(-\frac{\partial p}{\partial x} \cdot dx \right) 2\pi r \cdot dr - 2\pi \tau \cdot dr \cdot dx - 2\pi \frac{\partial \tau}{\partial r} dr \cdot r \cdot dx - 2\pi \frac{\partial \tau}{\partial r} \cdot dr \cdot dr \cdot dx = 0$$

Neglecting the last term which is of higher order, and dividing throughout by the volume of the element $2\pi r \cdot dr \cdot dx$, we get:

$$-\frac{\partial p}{\partial x} - \frac{\tau}{r} - \frac{\partial \tau}{\partial r} = 0$$

$$\text{or, } \frac{\partial p}{\partial x} + \frac{\partial \tau}{\partial r} + \frac{\tau}{r} = 0 \quad \dots[10.12 (a)]$$

Since p is dependent on x and τ on r , the above equation may be expressed as:

$$\frac{\partial p}{\partial x} + \frac{1}{r} \frac{\partial}{\partial r} (\tau r) = 0 \quad \dots[10.12 (b)]$$

Integrating w.r.t. r , we get:

$$\frac{\partial p}{\partial x} \int r \cdot dr + \int \frac{d}{dr} (\tau r) \cdot dr = C_1$$

(where, C_1 = constant of integration)

$$\left(\frac{\partial p}{\partial x} \right) \frac{r^2}{2} + \tau \cdot r = C_1$$

Since $\tau = -\mu \cdot \left(\frac{\partial u}{\partial r} \right)$, by substitution, we have:

$$\left(\frac{\partial p}{\partial x} \right) \frac{r^2}{2} - \mu \cdot \frac{\partial u}{\partial r} \cdot r = C_1$$

Dividing the expression throughout by r , we get:

$$\left(\frac{\partial p}{\partial x} \right) \frac{r}{2} - \mu \cdot \frac{\partial u}{\partial r} = \frac{C_1}{r}$$

Integrating w.r.t. r , we get:

$$\left(\frac{\partial p}{\partial x} \right) \frac{r^2}{4} - \mu u = C_1 \log_e r + C_2 \quad \dots(10.13)$$

(where, C_2 = second constant of integration).

The two constants of integration (*i.e.* C_1 and C_2) can be evaluated from the known boundary conditions;

i.e. at $r = R_1$ $u = 0$, and

$$r = R_2 \quad u = 0$$

After substituting these conditions and solving for C_1 and C_2 , we obtain the velocity distribution as:

$$u = -\frac{1}{4\mu} \left(\frac{\partial p}{\partial x} \right) \left[R_1^2 - r^2 - \frac{R_1^2 - R_2^2}{\ln\left(\frac{R_1}{R_2}\right)} \cdot \ln\left(\frac{R_1}{r}\right) \right] \quad \dots(10.14)$$

In order to locate the point where maximum velocity occurs we differentiate eqn. (10.14) w.r.t. r and equate it to zero. Thus, we have

$$\frac{\partial u}{\partial r} = 0 = -\frac{1}{4\mu} \left(\frac{\partial p}{\partial x} \right) \left[-2r + \frac{1}{r} \frac{(R_1^2 - R_2^2)}{\ln\left(\frac{R_1}{R_2}\right)} \right]$$

$$\therefore r = \left[\frac{R_1^2 - R_2^2}{2 \ln\left(\frac{R_1}{R_2}\right)} \right]^{1/2} \quad \dots(10.15)$$

By substituting this value of r in eqn. (10.14), the value of maximum velocity may be obtained. The discharge through the annulus,

$$\begin{aligned} Q &= \int_{R_2}^{R_1} 2\pi r \cdot dr \cdot u \\ &= -\frac{\pi}{8\mu} \left(\frac{\partial p}{\partial x} \right) \left[R_1^4 - R_2^4 - \frac{(R_1^2 - R_2^2)^2}{\ln\left(\frac{R_1}{R_2}\right)} \right] \end{aligned} \quad \dots(10.16)$$

The average velocity of flow through the annulus is given by,

$$\begin{aligned} \bar{u} &= \frac{Q}{\pi (R_1^2 - R_2^2)} \\ &= -\frac{1}{8\mu} \left(\frac{\partial p}{\partial x} \right) \left[(R_1^2 + R_2^2) - \frac{(R_1^2 - R_2^2)}{\ln\left(\frac{R_1}{R_2}\right)} \right] \end{aligned} \quad \dots(10.17)$$

The shear stress is given by,

$$\tau = -\mu \cdot \frac{du}{dr}$$

From eqn. (10.14), the velocity gradient may be obtained as:

$$\begin{aligned} \frac{du}{dr} &= -\frac{1}{4\mu} \left(\frac{\partial p}{\partial x} \right) \left[-2r - \frac{R_1^2 - R_2^2}{\ln\left(\frac{R_1}{R_2}\right)} \left(-\frac{1}{r} \right) \right] \\ &= -\frac{1}{4\mu} \left(-\frac{\partial p}{\partial x} \right) \left(2r - \frac{1}{r} \cdot \frac{R_1^2 - R_2^2}{\ln\left(\frac{R_1}{R_2}\right)} \right) \end{aligned}$$

$\therefore$ **Shear stress distribution** is given by:

$$\tau = -\mu \cdot \frac{\partial u}{\partial r}$$

$$\text{or,} \quad \tau = \frac{1}{4} \left(-\frac{\partial p}{\partial x} \right) \left[2r - \frac{1}{r} \cdot \frac{R_1^2 - R_2^2}{\ln\left(\frac{R_1}{R_2}\right)} \right] \quad \dots(10.18)$$

Example 10.23. A uniform circular tube of bore radius R_1 has a fixed co-axial cylindrical solid core of radius R_2 . An incompressible viscous fluid flows through the annular passage under a pressure gradient $\left(-\frac{\partial p}{\partial x}\right)$. Determine the radius at which shear stress in the stream is zero given that the flow is laminar under steady state condition. [GATE]

Solution. Bore radius of circular tube = R_1

Radius of the solid core = R_2

Pressure gradient = $\left(-\frac{\partial p}{\partial x}\right)$

Radius at which shear stress is zero:

The shear stress distribution is given by,

$$\tau = \frac{1}{4} \left(-\frac{\partial p}{\partial x}\right) \left[2r - \frac{1}{r} \cdot \frac{R_1^2 - R_2^2}{\ln\left(\frac{R_1}{R_2}\right)} \right] \quad \dots[\text{Eqn. (10.18)}]$$

For zero shear stress, $\tau = 0$ and we have:

$$2r - \frac{1}{r} \cdot \frac{R_1^2 - R_2^2}{\ln\left(\frac{R_1}{R_2}\right)} = 0$$

$$\text{or,} \quad 2r^2 = \frac{R_1^2 - R_2^2}{\ln\left(\frac{R_1}{R_2}\right)}$$

$$\text{or,} \quad r = \left[\frac{1}{2} \left(\frac{R_1^2 - R_2^2}{\ln\left(\frac{R_1}{R_2}\right)} \right) \right]^{1/2} \quad (\text{Ans.})$$

10.7. FLOW OF VISCOUS FLUID BETWEEN TWO PARALLEL PLATES

10.7.1. One Plate Moving and Other at Rest—Couette Flow

Let us consider laminar flow between two parallel flat plates located at a distance b apart such that the lower plate is at rest and the upper plate moves uniformly with a constant velocity U as shown in Fig. 10.20. A small rectangular element of fluid of length dx , thickness dy and unit width is considered as a free body (see Fig. 10.20). The forces acting on the fluid element are:

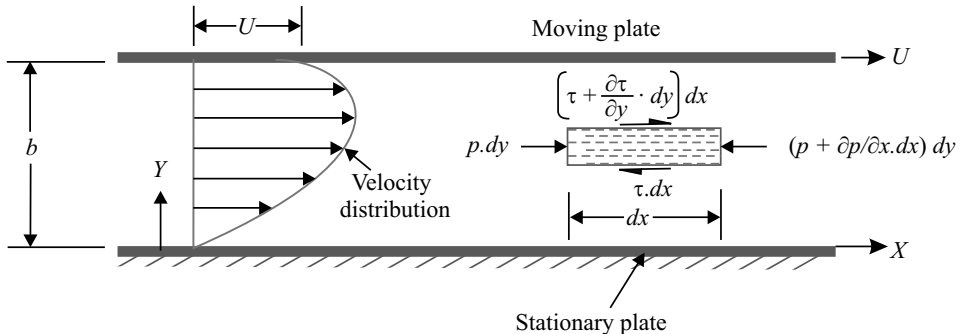


Fig. 10.20. Couette flow.

1. The pressure force, $p \cdot dy \times 1$ on the left end,
2. The pressure force, $\left(p + \frac{\partial p}{\partial x} \cdot dx\right) dy \times 1$ on the right end,
3. The shear force, $\tau \cdot dx \times 1$ on the lower surface, and
4. The shear force, $\left(\tau + \frac{\partial \tau}{\partial y} \cdot dy\right) dx \times 1$ on the upper surface.

For steady and uniform flow, there is no acceleration and hence the resultant force in the direction of flow is *zero*.

$$\therefore p \cdot dy - \left(p + \frac{\partial p}{\partial x} \cdot dx\right) dy - \tau \cdot dx + \left(\tau + \frac{\partial \tau}{\partial y} \cdot dy\right) dx = 0$$

$$\text{or,} \quad -\frac{\partial p}{\partial x} \cdot dx \cdot dy + \frac{\partial \tau}{\partial y} \cdot dy \cdot dx = 0$$

Dividing by the volume of the element $dx \cdot dy$, we get:

$$\frac{\partial p}{\partial x} = \frac{\partial \tau}{\partial y} \quad \dots(10.19)$$

Eqn. (10.19) shows the interdependence of shear and pressure gradients and is *applicable for laminar as well as turbulent flow*. Accordingly the pressure gradient, in the direction of flow, is *equal* to the shear gradient across the flow.

According to Newton's law of viscosity for laminar flow the shear stress, $\tau = \mu \cdot \frac{du}{dy}$. Substituting for τ in eqn. (10.19), we get:

$$\frac{\partial p}{\partial x} = \mu \cdot \frac{\partial^2 u}{\partial y^2}$$

Since $\frac{\partial p}{\partial x}$ is independent of y , integrating the above equation twice w.r.t. y gives:

$$u = \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} y^2 + C_1 y + C_2 \quad \dots(10.20)$$

where, C_1 and C_2 are the constants of integration to be evaluated from the known boundary conditions. In the present case the boundary conditions are:

$$\text{At} \quad y = 0, u = 0, \text{ and at } y = b, u = U$$

$$\therefore \quad C_2 = 0, \text{ and } C_1 = \frac{U}{b} - \frac{1}{2\mu} \left(\frac{\partial p}{\partial x}\right) b$$

Hence, substituting the values of C_1 and C_2 in eqn. (10.20), it yields the following equation for the *velocity distribution* for generalised *Couette flow*:

$$u = \frac{U}{b} y - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (by - y^2) \quad \dots(10.21)$$

The eqn. (10.21) indicates that the velocity distribution in Couette flow depends on both U and $\left(\frac{\partial p}{\partial x}\right)$. However, the pressure gradient $\left(\frac{\partial p}{\partial x}\right)$ in this case may be either positive or negative. In a particular case when $\left(\frac{\partial p}{\partial x}\right)$ equals *zero*, there is no pressure gradient in the direction of flow, then, we have $u = U \cdot \frac{y}{b}$ which indicates that the velocity distribution is *linear*. This particular case is known as *simple (or plain) Couette flow* or *simple shear flow*.

The discharge per unit width (q) may be obtained as follows:

$$\begin{aligned}
 q &= \int_0^b u \cdot dy = \int_0^b \left[\frac{U}{b} y - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (by - y^2) \right] dy \\
 &= U \cdot \frac{b}{2} - \frac{b^3}{12\mu} \cdot \frac{\partial p}{\partial x} \quad \dots(10.22)
 \end{aligned}$$

The *distribution of shear stress* across any section may be determined by using Newton's law of viscosity. Thus,

$$\begin{aligned}
 \tau &= \mu \cdot \frac{\partial u}{\partial y} = \mu \left[\frac{U}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (b - 2y) \right] \\
 &= \mu \cdot \frac{U}{b} - \frac{1}{2} \cdot \frac{\partial p}{\partial x} (b - 2y) \quad \dots(10.23)
 \end{aligned}$$

The type of flow discussed above (*i.e.* flow of viscous fluid between two plates-one *stationary* and the other *moving*) is known as **generalised Couette flow**.

10.7.2. Both Plates at Rest

In this case the equations for velocity, discharge q and the shear stress can be obtained from similar equations for generalised Couette flow by putting $U = 0$. Thus for flow between two stationary parallel plates, shown in Fig. 10.21, we have:

Velocity,

$$u = -\frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (by - y^2) \quad \dots(10.24)$$

[Eqn. (10.24) represents the plane *Poiseuille flow*]

Discharge per unit width,

$$q = -\frac{b^3}{12\mu} \cdot \frac{\partial p}{\partial x} \quad \dots(10.25)$$

$$\text{Shear stress, } \tau = -\frac{1}{2} \cdot \frac{\partial p}{\partial x} (b - 2y) \quad \dots(10.26)$$

10.7.3. Both Plates Moving in Opposite Directions

For flow between parallel plates, the velocity distribution is given by:

$$u = \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \cdot y^2 + C_1 y + C_2 \quad \dots \text{Eqn. (10.20)}$$

In the present case the boundary conditions are:

At, $y = 0, u = -V$, and

At, $y = b, u = U$

Substituting these boundary conditions in eqn. (10.20), we get:

$$-V = C_2 \quad \text{i.e. } C_2 = -V$$

and,

$$U = \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} b^2 + C_1 b - V$$

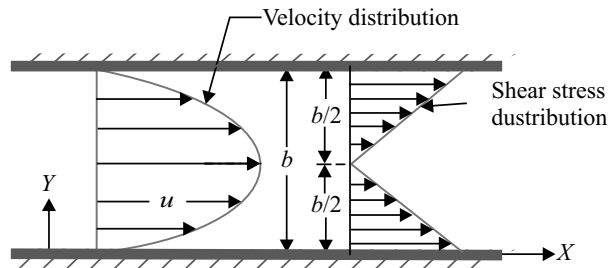


Fig. 10.21. Flow between stationary plates.

or,
$$U + V = \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} b^2 + C_1 b$$

$\therefore$
$$C_1 = (U + V) \frac{1}{b} - \frac{1}{2\mu} \frac{\partial p}{\partial x} \cdot b$$

Hence the eqn. (10.20) becomes:

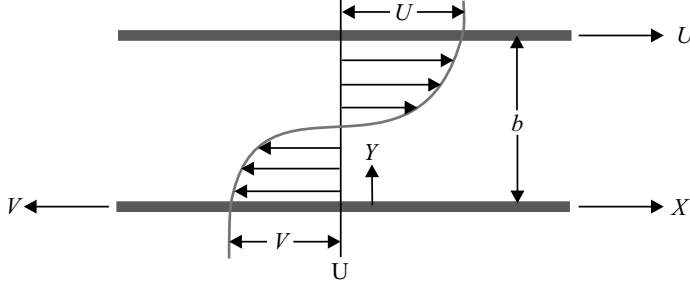


Fig. 10.22. Flow between parallel horizontal plates, both the plates moving in opposite directions.

$$u = \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \cdot y^2 + \left[(U + V) \frac{1}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \cdot b \right] y - V$$

$\therefore$
$$u = (U + V) \frac{y}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (by - y^2) - V \quad \dots(10.27)$$

The distance y at which the velocity u is zero may be determined as follows:

$$(U + V) \frac{y}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (by - y^2) - V = 0$$

Rearranging the above equation, we have:

$$\frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} y^2 + \left(\frac{U + V}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \cdot b \right) y - V = 0$$

Solving this quadratic equation, we have:

$$y = \frac{- \left[\frac{U + V}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \cdot b \right] \pm \sqrt{\left(\frac{U + V}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \cdot b \right)^2 + 4 \cdot \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \cdot V}}{2 \times \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x}}$$

$$= \frac{- \left[\frac{U + V}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \cdot b \right] \pm \sqrt{\left(\frac{U + V}{b} \right)^2 + \frac{1}{4\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 b^2 - \frac{(U - V)}{\mu} \cdot \frac{\partial p}{\partial x}}}{\frac{1}{\mu} \cdot \frac{\partial p}{\partial x}}$$

The above equation will yield two values of y , one which is +ve and less than b will be accepted and the other one rejected.

The discharge per unit width of plates is given by,

$$q = \int_0^b u \, dy$$

$$= \int_0^b \left[(U + V) \frac{y}{b} - \frac{1}{2\mu} \frac{\partial p}{\partial x} (by - y^2) - V \right] dy$$

$$\begin{aligned}
 &= (U + V) \left[\frac{y^2}{2b} \right]_0^b - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \left[b \cdot \frac{y^2}{2} - \frac{y^3}{3} \right] - V [y]_0^b \\
 &= (U + V) \frac{b}{2} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} \left(\frac{b^3}{2} - \frac{b^3}{3} \right) - Vb \\
 &= (U - V) \frac{b}{2} - \frac{1}{12\mu} \frac{\partial p}{\partial x} b^3 \quad \dots(10.28)
 \end{aligned}$$

The distribution of shear stress across any section may be determined by using Newton's law of viscosity. Thus,

$$\begin{aligned}
 \tau &= \mu \cdot \frac{du}{dy} \\
 &= \mu \frac{d}{dy} \left[(U + V) \frac{y}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (by - y^2) - V \right] \\
 &= \mu \left[\frac{U + V}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (b - 2y) \right] \\
 &= \mu \left(\frac{U + V}{b} \right) - \frac{1}{2} \frac{\partial p}{\partial x} (b - 2y) \\
 &= (U + V) \frac{\mu}{b} - \frac{\partial p}{\partial x} \left(\frac{b}{2} - y \right) \quad \dots(10.29)
 \end{aligned}$$

The distance y at which the shear stress will be zero is obtained by putting eqn. (10.29) to zero. Thus,

$$\begin{aligned}
 (U + V) \frac{\mu}{b} - \frac{\partial p}{\partial x} \left(\frac{b}{2} - y \right) &= 0 \\
 \text{or,} \quad \frac{\partial p}{\partial x} \left(\frac{b}{2} - y \right) &= (U + V) \frac{\mu}{b} \\
 \text{or,} \quad \frac{b}{2} - y &= \frac{(U + V) \frac{\mu}{b}}{\frac{\partial p}{\partial x}} \\
 \therefore y &= \frac{b}{2} - \frac{\mu}{b} \left(\frac{U + V}{\partial p / \partial x} \right) \quad \dots(10.30)
 \end{aligned}$$

Example 10.24. Determine the direction and amount of flow per metre width between two parallel plates when one is moving relative to the other with a velocity of 3 m/s in the negative direction, if $\frac{\partial p}{\partial x} = -100 \times 10^6 \text{ N/m}^3$ and $\mu = 0.4 \text{ poise}$ and distance between the plates is 1 mm. (MGU, Kerala)

Solution. Given: $U = -3 \text{ m/s}$; $\frac{dp}{dx} = -100 \times 10^6 \text{ N/m}^3$, $\mu = 0.4 \text{ poise} = 0.4 \times \frac{1}{10} = 0.04 \text{ Ns/m}^2$; $b = 1 \text{ mm} = 0.001 \text{ m}$.

We know that,
$$q = U \cdot \frac{b}{2} - \frac{b^3}{12\mu} \cdot \frac{\partial p}{\partial x} \quad [\text{Eqn. (10.22)}]$$

Substituting the values, we have:

$$q = -3 \times \frac{0.001}{2} - \frac{0.001^3}{12 \times 0.04} \times (-100 \times 10^6) = 0.2068 \text{ m}^3/\text{s}$$

Hence, amount of flow per metre width = **0.2068 m³/s. (Ans.)**

Positive direction (i.e. in the direction opposite to that of the moving plate). (Ans.)

Example 10.25. Two parallel plates kept 100 mm apart have laminar flow of oil between them with a maximum velocity of 1.5 m/s. Calculate:

- (i) The discharge per metre width,
- (ii) The shear stress at the plates,
- (iii) The difference in pressure between two points 20 m apart,
- (iv) The velocity gradient at the plates, and
- (v) The velocity at 20 mm from the plate.

Assume viscosity of oil to be 24.5 poise.

Solution. Distance between the parallel plates, $b = 100 \text{ mm} = 0.1 \text{ m}$

Maximum velocity of the oil, $u_{\max} = 1.5 \text{ m/s}$

Viscosity of the oil, $\mu = 24.5 \text{ poise} = 2.45 \text{ Ns/m}^2$

(i) **The discharge per metre width, q :**

In this case the average velocity of flow,

$$\bar{u} = \frac{2}{3} u_{\max} = \frac{2}{3} \times 1.5 = 1.0 \text{ m/s}$$

$$\therefore q = \bar{u} \times b = 1.0 \times 0.1 = \mathbf{0.1 \text{ m}^3/\text{s per m (Ans.)}$$

(ii) **The shear stress at the plates τ_0 :**

$$\text{We know, } q = \frac{b^3}{12\mu} \left(-\frac{\partial p}{\partial x} \right) \quad \dots[\text{Eqn. (10.25)}]$$

Substituting the values, we have:

$$0.1 = \frac{0.1^3}{12 \times 2.45} \left(-\frac{\partial p}{\partial x} \right)$$

$$\text{or, } \left(-\frac{\partial p}{\partial x} \right) = \frac{0.1 \times 12 \times 2.45}{0.1^3} = 2940 \text{ N/m}^2/\text{m}$$

The shear stress across any section is given by:

$$\tau = \frac{1}{2} \left(-\frac{\partial p}{\partial x} \right) (b - 2y) \quad \dots[\text{Eqn. (10.26)}]$$

The shear stress at the plates is obtained by putting $y = 0$ in the above equation. Thus,

$$\begin{aligned} \tau_0 &= \frac{1}{2} \left(-\frac{\partial p}{\partial x} \right) b \\ &= \frac{1}{2} \times 2940 \times 0.1 = \mathbf{147 \text{ N/m}^2 \text{ (Ans.)}} \end{aligned}$$

(iii) **Pressure difference between two points 20 m apart:**

$$\text{We know, } -\frac{\partial p}{\partial x} = 2940$$

$$\text{or, } -\partial p = 2940 \partial x$$

Integrating w.r.t. x , we get:

$$\int_{p_1}^{p_2} (-\partial p) = \int_{x_1}^{x_2} 2940 (\partial x)$$

$$\begin{aligned} \text{or, } p_1 - p_2 &= 2940 (x_2 - x_1) \\ &= 2940 \times 20 = 58800 \text{ N/m}^2 \text{ or } \mathbf{58.8 \text{ kN/m}^2 \text{ (Ans.)}} \end{aligned}$$

(iv) The velocity gradient at the plates, $\left(\frac{\partial u}{\partial y}\right)_{y=0}$:

$$\tau_0 = \mu \cdot \left(\frac{\partial u}{\partial y}\right)_{y=0}$$

or,
$$\left(\frac{\partial u}{\partial y}\right)_{y=0} = \frac{\tau_0}{\mu} = \frac{147}{2.45} = 60 \text{ s}^{-1} \text{ (Ans.)}$$

(v) The velocity at 20 mm from the plate:

$$u = \frac{1}{2\mu} \cdot \left(-\frac{\partial p}{\partial x}\right) (by - y^2) \quad \dots[\text{Eqn. (10.24)}]$$

$$= \frac{1}{2 \times 2.45} \times 2940 (0.1 \times 0.02 - 0.02^2)$$

$$(\because y = 20 \text{ mm} = 0.02 \text{ m})$$

$$= 0.96 \text{ m/s (Ans.)}$$

Example 10.26. A liquid of viscosity of 0.9 poise is filled between two horizontal plates 10 mm apart. If the upper plate is moving at 1 m/s with respect to the lower plate which is stationary and the pressure difference between two sections 60 m apart is 60 kN/m², determine:

- (i) The velocity distribution,
- (ii) The discharge per unit width, and
- (iii) The shear stress on the upper plate.

Solution. Viscosity of the liquid,

$$\mu = 0.9 \text{ poise} = 0.09 \text{ N s/m}^2$$

Distance between the plates,

$$b = 10 \text{ mm} = 0.01 \text{ m}$$

Velocity of the upper plate, $U = 1 \text{ m/s}$

Pressure difference between the sections 60 m apart = 60 kN/m²

$$\therefore \left(-\frac{\partial p}{\partial x}\right) = \frac{60 \times 10^3}{60} = 103 \text{ N/m}^2/\text{m}$$

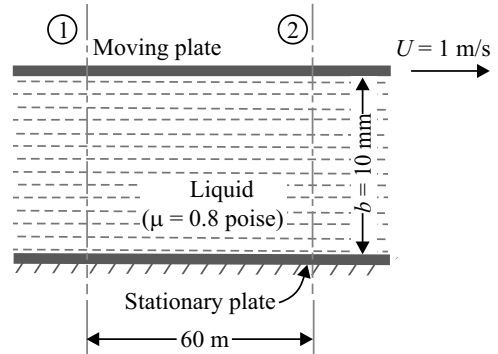


Fig. 10.23

(i) **The velocity distribution:**

The system corresponds to Couette flow for which the velocity distribution is given as:

$$u = \frac{U}{b} y + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x}\right) (by - y^2) \quad \dots[\text{Eqn. (10.21)}]$$

$$= \frac{1}{0.01} y + \frac{1}{2 \times 0.09} \times 10^3 (0.01 \times y - y^2)$$

$$= y (100 + 55.55 - 5555.55 y)$$

$$= y (155.55 - 5555.55 y)$$

Hence, the velocity distribution is: $u = y (155.55 - 5555.55 y)$ (Ans.)

(ii) Discharge per unit width, q :

$$\begin{aligned}
 q &= \int_0^b u \, dy \\
 &= \int_0^{0.01} (155.55 \, y - 5555.55 \, y^2) \, dy \\
 &= \left[155.55 \times \frac{y^2}{2} - 5555.55 \times \frac{y^3}{3} \right]_0^{0.01} \\
 &= \left[155.55 \times \frac{0.01^2}{2} - 5555.55 \times \frac{0.01^3}{3} \right] \\
 &= (0.007777 - 0.001852) = \mathbf{0.005925 \, m^3/s \, (Ans.)}
 \end{aligned}$$

(iii) The shear stress on the upper plate, τ_0 :

$$\begin{aligned}
 \text{Shear stress, } \tau &= \mu \cdot \left(\frac{\partial u}{\partial y} \right) = \mu \frac{\partial}{\partial y} (155.55 \, y - 5555.55 \, y^2) \\
 &= 0.09 (155.55 - 11111.1 \, y)
 \end{aligned}$$

For the top plate, $y = 0.01 \, \text{m}$

$$\therefore \tau_0 = 0.09 (155.55 - 11111.1 \times 0.01) = \mathbf{4 \, N/m^2 \, (Ans.)}$$

Example 10.27. Fluid is in laminar motion between two parallel plates under the action of motion of one of the plates and also under the presence of a pressure gradient in such a way that the net forward discharge across any section is zero.

(i) Find out the point where minimum velocity occurs and its magnitude.

(ii) Draw the velocity distribution graph across any section.

Solution. In the given case of flow, the velocity distribution is given by:

$$u = \frac{U}{b} y - \frac{1}{2\mu} \frac{\partial p}{\partial x} (by - y^2) \quad \dots[\text{Eqn. (10.21)}]$$

and, the discharge per unit width,

$$q = \frac{Ub}{2} - \frac{b^3}{12\mu} \cdot \frac{\partial p}{\partial x} \quad \dots[\text{Eqn. (10.22)}]$$

Net forward discharge,

$$q = 0 \quad \dots(\text{Given})$$

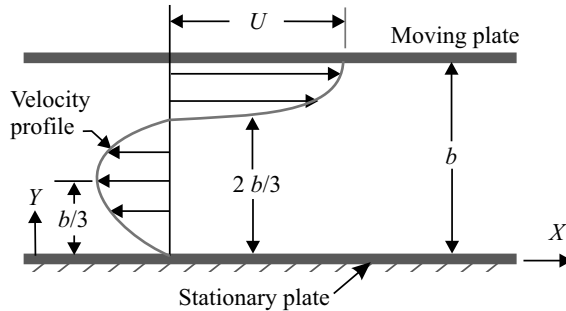


Fig. 10.24

$$\therefore 0 = \frac{Ub}{2} - \frac{b^3}{12\mu} \cdot \frac{\partial p}{\partial x}$$

$$\text{or, } \frac{\partial p}{\partial x} = \frac{Ub}{2} \times \frac{12\mu}{b^3} = \frac{6\mu U}{b^2}$$

Minimum velocity occurs where,

$$\frac{\partial u}{\partial y} = 0$$

$$\text{Thus, } \frac{\partial}{\partial y} \left[\frac{U}{b} \cdot y - \frac{1}{2\mu} \frac{\partial p}{\partial x} (by - y^2) \right] = 0$$

$$\text{or, } \frac{U}{b} - \frac{1}{2\mu} \frac{\partial p}{\partial x} (b - 2y) = 0$$

$$\text{or, } \frac{U}{b} = \frac{1}{2\mu} \frac{\partial p}{\partial x} (b - 2y)$$

$$\text{or, } (b - 2y) = \frac{(U/b) \times 2\mu}{(\partial p / \partial x)}$$

$$\text{or, } 2y = b - \frac{(U/b) \times 2\mu}{(\partial p / \partial x)}$$

$$\text{or, } y = \frac{b}{2} - \frac{(U/b) \times \mu}{6\mu U / b^2} = \frac{b}{3} \quad \left(\because \frac{\partial p}{\partial x} = \frac{6\mu U}{b^2} \right)$$

Hence, *minimum velocity occurs at a distance $\frac{b}{3}$ from the fixed plate. (Ans.)*

The *magnitude of minimum velocity* is obtained by putting $y = \frac{b}{3}$ in the equation of velocity distribution. Thus,

$$u = \frac{U}{b} \cdot y - \frac{1}{2\mu} \frac{\partial p}{\partial x} (by - y^2)$$

$$\begin{aligned} \text{or, } u_{\min} &= \frac{U}{b} \times \frac{b}{3} - \frac{1}{2\mu} \times \frac{6\mu U}{b^2} \left(b \times \frac{b}{3} - \frac{b^2}{9} \right) \\ &= \frac{U}{3} - \frac{2U}{3} = -\frac{U}{3} \quad (\text{Ans.}) \end{aligned}$$

(ii) Velocity distribution graph:

The velocity distribution graph may be drawn by substituting arbitrary values of y such as $0.1b$, $0.2b$, $0.3b$ etc. in the equation,

$$u = \frac{U}{b} y - \frac{3U}{b^2} (by - y^2),$$

and computing u in terms of U .

Also, when $u = 0$, $y = 0$ and $\frac{2}{3}b$

The velocity distribution graph is shown in Fig. 10.24.

Example 10.28. Show that the discharge per unit width between two parallel plates distance b apart, when one plate is moving at velocity U while the other one is held stationary, for the condition of zero shear stress at the fixed plate is $q = \frac{Ub}{3}$

Solution. The given case of flow corresponds to Couette flow for which the velocity distribution given by:

$$u = \frac{U}{b} \cdot y + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) (by - y^2)$$

$\therefore$ Discharge per unit width,

$$\begin{aligned} q &= \int_0^b u \cdot dy = \int_0^b \left[\frac{U}{b} \cdot y + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) (by - y^2) \right] dy \\ &= \left[\frac{U}{b} \cdot \frac{y^2}{2} + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) \left(b \cdot \frac{y^2}{2} - \frac{y^3}{3} \right) \right]_0^b \\ &= \frac{U}{b} \cdot \frac{b^2}{2} + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) \left(\frac{b^3}{2} - \frac{b^3}{3} \right) \\ &= \frac{Ub}{2} + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) \frac{b^3}{6} \quad \dots(i) \end{aligned}$$

$$\begin{aligned} \text{Stress, } \tau &= \mu \cdot \frac{\partial u}{\partial y} \\ &= \mu \frac{d}{dy} \left[\frac{Uy}{b} + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) (by - y^2) \right] \\ &= \mu \left[\frac{U}{b} + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) (b - 2y) \right] \end{aligned}$$

But shear stress at the surface of fixed plate ($y = 0$) = 0 ... (Given)

$$\therefore 0 = \mu \left[\frac{U}{b} + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) b \right]$$

$$\text{or, } \frac{U}{b} + \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) b = 0$$

$$\text{or, } \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) = -\frac{U}{b^2}$$

Making substitution for this expression in (i), we get:

$$q = \frac{Ub}{2} - \frac{U}{b^2} \times \frac{b^3}{6} = \frac{Ub}{3} \quad \dots(\text{Proved})$$

Example 10.29. Laminar flow of a fluid of viscosity 0.9 Ns/m^2 and specific gravity 1.26 occurs between a pair of parallel plates of extensive width, inclined at 45° to the horizontal, the plates being 10 mm apart. The upper plate moves with a velocity of 2.0 m/s relative to the lower plate and in a direction opposite to the fluid flow. Pressure gauges mounted at two points 1 m vertically apart on the upper plate record pressures of 250 kN/m^2 and 80 kN/m^2 respectively. Determine:

- (i) The velocity and shear stress distribution between the plates,
- (ii) The maximum flow velocity, and
- (iii) The shear stress on the upper plate.

[MU]

Solution. Viscosity of the fluid, $\mu = 0.9 \text{ Ns/m}^2$

Specific gravity of the fluid = 1.26

Distance between the plates, $b = 10 \text{ mm} = 0.01 \text{ m}$

Velocity of upper (moving) plate, $U = -2.0$ m/s

Pressure, $p_1 = 250$ kN/m²

Pressure, $p_2 = 80$ kN/m².

(i) **The velocity and shear stress distribution between the plates:**

Considering sections 1 and 2, we have:

$$\begin{aligned} h_1 - h_2 &= \left(\frac{p_1}{w} + z_1 \right) - \left(\frac{p_2}{w} + z_2 \right) \\ &= \left(\frac{250 \times 10^3}{1.26 \times 9810} + 1 \right) - \left(\frac{80 \times 10^3}{1.26 \times 9810} \right) + 0 \end{aligned}$$

or,

$$\begin{aligned} h_1 - h_2 &= 21.225 - 6.47 \\ &= 14.755 \text{ m in } \sqrt{2} \text{ m or } 1.414 \text{ m} \end{aligned}$$

Since $h_1 > h_2$, flow will be in *downward* direction.

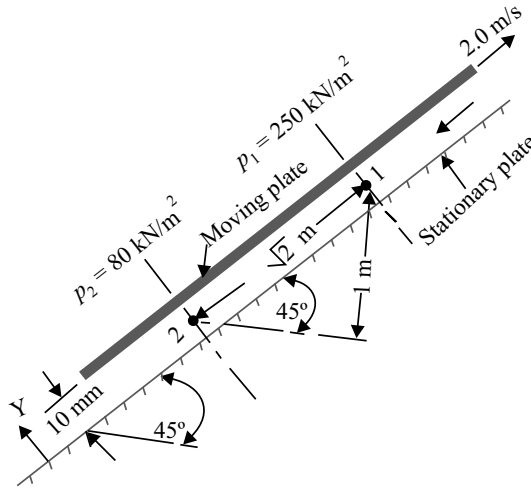


Fig. 10.25

$$\frac{\partial h}{\partial x} = -\frac{14.755}{1.414} = -10.435$$

$$\begin{aligned} \text{and, } \frac{\partial p}{\partial x} &= w \frac{\partial h}{\partial x} = (1.26 \times 9810) \times (-10.435) \\ &= -128983 \text{ N/m}^2 \text{ or } -128.983 \text{ kN/m}^2 \end{aligned}$$

The velocity distribution in this case of flow is given by:

$$u = \frac{U}{b} \cdot y - \frac{1}{2\mu} \cdot \left(\frac{\partial p}{\partial x} \right) (by - y^2) \quad [\text{Eqn. (10.21)}]$$

Substituting the values, we get:

$$\begin{aligned} u &= -\frac{2.0}{0.01} y - \frac{1}{2 \times 0.9} \times (-128.983 \times 10^3) (0.01 y - y^2) \\ &= -200y + 716.57y - 71657y^2 \\ &= 516.57y - 71657y^2 \end{aligned}$$

Hence, velocity distribution is given by:

$$u = 516.57y - 71657y^2 \text{ (Ans.)}$$

The shear stress distribution is given by:

$$\begin{aligned}\tau &= \mu \cdot \frac{U}{b} - \frac{1}{2} \frac{\partial p}{\partial x} (b - 2y) \quad \dots[\text{Eqn. (10.21)}] \\ &= 0.9 \times \left(\frac{-2}{0.01} \right) - \frac{1}{2} \times (-128.983 \times 10^3) (0.01 - 2y) \\ &= -180 + 644.91 - 128982y = 464.91 - 128982y\end{aligned}$$

Hence, the shear stress distribution is given by:

$$\tau = 464.91 - 128982y \text{ (Ans.)}$$

(ii) Maximum flow velocity, $u_{\max}$:

$$\text{For maximum velocity, } \frac{du}{dy} = 0$$

$$\text{or, } \frac{d}{dy} (516.57y - 71657y^2) = 0$$

$$\text{or, } 516.57 - 143314y = 0$$

$$\text{or, } y = 3.604 \times 10^{-3} \text{ m}$$

$$\begin{aligned}\therefore \text{Maximum velocity, } u_{\max} &= 516.57 \times 3.604 \times 10^{-3} - 71657 \times (3.604 \times 10^{-3})^2 \\ &= 1.862 - 0.931 = 0.931 \text{ m/s (Ans.)}\end{aligned}$$

(iii) The shear stress on the upper plate:

$$(\tau)_{0.01} = 464.91 - 128982 \times 0.01 = -824.91 \text{ N/m}^2 \text{ (Ans.)}$$

Example 10.30. A large thin plate is pulled at constant velocity U through a narrow gap of height h . On one side of the plate is oil of viscosity μ and on the other side oil of viscosity $\alpha\mu$, where α is a constant. Calculate the position of the plate so that the drag force on it will be minimum.

(UPTU)

Solution. Refer to Fig. 10.26. Let a be the distance of the thin plate from the top surface.

Shear stress on *top portion*;

$$\tau_1 = \mu_1 \frac{du}{dy} = \mu \frac{U}{a}$$

Shear stress on the *bottom portion*;

$$\tau_2 = \mu \frac{du}{dy} = \alpha\mu \frac{U}{(h-a)}$$

Total drag force on the plate,

$$\begin{aligned}F &= A (\tau_1 + \tau_2) \\ &= A \left[\mu \frac{U}{a} + \alpha\mu \frac{U}{(h-a)} \right] \\ &= AU \left(\frac{\mu}{a} + \frac{\alpha\mu}{h-a} \right) \\ &= AU\mu \left(\frac{1}{a} + \frac{\alpha}{h-a} \right)\end{aligned}$$

For F to be *minimum*:

$$\frac{dF}{da} = -\frac{1}{a^2} + \frac{\alpha}{(h-a)^2} = 0$$

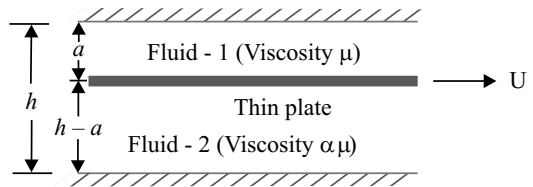


Fig. 10.26

$$\text{or,} \quad \frac{\alpha}{(h-a)^2} = \frac{1}{a^2}$$

$$\text{or,} \quad \frac{\sqrt{\alpha}}{h-a} = \frac{1}{a}$$

$$\text{or,} \quad a\sqrt{\alpha} = h-a$$

$$\text{or,} \quad a(1+\sqrt{\alpha}) = h$$

$$\text{or,} \quad a = \frac{h}{1+\sqrt{\alpha}}$$

which gives the position of the plate for *minimum drag* (Ans.)

10.8. LAMINAR FLOW THROUGH POROUS MEDIA

It was established by Darcy through experiments that the velocity of a fluid through a porous media varies *linearly* with the loss of head h_f , which indicates that the flow through the porous media is *laminar*.

Consider a circular pipe of length L and diameter D completely filled with porous material of grain diameter d_s . The flow takes places through the interstices of the porous material. If porosity is n , the diameter of the passage through the particles is nd_s . The loss of head when liquid flows through a porous media can be determined by using the general expression for head loss in laminar flow.

The loss of head for laminar/viscous flow *through a pipe* is given by:

$$h_f = \frac{32 \mu \bar{u} L}{w D^2}$$

Similarly, the loss of head for laminar flow *through parallel plates* is given by:

$$h_f = \frac{12 \mu \bar{u} L}{w b^2}$$

Hence, the general expression for laminar flow may be expressed as:

$$h_f = \frac{K \mu \bar{u} L}{w D^2} \quad \dots(10.31)$$

where, h_f = The loss of head in length L ,

K = A constant, the value of which depends on the shape of the passage,

μ = Dynamic viscosity of the fluid,

$\bar{u}$ = Average velocity of flow,

w = Weight density of the fluid, and

D = A characteristic length representing the geometry of the passage.

Eqn. (10.31) can be used for laminar flow through porous media. The *diameter of the passage through particles* is given by:

$$d = n d_s$$

Substituting this value of d for D in eqn. (10.31), we get:

$$h_f = \frac{K \mu \bar{u} L}{w n^2 d_s^2}$$

$$\begin{aligned} \text{or,} \quad \bar{u} &= \frac{wn^2d_s^2}{K\mu} \left(\frac{h_f}{L} \right) \\ \text{or,} \quad \bar{u} &= ki \quad \dots(10.32) \\ \text{where,} \quad k &= \text{A constant, called the co-efficient of permeability, and} \\ i &= \text{The hydraulic gradient} \left(= \frac{h_f}{L} \right) \end{aligned}$$

Eqn. (10.32) is the well known Darcy's equation for flow of water through soil.

— The equation is applicable for the Reynolds number < 1 .

— The equation is normally valid (according to Ehrenberger) for velocities $\bar{u}$ less than 3 to 4.5 mm/s.

Example 10.31. Water at a rate of 0.0006 litre/sec. is flowing through a sandy specimen of 8 cm height and 45 cm² cross-sectional area under a constant head of 7 cm. Determine the co-efficient of permeability.

Solution. Discharge through sandy specimen, $Q = 0.0006$ litre/sec.

Area of cross-section, $A = 45$ cm²

Height of specimen, $L = 8$ cm

Constant head, $h_f = 7$ cm

Co-efficient of permeability, k :

$$\text{Average velocity, } \bar{u} = \frac{Q}{A} = \frac{0.0006 \times 10^3}{45} = 0.0133 \text{ cm/s}$$

Using the relation :

$$\bar{u} = ki = k \left(\frac{h_f}{L} \right), \text{ we get:}$$

$$0.0133 = k \times \frac{7}{8}$$

$$\text{or} \quad k = \frac{0.0133 \times 8}{7} = \mathbf{0.0152 \text{ cm/s (Ans.)}}$$

10.9. POWER ABSORBED IN BEARINGS

In a bearing, a very thin film of lubricating oil is maintained between its stationary surface and the surface of the rotating shaft. The lubricating oils are viscous and hence the theory of laminar/viscous flow can be applied to the theory of lubrication. A very viscous oil leads to greater resistance and causes great power loss; a light oil, on the other hand, may not be able to maintain the required film between the metal surfaces and wear of the surfaces will take place. The expressions for power absorbed due to viscous resistance in different types of bearing are derived as given below.

10.9.1. Journal Bearing

Fig. 10.27 shows a journal bearing in which a shaft is rotating. A lubricating oil is filled in the annular space between the shaft and the bearing.

Let, D = Diameter of the shaft,
 t = Thickness of oil film,
 L = Length of the bearing, and
 N = Speed of the shaft in r.p.m.

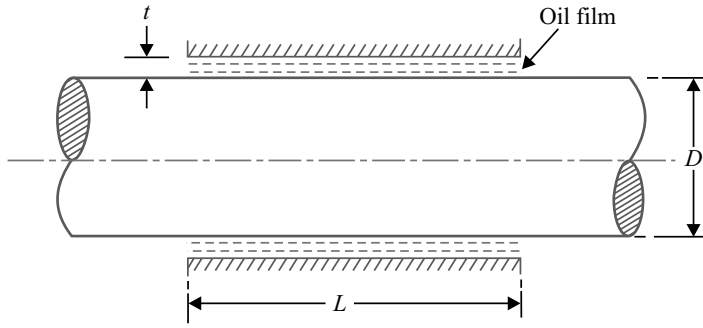


Fig. 10.27. Journal bearing.

∴ Angular speed,

$$\omega = \frac{2\pi N}{60} \text{ rad/s}$$

Tangential speed of the shaft,

$$V = \omega R$$

or,

$$V = \frac{2\pi N}{60} \times \frac{D}{2} = \frac{\pi DN}{60} \text{ m/s.}$$

As the thickness ' t ' of the oil film is very small, a linear velocity distribution can be presumed.

Hence,

$$\frac{du}{dy} = \frac{V - 0}{t} = \frac{V}{t} = \frac{\pi DN}{60 \times t}$$

∴ Shear stress, $\tau = \mu \times \frac{\pi DN}{60 \times t}$

Shear force or viscous resistance,

$$\begin{aligned} F &= \tau \times \text{area of surface of the shaft} \\ &= \frac{\mu \pi DN}{60t} \times \pi DL = \frac{\mu \pi^2 D^2 NL}{60t} \end{aligned}$$

∴ Torque required to overcome the viscous resistance of the whole of the bearing,

$$T = F \times \frac{D}{2} = \frac{\mu \pi^2 D^2 NL}{60t} \times \frac{D}{2} = \frac{\mu \pi^2 D^3 NL}{120t} \quad \dots(10.33)$$

∴ Power absorbed in overcoming the resistance,

$$P = T \cdot \omega = \frac{2\pi NT}{60} \text{ watts} \quad \dots(10.34)$$

$$\left(\text{where } T = \frac{\mu \pi^2 D^3 NL}{120t} \right)$$

Example 10.32. A shaft of 100 mm diameter rotates at 60 r.p.m. in a 200 mm long bearing. Taking that the two surfaces are uniformly separated by a distance of 0.5 mm and taking linear velocity distribution in the lubricating oil having dynamic viscosity 0.04 poise, find the power absorbed in bearing. [PTU]

Solution. Diameter of the shaft, $D = 100 \text{ mm} = 0.1 \text{ m}$

Speed of the shaft, $N = 60 \text{ r.p.m.}$

Length of the bearing, $L = 200 \text{ mm} = 0.2 \text{ m}$

Thickness of oil film, $t = 0.5 \text{ mm} = 0.0005 \text{ m}$

Dynamic viscosity, $\mu = 0.04 \text{ poise} = 0.004 \text{ Ns/m}^2$

Power absorbed, P:

The torque required to overcome viscous resistance in a journal bearing is given by:

$$T = \frac{\mu \pi^2 D^3 N L}{120 t} = \frac{0.004 \times \pi^2 \times 0.1^3 \times 60 \times 0.2}{120 \times 0.0005} = 0.00789 \text{ Nm}$$

$$\begin{aligned} \therefore \text{Power absorbed} &= \frac{2\pi NT}{60} = \frac{2\pi \times 60 \times 0.00789}{60} \\ &= \mathbf{0.0496 \text{ W (Ans.)}} \end{aligned}$$

10.9.2. Foot-step Bearing

Fig. 10.28 shows a foot-step bearing at the end of a vertical shaft. The space between the surface of the shaft and bearing is filled with oil of viscosity μ .

Let, R = Radius of the shaft,
 N = Speed of the shaft, and
 t = Thickness of oil film.

Consider an elementary circular ring of radius r and thickness dr as shown in the Fig. 10.28.

Area of the elementary ring = $2\pi r \cdot dr$

$$\text{The shear stress, } \tau = \mu \cdot \frac{\partial u}{\partial y} = \mu \cdot \frac{V}{t}$$

where, V is the tangential velocity of shaft at radius r and is equal to

$$\omega r = \frac{2\pi N}{60} \times r$$

$\therefore$ Shear force on the elementary ring,

$$dF = \tau \times \text{area of the ring}$$

$$= \mu \times \frac{2\pi N}{60} \times \frac{r}{t} \times 2\pi r \cdot dr$$

$$= \frac{\mu}{15} \times \frac{\pi^2 N r^2}{t} \cdot dr$$

$$\text{Torque on the ring, } dT = dF \times r = \frac{\mu}{15t} \cdot \pi^2 N r^2 \cdot dr \cdot r$$

$\therefore$ Total torque required to overcome viscous resistance,

$$\begin{aligned} T &= \int_0^R \frac{\mu}{15t} \cdot \pi^2 N r^3 \cdot dr = \frac{\mu}{15t} \cdot \pi^2 \cdot N \int_0^R r^3 dr \\ &= \frac{\mu}{15t} \cdot \pi^2 \cdot N \left[\frac{r^4}{4} \right]_0^R = \frac{\mu}{15t} \cdot \pi^2 \cdot N \cdot \frac{R^4}{4} \end{aligned}$$

$$\text{or, } T = \frac{\mu}{60t} \cdot \pi^2 \cdot N R^4 \quad \dots(10.35)$$

$\therefore$ Power absorbed by the entire bearing,

$$P = T \cdot \omega = \frac{2\pi NT}{60} \text{ watts}$$

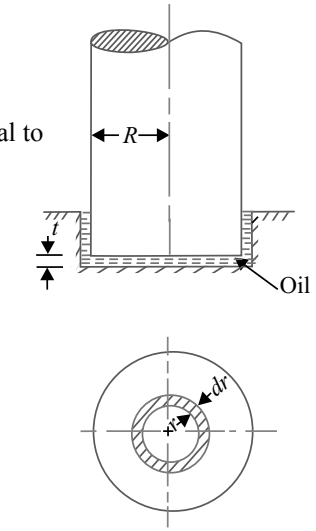


Fig. 10.28. Foot-step bearing.

$$\left(\text{where, } T = \frac{\mu}{60t} \cdot \pi^2 \cdot N \cdot R^4 \right)$$

Example 10.33. Find the power required to rotate a vertical shaft of diameter 100 mm at 750 r.p.m. The lower end of the shaft rests in a foot-step bearing. The end of the shaft and surface of the bearing are both flat and are separated by an oil film of thickness 0.5 mm. The viscosity of the oil is 1.5 poise. [Delhi University]

Solution. Diameter of the shaft, $D = 100 \text{ mm} = 0.1 \text{ m}$

Speed of the shaft, $N = 750 \text{ r.p.m.}$

Thickness of the film, $t = 0.5 \text{ mm} = 0.0005 \text{ m}$

The viscosity of the oil, $\mu = 1.5 \text{ poise} = 0.15 \text{ Ns/m}^2$

Power required, P:

The torque required to overcome the viscous resistance in a foot-step bearing is given by:

$$T = \frac{\mu}{60t} \cdot \pi^2 \cdot N \cdot R^4 = \frac{0.15}{60 \times 0.0005} \times \pi^2 \times 750 \times \left(\frac{0.1}{2}\right)^4$$

$$= 0.2313 \text{ Nm}$$

$\therefore$ Power required to rotate the shaft,

$$P = \frac{2\pi NT}{60} = \frac{2\pi \times 750 \times 0.2313}{60} = 18.16 \text{ W (Ans.)}$$

10.9.3. Collar Bearing

Fig. 10.29 shows a collar bearing which takes up an axial thrust of shaft. The face of the collar is separated from the surface of the bearing with a film of uniform thickness which is maintained by a forced lubrication system.

Let,

R_1 = Internal radius of the collar,

R_2 = External radius of the collar,

t = Thickness of oil film, and

N = Speed of the shaft in r.p.m.

Consider an elementary ring of bearing surface of radius r and thickness dr as shown in Fig. 10.29.

Area of elementary ring, $dA = 2\pi r \cdot dr$

$$\text{Viscous shear stress, } \tau = \mu \cdot \frac{\partial u}{\partial y} = \mu \cdot \frac{V}{t}$$

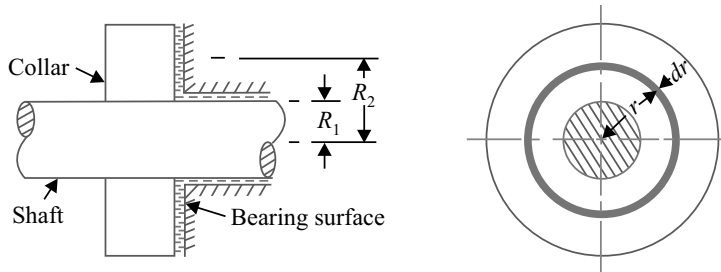


Fig. 10.29. Collar bearing.

where,

$$V = \omega r = \frac{2\pi N}{60} \cdot r$$

Shear force on the elementary ring,

$$dF = \text{Viscous shear stress} \times \text{area of the ring}$$

$$= \tau \times 2\pi r \cdot dr = \mu \times \frac{2\pi N r}{60t} \times 2\pi r \cdot dr$$

$$= \frac{\mu}{15} \frac{\pi^2 N r^2}{t} \cdot dr$$

Torque on the elementary ring, $dT = dF \times r$

$$= \frac{\mu}{15t} \cdot \pi^2 N r^2 \cdot dr \cdot r$$

$\therefore$ Total torque required to overcome the viscous resistance,

$$T = \int_{R_1}^{R_2} \frac{\mu}{15t} \cdot \pi^2 N r^3 dr = \frac{\mu}{15t} \pi^2 N \int_{R_1}^{R_2} r^3 dr$$

$$= \frac{\mu}{15t} \pi^2 N \left[\frac{r^4}{4} \right]_{R_1}^{R_2}$$

$$T = \frac{\mu}{60t} \pi^2 N (R_2^4 - R_1^4) \quad \dots(10.36)$$

$\therefore$ Power absorbed by the entire bearing,

$$P = \frac{2\pi NT}{60} \text{ watts.}$$

Example 10.34. A collar bearing having external and internal diameters 240 mm and 180 mm respectively is used to take the thrust of a shaft. An oil film of thickness 0.25 mm and of viscosity 0.8 poise is maintained between the collar surface and the bearing. Find the power lost in overcoming the viscous resistance of oil when the shaft is running at 300 r.p.m.

Solution. External radius, $R_2 = \frac{240}{2} = 120 \text{ mm} = 0.12 \text{ m}$

Internal radius, $R_1 = \frac{180}{2} = 90 \text{ mm} = 0.09 \text{ m}$

Thickness of oil film, $t = 0.25 \text{ mm} = 0.00025 \text{ m}$

Viscosity of the oil, $\mu = 0.8 \text{ poise} = 0.08 \text{ Ns/m}^2$

Speed of the shaft, $N = 300 \text{ r.p.m.}$

Power lost in viscous resistance, P:

Torque required to overcome the viscous resistance,

$$T = \frac{\mu}{60t} \pi^2 N (R_2^4 - R_1^4) \quad \dots[\text{Eqn. (10.36)}]$$

$$= \frac{0.08}{60 \times 0.00025} \pi^2 \times 300 (0.12^4 - 0.09^4) = 2.238 \text{ Nm}$$

$\therefore$ Power lost in viscous resistance,

$$P = \frac{2\pi NT}{60} = \frac{2\pi \times 300 \times 2.238}{60} = 70.31 \text{ W (Ans.)}$$

10.10. LOSS OF HEAD DUE TO FRICTION IN VISCOUS FLOW

In a pipe of diameter D in which a viscous fluid of viscosity μ is flowing with a velocity $\bar{u}$, the loss of pressure head, h_f , is given by eqn. (10.11) as:

$$h_f = \frac{32 \mu \bar{u} L}{w D^2} \quad \dots(i)$$

where,

L = Length of the pipe, and

w = Weight density of the fluid.

The loss of head due to friction is given by:

$$h_f = \frac{4fLV^2}{D \times 2g} = \frac{4fL\bar{u}^2}{D \times 2g} \quad \dots(ii)$$

where, f is the co-efficient of friction between pipe and fluid, and $V = \bar{u}$.

(Note : For derivation of this formula, please refer to Art 11.2)

From eqn. (i) and (ii), we have:

$$\frac{32 \mu \bar{u} L}{w D^2} = \frac{4 f L \bar{u}^2}{D \times 2g}$$

$$\text{or,} \quad f = \frac{32 \mu \bar{u} L \times D \times 2g}{4 L \bar{u}^2 \cdot \rho \cdot g D^2} = \frac{16 \mu}{\bar{u} \rho D} \quad (\because w = \rho \cdot g)$$

$$= \frac{16 \mu}{\rho V D} = 16 \times \frac{1}{Re} \quad (\because \bar{u} = V)$$

$$\left[\text{where, } Re \left(= \frac{\rho V D}{\mu} \right) \text{ is the Reynolds number} \right]$$

$$\text{i.e.} \quad f = \frac{16}{Re} \quad \dots(10.37)$$

Example 10.35. In a pipe of 200 mm diameter in which water is flowing, there is a shear stress of 0.12 kN/m² at a point distant 30 mm from the pipe axis. If the co-efficient of friction between the pipe and the fluid is 0.04, calculate the shear stress at the pipe wall. [Nagpur University]

Solution. Diameter of the pipe, $D = 200 \text{ mm} = 0.2 \text{ m}$

Co-efficient of friction, $f = 0.04$

Shear stress at $r = 30 \text{ mm}$, $\tau = 0.12 \text{ kN/m}^2$

Shear stress at the pipe wall, τ_0 :

$$\text{Co-efficient of friction, } f = \frac{16}{Re} \quad \dots[\text{Eqn. (10.37)}]$$

$$\text{or,} \quad Re = \frac{16}{f} = \frac{16}{0.04} = 400$$

Since $Re < 2000$, hence the flow is *viscous/ laminar*.

The shear stress in case of viscous flow through a pipe is given as:

$$\tau = -\frac{\partial p}{\partial x} \cdot \frac{r}{2} \quad \dots[\text{Eqn. (10.4)}]$$

But $\frac{\partial p}{\partial x}$ is constant across a section, therefore, $\tau \propto r$

Shear stress at the pipe wall ($r = 0.1 \text{ m}$) is τ_0 .

$$\therefore \quad \frac{\tau}{r} = \frac{\tau_0}{0.1} \quad \text{or} \quad \frac{0.12}{0.03} = \frac{\tau_0}{0.1}$$

$$\text{or,} \quad \tau_0 = \frac{0.012 \times 0.1}{0.03} = \mathbf{0.4 \text{ kN/m}^2 \text{ (Ans.)}}$$

Example 10.36. A pipe 240 mm in diameter and 10000 m long is laid at a slope of 1 in 180. An oil of specific gravity 0.85 and viscosity 1.5 poise is pumped up at the rate of 0.02 m³/s. Find:

(i) Head lost due to friction, and

(ii) Power required to pump the oil.

Solution. Diameter of the pipe, $D = 240 \text{ mm} = 0.24 \text{ m}$

$$\therefore \quad \text{Area, } A = (\pi/4) \times 0.24^2 = 0.04524 \text{ m}^2$$

Length of the pipe, $L = 10000$ m

Slope of the pipe, $i = 1$ in 180

Specific gravity of oil, $S = 0.85$

Viscosity of oil, $\mu = 1.5$ poise $= 0.15$ Ns/m²

Discharge, $Q = 0.02$ m³/s

(ii) Head lost due to friction, h_f :

$$\text{Velocity of flow, } \bar{u} = \frac{Q}{A} = \frac{0.02}{0.04524} = 0.442 \text{ m/s}$$

$$\text{Reynolds number, } Re = \frac{\rho V D}{\mu} = \frac{(0.85 \times 1000) \times 0.442 \times 0.24}{0.15} = 601$$

As $Re < 2000$, the flow is *viscous/laminar*.

$$\begin{aligned} \text{The co-efficient of friction, } f &= \frac{16}{Re} \quad \dots [\text{Eqn. (10.37)}] \\ &= \frac{16}{601} = 0.02662 \end{aligned}$$

$\therefore$ Head lost due to friction,

$$h_f = \frac{4 f L \bar{u}^2}{D \times 2g} = \frac{4 \times 0.02662 \times 10000 \times 0.442^2}{0.24 \times 2 \times 9.81} = 44.17 \text{ m}$$

Height through which oil is to be raised by the pump

= Slope $\times$ length of pipe

$$= i \times L = \frac{1}{180} \times 10000 = 55.55 \text{ m}$$

Total head against which pump is to work,

$$H = h_f + i \times L = 44.17 + 55.55 = 99.72 \text{ m}$$

$\therefore$ Power required to pump the oil,

$$\begin{aligned} P &= \frac{wQH}{1000} \text{ kW} = \frac{(0.85 \times 9810) \times 0.02 \times 99.72}{1000} \\ &= \mathbf{16.63 \text{ kW (Ans.)}} \end{aligned}$$

10.11. MOVEMENT OF PISTON IN DASHPOT

A **dashpot** is a device employed for damping vibrations of machines (Fig. 10.30). It consists of piston that moves in a concentric cylinder, the diameter of which is only slightly greater than that of the piston. The cylinder contains a viscous oil, the quantity of which should be sufficient to cover the top of the piston. The piston is connected with the machine element whose motion is to be restrained. On the downward movement of piston, under the load W , the oil is displaced from the underneath and it moves to the space above the piston through the small annular clearance between the piston and the cylinder. Conversely, during the upward movement of the piston the oil is displaced downwards. It is due to this aspect that the mechanical vibrations of the machine element (connected to the piston) are damped.

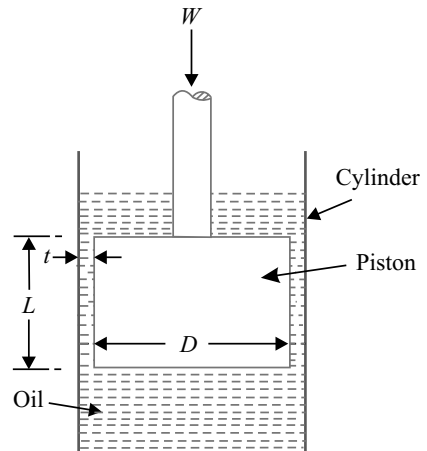


Fig. 10.30 Dashpot mechanism.

Let,

D = Diameter of the piston,

L = Length of the piston,

μ = Viscosity of the oil,

V = Velocity of the piston,

$\bar{u}$ = Average velocity of oil in clearance,

t = Clearance between dashpot and piston, and

Δp = Difference of pressure intensities between the two ends of the piston.

$$= \frac{W}{\frac{\pi}{4} D^2} = \frac{4W}{\pi D^2} \quad \dots(i)$$

The flow of oil through the clearance space is similar to the viscous/laminar flow between parallel plates and as such the following relation holds good,

$$\Delta p = \frac{12\mu\bar{u}L}{t^2} \quad \dots(ii)$$

From eqns. (i) and (ii), we have:

$$\frac{4W}{\pi D^2} = \frac{12\mu\bar{u}L}{t^2}$$

$$\text{or,} \quad \bar{u} = \frac{Wt^2}{3\pi\mu LD^2} \quad \dots(iii)$$

The rate of oil flow in dash pot

$$= V \times \frac{\pi}{4} D^2$$

(where, V = velocity of piston or the velocity of oil in dashpot in contact with piston)

Rate of flow through clearance

$$= \bar{u} \times \pi D \cdot t$$

By continuity equation, we have:

$$\bar{u} \times \pi D \cdot t = V \times \frac{\pi}{4} D^2$$

$$\text{or,} \quad \bar{u} = V \times \frac{\pi}{4} D^2 \times \frac{1}{\pi D \cdot t} = \frac{VD}{4t} \quad \dots(iv)$$

From eqns. (iii) and (iv), we have:

$$\frac{Wt^2}{3\pi\mu LD^2} = \frac{VD}{4t}$$

$$\text{or,} \quad \mu = \frac{4Wt^3}{3\pi LD^3 V} \quad \dots(10.38)$$

Example 10.37. An oil dashpot consists of a piston moving in a cylinder having oil. This arrangement is used to damp out the vibrations. The piston falls with uniform speed and covers 50 mm in 100 seconds. If an additional weight of 1.334 N is placed on the top of the piston, it falls through 50 mm in 86 seconds with uniform speed. The diameter of the piston is 75 mm and its length is 100 mm. The clearance between the piston and the cylinder is 1.2 mm which is uniform throughout. Find the viscosity of the oil. [UPTU]

Solution. Diameter of the piston, $D = 75 \text{ mm} = 0.075 \text{ m}$

Length of the piston, $L = 100 \text{ mm} = 0.1 \text{ m}$

Clearance, $t = 1.2 \text{ mm} = 0.0012 \text{ m}$

Additional weight = 1.334 N

Viscosity of oil, μ :

Let,

W = Weight of the piston,

V = Velocity of piston without additional weight, and

V' = Velocity of piston with the additional weight.

Using eqn. (10.38), we have:

$$\mu = \frac{4Wt^3}{3\pi LD^3 V} = \frac{4(W + 1.334)t^3}{3\pi LD^3 V'}$$

or,
$$\frac{W}{V} = \frac{W + 1.334}{V'}$$

or,
$$\frac{V}{V'} = \frac{W}{W + 1.334} \quad \dots(i)$$

But,
$$V = \frac{50}{100} = 0.50 \text{ mm/s}$$

and,
$$V' = \frac{50}{86} = 0.581 \text{ mm/s} \quad \left. \vphantom{\begin{matrix} V \\ V' \end{matrix}} \right\} \text{Data given}$$

$\therefore \frac{V}{V'} = \frac{0.5 \times 10^{-3}}{0.581 \times 10^{-3}} = 0.86 \quad \dots(ii)$

From eqns. (i) and (ii), we have:

$$\frac{W}{W + 1.334} = 0.86$$

or,
$$W = 0.86(W + 1.334) = 0.86W + 1.147$$

or,
$$W = 8.19 \text{ N}$$

Now,
$$\mu = \frac{4Wt^3}{3\pi LD^3 V} \quad \dots[\text{Eqn. (10.38)}]$$

$$= \frac{4 \times 8.19 \times (0.0012)^3}{3\pi \times 0.1 \times (0.075)^3 \times \left(\frac{50}{1000} \times \frac{1}{100}\right)}$$

$$= \mathbf{0.2847 \text{ Ns/m}^2 \text{ (Ans.)}}$$

10.12. MEASUREMENT OF VISCOSITY

To determine the co-efficient of viscosity of a liquid the following experimental methods are used:

1. Rotating cylinder method
2. Falling sphere method
3. Capillary tube
4. Efflux viscometers.

The devices used for measurement of viscosity are known as **Viscometers**.

10.12.1. Rotating Cylinder Method

In this method Newton's law of viscosity is used to measure the viscosity of a fluid. Fig. 10.31 shows a rotating cylinder viscometer. It consists of two concentric cylinders, the annular space between them is filled with the liquid whose viscosity is to be determined. The outer cylinder is rotated at a constant angular velocity ω with respect to the inner stationary cylinder. The torque transmitted by the enclosed liquid to the stationary cylinder is measured by the torsional strain of the restraining spring attached to the top of the inner cylinder.

According to Newton's law of viscosity,

$$\text{Shear stress } (\tau) = \mu \times \text{velocity gradient} \left(\frac{du}{dy} \right)$$

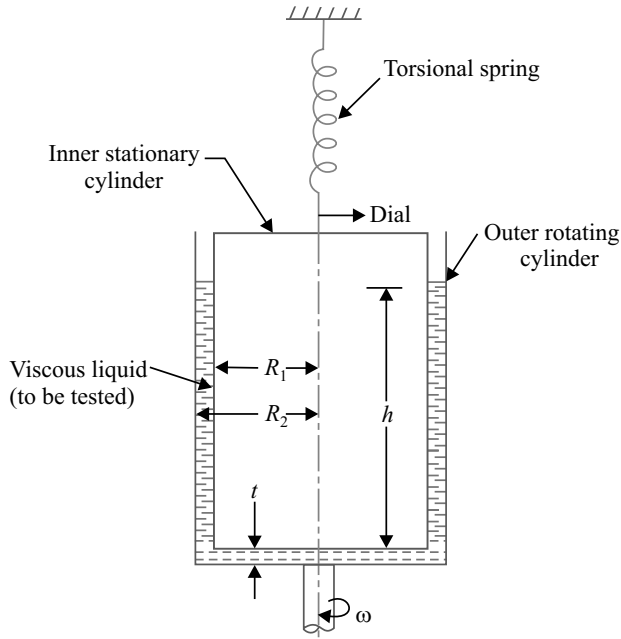


Fig. 10.31. Rotating cylinder viscometer.

Since the annular space $t = (R_2 - R_1)$ is quite small (where R_2 and R_1 are the radii of the outer and inner cylinders respectively), the velocity gradient,

$$\frac{du}{dy} = \frac{V}{t} = \frac{2\pi R_2 N}{60t}$$

(where, N = the rotational speed of the outer cylinder in r.p.m.)

$$\therefore \text{Shear stress } (\tau) = \mu \times \frac{2\pi R_2 N}{60t}$$

$$\begin{aligned} \text{Viscous drag} &= \text{Shear stress} \times \text{area} \\ &= \mu \times \frac{2\pi R_2 N}{60t} \times 2\pi R_1 h \end{aligned}$$

(where, h = height of liquid)

$$\begin{aligned} \text{Viscous torque} &= \text{Viscous drag} \times \text{radius} \\ &= \left(\mu \times \frac{2\pi R_2 N}{60t} \times 2\pi R_1 h \right) \times R_1 = \frac{\mu \pi^2 R_1^2 R_2 h N}{15t} \end{aligned}$$

Viscous torque must equal the torque T exerted by the torquemeter.

$$\therefore T = \frac{\mu \pi^2 R_1^2 R_2 \cdot h N}{15t}$$

$$\text{or, } \mu = \frac{15 T t}{\pi^2 R_1^2 R_2 h N} \quad \dots(10.37)$$

Thus a rotational type viscometer can be calibrated to directly give μ for given speed of rotation N .

Example 10.38. In a rotating cylinder viscometer, the radii of the cylinder are 32 mm and 30 mm, and the outer cylinder is rotated steadily at 200 r.p.m. For a certain liquid filled in the annular space to a depth 80 mm, the torque produced on the inner cylinder is 0.9×10^{-4} Nm. Calculate the viscosity of the liquid.

Assume the velocity distribution to be linear.

Solution. Radius of the outer rotating cylinder, $R_2 = 32 \text{ mm} = 0.032 \text{ m}$

Radius of the inner stationary cylinder, $R_1 = 30 \text{ mm} = 0.03 \text{ m}$

Height of liquid, $h = 80 \text{ mm} = 0.08 \text{ m}$

Speed of the outer cylinder, $N = 200 \text{ r.p.m.}$

Torque produced on the inner cylinder $T = 0.9 \times 10^{-4} \text{ Nm}$

Viscosity of the liquid, μ :

$$\text{The tangential velocity, } V = \frac{2\pi R_2 N}{60} = \frac{2\pi \times 0.032 \times 200}{60} = 0.67 \text{ m/s}$$

Since the velocity distribution is linear, therefore,

$$\frac{du}{dy} = \frac{V}{t} = \frac{0.67}{(R_2 - R_1)} = \frac{0.67}{(0.032 - 0.03)} = 335 \text{ s}^{-1}$$

$$\text{Viscous shear stress, } \tau = \mu \frac{V}{t} = 335 \mu$$

$$\begin{aligned} \text{Viscous force/drag} &= 335 \mu \times 2\pi R_1 h \\ &= 335 \mu \times 2\pi \times 0.03 \times 0.08 = 5.05 \mu \end{aligned}$$

$$\begin{aligned} \text{Viscous torque} &= 5.05 \mu \times R_1 \\ &= 5.05 \mu \times 0.03 = 0.1515 \mu \end{aligned}$$

But, Viscous torque = Torque (T) measured by the torquemeter

$$\therefore 0.1515 \mu = 0.9 \times 10^{-4}$$

$$\text{or, } \mu = \frac{0.9 \times 10^{-4}}{0.1515} = 5.94 \times 10^{-4} \text{ Ns/m}^2 \text{ (Ans.)}$$

Example 10.39. In a torsion viscometer, the outer cylinder of 150.5 mm diameter is rotated by turning the shaft at a constant speed of 100 r.p.m. Owing to its viscosity, the liquid under test transmits a torque of 540 Nm to the inner cylinder of 150 mm diameter, which is suspended by torsion wire fixed at its upper end. If the liquid is 130 mm deep, find its viscosity.

(MGU, Kerala)

Solution. Refer to Fig. 10.31.

Given : $R_2 = \frac{150.5}{2} = 75.25 \text{ mm} = 0.07525 \text{ m}$; $R_1 = \frac{150}{2} = 75 \text{ mm} = 0.075 \text{ m}$; $N = 100 \text{ r.p.m.}$; $T = 540 \text{ Nm}$; $h + t = 130 \text{ mm} = 0.13 \text{ m}$

Viscosity of the liquid, μ :

Let us assume that the clearance at the bottom of the two cylinder is 0.25 mm, i.e. the difference between radii of two cylinders, then,

$$t = 0.25 \text{ mm} = 0.00025 \text{ m}$$

$$\therefore h = 0.13 - 0.00025 = 0.12975 \text{ m}$$

$$\text{We know that, } \mu = \frac{15Tt}{\pi^2 R_1^2 R_2 h N} \quad \dots [\text{Eqn. (10.39)}]$$

Substituting the values, we get:

$$\mu = \frac{15 \times 540 \times 0.00025}{\pi^2 \times 0.075^2 \times 0.07525 \times 0.12975 \times 100} = 37.36 \text{ Ns/m}^2 \text{ (Ans.)}$$

10.12.2. Falling Sphere Method

Falling sphere method of measuring viscosity of a liquid is based on 'Stokes' law.

Fig. 10.32 shows a falling sphere viscometer. A small spherical ball is released into the liquid to be tested and it accelerates under the gravitational force until it reaches a maximum/terminal velocity V when the *buoyant and viscous drag forces balance the gravity force*.

Let,

d = Diameter of the spherical ball,

l = Distance travelled by the sphere in viscous liquid,

t = Time taken by the sphere to cover distance l ,

ρ_s = Density of sphere,

ρ_f = Density of fluid/liquid,

W = Weight of sphere,

F_B = Buoyant force, acting on the sphere

F_D = Drag force, and

μ = Dynamic viscosity of the liquid under test.

The forces acting on the sphere are:

1. Weight, W acting vertically *downwards*
2. Buoyant force, F_B acting vertically *upwards*
3. Drag force, F_D acting vertically *upwards*

$$W = \text{Volume} \times \text{density of sphere} \times g$$

$$= \frac{\pi}{6} d^3 \times \rho_s \times g \quad \left(\because \text{Volume of sphere} = \frac{\pi}{6} d^3 \right)$$

$$F_B = \text{Weight of liquid displaced}$$

$$= \text{Volume of liquid displaced} \times \text{density of liquid} \times g$$

$$= \frac{\pi}{6} d^3 \times \rho_f \times g$$

$$(\because \text{Volume of liquid displaced} = \text{volume of sphere})$$

$$F_D = 3\pi \mu VD$$

$$\dots \text{ (Stokes law)}$$

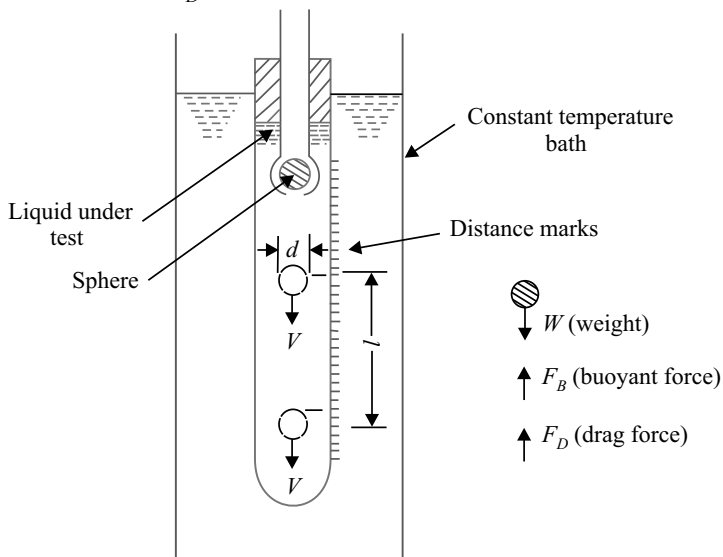


Fig. 10.32. Falling sphere viscometer.

For equilibrium,

$$F_D + F_B = W$$

or,

$$F_D = W - F_B$$

$$3\pi\mu Vd = \frac{\pi}{6}d^3 \times \rho_s \times g - \frac{\pi}{6}d^3 \times \rho_f \times g \quad (\text{substituting the values})$$

$$= \frac{\pi}{6}d^3 \times g (\rho_s - \rho_f)$$

$$\text{or, } \mu = \frac{\pi}{6} \cdot \frac{d^3 \times g (\rho_s - \rho_f)}{3\pi Vd} = \frac{gd^2}{18V} (\rho_s - \rho_f) \quad \dots(10.40)$$

From the above equation the value of dynamic viscosity μ can be determined.

Note. Stokes' law is essentially valid for Reynolds number below 0.1 where the wall has no effect on the terminal fall velocity V .

Example 10.40. In a falling sphere viscometer, a lubricating oil of density 900 kg/m^3 was placed in a 80 mm inside diameter tube. A 10 mm diameter steel ball of density 8000 kg/m^3 was found to travel a distance of 950 mm in 19 seconds. Determine the viscosity of the oil.

Solution. Density of lubricating oil, $\rho_f = 900 \text{ kg/m}^3$

Diameter of the sphere, $d = 10 \text{ mm} = 0.01 \text{ m}$

Density of steel ball, $\rho_s = 8000 \text{ kg/m}^3$

Distance travelled in 19 seconds = 950 mm = 0.95 m

Viscosity of the oil, μ :

$$\text{Weight of the ball, } W = \frac{\pi}{6}d^3 \times \rho_s \times g = \frac{\pi}{6} \times 0.01^3 \times 8000 \times 9.81 = 0.0411 \text{ N}$$

$$\text{Buoyant force, } F_B = \frac{\pi}{6}d^3 \times \rho_f \times g = \frac{\pi}{6} \times 0.01^3 \times 900 \times 9.81 = 0.00462 \text{ N}$$

$$\text{Drag force, } F_D = 3\pi\mu Vd = 3\pi \times \mu \times \left(\frac{0.95}{19}\right) \times 0.01 = 0.00471 \mu \text{ N}$$

For equilibrium:

$$F_D + F_B = W$$

or,

$$F_D = W - F_B$$

or,

$$0.00471 \mu = 0.0411 - 0.00462 = 0.0365$$

or,

$$\mu = \frac{0.0365}{0.00471} = 7.75 \text{ Ns/m}^2 \text{ (Ans.)}$$

Let us check the Reynolds number, Re .

$$Re = \frac{\rho Vd}{\mu} = \frac{900 \times (0.95/19) \times 0.01}{7.75} = 0.058 < 0.1$$

10.12.3. Capillary Tube Method

This method makes use of Hagen-Poiseuille equation for laminar flow through circular tubes.

Fig. 10.33 shows a capillary tube viscometer. It consists of a tank in which the liquid whose viscosity is to be determined is filled. A capillary tube of diameter D and length L is attached horizontally very close to the bottom of tank. The tube is allowed to discharge freely into the atmosphere. The liquid is collected in a measuring tank for a given time. Then the rate of liquid collected in the tank per second is determined. The pressure head is measured at a point far away from the tank.

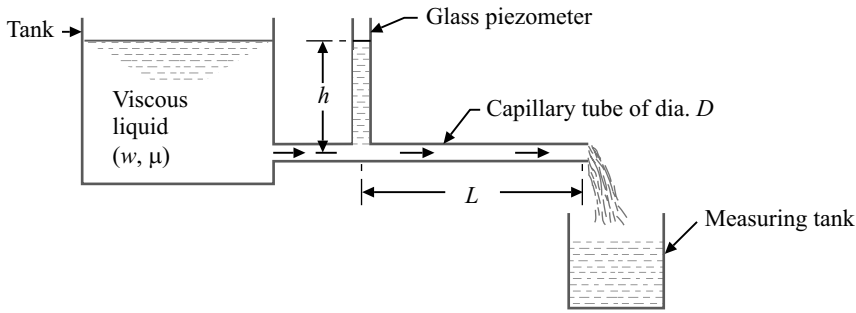


Fig. 10.33. Capillary tube viscometer.

Let, h = Difference of pressure head for length L ,
 μ = Co-efficient of viscosity, and
 w = Weight density of the liquid.

According to Hagen-Poiseuille's equation, we have:

$$h = \frac{32\mu\bar{u}L}{wD^2}$$

But, $\bar{u} = \frac{Q}{\frac{\pi}{4}D^2}$

(where, Q = discharge of liquid through the tube)

$$\therefore h = \frac{32\mu \times \frac{Q}{(\pi/4) \times D^2} \times L}{wD^2} = \frac{128\mu QL}{\pi wD^4}$$

$$\text{or, } \mu = \frac{\pi wh D^4}{128 QL} \quad \dots(10.41)$$

Example 10.41. The viscosity of an oil of specific gravity 0.8 is measured by a capillary tube of diameter 40 mm. The difference of pressure head between two points 1.2 m apart is 0.3 m of water. The weight of oil collected in a measuring tank is 400 N in 100 seconds. Find the viscosity of oil.

Solution. Sp. gr. of oil = 0.8

$\therefore$ Weight density of oil = $0.8 \times 9810 = 7848 \text{ N/m}^3$

Dia. of the capillary tube, $D = 40 \text{ mm} = 0.04 \text{ m}$

Length of the tube, $L = 1.2 \text{ m}$

Difference of pressure head, $h = 0.3 \text{ m of water}$

Weight of oil collected, $W = 400 \text{ N}$

Time, $t = 100 \text{ s.}$

Viscosity of the oil, μ :

$$\text{Discharge, } Q = \frac{\text{Weight of oil collected/sec}}{\text{Weight density}} = \frac{(400/100)}{7848} = 0.000509 \text{ m}^3/\text{s}$$

$$\begin{aligned} \text{We know, } \mu &= \frac{\pi wh D^4}{128 QL} \\ &= \frac{\pi \times 7848 \times 0.3 \times (0.04)^4}{128 \times 0.000509 \times 1.2} = \mathbf{0.242 \text{ Ns/m}^2 \text{ (Ans.)}} \end{aligned}$$

10.12.4. Efflux Viscometers

In Efflux viscometers the *viscosity is determined by noting the time of efflux, under specific conditions, for a fixed volume of fluid through a specific capillary or aperture* (standardized nozzle or orifice). The operation of *Saybolt, Redwood and Engler* is based on this principle.

Fig. 10.34 shows a **Saybolt viscometer**. It consists of a tank in the bottom of which is a short capillary tube. The tank is filled with the liquid whose viscosity is to be determined; a constant temperature both surrounds this tank to ensure that the liquid remains at uniform constant temperature during the test run. The viscosity is determined by measuring the time required for 60 cm³ of liquid at a known temperature to flow out of the reservoir through the tube. The initial level of the liquid in the reservoir is previously adjusted to a standard height. From the time measurement, the kinematic viscosity of the liquid can be determined by the use of empirical formula or the calibration chart.

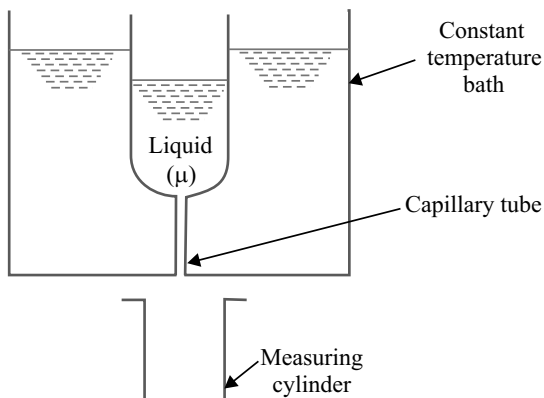


Fig. 10.34. Saybolt viscometer.

These viscometers (laboratory instruments) are widely employed in *petroleum and allied industries*.

HIGHLIGHTS

1. Reynolds number, $Re < 2000$...Laminar flow
Reynolds number, $Re > 4000$...Turbulent flow
2. In case of laminar flow : The loss of head $\propto V$, where V is the velocity of flow.
In case of turbulent flow : The loss of head $\propto V^2$ (approx).
 $\propto V^n$ (more exactly), where n varies from 1.75 to 2.0.
3. Relationship between shear stress and pressure gradient:

$$\frac{\partial \tau}{\partial y} = \frac{\partial p}{\partial x}$$

This equation indicates that the pressure gradient in the direction of flow is equal to the shear gradient in the direction normal to the direction of flow.

4. In case of viscous flow through circular pipes, we have:

$$(i) \quad \text{Shear stress, } \tau = -\frac{\partial p}{\partial x} \cdot \frac{r}{2}$$

$$(ii) \quad \text{Velocity, } u = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} (R^2 - r^2)$$

$$\text{Max. velocity, } u_{\max} = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} R^2$$

$$\text{Average velocity, } \bar{u} = \frac{u_{\max}}{2}$$

$$\text{Again, } u = u_{\max} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

(iii) Loss of pressure head, $h_f = \frac{32 \mu \bar{u} L}{w \cdot D^2}$

where,

r = Radius at any point,

R = Radius of the pipe,

D = Diameter of the pipe,

μ = Co-efficient of viscosity,

w = Weight density ($\rho \cdot g$), and

$\bar{u}$ = Average velocity $= \frac{Q}{\pi R^2}$.

5. For a circular pipe:

K.E. correction factor, $\alpha = 2.0$, and

Momentum correction factor, $\beta = \frac{4}{3}$.

6. For flow of viscous fluid through an annulus:

(i) The velocity distribution is given by:

$$u = -\frac{1}{4\mu} \left(\frac{\partial p}{\partial x} \right) \left[R_1^2 - r^2 - \frac{R_1^2 - R_2^2}{\ln \left(\frac{R_1}{R_2} \right)} \cdot \ln \left(\frac{R_1}{r} \right) \right]$$

Velocity will be maximum, when

$$r = \left[\frac{R_1^2 - R_2^2}{2 \ln \left(\frac{R_1}{R_2} \right)} \right]^{1/2}$$

The discharge through annulus,

$$Q = -\frac{\pi}{8\mu} \left(\frac{\partial p}{\partial x} \right) \left[R_1^4 - R_2^4 - \frac{(R_1^2 - R_2^2)^2}{\ln \left(\frac{R_1}{R_2} \right)} \right]$$

The average velocity of flow through the annulus is given by,

$$\bar{u} = -\frac{1}{8\mu} \left(\frac{\partial p}{\partial x} \right) \left[R_1^2 + R_2^2 - \frac{(R_1^2 - R_2^2)}{\ln \left(\frac{R_1}{R_2} \right)} \right]$$

(iii) The shear stress distribution is given by:

$$\tau = \frac{1}{4} \left(-\frac{\partial p}{\partial x} \right) \left[2r - \frac{1}{r} \cdot \frac{R_1^2 - R_2^2}{\ln \left(\frac{R_1}{R_2} \right)} \right]$$

(where, R_1 and R_2 are the outer and inner radii of the annulus respectively)

7. For the viscous flow between two parallel plates:

Case I. One plate moving and the other at rest - Couette flow:

Velocity distribution: $u = \frac{U}{b} y - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (by - y^2)$

The Discharge per unit width, $q = \frac{U \cdot b}{2} - \frac{b^3}{12\mu} \cdot \frac{\partial p}{\partial x}$

Shear stress distribution: $\tau = \mu \cdot \frac{U}{b} - \frac{1}{2} \cdot \frac{\partial p}{\partial x} (b - 2y)$

Case II. Both plates at rest:

$$u = -\frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (by - y^2)$$

$$q = -\frac{b^3}{12\mu} \cdot \frac{\partial p}{\partial x}, \text{ and}$$

$$\tau = -\frac{1}{2} \cdot \frac{\partial p}{\partial x} (b - 2y)$$

Case III. Both plates moving in opposite directions:

$$u = (U + V) \frac{y}{b} - \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (by - y^2) - V$$

$$q = (U - V) \frac{b}{2} - \frac{1}{12\mu} \cdot \frac{\partial p}{\partial x} \cdot b^3$$

$$\tau = (U + V) \frac{\mu}{b} - \frac{\partial p}{\partial x} \left(\frac{b}{2} - y \right)$$

The shear stress (τ) will be zero at,

$$y = \frac{b}{2} - \frac{\mu}{b} \left[\frac{U + V}{(\partial p / \partial x)} \right]$$

where, U and V = The velocities (in opposite directions) of the upper and lower plates respectively.

b = Distance between the parallel plates, and

μ = Co-efficient of viscosity of the fluid.

8. Laminar flow through porous media:

$$\text{Loss of head, } h_f = \frac{K\mu\bar{u}L}{wn^2d_s^2}$$

where, k = A constant, the value of which depends on the shape of the passage,

μ = Co-efficient of viscosity,

$\bar{u}$ = Average velocity,

L = Length,

w = Weight density of the fluid,

n = Porosity, and

d_s = Grain diameter-porosity material.

(nd_s = diameter of the passage through the particles)

From the above eqn. we have, $\bar{u} = ki$... known as Darcy's equation.

where, k = a constant, called the co-efficient of permeability, and

i = the hydraulic gradient $\left(\frac{h_f}{L} \right)$

9. Power absorbed in bearings (P) is given by:

$$P = T \cdot \omega = \frac{2\pi NT}{60} \text{ watts, where } T \text{ is in Nm.}$$

where,

T = Torque required to overcome the viscous resistance of the whole of bearing,

ω = Angular speed of the shaft $\left(= \frac{2\pi N}{60} \right)$, and

N = Speed of the shaft in r.p.m.

For various bearings, the values of T are given by:

$$T = \frac{\mu \pi^2 D^3 N L}{120t} \quad \dots \text{Journal bearing}$$

$$T = \frac{\mu}{60t} \pi^2 N R^4 \quad \dots \text{Foot-step bearing}$$

$$T = \frac{\mu}{60} \pi^2 N (R_2^4 - R_1^4) \quad \dots \text{Collar bearing}$$

where, t = thickness of oil film.

10. For the viscous flow the co-efficient of friction is given by:

$$f = \frac{16}{Re}$$

where, Re = Reynolds number $= \frac{\rho V D}{\mu} = \frac{V D}{\nu}$

11. In a dashpot arrangement, the co-efficient of viscosity is given by:

$$\mu = \frac{4Wt^3}{3\pi L D^3 V}$$

where, W = Weight of piston/force on piston,

t = Clearance between dashpot and piston,

L = Length of the piston,

D = Diameter of the piston, and

V = Velocity of the piston.

12. The co-efficient of viscosity may be determined by the following methods:

(i) **Rotating cylinder method:** $\mu = \frac{15Tt}{\pi^2 R_1^2 R_2 h N}$

where, T = Torque,

t = Annular space,

R_1 = Radius of inner stationary cylinder,

R_2 = Radius of the outer rotating cylinder,

h = Height of liquid, and

N = Speed of the rotating cylinder in r.p.m.

(ii) **Falling sphere method:** $\mu = \frac{gd^2}{18V} (\rho_s - \rho_f)$

where, d = Diameter of the sphere,

V = Velocity of sphere,

ρ_s = Density of sphere, and

ρ_f = Density of fluid/liquid.

(iii) **Capillary tube method:** $\mu = \frac{\pi w h D^4}{128 Q L}$

- where, w = Weight density of the liquid,
 h = Difference of pressure head for length L ,
 D = Diameter of the capillary tube, and
 Q = Discharge of liquid through the capillary tube.

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer:

- The laminar/viscous flow is characterised by Reynolds number which is
 - less than the critical value
 - equal to critical value
 - more than critical value
 - none of the above.
- The laminar flow is characterised by
 - existence of eddies
 - irregular motion of fluid particles
 - fluid particles moving in layers parallel to the boundary surface
 - none of the above.
- Which of the following is an example of laminar flow?
 - Underground flow
 - Flow past tiny bodies
 - Flow of oil in measuring instruments
 - All of the above
 - None of the above.
- In case of laminar flow, the loss of pressure head is proportional to
 - velocity,
 - velocity²
 - velocity³
 - none of the above.
- The pressure gradient in the direction of flow is equal to the shear gradient in the direction
 - parallel to the direction of flow
 - normal to the direction of flow
 - either of the above
 - none of the above.
- ... studied the laminar flow through a circular tube experimentally
 - Prandtl
 - Pascal
 - Hagen and Poiseuille
 - None of the above.
- ... is the most commonly used equation for the velocity distribution for laminar flow through pipes.

$$(a) u = u_{\max} \left[1 - \frac{r}{R} \right]$$

$$(b) u = u_{\max} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

$$(c) u = u_{\max} \left[1 - \left(\frac{r}{R} \right)^3 \right]$$

$$(d) u = u_{\max}^2 \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

- In laminar flow the pressure drop per unit length of pipe ($\Delta p/L$) is given as
 - $\frac{32\mu\bar{u}}{D^2}$
 - $\frac{2\mu\bar{u}}{D^2}$
 - $\frac{32\mu\bar{u}}{D^3}$
 - none of above.
- The K.E. correction factor a for a circular pipe is equal to
 - 2
 - 3
 - 4
 - 6.
- The momentum correction factor b for a circular pipe is to equal to
 - $\frac{1}{3}$
 - $\frac{2}{3}$
 - $\frac{4}{3}$
 - $\frac{5}{3}$.
- The flow through a porous media is governed by well known Darcy's law which relates the velocity with the head loss and is usually expressed as
 - $u = \frac{kh_f}{L}$
 - $u = \frac{kL}{h_f}$
 - $u = k \cdot \left(\frac{h_f}{L} \right)^2$
 - $u = k \left(\frac{L}{h_f} \right)^3$.
- The shear stress distribution in pipe flow is given as
 - $\tau = - \left(\frac{\partial p}{\partial x} \right) \frac{r}{2}$
 - $\tau = r \cdot \left(\frac{\partial p}{\partial x} \right)$
 - $\tau = - 2r \left(\frac{\partial p}{\partial x} \right)$
 - none of the above.

13. For viscous flow the co-efficient of friction is given by
 (a) $f = \frac{8}{Re}$ (b) $f = \frac{16}{Re}$
 (c) $f = \frac{32}{Re}$ (d) $f = \frac{60}{Re}$
14. In case of viscous flow through circular pipes
 (a) $\bar{u} = 2 u_{\max}$ (b) $\bar{u} = \frac{3}{2} u_{\max}$
- (c) $\bar{u} = \frac{u_{\max}}{2}$ (d) none of the above.
15. The maximum velocity in a circular pipe when flow is laminar occurs at
 (a) the top of the pipe
 (b) the bottom of the pipe
 (c) the centre of the pipe
 (d) not necessarily at the centre.

ANSWERS

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (c) | 3. (d) | 4. (a) | 5. (b) | 6. (c) |
| 7. (b) | 8. (a) | 9. (a) | 10. (c) | 11. (a) | 12. (a) |
| 13. (b) | 14. (b) | 15. (c) | | | |

THEORETICAL QUESTIONS

- What is the difference between a laminar flow and a turbulent flow?
- What are the characteristics of a laminar flow?
- Enumerate examples of laminar flow.
- Draw a neat sketch of the Reynolds apparatus, and explain how the laminar flow can be demonstrated with the help of the apparatus.
- Derive a relationship between shear stress and pressure gradient.
- Derive an expression for the velocity distribution for viscous flow through a circular pipe. Also sketch the distribution of velocity and shear stress across a section of the pipe.
- For a steady laminar flow through a circular pipe prove that the velocity distribution across the section is parabolic and the average velocity is half of the maximum local velocity.
- What factors account for the loss of energy in laminar flow? How does the energy loss vary with velocity of flow?
- Derive Hagen-Poiseuille equation and state the assumptions made.
- For flow of viscous fluid through an annulus derive expressions for the following:
 (i) Discharge through the annulus,
 (ii) Average velocity of flow, and
 (iii) Shear stress distribution.
- What is a Couette flow?
- For a viscous flow through a circular pipe prove that the kinetic energy correction factor is equal to 2.
- Find an expression for the power absorbed in overcoming viscous resistance in case of a collar bearing.
- Show that the value of co-efficient of friction for viscous flow through a circular pipe is given by,
 $f = \frac{16}{Re}$, where Re = Reynolds number.
- Derive an expression for the co-efficient of viscosity in case of a dashpot arrangement.
- Describe briefly any two methods of determining the co-efficient of viscosity of a liquid.

UNSOLVED EXAMPLES

- An oil of 8 poise and specific gravity 0.9 is flowing through a horizontal pipe of 50 mm diameter. If the pressure drop in 100 m length of the pipe is 2000 kN/m², determine: (i) Rate of flow of oil (ii) Centre-line velocity; (iii) Total frictional drag over 100 m length of pipe; (iv) Power required to maintain the flow; (v) Velocity gradient at the pipe wall; (vi) Velocity and shear stress at 10 mm from the wall.
 [Ans. (a) 3.83 lit./s; (ii) 3.9 m/s; (iii) 3.93 kN; (iv) 7.65 kW (v) 312 s⁻¹; (vi) 2.5 m/s; 150 N/m²]
- An oil of viscosity 1 poise and relative density 0.9 is flowing through a circular pipe of diameter 50 mm and of length 300 m. The rate of flow of liquid is 0.0035 m³/s. Find the pressure drop in a length of 300 m and shear stress at the wall.
 [Ans. 684.3 kN/m², 28.5 N/m²]

3. In a circular pipe of diameter 100 mm a fluid of viscosity 7 poise and sp. gr. 1.3 is flowing. If the maximum shear stress at the wall of the pipe is 196.2 N/m^2 , find:

(i) The pressure gradient,
 (ii) The average velocity, and
 (iii) Reynolds number of flow.

[Ans. (i) 7848 N/m^2 per m; (ii) 3.5 m/s ; (iii) 650]

4. An oil of viscosity 0.02 poise and sp. gr. 0.8 is flowing through 50 mm diameter pipe of length 500 m at the rate of 0.19 lit./sec. Determine:

(i) Reynolds number of flow,
 (ii) Centre-line velocity,
 (iii) Pressure gradient,
 (iv) Wall shear stress, and
 (v) Power required to maintain the flow.

[Ans. (i) 1936.3; (ii) 0.1936 m/s ; (iii) 2.478 N/m^2 ;
 (iv) 0.031 N/m^2 ; (v) 0.2354 W]

5. What power is required per km of a line to overcome the viscous resistance to the flow of glycerine through a horizontal pipe of diameter 100 mm at the rate of 10 lit./s? Take $\mu = 8$ poise and kinematic viscosity $(\nu) = 6.0$ stokes.

[Delhi University]

[Ans. 32.5 kW]

6. In a pipe of 200 mm diameter the maximum velocity of flow is found to be 1.5 m/s . If flow in the pipe is laminar, find:

(i) The average velocity and the radius at which it occurs, and
 (ii) The velocity at 40 mm from the wall of the pipe.

[Ans. (i) 0.75 m/s ; 70.7 mm ; (ii) 0.96 m/s]

7. An oil of viscosity 0.143 Ns/m^2 and specific gravity 0.9 is flowing through a circular pipe of diameter 25 mm and of length 3 m at $\frac{1}{10}$ th of critical velocity for which Reynolds number is 2500. Find:

(i) The velocity of flow through the pipe,
 (ii) The head in metres of oil across the pipe length required to maintain the flow, and
 (iii) The power required to overcome viscous resistance to flow of oil.

[Ans. (i) 1.589 m/s , (ii) 3.593 m , (iii) 24.73 W]

8. In a laboratory a horizontal pipe of 500 mm diameter was used to measure the viscosity of a crude oil having specific weight of 9 kN/m^3 . During the test a pressure difference of 18 kN/m^2 was recorded from two pressure gauges located 6 m apart on the pipe. The oil was allowed to

discharge into a weighing tank and 5 kN of oil was collected in 3 minutes. Find the dynamic viscosity of the oil. [Ans. 1.49 poise]

9. The fixed parallel plates kept at 80 mm apart have laminar flow of oil between them with a maximum velocity 1.5 m/s . Taking dynamic viscosity of oil to be $\mu = 19.62$ poise, calculate;

(i) The discharge per metre width,
 (ii) The shear stress at the plates,
 (iii) The pressure difference between two points 25 m apart,
 (iv) The velocity at 20 mm from the plate, and
 (v) The velocity gradient at the plates end.

[Ans. (i) $0.08 \text{ m}^3/\text{s}$; (ii) 147 N/m^2 ; (iii) 91.97 kN/m^2 ; (iv) 1.125 m/s ; (v) 75s^{-1}]

10. A liquid of viscosity 0.1 Ns/m^2 and sp. gr. 0.9 is filled between two horizontal plates 10 mm apart. If the upper plate is moving at 2 m/s and the pressure difference between two sections 10 m apart is 9.81 kN/m^2 , determine the shear stress on the plate. [Ans. 15 N/m^2]

11. Water at a rate of 0.0008 litre/sec. is flowing through a sandy specimen of 10 cm height and 50 cm^2 cross-sectional area under constant head of 8 cm. Calculate the co-efficient of permeability.

[Ans. 0.02 cm/s]

12. Find the torque and power absorbed to rotate a shaft of diameter 50 mm, at 1200 r.p.m. concentrically within a sleeve 50.17 mm in diameter and 90 mm long, flooded with oil for which $\mu = 0.8$ poise.

[Ans. 1.045 Nm , 0.1313 kW]

$$\left(\begin{array}{l} \text{Hint. Thickness of oil film,} \\ t = \frac{50.17 - 50}{2} = 0.085 \text{ mm} \end{array} \right)$$

13. Find the power required to rotate a circular disc of diameter 200 mm at 100 r.p.m. The circular disc has a clearance of 0.4 mm from the bottom flat plate and clearance contain oil of viscosity 0.11 Ns/m^2 [Ans. 473.48 W]

14. A collar bearing having external and internal diameters 150 mm and 100 mm respectively is used to take the thrust of a shaft. An oil film of thickness 0.25 mm and of viscosity 0.8 poise is maintained between the collar surface and the bearing. Find the power lost in overcoming the viscous resistance of oil when the shaft is running at 300 r.p.m. [Ans. 12.54 W]

15. Water is flowing through a 240 mm diameter pipe with co-efficient of friction $f = 0.042$. The shear stress at a point 48 mm from the pipe axis is 0.1 kN/m^2 . Calculate the shear stress at the pipe wall. [Ans. 0.25 kN/m^2]

16. A pipe 200 mm in diameter and 10000 m long is laid at a slope of 1 in 200. An oil of specific gravity 0.9 and viscosity 0.15 Ns/m^2 is pumped up at the rate of $0.02 \text{ m}^3/\text{s}$. Find:
(i) Head lost due to friction, and
(ii) Power required to pump the oil.
[Ans. (i) 86.5 m; (ii) 24.09 kW]
17. In a rotating cylinder viscometer, the radii of the cylinders are 32 mm and 30 mm, and the outer cylinder is rotated steadily at 180 r.p.m. For a certain liquid filled in the annular space to a depth 75 mm, the torque produced on the inner cylinder is $1.2 \times 10^{-4} \text{ Nm}$. Calculate the viscosity of the liquid. [Ans. $9.39 \times 10^{-4} \text{ Ns/m}^2$]
18. In order to determine the viscosity of a lubricating oil by falling-sphere method, a steel spherical ball of specific gravity 7.7 and diameter 2 mm is allowed to fall freely under gravity through a distance of 140 mm in 215 seconds. The sp. gr. of the oil is 0.8. Determine the viscosity of the oil. [Ans. 2.31 Ns/m^2]
19. A fluid of sp.gr. 1.02 was made to pass through an accurate tube, length 35 cm and bore 0.1 cm under a head of 18 cm. A discharge equivalent to 40 cm^3 was collected in a period of 400 seconds. Find the dynamic viscosity of the fluid.
[Ans. $1.266 \times 10^{-3} \text{ Ns/m}^2$]



TURBULENT FLOW IN PIPES

11.1.1. Introduction

11.1.2. Loss of head due to friction in pipe flow—Darcy equation

11.1.3. Characteristics of turbulent flow

11.1.4. Shear stresses in turbulent flow

11.1.5. Universal velocity distribution equation

11.1.6. Hydrodynamically smooth and rough boundaries—velocity distribution for turbulent flow in smooth pipes—velocity distribution for turbulent flow in rough pipes

11.1.7. Velocity distribution for both smooth and rough pipes

11.1.8. Velocity distribution for turbulent flow in smooth pipes by power law

11.1.9. Resistance to flow of fluid in smooth and rough pipes

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples

(ii) In turbulent flow the velocity gradients near the boundary shall be quite large resulting in more shear.

(iii) In turbulent flow the flatness of velocity distribution curve in the core region away from the wall is because of the *mixing of fluid layers and exchange of momentum between them*.

(iv) The velocity distribution which is paraboloid in laminar flow, tends to follow power law and logarithmic law in turbulent flow.

11.1. INTRODUCTION

In a pipe, a laminar flow occurs when Reynolds number (Re) is less than 2000 and a turbulent flow occurs when $Re > 4000$. In a turbulent flow, the fluid motion is irregular and chaotic and there is complete mixing of fluid due to collision of fluid masses with one another. The fluid masses are interchanged between adjacent layers. As the fluid masses in adjacent layers have different velocities, interchange of fluid masses between the adjacent layers is accompanied by a transfer of momentum which causes additional shear stresses of high magnitude between adjacent layers. *The shear in turbulent flow is mainly due to momentum transfer.* The contribution of fluid viscosity to total shear is small and is usually neglected. In case of laminar flow, because of definite functional relationship 'between shear stress due to viscosity and velocity' it has been possible to derive a mathematical relationship for evaluation of energy dissipation or frictional head but such a simple relationship does not exist for turbulent flow. However to solve some of the practical problems, efforts have been made to evolve semi-empirical theories of turbulence.

Following points are worth noting about turbulent flow:

- (i) The velocity distribution in turbulent flow is more uniform than in laminar flow.

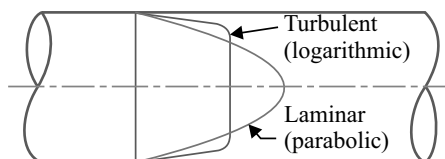


Fig. 11.1. Shows the velocity distribution curves for laminar and turbulent flows in a pipe.

- (v) Random orientation of fluid particles in a turbulent flow gives rise to additional stresses, called the **Reynolds stresses**.
- (vi) Formation of eddies, mixing and curving of path lines in a turbulent flow results in much greater frictional losses for the same rate of discharge, viscosity and pipe size.

The turbulent motion can be *classified* as follows:

1. **Wall turbulence.** It occurs in immediate vicinity of solid surfaces and in the boundary layer flows where the fluid has a negligible mean acceleration.
2. **Free turbulence.** It occurs in jets, wakes, mixing layers etc.
3. **Convective turbulence.** It takes place where there is conversion of P.E into K.E. by the process of mixing (e.g. the turbulent flow in the annular space between the concentric rotating cylinder, conventional flow between parallel horizontal plates etc.).

11.2. LOSS OF HEAD DUE TO FRICTION IN PIPE FLOW–DARCY EQUATION

In case of turbulent flow through pipes it has been observed through experiments that the viscous friction effects associated with fluid are proportional to:

- (i) The length of the pipe, L ,
- (ii) The wetted perimeter, P , and
- (iii) V^n , where V is the average velocity of flow and n is an index varying from 1.5 to 2 (depending on the material and nature of the pipe surface); for commercial pipes ≈ 2 (with turbulent flow).

Fig. 11.2. shows a horizontal pipe having steady flow. Consider control volume enclosed between sections 1 and 2 of the pipe.

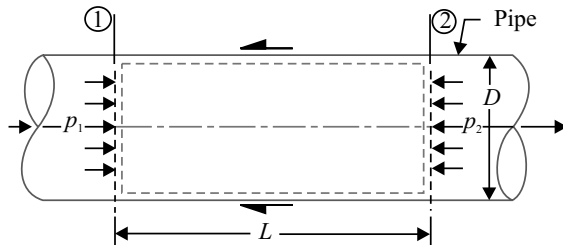


Fig. 11.2. Forces on a control volume in a pipe flow.

- Let,
- p_1 = Intensity of pressure at section 1,
 - p_2 = Intensity of pressure at section 2,
 - L = Length of the pipe, between sections 1 and 2,
 - D = Diameter of the pipe,
 - f' = Non-dimensional factor (whose value depends upon the material and nature of the pipe surface), and
 - h_f = Loss of head due to friction.

Propelling force on the flowing fluid between the two sections is

$$= (p_1 - p_2) A$$

(where, A = area of cross-section of the pipe)

Frictional resistance force = $f' PLV^2$

where, P = Wetted perimeter, and

V = Average flow velocity.

Under equilibrium conditions:

Propelling force = Frictional resistance force

$$\text{i.e.} \quad (p_1 - p_2) A = f' PLV^2$$

Dividing both sides by weight density w , we have:

$$\left(\frac{p_1 - p_2}{w} \right) A = \frac{f'}{w} PLV^2$$

$$\text{or,} \quad h_f = \frac{f'}{w} \left(\frac{P}{A} \right) LV^2$$

$$\text{or,} \quad h_f = \frac{2gf'}{w} \left(\frac{P}{A} \right) \frac{LV^2}{2g} = \frac{2gf'}{w} \times \frac{L}{m} \times \frac{V^2}{2g} \quad \dots(11.1)$$

The ratio $\frac{A}{P}$ is called the **hydraulic mean depth or hydraulic radius**, denoted by m (or R).

The term $\left(\frac{L}{m} \times \frac{V^2}{2g} \right)$ has dimensions of h_f and thus the term $\frac{2gf'}{w}$ is a non-dimensional quantity and let us replace it by another constant f .

$$\therefore \quad h_f = f \times \frac{L}{m} \times \frac{V^2}{2g} \quad \dots[11.1(a)]$$

In case of a circular pipe,

$$\text{Hydraulic mean depth, } m = \frac{A}{P} = \frac{\frac{\pi}{4} \times D^2}{\pi D} = \frac{D}{4}$$

Substituting this value in eqn. (11.1 (a)), we get:

$$h_f = f \times \frac{L}{D/4} \times \frac{V^2}{2g} = \frac{4fLV^2}{D \times 2g} \quad \dots(11.2)$$

(The factor f is known as *Darcy coefficient of friction*.)

Eqn. (11.2) is known as **Darcy-Weisbach equation** and it holds good for all types of flows provided a proper value of f is chosen.

Sometimes eqn. (11.2) is written as:

$$h_f = \frac{f_1 LV^2}{D \times 2g}$$

where, f_1 is known as **friction factor** ($f_1 = 4f$)

Expression for co-efficient of friction in terms of shear stress:

Refer to Fig. 11.2,

$$\begin{aligned} (p_1 - p_2) A &= \text{Force due to shear stress, } \tau_0 \\ (\text{where, } \tau_0 &= \text{shear stress at the pipe wall}) \\ &= \text{Shear stress } (\tau_0) \times \text{surface area} \\ &= \tau_0 \times \pi DL \end{aligned}$$

$$\text{or,} \quad (p_1 - p_2) \frac{\pi}{4} D^2 = \tau_0 \times \pi DL$$

$$\text{or,} \quad (p_1 - p_2) \frac{D}{4} = \tau_0 L$$

$$\text{or,} \quad (p_1 - p_2) = \frac{4\tau_0 \times L}{D} \quad \dots(11.3)$$

Eqn. (11.2) can be written as:

$$h_f = \frac{p_1 - p_2}{w} = \frac{4fLV^2}{D \times 2g}$$

or, $(p_1 - p_2) = \frac{4fLV^2}{D \times 2g} \times w$... (11.4)

Equating eqns. (11.3) and (11.4), we get:

$$\frac{4\tau_0 L}{D} = \frac{4fLV^2}{D \times 2g} \times w$$

or, $\tau_0 = \frac{fV^2 \times w}{2g} = \frac{fV^2 \times \rho g}{2g} = \frac{f\rho V^2}{2}$... [11.5 (a)]

or, $f = \frac{2\tau_0}{\rho V^2}$... [11.5 (b)]

11.3. CHARACTERISTICS OF TURBULENT FLOW

The turbulent flow is characterised by *random, irregular and haphazard movement of fluid particles*. It has been observed during experimentation that at any fixed point in turbulent field, the velocity and consequently the pressure fluctuates with time about a mean value.

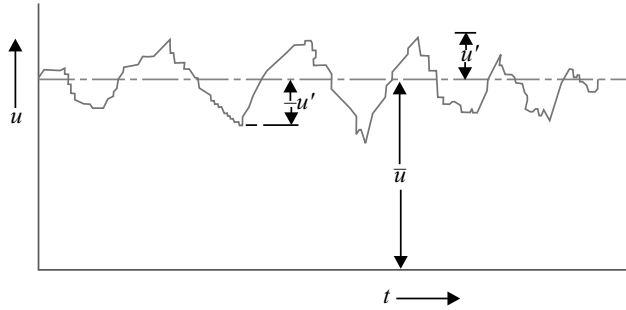


Fig. 11.3. Variation of u with time t at a point in turbulent flow.

Fig. 11.3 shows random velocity fluctuations at a point in turbulent flow.

The instantaneous velocity *i.e.* velocity at any time at the given point can be expressed as:

$$u = \bar{u} + u' \quad \dots (11.6)$$

where, u = Instantaneous velocity,
 $\bar{u}$ = Time average or temporal mean velocity, and
 u' = Velocity fluctuation (fluctuating component).

Similarly, $v = \bar{v} + v'$,
 $w = \bar{w} + w'$,
and, $p = \bar{p} + p'$... (11.7)

From the definition of average-velocities, we have:

$$\left\{ \begin{array}{l} \frac{1}{T} \int_0^T u dt = \bar{u}; \quad \frac{1}{T} \int_0^T v dt = \bar{v}; \\ \frac{1}{T} \int_0^T w dt = \bar{w}; \quad \frac{1}{T} \int_0^T p dt = \bar{p} \end{array} \right\} \quad \dots (11.8)$$

and,
$$\left\{ \begin{array}{l} \frac{1}{T} \int_0^T u' dt = \bar{u}' = 0; \quad \frac{1}{T} \int_0^T v' dt = \bar{v}' = 0; \\ \frac{1}{T} \int_0^T w' dt = \bar{w}' = 0; \quad \frac{1}{T} \int_0^T p' dt = \bar{p}' = 0 \end{array} \right\} \quad \dots(11.9)$$

where, T = Large interval of time.

Magnitude of turbulence = Arithmetic mean of root-mean square value of turbulent fluctuations in the three directions

$$= \sqrt{\left(\frac{\overline{u'^2} + \overline{v'^2} + \overline{w'^2}}{3} \right)} \quad \dots(11.10)$$

Intensity of turbulence

$$= \frac{\sqrt{\frac{\overline{u'^2} + \overline{v'^2} + \overline{w'^2}}{3}}}{\bar{V}} \quad \dots(11.11)$$

where, $\bar{V}$ = Line average resultant velocity at the point.

For describing the turbulence fully, besides the intensity of turbulence, the average size of the eddy is also necessary which can be obtained from the curve of velocity variation with time (as shown in Fig. 11.3) by multiplying the average time interval at which the curve crosses the mean value, with the average velocity of flow.

11.4. SHEAR STRESSES IN TURBULENT FLOW

In turbulent flow, as stated earlier, velocity fluctuations cause momentum transport which results in developing additional shear stresses of high magnitude between adjacent layers of the fluid. In order to determine the magnitude of the turbulent shear stress a number of semi-empirical theories have been developed some of which are discussed below.

11.4.1 Boussinesq's Theory

According to this theory (1877), the expression for the shear stress, τ_t for the turbulent flow can be written as :

$$\tau_t = \eta \cdot \frac{d\bar{u}}{dy} \quad \dots(11.12)$$

where η (eta) is called “*eddy*” *viscosity*, and $\bar{u}$ is the temporal mean velocity in the direction of flow at a point at distance y from the solid boundary.

Similar to kinematic viscosity $\nu = \frac{\mu}{\rho}$, the “*eddy*” *kinematic viscosity* ϵ (Greek ‘epsilon’) is also obtained by dividing eddy viscosity η , by the mass density of the fluid ρ , thus,

$$\epsilon = \frac{\eta}{\rho}$$

When viscous action is also included, the total shear stress may be expressed as :

$$\tau = \tau_v + \tau_t$$

(where τ_v = shear stress due to viscosity)

or,
$$\tau = \mu \frac{du}{dy} + \eta \frac{d\bar{u}}{dy} \quad \dots(11.13)$$

The magnitude of η may vary from zero (if the flow is laminar) to several thousand times that of μ . As the values of η and ϵ cannot be predicted, the Boussinesq's equation has a *limited use*.

11.4.2 Reynolds Theory

According to this theory (1886), the turbulent shear stress between two layers of a fluid at a small distance apart is given as:

$$\tau = \rho u' v' \quad \dots(11.14)$$

where u' and v' are the fluctuating components of velocity in the directions of x and y due to turbulence.

Since both u' and v' vary and subsequently τ also varies, therefore, to find the shear stress, the time average is taken and eqn. (11.14) becomes:

$$\bar{\tau} = \overline{\rho u' v'} \quad \dots(11.15)$$

11.4.3 Prandtl's Mixing Length Theory

According to Prandtl (1925), the **mixing length** (l) is defined as the average lateral distance through which a small mass of fluid particles would move from one layer to the other adjacent layers before acquiring the velocity of the new layer. He assumed that components u' and v' are of the same order and the velocity fluctuation in X -direction is related to the mixing length as:

$$u' = l \frac{du}{dy}$$

$$\therefore \overline{u' \times v'} = \overline{u' v'} = \left(l \frac{du}{dy} \right) \times \left(l \frac{du}{dy} \right) = l^2 \left(\frac{du}{dy} \right)^2 \quad \left(\because v' = l \frac{du}{dy} \right)$$

Substituting the value of $\overline{u' v'}$ in eqn. (11.15), we get:

$$\bar{\tau} = \rho l^2 \left(\frac{du}{dy} \right)^2 \quad \dots(11.16)$$

When the viscous action is also included the total shear stress may be expressed as :

$$\bar{\tau} = \mu \frac{du}{dy} + \rho l^2 \left(\frac{du}{dy} \right)^2 \quad \dots(11.17)$$

Eqn. (11.17) is used for most of the turbulent flow problems for determining the shear stress (viscous shear stress is negligible except near the boundary).

11.5. UNIVERSAL VELOCITY DISTRIBUTION EQUATION

Assuming the viscous shear stress to be negligible near the boundary the shear stress in turbulent flow is given by the eqn. (11.16).

$$i.e. \quad \bar{\tau} = \rho l^2 \left(\frac{du}{dy} \right)^2$$

From this equation, we can obtain velocity distribution if the relation between l , the mixing length, and y is known.

Also $l \propto y$ (from the pipe wall)

...Prandtl's hypothesis

or, $l = \lambda y$

where, λ = a constant of proportionality, known as '**Karman universal constant**' (= 0.4).

Substituting the values of l in eqn. (11.16), we get:

$$\bar{\tau} \text{ or } \tau = \rho \times (\lambda y)^2 \times \left(\frac{du}{dy} \right)^2 = \rho \lambda^2 y^2 \left(\frac{du}{dy} \right)^2 \quad \dots(i)$$

Assuming that the turbulent shear stress remains constant in the vicinity of wall, we have

$$\tau = \tau_0 \quad (\tau_0 = \text{the boundary shear stress})$$

The eqn. (i) becomes:

$$\tau_0 = \rho \lambda^2 y^2 \left(\frac{du}{dy} \right)^2$$

or,

$$\frac{du}{dy} = \frac{1}{\lambda y} \sqrt{\frac{\tau_0}{\rho}} = u_f \left(\frac{1}{\lambda y} \right) \quad \dots(ii)$$

$$\left[\text{where, } u_f = \text{shear friction velocity or } \textbf{shear velocity} = \sqrt{\frac{\tau_0}{\rho}} \right]$$

or,

$$du = u_f \left(\frac{1}{\lambda y} \right) dy \quad \dots(iii)$$

(u_f is constant for a given case of turbulent flow)

Integrating the other equation, we get:

$$u = \frac{u_f}{\lambda} \ln(y) + C \quad \dots(11.18)$$

(where, C = constant of integration)

Eqn. (11.18) shows that velocity distribution in turbulent flow is *logarithmic* in nature.

The constant of integration C is determined by the boundary condition.

At $y = R$ (radius of the pipe), $u = u_{\max}$

By substituting the above values in eqn. (11.18), we have:

$$u_{\max} = \frac{u_f}{\lambda} \ln(R) + C$$

or

$$C = u_{\max} - \frac{u_f}{\lambda} \ln(R)$$

Substituting this value of C in eqn. (11.18), we get:

$$\begin{aligned} u &= \frac{u_f}{\lambda} \ln(y) + u_{\max} - \frac{u_f}{\lambda} \ln(R) \\ &= u_{\max} + \frac{u_f}{\lambda} [\ln(y) - \ln(R)] \end{aligned}$$

or,

$$u = u_{\max} + \frac{u_f}{\lambda} \ln\left(\frac{y}{R}\right)$$

Taking

$$\lambda = 0.4, \text{ we get:}$$

$$u = u_{\max} + 2.5 u_f \ln\left(\frac{y}{R}\right) \quad \dots(11.19)$$

Eqn. (11.19) is called **Prandtl's universal distribution equation**. This equation is applicable to smooth as well as rough boundaries.

This equation (11.19) may be written in *non-dimensional form*:

$$\begin{aligned} \frac{u_{\max} - u}{u_f} &= 2.5 \ln\left(\frac{R}{y}\right) \\ &= 5.75 \log_{10}\left(\frac{R}{y}\right) \end{aligned}$$

$$\text{i.e.} \quad \frac{u_{\max} - u}{u_f} = 5.75 \log_{10} \left(\frac{R}{y} \right) \quad \dots(11.20)$$

The difference ($u_{\max} - u$) is known as the **velocity defect**.

Example 11.1. In a pipe of 360 mm diameter having turbulent flow, the centre-line velocity is 7 m/s and that at 60 mm from the pipe wall is 6 m/s. Calculate the shear friction velocity.

Solution. Radius of the pipe = $\frac{360}{2} = 180 \text{ mm} = 0.18 \text{ m}$

Centre-line velocity, $u_{\max} = 7 \text{ m/s}$

Velocity at 60 mm (i.e. distance y), $u = 6 \text{ m/s}$

Shear velocity, u_f :

We know,
$$\frac{u_{\max} - u}{u_f} = 5.75 \log_{10} \left(\frac{R}{y} \right) \quad \dots[\text{Eqn. (11.20)}]$$

$$\therefore \frac{7 - 6}{u_f} = 5.75 \log_{10} \left(\frac{0.18}{0.06} \right) = 2.743$$

$$\therefore u_f = 0.36 \text{ m/s (Ans.)}$$

Example 11.2. A pipe of 100 mm diameter is carrying water. If the velocities at the pipe centre and 30 mm from the pipe centre are 2.0 m/s and 1.5 m/s respectively and flow in the pipe is turbulent, calculate the wall shearing stress. (Anna University)

Solution. Given : $R = \frac{100}{2} = 50 \text{ mm} = 0.05 \text{ m}$; $u_{\max} = 2.0 \text{ m/s}$;

Velocity at $r = 30 \text{ mm}$ or $y = R - r = 50 - 30 = 20 \text{ mm}$, $u = 1.5 \text{ m/s}$.

Wall shearing stress, τ_0 :

$$\frac{u_{\max} - u}{u_f} = 5.75 \log_{10} \left(\frac{R}{y} \right) \quad \dots[\text{Eqn. (11.20)}]$$

(where, u_f = shear velocity)

Substituting the values, we get:
$$\frac{2.0 - 1.5}{u_f} = 5.75 \log_{10} \left(\frac{0.05}{0.02} \right) = 2.288$$

$$u_f = \frac{(2.0 - 1.5)}{2.288} = 0.218 \text{ m/s}$$

Using the relation:
$$u_f = \sqrt{\frac{\tau_0}{\rho}}$$

or,
$$0.218 = \sqrt{\frac{\tau_0}{1000}} \quad (\because \rho \text{ for water} = 1000 \text{ kg/m}^3)$$

or,
$$\tau_0 = 47.524 \text{ N/m}^2 \text{ (Ans.)}$$

11.6. HYDRODYNAMICALLY SMOOTH AND ROUGH BOUNDARIES

Refer to Fig. 11.4. If k is the average height of the irregularities of the surface of a boundary, then in general, the boundary is said to be **rough** if value of k is *high* and **smooth** if k is *low*. However, for proper classification of smooth and rough boundaries, besides the boundary characteristics, the flow and fluid characteristics need to be considered.

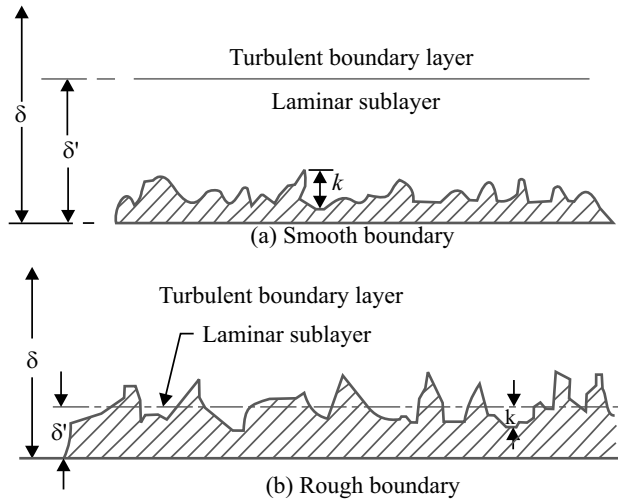


Fig. 11.4. Smooth and rough boundaries

As shown in Fig. 11.4 when the average height k of the irregularities (projecting from its surface) is *much less than* the thickness of the laminar sublayer δ' the flow outside the laminar sublayer is turbulent; the eddies of various sizes present, try to penetrate the laminar sublayer. *These eddies cannot reach the surface irregularities/projections due to greater thickness of the laminar sublayer and so the boundary acts as a **smooth boundary**.* As the Reynolds number (Re) increases the thickness of the sublayer decreases (can even become much less than k), the irregularities will then project through the laminar sublayer and eventually the laminar sublayer is destroyed completely. Subsequently, the eddies will come in contact with the surface irregularities and there will be a large amount of energy loss. This type of boundary is known as **hydrodynamically rough boundary**.

Through experiments Nikuradse found that the boundary behaves as:

- (i) Hydrodynamically smooth boundary ...when $\left(\frac{k}{\delta'}\right) < 0.25$,
- (ii) Hydrodynamically rough boundary ...when $\left(\frac{k}{\delta'}\right) > 6.0$, and
- (iii) Boundary in transition ...when $0.25 < \left(\frac{k}{\delta'}\right) < 6.0$.

In terms of roughness Reynolds number $\frac{u_f k}{\nu}$:

- (i) For smooth boundary ... $\frac{u_f k}{\nu} < 4$,
- (ii) For rough boundary ... $\frac{u_f k}{\nu} > 100$, and
- (iii) For boundary in transition stage ... $\frac{u_f k}{\nu}$ lies between 4 and 100.

Note : It may be noted that the thickness of laminar sublayer δ' is *not* a fixed quantity. It *depends upon the Reynolds number*; it is just possible that a pipe may behave as smooth at low Reynolds number and rough at high Reynolds number. *The height of roughness projection, however, remains constant.*

11.6.1. Velocity Distribution for Turbulent Flow in Smooth Pipes

The velocity distribution for turbulent flow in pipes is given by Eqn. 11.18 as :

$$u = \frac{u_f}{\lambda} \ln(y) + C$$

The peculiarity for this velocity distribution is that at the boundary, that is for $y = 0$, it gives velocity u equal to $-\infty$ (minus infinity). Thus it is only at a certain finite distance above the boundary say $y = y'$, that the velocity will be zero, hence the above equation becomes:

$$0 = \frac{u_f}{\lambda} \ln(y') + C$$

or,

$$C = -\frac{u_f}{\lambda} \ln(y')$$

Substituting the value of C in the above equation, we get:

$$u = \frac{u_f}{\lambda} \ln(y) - \frac{u_f}{\lambda} \ln(y') = \frac{u_f}{\lambda} \ln\left(\frac{y}{y'}\right)$$

Substituting the value of $\lambda = 0.4$, we have:

$$u = \frac{u_f}{0.4} \ln\left(\frac{y}{y'}\right) = 2.5 u_f \ln\left(\frac{y}{y'}\right)$$

or,

$$\frac{u}{u_f} = 2.5 \times 2.3 \log_{10}\left(\frac{y}{y'}\right)$$

or,

$$\frac{u}{u_f} = 5.75 \log_{10}\left(\frac{y}{y'}\right) \quad \dots(11.21)$$

It has been observed from Nikuradse's experimental studies of turbulent flow in smooth pipes that for turbulent flow in smooth pipes of any size the value of the parameter $\left(\frac{u_f y}{v}\right)$ for $y = \delta'$ is approximately 11.6 and for $y = y'$ it is approximately 0.108.

i.e.

$$\frac{u_f \delta'}{v} = 11.6 \quad \text{or} \quad \delta' = \frac{11.6v}{u_f} \quad \dots(11.22)$$

and,

$$\frac{u_f y'}{v} = 0.108$$

or,

$$y' = \frac{0.108v}{u_f} \left(= \frac{\delta'}{107}\right) \quad \dots(11.23)$$

Substituting the value of $y' = \frac{0.108v}{u_f}$ in eqn. 11.21, we get:

$$\begin{aligned} \frac{u}{u_f} &= 5.75 \log_{10}\left(\frac{y}{\frac{0.108v}{u_f}}\right) \\ &= 5.75 \log_{10}\left(\frac{u_f \cdot y}{0.108v}\right) = 5.75 \log_{10}\left(\frac{u_f \cdot y}{v}\right) = 5.75 \log_{10}(0.108) \end{aligned}$$

or,

$$\frac{u}{u_f} = 5.75 \log_{10}\left(\frac{u_f \cdot y}{v}\right) + 5.5 \quad \dots(11.24)$$

The eqn. (11.24) is known as **Karman-Prandtl equation for the velocity distribution near hydrodynamically smooth boundaries.**

11.6.2 Velocity Distribution for Turbulent Flow in Rough Pipes

As shown in Fig. 11.4 (b), the thickness of laminar sublayer is very small, the surface irregularities are above the laminar sublayer and hence the laminar sublayer is completely destroyed. From the experiments conducted by Nikuradse and others, using pipes artificially roughened by cemented coatings of sand grains (irregularities/projections) of diameter k , it has been found that y' is directly found proportional to k and $y' = \frac{k}{30}$.

Substituting this value of y' in eqn. (11.24), we get:

$$\begin{aligned}\frac{u}{u_f} &= 5.75 \log_{10} \left(\frac{y}{k/30} \right) = 5.75 [\log_{10} (y/k) \times 30] \\ &= 5.75 \log_{10} (y/k) + 5.75 \log_{10} 30 \\ \text{or,} \quad \frac{u}{u_f} &= 5.75 \log_{10} (y/k) + 8.5 \quad \dots(11.25)\end{aligned}$$

The eqn. (11.25) is known as **Karman-Prandtl equation for the velocity distribution near hydrodynamically rough boundaries.**

Example 11.3. The velocity of flow in a badly corroded 7.5 cm pipe is found to increase 20 percent as a pitot tube is moved from a point 1 cm from the wall to a point 2 cm from the wall. Estimate the height of roughness elements. [Roorkee University]

Solution. The velocity distribution near the rough boundaries is given by:

$$\frac{u}{u_f} = 5.75 \log_{10} \left(\frac{y}{k} \right) + 8.5 \quad \dots[\text{Eqn. (11.25)}]$$

where, k = Average height of roughness elements, and

u_f = Shear friction velocity.

Let, u = The velocity at a distance (y) of 1 cm from the pipe wall, and $1.2u$ = the velocity at a distance of 2 cm from the pipe wall (given),

$$\text{then,} \quad \frac{u}{u_f} = 5.75 \log_{10} \left(\frac{1}{k} \right) + 8.5 \quad \dots(i)$$

$$\text{and,} \quad \frac{1.2u}{u_f} = 5.75 \log_{10} \left(\frac{2}{k} \right) + 8.5 \quad \dots(ii)$$

Dividing (i) by (ii), we get:

$$\frac{1}{1.2} = \frac{5.75 \log_{10} \left(\frac{1}{k} \right) + 8.5}{5.75 \log_{10} \left(\frac{2}{k} \right) + 8.5}$$

$$\text{or,} \quad 5.75 \log_{10} \left(\frac{2}{k} \right) + 8.5 = 1.2 \left[5.75 \log_{10} \left(\frac{1}{k} \right) + 8.5 \right]$$

$$5.75 \log_{10} \left(\frac{2}{k} \right) + 8.5 = 6.9 \log_{10} \left(\frac{1}{k} \right) + 10.2$$

$$5.75 \log_{10} (2) - 5.75 \log_{10} k + 8.5 = 6.9 \log_{10} 1 - 6.9 \log_{10} k + 10.2$$

$$1.73 - 5.75 \log_{10} k + 8.5 = 0 - 6.9 \log_{10} k + 10.2$$

$$1.15 \log_{10} k = -0.03$$

$$\text{or,} \quad \log_{10} k = -\frac{0.03}{1.15} = -0.0261$$

$$\therefore k = 0.942 \text{ cm (Ans.)}$$

Example 11.4. A pipeline carrying water has surface protrusions of average height of 0.10 mm. If the shear stress developed is 8.2 N/m^2 determine whether the pipe surface acts as smooth, rough or in transition. For water take $\rho = 1000 \text{ kg/m}^3$ and kinematic viscosity $\nu = 0.0093 \text{ stokes}$.

[Bangalore University]

Solution. Average height of surface protrusions,

$$k = 0.10 \text{ mm} = 0.1 \times 10^{-3} \text{ m}$$

$$\text{Shear stress developed, } \tau_0 = 8.2 \text{ N/m}^2$$

$$\text{Density of water, } \rho = 1000 \text{ kg/m}^3$$

$$\text{Kinematic viscosity, } \nu = 0.0093 \text{ stokes} = 0.0093 \times 10^{-4} \text{ m}^2/\text{s}$$

Shear velocity is given by,

$$u_f = \sqrt{\frac{\tau_0}{\rho}}$$

$$\therefore u_f = \sqrt{\frac{8.2}{1000}} = 0.0906 \text{ m/s}$$

Roughness Reynolds number is

$$= \frac{u_f k}{\nu} = \frac{0.0906 \times (0.1 \times 10^{-3})}{0.0093 \times 10^{-4}} = 9.74$$

Since $\frac{u_f k}{\nu}$ lies between 4 and 100 the pipe surface behaves as in **transition**. (Ans.)

Example 11.5. In a pipe of diameter 100 mm, carrying water, the velocities at the pipe centre and 30 mm from the pipe centre are found to be 2.5 m/s and 2.2 m/s respectively. Find the wall shearing stress.

Solution. Radius of the pipe, $R = \frac{100}{2} = 50 \text{ mm} = 0.05 \text{ m}$

$$\text{Velocity at the centre, } u_{\max} = 2.5 \text{ m/s}$$

$$\text{Velocity at 30 mm from the centre} = 2.2 \text{ m/s.}$$

Wall shearing stress, τ_0 :

Using the equation:

$$\frac{u_{\max} - u}{u_f} = 5.75 \log_{10} \left(\frac{R}{y} \right)$$

(where, u_f = shear friction velocity)

$$\text{where, } u = 2.2 \text{ m/s at } y = (R - 30) \text{ mm} = (50 - 30) \text{ mm} = 0.02 \text{ m}$$

$$\therefore \frac{2.5 - 2.2}{u_f} = 5.75 \log_{10} \left(\frac{0.05}{0.02} \right) = 2.288$$

$$\text{or, } \frac{0.3}{u_f} = 2.88 \quad \text{or} \quad u_f = 0.1311$$

Now using the relation:

$$u_f = \sqrt{\frac{\tau_0}{\rho}}, \text{ we have:}$$

$$0.1311 = \sqrt{\frac{\tau_0}{1000}} \quad \text{or} \quad \frac{\tau_0}{1000} = (0.1311)^2 = 0.01719$$

$$\therefore \tau_0 = 17.19 \text{ N/m}^2 \text{ (Ans.)}$$

Example 11.6. A smooth pipe of 80 mm diameter and 1000 m long is carrying water at the rate of 8 litres/sec. If the kinematic viscosity of water is 0.015 stokes and the value of co-efficient of friction ' f ' is given by the relation $f = \frac{0.0791}{(Re)^{1/4}}$, where Re is Reynolds number, calculate:

- (i) Loss of head,
- (ii) Wall shearing stress,
- (iii) Centre-line velocity,
- (iv) Velocity and shear stress at 20 mm from the pipe wall, and
- (v) Thickness of laminar sublayer.

Solution. Diameter of the pipe, $D = 80 \text{ mm} = 0.08 \text{ m}$

Length of the pipe, $L = 1000 \text{ m}$

Discharge, $Q = 8 \text{ litres/sec.} = 0.008 \text{ m}^3/\text{s}$

Kinematic viscosity of water, $\nu = 0.015 \text{ stokes}$

$$= 0.015 \times 10^{-4} \text{ m}^2/\text{s}$$

$$\text{Mean velocity, } V = \frac{Q}{A} = \frac{0.008}{\frac{\pi}{4} \times (0.08)^2} = 1.59 \text{ m/s}$$

$$\therefore \text{ Reynolds number, } Re = \frac{VD}{\nu} = \frac{1.59 \times 0.08}{0.015 \times 10^{-4}} = 84800$$

$$\text{Co-efficient of friction, } f = \frac{0.0791}{(Re)^{1/4}} = \frac{0.0791}{(84800)^{1/4}} = 0.004635$$

(i) Loss of head, h_f :

$$h_f = \frac{4fLV^2}{D \times 2g} = \frac{4 \times 0.004635 \times 1000 \times 1.59^2}{0.08 \times 2 \times 9.81} = \mathbf{29.98 \text{ m (Ans.)}}$$

(ii) Wall shearing stress, τ_0 :

$$\begin{aligned} \tau_0 &= \frac{fV^2\rho}{2} && [\text{Eqn. 11.5 (a)}] \\ &= \frac{0.004635 \times 1.59^2 \times 1000}{2} = \mathbf{5.86 \text{ N/m}^2 \text{ (Ans.)}} \end{aligned}$$

(iii) Centre-line velocity, $u_{\max}$:

$$\frac{u}{u_f} = 5.75 \log_{10} \left(\frac{u_f y}{\nu} \right) + 5.5 \quad \dots(i) [\text{Eqn. (11.24)}]$$

where, u_f = shear friction velocity

$$= \sqrt{\frac{\tau_0}{\rho}} = \sqrt{\frac{5.86}{1000}} = 0.0765 \text{ m/s}$$

The velocity will be maximum where $y = \frac{D}{2} = \frac{0.08}{2} = 0.04 \text{ m}$

Hence, at $y = 0.04 \text{ m}$ $u = u_{\max}$; substituting these values in (i), we get:

$$\frac{u_{\max}}{0.0765} = 5.75 \log_{10} \left(\frac{0.0765 \times 0.04}{0.015 \times 10^{-4}} \right) + 5.5 = 24.53$$

$$\therefore u_{\max} = 0.0765 \times 24.53 = \mathbf{1.876 \text{ m/s (Ans.)}}$$

(iv) **Velocity and shear stress at 20 mm from the pipe wall:**

The shear stress (τ) at any point is given by:

$$\tau = -\frac{\partial p}{\partial x} \times \frac{r}{2} \quad \dots(ii)$$

where, r = distance from the centre of the pipe.

$\therefore$ Shear stress at pipe wall (where $r = R$),

$$\tau_0 = -\frac{\partial p}{\partial x} \times \frac{R}{2} \quad \dots(iii)$$

Dividing (ii) by (iii), we get:

$$\frac{\tau}{\tau_0} = \frac{r}{R}$$

$$\therefore \text{Shear stress, } \tau = \tau_0 \times \frac{r}{R}$$

where, $r = 40 - 20 = 20 \text{ mm} = 0.02 \text{ m}$

$$\therefore \tau = 5.86 \times \frac{0.02}{0.04} = \mathbf{2.93 \text{ N/m}^2} \text{ (Ans.)}$$

Again,
$$\frac{u}{u_f} = 5.75 \log_{10} \left(\frac{u_f \cdot y}{\nu} \right) + 5.5$$

where, $u_f = 0.0765 \text{ m/s}$ and $y = 0.2 \text{ m}$ from the pipe wall

$$\therefore \frac{u}{0.0765} = 5.75 \log_{10} \left(\frac{0.0765 \times 0.02}{0.015 \times 10^{-4}} \right) + 5.5 = 22.80$$

or $u = \mathbf{1.744 \text{ m/s (Ans.)}}$

(v) **Thickness of laminar sublayer, δ' :**

Thickness of laminar sublayer is given by,

$$\begin{aligned} \delta' &= \frac{11.6 \nu}{u_f} \\ &= \frac{11.6 \times 0.015 \times 10^{-4}}{0.765} \times 10^3 \text{ mm} = \mathbf{0.227 \text{ mm (Ans.)}} \end{aligned} \quad [\text{Eqn. (11.22)}]$$

11.7. COMMON EQUATION FOR VELOCITY DISTRIBUTION FOR BOTH SMOOTH AND ROUGH PIPES

Refer Fig. 11.5. Consider an elementary circular ring of radius r and thickness dr as shown in Fig. 11.5. The distance of the ring from the pipe wall,

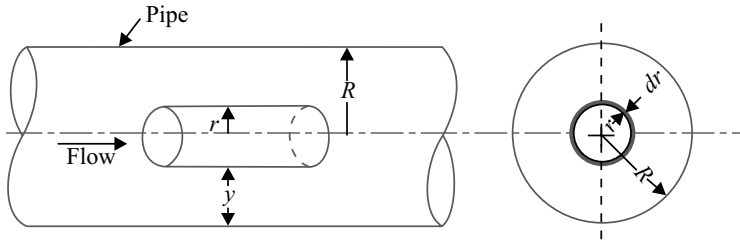


Fig. 11.5. Average velocity for turbulent flow.

$$y = R - r$$

(where, R = radius of the pipe).

The discharge through the ring is given by:

$$\begin{aligned} dQ &= \text{Area of the ring} \times \text{velocity} \\ &= 2\pi r \cdot dr \times u \end{aligned}$$

$$\therefore \quad \text{Total discharge, } Q = \int dQ = \int_0^R u \times 2\pi r \cdot dr$$

(i) For smooth pipes:

In the case of smooth pipes the velocity distribution is given by [Eqn. (11.24)] as:

$$\begin{aligned} \frac{u}{u_f} &= 5.75 \log_{10} \left(\frac{u_f \cdot y}{\nu} \right) + 5.5 \\ M &= \left[5.75 \log_{10} \frac{u_f (R-r)}{\nu} + 5.5 \right] \times u_f \end{aligned}$$

Substituting the value of u in eqn. (11.26), we get:

$$Q = \int_0^R \left[5.75 \log_{10} \frac{u_f (R-r)}{\nu} + 5.5 \right] u_f \times 2\pi r \cdot dr$$

$$\therefore \quad \text{Average velocity, } \bar{U} = \frac{Q}{A} = \frac{Q}{\pi R^2}$$

$$\bar{U} = \frac{1}{\pi R^2} \int_0^R \left[5.75 \log_{10} \frac{u_f (R-r)}{\nu} + 5.5 \right] u_f \times 2\pi r \cdot dr$$

After integration and simplification, we have:

$$\frac{\bar{U}}{u_f} = 5.75 \log_{10} \frac{u_f R}{\nu} + 1.75 \quad \dots(11.27)$$

(ii) For rough pipes:

In case of rough pipes, the velocity at any point in the turbulent flow is given by eqn. (11.25) as:

$$\frac{u}{u_f} = 5.75 \log_{10} (y/k) + 8.5 = 5.75 \log_{10} \left(\frac{R-r}{k} \right) + 8.5 \quad (\because y = R-r)$$

or,

$$u = u_f \left[5.75 \log_{10} \left(\frac{R-r}{k} \right) + 8.5 \right]$$

Substituting the value of u in eqn. (11.26), we get:

$$Q = \int_0^R u_f \left[5.75 \log_{10} \left(\frac{R-r}{k} \right) + 8.5 \right] 2\pi r \cdot dr$$

$$\therefore \quad \text{Average velocity, } \bar{U} = \frac{Q}{\pi R^2} = \frac{1}{\pi R^2} \int_0^R u_f \left[5.75 \log_{10} \left(\frac{R-r}{k} \right) + 8.5 \right] 2\pi r \cdot dr$$

After integration and simplification, we have:

$$\frac{\bar{U}}{u_f} = 5.75 \log_{10} \left(\frac{R}{k} \right) + 4.75 \quad \dots(11.28)$$

From eqns. (11.24) and (11.27) by subtraction, we have:

$$\frac{u}{u_f} - \frac{\bar{U}}{u_f} = \left[5.75 \log_{10} \left(\frac{u_f \cdot y}{\nu} \right) + 5.5 \right] - \left[5.75 \log_{10} \frac{u_f R}{\nu} + 1.75 \right]$$

$$\begin{aligned}\text{or, } \frac{u - \bar{U}}{u_f} &= 3.75 - 5.75 \log_{10} \left(\frac{u_f R}{v} \times \frac{v}{u_f y} \right) \\ &= 3.75 - 5.75 \log_{10} \left(\frac{R}{y} \right)\end{aligned}$$

$$\text{or, } \frac{u - \bar{U}}{u_f} = 5.75 \log_{10} \left(\frac{y}{R} \right) + 3.75 \quad \dots(i)$$

Similarly, from eqns. (11·25) and (11·28), we get:

$$\begin{aligned}\frac{u}{u_f} - \frac{\bar{U}}{u_f} &= \left[5.75 \log_{10} \left(\frac{y}{k} \right) + 8.5 \right] - \left[5.75 \log_{10} \left(\frac{R}{k} \right) + 4.75 \right] \\ \frac{u - \bar{U}}{u_f} &= 5.75 \log_{10} \left(\frac{y}{R} \right) + 3.75 \quad \dots(ii)\end{aligned}$$

As eqns. (i) and (ii) are identical, the **velocity distribution in both types of pipes is the same.**

$$\therefore \frac{u - \bar{U}}{u_f} = 5.75 \log_{10} \left(\frac{y}{R} \right) + 3.75 \quad \dots(11\cdot29)$$

The common equation holds good for both types of pipes due to the reason that the *velocity distribution for the turbulent core is identical in both cases.*

Example 11.7. Find the distance from the pipe wall at which the local velocity is equal to the average velocity for turbulent flow in pipes.

Solution. Local velocity at a point = Average velocity ...(Given)

$$\text{i.e. } u = \bar{U}$$

Using the relation:

$$\frac{u - \bar{U}}{u_f} = 5.75 \log_{10} \left(\frac{y}{R} \right) + 3.75 \quad [\text{Eqn. (11·29)}]$$

$$\text{or, } 5.75 \log_{10} \left(\frac{y}{R} \right) + 3.75 = 0 \quad (\because u = \bar{U})$$

$$\text{or, } 5.75 \log_{10} \left(\frac{y}{R} \right) = -3.75$$

$$\text{or, } \log_{10} \left(\frac{y}{R} \right) = -\frac{3.75}{5.75} = -0.652$$

$$\text{or, } \frac{y}{R} = 0.223 \quad \text{or } y = 0.223 R \text{ (Ans.)}$$

11.8. VELOCITY DISTRIBUTION FOR TURBULENT FLOW IN SMOOTH PIPES BY POWER LAW

The eqns. (11·20), (11·24) and (11·25) of velocity distribution for turbulent flow are inconvenient to use, being *logarithmic* in nature. Nikuradse, through experiments, established the following velocity distribution law (exponential form) for smooth pipes:

$$\frac{u}{u_{\max}} = \left(\frac{y}{R} \right)^{\frac{1}{n}} \quad \dots(11\cdot30)$$

where, exponent $\frac{1}{n}$ depends on Reynolds number (Re) and it *decreases* with the *increasing* Re .

For:

$$Re = 400, \frac{1}{n} = \frac{1}{6}$$

$$Re = 1.1 \times 10^5, \frac{1}{n} = \frac{1}{7}$$

$$Re \geq 2 \times 10^6, \frac{1}{n} = \frac{1}{10}$$

Therefore, for $\frac{1}{n} = \frac{1}{7}$, the velocity distribution law becomes:

$$\frac{u}{u_{\max}} = \left(\frac{y}{R} \right)^{1/7} \quad \dots(11-31)$$

Eqn. (11-31) is known as $\frac{1}{7}$ **th power law of velocity distribution for smooth pipes**.

11.9. RESISTANCE TO FLOW OF FLUID IN SMOOTH AND ROUGH PIPES

When a fluid flows through a pipe frictional resistance is offered to the motion of the fluid and the loss of head due to friction is expressed by Darcy-Weisbach equation, $h_f = \frac{4fLV^2}{D \times 2g}$. But the loss of head can be predicted correctly only if the friction co-efficient can be evaluated accurately. It can be shown by dimensional analysis that the friction co-efficient f depends upon the Reynolds number $\left(\frac{\rho VD}{\mu} \right)$ and the ratio k/D .

Thus,

$$f = \phi \left[\left(\frac{\rho VD}{\mu} \right), \frac{k}{D} \right] \quad \dots(11-32)$$

where, D = Diameter of the pipe,
 ρ = Density of the fluid,
 μ = Dynamic viscosity of the fluid, and
 k = Average height of pipe wall roughness protrusions.

(The term $\frac{k}{D}$ is commonly known as *relative roughness*).

The eqn. (11-32) is a *general equation which is applicable to laminar as well as turbulent flows in pipes*.

(a) Variation of friction co-efficient ' f ' for "laminar flow":

As derived in previous chapter the co-efficient of friction ' f ' for laminar flow in pipes is given by:

$$f = \frac{16}{Re} \quad \dots(11-33)$$

The eqn. (11-33) shows that for laminar flow the friction coefficient f varies inversely with Re and it is independent of $\left(\frac{k}{D} \right)$ ratio.

(b) Variation of ' f ' for "turbulent flow":

For the fully developed turbulent flow the friction coefficient ' f ' is a function of Re or k/D ratio or both, depending on whether the boundary is hydrodynamically smooth or rough or it is in transition.

(i) Variation of friction co-efficient ' f ' "for smooth pipes":

The coefficient of friction ' f ' for turbulent flow in smooth pipes is a function of Reynolds number (Re) only, and is independent of relative roughness k/D . The value of ' f ' for smooth pipes for Re varying from 4000 to 1×10^5 is given by the following empirical relation (*developed by Blasius*):

$$f = \frac{0.0791}{(Re)^{1/4}} \quad \dots(11.34)$$

The value of ' f ' for $Re > 10^5$ is obtained from eqn. (11.27),

$$\frac{\bar{U}}{u_f} = 5.75 \log_{10} \frac{u_f R}{\nu} + 1.75 \quad \dots(\text{Eqn. 11.27})$$

Also, $f = \frac{2\tau_0}{\rho V^2} \quad \dots[\text{Eqn. 11.5 (b)}]$

(where, V = average velocity)

$$\therefore f = \frac{2\tau_0}{\rho \bar{U}^2} = \frac{2}{\bar{U}^2} \times u_f^2 \quad \left[\because \sqrt{\frac{\tau_0}{\rho}} = u_f \right]$$

or, $u_f^2 = \frac{f \bar{U}^2}{2}$

or, $u_f = \bar{U} \sqrt{\frac{f}{2}}$

Substituting the value of u_f in eqn. (11.27), we get:

$$\frac{\bar{U}}{\bar{U} \frac{\sqrt{f}}{2}} = 5.75 \log_{10} \left(\frac{\bar{U}}{\nu} \sqrt{\frac{f}{2}} \right) R + 1.75$$

or, $\frac{1}{\sqrt{f/2}} = 5.75 \log_{10} \left(\frac{\bar{U} R}{\nu} \sqrt{f/2} \right) + 1.75$

Substituting $R = D/2$ and simplifying, we get:

$$\frac{1}{\sqrt{4f}} = 2.03 \log_{10} \left(\frac{\bar{U} D}{\nu} \sqrt{4f} \right) - 0.91$$

But, $\frac{\bar{U} D}{\nu} = Re$

$$\frac{1}{\sqrt{4f}} = 2.03 \log_{10} (Re \sqrt{4f}) - 0.91 \quad \dots(11.35)$$

Eqn. (11.35) is valid upto $Re = 4 \times 10^6$

Karman-Prandtl resistance equation for turbulent flow in smooth pipes is given by:

$$\frac{1}{\sqrt{4f}} = 2.0 \log_{10} (Re \sqrt{4f}) - 0.8 \quad \dots(11.36)$$

From Nikuradse's experimental measurements eqn. (11.36) has been found to be valid from $Re = 5 \times 10^4$ to Re as high as 4×10^7 . The eqn. (11.36) can be solved by hit and trial method. However the following empirical relationship given by Nikuradse for ' f ' can be used directly:

$$f = 0.0008 + \frac{0.05525}{(Re)^{0.237}} \quad \dots(11.37)$$

(ii) Variation of friction co-efficient ' f ' for "rough pipes":

For turbulent flow in rough pipes the friction co-efficient ' f ' depends only on relative roughness (k/D) and is independent of Reynolds number (Re). An expression for ' f ' is obtained as follows:

For turbulent flow in rough pipes the mean velocity $\bar{U}$ of flow has been expressed by the eqn. 11.28,

$$\frac{\bar{U}}{u_f} = 5.75 \log_{10} \left(\frac{R}{k} \right) + 4.75 \quad \dots(\text{Eqn. 11.28})$$

But, $u_f = \bar{U} \sqrt{\frac{f}{2}}$

$$\therefore \frac{\bar{U}}{\bar{U} \sqrt{f/2}} = 5.75 \log_{10} (R/k) + 4.75$$

On simplification, we get:

$$\frac{1}{\sqrt{4f}} = 2.03 \log_{10} (R/k) + 1.68 \quad \dots(11.38)$$

The experimental results obtained by Nikuradse follow closely the trend of the following equation, (instead of eqn. 11.38):

$$\frac{1}{\sqrt{4f}} = 2.0 \log_{10} (R/k) + 1.74 \quad \dots(11.39)$$

(iii) **Value of friction factor for “commercial pipes”:**

Colebrook and White developed an empirical equation of the following form to predict the friction factor for commercial pipes,

$$\frac{1}{\sqrt{f_1}} - 2.0 \log_{10} \left(\frac{R}{k} \right) = 1.74 - 2.0 \log_{10} \left[1 + 18.7 \frac{(R/k)}{Re \sqrt{f_1}} \right] \quad \dots(11.40)$$

where, f_1 (friction factor) = $4f$ (friction coefficient).

Example 11.8. In a rough pipe of diameter 0.6 m and length 4500 m water is flowing at the rate of $0.6 \text{ m}^3/\text{s}$. If the average height of roughness is 0.48 mm find the power required to maintain this flow.

Solution. Diameter of the pipe, $D = 0.6 \text{ m}$

$$\therefore \text{Radius, } R = \frac{0.6}{2} = 0.3 \text{ m}$$

Length of the pipe, $L = 4500 \text{ m}$

Discharge, $Q = 0.6 \text{ m}^3/\text{s}$

Average height of roughness, $k = 0.48 \text{ mm} = 0.48 \times 10^{-3} \text{ m}$

Power required to maintain the flow, P:

$$\text{Power required, } P = wQh_f \quad \dots(i)$$

where, $h_f = \frac{4fLV^2}{D \times 2g}$

where, f = Co-efficient of friction,

V = Average velocity of flow, and

w = Weight density of water ($= 9.81 \text{ kN/m}^3$).

Let us first calculate the value of ' f '.

For a rough pipe, the value of ' f ' is given by:

$$\begin{aligned} \frac{1}{\sqrt{4f}} &= 2.0 \log_{10} \left(\frac{R}{k} \right) + 1.74 \quad \dots[\text{Eqn. (11.39)}] \\ &= 2.0 \log_{10} \left(\frac{0.3}{0.48 \times 10^{-3}} \right) + 1.74 = 7.331 \end{aligned}$$

$$\text{or, } \sqrt{4f} = \frac{1}{7.331} = 0.1364$$

$$\text{or, } f = 0.00465$$

$$\text{Also, average velocity, } V = \frac{Q}{\left(\frac{\pi}{4} \times D^2\right)} = \frac{0.6}{\frac{\pi}{4} \times 0.6^2} = 2.122 \text{ m/s}$$

$$\therefore \text{ Head lost in friction, } h_f = \frac{4fLV^2}{D \times 2g} = \frac{4 \times 0.00465 \times 4500 \times (2.122)^2}{0.6 \times 2 \times 9.81} = 32 \text{ m}$$

Substituting the values in eqn. (i), we get:

$$P = 9.81 \times 0.6 \times 32 = \mathbf{188.35 \text{ kW (Ans.)}}$$

Example 11.9. The friction for turbulent flow through rough pipes can be determined by Kaman-Prandtl equation

$$\frac{1}{\sqrt{f}} = 2 \log_{10} (R_0 / k) + 1.74$$

where, f = friction factor, R_0 = pipe radius and k = average roughness.

Two reservoirs with a surface level difference of 20 metres are to be connected by 1 metre diameter pipe 6 km long.

- (i) What will be the discharge when a cast-iron pipe of roughness $k = 0.3 \text{ mm}$ is used ?
- (ii) What will be the percentage increase in discharge if the cast-iron pipe is replaced by a steel pipe of roughness $k = 0.1 \text{ mm}$?

Neglect all local losses.

[Delhi University]

Solution. Difference in levels, $h = 20 \text{ m}$

Diameter of the pipe, $D = 1 \text{ m}$

$\therefore$ Radius, $R_0 = 0.5 \text{ m}$

Length of the pipe, $L = 6 \text{ km} = 6 \times 1000 = 6000 \text{ m}$

Roughness of C.I. pipe, $k = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$

Roughness of steel pipe, $k = 0.1 \text{ mm} = 0.1 \times 10^{-3} \text{ m}$

(i) **Discharge with C.I. pipe, $Q_{C.I.}$:**

$$\frac{1}{\sqrt{f}} = 2 \log_{10} (R_0 / k) + 1.74 \quad \dots(\text{Given}) \quad \dots(i)$$

$$= 2 \log_{10} [0.5 / (0.3 \times 10^{-3})] + 1.74 = 8.1837$$

$$f = \left(\frac{1}{8.1837} \right)^2 = 0.0149$$

Head loss due to friction,

$$h_f = \frac{fLV^2}{D \times 2g}$$

[where, f = friction factor (= $4 \times$ friction coefficient)]

$$20 = \frac{0.0149 \times 6000 \times V^2}{1 \times 2 \times 9.81} \quad (\text{neglecting all local losses})$$

$$\text{or, } V = 2.095 \text{ m/s}$$

$\therefore$ Discharge through C.I. pipe,

$$Q_{C.I.} = (\pi/4) \times 1^2 \times 2.095 = \mathbf{1.645 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

(ii) Percentage increase in discharge :

For steel pipe : $k = 0.1 \times 10^{-3} \text{ m}$, $R_0 = 0.5 \text{ m}$

Substituting the values in eqn. (i), we get:

$$\frac{1}{\sqrt{f}} = 2 \log_{10} [0.5 / (0.1 \times 10^{-3})] + 1.74 = 9.1379 \quad \text{or} \quad f = 0.0119$$

Head lost due to friction,

$$20 = \frac{fLV^2}{D \times 2g}$$

$$\text{or,} \quad 20 = \frac{0.0119 \times 6000 \times V^2}{1 \times 2 \times 9.81}$$

$$\text{or,} \quad V = 2.344 \text{ m/s}$$

$\therefore$ Discharge through steel pipe,

$$Q_s = (\pi/4) \times 1^2 \times 2.344 = 1.841 \text{ m}^3/\text{s}$$

$\therefore$ % age increase in discharge

$$= \frac{Q_s - Q_{C.I.}}{Q_{C.I.}} \times 100 = \frac{1.841 - 1.645}{1.645} \times 100 = \mathbf{11.91\% \text{ (Ans.)}}$$

Example 11.10. A smooth pipeline of 100 mm diameter carries 2.27 m³ per minute of water at 20°C with kinematic viscosity of 0.009 stokes, calculate:

(i) Friction factor:

(ii) Maximum velocity;

(iii) Shear stress at the boundary.

(UPTU)

$$\text{Solution. Given: } R = \frac{100}{2} = 50 \text{ mm} = 0.05 \text{ m}; \quad Q = \frac{2.27}{60} = 0.0378 \text{ m}^3/\text{s};$$

$$\nu = 0.0098 \text{ stokes} = 0.0098 \text{ cm}^2/\text{s} = 0.0098 \times 10^{-4} \text{ m}^2/\text{s}$$

$$\text{Average velocity, } \bar{U} = \frac{Q}{\text{Area}} = \frac{0.0378}{\pi R^2} = \frac{0.0378}{\pi \times 0.05^2} = 4.81 \text{ m/s}$$

$$\therefore \text{ Reynolds number, } Re = \frac{\bar{U}D}{\nu} = \frac{4.81 \times 0.1}{0.0098 \times 10^{-4}} = 4.91 \times 10^5$$

The flow is *turbulent* as $Re > 10^5$. Hence for smooth pipe, the coefficient of friction ' f ' is obtained from the equation,

$$\frac{1}{\sqrt{4f}} = 2.0 \log_{10} (Re \sqrt{4f}) - 0.8 \quad \dots [\text{Eqn. (11.36)}]$$

$$\begin{aligned} \text{or,} \quad \frac{1}{\sqrt{4f}} &= 2.0 \log_{10} (4.9 \times 10^5 \times \sqrt{4f}) - 0.8 \\ &= 2.0 [\log_{10} (4.91 \times 10^5) + \log_{10} \sqrt{4f}] - 0.8 \\ &= 2.0 [5.6911 + \log_{10} \sqrt{4f}] - 0.8 \\ &= 2 \times 5.6911 + 2 \log_{10} \sqrt{4f} - 0.8 \\ &= 11.382 + \log_{10} (\sqrt{4f})^2 - 0.8 = 10.582 + \log_{10} (4f) \end{aligned}$$

$$\text{or,} \quad \frac{1}{\sqrt{4f}} - \log_{10} (4f) = 10.582 \quad \dots(i)$$

(i) **Friction factor, f_1 :**

Friction factor $f_1 = 4 \times \text{coefficient of friction} = 4f$

Substituting the value of ' $4f$ ' in (i), we get:

$$\frac{1}{\sqrt{f_1}} - \log_{10}(f_1) = 10.582 \quad \dots(ii)$$

Solving by *hit and trial method*, we get $f_1 = \mathbf{0.013}$ (Ans.)

(ii) **Maximum velocity, $u_{\max}$:**

$$f = \frac{f_1}{4} = \frac{0.013}{4} = 0.00325$$

Also,

$$u_f = \frac{\bar{U} \sqrt{f}}{2} = 4.81 \times \frac{\sqrt{0.00325}}{2} = 0.194 \text{ m/s}$$

For smooth pipe, the velocity at any point is given by:

$$u = u_f \left[5.75 \log_{10} \left(\frac{u_f \times y}{\nu} \right) + 5.5 \right] \quad \dots[\text{Eqn. (11.24)}]$$

The velocity will be maximum at the centre of the pipe where $y = R = 0.05 \text{ m}$ i.e. radius of the pipe. Hence, the above equation becomes:

$$\begin{aligned} u_{\max} &= u_f \left[5.75 \log_{10} \left(\frac{u_f \times R}{\nu} \right) + 5.5 \right] \\ &= 0.194 \left[5.75 \log_{10} \left(\frac{0.194 \times 0.05}{0.0098 \times 10^{-4}} \right) + 5.5 \right] = \mathbf{5.524 \text{ m/s (Ans.)}} \end{aligned}$$

(iii) **Shear stress at the boundary, τ_0 :**

$$\begin{aligned} u_f &= \frac{\sqrt{\tau_0}}{\rho} \quad \text{or} \quad (u_f)^2 = \frac{\tau_0}{\rho} \\ \tau_0 &= \rho (u_f)^2 = 1000 \times (0.194)^2 = \mathbf{37.64 \text{ N/m}^2 \text{ (Ans.)}} \end{aligned}$$

Example 11.11. In a pipe of diameter 300 mm the centre-line velocity and the velocity at a point 100 mm from the centre, as measured by pitot tube, are 2.4 m/s and 2.0 m/s respectively. Assuming the flow in the pipe to be turbulent, find:

- (i) Discharge through the pipe, (ii) Co-efficient of friction, and
(iii) Height of roughness projections.

Solution. Diameter of the pipe, $D = 300 \text{ mm} = 0.3 \text{ m}$

$\therefore$ Radius, $R = \frac{0.3}{2} = 0.15 \text{ m}$

Centre-line velocity, $u_{\max} = 2.4 \text{ m/s}$

Velocity at $r = 100 \text{ mm}$ or $y = 150 - 100 = 50 \text{ mm}$, $u = 2.0 \text{ m/s}$

(i) **Discharge through the pipe, Q :**

$$\frac{u_{\max} - u}{u_f} = 5.75 \log_{10} \left(\frac{R}{y} \right) \quad \dots[\text{Eqn. (11.20)}]$$

(where u_f is shear velocity)

Substituting the values, we get:

$$\frac{2.4 - 2.0}{u_f} = 5.75 \log_{10} \left(\frac{0.15}{0.05} \right) = 2.743$$

or, $u_f = 0.146 \text{ m/s}$

Using the equation :

$$\frac{u - \bar{U}}{u_f} = 5.75 \log_{10} \left(\frac{y}{R} \right) + 3.75, \text{ we have:} \quad \dots[\text{Eqn. (11.29)}]$$

At,

$$y = R, \quad u = u_{\max}$$

$$\therefore \frac{u_{\max} - \bar{U}}{u_f} = 5.75 \log_{10} \left(\frac{R}{R} \right) + 3.75 = 5.75 \times 0 + 3.75 = 3.75$$

But,

$$u_{\max} = 2.4 \text{ m/s} \quad \text{and} \quad u_f = 0.146 \text{ m/s}$$

$$\therefore \frac{2.4 - \bar{U}}{0.146} = 3.75$$

or,

$$\bar{U} = 2.4 - 0.146 \times 3.75 = 1.85 \text{ m/s}$$

$\therefore$

$$\begin{aligned} \text{Discharge, } Q &= \text{Area} \times \text{average velocity} \\ &= (\pi/4) \times 0.3^2 \times 1.85 = \mathbf{0.1307 \text{ m}^3/\text{s}} \text{ (Ans.)} \end{aligned}$$

(ii) Co-efficient of friction, ' f ' :

$$\text{We have,} \quad u_f = \bar{U} \sqrt{\frac{f}{2}}$$

$$\text{or,} \quad 0.146 = 1.85 \sqrt{\frac{f}{2}}$$

$$\text{or,} \quad \sqrt{\frac{f}{2}} = \frac{0.146}{1.85} = 0.0789$$

$$\therefore f = \mathbf{0.0124 \text{ (Ans.)}}$$

(iii) Height of roughness projections, k :

$$\text{We know that} \quad \frac{1}{\sqrt{4f}} = 2.0 \log_{10} (R/k) + 1.74 \quad \dots[\text{Eqn. (11.39)}]$$

$$\frac{1}{\sqrt{4 \times 0.0124}} = 2.0 \log_{10} \left(\frac{0.15}{k} \right) + 1.74$$

$$4.49 = 2.0 \log_{10} \left(\frac{0.15}{k} \right) + 1.74$$

$$\text{or,} \quad \log_{10} \left(\frac{0.15}{k} \right) = \frac{4.49 - 1.74}{2} = 1.375$$

$$\text{or,} \quad \frac{0.15}{k} = 23.71$$

$$\text{or,} \quad k = 0.00633 \text{ m or } \mathbf{6.33 \text{ mm}} \text{ (Ans.)}$$

Example 11.12. A rough plastic pipe 500 mm in diameter and 300 m in length carrying water with a velocity of 3 m/s, has an absolute roughness of 0.25 mm and a kinematic viscosity of 0.9 centistokes,

(i) Is the flow turbulent or laminar? (ii) What is the head lost in friction?

$$\text{For laminar flow,} \quad f = \frac{64}{Re}$$

$$\text{For turbulent flow,} \quad \frac{1}{\sqrt{f}} = 2 \log_{10} \frac{R}{k} + 1.74 \quad [\text{UPSC Civil Services (IAS) Exams.}]$$

(f = friction factor)

Solution. Diameter of the pipe, $D = 500 \text{ mm} = 0.5 \text{ m}$

Length of the pipe, $L = 300 \text{ m}$

Velocity of water, $V = 3 \text{ m/s}$

Absolute roughness, $k = 0.25 \text{ mm} = 0.25 \times 10^{-3} \text{ m}$

Kinematic viscosity, $\nu = 0.9 \text{ centistokes} = 0.9 \times 10^{-6} \text{ m}^2/\text{s}$.

(i) **Is the flow turbulent or laminar?**

$$\text{Reynolds number of flow, } Re = \frac{VD}{\nu} = \frac{3 \times 0.5}{0.9 \times 10^{-6}} = 1.667 \times 10^6$$

Since $Re > 2000$, therefore, the flow is **turbulent**. (Ans.)

(ii) **Head lost in friction:**

Friction factor (f) for turbulent flow in rough pipes is given by :

$$\frac{1}{\sqrt{f}} = 2 \log_{10} \frac{R}{k} + 1.74 \quad \dots(\text{Given})$$

$$= 2 \log_{10} \frac{0.25}{0.25 \times 10^{-3}} + 1.74 = 7.74$$

$$\therefore f = \left(\frac{1}{7.74} \right)^2 = 0.0167$$

Head lost in friction,

$$h_f = \frac{fLV^2}{D \times 2g} = \frac{0.0167 \times 300 \times 3^2}{0.5 \times 2 \times 9.81} = 4.59 \text{ m (Ans.)}$$

Example. 11.13. In a smooth pipe of diameter 0.5 m and length 1000 m water is flowing at the rate of $0.05 \text{ m}^3/\text{s}$. Assuming the kinematic viscosity of water as 0.02 stokes, find:

- (i) Head lost due to friction,
- (ii) Wall shear stress,
- (iii) Centre-line velocity, and
- (iv) Thickness of laminar sublayer.

Solution. Diameter of smooth pipe, $D = 0.5 \text{ m}$

$$\therefore \text{Radius, } R = \frac{0.5}{2} = 0.25 \text{ m}$$

Length of the pipe, $L = 1000 \text{ m}$

Discharge through the pipe, $Q = 0.05 \text{ m}^3/\text{s}$

Kinematic viscosity of water, $\nu = 0.02 \times 10^{-4} \text{ m}^2/\text{s}$

$$\text{Average velocity, } \bar{U} = \frac{Q}{\text{Area}} = \frac{0.05}{\frac{\pi}{4} \times (0.5)^2} = 0.2546 \text{ m/s}$$

$$\therefore \text{Reynolds number, } Re = \frac{V \times D}{\nu} = \frac{\bar{U} \times D}{\nu} = \frac{0.2546 \times 0.5}{0.02 \times 10^{-4}} = 6.365 \times 10^4$$

Since $Re > 4000$, the flow is *turbulent*.

$$\begin{aligned} \text{We know that, } f &= \frac{0.0791}{(Re)^{1/4}} \quad \dots[\text{Eqn. (11.34)}] \\ &= \frac{0.0791}{(6.365 \times 10^4)^{1/4}} = 0.00498 \end{aligned}$$

(i) Head lost due to friction, h_f :

$$h_f = \frac{4fL\bar{V}^2}{D \times 2g} = \frac{4fL\bar{U}^2}{D \times 2g} = \frac{4 \times 0.00498 \times 1000 \times (0.2546)^2}{0.5 \times 2 \times 9.81}$$

$$= \mathbf{0.1316 \text{ m (Ans.)}}$$

(ii) Wall shear stress, τ_0 :

We know that,

$$\tau_0 = \frac{f\rho V^2}{2} = \frac{f\rho\bar{U}^2}{2} \quad \dots[\text{Eqn. 11.5 (a)}]$$

$$= \frac{0.00498 \times 1000 \times (10.2546)^2}{2} = \mathbf{0.1614 \text{ N/m}^2 \text{ (Ans.)}}$$

(iii) Centre-line velocity, $u_{\max}$:

We know that,

$$\frac{u}{u_f} = 5.75 \log_{10} \left(\frac{u_f y}{\nu} \right) + 5.5 \quad \dots[\text{Eqn. (11.24)}]$$

At,

$$y = R, \quad u = u_{\max}$$

$$\therefore \frac{u_{\max}}{u_f} = 5.75 \log_{10} \left(\frac{u_f \cdot R}{\nu} \right) + 5.5$$

where,

$$u_f = \sqrt{\frac{\tau_0}{\rho}} = \sqrt{\frac{0.1614}{1000}} = 0.0127 \text{ m/s}$$

Substituting the values in the above eqn., we get:

$$\frac{u_{\max}}{0.0127} = 5.75 \log_{10} \left(\frac{0.0127 \times 0.25}{0.02 \times 10^{-4}} \right) + 5.5 = 23.9$$

or,

$$u_{\max} = \mathbf{0.303 \text{ m/s (Ans.)}}$$

(iv) Thickness of laminar sublayer, δ' :

We know that,

$$\delta' = \frac{11.6 \times \nu}{u_f} \quad \dots[\text{Eqn (11.22)}]$$

$$= \frac{11.6 \times 0.02 \times 10^{-4}}{0.0127} = 0.001826 \text{ m} = \mathbf{1.826 \text{ mm (Ans.)}}$$

Example 11.14. Water is flowing in a rough pipe of 0.5 m diameter and 800 m length at the rate of $0.5 \text{ m}^3/\text{s}$. Assuming the average height of roughness as 0.15 mm, determine:

(i) Co-efficient of friction,

(ii) Wall shear stress, and

(iii) Centre-line velocity and velocity at a distance of 200 mm from the pipe wall.

Solution. Diameter of the pipe, $D = 0.5 \text{ m}$

$$\text{Radius, } R = \frac{0.5}{2} = 0.25 \text{ m}$$

Length of the pipe, $L = 800 \text{ m}$

Discharge, $Q = 0.5 \text{ m}^3/\text{s}$

Average height of roughness, $k = 0.15 \text{ mm} = 0.15 \times 10^{-3} \text{ m}$

(i) Co-efficient of friction, f :

Using the equation:

$$\frac{1}{\sqrt{4f}} = 2.0 \log_{10} \left(\frac{R}{k} \right) + 1.74 \quad \dots[\text{Eqn. (11.39)}]$$

or,

$$\frac{1}{\sqrt{4f}} = 2.0 \log_{10} \left(\frac{0.25}{0.015 \times 10^{-3}} \right) + 1.74 = 10.184$$

$$\sqrt{4f} = \frac{1}{10.184} = 0.0982$$

$$f = \mathbf{0.00241 \text{ (Ans.)}}$$

(ii) Wall shear stress, τ_0 :

$$\tau_0 = \frac{f \rho V^2}{2} \quad \dots [\text{Eqn. 11.5 (a)}]$$

where,

$$V = \frac{\text{Discharge}}{\text{Area}} = \frac{0.5}{\frac{\pi}{4} \times 0.5^2} = 2.546 \text{ m/s}$$

$$\therefore \tau_0 = \frac{0.00241 \times 1000 \times (2.546)^2}{2} = \mathbf{7.81 \text{ N/m}^2 \text{ (Ans.)}}$$

(iii) Centre-line velocity, $u_{\max}$:

For rough pipe:

$$\frac{u}{u_f} = 5.75 \log_{10} \left(\frac{y}{k} \right) + 8.5 \quad \dots [\text{Eqn. (11.25)}]$$

At $y = R$, $u = u_{\max}$

$$\therefore \frac{u_{\max}}{u_f} = 5.75 \log_{10} \left(\frac{R}{k} \right) + 8.5 \quad \dots (i)$$

where,

$$u_f = \sqrt{\frac{\tau_0}{\rho}} = \sqrt{\frac{7.81}{1000}} = 0.0884 \text{ m/s}$$

Substituting the values of u_f , R and k in eqn. (i), we get:

$$\frac{u}{0.0884} = 5.75 \log_{10} \left(\frac{0.2}{0.015 \times 10^{-3}} \right) + 8.5 = 32.77$$

$$\therefore u_{\max} = 0.0884 \times 32.77 = \mathbf{2.897 \text{ m/s (Ans.)}}$$

Velocity at a distance of 200 mm from the pipe wall, u :

For a rough pipe:

$$\frac{u}{u_f} = 5.75 \log_{10} \left(\frac{y}{k} \right) + 8.5$$

where,

$$u_f = 0.0884 \text{ m/s}, y = 200 \text{ mm} = 0.2 \text{ m} \text{ and } k = 0.015 \times 10^{-3} \text{ m}$$

$$\therefore \frac{u}{0.0884} = 5.75 \log_{10} \left(\frac{0.2}{0.015 \times 10^{-3}} \right) + 8.5 = 32.22$$

or,

$$u = 0.0884 \times 32.22 = \mathbf{2.848 \text{ m/s (Ans.)}}$$

Example 11.15. *Hydrodynamically smooth pipe carries water at the rate of 300 l/s at 20° C ($\rho = 1000 \text{ kg/m}^3$, $\nu = 10^{-6} \text{ m}^2/\text{s}$) with a head loss of 3 m in 100 m length of pipe. Determine the pipe diameter. Use $f = 0.0032 + \frac{0.221}{(Re)^{0.237}}$ equation for f , where $h_f = \frac{f \times L \times V^2}{D \times 2g}$ and $Re = \frac{\rho V D}{\mu}$.*

(Anna University)

Solution. Given: $Q = 300 \text{ l/s} = 0.3 \text{ m}^3/\text{s}$; $\rho = 1000 \text{ kg/m}^3$; $\nu = 10^{-6} \text{ m}^2/\text{s}$; $h_f = 3 \text{ m}$;

$L = 100 \text{ m}$; Friction factor, $f = 0.0032 + \frac{0.221}{(Re)^{0.237}}$.

Diameter of the pipe, D:

$$h_f = \frac{f L V^2}{D \times 2g}$$

$$\text{or,} \quad 3 = \frac{f \times 100 \times V^2}{D \times 2 \times 9.81} \quad \text{or} \quad f = \frac{0.5886D}{V^2}$$

$$\text{Now,} \quad Q = A \times V$$

$$\text{or,} \quad 0.3 = \frac{\pi}{4} \times D^2 \times V \quad \text{or} \quad D^2 V = 0.382$$

$$\therefore \quad V = \frac{0.382}{D^2}$$

$$\text{Also,} \quad f = 0.0032 + \frac{0.221}{(Re)^{0.237}} \quad \dots(\text{Given})$$

$$\text{or,} \quad \frac{0.5886D}{V^2} = 0.0032 + \frac{0.221}{(V \times D \times 10^6)^{0.237}}$$

$$\text{or,} \quad \frac{0.5886D}{(0.382/D^2)^2} = 0.0032 + \frac{0.221}{\left[\frac{0.382}{D^2} \times D \times 10^6\right]^{0.237}}$$

$$\text{or,} \quad \frac{0.5886D^5}{(0.382)^2} = 0.0032 + \frac{0.221}{\left(\frac{0.382 \times 10^6}{D}\right)^{0.237}}$$

$$\text{or,} \quad 4.034 D^5 = 0.0032 + 0.0105 \times D^{0.237}$$

$$\text{or,} \quad 4.034 D^5 - 0.0105 D^{0.237} - 0.0032 = 0$$

Solving by hit and trial method, we get:

$$D \approx 0.308 \text{ m (Ans.)}$$

Example 11.16. Design the diameter of a steel pipe to carry water having kinematic viscosity $\nu = 10^{-6} \text{ m}^2/\text{s}$ with a mean velocity of 1 m/s. The head loss is to be limited to 5 m per 100 m length of pipe. Consider the equivalent sand roughness height of pipe, $k_s = 45 \times 10^{-4} \text{ cm}$. Assume that the Darcy-Weisbach friction co-efficient over the whole range of turbulent flow can be expressed as

$$f = 0.0055 \left[1 + \left(20 \times 10^3 \frac{k_s}{D} + \frac{10^6}{Re} \right)^{1/3} \right]$$

where, D = Diameter of pipe and Re = Reynolds number.

(Delhi University)

Solution. Given : $\nu = 10^{-6} \text{ m}^2/\text{s}$; $\bar{U} = 1 \text{ m/s}$; $h_f = 5 \text{ m}$ in a length 100 m (L);

$$k = 45 \times 10^{-4} \text{ cm} = 45 \times 10^{-6} \text{ m}$$

$$\text{Friction factor, } f = 0.0055 \left[1 + \left(20 \times 10^3 \frac{k_s}{D} + \frac{10^6}{Re} \right)^{1/3} \right] \quad \dots(i)$$

Diameter of the steel pipe, D:

Using Darcy-Weisbach equation: $h_f = \frac{4fL\bar{U}^2}{D \times 2g}$, we have:

$$f = \frac{h_f \times D \times 2g}{4L\bar{U}^2} = \frac{5 \times D \times 2 \times 9.81}{4 \times 100 \times 1^2} = 0.2452 D$$

$$\text{Reynolds number, } Re = \frac{\rho \bar{U} D}{\mu} = \frac{\bar{U} D}{\nu} = \frac{1 \times D}{10^{-6}} = 10^6 D$$

Substituting the values in (i), we get:

$$0.2452 D = 0.0055 \left[1 + \left(20 \times 10^3 \times \frac{45 \times 10^{-6}}{D} + \frac{10^6}{10^6 D} \right)^{1/3} \right]$$

$$\text{or, } \frac{0.2452 D}{0.0055} = \left[1 + \left(\frac{0.9}{D} + \frac{1}{D} \right)^{1/3} \right]$$

$$\text{or, } 44.58 D = \left[1 + \left(\frac{1.9}{D} \right)^{1/3} \right] \quad \text{or} \quad (44.58 D - 1) = \left(\frac{1.9}{D} \right)^{1/3}$$

$$\text{or, } (44.58 D - 1)^3 = \frac{1.9}{D} \quad \dots(ii)$$

$$\text{or, } D (44.58 D - 1)^3 = 1.9$$

Solving by *hit and trial method*, we get $D = 0.0854 \text{ m (Ans.)}$

Example 11.17. Water flows through a horizontal conical pipe, 2 m long and having a diameter of 200 mm at the inlet and 150 mm at the discharge end. A constant discharge of $0.4 \text{ m}^3/\text{s}$ flows through the pipe. Starting from first principles determine the loss of head due to pipe friction. Take friction factor = 0.04. [UPSC Exams, Hydraulic and Hydraulic m/cs]

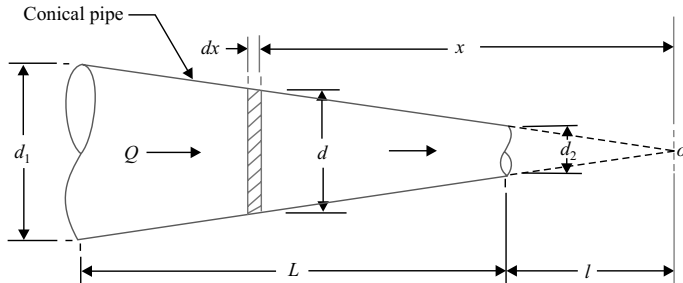


Fig. 11.6

Solution. Diameter at the inlet, $d_1 = 200 \text{ mm} = 0.2 \text{ m}$

Diameter at the outlet, $d_2 = 150 \text{ mm} = 0.15 \text{ m}$

Length of the pipe, $L = 2 \text{ m}$

Discharge through the pipe, $Q = 0.04 \text{ m}^3/\text{s}$

Friction factor, $f_1 = 0.04$

Let us first derive the expression for loss of head due to friction in a tapering pipe as follows:

The Darcy-Weisbach equation in differential form can be written as:

$$dh_f = \frac{f_1 \cdot dx \cdot V^2}{d \times 2g} \quad \dots(i)$$

where,

f_1 = Friction factor, and

V = Velocity of flow (at the section considered)

Refer to Fig. 11.6. From the geometry of the cone we can write,

$$\frac{d_1}{L + l} = \frac{d_2}{l} = \frac{d}{x}$$

(where, d = diameter at a distance x from O)

or,
$$d = \frac{x}{L} (d_1 - d_2)$$

The velocity,
$$V = \frac{Q}{\frac{\pi}{4} d^2} = \frac{4Q}{\pi d^2} = \frac{4Q}{\pi \left\{ \frac{x}{L} (d_1 - d_2) \right\}^2}$$

Substituting for d and V in eqn. (i), we get:

$$\begin{aligned} dh_f &= \frac{f_1 L \cdot dx}{x (d_1 - d_2)} \cdot \frac{1}{2g} \left\{ \frac{4QL^2}{\pi x^2 (d_1 - d_2)^2} \right\}^2 \\ &= \left\{ \frac{8f_1 Q^2 L^5}{g \pi^2 (d_1 - d_2)^5} \right\} \frac{dx}{x^5} \end{aligned}$$

Assuming f_1 to be constant, the friction loss in the conical pipe,

$$\begin{aligned} h_f &= \frac{8f_1 Q^2 L^5}{g \pi^2 (d_1 - d_2)^5} \int_1^{(L+l)} \frac{d}{x^5} = \frac{8f_1 Q^2 L^5}{g \pi^2 (d_1 - d_2)^5} \left[-\frac{1}{4x^4} \right]_1^{L+l} \\ h_f &= \frac{2f_1 Q^2 L}{g \pi^2 (d_1 - d_2)} \left(\frac{1}{d_2^4} - \frac{1}{d_1^4} \right) \end{aligned}$$

Substituting the values, we get:

$$\begin{aligned} h_f &= \frac{2 \times 0.04 \times 0.04^2 \times 2}{9.81 \times \pi^2 (0.2 - 0.15)} \left(\frac{1}{0.15^4} - \frac{1}{0.20^4} \right) \\ &= \mathbf{0.0714 \text{ m (Ans.)}} \end{aligned}$$

HIGHLIGHTS

1. The flow is turbulent when Reynolds number (Re) is more than 4000. The turbulent flow is characterised by random, irregular and haphazard movement of fluid particles.
2. The shear in turbulent flow is mainly due to momentum transfer.
3. Loss of head due to friction in pipe flow is given by

$$\begin{aligned} h_f &= \frac{4fLV^2}{D \times 2g}, \text{ where } f \text{ is the friction co-efficient} \\ &= \frac{f_1 LV^2}{D \times 2g}, \text{ where } f_1 (= 4f) \text{ is the friction factor.} \end{aligned}$$

4. Co-efficient of friction in terms of shear stress is expressed as $f = \frac{2\tau_0}{\rho V^2}$

where,

τ_0 = Shear stress,

ρ = Mass density of fluid, and

V = Average/mean velocity of flow.

5. In turbulent flow the shear stress (τ) is the sum of shear stress due to viscosity (τ_v) and shear stress due to turbulence (τ_t) i.e.,

$$\tau = \tau_v + \tau_t = \mu \frac{du}{dy} + \eta \frac{d\bar{u}}{dy}$$

6. According to Reynolds theory the shear stress (τ) is given as:

$$\tau = \rho u' v'$$

where u' and v' are the fluctuating components of velocity.

7. According to Prandtl's mixing theory the shear stress (τ) is given as:

$$\tau = \rho l^2 \left(\frac{du}{dy} \right)^2$$

8. Prandtl's universal velocity distribution equation is expressed as:

$$u = u_{\max} + 2.5 u_f \ln \left(\frac{y}{R} \right)$$

where,

$u_{\max}$ = Centre-line velocity,

y = Distance from the pipe wall,

R = Radius of the pipe, and

u_f = Shear friction velocity $\sqrt{\frac{\tau_0}{\rho}}$

9. *Velocity defect* is the difference between the maximum velocity ($u_{\max}$) and local velocity (u) at any point and is given by:

$$(u_{\max} - u) = u_f \times 5.75 \log_{10} (R/y)$$

10. If ' k ' is the average height of the irregularities of the surface of a boundary, then in general, the boundary is said to be *rough* if the value of ' k ' compared to the thickness of the laminar sublayer δ' is high and *smooth* if ' k ' is low (in comparison with δ').

Boundary is smoothwhen $\frac{k}{\delta'} < 0.25$

Boundary is roughwhen $\frac{k}{\delta'} > 6.0$

Boundary is in transitionwhen $\frac{k}{\delta'}$ lies between 0.25 to 6.0.

11. For turbulent flow, the velocity distribution is given as:

$$\text{For smooth pipes: } \frac{u}{u_f} = 5.75 \log_{10} \left(\frac{u_f y}{\nu} \right) + 5.5$$

$$\text{For rough pipes: } \frac{u}{u_f} = 5.75 \log_{10} \left(\frac{y}{k} \right) + 8.5$$

where,

u_f = Shear friction velocity $= \sqrt{\frac{\tau_0}{\rho}}$,

ν = Kinematic viscosity of the fluid,

y = Distance from the pipe wall, and

k = Roughness factor.

12. For turbulent flow the velocity distribution in terms of average velocity is given as:

$$\text{For smooth pipes: } \frac{\bar{U}}{u_f} = 5.75 \log_{10} \left(\frac{u + R}{\nu} \right) + 1.75$$

$$\text{For rough pipes: } \frac{\bar{U}}{u_f} = 5.75 \log_{10} (R/k) + 4.75$$

13. The difference of local velocity and average velocity for smooth and rough pipes is given by:

$$\frac{u - \bar{U}}{u_f} = 5.75 \log_{10} (y/R) + 3.75$$

14. Velocity distribution for turbulent flow in smooth pipes by *power law* is given as :

$$\frac{u}{u_{\max}} = \left(\frac{y}{R} \right)^{1/n} \quad \text{where exponent } \frac{1}{n} \text{ depends on Reynolds number } (Re) \text{ and it decreases with the increasing } Re.$$

15. The co-efficient of friction is given by:

$$f = \frac{16}{Re} \quad \dots \text{for laminar flow}$$

$$= \frac{0.0791}{(Re)^{1/4}} \quad \text{for turbulent flow in smooth pipes}$$

$$\text{for } Re \geq 4000 \text{ but } \leq 10^5$$

$$= 0.0008 + \frac{0.05525}{(Re)^{0.237}} \quad \text{for } Re \geq 10^5 \text{ but } \leq 4 \times 10^7$$

$$\frac{1}{\sqrt{4f}} = 2 \log_{10} (R/k) + 1.74 \quad \text{for rough pipes (where, } Re = \text{Reynolds number)}$$

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answers:

- The flow is said to be turbulent when Reynolds number is
 - less than 1000
 - equal to 2000
 - greater than 4000
 - between 1000 to 4000.
- The shear in turbulent flow is mainly due to
 - heat transfer
 - mass transfer
 - momentum transfer
 - all of the above.
- Which of the following statements is *correct*?
Wall turbulence occurs
 - in immediate vicinity of solid surfaces and in the boundary layer flows where the fluid has a negligible mean acceleration
 - in jets, wakes, mixing layer etc.
 - where there is conversion of potential energy into kinetic energy by the process of mixing
 - none of the above.
- Turbulence in flow is characterised by which of the following?
 - Fluctuating components of velocities
 - High Reynolds number
 - Cross currents and eddies with intermixing of particles
 - Excess energy dissipation with rise in temperature
 - All to the above.
- The flow in town water supply pipes is generally
 - laminar
 - turbulent
 - transition
 - any of the above.
- The most essential feature of a turbulent flow is
 - high velocity
 - velocity at a point remains constant with time
 - large discharge
 - Velocity and pressure at a point exhibit irregular fluctuations of high frequency.
- In turbulent flow the velocity distribution is a function of the distance y measured from the boundary surface and shear friction velocity u_f and follows a
 - linear law
 - hyperbolic law
 - parabolic law
 - logarithmic law.
- A turbulent flow is considered steady when
 - the algebraic sum of velocity fluctuations is zero
 - the velocity at a point does not change with time

- (c) temporal mean velocity at a point remains constant with time
 (d) the discharge remains constant.
9. The Darcy-Weisbach friction factor f which is a direct measure of resistance to flow in pipes depends on which of the following?
 (a) Relative roughness, velocity and viscosity
 (b) Relative roughness, diameter and viscosity
 (c) Roughness height, diameter and velocity
 (d) Roughness height, diameter, velocity and kinematic viscosity.
10. Commercial cast-iron and steel pipes carrying fluids under pressure are regarded as hydraulically smooth when
 (a) the laminar layer is thin as compared to the average height of roughness elements
 (b) the height of the roughness projections is low
 (c) the roughness elements are all completely covered by the laminar sublayer
 (d) none of the above.
11. Intensity of turbulence is
 (a) the average K.E. of turbulence
 (b) the violence of turbulent fluctuations and is measured by the root mean square value of velocity fluctuations
 (c) the mean time interval between the reversals in the sign of velocity fluctuation
 (d) none of the above.
12. Which of the following factors determine the friction factor for turbulent flow in a rough pipe?
 (a) Mach number and relative roughness
 (b) Froude's number and Mach's number
 (c) Reynolds number and relative roughness
 (d) Froude's number and relative roughness.
13. In case of turbulent flow of a fluid through a circular tube (as compared to the case of laminar flow at the same flow rate) the maximum velocity is, shear stress at the wall is, and the pressure drop across a given length is, the correct words for the blanks are, respectively
 (a) lower, higher, lower
 (b) lower, higher, higher
 (c) higher, lower, lower
 (d) higher, higher, higher.
14. Prandtl's universal equation is given as:
 (a) $u = u_{\max} + 2.5 u_f \log_e \left(\frac{y}{R} \right)$
 (b) $u = u_{\max} + 3.5 u_f \log_e \left(\frac{y}{R} \right)$
 (c) $u = u_{\max} + 4.5 u_f \log_e \left(\frac{y}{R} \right)$
 (d) $u = u_{\max} + 5.5 u_f \log_e \left(\frac{y}{R} \right)$
- where, u_f = Shear friction velocity,
 y = Distance from the pipe wall, and
 R = Radius of the pipe.

ANSWERS

- | | | | | | |
|---------|----------|--------|---------|---------|---------|
| 1. (c) | 2. (c) | 3. (a) | 4. (e) | 5. (b) | 6. (d) |
| 7. (d) | 8. (c) | 9. (d) | 10. (c) | 11. (b) | 12. (c) |
| 13. (b) | 14. (a). | | | | |

THEORETICAL QUESTIONS

- Enumerate the factors which influence the stability of laminar flow.
- What is turbulence ?
- How is turbulent motion classified?
- What are the characteristics of a turbulent flow?
- What do you understand by wall turbulent and free turbulence?
- Derive an expression for the loss of head due to friction in pipes.
- Define 'shear velocity' for turbulent flow in circular pipes.
- Obtain an expression for the co-efficient of friction in terms of shear stress.
- Derive an expression for shear stress on the basis of 'Prandtl Mixing Length Theory'.
- Obtain an expression for the Prandtl's universal velocity distribution for turbulent flow in pipes. Why this velocity distribution is called universal?
- In what way does the flow through a rough pipe differ from that in smooth pipe?
- What is meant by a smooth boundary and a rough boundary?
- Explain why the hydraulic loss in a pipe is influenced by the surface roughness only at higher Reynolds numbers.

14. Derive an expression for the velocity distribution for turbulent flow in smooth pipes.
15. Show that velocity distribution for turbulent flow through rough pipes is given as:

$$\frac{u}{u_f} = 5.75 \log_{10} (y/k) + 8.5$$

where, u_f = Shear velocity,
 y = Distance from pipe wall, and
 k = Roughness factor.

16. Derive expressions for velocity distribution in terms of average velocity for (i) smooth pipe and, (ii) rough pipe.
17. Prove that the difference of local velocity (u) and average velocity ($\bar{U}$) for turbulent flow through rough or smooth pipes is given by:

$$\frac{u - \bar{U}}{u_f} = 5.75 \log_{10} (y/R) + 3.75.$$

UNSOLVED EXAMPLES

- In a pipe of 300 mm diameter having turbulent flow, the centre-line velocity is 6 m/s and that at 50 mm from the pipe wall is 5 m/s. Calculate the shear friction velocity. [Ans. 0.36 m/s]
- A pipeline carrying water has surface protrusions of average height of 0.15 mm. If the shear stress developed is 4.9 N/m² determine whether the pipe surface acts as smooth, rough or in transition. The kinematic viscosity of water may be taken as 0.01 stokes. [Ans. Transition]
- The velocity of flow in a badly corroded 8 cm pipe is found to increase 30 percent as a pitot tube is moved from a point 1 cm from the wall to 3 cm from the wall. Estimate the height of the roughness elements. [Ans. 0.773 cm]
- Motor having dynamic viscosity of 0.01 poise flows in a 75 mm diameter smooth pipe at the rate of 0.007 m³/sec. Calculate:
 - The shear friction velocity,
 - The velocity at 25 mm from the pipe centre, and
 - The thickness of laminar sublayer.
 Take friction factor = 0.018.
 [Ans. (i) 0.0752 m/s; (ii) 1.7 m/s; (iii) 0.155 mm]
- A smooth pipe 100 mm in diameter and 1000 m long carries water at the rate of 0.0075 m³/s. If the kinematic viscosity of water is 0.02 stokes, calculate:
 - Head lost,
 - Wall shearing stress,
 - Centre-line velocity,
 - Shear stress and velocity at 40 mm from the centre-line, and
 - Thickness of the laminar sublayer.
 [Ans. (i) 9.96 m; (ii) 2.45 N/m²; (iii) 1.15 m/s; (iv) 1.96 N/m²; 0.95 m/s; (v) 0.47 mm]
- A 300 mm diameter pipe is carrying water. If the velocities at the pipe centre and at a point 100 mm from the pipe centre are respectively 3 m/s and 2.5 m/s, determine the wall shearing stress. Assume the flow to be turbulent.
 [Ans. 32.96 N/m²]
- In a rough pipe of diameter 500 mm and length 3500 m water is flowing at the rate of 0.5 m³/s. If the average height of roughness is 0.40 mm find the power required to maintain this flow.
 [Ans. 210.9 kW]
- In a pipe of diameter 300 mm the centre-line velocity and the velocity at a point 100 mm from the centre as measured by pitot-tube are 2.0 m/s and 1.6 m/s respectively. Assuming the flow in the pipe to be turbulent, find:
 - Discharge through the pipe,
 - Co-efficient of friction, and
 - Height of roughness projections.
 [Ans. (i) 0.1027 m³/s; (ii) 0.02; (iii) 18.98 mm]
- In a smooth pipe of diameter 0.4 m and length 800 m water is flowing at the rate of 0.04 m³/s. Assuming the kinematic viscosity of water as 0.018 stokes, find:
 - Head lost due to friction,
 - Wall shear stress,
 - Centre-line velocity, and
 - Thickness of laminar sublayer.
 [Ans. (i) 0.2 m; (ii) 0.245 N/m²; (iii) 0.0377 m/s; (iv) 1.338 mm]
- Water is flowing in a rough pipe of 400 mm diameter and 1000 m long at the rate of 400 litres/sec. Assuming the average height of roughness as 0.012 mm, determine:
 - Co-efficient of friction,
 - Wall shear stress,
 - Centre-line velocity and velocity at a distance of 150 mm from the pipe wall.
 [Ans. (i) 0.00241; (ii) 12.2 N/m²; (iii) 3.6 m/s; 3.52 m/s]



FLOW THROUGH PIPES

12.1. Introduction

12.2. Loss of energy (or head) in pipes

12.3. Major energy losses—Darcy-Weisbach formula—Chezy's formula for loss of head due to friction

12.4. Minor energy losses—loss of head due to sudden enlargement—loss of head due to sudden contraction—loss of head due to obstruction in pipe—loss of head due to entrance to pipe—loss of head at the exit of a pipe—loss of head due to bend in the pipe—loss of head in various pipe fittings

12.5. Hydraulic and total energy line

12.6. Pipes in series or compound pipes

12.7. Equivalent pipe

12.8. Pipes in parallel

12.9. Syphon

12.10. Power transmission through pipes

12.11. Flow through nozzle at the end of a pipe—power transmitted through the nozzle—condition for transmission of maximum power

12.12. Water hammer in pipes

Highlights

Objective Type Questions

Answers

Theoretical Questions

Unsolved Examples

12.1. INTRODUCTION

A **pipe** is a closed conduit (generally of circular section) which is used for carrying fluids under pressure. The flow in a pipe is termed *pipe flow* only when the fluid *completely fills the cross-section and there is no free surface of fluid*. The pipe running partially full (in such a case *atmospheric pressure exists* inside the pipe) behaves like an **open channel**.

12.2. LOSS OF ENERGY (OR HEAD) IN PIPES

When water flows in a pipe, it experiences some resistance to its motion, due to which its velocity and ultimately the head of water available is reduced. This loss of energy (or head) is classified as follows :

A. Major Energy Losses

This loss is due to *friction*.

B. Minor Energy Losses

These losses are due to :

1. Sudden enlargement of pipe,
2. Sudden contraction of pipe,
3. Bend of pipe,
4. An obstruction in pipe,
5. Pipe fittings, etc.

12.3. MAJOR ENERGY LOSSES

These losses which are *due to friction* are calculated by :

1. Darcy-Weisbach formula
2. Chezy's formula.

12.3.1 Darcy-Weisbach Formula

The loss of head (or energy) in pipes due to friction is calculated from Darcy-Weisbach formula (derived in chapter 11 Art. 11.2) which is given by:

$$h_f = \frac{4fLV^2}{D \times 2g} \quad \dots(12.1)$$

where,

h_f = Loss of head due to friction,

f = Co-efficient of friction, (a function of Reynolds number, Re)

$$h = \frac{0.0791}{(Re)^{1/4}} \text{ for } Re \text{ varying from } 4000 \text{ to } 10^6$$

$$= \frac{16}{Re} \text{ for } Re < 2000 \text{ (laminar/viscous flow)}$$

L = Length of the pipe,

V = Mean velocity of flow, and

D = Diameter of the pipe.

12.3.2 Chezy's Formula for Loss of Head due to Friction

Refer to Fig. 11.2. An equilibrium between the propelling force due to pressure difference and the frictional resistance gives :

$$(p_1 - p_2) A = f' PLV^2$$

$$\text{or} \quad \frac{(p_1 - p_2)}{w} \cdot A = \frac{f'}{w} PLV^2 \quad [\text{Refer to Art. 11.2}]$$

$$\text{or} \quad h_f = \frac{f'}{w} \frac{P}{A} LV^2$$

$$\therefore \text{Mean velocity, } V = \sqrt{\frac{w}{f'}} \times \sqrt{\frac{A}{P} \times \frac{h_f}{L}}$$

where, the factor $\sqrt{\frac{w}{f'}}$, is called the Chezy's constant, C ;

the ratio $\frac{A}{P} \left(= \frac{\text{area of flow}}{\text{wetted perimeter}} \right)$ is called the **hydraulic mean depth** or **hydraulic radius** and denoted by m (or R);

the ratio $\frac{h_f}{L}$ prescribes the *loss of head per unit length of pipe* and is denoted by **i** or **S** (slope).

$$\therefore \text{Mean velocity, } V = C \sqrt{m i} \quad \dots(12.2)$$

Eqn. (12.2) is known as **Chezy's formula**. This formula helps to find the head loss due to friction if the mean flow velocity through the pipe and also the value of Chezy's constant C are known.

Note : (i) Darcy-Weisbach formula (for loss of head) is generally used for the flow through *pipes*.

(ii) Chezy's formula (for loss of head) is generally used for the flow through *open channels*.

(iii) The values of hydraulic mean depth for a *circular pipe*,

$$m = \frac{D}{4} \left[\because m = \frac{\text{Area}}{\text{Perimeter}} = \frac{\frac{\pi}{4} \times D^2}{\pi D} = \frac{D}{4} \right]$$

Example 12.1. In a pipe of diameter 350 mm and length 75 m water is flowing at a velocity of 2.8 m/s. Find the head lost due to friction using :

(i) Darcy-Weisbach formula; (ii) Chezy's formula for which $C = 55$.

Assume kinematic viscosity of water as 0.012 stoke.

Solution. Diameter of the pipe, $D = 350 \text{ mm} = 0.35 \text{ m}$

Length of the pipe, $L = 75 \text{ m}$

Velocity of flow, $V = 2.8 \text{ m/s}$

Chezy's constant, $C = 55$

Kinematic viscosity of water, $\nu = 0.012 \text{ stoke} = 0.012 \times 10^{-4} \text{ m}^2/\text{s}$.

Head lost due to friction, h_f :

(i) **Darcy-Weisbach formula :**

Darcy-Weisbach formula is given by:

$$h_f = \frac{4fLV^2}{D \times 2g}$$

where, f = coefficient of friction (a function of Reynolds number, Re)

$$Re = \frac{V \times D}{\nu} = \frac{2.8 \times 0.35}{0.012 \times 10^{-4}} = 8.167 \times 10^5$$

$$\therefore f = \frac{0.0791}{(Re)^{1/4}} = \frac{0.0791}{(8.167 \times 10^5)^{1/4}} = 0.00263$$

$\therefore$ Head lost due to friction,

$$h_f = \frac{4 \times 0.00263 \times 75 \times (2.8)^2}{0.35 \times 2 \times 9.81} = \mathbf{0.9 \text{ m (Ans.)}}$$

(ii) **Chezy's formula :**

$$V = C\sqrt{mi}$$

$$\text{where, } C = 55, m = \frac{A}{p} = \frac{\frac{\pi}{4} \times D^2}{\pi D} = \frac{D}{4} = \frac{0.35}{4} = 0.0875 \text{ m}$$

$$\therefore 2.8 = 55 \sqrt{0.0875 \times i}$$

$$\text{or, } 0.0875 \times i = \left(\frac{2.8}{55}\right)^2 = 0.00259$$

$$\text{or, } i = 0.0296$$

$$\text{But, } i = \frac{h_f}{L} = 0.0296$$

$$\therefore \frac{h_f}{75} = 0.0296$$

$$\text{or, } h_f = 75 \times 0.0296 = \mathbf{2.22 \text{ m (Ans.)}}$$

Example 12.2. Water flows through a pipe of diameter 300 mm with a velocity of 5 m/s. If the co-efficient of friction is given by $f = 0.015 + \frac{0.08}{Re^{0.3}}$ where Re is the Reynolds number, find the head lost due to friction for a length of 10 m. Take kinematic viscosity of water as 0.01 stoke.

Solution. Diameter of the pipe, $D = 300 \text{ mm} = 0.3 \text{ m}$

Velocity of water $V = 5 \text{ m/s}$

Length of the pipe, $L = 10 \text{ m}$

Viscosity of water, $\nu = 0.01 \text{ stoke} = 0.01 \times 10^{-4} \text{ m}^2/\text{s}$. ($\because 1 \text{ stoke} = 1 \text{ cm}^2/\text{s} = 1 \times 10^{-4} \text{ m}^2/\text{s}$)

Head lost due to friction h_f :

$$\text{Co-efficient of friction, } f = 0.015 + \frac{0.08}{(Re)^{0.3}} \quad \dots(\text{given})$$

$$\text{But, Reynolds number, } Re = \frac{\rho VD}{\mu} = \frac{VD}{\nu} = \frac{5 \times 0.3}{0.01 \times 10^{-4}} = 1.5 \times 10^6$$

$$\therefore f = 0.015 + \frac{0.08}{(1.5 \times 10^6)^{0.3}} = 0.0161$$

$\therefore$ Head lost due to friction,

$$h_f = \frac{4fLV^2}{D \times 2g} = \frac{4 \times 0.0161 \times 10 \times 5^2}{0.3 \times 2 \times 9.81}$$

$$= 2.735 \text{ m (Ans.)}$$

Example 12.3. In a pipe of 300 mm diameter and 800 m length an oil of specific gravity 0.8 is flowing at the rate of $0.45 \text{ m}^3/\text{s}$. Find :

(i) Head lost due to friction, and

(ii) Power required to maintain the flow.

Take kinematic viscosity of oil as 0.3 stoke.

Solution. Diameter of the pipe, $D = 300 \text{ mm} = 0.3 \text{ m}$

Length of the pipe, $L = 800 \text{ m}$

Specific gravity of oil = 0.8

Kinematic viscosity of oil, $\nu = 0.3 \text{ stoke} = 0.3 \times 10^{-4} \text{ m}^2/\text{s}$

Discharge, $Q = 0.45 \text{ m}^3/\text{s}$.

(i) Head lost due to friction, h_f :

$$\text{Velocity, } V = \frac{Q}{\text{Area}} = \frac{0.45}{\frac{\pi}{4} \times 0.3^2} = 6.366 \text{ m/s}$$

$$\therefore \text{ Reynolds number, } Re = \frac{V \times D}{\nu} = \frac{6.366 \times 0.3}{0.3 \times 10^{-4}} = 6.366 \times 10^4$$

$$\therefore \text{ Co-efficient of friction, } f = \frac{0.0791}{(Re)^{1/4}} = \frac{0.0791}{(6.366 \times 10^4)^{1/4}} = 0.00498$$

$$\therefore h_f = \frac{4fLV^2}{D \times 2g} = \frac{4 \times 0.00498 \times 800 \times (6.366)^2}{0.3 \times 2 \times 9.81}$$

$$= 109.72 \text{ m (Ans.)}$$

(ii) Power required, P :

Power required to maintain the flow, $P = wQh_f$

where, $w = 0.8 \times 9.81 = 7.848 \text{ kN/m}^3$

$h_f = 109.72 \text{ m}$, $Q = 0.45 \text{ m}^3/\text{s}$

$$\therefore P = 7.848 \times 0.45 \times 109.72 = 387.48 \text{ kW (Ans.)}$$

Example 12.4. Water is to be supplied to the inhabitants of a college campus through a supply main. The following data is given :

Distance of the reservoir from the campus = 3000 m

Number of inhabitants = 4000

Consumption of water per day of each inhabitant = 180 litres

Loss of head due to friction = 18 m

Co-efficient of friction for the pipe, $f = 0.007$

If the half of the daily supply is pumped in 8 hours, determine the size of the supply main.

Solution. Distance of the reservoir from the college campus = 3000 m

Number of inhabitants = 4000

Consumption per day per inhabitant = 180 litres = 0.18 m^3

$\therefore$ Total supply per day = $4000 \times 0.18 = 720 \text{ m}^3$

Since half of this supply is to be pumped in 8 hours, therefore maximum flow for which the pipe is to be designed,

$$Q = \frac{720}{2 \times 8 \times 3600} = 0.0125 \text{ m}^3/\text{s}$$

Loss of head due to friction, $h_f = 18 \text{ m}$

Co-efficient of friction, $f = 0.007$

Diameter of the supply line, D :

Using the relation :

$$h_f = \frac{4fLV^2}{D \times 2g}$$

where,

$$V = \frac{Q}{A} = \frac{0.0125}{\frac{\pi}{4} \times D^2} = \frac{0.0159}{D^2}$$

$$\therefore 18 = \frac{4 \times 0.007 \times 3000 \times (0.0159 / D^2)^2}{D \times 2 \times 9.81}$$

$$\text{or, } D^5 = \frac{4 \times 0.007 \times 3000 \times 0.0159^2}{18 \times 2 \times 9.81} = 6.013 \times 10^{-5}$$

$$\therefore D = 0.143 \text{ m or } 143 \text{ mm (Ans.)}$$

Example 12.5. Water flows through a pipeline whose diameter varies from 25 cm to 15 cm in a length of 10 m. If the Darcy-Weisbach friction factor is assumed constant at 0.018 for the whole pipe, determine the head loss in friction when the pipe is flowing full with a discharge of $0.06 \text{ m}^3/\text{s}$.

Solution. Given : $D_1 = 25 \text{ cm} = 0.25 \text{ m}$; $D_2 = 15 \text{ cm} = 0.15 \text{ m}$, $L = 10 \text{ m}$; $f = 0.018$; $Q = 0.06 \text{ m}^3/\text{s}$

Consider a stretch of length dx at a distance x from the 25 cm diameter end (Fig. 12.1).

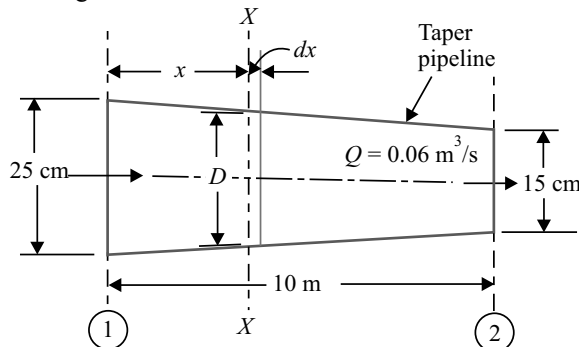


Fig. 12.1

$$\begin{aligned}
 dh_{fx} &= \frac{f dx V^2}{D \times 2g} \\
 &= \frac{f dx \left[Q \div \left(\frac{\pi}{4} \times D^2 \right) \right]^2}{D \times 2g} = \frac{f Q^2 dx}{2g \left(\frac{\pi}{4} \right)^2 D^5} \\
 &= 0.08263 \times \frac{f Q^2 dx}{D^5}
 \end{aligned}$$

where,

D = Diameter at the section XX

$$D = \left[0.25 - \left(\frac{0.25 - 0.15}{10} \right) x \right] = \frac{1}{100} (25 - x) \text{ m}$$

Hence,

$$\begin{aligned}
 dh_{fx} &= 0.08263 \times 0.018 (0.06)^2 \times (100)^5 \times \frac{dx}{(25 - x)^5} \\
 &= 53544 \times \frac{dx}{(25 - x)^5}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total head loss, } h_f &= \int_0^{10} dh_{fx} \\
 &= 53544 \int_0^{10} (25 - x)^{-5} dx = 53544 \left[\frac{1}{4 (25 - x)^4} \right]_0^{10} \\
 &= \frac{53544}{4} \left[\frac{1}{(15)^4} - \frac{1}{(25)^4} \right] = \mathbf{0.23 \text{ m (Ans.)}}
 \end{aligned}$$

Example. 12.6. A pipeline 50 cm diameter takes off from a reservoir whose water surface elevation is 145 m above datum. The pipe is 4500 m long and is laid completely at the datum level. In the last 1000 m of the pipe, water is withdrawn by a series of pipes at a uniform rate of 0.075 m³/s per 250 m. Find the pressure at the end of the pipeline.

Assume f (friction factor) = 0.018 and the pipe to have a dead end.

(UPSC)

Solution. Given : Diameter of pipeline, $D = 50 \text{ cm} = 0.5 \text{ m}$; $L = 4500 \text{ m}$; $L_0 = 1000 \text{ m}$; $f = 0.018$.

First an expression for loss of head in a pipe having a uniform withdrawal of $q^* \text{ m}^3/\text{s}$ per metre length is derived.

Refer to Fig. 12.2. Consider a section at a distance x from the start of the uniform withdrawal at q^* per metre length.

$$\text{Discharge, } Q_x = Q_0 - q^*x$$

In a small distance dx ,

$$dh_f = \frac{f L V^2}{D \times 2g} = \frac{f}{2g} \times \left(\frac{Q_0 - q^*x}{\frac{\pi}{4} D^2} \right)^2 \times \frac{1}{D} \times dx = \frac{8f}{\pi^2 g D^5} (Q_0 - q^*x)^2 dx$$

$$\therefore h_f = \int_0^{L_0} dh_f = \frac{-8f}{3\pi^2 g D^5} \times \frac{1}{q^*} \times \left[(Q_0 - q^*x)^3 \right]_0^{L_0}$$

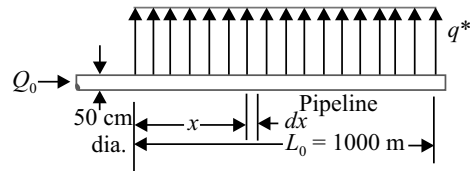


Fig. 12.2

or,
$$h_f = \frac{8f}{3\pi^2 g D^5} \times \frac{1}{q^*} \times [Q_0^3 - (Q_0 - q^* L_0)^3]$$

Here,
$$Q_0 = \frac{1000}{250} \times 0.075 = 0.3 \text{ m}^3/\text{s}$$

$$q^* = \frac{0.075}{250} = 0.0003 \text{ m}^3/\text{s}$$

$$L_0 = 1000 \text{ m}$$

H_L = Total head lost = [Head lost in first (4500 – 1000) m with a discharge $Q_d = 0.3 \text{ m}^3/\text{s}$]
+ [Head lost in 1000 m with a uniform withdrawal of q^*]

$$= h_{f1} + h_{f2}$$

$$h_{f1} = \frac{0.018 \times 3500 \times \left[0.3 \div \left(\frac{\pi}{4} \times 0.5 \right)^2 \right]^2}{0.5 \times 2 \times 9.81} = 24.3 \text{ m}$$

$$\begin{aligned} h_{f2} &= \frac{8f}{3\pi^2 g D^5} \times \frac{1}{q^*} \times [Q_0^3 - (Q_0 - q^* L_0)^3] \\ &= \frac{8 \times 0.018}{3\pi^2 \times 9.81 \times (0.5)^5} \times \frac{1}{0.0003} [(0.3)^3 - (0.3 - 0.0003 \times 1000)^3] \\ &= 1.43 \text{ m} \end{aligned}$$

Total head loss = 24.3 + 1.43 = 25.73 m

Residual head at the dead end = 145 – 25.73 = **119.27 m (Ans.)**

Example 12.7. A pump delivers water from a tank A (water surface elevation = 110 m) to tank B (water surface elevation = 170 m). The suction pipe is 45 m long (friction factor, $f = 0.024$) and 35 cm in diameter. The delivery pipe is 950 m long ($f = 0.022$) and 25 cm in diameter. The head discharge relationship for the pump is given by $H_p = (90 - 8000 Q^2)$, where H_p is in metres and Q in m^3/s . Calculate :

- (i) The discharge in the pipeline. (ii) The power delivered by the pump.

Solution. Refer to Fig. 12.3.

Given: $D_1 = 35 \text{ cm} = 0.35 \text{ m}$; $L_1 = 45 \text{ m}$; $D_2 = 25 \text{ cm} = 0.25 \text{ m}$; $L_2 = 950 \text{ m}$; $f_1 = 0.024$; $f_2 = 0.022$;
 $H_p = 90 - 8000 Q^2$

Suction pipe :

$$\begin{aligned} \text{Head loss, } h_{L1} &= \frac{f_1 L_1 V_1^2}{D_1 \times 2g} \\ &= \frac{0.024 \times 45}{0.35} \times \frac{V_1^2}{2g} = 3.086 \frac{V_1^2}{2g} \end{aligned}$$

Delivery pipe :

$$\begin{aligned} \text{Head loss, } h_{L2} &= \frac{f_2 L_2 V_2^2}{D_2 \times 2g} \\ &= \frac{0.022 \times 950}{0.25} \times \frac{V_2^2}{2g} = 83.6 \frac{V_2^2}{2g} \end{aligned}$$

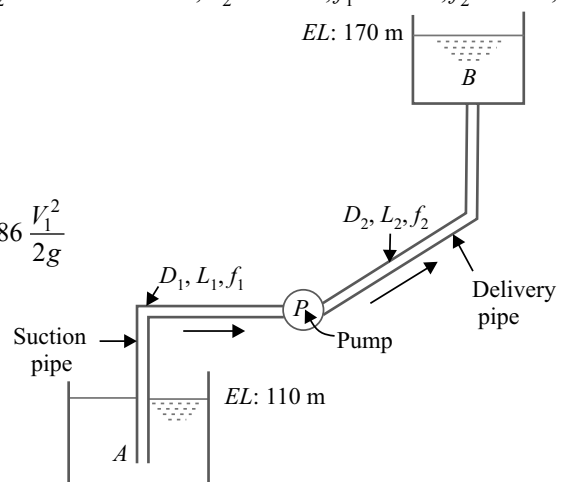


Fig. 12.3

$$\text{Total head loss, } H_L = 3.086 \frac{V_1^2}{2g} + 83.6 \frac{V_2^2}{2g}$$

By continuity equation,

$$A_1 V_1 = A_2 V_2$$

$$\frac{\pi}{4} \times (0.35)^2 \times V_1 = \frac{\pi}{4} \times (0.25)^2 \times V_2$$

or,

$$V_1 = 0.51 V_2$$

$$\frac{V_1^2}{2g} = 0.26 \frac{V_2^2}{2g}$$

$$\therefore H_L = 3.086 \times 0.26 \frac{V_2^2}{2g} + 83.6 \frac{V_2^2}{2g} = 84.4 \frac{V_2^2}{2g}$$

$$\text{Static head} = 170 - 110 = 60 \text{ m}$$

H_p = Head delivered by the pump = Static head + friction head

$$= 60 + 84.4 \frac{V_2^2}{2g}$$

$$= 60 + 84.4 \times \left(\frac{Q}{\frac{\pi}{4} \times (0.25)^2} \right)^2 \times \frac{1}{2 \times 9.81}$$

$$= 60 + 1785.3 Q^2$$

Also,

$$H_p = 90 - 8000 Q^2$$

...(Given)

$$\therefore 90 - 8000 Q^2 = 60 + 1785.3 Q^2$$

or,

$$Q = 0.05537 \text{ m}^3/\text{s}$$

$\therefore$

$$H_p = 60 + 1785.3 \times (0.05537)^2 = 65.47 \text{ m}$$

Hence, power delivered by the pump,

$$P = \rho Q H_p = 9.81 \times 0.05537 \times 65.47 = \mathbf{35.56 \text{ kW (Ans.)}}$$

12.4. MINOR ENERGY LOSSES

Whereas the major loss of energy or head is due to friction, the minor loss of energy (or head) includes the following cases :

1. Loss of head due to sudden enlargement,
2. Loss of head due to sudden contraction,
3. Loss of head due to an obstruction in the pipe,
4. Loss of head at the entrance to a pipe,
5. Loss of head at the exit of a pipe,
6. Loss of head due to bend in the pipe, and
7. Loss of head in various pipe fittings.

12.4.1 Loss of Head due to Sudden Enlargement

Fig. 12.4. shows a liquid flowing through a pipe which has *sudden enlargement*. Due to sudden enlargement, the flow is decelerated abruptly and eddies are developed resulting in loss of energy (or head).

Consider two sections 1 – 1 (before enlargement) and 2 – 2 (after enlargement).

Let, A_1 = Area of pipe at section 1-1.

$$= \frac{\pi}{4} D_1^2 \text{ (where } D_1 \text{ is the diameter of the pipe),}$$

p_1 = Intensity of pressure at section 1-1,

V_1 = Velocity of flow at section 1-1,

$$A_2 \left(= \frac{\pi}{4} D_2^2 \right), p_2 \text{ and } V_2 = \text{Corresponding values at section 2-2,}$$

p_0 = Intensity of pressure of the liquid eddies on the area $(A_2 - A_1)$, and

h_e = Loss of head due to sudden enlargement.

Applying Bernoulli's equation to sections 1-1 and 2-2, we have:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + \text{Loss of head due to sudden enlargement } (h_e)$$

But, $z_1 = z_2$...pipe being horizontal

$$\therefore \frac{p_1}{w} + \frac{V_1^2}{2g} = \frac{p_2}{w} + \frac{V_2^2}{2g} + h_e$$

$$\text{or, } h_e = \left(\frac{p_1}{w} - \frac{p_2}{w} \right) + \left(\frac{V_1^2}{2g} - \frac{V_2^2}{2g} \right) \quad \dots(i)$$

Now, the force acting on liquid in the control volume (between sections 1-1 and 2-2) in the flow direction is given by :

$$F_x = p_1 \cdot A_1 + p_0 (A_2 - A_1) - p_2 \cdot A_2$$

Assuming $p_0 = p_1$, we have:

$$\begin{aligned} F_x &= p_1 \cdot A_1 + p_1 (A_2 - A_1) - p_2 \cdot A_2 \\ &= p_1 A_2 - p_2 A_2 = (p_1 - p_2) A_2 \end{aligned} \quad \dots(ii)$$

Consider **momentum** of liquid at the sections 1-1 and 2-2; momentum of liquid /sec at section 1-1 = Mass $\times$ velocity.

$$= \rho A_1 V_1 \times V_1 = \rho A_1 V_1^2$$

$$\text{Momentum of liquid/sec. at section 2-2} = \rho A_2 V_2 \times V_2 = \rho A_2 V_2^2$$

$\therefore$ Change of momentum of liquid/sec.

$$= \rho A_2 V_2^2 - \rho A_1 V_1^2$$

But from continuity equation, we have:

$$A_1 V_1 = A_2 V_2$$

$$\text{or, } A_1 = \frac{A_2 V_2}{V_1}$$

$\therefore$ Change of momentum/sec.

$$= \rho A_2 V_2^2 - \rho \times \frac{A_2 V_2}{V_1} \times V_1^2$$

$$= \rho A_2 V_2^2 - \rho A_2 V_1 V_2$$

$$= \rho A_2 (V_2^2 - V_1 V_2) \quad \dots(iii)$$

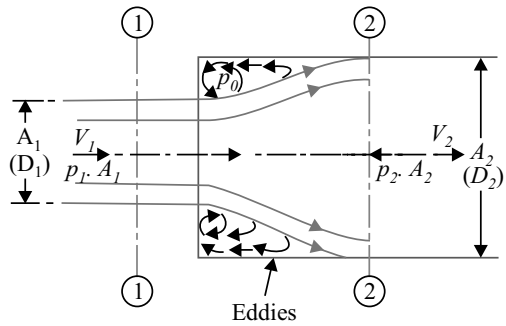


Fig. 12.4. Loss of head due to sudden enlargement.

Now, Net force = Change of momentum

$$\therefore (p_1 - p_2) A_2 = \rho A_2 (V_2^2 - V_1 V_2)$$

$$\text{or, } \frac{p_1 - p_2}{\rho} = V_2^2 - V_1 V_2$$

Dividing both sides by g , we get:

$$\frac{p_1 - p_2}{\rho g} = \frac{V_2^2 - V_1 V_2}{g}$$

$$\text{or, } \frac{p_1}{w} - \frac{p_2}{w} = \frac{V_2^2 - V_1 V_2}{g} \quad (\because \rho g = w)$$

Substituting the value of $\left(\frac{p_1}{w} - \frac{p_2}{w}\right)$ in eqn. (i), we get:

$$\begin{aligned} h_e &= \frac{V_2^2 - V_1 V_2}{g} + \frac{V_1^2}{2g} - \frac{V_2^2}{2g} \\ &= \frac{2V_2^2 - 2V_1 V_2 + V_1^2 - V_2^2}{2g} = \frac{V_1^2 + V_2^2 - 2V_1 V_2}{2g} = \frac{(V_1 - V_2)^2}{2g} \end{aligned}$$

$$\therefore h_e = \frac{(V_1 - V_2)^2}{2g} \quad \dots(12.2)$$

Example 12.8. At a sudden enlargement of a water main from 240 mm to 480 mm diameter, the hydraulic gradient rises by 10 mm. Calculate the rate of flow. [MDU, Haryana]

Solution. Diameter of the smaller pipe, $D_1 = 240 \text{ mm} = 0.24 \text{ m}$

Diameter of larger pipe, $D_2 = 480 \text{ mm} = 0.48 \text{ m}$

Rise of hydraulic gradient, i.e.

$$\left(\frac{p_2}{w} + z_2\right) - \left(\frac{p_1}{w} + z_1\right) = 10 \text{ mm} = 0.01 \text{ m}$$

$$\left[\text{The term } \left(\frac{p}{w} + z\right) \text{ prescribes the hydraulic gradient} \right]$$

Rate of flow, Q :

Applying Bernoulli's equation to small and large pipe sections (1-1 and 2-2), we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_e \text{ (i.e., head lost due to sudden enlargement)} \quad \dots(i)$$

$$\text{But, } h_e = \frac{(V_1 - V_2)^2}{2g} \quad \dots(ii)$$

From continuity equation, we have:

$$A_1 V_1 = A_2 V_2$$

$$\therefore V_1 = \frac{A_2 V_2}{A_1} = \frac{\frac{\pi}{4} \times D_2^2}{\frac{\pi}{4} \times D_1^2} \times V_2 = \left(\frac{D_2}{D_1}\right)^2 \times V_2$$

$$\text{or, } V_1 = \left(\frac{0.48}{0.24}\right)^2 \times V_2 = 4V_2$$

Substituting this value of V_1 in eqn. (ii), we get:

$$h_e = \frac{(4V_2 - V_2)^2}{2g} = \frac{9V_2^2}{2g}$$

Now, substituting the values of h_e and V_1 in eqn. (i), we have:

$$\frac{p_1}{w} + \frac{(4V_2)^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + \frac{9V_2^2}{2g}$$

$$\text{or, } \frac{16V_2^2}{2g} - \frac{V_2^2}{2g} - \frac{9V_2^2}{2g} = \left(\frac{p_2}{w} + z_2 \right) - \left(\frac{p_1}{w} + z_1 \right)$$

$$\text{or, } \frac{6V_2^2}{2g} = 0.01$$

$$\text{or, } V_2 = \left(\frac{0.01 \times 2 \times 9.81}{6} \right)^{1/2} = 0.181 \text{ m/s}$$

$$\therefore \text{Rate of flow, } Q = A_2 V_2 = \frac{\pi}{4} \times 0.48^2 \times 0.181 = \mathbf{0.03275 \text{ m}^3/\text{s}} \text{ (Ans.)}$$

Example 12.9. A horizontal pipe 150 mm in diameter is joined by sudden enlargement to a 225 mm diameter pipe. Water is flowing through it at the rate of $0.05 \text{ m}^3/\text{s}$. Find :

- (i) Loss of head due to abrupt expansion,
- (ii) Pressure difference in the two pipes, and
- (iii) Change in pressure if the change of section is gradual without any loss.

Solution. Diameter of the smaller pipe, $D_1 = 150 \text{ mm} = 0.15 \text{ m}$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

Diameter of the larger pipe, $D_2 = 225 \text{ mm} = 0.225 \text{ m}$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.225^2 = 0.03976 \text{ m}^2$$

$$\text{Discharge, } Q = 0.05 \text{ m}^3/\text{s}$$

- (i) **Loss of head due to abrupt expansion, h_e :**

$$h_e = \frac{(V_1 - V_2)^2}{2g}$$

where, V_1 and V_2 are the velocities of flow in the smaller and larger diameter pipes respectively.

$$V_1 = \frac{Q}{A_1} = \frac{0.05}{0.01767} = 2.83 \text{ m/s}$$

$$V_2 = \frac{Q}{A_2} = \frac{0.05}{0.03976} = 1.26 \text{ m/s}$$

$$\text{Hence, } h_e = \frac{(2.83 - 1.26)^2}{2 \times 9.81} = \mathbf{0.1256 \text{ m}} \text{ (Ans.)}$$

- (ii) **Pressure difference in the two pipes :**

Applying Bernoulli's equation at the smaller and the larger pipe sections, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_e$$

$$\text{or, } \left(\frac{p_2 - p_1}{w} \right) = \frac{V_1^2 - V_2^2}{2g} - h_e \quad [\because z_1 = z_2, \text{ the pipe being horizontal}]$$

$$\text{or, } \left(\frac{p_2 - p_1}{w} \right) = \frac{2.83^2 - 1.26^2}{2 \times 9.81} - 0.1256 = \mathbf{0.202 \text{ m of water (Ans.)}}$$

The positive sign indicates that there is *gain* in pressure. Thus, although there is an energy loss, the pressure increases across a sudden flow of expansion.

(iii) **Change in pressure with gradual change of section :**

If the change of section is gradual *without loss*, then, gain in pressure,

$$\frac{p_2 - p_1}{w} = \frac{V_1^2 - V_2^2}{2g} = \frac{2.83^2 - 1.26^2}{2 \times 9.81} = \mathbf{0.327 \text{ m of water (Ans.)}}$$

Example 12.10. The diameter of a horizontal pipe which is 300 mm is suddenly enlarged to 600 mm. The rate of flow of water through this pipe is $0.4 \text{ m}^3/\text{s}$. If the intensity of pressure in the smaller pipe is 125 kN/m^2 , determine.

- (i) Loss of head, due to sudden enlargement,
- (ii) Intensity of pressure in the larger pipe, and
- (iii) Power lost due to enlargement.

Solution. Diameter of the smaller pipe, $D_1 = 300 \text{ mm} = 0.3 \text{ m}$

$$\text{Area, } A_1 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$$

Diameter of the longer pipe, $D_2 = 600 \text{ mm} = 0.6 \text{ m}$

$$\text{Area, } A_2 = \frac{\pi}{4} \times 0.6^2 = 0.2828 \text{ m}^2$$

Rate of flow of water, $Q = 0.4 \text{ m}^3/\text{s}$

Intensity of pressure in the smaller pipe, $p_1 = 125 \text{ kN/m}^2$

$$\text{Now velocity, } V_1 = \frac{Q}{A_1} = \frac{0.4}{0.0707} = 5.66 \text{ m/s}$$

$$\text{Velocity, } V_2 = \frac{Q}{A_2} = \frac{0.4}{0.2828} = 1.414 \text{ m/s}$$

(i) **Loss of head due to sudden enlargement, h_e :**

Loss of head due to sudden enlargement,

$$\begin{aligned} h_e &= \frac{(V_1 - V_2)^2}{2g} \\ &= \frac{(5.66 - 1.414)^2}{2 \times 9.81} = \mathbf{0.918 \text{ m (Ans.)}} \end{aligned}$$

(ii) **Intensity of pressure in the large pipe, p_2 :**

Applying Bernoulli's equation before and after sudden enlargement, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_e$$

But,

$$z_1 = z_2$$

...because pipe is horizontal

$$\therefore \frac{p_1}{w} + \frac{V_1^2}{2g} = \frac{p_2}{w} + \frac{V_2^2}{2g} + h_e$$

$$\text{or, } \frac{p_2}{w} = \frac{p_1}{w} + \frac{V_1^2}{2g} - \frac{V_2^2}{2g} - h_e$$

$$= \frac{125}{9.81} + \frac{5.66^2}{2 \times 9.81} - \frac{1.414^2}{2 \times 9.81} - 0.918$$

$$= 12.74 + 1.63 - 0.1 - 0.918 = 13.35 \text{ m}$$

∴

$$p_2 = w \times 13.35 = 9.81 \times 13.35$$

$$= \mathbf{130.9 \text{ kN/m}^2 \text{ (Ans.)}}$$

(iii) Power lost due to sudden enlargement, P_{lost} :

$$P_{\text{lost}} = \frac{wQh_e}{1000} \text{ kW}$$

where,

$$w = 9.81 \times 1000 \text{ N/m}^2,$$

$$Q = 0.4 \text{ m}^3/\text{s}, \text{ and}$$

$$h_e = 0.918 \text{ m}$$

∴

$$P_{\text{lost}} = \frac{(9.81 \times 1000) \times 0.4 \times 0.918}{1000}$$

$$= \mathbf{3.6 \text{ kW (Ans.)}}$$

Example 12.11. In a 80 mm diameter pipeline an oil of specific gravity 0.8 is flowing at the rate of $0.0125 \text{ m}^3/\text{s}$. A sudden expansion takes place into a second pipeline of such diameter that maximum pressure rise is obtained. Find :

(i) Loss of energy in sudden expansion,

(ii) Differential gauge length indicated by an oil-mercury manometer connected between the two pipes. [PTU]

Solution. Diameter of the smaller pipe, $D_1 = 80 \text{ mm} = 0.08 \text{ m}$

Specific gravity of oil, $S = 0.8$

Discharge, $Q = 0.0125 \text{ m}^3/\text{s}$.

(i) Loss of energy in sudden expansion :

$$\text{Velocity of flow, } V = \frac{Q}{\text{Area}} = \frac{0.0125}{\frac{\pi}{4} \times 0.08^2} = 2.49 \text{ m/s}$$

The pressure rise will be maximum when:

$$\frac{D_1}{D_2} = \frac{1}{\sqrt{2}} \quad (\text{where, } D_2 = \text{diameter of the larger pipe})$$

[For derivation of the formula, refer to Example 12.12]

$$\text{or, } D_2 = \sqrt{2} D_1 = \sqrt{2} \times 0.08 = 0.1131 \text{ m}$$

$$\therefore V_2 = \frac{0.0125}{\frac{\pi}{4} \times (0.1131)^2} = 1.244 \text{ m/s}$$

Loss of energy (or head) in sudden expansion,

$$h_2 = \frac{(V_1 - V_2)^2}{2g} = \frac{(2.49 - 1.244)^2}{2 \times 9.81} = \mathbf{0.079 \text{ m of oil (Ans.)}}$$

(iii) Reading of the manometer :

The energy equation is given as:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_e \quad (z_1 = z_2, \text{ the pipe being horizontal})$$

$$\begin{aligned}\left(\frac{p_2 - p_1}{w}\right) &= \frac{V_1^2}{2g} - \frac{V_2^2}{2g} - h_e \\ &= \frac{2.49^2}{2 \times 9.81} - \frac{1.244^2}{2 \times 9.81} - 0.079 = 0.158 \text{ of oil}\end{aligned}$$

Let, h = Reading of the U-tube oil-mercury manometer where limbs are connected across the expanded transition:

$$\text{Then, } \frac{p_2 - p_1}{w} = h \left(\frac{S_m}{S_0} - 1 \right)$$

[where, S_m = specific gravity of mercury (= 13.6)]

$$\text{or, } 0.158 = h \left(\frac{13.6}{0.8} - 1 \right) = 16 h$$

$$\text{or, } h = \frac{0.158}{16} = 0.009875 \text{ m or } \mathbf{9.875 \text{ mm (Ans.)}}$$

Example 12.12. For sudden expansion in a pipe flow, work out the optimum ratio between the diameter of the pipe before expansion and the diameter of pipe after expansion so that pressure rise is maximum ? Also find the maximum pressure rise. (UPSC Exams.)

Solution. Applying Bernoulli's equation at sections 1 – 1 and 2 – 2, we have:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_e$$

$$\text{where, } h_e \text{ (energy loss due to sudden expansion)} = \frac{(V_1 - V_2)^2}{2g}$$

$$\therefore \frac{p_1}{w} + \frac{V_1^2}{2g} = \frac{p_2}{w} + \frac{V_2^2}{2g} + \frac{(V_1 - V_2)^2}{2g} \quad (\because z_1 = z_2, \text{ the pipe being horizontal})$$

$$\text{Pressure rise, } \Delta p = (p_2 - p_1) = w \left[\frac{V_1^2}{2g} - \frac{V_2^2}{2g} - \frac{(V_1 - V_2)^2}{2g} \right]$$

From continuity consideration,

$$A_1 V_1 = A_2 V_2$$

$$V_2 = \frac{A_1 V_1}{A_2} = \left(\frac{D_1}{D_2} \right)^2 V_1$$

$$\therefore \Delta p = w \times \frac{V_1^2}{2g} \left[1 - \left(\frac{D_1}{D_2} \right)^4 - \left\{ 1 - \left(\frac{D_1}{D_2} \right)^2 \right\}^2 \right]$$

$$\text{or } \Delta p = w \times \frac{V_1^2}{2g} \left[2 \left(\frac{D_1}{D_2} \right)^2 - 2 \left(\frac{D_1}{D_2} \right)^4 \right]$$

There is only one value or ratio $\left(\frac{D_1}{D_2} \right)$ which will provide the maximum pressure rise.

$$\therefore \text{ For maximum pressure rise, } \frac{d(\Delta p)}{d(D_1/D_2)} = 0$$

$$\text{or, } \frac{d(\Delta p)}{d(D_1/D_2)} = \left[4 \left(\frac{D_1}{D_2} \right) - 8 \left(\frac{D_1}{D_2} \right)^3 \right] = 0$$

From which $\left(\frac{D_1}{D_2}\right)^2 = \frac{1}{2}$ or $\frac{D_1}{D_2} = \frac{1}{\sqrt{2}}$

$\therefore$ Diameter ratio for the maximum pressure rise is:

$$\frac{D_1}{D_2} = \frac{1}{\sqrt{2}} \text{ (Ans.)}$$

Maximum pressure rise is:

$$\begin{aligned} (\Delta p)_{\max} &= w \times \frac{V_1^2}{2g} \left[2 \times \left(\frac{1}{\sqrt{2}}\right)^2 - 2 \left(\frac{1}{\sqrt{2}}\right)^4 \right] \\ &= \frac{wV_1^2}{2g} (1 - 0.5) = \frac{0.5wV_1^2}{2g} \text{ (Ans.)} \end{aligned}$$

12.4.2. Loss of Head due to Sudden Contraction

Due to sudden contraction, the stream lines converge to a minimum cross-section called the *vena-contracta* and then expand to fill the downstream pipe (Fig. 12.5.)

Let, A_c = Area of flow at section C-C,

V_c = Velocity of flow at section C-C,

A_2 = Area of flow at section 2-2,

V_2 = Velocity of flow at section 2-2, and

h_c = Loss of head due to sudden contraction.

Loss of head due to sudden contraction
= Loss up to vena-contracta + loss due to sudden enlargement beyond vena-contracta

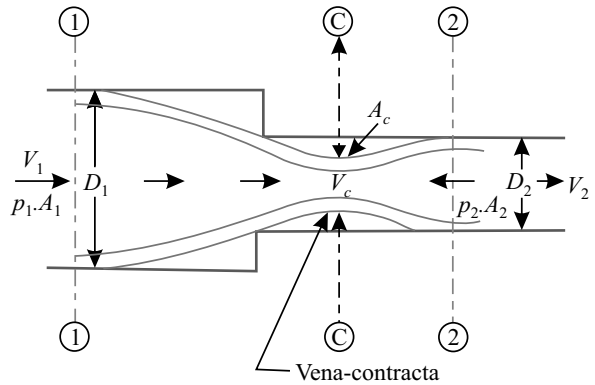


Fig. 12.5

or, $h_c = \text{Negligibly small} + \frac{(V_c - V_2)^2}{2g} \quad \dots (i)$

From continuity equation, we have:

$$A_c V_c = A_2 V_2$$

or, $\frac{V_c}{V_2} = \frac{A_2}{A_c} = \frac{1}{(A_c/A_2)} = \frac{1}{C_c} \quad \left(\because C_c = \frac{A_c}{A_2} \right)$

or, $V_c = \frac{V_2}{C_c}$

Substituting the value of V_c in eqn. (i), we get:

$$h_c = \frac{\left(\frac{V_2}{C_c} - V_2\right)^2}{2g} = \frac{V_2^2}{2g} \left(\frac{1}{C_c} - 1\right)^2$$

i.e., $h_c = \frac{V_2^2}{2g} \left(\frac{1}{C_c} - 1\right)^2 \quad \dots (12.3)$

In general,
$$h_c = k \frac{V_2^2}{2g}$$

where,
$$k = \left(\frac{1}{C_c} - 1 \right)^2$$

From experiments : $C_c = 0.62 + 0.38 \left(\frac{A_2}{A_1} \right)^3$

and thus the loss co-efficient k is a function of ratio

$$\frac{A_1}{A_2} \text{ or } \frac{D_2}{D_1}$$

and, $k = 0.375$ for $C_c = 0.62$.

For gradual contraction (conical reducers) k is a function of cone angle and ≈ 0.1 .

Note : If the value of C_c is not given then loss of head due to contraction may be taken as $0.5 \frac{V_2^2}{2g}$

i.e.,
$$h_e = 0.5 \frac{V_2^2}{2g} \quad \dots(12.4)$$

Example 12.13. A horizontal pipe carries water at the rate of $0.04 \text{ m}^3/\text{s}$. Its diameter, which is 300 mm reduces abruptly to 150 mm. Calculate the pressure loss across the contraction. Take the co-efficient of contraction = 0.62.

Solution. Diameter of the large pipe, $D_1 = 300 \text{ mm} = 0.3 \text{ m}$

$\therefore$ Area, $A_1 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$

Diameter of the small pipe, $D_2 = 150 \text{ mm} = 0.15 \text{ m}$

$\therefore$ Area, $A_2 = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$

Discharge, $Q = 0.04 \text{ m}^3/\text{s}$.

Co-efficient of contraction, $C_c = 0.62$

Pressure loss across the contraction, $(p_1 - p_2)$:

From continuity equation, we have:

$$A_1 V_1 = A_2 V_2 = Q$$

$\therefore V_1 = \frac{Q}{A_1} = \frac{0.04}{0.0707} = 0.566 \text{ m/s}$

and, $V_2 = \frac{Q}{A_2} = \frac{0.04}{0.01767} = 2.26 \text{ m/s}$

Applying Bernoulli's equation before and after contraction, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_c \quad \dots(i)$$

But, $z_1 = z_2$...because the pipe is horizontal and head loss due to contraction (h_c) is given as :

$$h_c = \left[\frac{1}{C_c} - 1 \right]^2 \frac{V_2^2}{2g} = \left[\frac{1}{0.62} - 1 \right]^2 \times \frac{2.26^2}{2 \times 9.81} = 0.0978$$

Substituting these values in eqn. (i), we get:

$$\frac{p_1}{w} + \frac{0.566^2}{2 \times 9.81} = \frac{p_2}{w} + \frac{2.26^2}{2 \times 9.81} + 0.0978$$

$$\therefore \frac{p_1}{w} - \frac{p_2}{w} = \frac{2.26^2}{2 \times 9.81} + 0.0978 - \frac{0.566^2}{2 \times 9.81}$$

$$= 0.26 + 0.0978 - 0.016 = 0.3418$$

Hence, $p_1 - p_2 = w \times 0.3418 = 9.81 \times 0.3418$

$$= \mathbf{3.35 \text{ kN/m}^2} \quad (\text{Ans.})$$

Example 12.14. A pipe of diameter 225 mm is attached to a 150 mm diameter pipe by means of a flange in such a manner that the axes of the two pipes are in a straight line. Water flows through the arrangement at the rate of $0.05 \text{ m}^3/\text{s}$. The pressure loss at the transition as indicated by differential gauge length on a water-mercury manometer connected between two pipes equals 35 mm. Calculate :

- (i) The loss of head due to contraction, and
(ii) The co-efficient of contraction.

Solution. Diameter of the large pipe, $D_1 = 225 \text{ mm} = 0.225 \text{ m}$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.225^2 = 0.03976 \text{ m}^2$$

Diameter of the small pipe, $D_2 = 150 \text{ mm} = 0.15 \text{ m}$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$$

Discharge, $Q = 0.05 \text{ m}^3/\text{s}$

Reading of the differential gauge, $h = 35 \text{ mm} = 0.035 \text{ m}$

- (i) **Loss of head due to contraction h_c :**

When the water-mercury manometer is connected across the contracted transition, then

$$\frac{p_1 - p_2}{w} = h \left(\frac{S_m}{S_w} - 1 \right)$$

where, S_m = Sp. gr. of mercury (= 13.6), and

S_w = Sp. gr. of water (= 1).

Substituting the values in the above eqn., we get:

$$\frac{p_1 - p_2}{w} = 0.035 \left(\frac{13.6}{1.0} - 1 \right) = 0.441 \text{ m}$$

Let V_1 and V_2 be the velocities of flow in the large diameter and small diameter pipes respectively, then:

$$V_1 = \frac{Q}{A_1} = \frac{0.05}{0.03976} = 1.26 \text{ m/s, and}$$

$$V_2 = \frac{Q}{A_2} = \frac{0.05}{0.01767} = 2.83 \text{ m/s}$$

Invoking Bernoulli's equation, we have:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_c$$

$$\left[\begin{array}{l} \text{where, } h_c = \text{head loss due to contraction, and} \\ z_1 = z_2 \dots\dots\dots \text{the pipe being horizontal} \end{array} \right]$$

or,
$$h_c = \left(\frac{p_1 - p_2}{w} \right) + \frac{V_1^2 - V_2^2}{2g}$$

$$= 0.441 + \frac{(1.26)^2 - (2.83)^2}{2 \times 9.81} = \mathbf{0.114 \text{ m of water (Ans.)}$$

(ii) **The co-efficient of contraction, C_c :**

The loss of head due to contraction is given by:

$$h_c = \left(\frac{1}{C_c} - 1 \right)^2 \times \frac{V_2^2}{2g}$$

$$\text{or,} \quad 0.114 = \left(\frac{1}{C_c} - 1 \right)^2 \times \frac{2.83^2}{2 \times 9.81}$$

From which, $C_c = \mathbf{0.65 \text{ (Ans.)}}$

Example 12.15. When a sudden contraction is introduced in a horizontal pipeline from 500 mm diameter to 250 mm diameter, the pressure changes from 105 kN/m² to 69 kN/m². If the co-efficient of contraction is assumed to be 0.65, calculate the water flow rate.

Following this if there is sudden enlargement from 250 mm to 500 mm and if the pressure at the 250 mm section is 69 kN/m², what is the pressure at the 500 mm enlarged portion ? [Roorkee University]

Solution. Diameter of the large pipe, $D_1 = 500 \text{ mm} = 0.5 \text{ m}$

$$\therefore \quad \text{Area, } A_1 = \frac{\pi}{4} \times 0.5^2 = 0.1963 \text{ m}^2$$

Diameter of the small pipe, $D_2 = 250 \text{ mm} = 0.25 \text{ m}$

$$\therefore \quad \text{Area, } A_2 = \frac{\pi}{4} \times 0.25^2 = 0.04908 \text{ m}^2$$

Pressure inside the large pipe, $p_1 = 105 \text{ kN/m}^2$

Pressure inside the small pipe, $p_2 = 69 \text{ kN/m}^2$

Co-efficient of contraction, $C_c = 0.65$

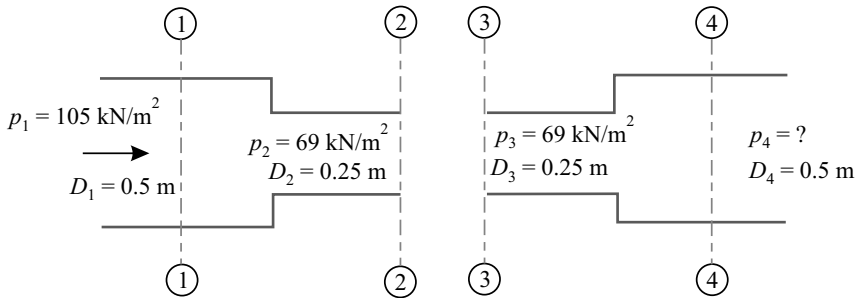


Fig. 12.6

(i) **Flow rate, Q :**

Head lost due to contraction is given by:

$$h_c = \frac{V_2^2}{2g} \left(\frac{1}{0.65} - 1.0 \right)^2 = \frac{V_2^2}{2g} \left[\frac{1}{0.65} - 1.0 \right]^2 \quad [\text{Eqn. (12.3)}]$$

$$= 0.2899 \frac{V_2^2}{2g} \quad \dots(i)$$

From continuity considerations, we have:

$$A_1 V_1 = A_2 V_2$$

$$V_1 = \frac{A_2}{A_1} \times V_2 = \frac{(\pi/4) \times D_2^2}{(\pi/4) \times D_1^2} \times V_2$$

$$\text{or,} \quad V_1 = \left(\frac{0.25}{0.50} \right)^2 \times V_2 = \frac{V_2}{4}$$

Applying Bernoulli's equation at 1-1 and 2-2, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_c$$

But, $z_1 = z_2$...the pipe being horizontal.

$$\therefore \frac{p_1}{w} + \frac{V_1^2}{2g} = \frac{p_2}{w} + \frac{V_2^2}{2g} + h_c$$

Substituting the values, we get:

$$\frac{105}{9.81} + \frac{(V_2/4)^2}{2 \times 9.81} = \frac{69}{9.81} + \frac{V_2^2}{2 \times 9.81} + 0.2899 \frac{V_2^2}{2 \times 9.81}$$

$$\text{or,} \quad 210 + \frac{V_2^2}{16} = 138 + V_2^2 + 0.2899 V_2^2$$

$$\text{or,} \quad 72 = 1.2899 V_2^2 - \frac{V_2^2}{16} = 1.2274 V_2^2$$

$$V_2 = 7.66 \text{ m/s}$$

Hence, rate of flow, $Q = A_2 V_2 = 0.04908 \times 7.66 = \mathbf{0.376 \text{ m}^3/\text{s}}$ (Ans.)

(ii) **Pressure at the enlarged section, p_4 :**

Applying Bernoulli's equation at the sections 3-3 and 4-4, we get:

$$\frac{p_3}{w} + \frac{V_3^2}{2g} + z_3 = \frac{p_4}{w} + \frac{V_4^2}{2g} + z_4 + h_e \text{ (loss of head due to sudden enlargement)}$$

But, $p_3 = 69 \text{ kN/m}^2$

$$V_3 = V_2 = 7.66 \text{ m/s}$$

$$V_4 = V_1 = \frac{V_2}{4} = \frac{7.66}{4} = 1.915 \text{ m/s}$$

$$z_3 = z_4$$

$$\text{And,} \quad h_e = \frac{(V_3 - V_4)^2}{2g} = \frac{(7.66 - 1.915)^2}{2 \times 9.81} \text{ m} = 1.68 \text{ m}$$

Substituting the values in the above equation, we get:

$$\frac{69}{9.81} + \frac{7.66^2}{2 \times 9.81} = \frac{p_4}{9.81} + \frac{(1.915)^2}{2 \times 9.81} + 1.68$$

$$7.033 + 2.99 = \frac{p_4}{9.81} + 0.187 + 1.68$$

$$\text{or,} \quad p_4 = \mathbf{80 \text{ kN/m}^2} \text{ (Ans.)}$$

12.4.3 Loss of Head due to Obstruction in Pipe

Refer to Fig. 12.7. The loss of energy due to an obstruction in pipe takes place on account of the reduction in the cross-sectional area of the pipe by the presence of obstruction which is followed by an abrupt enlargement of the stream beyond the obstruction.

Head loss due to obstruction ($h_{\text{obs.}}$) is given by the relation :

$$h_{obs} = \left[\frac{A}{C_c (A - a)} \right]^2 \frac{V^2}{2g} \quad \dots(12.5)$$

where, A = Area of the pipe,
 a = Maximum area of obstruction, and
 V = Velocity of liquid in pipe.

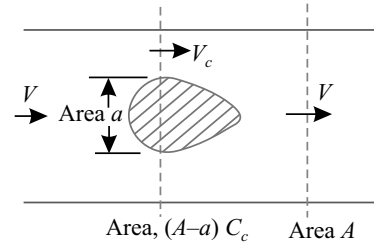


Fig. 12.7

12.4.4 Loss of Head at the Entrance to Pipe

Loss of head at the entrance to pipe (h_i) is given by the relation :

$$h_i = 0.5 \frac{V^2}{2g} \quad \dots(12.6)$$

where, V = Velocity of liquid in pipe.

12.4.5 Loss of Head at the Exit of a Pipe

Loss of head at the exit of a pipe is denoted by h_0 and is given by the relation:

$$h_0 = \frac{V^2}{2g} \quad \dots(12.7)$$

where, V = Velocity at outlet of pipe.

12.4.6 Loss of Head due to Bend in the Pipe

In general the loss of head in bends (h_b) provided in pipes may be expressed as :

$$h_0 = k \frac{V^2}{2g} \quad \dots(12.8)$$

where, V = Mean velocity of flow of fluid, and

and, k = Co-efficient of bend; it depends upon *angle of bend, radius of curvature of bend and diameter of pipe*.

12.4.7 Loss of Head in Various Pipe Fittings

The loss of head in the various pipe fittings (such as valves, couplings, etc.) may also be represented as :

$$h_{\text{fittings}} = k \frac{V^2}{2g}$$

where, V = Mean velocity flow in the pipe, and k = value of the co-efficient; it depends on the type of the pipe fitting.

12.5. HYDRAULIC GRADIENT AND TOTAL ENERGY LINES

The concept of hydraulic gradient line and total energy line is quite useful in the study of flow of fluid in pipes. These lines may be obtained as indicated below.

Total Energy Line (T.E.L. or E.G.L.):

It is known that the *total head* (which is also total energy per unit weight) with respect to any arbitrary datum, is the sum of the elevation (potential) head, pressure head and velocity head, *i.e.*,

$$\text{Total head} = \frac{p}{w} + z + \frac{V^2}{2g}$$

When the fluid flows along the pipe, there is loss of head (energy) and the total energy decreases in the direction of flow. If the total energy at various points along the axis of the pipe is plotted and joined by a line, the line so obtained is called the '**Energy gradient line**' (E.G.L.).

In literature, energy gradient line (E.G.L.) is also known as '**Total energy line**' (T.E.L.).

Hydraulic Gradient Line (H.G.L.):

The sum of potential (or elevation) head and the pressure head $\left(\frac{p}{w} + z\right)$ at any point is called the *piezometric head*. If a line is drawn joining the piezometric levels at various points, the line so obtained is called the '**Hydraulic gradient line.**'

The following points are worth noting :

1. Energy gradient line (E.G.L.) *always drops in the direction of flow because of loss of head.*
2. Hydraulic gradient line (H.G.L.) may *rise or fall* depending on the pressure changes.
3. Hydraulic gradient line (H.G.L.) is *always below the energy gradient line (E.G.L.)* and the vertical intercept between the two is equal to the velocity head $\left(\frac{V^2}{2g}\right)$.
4. For a pipe of *uniform cross-section the slope of the hydraulic gradient line is equal to the slope of energy gradient line.*
5. *There is no relation whatsoever between the slope of energy gradient line and the slope of the axis of the pipe.*

Example 12.16. A horizontal pipe line 40 m long is connected to a water tank at one end and discharges freely into the atmosphere at the other end. For the first 25 m of its length from the tank, the pipe is 150 mm diameter and its diameter is suddenly enlarged to 300 mm. The height of water level in the tank is 8 m above the centre of the pipe. Considering all losses of head which occur,

(i) Determine the rate of flow.

(ii) Draw the hydraulic gradient and energy gradient lines. Take $f = 0.01$ for both sections of the pipe. [M.U.]

Solution. Total length of the horizontal pipeline, $L = 40$ m.

Length of first pipe $L_1 = 25$ m

Diameter of first pipe $D_1 = 50$ mm = 0.15 m

Length of second pipe, $L_2 = 40 - 25 = 15$ m

Diameter of second pipe, $D_2 = 300$ mm = 0.3 m

Height of water, $H = 8$ m

Co-efficient of friction, $f = 0.01$

(i) **Rate of flow, Q :**

Applying Bernoulli's equation to the free water surface (F.W.S.) in the tank and outlet of the pipe as shown in Fig. 12.8., we get:

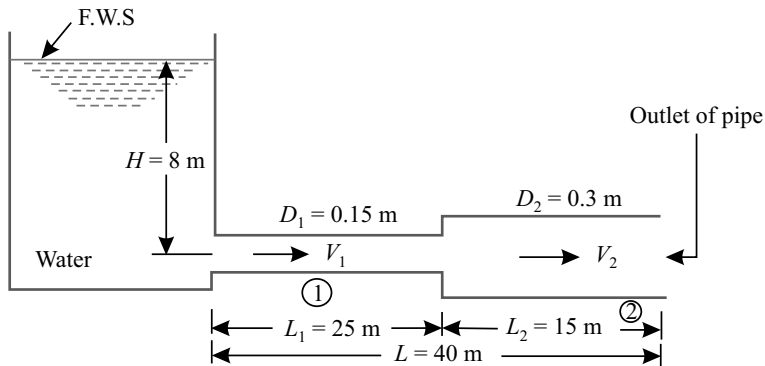


Fig. 12.8

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + \text{all losses}$$

$$0 + 0 + 8.0 = 0 + \frac{V_2^2}{2g} + 0 + h_i + h_{f_1} + h_e + h_{f_2} \quad \dots(i)$$

where,

V_2 = Velocity of water at the outlet of pipe,

h_i = Loss of head at entrance = $0.5 \frac{V_1^2}{2g}$,

h_{f_1} = Head lost due to friction in pipe 1 = $\frac{4fL_1V_1^2}{D_1 \times 2g}$,

h_e = Loss of head due to sudden enlargement = $\frac{(V_1 - V_2)^2}{2g}$, and

h_{f_2} = Head lost due to friction in pipe 2 = $\frac{4fL_2V_2^2}{D_2 \times 2g}$

From continuity equation, we have:

$$A_1V_1 = A_2V_2$$

$\therefore$

$$\begin{aligned} V_1 &= \frac{A_2V_2}{A_1} = \frac{(\pi/4) \times D_2^2 \times V_2}{(\pi/4) \times D_1^2} \\ &= \left(\frac{D_2}{D_1}\right)^2 \times V_2 = \left(\frac{0.3}{0.15}\right)^2 \times V_2 = 4V_2 \end{aligned}$$

Substituting the value of V_1 in different head losses, we have:

$$h_i = \frac{0.5 V_1^2}{2g} = \frac{0.5 (4 \times V_2)^2}{2g} = \frac{8V_2^2}{2g}$$

$$h_{f_1} = \frac{4fL_1V_1^2}{D_1 \times 2g} = \frac{4 \times 0.01 \times 25 \times (4 \times V_2)^2}{0.15 \times 2g} = 106.6 \frac{V_2^2}{2g}$$

$$h_e = \frac{(V_1 - V_2)^2}{2g} = \frac{(4V_2 - V_2)^2}{2g} = \frac{9V_2^2}{2g}$$

$$h_{f_2} = \frac{4fL_2V_2^2}{D_2 \times 2g} = \frac{4 \times 0.01 \times 15 \times V_2^2}{0.3 \times 2g} = \frac{2V_2^2}{2g}$$

Substituting the values of these losses in eqn. (i), we get:

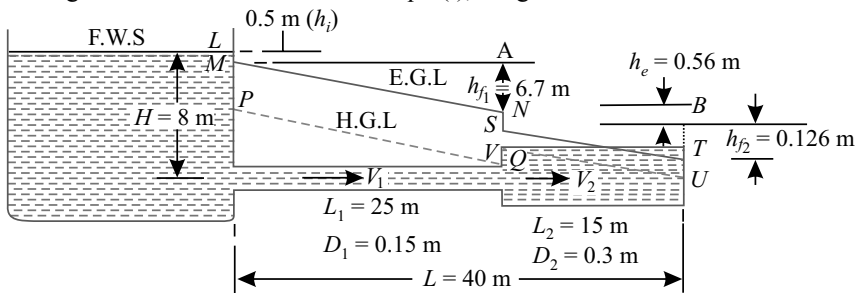


Fig. 12.9

$$\begin{aligned} 8 &= \frac{V_2^2}{2g} + \frac{8V_2^2}{2g} + \frac{106.6 V_2^2}{2g} + \frac{9V_2^2}{2g} + \frac{2V_2^2}{2g} \\ &= \frac{V_2^2}{2g} (1 + 8 + 106.6 + 9 + 2) = 126.6 \frac{V_2^2}{2g} \end{aligned}$$

$$\therefore V_2 = \sqrt{\frac{8 \times 2g}{126.6}} = \sqrt{\frac{8 \times 2 \times 9.81}{126.6}} = 1.11 \text{ m/s}$$

Hence, Rate of flow, $Q = A_2 V_2 = \frac{\pi}{4} \times 0.3^2 \times 1.11 = \mathbf{0.078 \text{ m}^3/\text{s}}$ (Ans.)

(ii) **E.G.L. and H.G.L. :**

The various head losses are : (Refer to Fig. 12-9)

$$\begin{aligned} h_i &= \frac{8V_2^2}{2g} = \frac{8 \times (1.11)^2}{2 \times 9.81} = 0.5 \text{ m} \\ h_{f1} &= 106.6 \frac{V_2^2}{2g} = \frac{106.6 \times (1.11)^2}{2 \times 9.81} = 6.7 \text{ m} \\ h_e &= \frac{9V_2^2}{2g} = \frac{9 \times (1.11)^2}{2 \times 9.81} = 0.56 \text{ m} \\ h_{f2} &= \frac{2V_2^2}{2g} = \frac{2 \times (1.11)^2}{2 \times 9.81} = 0.126 \text{ m} \end{aligned}$$

To draw E.G.L. and H.G.L. the following *procedure* is followed.

E.G.L. (Energy gradient line) :

The point L lies on F.W.S. (free water surface).

- Take $LM = h_i = 0.5 \text{ m}$
- From M draw a horizontal line. Taking MA equal to the length of the pipe (*i.e.*, L_1) draw a vertical line downward from the point A . Cut $AN = h_{f1} = 6.7 \text{ m}$
- Join MN
- From N , draw a line NS vertically downward equal to $h_e (= 0.56 \text{ m})$
- From S , draw SB horizontal and from point U (which is lying on the centre of the pipe) draw a vertical line in the upward direction, meeting at B . From B take $BT = h_{f2} = 0.126 \text{ m}$.
- Join ST
- The line $LMNST$ represents the energy gradient line (E.G.L.)

H.G.L. (Hydraulic gradient line) :

- From M , take $MP = \frac{V_1^2}{2g} = \frac{(4 \times 1.11)^2}{2 \times 9.81} = 1.0 \text{ m}$ ($\because V_1 = 4V_2$)
- Draw the line PQ parallel to the line MN
- From the point U , draw a line UV parallel to the line TS
- Join QV
- The line $PQVU$ represents the hydraulic gradient line (H.G.L.).

Example 12.17. Two reservoirs are connected by a pipeline consisting of two pipes, one of 15 cm diameter and length 6 m and the other of diameter 22.5 cm and 16 m length. If the difference of water levels in the two reservoirs is 6 m, calculate the discharge and draw the energy gradient line. Take $f = 0.04$. (Delhi University)

Solution. Refer to Fig. 12.10. Given : $D_1 = 15 \text{ cm} = 0.15 \text{ m}$; $L_1 = 6 \text{ m}$; $D_2 = 22.5 \text{ cm} = 0.225 \text{ m}$; $L_2 = 16 \text{ m}$; total losses = 6 m; $f = 0.04$.

From the continuing equation, we have: $A_1 V_1 = A_2 V_2$

or, $\frac{\pi}{4} \times 0.15^2 \times V_1 = \frac{\pi}{4} \times 0.225^2 \times V_2 \quad \therefore V_1 = 2.25 V_2 \quad \dots(i)$

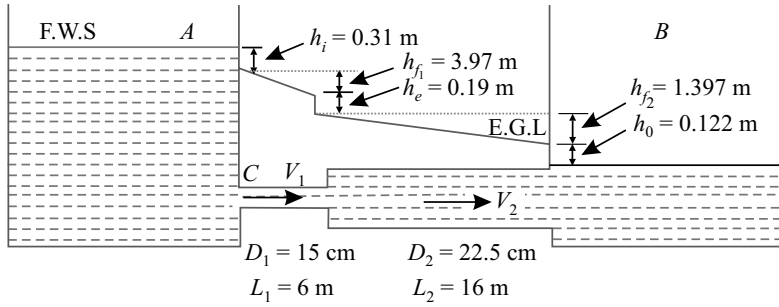


Fig. 12.10

Loss of head at entrance to a pipe, $h_i = \frac{0.5 V_1^2}{2g}$

Loss of head due to friction in pipe AB,

$$h_{f1} = \frac{4fL_1V_1^2}{D_1 \times 2g} = \frac{4 \times 0.04 \times 6 \times V_1^2}{0.15 \times 2g} = 6.4 \frac{V_1^2}{2g}$$

Loss of head due to sudden enlargement,

$$h_e = \frac{(V_1 - V_2)^2}{2g} = \frac{\left(V_1 - \frac{V_1}{2.25}\right)^2}{2g} = 0.308 \frac{V_1^2}{2g}$$

Loss of head due to friction in the pipe, BC,

$$h_{f2} = \frac{4fL_2V_2^2}{D_2 \times 2g} = \frac{4 \times 0.04 \times 16 \times \left(\frac{V_1}{2.25}\right)^2}{0.225 \times 2g} = 2.25 \frac{V_1^2}{2g}$$

Loss of head due to friction in the pipe BC,

$$h_0 = \frac{V_2^2}{2g} = \left(\frac{V_1}{2.25}\right)^2 \times \frac{1}{2g} = 0.197 \frac{V_1^2}{2g}$$

Applying Bernoulli's equation to free water surface (F.W.S.) in the two tanks, we have:

$$\frac{p_A}{w} + \frac{V_A^2}{2g} + z_A = \frac{p_B}{w} + \frac{V_B^2}{2g} + z_B + \text{losses}$$

$$\text{i.e., } p_A = p_B = 0, V_A = V_B = 0, z_A - z_B = 6 \text{ m}$$

Hence, Total losses = 6 m

$$\text{i.e., } h_i + h_{f1} + h_e + h_{f2} + h_0 = 6$$

$$\frac{0.5V_1^2}{2g} + 6.4 \frac{V_1^2}{2g} + 0.308 \frac{V_1^2}{2g} + 2.25 \frac{V_1^2}{2g} + 0.197 \frac{V_1^2}{2g} = 6$$

$$V_1 = 3.49 \text{ m/s}$$

$$\text{Discharge, } Q = A_1 V_1 = \frac{\pi}{4} \times 0.15^2 \times 3.49 = 0.0617 \text{ m}^3/\text{s (Ans.)}$$

Energy gradient line is shown in the Fig. 12-10.

Example 12.18. A pipe ABC connecting two reservoirs is 80 mm in diameter. From A to B the pipe is horizontal as shown in Fig. 12-11. and from B to C it falls by 3.5 metres. The lengths AB and BC are 25 m and 15 m respectively. If the water surface in the reservoir at A is 4 m above the centre-line of the pipe and at C 1 m above the line of the pipe, calculate :

(i) The rate of flow, and

(ii) The pressure head in the pipe at B.

Neglect the loss at the bend but consider all other losses. Also draw the energy and hydraulic gradient lines. Take Darcy friction factor = 0.024 and $K_{\text{entrance}} = 0.5$. [IIT Delhi]

Solution. Diameter of the pipe, $D = 80 \text{ mm} = 0.08 \text{ m}$

$$\text{Area, } A = \frac{\pi}{4} \times 0.08^2 = 0.005026 \text{ m}^2$$

$$\text{Friction factor } (= 4f) = 0.024$$

$$K_{\text{entrance}} = 0.5$$

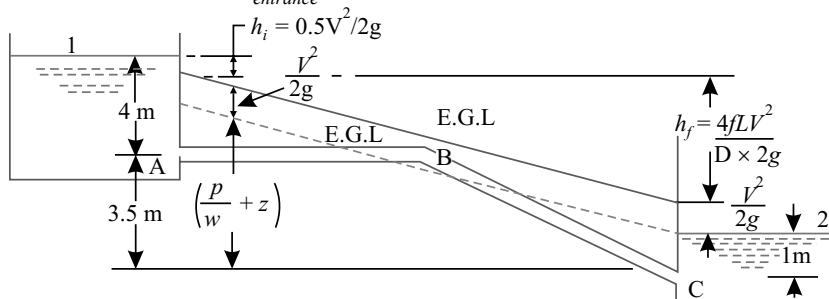


Fig. 12.11

(i) The rate of flow, Q :

Applying Bernoulli's equation between the water surfaces 1 and 2 in the two reservoirs (considering horizontal plane through C as datum), we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + \text{loss at entrance} + h_f \text{ (loss due to friction)} + \frac{V^2}{2g}$$

$$0 + 0 + (4 + 3.5 - 1) = 0 + 0 + 0 + \frac{0.5 V^2}{2g} + \frac{4fLV^2}{D \times 2g} + \frac{V^2}{2g}$$

(where, V = velocity of flow in the pipe)

$$\text{or, } 6.5 = \frac{0.5 V^2}{2g} + \frac{0.024 \times (25 + 15) \times V^2}{0.08 \times 2g} + \frac{V^2}{2g}$$

$$\text{or, } = \frac{V^2}{2g} (0.5 + 12 + 1) + 13.5 \frac{V^2}{2g}$$

$$\text{or, } V^2 = \frac{6.5 \times 2 \times 9.81}{13.5} = 9.446$$

$$\text{or, } V = 3.073 \text{ m/s (Ans.)}$$

$$\therefore \text{Flow rate} = A \times V = 0.005026 \times 3.073 = 0.01544 \text{ m}^3/\text{s (Ans.)}$$

(ii) Pressure head in the pipe at B, $\frac{p_B}{w}$:

Applying Bernoulli's equation at A and B, we get:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_B}{w} + \frac{V_B^2}{2g} + z_B + \frac{0.5 V_B^2}{2g} + h_f$$

$$0 + 0 + = \frac{p_B}{w} + \frac{V^2}{2g} + 0 + \frac{0.5 V^2}{2g} + \frac{4fL_{AB}V^2}{D \times 2g}$$

$$4 = \frac{p_B}{w} + \frac{V^2}{2g} + \frac{0.5 V^2}{2g} + \frac{0.024 \times 25 \times V^2}{0.08 \times 2g} \quad (\because V_B = V = 3.073 \text{ m/s})$$

$$4 = \frac{p_B}{w} + \frac{V^2}{2g} + \frac{0.5V^2}{2g} + \frac{7.5V^2}{2g}$$

$$= \frac{p_B}{w} + \frac{9V^2}{2g}$$

or,
$$\frac{p_B}{w} = 4 - \frac{9V^2}{2g} = 4 - \frac{9 \times (3.073)^2}{2 \times 9.81}$$

= **-0.33 m of water** (below atmosphere) (Ans.)

Energy gradient and hydraulic gradient lines (E.G.L. and H.G.L.) :

— For plotting E.G.L. and H.G.L., we require the velocity head, (same throughout)

$$\frac{V^2}{2g} = \frac{(3.073)^2}{2 \times 9.81} = 0.481 \text{ m}$$

— Total energy at B w.r.t. horizontal datum through C

$$= 3.5 + \frac{p_B}{w} + \frac{V^2}{2g} = 3.5 - 0.33 + \frac{(3.073)^2}{2 \times 9.81} = 3.65 \text{ m}$$

Energy gradient and hydraulic gradient lines are shown firm and dotted respectively in the Fig. 12.11; H.G.L. below the pipeline near B indicates that pressure is negative.

Example 12.19. Two reservoirs A and C having a difference of level of 15.5 m are connected by a pipeline ABC, the elevation of point B being 4.0 m below the level of water in reservoir A. The length AB of the pipeline is 250 m, the pipe being made of mild steel having a friction co-efficient f_1 , while the length BC is 450 m, the pipe having made of cast-iron having a friction co-efficient f_2 . Both the lengths AB and BC have a diameter of 200 mm. A partially closed valve is located in the length BC at a distance of 150 m from reservoir C.

If the flow through the pipeline is $3 \text{ m}^3/\text{min}$, the pressure head at B is 0.5 m and the head loss at the valve is 5.0 m.

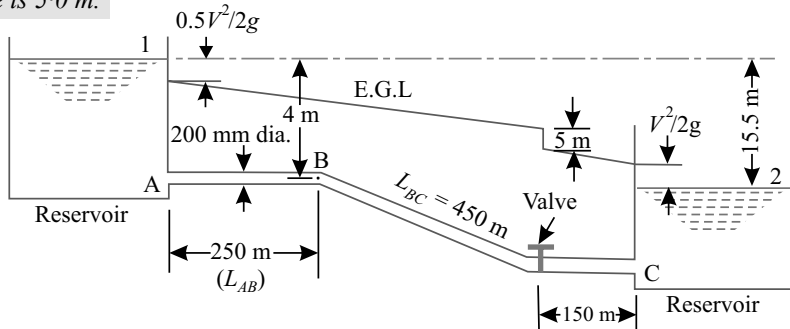


Fig. 12.12

(i) Find the friction coefficients f_1 and f_2 .

(ii) Draw the hydraulic grade line of the pipeline and indicate on the diagram head loss values at significant points. Take into account head loss at entrance and exit points of the pipeline.

[UPTU]

Solution. Difference of water level between two reservoirs = 15.5 m

Diameter of the pipe ABC, $D = 200 \text{ mm} = 0.2 \text{ m}$

Length AB, $L_{AB} = 250 \text{ m}$

Length BC, $L_{BC} = 450 \text{ m}$

Discharge through the pipe, $Q = 3 \text{ m}^3/\text{min} = 0.05 \text{ m}^3/\text{s}$

$$\text{Pressure head at } B, h_B = \left(= \frac{p_B}{w} \right) = 0.5 \text{ m}$$

Head loss at the valve = 5.0 m

(i) **Friction co-efficients f_1 and f_2 :**

$$\text{Velocity in the pipe } ABC, V = \frac{Q}{\text{Area}} = \frac{0.05}{\frac{\pi}{4} \times 0.2^2} = 1.59 \text{ m/s}$$

Applying Bernoulli's equation at '1' and at 'B', we get:

$$\begin{aligned} \frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 &= \frac{p_B}{w} + \frac{V_B^2}{2g} + z_2 + \frac{0.5V_B^2}{2g} + (h_f)_{AB} \\ 0 + 0 + 4 &= 0.5 + \frac{V_B^2}{2g} + 0 + \frac{0.5V_B^2}{2g} + \frac{4f_1L_{AB}V_B^2}{D \times 2g} \\ 4 &= 0.5 + \frac{V^2}{2g} + \frac{0.5V^2}{2g} + \frac{4f_1 \times 250 \times V^2}{0.2 \times 2g} \quad (\because V_B = V) \end{aligned}$$

$$\text{or,} \quad 4 = 0.5 + \frac{(1.59)^2}{2 \times 9.81} + \frac{0.5 \times (1.59)^2}{2 \times 9.81} + \frac{4f_1 \times 250 \times (1.59)^2}{0.2 \times 2 \times 9.81}$$

$$= 0.5 + 0.129 + 0.0644 + 644.3 f_1$$

$$\text{or,} \quad f_1 = \mathbf{0.0051 \text{ (Ans.)}}$$

Applying Bernoulli's equation between '1' and '2' and considering all losses in the pipeline ABC including the exit loss, we have:

$$\begin{aligned} \frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 &= \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + \frac{0.5V^2}{2g} + \frac{4f_1L_{AB}V^2}{D \times 2g} + \frac{4f_2L_{BC}V^2}{D \times 2g} + 5.0 + \frac{V^2}{2g} \\ 0 + 0 + 15.5 &= 0 + 0 + 0 + \frac{0.5 \times (1.59)^2}{2 \times 9.81} + \frac{4 \times 0.0051 \times 250 \times (1.59)^2}{0.2 \times 2 \times 9.81} \\ &\quad + \frac{4f_2 \times 450 \times (1.59)^2}{0.2 \times 2 \times 9.81} + 5.0 + \frac{(1.59)^2}{2 \times 9.81} \end{aligned}$$

$$\text{or,} \quad 15.5 = 0.0644 + 3.28 + 1159.6 f_2 + 5.0 + 0.1288$$

$$\text{or,} \quad f_2 = \mathbf{0.00606 \text{ (Ans.)}}$$

(ii) **H.G.L. (hydraulic gradient line) :**

Fig. 12.9 shows the E.G.L. (energy gradient line), H.G.L. will be $\frac{V^2}{2g}$ below the E.G.L.

Example 12.20. Water is being pumped at the rate of $0.02 \text{ m}^3/\text{s}$ to an overhead tank through a 150 mm diameter 300 m long delivery pipe. In the tank, the pipe discharges freely at height of 15 m above the pump. If the Darcy-Weisbach friction factor = 0.03 for the pipe, determine :

(i) The pressure developed by the pump on its delivery side, and

(ii) The power delivered to water by the pump.

Draw also the hydraulic gradient from the pump to the tank. Assume that the first 285 m of the delivery pipe is horizontal and the rest is vertical. [Delhi University]

Solution. Rate of flow, $Q = 0.02 \text{ m}^3/\text{s}$

Diameter of the pipe, $D = 150 \text{ mm} = 0.15 \text{ m}$

Length of the pipe, $L = 300 \text{ m}$

Darcy-Weisbach friction factor ($4f$) = 0.03

(i) The pressure developed by the pump on its delivery side :

$$\text{Velocity of flow, } V = \frac{Q}{\text{Area}} = \frac{0.02}{\frac{\pi}{4} \times 0.15^2} = 1.132 \text{ m/s}$$

Let, p = Pressure (gauge) just on the delivery side of the pump.

Applying Bernoulli's equation to the section just on the delivery side of the pump and to the discharge end of the pipeline where the gauge pressure is zero *i.e.*, sections 1 and 2, we have:

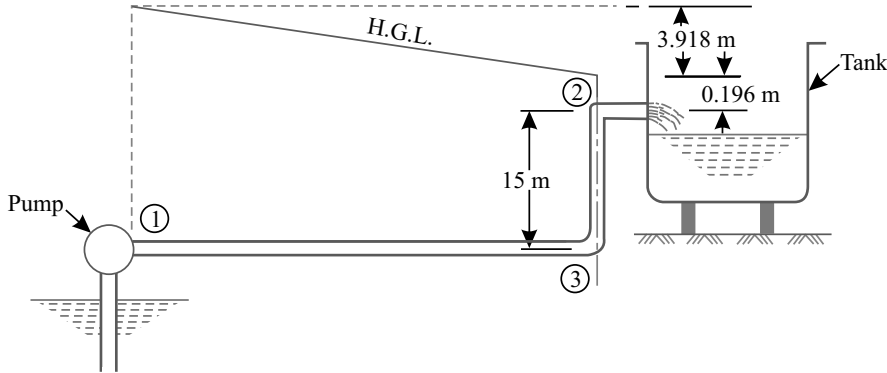


Fig. 12.13

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + h_f \quad (h_f = \text{head loss due to friction})$$

$$\frac{p_1}{w} + \frac{(1.132)^2}{2 \times 9.81} + 0 = 0 + \frac{(1.132)^2}{2 \times 9.81} + 15 + \frac{0.03 \times 300 \times (1.132)^2}{0.15 \times 2 \times 9.81}$$

$$\left(\because h_f = \frac{fLV^2}{D \times 2g} \text{ where, } L = 285 + 15 = 300 \text{ m} \right)$$

$$\frac{p_1}{w} + 0.0653 = 0.0653 + 15 + 3.918$$

or, $\frac{p_1}{w} = 18.918 \text{ m}$

Hence, the pressure developed by the pump on the delivery side
= **18.918 m of water (Ans.)**

(ii) The power delivered to water by the pump, P :

$$P = wQh_f = 9.81 \times 0.02 \times 3.918 \text{ kW}$$

$$= \mathbf{0.768 \text{ kW (Ans.)}}$$

[Note: The power required to drive the pump = $\frac{wQh_f}{\eta}$, where η is the efficiency of the pump.]

H.G.L. (hydraulic gradient line) :

Head loss in 15 m vertical length of pipeline

$$= \frac{h_f}{300} \times 15 = \frac{3.918}{300} \times 15 = 0.196 \text{ m}$$

Now piezometric head $\left(\frac{p}{w} + z \right)$ at:

Section 1 = 19.114 m (*i.e.*, $18.918 + 0.196 = 19.114 \text{ m}$)

Section 2 = 15.196 m (*i.e.*, $19.114 - 3.918 = 15.196 \text{ m}$)

Section 3 = 15 m (*i.e.*, $15.196 - 0.196 = 15 \text{ m}$)

The HGL from the pump to the overhead tank is plotted by marking the ordinates of piezometric heads at 1, 2 and 3 (as above) and joining these by straight lines as shown in Fig. 12.13.

Example 12.21. A pipeline ABC 200 m long, is laid on an upward slope of 1 in 40. The length of the portion AB is 100 m and its diameter is 100 mm. At B the pipe section suddenly enlarges to 200 mm diameter and remains so for the remainder of its length BC, 100 m. A flow of $0.02 \text{ m}^3/\text{s}$ is pumped into the pipe at its lower end A and is discharged at the upper end C into a closed tank. The pressure at the supply end A is 200 kN/m^2 .

(i) What is the pressure at C?

(ii) Draw the energy gradient and hydraulic gradient lines.

Assume co-efficient of friction $f = 0.008$.

Solution. Length of pipe, $ABC = 200 \text{ m}$
 Slope of the pipe, $= 1 \text{ in } 40$
 Length of pipe AB, $L_{AB} = 100 \text{ m}$
 Diameter of the pipe AB, $D_{AB} = 100 \text{ mm} = 0.1 \text{ m}$
 Length of pipe BC, $L_{BC} = 100 \text{ m}$
 Diameter of the pipe BC, $D_{BC} = 200 \text{ mm} = 0.2 \text{ m}$
 Co-efficient of friction, $f = 0.008$
 Discharge, $Q = 0.02 \text{ m}^3/\text{s}$
 The pressure at the supply end, $p_A = 200 \text{ kN/m}^2$

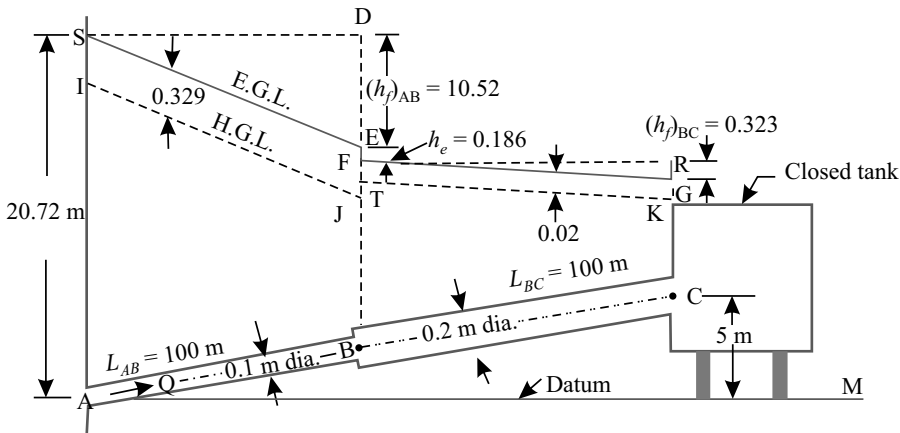


Fig. 12.14

(i) Pressure at C, p_c :

$$\text{Velocity of flow in pipe AB, } V_{AB} = \frac{0.02}{(\pi/4) \times 0.1^2} = 2.54 \text{ m/s}$$

$$\text{Velocity of flow in pipe BC, } V_{BC} = \frac{0.02}{(\pi/4) \times 0.2^2} = 0.63 \text{ m/s}$$

Invoking Bernoulli's equation at points A and C, we have:

$$\frac{p_A}{w} + \frac{V_A^2}{2g} + z_A = \frac{p_C}{w} + \frac{V_C^2}{2g} + z_C + (h_f)_{AB} + h_e + (h_f)_{BC} \quad \dots(i)$$

where,
$$(h_f)_{AB} = \frac{4fL_{AB}V_{AB}^2}{D_{AB} \times 2g} = \frac{4 \times 0.008 \times 100 \times 2.54^2}{0.1 \times 2 \times 9.81} = 10.52 \text{ m}$$

Loss of head due to sudden enlargement,

$$h_e = \frac{(V_{AB} - V_{BC})^2}{2g} = \frac{(2.54 - 0.63)^2}{2 \times 9.81} = 0.186 \text{ m}$$

$$(h_f)_{BC} = \frac{4fL_{BC}V_{BC}^2}{D_{BC} \times 2g} = \frac{4 \times 0.008 \times 100 \times 0.63^2}{0.2 \times 2 \times 9.81} = 0.323 \text{ m}$$

Substituting the values in eqn. (i), we get:

$$\frac{200}{9.81} + \frac{(2.54)^2}{2 \times 9.81} + 0 = \frac{p_C}{w} + \frac{0.63^2}{2 \times 9.81} + 5.0 + 10.52 + 0.186 + 0.323$$

$$20.38 + 0.329 = \frac{p_C}{w} + 0.02 + 16.03$$

or,
$$\frac{p_C}{w} = 4.659 \text{ m}$$

or,
$$p_C = 9.81 \times 4.659 = \mathbf{45.7 \text{ kN/m}^2 \text{ (Ans.)}}$$

(ii) **Energy gradient and hydraulic gradient lines :**

Pipe AB : Assuming the datum line passing through *A*, then total energy at *A*

$$= \frac{p_A}{w} + \frac{V_A^2}{2g} + z_A = \frac{200}{9.81} + \frac{(2.54)^2}{2 \times 9.81} + 0 = 20.72 \text{ m}$$

$$\begin{aligned} \text{Total energy at } B &= \text{Total energy at } A - (h_f)_{AB} \\ &= 20.72 - 10.52 = 10.2 \text{ m} \end{aligned}$$

Also,
$$\frac{V_C^2}{2g} = \frac{(0.63)^2}{2 \times 9.81} = 0.02 \text{ m}$$

Energy gradient line (E.G.L.)

— Draw a horizontal line *AM* as shown in Fig. 12.14.

— Draw the centreline of the pipe in such a way that slope of the pipe is 1 in 40. Thus, point *C* will be at a height of $\frac{1}{40} \times 200 = 5 \text{ m}$ from the line *AM*.

— Draw a vertical line *AS* equal to total energy at *A* i.e., *AS* = 20.72 m

— From point *S*, draw a horizontal line and from point *B*, a vertical line, meeting at *D*.

— From *D*, take vertical distance *DE* = $(h_f)_{AB} = 10.52 \text{ m}$. Join *SE*.

— From *E* take *EF* = $h_e = 0.186 \text{ m}$

— From *F* draw a horizontal line and from *C*, a vertical line meeting at *R*. From *R* take *RG* = $(h_f)_{BC} = 0.323 \text{ m}$. Join *F* to *G*. Then *SEFG* represents the **energy gradient or total energy line**.

Hydraulic gradient line (H.G.L.)

— Draw the line *IJ* parallel to the line *SE* at a distance of $\frac{V_{AB}^2}{2g} = \frac{(2.54)^2}{2 \times 9.81} = 0.329 \text{ m}$ in the downward direction.

— Draw the line *KT* parallel to the line *GF* at a distance of $\frac{V_C^2}{2g} = \frac{(0.63)^2}{2 \times 9.81} = 0.02 \text{ m}$. Join *J* to *T*.

The line *IJTK* represents the **hydraulic gradient line**.

12.6. PIPES IN SERIES OR COMPOUND PIPES

Fig. 12-15 shows a system of pipes in series.

Let, D_1, D_2, D_3 = Diameters of pipes 1, 2 and 3 respectively,

L_1, L_2, L_3 = Lengths of pipes 1, 2 and 3 respectively,

V_1, V_2, V_3 = Velocities of flow through pipes 1, 2 and 3 respectively

f_1, f_2, f_3 = Co-efficients of friction for pipes 1, 2 and 3 respectively, and

H = Difference of water level in the two tanks.

As the rate of flow (Q) of water through each pipe is same, therefore,

$$Q = A_1 V_1 = A_2 V_2 = A_3 V_3$$

Also, The difference in liquid surface levels = Sum of the various head losses in the pipes

$$\text{i.e.,} \quad H = h_i + h_{f_1} + h_c + h_{f_2} + h_e + h_{f_3} + \frac{V_3^2}{2g} \quad \dots(i)$$

$$\text{where,} \quad h_i = \text{Head loss at entrance} = \frac{0.5V_1^2}{2g}$$

$$h_{f_1} = \text{Head loss due to friction in pipe 1} = \frac{4f_1 L_1 V_1^2}{D_1 \times 2g}$$

$$h_c = \text{Head loss at contraction} = \frac{0.5V_2^2}{2g}$$

$$h_{f_2} = \text{Head loss due to friction in pipe 2} = \frac{4f_2 L_2 V_2^2}{D_2 \times 2g}$$

$$h_e = \text{Head loss due to enlargement} = \frac{(V_2 - V_3)^2}{2g}$$

$$h_{f_3} = \text{Head loss due to friction in pipe 3} = \frac{4f_3 L_3 V_3^2}{D_3 \times 2g}$$

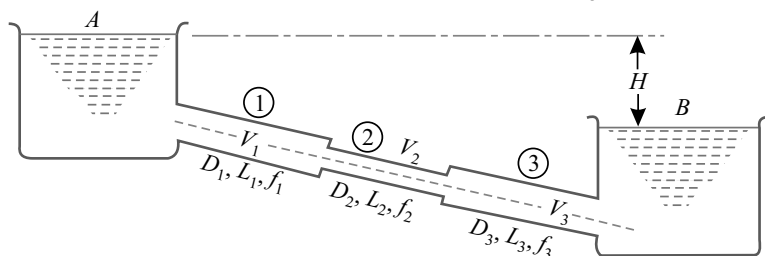


Fig. 12.15. Pipes in series.

Substituting the values in (i), we have:

$$\begin{aligned} H &= h_i + h_{f_1} + h_c + h_{f_2} + h_e + h_{f_3} + \frac{V_3^2}{2g} \\ &= \frac{0.5V_1^2}{2g} + \frac{4f_1 L_1 V_1^2}{D_1 \times 2g} + \frac{0.5V_2^2}{2g} + \frac{4f_2 L_2 V_2^2}{D_2 \times 2g} + \frac{(V_2 - V_3)^2}{2g} + \frac{4f_3 L_3 V_3^2}{D_3 \times 2g} + \frac{V_3^2}{2g} \quad \dots(12-9) \end{aligned}$$

If *minor losses are neglected*, then above equation becomes:

$$H = \frac{4f_1 L_1 V_1^2}{D_1 \times 2g} + \frac{4f_2 L_2 V_2^2}{D_2 \times 2g} + \frac{4f_3 L_3 V_3^2}{D_3 \times 2g} \quad \dots(12-10)$$

If, $f_1 = f_2 = f_3 = f$, then:

$$\begin{aligned}
 H &= \frac{4fL_1V_1^2}{D_1 \times 2g} + \frac{4fL_2V_2^2}{D_2 \times 2g} + \frac{4fL_3V_3^2}{D_3 \times 2g} \\
 &= \frac{4f}{2g} \left[\frac{L_1V_1^2}{D_1} + \frac{L_2V_2^2}{D_2} + \frac{L_3V_3^2}{D_3} \right] \quad \dots(12.11)
 \end{aligned}$$

Example 12.22. Three pipes of diameters 300 mm, 200 mm and 400 mm and lengths 450 m, 255 m and 315 m respectively are connected in series. The difference in water surface levels in two tanks is 18 m. Determine the rate of flow of water if co-efficients of friction are 0.0075, 0.0078 and 0.0072 respectively considering :

(i) Minor losses also, and

(ii) Neglecting minor losses.

Solution. Pipe 1 : $L_1 = 450$ m, $D_1 = 300$ mm = 0.3 m, $f_1 = 0.0075$

Pipe 2 : $L_2 = 255$ m, $D_2 = 200$ mm = 0.2 m, $f_2 = 0.0078$

Pipe 3 : $L_3 = 315$ m, $D_3 = 400$ mm = 0.4 m, $f_3 = 0.0072$

Difference of water level, $H = 18$ m.

(i) **Considering minor losses :**

Let V_1 , V_2 and V_3 be the velocities in 1st, 2nd, and 3rd pipe respectively.

From continuity considerations, we have:

$$A_1V_1 = A_2V_2 = A_3V_3$$

$$\therefore V_2 = \frac{A_1V_1}{A_2} = \frac{(\pi/4) \times D_1^2}{(\pi/4) \times D_2^2} \times V_1 = \frac{D_1^2}{D_2^2} \times V_1 = \left(\frac{0.3}{0.2}\right)^2 V_1 = 2.25 V_1$$

$$\text{and, } V_3 = \frac{A_1V_1}{A_3} = \frac{(\pi/4) \times D_1^2}{(\pi/4) \times D_3^2} \times V_1 = \frac{D_1^2}{D_3^2} \times V_1 = \left(\frac{0.3}{0.4}\right)^2 V_1 = 0.5625 V_1$$

$$\begin{aligned}
 \text{We know that: } H &= \frac{0.5V_1^2}{2g} + \frac{4f_1L_1V_1^2}{D_1 \times 2g} + \frac{0.5V_2^2}{2g} + \frac{4f_2L_2V_2^2}{D_2 \times 2g} + \frac{(V_2 - V_3)^2}{2g} \\
 &\quad + \frac{4f_3L_3V_3^2}{D_3 \times 2g} + \frac{V_3^2}{2g} \quad \dots[\text{Eqn. (12.9)}]
 \end{aligned}$$

$$\begin{aligned}
 18 &= \frac{0.5V_1^2}{2g} + \frac{4 \times 0.0075 \times 450 \times V_1^2}{0.3 \times 2g} + \frac{0.5 \times (2.25 V_1)^2}{2g} + \frac{4 \times 0.0078 \times 255 \times (2.25 V_1)^2}{0.2 \times 2g} \\
 &\quad + \frac{(2.25 V_1 - 0.5625 V_1)^2}{2g} + \frac{4 \times 0.0072 \times 315 \times (0.5625 V_1)^2}{0.4 \times 2g} + \frac{(0.5625 V_1)^2}{2g}
 \end{aligned}$$

$$\begin{aligned}
 18 &= \frac{V_1^2}{2g} (0.5 + 45 + 2.53 + 201.4 + 2.847 + 7.176 + 0.316) \\
 &= 259.77 \frac{V_1^2}{2g}
 \end{aligned}$$

$$\text{or, } V_1 = \sqrt{\frac{18 \times 2 \times 9.81}{259.77}} = 1.166 \text{ m/s}$$

$$\therefore \text{Rate of flow, } Q = A_1 \times V_1 = (\pi/4) \times 0.3^2 \times 1.166 = \mathbf{0.0824 \text{ m}^3/\text{s}} \quad (\text{Ans.})$$

(ii) **Neglecting minor losses :**

$$\text{We know that, } H = \frac{4f_1L_1V_1^2}{D_1 \times 2g} + \frac{4f_2L_2V_2^2}{D_2 \times 2g} + \frac{4f_3L_3V_3^2}{D_3 \times 2g} \quad \dots[\text{Eqn. (12.10)}]$$

$$18 = \frac{V_1^2}{2g} \left(\frac{4 \times 0.0075 \times 450}{0.3} + \frac{4 \times 0.0078 \times 255 \times 2.25^2}{0.2} + \frac{4 \times 0.0072 \times 315 \times (0.5625)^2}{0.4} \right)$$

$$= \frac{V_1^2}{2g} (45 + 201.4 + 7.176) = 253.57 \times \frac{V_1^2}{2g}$$

or,
$$V_1 = \sqrt{\frac{18 \times 2 \times 9.81}{253.57}} = 1.18 \text{ m}$$

$\therefore$ Discharge, $Q = A_1 V_1 = (\pi/4) \times 0.3^2 \times 1.18 = 0.0834 \text{ m}^3/\text{s}$ (Ans.)

Example 12.23. Two reservoirs with a difference in elevation of 15 m are connected by the three pipes in series. The pipes are 300 m long of diameter 30 cm, 150 m long of 20 cm diameter; and 200 m long of 25 cm diameter respectively. The friction factors (f) in the relation

$$h_f = \frac{fLV^2}{D \times 2g}$$

for the three pipes are, respectively, 0.018, 0.020 and 0.019, and which account for friction and all losses. Further the contractions and expansions are sudden. Determine the flow rate in l/s. The loss co-efficient for sudden contraction from dia. 30 cm to 20 cm = 0.24. (PTU)

Solution. Refer to Fig. 12.16. Given : $D_1 = 30 \text{ cm} = 0.3 \text{ m}$; $L_1 = 300 \text{ m}$; $D_2 = 20 \text{ cm} = 0.2 \text{ m}$; $L_2 = 150 \text{ m}$; $D_3 = 25 \text{ cm} = 0.25 \text{ m}$; $L_3 = 200 \text{ m}$; $f_1 = 0.018$; $f_2 = 0.020$; $f_3 = 0.019$.

Loss co-efficient for sudden contraction = 0.24

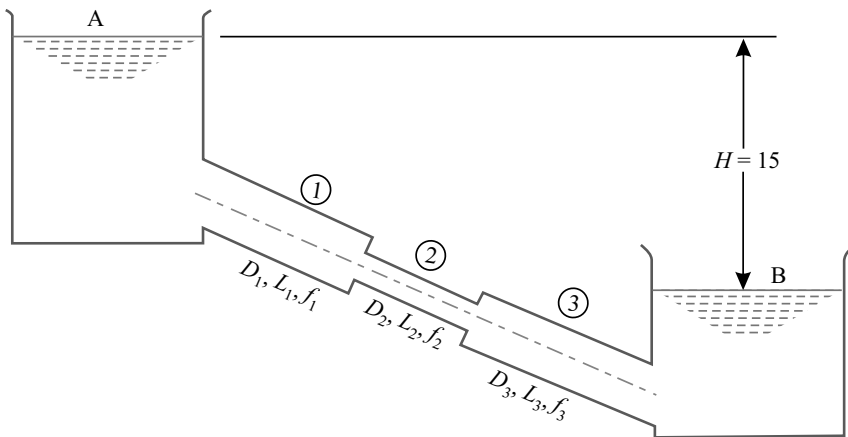


Fig. 12.16

Flow rate in l/s, Q :

Various types of losses which occur in the pipelines 1, 2 and 3 are :

(i) Head loss at entrance, $h_i = 0.5 \times \frac{V_1^2}{2g} = \frac{0.5}{2 \times 9.81} \times \left[\frac{4Q}{\pi \times 0.30^2} \right]^2 = 5.1 Q^2$

(ii) Head loss due to friction in pipe 1, $h_{f1} = \frac{f_1 L_1 V_1^2}{D_1 \times 2g} = \frac{f_1 \times L_1}{D_1 \times 2g} \left[\frac{4Q}{\pi D_1^2} \right]^2$

$$= \frac{0.018 \times 300}{0.3 \times 2 \times 9.81} \left[\frac{4Q}{\pi \times 0.3^2} \right]^2 = 183.6 Q^2$$

(iii) Head loss at contraction, $h_c = 0.24 \frac{V_2^2}{2g} = \frac{0.24}{2g} \left[\frac{4Q}{\pi D_2^2} \right]^2$

$$= \frac{0.24}{2 \times 9.81} \left[\frac{4Q}{\pi \times 0.2^2} \right]^2 = 12.394 Q^2$$

$$(iv) \text{ Head loss due to friction in pipe 2, } h_{f_2} = \frac{f_2 L_2 V_2^2}{D_2 \times 2g} = \frac{f_2 \times L_2}{D_2 \times 2g} \left[\frac{4Q}{\pi D_2^2} \right]^2$$

$$= \frac{0.02 \times 150}{0.2 \times 2 \times 9.81} \left[\frac{4Q}{\pi \times 0.2^2} \right]^2 = 774.267 Q^2$$

$$(v) \text{ Head loss due to sudden enlargement, } h_e = \frac{(V_2 - V_3)^2}{2g} = \frac{1}{2g} \left[\frac{4Q}{\pi D_2^2} - \frac{4Q}{\pi D_3^2} \right]^2$$

$$= \frac{16Q^2}{2 \times 9.81 \times \pi^2} \left[\frac{1}{0.2^2} - \frac{1}{0.25^2} \right]^2 = 6.69 Q^2$$

$$(vi) \text{ Head loss due to friction in pipe 3, } h_{f_3} = \frac{f_3 L_3 V_3^2}{D_3 \times 2g} = \frac{f_3 L_3}{D_3 \times 2g} \left[\frac{4Q}{\pi D_3^2} \right]^2$$

$$= \frac{0.019 \times 200}{0.25 \times 2 \times 9.81} \left[\frac{4Q}{\pi \times 0.25^2} \right]^2 = 321.518 Q^2$$

$$(vii) \text{ Head loss at the exit, } h_0 = \frac{V_3^2}{2g} = \frac{1}{2g} \left[\frac{4Q}{\pi D_3^2} \right]^2 = \frac{1}{2 \times 9.81} \left[\frac{4Q}{\pi \times 0.25^2} \right]^2 = 21.152 Q^2$$

Applying the Bernoulli's equation between the water surfaces of the two reservoirs, we get:

$$\frac{p_A}{w} + \frac{V_A^2}{2g} + z_A = \frac{p_B}{w} + \frac{V_B^2}{2g} + z_B + \text{all losses.}$$

$$0 + 0 + 15 = 0 + 0 + 0 + (5.1 + 183.6 + 12.394 + 774.267 + 6.69 + 321.518 + 21.152) Q^2$$

or, $Q = 0.1064 \text{ m}^3/\text{s}$ or **106.4 l/s (Ans.)**

12.7. EQUIVALENT PIPE

An **equivalent pipe** is defined as the pipe of uniform diameter having loss of head and discharge equal to the loss of head and discharge of a compound pipe consisting of several pipes of different lengths and diameters. The uniform diameter of the equivalent pipe is known as the *equivalent diameter* of the series or compound pipe.

Let, L_1, L_2, L_3 , etc. = Lengths of pipes 1, 2, 3, etc.

D_1, D_2, D_3 , etc. = Diameters of pipes 1, 2, 3, etc.,

H = Total head loss,

L = Length of the equivalent pipe, and

D = Diameter of the equivalent pipe.

Then, neglecting minor losses, total head loss,

$$h_f = h_{f_1} + h_{f_2} + h_{f_3} + \dots$$

$$\text{or, } H = \frac{4f_1 L_1 V_1^2}{D_1 \times 2g} + \frac{4f_2 L_2 V_2^2}{D_2 \times 2g} + \frac{4f_3 L_3 V_3^2}{D_3 \times 2g} + \dots \quad \dots(12.12)$$

(where, f_1, f_2 and f_3 , etc. are co-efficients of friction)

Also, from continuity considerations:

$$Q = A_1 V_1 = A_2 V_2 = A_3 V_3$$

$$= \frac{\pi}{4} \times D_1^2 V_1 = \frac{\pi}{4} \times D_2^2 V_2 = \frac{\pi}{4} \times D_3^2 V_3$$

$$\therefore V_1 = \frac{4Q}{\pi D_1^2}, V_2 = \frac{4Q}{\pi D_2^2}, V_3 = \frac{4Q}{\pi D_3^2}$$

Substituting these values in eqn. (12.12), assuming $f_1 = f_2 = f_3$, etc. = f , we get:

$$H = \frac{4fL_1 \times \left(\frac{4Q}{\pi D_1^2}\right)^2}{D_1 \times 2g} + \frac{4fL_2 \times \left(\frac{4Q}{\pi D_2^2}\right)^2}{D_2 \times 2g} + \frac{4fL_3 \times \left(\frac{4Q}{\pi D_3^2}\right)^2}{D_3 \times 2g} + \dots$$

$$= \frac{4 \times 16fQ^2}{\pi^2 \times 2g} \left(\frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5} + \dots \right) \quad \dots(12.13)$$

Head loss in the equivalent pipe,

$$H = \frac{4fLV^2}{D \times 2g} \quad (\text{assuming the same value of } f \text{ as in compound pipe})$$

where,

$$V = \frac{Q}{A} = \frac{Q}{\frac{\pi}{4} \times D^2} = \frac{4Q}{\pi D^2}$$

$$\therefore H = \frac{4fL \left(\frac{4Q}{\pi D^2}\right)^2}{D \times 2g} = \frac{4 \times 16fQ^2}{\pi^2 \times 2g} \left[\frac{L}{D^5} \right] \quad \dots(12.14)$$

From eqns. (12.13) and (12.14), we have:

$$\frac{4 \times 16fQ^2}{\pi^2 \times 2g} \left(\frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5} + \dots \right) = \frac{4 \times 16fQ^2}{\pi^2 \times 2g} \left(\frac{L}{D^5} \right)$$

$$\text{or,} \quad \frac{L}{D^5} = \frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5} + \dots \quad \dots(12.15)$$

Eqn. 12.15 is known as **Dupit's equation**. If the length of the equivalent pipe is equal to the length of the compound pipe i.e., $L = (L_1 + L_2 + L_3 + \dots)$, the diameter D of the equivalent pipe may be determined by using this equation. Sometimes a pipe of a given diameter D which is available may be required to be used as equivalent pipe to replace a compound pipe; in this case the length of the equivalent pipe may be required to be determined and the same may also be determined by using eqn. (12.15).

Example 12.24. A piping system consists of three pipes arranged in series; the lengths of the pipes are 1200 m, 750 m and 600 m and diameters 750 mm, 600 mm and 450 mm respectively.

- (i) Transform the system to an equivalent 450 mm diameter pipe, and
- (ii) Determine an equivalent diameter for the pipe, 2550 m long.

Solution. Pipe 1: $L_1 = 1200 \text{ m}$; $D_1 = 750 \text{ mm} = 0.75 \text{ m}$
 Pipe 2: $L_2 = 750 \text{ m}$; $D_2 = 600 \text{ mm} = 0.6 \text{ m}$
 Pipe 3: $L_3 = 600 \text{ m}$; $D_3 = 450 \text{ mm} = 0.45 \text{ m}$

(i) **Equivalent length, L :**

Diameter of the equivalent pipe, $D = 450 \text{ mm} = 0.45 \text{ m}$ (Given)

Using the relation :

$$\frac{L}{D^5} = \frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5}$$

$$= \frac{1200}{(0.75)^5} + \frac{750}{(0.6)^5} + \frac{600}{(0.45)^5}, \text{ we have:}$$

$$\frac{L}{(0.45)^5} = 5056.8 + 9645 + 32515.4 = 47217.2$$

$$\text{or, } L = 47217.2 \times (0.45)^5 = \mathbf{871.3 \text{ m (Ans.)}}$$

(ii) **Equivalent diameter, D :**

Length of the equivalent pipe, $L = 2550 \text{ m}$ (Given)

$$\text{Now, } \frac{L}{D^5} = \frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5}$$

$$\text{or, } \frac{2550}{D^5} = \frac{1200}{(0.75)^5} + \frac{750}{(0.6)^5} + \frac{600}{(0.45)^5}$$

$$= 5056.8 + 9645 + 32515.4 = 47217.2$$

$$\text{or, } D = \left(\frac{2550}{47217.2} \right)^{1/5} = 0.5578 \text{ m or } \mathbf{557.8 \text{ mm (Ans.)}}$$

Example. 12.25. A compound piping system consists of 1800 m of 50 cm diameter, 1200 m of 40 cm diameter and 600 m of 30 cm diameter pipes of the same material connected in series. What is the equivalent length of a 40 cm diameter pipe of the same material ? State clearly the assumption (s) made. (PEC)

Solution. Given : $L_1 = 1800 \text{ m}$; $D_1 = 50 \text{ cm} = 0.5 \text{ m}$; $L_2 = 1200 \text{ m}$; $D_2 = 40 \text{ cm} = 0.4 \text{ m}$; $L_3 = 600 \text{ m}$; $D_3 = 30 \text{ cm} = 0.3 \text{ m}$.

Equivalent length of a 0.4 m diameter pipe of the same material, L :

Refer to Fig. 12-17.

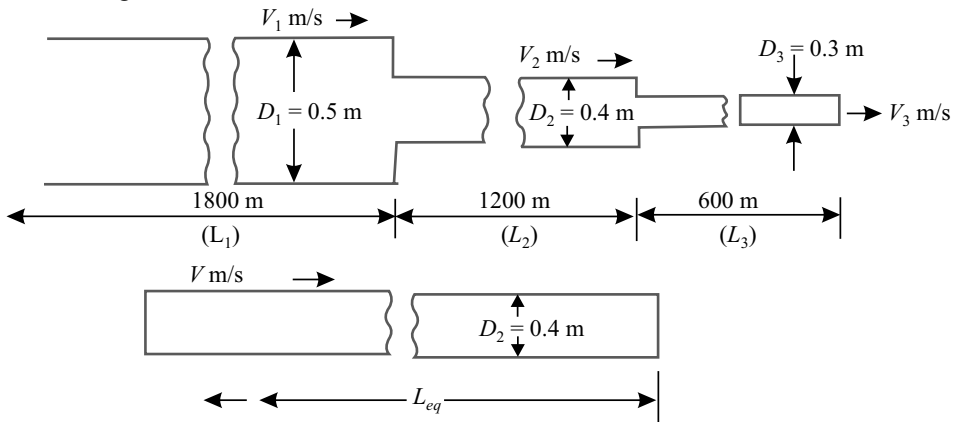


Fig. 12.17

Assumptions : 1. f is constant, and is the same for all the pipes.

2. The head loss due to contraction is ignored.

$$\text{Using the relation : } \frac{L}{D^5} = \frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5} \quad \dots [\text{Eqn. (12.15)}]$$

$$\text{or, } \frac{L}{(0.4)^5} = \frac{1800}{(0.5)^5} + \frac{1200}{(0.4)^5} + \frac{600}{(0.3)^5}$$

$$L = 1800 \times \left(\frac{0.4}{0.5}\right)^5 + 1200 + 600 \left(\frac{0.4}{0.3}\right)^5$$

$$= 589.82 + 1200 + 2528.39 = \mathbf{4318.2 \text{ m (Ans.)}}$$

Example 12.26. A pipe 150 mm in diameter and 15 m long is connected to the bottom of a tank, 15 metres long by 12 metres wide. The original head over the open end of the pipe is 5 metres. Find the time of emptying the tank, assuming the entrance to the pipe is sharp-edged.

Assume $f = 0.01$ in $h_f = \frac{fLV^2}{D \times 2g}$.

[GATE]

Solution. Diameter of the pipe, $D = 150 \text{ mm} = 0.15 \text{ m}$

Length of the pipe, $L = 15 \text{ m}$

Area of the tank, $= 12 \times 15 = 180 \text{ m}^2$

Friction factor, $f = 0.01$

Writing the Bernoulli's equation between the liquid surface at height ' h ' and the lower end of the pipe, considering the entrance and the friction losses, we get:

$$h + 15 = 0.5 \frac{V^2}{2g} + \frac{V^2}{2g} + h_f$$

$$= 0.5 \frac{V^2}{2g} + \frac{V^2}{2g} + \frac{0.01 \times 15 \times V^2}{0.15 \times 2g}$$

(where, V = velocity of flow in pipe)

$$= \frac{2.5V^2}{2g}$$

or,
$$V = \sqrt{\frac{2g(h+15)}{2.5}}$$

Let us assume that the liquid surface falls a distance dh in time dt , then:

$$12 \times 15 \times (-dh) = Q \cdot dt$$

or,
$$180 \times (-dh) = (\pi/4) \times 0.15^2 \times \sqrt{\frac{2g(h+15)}{2.5}} \cdot dt$$

$$= 0.0495 \sqrt{(h+15)} dt$$

or,
$$\int_0^T dt = -\frac{180}{0.0495} \int_5^0 \frac{dh}{\sqrt{h+15}}$$

Let, $h + 15 = H$, then:

$$T = -3636 \int_{20}^{15} \frac{dh}{\sqrt{H}} = -3636 \times 2 [\sqrt{H}]_{20}^{15} = -3636 \times 2 (\sqrt{15} - \sqrt{20})$$

$$= 4357 \text{ s or } \mathbf{1.21 \text{ hours (Ans.)}}$$

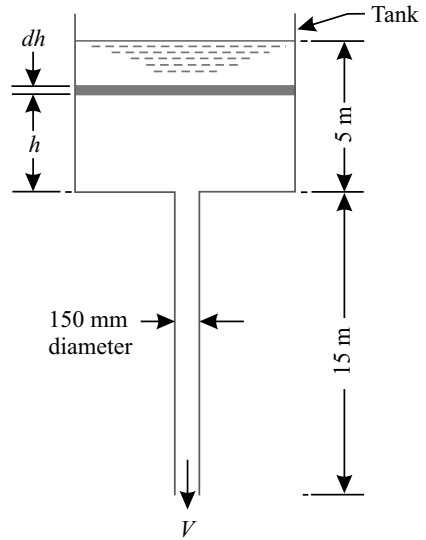


Fig. 12.18

12.8. PIPES IN PARALLEL

The pipes are said to be in *parallel* (Fig. 12.19) when a main line divides into two or more parallel pipes which again join together downstream and continues as a main line.

It may be seen from Fig. 12.19 that the rate of discharge in the main line is equal to the pipes.

Thus,
$$Q = Q_1 + Q_2 \quad \dots(12.16)$$

When the pipes are arranged in parallel, the *loss of head in each pipe (branch) is same*.

∴ Loss of head in pipe 1 = Loss of head in pipe 2.

$$\text{or, } h_f = \frac{4f_1 L_1 V_1^2}{D_1 \times 2g} = \frac{4f_2 L_2 V_2^2}{D_2 \times 2g} \quad \dots(12.17)$$

When, $f_1 = f_2$, then:

$$\frac{L_1 V_1^2}{D_1 \times 2g} = \frac{L_2 V_2^2}{D_2 \times 2g} \quad \dots(12.18)$$

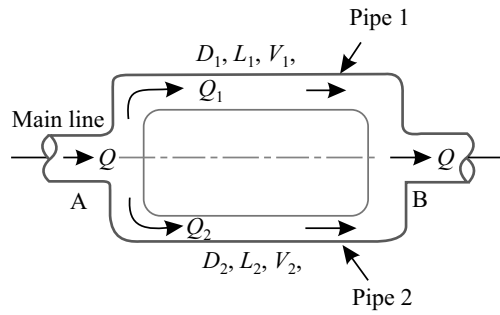


Fig. 12.19

Example 12.27. The main pipe divides into two parallel pipes which again forms one pipe as shown in Fig. 12.19. The data is as follows :

First parallel pipe : Length = 1000 m, diameter = 0.8 m

Second parallel pipe : Length = 1000 m, diameter = 0.6 m

Co-efficient of friction for each parallel pipe = 0.005

If the total rate of flow in the main is $2 \text{ m}^3/\text{s}$ find the rate of flow in each parallel pipe.

Solution. Length of pipe 1, $L_1 = 1000 \text{ m}$

Diameter of pipe 1, $D_1 = 0.8 \text{ m}$

Length of pipe 2, $L_2 = 1000 \text{ m}$

Diameter of pipe 2, $D_2 = 0.6 \text{ m}$

Total rate of flow, $Q = 2 \text{ m}^3/\text{s}$

Co-efficients of friction, $f_1 = f_2 = 0.005$

Rate of flow in each pipe :

Let, $Q_1 =$ Rate of flow in pipe 1,

$Q_2 =$ Rate of flow in pipe 2, and

$Q =$ Total rate of flow (in main line).

Then, $Q = Q_1 + Q_2$ (Eqn. 12.16)

$$\text{Also, } h_f = \frac{4f_1 L_1 V_1^2}{D_1 \times 2g} = \frac{4f_2 L_2 V_2^2}{D_2 \times 2g} \quad \dots[\text{Eqn. 12.17}]$$

$$f_1 = f_2 (= 0.005) \text{ and } L_1 = L_2 (= 1000 \text{ m})$$

The above equation reduces to :

$$\frac{V_1^2}{D_1} = \frac{V_2^2}{D_2} \quad \text{or} \quad \frac{V_1^2}{0.8} = \frac{V_2^2}{0.6}$$

$$\text{or, } V_1 = \sqrt{\frac{0.8}{0.6}} \times V_2 = 1.15 V_2 \quad \dots(ii)$$

$$\begin{aligned} \text{Now, } Q_1 &= \frac{\pi}{4} \times D_1^2 \times V_1 \\ &= \frac{\pi}{4} \times 0.8^2 \times 1.15 V_2 = 0.578 V_2 \end{aligned}$$

$$\text{and, } Q_2 = \frac{\pi}{4} \times D_2^2 \times V_2 = \frac{\pi}{4} \times 0.6^2 \times V_2 = 0.283 V_2$$

Substituting the values of Q_1 and Q_2 in eqn. (i), we get:

$$2 = 0.578 V_2 + 0.283 V_2$$

$$\text{or, } V_2 = \frac{2}{(0.578 + 0.283)} = 2.32 \text{ m/s}$$

Substituting the values of V_2 in eqn. (ii), we get:

$$V_1 = 1.15 \times 2.32 = 2.67 \text{ m/s}$$

$$\text{Hence, } Q_1 = A_1 V_1 = \frac{\pi}{4} \times 0.8^2 \times 2.67 = 1.342 \text{ m}^3/\text{s} \text{ (Ans.)}$$

$$\therefore Q_2 = Q - Q_1 = 2 - 1.342 \\ = 0.658 \text{ m}^3/\text{s} \text{ (Ans.)}$$

Example 12.28. A pipeline of 600 mm diameter is 1.5 km long. To increase the discharge another line of the same diameter is introduced parallel to the first in the second-half of the length. If $f = 0.01$ and head at inlet is 300 mm calculate the increase in discharge.

Neglect minor losses.

[M.U.]

Solution. Diameter of the pipeline, $D = 0.6 \text{ m}$

Length of the pipeline, $L = 1.5 \text{ km} = 1.5 \times 1000 = 1500 \text{ m}$.

Co-efficient of friction, $f = 0.01$

Head at inlet, $h = 0.3 \text{ m}$

Head at outlet (= atmospheric head) = 0

$\therefore$ Head lost, $h_f = 0.3$

Length of another parallel pipe, $L_2 (= L_1) = \frac{1500}{2} = 750 \text{ m}$

Diameter of another parallel pipe, $D_2 (= D_1) = 0.6 \text{ m}$

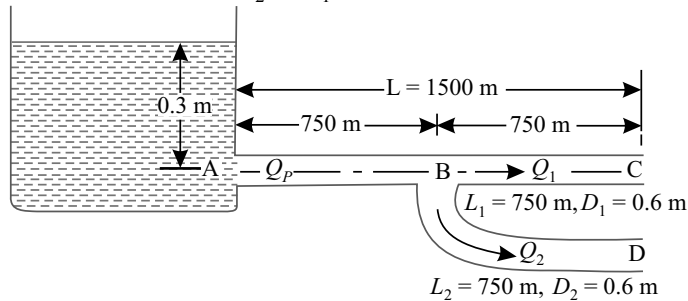


Fig. 12.20

The arrangement of the pipe system is shown in Fig. 12-20.

Increase in discharge :

Case. I. Discharge (Q) for a single pipe of length 1500 m and diameter 0.6 m:

The head lost due to friction in single pipe is given as :

$$h_f = \frac{4fLV^2}{D \times 2g}$$

(where, V = velocity of flow for a single pipe)

$$\therefore 0.3 = \frac{4 \times 0.01 \times 1500 \times V^2}{0.6 \times 2 \times 9.81}$$

$$\text{or, } V = \left[\frac{0.3 \times 0.6 \times 2 \times 9.81}{4 \times 0.01 \times 1500} \right]^{\frac{1}{2}} = 0.243 \text{ m/s}$$

$$\therefore \text{Discharge} \quad Q = A \times V = \frac{\pi}{4} \times 0.6^2 \times 0.243 = 0.0687 \text{ m}^3/\text{s} \quad \dots(i)$$

Case. II. When an additional pipe of length 750 m and diameter 0.6 m is connected in parallel with the last half length of the pipe:

Let, Q_1 = Discharge in first parallel pipe,

Q_2 = Discharge in second parallel pipe,

Q_p = Discharge in the main pipe (when pipes are connected in parallel).

$$\text{Then,} \quad Q_p = Q_1 + Q_2 \quad \dots(\text{Fig. 12-20})$$

As the pipes in parallel have the same diameter and length,

$$\therefore \quad Q_1 = Q_2 = \frac{Q_p}{2}$$

Consider the flow through ABC or ABD.

Head lost (due to friction) in ABC

$$= \text{Head lost in AB} + \text{head lost in BC} \quad \dots(ii)$$

$$\text{Head lost in ABC} = 0.3 \text{ m (given)}$$

$$\text{Now,} \quad \text{Head lost in AB} = \frac{4 \times 0.01 \times 750 \times V_{AB}^2}{0.6 \times 2 \times 9.81}$$

$$\text{But,} \quad V_{AB} = \frac{Q_p}{\text{Area}} = \frac{Q_p}{(\pi/4) \times 0.6^2} = 3.54 Q_p$$

$\therefore$ Head Lost (due to friction) in AB

$$= \frac{4 \times 0.01 \times 750 \times (3.54 Q_p)^2}{0.6 \times 2 \times 9.81} = 31.9 Q_p^2$$

Head lost due to friction through BC

$$= \frac{4fL_1 V_{BC}^2}{D_1 \times 2g}$$

$$= \frac{4 \times 0.01 \times 750 \times (1.77 Q_p)^2}{0.6 \times 2 \times 9.81}$$

$$= 7.98 Q_p^2 \quad \left[\because V_{BC} = \frac{(Q_p/2)}{\text{area}} = \frac{(Q_p/2)}{(\pi/4) \times 0.6^2} = 1.77 Q_p \right]$$

Substituting these values in eqn. (ii), we get:

$$0.3 = 31.9 Q_p^2 + 7.98 Q_p^2$$

$$\text{or,} \quad Q_p = \left[\frac{0.3}{31.9 + 7.98} \right]^{1/2} = 0.087 \text{ m}^3/\text{s}$$

$$\therefore \quad \text{Increase in discharge} = Q_p - Q \\ = 0.087 - 0.0687 = \mathbf{0.0183 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

Example 12.29. Two sharp ended pipes of diameters 50 mm and 100 mm respectively, each of length 100 m respectively, are connected in parallel between two reservoirs which have a difference of level of 10 m. If the friction factor for each pipe is 0.32, calculate :

(i) Rate of flow for each pipe, and

(ii) The diameter of a single pipe 100 m long which would give the same discharge, if it were substituted for the original two pipes.

[Allahabad University]

Solution. Diameter of pipe 1, $D_1 = 50 \text{ mm} = 0.05 \text{ m}$

Diameter of pipe 2, $D_2 = 100 \text{ mm} = 0.1 \text{ m}$

Length of pipe 1, $L_1 = 100 \text{ m}$

Length of pipe 2, $L_2 = 100 \text{ m}$

Difference in level, $h = 10 \text{ m}$

Friction factor, $(4f) = 0.32$

(i) **Rate of flow for each pipe :**

Let, $V_1 =$ Velocity of flow in pipe 1, and

$V_2 =$ Velocity of flow in pipe 2.

Since the pipes are connected in *parallel*, therefore the *loss of head will be same* in both the pipes.

For the *pipe 1*, the loss of head,

$$10 = h_f = \frac{4fL_1V_1^2}{D_1 \times 2g} = \frac{0.32 \times 100 \times V_1^2}{0.05 \times 2 \times 9.81} = 32.62 V_1^2 \quad (\because 4f = 0.32)$$

or, $10 = 32.62 V_1^2$

or, $V_1 = \left(\frac{10}{32.62} \right)^{1/2} = 0.55 \text{ m/s}$

$\therefore$ Rate of flow in pipe 1,

$$Q_1 = A_1 V_1 = \frac{\pi}{4} \times (0.05)^2 \times 0.55$$

$$= \mathbf{0.00108 \text{ m}^3/\text{s}} \quad (\text{Ans.})$$

For the *pipe 2* the loss of head is given by:

$$10 = \frac{4fL_2V_2^2}{D_2 \times 2g} = \frac{0.32 \times 100 \times V_2^2}{0.1 \times 2 \times 9.81} = 16.31 V_2^2$$

or, $V_2 = \left(\frac{10}{16.31} \right)^{1/2} = 0.78 \text{ m/s}$

$\therefore$ Rate of flow in pipe 2,

$$Q_2 = A_2 V_2 = \frac{\pi}{4} \times 0.1^2 \times 0.78 = \mathbf{0.00613 \text{ m}^3/\text{s}} \quad (\text{Ans.})$$

(ii) **Diameter of the single pipe, D :**

Let, $D =$ Diameter of the single pipe,

$L =$ Length of the single pipe = 100 m,

$V =$ Velocity of liquid in the single pipe, and

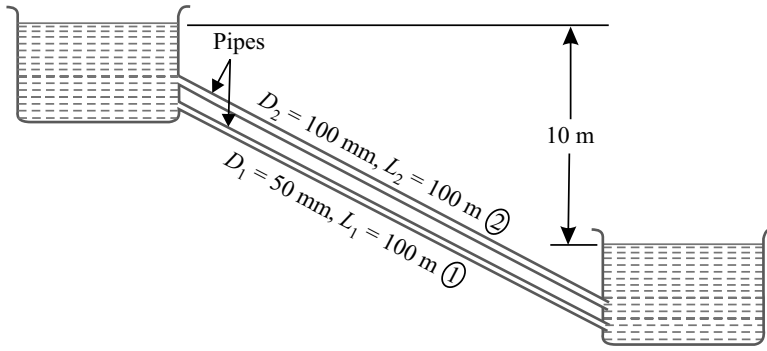
$Q =$ Discharge through the single pipe.

Now, $Q = Q_1 + Q_2$
 $= 0.00108 + 0.00613 = 0.00721 \text{ m}^3/\text{s}$

$\therefore \quad V = \frac{Q}{A} = \frac{0.00721}{(\pi/4) \times D^2} = \frac{0.00918}{D^2} \text{ m/s}$

Loss of head through the single pipe,

$$10 = h_f = \frac{4fLV^2}{D \times 2g} = \frac{0.32 \times 100 \times \left(\frac{0.00918}{D^2} \right)^2}{D \times 2 \times 9.81}$$

**Fig. 12.21**

$$\text{or,} \quad 10 = \frac{0.32 \times 100 \times (0.00918)^2}{2 \times 9.81 \times D^5} = \frac{0.0001375}{D^5}$$

$$\text{or} \quad D = \left[\frac{0.0001375}{10} \right]^{1/5} = 0.1066 \text{ m} = 106.6 \text{ mm}$$

$$\text{i.e.,} \quad D = \mathbf{106.6 \text{ mm} \quad (\text{Ans.})}$$

Example 12.30. A 250 mm diameter, 3 km long straight pipe runs between two reservoirs of surface elevations 135 m and 60 m. A 1.5 km long, 300 mm diameter pipe is laid parallel to the 250 mm diameter pipe from its mid-point to the lower reservoir. Neglecting all minor losses and assuming a friction factor of 0.02 for both pipes, find the increase in discharge caused by addition of 300 mm diameter pipe. (Anna University)

Solution. Neglecting minor losses, the application of Bernoulli's equation between the water surfaces of the two reservoirs yields:

$$(135 - 60) = \frac{fLV^2}{D \times 2g} = \frac{0.02 \times 3000 \times V^2}{0.25 \times 2 \times 9.81}$$

$$\text{or,} \quad V = \left[\frac{(135 - 60) \times 0.25 \times 2 \times 9.81}{0.02 \times 3000} \right]^{1/2} = 2.476 \text{ m/s}$$

The discharge through the pipeline,

$$Q = \frac{\pi}{4} \times (0.25)^2 \times 2.476 = 0.1215 \text{ m}^3/\text{s}$$

In case of altered pipeline (see fig. 12.22) the discharge through pipe section AB is the sum of the discharges through sections BC and BD , or

$$\begin{aligned} Q_1 &= Q_2 + Q_3 \\ \text{or,} \quad \frac{\pi}{4} D_1^2 V_1 &= \frac{\pi}{4} D_2^2 V_2 + \frac{\pi}{4} D_3^2 V_3 \\ D_1^2 V_1 &= D_2^2 V_2 + D_3^2 V_3 \\ &= D_1^2 V_2 + D_3^2 V_3 \end{aligned} \quad \dots(i)$$

Also, as the end points of sections BC and CD are same (they are in parallel.),

$$\begin{aligned} h_{f_2} &= h_{f_3} \\ \text{or,} \quad \frac{f(L/2)V_2^2}{D_2 \times 2g} &= \frac{f(L/2)V_3^2}{D_3 \times 2g} \end{aligned}$$

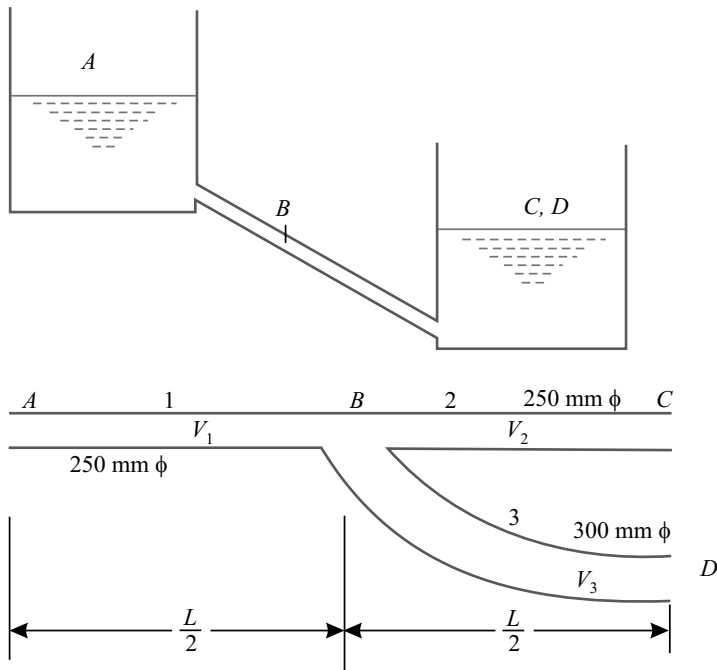


Fig. 12.22

$$\text{or,} \quad \frac{V_2^2}{D_2} = \frac{V_3^2}{D_3}$$

$$\begin{aligned} \text{or,} \quad V_3 &= \left[V_2^2 \times \frac{D_3}{D_2} \right]^{1/2} \\ &= \left[V_2^2 \times \frac{300}{250} \right]^{1/2} \\ &= \sqrt{\frac{300}{250}} V_2 \end{aligned} \quad \dots(ii)$$

Substituting for V_3 in (i), we get:

$$\text{or,} \quad (250)^2 V_1 = (250)^2 V_2 + (300)^2 \times \sqrt{\frac{300}{250}} V_2$$

$$\text{or,} \quad (250)^2 (V_1 - V_2) = (300)^2 \times \sqrt{\frac{300}{250}} V_2$$

$$\text{or,} \quad V_1 - V_2 = 1.5774 V_2$$

$$\text{or,} \quad V_1 = 2.5774 V_2 \quad \text{or} \quad V_2 = 0.388 V_1$$

Again, applying Bernoulli's equation between the water surfaces of the two reservoirs through ABC , we get:

$$\begin{aligned} (135 - 60) &= \frac{f(L/2)V_1^2}{D_1 \times 2g} + \frac{f(L/2)V_2^2}{D_2 \times 2g} \\ 75 &= \frac{f(L/2)}{D_1 \times 2g} (V_1^2 + V_2^2) \quad (\because D_1 = D_2) \end{aligned}$$

$$= \frac{0.02 \times 1500}{0.25 \times 2 \times 9.81} \{V_1^2 + (0.388 V_1)^2\} = 6.116 \times 1.1505 V_1^2$$

or,
$$V_1 = \left(\frac{75}{6.116 \times 1.1505} \right)^{1/2} = 3.26 \text{ m/s}$$

$\therefore$ Discharge $= \frac{\pi}{4} D_1^2 V_1 = \frac{\pi}{4} \times 0.25^2 \times 3.26 = 0.16 \text{ m}^3/\text{s}$

$\therefore$ Increase in discharge $= 0.16 - 0.1215 = 0.0385 \text{ m}^3/\text{s}$ or **31.7 % (Ans.)**

Example 12.31. A farmer wishes to connect two pipes of different lengths and diameters to a common header supplied with $8 \times 10^{-3} \text{ m}^3/\text{s}$ of water from a pump. One pipe is 100 m long and 5 cm in diameter. The other pipe is 800 m long. Determine the diameter of the second pipe such that both pipes have the same flow rate. Assume the pipes to be laid on level ground and friction co-efficient for both pipes as 0.02. Also determine the head loss in metres of water in the pipes.

(GATE)

Solution. Refer to Fig. 12.23. Given : $Q = 8 \times 10^{-3} \text{ m}^3/\text{s}$; $D_1 = 5 \text{ cm} = 0.05 \text{ m}$; $L_1 = 100 \text{ m}$; $L_2 = 800 \text{ m}$; Friction co-efficient, $f = 0.02$.

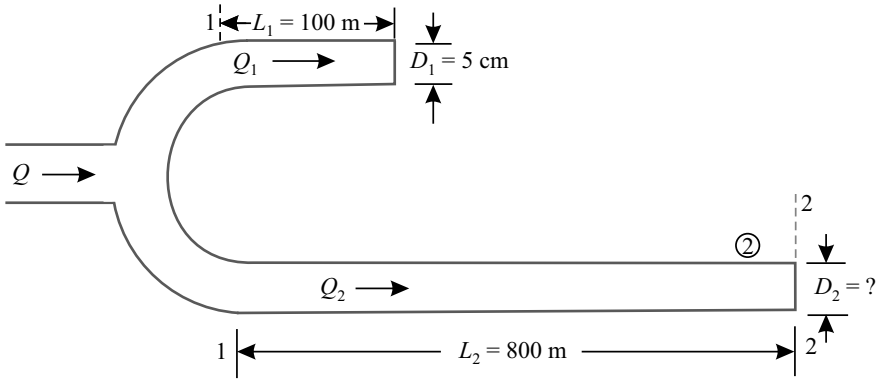


Fig. 12.23

Diameter, D_2 :

$$Q = Q_1 + Q_2 \quad [\text{where, } Q_1 = Q_2 = \frac{8 \times 10^{-3}}{2} = 4 \times 10^{-3} \text{ m}^3/\text{s} \text{ (Given)}]$$

For pipe 1,
$$h_{f_1} = \frac{4fL_1V_1^2}{D_1 \times 2g}$$

where,
$$V_1 = \frac{Q_1}{(\pi/4) \times D_1^2} = \frac{4Q_1}{\pi D_1^2}$$

$$\therefore h_{f_1} = \frac{4fL_1 \times \left[\frac{4Q_1}{\pi D_1^2} \right]^2}{D_1 \times 2g} = \frac{32fL_1Q_1^2}{\pi^2 \times D_1^5 \times g}$$

Similarly, for pipe 2,
$$h_{f_2} = \frac{32fL_2Q_2^2}{\pi^2 \times D_2^5 \times g} \quad \dots(ii)$$

Equating (i) and (ii) [since $h_{f_1} = h_{f_2}$], we get:

$$\frac{32fL_1Q_1^2}{\pi^2 \times D_1^5 \times g} = \frac{32fL_2Q_2^2}{\pi^2 \times D_2^5 \times g}$$

$$\text{or, } \frac{Q_1^2}{Q_2^2} = \frac{L_2 D_1^5}{L_1 D_2^5}$$

$$\text{But, } Q_1 = Q_2 \quad \dots(\text{Given})$$

$$\therefore L_2 D_1^5 = L_1 D_2^5$$

$$\text{or, } D_2 = \left(\frac{L_2 D_1^5}{L_1} \right)^{1/5} = \left[\frac{800 \times (0.05)^5}{100} \right]^{1/5}$$

$$= 0.7578 \text{ m} = \mathbf{7.578 \text{ cm (Ans.)}}$$

Head loss :

$$h_{f1} (= h_{f2}) = \frac{32 f L_1 Q_1^2}{\pi^2 \times D_1^5 \times g} = \frac{32 \times 0.02 \times 100 \times (4 \times 10^{-3})^2}{\pi^2 \times (0.05)^5 \times 9.81} = \mathbf{33.84 \text{ m (Ans.)}}$$

Example 12.32. A pipeline with diameter 0.8 m and length 3000 m connects two open reservoirs of water which have their water surfaces of elevations of 100 m and 70 m above a datum. In order to increase the rate of flow between the reservoirs by 20 % it is decided to lay an additional 0.8 m diameter pipeline from the upper reservoir. The second pipeline is to be parallel to the original pipeline and is to be connected to the latter at some suitable point. Determine the point of connection, assuming that the friction factor is 0.04 for each pipeline. Neglect minor losses. (PEC)

Solution. Refer to Fig. 12.24. Given : $L = 3000 \text{ m}$; $D = 0.8 \text{ m}$; $f = 0.04$

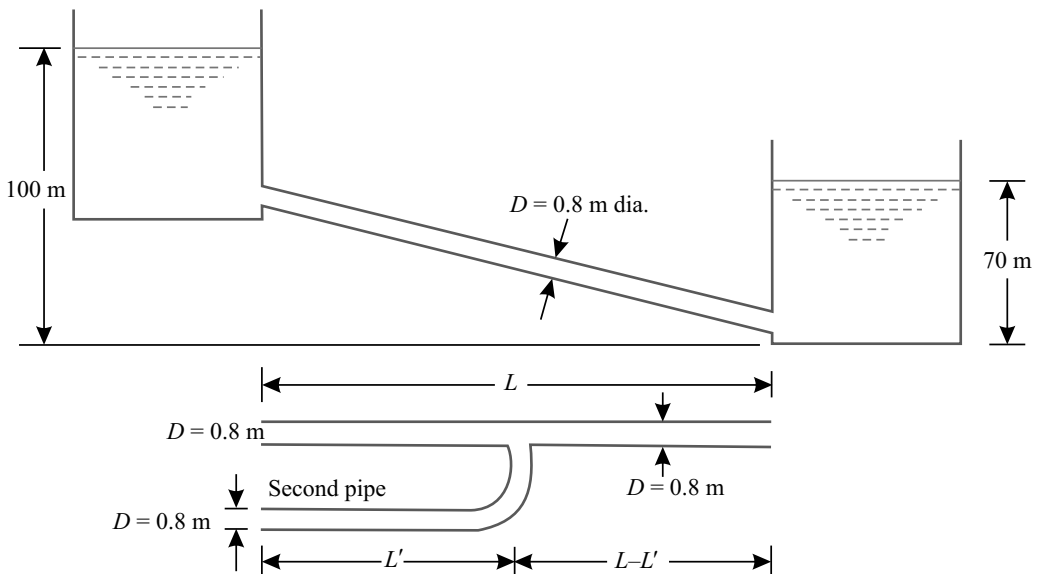


Fig. 12.24

Point of connection of second pipe, L' :

$$\text{Head at inlet of pipe} = 100 - 70 = 30 \text{ m}$$

Case I. Discharge (Q_1) for a single pipe length 3000 m and diameter 0.8 m:

The head lost due to friction in single pipe is given as :

$$h_f = \frac{f L V_1^2}{D \times 2g}$$

where,

$$V_1 = \frac{Q_1}{A_1} = \frac{Q_1}{\frac{\pi}{4} \times D^2} = \frac{4Q_1}{\pi D^2} \quad \left[\text{where, } f = \text{friction factor} \right. \\ \left. (= 4 \times \text{co-efficient of friction}) \right]$$

$$\therefore h_f = \frac{fL \times \left(\frac{4Q_1}{\pi D^2}\right)^2}{D \times 2g} = \frac{8fLQ_1^2}{\pi^2 D^5 \times g} = \frac{fLQ_1^2}{12D^5}$$

Substituting the values, we get:

$$30 = \frac{0.04 \times 3000 \times Q_1^2}{12 \times (0.8)^5}$$

$$\text{or, } Q_1 = \left[\frac{30 \times 12 \times (0.8)^5}{0.04 \times 3000} \right]^{1/2} = 0.99 \text{ m}^3/\text{s}$$

Case II. When another pipeline of length L' is added :

$$\text{Total discharge, } Q_2 = 1.2 Q_1 = 1.2 \times 0.99 = 1.188 \text{ m}^3/\text{s}$$

$$\text{Discharge through each pipeline} = \frac{Q_2}{2}$$

In this case, Total head loss = Sum of head losses in two pipes

$$\text{i.e., } h_f = \frac{fL'V_1^2}{D \times 2g} + \frac{f(L-L')V_2^2}{D \times 2g}$$

$$\text{or, } 30 = \frac{fL' \times \left(\frac{Q_2}{2}\right)^2}{12 D^5} + \frac{f(L-L')Q_2^2}{12 D^5}$$

$$\text{or, } 30 = \frac{fL'Q_2^2}{4 \times 12 D^5} + \frac{f(L-L')Q_2^2}{12 D^5}$$

Substituting the values, we get:

$$\text{or, } \frac{30 \times 12 \times (0.8)^5}{0.04 \times (1.188)^2} = \frac{L'}{4} - L' + 3000$$

$$2089.6 = \frac{L'}{4} - L' + 3000$$

$$\text{or, } L' - \frac{L'}{4} = 3000 - 2089.7 = 910.4$$

$$\therefore L' = 1213.87 \text{ m (Ans.)}$$

Example 12.33. Two reservoirs have a constant difference of levels of 70 m and are connected by a 250 mm diameter pipe which is 4 km long. The pipe is tapped mid-way between the reservoirs and water is drawn at the rate of $0.04 \text{ m}^3/\text{s}$. Assuming friction factor = 0.04, determine the rate at which water enters the lower reservoir.

Solution. Diameter of the pipe, $D = 250 \text{ mm} = 0.25 \text{ m}$

Difference of level, $h = 70 \text{ m}$

Friction factor, $4f = 0.04$

Rate at which water enters the lower reservoir :

Let, $Q =$ Discharge entering the lower reservoir.

Then, Discharge at the inlet = $(Q + 0.04) \text{ m}^3/\text{s}$.

Now, from the application of Bernoulli's equation, we have:

$$h = h_{f_1} + h_{f_2}$$

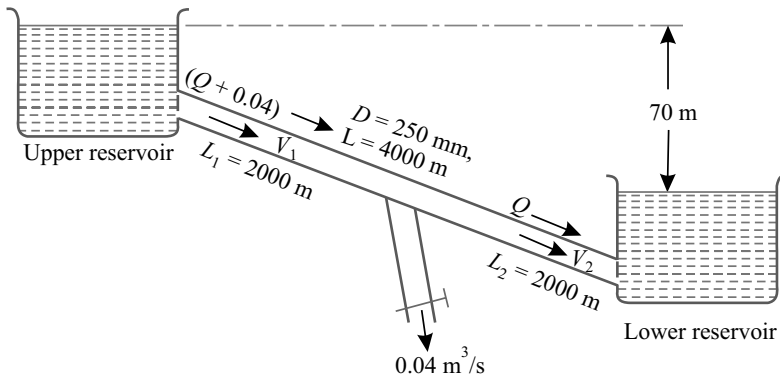


Fig. 12.25

$$\begin{aligned}
 70 &= \frac{4fL_1V_1^2}{D \times 2g} + \frac{4fL_2V_2^2}{D \times 2g} \\
 &= \frac{0.04 \times 2000 \times V_1^2}{0.25 \times 2g} + \frac{0.04 \times 2000 \times V_2^2}{0.25 \times 2g} \quad (\because 4f = 0.04)
 \end{aligned}$$

or,

$$70 = \frac{320}{2g} (V_1^2 + V_2^2)$$

Substituting :

$$V_1 = \frac{Q + 0.04}{A} = \frac{(Q + 0.04)}{(\pi/4) \times 0.25^2} = 20.37 (Q + 0.04)$$

and,

$$V_2 = \frac{Q}{A} = \frac{Q}{(\pi/4) \times 0.25^2} = 20.37 Q$$

Eqn. (i) becomes :

$$70 = \frac{320}{2 \times 9.81} \left[\{20.37 (Q + 0.04)\}^2 + (20.37 Q)^2 \right]$$

or,

$$70 = \frac{320 \times 20.37^2}{2 \times 9.81} [(Q + 0.04)^2 + Q^2]$$

or,

$$70 = 6768 (Q^2 + 0.0016 + 0.08Q + Q^2)$$

$$\text{or, } 2Q^2 + 0.08Q + 0.0016 = 0.0103$$

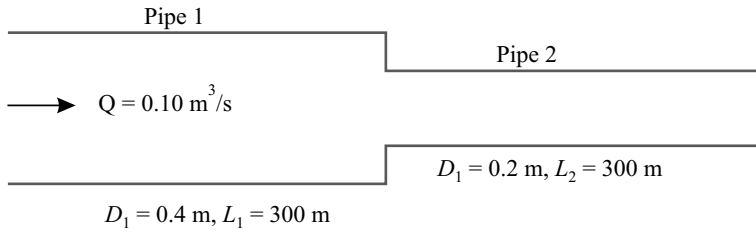
$$\text{or, } Q^2 + 0.04Q - 0.0043 = 0$$

or,

$$\begin{aligned}
 Q &= \frac{-0.04 \pm \sqrt{0.0016 + 0.0172}}{2} \\
 &= \frac{-0.04 \pm 0.137}{2} \\
 &= \frac{0.097}{2} = 0.0485 \text{ m}^3/\text{s}
 \end{aligned}$$

Hence, the rate at which water enters the lower reservoir = **0.0485 m³/s (Ans.)**

Example 12.34. Two pipes of diameters 400 mm and 200 mm are each 300 m long. When the pipes are connected in series the discharge through the pipeline is 0.10 m³/s, find the loss of head incurred. What would be the loss of head in the system to pass the same total discharge when the pipes are connected in parallel? Take friction factor = 0.0075 for each pipe. [Nagpur University]

**Fig. 12.26**

Solution. Diameter of the pipe 1, $D_1 = 400 \text{ mm} = 0.4 \text{ m}$

Length of the pipe 1, $L_1 = 300 \text{ m}$

Diameter of the pipe 2, $D_2 = 200 \text{ mm} = 0.2 \text{ m}$

Length of the pipe 2, $L_2 = 300 \text{ m}$

Friction factor for each pipe ($4f$) = 0.0075

Discharge, $Q = 0.1 \text{ m}^3/\text{s}$

(i) **Pipes connected in series – Loss of head :**

$$\text{Velocity of flow in pipe 1, } V_1 = \frac{0.1}{(\pi/4) \times 0.4^2} = 0.796 \text{ m/s}$$

$$\text{Velocity of flow in pipe 2, } V_2 = \frac{0.1}{(\pi/4) \times 0.2^2} = 3.183 \text{ m/s}$$

$$\text{Head lost due to friction in pipe 1} = \frac{4f_1 L_1 V_1^2}{D_1 \times 2g} = \frac{0.0075 \times 300 \times 0.796^2}{0.4 \times 2 \times 9.81} = 0.1816 \text{ m}$$

Assuming head lost due to contraction,

$$h_c = k \frac{V_2^2}{2g}$$

$$\begin{aligned} \text{or, } h_c &= 0.33 \frac{V_2^2}{2g} \left[\text{for } \frac{D_2}{D_1} = 0.5, k = 0.33 \dots (\text{from tables}) \right] \\ &= \frac{0.33 \times 3.183^2}{2 \times 9.81} = 0.17 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Head lost due to friction in pipe 2} &= \frac{4f_2 L_2 V_2^2}{D_2 \times 2g} = \frac{0.0075 \times 300 \times 3.183^2}{0.2 \times 2 \times 9.81} \\ &= 5.809 \text{ m} \end{aligned}$$

$\therefore$ Head lost in the pipeline = $0.1816 + 0.17 + 5.809 = 6.16 \text{ m}$ (Ans.)

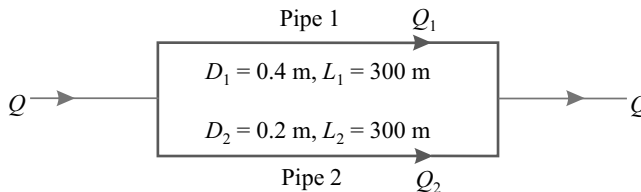
(ii) **Pipes in Parallel–Loss of head :**

From continuity consideration, we have:

$$Q = Q_1 + Q_2$$

$$0.1 = Q_1 + Q_2$$

...(Given)

**Fig. 12.27**

$$\begin{aligned} \text{or, } 0.1 &= \frac{\pi}{4} \times 0.4^2 \times V_1 + \frac{\pi}{4} \times 0.2^2 \times V_2 \\ &= 0.1257 V_1 + 0.0314 V_2 \end{aligned} \quad \dots(1)$$

Also, head lost will be same, since the pipes are connected in parallel.

$$\therefore h_f = \frac{4f_1 L_1 V_1^2}{D_1 \times 2g} = \frac{4f_2 L_2 V_2^2}{D_2 \times 2g}$$

But, $f_1 = f_2$ and $L_1 = L_2$

$$\therefore \frac{V_1^2}{D_1} = \frac{V_2^2}{D_2} \quad \text{or} \quad \frac{V_1^2}{V_2^2} = \frac{D_1}{D_2} = \frac{0.4}{0.2} = 2$$

$$\text{or, } V_1^2 = 2V_2^2 \quad \dots(2)$$

Substituting the value of V_2 from (1) in (2), we get:

$$V_1^2 = 2 \left[\frac{0.1 - 0.1257 V_1}{0.0314} \right]^2 = 2 (3.185 - 4V_1)^2 = 20.29 + 32V_1^2 - 50.96V_1$$

$$\text{or, } 31V_1^2 - 50.96V_1 + 20.29 = 0$$

$$\text{or, } V_1 = \frac{50.96 \pm \sqrt{50.96^2 - 4 \times 31 \times 20.29}}{2 \times 31} = 0.97 \text{ m/s, } 0.677 \text{ m/s}$$

Using $V_1 = 0.97 \text{ m/s}$, we have:

$$Q_1 = (\pi/4) \times 0.4^2 \times 0.97 = 0.1219 \text{ m}^3/\text{s}$$

Since $Q_1 > Q$, $V_1 = 0.97 \text{ m/s}$ is not realistic.

Using $V_1 = 0.677 \text{ m/s}$, we have:

$$Q_1 = (\pi/4) \times 0.4^2 \times 0.677 = 0.085 \text{ m}^3/\text{s}$$

$$Q_2 = 0.1 - 0.085 = 0.015 \text{ m}^3/\text{s}$$

$$\text{Head lost} = \frac{4f_1 L_1 V_1^2}{D_1 \times 2g} = \frac{0.0075 \times 300 \times 0.677^2}{0.4 \times 2 \times 9.81} = \mathbf{0.131 \text{ m (Ans.)}}$$

Example 12.35. The pipes of diameter D and d of equal length L are considered. If the pipes are arranged in parallel, the loss of head for either pipe for a flow of Q is h . If the pipes are arranged in series and the same quantity Q flows through them, the loss of head is H . If $d = 0.5 D$, find the percentage of total flow through each pipe when placed in parallel and the ratio of H to h neglecting minor losses and assuming friction co-efficient to be constant. [UPSC Exams.]

Solution. Diameter of pipe 1, $D_1 = D$

Length of pipe 1, $L_1 = L$

Diameter of pipe 2, $D_2 = d$

Length of pipe 2, $L_2 = L$

Total discharge = Q

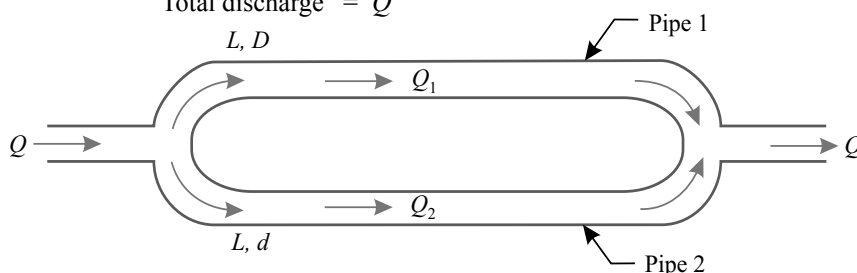


Fig. 12.28. Pipes connected in parallel.

Head lost when pipes are arranged in parallel = h

Head lost when pipes are arranged in series = H

$d = 0.5 D$ and f is constant.

Case I. Pipes connected in “parallel” :

When pipes are connected in *parallel*,

$$Q = Q_1 + Q_2 \quad \dots(i)$$

Loss of head in each pipe = h

$$\text{For pipe 1 :} \quad h = \frac{4fL_1V_1^2}{D \times 2g}$$

$$\text{where,} \quad V_1 = \frac{Q_1}{A_1} = \frac{Q_1}{(\pi/4) \times D^2} = \frac{4Q_1}{\pi D^2}$$

$$\therefore \quad h = \frac{4fL \times \left(\frac{4Q_1}{\pi D^2}\right)^2}{D \times 2g} = \frac{32fLQ_1^2}{\pi^2 D^5 \times g} \quad \dots(ii) \quad (\because L_1 = L)$$

$$\text{For pipe 2 :} \quad h = \frac{32fLQ_2^2}{\pi^2 d^5 \times g} \quad \dots(iii)$$

From eqns. (ii) and (iii), we have:

$$\frac{32fLQ_1^2}{\pi^2 D^5 \times g} = \frac{32fLQ_2^2}{\pi^2 d^5 \times g}$$

$$\text{or,} \quad \frac{Q_1^2}{D^5} = \frac{Q_2^2}{d^5}$$

$$\text{or,} \quad \left(\frac{Q_1}{Q_2}\right)^2 = \left(\frac{D}{d}\right)^5 = \left(\frac{D}{0.5D}\right)^5 = 32 \quad (\because d = 0.5D \dots \text{Given})$$

$$\text{or,} \quad \frac{Q_1}{Q_2} = 5.567 \quad \text{or} \quad Q_1 = 5.567 Q_2$$

Substituting the value of Q_1 in eqn. (i), we get:

$$Q = 5.567Q_2 + Q_2 = 6.567Q_2$$

$$\therefore \quad Q_2 = \frac{Q}{6.567} = 0.15Q \quad \dots(iv)$$

$$\text{and,} \quad Q_1 = Q - 0.15Q = 0.85Q \quad \dots[\text{From (i)}] \quad \dots(v)$$

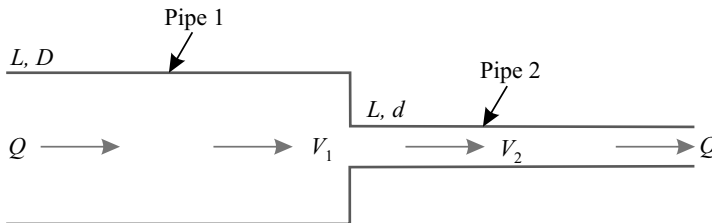


Fig. 12.29. Pipes connected in series.

Case II. Pipes connected in “series” :

In this case, Total loss = Sum of head losses in the two pipes

$$\therefore \quad H = \frac{4fLV_1^2}{D \times 2g} + \frac{4fLV_2^2}{d \times 2g}$$

where,

$$V_1 = \frac{Q}{(\pi/4) \times D^2} = \frac{4Q}{\pi D^2}$$

$$V_2 = \frac{Q}{(\pi/4) \times d^2} = \frac{4Q}{\pi d^2}$$

$$\therefore H = \frac{4fL \times \left(\frac{4Q}{\pi D^2}\right)^2}{D \times 2g} + \frac{4fL \times \left(\frac{4Q}{\pi d^2}\right)^2}{d \times 2g} \quad \dots(vi)$$

or,
$$H = \frac{32fLQ^2}{\pi^2 D^5 \times g} + \frac{32fLQ^2}{\pi^2 d^5 \times g}$$

From eqn. (ii)
$$\frac{32fL}{\pi^2 D^5 \times g} = \frac{h}{Q_1^2}$$

and, from eqn. (iii)
$$\frac{32fL}{\pi^2 d^5 \times g} = \frac{h}{Q_2^2}$$

Substituting these values in eqn. (vi), we get:

$$H = Q^2 \times \frac{h}{Q_1^2} + Q^2 \times \frac{h}{Q_2^2} = h \left(\frac{Q^2}{Q_1^2} + \frac{Q^2}{Q_2^2} \right)$$

$$\therefore \frac{H}{h} = \frac{Q^2}{Q_1^2} + \frac{Q^2}{Q_2^2}$$

But from eqns. (iv) and (v),

$$Q_1 = 0.85 Q \text{ and } Q_2 = 0.15 Q$$

$$\therefore \frac{H}{h} = \frac{Q^2}{(0.85Q)^2} + \frac{Q^2}{(0.15Q)^2} = 45.828 \text{ (Ans.)}$$

Example 12.36. A pumping plant forces water through a 600 mm diameter main, the friction head being 27 m. In order to reduce the power consumption, it is proposed to lay another main of appropriate diameter along the side of the existing one, so that the two pipes may work in parallel for the entire length and reduce the friction head to 9.6 m only. Find the diameter of the new main if with the exception of diameter, it is similar to the existing one in every respect. [Delhi University]

Solution. Diameter of single main pipe, $D = 600 \text{ mm} = 0.6 \text{ m}$

Friction head, $h_f = 27 \text{ m}$

Friction head for two parallel pipes = 9.6 m

Diameter of the new main :

Case I. Single main :

$$h_f = \frac{4fLV^2}{D \times 2g}$$

$$27 = \frac{4fLV^2}{0.6 \times 2 \times 9.81}$$

$$fLV^2 = \frac{27 \times 0.6 \times 2 \times 9.81}{4} = 79.461$$

But,
$$V = \frac{Q}{A}$$

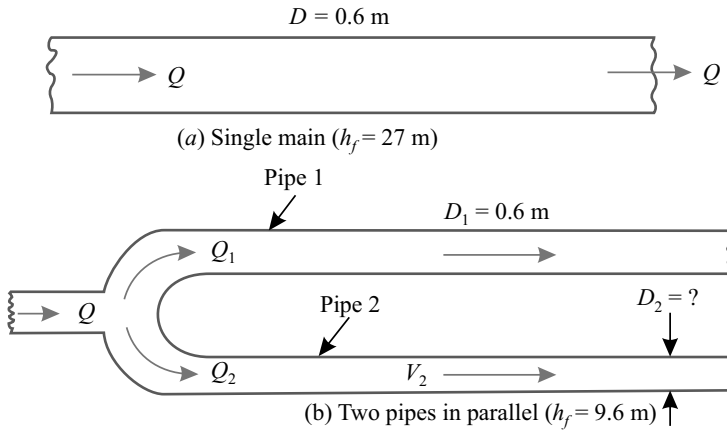


Fig. 12.30

$$\therefore fL \frac{Q^2}{A^2} = 79.461 \quad \dots(i)$$

Case II. Two pipes in parallel :

Loss of head, $h_f = 9.6$ m

For pipe 1 :

$$h_{f1} = \frac{4fL_1V_1^2}{D_1 \times 2g} = 9.6$$

But, $L_1 = L, V_1 = \frac{Q_1}{A_1} = \frac{Q_1}{A} \quad (\because A_1 = A)$

$$D_1 = D = 0.6 \text{ m}$$

$$\therefore \frac{4fL}{0.6 \times 2 \times 9.81} \times \frac{Q_1^2}{A^2} = 9.6$$

or, $fL \frac{Q_1^2}{A^2} = \frac{9.6 \times 0.6 \times 2 \times 9.81}{4} = 28.25 \quad \dots(ii)$

For pipe 2 :

$$h_{f2} = \frac{4fL_2V_2^2}{D_2 \times 2g} = 9.6$$

where, $L_2 = L, V_2 = \frac{Q_2}{A_2}$

$$\therefore \frac{4fLQ_2^2}{D_2 \times 2g \times A_2^2} = 9.6$$

or, $\frac{fLQ_2^2}{D_2A_2^2} = \frac{9.6 \times 2 \times 9.81}{4} = 47.09 \quad \dots(iii)$

Dividing (i) by (iii), we get:

$$\frac{Q^2}{Q_1^2} = \frac{79.461}{28.25} = 2.81$$

or, $\frac{Q}{Q_1} = 1.67$

$$\text{or, } Q_1 = \frac{Q}{1.67} = 0.59 Q$$

$$\text{But, } Q_1 + Q_2 = Q$$

$$\therefore Q_2 = Q - Q_1 = Q - 0.59Q = 0.41Q$$

Dividing (ii) by (iii), we get:

$$\frac{Q_1^2 \times D_2 \times A_2^2}{Q_2^2 \times A^2} = \frac{28.25}{47.09} = 0.6$$

$$\text{or, } \frac{Q_1 \times D_2 \times (\pi/4 \times D_2^2)^2}{Q_2^2 \times [\pi/4 \times (0.6)^2]^2} = 0.6$$

$$\left(\frac{0.59Q}{0.41Q} \right)^2 \times \frac{D_2^5}{(0.36)^2} = 0.6$$

$$\text{or, } D_2^5 = 0.6 \times (0.36)^2 \times \left(\frac{0.41}{0.59} \right)^2 = 0.03755$$

$$\text{or, } D_2 = 0.518 \text{ m} = \mathbf{518 \text{ mm (Ans.)}}$$

Example 12.37. Two pipes A and B are connected in parallel between two points. Pipe A is 180 m long and has a diameter of 12 cm. Pipe B is 120 m long and has a diameter of 10 cm. Both the pipes have the same friction factor of 0.017. A partially closed valve in pipe A causes the discharge in the two pipes to be the same (Fig. 12.31.). Neglecting all other minor losses, calculate the value of the valve coefficient.

Solution. Given : $L_A = 180 \text{ m}$; $D_A = 12 \text{ cm} = 0.12 \text{ m}$; $L_B = 120 \text{ m}$; $D_B = 10 \text{ cm} = 0.1 \text{ m}$; Friction factor; $f = 0.017$.

Value of the valve co-efficient, K_v :

Since the discharges are same in both the pipes,

$$A_A V_A = A_B V_B$$

$$\text{or, } \frac{\pi}{4} \times (0.12)^2 \times V_A = \frac{\pi}{4} \times (0.1)^2 \times V_B$$

$$\therefore V_B = 1.44 V_A$$

$$\text{Let the losses in the valve be } K_v \frac{V_A^2}{2g}.$$

Head losses in both the pipes are same.

$$\text{Hence, } \frac{f_A L_A V_A^2}{D_A \times 2g} + \frac{K_v V_A^2}{2g} = \frac{f_B L_B V_B^2}{D_B \times 2g}$$

$$\frac{0.017 \times 180}{0.12} \times \frac{V_A^2}{2g} + K_v \frac{V_A^2}{2g} = \frac{0.017 \times 120 \times (1.44)^2}{0.10} \times \frac{V_A^2}{2g}$$

$$25.5 + K_v = 42.30$$

$$\therefore K_v = \mathbf{16.8 \text{ (Ans.)}}$$

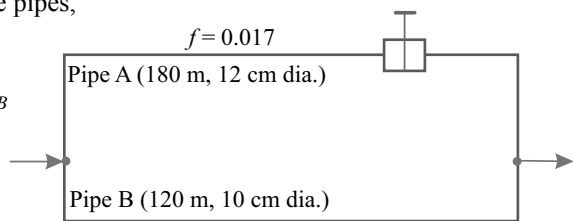


Fig 12.31

Example 12.38. Two pipes 1 and 2, each of 12 cm diameter branch off from a point A in a pipeline and rejoin at B. Pipe 1 is 480 m long and pipe 2 is 720 m long. So total head at A is 36 m. A short pipe 10 cm diameter is fitted at B and the flow is discharged into atmosphere through it as shown in Fig. 12.32. Assuming $f = 0.018$ for both the pipes, Calculate :

(i) Total discharge, and

(ii) Distribution of discharge in pipes 1 and 2.

Solution. Given : $D_1 = 12 \text{ cm} = 0.12 \text{ m}$; $L_1 = 480 \text{ m}$; $D_2 = 12 \text{ cm} = 0.12 \text{ m}$, $L_2 = 720 \text{ m}$; $D_3 = 10 \text{ cm} = 0.1 \text{ m}$; $f = 0.018$.

(i) **Total discharge, Q :**

As 10 cm diameter pipe is short, the friction loss in it can be neglected.

$$H_B (= \text{Head at } B) = \frac{V_3^2}{2g}$$

$$H_A - H_B = 36 - \frac{V_3^2}{2g}$$

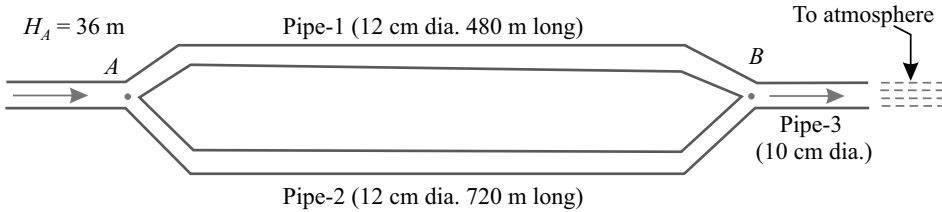


Fig. 12.32

Consider an equivalent pipe $D_{eq.} = 0.1 \text{ m}$ and $f_{eq.} = 0.018$ to replace the parallel pipes 1 and 2. Then,

$$\left(\frac{D_{eq.}^5}{f_{eq.} L_{eq.}} \right)^{\frac{1}{2}} = \left(\frac{D_1^5}{f_1 L_1} \right)^{\frac{1}{2}} + \left(\frac{D_2^5}{f_2 L_2} \right)^{\frac{1}{2}}$$

Since,

$$f_{eq.} = f_1 = f_2 \text{ and } D_1 = D_2 = 0.12 \text{ m}$$

$$\begin{aligned} \therefore \frac{(0.10)^{5/2}}{(L_{eq.})^{1/2}} &= (0.12)^{5/2} \left[\frac{1}{\sqrt{480}} + \frac{1}{\sqrt{720}} \right] \\ &= 0.004988 (0.04564 + 0.03727) = 0.0004136 \end{aligned}$$

$$\therefore L_{eq.} = \left[\frac{(0.10)^{5/2}}{0.0004136} \right]^2 = 58.46 \text{ m}$$

As

$$D_{eq.} = 0.1 \text{ m, velocity in this pipe} = V_3 = V_{eq.}$$

$$\begin{aligned} \therefore \text{Head loss} = H_A - H_B &= 36 - \frac{V_3^2}{2g} = \frac{f_{eq.} L_{eq.} V_{eq.}^2}{D_{eq.} \times 2g} \\ &= \frac{0.018 \times 58.46}{0.1} \times \frac{V_3^2}{2g} = 10.52 \frac{V_3^2}{2g} \\ \therefore \frac{V_3^2}{2g} &= (10.52 + 1) = 36 \end{aligned}$$

$$V_3 = \left(\frac{36}{11.52} \times 2 \times 9.81 \right)^{1/2} = 7.83 \text{ m/s}$$

$$\text{Total discharge, } Q = \frac{\pi}{4} \times (0.10)^2 \times 7.83 = \mathbf{0.06149 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

(ii) **Division of discharge in pipes 1 and 2; Q_1 , Q_2 :**

$$H_A - H_B = 36 - \frac{(7.83)^2}{2 \times 9.81} = 32.875 \text{ m} = h_{f_1} = h_{f_2}$$

$$\therefore \frac{f_1 L_1 V_1^2}{D_1 \times 2g} = \frac{0.018 \times 480}{0.12} \times \frac{V_1^2}{2g} = 32.875$$

$$\therefore V_1 = \left(\frac{32.875 \times 2 \times 9.81 \times 0.12}{0.018 \times 480} \right)^{1/2} = 2.993 \text{ m/s}$$

$$\therefore Q_1 = \frac{\pi}{4} \times (0.12)^2 \times 2.993 = \mathbf{0.03385 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

Again,

$$\frac{f_2 L_2}{D_2} \times \frac{V_2^2}{2g} = \frac{0.018 \times 720}{0.12} \times \frac{V_2^2}{2g} = 32.875$$

or,

$$V_2 = \left(\frac{32.875 \times 2 \times 9.81 \times 0.12}{0.018 \times 720} \right)^{1/2} = 2.444 \text{ m/s}$$

$$\therefore Q_2 = \frac{\pi}{4} \times (0.12)^2 \times 2.444 = \mathbf{0.02764 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

$$[\text{Check : } Q_1 + Q_2 = 0.03385 + 0.02764 = 0.06149 \text{ m}^3/\text{s}]$$

Example 12.39. Two reservoirs A and B are connected through a piping system consisting of 50 cm diameter pipe, 450 m long branching two pipes of 35 cm diameter and 25 cm diameter, each 650 m long. A pump situated at reservoir A pumps 0.35 m³/s of water through this pipe system to reservoir B whose water surface elevation is 50 m above that of A. Assuming pump efficiency as 60 percent and $f = 0.018$, determine the input power for the pump.

Solution. Refer to Fig. 12.33. Given : $D_1 = 50 \text{ cm} = 0.5 \text{ m}$, $L_1 = 450 \text{ m}$; $D_2 = 35 \text{ cm} = 0.35 \text{ m}$, $L_2 = 650 \text{ m}$; $D_3 = 25 \text{ cm} = 0.25 \text{ m}$, $L_3 = 650 \text{ m}$, $Q = 0.35 \text{ m}^3/\text{s}$; $\eta_{\text{pump}} = 60\%$; $f = 0.018$.

Consider equivalent pipe of diameter 0.5 m to replace the two parallel pipes. The equivalent pipe (D_{eq} , L_{eq} , f_{eq}) to replace a set of parallel pipes (D_2 , L_2 , f_2) and (D_3 , L_3 , f_3) is given by:

$$\left(\frac{D_{eq}^5}{f_{eq} L_{eq}} \right)^{1/2} = \left(\frac{D_2^5}{f_2 L_2} \right)^{1/2} + \left(\frac{D_3^5}{f_3 L_3} \right)^{1/2}$$

Here, $f_{eq} = f_1 = f_2$

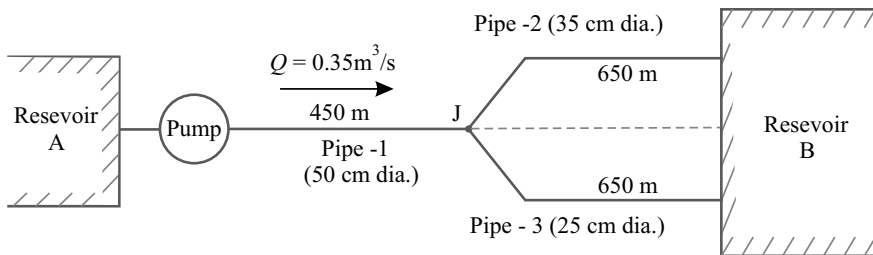


Fig. 12.33

Substituting the various values in the above eqn., we get:

$$\left[\frac{(0.5)^5}{L_{eq}} \right]^{1/2} = \left[\frac{(0.35)^5}{650} \right]^{1/2} + \left[\frac{(0.25)^5}{650} \right]^{1/2}$$

or,

$$\frac{0.1768}{(L_{eq})^{1/2}} = 0.002843 + 0.001226$$

or, $L_{eq} = 1888 \text{ m} = \text{Equivalent length of } 0.5 \text{ m diameter pipe to replace the parallel pipes.}$

The total equivalent length of 0.5 m pipe is now

$$= 450 + 1888 = 2338 \text{ m}$$

$$V = \frac{Q}{A} = \frac{0.35}{\frac{\pi}{4} \times (0.5)^2} = 1.78 \text{ m/s}$$

$$h_f = \frac{f_{eq} \cdot L_{eq} \cdot V^2}{D \times 2g} = \frac{0.018 \times 2338 \times (1.78)^2}{0.5 \times 2 \times 9.81} = 13.59 \text{ m}$$

$$h_t = \text{Total pumping head} = 50 + 13.59 = 63.59 \text{ m}$$

Power input for the pump,

$$P = \frac{wQh_f}{\eta_{\text{pump}}} = \frac{9.81 \times 0.35 \times 63.59}{0.6} = 363.9 \text{ kW (Ans.)}$$

Note : This question could also be solved without considering the equivalent pipe. First the discharge through the each pipe is determined and then the total frictional loss is calculated. However, the calculations are definitely less with the equivalent pipe method.

Example 12.40. (Flow through branched pipes). The water levels in the two reservoirs A and B are 104.5 m and 100 m respectively above the datum. A pipe joins each to a common point D, where pressure is 98.1 kN/m² gauge and height is 83.5 m above datum. Another pipe connects D to another tank C. What will be the height of water level in C assuming the same value of 'f' for all pipes. Take friction co-efficient = 0.0075. The diameters of the pipes AD, BD and CD are 300 mm, 450 mm, 600 mm respectively and their lengths are 240 m, 270 m, 300 m respectively. [IIT Delhi]

Solution.

$$\text{For pipe AD : } D_{AD} = 300 \text{ mm} = 0.3 \text{ m}$$

$$L_{AD} = 240 \text{ m}$$

$$\text{For pipe BD : } D_{BD} = 450 \text{ mm} = 0.45 \text{ m}$$

$$L_{BD} = 270 \text{ m}$$

$$\text{For pipe CD : } D_{CD} = 600 \text{ mm} = 0.6 \text{ m}$$

$$L_{CD} = 300 \text{ m}$$

Friction co-efficient for each pipe, $f = 0.0075$

$$\text{Pressure at D, } p_0 = 98.1 \text{ kN/m}^2$$

Height of water level in tank C :

$$\text{The pressure head at D} = \frac{p_D}{w} = \frac{98.1}{9.81} = 10 \text{ m of water}$$

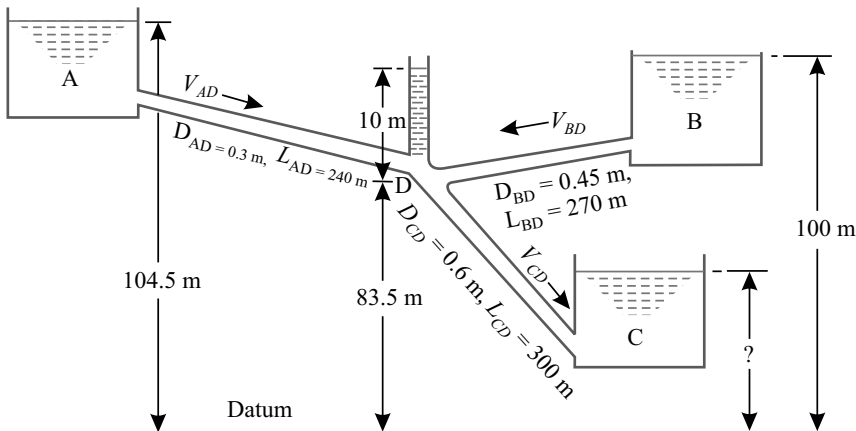


Fig. 12.34

∴ The piezometric head at $D = 83.5 + 10 = 93.5$ m

Head loss between A and $D = 104.5 - 93.5 = 11.0$ m

Head loss between B and $D = 100 - 93.5 = 6.5$ m

Using Darcy-Weisbach equation, we get:

$$\text{For pipe AD :} \quad 11 = \frac{4fL_{AD}V_{AD}^2}{D_{AD} \times 2g} = \frac{4 \times 0.0075 \times 240 \times V_{AD}^2}{0.3 \times 2 \times 9.81}$$

$$\text{or,} \quad V_{AD}^2 = \frac{11 \times 0.3 \times 2 \times 9.81}{4 \times 0.0075 \times 240} = 8.99$$

$$\text{or,} \quad V_{AD} = 3 \text{ m/s}$$

$$\text{For pipe BD :} \quad 6.5 = \frac{4fL_{BD}V_{BD}^2}{D_{BD} \times 2g} = \frac{4 \times 0.0075 \times 270 \times V_{BD}^2}{0.45 \times 2 \times 9.81}$$

$$\text{or,} \quad V_{BD}^2 = \frac{6.5 \times 0.45 \times 2 \times 9.81}{4 \times 0.0075 \times 270} = 7.085$$

$$\text{or,} \quad V_{BD} = 2.66 \text{ m/s}$$

From continuity considerations, we have:

$$Q_{AD} + Q_{BD} = Q_{CD}$$

$$\begin{aligned} \text{or,} \quad Q_{CD} &= (\pi/4) \times D_{AD}^2 \times V_{AD} + (\pi/4) \times D_{BD}^2 \times V_{BD} \\ &= (\pi/4) \times (0.3)^2 \times 3 + (\pi/4) \times (0.45)^2 \times 2.66 = 0.635 \text{ m}^3/\text{s} \end{aligned}$$

∴ Velocity of flow in pipe CD ,

$$V_{CD} = \frac{Q_{CD}}{(\pi/4) \times D_{CD}^2} = \frac{0.635}{(\pi/4) \times 0.6^2} = 2.24 \text{ m/s}$$

$$\text{Head loss in pipe } CD = \frac{4fL_{CD}V_{CD}^2}{D_{CD} \times 2g} = \frac{4 \times 0.0075 \times 300 \times 2.24^2}{0.6 \times 2 \times 9.81} = 3.84 \text{ m}$$

∴ Water level in tank $C = 93.5 - 3.84 = 89.66$ m (Ans.)

Example 12.41. (Flow through branched pipes). Fig. 12.35 shows three reservoirs connected by pipes. Each pipe is 300 mm in diameter and 1500 m long. Assuming co-efficient of friction for each pipe, $f = 0.01$ find the discharge in each pipe.

Solution. Diameter of each pipe, $D_1 = D_2 = D_3 = 300 \text{ mm} = 0.3 \text{ m}$

Length of each pipe, $L_1 = L_2 = L_3 = 1500 \text{ m}$

Co-efficient of friction for each pipe, $f = 0.01$

Discharge in each pipe :

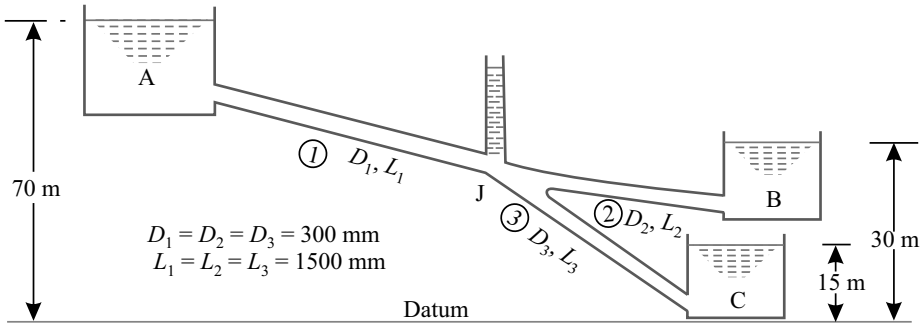
To find out the direction of flow in pipe 2, let us *assume* that no flow occurs in pipe 2. That is, the piezometric level is 30 m.

$$\therefore \text{Head loss in pipe 1, } h_{f_1} = 70 - 30 = 40 \text{ m}$$

$$\text{Also,} \quad h_{f_1} = \frac{4fL_1V_1^2}{D_1 \times 2g}$$

$$\therefore \quad 40 = \frac{4 \times 0.01 \times 1500V_1^2}{0.3 \times 2 \times 9.81}$$

$$\text{or,} \quad V_1^2 = \frac{40 \times 0.3 \times 2 \times 9.81}{4 \times 0.01 \times 1500} = 3.924$$

**Fig. 12.35**

or, $V_1 = 1.981 \text{ m/s}$

$\therefore$ Discharge through the pipe 1,

$$Q_1 = A_1 V_1 = \frac{\pi}{4} \times 0.3^2 \times 1.981 = 0.14 \text{ m}^3/\text{s}$$

Again, head loss in pipe 3, $h_{f_3} = 30 - 15 = 15 \text{ m}$

But,
$$h_{f_3} = \frac{4fL_3 V_3^2}{D_3 \times 2g}$$

$$\therefore 15 = \frac{4 \times 0.01 \times 1500 \times V_3^2}{0.3 \times 2 \times 9.81}$$

$$\text{or, } V_3^2 = \frac{15 \times 0.3 \times 2 \times 9.81}{4 \times 0.01 \times 1500} = 1.471$$

or, $V_3 = 1.213 \text{ m/s}$

$\therefore$ Discharge through the pipe 3,

$$Q_3 = A_3 V_3 = \frac{\pi}{4} \times 0.3^2 \times 1.213 = 0.0857 \text{ m}^3/\text{s}$$

Since $Q_1 > Q_3$, the direction of flow is from J to B.

Considering the flow from reservoir A and B, we have:

$(70 - 30) =$ Head loss in pipe 1 + head loss in pipe 2

$$\text{or, } 40 = h_{f_1} + h_{f_2} = \frac{4fL_1 V_1^2}{D_1 \times 2g} + \frac{4fL_2 V_2^2}{D_2 \times 2g}$$

$$40 = \frac{4 \times 0.01 \times 1500 \times V_1^2}{0.3 \times 2 \times 9.81} + \frac{4 \times 0.01 \times 1500 \times V_2^2}{0.3 \times 2 \times 9.81}$$

$$\text{or, } 40 = 10.2 (V_1^2 + V_2^2)$$

$$\text{or, } V_1^2 + V_2^2 = \frac{40}{10.2} = 3.92$$

$$\text{or, } V_2 = \sqrt{3.92 - V_1^2} \quad \dots(i)$$

Similarly, considering the flow from reservoir A to C, we have:

$$70 - 15 = h_{f_1} + h_{f_3}$$

$$= \frac{4fL_1 V_1^2}{D_1 \times 2g} + \frac{4fL_3 V_3^2}{D_3 \times 2g} = \frac{4 \times 0.01 \times 1500 \times V_1^2}{0.3 \times 2 \times 9.81} + \frac{4 \times 0.01 \times 1500 \times V_3^2}{0.3 \times 2 \times 9.81}$$

$$\text{or, } 55 = 10.2 (V_1^2 + V_3^2)$$

or,
$$V_1^2 + V_3^2 = \frac{55}{10.2} = 5.39$$

or,
$$V_3 = \sqrt{5.39 - V_1^2} \quad \dots(ii)$$

From continuity considerations, we have:

$$Q_1 = Q_2 + Q_3$$

$$A_1 V_1 = A_2 V_2 + A_3 V_3$$

But, $A_1 = A_2 = A_3 \quad (\because D_1 = D_2 = D_3)$

$\therefore V_1 = V_2 + V_3 \quad \dots(iii)$

From eqns. (i), (ii) and (iii), we have:

$$V_1 = \sqrt{3.92 - V_1^2} + \sqrt{5.39 - V_1^2} \quad \dots(iv)$$

By trial and error, we get, $V_1 \approx 1.9 \text{ m/s}$

From eqn. (i) : $V_2 = \sqrt{3.92 - 1.9^2} = 0.56 \text{ m/s}$

From (ii) : $V_3 = \sqrt{5.39 - 1.9^2} = 1.34 \text{ m/s}$

Thus,

$$Q_1 = (\pi/4) \times 0.3^2 \times 1.9 = \mathbf{0.134 \text{ m}^3/\text{s}} \quad (\text{Ans.})$$

$$Q_2 = (\pi/4) \times 0.3^2 \times 0.56 = \mathbf{0.0396 \text{ m}^3/\text{s}} \quad (\text{Ans.})$$

$$Q_3 = (\pi/4) \times 0.3^2 \times 1.34 = \mathbf{0.0947 \text{ m}^3/\text{s}} \quad (\text{Ans.})$$

Example 12.42. Fig. 12-36 shows a pump supplying water from a sump at elevation 20 m to a reservoir at elevation 30 m through a pipeline of 0.5 m diameter and length 1000 m, $f = 0.005$. At mid-length a branch pipe 0.3 diameter, 500 m long, $f = 0.005$, discharges free at elevation 25 m at the rate of $0.25 \text{ m}^3/\text{s}$. Determine :

- The discharge into the reservoir;
- The pressure to be maintained by the pump, and
- The power of the pump assuming an overall efficiency of 70 per cent.

Solution. Refer to Fig. 12-36.

Given :

$D_1 = 0.5 \text{ m},$	$L_1 = 500 \text{ m}$
$D_2 = 0.5 \text{ m},$	$L_2 = 500 \text{ m}$
$D_3 = 0.3 \text{ m},$	$L_3 = 500 \text{ m}$

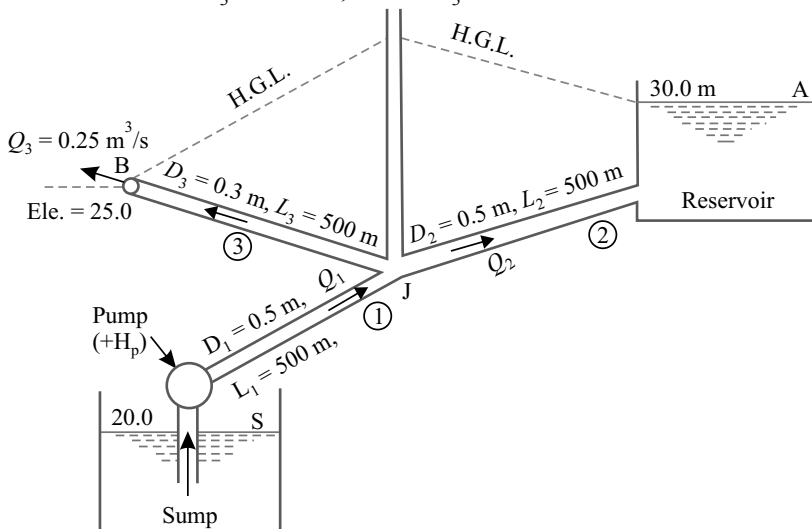


Fig. 12.36

Co-efficient of friction, $f_1 = f_2 = f_3 = f = 0.005$

Discharge through pipe 3, $Q_3 = 0.25 \text{ m}^3/\text{s}$

Overall efficiency, $\eta_0 = 70\%$.

(i) The discharge into the reservoir, Q_2 :

Energy at the joint J,

$$\begin{aligned}
 E_J &= E_B \text{ (energy at B)} + (h_f)_{JB} \\
 &= \left(\frac{p_B}{w} + \frac{V_3^2}{2g} + z_B \right) + (h_f)_{JB} \\
 &= \frac{p_B}{w} + \frac{V_3^2}{2g} + z_B + \frac{4f_3 L_3 V_3^2}{D_3 \times 2g} \\
 &= 0 + \frac{V_3^2}{2g} \left(1 + \frac{4f_3 L_3}{D_3} \right) + 25 \\
 &= \left(\frac{0.25}{(\pi/4) \times 0.3^2} \right)^2 \times \frac{1}{2 \times 9.81} \left(1 + \frac{4 \times 0.005 \times 500}{0.3} \right) + 25 \\
 &= 0.637 (1 + 33.33) + 25 = 46.87 \text{ m}
 \end{aligned}$$

$$\text{Head loss, } h_{f_2} = 46.87 - 30 = 16.87 \text{ m}$$

$$\text{i.e.} \quad 16.87 = \frac{4f_2 L_2 V_2^2}{D_2 \times 2g} = \frac{4 \times 0.005 \times 500 \times V_2^2}{0.5 \times 2 \times 9.81}$$

$$\text{or,} \quad V_2 = \left(\frac{16.87 \times 0.5 \times 2 \times 9.81}{4 \times 0.005 \times 500} \right)^{1/2} = 4.07 \text{ m/s}$$

$$\begin{aligned}
 \therefore \quad Q_2 &= (\pi/4) \times 0.5^2 \times 4.07 = \mathbf{0.8 \text{ m}^3/\text{s}} \quad (\text{Ans.}) \\
 Q_1 &= Q_2 + Q_3 = 0.8 + 0.25 = 1.05 \text{ m}^3/\text{s}
 \end{aligned}$$

$$\text{Head loss, } h_{f_1} = \frac{4f_1 L_1 V_1^2}{D_1 \times 2g} = \frac{4 \times 0.005 \times 500 \times (5.35)^2}{0.5 \times 2 \times 9.81} = 29.17 \text{ m}$$

$$\left(\because V_1 = \frac{Q_1}{A_1} = \frac{1.05}{(\pi/4) \times 0.5^2} = 5.35 \text{ m/s} \right)$$

Applying energy equation between the sump (S) and the junction (J), we have:

$$E_S + H_p = h_{f_1}$$

$$0 + 0 + 20 + H_p = 46.87 + 29.17 = 76.04 \text{ m}$$

$$H_p = 56.04 \text{ m}$$

(i) The pressure to be maintained by the pump, p :

$$p = wH_p = 9.81 \times 56.04 = \mathbf{549.7 \text{ kN/m}^2} \quad (\text{Ans.})$$

(ii) The power of pump, P :

$$P = \frac{wQ_1 H_p}{\eta_0} = \frac{9.81 \times 1.05 \times 56.04}{0.7} = \mathbf{824.6 \text{ kW}} \quad (\text{Ans.})$$

Example 12.43. (Pipe networks). Fig. 12-37 shows a network in which Q and h_f refer to discharges and pressure drops respectively. Subscripts 1, 2, 3, 4 and 5 designate respective values in pipe lengths AC, BC, CD, DA and AC. Subscripts A, B, C and D designate discharges entering or leaving the junction points A, B, C and D respectively.

By sticking to the values given in the figure find the following discharges Q_B , Q_2 , Q_4 and Q_5 , and pressure drops h_{f_4} and h_{f_5} and give these computed values at their respective places on a neat sketch of the network along with flow directions. [GATE]

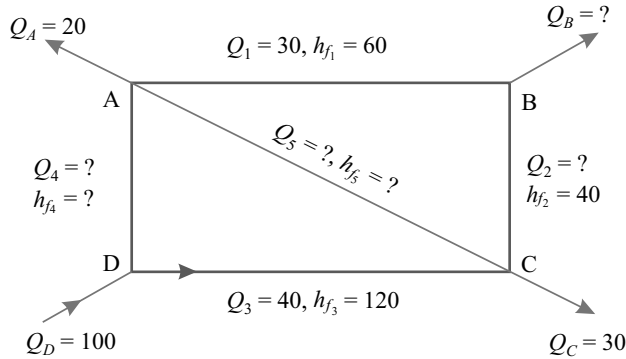


Fig. 12.37

Solution. At junctions, $\Sigma Q = 0$

i.e., Discharge entering the junction = Discharge leaving the junction

At junction D :

$$Q_D = Q_3 + Q_4$$

or,

$$100 = 40 + Q_4$$

or,

$$Q_4 = 100 - 40 = 60$$

...leaving the junction

At junction A :

$$Q_4 = Q_A + Q_1 + Q_5$$

$$60 = 20 + 30 + Q_5$$

$\therefore$

$$Q_5 = 60 - 20 - 30 = 10$$

...leaving the junction

At junction C :

$$Q_3 + Q_5 + Q_2 = Q_C$$

$$40 + 10 + Q_2 = 30$$

$\therefore$

$$Q_2 = 30 - 40 - 10 = -20$$

...leaving the junction C

At junction B :

$$Q_1 + Q_2 = Q_B$$

$$30 + 20 = Q_B$$

i.e.,

$$Q_B = 50$$

...leaving the junction B

For each elementary circuit, $\Sigma h_f = 0$

Circuit ABC :

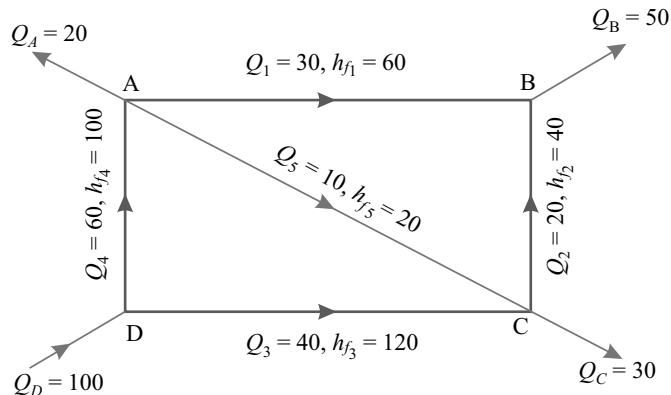


Fig. 12.38

$$h_{f_1} - h_{f_2} - h_{f_5} = 0$$

$$60 - 40 - h_{f_5} = 0$$

$$\therefore h_{f_5} = 20$$

Circuit ACD :

$$h_{f_5} - h_{f_3} + h_{f_4} = 0$$

$$20 - 120 + h_{f_4} = 0$$

$$\therefore h_{f_4} = 100$$

The calculated values and the flow directions are shown in Fig. 12.38.

12.9. SYPHON

A **syphon** is a long bent pipe employed for carrying water from a reservoir at a higher elevation to another reservoir at a lower elevation when the two reservoirs are separated by a hill or high level ground in between as shown in Fig. 12.39.

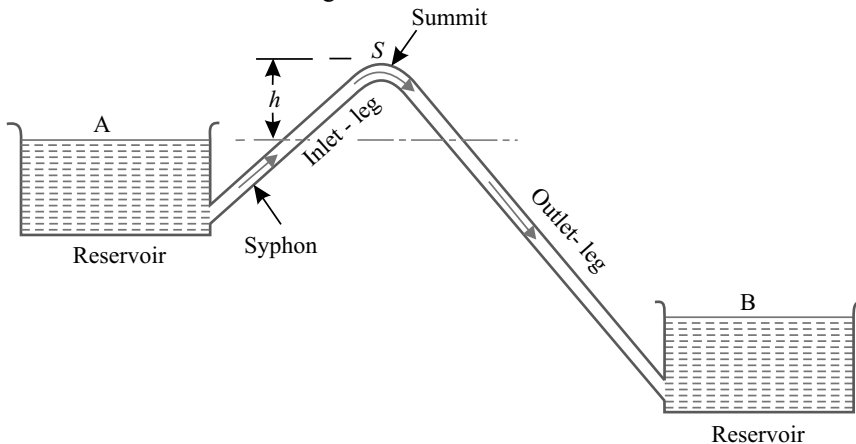


Fig. 12.39. Syphon.

The highest point (S) of the syphon is called the **summit**. The pressure at the point S is **less than atmospheric pressure** (since S lies above the free water surface in the tank A). The pressure at S can be reduced theoretically to -10.3 m of water but in actual practice this pressure is only -7.6 m of water (or $10.3 - 7.6 = 2.7$ m of water *absolute*). When the pressure at S becomes less than 2.7 m of water absolute, the dissolved air and other gases would come out from water and collect at the summit. *Therefore syphon should be so laid that no section of the pipe will be more than 7.6 m above the hydraulic gradient at that section.* Moreover, in order to limit the reduction of the pressure at the summit the length of the *inlet-leg* (rising portion of the syphon) of the syphon is also required to be limited (this is so because, if the *inlet leg* is very long a considerable loss of head due to friction is caused, resulting in further reduction of the pressure at the summit).

Example 12.44. Two reservoirs, having a difference in elevation of 15 m, are connected by a 200 mm diameter syphon. The length of the syphon is 400 m and the summit is 3 m above the water level in the upper reservoir. The length of the pipe from upper reservoir to the summit is 120 m. If the co-efficient of friction is 0.005 , determine :

- (i) Discharge through the syphon, and
- (ii) Pressure at the summit.

Neglect minor losses.

Solution. Diameter of the syphon, $D = 200 \text{ mm} = 0.2 \text{ m}$

Length of the syphon, $L = 400 \text{ m}$

Difference in level of the two reservoirs, $H = 15 \text{ m}$

Height of the summit from upper reservoir, $h = 3 \text{ m}$

Co-efficient of friction, $f = 0.005$

(i) **Discharge through the syphon, Q :**

Applying Bernoulli's equation to points A and B , we get:

$$\frac{p_A}{w} + \frac{V_A^2}{2g} + z_A = \frac{p_B}{w} + \frac{V_B^2}{2g} + z_B + \text{loss of head due to friction from } A \text{ to } B$$

$$\text{or,} \quad 0 + 0 + z_A = 0 + 0 + z_B + h_f$$

$$[\because p_A = p_B = \text{atmospheric pressure, and } V_A = V_B = 0]$$

$$\text{or,} \quad z_A - z_B = h_f = 15$$

$$\text{But,} \quad h_f = \frac{4fLV^2}{D \times 2g} \quad (\text{where, } V = \text{velocity of water in the pipe})$$

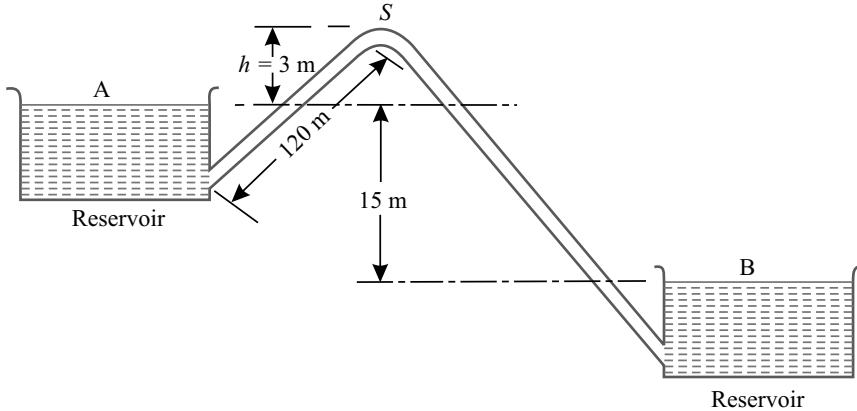


Fig. 12.40

$$\therefore \quad 15 = \frac{4 \times 0.005 \times 400 \times V^2}{0.2 \times 2 \times 9.81}$$

$$\text{or,} \quad V^2 = \frac{15 \times 0.2 \times 2 \times 9.81}{4 \times 0.005 \times 400}$$

$$\text{or,} \quad V = 2.7 \text{ m/s}$$

$$\begin{aligned} \therefore \quad \text{Discharge, } Q &= \text{Area} \times \text{velocity} \\ &= \frac{\pi}{4} \times 0.2^2 \times 2.7 = \mathbf{0.0848 \text{ m}^3/\text{s}} \quad (\text{Ans.}) \end{aligned}$$

(ii) **Pressure at summit :**

Applying Bernoulli's equation to points A and S , we get:

$$\frac{p_A}{w} + \frac{V_A^2}{2g} + z_A = \frac{p_S}{w} + \frac{V_S^2}{2g} + z_S$$

+ loss of head due to friction between A and S

(... assuming datum line passing through A)

$$0 + 0 + 0 = \frac{p_s}{w} + \frac{2.7^2}{2 \times 9.81} + 3 + \frac{4 \times 0.005 \times 120 \times (2.7)^2}{0.2 \times 2 \times 9.81}$$

$$0 = \frac{p_s}{w} + 0.37 + 3 + 4.46 = \frac{p_s}{w} + 7.83$$

$$\frac{p_s}{w} = 7.83 \text{ m of water (Ans.)}$$

Example 12.45. A 200 mm diameter pipe, 4000 m long connects two reservoirs whose surface levels differ by 40 m. At a distance of 400 m from the upper reservoir, the pipe crosses a ridge the summit of which is 9 m above the level of water in the upper reservoir. Determine :

(i) The minimum depth of the pipe below the summit of the ridge, if the absolute pressure head at the summit of syphon is not to fall below 3.0 m of the water (absolute).

(ii) The discharge through the pipe.

Take co-efficient of friction $f = 0.006$ and atmospheric head = 10.3 m of water. Neglect minor losses.

Solution.

Diameter of the pipe, $D = 200 \text{ mm} = 0.2 \text{ m}$

Total length of the pipe, $L = 4000 \text{ m}$

Length of syphon from upper reservoir to the summit, $L_1 = 400 \text{ m}$

Difference in levels of two reservoirs = 40 m

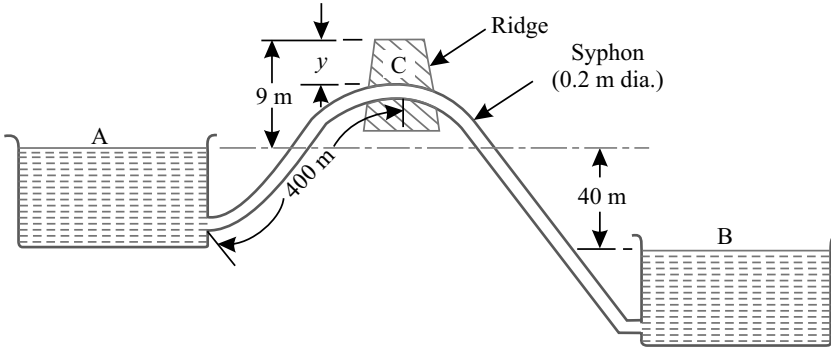


Fig. 12.41

Friction co-efficient, $f = 0.006$

Atmospheric pressure head = 10.3 m of water

Pressure head at C, $\frac{p_s}{w} = 3.0 \text{ m of water (absolute)}$

(i) **Minimum depth of pipe below the summit, y :**

Let, y = Depth of the pipe below the summit of the ridge.

Then, height of syphon from the water surface in the upper reservoir = $(9 - y)$

Applying Bernoulli's equation at A and B (taking datum line passing through B), we have:

$$\frac{p_A}{w} + \frac{V_A^2}{2g} + z_A = \frac{p_B}{w} + \frac{V_B^2}{2g} + z_B + (h_f)_{A-B}$$

$$0 + 0 + 40 = 0 + 0 + 0 + \frac{4fLV^2}{D \times 2g} \quad (\because V_A = V_B = 0)$$

$$\text{or,} \quad 40 = \frac{4 \times 0.006 \times 4000 \times V^2}{0.2 \times 2 \times 9.81}$$

$$\therefore V = \left(\frac{40 \times 0.2 \times 2 \times 9.81}{40 \times 0.006 \times 4000} \right)^{1/2} = 1.278 \text{ m/s}$$

Now applying Bernoulli's equation at A and C (assuming datum line passing through A), we have:

$$\frac{p_A}{w} + \frac{V_A^2}{2g} + z_A = \frac{p_C}{w} + \frac{V_C^2}{2g} + z_C + (h_f)_{A-C}$$

$$10.3 + 0 + 0 = 3.0 + \frac{V^2}{2g} + (9 - y) + \frac{4fL_1V^2}{D \times 2g}$$

$$\text{or,} \quad 10.3 = 3 + \frac{(1.278)^2}{2 \times 9.81} + (9 - y) + \frac{4 \times 0.006 \times 400 \times (1.278)^2}{0.2 \times 2 \times 9.81}$$

$$= 3 + 0.0832 + (9 - y) + 3.99$$

$$\text{or,} \quad y = 5.77 \text{ m (Ans.)}$$

(ii) The discharge through the pipe, Q :

$$Q = A \times V = \frac{\pi}{4} \times 0.2^2 \times 1.278$$

$$= 0.04 \text{ m}^3/\text{s (Ans.)}$$

Example 12.46. Water from a main canal is syphoned to a branch canal over an embankment by means of wrought iron pipes of 90 mm diameter. The length of pipeline up to the summit is 25 m and the total length is 65 m. Entry loss may be assumed as one-half of the velocity head in the pipe. Assume friction factor, $f = 0.03$. Water surface elevation in the branch canal is 10 m below that of the main canal.

- (i) If the total quantity of water required to be conveyed is $0.06 \text{ m}^3/\text{s}$, how many pipelines are needed?
- (ii) What is the maximum permissible height of the summit above the water level in the main canal so that the water pressure at summit may not fall below 20 kN/m^2 absolute, the barometric reading being 10 m of water? (UPSC Exams.)

Solution.

Diameter of the pipe, $D = 90 \text{ mm} = 0.9 \text{ m}$

Total length of the pipeline, $L = 65 \text{ m}$

The length of the pipeline up to the summit, $L_1 = 25 \text{ m}$

$$\text{Entry loss} = 0.5 \frac{V^2}{2g}$$

$$\text{Friction Factor, } f = 0.03$$

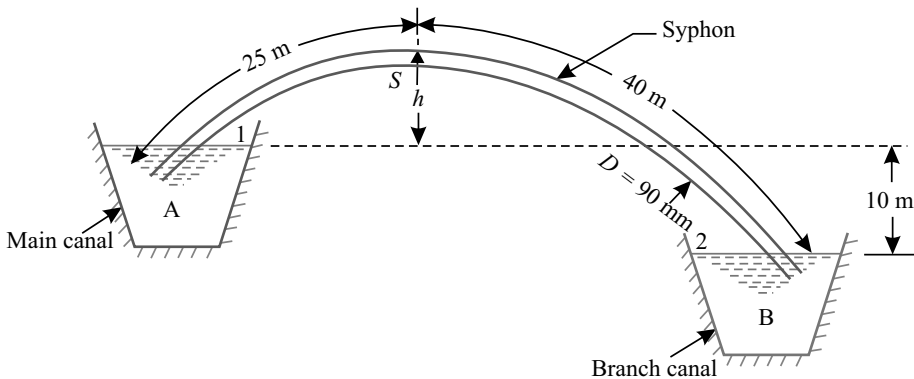


Fig. 12.42

Total discharge $Q = 0.06 \text{ m}^3/\text{s}$

Pressure at the summit, $p_s = 20 \text{ kN/m}^2$ absolute

Atmospheric pressure head = 10 m of water.

(i) Number of pipelines needed :

Applying Bernoulli's equation between water surfaces (1 and 2) of two canals, we have:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 + 0.5 \frac{V^2}{2g} + \frac{fLV^2}{D \times 2g} + \frac{V^2}{2g}$$

$$0 + 0 + 10 = 0 + 0 + 0 + 1.5 \frac{V^2}{2g} + \frac{0.03 \times 65 \times V^2}{0.09 \times 2g}$$

$$\text{or,} \quad 10 = \frac{23.17 V^2}{2g} \quad (\because V_1 = V_2 = 0)$$

$$\text{or,} \quad V = \left(\frac{10 \times 2 \times 9.81}{23.17} \right)^{1/2} = 2.91 \text{ m/s}$$

(where, V = velocity of flow through the pipe)

Discharge through a 90 mm diameter pipe,

$$= \frac{\pi}{4} \times 0.09^2 \times 2.91 = 0.0185 \text{ m}^3/\text{s}$$

Number of 90 mm diameter pipes required to convey $0.06 \text{ m}^3/\text{s}$

$$= \frac{0.06}{0.0185} = 3.24, \text{ say } \mathbf{4} \text{ (Ans.)}$$

(ii) Height of the summit, h :

Invoking Bernoulli's equation between water surface 1 and the summit point S, we have:

$$\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 = \frac{p_s}{w} + \frac{V_s^2}{2g} + z_s + \frac{0.5V^2}{2g} + \frac{fL_1V^2}{D \times 2g}$$

$$10 + 0 + 0 = \frac{20}{9.81} + \frac{V^2}{2g} + h + \frac{0.5V^2}{2g} + \frac{0.03 \times 25 \times V^2}{0.09 \times 2g}$$

$$\text{or,} \quad 10 = 2.038 + \frac{1.5 \times 2.91^2}{2 \times 9.81} + h + \frac{0.03 \times 25 \times (2.91)^2}{0.09 \times 2 \times 9.81}$$

$$\text{or,} \quad 10 = 2.038 + 0.647 + h + 3.596$$

$$\text{or,} \quad h = \mathbf{3.72 \text{ m}} \text{ (Ans.)}$$

12.10. POWER TRANSMISSION THROUGH PIPES

The transmission of power through pipes carrying water or other liquids is commonly used for working of several hydraulic machines. The hydraulic power transmitted by a pipe however depends on (i) the *discharge* passing through the pipe and (ii) the *total head of water* (or liquid).

Consider a pipe AB connected to a high level storage tank as shown in Fig. 12-43.

Let, H = Head of water available at the inlet of pipe, m,

L = Length of the pipe, m,

D = Diameter of the pipe, m,

V = Velocity of water in the pipe m/s,

f = Co-efficient of friction, and

h_f = Loss of head in the pipe AB, due to friction, m.

Weight of water flowing through the pipe per second

$$= wQ = wAV \quad \dots(i)$$

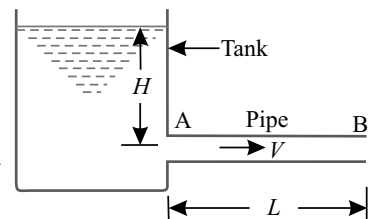


Fig. 12.43

(where, Q = discharge of water through the pipe, m^3/s)

and, net head of water available at B (neglecting minor losses)

$$= H - h_f = H - \frac{4fLV^2}{D \times 2g}$$

Also, The **efficiency of transmission**,

$$\eta = \frac{H - h_f}{H}$$

And, **Power, P** = $\frac{\left\{ \begin{array}{l} \text{Weight of water flowing/sec.} \\ \times \text{head of water} \end{array} \right\}}{1000} \text{ kW}$

$$= wQ(H - h_f) \text{ kW}$$

(where, $w = 9.81 \text{ kN/m}^3$ for water)

$$= wAV \left(H - \frac{4fLV^2}{D \times 2g} \right) \text{ kW}$$

$$= wA \left(HV - \frac{4fLV^3}{D \times 2g} \right) \text{ kW} \quad \dots(iii)$$

It is evident from eqn. (iii) that power transmitted depends upon the velocity of water (V), as the other things are constant.

$\therefore$ Power transmitted will be *maximum*, when:

$$\frac{d(P)}{dV} = 0$$

or, $\frac{d}{dV} \left[wA \left(HV - \frac{4fLV^3}{D \times 2g} \right) \right] = 0$

or, $wA \left(H - \frac{4 \times 3fLV^2}{D \times 2g} \right) = 0$

or, $H - 3 \times \frac{4fLV^2}{D \times 2g} = 0$

or, $H - 3h_f = 0 \quad \left[\because h_f = \frac{4fLV^2}{D \times 2g} \right]$

or, $H = 3h_f$

or, $h_f = \frac{H}{3}$

It means that *power transmitted through the pipe is maximum, when head lost due to friction in the pipe is equal to $\frac{1}{3}$ of the total supply head.*

The **maximum efficiency** would correspond to the maximum power transmitted and hence maximum efficiency,

$$\eta = \frac{H - \frac{H}{3}}{H} = \frac{\frac{2}{3}H}{H} = \frac{2}{3} \text{ or } 66.7\%$$

Example 12.47. A 2500 m long pipeline is used for transmission of power. 120 kW power is to be transmitted through the pipe in which water having a pressure of 4000 kN/m² at inlet is flowing. If the pressure drop over the length of pipe is 800 kN/m² and $f = 0.006$, find :

(i) Diameter of the pipe, and

(ii) Efficiency of transmission.

Solution. Length of the pipeline, $L = 2500$ m

Power transmitted, $P = 120$ kW

Pressure at inlet, $p = 4000$ kN/m²

$$H = \frac{p}{w} = \frac{4000}{9.81} = 407.7 \text{ m}$$

Pressure drop = 800 kN/m²

$$\therefore \text{Loss of head, } h_f = \frac{800}{9.81} = 81.5 \text{ m}$$

$$\therefore \text{Co-efficient of friction, } f = 0.006.$$

(i) **Diameter of the pipe, D :**

Head available at the end of the pipe, $H - h_f = 407.7 - 81.5 = 326.2$ m

Now, power transmitted is given by : $P = wQ(H - h_f)$ kW

$$120 = 9.81 \times Q \times 326.2$$

where, Q = Discharge through the pipe in m³/s, and

and, w = Specific weight of water = 9.81 kN/m³

$$\therefore Q = \frac{120}{9.81 \times 326.2} = 0.0375 \text{ m}^3/\text{s}$$

$$\text{But, } Q = \frac{\pi}{4} D^2 \times V$$

$$\therefore 0.0375 = \frac{\pi}{4} D^2 \times V$$

$$\text{or, } V = \frac{0.0375 \times 4}{\pi D^2} = \frac{0.0477}{D^2} \quad \dots(i)$$

$$\text{The head lost due to friction, } h_f = \frac{4fLV^2}{D \times 2g}$$

$$\text{But, } h_f = 81.5 \text{ m (calculated above)}$$

$$\therefore 81.5 = \frac{4 \times 0.006 \times 2500 \times (0.0477/D^2)^2}{D \times 2 \times 9.81}$$

$$\text{or, } 81.5 = \frac{4 \times 0.006 \times 2500 \times (0.0477)^2}{D^5 \times 2 \times 9.81}$$

$$\text{or, } D^5 = \frac{4 \times 0.006 \times 2500 \times (0.0477)^2}{81.5 \times 2 \times 9.81}$$

$$\text{or, } D = 0.1535 \text{ m or } \mathbf{153.5 \text{ mm (Ans.)}}$$

(ii) **Efficiency of transmission, η :**

$$\eta = \frac{H - h_f}{H} = \frac{407.7 - 81.5}{407.7} = 0.8 \text{ or } \mathbf{80\% (Ans.)}$$

12.11. FLOW THROUGH NOZZLE AT THE END OF A PIPE

Refer to Fig. 12-44. A **nozzle** is a tapering mouthpiece, which is fitted to the outlet end of a pipe. The total energy at the end of the pipe consists of pressure energy and kinetic energy. By fitting the nozzle at the end of a pipe, the total energy is converted into *kinetic energy*. A high velocity is required in the fields of power development, fire fighting, mining, etc.

Fig. 12-44 shows a nozzle fitted at the end of a pipe connected to a reservoir.

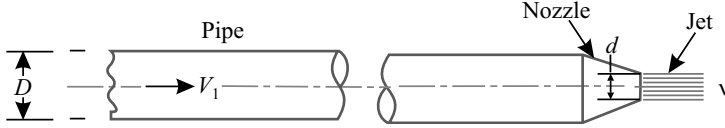


Fig. 12.44

- Let,
- D = Diameter of the pipe,
 - L = Length of the pipe,
 - d = Diameter of the nozzle,
 - V = Velocity of flow in pipe,
 - v = Velocity of flow at the outlet of the nozzle,
 - f = Co-efficient of friction for the pipe, and
 - H = Height of water level in the reservoir above the centre-line of the nozzle.

Head lost due to friction in pipe,

$$h_f = \frac{4fLV^2}{D \times 2g}$$

$\therefore$ Head available at the base of the nozzle

$$= H - h_f = H - \frac{4fLV^2}{D \times 2g}$$

Assuming the minor losses and losses in the nozzle to be negligible, we have:

$$\text{Total head at the nozzle outlet} = \frac{v^2}{2g}$$

$$\therefore H = h_f + \frac{v^2}{2g} = \frac{4fLV^2}{D \times 2g} + \frac{v^2}{2g} \quad \dots(i)$$

From continuity consideration, we have:

$$AV = av$$

(where A and a are the areas of the pipe and area of the nozzle at outlet respectively)

$$\text{or,} \quad V = \frac{av}{A}$$

Substituting the value of V in eqn. (i), we get:

$$\begin{aligned} H &= \frac{4fLa^2v^2}{D \times 2g \times A^2} + \frac{v^2}{2g} \\ &= \frac{v^2}{2g} \left(\frac{1 + 4fLa^2}{D \times A^2} \right) \end{aligned}$$

$$\therefore v = \sqrt{\frac{2gH}{1 + \frac{4fL}{D} \times \frac{a^2}{A^2}}} \quad \dots(12.20)$$

$\therefore$ Discharge through the nozzle = $a \times v$

12.11.1 Power Transmitted through the Nozzle

Mass of liquid flowing per second at the outlet of the nozzle, $m = \rho av$

The K.E. of the jet at outlet of the nozzle

$$= \frac{1}{2} mv^2 = \frac{1}{2} \times \rho av \times v^2 = \frac{1}{2} \rho av^3$$

$$\therefore \text{Power available at the outlet of nozzle} = \frac{1}{2} \rho av^3 \text{ watts}$$

Also, power available at the inlet of pipe = wQH

$\therefore$ Efficiency of power transmission through the nozzle,

$$\eta = \frac{\text{Power available at the outlet of nozzle}}{\text{Power available at the inlet of pipe}} = \frac{\frac{1}{2} \rho av^3}{wQH}$$

But,

$$w = \rho g \quad \text{and} \quad Q = av$$

$$\therefore \eta = \frac{\frac{1}{2} \rho av^3}{\rho g \times av \times H} = \frac{v^2}{2gH} = \left[\frac{1}{1 + \frac{4fL}{D} \times \frac{a^2}{A^2}} \right] \quad \dots(12.21)$$

$$\left[\because v = \sqrt{\frac{2gH}{1 + \frac{4fL}{D} \times \frac{a^2}{A^2}}} \quad \dots \text{eqn. (12.20)} \right]$$

12.11.2 Condition for Transmission of Maximum Power Through Nozzle

$$\text{We know that,} \quad H = h_f + \frac{v^2}{2g} = \frac{4fLV^2}{D \times 2g} + \frac{v^2}{2g}$$

$$\text{or,} \quad \frac{v^2}{2g} = H - \frac{4fLV^2}{D \times 2g}$$

But power transmitted through the nozzle,

$$\begin{aligned} P &= \frac{1}{2} \rho av^3 = \frac{1}{2} \rho av \times v^2 \\ &= \frac{1}{2} \rho av \left[2g \left(H - \frac{4fLV^2}{D \times 2g} \right) \right] \\ &= w av \left(H - \frac{4fLV^2}{D \times 2g} \right) \quad \dots(12.22) \end{aligned}$$

From continuity consideration, we have:

$$AV = av \quad \text{or} \quad V = \frac{av}{A}$$

Substituting the value of V in eqn. (12.22), we get:

$$\text{Power transmitted through nozzle, } P = w av \left(H - \frac{4fL \times a^2 v^2}{D \times 2g \times A^2} \right) \quad \dots[12.22(a)]$$

Power transmitted will be *maximum*, when $\frac{dP}{dv} = 0$

$$\frac{d}{dv} \left[wav \left(H - \frac{4fL}{D \times 2g} \times \frac{a^2 v^2}{A^2} \right) \right] = 0$$

$$\text{or, } \frac{d}{dv} \left[wa \left(Hv - \frac{4fL}{D \times 2g} \times \frac{a^2 v^3}{A^2} \right) \right] = 0$$

$$\text{or, } H - 3 \times \frac{4fL}{D \times 2g} \times V^2 = 0 \quad \left(\because \frac{a^2 v^2}{A^2} = V^2 \right)$$

$$\text{or, } H - 3h_f = 0 \quad \left(\because h_f = \frac{4fLV^2}{D \times 2g} \right)$$

$$\text{or, } h_f = \frac{H}{3} \quad \dots(12.23)$$

The eqn. (12.23) indicates that the power transmitted by a nozzle is maximum when the head lost due to friction in pipe is equal to one-third the total head supplied at the inlet of pipe.

12.11.3 Diameter of the Nozzle for Transmitting Maximum Power

$$\text{We know that, } H = h_f + \frac{v^2}{2g}$$

$$\text{But, } H = 3h_f \quad [\text{From eqn. (12.22)}]$$

$$\therefore 3h_f = h_f + \frac{v^2}{2g} \quad \text{or} \quad 2h_f = \frac{v^2}{2g}$$

$$\frac{2 \times 4fLV^2}{D \times 2g} = \frac{v^2}{2g}$$

For continuity considerations, we have:

$$AV = av \quad \text{or} \quad V = \frac{av}{A}$$

$$\therefore \frac{2 \times 4fL \times a^2 v^2}{D \times 2g \times A^2} = \frac{v^2}{2g}$$

$$\text{or, } \frac{A^2}{a^2} = \frac{8fL}{D} \quad \text{or} \quad \frac{A}{a} = \sqrt{\frac{8fL}{D}} \quad \dots(12.24)$$

Eqn. (12.24) gives the ratio between the areas of the supply pipe and the nozzle for maximum power transmission.

Substituting the values of A and a in eqn. (12.24) and squaring both sides, we have:

$$\left(\frac{\frac{\pi}{4} \times D^2}{\frac{\pi}{4} \times d^2} \right)^2 = \frac{8fL}{D}$$

$$\text{or, } \frac{D^4}{d^4} = \frac{8fL}{D} \quad \text{or} \quad D^5 = 8fLd^4$$

$$\therefore d = \left(\frac{D^5}{8fL} \right)^{1/4} \quad \dots(12.25)$$

Example 12.48. A nozzle is fitted to a pipe 120 mm in diameter and 250 m long, with co-efficient of friction as 0.01. If the available head at the nozzle is 100 m find the diameter of the nozzle and the maximum power transmitted by a jet of water discharging freely out of a nozzle.

Solution. Diameter of the pipe, $D = 120 \text{ mm} = 0.12 \text{ m}$

Length of the pipe, $L = 250 \text{ m}$

Co-efficient of friction, $f = 0.01$

Head of water, $H = 100 \text{ m}$.

(i) **Diameter of the nozzle for maximum power, d :**

Using the relation :

$$d = \left[\frac{D^5}{8fL} \right]^{1/4} \quad \dots [\text{Eqn. (12.25)}]$$

$$= \left[\frac{0.12^5}{8 \times 0.01 \times 250} \right]^{1/4} = 0.0334 \text{ m or } \mathbf{33.4 \text{ mm}}$$

i.e., $d = \mathbf{33.4 \text{ mm}}$ (Ans.)

(ii) **Maximum power transmitted by the jet, P :**

We know that for the maximum transmission of power, the head lost due to friction $= \frac{H}{3}$

$$\therefore \text{Available head, } h = 100 - \frac{100}{3} = \frac{200}{3} = 66.67 \text{ m}$$

$\therefore$ Velocity of water through the nozzle,

$$v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 66.67} = 36.2 \text{ m/s}$$

Now using the relation, $P = wQH$, we have:

($\because Q = a.v$)

$$P = w a v H$$

$$= 9.81 \times \frac{\pi}{4} \times 0.0334^2 \times 36.2 \times 66.67 = 20.74 \text{ kW}$$

i.e., $P = \mathbf{20.74 \text{ kW}}$ (Ans.)

Example 12.49. Find the maximum power transmitted by a jet of water discharging freely out of nozzle fitted to a pipe 300 m long and 100 mm diameter with co-efficient of friction as 0.01. The available head at the nozzle is 90 m. [Delhi University]

Solution. Length of the pipe, $L = 300 \text{ m}$

Diameter of the pipe, $D = 100 \text{ mm} = 0.1 \text{ m}$

$$\text{Area of the pipe, } A = \frac{\pi}{4} \times 0.1^2 = 0.007854 \text{ m}^2$$

Co-efficient of friction, $f = 0.01$

Head available at the nozzle, $h = 90 \text{ m}$

Maximum power transmitted, P :

Let, a = Area of the nozzle.

$$\text{Also, } \frac{A}{a} = \sqrt{\frac{8fL}{D}}$$

$$\therefore \frac{0.007854}{a} = \sqrt{\frac{8 \times 0.01 \times 300}{0.1}} = 15.492$$

or,
$$a = \frac{0.007854}{15.492} = 0.0005069 \text{ m}^2$$

Also,
$$h = \frac{v^2}{2g}$$

$$\therefore 90 = \frac{v^2}{2g} \quad \text{or} \quad v = \sqrt{90 \times 2 \times 9.81} = 42.02 \text{ m/s}$$

Discharge through the nozzle, $Q = av = 0.0005069 \times 42.02 = 0.0213 \text{ m}^3/\text{s}$

$\therefore$ Maximum power transmitted,

$$P = wQh = 9.81 \times 0.0213 \times 90 \text{ kW} = \mathbf{18.8 \text{ kW}} \quad (\text{Ans.})$$

($\because w = 9.81 \text{ kN/m}^3$)

Example 12.50. A fire engine supplies water to a hosepipe, 75 m long and 0.075 m in diameter, at a pressure of 294 kN/m² (gauge). The discharge end of the hosepipe has a nozzle of diameter d fixed to it. Taking friction factor as 0.032, determine the diameter d of the nozzle, so that the momentum of the issuing jet may be a maximum. (UPSC Exams.)

Solution. Diameter of the hosepipe, $D = 0.075 \text{ m}$

Length of the hosepipe, $L = 75 \text{ m}$

Pressure of water, $p = 294 \text{ kN/m}^2$ (gauge)

Friction factor ($= 4f$) = 0.032

Diameter of the nozzle, d :

$$\text{Head lost in the hosepipe, } h_f = \frac{4fLV^2}{D \times 2g} = \frac{0.032 \times 75 \times V^2}{0.075 \times 2g} = \frac{32V^2}{2g}$$

Applying the continuity equation to the pipe and the jet, we get:

$$Q = AV = av$$

or,
$$v = \frac{AV}{a} = \frac{\frac{\pi}{4} \times D^2 \times V}{\frac{\pi}{4} \times d^2} = \left(\frac{0.0075}{d} \right)^2 V$$

where, d = Diameter of the nozzle,

a = Area of the jet, and

A = Area of the hosepipe.

Applying Bernoulli's equation to the hosepipe at the fire engine and to the nozzle jet, considering the hosepipe and the nozzle to be in the horizontal plane, we have:

$$\frac{294}{9.81} + \frac{V^2}{2g} + 0 = 0 + \frac{v^2}{2g} + 0 + h_f \quad (\text{head lost in the hosepipe})$$

(Neglecting the energy loss in the nozzle, being very small)

or,
$$30 + \frac{V^2}{2g} = \left(\frac{0.075}{d} \right)^4 \frac{V^2}{2g} + \frac{32V^2}{2g}$$

or,
$$30 = \frac{V^2}{2g} \left[32 + \left(\frac{0.075}{d} \right)^4 - 1 \right]$$

or,
$$V = \sqrt{\frac{30 \times 2 \times 9.81}{31 + \left(\frac{0.075}{d} \right)^4}}$$

Momentum of issuing jet,

$$M = \rho Qv = 1000 \times \left\{ \left(\frac{\pi}{4} \right) \times (0.075)^2 \times V \right\} \left(\frac{0.075}{d} \right)^2 V = 0.0248 \frac{V^2}{d^2}$$

Substituting for V , we get:

$$M = \frac{0.0248}{d^2} \left[\frac{30 \times 2 \times 9.81}{31 + (0.075/d)^4} \right] = \frac{14.6}{d^2 [31 + (0.075/d)^4]}$$

For momentum to be maximum, $\frac{dM}{dd} = 0$

$$\frac{d}{dd} \left[d^2 \left\{ 31 + \left(\frac{0.075}{d} \right)^4 \right\} \right] = 0$$

$$\text{or,} \quad \frac{d}{dd} \left[31 d^2 + \frac{0.075^4}{d^2} \right] = 0$$

$$\text{or,} \quad 62d - \frac{2 \times 0.075^4}{d^3} = 0$$

$$\text{or,} \quad 62d^4 - 6.328 \times 10^{-5} = 0$$

$$\text{or,} \quad d = \left(\frac{6.328 \times 10^{-5}}{62} \right)^{1/4} = 0.03178 \text{ m} \quad \text{or} \quad \mathbf{31.78 \text{ mm (Ans.)}}$$

12.12. WATER HAMMER IN PIPES

In a long pipe, when the *flowing water is suddenly brought to rest by closing the valve or by any similar cause*, there will be a *sudden rise in pressure* due to the momentum of water being destroyed. A pressure wave is transmitted along the pipe. A sudden rise in pressure has the effect of hammering action on the walls of the pipe. This *phenomenon of sudden rise in pressure is known as water hammer or hammer blow*. The magnitude of pressure rise depends on :

- (i) The speed at which valve is closed,
- (ii) The velocity of flow,
- (iii) The length of pipe, and
- (iv) The elastic properties of the pipe material as well as that of the flowing fluid.

The rise in pressure in some cases may be so large that the pipe may even burst and therefore it is essential to take into account this pressure rise in the design of the pipes.

12.12.1 Gradual Closure of Valve

Consider a long pipe carrying liquid (Fig. (12.45)) and provided with a valve which is closed gradually.

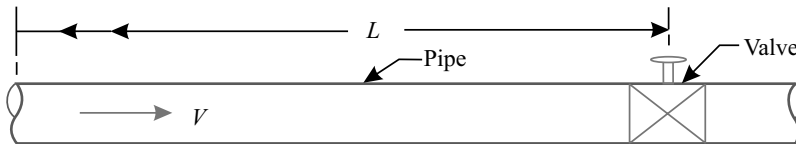


Fig. 12.45. Water hammer.

- Let,
- A = Area of cross-section of the pipe,
 - L = Length of the pipe,
 - V = Velocity of flow of water in the pipe,
 - t = Time required to close the valve (in seconds), and
 - p = Intensity of pressure wave produced.

The mass of liquid contained in the pipe is $= \rho AL$

Assuming that the rate of closure of the valve is so adjusted that the liquid column in the pipe is brought to rest with a uniform retardation; from an initial velocity V to zero in time t seconds, we have:

$$\text{Retardation of water} = \frac{V - 0}{t} = \frac{V}{t}$$

$\therefore$ The axial force available for producing retardation

$$= \text{Mass} \times \text{retardation}$$

$$= \rho AL \times \frac{V}{t} \quad \dots(i)$$

Also, force due to pressure wave is $= p.A$

$\dots(ii)$

Equating the two forces given by eqns. (i) and (ii), we have:

$$\rho AL \times \frac{V}{t} = p \times A$$

$$\text{or,} \quad p = \frac{\rho LV}{t} \quad \dots(12-26)$$

$$\therefore \quad \text{Head of pressure, } H = \frac{p}{w} = \frac{\rho LV}{w \times t} = \frac{\rho LV}{\rho \cdot g \cdot t} = \frac{LV}{gt}$$

$$\text{i.e.,} \quad H = \frac{LV}{gt} \quad \dots(12-27)$$

$$(i) \text{ The closure of valve is said to be } \textit{gradual} \text{ when } t > \frac{2L}{C} \quad \dots(12-28)$$

$$(ii) \text{ The closure of valve is said to be } \textit{instantaneous} \text{ when } t < \frac{2L}{C} \quad \dots(12-29)$$

where, C = velocity of the pressure wave.

12.12.2 Instantaneous Closure of Valve in Rigid Pipes

Eqn. (11-26) indicates that when the valve is closed instantaneously (*i.e.*, $t = 0$), the inertia head should rise to infinity. However, in practice, it is not possible to close the valve instantaneously, as it always takes some time. Thus, even for a very rapid closure of the valve, as observed during experimentation, the pressure rise is quite finite and measurable. Moreover, eqn. (12-26) has been derived on the *assumption that the liquid is incompressible*. This assumption is *incorrect*, because at very high pressures even liquids get compressed to *some extent* and *behave like compressible fluids*.

Consider a pipe of length L and area of cross-section A (Fig. 12-45) carrying water which is flowing through it at a velocity V . When the valve is closed instantaneously the K.E. of the flowing water is converted into strain energy of water (neglecting effect of friction and assuming the pipe wall to be perfectly rigid).

$$\text{Loss of K.E.} = \frac{1}{2} m V^2 = \frac{1}{2} \rho AL \times V^2 \quad (\because m = \rho \times A \times L)$$

$$\text{Gain of strain energy} = \frac{1}{2} \left(\frac{p^2}{K} \right) \times \text{volume} = \frac{1}{2} \frac{p^2}{K} \times AL$$

$$\left[\begin{array}{l} \text{where, } k = \text{Bulk modulus of elasticity of water, and} \\ p = \text{Intensity of pressure wave produced.} \end{array} \right]$$

Equating the loss of K.E. to the gain of strain energy, we get:

$$\frac{1}{2} \rho AL \times V^2 = \frac{1}{2} \frac{p^2}{K} \times AL$$

$$\text{or,} \quad p^2 = \frac{1}{2} \rho AL V^2 \times \frac{2K}{AL} = \rho K V^2$$

$$\therefore p = \sqrt{\rho K V^2} = V \sqrt{\rho K} = V \sqrt{\frac{K \rho^2}{\rho}}$$

or, $p = V \rho C$... (12-30)

$$\left(\text{where, } C = \sqrt{\frac{K}{\rho}}, C \text{ being the velocity of pressure wave.} \right)$$

12.12.3 Instantaneous Closure of Valve in Elastic Pipes

As shown in Fig. 12.45, consider a pipe of length L , diameter D , thickness t (small compared to diameter).

Let, p = Increase of pressure due to water hammer,

E = Modulus of elasticity of pipe material, and

$\frac{1}{m}$ = Poisson's ratio for pipe material.

When the valve is closed intantaneously, rise of pressure takes place due to which circumferential and longitudinal stresses are produced in the pipe wall; these stresses are given as (from knowledge of strength of materials):

$$\sigma_c = \frac{pD}{2t} \quad \text{and} \quad \sigma_l = \frac{pD}{4t}$$

where,

σ_c = Circumferential stress, and

σ_l = Longitudinal stress.

Also, strain energy stored in the pipe material per unit volume is

$$\begin{aligned} &= \frac{1}{2E} \left(\sigma_c^2 + \sigma_l^2 - \frac{2\sigma_c \sigma_l}{m} \right) \\ &= \frac{1}{2E} \left[\left(\frac{pD}{2t} \right)^2 + \left(\frac{pD}{4t} \right)^2 - \frac{2 \times \frac{pD}{2t} \times \frac{pD}{4t}}{m} \right] \\ &= \frac{1}{2E} \left[\frac{p^2 D^2}{4t^2} + \frac{p^2 D^2}{16t^2} - \frac{p^2 D^2}{4mt^2} \right] \end{aligned}$$

Assuming, $\frac{1}{m} = 1/4$, we have:

$$\text{Strain energy per unit volume} = \frac{1}{2E} \left[\frac{p^2 D^2}{4t^2} + \frac{p^2 D^2}{16t^2} - \frac{p^2 D^2}{16t^2} \right] = \frac{p^2 D^2}{8Et^2}$$

Total strain energy stored in pipe material

$$\begin{aligned} &= \frac{p^2 D^2}{8Et^2} \times \text{total volume of pipe material} \\ &= \frac{p^2 D^2}{8Et^2} \times \pi D t \times L = \frac{p^2 \times D^3 L}{8Et} \\ &= \frac{p^2 \times \pi D^2 \times DL}{8Et} = \frac{p^2 ADL}{2Et} \quad \left[\because A \text{ (area of the pipe)} = \frac{\pi}{4} \times D^2 \right] \end{aligned}$$

$$\text{Loss of K.E. of water} = \frac{1}{2} m V^2 = \frac{1}{2} \rho A L \times V^2$$

$$\text{Gain of strain energy in water} = \frac{1}{2} \left(\frac{p^2}{K} \right) \times \text{volume} = \frac{1}{2} \frac{p^2}{K} \times A L$$

Also, The loss of K.E. of water = Gain of strain energy in water + strain energy stored in material.

$$\therefore \frac{1}{2} \rho AL \times V^2 = \frac{1}{2} \frac{p^2}{K} \times AL + \frac{p^2 ADL}{2Et}$$

Dividing both sides by $\frac{AL}{2}$, we get:

$$\rho V^2 = \frac{p^2}{K} + \frac{p^2 D}{Et} = p^2 \left(\frac{1}{K} + \frac{D}{Et} \right)$$

$$\therefore p^2 = \frac{\rho V^2}{\left(\frac{1}{K} + \frac{D}{Et} \right)}$$

$$\text{or, } p = \sqrt{\frac{\rho V^2}{\left(\frac{1}{K} + \frac{D}{Et} \right)}} = V \times \sqrt{\frac{\rho}{\left(\frac{1}{K} + \frac{D}{Et} \right)}} \quad \dots(12.31)$$

12.12.4 Time required by Pressure Wave to travel from the Valve to the Tank and from Tank to Valve

$$\begin{aligned} \text{Time taken, } t &= \frac{\text{Distance travelled from valve to tank and back}}{\text{Velocity of pressure wave}} \\ &= \frac{L + L}{C} = \frac{2L}{C} \quad \text{i.e., } t = \frac{2L}{C} \quad \dots(12.32) \end{aligned}$$

where,

L = Length of the pipe, and

C = Velocity of pressure wave.

Example 12.51. In a pipe 600 mm diameter and 3000 m length, provided with a valve at its end, water is flowing with a velocity of 2 m/s. Assuming velocity of pressure wave $C = 1500$ m/s, find :

- (i) The rise in pressure if the valve is closed in 20 seconds, and
- (ii) The rise in pressure if the valve is closed in 2.5 seconds. Assume the pipe to be rigid one and take bulk modulus of water as 2 GN/m^2 .

Solution. Diameter of the pipe, $D = 600 \text{ mm} = 0.6 \text{ m}$

Length of the pipe, $L = 3000 \text{ m}$

Velocity of water, $V = 2 \text{ m/s}$

Velocity of pressure wave, $C = 1500 \text{ m/s}$.

(i) **Rise in pressure, p :**

Time taken to close the valve, $t = 20 \text{ s}$

$$\text{Now, The ratio, } \frac{2L}{C} = \frac{2 \times 3000}{1500} = 4$$

The close of valve is said to be *gradual* if,

$$t > \frac{2L}{C} \quad \dots[\text{Eqn. (12.28)}]$$

Hence, the valve is closed *gradually*.

The rise in pressure (p), for gradual closure of valve, is given by:

$$\begin{aligned} p &= \frac{\rho LV}{t} \quad \dots[\text{Eqn. (12.26)}] \\ &= \frac{1000 \times 3000 \times 2}{20} = 300 \times 10^3 \text{ N/m}^2 \text{ or } \mathbf{300 \text{ kN/m}^2} \text{ (Ans.)} \end{aligned}$$

(ii) **Rise in pressure, p :**Time taken to close the valve, $t = 2.5$ sBulk modulus of water, $K = 2$ GN/m²

Velocity of pressure wave is given by,

$$C = \sqrt{\frac{K}{\rho}} = \sqrt{\frac{2 \times 10^9}{1000}} = 1414.2 \text{ m/s}$$

$$\text{The ratio, } \frac{2L}{C} = \frac{2 \times 3000}{1414.2} = 4.24 \text{ s}$$

$$\therefore t < \frac{2L}{C}$$

Thus, the valve is closed *instantaneously* [From eqn. (12.29)]

When pipe is rigid, the rise in pressure due to instantaneous closure of the valve is given by (eqn. 12.30),

$$p = V\rho C = 2 \times 1000 \times 1414.2 \text{ N/m}^2 \text{ or } \mathbf{2828.4 \text{ kN/m}^2 \text{ (Ans.)}}$$

Example 12.52. Water is flowing in a pipe of 150 mm diameter with a velocity of 2.5 m/s when it is suddenly brought to rest by closing the valve. Find the pressure rise assuming pipe is elastic, $E = 206$ GN/m², Poisson's ratio = 0.25 and K for water = 2.06 GN/m². Pipe wall is 5 mm thick.

Solution. Diameter of the pipe, $D = 150$ mm = 0.15 mThickness of the pipe, $t = 5$ mm = 0.005 mVelocity of water, $V = 2.5$ m/sModulus of elasticity, $E = 206$ GN/m²Bulk modulus of water, $K = 2.06$ GN/m²

$$\text{Poisson's ratio, } \frac{1}{m} = 1/4$$

Pressure rise, p :

Using the relation :

$$\begin{aligned}
 p &= V \sqrt{\frac{\rho}{\left(\frac{1}{K} + \frac{D}{Et}\right)}} \quad \dots[\text{Eqn. (12.31)}] \\
 &= 2.5 \sqrt{\frac{1000}{\left(\frac{1}{2.06 \times 10^9} + \frac{0.15}{2.06 \times 10^9 \times 0.005}\right)}} = 2.5 \sqrt{\frac{10^{12}}{0.485 + 0.1456}} \\
 &= \mathbf{3148 \text{ kN/m}^2 \text{ (Ans.)}}
 \end{aligned}$$

Example 12.53. In a pressure penstock 4500 m long water is flowing at 4 m/s. If the velocity of the pressure wave travelling in the pipe due to sudden complete closure of a valve at the downstream end is given as 1500 m/s, find :

(i) The maximum pressure rise, and

(ii) The period of oscillation.

Show how the pressure changes with time at the middle point of the penstock length. All friction losses may be neglected. (UPSC Exams.)

Solution. Length of the penstock, $L = 4500$ mVelocity of water, $V = 4$ m/sVelocity of the pressure wave, $C = 1500$ m/s

(i) **The maximum pressure rise :**

Maximum pressure is given by:

$$p = V\rho C \quad \dots[\text{Eqn. (12.39)}]$$

$$= 4 \times 1000 \times 1500 \text{ N/m}^2 \text{ or } 6 \text{ MN/m}^2 \text{ (Ans.)}$$

(ii) **The period of oscillation :**

$$\text{The period of oscillation} = \frac{2L}{C} = \frac{2 \times 4500}{1500} = 6 \text{ seconds (Ans.)}$$

Pressure changes with time at the middle point of the penstock length :

The pressure wave reaches at the middle point of the penstock length in $1.5 \text{ s} \left(\frac{4500}{2 \times 1500} = 1.5 \text{ s} \right)$;

at this instant the pressure at the middle point rises by 6 MN/m^2 and remains unchanged until the pressure wave returns as a wave of rarefaction of negative pressure. This happens at 4.5 s after the closure of the valve when the pressure in the penstock at the middle point is reduced by an equal amount. Fig. 12.46 shows the pressure changes at the middle point with respect to time.

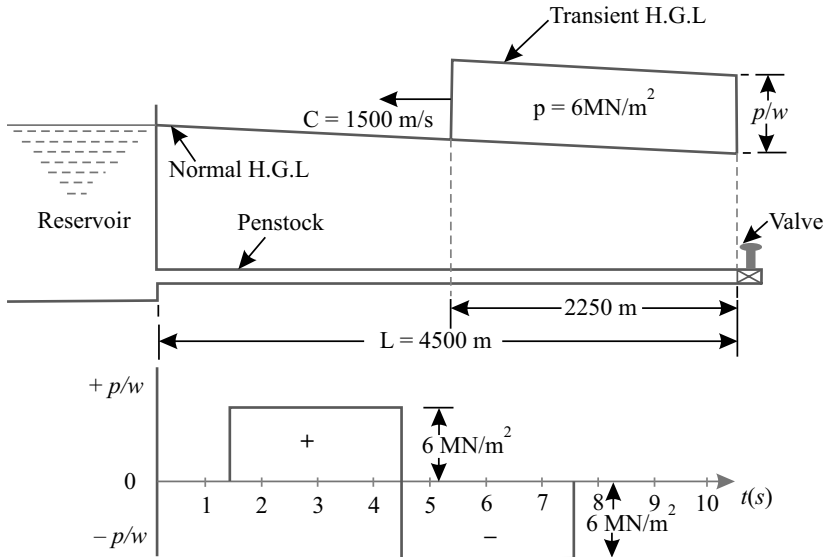


Fig. 12.46. Pressure changes at the middle point of the penstock length.

HIGHLIGHTS

1. A pipe is a closed conduit (generally of circular section) which is used for carrying fluids under pressure.
2. On the basis of experiments Reynolds discovered that :
 - (i) In case of *laminar flow* : The loss of pressure head $\propto$ velocity (V)
 - (ii) In case of *turbulent flow* : the loss of pressure head $\propto V^2$ (approximately)
3. *Energy (or head) losses* :
 - A. *Major energy losses.....due to friction.*
 - B. *Minor energy losses.*
 These losses are due to:

- (i) Sudden enlargement of pipe
- (ii) Sudden contraction of pipe,
- (iii) Bend in pipe,
- (iv) An obstruction in pipe, and
- (v) Pipe fittings, etc.

4. Major energy losses (due to friction), Important formulae :

- (i) *Darcy-Weisbach formula* (for loss of head due to friction)

$$h_f = \frac{4fLV^2}{D \times 2g} = \frac{f_1LV^2}{D \times 2g}$$

where, f = co-efficient of friction, f_1 = friction factor ($= 4f$)

- (ii) *Chezy's formula* (for loss of head due to friction)

$$V = C \sqrt{mi} \left(\text{where, } i = \frac{h_f}{L} \right)$$

[$\therefore h_f = i \times L$, where i is obtained from Chezy's formula.]

5. Minor energy losses; Important formulae :

- (i) Loss of head due to *sudden enlargement*

$$h_e = \frac{(V_1 - V_2)^2}{2g}$$

- (ii) Loss of head due to *sudden contraction*.

$$h_c = \frac{V_2^2}{2g} \left[\frac{1}{C_c} - 1 \right]^2$$

- (iii) Loss of head due to *obstruction in pipe*,

$$h_{obs.} = \left[\frac{A}{C_c (A - a)} - 1 \right]^2 \frac{V^2}{2g}$$

where, a = Maximum area of obstruction, and
 A = Area of the pipe.

- (iv) Loss of head at the entrance to pipe,

$$h_i = 0.5 \frac{V^2}{2g}$$

where, V = velocity of liquid in pipe.

- (v) Loss of head at the *exit of pipe*,

$$h_0 = \frac{V^2}{2g}$$

where, V = velocity at outlet of pipe.

- (vi) Loss of head due to *bend in the pipe*,

$$h_b = k \frac{V^2}{2g}$$

where, k = co-efficient of bend.

- (vii) Loss of head in *various pipe fittings*,

$$h_{\text{fittings}} = k \frac{V^2}{2g}$$

where, k = value of co-efficient; it depends on the type of pipe fittings.

6. *Energy gradient line* (E.G.L.). If the total energy at various points along the axis of the pipe is plotted and joined by a line, the line so obtained is called the '*Energy gradient line*'.

7. *Hydraulic gradient line* (H.G.L.). If a line is drawn joining the piezometric levels at various points, the line so obtained is called the '*Hydraulic gradient line*'.
8. *Equivalent pipe*. It is defined as the pipe of uniform diameter having loss of head and discharge equal to the loss of head and discharge of a compound pipe consisting of several pipes of different lengths and diameters. To determine the size of the equivalent pipe Dupit's equation, given below, is used :

$$\frac{L}{D^5} = \left| \frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5} + \dots \right|$$

9. In case of *parallel pipes* :

(i) Rate of discharge in the main line = Sum of the discharges in each of the parallel pipes.

$$\text{i.e.,} \quad Q = Q_1 + Q_2 + \dots$$

(ii) The loss of head in each pipe is same.

10. A *siphon* is a long bent pipe employed for carrying water from a reservoir at a higher elevation to another reservoir at lower elevation when the two reservoirs are separated by a hill or high level ground in between.

11. *Power transmission through pipes* :

$$\text{Efficiency, } \eta = \frac{H - h_f}{H}$$

$$\text{Power, } P = wQ(H - h_f) \text{ kW} \quad (\text{where, } w = 9.81 \text{ kN/m}^3 \text{ for water})$$

$$\text{Power transmitted will be maximum when, } h_f = \frac{H}{3}$$

$$\text{Then,} \quad \eta_{\max} = \frac{H - H/3}{H} = 66.7\%$$

12. *Flow through nozzles*; Important formulae :

$$(i) \text{ Velocity, } v = \sqrt{1 + \frac{4fL}{D} \cdot \frac{a^2}{A^2}} \cdot \frac{2gh}{2}$$

$$(ii) \text{ Power, } P = wav \left[H - \frac{4fL}{D \times 2g} \left(\frac{a^2 v^2}{A^2} \right) \right] \text{ kW} \quad (\text{where, } w = 9.81 \text{ kN/m}^3 \text{ for water})$$

$$(iii) \text{ Condition for maximum power transmission : } h_f = \frac{H}{3}$$

(iv) Diameter of nozzle for maximum power transmission,

$$d = \left(\frac{D^5}{8fL} \right)^{1/4}$$

13. *Water hammer in pipes*. The phenomenon of sudden rise in pressure in a pipe when water flowing in it is suddenly brought to rest by closing the valve is known as *water hammer* or *hammer blow*.

14. Valve closure is *gradual* when $t > \frac{2L}{C}$

$$\text{Valve closure is } \textit{sudden} \text{ when } t < \frac{2L}{C}$$

where, $C = \sqrt{\frac{K}{\rho}}$, C being the velocity of pressure wave produced due to water hammer.

15. The intensity of pressure rise due to water hammer is given by, $p = \frac{\rho L V}{t}$... when valve is

closed gradually (where, t = time required to close the valve),

$p = V \sqrt{\rho K}$...when the valve is *closed suddenly* and pipe is assumed *rigid*, and

$p = V \times \sqrt{\frac{\rho}{\left(\frac{1}{K} + \frac{D}{Et}\right)}}$...when valve is *closed suddenly* and the pipe is *elastic*.

(where, t = Thickness of pipewall)

where,

L = Length of pipe,

V = Velocity of flow,

K = Bulk modulus of water, and

E = Modulus of elasticity for pipe material.

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer :

- The pipe running partially/completely full behaves like an open channel.
- In a laminar flow, Reynold's number is
 - less than 2000
 - more than 2000
 - more than 2000 but less than 4000
 - none of the above.
- In a turbulent flow, Reynold's number is
 - less than 4000
 - more than 4000
 - between 2000 and 4000
 - none of the above.
- In case of a laminar flow, the loss of pressure head is
 - proportional to (velocity)²
 - proportional to velocity
 - proportional to (velocity)^{1/2}
 - none of the above.
- In case of a turbulent flow, the loss of head is approximately proportional to
 - velocity
 - (velocity)^{1/2}
 - (velocity)^{3/4}
 - (velocity)²
- Darcy-Weisbach equation is used to find loss of head due to :
 - sudden enlargement
 - sudden contraction
 - friction
 - none of the above.
- Chezy's formula is given as
 - $V = C \sqrt{m^2 i}$
 - $V = C^2 \sqrt{m i^2}$
 - $V = C \sqrt{m i}$
 - $V = C \sqrt{m^2 i^3}$
- Loss of head due to sudden enlargement is given as
 - $\frac{(V_1 - V_2)^3}{2g}$
 - $\frac{(V_1 - V_2)^2}{2g}$
 - $\frac{V_1^2 - V_2^2}{2g}$
 - $\frac{\sqrt{V_1 - V_2}}{2g}$
- Loss of head due to sudden contraction is given as
 - $\frac{V^2}{g} \left(\frac{1}{C_c} - 1 \right)^2$
 - $\frac{V_2^2}{2g} \left(\frac{1}{C_c} - 1 \right)^2$
 - $\frac{V_2}{g^2} \left(\frac{1}{C_c} - 1 \right)^2$
 - $\frac{V_2^2}{2g} \left(\frac{1}{C_c} - 1 \right)$
- Loss of head due to an obstruction is given as
 - $\left[\frac{A}{A-a} - 1 \right]^2 \frac{V^2}{2g}$
 - $\left[\frac{A}{C_c(A-a)} - 1 \right] \frac{V^2}{2g}$
 - $\left[\frac{A}{C_c a} - 1 \right]^2 \frac{V^2}{2g}$
 - $\left[\frac{A^2}{A-a} - 1 \right]^2 \frac{V^2}{2g}$

11. Loss of head at entrance to a pipe is given as
 - (a) $\frac{V^2}{2g}$
 - (b) $\frac{V}{g}$
 - (c) $0.5 \frac{V^2}{2g}$
 - (d) $\frac{V^3}{2g}$
12. Loss of head at exit of a pipe is given as
 - (a) $\frac{V^2}{2g}$
 - (b) $\frac{V^2}{g}$
 - (c) $\frac{V^3}{g}$
 - (d) $\frac{V^3}{2g}$
13. There is specific relation/no relation between the slope of the energy gradient line and the slope of the axis of the pipe.
14. The pipes are said to be in series/parallel when a main pipe line divides into two or more parallel pipes which again join together downstream and continue as a mainline.
15. The power transmitted through the pipe is maximum when head lost due to friction in the pipe is equal to
 - (a) $\frac{1}{3}$ rd of the total supply head
 - (b) $\frac{1}{4}$ th of the total supply head
 - (c) $\frac{1}{5}$ th of the total supply head
 - (d) $\frac{1}{8}$ th of the total supply head.
16. Diameter of nozzle (d) for maximum transmission of power is equal to
 - (a) $\left(\frac{D^5}{4fL}\right)^{1/4}$
 - (b) $\left(\frac{D^5}{8fL}\right)^{1/4}$
 - (c) $\left(\frac{D^4}{8fL}\right)^{1/4}$
 - (d) $\left(\frac{D^3}{8fL}\right)^{1/4}$
17. The energy loss in a pipeline is due to
 - (a) surface roughness only
 - (b) viscous action only
 - (c) friction offered by pipe wall as well as by viscous function
 - (d) none of the above.
18. In a pipe flow the minor losses are those
 - (a) which depend on the length of the pipeline
 - (b) caused by friction and are thus also called friction losses.
 - (c) which have a small magnitude
 - (d) which are caused on account of total disturbance produced by such fittings as valves, bends, etc.
19. In flow through pipe bends the pressures on inner and out radii
 - (a) stand at the same level increasing gradually towards the pipe centre.
 - (b) vary, it being more on the inner core.
 - (c) are different, pressure increases with increase in radius and is, therefore, more at the outer radius.
 - (d) do not vary and are the same as at the centre of the pipe.
20. The condition for maximum transmission of power through a pipeline is that one-third of the available head must be consumed in friction. The corresponding efficiency of the pipeline is
 - (a) 33.3%
 - (b) 66.67%
 - (c) 90%
 - (d) 100%.
21. For achieving continuous flow through a system, no position of the pipe should be higher than
 - (a) 20
 - (b) 6 m
 - (c) 7.6 m
 - (d) 10 m.
 measured above the hydraulic gradient line.
22. For turbulent flow in smooth pipes, the entrance length is taken as
 - (a) 20
 - (b) 50
 - (c) 80
 - (d) 115.
23. The entrance length or length of establishment of flow is
 - (a) the length in which the boundary layer remains uniform
 - (b) the pipe length inside the reservoir
 - (c) the length of pipe from its entrance in which the flow may be assumed irrotational
 - (d) the initial length in which the flow develops fully such that the velocity profile does not change downstream.
24. Due to which of the following phenomena water hammer is caused ?
 - (a) Incompressibility of fluid
 - (b) Sudden opening of a valve in a pipeline
 - (c) The material of the pipe being elastic
 - (d) Sudden closure (partial or complete) of a valve in pipe flow.
25. Under which of the following conditions the closure of valve is considered rapid ?
 - (a) the duration of valve closure is greater than $\frac{2L}{C}$
 - (b) the duration of valve closure is less than $\frac{L}{C}$
 - (c) the duration of valve closure is less than $\frac{2L}{C}$
 - (d) none of the above.
 (where L = length of pipe, C = velocity of pressure wave produced due to water hammer.)

ANSWERS

- | | | | | | |
|-----------------|--------------|---------|---------|---------|---------|
| 1. Partially | 2. (a) | 3. (b) | 4. (b) | 5. (d) | 6. (c) |
| 7. (c) | 8. (b) | 9. (b) | 10. (b) | 11. (c) | 12. (a) |
| 13. No relation | 14. Parallel | 15. (a) | 16. (b) | 17. (c) | 18. (d) |
| 19. (d) | 20. (b) | 21. (c) | 22. (b) | 23. (d) | 24. (d) |
| 25. (c). | | | | | |

THEORETICAL QUESTIONS

- Differentiate between a laminar flow and a turbulent flow.
- Define the terms : Major energy losses and minor energy losses in pipe.
- Derive Darcy-Weisbach formula for calculating loss of head due to friction in a pipe.
- Derive Chezy's formulae for loss of head due to friction in a pipe
- Derive formulae for calculating loss of head due to
 - Sudden enlargement, and
 - Sudden contraction.
- Explain briefly the following :
 - Hydraulic gradient line (H.G.L.)
 - Energy gradient line (E.G.L.)
- What is an equivalent pipe ?
- What is syphon ? Where is it used ?
- Derive an expression for the power transmission through the pipes. Find also the condition

for maximum transmission of power and corresponding efficiency of transmission.

- Find an expression for the ratio of the outlet area of the nozzle to the area of the pipe for maximum transmission of power.
- Show that the diameter of the nozzle for maximum transmission of power is given by

$$d = \left(\frac{D^5}{8fL} \right)^{1/4}$$

where, D = Diameter of the pipe

L = Length of the pipe, and

f = Friction co-efficient.

- What is meant by water hammer ? Derive an expression for the rise of pressure when the flowing water in a pipe is brought to rest by closing the valve gradually.
- Obtain a formula for rise in pressure in a thin plastic pipe of circular section in which the flow of water is stopped by sudden closure of a valve.

UNSOLVED EXAMPLES

- Find the head lost due to friction in a pipe of diameter 200 mm and length 60 m, through which water is flowing at a velocity of 2.5 m/s using :
 - Darcy-Weisbach formula (assuming $f = 0.005$), and
 - Chezy's formula for which $C = 55$.

[Ans. (i) 1.91 m; (ii) 2.48 m]

- Water is flowing through a pipe of diameter 200 mm with a velocity of 3 m/s. If the co-efficient of friction is given by $f = 0.002 + \frac{0.09}{(Re)^{0.3}}$,

where Re is Reynolds number, find the head lost due to friction for a length of 5 m. Take kinematic viscosity of water = 0.01 stoke.

[Ans. (0.99 m)]

- In a pipe of diameter 200 mm and length 500 m, an oil of specific gravity 0.9 and viscosity 0.06 poise is flowing at the rate of 0.06 m³/s. Find :
 - The head lost due to friction, and
 - Power required to maintain the flow.

[Ans. (i) 9.48 m of water; (ii) 5.016 kW]

- A horizontal pipe of 100 mm diameter is joined by sudden enlargement to a 150 mm diameter pipe. Water is flowing through it at the rate of 2 m³/min. Find :
 - Loss of head due to abrupt expansion,
 - Pressure difference in two pipes, and
 - Change in pressure if the change of section is gradual, without any loss.

[Ans. (i) 0.286 m of water;

(ii) 0.455 m of water; (iii) 0.74 m of water]

5. The discharge of water through a horizontal pipe is $0.25 \text{ m}^3/\text{s}$. Its diameter, which is 200 mm, suddenly enlarges to 400 mm. If the intensity of pressure of water in the smaller pipe is 120 kN/m^2 , determine :
 - (i) Loss of head due to sudden enlargement,
 - (ii) Intensity of pressure in the large pipe, and
 - (iii) Power lost due to enlargement.

[Ans. (i) 1.816 m ; (ii) 132 kN/m^2 ; (iii) 4.45 kW]
6. A horizontal pipe carries water at a rate of $0.03 \text{ m}^3/\text{s}$. Its diameter reduces abruptly from 150 mm to 100 mm. If the co-efficient of friction is 0.6 find the pressure loss across the contraction.

[Ans. 9.3 kN/m^2]
7. A pipe of 150 mm diameter is attached to a 100 mm diameter pipe by means of a flange in such a manner that axes of the two pipes are in a straight line. Water flows through the arrangement at a rate of $2 \text{ m}^3/\text{min}$. The pressure loss at the transition as indicated by differential gauge length on a water-mercury manometer connected between the two pipes equals 80 mm. Calculate :
 - (i) The loss of head due to contraction, and
 - (ii) The co-efficient of contraction.

[Ans. (i) 0.268 m of water; (ii) 0.65]
8. Three pipes of diameters 300 mm, 200 mm and 400 mm and lengths 300 m, 170 m and 210 m respectively are connected in series. The difference in water surface levels in two tanks is 12 m. Determine the rate of flow if co-efficients of frictions are 0.005, 0.0052 and 0.0048 respectively, considering :
 - (i) Minor losses also, and
 - (ii) Neglecting minor losses.

[Ans. (i) $0.9945 \text{ m}^3/\text{s}$, (ii) $0.1021 \text{ m}^3/\text{s}$]
9. A pipeline ABC 180 m long is laid on an upward slope of 1 in 60. The length of the portion AB is 90 m and its diameter is 150 mm. At B the pipe section suddenly enlarges to 300 mm diameter and remains so for the remainder of its length BC , 90 m. A flow of $0.05 \text{ m}^3/\text{s}$ is pumped into the pipe at its lower end A and is discharged at the upper end C into a closed tank. The pressure at the supply end A is 140 kN/m^2 .
 - (i) Find the pressure at the discharge end C ;
 - (ii) Draw energy gradient line and hydraulic gradient line.

[Ans. 61 kN/m^2]
10. Two reservoirs are connected by two pipes of the same length laid in parallel. The diameters of the pipes are 100 mm and 300 mm respectively. If the discharge through 100 mm diameter pipe is $0.01 \text{ m}^3/\text{s}$, what will be the discharge through 300 mm pipe ? Assume that f is same for both pipes.

[Ans. $0.156 \text{ m}^3/\text{s}$]
11. A main pipe divides into two parallel pipes which again forms one pipe. The length and diameter for the first parallel pipe are 2000 m and 1.0 m respectively, while the length and diameter of the second pipe are 2000 m and 0.8 m respectively. If the total flow in the main is $3 \text{ m}^3/\text{s}$ and the co-efficient of friction for each parallel pipe is same and equal to 0.005, find the rate of flow in each parallel pipe.

[Ans. (i) $1.906 \text{ m}^3/\text{s}$; $1.094 \text{ m}^3/\text{s}$]
12. Two reservoirs are connected by a pipeline consisting of two pipes, one of 150 mm diameter and length 6 m and the other of diameter 225 mm and 16 m length. If the difference of levels in the reservoir is 6 m, calculate the discharge and draw the energy gradient line. Take friction factor = 0.04.

[Ans. $0.018 \text{ m}^3/\text{s}$]
13. Two reservoirs have difference of water levels of 6 m. They are connected by a pipe system which consists of a single pipe of 600 mm diameter for the first 3000 m and then two pipes in parallel, each of 300 mm diameter and 3000 m in length. Calculate the rate of flow. Assume friction factor = 0.04.

[Ans. $0.0725 \text{ m}^3/\text{s}$]
14. A compound pipeline 1650 m long made up of pipes 450 mm diameter for 900 m, 375 mm for 450 m and 300 mm for 300 m, is required to be replaced by a pipe of uniform diameter. Find the diameter of the new pipe, assuming the length to remain the same.

[Ans. 372 mm]
15. A pipeline of 600 mm diameter and 4 km length connects two reservoirs. The difference of water levels in the reservoirs is 20 m. At a distance of 1 km from the upper reservoir, a small pipe is connected to the pipeline. The water can be taken from the small pipe. Find the discharge to the lower reservoir if (i) No water is taken from the small pipe, and (ii) $0.1 \text{ m}^3/\text{s}$ of water is taken from small pipe. Take co-efficient of friction = 0.005 and neglect minor losses.

[Ans. (i) $0.485 \text{ m}^3/\text{s}$; (ii) $0.458 \text{ m}^3/\text{s}$]
16. An existing 300 mm diameter pipeline of 3200 length connects two reservoirs with 13 m difference in their water levels. Calculate the discharge Q_1 . If a parallel pipe 300 mm diameter is attached

to last 1600 m length of the existing pipeline, find the new discharge Q_2 . Take only wall friction into account. Assume friction factor = 0.04 in Darcy-Weibach formula.

[Ans. 0.547 m³/s; 0.698 m³/s]

[IIT Madras]

17. A pipeline of 0.6 m diameter is 1.5 km long. To augment the discharge, another pipeline of the same diameter is introduced parallel to the first in the second half of its length. Find the increase in discharge if the friction factor is 0.04 and head at the inlet is 30 m. (Ans. 0.1808 m³/s)

[UPSC Exams.]

18. A pipe of 400 mm diameter and 2 km long is connected to a reservoir at one end. The other end of the pipe is connected to junction from which two pipes each of length 1 km and diameter 300 mm run in parallel. These parallel pipes are connected to another reservoir, which is having level of water 10 m below the water level of the above reservoir. Determine the total discharge if the friction factor = 0.06.

Neglect minor losses. [Ans. 0.0822 m³/s]

19. The rate of flow of water pumped into a pipe ABC, which is 180 m long, is 0.05 m³/s. The pipe is laid on an upward slope of 1 in 60. The length of the portion AB is 90 m and its diameter is 150 mm, while the length of the portion BC is also 90 m but its diameter is 300 mm. The change of diameter at B is sudden. The flow is taking place from A to C where the pressure at A is 137.34 kN/m² and end C is connected to a closed end tank. Find the pressure at the discharge end C and sketch the total energy and the hydraulic gradient lines.

[Ans. 59.84 kN/m²]

20. A pipeline of 500 mm diameter and 4.5 km length connects two reservoirs whose constant difference of water level is 12 m. A branch pipe 1.25 km long and taken from a point distance 1.5 km from reservoir A, leads to the reservoir C whose water level is 15 m below that of reservoir A. Find the diameter of the branch pipe, so that the flow into both the reservoirs is same.

Assume co-efficient of friction for each pipe, $f = 0.0075$.

[Ans. 375 mm]

21. A 200 mm diameter pipe, 8000 m long connects two reservoirs whose surface levels differ by 40 m. At a distance of 500 m from the upper reservoir, the pipe crosses a ridge the summit of which is 8 m above the level of water in the upper reservoir. Determine :

- (i) The minimum depth of the pipe below the summit of the ridge, if the absolute pressure

head at the summit of system is not to fall below 3.0 m of water, and

- (ii) The discharge through the pipe.

Take friction co-efficient, $f = 0.006$ and atmospheric head = 10.3 m of water, Neglect minor losses.

[Ans. (i) 3.24 m, (ii) 0.02883 m³/s]

22. A pipe of 1 m diameter connects two reservoirs having a difference of levels of 6 m. The total length of the pipe is 800 m and rises to a maximum height of 3 m above the level of water in the higher reservoir at a distance of 200 m from the entrance. Find :

- (i) Discharge in the pipe, and

- (ii) Pressure at the highest point.

Take friction factor = 0.04, and neglect minor losses.

[Ans. - 4.69 m of water]

23. A syphon of diameter 200 mm connects two reservoirs having a difference in elevation of 15 m. The total length of the syphon is 600 m and the summit is 4 m above the water level in the upper reservoir. If the separation takes place at 2.8 m of water absolute, find the maximum length of the syphon from the upper reservoir to the summit.

Take friction factor = 0.016 and atmospheric pressure head = 10.3 m of water.

[Ans. 128.6 m]

24. A pipeline 2000 m long is used for power transmission. 110 kW is to be transmitted through the pipe in which water having a pressure of 5000 kN/m² at inlet is flowing. If the pressure drop over the length of the pipe is 1000 kN/m² and co-efficient of friction is 0.0065, find :

- (i) The diameter of the pipe, and

- (ii) Efficiency of transmission.

[Ans. (i) 128 mm (ii) 80%]

25. Calculate the diameter of the nozzle and the maximum power transmitted by a jet of water discharging freely out of a nozzle, fitted to a pipe 300 m long and 100 mm diameter with co-efficient of friction as 0.01. The available head at the nozzle is 90 m.

[Ans. (i) 25.4 mm; (ii) 10.29 kW]

26. A horizontal pipe of 150 mm diameter and 200 m length conveys water from a reservoir to a nozzle 50 mm in diameter. What would be the power of the jet if the level of water in the reservoir is

15 m above the axis of the pipe ? Take friction co-efficient = 0.01.

Neglect losses in the nozzle. [Ans. 2.31 kW]

27. In a pipe of 500 mm diameter and 2500 m length, provided with a valve at its end, water is flowing with a velocity of 1.5 m/s. Assuming velocity of pressure wave = 1460 m/s, find :

- (i) The rise in pressure if the valve is closed in 25 seconds, and
- (ii) The rise in pressure if the valve is closed in 2 seconds. Assume the pipe to be rigid one and

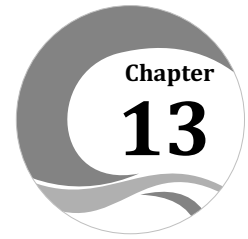
take bulk modulus of water as 1.962 GN/m^2 .

[Ans. (i) 150 kN/m^2 ; (ii) 2101 kN/m^2]

28. Water is flowing through a cast-iron pipe of diameter 150 mm and thickness 10 mm which is provided with a valve at its end. Water is suddenly stopped by closing the valve. Find the maximum velocity of water, when the rise of pressure due to sudden closure of valve is 1.962 MN/m^2 .

Take K for water = 1.962 GN/m^2 and E for cast-iron pipe = 117.7 GN/m^2 .

[Ans. 1.57 m/s]



BOUNDARY LAYER THEORY

13.1. Introduction

13.2. Boundary layer definitions and characteristics- boundary layer thickness (δ)—displacement thickness (δ^*)—momentum thickness (θ)— energy thickness (δ_e).

13.3. Momentum equation for boundary layer by Vonkarman.

13.4. Laminar boundary layer.

13.5. Turbulent boundary layer.

13.6. Total drag due to laminar and turbulent layers.

13.7. Boundary layer separation and its control

Highlights

Objective Type Questions

Theoretical Questions

Unsolved Examples.

13.1. INTRODUCTION

The concept of boundary layer was first introduced by L. Prandtl in 1904 and since then it has been applied to several fluid flow problems.

When a real fluid (viscous fluid) flows past a stationary solid boundary, a layer of fluid which comes in contact with the boundary surface, adheres to it (on account of viscosity) and condition of no slip occurs (The *no-slip* condition implies that the velocity of fluid at a solid boundary must be same as that of boundary itself). Thus the layer of fluid which cannot slip away from the boundary surface undergoes retardation; this retarded layer further causes retardation for the adjacent layers of the fluid, thereby developing a small region in the immediate vicinity of the boundary surface in which the velocity of the flowing fluid increases rapidly from *zero at the boundary surface and approaches the velocity of main stream*. The layer adjacent to the boundary is known as **boundary layer**. Boundary layer is formed whenever there is relative motion between the boundary

and the fluid. Since $\tau_0 = \mu \left(\frac{\partial u}{\partial y} \right)_{y=0}$, the fluid exerts a

shear stress on the boundary and boundary exerts an equal and opposite force on fluid known as the *shear resistance*.

According to boundary layer theory the extensive fluid medium around bodies moving in fluids can be divided into following two regions:

- (i) A thin layer adjoining the boundary is called the *boundary layer* where the *viscous shear takes place*.
- (ii) A region outside the boundary layer where the flow behaviour is quite like that of an *ideal fluid and the potential flow theory is applicable*.

13.2. BOUNDARY LAYER DEFINITIONS AND CHARACTERISTICS

Consider the boundary layer formed on a flat plate kept parallel to flow of fluid of velocity U (Fig. 13.1) (Though the growth of a boundary layer depends upon the *body shape*, flow over a flat plate aligned in the direction of flow is considered, since most of the flow surfaces can be approximated to a flat plate and for simplicity).

- The edge facing the direction of flow is called *leading edge*.
- The rear edge is called the *trailing edge*.
- Near the leading edge of a flat plate, the boundary layer is wholly *laminar*. For a laminar boundary layer, the velocity distribution is *parabolic*.
- The thickness of the boundary layer (δ) increases with distance from the leading edge x , as more and more fluid is slowed down by the viscous boundary, becomes unstable and breaks into *turbulent boundary layer* over a transition region.

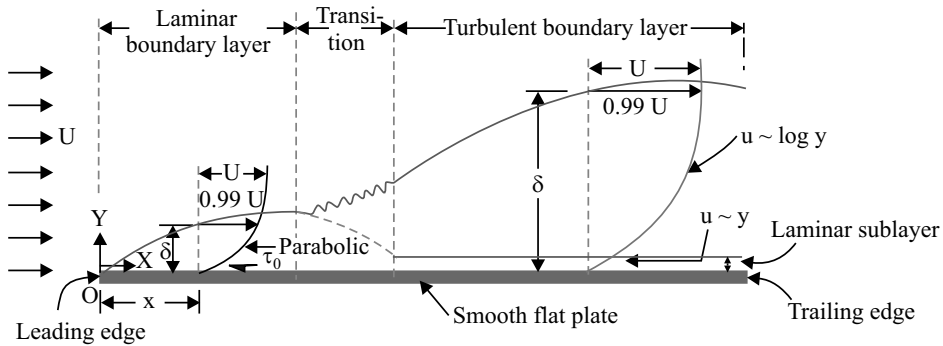


Fig. 13.1. Boundary layer on a flat plate.

For a turbulent boundary layer, if the boundary is smooth, the roughness projections are covered by a very thin layer which remains laminar, called *laminar sublayer*. The velocity distribution in the turbulent boundary layer is given by *Log law* or *Prandtl's one-seventh power law*.

The *characteristics* of a boundary layer may be summarised as follows:

- (i) δ (thickness of boundary layer) increases as distance from leading edge x increases.
- (ii) δ decreases as U increases.
- (iii) δ increases as kinematic viscosity (ν) increases.
- (iv) $\tau_0 \approx \mu \left(\frac{U}{\delta} \right)$; hence τ_0 decreases as x increases. However, when boundary layer becomes turbulent, it shows a sudden increase and then decreases with increasing x .
- (v) When U increases in the downward direction, boundary layer growth is reduced.
- (vi) When U decreases in the downward direction, flow near the boundary is further retarded, boundary layer growth is faster and is susceptible to separation.
- (vii) The various characteristics of the boundary layer on flat plate (e.g. variation of δ , τ_0 or force F) are governed by inertial and viscous forces; hence they are functions of either $\frac{Ux}{\nu}$ or $\frac{UL}{\nu}$.
- (viii) If $\frac{Ux}{\nu} < 5 \times 10^5$... boundary layer is *laminar* (velocity distribution is *parabolic*).
- If $\frac{Ux}{\nu} > 5 \times 10^5$... boundary layer is *turbulent* on that portion (velocity distribution follows *Log law* or a *power law*).

(ix) Critical value of $\frac{Ux}{\nu}$ at which boundary layer changes from laminar to turbulent depends on:

- turbulence in ambient flow,
- surface roughness,
- pressure gradient,
- plate curvature, and
- temperature difference between fluid and boundary.

(x) Though the velocity distribution would be a parabolic curve in the laminar sub-layer zone, but in view of the very small thickness we can reasonably assume that velocity distribution is linear and so the velocity gradient can be considered constant.

13.2.1. Boundary Layer Thickness (δ)

The velocity within the boundary layer increases from zero at the boundary surface to the velocity of the main stream asymptotically. Therefore the thickness of the boundary layer is arbitrarily defined as *that distance from the boundary in which the velocity reaches 99 per cent of the velocity of the free stream ($u = 0.99U$)*. It is denoted by the symbol δ . This definition however gives an approximate value of the boundary layer thickness and hence δ is generally termed as **nominal thickness** of the boundary layer.

The boundary layer thickness for *greater accuracy* is defined in terms of certain mathematical expressions which are the measure of the boundary layer on the flow. The commonly adopted definitions of the boundary layer thickness are:

1. Displacement thickness (δ^*)
2. Momentum thickness (θ)
3. Energy thickness (δ_e).

13.2.2. Displacement Thickness (δ^*)

The *displacement thickness* can be defined as follows:

“It is the distance, measured perpendicular to the boundary, by which the main/free stream is displaced on account of formation of boundary layer.”

Or

“It is an additional “wall thickness” that would have to be added to compensate for the reduction in flow rate on account of boundary layer formation”.

The displacement thickness is denoted by δ^* .

Let fluid of density ρ flow past a stationary plate with velocity U as shown in the Fig. 13.2. Consider an elementary strip of thickness dy at a distance y from the plate.

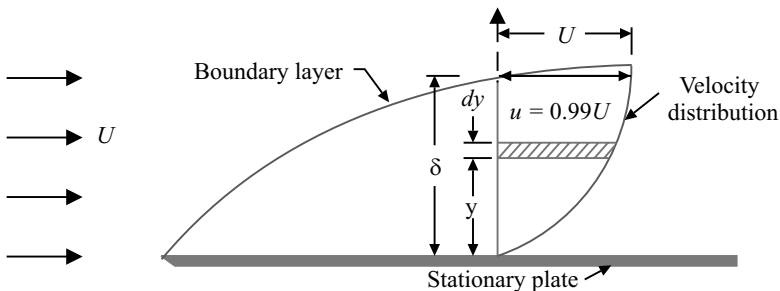


Fig. 13.2. Displacement thickness.

Assuming *unit width*, the mass flow per second through the elementary strip

$$= \rho u dy$$

Mass flow per second through the elementary strip (unit width) if the plate were not there

$$= \rho U \cdot dy \quad \dots(ii)$$

Reduction of mass flow rate through the elementary strip

$$= \rho (U - u) dy$$

[The difference $(U - u)$ is called **velocity of defect**].

Total reduction of mass flow rate due to introduction of plate

$$= \int_0^{\delta} \rho (U - u) dy \quad \dots(iii)$$

(if the fluid is incompressible)

Let the plate is displaced by a distance δ^* and velocity of flow for the distance δ^* is equal to the main/free stream velocity (*i.e.* U). Then, loss of the mass of the fluid/sec. flowing through the distance δ^*

$$= \rho U \delta^* \quad \dots(iv)$$

Equating eqns. (iii) and (iv), we get:

$$\rho U \delta^* = \int_0^{\delta} \rho (U - u) dy$$

or,

$$\delta^* = \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy \quad \dots(13.1)$$

13.2.3. Momentum Thickness (θ)

Momentum thickness is defined as the distance through which the total loss of momentum per second be equal to if it were passing a stationary plate. It is denoted by θ .

It may also be defined as the distance, measured perpendicular to the boundary of the solid body, by which the boundary should be displaced to compensate for reduction in momentum of the flowing fluid on account of boundary layer formation.

Refer to fig. 13.2. Mass of flow per second through the elementary strip = $\rho u dy$

Momentum/sec. of this fluid inside the boundary layer = $\rho u dy \times u = \rho u^2 dy$

Momentum/sec. of the same mass of fluid before entering boundary layer = $\rho u U dy$

$$\text{Loss of momentum/sec.} = \rho u U dy - \rho u^2 dy = \rho u (U - u) dy$$

$\therefore$ Total loss of momentum/sec.

$$= \int_0^{\delta} \rho u (U - u) dy \quad \dots(i)$$

Let, θ = Distance by which plate is displaced when the fluid is flowing with a constant velocity U .

Then loss of momentum/sec. of fluid flowing through distance θ with a velocity U

$$= \rho \theta U^2 \quad \dots(ii)$$

Equating eqns. (i) and (ii), we have:

$$\rho \theta U^2 = \int_0^{\delta} \rho u (U - u) dy$$

or,

$$\theta = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \quad \dots(13.2)$$

The momentum thickness is useful in *kinetics*.

13.2.4. Energy Thickness (δ_e)

Energy thickness is defined as the distance, measured perpendicular to the boundary of the solid body, by which the boundary should be displaced to compensate for the reduction in K.E. of the flowing fluid on account of boundary layer formation. It is denoted by δ_e .

Refer to Fig. 13.2. Mass of flow per second through the elementary strip = $\rho u dy$

K.E. of this fluid inside the boundary layer

$$= \frac{1}{2} m u^2 = \frac{1}{2} (\rho u dy) u^2$$

K.E. of the same mass of fluid before entering the boundary layer

$$= \frac{1}{2} (\rho u dy) U^2$$

Loss of K.E. through elementary strip

$$= \frac{1}{2} (\rho u dy) U^2 - \frac{1}{2} (\rho u dy) u^2 = \frac{1}{2} \rho u (U^2 - u^2) dy \quad \dots(i)$$

$$\therefore \text{Total loss of K.E. of fluid} = \int_0^{\delta} \frac{1}{2} \rho u (U^2 - u^2) dy$$

Let, δ_e = Distance by which the plate is displaced to compensate for the reduction in K.E.

Then, loss of K.E. through δ_e of fluid flowing with velocity U

$$= \frac{1}{2} (\rho U \delta_e) U^2 \quad \dots(ii)$$

Equating eqns. (i) and (ii), we have:

$$\frac{1}{2} (\rho U \delta_e) U^2 = \int_0^{\delta} \frac{1}{2} \rho u (U^2 - u^2) dy$$

or,

$$\delta_e = \frac{1}{U^3} \int_0^{\delta} u (U^2 - u^2) dy$$

$$\therefore \delta_e = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u^2}{U^2} \right) dy \quad \dots(13.3)$$

Example 13.1. The velocity distribution in the boundary layer is given by: $\frac{u}{U} = \frac{y}{\delta}$, where u is the velocity at a distance y from the plate and $u = U$ at $y = \delta$, δ being boundary layer thickness. Find :

(i) The displacement thickness, (ii) The momentum thickness,

(iii) The energy thickness, and (iv) The value of $\frac{\delta^*}{\theta}$.

Solution. Velocity distribution: $\frac{u}{U} = \frac{y}{\delta}$...(Given)

(i) The displacement thickness, δ^*

$$\delta^* = \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy \quad \dots[\text{Eqn. (13.1)}]$$

$$= \int_0^{\delta} \left(1 - \frac{y}{\delta}\right) dy \quad \left(\because \frac{u}{U} = \frac{y}{\delta}\right)$$

$$= \left[y - \frac{y^2}{2\delta} \right]_0^{\delta}$$

$$\delta^* = \left(\delta - \frac{\delta^2}{2\delta} \right) = \delta - \frac{\delta}{2} = \frac{\delta}{2} \quad (\text{Ans.})$$

(ii) The momentum thickness, θ

$$\theta = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U}\right) dy \quad \dots[\text{Eqn. (13.2)}]$$

$$= \int_0^{\delta} \frac{y}{\delta} \left(1 - \frac{y}{\delta}\right) dy = \int_0^{\delta} \left(\frac{y}{\delta} - \frac{y^2}{\delta^2} \right) dy$$

or,

$$\theta = \left[\frac{y^2}{2\delta} - \frac{y^3}{3\delta^2} \right]_0^{\delta} = \frac{\delta^2}{2\delta} - \frac{\delta^3}{3\delta^2} = \frac{\delta}{2} - \frac{\delta}{3} = \frac{\delta}{6} \quad (\text{Ans.})$$

(iii) The energy thickness, δ_e

$$\delta_e = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u^2}{U^2}\right) dy \quad \dots[\text{Eqn. (13.3)}]$$

$$= \int_0^{\delta} \frac{y}{\delta} \left(1 - \frac{y^2}{\delta^2}\right) dy = \int_0^{\delta} \left(\frac{y}{\delta} - \frac{y^3}{\delta^3} \right) dy$$

$$= \left[\frac{y^2}{2\delta} - \frac{y^4}{4\delta^3} \right]_0^{\delta} = \frac{\delta^2}{2\delta} - \frac{\delta^4}{4\delta^3} = \frac{\delta}{2} - \frac{\delta}{4} = \frac{\delta}{4}$$

i.e.

$$\delta_e = \frac{\delta}{4} \quad (\text{Ans.})$$

(iv) The value of $\frac{\delta^*}{\theta}$:

$$\frac{\delta^*}{\theta} = \frac{\delta/2}{\delta/6} = 3.0 \quad (\text{Ans.})$$

Example 13.2. The velocity distribution in the boundary layer is given by $\frac{u}{U} = \frac{3}{2} \frac{y}{\delta} - \frac{1}{2} \frac{y^2}{\delta^2}$, δ being boundary layer thickness.

Calculate the following:

(i) The ratio of displacement thickness to boundary layer thickness $\left(\frac{\delta^*}{\delta} \right)$,

(ii) The ratio of momentum thickness to boundary layer thickness $\left(\frac{\theta}{\delta}\right)$.

Solution. Velocity distribution : $\frac{u}{U} = \frac{3}{2} \frac{y}{\delta} - \frac{1}{2} \frac{y^2}{\delta^2}$... (Given)

(i) δ^*/δ :

$$\begin{aligned}\delta^* &= \int_0^\delta \left(1 - \frac{u}{U}\right) dy = \int_0^\delta \left(1 - \frac{3}{2} \frac{y}{\delta} + \frac{1}{2} \frac{y^2}{\delta^2}\right) dy \\&= \left[y - \frac{3}{2} \times \frac{y^2}{2\delta} + \frac{1}{2} \times \frac{y^3}{3\delta^2} \right]_0^\delta \\&= \left[\delta - \frac{3}{4} \cdot \frac{\delta^2}{\delta} + \frac{1}{2} \times \frac{\delta^3}{3\delta^2} \right] = \left(\delta - \frac{3}{4} \delta + \frac{\delta}{6} \right) = \frac{5}{12} \delta \\ \therefore \quad \frac{\delta^*}{\delta} &= \frac{5}{12} \text{ (Ans.)}\end{aligned}$$

(ii) θ/δ :

$$\begin{aligned}\theta &= \int_0^\delta \frac{u}{U} \left(1 - \frac{u}{U}\right) dy \\&= \int_0^\delta \left(\frac{3}{2} \frac{y}{\delta} - \frac{1}{2} \frac{y^2}{\delta^2} \right) \left(1 - \frac{3}{2} \frac{y}{\delta} + \frac{1}{2} \frac{y^2}{\delta^2} \right) dy \\&= \int_0^\delta \left(\frac{3}{2} \frac{y}{\delta} - \frac{9}{4} \frac{y^2}{\delta^2} + \frac{3}{4} \cdot \frac{y^3}{\delta^3} - \frac{1}{2} \frac{y^2}{\delta^2} + \frac{3}{4} \frac{y^3}{\delta^3} - \frac{1}{4} \frac{y^4}{\delta^4} \right) dy \\&= \int_0^\delta \left[\frac{3}{2} \frac{y}{\delta} - \left(\frac{9}{4} \frac{y^2}{\delta^2} + \frac{1}{2} \frac{y^2}{\delta^2} \right) + \left(\frac{3}{4} \frac{y^3}{\delta^3} + \frac{3}{4} \frac{y^3}{\delta^3} \right) - \frac{1}{4} \frac{y^4}{\delta^4} \right] dy \\&= \int_0^\delta \left[\frac{3}{2} \frac{y}{\delta} - \frac{11}{4} \frac{y^2}{\delta^2} + \frac{3}{2} \frac{y^3}{\delta^3} - \frac{1}{4} \frac{y^4}{\delta^4} \right] dy \\&= \left[\frac{3}{2} \times \frac{y^2}{2\delta} - \frac{11}{4} \times \frac{y^3}{3\delta^2} + \frac{3}{2} \times \frac{y^4}{4\delta^3} - \frac{1}{4} \times \frac{y^5}{5\delta^4} \right]_0^\delta \\&= \left[\frac{3}{2} \times \frac{\delta^2}{2\delta} - \frac{11}{4} \times \frac{\delta^3}{3\delta^2} + \frac{3}{2} \times \frac{\delta^4}{4\delta^3} - \frac{1}{4} \times \frac{\delta^5}{5\delta^4} \right]_0^\delta \\&= \left(\frac{3}{4} \delta - \frac{11}{12} \delta + \frac{3}{8} \delta - \frac{1}{20} \delta \right) = \frac{19}{120} \delta\end{aligned}$$

or, $\frac{\theta}{\delta} = \frac{19}{120} \text{ (Ans.)}$

Example 13.3. The velocity distribution in the boundary layer is given by

$$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2, \delta \text{ being boundary layer thickness.}$$

Calculate the following:

- (i) Displacement thickness,
- (ii) Momentum thickness, and
- (iii) Energy thickness.

Solution. (i) Displacement thickness, δ^*

$$\begin{aligned}
 \delta^* &= \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy \\
 &= \int_0^{\delta} \left[1 - \left\{2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2\right\}\right] dy \\
 &= \int_0^{\delta} \left[1 - 2\left(\frac{y}{\delta}\right) + \left(\frac{y}{\delta}\right)^2\right] dy \\
 &= \left[y - \frac{2}{2} \times \frac{y^2}{\delta} + \frac{y^3}{3\delta^2}\right]_0^{\delta} = \left[\delta - \frac{\delta^2}{\delta} + \frac{\delta^3}{3\delta^2}\right] \\
 &= \delta - \delta + \frac{\delta}{3} = \frac{\delta}{3} \text{ (Ans.)}
 \end{aligned}$$

(ii) Momentum thickness, θ

$$\begin{aligned}
 \theta &= \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U}\right) dy \\
 &= \int_0^{\delta} \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2}\right) \left[1 - \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2}\right)\right] dy \\
 &= \int_0^{\delta} \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2}\right) \left[1 - \frac{2y}{\delta} + \frac{y^2}{\delta^2}\right] dy \\
 &= \int_0^{\delta} \left[\frac{2y}{\delta} - \frac{4y^2}{\delta^2} + \frac{2y^3}{\delta^3} - \frac{y^2}{\delta^2} + \frac{2y^3}{\delta^3} - \frac{y^4}{\delta^4}\right] dy \\
 &= \int_0^{\delta} \left[\frac{2y}{\delta} - \frac{5y^2}{\delta^2} + \frac{4y^3}{\delta^3} - \frac{y^4}{\delta^4}\right] dy \\
 &= \left[\frac{2}{2} \times \frac{y^2}{\delta} - \frac{5}{3} \times \frac{y^3}{\delta^2} + \frac{4}{4} \times \frac{y^4}{\delta^3} - \frac{1}{5} \times \frac{y^5}{\delta^4}\right]_0^{\delta} \\
 &= \left[\delta - \frac{5}{3} \delta + \delta - \frac{1}{5} \delta\right] = \frac{2}{15} \delta \text{ (Ans.)}
 \end{aligned}$$

(iii) Energy thickness, δ_e

$$\begin{aligned}
 \delta_e &= \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u^2}{U^2}\right) dy \\
 &= \int_0^{\delta} \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2}\right) \left[1 - \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2}\right)^2\right] dy
 \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\delta} \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2} \right) \left[1 - \left(\frac{4y^2}{\delta^2} + \frac{y^4}{\delta^4} - \frac{4y^3}{\delta^3} \right) \right] dy \\
&= \int_0^{\delta} \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2} \right) \left[1 - \frac{4y^2}{\delta^2} - \frac{y^4}{\delta^4} + \frac{4y^3}{\delta^3} \right] dy \\
&= \int_0^{\delta} \left[\frac{2y}{\delta} - \frac{8y^3}{\delta^3} - \frac{2y^5}{\delta^5} + \frac{8y^4}{\delta^4} - \frac{y^2}{\delta^2} + \frac{4y^4}{\delta^4} + \frac{y^6}{\delta^6} - \frac{4y^5}{\delta^5} \right] dy \\
&= \int_0^{\delta} \left[\frac{2y}{\delta} - \frac{y^2}{\delta^2} - \frac{8y^3}{\delta^3} + 12 \times \frac{y^4}{\delta^4} - 6 \times \frac{y^5}{\delta^5} + \frac{y^6}{\delta^6} \right] dy \\
&= \int_0^{\delta} \left[\frac{2y}{\delta} - \frac{y^2}{\delta^2} - \frac{8y^3}{\delta^3} + 12 \times \frac{y^4}{\delta^4} - 6 \times \frac{y^5}{\delta^5} + \frac{y^6}{\delta^6} \right] dy \\
&= \int_0^{\delta} \left[\frac{2}{2} \times \frac{y^2}{\delta} - \frac{1}{3} \times \frac{y^3}{\delta^2} - \frac{8}{4} \times \frac{y^4}{\delta^3} + \frac{12}{5} \times \frac{y^5}{\delta^4} - \frac{6}{6} \times \frac{y^6}{\delta^5} + \frac{1}{7} \times \frac{y^7}{\delta^6} \right]_0^{\delta} \\
&= \left(\delta - \frac{\delta}{3} - 2\delta + \frac{12\delta}{5} - \delta + \frac{\delta}{7} \right) = \frac{22\delta}{105} \text{ (Ans.)}
\end{aligned}$$

Example 13.4. If velocity distribution in laminar boundary layer over a flat plate is assumed to be given by second order polynomial $u = a + by + cy^2$, determine its form using the necessary boundary conditions.

Solution. Velocity distribution: $u = a + by + cy^2$

The following *boundary conditions* must be satisfied:

(i) At $y = 0, u = 0$

$\therefore u = a + by + cy^2$

$0 = a + 0 + 0$

$\therefore a = 0$

(ii) At $y = \delta, u = U$

$\therefore U = b\delta + c\delta^2$

...(i)

(iii) At $y = \delta, \frac{du}{dy} = 0$

$\therefore \left(\frac{du}{dy} \right)_{y=\delta} = \frac{d}{dy} (a + by + cy^2) = b + 2cy = b + 2c\delta = 0$

...(ii)

Substituting the value of b ($= -2c\delta$) from (ii) in (i), we get:

$$U = (-2c\delta)\delta + c\delta^2 = -2c\delta^2 + c\delta^2 = -c\delta^2$$

or, $c = -\frac{U}{\delta^2}$

$\therefore b = -2c\delta = -2 \times \left(-\frac{U}{\delta^2} \right) \delta = \frac{2U}{\delta}$

Hence, form of the velocity distribution is:

$$u = \frac{2U}{\delta} y - \frac{U}{\delta^2} y^2$$

or,
$$\frac{u}{U} = 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \quad (\text{Ans.})$$

Example 13.5. The velocity distribution in the boundary layer is given by

$$\frac{u}{U} = \left(\frac{y}{\delta}\right)^{1/7}$$

Calculate the following

- (i) Displacement thickness,
- (ii) Momentum thickness,
- (iii) Shape factor,
- (iv) Energy thickness, and
- (v) Energy loss due to boundary layer if at a particular section, the boundary layer thickness is 25 mm and the free stream velocity is 15 m/s. If the discharge through the boundary layer region is 6 m³/s per metre width, express this energy loss in terms of metres of head. Take $\rho = 1.2 \text{ kg/m}^3$.

Solution. Velocity distribution: $\frac{u}{U} = \left(\frac{y}{\delta}\right)^{1/7} \quad \dots(\text{Given})$

(i) Displacement thickness, δ^* :

$$\begin{aligned} \delta^* &= \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy \\ &= \int_0^{\delta} \left[1 - \left(\frac{y}{\delta}\right)^{1/7}\right] dy = \left[y - \frac{7}{8} \frac{y^{8/7}}{\delta^{1/7}}\right]_0^{\delta} = \left[\delta - \frac{7}{8} \times \frac{\delta^{8/7}}{\delta^{1/7}}\right] \\ &= \frac{\delta}{8} \quad (\text{Ans.}) \end{aligned}$$

(ii) Momentum thickness, θ :

$$\begin{aligned} \theta &= \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U}\right) dy \\ &= \int_0^{\delta} \left(\frac{y}{\delta}\right)^{1/7} \left[1 - \left(\frac{y}{\delta}\right)^{1/7}\right] dy = \int_0^{\delta} \left[\left(\frac{y}{\delta}\right)^{1/7} - \left(\frac{y}{\delta}\right)^{2/7}\right] dy \\ &= \left[\frac{7}{8} \frac{y^{8/7}}{\delta^{1/7}} - \frac{7}{9} \cdot \frac{y^{9/7}}{\delta^{2/7}}\right]_0^{\delta} = \left[\frac{7}{8} \frac{\delta^{8/7}}{\delta^{1/7}} - \frac{7}{9} \frac{\delta^{9/7}}{\delta^{2/7}}\right] \\ &= \left(\frac{7}{8} \delta - \frac{7}{9} \delta\right) = \frac{7}{72} \delta \quad (\text{Ans.}) \end{aligned}$$

(iii) Shape factor:

$$\text{Shape factor} : = \frac{\delta^*}{\theta} = \frac{\delta/8}{7\delta/72} = \frac{\delta}{8} \times \frac{72}{7\delta} = 1.286 \quad (\text{Ans.})$$

(iv) Energy thickness, δ_e :

$$\delta_e = \int_0^{\delta} \frac{u}{U} \left[1 - \frac{u^2}{U^2}\right] dy$$

$$\begin{aligned}
 &= \int_0^{\delta} \left(\frac{y}{\delta} \right)^{1/7} \left[1 - \left(\frac{y}{\delta} \right)^{2/7} \right] dy = \int_0^{\delta} \left[\left(\frac{y}{\delta} \right)^{1/7} - \left(\frac{y}{\delta} \right)^{3/7} \right] dy \\
 &= \left[\frac{7}{8} \times \frac{y^{8/7}}{\delta^{1/7}} - \frac{7}{10} \times \frac{y^{10/7}}{\delta^{3/7}} \right]_0^{\delta} = \left[\frac{7}{8} \delta - \frac{7}{10} \delta \right] = \frac{7}{40} \delta \quad (\text{Ans.})
 \end{aligned}$$

(v) Energy loss

Given: Boundary layer thickness, $\delta^* = 25$ mm

Free stream velocity, $U = 15$ m/s

Discharge per metre width, $q = 6$ m³/s

Density of fluid, $\rho = 1.2$ kg/m³

$$\text{Now, energy thickness, } \delta_e = \frac{7}{40} \delta = \frac{7}{40} \times 25 = 4.375 \text{ mm}$$

Energy loss per unit width, due to boundary layer

$$= \frac{1}{2} (\rho \delta_e U) \times U^2 = \frac{1}{2} \times 1.2 \times \left(\frac{4.375}{1000} \right) \times 15^3 = 8.859 \text{ Nm/s} \quad (\text{Ans.})$$

Let, h_1 = Energy loss (per unit width) in terms of metres of head.

$$\text{Then,} \quad \rho q h_1 = 8.859$$

$$\text{or,} \quad h_1 = \frac{8.859}{\rho q} = \frac{8.859}{1.2 \times 6} = 1.23 \text{ m} \quad (\text{Ans.})$$

Example 13.6. Explain what you understand by boundary layer thickness and displacement thickness. Determine the relationship between the two for a boundary layer which is (i) laminar throughout, and (ii) turbulent throughout. Assume (1) in the laminar boundary layer the flow obeys the law, shear stress $\tau = \mu \frac{du}{dy}$, where μ is the viscosity, which leads to the velocity profile $(U - u) = k(\delta - y)^2$ where U is the free stream velocity, u is the velocity at a distance y above the plate and k is a constant; (2) the velocity distribution in the turbulent boundary layer is given by $\frac{u}{U} = \left(\frac{y}{\delta} \right)^{1/7}$.
(UPSC)

Solution. (a) **Boundary layer thickness.** It is defined as the distance from the boundary of solid body measured in Y-direction to the point, where the velocity of fluid is approximately equal to 0.99 times the free stream (U) velocity of fluid.

Displacement thickness. It is defined as the distance, measured perpendicular to the boundary of the solid body by which the boundary should be displaced to compensate for the reduction in flow rate on account of boundary layer formation.

(i) **When the flow is laminar throughout:**

$$\text{Velocity profile:} \quad (U - u) = k(\delta - y)^2 \quad \dots(1)$$

$$\tau = \mu \frac{du}{dy}$$

where,

U = Free stream velocity,

u = Velocity at a distance y above plate, and

k = Constant.

Let, δ^* = Displacement thickness, and
 δ = Boundary layer thickness.

$$\delta^* = \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy$$

Dividing eqn. (1) by U , we get:

$$1 - \frac{u}{U} = \frac{k}{U} (\delta - y)^2$$

$$\therefore \delta^* = \int_0^{\delta} \frac{k}{U} (\delta - y)^2 dy = \left[-\frac{k}{U} \times \frac{(\delta - y)^3}{3} \right]_0^{\delta} = \left[\frac{k (\delta - y)^3}{3U} \right]_{\delta}^0 = \frac{k\delta^3}{3U} \quad \dots(2)$$

When $y = 0$, $u = 0$ i.e. on the surface of the plate.

Substituting these parameters in eqn. (1), we get $U = k\delta^2$...(3)

Substituting the value of U from (3) in (2), we get:

$$\delta^* = \frac{k\delta^3}{3k\delta^2} = \frac{\delta}{3} \text{ (Ans.)}$$

(ii) *When the flow is turbulent throughout*

Velocity profile: $\frac{u}{U} = \left(\frac{y}{\delta}\right)^{1/7}$

$$\delta^* = \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy = \int_0^{\delta} \left[1 - \left(\frac{y}{\delta}\right)^{1/7}\right] dy$$

$$\left[y - \frac{7}{8} \frac{y^{8/7}}{\delta^{1/7}} \right]_0^{\delta} = \frac{1}{8} \delta \text{ (Ans.)}$$

Example 13.7. *In the boundary layer over the face of a high spillway, the velocity distribution was observed to have the following form :*

$$\frac{u}{U} = \left(\frac{y}{\delta}\right)^{0.22}$$

The free stream velocity U at a certain section was observed to be 30 m/s and a boundary layer thickness of 60 mm was estimated from the velocity distribution measured at the section. The discharge passing over the spillway was 6 m³/s per metre length of spillway. Calculate :

- (i) *The displacement thickness,*
- (ii) *The energy thickness, and*
- (iii) *The loss of energy upto the section under consideration.*

Solution. Velocity distribution: $\frac{u}{U} = \left(\frac{y}{\delta}\right)^{0.22}$

The free stream velocity at a certain section, $U = 30$ m/s

Thickness of the boundary layer, $\delta = 60$ mm

The discharge passing over the spillway, $q = 6$ m³/s per metre length of the spillway.

(i) The displacement thickness, δ^*

$$\begin{aligned}\delta^* &= \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy = \int_0^{\delta} \left[1 - \left(\frac{y}{\delta}\right)^{0.22}\right] dy \\ &= \left[y - \frac{y^{1.22}}{1.22 \times \delta^{0.22}}\right]_0^{\delta} = \left(\delta - \frac{\delta^{1.22}}{1.22 \times \delta^{0.22}}\right) = \delta \left(1 - \frac{1}{1.22}\right) \\ &= 0.18\delta = 0.18 \times 60 \text{ mm} = \mathbf{10.8 \text{ mm (Ans.)}}\end{aligned}$$

(ii) The energy thickness, δ_e

$$\begin{aligned}\delta_e &= \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u^2}{U^2}\right) dy \\ &= \int_0^{\delta} \left(\frac{y}{\delta}\right)^{0.22} \left[1 - \left(\frac{y}{\delta}\right)^{0.44}\right] dy \\ &= \int_0^{\delta} \left[\left(\frac{y}{\delta}\right)^{0.22} - \left(\frac{y}{\delta}\right)^{0.66}\right] dy \\ &= \left[\frac{y^{1.22}}{1.22 \times \delta^{0.22}} - \frac{\delta^{1.66}}{1.66 \delta^{0.66}}\right]_0^{\delta} = \left[\frac{\delta^{1.22}}{1.22 \times \delta^{0.22}} - \frac{\delta^{1.66}}{1.66 \times \delta^{0.66}}\right] \\ &= \left[\frac{\delta}{1.22} - \frac{\delta}{1.66}\right] = 0.217 \delta = 0.217 \times 60 = \mathbf{13.02 \text{ mm (Ans.)}}\end{aligned}$$

(iii) The loss of energy:

The energy loss *per metre length* of spillway is,

$$\begin{aligned}E_L &= \frac{1}{2} \times (\rho \times \delta_e \times U) \times U^2 = \frac{1}{2} \rho \cdot \delta_e U^3 = \frac{1}{2} \times 1000 \times \frac{13.02}{1000} \times (30)^3 \\ &= \mathbf{175.77 \text{ kNm/s (Ans.)}}\end{aligned}$$

Energy loss (*per metre length*) in terms of metres of head

$$= \frac{E_L}{wq} = \frac{175.77}{9.81 \times 6} = \mathbf{2.986 \text{ m (Ans.)}}$$

Example 13.8. The velocity distribution in the boundary layer over the face of a spillway was observed to be:

$$\frac{u}{U} = \left(\frac{y}{\delta}\right)^{0.22}$$

The free stream velocity U is 20 m/s and boundary layer thickness 5 cm at a certain section. The discharge is 5 m³/s per meter length of spillway. Calculate displacement thickness, energy thickness and loss of energy up to section under consideration. (Delhi University)

Solution. Given : Velocity distribution: $\frac{u}{U} = \left(\frac{y}{\delta}\right)^{0.22}$

$U = 20 \text{ m/s}$; $\delta = 5 \text{ cm}$; $Q = 5 \text{ m}^3/\text{s}$ per metre length of spillway,

δ^*, δ_e, E_L

The displacement thickness is given by the equation:

$$\begin{aligned}\delta^* &= \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy = \int_0^{\delta} \left\{1 - \left(\frac{y}{\delta}\right)^{0.22}\right\} dy = \left[y - \frac{1}{(\delta)^{0.22}} \times \frac{(y)^{0.22+1}}{1.22} \right]_0^{\delta} \\ &= \delta - \frac{\delta}{1.22} = \frac{0.22}{1.22} \delta = \frac{0.22}{1.22} \times 5 = \mathbf{0.9016 \text{ cm (Ans.)}}\end{aligned}$$

The *energy thickness* is given by the equation:

$$\begin{aligned}\delta_e &= \int_0^{\delta} \frac{u}{U} \left\{1 - \left(\frac{u}{U}\right)^2\right\} dy = \int_0^{\delta} \left\{\frac{u}{U} - \left(\frac{u}{U}\right)^3\right\} dy = \int_0^{\delta} \left\{\left(\frac{y}{\delta}\right)^{0.22} - \left(\frac{y}{\delta}\right)^{0.66}\right\} dy \\ &= \int_0^{\delta} \left[\frac{1}{(\delta)^{0.22}} \times \frac{(y)^{0.22+1}}{1.22} - \frac{1}{(\delta)^{0.66}} \times \frac{(y)^{0.66+1}}{1.66} \right] dy \\ &= \frac{8}{1.22} - \frac{\delta}{1.66} = \delta \left(\frac{1}{1.22} - \frac{1}{1.66} \right) = 5 \left(\frac{1.66 - 1.22}{1.22 \times 1.66} \right) = \mathbf{1.086 \text{ cm (Ans.)}}\end{aligned}$$

The *energy loss per m length* of spillway is,

$$E_L = \frac{1}{2} (\rho \times \delta_e \times U) \times U^2 = \frac{1}{2} \rho \delta_e U^3$$

$$\therefore E_L = \frac{1}{2} \times 1000 \times \left(\frac{1.086}{100} \right) \times (20)^3 \times 10^{-3} \text{ kNm/s} = 43.44 \text{ kNm/s}$$

Energy loss in terms of *m* of head

$$= \frac{E_L}{\rho Q g} = \frac{43.44 \times 1000}{1000 \times 5 \times 9.81} = \mathbf{0.8856 \text{ m (Ans.)}}$$

Example 13.9. For steady Poiseuille flow in a pipe of radius *R*, obtain an expression for ratio of the displacement thickness (δ^*) to momentum thickness (θ). [Roorkee University]

Solution. Radius of the pipe = *R*

$$\frac{\delta^*}{\theta} :$$

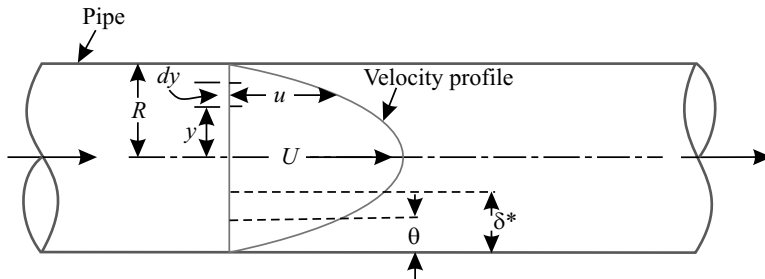


Fig. 13.3

For steady Poiseuille flow in a circular pipe, the velocity distribution is given by,

$$u = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} (R^2 - y^2) \quad \dots(i)$$

where *y* being measured from the centre of the pipe.

At, $y = 0$, $u = U$ (maximum velocity)

$$\therefore U = -\frac{1}{4\mu} \cdot \frac{\partial p}{\partial x} R^2 \quad \dots(ii)$$

From the definition of displacement thickness, we have:

$$[\pi R^2 - \pi(R - \delta^*)^2]U = \int_0^R 2\pi y (U - u) dy$$

$$\text{or,} \quad 2\pi R\delta^*U = \int_0^R 2\pi y (U - u) dy$$

[neglecting the term containing $(\delta^*)^2$, since δ^* is very small.]

$$\text{or,} \quad \delta^* = \frac{1}{R} \int_0^R \left(1 - \frac{u}{U}\right) y dy = \frac{1}{R} \int_0^R \frac{y^2}{R^2} y dy = \frac{R}{4}$$

$$\left[\text{Dividing (i) by (ii), we have:} \quad \frac{u}{U} = \frac{R^2 - y^2}{R^2} = 1 - \frac{y^2}{R^2} \text{ or } \left(1 - \frac{u}{U}\right) = \frac{y^2}{R^2} \right]$$

From the definition of momentum thickness, we have:

$$[\rho \{ \pi R^2 - \pi(R - \theta)^2 \}]U = \int_0^R \rho 2\pi y \cdot dy \cdot u (U - u)$$

On simplification, we get:

$$2\pi R\theta U^2 = \int_0^R 2\pi y u (U - u) dy$$

$$\theta = \frac{1}{R} \int_0^R \frac{u}{U} \left(1 - \frac{u}{U}\right) y dy$$

$$= \frac{1}{R} \int_0^R \left(1 - \frac{y^2}{R^2}\right) \frac{y^2}{R^2} y dy = \frac{1}{R} \int_0^R \left(\frac{y^3}{R^2} - \frac{y^5}{R^4}\right) dy$$

$$= \frac{1}{R} \left[\frac{y^4}{4R^2} - \frac{y^6}{6R^4} \right]_0^R = \frac{1}{R} \left[\frac{R^4}{4R^2} - \frac{R^6}{6R^4} \right] = \frac{1}{R} \left[\frac{R^2}{4} - \frac{R^2}{6} \right] = \frac{R}{12}$$

$$\therefore \frac{\delta^*}{\theta} = \frac{R/4}{R/12} = 3 \text{ (Ans.)}$$

13.3. MOMENTUM EQUATION FOR BOUNDARY LAYER BY VON KARMAN

Von Karman suggested a method based on the *momentum equation* by the use of which the growth of a boundary layer along a flat plate, the wall shear stress and the drag force could be determined (when the velocity distribution in the boundary layer is known). Starting from the beginning of the plate, the method can be used for *both laminar and turbulent boundary layers*.

Fig. 13.4. shows a fluid flowing over a thin plate (placed at zero incidence) with a free stream velocity equal to U . Consider a small length dx of the plate at a distance x from the leading edge as shown in Fig. 13.4 (a); the enlarged view of the small length of the plate is shown in Fig. 13.4 (b). Consider *unit width* of plate perpendicular to the direction of flow.

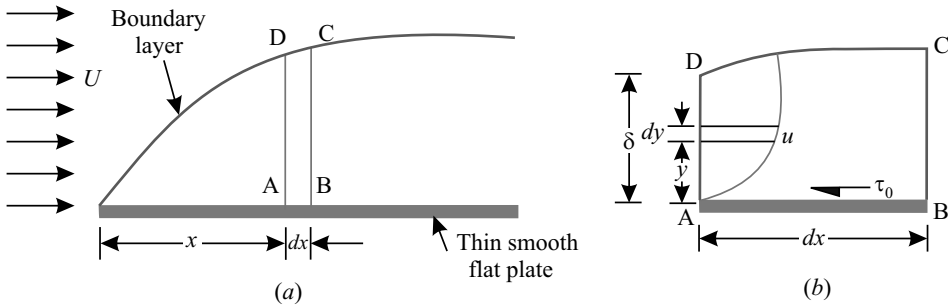


Fig. 13.4. Momentum equation for boundary layer by Von Karman.

Let $ABCD$ be a small element of a boundary layer (the edge DC represents the outer edge of the boundary layer).

Mass rate of fluid entering through AD

$$= \int_0^{\delta} \rho u dy$$

Mass rate of fluid leaving through BC

$$= \int_0^{\delta} \rho u dy + \frac{d}{dx} \left[\int_0^{\delta} \rho u dy \right] dx$$

$\therefore$ Mass rate of fluid entering the control volume through the surface DC

= Mass rate of fluid through BC – mass rate of fluid through AD

$$= \int_0^{\delta} \rho u dy + \frac{d}{dx} \left[\int_0^{\delta} \rho u dy \right] dx - \int_0^{\delta} \rho u dy = \frac{d}{dx} \left[\int_0^{\delta} \rho u dy \right] dx$$

The fluid is entering through DC with a uniform velocity U .

Momentum rate of fluid entering the control volume in X -direction through AD

$$= \int_0^{\delta} \rho u^2 dy$$

Momentum rate of fluid leaving the control volume in X -direction through BC

$$= \int_0^{\delta} \rho u^2 dy + \frac{d}{dx} \left[\int_0^{\delta} \rho u^2 dy \right] dx$$

Momentum rate of fluid entering the control volume through DC in X -direction

$$= \frac{d}{dx} \left[\int_0^{\delta} \rho u dy \right] dx \times U \quad (\because \text{Velocity} = U)$$

$$= \frac{d}{dx} \left[\int_0^{\delta} \rho u U dy \right] dx$$

$\therefore$ Rate of change of momentum of control volume

= Momentum rate of fluid through BC – momentum rate of fluid through AD
– momentum rate of fluid through DC

$$= \int_0^{\delta} \rho u^2 dy + \frac{d}{dx} \left[\int_0^{\delta} \rho u^2 dy \right] dx - \int_0^{\delta} \rho u^2 dy - \frac{d}{dx} \left[\int_0^{\delta} \rho u U dy \right] dx$$

$$\begin{aligned}
&= \frac{d}{dx} \left[\int_0^{\delta} \rho u^2 dy - \int_0^{\delta} \rho u U dy \right] dx \\
&= \frac{d}{dx} \left[\int_0^{\delta} (\rho u^2 dy - \rho u U dy) \right] dx \\
&= \frac{d}{dx} \left[\rho \int_0^{\delta} (u^2 - uU) dy \right] dx \\
&\quad (\rho \text{ is constant for incompressible fluid}) \\
&= \rho \frac{d}{dx} \left[\int_0^{\delta} (u^2 - uU) dy \right] dx \quad \dots(13.4)
\end{aligned}$$

As per momentum principle the rate of change of momentum on the control volume $ABCD$ must be equal to the total force on the control volume in the same direction. The only external force acting on the control volume is the shear force acting on the side AB in the direction B to A (Fig. 13.4 *b*). The value of this force (*drag force*) is given by,

$$\Delta F_D = \tau_0 \times dx$$

Thus the total external force in the direction of rate of change of momentum

$$= -\tau_0 \times dx \quad \dots(13.5)$$

Equating the eqns. (13.4) and (13.5), we have:

$$-\tau_0 \times dx = \rho \frac{d}{dx} \left[\int_0^{\delta} (u^2 - uU) dy \right] dx$$

$$\text{or,} \quad \tau_0 = -\rho \frac{d}{dx} \left[\int_0^{\delta} (u^2 - uU) dy \right]$$

$$\begin{aligned}
\text{or,} \quad &= \rho \frac{d}{dx} \left[\int_0^{\delta} (uU - u^2) dy \right] \\
&= \rho \frac{d}{dx} \left[\int_0^{\delta} U^2 \left(\frac{u}{U} - \frac{u^2}{U^2} \right) dy \right] \\
&= \rho U^2 \frac{d}{dx} \left[\int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \right]
\end{aligned}$$

$$\text{or,} \quad \frac{\tau_0}{\rho U^2} = \frac{d}{dx} \left[\int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \right] \quad \dots(13.6)$$

$$\text{But,} \quad \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy = \text{momentum thickness } (\theta)$$

$$\therefore \quad \frac{\tau_0}{\rho U^2} = \frac{d\theta}{dx} \quad \dots(13.7)$$

Eqn. (13.7) is known as **Von Karman momentum equation** for boundary layer flow, and is used to find out the frictional drag on smooth flat plate for both laminar and turbulent boundary layers.

The following *boundary conditions* must be satisfied for any *assumed velocity distribution*:

(i) At the surface of the plate: $y = 0$, $u = 0$, $\frac{du}{dy} = \text{finite value}$

(ii) At the outer edge of boundary layer: $y = \delta$, $u = U$

$$y = \delta, \frac{du}{dy} = 0$$

The shear stress, τ_0 for a given velocity profile in laminar, transition or turbulent zone is obtained from eqn. (13.6) or (13.7). Then drag force on a small distance dx of a plate is given by,

$$\begin{aligned}\Delta F_D &= \text{Shear stress} \times \text{area} \\ &= \tau_0 \times (B \times dx) = \tau_0 \times B \times dx [\text{assuming width of plate as unity}] \\ &\quad (\text{where, } B = \text{width of the plate})\end{aligned}$$

$\therefore$ Total drag on the plate of length L one side,

$$F_D = \int \Delta F_D = \int_0^L \tau_0 \times B \times dx \quad \dots(13.8)$$

—The ratio of the shear stress τ_0 to the quantity $\frac{1}{2} \rho U^2$ is known as the “**Local co-efficient of drag**” (or **co-efficient of skin friction**) and is denoted by C_D^*

$$\text{i.e.} \quad C_D^* = \frac{\tau_0}{\frac{1}{2} \rho U^2} \quad \dots(13.9)$$

— The ratio of the total drag force to the quantity $\frac{1}{2} \rho A U^2$ is called ‘**Average co-efficient of drag**’ and is denoted by C_D

$$\text{i.e.} \quad C_D = \frac{F_D}{\frac{1}{2} \rho A U^2} \quad \dots(13.10)$$

where,

ρ = Mass density of fluid,

A = Area of surface/plate, and

U = Free stream velocity.

13.4. LAMINAR BOUNDARY LAYER

Let us find out boundary layer thickness (δ), shear stress (τ_0), local co-efficient of drag (C_D^*) and co-efficient of drag (C_D) for the following velocity distribution in the laminar boundary layer:

$$1. \quad \frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$$

$$2. \quad \frac{u}{U} = \frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3$$

...Prandtl’s velocity distribution.

$$3. \quad \frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - 2 \left(\frac{y}{\delta} \right)^3 + \left(\frac{y}{\delta} \right)^4$$

$$4. \quad \frac{u}{U} = \sin \left(\frac{\pi}{2} \frac{y}{\delta} \right)$$

Case 1. Velocity distribution: $\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$... (i)

(i) **Boundary layer thickness**

We know, $\frac{\tau_0}{\rho U^2} = \frac{d}{dx} \left[\int_0^\delta \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \right]$ [Eqn. (13.6)]

Substituting the value of $\frac{u}{U}$, we get:

$$\begin{aligned} \frac{\tau_0}{\rho U^2} &= \frac{d}{dx} \left[\int_0^\delta \left\{ \frac{2y}{\delta} - \frac{y^2}{\delta^2} \right\} \left\{ 1 - \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2} \right) \right\} dy \right] \\ &= \frac{d}{dx} \left[\int_0^\delta \left\{ \frac{2y}{\delta} - \frac{y^2}{\delta^2} \right\} \left\{ 1 - \frac{2y}{\delta} + \frac{y^2}{\delta^2} \right\} dy \right] \\ &= \frac{d}{dx} \left[\int_0^\delta \left\{ \frac{2y}{\delta} - \frac{4y^2}{\delta^2} + \frac{2y^3}{\delta^3} - \frac{y^2}{\delta^2} + \frac{2y^3}{\delta^3} - \frac{y^4}{\delta^4} \right\} dy \right] \\ &= \frac{d}{dx} \left[\int_0^\delta \left\{ \frac{2y}{\delta} - \frac{5y^2}{\delta^2} + \frac{4y^3}{\delta^3} - \frac{y^4}{\delta^4} \right\} dy \right] \\ &= \frac{d}{dx} \left[\frac{2}{\delta} \frac{y^2}{2} - \frac{5}{\delta^2} \frac{y^3}{3} + \frac{4}{\delta^3} \frac{y^4}{4} - \frac{1}{\delta^4} \frac{y^5}{5} \right]_0^\delta \\ &= \frac{d}{dx} \left[\delta - \frac{5}{3} \delta + \delta - \frac{1}{5} \delta \right] = \frac{d}{dx} \left(\frac{2}{15} \delta \right) \\ \therefore \tau_0 &= \rho U^2 \times \frac{d}{dx} \left(\frac{2}{15} \delta \right) = \frac{2}{15} \rho U^2 \frac{d\delta}{dx} \end{aligned} \quad \dots (13.11)$$

Also, according to Newton's law of viscosity,

$$\tau_0 = \mu \left(\frac{du}{dy} \right)_{y=0} \quad \dots (ii)$$

But, $u = U \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2} \right)$... [From eqn. (i)]

and, $\frac{du}{dy} = U \left(\frac{2}{\delta} - \frac{2y}{\delta^2} \right)$, U being constant

$$\therefore \left(\frac{du}{dy} \right)_{y=0} = U \left(\frac{2}{\delta} - 0 \right) = \frac{2U}{\delta}$$

Substituting this value in (ii), we get:

$$\tau_0 = \frac{2\mu U}{\delta} \quad \dots (13.12)$$

Equating the values of τ_0 given by eqns. (13.11) and (13.12), we get:

$$\frac{2}{15} \rho U^2 \frac{d\delta}{dx} = \frac{2\mu U}{\delta}$$

$$\text{or,} \quad \delta \cdot \frac{d\delta}{dx} = \frac{15\mu U}{\rho U^2} = \frac{15\mu}{\rho U}$$

$$\text{or,} \quad \delta \cdot d\delta = \frac{15\mu}{\rho U} dx$$

Integrating both sides, we get:

$$\frac{\delta^2}{2} = \frac{15\mu}{\rho U} x + C \quad (\text{where, } C = \text{constant of integration})$$

$$\text{At,} \quad x = 0, \delta = 0 \quad \therefore C = 0$$

$$\therefore \quad \frac{\delta^2}{2} = \frac{15\mu x}{\rho U}$$

$$\begin{aligned} \text{or,} \quad \delta &= \sqrt{\frac{2 \times 15\mu x}{\rho U}} = 5.48 \sqrt{\frac{\mu x}{\rho U}} \\ &= 5.48 \sqrt{\frac{\mu x \times x}{\rho U \times x}} = 5.48 \sqrt{\frac{x^2}{Re_x}} \end{aligned}$$

$$\left(\text{where, } Re_x = \frac{\rho U x}{\mu} \right)$$

$$\text{or,} \quad \delta = 5.48 \frac{x}{\sqrt{Re_x}} \quad \dots(13.13)$$

(ii) **Shear stress τ_0**

From eqn. (13.12), we have:

$$\tau_0 = \frac{2\mu U}{\delta}$$

$$\text{But,} \quad \delta = 5.48 \frac{x}{\sqrt{Re_x}} \quad [\text{Eqn. (13.12)}]$$

$$\therefore \quad \tau_0 = \frac{2\mu U}{5.48 \frac{x}{\sqrt{Re_x}}} = \frac{2\mu U \sqrt{Re_x}}{5.48 x} = 0.365 \frac{\mu U}{x} \sqrt{Re_x} \quad \dots(13.14)$$

(iii) **Local co-efficient of drag, C_D^***

$$\tau_0 = 0.365 \frac{\mu U}{x} \sqrt{Re_x} \quad \dots[\text{Eqn. (13.14)}]$$

$$\text{Also,} \quad \tau_0 = C_D^* \frac{\rho U^2}{2} \quad \dots[\text{Eqn. (13.9)}]$$

(where, C_D^* = local co-efficient of drag)

Equating the two values of τ_0 , given by eqns. (13.14) and (13.9), we get:

$$C_D^* = 0.365 \frac{\mu U}{x} \sqrt{Re_x} \quad \text{or} \quad C_D^* = 0.365 \times 2 \times \frac{\sqrt{Re_x}}{\frac{\rho U x}{\mu}} = \frac{0.73}{\sqrt{Re_x}}$$

Hence, $C_D^* = \frac{0.73}{\sqrt{Re_x}} \quad \dots(13.15)$

(iv) Co-efficient of drag, C_D :

...[Eqn. (13.10)]

We know that, $C_D = \frac{F_D}{\frac{1}{2} \rho A U^2}$

where, $F_D = \int_0^L \tau_0 \times B \times dx \quad \dots[\text{Eqn. (13.8)}]$

$$\begin{aligned} &= \int_0^L 0.365 \frac{\mu U}{x} \sqrt{Re_x} \times B \times dx = \int_0^L \frac{\mu U}{x} \sqrt{\frac{\rho U x}{\mu}} \times B \times dx \quad \left(\because Re_x = \frac{\rho U x}{\mu} \right) \\ &= 0.365 \int_0^L \mu U \sqrt{\frac{\rho U}{\mu}} \times \frac{1}{\sqrt{x}} \times B \times dx = 0.365 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \int_0^L x^{-1/2} \times dx \\ &= 0.365 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \left[\frac{x^{1/2}}{1/2} \right]_0^L = 0.365 \times 2 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \sqrt{L} \end{aligned}$$

or, $F_D = 0.73 \mu U B \sqrt{\frac{\rho U L}{\mu}} \quad \dots(13.16)$

$\therefore C_D = \frac{0.73 \mu U B \sqrt{\frac{\rho U L}{\mu}}}{\frac{1}{2} \rho A U^2}$

(where, $A = \text{area of plate} = L \times B$, L and B being length and width of the plate respectively)

$$\begin{aligned} \therefore C_D &= \frac{0.73 \mu U B \sqrt{\frac{\rho U L}{\mu}}}{\frac{1}{2} \rho \times L \times B \times U^2} = \frac{1.46 \mu}{\rho L U} \sqrt{\frac{\rho U L}{\mu}} \\ &= \frac{1.46 \sqrt{\mu}}{\sqrt{\rho U L}} = 1.46 \sqrt{\frac{\mu}{\rho U L}} = \frac{1.46}{\sqrt{Re_L}} \quad \dots(13.17) \end{aligned}$$

Case 2. Velocity distribution: $\frac{u}{U} = \frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3$

(i) Boundary layer thickness, δ :

$$\frac{\tau_0}{\rho U^2} = \frac{d}{dx} \left[\int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \right] \quad [\text{Eqn. (13.6)}]$$

Substituting the value of $\frac{u}{U}$, we get:

$$\begin{aligned}
\frac{\tau_0}{\rho U^2} &= \frac{d}{dx} \left[\int_0^\delta \left(\frac{3}{2} \frac{y}{\delta} - \frac{1}{2} \frac{y^3}{\delta^3} \right) \left\{ 1 - \frac{3}{2} \frac{y}{\delta} + \frac{1}{2} \frac{y^3}{\delta^3} \right\} dy \right] \\
&= \left[\int_0^\delta \left(\frac{3}{2} \frac{y}{\delta} - \frac{9}{4} \frac{y^2}{\delta^2} + \frac{3}{4} \frac{y^4}{\delta^4} - \frac{1}{2} \frac{y^3}{\delta^3} + \frac{3}{4} \frac{y^4}{\delta^4} - \frac{1}{4} \frac{y^6}{\delta^6} \right) dy \right] \\
&= \frac{d}{dx} \left[\frac{3}{2} \times \frac{1}{2} \frac{y^2}{\delta} - \frac{9}{4} \times \frac{1}{3} \frac{y^3}{\delta^2} + \frac{3}{4} \times \frac{1}{5} \frac{y^5}{\delta^4} - \frac{1}{2} \times \frac{1}{4} \frac{y^4}{\delta^3} + \frac{3}{4} \times \frac{1}{5} \frac{y^5}{\delta^4} - \frac{1}{4} \times \frac{1}{7} \frac{y^7}{\delta^6} \right]_0^\delta \\
&= \frac{d}{dx} \left[\frac{3}{4} \delta - \frac{3}{4} \delta + \frac{3}{20} \delta + \frac{3}{20} \delta - \frac{1}{8} \delta - \frac{1}{28} \delta \right] = \frac{39}{280} \frac{d\delta}{dx}
\end{aligned}$$

$$\text{or, } \tau_0 = \rho U^2 \times \frac{39}{280} \frac{d\delta}{dx} = \frac{39}{280} \rho U^2 \frac{d\delta}{dx} \quad \dots(13.18)$$

$$\text{Also, } \tau_0 = \mu \left(\frac{du}{dy} \right)_{y=0}$$

$$\text{But, } u = U \left[\frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3 \right]$$

$$\text{and, } \frac{du}{dy} = U \left(\frac{3}{2\delta} - \frac{3}{2} \frac{y^2}{\delta^3} \right)$$

$$\therefore \tau_0 = \mu \left(\frac{du}{dy} \right)_{y=0} = \mu U \left(\frac{3}{2\delta} - 0 \right) = \frac{3\mu U}{2\delta} \quad \dots(13.19)$$

Equating the two values of τ_0 given by eqns. (13.18) and (13.19), we get:

$$\begin{aligned}
\frac{3\mu}{280} \rho U^2 \frac{d\delta}{dx} &= \frac{3\mu U}{2\delta} \\
\therefore \delta \cdot d\delta &= \frac{3}{2} \mu U \times \frac{280}{39} \times \frac{dx}{\rho U^2} = \frac{420}{39} \frac{\mu}{\rho U} dx
\end{aligned}$$

Integrating both sides, we get:

$$\delta^2 = \frac{420}{39} \frac{\mu}{\rho U} x + C$$

(where, C = constant of integration)

When, $x = 0, \delta = 0 \therefore C = 0$

$$\therefore \frac{\delta^2}{2} = \frac{420}{39} \frac{\mu}{\rho U} x$$

$$\begin{aligned}
\text{or, } \delta &= \sqrt{\frac{420 \times 2}{39} \cdot \frac{\mu}{\rho U} x} = 4.64 \sqrt{\frac{\mu x}{\rho U}} \\
&= 4.64 \sqrt{\frac{\mu x}{\rho U} \times \frac{x}{x}} = 4.64 \sqrt{\frac{\mu}{\rho U x}} \cdot x = \frac{4.64 x}{\sqrt{Re_x}} \quad \dots(13.20)
\end{aligned}$$

(ii) Shear stress, τ_0

$$\tau_0 = \frac{3\mu U}{2\delta}$$

But,

$$\delta = \frac{4.64 x}{\sqrt{Re_x}} \quad [\text{Eqn. (13.19)}]$$

 $\therefore$

$$\tau_0 = \frac{3\mu U}{2 \times \frac{4.64x}{\sqrt{Re_x}}} = \frac{3}{9.28} \frac{\mu U \sqrt{Re_x}}{x} = 0.323 \frac{\mu U}{x} \sqrt{Re_x} \quad \dots(13.21)$$

(iii) Local co-efficient of drag, C_D^*

$$\tau_0 = 0.323 \frac{\mu U}{x} \sqrt{Re_x}$$

Also,

$$\tau_0 = C_D^* \frac{\rho U^2}{2}$$

$$\therefore C_D^* \frac{\rho U^2}{2} = 0.323 \frac{\mu U}{x} \sqrt{Re_x} \quad \text{or} \quad C_D^* = \frac{0.646}{\sqrt{Re_x}} \quad \dots(13.22)$$

(iv) Co-efficient of drag (C_D):

$$C_D = \frac{F_D}{\frac{1}{2} \rho A U^2} \quad [\text{Eqn. (13.10)}]$$

$$\text{where, } F_D = \int_0^L \tau_0 \times B \times dx \quad [\text{Eqn. (13.8)}]$$

$$= \int_0^L \frac{\mu U}{x} Re_x \times B \times dx = 0.0323 \int_0^L \frac{\mu U}{x} \sqrt{\frac{\rho U x}{\mu}} \times B \times dx$$

$$= 0.323 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \int_0^L x^{-1/2} dx = 0.323 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \left[\frac{x^{1/2}}{1/2} \right]_0^L$$

$$= 0.323 \times 2 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \times \sqrt{L}$$

$$\text{or, } F_D = 0.646 \mu U \sqrt{\frac{\rho UL}{\mu}} \times B \quad \dots(13.23)$$

$$\therefore C_D = \frac{0.646 \mu UB \sqrt{\frac{\rho UL}{\mu}}}{\frac{1}{2} \rho A U^2} \quad (\text{where, } A = L \times B)$$

$$= \frac{0.646 \mu UB \sqrt{\frac{\rho UL}{\mu}}}{\frac{1}{2} \rho \times L \times B \times U^2} = 0.646 \times 2 \times \frac{\mu}{\rho UL} \times \sqrt{\frac{\rho UL}{\mu}} = \frac{1292}{\sqrt{\frac{\rho UL}{\mu}}}$$

$$\text{or, } C_D = \frac{1.292}{\sqrt{Re_L}} \quad \dots(13.24)$$

$$\left(\text{where, } Re_L = \sqrt{\frac{\rho UL}{\mu}} \right)$$

$$\text{Case 3. Velocity distribution: } \frac{\tau_0}{\rho U^2} = \frac{d}{dx} \left[\int_0^\delta \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \right]$$

(i) Boundary layer thickness, δ

$$\frac{\tau_0}{\rho U^2} = \frac{d}{dx} \left[\int_0^\delta \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \right] \quad [\text{Eqn. (13.6)}]$$

Substituting the value of $\frac{u}{U}$, we get:

$$\begin{aligned} \tau_0 &= \frac{d}{dx} \left[\int_0^\delta \left(\frac{2y}{\delta} - \frac{2y^3}{\delta^3} + \frac{y^4}{\delta^4} \right) \left(1 - \frac{2y}{\delta} + \frac{2y^3}{\delta^3} - \frac{y^4}{\delta^4} \right) dy \right] \\ &= \frac{d}{dx} \left[\int_0^\delta \left(\frac{2y}{\delta} - \frac{4y^2}{\delta^2} + \frac{4y^4}{\delta^4} - \frac{2y^5}{\delta^5} - \frac{2y^3}{\delta^3} + \frac{4y^4}{\delta^4} - \frac{4y^6}{\delta^6} + \frac{2y^7}{\delta^7} + \frac{y^4}{\delta^4} - \frac{2y^5}{\delta^5} + \frac{2y^7}{\delta^7} - \frac{y^8}{\delta^8} \right) dy \right] \\ &= \frac{d}{dx} \left[\frac{2}{2} \times \frac{y^2}{\delta} - \frac{4}{3} \frac{y^3}{\delta^2} + \frac{4}{5} \frac{y^5}{\delta^4} - \frac{2}{6} \times \frac{y^6}{\delta^5} - \frac{2}{4} \frac{y^4}{\delta^3} + \frac{4}{5} \frac{y^5}{\delta^4} - \frac{4}{7} \frac{y^7}{\delta^6} + \frac{2}{8} \frac{y^8}{\delta^7} + \frac{1}{5} \frac{y^5}{\delta^4} - \frac{2}{6} \frac{y^6}{\delta^5} + \frac{2}{8} \frac{y^8}{\delta^7} - \frac{1}{9} \frac{y^9}{\delta^8} \right]_0^\delta \\ &= \frac{d}{dx} \left[\delta - \frac{4}{3} \delta + \frac{4}{5} \delta - \frac{1}{3} \delta - \frac{1}{2} \delta + \frac{4}{5} \delta - \frac{4}{7} \delta + \frac{1}{4} \delta + \frac{1}{5} \delta - \frac{1}{3} \delta + \frac{1}{4} \delta - \frac{1}{9} \delta \right] \\ &= \frac{d\delta}{dx} \left[1 - \frac{4}{3} + \frac{4}{5} - \frac{1}{3} - \frac{1}{2} + \frac{4}{5} - \frac{4}{7} + \frac{1}{4} + \frac{1}{5} - \frac{1}{3} + \frac{1}{4} - \frac{1}{9} \right] \\ &= \frac{d\delta}{dx} \left[1 - \left(\frac{4}{3} + \frac{1}{3} + \frac{1}{3} \right) + \left(\frac{4}{5} + \frac{4}{5} + \frac{1}{5} \right) + \left(\frac{1}{4} + \frac{1}{4} - \frac{1}{2} \right) - \left(\frac{4}{7} + \frac{1}{9} \right) \right] \\ &= \frac{d\delta}{dx} \left(1 - 2 + \frac{9}{5} + 0 - \frac{43}{63} \right) = \frac{d\delta}{dx} \left(-1 + \frac{9}{5} - \frac{43}{63} \right) \\ &= \frac{d\delta}{dx} \left(\frac{-315 + 567 - 215}{315} \right) = \frac{37}{315} \frac{d\delta}{dx} \\ \therefore \tau_0 &= \frac{37}{315} \rho U^2 \frac{d\delta}{dx} \quad \dots(13.25) \end{aligned}$$

$$\text{Also, } \tau_0 = \mu \left(\frac{du}{dy} \right)_{y=0}$$

$$\text{But, } u = U \left[\frac{2y}{\delta} - \frac{2y^3}{\delta^3} + \frac{y^4}{\delta^4} \right]$$

$$\text{and, } \frac{du}{dy} = U \left(\frac{2}{\delta} - \frac{6y^2}{\delta^3} + \frac{4y^3}{\delta^4} \right)$$

$$\therefore \tau_0 = \mu \left(\frac{du}{dy} \right)_{y=0} = \mu U \left(\frac{2}{\delta} \right) = \frac{2\mu U}{\delta} \quad \dots(13.26)$$

Equating the two values of τ_0 given by eqns. (13.25) and (13.26), we get:

$$\frac{37}{315} \rho U^2 \frac{d\delta}{dx} = \frac{2\mu U}{\delta}$$

$$\text{or,} \quad \delta \cdot d\delta = \frac{315}{37} \times \frac{2\mu U}{\rho U^2} dx = \frac{630}{37} \frac{\mu}{\rho U} dx$$

Integrating both sides, we get:

$$\frac{\delta^2}{2} = \frac{630}{37} \frac{\mu}{\rho U} x + C$$

(where, C = constant of integration)

$$\text{At,} \quad x = 0, \delta = 0 \therefore C = 0$$

$$\therefore \frac{\delta^2}{2} = \frac{630}{37} \frac{\mu}{\rho U} x$$

$$\therefore \delta = \sqrt{\frac{630 \times 2}{37} \frac{\mu}{\rho U} x} = 5.84 \sqrt{\frac{\mu x}{\rho U}} = 5.84 \sqrt{\frac{\mu x}{\rho U} \times \frac{x}{x}}$$

$$\text{or,} \quad \delta = 5.84 \sqrt{\frac{\mu}{\rho U x}} \times x = \frac{5.84 x}{\sqrt{Re_x}} \quad \dots(13.27)$$

(ii) **Shear stress, τ_0 :**

$$\tau_0 = \frac{2\mu U}{\delta}$$

$$\text{But,} \quad = \frac{5.84 x}{\sqrt{Re_x}} \quad \dots[\text{Eqn. (13.27)}]$$

$$\therefore \tau_0 = \frac{2\mu U}{\frac{5.84 x}{\sqrt{Re_x}}} = \frac{2\mu U}{5.84 x} \sqrt{Re_x} = 0.343 \frac{\mu U}{x} \sqrt{Re_x} \quad \dots(13.28)$$

(iii) **Local co-efficient of drag, C_D^* :**

$$\tau_0 = 0.343 \frac{\mu U}{x} \sqrt{Re_x}$$

$$\text{Also,} \quad \tau_0 = C_D^* \frac{\rho U^2}{2}$$

$$\therefore C_D^* \frac{\rho U^2}{2} = 0.343 \frac{\mu U}{x} \sqrt{Re_x}$$

$$\text{or,} \quad C_D^* = \frac{0.686}{\sqrt{Re_x}} \quad \dots(13.29)$$

(iv) **Co-efficient of drag (C_D):**

$$C_D = \frac{F_D}{\frac{1}{2} \rho A U^2} \quad [\text{Eqn. (13.10)}]$$

where,
$$F_D = \int_0^L \tau_0 \times B \times dx \quad [\text{Eqn. (13.8)}]$$

$$= \int_0^L 0.343 \frac{\mu U}{x} \sqrt{Re_x} \times B \times dx = 0.343 \int_0^L \frac{\mu U}{x} \sqrt{\frac{\rho U x}{\mu}} \times B \times dx$$

$$= 0.343 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \int_0^L x^{-1/2} dx = 0.343 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \left[\frac{x^{1/2}}{1/2} \right]_0^L$$

$$= 0.343 \times 2 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \sqrt{L}$$

or,
$$F_D = 0.686 \mu U B \sqrt{\frac{\rho U L}{\mu}} \quad \dots(13.30)$$

$$\therefore C_D = \frac{0.686 \mu U B \sqrt{\frac{\rho U L}{\mu}}}{\frac{1}{2} \rho A U^2} = \frac{0.686 \mu U B \sqrt{\frac{\rho U L}{\mu}}}{\frac{1}{2} \rho \times L \times B \times U^2}$$

(where, $A = L \times B$)

$$= 0.686 \times 2 \times \frac{\mu}{\rho U L} \times \sqrt{\frac{\rho U L}{\mu}} = 1.372 \frac{1}{\sqrt{\frac{\rho U L}{\mu}}}$$

or,
$$C_D = \frac{1.372}{\sqrt{Re_L}} \quad \dots(13.31)$$

Case 4. Velocity distribution : $\frac{u}{U} = \sin \left(\frac{\pi}{2} \frac{y}{\delta} \right)$

(i) **Boundary layer thickness, δ :**

$$\frac{\tau_0}{\rho U^2} = \frac{d}{dx} \left[\int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \right] \quad [\text{Eqn. (13.6)}]$$

Substituting the value of $\frac{u}{U}$, we get:

$$\begin{aligned} \frac{\tau_0}{\rho U^2} &= \frac{d}{dx} \left[\int_0^{\delta} \sin \left(\frac{\pi}{2} \frac{y}{\delta} \right) \left\{ 1 - \sin \left(\frac{\pi}{2} \frac{y}{\delta} \right) \right\} dy \right] \\ &= \frac{d}{dx} \left[\int_0^{\delta} \left\{ \sin \left(\frac{\pi}{2} \frac{y}{\delta} \right) - \sin^2 \left(\frac{\pi}{2} \frac{y}{\delta} \right) \right\} dy \right] \\ &= \frac{d}{dx} \left[\int_0^{\delta} \left\{ \sin \left(\frac{\pi}{2} \frac{y}{\delta} \right) - \frac{1 - \cos \left(\frac{\pi y}{\delta} \right)}{2} \right\} dy \right] \\ &\quad \left(\because \sin^2 \theta = \frac{1 - \cos 2\theta}{2} \right) \end{aligned}$$

$$\begin{aligned}
&= \frac{d}{dx} \left[\frac{-\cos\left(\frac{\pi y}{2\delta}\right)}{\frac{\pi}{2\delta}} - \frac{y}{2} + \frac{\sin\left(\frac{\pi y}{\delta}\right)}{\frac{\pi}{\delta}} \right]_0^\delta \\
&= \frac{d}{dx} \left[\left\{ \frac{-\cos\left(\frac{\pi\delta}{2\delta}\right)}{\frac{\pi}{2\delta}} + \frac{\cos\left(\frac{\pi}{2\delta} \times 0\right)}{\frac{\pi}{2\delta}} \right\} - \frac{\delta}{2} + \left\{ \frac{\sin\left(\frac{\pi}{\delta} \times \delta\right)}{\frac{\pi}{\delta}} - \frac{\sin\left(\frac{\pi}{\delta} \times 0\right)}{\frac{\pi}{\delta}} \right\} \right] \\
&= \frac{d}{dx} \left[\left(0 + \frac{1}{\frac{\pi}{2\delta}} \right) - \frac{\delta}{2} + 0 \right] = \frac{d}{dx} \left(\frac{2\delta}{\pi} - \frac{\delta}{2} \right) = \left(\frac{4-\pi}{2\pi} \right) \frac{d\delta}{dx}
\end{aligned}$$

$$\therefore \tau_0 = \left(\frac{4-\pi}{2\pi} \right) \rho U^2 \frac{d\delta}{dx} \quad \dots(13.32)$$

$$\text{Also, } \tau_0 = \mu \left(\frac{du}{dy} \right)_{y=0}$$

$$\text{But, } u = U \left[\sin \left(\frac{\pi y}{2\delta} \right) \right]$$

$$\text{and, } \frac{du}{dy} = U \left[\cos \left(\frac{\pi y}{2\delta} \right) \right] \times \frac{\pi}{2\delta}$$

$$\therefore \tau_0 = \mu \left(\frac{du}{dy} \right)_{y=0} = \mu U \left[\cos \left(\frac{\pi}{2} \times \frac{0}{\delta} \right) \right] \times \frac{\pi}{2\delta} = \frac{\mu U \pi}{2\delta} \quad \dots(13.33)$$

Equating the two values of τ_0 given by eqns. (13.32) and (13.33), we get:

$$\left(\frac{4-\pi}{2\pi} \right) \rho U^2 \frac{d\delta}{dx} = \frac{\mu U \pi}{2\delta}$$

$$\text{or, } \delta \cdot d\delta = \frac{\mu U \pi}{2} \times \left(\frac{2\pi}{4-\pi} \right) \times \frac{1}{\rho U^2} \times dx = \frac{\pi^2}{(4-\pi)} \cdot \frac{\mu U}{\rho U^2} \cdot dx$$

$$\text{or, } \delta \cdot d\delta = 11.4975 \frac{\mu}{\rho U} dx$$

Integrating both sides, we get:

$$\frac{\delta^2}{2} = 11.4975 \frac{\mu}{\rho U} x + C$$

(where, C = constant of integration)

$$\text{At, } x = 0, \delta = 0 \quad \therefore C = 0$$

$$\therefore \frac{\delta^2}{2} = 11.4975 \frac{\mu}{\rho U} x$$

$$\begin{aligned}
 \therefore \delta &= \sqrt{2 \times 11.4975 \times \frac{\mu}{\rho U}} x = 4.795 \sqrt{\frac{\mu}{\rho U}} x = 4.795 \sqrt{\frac{\mu}{\rho U x}} \times x \\
 &= \frac{4.795 x}{\sqrt{Re_x}} \quad \dots(13.34)
 \end{aligned}$$

(ii) Shear stress, τ_0 :

$$\tau_0 = \frac{\mu U \pi}{2\delta}$$

But,
$$\delta = \frac{4.795x}{\sqrt{Re_x}} \quad [\text{Eqn. (13.31)}]$$

$$\therefore \tau_0 = \frac{\mu U \pi}{2 \times \frac{4.795x}{\sqrt{Re_x}}} = \frac{\mu U \pi \sqrt{Re_x}}{2 \times 4.795x} = 0.327 \frac{\mu U}{x} \sqrt{Re_x} \quad \dots(13.35)$$

(iii) Local co-efficient of drag, C_D :

$$\tau_0 = 0.327 \frac{\mu U}{x} \sqrt{Re_x}$$

Also,
$$\tau_0 = C_D^* \cdot \frac{\rho U^2}{2}$$

$$\therefore C_D^* \frac{\rho U^2}{2} = 0.327 \frac{\mu U}{x} \sqrt{Re_x}$$

or,
$$C_D^* = \frac{0.654}{\sqrt{Re_x}} \quad \dots(13.36)$$

(iv) Co-efficient of drag, C_D :

$$C_D = \frac{F_D}{\frac{1}{2} \rho A U^2} \quad [\text{Eqn. (13.10)}]$$

where,
$$F_D = \int_0^L \tau_0 \times B \times dx \quad [\text{Eqn. (13.8)}]$$

$$\begin{aligned}
 &= \int_0^L 0.327 \frac{\mu U}{x} \sqrt{Re_x} \times B \times dx = 0.327 \int_0^L \frac{\mu U}{x} \sqrt{\frac{\rho U x}{\mu}} \times B \times dx \\
 &= 0.327 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \int_0^L x^{-1/2} dx = 0.327 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \left[\frac{x^{1/2}}{1/2} \right]_0^L \\
 &= 0.327 \times 2 \mu U \sqrt{\frac{\rho U}{\mu}} \times B \times \sqrt{L}
 \end{aligned}$$

or,
$$F_D = 0.654 \mu U B \sqrt{\frac{\rho U L}{\mu}} \quad \dots(13.37)$$

$$\therefore C_D = \frac{0.654 \mu UB \sqrt{\frac{\rho UL}{\mu}}}{\frac{1}{2} \rho AU^2} = \frac{0.654 \mu UB \sqrt{\frac{\rho UL}{\mu}}}{\frac{1}{2} \rho \times L \times B \times U^2} = 1.31 \frac{\mu}{\rho LU} \sqrt{\frac{\rho UL}{\mu}}$$

(where, $A = L \times B$)

or, $C_D = 1.31 \times \frac{1}{\sqrt{\frac{\rho UL}{\mu}}} = \frac{1.31}{\sqrt{Re_L}} \quad \dots(13.38)$

Table 13.1 shows the values of δ (boundary layer thickness), C_D^* (local co-efficient of drag), C_D (average co-efficient of drag) in terms of Reynolds number (Re) for various velocity profiles/distributions.

Table 13.1. Values of δ , C_D^* and C_D in terms of Re

S. No.		δ	C_D^*	C_D
1.	$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$	$\frac{5.48x}{\sqrt{Re_x}}$	$\frac{0.73}{\sqrt{Re_x}}$	$\frac{1.46}{\sqrt{Re_L}}$
2.	$\frac{u}{U} = \frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3$	$\frac{4.64x}{\sqrt{Re_x}}$	$\frac{0.646}{\sqrt{Re_x}}$	$\frac{1.292}{\sqrt{Re_L}}$
3.	$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - 2 \left(\frac{y}{\delta} \right)^3 + \left(\frac{y}{\delta} \right)^4$	$\frac{5.84x}{\sqrt{Re_e}}$	$\frac{0.686}{\sqrt{Re_x}}$	$\frac{1.372}{\sqrt{Re_L}}$
4.	$\frac{u}{U} = \sin \left(\frac{\pi y}{2 \delta} \right)$	$\frac{4.795 x}{\sqrt{Re_x}}$	$\frac{0.654}{\sqrt{Re_x}}$	$\frac{1.31 x}{\sqrt{Re_L}}$
5.	Blasius results ($Re < 3.2 \times 10^5$)	$\frac{5x}{\sqrt{Re_x}}$	$\frac{0.664}{\sqrt{Re_x}}$	$\frac{1.328}{\sqrt{Re_L}}$

Example 13.10. The boundary layer thickness at a distance of 1 m from the leading edge of a flat plate kept over zero angle of incidence to the flow direction is 1 mm. The velocity outside the boundary layer is 25 m/s. The boundary layer thickness at a distance of 4 m is (i) 4 mm, (ii) 2 mm, (iii) 1 mm

Select the correct answer. Assume that the boundary layer is entirely laminar

[UPTU]

Solution. Free stream velocity, $U = 25$ m/s

The boundary layer thickness at $x_1 = 1$ m, $\delta_1 = 1$ mm

The boundary layer thickness at a distance of 4 m, δ_2 :

Thickness of boundary layer is given by,

$$\delta = \frac{5x}{\sqrt{Re_x}} = \frac{5x}{\sqrt{\frac{Ux}{\nu}}} = 5 \sqrt{\frac{\nu x}{U}}$$

$$\therefore \delta_1 = 5 \sqrt{\frac{\nu x_1}{U}}$$

$$\begin{aligned} \text{and,} \quad \delta_2 &= 5 \sqrt{\frac{\nu x_2}{U}} \\ \text{or,} \quad \frac{\delta_1}{\delta_2} &= \sqrt{\frac{x_1}{x_2}}, \\ \text{or,} \quad \frac{1}{\delta_2} &= \sqrt{1/4} \text{ or } \delta_2 = \mathbf{2 \text{ mm (Ans.)}} \end{aligned}$$

Example 13.11. A smooth plate 2 m wide and 2.5 m long is towed in oil (sp. gr. = 0.8) at a velocity of 1.5 m/s along its length. Find the thickness of boundary layer and shear stress at the trailing edge of the plate. $\nu_{oil} = 10^{-4} \text{ m}^2/\text{s}$.

Solution. Given : $B = 2 \text{ m}$; $L = 2.5 \text{ m}$; Sp. gravity = 0.8; $U = 1.5 \text{ m/s}$, $\nu_{oil} = 10^{-4} \text{ m}^2/\text{s}$.

Thickness of boundary layer, δ_L :

$$Re_L = \frac{UL}{\nu} = \frac{1.5 \times 2.5}{10^{-4}} = 37500$$

Since Re_L is less than 5×10^5 , so the boundary layer at the trailing edge is *laminar*, and Blasius equation gives,

$$\delta_x = \frac{5x}{Re_x}$$

At the trailing edge $x = L$,

$$\therefore \delta_L = \frac{5 \times 2.5}{\sqrt{37500}} = 0.0645 \text{ m} = \mathbf{64.55 \text{ mm (Ans.)}}$$

Shear stress at the trailing edge, τ_L :

According to Blasius average co-efficient of drag (C_D^*) is given by:

$$C_D^* = \frac{0.664}{\sqrt{Re_x}} = \frac{0.664}{\sqrt{Re_L}} = \frac{0.664}{\sqrt{37500}}$$

$$\begin{aligned} \therefore \tau_L &= C_D^* \times \frac{1}{2} \rho U^2 \\ &= \frac{0.664}{\sqrt{37500}} \times \frac{1}{2} \times (0.8 \times 1000) \times 1.5^2 = \mathbf{3.086 \text{ N/m}^2 \text{ (Ans.)}} \end{aligned}$$

Example 13.12. For the velocity profile in laminar boundary layer as,

$$\frac{u}{U} = \frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3$$

find the thickness of the boundary layer and the shear stress 1.5 m from the leading edge of a plate. The plate is 2 m long and 1.4 m wide and is placed in water which is moving with a velocity of 200 mm per second. Find the total drag force on the plate if μ for water = .01 poise.

(Delhi University)

Solution. Given : $x = 1.5 \text{ m}$; $L = 2 \text{ m}$; $B = 1.4 \text{ m}$; $U = 200 \text{ mm/s} = 0.2 \text{ m/s}$;

$$\mu = 0.01 \text{ poise} = \frac{0.01}{10} = 0.001 \text{ Ns/m}^2.$$

Velocity profile:
$$\frac{u}{U} = \frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3$$

For the given profile, $\delta = \frac{4.64x}{\sqrt{Re_x}}$...[Eqn. (13.20)]

$$\left[\text{Here, } Re_x = \frac{\rho Ux}{\mu} = \frac{1000 \times 0.2 \times 1.5}{0.001} = 3 \times 10^5 \right]$$

$$\delta = \frac{4.64 \times 1.5}{\sqrt{3 \times 10^5}} = 0.0127 \text{ m} = \mathbf{12.7 \text{ mm (Ans.)}}$$

Shear stress (τ_0) is given by,

$$\tau_0 = 0.323 \frac{\mu U}{x} \sqrt{Re_x} \quad \dots[\text{Eqn. (3.21)}]$$

or

$$\tau_0 = 0.323 \times \frac{0.001 \times 0.2}{1.5} \times \sqrt{3 \times 10^5} = \mathbf{0.0236 \text{ N/m}^2 \text{ (Ans.)}}$$

Drag force (F_D) on one side of the plate is given as:

$$\begin{aligned} F_D &= 0.646 \mu U \sqrt{\frac{\rho U L}{\mu}} \times B \quad \dots[\text{Eqn. (13.23)}] \\ &= 0.646 \times 0.001 \times 0.2 \sqrt{\frac{1000 \times 0.2 \times 2}{0.001}} \times 1.4 = 0.114 \text{ N} \end{aligned}$$

$\therefore$ **Total Drag force** = Drag force on both sides of the plate = $2 \times 0.114 = \mathbf{0.228 \text{ N (Ans.)}}$

Example 13.13. A plate 450 mm $\times$ 150 mm has been placed longitudinally in a stream of crude oil (specific gravity 0.925 and kinematic viscosity of 0.9 stoke) which flows with velocity of 6 m/s. Calculate:

- (i) The friction drag on the plate,
- (ii) Thickness of the boundary layer at the trailing edge, and
- (iii) Shear stress at the trailing edge.

[PTU]

Solution. Length of the plate, $L = 450 \text{ mm} = 0.45 \text{ m}$

Width of the plate, $B = 150 \text{ mm} = 0.15 \text{ m}$

Specific gravity of oil, $S = 0.925$

Kinematic viscosity of oil, $\nu = 0.9 \text{ stoke} = 0.9 \times 10^{-4} \text{ m}^2/\text{s}$

Velocity of oil, $U = 6 \text{ m/s}$

(i) **The friction drag on the plate, F_D :**

Reynolds number at the end of plate,

$$Re_L = \frac{UL}{\nu} = \frac{6 \times 0.45}{0.9 \times 10^{-4}} = 30000$$

Since $Re_L < 5 \times 10^5$, the flow over the plate is *entirely laminar*.

$\therefore$ Average co-efficient of drag,

$$C_D = \frac{1.328}{\sqrt{Re_L}} = \frac{1.328}{\sqrt{30000}} = 0.007667$$

Drag on the *one side* of the plate,

$$F_D = C_D \times \frac{1}{2} \rho A U^2 = 0.007667 \times \frac{1}{2} \times (0.925 \times 1000) \times (0.45 \times 0.15) \times 6^2$$

(where, A = area of the plate)

$$= 8.62 \text{ N (Ans.)}$$

(ii) **The thickness of boundary layer at the trailing edge, δ :**

The thickness of boundary in the laminar range is given by,

$$\delta = \frac{5x}{\sqrt{Re_x}}$$

$\therefore$ Thickness at the trailing edge ($x = 0.45 \text{ m}$),

$$\delta = \frac{5 \times 0.45}{\sqrt{30000}} = 0.013 \text{ m} = \mathbf{13 \text{ mm (Ans.)}}$$

(iii) **Shear stress at the trailing edge, τ_0 :**

$$\text{Local co-efficient of drag, } C_D^* = \frac{0.664}{\sqrt{Re_x}} = \frac{0.664}{\sqrt{30000}} = 0.00383$$

$$\text{By definition, } C_D^* = \frac{\tau_0}{\frac{1}{2} \rho U^2}$$

$$\begin{aligned} \therefore \tau_0 &= C_D^* \times \frac{1}{2} \rho U^2 = 0.00383 \times \frac{1}{2} \times (0.925 \times 1000) \times 6^2 \\ &= \mathbf{63.77 \text{ N/m}^2 \text{ (Ans.)}} \end{aligned}$$

Example 13.14. *The velocity profile for laminar boundary is in the form given below:*

$$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$$

Find the thickness of boundary layer at the end of the plate and the drag force on one side of a plate 1.5 m long and 1 m wide when placed in water flowing with a velocity of 0.12 m/s. Calculate the value of co-efficient of drag also.

Take μ for water = 0.001 Ns/m²

$$\text{Solution. Velocity distribution } \frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2 \quad \dots(i) \text{ (Given)}$$

The length of the plate, $L = 1.5 \text{ m}$

The width of the plate, $B = 1 \text{ m}$

Free stream velocity, $U = 0.12 \text{ m/s}$.

μ for water = 0.001 Ns/m²

Thickness of the boundary layer, δ :

Reynolds number at the end of the plate (*i.e.* at a distance of 1.5 m from the leading edge) is given by,

$$Re_L = \frac{\rho UL}{\mu} = \frac{1000 \times 0.12 \times 1.5}{0.001} = 180000$$

Since $Re_L < 5 \times 10^5$, therefore, this is the case of *laminar* boundary layer. Thickness of boundary layer at a distance of 1.5 m is given by:

$$\begin{aligned} \delta &= \frac{5.48x}{\sqrt{Re_x}} \quad \dots[\text{Eqn. (13.13)}] \\ &= \frac{5.48 \times 1.5}{\sqrt{180000}} = 0.01937 \text{ m} \quad \text{or} \quad \mathbf{19.37 \text{ mm (Ans.)}} \end{aligned}$$

Drag force on one side of the plate, F_D :

$$F_D = 0.73 \mu U B \sqrt{\frac{\rho U L}{\mu}} \quad \dots[\text{Eqn. (13.16)}]$$

$$= 0.73 \times 0.001 \times 0.12 \times 1 \times \sqrt{\frac{1000 \times 0.12 \times 1.5}{0.001}} = \mathbf{0.0372 \text{ N (Ans.)}}$$

Co-efficient of drag, C_D :

$$= \frac{1.46}{\sqrt{Re_L}} = \frac{1.46}{\sqrt{180000}} = \mathbf{0.00344 \text{ (Ans.)}} \quad \dots[\text{Eqn. (13.17)}]$$

Example 13.15. Air is flowing over a smooth flat plate with a velocity of 12 m/s. The velocity profile is in the form:

$$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$$

The length of the plate is 1.1 m and width 0.9 m. If laminar boundary layer exists upto a value of $Re = 2 \times 10^5$ and kinematic viscosity of air is 0.15 stoke, find:

- (i) The maximum distance from the leading edge upto which laminar boundary layer exists, and
- (ii) The maximum thickness of boundary layer.

Solution. Velocity distribution: $\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$

Velocity of air, $U = 12 \text{ m/s}$

Length of plate, $L = 1.1 \text{ m}$

Width of plate, $B = 0.9 \text{ m}$

Reynolds number upto which laminar boundary exists, $Re = 2 \times 10^5$

Kinematic viscosity of air, $\nu = 0.15 \text{ stokes} = 0.15 \times 10^{-4} \text{ m}^2/\text{s}$

- (i) The maximum distance from the leading edge upto which laminar boundary layer exists, x :

$$Re_x = \frac{Ux}{\nu} \quad \text{or} \quad 2 \times 10^5 = \frac{12 \times x}{0.15 \times 10^{-4}}$$

or,
$$x = \frac{2 \times 10^5 \times 0.15 \times 10^{-4}}{12} = \mathbf{0.25 \text{ m (Ans.)}}$$

- (ii) The maximum thickness of boundary layer, δ

For the given velocity profile, the maximum thickness of boundary layer is given by:

$$\delta = \frac{5.48x}{\sqrt{Re_x}} \quad \dots[\text{Eqn. (13.13)}]$$

$$= \frac{5.48 \times 0.25}{\sqrt{2 \times 10^5}} = 0.00306 \text{ m} \quad \text{or} \quad \mathbf{3.06 \text{ mm (Ans.)}}$$

Example 13.16. Atmospheric air at 20°C is flowing parallel to a flat plate at a velocity of 2.8 m/s. Assuming cubic velocity profile and using exact Blasius solution estimate the boundary layer thickness and the local co-efficient of drag (or skin friction) at $x = 1.2 \text{ m}$ from the leading edge of the plate. Also find the deviation of the approximate solution from the exact solution.

Take the kinematic viscosity of air at $20^\circ\text{C} = 15.4 \times 10^{-6} \text{ m}^2/\text{s}$.

Solution. Velocity of air, $U = 2.8 \text{ m/s}$

Distance from the leading edge of the plate, $x = 1.2 \text{ m}$

$$\text{Reynolds number, } R_{e_x} = \frac{Ux}{\nu} = \frac{2.8 \times 1.2}{15.4 \times 10^{-6}} = 2.18 \times 10^5$$

Blasius solution:

$$\text{Boundary layer thickness, } \delta = \frac{5x}{\sqrt{R_{ex}}} = \frac{5 \times 1.2}{\sqrt{2.18 \times 10^5}} = 0.01285 \text{ m} = \mathbf{12.85 \text{ mm (Ans.)}}$$

$$\text{Local co-efficient of drag, } C_D^* = \frac{0.664}{\sqrt{R_{ex}}} = \frac{0.664}{\sqrt{2.18 \times 10^5}} = 0.001422$$

Approximate solution (with assumption of cubic velocity profile):

$$\delta = \frac{4.64x}{\sqrt{R_{ex}}} = \frac{4.64 \times 1.2}{\sqrt{2.18 \times 10^5}} = 0.0119 \text{ m} = 11.92 \text{ mm}$$

$$C_D^* = \frac{0.646}{\sqrt{R_{ex}}} = \frac{0.646}{\sqrt{2.18 \times 10^5}} = 0.001383$$

The approximate solution deviates from the exact solution by:

$$\text{Deviation for } \delta: \frac{12.85 - 11.92}{12.85} \times 100 = \mathbf{7.24\% \text{ (Ans.)}}$$

$$\text{Deviation for } C_D^* = \frac{0.001422 - 0.001383}{0.001422} \times 100 = \mathbf{2.74\% \text{ (Ans.)}}$$

Example 13.17. Air is flowing over a flat plate 5 m long and 2.5 m wide with a velocity of 4 m/s at 15°C. If $\rho = 1.208 \text{ kg/m}^3$ and $\nu = 1.47 \times 10^{-5} \text{ m}^2/\text{s}$, calculate:

- (i) Length of plate over which the boundary layer is laminar, and thickness of the boundary layer (laminar),
- (ii) Shear stress at the location where boundary layer ceases to be laminar, and
- (iii) Total drag force on both sides on that portion of plate where boundary layer is laminar.

Solution. Length of the plate, $L = 5 \text{ m}$

Width of the plate, $B = 2.5 \text{ m}$

Velocity of air, $U = 4 \text{ m/s}$

Density of air, $\rho = 1.208 \text{ kg/m}^3$

Kinematic viscosity of air, $\nu = 1.47 \times 10^{-5} \text{ m}^2/\text{s}$

(i) Length of plate over which the boundary layer is laminar:

$$\text{Reynolds number, } R_{e_L} = \frac{UL}{\nu} = \frac{4 \times 5}{1.47 \times 10^{-5}} = 1.361 \times 10^6$$

Hence on the front portion, boundary layer is laminar and on the rear, it is turbulent.

$$R_{e_x} = \frac{Ux}{\nu} = 5 \times 10^5$$

$$\therefore \frac{4 \times x}{1.47 \times 10^{-5}} = 5 \times 10^5$$

or
$$x = \frac{5 \times 10^5 \times 1.47 \times 10^{-5}}{4} = 1.837 \text{ m}$$

Hence the boundary layer is **laminar on 1.837 m length of the plate.** (Ans.)

Thickness of the boundary layer (laminar), δ

$$\delta = \frac{5x}{\sqrt{Re_x}} = \frac{5 \times 1.837}{\sqrt{5 \times 10^5}} = 0.01299 \text{ m or } \mathbf{12.99 \text{ mm}} \text{ (Ans.)}$$

(ii) Shear stress at the location where boundary layer ceases to be laminar, τ_0 :

Local co-efficient of drag, $C_d^* = \frac{0.664}{\sqrt{5 \times 10^5}} = 0.000939$

$$\therefore = C_D^* \times \frac{1}{2} \rho U^2 = 0.000939 \times \frac{1}{2} \times 1.208 \times 4^2 = \mathbf{0.00907 \text{ N/m}^2} \text{ (Ans.)}$$

(iii) Total drag force on both sides of plate, F_D :

$$F_D = 2 \times C_D \times \frac{1}{2} \times \rho A U^2$$

where, $C_D = \text{Average co-efficient of drag} = \frac{1.328}{\sqrt{5 \times 10^5}} = 1.878 \times 10^{-3}$

and, $A = \text{Area of the plate} = 1.837 \times 2.5 = 4.59 \text{ m}^2$

$$\therefore F_D = 2 \times 1.878 \times 10^{-3} \times \frac{1}{2} \times 1.208 \times 4.59 \times 4^2 = \mathbf{0.167 \text{ N}} \text{ (Ans.)}$$

Example 13.18. Air flows over a plate 0.5 m long and 0.6 m wide with a velocity of 4 m/s. The velocity profile is in the form.

$$\frac{u}{U} = \sin \left(\frac{\pi y}{2 \delta} \right)$$

If $\rho = 1.24 \text{ kg/m}^3$ and $\nu = 0.15 \times 10^{-4} \text{ m}^2/\text{s}$, calculate:

- (i) Boundary layer thickness at the end of the plate,
- (ii) Shear stress at 250 mm from the leading edge, and
- (iii) Drag force on one side of the plate.

[Delhi University]

Solution. Length of plate, $L = 0.5 \text{ m}$

Width of plate, $B = 0.6 \text{ m}$

Velocity of air, $U = 4 \text{ m/s}$

Density of air, $\rho = 1.24 \text{ kg/m}^3$

Kinematic viscosity of air, $\nu = 0.15 \times 10^{-4} \text{ m}^2/\text{s}$

Velocity profile :
$$\frac{u}{U} = \sin \left(\frac{\pi y}{2 \delta} \right)$$

(i) Boundary layer thickness at the end of the plate, δ :

$$\text{Reynolds number, } Re_x = \frac{Ux}{\nu} = \frac{4 \times 0.5}{0.15 \times 10^{-4}} = 1.33 \times 10^5$$

Since $Re_x < 5 \times 10^5$, therefore, the boundary layer is *laminar over the entire length of the plate.*

We know,
$$\delta = \frac{4.795x}{\sqrt{Re_x}} = \frac{4.795L}{\sqrt{Re_x}} = \frac{4.795 \times 0.5}{\sqrt{1.33 \times 10^5}}$$

$$= 0.00657 \text{ m} = \mathbf{6.57 \text{ mm (Ans.)}}$$

(ii) Shear stress at 250 mm from the leading edge, τ_0 :

$$\tau_0 = C_D^* \times \frac{\rho U^2}{2} \quad \dots[\text{Eqn. (13.9)}]$$

But,

$$C_D^* = \frac{0.654}{\sqrt{Re_x}} = \frac{0.654}{\sqrt{\frac{Ux}{\nu}}} = \frac{0.654}{\sqrt{\frac{4 \times 0.25}{0.15 \times 10^{-4}}}} = 0.002533$$

$$\therefore (\tau_0)_{x=0.25 \text{ m}} = 0.002533 \times \frac{1.24 \times 4^2}{2} = \mathbf{0.025 \text{ N/m}^2 \text{ (Ans.)}}$$

(iii) Drag force on one side of the plate, F_D :

$$F_D = C_D \times \frac{1}{2} \rho A U^2$$

where,

$$C_D = \frac{1.31}{\sqrt{Re_L}} = \frac{1.31}{\sqrt{1.33 \times 10^5}} = 0.003592$$

and, $A = \text{area of the plate} = L \times B = 0.5 \times 0.6 = 0.3 \text{ m}^2$

$$\therefore F_D = 0.003592 \times \frac{1}{2} \times 1.24 \times 0.3 \times 4^2 = \mathbf{0.01069 \text{ N (Ans.)}}$$

Example 13.19. Find the ratio of friction drag on the front half and rear half of the flat plate kept at zero incidence in a stream of uniform velocity, if the boundary layer is laminar over the whole plate.

Solution. Let, L = Length of the plate,
 U = Velocity of the fluid, and
 ν = Kinematic viscosity of the fluid.

Ratio of friction drag on the front half and rear half, $\frac{F_{D_1}}{F_{D_2}}$:

$$\text{Reynolds number of whole plate} = \frac{UL}{\nu}$$

$$\text{Reynolds number for the front half} = \frac{UL}{2\nu}$$

$$\text{Average co-efficient of drag for total plate, } C_D = \frac{1.328}{\sqrt{\frac{UL}{\nu}}}$$

$$\therefore C_{D_1} \text{ for front half} = \frac{1.328}{\sqrt{\frac{UL}{2\nu}}} = \frac{1.328 \times \sqrt{2}}{\sqrt{\frac{UL}{\nu}}} = \frac{1.878}{\sqrt{\frac{UL}{\nu}}}$$

$$\begin{aligned} \text{Friction drag on total plate, } F_D &= C_D \times \frac{1}{2} \rho A U^2 = C_D \times \frac{1}{2} \rho \times L B U^2 \\ &= \frac{1.328}{\sqrt{\frac{UL}{\nu}}} \times \frac{1}{2} \rho L B U^2 = \frac{0.664}{\sqrt{\frac{UL}{\nu}}} \times \rho L B U^2 \end{aligned}$$

$$\begin{aligned}\text{Friction drag on front half, } F_{D_1} &= C_{D_1} \times \frac{1}{2} \rho A U^2 = C_{D_1} \times \frac{1}{2} \rho \times \frac{L}{2} B U^2 \\ &= \frac{1.878}{\sqrt{\frac{UL}{\nu}}} \times \frac{1}{4} \times L B U^2 = \frac{0.4695}{\sqrt{\frac{UL}{\nu}}} \times \rho L B U^2\end{aligned}$$

$$\begin{aligned}\therefore \text{Friction drag on rear half, } F_{D_2} &= F_D - F_{D_1} \\ &= \frac{0.664}{\sqrt{\frac{UL}{\nu}}} \times \rho L B U^2 - \frac{0.4695}{\sqrt{\frac{UL}{\nu}}} \times \rho L B U^2 \\ &= \frac{0.1945}{\sqrt{\frac{UL}{\nu}}} \times \rho L B U^2 \\ \therefore \text{Ratio, } \frac{F_{D_1}}{F_{D_2}} &= \frac{\frac{0.4695}{\sqrt{UL/\nu}} \times \rho L B U^2}{\frac{0.1945}{\sqrt{UL/\nu}} \times \rho L B U^2} = \mathbf{2.414 \text{ (Ans.)}}\end{aligned}$$

Example 13.20. Air at standard conditions is flowing over a flat plate which is 1 m long and 0.3 m wide. The flow is uniform at the leading edge of the plate. The velocity profile in the boundary layer is assumed to be $\frac{u}{U} = 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2$ as the free stream velocity is $U = 30$ m/s. Assume that the flow conditions are independent of Z . Using control volume $abcd$, shown by dashed line, calculate the mass flow rate across the surface ab . [Density of air may be taken as 1.23 kg/m^3 , refer to Fig. 13.5] (GATE)

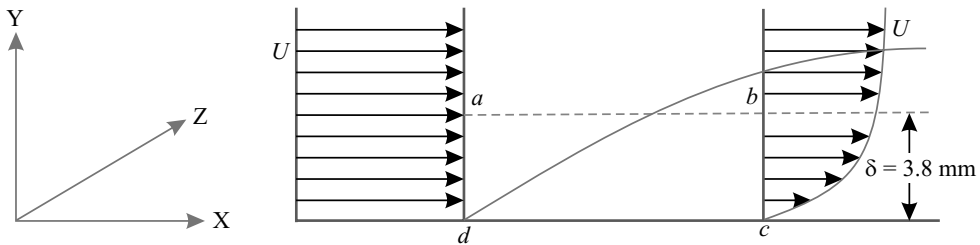


Fig. 13.5

Solution. By continuity equation for a volume $abcd$,

{Mass rate of flow across 'ad' – mass rate of flow across 'bc'} = mass rate of flow across 'ab'.

$$\begin{aligned}&= \rho U \times 3.8 \times 10^{-3} \times 0.3 - \rho U \times 0.3 \times \int_0^\delta \left\{ 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \right\} dy \\ &= \rho U \times 3.8 \times 10^{-3} \times 0.3 - \rho U \times 0.3 \left[\frac{2}{\delta} \cdot \frac{y^2}{2} - \frac{y^3}{3\delta^2} \right]_0^\delta \\ &= \rho U \times 3.8 \times 10^{-3} \times 0.3 - \rho U \times 0.3 \left[\delta - \frac{\delta}{3} \right]_0^\delta \\ &= \rho U \times 3.8 \times 10^{-3} \times 0.3 \left(1 - \frac{2}{3} \right) \\ &= 1.23 \times 30 \times 3.8 \times 10^{-3} \times 0.3 \times 0.3333 = \mathbf{0.014 \text{ kg/s (Ans.)}}\end{aligned}$$

Example 13.21. Air at 20°C and 1 bar flows over a flat plate at 1.5 m/s. The velocity profile for the laminar boundary layer is in the form

$$\frac{u}{U} = \frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3$$

If the kinematic viscosity of air at 20°C = $15.5 \times 10^{-6} \text{ m}^2/\text{s}$, calculate:

- (i) The boundary layer thickness at distances of 200 mm and 350 mm, and
(ii) The mass entrainment between the above two sections.

Solution. Temperature of air, $T = 20 + 273 = 293 \text{ K}$

Pressure of air, $p = 1 \text{ bar} = 1 \times 10^5 \text{ N/m}^2$

Velocity of air, $U = 1.5 \text{ m/s}$

Kinematic viscosity of air, $\nu = 15.5 \times 10^{-6} \text{ m}^2/\text{s}$

We know that, $pV = mRT$

or,
$$p = \frac{m}{V} RT = \rho RT$$

$$\therefore \rho = \frac{p}{RT} = \frac{1 \times 10^5}{287 \times 293} = 1.189 \text{ kg/m}^3$$

(The characteristic gas constant, $R = 287 \text{ J/kg K}$)

- (i) **The boundary layer thickness (at $x = 200 \text{ mm}$ and $x = 350 \text{ mm}$) δ_1, δ_2 :**

The Reynolds number, $Re_x = \frac{Ux}{\nu}$

$$\therefore Re_{x_1} = \frac{1.5 \times (200/1000)}{15.5 \times 10^{-6}} = 19355$$

(where, $x_1 = 200 \text{ mm}$)

$$Re_{x_2} = \frac{1.5 \times (350/1000)}{15.5 \times 10^{-6}} = 33871$$

(where, $x_2 = 350 \text{ mm}$)

For the given velocity profile, the boundary layer thickness is given by:

$$\delta = \frac{4.64x}{\sqrt{Re_x}}$$

$$\therefore \delta_1 = \frac{4.64 \times 200}{\sqrt{Re_{x_1}}} = \frac{4.64 \times 200}{\sqrt{19355}} = \mathbf{6.67 \text{ mm (Ans.)}}$$

$$\text{and, } \delta_2 = \frac{4.64 \times 300}{\sqrt{Re_{x_2}}} = \frac{4.64 \times 300}{\sqrt{33871}} = \mathbf{8.824 \text{ mm (Ans.)}}$$

- (ii) **The mass entrainment between the two sections:**

The mass flow in the boundary layer, at any position, is given by the integral

$$m_x = \int_0^{\delta} \rho u dy \quad \dots(i)$$

Also,
$$u = U \left[\frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3 \right]$$

(From the given velocity profile)

Substituting the value of u in eqn. (i) and integrating, we get:

$$\begin{aligned} m_x &= \int_0^{\delta} \rho \left[U \left\{ \frac{3y}{2\delta} - \frac{1}{2} \frac{y^3}{\delta^3} \right\} \right] dy = \rho U \left[\frac{3}{2} \times \frac{y^2}{2\delta} - \frac{1}{2} \times \frac{y^4}{4\delta^3} \right]_0^{\delta} \\ &= \rho U \left[\frac{3}{4} \times \frac{\delta^2}{\delta} - \frac{1}{8} \times \frac{\delta^4}{\delta^3} \right] = \rho U \left(\frac{3}{4} \delta - \frac{1}{8} \delta \right) = \frac{5}{8} \rho U \delta \end{aligned}$$

$\therefore$ The mass entrainment between the two sections

$$\begin{aligned} &= \frac{5}{8} \rho U (\delta_2 - \delta_1) = \frac{5}{8} \times 1.189 \times 1.5 (8.824 \times 10^{-3} - 6.67 \times 10^{-3}) \\ &= 0.0024 \text{ kg/s or } \mathbf{8.64 \text{ kg/h (Ans.)}} \end{aligned}$$

Example 13.22. (a) Water flows over a flat plate at a free stream velocity of 0.15 m/s. There is no pressure gradient and laminar boundary layer at a location is 6 mm thick. Assume a sinusoidal velocity profile given by

$$\frac{u}{U} = \sin \frac{\pi}{2} \left(\frac{y}{\delta} \right)$$

where, δ is the boundary layer thickness, U is the free stream velocity, and u is the velocity at a distance y from the wall. Calculate the local shear stress and the skin friction co-efficient on the plate if

$$\begin{aligned} \mu &= 1.02 \times 10^{-3} \text{ Ns/m}^2 \\ \rho &= 1000 \text{ kg/m}^3. \end{aligned}$$

(b) During flow over a flat plate, the laminar boundary layer undergoes a transition to turbulent boundary layer as the flow proceeds in the downstream. If the 1/7th power law turbulent velocity profile at a section is given by

$$\frac{\bar{u}}{U} = \left(\frac{y}{\delta} \right)^{1/7} = (\eta)^{1/7}$$

find out the momentum flux within the turbulent boundary layer. The width of the boundary layer is a . The boundary layer thickness is δ . (Delhi University)

Solution. (a) Given : $U = 0.15 \text{ m/s}$; $\delta = 6 \text{ mm} = 0.006 \text{ m}$; $\mu = 1.02 \times 10^{-3}$; $\rho = 1000 \text{ kg/m}^3$

Velocity profile:
$$\frac{u}{U} = \sin \left[\frac{\pi}{2} \left(\frac{y}{\delta} \right) \right]$$

Local shear stress, τ_0 :

From the given profile, we have:

$$u = U \sin \left[\frac{\pi}{2} \left(\frac{y}{\delta} \right) \right]$$

Differentiating the above equation w.r.t y , we have:

$$\frac{du}{dy} = U \cos \left[\frac{\pi}{2} \left(\frac{y}{\delta} \right) \right] \times \frac{\pi}{2\delta}$$

Local shear stress,
$$\tau_0 = \mu \left. \frac{du}{dy} \right|_{y=0}$$

or,
$$\tau_0 = \mu U \frac{\pi}{2\delta} = \frac{(1.02 \times 10^{-3}) \times 0.15 \times \pi}{2 \times 0.006} = \mathbf{0.04 \text{ N/m}^2 \text{ (Ans.)}}$$

Skin friction co-efficient, C_D^*

$$C_D^* = \frac{\tau_0}{\frac{1}{2} \rho U^2} \quad \dots [\text{Eqn. (13.9)}]$$

$$= \frac{0.04}{\frac{1}{2} \times 1000 \times (0.15)^2} = \mathbf{0.00356 \text{ (Ans.)}}$$

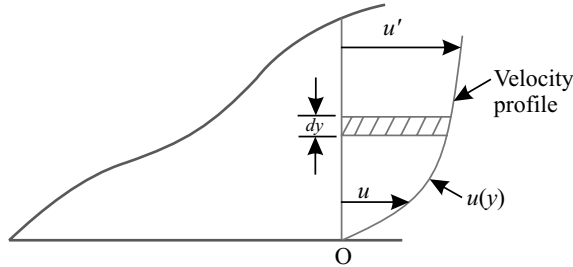


Fig. 13.6

(b) Given: Velocity profile: $\frac{u}{U} = \left(\frac{y}{\delta}\right)^{1/7}$; width of boundary layer = a ; thickness of boundary layer = δ .

Momentum flux within the turbulent boundary layer:

$$\begin{aligned} \text{The momentum flux} &= \int_0^\delta (\text{Mass flow through the strip } dy) \times u \\ &= \int_0^\delta (\rho a \, dy \cdot u) u = a\rho \int_0^\delta u^2 \cdot dy = a\rho \left[\int_0^\delta U \left(\frac{y}{\delta}\right)^{1/7} dy \right]^2 \\ &= a\rho U^2 \int_0^\delta \left(\frac{y}{\delta}\right)^{2/7} dy = a\rho \frac{U^2}{(\delta)^{2/7}} \left[\frac{(y)^{\frac{2}{7}+1}}{\left(\frac{2}{7}+1\right)} \right]_0^\delta \\ &= \frac{a\rho U^2}{(\delta)^{2/7}} \times \frac{(\delta)^{2/7+1}}{\left(\frac{9}{7}\right)} = \frac{7}{9} a\rho U^2 \delta \quad \mathbf{(Ans.)} \end{aligned}$$

Example 13.23. A plate 25 m long $\times$ 1.25 m wide is moving under water in the direction of its length. The drag force on the two sides of the plate is estimated to be 8500 N. Calculate:

- The velocity of the plate,
- The boundary layer thickness at the trailing edge of the plate, and
- The distance x_c at which the laminar boundary layer existing at the leading edge transforms into turbulent boundary layer. Take for water: $\rho = 1000 \text{ kg/m}^3$; $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$.

Solution. Length of the plate, $L = 25 \text{ m}$
Width of the plate, $B = 1.25 \text{ m}$

Drag force on the two sides of the plate, $F_D = 8500 \text{ N}$

Density of water, $\rho = 1000 \text{ kg/m}^3$

Kinematic viscosity of water, $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$

(i) **Velocity of the plate, U :**

Since U is not known, Re_L cannot be computed. Hence assume any reasonable value of C_D between 0.005 and 0.001. Let us assume $C_D = 0.002$.

$$\text{Drag force (both sides), } F_D = C_D \times \frac{1}{2} \rho A U^2 = C_D \times \frac{1}{2} \rho (2 \times L \times B) U^2$$

$$8500 = C_D \times \frac{1}{2} \times 1000 \times (2 \times 25 \times 1.25) U^2$$

$$\text{or, } U^2 = \frac{0.272}{C_D} = \frac{0.272}{0.002} = 136$$

$$\therefore U = 11.66 \text{ m/s}$$

$$\text{Reynolds number, } Re_L = \frac{UL}{\nu} = \frac{11.66 \times 25}{1 \times 10^{-6}} = 291.5 \times 10^6 \text{ (turbulent range assumed)}$$

$$C_D = \frac{0.455}{[\log_{10}(291.5 \times 10^6)]^{2.58}} = 0.00184$$

$$\text{Thus recalculation gives: } U^2 = \frac{0.272}{0.00184} = 14782 \text{ or } U = 12.16 \text{ m/s}$$

By another trial, we get:

$$Re_L = \frac{12.16 \times 25}{1 \times 10^{-6}} = 304 \times 10^6$$

$$C_D = \frac{0.455}{[\log_{10}(304 \times 10^6)]^{2.58}} = 0.001829$$

$$U^2 = \frac{0.272}{0.001829} = 148.715 \text{ or } U = 12.19 \text{ m/s}$$

i.e., $U = 12.19 \text{ m/s}$ which is within reasonable accuracy.

(ii) **The boundary layer thickness, δ :**

The thickness of boundary layer for turbulent flow is given by eqn. (13.43),

$$\delta = \frac{0.371L}{(Re_L)^{1/5}} = \frac{0.371 \times 25}{(304 \times 10^6)^{1/5}} = \mathbf{0.1865 \text{ m (Ans.)}}$$

(iii) **The distance, x_c :**

Transition from laminar to turbulent boundary may be assumed to occur at

$$(Re)x_c = 5 \times 10^5$$

$$(Re)x_c = \frac{Ux_c}{\nu} = 5 \times 10^5$$

$$\therefore x_c = \frac{5 \times 10^5 \times \nu}{U} = \frac{5 \times 10^5 \times 1 \times 10^{-6}}{12.19} = 0.041 \text{ m or } \mathbf{41 \text{ mm (Ans.)}}$$

13.5. TURBULENT BOUNDARY LAYER

As compared to laminar boundary layers, the turbulent boundary layers are *thicker*. Further in a turbulent boundary layer the velocity distribution is much more *uniform*, than in a laminar boundary layer, due to intermingling of fluid particles between different layers of the fluid. The velocity distribution in a turbulent boundary layer follows a logarithmic law *i.e.* $u \sim \log y$, which can also be represented by a *power law of the type*,

$$\frac{u}{U} = \left(\frac{y}{\delta} \right)^n \quad \dots(13.39)$$

where, $n = \frac{1}{7}$ (approx.) for $Re < 10^7$ but $> 5 \times 10^5$

$$\therefore \frac{u}{U} = \left(\frac{y}{\delta} \right)^{1/7} \quad \dots(13.40)$$

This is known as **one-seventh power law**.

The eqn. (13.40), however, cannot be applied at the boundary itself because at $y = 0$, $\left(\frac{\delta u}{\delta y} \right) = \frac{1}{7} U \delta^{-1/7} y^{-6/7} = \infty$. This difficulty is circumvented by considering the *velocity in the viscous laminar sublayer to be linear and tangential to the seventh-root profile* at the point, where the laminar sublayer merges with the turbulent part of the boundary layer.

Blasius suggested the following relation for viscous shear stress:

$$\tau_0 = 0.0226 \rho U^2 \left(\frac{\mu}{\rho U \delta} \right)^{1/4} \quad \dots(13.41)$$

(for Re ranging from 5×10^5 to 10^7)

Let us now find the values of δ , τ_0 , C_D^* , F_D and C_d for the velocity distribution

given by eqn. (13.40) $\left[i.e. \frac{u}{U} = \left(\frac{y}{\delta} \right)^{1/7} \right]$

(i) **Boundary layer thickness, δ :**

Substituting the value of $\frac{u}{U}$ in Von Karman integral eqn. (13.6), we have:

$$\begin{aligned} \frac{\tau_0}{\rho U^2} &= \frac{d}{dx} \left[\int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \right] \\ &= \frac{d}{dx} \left[\int_0^{\delta} \left\{ \left(\frac{y}{\delta} \right)^{1/7} \left[1 - \left(\frac{y}{\delta} \right)^{1/7} \right] \right\} dy \right] \\ &= \frac{d}{dx} \left[\int_0^{\delta} \left\{ \left(\frac{y}{\delta} \right)^{1/7} \left(\frac{y}{\delta} \right)^{2/7} \right\} dy \right] \\ &= \frac{d}{dx} \left[\frac{7}{8} \frac{y^{8/7}}{\delta^{1/7}} - \frac{7}{9} \frac{y^{9/7}}{\delta^{2/7}} \right]_0^{\delta} = \frac{d}{dx} \left[\frac{7}{8} \delta - \frac{7}{9} \delta \right] = \frac{7}{72} \frac{d\delta}{dx} \end{aligned}$$

[In the expression above, the limits have been taken from 0 to δ instead of δ' to δ since the laminar sublayer (δ') is very thin]

$$\therefore \tau_0 = \frac{7}{72} \rho U^2 \frac{d\delta}{dx} \quad \dots(13.42)$$

$$\text{Also,} \quad \tau_0 = 0.0225 \rho U^2 \left(\frac{\mu}{\rho U \delta} \right)^{1/4} \quad \dots[\text{Eqn. (13.27)}]$$

Now equating the eqns. (13.42) and (13.41), we have:

$$\frac{7}{72} \rho U^2 \frac{d\delta}{dx} = 0.0225 \rho U^2 \left(\frac{\mu}{\rho U \delta} \right)^{1/4}$$

$$\text{or,} \quad \frac{7}{72} \frac{d\delta}{dx} = 0.0225 \left(\frac{\mu}{\rho U} \right)^{1/4} \times \frac{1}{(\delta)^{1/4}} \quad (\text{cancelling } \rho U^2 \text{ on both sides})$$

$$\text{or,} \quad \delta^{1/4} d\delta = 0.0225 \times \frac{72}{7} \times \left(\frac{\mu}{\rho U} \right)^{1/4} dx$$

$$\text{or,} \quad \delta^{1/4} d\delta = 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} dx$$

Integrating both sides, we have:

$$\frac{4}{5} \delta^{5/4} = 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} x + C \quad (\text{where, } C = \text{constant of integration})$$

Let boundary layer be *assumed to be turbulent over the entire length of plate.*

Hence, at $x = 0, \delta = 0 \quad \therefore C = 0$

$$\therefore \frac{4}{5} \delta^{5/4} = 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} \times x$$

$$\text{or,} \quad \delta^{5/4} = (5/4 \times 0.2314) \left(\frac{\mu}{\rho U} \right)^{1/4} \times x$$

$$\begin{aligned} \text{or,} \quad \delta &= [5/4 \times 0.2314]^{4/5} \left(\frac{\mu}{\rho U} \right)^{1/5} \times x^{4/5} \\ &= 0.371 \left(\frac{\mu}{\rho U x} \right)^{1/5} x^{1/5} \times x^{4/5} = 0.371 \left(\frac{1}{Re_x} \right)^{1/5} \times x = \frac{0.71x}{(Re_x)^{1/5}} \end{aligned}$$

$$\text{i.e.} \quad \delta = \frac{0.371x}{(Re_x)^{1/5}} \quad \dots(13.43)$$

(ii) **Shear stress, τ_0 :**

$$\tau_0 = 0.0225 \rho U^2 \left(\frac{\mu}{\rho U \delta} \right)^{1/4} \quad [\text{Eqn. (13.41)}]$$

Substituting the value of δ from eqn. (13.43), we get:

$$\tau_0 = 0.0225 \rho U^2 \left(\frac{\mu}{\rho U \times \frac{0.371x}{(Re_x)^{1/5}}} \right)^{1/4}$$

$$= \frac{0.0225}{(0.371)^{1/4}} \rho U^2 \left[\frac{\mu}{\rho U x} \times (Re_x)^{1/5} \right]^{1/4} = 0.0288 \rho U^2 \left[\frac{(Re_x)^{1/5}}{Re_x} \right]^{1/4}$$

$$\left(\because Re_x = \frac{\mu}{\rho U x} \right)$$

or, $\tau_0 = \frac{\rho U^2}{2} \times \frac{0.0576}{(Re_x)^{1/5}}$... (13.44)

(iii) **Local co-efficient of drag, C_D^* :**

We know $\tau_0 = \frac{\rho U^2}{2} \times \frac{0.0576}{(Re_x)^{1/5}}$ [Eqn. (13.44)]

Also, $\tau_0 = C_D^* \times \frac{1}{2} \rho U^2$ [Eqn. (13.9)]

Now equating the eqns. (13.44) and (13.9), we have:

$$C_D^* \times \frac{1}{2} \rho U^2 = \frac{\rho U^2}{2} \times \frac{0.0576}{(Re_x)^{1/5}}$$

or, $C_D^* = \frac{0.0576}{(Re_x)^{1/5}}$... (13.45)

(iv) **Drag force, F_D :**

The total drag force (F_D) on one side of the plate of width B and length L is given by,

$$F_D = \int_0^L \tau_0 \times B \times dx \quad \dots [\text{Eqn. (13.8)}]$$

$$= \int_0^L \frac{\rho U^2}{2} \times \frac{0.0576}{(Re_x)^{1/5}} \times B \times dx = \int_0^L \frac{\rho U^2}{2} \times \frac{0.0576}{\left(\frac{\rho U x}{\mu}\right)^{1/5}} B \times dx$$

$$= \frac{\rho U^2}{2} \times \frac{0.0576}{\left(\frac{\rho U}{\mu}\right)^{1/5}} B \int_0^L x^{-1/5} dx = \frac{\rho U^2}{2} \times \frac{0.0576}{\left(\frac{\rho U}{\mu}\right)^{1/5}} B \left[\frac{5}{4} \times x^{4/5} \right]_0^L$$

$$= \frac{\rho U^2}{2} \times \frac{0.0576}{\left(\frac{\rho U}{\mu}\right)^{1/5}} \times B \times \frac{5}{4} (L)^{4/5} = \frac{\rho U^2}{2} \times \frac{0.072}{\left(\frac{\rho U L}{\mu}\right)^{1/5}} \times B \times L$$

or, $F_D = \frac{\rho U^2}{2} \times \frac{0.072}{(Re_L)^{1/5}} \times B \times L$... (13.46)

(v) **Co-efficient of drag, C_D :**

We know, $F_D = C_D \times \frac{1}{2} \rho A V^2$... [Eqn. (13.10)]

Now equating eqns. (13.10) and (13.46), we have:

$$C_D \times \frac{1}{2} \rho \times B \times L \times U^2 = \frac{\rho U^2}{2} \times \frac{0.072}{(Re_L)^{1/5}} \times B \times L \quad (\because \text{Area of the plate, } A = B \times L)$$

$$\text{or,} \quad C_D = \frac{0.072}{(Re_L)^{1/5}} \quad \dots(13.42)$$

This is valid for $5 \times 10^5 < Re_L < 10^7$.

For Reynolds number between 10^7 and 10^9 the following relationship suggested by Prandtl and Schlichting holds good,

$$C_D = \frac{0.455}{(\log_{10} Re_L)^{2.58}} \quad \dots(13.48)$$

13.6. TOTAL DRAG DUE TO LAMINAR AND TURBULENT LAYERS

When the leading edge is not very rough, the turbulent boundary layer does not begin at the leading edge, it is usually preceded by the laminar boundary layer. *The point of transition from laminar to turbulent layer depends upon the intensity of turbulence.* The distance x_c (Fig. 13.7) of the transition from the leading edge can be obtained from critical Reynolds number which normally ranges from 3×10^5 to 3×10^6 .

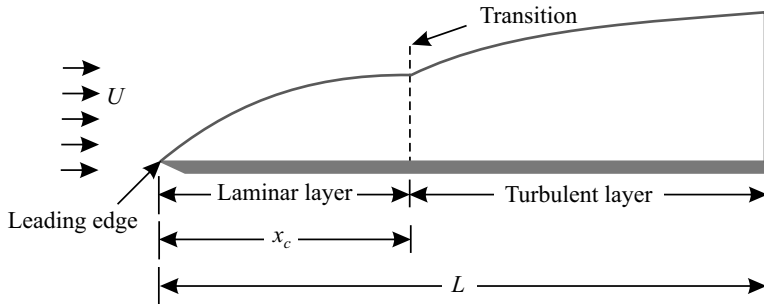


Fig. 13.7. Drag due to laminar and turbulent boundary layers.

Drag force ($F_D = F$) for the turbulent boundary layer can be estimated from the following relation:

$$F_{turb.} = (F_{turb.})_{total} - (F_{turb.})_{x_c}$$

where,

$$(F_{turb.})_{total} = \text{The drag which would occur if a turbulent boundary extends along the entire length of the plate, and}$$

$$(F_{turb.})_{x_c} = \text{The drag due to fictitious turbulent boundary layer from the leading edge to a distance } x_c.$$

Let us assume that the plate is long enough so that Reynolds number is greater than 10^7 , then the turbulent drag is given by,

$$F_{turb} = \frac{0.455}{(\log_{10} Re_L)^{2.58}} \times \frac{\rho U^2}{2} \times L \times B - \frac{0.072}{(Re_c)^{1/5}} \times \frac{\rho U^2}{2} \times x_c \times B \quad \dots(i)$$

where,

L = Length of the plate,

B = Width of the plate, and

U = Free stream velocity.

The laminar boundary layer prevails within the length x_c and its contribution to drag force is given by:

$$F_{\text{laminar}} = \frac{1.328}{\sqrt{Re_c}} \times \frac{\rho U^2}{2} \times x_c \times B = \frac{1.328 x_c}{\sqrt{Re_c}} \times B \times \frac{\rho U^2}{2} \quad \dots(ii)$$

$$\therefore F_{\text{total}} = F_{\text{laminar}} + F_{\text{turb.}}$$

$$\begin{aligned} F_{\text{total}} &= \frac{1.328 x_c}{\sqrt{Re_c}} \times B \times \frac{\rho U^2}{2} + \left[\frac{0.455 L}{(\log_{10} Re_L)^{2.58}} \times B \times \frac{\rho U^2}{2} - \frac{0.072 x_c}{(Re_c)^{1/5}} \times B \times \frac{\rho U^2}{2} \right] \\ &= \left[\frac{1.328 x_c}{\sqrt{Re_c}} + \frac{0.455 L}{(\log_{10} Re_L)^{2.58}} - \frac{0.072 x_c}{(Re_c)^{1/5}} \right] \frac{B \rho U^2}{2} \quad \dots(iii) \end{aligned}$$

$$\text{Also, } \frac{Re_c}{Re_L} = \frac{(\rho U x_c / \mu)}{(\rho U L / \mu)} = \frac{x_c}{L}$$

$$\text{or, } x_c = \frac{Re_c L}{Re_L}$$

Substituting the value of x_c in eqn (iii), we have:

$$F_{\text{total}} = \left[\frac{1.328 \sqrt{Re_c}}{Re_L} + \frac{0.455}{(\log_{10} Re_L)^{2.58}} - \frac{0.072 Re_c^{0.8}}{Re_L} \right] \frac{L B \rho U^2}{2}$$

Assuming that transition occurs at $Re_c = 5 \times 10^5$,

$$F_{\text{total}} = \left[\frac{0.455}{(\log_{10} Re_L)^{2.58}} - \frac{1700}{Re_L} \right] \frac{L B \rho U^2}{2} \quad \dots(13.49)$$

$$\text{Also, } F_{\text{total}} = C_D \times \frac{1}{2} \rho A U^2 = C_D \times \frac{L B \rho U^2}{2} \quad \dots[\text{Eqn. (13.10)}]$$

(where, C_D = average co-efficient of drag.)

Equating the above two equations, we have:

$$C_D = \frac{0.455}{(\log_{10} Re_L)^{2.58}} - \frac{1700}{Re_L} \quad \dots(13.50)$$

Example 13.24. A submarine can be assumed to have cylindrical shape with rounded nose. Assuming its length to be 50 m and diameter 5.0 m, determine the total power required to overcome boundary friction if it cruises at 8 m/s velocity in sea water at 20°C ($\rho = 1030 \text{ kg/m}^3$), $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$.

Solution. Length of submarine, $L = 50 \text{ m}$

Diameter of submarine, $D = 5.0 \text{ m}$

Velocity of submarine, $U = 8 \text{ m/s}$

Density of sea water, $\rho = 1030 \text{ kg/m}^3$

Kinematic viscosity of sea water, $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$

Total power required to overcome boundary friction, P :

$$\text{Reynolds number, } Re_L = \frac{UL}{\nu} = \frac{8 \times 50}{1 \times 10^{-6}} = 4 \times 10^8$$

The length over which boundary layer will be laminar is given by,

$$\frac{Ux}{\nu} = 5 \times 10^5 \text{ or } x = \frac{5 \times 10^5 \times \nu}{U}$$

or,
$$x = \frac{5 \times 10^5 \times 1 \times 10^{-6}}{8} = 0.0625 \text{ m}$$

This being very small, contribution to total drag from laminar boundary layer is negligible; hence C_D is given by,

$$C_D = \frac{0.455}{(\log_{10} Re_L)^{2.58}} = \frac{0.455}{[\log_{10} (4 \times 10^8)]^{2.58}} = 0.001765$$

$$\text{Area, } A = \pi DL = \pi \times 5 \times 50 = 785.4 \text{ m}^2$$

$$\therefore \text{ Drag force, } F_D = C_D \times \frac{1}{2} \rho A U^2 = 0.001765 \times \frac{1}{2} \times 1030 \times 785.4 \times 8^2 = 45690.2 \text{ N}$$

Hence, total power required to overcome boundary friction,

$$P = \frac{F_D U}{1000} \text{ kW} = \frac{45690.2 \times 8}{1000} = \mathbf{365.52 \text{ kW (Ans.)}}$$

Example 13.25. 12000 kW power is required to cruise a passenger ship of 300 m length and 12.0 m draft at 40 km/h. If $\rho = 1030 \text{ kg/m}^3$ and $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$, determine the combined form and wave resistance of the ship.

Solution. Power required to cruise the ship, $P = 12000 \text{ kW}$

Length of the ship = 300 m

Draft of the ship = 12 m

$$\text{Speed of the ship, } U = 40 \text{ km/h} = \frac{40 \times 1000}{3600} = 11.11 \text{ m/s}$$

Density of water, $\rho = 1030 \text{ kg/m}^3$

Kinematic viscosity of water, $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$

Combined form and wave resistance:

$$\text{Reynolds number, } Re_L = \frac{UL}{\nu} = \frac{11.11 \times 300}{1 \times 10^{-6}} = 3.333 \times 10^9$$

At this Reynolds number, the boundary layer will be turbulent on almost the whole length; hence, C_D is given by:

$$C_D = \frac{0.455}{(\log_{10} Re_L)^{2.58}} = \frac{0.455}{[\log_{10} (3.333 \times 10^9)]^{2.58}} = 0.001358$$

$$\begin{aligned} F_{\text{friction}} &= 2 \times C_D \times \frac{1}{2} \rho A U^2 = 2 \times 0.001358 \times \frac{1}{2} \times 1030 \times 300 \times 12 \times (11.11)^2 \\ &= 621538 \text{ N or } 621.54 \text{ kN} \end{aligned}$$

Total power required, $P = FU$

$$\therefore \text{ Total force, } F = \frac{P}{U} = \frac{12000}{11.11} = 1080 \text{ kN}$$

Also, $F = F_{\text{friction}} + (F_{\text{form}} + F_{\text{wave}})$

$$\therefore (F_{\text{form}} + F_{\text{wave}}) = 1080 - F_{\text{friction}} = 1080 - 621.54 = \mathbf{458.46 \text{ kN (Ans.)}}$$

Example 13.26. Find the ratio of friction drag on the front half and rear half of the flat plate kept at zero incidence in a stream of uniform velocity, if the boundary layer is turbulent over the whole plate.

Solution. The average co-efficient of drag (C_D) for turbulent boundary layer is given by:

$$C_D = \frac{0.072}{(Re_L)^{1/5}} \quad \dots[\text{Eqn. (13.47)}]$$

$$\text{For the entire plate, } Re_L = \frac{UL}{\nu}$$

$$\text{For the first half of the plate, } Re_x = \frac{Ux}{\nu} = \frac{UL}{2\nu}$$

Drag force per unit width for the entire plate is,

$$\begin{aligned} F_D &= C_D \times \frac{\rho U^2}{2} \times \text{area per unit width} \\ &= \frac{0.072}{\left(\frac{UL}{\nu}\right)^{1/5}} \times \frac{\rho U^2}{2} \times L \end{aligned}$$

Similarly the drag force per unit width for the front half portion of the plate is,

$$F_1 = \frac{0.072}{\left(\frac{UL}{2\nu}\right)^{1/5}} \times \frac{\rho U^2}{2} \times \frac{L}{2} = \frac{0.072}{\left(\frac{UL}{\nu}\right)^{1/5}} \times \frac{\rho U^2}{2} \times \frac{L}{2} (2)^{1/5}$$

$\therefore$ Drag force for the rear half portion of the plate is,

$$F_2 = F - F_1 = \frac{0.072L}{\left(\frac{UL}{\nu}\right)^{1/2}} \times \frac{\rho U^2}{2} \left[1 - \frac{1}{2} (2)^{1/5}\right]$$

$$\text{Hence, } \frac{F_1}{F_2} = \frac{\frac{1}{2} \times (2)^{1/5}}{1 - \frac{1}{2} (2)^{1/5}} = \frac{0.574}{1 - 0.574} = \mathbf{1.347 \text{ (Ans.)}}$$

Example 13.27. A streamlined train is 200 m long with a typical cross-section having a perimeter of 9 m above the wheels. If the kinematic viscosity of air at the prevailing temperature is $1.5 \times 10^{-5} \text{ m}^2/\text{s}$ and density 1.24 kg/m^3 , determine the approximate surface drag (friction drag) of the train when running at 90 km/h.

Make allowance for the fact that boundary layer changes from laminar to turbulent on the train surface.

Solution. Length of the train, $L = 200 \text{ m}$

Perimeter of cross-section of the train above wheels, $P = 9 \text{ m}$

$\therefore$ Surface area, $A = L \times P = 200 \times 9 = 1800 \text{ m}^2$

Kinematic viscosity of air, $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$

Density of air, $\rho = 1.24 \text{ kg/m}^3$

Free stream velocity, $U = 90 \text{ km/h} = \frac{90 \times 1000}{3600} = 25 \text{ m/s}$

Approximate friction drag, F_D

The Reynolds number with length of the train as the characteristic length,

$$Re_L = \frac{UL}{\nu} = \frac{25 \times 200}{1.5 \times 10^{-5}} = 3.333 \times 10^8$$

Obviously the boundary layer is *turbulent*.

Assuming that the abrupt transition from laminar to turbulent flow occurs at a Reynolds number of 5×10^5 , the average co-efficient of drag,

$$\begin{aligned} C_D &= \frac{0.455}{(\log_{10} Re_L)^{2.58}} - \frac{1700}{Re_L} \quad [\text{Eqn. (13.50)}] \\ &= \frac{0.455}{[\log_{10} (3.333 \times 10^8)]^{2.58}} - \frac{1700}{3.333 \times 10^8} \\ &= 0.001807 - 5.1 \times 10^{-6} = 0.0018 \end{aligned}$$

The approximate friction drag over the train surface,

$$F_D = C_D \times \frac{1}{2} \rho A U^2 = 0.0018 \times \frac{1}{2} \times 1.24 \times 1800 \times 25^2 = \mathbf{1255.5 \text{ N (Ans.)}}$$

Example 13.28. A barge with a rectangular bottom surface 30 m long $\times$ 10 m wide is travelling down a river with a velocity of 0.6 m/s. A laminar boundary layer exists upto a Reynolds number equivalent to 5×10^5 and subsequently abrupt transition occurs to turbulent boundary layer. Calculate:

- (i) The maximum distance from the leading edge upto which laminar boundary layer persists and the maximum boundary layer thickness at that point.
- (ii) The total drag force on the flat bottom surface of the barge, and
- (iii) The power required to push the bottom surface through water at the given velocity.

For water $\rho = 998 \text{ kg/m}^3$ and $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$.

Solution. Length of the bottom surface, $L = 30 \text{ m}$

Width of the bottom surface, $B = 10 \text{ m}$

$$\therefore \text{Area, } A = L \times B = 30 \times 10 = 300 \text{ m}^2$$

$$\text{Velocity, } U = 0.6 \text{ m/s}$$

$$\text{Density of water, } \rho = 998 \text{ kg/m}^3$$

$$\text{kinematic viscosity, } \nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$$

- (i) The maximum distance up to which laminar boundary layer persists, x_c :

$$(Re)_{x_c} = \frac{Ux_c}{\nu} = 5 \times 10^5$$

$$\therefore x_c = \frac{5 \times 10^5 \times \nu}{U} = \frac{5 \times 10^5 \times 1 \times 10^{-6}}{0.6} = \mathbf{0.833 \text{ m (Ans.)}}$$

Maximum boundary layer (laminar) thickness, δ :

$$\delta = \frac{5x_c}{\sqrt{(Re)_{x_c}}} = \frac{5 \times 0.833}{\sqrt{5 \times 10^5}} = 5.89 \times 10^{-3} \text{ m or } \mathbf{5.89 \text{ mm (Ans.)}}$$

- (ii) The total drag force F_D :

$$Re_L = \frac{UL}{\nu} = \frac{0.6 \times 30}{1 \times 10^{-6}} = 1.8 \times 10^7$$

The average co-efficient of drag,

$$C_D = \frac{0.455}{(\log_{10} R_{eL})^{2.58}} - \frac{1700}{R_{eL}} \quad \dots[\text{Eqn. (13.50)}]$$

$$= \frac{0.455}{[\log_{10} (1.8 \times 10^7)]^{2.58}} - \frac{1700}{1.8 \times 10^7} = 0.002644$$

∴ Drag force on the bottom surface of the barge,

$$F_D = C_D \times \frac{1}{2} \rho A U^2 = 0.002644 \times \frac{1}{2} \times 998 \times 300 \times (0.6)^2$$

$$= 142.49 \text{ N (Ans.)}$$

(iii) The power required, P :

The power required to push the bottom surface through water at the given velocity,

$$P = F_D \times U = 142.49 \times 0.6 = 85.49 \text{ W (Ans.)}$$

13.7. BOUNDARY LAYER SEPARATION AND ITS CONTROL

In a flowing fluid when a solid body is immersed, a thin layer of fluid called the boundary layer is formed adjacent to the solid body. The forces acting on the fluid in the boundary layer are:

(i) Inertia forces, (ii) Viscous forces, and (iii) Pressure forces.

— When the pressure gradient in the direction of flow is negative $\left(\frac{dp}{dx} < 0\right)$ i.e. when the pres-

sure *decreases* in the direction of flow, the flow is accelerated. In this case, the pressure force and inertia force add together and jointly tend to reduce the effect of viscous forces in the boundary layer. This results in a decrease in the thickness of boundary layer in the direction of flow, as a consequence of which there are *low losses* and *high efficiencies in accelerating flows*.

— When the pressure increases in the direction of flow $\left(\frac{dp}{dx} > 0\right)$, the pressure forces act *op-*

posite to the direction of flow and further increase the retarding effect of the viscous forces. Subsequently the thickness of the boundary layer increases rapidly in the direction of flow. *If these forces act over a long stretch, the boundary layer gets separated from the surface and moves into the main stream. This phenomenon is called **separation**. The point of the body at which the boundary layer is on the verge of separation from the surface is called “point of separation”.*

Consider a flow of fluid over a curved surface as shown in Fig. 13-8.

— As the fluid flows round the surface (the area of flow decreases) it is accelerated over the left hand section until at point B the velocity just outside the boundary is maximum and the pressure is minimum (as shown by the graph below the surface). Thus from A and B the pressure gradient is *negative*. As long as $\frac{dp}{dx} < 0$, the entire boundary layer *moves forward*.

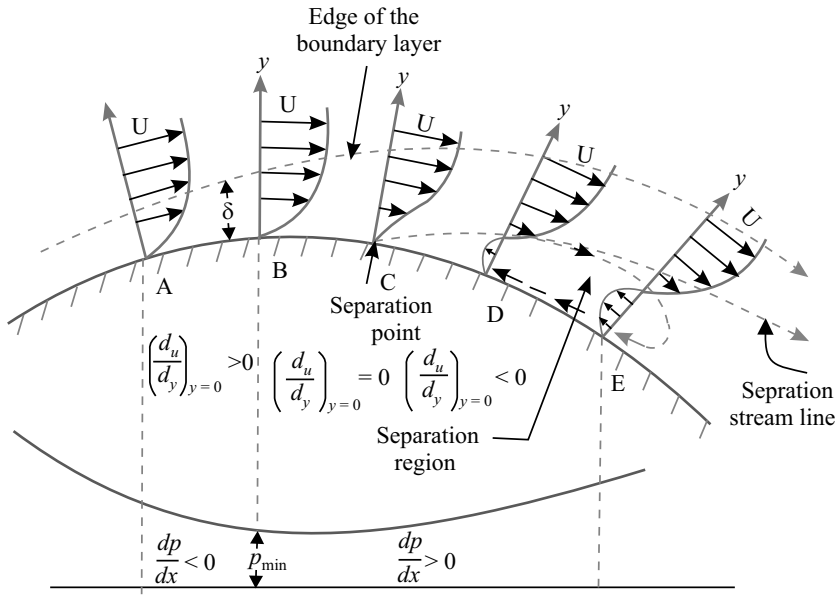


Fig. 13.8. Separation of boundary layer.

— Beyond B (i.e. along the region $BCDE$), the area of flow increases and hence velocity of flow decreases; due to decrease of velocity the pressure increases (in the direction of flow) and hence the pressure gradient $\frac{dp}{dx}$ is *positive* i.e. $\frac{dp}{dx} > 0$. The value of the velocity gradient $\left(\frac{du}{dy}\right)$ at the boundary is *zero* at the point C , this point is known as a **separation point** (the

boundary layer starts separating from the surface because further retardation of flow near the surface is physically impossible). Large *turbulent eddies* are formed downstream of the point of separation. The *disturbed region in which the eddies are formed* is called **turbulent wake**.

The *flow separation* depends upon *factors* such as:

- (i) *The curvature of the surface;*
- (ii) *The Reynolds number of flow;*
- (iii) *The roughness of the surface.*

The velocity gradient, for a given velocity profile, exhibits the following characteristics for the flow to remain *attached*, get *detached* or be on the *verge of separation*:

1. $\left(\frac{\partial u}{\partial y}\right)_{y=0}$ is +ve ...*Attached flow* (The flow will not separate)
2. $\left(\frac{\partial u}{\partial y}\right)_{y=0}$ is zero ...The flow is on the *verge of separation*
3. $\left(\frac{\partial u}{\partial y}\right)_{y=0}$ is -ve ...*Separated flow*.

— *Boundary layer separation is unstable, inefficient process and entails large losses due to appreciable eddying region.*

- **Separation** occurs in the following cases:

- (i) Diffusers,
- (ii) Open channel transitions,
- (iii) Pumps,
- (iv) Fans,
- (v) Aerofoils,
- (vi) Turbine blades etc.

Methods of preventing the separation of boundary layer:

Following are some of the methods generally adopted to *retard or arrest the flow separation*:

1. Streamlining the body shape.
2. Tripping the boundary layer from laminar to turbulent by provision of surface roughness.
3. Sucking the retarded flow.
4. Injecting high velocity fluid in the boundary layer.
5. Providing slots near the leading edge.
6. Guidance of flow in a confined passage.
7. Providing a rotating cylinder near the leading edge.
8. Energising the flow by introducing optimum amount of swirl in the incoming flow.

Note: Refer to Example 13.29 also.

Example 13.29. Explain what is meant by separation of boundary layer. Describe with sketches the methods to control separation. (PEC)

Solution. When a solid body is immersed in a flowing fluid, a thin layer of fluid called *boundary layer* is formed, adjacent to the solid body. In this thin layer of fluid, the velocity varies from *zero to free stream velocity* in the direction normal to the solid body. Along the length of the solid body, the thickness of the boundary layer increases. *The fluid layer adjacent to the solid surface has to do work against surface friction at the expense of kinetic energy. This loss of kinetic energy is recovered from the immediate fluid layer in contact with the layer adjacent to the solid surface through momentum exchange process. Thus the velocity of the layer goes on decreasing.* Along the length of solid body, at a certain point a stage may come when the boundary layer may not be able to keep sticking to the solid body, if it cannot provide kinetic energy to overcome the resistance offered by the solid body. In other words, the boundary layer will be *separated from the surface*. This phenomenon is called the **boundary layer separation**. The point on the body at which the boundary layer is on the verge of separation from the surface is called **point of separation**.

Methods to control separation

1. Motion of solid boundary:

By rotating a circular cylinder lying in a stream of fluid, so that the upper side of cylinder where the fluid as well as the cylinder move in the *same direction*, the boundary layer does not form. However on the lower side of cylinder where the fluid motion is *opposite* to that of cylinder separation would occur (Fig. 13.9).

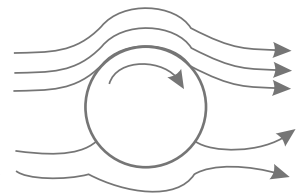
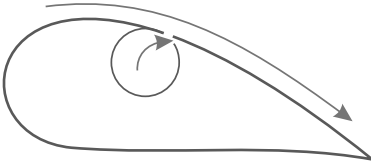
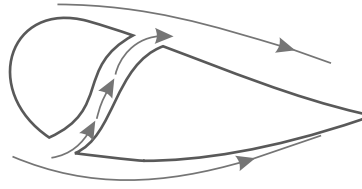


Fig. 13.9

2. Acceleration of fluid in the boundary layer:

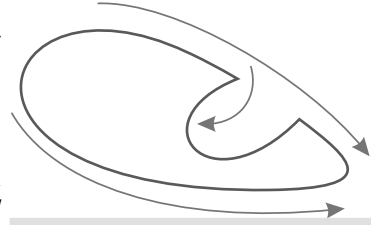
This method of controlling separation consists of *supplying additional energy to particles of fluid which are being retarded in the boundary layer*. This may be achieved either by injecting the fluid into the region of boundary layer from the interior of the body with the help of some available device as shown in Fig. 13.10 or by diverting a portion of fluid of the main stream from the region of high pressure to the retarded region of boundary layer through a slot provided in the body (Fig. 13.11)

**Fig. 13.10.** Injecting fluid into boundary layer.**Fig. 13.11.** Slotting wing.**3. Suction of fluid from the boundary layer:**

In this method, the slow moving fluid in the boundary layer is removed by suction through slots or through a porous surface as shown in the Fig. 13.12.

4. Streamlining of body shapes:

By the use of suitably shaped bodies, the point of transition of the boundary layer from laminar to turbulent can be moved downstream which results in the reduction of the skin friction drag. Further more by streamlining of body shapes, the separation may be eliminated.

**Fig. 13.12**

Example 13.30. For the following velocity profiles determine whether flow is attached or detached or on the verge of separation:

$$(i) \frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$$

$$(ii) \frac{u}{U} = -2 \left(\frac{y}{\delta} \right) + \left(\frac{y}{\delta} \right)^3 + 2 \left(\frac{y}{\delta} \right)^4$$

$$(iii) \frac{u}{U} = 2 \left(\frac{y}{\delta} \right)^2 + \left(\frac{y}{\delta} \right)^3 - 2 \left(\frac{y}{\delta} \right)^4$$

[Nagpur University]

Solution. (i) $\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$ or $u = 2U \left(\frac{y}{\delta} \right) - U \left(\frac{y}{\delta} \right)^2$

Differentiating w.r.t. y the above equation, we get:

$$\frac{du}{dy} = 2U \left(\frac{1}{\delta} \right) - 2U \left(\frac{y}{\delta} \right) \times \frac{1}{\delta}$$

At, $y = 0, \left(\frac{\partial u}{\partial y} \right)_{y=0} = \frac{2U}{\delta}$

As $\left(\frac{\partial u}{\partial y} \right)_{y=0}$ is +ve, the given flow is **attached (Ans.)**

$$(ii) \frac{u}{U} = -2 \left(\frac{y}{\delta} \right) + \left(\frac{y}{\delta} \right)^3 + 2 \left(\frac{y}{\delta} \right)^4$$

or, $u = -2U \left(\frac{y}{\delta} \right) + U \left(\frac{y}{\delta} \right)^3 + 2U \left(\frac{y}{\delta} \right)^4$

$$\therefore \frac{\partial u}{\partial y} = -2U \left(\frac{1}{\delta} \right) + 3U \left(\frac{y}{\delta} \right)^2 \times \frac{1}{\delta} + 8U \left(\frac{y}{\delta} \right)^3 \times \frac{1}{\delta}$$

At,
$$y = 0, \left(\frac{\partial u}{\partial y} \right)_{y=0} = -\frac{2U}{\delta}$$

As $\left(\frac{\partial u}{\partial y} \right)_{y=0}$ is $-ve$, the given flow is **detached** (i.e. the flow has *separated*). (Ans.)

(iii)
$$\frac{u}{U} = \left(\frac{y}{\delta} \right)^2 + \left(\frac{y}{\delta} \right)^3 - 2 \left(\frac{y}{\delta} \right)^4$$

or,
$$u = 2U \left(\frac{y}{\delta} \right)^2 + U \left(\frac{y}{\delta} \right)^3 - 2U \left(\frac{y}{\delta} \right)^4$$

$\therefore \frac{\partial u}{\partial y} = 4U \left(\frac{y}{\delta} \right) \times \frac{1}{\delta} + 3U \left(\frac{y}{\delta} \right)^2 \times \frac{1}{\delta} - 8U \left(\frac{y}{\delta} \right)^3 \times \frac{1}{\delta}$

At,
$$y = 0, \left(\frac{\partial u}{\partial y} \right)_{y=0} = 0$$

As $\left(\frac{\partial u}{\partial y} \right)_{y=0} = 0$, the given flow is **on the verge of separation**. (Ans.)

HIGHLIGHTS

1. When a viscous fluid flows past an immersed body, a thin boundary layer is formed in the immediate neighbourhood of solid surface. In the boundary layer, the velocity gradient $\left(\frac{\partial u}{\partial y} \right)$ is very high.
2. The resistance due to viscosity is confined only in the boundary layer. The fluid outside the boundary layer may be considered as ideal.
3. Near the leading edge of a flat plate, the boundary layer is wholly laminar. For a boundary layer, the velocity distribution is parabolic. The thickness of the boundary layer (δ) increases with distance from the leading edge, as more and more fluid is slowed down by the viscous boundary, becomes unstable and breaks into turbulent boundary layer over a transition region.
4. For a turbulent boundary layer, if the boundary is smooth, the roughness projections are covered by a very thin layer which remains laminar, called *laminar sublayer*. The velocity distribution in the turbulent boundary layer is given by *Log law* or *Prandtl's one-seventh power law*.
5. For a flow, when $Re = \frac{Ux}{\nu} < 5 \times 10^5$... boundary layer is laminar, and
 when, $Re = \frac{Ux}{\nu} > 5 \times 10^5$... boundary layer is called turbulent.
 where, U = Free stream velocity,
 x = Distance from the leading edge, and
 ν = Kinematic viscosity of fluid.
6. The *thickness of the boundary layer* is arbitrarily defined as that distance from the boundary in which the velocity reaches 99 percent of the velocity of the free stream. It is denoted by the symbol δ .

7. Displacement thickness, $\delta^* = \int_0^{\delta} \left(1 - \frac{u}{U}\right) dy$

8. Momentum thickness, $\delta = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U}\right) dy$

9. Energy thickness, $\delta_e = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u^2}{U^2}\right) dy$

10. Von Karman momentum integral equation is given as:

$$\frac{\tau_0}{\rho U^2} = \frac{d\theta}{dx}$$

where, $\theta = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U}\right) dy$, and

τ_0 = Shear stress at surface.

This equation is applicable to laminar, transition and turbulent boundary layer flows.

11. As per Blasius results:

The thickness of laminar boundary layer, $\delta = \frac{5x}{\sqrt{Re_x}}$ (where, Re_x = Reynolds number)

$$\text{Average co-efficient of drag, } C_D = \frac{1.328}{\sqrt{Re_x}}$$

12. For turbulent boundary layer, the velocity profile is given as :

$$\frac{u}{U} = \left(\frac{y}{\delta}\right)^{1/7}$$

This equation is *not valid very near the boundary where laminar sublayer exists*.

13. For turbulent boundary layer over a flat plate, the shear stress at the boundary is given as

$$\tau_0 = 0.0225 \rho U^2 \left(\frac{\mu}{\rho U \delta}\right)^{1/4}$$

14. In case of a turbulent boundary layer:

For $5 \times 10^5 < Re < 10^7$:

$$\delta = \frac{0.371x}{(Re_x)^{1/5}}, \quad \text{and} \quad C_D = \frac{0.072}{(Re_L)^{1/5}}$$

For $10^7 < Re < 10^9$:

$$C_D = \frac{0.455}{(\log_{10} Re_L)^{2.58}} \dots \text{Prandtl-Schlichting empirical equation}$$

15. Total drag on a flat plate due to laminar and turbulent layers:

$$F_{\text{total}} = \left[\frac{0.455}{(\log_{10} Re_L)^{2.58}} - \frac{1700}{Re_x} \right] \frac{LB\rho U^2}{2}$$

$$\text{Average co-efficient of drag, } C_D = \frac{0.455}{(\log_{10} Re_L)^{2.58}} - \frac{1700}{Re_L}$$

16. The velocity gradient, for a given velocity profile, exhibits the following characteristic for the flow to remain attached, get detached or be on the verge of separation.

- (i) $\left(\frac{\partial u}{\partial y}\right)_{y=0}$ is + ve ... *Attached flow* (The flow will not separate)
- (ii) $\left(\frac{\partial u}{\partial y}\right)_{y=0}$ is zero ... The flow is *on the verge of separation*
- (iii) $\left(\frac{\partial u}{\partial y}\right)_{y=0}$ is - ve ... *Separated flow*.

OBJECTIVE TYPE QUESTIONS

- In turbulent flow the velocity
 - varies with time and space
 - varies with time only, the patterns of fluctuation, with respect to time, being same at all points
 - is constant at every point
 - none of the above.
- In a turbulent flow the shear stress is mainly due to the
 - density of the fluid
 - dynamic viscosity of the fluid
 - kinematic viscosity of the fluid
 - eddy viscosity of the fluid
 - all of the above.
- For which of the following flows Blasius equation is used?
 - Laminar flow
 - Turbulent flow in rough pipes
 - Turbulent flow in smooth pipes for any Reynolds number
 - Turbulent flow in smooth pipes for $Re < 10^5$
 - none of the above.
- In turbulent flow, which of the following gives the exact velocity distribution?
 - Logarithmic distribution
 - Blasius equation
 - Power law with index varying
 - Prandtl's one-seventh power.
- The boundary layer exists in which of the following?
 - Flow of real fluids
 - Flow of ideal fluids
 - Flow over flat surfaces only
 - Pipe-flow only.
- On account of which of the following boundary layer exists?
 - Surface tension
 - Gravitational effect
 - Viscosity of fluid
 - None of the above.
- On account of which of the following L. Prandtl is regarded as the father of modern fluid mechanics?
 - His pioneering research on flow of low-viscosity fluids bringing forward a new concept of boundary layer
 - His new interpretations on fluid resistance
 - His fundamental research in the field of aircraft-engineering
 - None of the above.
- The displacement thickness is
 - the layer in which the loss of energy is minimum
 - the layer which represents reduction in momentum caused by the boundary layer
 - the thickness upto which the velocity approaches 99% of the free-stream velocity
 - the distance measured perpendicular to the boundary by which the free-stream is displaced on account of formation of boundary layer.
- Over a long flat plate, the laminar boundary-layer becomes unstable and changes flow characteristics from laminar to turbulent when the plate Reynolds number approaches a value between
 - 3×10^4 to 5×10^4
 - 3×10^5 to 6×10^5
 - 2×10^6 to 5×10^6
 - 5×10^6 to 8×10^6 .

10. When the fluid flows along the solid boundary, more and more fluid gets retarded in the vicinity of the boundary; this deceleration is due to
 (a) high velocity of the fluid
 (b) high velocity flow outside the boundary layer
 (c) high velocity gradients which exist at and near the boundary
 (d) none of the above.
11. Momentum thickness is given by which of the following relations?
 (a) $\int_0^{\delta} \left(1 - \frac{u}{U}\right) dy$ (b) $\int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U}\right) dy$
 (c) $\int_0^{\delta} \frac{u}{U} \left(1 - \frac{u^2}{U^2}\right) dy$ (d) none of the above.
12. The boundary layer separation occurs when
 (a) $\frac{dp}{dx} < 0$ (b) $\left(\frac{\partial u}{\partial y}\right)_{y=0} = 0$
 (c) $\left(\frac{\partial u}{\partial y}\right)_{y=0} > 0$ (d) none of the above.
13. Which of the following is the condition for detached flow?
 (a) $\left(\frac{\partial u}{\partial y}\right)_{y=0} = 0$ (b) $\left(\frac{\partial u}{\partial y}\right)_{y=0} > 0$
 (c) $\left(\frac{\partial u}{\partial y}\right)_{y=0} < 0$ (d) none of the above.
14. Von Karman momentum integral equation $\left(\frac{\tau_0}{\rho U^2} = \frac{\partial \theta}{\partial x}\right)$ is applicable to
 (a) laminar boundary layer flow only
 (b) turbulent boundary layer flow only
 (c) transition boundary layer flow only
 (d) laminar, transition and turbulent boundary layer flows.
15. If the Reynolds number is more than 5×10^5 , the boundary layer is called
 (a) laminar boundary layer
 (b) turbulent boundary layer
 (c) either of the above
 (d) none of the above.
16. The separation of boundary can be prevented by
 (a) providing small divergence in a diffuser
 (b) providing a trip-wire ring in the laminar region for the flow over a sphere
 (c) providing a bypass in the slotted wing
 (d) suction of the slow moving fluid by a suction slot
 (e) all of the above.
17. The ratio of mean velocity to the maximum velocity in a pipe depends on which of the following factors?
 (a) Reynolds number of flow
 (b) The pressure drop in the pipe
 (c) The friction factor
 (d) The density of the fluid
 (e) The relative roughness of pipe
 (f) All of the above.
18. The critical value of Reynolds number at which boundary layer changes from laminar to turbulent depends on which of the following?
 (a) Turbulence in ambient flow
 (b) Surface roughness
 (c) Pressure gradient
 (d) Plate curvature
 (e) All of the above.
19. thickness is the distance through which the total loss of momentum per second be equal to if it were passing a stationary plate.
 (a) Displacement (b) Momentum
 (c) Energy (d) None of above.
20. Ageing of pipes implies which of the following?
 (a) A decrease in the value of friction factor
 (b) Increase in absolute roughness linearly with time and hence friction factor
 (c) Pipe becoming smoother with time
 (d) None of the above.

ANSWERS

- | | | | | | |
|---------|----------|---------|---------|---------|---------|
| 1. (b) | 2. (d) | 3. (d) | 4. (a) | 5. (a) | 6. (c) |
| 7. (a) | 8. (d) | 9. (b) | 10. (c) | 11. (b) | 12. (b) |
| 13. (c) | 14. (d) | 15. (b) | 16. (e) | 17. (f) | 18. (e) |
| 19. (b) | 20. (b). | | | | |

THEORETICAL QUESTIONS

1. Explain briefly the term boundary layer.
2. Give four examples in everyday life where formation of boundary layer is important.
3. What is the 'slip condition' at the boundary?
4. What boundary condition must be satisfied by the velocity distribution in laminar boundary layer over a plate?
5. Why is the flow in the boundary layer analysed on the principles of viscous flow theory?
6. What is the physical significance of displacement thickness of boundary layer?
7. Define momentum thickness and energy thickness.
8. Define boundary layer and explain the fundamental causes of its existence.
9. Explain the characteristics of laminar and turbulent boundary layers.
10. Will the laminar boundary layer on a flat plate held at zero incidence always turn into turbulent boundary layer at $Re_x = 5 \times 10^5$? Explain.
11. Why is it necessary to control the growth of boundary layer on most of the bodies? What methods are used for such a control?
12. Is the flow within the boundary layer rotational or irrotational?
13. It is stated that the pressure distribution within the boundary layer is determined by the outside flow which can be treated as inviscid. Explain.
14. How are the thickness of boundary layer, shear stress and the drag force along the flat plate determined by Von Karman momentum equation?
15. Obtain an expression for the boundary shear stress in terms of momentum thickness.
16. How will you find the drag on a flat plate due to laminar and turbulent boundary layers?
17. How will you determine whether a boundary layer flow is attached flow, detached flow or on the verge of separation?
18. What is a boundary layer? Why does it increase with distance from the upstream edge?
19. Obtain Von Karman momentum integral equation.
20. Prove that the momentum thickness and energy thickness for boundary layer flows are given by

$$\theta = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy$$
 and,

$$\delta_e = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u^2}{U^2} \right) dy$$
21. Define the following terms:
 - (i) Laminar boundary layer
 - (ii) Turbulent boundary layer
 - (iii) Laminar sublayer
 - (iv) Boundary layer thickness.
22. What is Blasius one-seventh power law of velocity distribution?
23. What is mean by average drag co-efficient C_D ? How does it differ from the local drag co-efficient, C_D^* ?
24. What is laminar sublayer? How is the concept of laminar sublayer useful?

UNSOLVED EXAMPLES

1. Show that for velocity distribution,

$$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2,$$

the ratio of $\delta/\delta^* = 3$

2. The velocity distribution in the boundary layer over a high spillway face was found to have the following form:

$$\frac{u}{U} = \left(\frac{y}{\delta} \right)^{0.22}$$

Prove that the displacement thickness, the momentum thickness and the energy thickness in terms of δ , the boundary layer thickness, can be expressed as

$$\frac{\delta^*}{\delta} = 0.18; \frac{\theta}{\delta} = 0.125; \frac{\delta_e}{\delta} = 0.127$$

(Ravi Shanker University)

3. Find the ratio of displacement thickness to momentum thickness and momentum thickness to energy thickness for the velocity distribution in the boundary layer given by

$$\frac{u}{U} = 2 (y/\delta) - (y/\delta)^2 \quad [\text{Ans. 2.5, 7/11}]$$

4. For the velocity profile in laminar layer given as

$$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$$

find the thickness of boundary layer at the end of the plate and the drag force on the side of the plate 1 m long and 0.8 m wide when placed in water flowing with a velocity of 0.15 m/s. Calculate the value of co-efficient of drag also. Take μ for water = 0.001 Ns/m²

[Ans. 14.15 mm, 0.0338 N, 0.00376]

5. Air flows over a smooth flat plate with a velocity of 10 m/s. The velocity profile is in the form

$$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$$

The length of the plate is 1.2 m and width 0.9 m. If laminar boundary layer exists upto a value of $Re = 2 \times 10^5$ and kinematic viscosity of air = 0.15 stokes, find: (i) The maximum distance from the leading edge upto which laminar boundary layer exists, and (ii) The maximum thickness of boundary layer.

[Ans. (i) 0.3 m; (ii) 3.67 mm]

6. A plate of length 500 mm and width 250 mm has been placed longitudinally in a stream of crude oil which flows with a velocity of 6 m/s. If the oil has a specific gravity of 0.9 and kinematic viscosity of 1 stoke, calculate:

- (i) Boundary layer thickness at the middle of plate,
(ii) Shear stress at the middle of plate, and
(iii) Friction drag on one side of the plate.

[Ans. (i) 10.5 mm, (ii) 87.8 N/m², (iii) 12.36 N]

7. Atmospheric air at 20°C is flowing parallel to a flat plate at a velocity of 3 m/s. Assuming cubic velocity profile and using exact Blasius solution estimate the boundary layer thickness and the local co-efficient of drag at $x = 1$ m from the leading edge of the plate. Also find the deviation of the approximate solution from the exact solution.

[Ans. 11.376 mm; 1.511×10^{-3} ; 7.2%, 2.78%]

8. Air is flowing over a plate 4 m $\times$ 2 m with a velocity of 5 m/s at 15°C. If $\rho = 1.208$ kg/m³ and $\nu = 1.47 \times 10^{-5}$ m²/s, calculate:

- (i) Length of plate over which the boundary layer is laminar and thickness of the laminar boundary layer,
(ii) Shear stress at the location where boundary layer ceases to be laminar, and
(iii) Total force on both sides on that portion of plate where boundary layer is laminar.

[Ans. (i) 1.47 m; 10.39 mm;

(ii) 0.01418 N/m²; (iii) 0.1662 N]

9. A submarine can be assumed to have cylindrical shape with rounded nose. Assuming its length to

be 55 m and diameter, 6.0 m, determine the total power required to overcome boundary friction if it cruises at 8.0 m/s velocity in water at 20°C. Take $\rho = 1030$ kg/m³; $\nu = 1 \times 10^{-6}$ m²/s

[Ans. 476.5 kW]

10. A 2 m wide and 5.0 m long plate when towed through water at 20°C experiences a drag of 30.38 N on both the sides. Determine the velocity of the plate and the length over which the boundary layer is laminar.

[Ans. 1.0 m/s; 0.5 m]

11. If the velocity distribution in laminar boundary layer is given by $\frac{u}{U} = \frac{y}{\delta}$, obtain values/

expressions for $\frac{\theta}{\delta}$, $\frac{\delta^*}{x}$, $\frac{\delta^*}{\theta}$

[Ans. 0.333, $\frac{3.64}{\sqrt{Re_x}}$, 2.998]

12. Assume velocity distribution in laminar boundary layer over a flat plate to follow the law:

$$\frac{u}{U} = \sin \left(\frac{\pi y}{2 \delta} \right)$$

Obtain expressions for $\frac{\delta^*}{\theta}$ and $\frac{\delta}{x}$.

[Ans. $\frac{2(\pi - 2)}{(4 - \pi)}$, $\frac{4.80}{\sqrt{Re_x}}$]

13. A plate 300 mm $\times$ 100 mm is immersed in a liquid of density 998 kg/m³ and kinematic viscosity 1×10^{-6} m²/s. The water is moving with a velocity of 15.0 m/s parallel to it. Calculate:

- (i) Drag force on that portion of the plate over which the boundary layer is laminar,
(ii) Total drag force on both sides of plate.

[Ans. (i) 1.39 N; (ii) 20.73 N]

14. A passenger ship of 300 m length and 12 m draft is travelling at 45 km/h. Assuming the ship's surface to act as flat plate, determine:

- (i) The total friction drag, and
(ii) The power required to overcome this resistance.

Take $\rho = 1000$ kg/m³ and $\nu = 1 \times 10^{-6}$ m²/s.

[Ans. (i) 751.68 kN, (ii) 9396 kW]

15. The average drag co-efficient for turbulent layer flow past a thin plate is given by:

$$C_D = \frac{0.455}{(\log_{10} Re_L)^{2.58}}$$

where Re_L is the Reynolds number based on plate length. A plate 500 mm wide and 5 m long is kept parallel to the flow of water with free stream velocity 3 m/s. Calculate the drag force on both sides of the plate. Total $\nu = 0.01$ stoke. [Ans. 63.37 N]

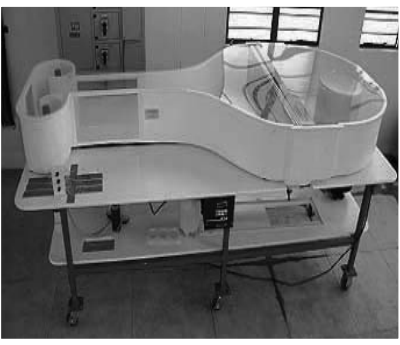
16. A streamlined train is 250 m long with a typical cross-section having a perimeter of 8 m above the wheels. If the kinematic viscosity of air at the prevailing temperature is $1.5 \times 10^{-5} \text{ m}^2/\text{s}$ and density is 1.24 kg/m^3 determine the appropriate surface drag (friction drag) of the train when running at 80 km/h. Make allowance for the fact that boundary layer changes from laminar to turbulent on the train surface. [Ans. 1088.5 N]

17. A barge with a rectangular bottom surface 25 m long $\times$ 8 m wide is travelling down a river with a velocity of 0.5 m/s. A laminar boundary layer exists upto Reynolds number equivalent to 5×10^5 and subsequently abrupt transition occurs to turbulent boundary layer. Calculate:

- (i) The maximum distance from the leading edge upto which laminar boundary layer persists and maximum boundary layer thickness at that point,
- (ii) The total drag force on the flat bottom surface of the barge, and
- (iii) The power required to push the bottom surface through water, at the given velocity.

For water $\rho = 998 \text{ kg/m}^3$ and $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$.

[Ans. (i) 1 m; 7.07 mm (ii) 69.68 N; (iii) 34.84 W]



FLOW AROUND SUBMERGED BODIES-DRAG AND LIFT

14.1. Introduction.

14.2. Force exerted by flowing fluid on a body.

14.3. Expression for drag and lift.

14.4. Dimensional analysis of drag and lift.

14.5. Streamlined and bluff bodies.

14.6. Drag on a sphere—terminal velocity of a body—applications of Stoke's law.

14.7. Drag of a cylinder.

14.8. Circulation and lift on a circular cylinder—flow patterns and development of lift—Position of stagnation points—Pressure at any point on the cylinder surface—expression for lift on the cylinder—expression for lift co-efficient for rotating cylinder—Magnus effect.

14.9. Lift on an airfoil.

Highlights

Objective type Questions

Theoretical questions

Unsolved examples.

14.1. INTRODUCTION

In various engineering fields we encounter with the problems which involve the flow of fluid around submerged bodies/objects. In such problems either a fluid may be flowing around a stationary submerged body or a body may be flowing through a large mass of stationary fluid or both the body and the fluid may be in motion. Some of the *examples* are:

- (i) Motion of very small objects/bodies such as fine sand particles in air or water,
- (ii) Very large bodies such as airplanes, submarines, automobiles, ships etc. moving through air or water, and
- (iii) The structures such as buildings, bridges etc. which are submerged in air or water,

14.2. FORCE EXERTED BY A FLOWING FLUID ON A BODY

Whenever there is relative motion between a real fluid and a body, the fluid exerts a force on the body. The body exerts an equal and opposite force on the fluid. If the body is moving at a constant velocity in a stationary fluid, the fluid motion is unsteady, because at a given point in space, the velocity changes with time. However if the body is stationary and fluid flows at a constant velocity, it is steady motion. *The magnitude of the force is same in both the cases.*

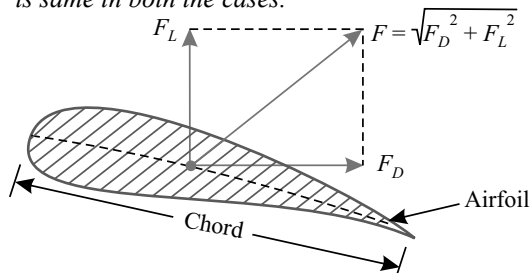
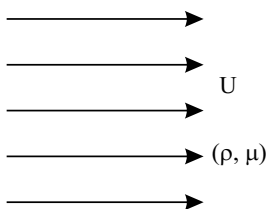


Fig. 14.1. Lift and drag on an airfoil.

A body wholly immersed in a real fluid may be subjected to flowing two kinds of forces due to relative motion between the body and the fluid (Fig. 14.1); these are:

- (i) **Drag force.** *The component of force in the direction of flow (free stream) on a submerged body is called the drag force, F_D ;*
- (ii) **Lift force.** *The component of force at right angles to the direction of flow is called the lift force, F_L .*

When a free stream approaches the body *along* the axis of symmetry, the force acting on the body is only the drag force, in the direction of flow and there is no lift force. The *production of lift force requires asymmetry of flow, while drag force exists always. It is possible to create drag without lift but impossible to create lift without drag.*

The fluid *viscosity* affects the flow around the body in three ways to cause the force on the body:

- (i) At low Reynolds number (Re) the fluid is deformed in very wide zone around the body causing pressure force and frictional force.
- (ii) As Reynolds number increases, viscous effects are confined to the boundary layers causing predominantly only friction force on the boundary.
- (iii) For certain body shapes, the boundary layer can separate causing additional pressure.

14.3. EXPRESSIONS FOR DRAG AND LIFT

Fig. 14.2 shows a body held stationary in a stream of real fluid moving at a uniform velocity, U .

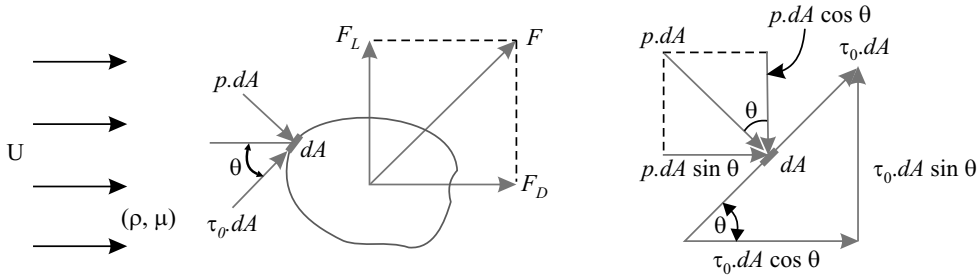


Fig. 14.2. Pressure and frictional forces on an elementary surface of an immersed body.

On an element of area dA on the surface of the body, let P and τ represent the static pressure and shear stress, and let θ be the inclination of the tangent to the element with the direction of flow. The component of the force, due to p and τ , along the direction of motion is known as *Drag force F_D* , while the component perpendicular to the direction of motion is known as *Lift force F_L* . Considering the Fig. 14.2 (b), we can write:

$$F_D = \int p.dA \sin \theta + \int \tau_0.dA \cos \theta \quad (14.1)$$

$$F_L = \int_A \tau_0.dA \sin \theta - \int_A p.dA \cos \theta \quad (14.2)$$

where the symbol $\int_A$ represents the integration over the entire body surface.

— The term $\int_A p.dA \cos \theta$ is called **pressure drag**.

— The term $\int_A \tau_0.dA \sin \theta$ is called **friction drag** or **skin drag** or **shear drag**.

The contribution of *shear stresses to the lift may be neglected since shear stresses are small as compared to the pressure* and act in direction roughly perpendicular to F_L . For a body moving through a fluid of mass density ρ , at a uniform velocity U , the mathematical expression for the calculation of the drag and the lift may also be written as follows:

$$F_D = C_D A \frac{\rho U^2}{2} \quad (14.3)$$

$$F_L = C_L A \frac{\rho U^2}{2} \quad (14.4)$$

where,

C_D = Co-efficient of drag (dimensionless),

C_L = Co-efficient of lift (dimensionless),

ρ = Density of fluid,

U = Relative velocity of fluid w.r.t. the body

A = Some characteristic area.

- For calculating the drag force (F_D), usually the *area A* is taken as the *area projected on the plane perpendicular to the relative motion of the fluid*.
- For calculating the lift force (F_L), the area A is taken as the *projected area of the body on a plane at right angles to the direction of lift force*.
- In the case of airfoil, the projection is conventionally taken on the plane of the chord, *i.e.* the area of the wing itself, independent of its inclination to the direction of flow.

$$\text{Area, } (A) = \text{Span } (l) \times \text{mean chord } (c) \quad (14.5)$$

Examples of immersed bodies having drag and/or lift forces:

- (i) A tall chimney exposed to wind;
- (ii) Flow of water past a bridge pier;
- (iii) Flow of fluids past blades in fans, blowers, compressors, turbines etc.;
- (iv) Motion of aeroplanes, submarines, torpedoes etc.

Examples of bodies where both drag and lift forces are produced:

- (i) Propeller blades; (ii) Aerofoils;
- (iii) Hydrofiles; (iv) Rotating cylindrical bodies;
- (v) Kites etc.

Following points are worth noting:

1. In contrast to drag, the lift forces may exist even in ideal fluids by the presence of circulation.
2. Real fluids also require vortices or circulation around the body for producing lift.
3. In motion of airfoils with finite spans, there is another kind of drag force associated with the lift force, called the “*induced drag*”.

Pressure drag and friction drag:

The relative contribution of *pressure drag* $\left(\int p \cdot dA \cos \theta \right)$ and *friction drag* $\left(\int \tau_0 \cdot dA \sin \theta \right)$ to the total drag depends on the following:

- (i) Characteristics of fluid,
- (ii) Shape of body, and
- (iii) Orientation of the body immersed in the fluid.

- When a thin plate is placed *parallel* to the direction of flow (Fig. 14.3), the pressure drag will be zero ($\theta = 90^\circ$) and the total drag is *entirely due to shear stresses* (and thus equal to friction drag or shear drag).
- When the same plate is held with its axis *normal* to flow direction (Fig. 14.4), the friction drag will be zero ($\theta = 0$) and *the flow separates at the edge forming a turbulent wake behind the plate*. In this case the total drag will be *due to the pressure force only*.
- When the plate is held *at an angle* with the direction of flow, the total drag will be equal to the *sum of pressure drag and friction drag*.

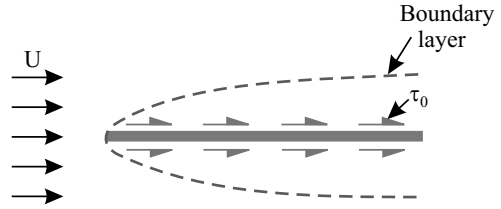


Fig. 14.3. Thin plate parallel to flow.

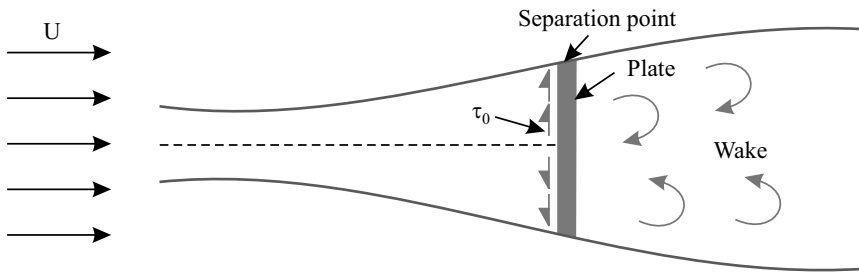


Fig. 14.4. Thin plate placed perpendicular to flow.

14.4. DIMENSIONAL ANALYSIS OF DRAG AND LIFT

For any given body shape, it is not possible to calculate the magnitude of the force, F , or its components F_D and F_L , theoretically. As such in almost all the cases the general practice is to evaluate forces experimentally. In order to plan the experiments properly and to analyse the results correctly, dimensional analysis of the problem is carried out as given below.

Let us consider an object of characteristic length L be placed in a fluid stream of velocity U , of density ρ , of viscosity μ and modulus of elasticity E . Then the force F exerted on the body could be written as :

$$F = f(L, \rho, \mu, E, U, g) \quad (14.6)$$

By adopting any of the methods of dimensional analysis eqn. (14.6) may be transformed to the following dimensionless form:

$$\frac{F}{\rho L^2 U^2} = f\left(\frac{\rho UL}{\mu}, \frac{U}{\sqrt{E/\rho}}, \frac{U}{\sqrt{gL}}\right)$$

But, $\frac{\rho UL}{\mu} = Re$, Reynolds number of flow

$\frac{U}{\sqrt{E/\rho}} = M$, Mach's number

$\frac{U}{\sqrt{gL}} = Fr$, Froude's number

$$\therefore \frac{F}{\rho L^2 U^2} = f(Re, M, Fr) \quad (14.7)$$

- At low speeds ($M \leq 0.3$) the change in density is insignificant and so the *effect of Mach's number is negligible*.
 - When the body is submerged fully in surrounding fluid the *Froude's number has no effect*.
- Under the above circumstances the eqn. (14.7) can be rewritten as:

$$\frac{F}{\rho L^2 U^2} = f(Re) \quad (14.8)$$

The parameter L^2 represents area of the body, and parameter $\frac{\rho U^2}{2}$ represents the dynamic pressure of the undisturbed flow stream.

Further, eqn. (14.8) applies equally to both lift and drag which can thus be expressed in dimensionless terms by the definition of drag and lift co-efficients as :

$$\text{Co-efficient of drag, } C_D = \frac{F_D}{\frac{1}{2} \rho U^2 \times A} \quad (14.9)$$

$$\text{Co-efficient of lift, } C_L = \frac{F_L}{\frac{1}{2} \rho U^2 \times A} \quad (14.10)$$

The co-efficients C_D and C_L are of paramount importance and are invariably used for *correlating aerodynamic lift forces*.

Example 14.1. A truck having a projected area of 6.5 m^2 travelling at 70 km/h has a total resistance of 2000 N . Of this 20 percent is due to rolling friction and 10 percent due to surface friction. The rest is due to form drag. Make calculations for the co-efficients of form drag. Take $\rho = 1.22 \text{ kg/m}^3$ for air. [PTU]

Solution. Projected area, $A = 6.5 \text{ m}^2$

$$\text{Speed of the truck, } U = 70 \text{ km/h} = \frac{70 \times 1000}{60 \times 60} = 19.44 \text{ m/s}$$

$$\text{Density of air, } \rho = 1.22 \text{ kg/m}^3$$

$$\text{Total resistance} = 2000 \text{ N}$$

$$\text{Resistance due to rolling friction} = \frac{20}{100} \times 2000 = 400 \text{ N}$$

$$\text{Resistance due to surface friction} = \frac{10}{100} \times 2000 = 200 \text{ N}$$

$$\text{Resistance due to form drag} = 2000 - 400 - 200 = 1400 \text{ N}$$

Co-efficient of form drag, C_D :

$$\text{Now, form drag} = C_D \times \frac{\rho U^2}{2} \times A$$

$$1400 = C_D \times \frac{1.22 \times (19.44)^2}{2} \times 6.5$$

$$\therefore C_D = \frac{1400 \times 2}{1.22 \times (19.44)^2 \times 6.5} = \mathbf{0.934 \text{ (Ans.)}}$$

Example 14.2. On a flat plate of 2 m (length) $\times$ 1 m (width), experiments were conducted in a wind tunnel with a wind speed of 50 km/h . The plate is kept at such an angle that the co-efficients of drag and lift are 0.18 and 0.9 respectively.

Determine:

- (i) Drag force,
- (ii) Lift force,
- (iii) Resultant force, and
- (iv) Power exerted by the air stream on the plate.

Take density of air = 1.15 kg/m^3

Solution. Area of the plate, $A = 2 \times 1 = 2 \text{ m}^2$

$$\text{Speed of wind, } U = 50 \text{ km/h} = \frac{50 \times 1000}{60 \times 60} = 13.89 \text{ m/s}$$

$$\text{Density of air, } \rho = 1.15 \text{ kg/m}^3$$

$$\text{Co-efficient of drag, } C_D = 0.18$$

$$\text{Co-efficient of lift, } C_L = 0.9.$$

(i) Drag force, F_D :

$$\begin{aligned} F_D &= C_D \times \frac{\rho U^2}{2} \times A \\ &= 0.18 \times \frac{1.15 \times 13.89^2}{2} \times 2 = 39.94 \text{ N (Ans.)} \end{aligned}$$

(ii) Lift force, F_L :

$$\begin{aligned} F_L &= C_L \times \frac{\rho U^2}{2} \times A \\ &= 0.9 \times \frac{1.15 \times 13.89^2}{2} \times 2 = 199.7 \text{ N (Ans.)} \end{aligned}$$

$$(iii) \text{ Resultant force, } F: \quad F = \sqrt{F_D^2 + F_L^2} = \sqrt{(39.94)^2 + (199.7)^2} = 203.65 \text{ N (Ans.)}$$

$$\text{Inclination with the velocity of air, } \theta = \tan^{-1} \frac{F_L}{F_D} = \tan^{-1} \frac{199.7}{39.94} = 78.69^\circ \text{ (Ans.)}$$

(iv) Power exerted by the air stream on the plate, P :

$$\begin{aligned} P &= F_D \times U \\ &= 39.94 \times 13.89 = 554.7 \text{ W (Ans.)} \end{aligned}$$

Example 14.3. Assuming the cross-sectional area of a passenger car to be 2.7 m^2 with a drag co-efficient of 0.6, estimate the energy requirement at a speed of 60 km/h. Assume the weight of car to be 30 kN and co-efficient of friction 0.012. Assume ρ to be 1.208 kg/m^3 .

Solution. Cross-sectional area of passenger car, $A = 2.7 \text{ m}^2$

$$\text{Co-efficient of drag, } C_D = 0.6$$

$$\text{Speed of the passenger car, } U = 60 \text{ km/h} = \frac{60 \times 1000}{60 \times 60} = 16.67 \text{ m/s}$$

$$\text{Weight of the car, } W = 30 \text{ kN}$$

$$\text{Co-efficient of friction, } \mu' = 0.012$$

Energy requirement, P :

Total resisting force, $(F) = \text{Aerodynamic drag on the car} + \text{friction at the road surface}$

$$\therefore F = C_D \frac{\rho U^2}{2} \times A + \mu' W$$

$$= 0.6 \times \frac{1.208 \times 16.67^2}{2} \times 2.7 + 0.012 \times (30 \times 1000)$$

$$= 217.9 + 360 = 631.9 \text{ N}$$

Now,

$$P = F \times U$$

$$= 631.9 \times 16.67 = 10553 \text{ W or } \mathbf{10.53 \text{ kW (Ans.)}}$$

(This is the power required at the wheels)

Example 14.4. The vertical component of the landing speed of a parachute is 6 m/s. Treat the parachute as an open hemisphere (Fig. 14.5) and determine its diameter if the total weight to be carried is 1200 N. Take $\rho = 1.208 \text{ kg/m}^3$ and $C_D = 1.33$.

Solution. Speed of parachute, $U = 6 \text{ m/s}$

Weight to be carried, $W = 1200 \text{ N}$

Density of air, $\rho = 1.208 \text{ kg/m}^3$

Co-efficient of drag, $C_D = 1.33$

Diameter of parachute, D :

Projected area of the hemispherical parachute,

$$A = \frac{\pi}{4} D^2$$

Drag force, $F_D = W = 1200 \text{ N}$

Using the equation:

$$F_D = C_D \times \frac{\rho U^2}{2} \times A, \text{ we get:}$$

$$1200 = 1.33 \times \frac{1.208 \times 6^2}{2} \times \frac{\pi}{4} D^2$$

or,

$$D^2 = \frac{1200 \times 2 \times 4}{1.33 \times 1.208 \times \pi \times 36} = 52.83$$

or,

$$D = \mathbf{7.72 \text{ m/s (Ans.)}}$$

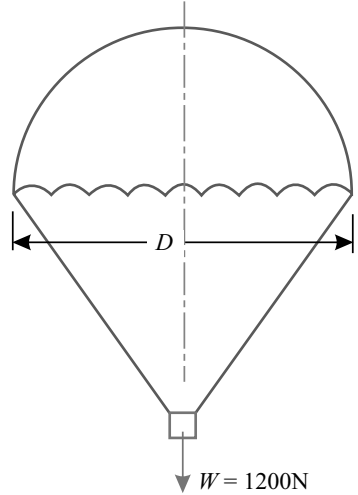


Fig. 14.5. Parachute.

Example 14.5. Experiments were conducted in a wind tunnel at 50 kmph on a flat plate of size $2 \text{ m} \times 1 \text{ m}$. The specific weight of air is 11.28 N/m^3 . The plate is kept at such an angle that the co-efficients of lift and drag are 0.75 and 0.15, respectively. Determine lift force, drag force, resulting force and power exerted by air stream on the plate. (Delhi University)

Solution. Given : $U = 50 \text{ kmph} = \frac{50 \times 100}{60 \times 60} = 13.89 \text{ m/s}; A = 2 \times 1 = 2 \text{ m}^2;$

$w = 11.28 \text{ N/m}^3; C_L = 0.75; C_D = 0.15$

$F_D, F_L, F_R, P:$

$$\rho = \frac{w}{g} = \frac{11.28}{9.8} = 1.15 \text{ kg/m}^3$$

$$\text{Lift force, } F_L = C_L \times \frac{\rho U^2}{2} \times A \quad [\text{Eqn. (14.3)}]$$

$$= 0.75 \times \frac{1.15 \times (13.89)^2}{2} \times 2 = \mathbf{166.4 \text{ N (Ans.)}}$$

$$\text{Drag force, } F_D = C_D \times \frac{\rho U^2}{2} \times A = 0.15 \times \frac{1.15 \times (13.89)^2}{2} \times 2 = \mathbf{33.28 \text{ N (Ans.)}}$$

$$\text{Resultant force, } F_R = \sqrt{F_L^2 + F_D^2} = \sqrt{(166.4)^2 + (33.28)^2} = \mathbf{169.69 \text{ N (Ans.)}}$$

$$\text{Inclination with the main stream, } \theta = \tan^{-1} \left(\frac{166.4}{33.28} \right) = \mathbf{78.69^\circ \text{ (Ans.)}}$$

Power exerted by the air stream on the plate,

$$P = F_D \times U = 33.28 \times 13.89 = 462.26 \text{ Nm/s} = \mathbf{462.26 \text{ W (Ans.)}}$$

Example 14.6. A kite weighing 9.8 N and having an area 1 m² makes an angle of 7° to horizontal when flying in a wind of 36 km/h. If pull on the string attached to the kite is 49 N and it is inclined to the horizontal at 45°, calculate the lift and drag co-efficients. Take ρ for air = 1.2 kg/m³. [Anna University]

Solution.

Weight of the kite = 9.8 N

Projected area of the kite, $A = 1 \text{ m}^2$

Angle made by the kite with the horizontal = 7°

Angle made by the string with the horizontal = 45°

Pull on the string, $P = 49 \text{ N}$.

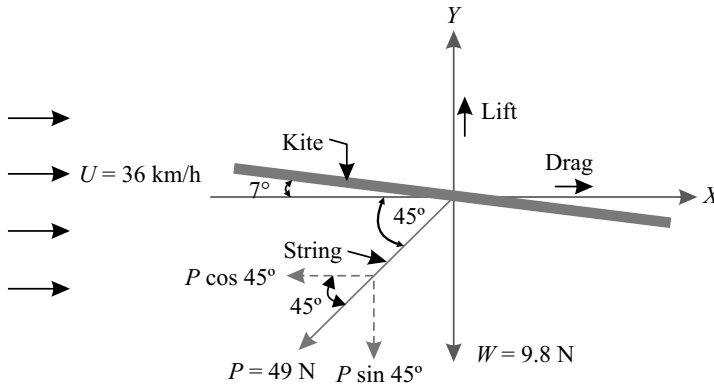


Fig. 14.6. Forces on a flying kite.

$$\text{Speed of the wind, } U = 36 \text{ km/h} = \frac{36 \times 1000}{60 \times 60} = 10 \text{ m/s}$$

$$\text{Density of air, } \rho = 1.0 \text{ kg/m}^3$$

The forces acting on the kite taken as free body are shown in Fig 14.6.

$$\begin{aligned} \text{Drag forces, } F_D &= \text{Force exerted by wind on the kite in the direction of motion,} \\ &\quad \text{i.e. in the X-direction} \\ &= \text{Components of pull along X- direction} \\ &= P \cos 45^\circ = 49 \cos 45^\circ = 34.65 \text{ N} \end{aligned}$$

Drag co-efficient, C_D :

$$\begin{aligned} F_D &= C_D \times \frac{\rho U^2}{2} \times A \\ 34.65 &= C_D \times \frac{1.2 \times 10^2}{2} \times 1 = 60 C_D \end{aligned}$$

or

$$C_D = \mathbf{0.577 \text{ (Ans.)}}$$

Lift co-efficient, C_L :

Lift force, F_L = Force exerted by wind on the kite perpendicular to the direction of motion i.e., along Y-direction

= Component of P in vertically downward direction + weight of kite

$$= P \sin 45^\circ + 9.8 = 49 \sin 45^\circ + 9.8 = 44.45 \text{ N}$$

Also,
$$F_L = C_L \times \frac{\rho U^2}{2} \times A$$

$$\therefore 44.45 = C_L \times \frac{1.2 \times 10^2}{2} \times 1 = 60 C_L$$

or,
$$C_L = \frac{44.45}{60} = \mathbf{0.741 \text{ (Ans.)}}$$

Example 14.7. A kite of dimensions $0.8 \text{ m} \times 0.8 \text{ m}$ and weighing 6 N is maintained in air at an angle of 10° to the horizontal. The string attached to the kite makes an angle at 45° to the horizontal and at this position, the drag and lift co-efficients are estimated to be 0.6 and 0.8 respectively. Determine;

(i) Wind speed, and (ii) Tension in the string.

Take ρ for air = 1.2 kg/m^3

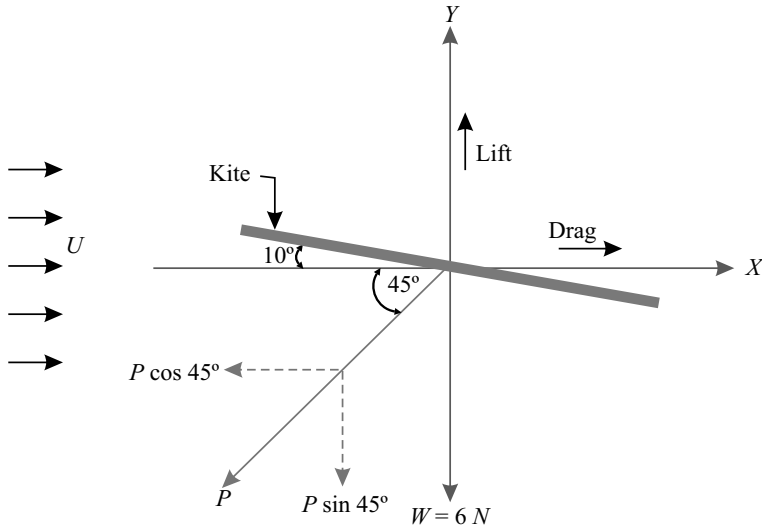


Fig. 14.7

Solution. Projected area of the kite, $A = 0.8 \times 0.8 = 0.64 \text{ m}^2$,

Weight of kite, $W = 6 \text{ N}$,

Angle made by kite with horizontal = 10° ,

Angle made by the string with horizontal = 45° ,

Co-efficient of drag, $C_D = 0.6$,

Co-efficient of lift, $C_L = 0.8$.

(i) Wind speed, U :

Drag force, F_D = Component of string force P in the X-direction
 $= P \cos 45^\circ$

Also,
$$F_D = C_D \times \frac{\rho U^2}{2} \times A$$

$$\therefore P \cos 45^\circ = 0.6 \times \frac{1.2U^2}{2} \times 0.64$$

$$\text{or, } P \cos 45^\circ = 0.2304 U^2 \quad (i)$$

$$\begin{aligned} \text{Lift force, } F_L &= \text{Component of string force } P \text{ vertically downwards} + \text{weight of kite} \\ &= P \sin 45^\circ + 6 \end{aligned}$$

$$\text{Also, } F_L = C_L \times \frac{\rho U^2}{2} \times A$$

$$\text{or, } P \sin 45^\circ + 6 = 0.8 \times \frac{1.2 \times U^2}{2} \times 0.64 = 0.3071 U^2$$

$$\text{or, } P \sin 45^\circ = 0.3071 U^2 - 6 \quad (ii)$$

$$\text{But, } P \sin 45^\circ = P \cos 45^\circ \quad (\because \sin 45^\circ = \cos 45^\circ)$$

$$\therefore 0.2304 U^2 = 0.3071 U^2 - 6 \quad [\text{equating eqns. (i) and (ii)}]$$

$$\text{or, } U^2 (0.3071 - 0.2304) = 6 \quad \text{or } U^2 = \frac{6}{0.3071 - 0.2304} = 78.125$$

$$\therefore U = 8.84 \text{ m/s} \quad \text{or } \frac{8.84 \times 3600}{1000} = 31.8 \text{ km/h (Ans.)}$$

(ii) Tension in the string, $T (= P)$:

Here tension in the string, $T = P$

Substituting the value of U in eqn (i), we get:

$$P \cos 45^\circ = 0.2304 U^2 = 0.2304 \times 78.125 = 18$$

$$P (= T) = \frac{18}{\cos 45^\circ} = 25.456 \text{ N (Ans.)}$$

Example 14.8. A sphere of 4 cm diameter made of aluminium (sp.gr. = 2.8) is attached to a string and suspended from the roof of a wind tunnel test section. If an air stream of 30 m/s flows past the sphere, find the inclination of the string and tension in the string.

$$\rho_a = 1.2 \text{ kg/m}^3$$

$$v_a = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$$

$$C_D = 0.5, 10^4 < Re \leq 3 \times 10^5, = 0.2, Re > 3 \times 10^5$$

(Neglect drag on string). (UPTU)

Solution. Given: Dia. of sphere,

$$D = 4 \text{ cm} = 0.04 \text{ m,}$$

$$\text{sp. gravity} = 2.8; U = 30 \text{ m/s; } \rho_a = 1.2 \text{ kg/m}^3;$$

$$v_a = 1.5 \times 10^{-5} \text{ m}^2/\text{s};$$

$$C_D = 0.5, 10^4 < Re \leq 3 \times 10^5, = 0.2, Re > 3 \times 10^5$$

θ, T :

$$\text{Weight} = \frac{4}{3} \pi R^3 \times \rho \times g = \frac{4}{3} \pi \times \left(\frac{0.04}{2} \right)^3 \times (2.8 \times 1000) \times 9.81 = 0.92 \text{ N}$$

$$Re = \frac{UD}{v_a} = \frac{30 \times 0.04}{1.5 \times 10^{-5}} = 0.8 \times 10^5$$

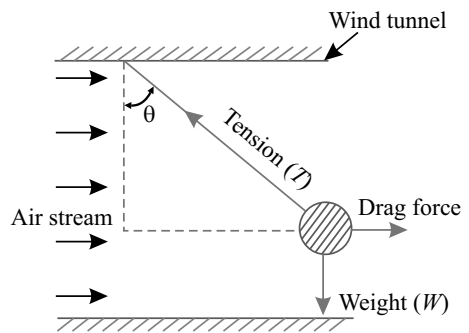


Fig. 14.8

∴

$$C_D = 0.5$$

$$\text{Drag force, } F_D = C_D \times \frac{\rho_a U^2}{2} \times A = 0.5 \times \frac{1.2 \times (30)^2}{2} \times \left[\pi \times \left(\frac{0.04}{2} \right)^2 \right]$$

$$= \mathbf{0.339 \text{ N (Ans.)}}$$

$$\tan \theta = \frac{\text{Drag force}}{\text{Weight}} = \frac{F_D}{W} = \frac{0.339}{0.92}$$

∴

$$\theta = \tan^{-1} \left(\frac{0.339}{0.92} \right) = \mathbf{20.22^\circ \text{ (Ans.)}}$$

$$\text{Tension in the string, } T = \sqrt{F_D^2 + W^2} = \sqrt{0.339^2 + 0.92^2} = \mathbf{0.98 \text{ N (Ans.)}}$$

Example 14.9. Air blows over a cylinder of diameter 60 mm and finite length with a velocity of 0.12 m/s. Find the total drag, shear drag and pressure drag on 1 m length of the cylinder if the total drag and shear drag co-efficients are 1.25 and 0.18 respectively. Take ρ for air = 1.25 kg/m³.

Solution. Diameter of cylinder, $D = 60 \text{ mm} = 0.06 \text{ m}$

Length of cylinder, $L = 1 \text{ m}$

Velocity of air, $U = 0.12 \text{ m/s}$

Total drag co-efficient, $C_{DT} = 1.25$

Shear drag co-efficient, $C_{DS} = 0.18$

Density of air, $\rho = 1.25 \text{ kg/m}^3$

Total drag,
$$F_{DT} = C_{DT} \times \frac{\rho U^2}{2} \times A$$

(where, $A = \text{projected area} = 1 \times 0.06 = 0.06 \text{ m}^2$)

or,
$$F_{DT} = 1.25 \times \frac{1.25 \times 0.12^2}{2} \times 0.06 = \mathbf{6.75 \times 10^{-4} \text{ N (Ans.)}}$$

Shear drag,
$$F_{DS} = C_{DS} \times \frac{\rho U^2}{2} \times A = 0.18 \times \frac{1.25 \times 0.12^2}{2} \times 0.06$$

$$= \mathbf{9.72 \times 10^{-5} \text{ N (Ans.)}}$$

Also, **Total drag = Pressure drag + shear drag**

∴ **Pressure drag = Total drag – shear drag**

$$= 6.75 \times 10^{-4} - 9.72 \times 10^{-5} = \mathbf{5.778 \times 10^{-4} \text{ N (Ans.)}}$$

Example 14.10. A 2.5 m long body having a projected area of 2.4 m² normal to the direction of motion, is moving through water which is having a viscosity of 0.0012 Ns/m². Find the drag on the body if it has drag co-efficient 0.45 for Reynolds number of 7×10^6 .

Solution. Length of body, $L = 2.5 \text{ m}$

Projected area of body, $A = 2.4 \text{ m}^2$

Viscosity of water, $\mu = 0.0012 \text{ Ns/m}^2$

Drag co-efficient, $C_D = 0.45$

Reynolds number, $Re = 7 \times 10^6$

Drag on the body, F_D :

Let us first find from the given Reynolds number the velocity with which the body is moving in water.

Now,
$$Re = \frac{\rho UL}{\mu}$$

or, $7 \times 10^6 = \frac{1000 \times U \times 2.5}{0.0012}$ or $U = \frac{7 \times 10^6 \times 0.0012}{1000 \times 2.5} = 3.36 \text{ m/s}$

Drag on the body (F_D) is given by:

$$F_D = C_D \times \frac{\rho U^2}{2} \times A = 0.45 \times \frac{1000 \times 3.36^2}{2} \times 2.4$$

$$= \mathbf{6096.4 \text{ N (Ans.)}}$$

Example 14.11. A cup anemometer shown in Fig. 14.9 rotates freely without air friction. Calculate the speed of rotation against a wind speed of 54 km/h. Take for hemisphere: For hollow upstream, $C_D = 1.33$; for hollow downstream, $C_D = 0.34$.

Solution. Wind speed,

$$U = 54 \text{ km/h} = \frac{54 \times 1000}{60 \times 60} = 15 \text{ m/s}$$

Speed of rotation, N (r.p.m.):

If the anemometer revolves at a uniform angular velocity ω , for steady rotation, net torque about the axis of rotation must be zero.

Fluid velocity relative to cup 1 = $15 - 0.2 \omega$

Fluid velocity relative to cup 2 = $15 + 0.2 \omega$

Corresponding drag forces on 1 and 2 are:

$$F_{D1} = 1.33 \times \frac{A\rho}{2} (15 - 0.2\omega)^2$$

$$F_{D2} = 0.34 \times \frac{A\rho}{2} (15 + 0.2\omega)^2$$

$$\left[\because F_D = C_D \times \frac{A\rho U^2}{2} \right]$$

Now, Torque = $(F_{D1} - F_{D2})r = 0$

or, $F_{D1} = F_{D2}$ ($\because r \neq 0$)

or, $1.33 \times \frac{A\rho}{2} (15 - 0.2\omega)^2 = 0.34 \times \frac{A\rho}{2} (15 + 0.2\omega)^2$

or, $\left(\frac{15 - 0.2\omega}{15 + 0.2\omega} \right)^2 = \frac{0.34}{1.33}$ or $\frac{15 - 0.2\omega}{15 + 0.2\omega} = 0.5056$

or, $15 - 0.2\omega = 0.5056 (15 + 0.2\omega) = 7.584 + 0.101\omega$

or, $0.301\omega = 7.146$, or $\omega = \frac{7.146}{0.301} = 24.64 \text{ rad/s.}$

But, $\omega = \frac{2\pi N}{60} = 24.64$

$\therefore N = \frac{24.64 \times 60}{2\pi} = \mathbf{235.3 \text{ r.p.m. (Ans.)}}$

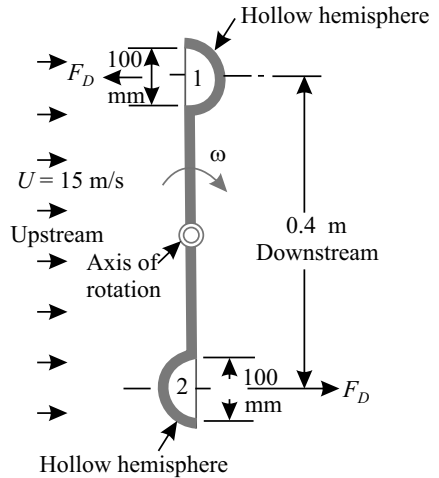


Fig. 14.9

Example 14.12. A mixer consists of two circular discs each 120 mm in diameter. These discs are spaced 1.2 m apart on the two ends of a horizontal rod whose centre has a vertical shaft attachment to it. The mixer is used to rotate in a solution having a density $\rho = 930 \text{ kg/m}^3$ and kinematic viscosity $\nu = 0.8 \text{ stoke}$. Neglecting the resistance of the rod and shaft, find the power required by the shaft revolving at 50 r.p.m. For $3000 < Re < 5000$, take $C_D = 1.15$.

Solution. Diameter of each circular disc, $D = 120 \text{ mm} = 0.12 \text{ m}$

Distance between the disc = 1.2 m

Density of solution, $\rho = 930 \text{ kg/m}^3$

Kinematic viscosity of the solution,

$$\nu = 0.8 \text{ stoke} = 0.8 \times 10^{-4} \text{ m}^2/\text{s}$$

Speed of the shaft, $N = 50 \text{ r.p.m.}$

Co-efficient of drag, $C_D = 1.15$

(for $3000 < Re < 5000$)

Power required, P :

Linear velocity of each disc,

$$U = \frac{\pi DN}{60} = \frac{\pi \times 1.2 \times 50}{60} = 3.14 \text{ m/s}$$

$$\text{Reynolds number, } Re = \frac{UD}{\nu}$$

$$= \frac{3.14 \times 0.12}{0.8 \times 10^{-4}} = 4710$$

As Re lies between 3000 and 5000, therefore, coefficient of drag, $C_D = 1.15$

$$\text{The drag on each disc, } F_D = C_D \times \frac{\rho U^2}{2} \times A$$

$$= 1.15 \times \frac{930 \times 3.14^2}{2} \times (\pi/4) \times 0.12^2 = 59.63 \text{ N}$$

$\therefore$ Torque produced by the drag on the two discs

$$= 2 F_D \times R = 2 \times 59.63 \times 0.6 = 71.55 \text{ Nm}$$

$$\text{Power required, } P = \frac{2\pi NT}{60} \text{ watts} = \frac{2\pi \times 50 \times 71.55}{60} = 374.6 \text{ W (Ans.)}$$

[Alternatively: $P = 2 \times F_D \times U = 2 \times 59.63 \times 3.14 = 374.5 \text{ W (Ans.)}$]

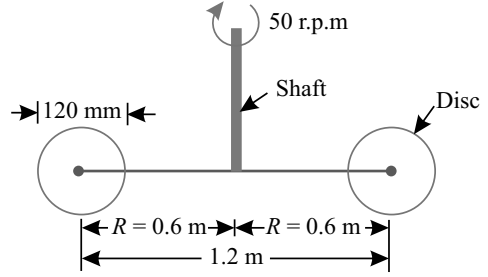


Fig. 14.10

Example 14.13. A ship is propelled by two cylindrical rotors each of diameter 2.5 m and length 7.5 m revolving at 150 r.p.m. about their axes which are horizontal. Estimate the force exerted upon the rotors in the direction of motion when the relative wind velocity is 40 km/h at an angle of 30° to the horizontal. Assume ρ for air as 1.22 kg/m^3 .

Solution. Diameter of each rotor, $D = 2.5 \text{ m}$

Length of each rotor, $L = 7.5 \text{ m}$

Speed of each rotor,

$N = 150 \text{ r.p.m.}$

Relative wind velocity,

$U = 40 \text{ km/h}$

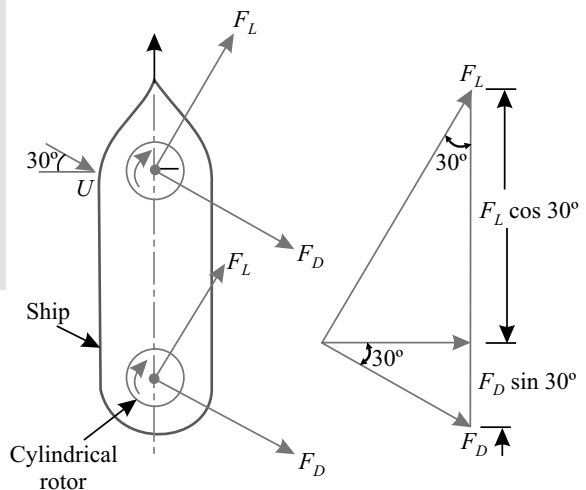


Fig. 14.11. Estimation of force on a ship propelled by two rotors.

$$= \frac{40 \times 1000}{600 \times 60} = 11.11 \text{ m/s}$$

Density of air, $\rho = 1.22 \text{ kg/m}^3$

Circumferential velocity of the rotors,

$$u_c = \frac{\pi DN}{60} = \frac{\pi \times 2.5 \times 150}{60} = 19.63 \text{ m/s}$$

$$\therefore \text{Ratio, } \frac{u_c}{U} = \frac{19.63}{11.11} = 1.77$$

From Fig. 14.20, the corresponding values of drag and lift co-efficients are:

$$C_L \approx 4.4 \quad \text{and} \quad C_D \approx 1.5$$

Lift force for each rotor,

$$F_L = C_L \times \frac{\rho U^2}{2} \times A = 4.4 \times \frac{1.22 \times 11.11^2}{2} \times (18.75) = 6211.7 \text{ N}$$

(where, A = Projected area = $7.5 \text{ m} \times 2.5 \text{ m} = 18.75 \text{ m}^2$)

Drag force for each rotor,

$$F_D = C_D \times \frac{\rho U^2}{2} \times A = 1.5 \times \frac{1.22 \times 11.11^2}{2} \times 18.75 = 2117.6 \text{ N}$$

Total force in the direction of motion,

$$\begin{aligned} F &= 2 (F_L \cos 30^\circ - F_D \sin 30^\circ) \\ &= 2 (6211.7 \cos 30^\circ - 2117.6 \sin 30^\circ) = \mathbf{8641 \text{ N (Ans.)}} \end{aligned}$$

14.5. STREAMLINED AND BLUFF BODIES

Streamlined body. A body whose surface coincides with the stream lines when placed in a flow, is called a **streamlined body** (Fig. 14.12). In this case flow separation takes place only at the trailing edge or rearmost part of the body. The wake formation zone behind a streamlined body is very small, as a consequence of which the pressure drag will be very small. In such a body although due to greater surface of the body the skin friction increases but the net effect is a significant reduction of total drag. A body may be streamlined at low velocities but may not be so at higher velocities, also when placed in a particular position in flow but may not be so when placed in another position.

Streamlined shapes are used for the wings of aeroplanes and for the blades of marine propellers and rotary axial flow machines.

Bluff body. A body whose surface does not coincide with streamlines when placed in a flow, is called a **bluff body** (Fig. 14.13). In this case there is extensive boundary layer separation accompanied by a wake with large scale eddies. Due to large wake formation, the resulting pressure drag is very large as compared to the drag due to friction on the body.

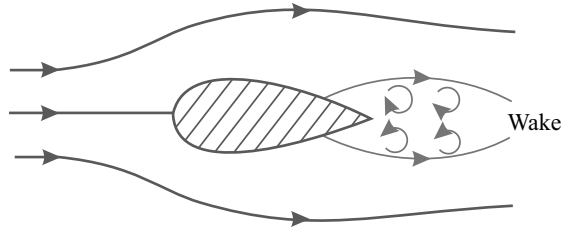


Fig. 14.12. Streamlined body.

14.6. DRAG ON A SPHERE

It has been observed that in the case of an ideal fluid flowing past a sphere (or any other object) there is no drag.

Let us consider a case when real fluid flows past a sphere. Let D be the diameter of the sphere, V be the velocity of flow of fluid of mass density ρ and viscosity μ .

- (i) **For $Re \leq 0.2$:** When the velocity of flow is very small or the fluid is very viscous such that the Reynolds number is very small, being as low as 0.2 $\left(i.e. Re = \frac{\rho U D}{\mu} \leq 0.2 \right)$ or even less then

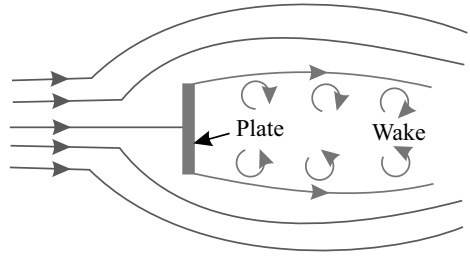


Fig. 14.13. Bluff body.

the viscous forces are much more predominant than the inertial forces. C.G. Stokes analysed theoretically the flow around a sphere under very low velocities, such that $Re < 0.2$. Stokes found that the total drag force is given by:

$$F_D = 3 \pi \mu D U \quad (14.11)$$

He further found out that out of the total drag given by eqn. (14.11), *two-thirds* is contributed by *skin friction* and *one-third* by the *pressure difference*. Thus:

$$\text{Skin friction drag} = \frac{2}{3} F_D = \frac{2}{3} \times 3 \pi \mu D U = 2 \pi \mu D U$$

$$\text{Pressure drag} = \frac{1}{3} F_D = \frac{1}{3} \times 3 \pi \mu D U = \pi \mu D U$$

Also, the total drag is given by:

$$F_D = C_D \times \frac{\rho U^2}{2} \times A \quad [\text{Eqn. (14.9)}]$$

$$(\text{where, } A = \text{projected area of the sphere} = \frac{\pi}{4} D^2)$$

From eqns. (14.11) and (14.9), we have:

$$\begin{aligned} 3 \pi \mu D U &= C_D \times \frac{\rho U^2}{2} \times \frac{\pi}{4} \times D^2 \\ \therefore C_D &= \frac{3 \pi \mu D U}{\frac{\rho U^2}{2} \times \frac{\pi}{4} D^2} = \frac{24 \mu}{\rho U D} = \frac{24}{Re} \end{aligned} \quad (14.12)$$

Eqn. (14.12) is generally called '**Stokes law**'.

- (ii) **For Re between 0.2 and 5:** Oseen made an improvement to the Stokes' solution by *partly* taking into account the *effect of inertial terms*. He found that

$$C_D = \frac{24}{Re} \left(1 + \frac{3}{16 Re} \right) \quad (14.13)$$

- (iii) **For $5 \leq Re \leq 1000$:** The value of C_D for Re between 5 to 1000 is equal to 0.4
 (iv) **For $1000 \leq Re \leq 100000$:** The value of C_D in the range is more or less independent of Reynolds number, and may be taken as 0.5.
 (v) **For $Re > 10^5$:** For Reynolds number greater than 10^5 the value of C_D is approximately equal to 0.2.

14.6.1. Terminal velocity of a body

The **terminal velocity** is the maximum velocity attained by a falling body. When a body is allowed to fall from rest in the atmosphere its velocity increases due to gravitational acceleration.

As the velocity increases the drag force (opposing the motion of the body) also increases. When the drag force becomes equal to the weight of the body, the acceleration ceases and the net external force acting in the body becomes zero and the body will move at constant speed (called terminal velocity).

The terminal velocity of a body falling through a liquid at rest is calculated from the following relation:

$$W = F_D + F_B \quad (14.14)$$

where,

W = Weight of the body, acting downward,

F_D = Drag force, acting vertically upward, and

F_B = Buoyant force, acting vertically upward.

The terminal velocity of a sphere falling through a liquid at rest is calculated as follows:

$$W = F_D + F_B$$

$$\frac{\pi}{6} D^3 \times w_s = 3\pi\mu DU + \frac{\pi}{6} D^3 \times w_f$$

where,	D = Diameter of the sphere, w_s = Specific weight of the material of sphere, w_f = Specific weight of the fluid, D = Diameter of the sphere, and U = Terminal velocity.
--------	---

or, $3\pi\mu DU = \frac{\pi}{6} D^3 (w_s - w_f)$

or, $U = \frac{D^2}{18\mu} (w_s - w_f) \quad (14.15)$

$$\left[\text{or, } \mu = \frac{D^2}{18U} (w_s - w_f) \right] \quad [14.15(a)]$$

14.6.2. Applications of Stokes' Law

The following are the *applications of Stokes' law*:

1. To calculate the terminal velocity of a falling sphere and hence the viscosity of the fluid.
2. Desilting river flow.
3. Separating the coolant from metal chips in machining operations.
4. Sanitary engineering—treatment of raw water and sewerage etc.

Example 14.14. A ball of 70 mm diameter is supported in a vertical air stream which is flowing at a velocity of 6.5 m/s. Calculate the weight of the ball. Take for air: $\rho = 1.25 \text{ kg/m}^3$ and $\nu = 1.4 \text{ stokes}$.

Solution. Diameter of the ball, $D = 70 \text{ mm} = 0.07 \text{ m}$.

Velocity of air, $U = 6.5 \text{ m/s}$

Density of air, $\rho = 1.25 \text{ kg/m}^3$

Kinematic viscosity of air,

$$\nu = 1.4 \text{ stokes} = 1.4 \times 10^{-4} \text{ m}^2/\text{s}.$$

Weight of the ball, W :

$$\text{Reynolds number, } Re = \frac{UD}{\nu} = \frac{6.5 \times 0.07}{1.4 \times 10^{-4}} = 3250$$

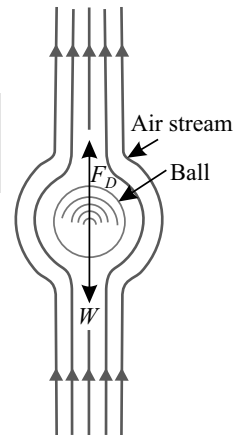


Fig. 14.14

Thus the value of Re lies between 1000 and 10000 and hence $C_D = 0.5$.

When the ball is just supported in stream, its weight is equal to the drag force (Fig. 14.14), neglecting buoyant force being very small.

But, Drag force, $F_D = C_D \times \frac{\rho U^2}{2} \times A$

where, $A =$ Projected area of the ball

$$= \frac{\pi}{4} \times D^2 = \frac{\pi}{4} \times 0.07^2 = 0.003848 \text{ m}^2$$

$$\therefore F_D = 0.5 \times \frac{1.25 \times 6.5^2}{2} \times 0.003848 = 0.0508 \text{ N}$$

Hence, weight of the ball, $W = F_D = 0.0508 \text{ N (Ans.)}$

Example 14.15. A steel sphere of 4 mm diameter falls in glycerine at a terminal velocity of 0.04 m/s. Assuming Stokes' law is applicable, determine:

- (i) Dynamic viscosity of glycerine,
- (ii) Drag force, and
- (iii) Drag co-efficient for the sphere.

Take specific weights of steel and glycerine as 75 kN/m^3 and 12.5 kN/m^3 respectively.

Solution. Diameter of the sphere, $D = 4 \text{ mm} = 0.004 \text{ m}$

Terminal velocity, $U = 0.04 \text{ m/s}$

Specific weight of steel, $w_s = 75 \text{ kN/m}^3$

Specific weight of glycerine, $w_f = 12.5 \text{ kN/m}^3$

(i) Dynamic viscosity of glycerine, μ :

$$\text{Weight of sphere, } W = \frac{\pi}{6} D^3 \times w_s = \frac{\pi}{6} \times (0.004)^3 \times (75 \times 10^3) = 0.002513 \text{ N}$$

$$\text{Buoyant force on sphere, } F_B = \frac{\pi}{6} D^3 \times w_f = \frac{\pi}{6} \times (0.004)^3 \times (12.5 \times 10^3) = 0.0004188 \text{ N}$$

$$\text{Drag force on sphere, } F_D = 3\pi\mu DV = 3\pi \times \mu \times 0.004 \times 0.04 = 0.001508 \mu \text{ N}$$

But, $W = F_D + F_B$

$$\therefore 0.002513 = 0.001508\mu + 0.0004188$$

$$\text{or, } \mu = \left(\frac{0.002513 - 0.0004188}{0.001508} \right) = 1.388 \text{ Ns/m}^2$$

$$\left(Re = \frac{\rho U D}{\mu} = \frac{12.5 \times 10^3 \times 0.04 \times 0.004}{9.81 \times 1.388} \right)$$

$$= 0.147; \text{ Since } Re < 0.2, \text{ therefore, the expression } F_D = 3\pi\mu DU \text{ is valid}$$

$$\left[\text{Also, } \mu = \frac{D^2}{18 U} (w_s - w_f) = \frac{0.004^2}{18 \times 0.04} (75 - 12.5) \times 10^3 = 1.388 \text{ Ns/m}^2 \right]$$

[Eqn. 14.15 (a)]

(ii) Drag force, F_D :

$$F_D = 3\pi\mu DU = 3\pi \times 1.388 \times 0.004 \times 0.04 = 0.00209 \text{ N (Ans.)}$$

(ii) Drag co-efficient for sphere, C_D :

$$C_D = \frac{24}{Re} \quad [\text{Eqn. 14.12}]$$

$$= \frac{24}{0.147} = 163.26 \quad (\text{Ans.})$$

Example 14.16. Determine the velocity of fall of rain drops of 0.3 mm diameter in atmospheric air having density 12 kg/m^3 and kinematic viscosity 0.15 stokes. Assume stokes' law holds good.

Solution. Diameter of the rain drop, $D = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$

Density of water (rain drops), $\rho = 1.2 \text{ kg/m}^3$

Kinematic viscosity, $\nu = 0.15 \text{ stokes} = 0.15 \times 10^{-4} \text{ m}^2/\text{s}$

$\therefore$ Dynamic viscosity, $\mu = \rho\nu = 1.2 \times 0.15 \times 10^{-4} = 1.8 \times 10^{-5} \text{ Ns/m}^2$

The terminal velocity of a spherical rain drop in laminar motion in air of large extent is given by (Stokes' law),

$$U = \frac{D^2}{18\mu} (w_s - w_f) \quad [\text{Eqn. (14.15)}]$$

where,

w_s = Specific weight of spherical rain drop ($= 9810 \text{ N/m}^3$), and

w_f = Specific weight of fluid (air) ($= 1.2 \times 9.81 \text{ N/m}^3$)

$$\therefore U = \frac{(0.3 \times 10^{-3})^2}{18 \times 1.8 \times 10^{-5}} (9810 - 1.2 \times 9.81) = 2.72 \text{ m/s} \quad (\text{Ans.})$$

Example 14.17. A 2 mm diameter metallic ball of specific gravity 11 is allowed to fall in a fluid of specific gravity 0.9 and viscosity 1.4 Ns/m^2 . Determine:

(i) Drag force (exerted by the fluid on the ball),

(ii) Pressure drag and skin friction drag, and

(iii) Terminal velocity of ball in fluid.

Solution. Diameter of the ball, $D = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

Specific gravity of metallic ball = 11

$\therefore$ Specific weight, $w_s = 11 \times 9.81 \text{ kN/m}^3 = 107.91 \text{ kN/m}^3$

Specific gravity of fluid, = 0.9

$\therefore$ Specific weight, $w_f = 0.9 \times 9.81 = 8.83 \text{ kN/m}^3$

Dynamic viscosity of fluid, $\mu = 1.4 \text{ Ns/m}^2$.

(i) Drag force, F_D :

$$\begin{aligned} \text{Weight of the ball, } W &= \frac{\pi}{6} D^3 \times w_s = \frac{\pi}{6} \times (2 \times 10^{-3})^3 \times (107.91 \times 10^3) \\ &= 4.52 \times 10^{-4} \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Buoyant force, } F_B &= (\pi/6) \times D^3 \times w_f = (\pi/6) \times (2 \times 10^{-3})^3 \times (8.83 \times 10^3) \\ &= 3.69 \times 10^{-5} \text{ N} \end{aligned}$$

When the ball attains the terminal velocity, we have:

$$W = F_D + F_B$$

$$\text{or, } F_D = W - F_B = 4.52 \times 10^{-4} - 3.69 \times 10^{-5} = 4.151 \times 10^{-4} \text{ N} \quad (\text{Ans.})$$

(ii) Pressure drag, skin friction drag:

$$\text{Pressure drag} = \frac{1}{3} F_D = \frac{1}{3} \times 4.151 \times 10^{-4} = 1.384 \times 10^{-4} \text{ N} \quad (\text{Ans.})$$

$$\text{Skin friction drag} = \frac{2}{3} F_D = \frac{2}{3} \times 4.151 \times 10^{-4} = 2.767 \times 10^{-4} \text{ N (Ans.)}$$

(iii) Terminal velocity of the ball U:

$$F_D = 3 \pi \mu D U \quad [\text{Eqn. (14.11)}]$$

$$\therefore 4.151 \times 10^{-4} = 3 \pi \times 1.4 \times 2 \times 10^{-3} \times U$$

$$\text{or,} \quad U = \frac{4.151 \times 10^{-4}}{3 \pi \times 1.4 \times 2 \times 10^{-3}} = 0.0157 \text{ m/s (Ans.)}$$

Now let us check the Reynolds number, Re .

$$Re = \frac{\rho U D}{\mu} = \frac{(8.83 \times 10^3) \times 0.0157 \times 2 \times 10^{-3}}{9.81 \times 1.4} = 0.02$$

Since $Re < 0.2$, therefore, the expression $F_D = 3 \pi \mu D U$ for the calculating the terminal velocity is valid,

Example 14.18. Stokes derived the drag F_D experienced by a sphere of diameter D moving at a uniform velocity U through a fluid of viscosity μ to be $F_D = 3 \pi \mu D U$. State the validity of this expression in relation to the particular Reynolds number. Derive the co-efficient of drag C_D from Stokes' law. [MDU, Haryana]

Solution. Stokes derived the expression for total drag,

$$F_D = 3 \pi \mu D U \quad (i)$$

on a sphere immersed in a flowing fluid for which Reynolds number is upto 0.2, so that inertia forces may be assumed negligible. In that case the various forces are much more important and predominant than inertia forces.

Drag is also given by the expression:

$$F_D = C_D \times A \times \rho \frac{U^2}{2} \quad (ii)$$

where, A (= projected area of sphere) = $\frac{\pi}{4} D^2$, and C_D = co-efficient of drag.

From (i) and (ii), we have:

$$3 \pi \mu D U = C_D \times \frac{\pi}{4} D^2 \times \frac{\rho U^2}{2}$$

$$\text{or,} \quad C_D = \frac{3 \pi \mu D U}{\frac{\pi}{4} D^2 \times \frac{\rho U^2}{2}} = \frac{24 \mu}{\rho U D} = \frac{24}{Re} \text{ (Ans.)}$$

Example 14.19. Determine the largest diameter and corresponding terminal velocity of a polystyrene spherical particle settling in air. It obeys Stokes' law.

Density of polystyrene particle = 1047.9 kg/m^3

Density of air = 1.2 kg/m^3

Kinematic viscosity of air = $1.5 \times 10^{-5} \text{ m}^2/\text{s}$

(M.U.)

Solution. Given: $\rho_s = 1047.9 \text{ kg/m}^3$; $\rho_a = 1.2 \text{ kg/m}^3$, $\nu_a = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$

D, U :

The Stokes' law is valid upto $Re = 1.0$. For maximum size particle that obeys Stokes' law,

$$Re_{\max} = 1 = \frac{UD}{\nu_a} \quad \text{or} \quad U = \frac{\nu_a}{D}$$

Stokes' law would give: $U = \frac{D^2}{18\mu} (w_s - w_f)$ [Eqn. (14.15)]

(where, suffices s and f stand for sphere and fluid respectively)

$$\frac{v_a}{D} = \frac{D^2}{18\mu} (\rho_s \times g - \rho_a \times g)$$

$$\begin{aligned}\mu &= \mu_a = v_a \times \rho_a \\ &= 1.5 \times 10^{-5} \times 1.2 \\ &= 1.8 \times 10^{-5}\end{aligned}$$

Substituting the values, we get:

$$\frac{1.5 \times 10^{-5}}{D} = \frac{D^2}{18 \times 1.8 \times 10^{-5}} (1047.9 \times 9.81 - 1.2 \times 9.81)$$

or,
$$D = \left[\frac{1.5 \times 10^{-5} \times 18 \times 1.8 \times 10^{-5}}{9.81 (1047.9 \times 1.2)} \right]^{\frac{1}{3}}$$

$$= 7.793 \times 10^{-5} \text{ m} = \mathbf{0.0779 \text{ mm (Ans)}}$$

$\therefore$ Terminal Velocity, $U = \frac{v_a}{D} = \frac{1.5 \times 10^{-5}}{0.0779 \times 10^{-3}} = \mathbf{0.1926 \text{ m/s (Ans.)}}$

14.7. DRAG ON A CYLINDER

Consider a real fluid flowing past a cylinder of a diameter D and length L (length being perpendicular to the direction of flow), with a uniform velocity U .

(i) For $Re < 1$:

— When $Re < 0.2$, the inertia force is negligibly small as compared to viscous force and hence flow pattern will be *symmetrical*.

— Also, when $Re < 1$, $F_D \propto U$ and $C_D \propto \frac{1}{Re}$.

(ii) For $Re = 1$ to 2000; C_D decreases and attains a minimum value of 0.95 at $Re = 2000$.

(iii) For $Re = 2000$ to 3×10^4 : C_D increase and becomes 1.2 at $Re = 3 \times 10^4$.

(iv) For $Re = 3 \times 10^4$ to 3×10^5 : C_D decreases and its value becomes 0.3 at $Re = 3 \times 10^5$

(v) For $Re = 3 \times 10^6$: C_D increases and attains a value of 0.7 in the end.

14.8. CIRCULATION AND LIFT ON A CIRCULAR CYLINDER

14.8.1. Flow Patterns and Development of Lift

Case I. Stationary cylinder. Consider an ideal fluid flowing over a *stationary cylinder* of radius R , with a uniform velocity U . In this case flow pattern will be symmetrical (Fig.14.15) and the velocity u at any point on the surface of the cylinder is given by,

$$u_\theta = 2U \sin \theta \quad (14.16)$$

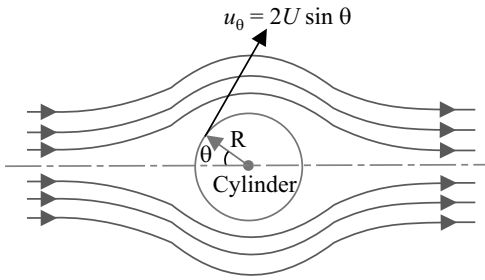


Fig. 14.15. Flow of ideal fluid over stationary cylinder.

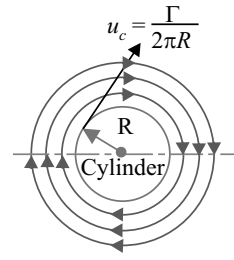


Fig. 14.16. Stream lines for free vortex.

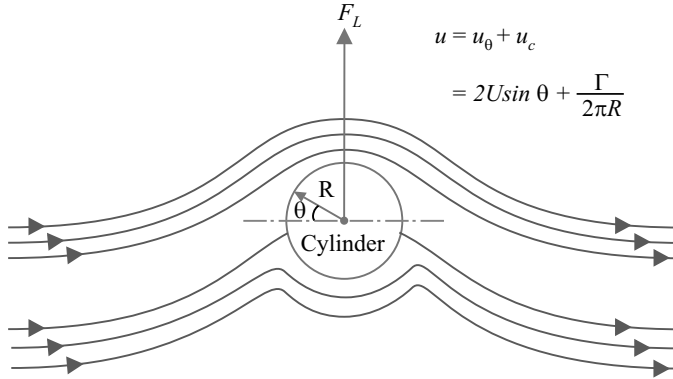


Fig. 14.17. Flow pattern over a rotating cylinder and development of lift on cylinder due to circulation.

where, θ = The angular distance of the point from the forward stagnation point.

In this case, since the flow pattern is symmetrical about the horizontal axis, the pressure distributions on the upper and lower halves of the cylinder are *identical*, and hence there is *no lift* acting on the cylinder.

Case II. Constant circulation imparted to the cylinder. When a constant circulation Γ is imparted to the same cylinder, the flow pattern around the cylinder consists of stream lines which are series of *concentric circles* (Fig. 14.16). The peripheral velocity on the surface of the cylinder due to circulation is given by,

$$u_c = \frac{\Gamma}{2\pi R} \quad (14.17)$$

Case III. Composite flow pattern. If the above two flow patterns are superimposed one over the other, then a composite flow pattern as shown in Fig. 14.17 will be obtained. The flow pattern is now unsymmetrical about the horizontal axis. The velocity at any point on the surface of the cylinder is obtained by adding eqns. (14.16) and (14.17) as:

$$u = u_\theta + u_c = 2U \sin \theta + \frac{\Gamma}{2\pi R} \quad (14.18)$$

As the circulation Γ has been taken as *clockwise*, the superimposition causes the velocity around the *upper half* portion of the cylinder to be *higher* than that around the lower half portion (this is so because around the upper half portion of the cylinder the velocity of flow and the velocity due to circulation being in the same direction are *added* together, while around the lower half portion of the cylinder both velocities being in opposite direction are *subtracted*). Hence on the lower half portion of the cylinder, where velocity is less, pressure will be more than the pressure on the upper half portion of the cylinder. As such a pressure force acts on the upward direction and obviously a *lift force* is exerted on the cylinder. However, *since the flow is symmetrical about the vertical axis, the cylinder is not subjected to any drag.*

14.8.2. Position of Stagnation Points

The **stagnation points** are those points on the surface of the cylinder, where velocity is zero. The velocity at any point on the surface of the rotating cylinder is given by eqn. 14.18 as:

$$u = 2U \sin \theta + \frac{\Gamma}{2\pi R}$$

For stagnation point, $u = 0$

$$\therefore 2U \sin \theta + \frac{\Gamma}{2\pi R} = 0$$

$$\text{or,} \quad \sin \theta = -\frac{\Gamma}{4\pi UR} \quad (14.19)$$

From this eqn. (14.19), we can find out the location of the stagnation points on the surface of the cylinder, as follows: (Fig. 14.18)

- (i) For Γ (circulation) = 0; Refer to Fig. 14.18 (i.) $\sin \theta = 0$ and $\theta = 0^\circ, 180^\circ$; S_1 and S_2 are the stagnation points [Fig. 14.18(i)]
- (ii) For $\Gamma < 4\pi RU$: $\sin \theta < -1$ and $\theta = <-90^\circ$ and $\theta > 180^\circ$; S_1 and S_2 are the stagnation points [Fig. 14.18(ii)].
- (iii) For $\Gamma = 4\pi RU$: $\sin \theta = -1$ and $\theta = -90^\circ$ and $\theta = 270^\circ$; the two stagnation points coincide and lie at the bottom of the cylinder, as stagnation point S [Fig. 14.18 (iii)].

By substituting for Γ from eqn. 14.17 in eqn. (14.19), we have:

$$\sin \theta = -1 = -\frac{2\pi R u_c}{4\pi UR} = -\frac{u_c}{2U} \text{ or } \frac{u_c}{U} = 2$$

- (iv) For $\Gamma > 4\pi RU$: $\sin \theta > -1$ which is not feasible. In this case stagnation points do not occur on the cylinder surface, they detach from the cylinder and lie into the fluid stream below the point -90° and 270° [Fig. 14.18 (iv)]

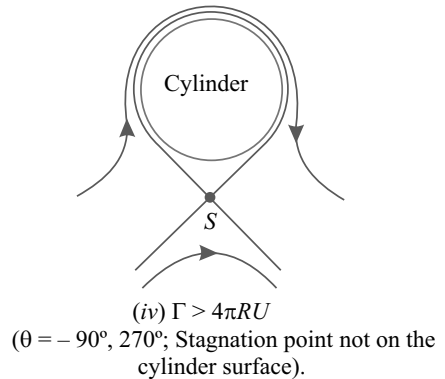
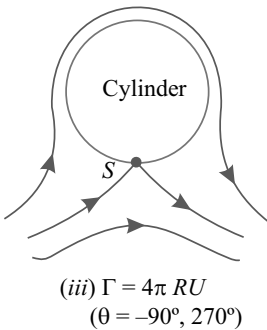
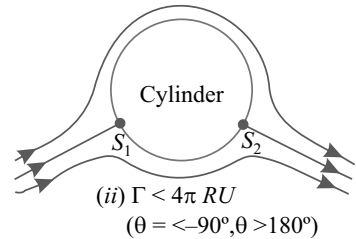
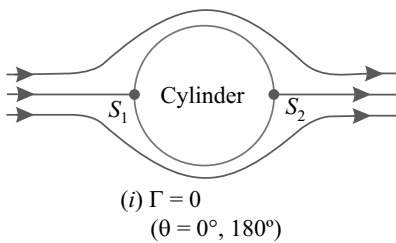


Fig. 14.18. Location of stagnation points.

14.8.3. Pressure at any Point on the Cylinder Surface

The magnitude of the lift exerted on the cylinder due to the composite flow pattern may be determined by integrating over the entire surface of the cylinder, the components of the pressure forces on elementary surface areas normal to the direction of uniform flow. The pressure at any point on the cylinder is obtained by applying Bernoulli's equation between any point in the unaffected flow (upstream condition) and any point on the surface of the cylinder. Thus,

$$p_0 + \frac{1}{2} \rho U^2 = p + \frac{1}{2} \rho u^2 \quad (14.20)$$

where,

P_0 = Pressure in the uniform flow at some distance ahead of cylinder ,

U = Velocity of uniform flow,

p = Pressure at any point on the cylinder, and

u = The velocity at any point on the surface of the cylinder
 $= 2U \sin \theta + \frac{\Gamma}{2\pi R}$.

Substituting the value of u in eqn. (14.20), we get:

$$p_0 + \frac{1}{2} \rho U^2 = p + \frac{1}{2} \rho \left(2U \sin \theta + \frac{\Gamma}{2\pi R} \right)^2$$

or,

$$p = p_0 + \frac{1}{2} \rho U^2 \left[1 - \left(2 \sin \theta + \frac{\Gamma}{2\pi UR} \right)^2 \right] \quad (14.21)$$

14.8.4. Expression for Lift on Cylinder (Kutta- Joukowski Theorem)

Consider a cylinder rotating in a uniform flow field. To determine the lift force on the cylinder, consider a small length of the element on the surface of the cylinder as shown in Fig. 14.19.

Let,

R = Radius of the cylinder,

ds = Length of the element ($= R d\theta$),

$d\theta$ = Angle made by the length ds at the centre of the cylinder,

p = Pressure on the surface of the element on cylinder,

p_0 = Pressure of the fluid (in the uniform flow) at some distance ahead of cylinder,

U = Velocity of uniform flow,

u = Velocity of fluid on the surface of the cylinder, and

L = Length of the cylinder.

Area of the small element *per unit length* of the cylinder, $dA = R d\theta \cdot L$

$\therefore$ Force acting on the element (directed towards the centre), $dF = p \cdot R d\theta \cdot L$

Resolving this force into the horizontal and vertical directions, we get the drag (dF_D) and lift (dF_L) components as follows:

$$dF_D = p \cdot R d\theta \cdot L \cdot \cos \theta; \quad dF_L = -p \cdot R d\theta \cdot L \sin \theta$$

By integrating the respective differential forces over the entire surface of the cylinder, we obtain the total drag and lift on the cylinder.

Thus,

$$F_L = - \int_0^{2\pi} p R L \sin \theta \cdot d\theta$$

But,

$$p = p_0 + \frac{1}{2} \rho U^2 \left[1 - \left(2 \sin \theta + \frac{\Gamma}{2\pi UR} \right)^2 \right] \quad [\text{Eqn. (14.21)}]$$

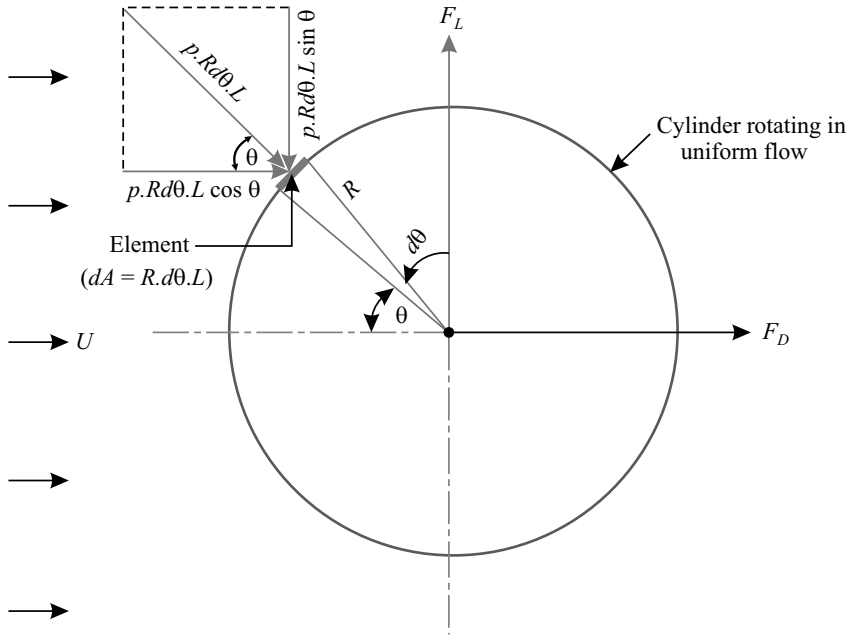


Fig. 14.19. Lift on a rotating cylinder.

$$\begin{aligned} \therefore F_L &= -R_L \left[\int_0^{2\pi} p_0 \sin \theta \cdot d\theta + \int_0^{2\pi} \frac{1}{2} \rho U^2 \sin \theta \cdot d\theta - \int_0^{2\pi} \frac{1}{2} \rho U^2 \left(2 \sin \theta + \frac{\Gamma}{2\pi UR} \right)^2 \sin \theta \cdot d\theta \right] \\ &= -RL \left[\int_0^{2\pi} p_0 \sin \theta \cdot d\theta + \int_0^{2\pi} \frac{1}{2} \rho U^2 \sin \theta \cdot d\theta - \frac{1}{2} \rho U^2 \left\{ \int_0^{2\pi} 4 \sin^3 \theta \cdot d\theta \right. \right. \\ &\quad \left. \left. + \int_0^{2\pi} \frac{2\Gamma}{\pi UR} \sin^2 \theta \cdot d\theta + \int_0^{2\pi} \frac{\Gamma^2}{4\pi^2 U^2 R^2} \sin \theta \cdot d\theta \right\} \right] \end{aligned}$$

But $\int_0^{2\pi} \sin^n \theta \cdot d\theta = 0$ when n is odd, therefore, the above expression reduces to:

$$\begin{aligned} F_L &= R.L \left(\frac{1}{2} \rho U^2 \times \frac{2\Gamma}{\pi UR} \right) \int_0^{2\pi} \sin^2 \theta \cdot d\theta \\ &= \frac{\rho L U \Gamma}{\pi} \int_0^{2\pi} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta \\ &= \frac{\rho U \Gamma}{\pi} \left[\frac{1}{2} \theta - \frac{\sin \theta}{2} \right]_0^{2\pi} = \rho U L \Gamma \text{ per unit length of cylinder.} \end{aligned}$$

Hence the total lift of a cylinder of length L is given by:

$$F_L = \rho L U \Gamma \quad (14.22)$$

Eqn. (14.22) is known as **Kutta-Joukowski equation**; it applies *not only to circular cylinder but also to other bodies of any shape* (including an airfoil) as well.

The resulting flow pattern for a rotating cylinder in a uniform flow field is shown in Fig. 14.17; it is symmetrical about the vertical axis of the cylinder. Hence the velocity distribution and pressure distribution is symmetrical about the vertical axis and as such *there will be no drag on the cylinder*

$\left(\text{i.e. } \int_0^{2\pi} p R d\theta \cdot L \cdot \cos \theta = 0 \right)$. The concept of zero drag on bodies immersed in a steady flow of ideal fluid is called **D'Alembert's paradox**.

14.8.5. Expression for Lift Coefficient for Rotating Cylinder

The lift coefficient (C_L) defined by eqn. (14.4) is given as:

$$F_L = C_L \frac{\rho U^2}{2} \times A \quad \text{or} \quad C_L = \frac{2F_L}{\rho U^2 A}$$

Also,

$$F_L = \rho L U \Gamma$$

Eqn. (14.22)

$\therefore$

$$C_L = \frac{2 \times \rho L U \Gamma}{\rho U^2 A} = \frac{2 \rho L U \Gamma}{\rho U^2 \times 2RL} = \frac{\Gamma}{UR} \quad (14.23)$$

(where,

$$A = \text{projected area} = 2RL)$$

From, eqn. (14.17), we have:

$$u_c = \frac{\Gamma}{2\pi R} \quad \text{or} \quad \frac{\Gamma}{R} = 2\pi u_c$$

Substituting this value of $\frac{\Gamma}{R}$ in eqn. (14.22), we get:

$$C_L = 2\pi \frac{u_c}{U} \quad (14.24)$$

(where, u_c = peripheral speed of the cylinder due to circulation)

Thus lift co-efficient depends on the ratio $\frac{u_c}{U}$.

Fig. 14.20 shows the variation of C_L and C_D with $\frac{u_c}{U}$ for a rotating circular cylinder in a fluid. The following observations are worth nothing:

1. For a circular cylinder rotating in an *ideal fluid*, the theoretical function

$$C_L = 2\pi \frac{u_c}{U} \text{ is shown as a broken line;}$$

when $\frac{u_c}{U} = 2$, the stagnation points

meet at the cylinder's bottom (Fig. 14.18 (iii)) and $C_L = 2\pi \times 2 = 4\pi = 12.56$ which is the *theoretical maximum* of the lift co-efficient.

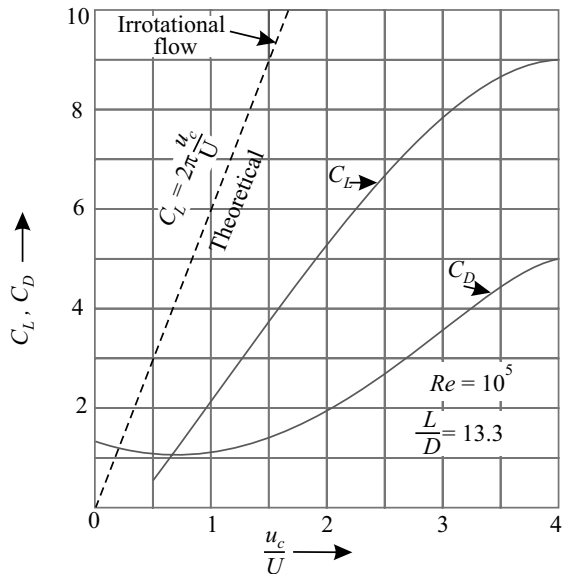


Fig. 14.20. C_L, C_D for a rotating cylinder.

2. For a circular cylinder rotating in a *real fluid* it can be seen that:

- (i) C_L becomes maximum $\frac{u_0}{U} = 4$ and C_D increases quite rapidly beyond $\frac{u_c}{U} = 1.5$.

The experimental values of C_L differ from the theoretical values due to the following factors:

- (a) Effect of viscosity,
- (b) Circulation around a rotating cylinder not being exactly the same as that due to an irrotational vortex, and
- (c) Effect of length.

- (ii) For a short cylinder, $\frac{L}{D} < 10$, C_L is reduced;

For $\frac{L}{D} < 5$, C_L is about half of that for a long cylinder, due to flow around ends.

14.8.6. Magnus Effect

The generation of lift by spinning cylinder in a fluid stream is called **Magnus effect**. The phenomenon of the lift produced by circulation around a cylinder of circular cross-section placed in a uniform stream of fluid, was first investigated experimentally by a German Physicist H.G. Magnus in 1852 and hence the name is given as *Magnus effect*.

- This effect has been successfully employed in the *propulsion of ships*.
- The Magnus effect may also be used with advantage in the *games like table tennis, golf, cricket etc.*

Example 14.20. A cylinder 1.8 m in diameter and 12 m long rotates at 240 r.p.m. with its axis perpendicular to the stream of water flowing at a velocity of 15 m/s. Assuming no slip between the cylinder and the circulating flow, determine:

- (i) The circulation,
- (ii) The theoretical lift,
- (iii) The position of stagnation points, and
- (iv) The r.p.m. of the cylinder for a single stagnation point.

Solution. Diameter of cylinder, $D = 1.8$ m

Length of cylinder, $L = 12$ m

Speed of rotation of cylinder, $N = 240$ r.p.m.

Velocity of water, $U = 15$ m/s

- (i) The circulation, Γ :

Velocity at the surface of the cylinder due to circulation alone,

$$u_c = \frac{\pi DN}{60} = \frac{\pi \times 1.8 \times 240}{60} = 22.6 \text{ m/s}$$

$$\text{Circulation, } \Gamma = 2\pi R u_c = \pi D u_c = \pi \times 1.8 \times 22.6 = \mathbf{127.8 \text{ m}^2/\text{s}}$$

- (ii) The theoretical, lift:

The theoretical lift, $F_L = \rho L U \Gamma$

$$= 1000 \times 12 \times 22.6 \times 127.8 = \mathbf{34.66 \times 10^6 \text{ N (Ans.)}}$$

- (iii) The position of stagnation points:

Net velocity on the cylinder surface due to combination of circulation and free stream velocity field is,

$$u = 2U \sin \theta + \frac{\Gamma}{2\pi R}$$

At stagnation points, $u = 0$

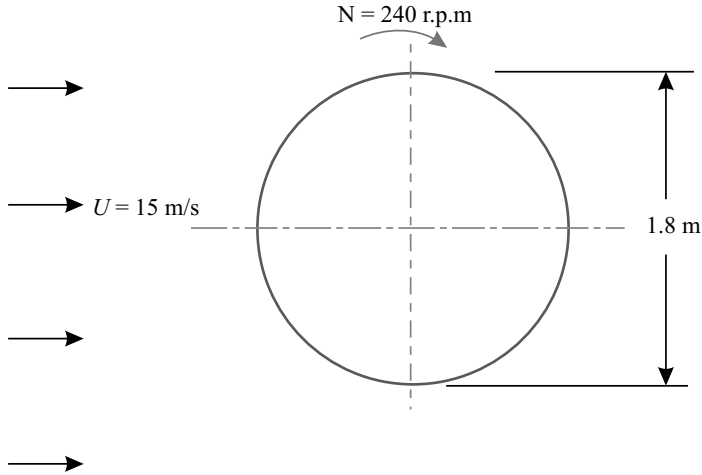


Fig. 14.21

$$2U \cdot \sin \theta + \frac{\Gamma}{2\pi R} = 0 \quad \text{or} \quad 2U \sin \theta = -\frac{\Gamma}{2\pi R}$$

or, $\Gamma = -4\pi RU \sin \theta$

or, $\sin \theta = -\frac{\Gamma}{4\pi RU} = -\frac{127.8}{4\pi \times 0.9 \times 15} = -0.753$

$$= -\sin (48.85^\circ)$$

$$= \sin (180^\circ + 48.85^\circ) \text{ and } \sin (360^\circ - 48.85^\circ)$$

$\therefore \theta = (180^\circ + 48.85^\circ) \text{ and } (360^\circ - 48.85^\circ)$

$$= 228.85^\circ \text{ and } 311.15^\circ \text{ (Ans.)}$$

The position of stagnation points is shown in Fig. 14.22

(iv) The r.p.m. of the cylinder for single stagnation point, N:

For a single stagnation point, we have:

$$\Gamma = 4\pi UR = 4\pi \times 15 \times 0.9 = 169.65 \text{ m}^2/\text{s}$$

Also, $u_c = \frac{\Gamma}{2\pi R}$

$$= \frac{169.65}{2\pi \times 0.9} = 30 \text{ m/s}$$

But, $u_c = \frac{\pi DN}{60}$ or $N = \frac{60u_c}{\pi D} = \frac{60 \times 30}{\pi \times 1.8}$

$$= 318.3 \text{ r.p.m. (Ans.)}$$

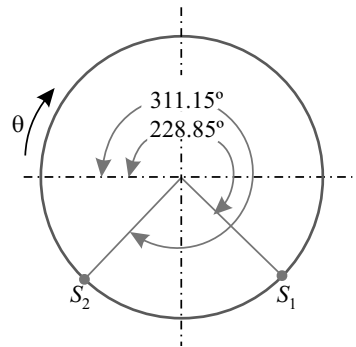


Fig. 14.22

Example 14.21. A cylinder whose axis is perpendicular to the stream of air having a velocity of 20 m/s rotates at 300 r.p.m. The cylinder is 2 m in diameter and 10 m long. Find:

- The circulation,
- The theoretical lift force per unit length,
- The position of stagnation points, and
- The actual lift, drag and direction of resultant force.

Take density of air = 1.24 kg/m^3 . For actual drag and lift, take $C_L = 3.4$, $C_D = 0.65$ and $\frac{u_c}{U} = 1.57$.

(v) Find also the speed of rotation of the cylinder, which will give only a single stagnation point. [PTU]

Solution. Velocity of air, $U = 20 \text{ m/s}$

Speed of rotation, $N = 300 \text{ r.p.m.}$

Diameter of cylinder, $D = 2 \text{ m}$

Length of cylinder, $L = 10 \text{ m}$

Density of air, $\rho = 1.24 \text{ kg/m}^3$

Peripheral velocity due to cylinder rotation,

$$u_c = \frac{\pi DN}{60} = \frac{\pi \times 2 \times 300}{60} = 31.4 \text{ m/s}$$

(i) The circulation, Γ :

Circulation = Circumference $\times$ peripheral velocity

$$\text{or, } \Gamma = 2\pi R u_c \quad [\text{Eqn. (14.17)}]$$

$$\text{or, } \Gamma = 2\pi \times \left(\frac{2}{2}\right) \times 31.4 = 197.3 \text{ m}^2/\text{s (Ans.)}$$

(ii) Theoretical lift force per unit length:

Theoretical lift is given by:

$$F_L = \rho L U \Gamma \quad [\text{Eqn. (14.22)}]$$

$\therefore$ Theoretical lift per unit length

$$\begin{aligned} &= \frac{F_L}{L} = \frac{\rho L U \Gamma}{L} = \rho U \Gamma = 1.24 \times 20 \times 197.3 \\ &= 4893 \text{ N/m length (Ans.)} \end{aligned}$$

(iii) The position of stagnation points:

Net velocity on the cylinder surface (u) due to combination of circulation and free stream velocity field is given by:

$$u = 2U \sin \theta + \frac{\Gamma}{2\pi R}$$

At stagnation points, $u = 0$

$$\therefore 0 = 2U \sin \theta + \frac{\Gamma}{2\pi R}$$

$$\text{or, } \sin \theta = -\frac{\Gamma}{2\pi R U} = -\frac{197.3}{4\pi \times \left(\frac{2}{2}\right) \times 20}$$

$$= -0.785 = -\sin(51.72^\circ)$$

$$= \sin(180^\circ + 51.72^\circ) \quad \text{and} \quad \sin(360^\circ - 51.72^\circ)$$

$$\therefore \theta = 231.72^\circ \text{ and } 308.28^\circ \text{ (Ans.)}$$

The position of stagnation points is shown in Fig. 14.23.

(iv) The actual lift, drag and direction of resultant force:

For actual lift and drag, $C_L = 3.4$, $C_D = 0.65$ and $\frac{u_c}{U} = 1.57$

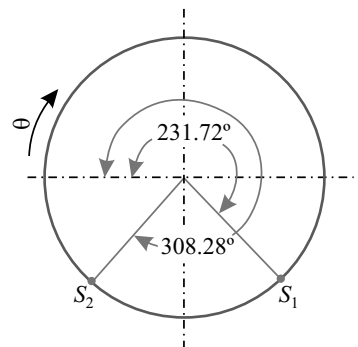


Fig. 14.23

The ratio of $\frac{u_c}{U}$ from theoretical consideration is given as:

$$\frac{u_c}{U} = \frac{31.4}{20} = 1.57$$

Now actual lift is given by:

$$F_L = C_L \times \frac{\rho U^2}{2} \times A = 3.4 \times \frac{1.24 \times 20^2}{2} \times 20 = \mathbf{16864 \text{ N (Ans.)}}$$

(where, A = projected area of cylinder = length $\times$ diameter = $10 \text{ m} \times 2 \text{ m} = 20 \text{ m}^2$)

Actual drag force,

$$F_D = C_D \times \frac{\rho U^2}{2} \times A = 0.65 \times \frac{1.24 \times 20^2}{2} \times 20 = \mathbf{3224 \text{ N (Ans.)}}$$

$$\text{Resultant force, } F = \sqrt{F_L^2 + F_D^2} = \sqrt{16864^2 + 3224^2} = \mathbf{17169.4 \text{ N (Ans.)}}$$

The inclination (α) of the resultant force with the horizontal is given by:

$$\tan \alpha = \frac{F_L}{F_D} = \frac{16864}{3224} = 5.23 \quad \text{or} \quad \alpha = \mathbf{79.2^\circ \text{ (Ans.)}}$$

(v) Speed of rotation of the cylinder for single stagnation point, N:

For a single stagnation point, we have:

$$\begin{aligned} \Gamma &= 4\pi UR \\ &= 4\pi \times 20 \times \left(\frac{2}{2}\right) = 251.32 \text{ m}^2/\text{s} \end{aligned}$$

Also,

$$u_c = \frac{\Gamma}{2\pi R} = \frac{251.32}{2\pi \times \left(\frac{2}{2}\right)} = 40 \text{ m/s}$$

But,

$$u_c = \frac{\pi DN}{60}$$

$\therefore$

$$40 = \frac{\pi \times 2 \times N}{60} \quad \text{or} \quad N = \frac{40 \times 60}{\pi \times 2} = \mathbf{382 \text{ r.p.m. (Ans.)}}$$

Example 14.22. Air having a velocity of 40 m/s is flowing over a cylinder of diameter 1.5 m and length 10 m , when the axis of the cylinder is perpendicular to the air stream. Find the speed at which the cylinder is to be rotated about its axis so that a lift force of 7 kN/m length of the cylinder is developed. Also determine the location of the stagnation points. Assume density of air as 1.25 kg/m^3 . (UPTU)

Solution. Given: $U = 40 \text{ m/s}$; $D = 1.5 \text{ m}$; $L = 10 \text{ m}$; $F_L = 7 \text{ kN/m}$ length; $\rho = 1.25 \text{ kg/m}^3$.

Speed N:

Using the relation: $F_L = \rho L U \Gamma$ [Eqn. (14.22)]

$$\text{or,} \quad \text{Circulation, } \Gamma = \frac{(F_L / L)}{\rho U} = \frac{(7 \times 1000)}{1.25 \times 40} = 140 \text{ m}^2/\text{s}$$

Circulation = Circumference $\times$ peripheral velocity

$$\text{i.e.} \quad \Gamma = 2\pi R \times u_c \quad \text{[Eqn. (14.17)]}$$

$$\text{or,} \quad 140 = 2\pi \times \frac{1.5}{2} \times u_c$$

or,

$$u_c = 29.71 \text{ m/s} = \omega R$$

 $\therefore$

$$\omega = \frac{u_c}{R} = \frac{29.71}{(1.5/2)} = 39.61 \text{ rad/s} = \frac{2\pi N}{60}$$

or,

$$N = \frac{39.61 \times 60}{2\pi} = \mathbf{378.2 \text{ r.p.m (Ans.)}}$$

Position of stagnation points :

The net velocity on the cylinder surface (u) due to combination of circulation and free stream velocity field is given by :

$$u = 2U \sin \theta + \frac{\Gamma}{2\pi R}$$

At stagnation point, $u = 0$

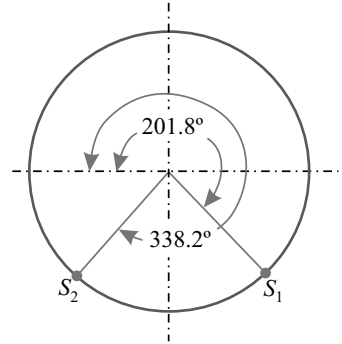
$$0 = 2U \sin \theta + \frac{\Gamma}{2\pi R}$$

$$\text{or, } \sin \theta = -\frac{\Gamma}{4\pi RU} = -\frac{140}{4\pi \times 0.75 \times 40} = -0.3714$$

$$= -\sin (21.8^\circ)$$

$$= \sin (180 + 21.8^\circ) \text{ and } \sin (360 - 21.8^\circ)$$

$$\theta = \mathbf{201.8^\circ \text{ and } 338.2^\circ \text{ (Ans.)}}$$

**Fig. 14.24**

The position of the stagnation points (S_1 and S_2) is shown in Fig. 14.24.

Example 14.23. As an application of the Magnus effect, a ship is built having two vertical rotors 10 m high and 3 m in diameter. The rotors are spun at 250 r.p.m. On a day when the air temperature is 20°C and the relative motion of the air to the ship results in 54 kmph wind, calculate the force emitted by the spinning rotors on the ship. Take ρ for air as 1.25 kg/m^3 . [IIT Madras]

Solution. Diameter of each rotor, $D = 3 \text{ m}$

Height of each rotor, $H (= L) = 10 \text{ m}$

Speed of each rotor, $N = 250 \text{ r.p.m.}$

$$\text{Wind velocity, } U = 54 \text{ km/h} = \frac{54 \times 1000}{60 \times 60} = 15 \text{ m/s}$$

Force emitted by the spinning rotors, F_L :

Circulation (Γ) generated by the rotation of cylinder is given by:

$$\Gamma = 2\pi R u_c \quad [\text{Eqn. (14.17)}]$$

where,

$$u_c = \frac{\pi D N}{60} = \frac{\pi \times 3 \times 250}{60} = 39.27 \text{ m/s}$$

 $\therefore$

$$\Gamma = 2\pi \times (3/2) \times 39.27 = 370.1 \text{ m}^2/\text{s}$$

Transverse force (perpendicular to the wind direction) developed by each rotating cylinder is given by:

$$F_L = \rho L U \Gamma$$

or,

$$\begin{aligned} F_L &= 1.25 \times 10 \times 15 \times 370.1 \\ &= 69393.75 \text{ N} \approx 69.4 \text{ kN} \end{aligned}$$

$\therefore$ Total force emitted by the spinning rotors

$$= 2 \times 69.4 = \mathbf{138.8 \text{ kN (Ans.)}}$$

14.9. LIFT ON AN AIRFOIL

An airfoil or aerofoil is a streamlined body which may be either symmetrical or unsymmetrical, as shown in Fig. 14.25.

Some of the *definitions* relating airfoil are given below:

1. **Chord line.** It is the line joining the leading and trailing edges of the airfoil. The length of the line is known as “chord of airfoil.”

2. **Profile centreline.** It is the line joining the midpoints of the profile.

3. **Angle of attack.** It is angle between the chordline and direction of the fluid stream.

4. **Camber.** It is the curvature of an airfoil.

5. **Stall.** An airfoil is said to be in *stall condition* when the angle of attack of an airfoil is *greater* than the angle of attack at maximum lift. At *stall* the air separates from the airfoil or wing and eddies are formed, as a consequence of which there is a considerable increase in the drag co-efficient.

6. **Aspect ratio (A.R.)** The ratio of span of the wing to its mean chord is called the **aspect ratio** of a wing.

$$\text{i.e.} \quad A.R. = \frac{l}{c} \quad (14.25)$$

where, l = Span of the wing, and
 c = Mean chord.

Normally the wings are not rectangle when viewed from the above ; in this case aspect ratio is given by:

$$A.R. = \frac{l}{c} = \frac{l}{A/l} = \frac{l^2}{A} \quad (14.26)$$

(when $A = l \times c$ or $c = A/l$)

— From the theoretical analysis, the circulation Γ developed on the airfoil so that the stream line at the trailing edge of the airfoil is tangential to the airfoil is given as:

$$\Gamma = \pi c U \sin \alpha \quad (14.26)$$

where,

c = Chord length,

α = Angle of attack, and

U = Free stream velocity of airfoil.

Also lift force, F_L is given as:

$$F_L = \rho L U \Gamma$$

$$\therefore F_L = \rho L U \times \pi c U \sin \alpha = \pi \rho c L U^2 \sin \alpha \quad (\because \Gamma = \pi c U \sin \alpha)$$

The lift force is also given by:

$$F_L = C_L \times \frac{\rho U^2}{2} \times A = C_L \times \frac{\rho U^2}{2} \times c \times L \quad (14.27)$$

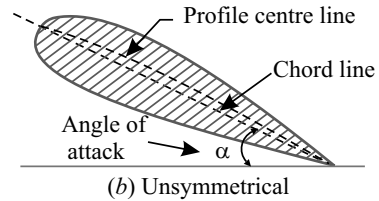
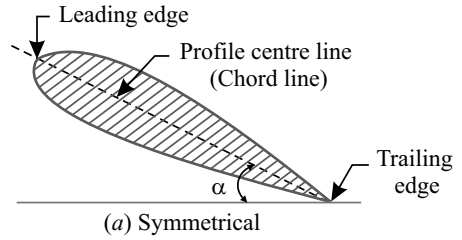


Fig. 14.25 Airfoil.

Equating the two values of lift force given by eqns. (14.26), and (14.27), we get:

$$\pi \rho c L U^2 \sin \alpha = C_L \times \frac{\rho U^2}{2} \times c \times L$$

$$\therefore C_L = \frac{\pi \rho c L U^2 \sin \alpha \times 2}{\rho U^2 \times c \times L} = 2\pi \sin \alpha$$

$$\text{i.e. } C_L = 2\pi \sin \alpha \quad (14.28)$$

From eqn. (14.28), we observe that co-efficient of lift depends upon the angle of attack. The actual lift co-efficient for an airfoil, in normal range of operation, is about 95 per cent of the theoretical value computed from the eqn. (14.28).

With real fluids, the airfoil creates its own circulation or vortex field in order to experience lift.

— When a flying object such as an airplane is in a *steady-state*, then:

$$(i) \text{ The weight of the airplane, } (W) = \text{The lift force} \left(C_L \times \frac{\rho U^2}{2} \times A \right),$$

$$(ii) \text{ The thrust developed by the engine} = \text{The drag force.}$$

Example 14.24. A jet plane weighing 24.5 kN and having a wing area of 16.7 m² flies at a velocity of 950 km/h. When the engine delivers 6125 kW, 65 percent of the power is used to overcome the drag resistance of the wing. Calculate the co-efficients of lift and drag for the wing. Take density of the atmospheric air = 1.208 kg/m³

Solution. Weight of the jet plane, $W = 24.5 \text{ kN}$

Wing area, $A = 16.7 \text{ m}^2$

$$\text{Velocity of the plane, } U = 950 \text{ km/h} = \frac{950 \times 1000}{60 \times 60} \approx 264 \text{ m/s.}$$

Power delivered by the engine = 6125 kW

Percentage of the power used to overcome the drag resistance = 65%

Density of atmospheric air, $\rho = 1.208 \text{ kg/m}^3$

Co-efficients of lift and drag, C_L and C_D :

Lift force, $F_L = \text{Weight of the jet plane}$

$$\text{or, } C_L \times \frac{\rho U^2}{2} \times A = 24.5 \times 10^3$$

$$\text{or, } C_L \times \frac{1.208 \times 264^2}{2} \times 16.7 = 24.5 \times 10^3 \text{ N}$$

$$\text{or, } C_L \times \frac{2 \times 24.5 \times 10^3}{1.208 \times 264^2 \times 16.7} = \mathbf{0.0348 \text{ (Ans.)}}$$

Power required to overcome drag resistance

$$= F_D \times U = 0.65 \times (6125 \times 10^3) \quad (\text{Given})$$

$$\text{But, } F_D \text{ (drag force)} = C_D \times \frac{\rho U^2}{2} \times A$$

$$\therefore C_D \times \frac{\rho U^2}{2} \times A \times U = 0.65 \times (6125 \times 10^3)$$

$$\text{or, } C_D \times \frac{1.208 \times 264^2}{2} \times 16.7 \times 264 = 0.65 \times (6125 \times 10^3)$$

$$\therefore C_D = \frac{2 \times 0.65 \times (6125 \times 10^3)}{1.208 \times 264^2 \times 16.7 \times 264} = \mathbf{0.0214 \text{ (Ans.)}}$$

Example 14.25. An aeroplane weighing 39.24 kN is flying in a horizontal direction at 360 km/h. The plane spans 15 m and has a wing surface area of 35 m². If drag co-efficient $C_D = 0.03$ and for air $\rho = 1.22 \text{ kg/m}^3$, determine:

- (i) Co-efficient of lift,
- (ii) Power required to drive the plane, and
- (iii) Theoretical value of the boundary layer circulation.

[Delhi University]

Solution. Weight of the aeroplane, $W = 39.24 \text{ kN}$

$$\text{Speed of the aeroplane, } U = 360 \text{ km/h} = \frac{360 \times 1000}{60 \times 60} = 100 \text{ m/s}$$

$$\text{Span of the aeroplane, } L = 15 \text{ m}$$

$$\text{Wing surface area, } A = 35 \text{ m}^2$$

$$\text{Co-efficient of drag, } C_D = 0.03$$

$$\text{Density of air, } \rho = 1.22 \text{ kg/m}^3.$$

(i) Co-efficient of lift, C_L :

For equilibrium in vertical direction, lift equals the weight of the aeroplane.

$$\therefore W = C_L \times \frac{\rho U^2}{2} \times A$$

$$\text{or, } 39.24 \times 10^3 = C_L \times \frac{1.2 \times 100^2}{2} \times 35$$

$$\text{or, } C_L = \frac{2 \times 39.24 \times 10^3}{1.2 \times 100^2 \times 35} = \mathbf{0.187 \text{ (Ans.)}}$$

(ii) Power required to drive the plane, P :

$$\begin{aligned} \text{Drag force, } F_D &= C_D \times \frac{\rho U^2}{2} \times A \\ &= 0.03 \times \frac{1.20 \times 100^2}{2} \times 35 = 6300 \text{ N} \end{aligned}$$

$$\text{Power required, } P = F_D \times U = 6300 \times 100 \times 10^{-3} \text{ kW} = \mathbf{630 \text{ kW (Ans.)}}$$

(iii) Theoretical value of boundary layer circulation, Γ :

$$\text{Lift force, } F_L = \rho L U \Gamma$$

$$\text{or, } 39.24 \times 10^3 = 1.22 \times 15 \times 100 \times \Gamma$$

$$\text{or, } \Gamma = \frac{39.24 \times 10^3}{1.22 \times 15 \times 100} = \mathbf{21.4 \text{ m}^2/\text{s} \text{ (Ans.)}}$$

Example 14.26. A wing of a small aeroplane is rectangular in plan having a span of 12 m and chord of 1.8 m. In a horizontal flight at 200 km/h the total aerodynamic force acting on the wing is 28 kN. If the lift-drag ratio is 10, determine:

- (i) Co-efficients of lift and drag,
- (ii) Total weight the aeroplane can carry, and

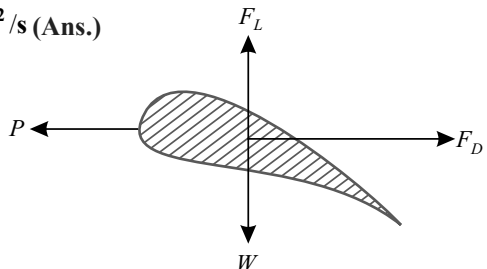


Fig. 14.26

(iii) Power required for the flight.

Take ρ for air = 1.2 kg/m^3

Solution. Span, $l = 12 \text{ m}$

Chord, $c = 1.8 \text{ m}$

Speed of the aeroplane,

$$U = 200 \text{ km/h} = \frac{200 \times 1000}{60 \times 60} = 55.55 \text{ m/s}$$

Aerodynamic force, $F_L = 28 \text{ kN}$

(i) Co-efficients of lift and drag:

For horizontal flight (Fig. 14.26):

$$F_L \uparrow = W \downarrow, F_D = C_D A \frac{\rho U^2}{2}, A = l \times c$$

$$\text{Now, } \frac{F_L}{F_D} = 10 \quad (\text{Given})$$

$$\therefore \frac{F_L}{F_D} = 10 = \frac{28 \times 1000}{C_D \times (12 \times 1.8) \times \frac{1.2 \times 55.55^2}{2}}$$

$$\text{or, } C_D = \frac{28 \times 1000 \times 2}{10 \times (12 \times 1.8) \times 1.2 \times 55.55^2} = \mathbf{0.07 \text{ (Ans.)}}$$

(ii) Total weight the aeroplane can carry, W :

$$W = F_L = \mathbf{28 \text{ kN (Ans.)}}$$

(iii) Power required for flight, P :

$$P = F_D \times U = \frac{F_L}{10} \times U = \frac{28}{10} \times 55.55 \quad \left(\because \frac{F_L}{F_D} = 10 \right)$$

$$= \mathbf{155.54 \text{ kN (Ans.)}}$$

HIGHLIGHTS

1. A body wholly immersed in a real fluid may be subjected to the following forces:
Drag force (F_D): It is the force exerted by fluid in the direction of flow (free stream).
Lift force (F_L): It is the force exerted by fluid at right angles to the direction of flow.
2. The mathematical expressions for F_D and F_L are:

$$F_D = C_D \times \frac{\rho U^2}{2} \times A \text{ and } F_L = C_L \times \frac{\rho U^2}{2} \times A$$

where, C_D = Co-efficient of drag,
 C_L = Co-efficient of lift,
 U = Free stream velocity of fluid,
 ρ = Density of fluid, and
 A = Projected area of the body.

$$\text{Resultant force, } F = \sqrt{F_D^2 + F_L^2}$$

3. Total drag on a body = Pressure drag + friction drag

4. A body whose surface coincides with the stream lines when placed in a flow, is called a *streamlined* body. If the surface of the body does not coincide with the streamlines, the body is called *bluff* body.

5. Stokes found out that for $Re < 0.2$ the total drag on a sphere is given by

$$F_D = 3\pi\mu DU; \text{ and of the total drag}$$

$$\text{Skin Friction drag} = \frac{2}{3} \times 3\pi\mu Du = 2\pi\mu DU, \text{ and}$$

$$\text{Pressure drag} = \frac{1}{3} \times 3\pi\mu DU = \pi\mu DU$$

6. For sphere, the values of C_D for different Reynolds number are:

Reynolds number (Re) C_D

(i) Less than 0.2	$\frac{24}{Re}$
(ii) Between 0.2 and 5.0	$\frac{24}{Re} \left(1 + \frac{3}{16Re} \right)$
(iii) Between 5 and 1000	0.4
(iv) Between 1000 and 10^5	0.5
(v) Greater than 10^5	0.2

7. The *terminal velocity* is the maximum velocity attained by a falling body. The terminal velocity of a body falling through a liquid at rest is calculated from the following relation:

$$W = F_D + F_B$$

where, F_D and F_B are the drag force and buoyant force respectively, acting vertically upward.

8. The velocity of ideal fluid at any point on the surface of the cylinder is given by $u_\theta = 2U \sin \theta$

where, u_θ = Tangential velocity on the surface of the cylinder,

U = Uniform velocity (or free stream velocity),

θ = The angular distance of the point from the forward stagnation point.

9. The peripheral velocity on the surface of the cylinder due to circulation (u_c) is given by :

$$u_c = \frac{\Gamma}{2\pi R}$$

where, Γ = circulation, and R = radius of the cylinder.

10. The resultant velocity in a circular cylinder which is rotated at a constant speed in a uniform flow field is given by:

$$u = u_\theta + u_c = 2U \sin \theta + \frac{\Gamma}{2\pi R}$$

11. The position of stagnation points is given by:

$$\sin \theta = \frac{\Gamma}{4\pi R}$$

For a single stagnation point, the condition is:

$$\Gamma = 4\pi UR \quad \dots \text{in terms of circulation}$$

$$u_c = 2U \quad \dots \text{in terms of tangential velocity.}$$

12. The pressure at any point on the cylinder surface (p) is given by:

$$p = p_o + \frac{1}{2} \rho U^2 \left[1 - \left(2 \sin \theta + \frac{\Gamma}{2\pi UR} \right)^2 \right]$$

where, p_o = The pressure in the uniform flow at some distance ahead of cylinder.

13. When a circular cylinder is rotated in a uniform flow, a lift force (F_L) is produced on the cylinder, the magnitude of which is given by:

$$F_L = \rho L U \Gamma$$

This equation is known as **Kutta - Joukowski equation**.

14. The expression for lift co-efficient for a rotating cylinder in a uniform flow is given by:

$$C_L = \frac{\Gamma}{UR} \quad \dots \text{in terms of circulation}$$

$$C_L = 2\pi \frac{u_c}{U} \quad \dots \text{in terms of tangential velocity.}$$

15. The generation of lift by spinning cylinder in a fluid stream is called **Magnus effect**.

16. Circulation developed on the airfoil is given by:

$$\Gamma = \pi c U \sin \alpha$$

where, c = Chord length, α = Angle of attack.

17. The expression for co-efficient of lift for an airfoil is given by:

$$C = 2\pi \sin \alpha$$

18. When an aeroplane is in steady-state:

$$(i) \text{ The weight of aeroplane (W) = The lift force } \left(C_L \times \frac{\rho U^2}{2} \times A \right)$$

$$(ii) \text{ The thrust developed by the engine = The drag force.}$$

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer:

- The pressure drag depends upon
 - the characteristics of the oncoming flow
 - the boundary layer formation
 - the separation of boundary layer and the size of the wake
 - the shear stresses generated on the body surface.
- In case of airfoils, the profile drag is one which is caused by
 - the compressibility effects
 - the shape and orientation of airfoil
 - the circulation induced around the airfoil
 - none of the above.
- The friction drag is primarily due to
 - separation of boundary layer
 - weight component in the direction of flow
 - shear stresses generated due to viscous action
 - none of the above.
- For a perfectly streamlined body which of the following statements is *incorrect*?
 - The pressure drag is very small
 - The boundary layer remains thin over the entire surface and does not separate
 - The flow separation points and a wake region is formed
 - The streamline pattern and pressure distribution are nearly the same as for an irrotational flow.
- The drag force is given by
 - $C_D \rho U^2 A$
 - $C_D \rho^2 U^2 A$
 - $C_D \rho U^2 A$
 - $C_D \frac{\rho U^2}{2} A$
- The shape of a streamlined body is such as to
 - fix the separation points as much ahead as possible
 - shift the boundary layer separation to the rearmost part thereby considerably reducing the wake-size
 - make the streamline pattern symmetrical
 - none of the above.
- Which of following statements is *correct* for bluff bodies?
 - The total drag is considerably larger as compared to that for streamlined bodies

- (b) The friction drag is greater than the pressure drag
 (c) The total drag is much less as compared to that for streamlined bodies.
 (d) None of the above.
8. The drag force experienced by an object is
 (a) the component of resultant fluid dynamic force in the flow direction
 (b) the horizontal force due to pressure variation over the surface of object
 (c) the resultant fluid dynamics force acting on the object
 (d) none of the above.
9. The drag and lift forces experienced by an object placed in a fluid stream are due to
 (a) pressure and turbulence
 (b) viscosity and turbulence
 (c) pressure and viscosity
 (d) pressure and gravity.
10. The lift force that may act on an object is
 (a) the component force due to the fluid displaced by the body
 (b) the component of resultant fluid dynamic force in a direction normal to the general direction of flow
 (c) the force due to shear stress that acts on the body surface
 (d) none of the above.
11. Which of the following conditions/requirements is necessary to induce lift on an object ?
 (a) The object should be so shaped that there are zones of high and low velocities resulting in pressure difference between upper and bottom surfaces of the object.
 (b) The object should be so designed that pressure distribution over its surface is symmetrical
 (c) The shape of the object should be symmetrical and the axis of the symmetry be aligned parallel to the flow direction
 (d) None of the above.
12. A streamlined body is defined as a body about which
 (a) the drag is zero
 (b) the flow is laminar
 (c) the flow is along streamlines
 (d) the flow separation is suppressed.
13. When a circular cylinder is rotated in a uniform flow, a lift force is produced on the cylinder which is caused by
 (a) the pressure difference between the two halves, the bottom-half being subjected to a higher pressure
 (b) the symmetrical streamline patterns
 (c) the shear stresses due to viscous action
 (d) none of the above.
14. The location of stagnation points is found from the relation
 (a) $\sin \theta = -\frac{\Gamma}{4\pi UR^2}$
 (b) $\sin \theta = -\frac{\Gamma^2}{4\pi UR}$
 (c) $\sin \theta = -\frac{\Gamma^2}{4\pi UR}$
 (d) $\sin \theta = -\frac{\Gamma}{4\pi UR}$.
15. For a single stagnation point, the condition is
 (a) $\Gamma = 4\pi U R$ (b) $\Gamma = 2\pi U R$
 (c) $\Gamma = 4\pi U^2 R$ (d) $\Gamma = 4\pi U R^2$.
16. The expression for co-efficient of lift for an airfoil is given by
 (a) $C_L = 2\pi \sin^2 \alpha$ (b) $C_L = 4\pi \sin \alpha$
 (c) $C_L = 2\pi \sin \alpha$ (d) none of the above.
17. The expression for lift co-efficient for a rotating cylinder in a uniform flow is given by
 (a) $C_L = \frac{\Gamma}{R^2 U}$ (b) $C_L = \frac{\Gamma}{RU^2}$
 (c) $C_L = \frac{\Gamma}{RU}$ (d) none of the above.
18. The velocity of ideal fluid at any point on the surface of the cylinder is given by
 (a) $U \sin \theta$ (b) $2U \sin \theta$
 (c) $3U \sin \theta$ (d) $4U \sin \theta$.
19. The drag on a sphere for Reynolds number less than 0.2 is given by
 (a) $\pi \mu D U$ (b) $2\pi \mu D U$
 (c) $3\pi \mu D U$ (d) none of the above.
20. The mathematical expression for lift force is given by
 (a) $F_L = C_L \rho A U$
 (b) $F_L = C_L \frac{\rho U^2}{2} A$
 (c) $F_L = C_L \rho A U^2 A$
 (d) none of the above.
21. The lift force on an airfoil is due to
 (a) the circulation of air around it
 (b) the pressure difference of the top and bottom surface
 (c) the formation of tip vortices
 (d) the angle of attack.

22. the terminal velocity of a small sphere settling in a viscous fluid varies as
 - (a) the Reynolds number
 - (b) the square of its diameter
 - (c) directly proportional to the viscosity of the fluid
 - (d) its diameter.
23. At the stall point for the airfoil
 - (a) the boundary layer separates at the leading edge
 - (b) the lift is maximum and the drag is minimum
 - (c) the lift is zero and the drag is maximum
 - (d) the lift is maximum and the drag increases sharply beyond it.
24. The velocity at the top of spinning ball is
 - (a) less than that at bottom
 - (b) greater than that at bottom
 - (c) equal to that at the bottom
 - (d) independent of spinning.
25. The circulation around an airfoil required for lift is produced
 - (a) when the airfoil is kept inclined to flow direction
 - (b) due to tip vortices
 - (c) by rotation of airfoil
 - (d) because of surface discontinuity formed at the trailing edge.
26. The terminal velocity of a body in a stationary mass of fluid corresponds to the situation when the
 - (a) body acquires a constant velocity in any direction
 - (b) net force acting on the body equals zero
 - (c) weight of the body equals the buoyancy force acting on it
 - (d) net force acting on the body acts in vertical direction.

ANSWERS

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (b) | 3. (c) | 4. (c) | 5. (d) | 6. (b) |
| 7. (a) | 8. (a) | 9. (c) | 10. (b) | 11. (a) | 12. (d) |
| 13. (a) | 14. (d) | 15. (a) | 16. (c) | 17. (c) | 18. (b) |
| 19. (c) | 20. (b) | 21. (a) | 22. (b) | 23. (d) | 24. (b) |
| 25. (b) | 26. (b) | | | | |

THEORETICAL QUESTIONS

1. Define drag force and lift force of an object immersed in a fluid.
2. Distinguish between the friction drag and the pressure drag.
3. When are the factors on which the total drag of a body fully immersed in a fluid depend?
4. Define co-efficient of drag and lift and state the factors on which these co-efficients depend.
5. Differentiate between a streamlined body and bluff body.
6. What is the expression for the drag on a sphere, when Reynolds number (Re) is upto 0.2 ? Hence prove that the drag co-efficient for sphere for this range of Re is given by

$$C_D = \frac{24}{Re}$$
7. What do you mean by terminal velocity of a body ?
8. Describe with the help of a sketch, the variation of drag co-efficient for a cylinder over a wide range of Reynolds number.
9. Why should circulation superimposed on flow past a body cause a lift ?
10. Draw and explain the approximate flow pattern and the pressure distribution around a flat plate placed perpendicular in a stream flow.
11. What is meant by Magnus effect ?
12. Derive an expression for the lift produced on a rotating cylinder placed in a uniform flow field such that the axis of the cylinder is perpendicular to the direction of flow.
13. Obtain an expression for co-efficient of lift for a rotating cylinder placed in a uniform flow.
14. Define stagnation points.
15. How is the position of the stagnation points for a rotating cylinder in a uniform flow determined?
16. Define the following terms for an airfoil.
 - (i) Chord line
 - (ii) Angle of attack
 - (iii) Camber
 - (iv) Profile centre line.

UNSOLVED EXAMPLES

- In a wind tunnel experiments were conducted with a wind speed of 50 km/h on a flat plate of size 2 m long and 1 m wide. The mass density of air is 1.15 kg/m^3 . The plate is kept at such an angle that co-efficients of lift and drag are 0.75 and 0.15 respectively. Determine: (i) Lift force, (ii) Drag force, (iii) Resultant Force, and (iv) Power exerted by the airstream on the plate.
[Ans. (i) 166.28 N; (ii) 33.25 N; (iii) 169.6 N; $\theta = 78.69^\circ$; (iv) 461.89 W]
- A passenger car has a weight of 30 kN and the co-efficient of friction between the road and tyres is 0.01. Assuming the cross-sectional area of the car to be 2.25 m^2 with a drag co-efficient of 0.60 estimate the energy requirement at a speed of 60 km/h.
[Ans. 8.78 kW]
- A man weighing 981 N descends to the ground from an aeroplane with the help of a parachute against the resistance of air. The shape of the parachute is hemispherical of 2 m diameter. Find the velocity of the parachute with which it comes down. Assume $C_D = 0.5$ and ρ for air = 1.25 kg/m^3 and $\nu = 0.015$ stoke
[Ans. 31.6 m/s]
- A kite $0.8 \times 0.8 \text{ m}$ weighing 3.924 N assumes an angle of 12° to the horizontal. The string attached to the kite makes an angle of 45° to the horizontal. The pull on the string is 24.525 N when the wind is flowing at a speed of 30 km/h. Find the corresponding co-efficients of drag and lift, Take ρ for air = 1.25 kg/m^3 .
[Ans. $C_D = 0.624$, $C_L = 0.765$]
- A kite weighing 7.85 N has an effective area of 0.8 m^2 . It is maintained in air at an angle of 10° to the horizontal. The string attached to the kite makes an angle of 45° to the horizontal and at this position the values of co-efficients of drag and lift are 0.6 and 0.8 respectively. Determine:
(i) The speed of wind, and
(ii) The tension in the string.
Take ρ for air = 1.25 kg/m^3
[Ans. 31.9 km/h; 33.25 N]
- A submarine assumed to approximate a cylinder 4 m in diameter and 20 m long is travelling submerged at 1.3 m/s in sea water. Find the drag exerted on it, if the drag co-efficient for Reynolds number greater than 10^5 may be taken as 0.75. Take for water, $\rho = 1035 \text{ kg/m}^3$ and $\nu = 0.015$ stoke.
[Ans. 52.47 kN]
- A ball of 80 mm diameter is supported in vertical air stream which is flowing at a velocity of 7 m/s. If the density and kinematic viscosity of air are 1.25 kg/m^3 and 1.5 stokes respectively, calculate the weight of the ball.
[Ans. 0.0769 N]
- A steel sphere of 3 mm diameter falls in glycerine at a terminal velocity of 0.035 m/s. If specific weights of steel and glycerine are 75 kN/m^3 and 12.5 kN/m^3 determine: (i) Dynamic viscosity of glycerine, (ii) Drag force, and (iii) Drag co-efficient for the sphere.
[Ans. (i) 0.893 Ns/m^2 , (ii) $8.837 \times 10^{-4} \text{ N}$, (iii) 160.2]
- A metallic ball (sp. gr. = 12) of 2 mm diameter is allowed to fall in fluid of specific gravity 0.95 and kinematic viscosity 1.5 Ns/m^2 . Determine: (i) Drag force, (ii) Pressure drag and skin friction drag, and (iii) terminal velocity of ball in fluid.
[Ans. (i) $4.54 \times 10^{-4} \text{ N}$; (ii) $1.514 \times 10^{-4} \text{ N}$; $3.026 \times 10^{-4} \text{ N}$; (iii) 0.016 m/s]
- A wing of a small aeroplane is rectangular in plan having a span of 10 m and chord of 1.6 m. In a horizontal flight at 2.2 km/h the total aerodynamic force acting of the wing is 25 kN. If the lift-drag ratio is 10, determine: (i) The co-efficient of lift and drag; (ii) The total weight the aeroplane can carry, and (iii) The power required for the flight.
[Ans. (i) 0.7, 0.07; (ii) 25 kN; (iii) 153 kW]
- A cylinder 1.2 m in diameter and 8 m long rotates at 90 r.p.m. with its axis perpendicular to an air stream with a wind velocity of 30 m/s. Assuming no slip condition between the cylinder and circulatory flow, find: (i) The magnitude of circulation, (ii) the transverse or lift force, and (iii) the position of stagnation points. Take specific weight of air as 12.25 N/m^3 .
[Ans. (i) $21.3 \text{ m}^2/\text{s}$; (ii) 6.33 kN; (iii) 185.4° ; 354.6°]
- A cylinder 1.2 m diameter and 8 m long rotates at 160 r.p.m. with its axis perpendicular to the stream of water flowing at a velocity of 10 m/s. Assuming no slip between the cylinder and the circulatory flow, determine:
(i) The circulation,
(ii) The theoretical lift,
(iii) The position of stagnation points, and

- (iv) The speed of cylinder (in r.p.m.) for a single stagnation point:
 [Ans. (i) $38 \text{ m}^2/\text{s}$; (ii) $3.04 \times 10^6 \text{ N}$; (iii) 210.2° ; 329.8° ; (iv) 318 r.p.m.]
13. A circular cylinder of 1.0 m diameter and 10 m length is rotated at 420 r.p.m. about its axis when it is kept in air stream with 11.0 m/s velocity perpendicular to its axis. Determine:
 (i) Circulation around the cylinder,
 (ii) Theoretical lift and lift co-efficient,
 (iii) Position of stagnation points,
 (iv) Actual drag and lift force on the cylinder, and
 (v) Actual resultant force and its direction.
 Take $\rho = 1.208 \text{ kg/m}^3$, and experimental values of C_D and C_L as 1.5 and 5.1 respectively.
 [Ans. (i) $69.1 \text{ m}^2/\text{s}$; (ii) 9182.67 N ; 12.565 (iii) -90° ; (iv) 1096.26 N ; 3727.28 N ; (v) 3885.15 N ; 73.61°]
14. A ship is propelled by two cylindrical rotors turning at 250 r.p.m. about their axes which are vertical. Each rotor is 6 m long and 2 m in diameter. Estimate the force exerted upon the rotors in the direction of motion when the relative wind velocity is 60 km/h at an angle of 60° opposing the direction of motion. Assume ρ for air is 1.25 kg/m^3 .
 [Ans. 10098 N]
15. A jet plane weighing 29.4 kN and having a wing area of 20 m^2 flies at a velocity of 950 km/h. When the engine delivers 7350 kW, 65 percent of the power is used to overcome the drag resistance of the wing. Calculate the co-efficient of lift and drag for the wing. Take density of atmospheric air 1.208 kg/m^3 .
 [Ans. 0.0349; 0.0215]



Chapter **15**

COMPRESSIBLE FLOW

- 15.1. Introduction
- 15.2. Basic thermodynamic relations
- 15.3. Basic thermodynamic processes
- 15.4. Basic equation of compressible fluid flow
- 15.5. Propagation of disturbances in fluid and velocity of sound
- 15.6. Mach's number
- 15.7. Propagation of disturbance in compressible fluid
- 15.8. Stagnation properties
- 15.9. Area-velocity relationship and effect of variation of area for subsonic, sonic and supersonic flows
- 15.10. Flow of compressible fluid through a convergent nozzle
- 15.11. Variables of flow in terms of Mach's number
- 15.12. Flow through Laval nozzle (convergent-divergent nozzle)
- 15.13. Shock waves
- 15.14. Measurement of compressible flow
- 15.15. Flow of compressible fluid through venturimeter

Highlight

Objective Type Questions

Theoretical Questions

Unsolved Examples

15.1. INTRODUCTION

A **compressible flow** is that flow in which the density of the fluid changes during flow. All real fluids are compressible to some extent and therefore their density will change with change in pressure or temperature. If the relative change in density $\Delta \rho / \rho$ is small, the fluid can be treated as incompressible. A compressible fluid, such as air, can be considered as incompressible with constant ρ if *changes in elevation are small, acceleration is small, and/or temperature changes are negligible*. In other words, if Mach's number U/C , where C is the sonic velocity, is small, compressible fluid can be treated as incompressible.

The gases are treated as compressible fluids and study of this type of flow is often referred to as '*Gas dynamics*'.

Some important problems where *compressibility effect* has to be considered are :

- (i) Flow of gases through nozzles, orifices ;
- (ii) Compressors ;
- (iii) Flight of aeroplanes and projectiles moving at higher altitudes;
- (iv) Water hammer and accoustics.

'Compressibility' affects the drag co-efficients of bodies by formation of shock waves, discharge co-efficients of measuring devies such as orificemeters, venturimeters and pitot tubes, stagnation pressure and flows in converging-diverging sections.

15.2. BASIC THERMODYNAMIC RELATIONS

15.2.1. The Characteristics Equation of State

At temperatures that are considerably in excess of critical temperature of a fluid, and also at very low pressure, the vapour of fluid tends to obey the equation:

$$\frac{pv}{T} = \text{constant} = R \left(\text{or } \frac{p}{\rho} = RT \right)$$

In practice, no gas obeys this law rigidly, but many gases tend towards it. An imaginary ideal gas which obeys this law is called a *perfect gas*, and the equation $\frac{pv}{T} = R$, is called the *characteristic equation of a state of a perfect gas*. The constant R is called the *gas constant*. Each perfect gas has a different gas constant.

Units of R are Nm/kg K or kJ/kg K

Usually, the characteristic equation is written as :

$$pv = RT \quad \dots(15.1)$$

or, for m kg, occupying $V \text{ m}^3$,

$$pV = mRT \quad \dots(15.2)$$

or,

$$p = \frac{m}{V} RT = \rho RT \quad \dots(15.2 (a))$$

Taking log on both sides, we get:

$$\ln(p) = \ln(\rho) + \ln(R) + \ln(T)$$

Upon differentiation, we have:

$$\frac{dp}{p} = \frac{d\rho}{\rho} + \frac{dR}{R} + \frac{dT}{T}$$

Since R is constant for a particular gas, its derivative is zero.

$$\therefore \frac{dp}{p} - \frac{d\rho}{\rho} - \frac{dT}{T} = 0 \quad \dots(15.3)$$

Eqn. (15.3) is the *differential equation of a perfect gas*.

15.2.2. Specific Heats

— The specific heat of a solid or liquid is usually defined as *the heat required to raise unit mass through one degree temperature rise*.

— For a gas there are an infinite number of ways in which heat may be added between any two temperatures, and hence a gas could have an infinite number of specific heats. However, only two specific heats for gases are defined.

(i) Specific heat at constant volume, c_v

(ii) Specific head at constant pressure, c_p .

(In case of real gases, c_p and c_v vary with temperature, but a suitable average value may be used for most practical purposes.)

$$c_p = c_v + R \quad \dots(15.4)$$

$$\frac{c_p}{c_v} = \gamma \text{ (gamma)} \quad \dots(15.5)$$

15.2.3. Internal Energy

It is the heat energy stored in a gas. If a certain amount of heat is supplied to a gas the result is that temperature of gas may increase or volume of gas may increase thereby doing some external work or both temperature and volume may increase. *If during heating of a gas the temperature increases its internal energy will also increase.*

Joule's law of internal energy states that the internal energy of a perfect gas is a function of temperature only. In other words, internal energy of a gas is dependent on the temperature change only and is not affected by the change in pressure and volume.

We do not know how to find the absolute quantity of internal energy in any substance, however, what is needed in engineering is the change of internal energy (ΔU).

15.2.4. Enthalpy

One of the fundamental quantities which occurs invariably in thermodynamics is the *sum of internal energy (u) and pressure volume product (pv)*. This sum is called **Enthalpy (h)**.

$$\text{i.e.} \quad h = u + pu$$

The total enthalpy of mass, m , of a fluid is given by,

$$H = U + pV, \text{ where } H = mh$$

15.2.5. Energy, Work and Heat

Energy. Energy is a general term embracing *energy in transition and stored energy*. The stored energy of a substance may be in the forms of *mechanical energy* and *internal energy* (other forms of stored energy may be chemical energy and electrical energy). Part of the stored energy may take the form of either potential energy or kinetic energy due to velocity. The balance part of the energy is known as *internal energy*.

Heat and work. These are the forms of energy in transition and are the only forms in which energy can cross the boundaries of a system. Neither heat nor work can exist as stored energy.

Work. Work is said to be done when a *force moves through a distance*. If a part of the boundary of a system undergoes a displacement under the action of a pressure, the work done W is the product of the force (pressure $\times$ area) and the distance it moves in the direction of the force.

Work is a transient quantity which only appears at the boundary while a change of state is taking place within a system. Work is ‘something’ which appears at the boundary when a system changes its state due to the movement of a part of the boundary under the action of a force.

$$\text{Work output of the system} = +W$$

$$\text{Work input to system} = -W$$

Heat. Heat (denoted by the symbol Q) may be defined in an analogous way to work as follows:

“*Heat is something which appears at the boundary when a system changes its state due to a difference in temperature between the system and its surroundings*”.

Heat, like work, is a transient quantity which only appears at the boundary while a change is taking place within the system.

$$\text{Heat received by the system} = +Q$$

$$\text{Heat rejected or given up by the system} = -Q$$

15.3. BASIC THERMODYNAMIC PROCESSES

The basic thermodynamic processes are given below :

1. **Constant volume (isochoric) process ($v = \text{constant}$).** A change in the state of system at constant volume is called *isochoric process*. An isochoric process results when the gas system is heated or cooled in an *enclosed space* (e.g. a rigid vessel).

Formulae (for unit mass) :

$$\text{Heat added, } Q = c_v (T_2 - T_1) \quad \dots(15.7)$$

$$\text{Work done, } W = 0 \quad \dots(15.8)$$

$$p, v, T \text{ relations : } \frac{T_2}{T_1} = \frac{p_2}{p_1} \quad \dots(15.9)$$

where suffix 1 and suffix 2 represent the ‘start’ and ‘finish’ points of the process respectively.

2. **Constant pressure (isobaric) process ($p = \text{constant}$).** In this process a change in the state of the gas (working fluid) takes place at *constant pressure*. For a constant pressure process, the boundary must move against an external resistance as heat is supplied; for instance a gas in a cylinder behind a piston can be made to undergo a constant pressure process. Since the piston is pushed through a certain distance by the force exerted by the gas, then the work is done on its surroundings.

Formulae (for unit mass) :

$$\text{Head added, } Q = c_p (T_2 - T_1) \quad \dots(15.10)$$

$$\text{Work done, } W = p (v_2 - v_1) \quad \dots(15.11)$$

$$p, v, T \text{ relations : } \frac{T_2}{T_1} = \frac{v_2}{v_1} \quad \dots(15.12)$$

3. **Isothermal process $p v$ or $\frac{p}{\rho} = \text{constant}$, $T = \text{constant}$).** A process at a constant temperature is called an *isothermal process*. When a working substance in a cylinder behind a piston expands from a high pressure there is a tendency for the temperature to fall. In an isothermal expansion heat must be added continuously in order to keep the temperature at the initial value. Similarly in an isothermal compression heat must be removed from the working substance continuously during the process.

Formulae (for unit mass) :

$$\text{Heat added, } Q = p_1 v_1 \ln \left(\frac{v_2}{v_1} \right) = R T_1 \ln \left(\frac{\rho_1}{\rho_2} \right) \quad \dots(15.13)$$

$$\text{Work done, } W = p_1 v_1 \ln \left(\frac{v_2}{v_1} \right) = R T_1 \ln \left(\frac{\rho_1}{\rho_2} \right) \quad \dots(15.14)$$

$$p, v, T, \text{ relations : } p_1 v_1 = \left(\text{or } \frac{p_1}{\rho_1} = \frac{p_2}{\rho_2} \right) \quad \dots(15.15)$$

4. **Adiabatic process ($p v^\gamma$ or $\frac{p}{\rho^\gamma} = \text{constant}$).** An adiabatic process is one in which no heat is transferred to or from the gas during the process. Such a process can be reversible or irreversible. For an adiabatic process to take place, perfect thermal insulation for the system must be available.

Formulae (for unit mass) :

$$\text{Heat added, } Q = 0 \quad \dots(15.16)$$

$$\text{Work done, } W = \frac{p_1 v_1 - p_2 v_2}{\gamma - 1} = \frac{R (T_1 - T_2)}{\gamma - 1} \quad \dots(15.17)$$

$$p, v, T, \text{ relations : } p_1 v_1^\gamma = p_2 v_2^\gamma \quad \dots(15.18)$$

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{\gamma-1} = \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \quad \dots(15.19)$$

If the adiabatic process is *reversible* (or frictionless), it is known as *isentropic process*. In case the pressure and density are related in such a way that $\gamma \neq \frac{c_p}{c_v}$ but is equal to some positive value then the process is known as *polytropic*, according to which $\frac{p}{\rho^n} = \text{constant}$ ($n \neq \gamma$).

15.4. BASIC EQUATIONS OF COMPRESSIBLE FLUID FLOW

The basic equations of compressible fluid flow are : (i) Continuity equation, (ii) Momentum equation, (iii) Energy equation, and (iv) Equation of state.

The only change from incompressible fluid cases is that thermodynamic laws are applied in addition to the basic principle of conservation of mass, energy and momentum.

15.4.1. Continuity Equation

In case of *one-dimensional flow*, mass per second = ρAV

(where, ρ = mass density, A = area of cross-section, V = velocity)

Since the mass or mass per second is constant according to law of conservation of mass, therefore,

$$\rho AV = \text{Constant} \quad \dots(15.20)$$

Differentiating the above equation, we get:

$$d(\rho AV) = 0 \quad \text{or} \quad \rho d(AV) + AVd\rho = 0$$

$$\text{or, } \rho (AdV + VdA) + AVd\rho = 0 \quad \text{or} \quad \rho AdV + \rho VdA + AVd\rho = 0$$

Dividing both sides by ρAV , and rearranging we get:

$$\frac{d\rho}{\rho} + \frac{dA}{A} + \frac{dV}{V} = 0 \quad \dots(15.21)$$

Eqn. (15.18) is also known as equation of continuity in *differential form*.

15.4.2. Momentum Equation

The momentum equation for compressible fluids is similar to the one for incompressible fluids. This is because in momentum equation the *change in momentum flux is equated to force required to cause this change*.

Momentum flux = Mass flux $\times$ velocity = $\rho AV \times V$

But the mass flux i.e. ρAV = constant

...By continuity equation

Thus the momentum equation is completely independent of the compressibility effects and hence for compressible fluids too the momentum equation, say X -direction, may be expressed as :

$$\Sigma F_x = (\rho AVV_x)_2 - (\rho AVV_x)_1 \quad \dots(15.22)$$

15.4.3. Bernoulli's or Energy Equation

In chapter 6 Bernoulli's equation for an incompressible fluid has been derived and the same procedure is followed. As the flow of compressible fluid is steady, the same Euler equation (Eqn. 6.) is obtained as :

$$\frac{dp}{\rho} + VdV + gdz = 0 \quad \dots(15.23)$$

Integrating both sides, we get:

$$\int \frac{dp}{\rho} + \int VdV + \int gdz = \text{constant}$$

$$\text{or, } \int \frac{dp}{\rho} + \frac{V^2}{2} + gz = \text{constant} \quad \dots(15.24)$$

In compressible flow since ρ is not constant it cannot be taken outside the integration sign. In compressible fluids the pressure (p) changes with change of density (ρ), depending on the type of process. Let us find out the Bernoulli's equation for isothermal and adiabatic processes.

(a) Bernoulli's equation for isothermal process :

In case of an *isothermal process*

$$pv = \text{constant} \quad \text{or} \quad \frac{p}{\rho} = \text{constant} = c_1 \text{ (say)}$$

(where $v = \text{specific volume} = 1/\rho$)

$$\therefore \quad \rho = \frac{p}{c_1}$$

$$\text{Hence,} \quad \int \frac{dp}{\rho} = \int \frac{dp}{p/c_1} = \int \frac{c_1 dp}{p} = c_1 \int \frac{dp}{p} = c_1 \ln(p) = \frac{p}{\rho} \ln(p) \quad \left(\because c_1 = \frac{p}{\rho} \right)$$

Substituting the value of $\int \frac{dp}{\rho}$ in eqn. (15.24), we get

$$\frac{p}{\rho} \ln(p) + \frac{V^2}{2} + gz = \text{constant}$$

Dividing both sides by g , we get

$$\frac{p}{\rho g} \ln(p) + \frac{V^2}{2g} + z = \text{constant} \quad \dots(15.25)$$

Eqn. (15.25) is the *Bernoulli's equation for compressible flow undergoing isothermal process*.

(b) Bernoulli's equation for adiabatic process :

In case of an *adiabatic process*:

$$pv^\gamma = \text{constant} \quad \text{or} \quad \frac{p}{\rho^\gamma} = \text{constant} = c_2 \text{ (say)}$$

$$\therefore \quad \rho^\gamma = \frac{p}{c_2} \quad \text{or} \quad \rho = \left(\frac{p}{c_2} \right)^{1/\gamma}$$

$$\text{Hence,} \quad \int \frac{dp}{\rho} = \int \frac{dp}{(p/c_2)^{1/\gamma}} = (c_2)^{1/\gamma} \int \frac{1}{p^{1/\gamma}} dp = (c_2)^{1/\gamma} \int p^{-1/\gamma} dp$$

$$= (c_2)^{1/\gamma} \left[\frac{p^{-\frac{1}{\gamma}+1}}{\left(-\frac{1}{\gamma}+1\right)} \right] = \frac{(c_2)^{1/\gamma} (p)^{\left(\frac{\gamma-1}{\gamma}\right)}}{\left(\frac{\gamma-1}{\gamma}\right)} = \frac{\gamma}{\gamma-1} c_2^{1/\gamma} (p)^{\left(\frac{\gamma-1}{\gamma}\right)}$$

$$= \left(\frac{\gamma}{\gamma-1} \right) \left(\frac{p}{\rho^\gamma} \right)^{1/\gamma} (p)^{\left(\frac{\gamma-1}{\gamma}\right)} \quad \left(\because c_2 = \frac{p}{\rho^\gamma} \right)$$

$$= \left(\frac{\gamma}{\gamma-1} \right) \left(\frac{p^{1/\gamma}}{\rho^{\frac{1}{\gamma} \times \frac{1}{\gamma}}} \right) (p)^{\left(\frac{\gamma-1}{\gamma}\right)}$$

$$= \left(\frac{\gamma}{\gamma-1} \right) \frac{p^{\left(\frac{1}{\gamma} + \frac{\gamma-1}{\gamma}\right)}}{\rho} = \left(\frac{\gamma}{\gamma-1} \right) \frac{p}{\rho}$$

Substituting the value of $\int \frac{dp}{\rho}$ in eqn. (15.24), we get

$$\left(\frac{\gamma}{\gamma - 1} \right) \frac{p}{\rho} + \frac{V^2}{2} + gz = \text{constant}$$

Dividing both sides by g , we get

$$\left(\frac{\gamma}{\gamma - 1} \right) \frac{p}{\rho g} + \frac{V^2}{2g} + z = \text{constant} \quad \dots(15.26)$$

Eqn. (15.26) is the *Bernoulli's equation for compressible flow undergoing adiabatic process*.

Example 15.1. A gas is flowing through a horizontal pipe. On a section where cross-section area is 50 cm^2 , the pressure and temperature are found to be 3 bar (gauge) and 20°C respectively. At another section where the area of cross-section is 25 cm^2 , the pressure is recorded 2 bar (gauge). If the mass rate of flow of gas through the pipe is 0.6 kg/s find the velocities of the gas at these sections, assuming an isothermal change.

Take $R = 287 \text{ J/kg K}$, and atmospheric pressure = 1 bar.

Solution.

Section 1: Area, $A_1 = 50 \text{ cm}^2 = 50 \times 10^{-4} \text{ m}^2$

Pressure, $p_1 = 3 \text{ bar (gauge)} = 3 + 1 = 4 \text{ bar} = 4 \times 10^5 \text{ N/m}^2 \text{ (abs.)}$

Temperature, $T_1 = 20 + 273 = 293 \text{ K}$

Section 2: Area, $A_2 = 25 \text{ cm}^2 = 25 \times 10^{-4} \text{ m}^2$

Pressure, $p_2 = 2 \text{ bar (gauge)} = 2 + 1 = 3 \text{ bar} = 3 \times 10^5 \text{ N/m}^2 \text{ (abs.)}$

Mass rate of flow of gas, $m = 0.6 \text{ kg/s}$

Characteristic gas constant, $R = 287 \text{ J/kg K}$

Atmospheric pressure = 1 bar

Velocities at the sections V_1, V_2 :

The characteristic equation is written as :

$$p = \rho RT \quad \dots[\text{Eqn. 15.2 (a)}]$$

$$\text{Section 1: } p_1 = \rho_1 RT_1 \quad \text{or} \quad \rho_1 = \frac{p_1}{RT_1} = \frac{4 \times 10^5}{287 \times 293} = 4.757 \text{ kg/m}^3$$

$$\text{Also, } m = \rho_1 A_1 V_1$$

$$\text{or, } V_1 = \frac{m}{\rho_1 A_1} = \frac{0.6}{4.757 \times 50 \times 10^{-4}} = \mathbf{25.22 \text{ m/s}}$$

$$\text{Section 2: } \rho^2 = \frac{p_2}{RT_2} = \frac{3 \times 10^5}{287 \times 293} = 3.567 \text{ kg/m}^3 \quad (\because T_1 = T_2 = 293 \text{ K})$$

$$\text{Also, } m = \rho_2 A_2 V_2$$

$$\therefore V_2 = \frac{m}{\rho_2 A_2} = \frac{0.6}{3.567 \times 25 \times 10^{-4}} = \mathbf{67.28 \text{ m/s (Ans.)}}$$

Example 15.2. Fig. 15.1 shows a horizontal pipe in which gas is flowing at a temperature of 6°C . The pressures at the sections 1 and 2 are 4 bar (gauge) and 3 bar (gauge) respectively. If $R = 287 \text{ J/kg K}$ and atmospheric pressure is 1 bar find the velocities of the gas at these sections.

Solution. Refer to Fig. 15.1.

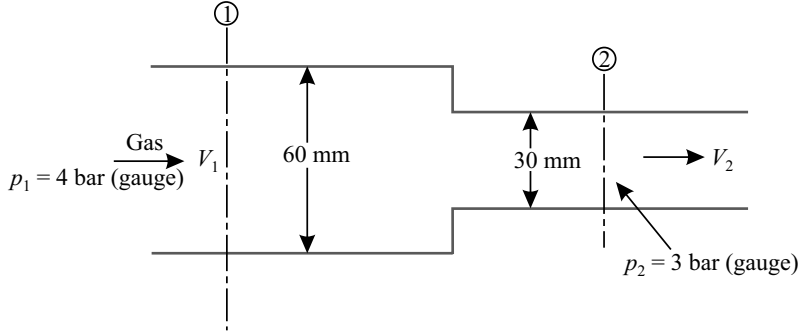


Fig. 15.1

Section 1: Diameter of pipe, $D_1 = 60 \text{ mm} = 0.06 \text{ m}$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.06^2 = 2.827 \times 10^{-3} \text{ m}^2$$

$$\text{Pressure, } p_1 = 4 \text{ bar (gauge)} = 4 + 1 = 5 \text{ bar (abs.)} = 5 \times 10^5 \text{ N/m}^2 \text{ (abs.)}$$

$$\text{Temperature, } T_1 = 6 + 273 = 279 \text{ K}$$

Section 2: Diameter of pipe, $D_2 = 30 \text{ mm} = 0.03 \text{ m}$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.03^2 = 7.0686 \times 10^{-4} \text{ m}^2$$

$$\text{Pressure, } p_2 = 3 \text{ bar (gauge)}$$

$$= 3 + 1 = 4 \text{ bar (abs.)} = 4 \times 10^5 \text{ N/m}^2 \text{ (abs.)}$$

$$\text{Gas constant, } R = 287 \text{ J/kg K}$$

Velocities of the gas at sections 1 and 2, V_1, V_2 :

Applying continuity equation at 1 and 2, we get:

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad \text{or} \quad \frac{V_2}{V_1} = \frac{\rho_1 A_1}{\rho_2 A_2} = \frac{\rho_1 \times 2.827 \times 10^{-3}}{\rho_2 \times 7.0686 \times 10^{-4}} = 4 \times \frac{\rho_1}{\rho_2} \quad \dots(i)$$

For an isothermal process, we have:

$$\frac{p_1}{\rho_1} = \frac{p_2}{\rho_2} \quad \text{or} \quad \frac{\rho_1}{\rho_2} = \frac{p_1}{p_2} = \frac{5 \times 10^5}{4 \times 10^5} = 1.25$$

Substituting the value of $\frac{\rho_1}{\rho_2} = 1.25$ in eqn. (i), we get:

$$\frac{V_2}{V_1} = 4 \times 1.25 = 5 \quad \text{or} \quad V_2 = 5V_1 \quad \dots(ii)$$

Applying Bernoulli's equation at sections 1 and 2 for isothermal process (Eqn. 15.25), we have:

$$\frac{p_1}{\rho_1 g} \ln(p_1) + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho_2} \ln(p_2) + \frac{V_2^2}{2g} + z_2$$

But, $z_1 = z_2$, since the pipe is horizontal,

$$\therefore \frac{p_1}{\rho_1 g} \ln(p_1) + \frac{V_1^2}{2g} = \frac{p_2}{\rho_2} \ln(p_2) + \frac{V_2^2}{2g}$$

$$\text{or, } \frac{p_1}{\rho_1 g} \ln(p_1) - \frac{p_2}{\rho_2 g} \ln(p_2) = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

But, $\frac{p_1}{\rho_1} = \frac{p_2}{\rho_2}$ (for an *isothermal process*),

$$\therefore \frac{p_1}{\rho_1 g} \ln(p_1) - \frac{p_1}{\rho_1 g} \ln(p_2) = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

$$\text{or, } \frac{p_1}{\rho_1 g} \ln\left(\frac{p_1}{p_2}\right) = \frac{(5V_1)^2}{2g} - \frac{V_1^2}{2g} = \frac{24V_1^2}{2g} = \frac{12V_1^2}{g} \quad (\because V_2 = 5V_1)$$

$$\text{or, } \frac{p_1}{\rho_1 g} \ln\left(\frac{5 \times 10^5}{4 \times 10^5}\right) = \frac{12V_1^2}{g} \quad \text{or} \quad 0.223 \frac{p_1}{\rho_1 g} = \frac{12V_1^2}{g}$$

$$\text{or, } \frac{p_1}{\rho_1} = \frac{12V_1^2}{0.223} = 53.8V_1^2 \quad \dots(iii)$$

From equation of state, we have:

$$p_1 = \rho_1 R T_1 \quad \dots \text{Section 1} \quad \text{or} \quad \frac{p_1}{\rho_1} = R T_1 = 287 \times 279 = 80073$$

Substituting the value of $\frac{p_1}{\rho_1}$ in eqn. (iii), we get:

$$53.8 V_1^2 = 80073 \quad \text{or} \quad V_1^2 = \frac{80073}{53.8} = 1488.34$$

$$\text{or, } V_1 = \mathbf{38.58 \text{ m/s (Ans.)}}$$

$$\text{From eqn. (ii) we have: } V_2 = 5 V_1 = 5 \times 38.58 = \mathbf{192.9 \text{ m/s (Ans.)}}$$

Example 15.3. A 120 mm diameter pipe reduces to 60 mm diameter through a sudden contraction. When it carries air at 25°C under isothermal condition, the absolute pressures observed in the two pipes just before and after the contraction are 480 kN/m² and 384 kN/m² respectively. Determine:

- (i) Densities at the two sections,
- (ii) Velocities at the two sections, and
- (iii) Mass rate of flow through the pipe.

Take $R = 287 \text{ J/kg K}$

Solution. Section 1: Diameter of the pipe,

$$D_1 = 120 \text{ mm} = 0.12 \text{ m}$$

$$\therefore \text{Area, } A_1 = \frac{\pi}{4} \times 0.12^2 = 0.01131 \text{ m}^2$$

$$\text{Pressure, } p_1 = 480 \text{ kN/m}^2$$

$$\text{Temperature, } T_1 = 25 + 273 = 298 \text{ K}$$

Section 2: Diameter of the pipe,

$$D_2 = 60 \text{ mm} = 0.06 \text{ m}$$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.06^2 = 2.827 \times 10^{-3} \text{ m}^2$$

$$\text{Pressure, } p_2 = 384 \text{ kN/m}^2$$

$$\text{Temperature, } T_2 = T_1 = 298 \text{ K (since the condition is isothermal)}$$

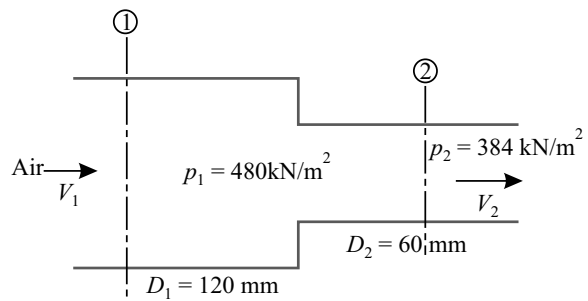


Fig. 15.2

(i) Densities, ρ_1 and ρ_2 :

For isothermal condition, $\frac{p_1}{\rho_1} = \frac{p_2}{\rho_2}$ [Eqn. (15.15)]

$$\text{or,} \quad \frac{p_1}{p_2} = \frac{\rho_1}{\rho_2} \quad \text{or} \quad \frac{480}{384} = \frac{\rho_1}{\rho_2} = 1.25$$

$$\text{Also} \quad p_1 = \rho_1 RT_1 \quad \text{or} \quad \rho_1 = \frac{p_1}{RT_1}$$

$$\text{or,} \quad \rho_1 = \frac{480 \times 10^3}{287 \times 298} = \mathbf{5.61 \text{ kg/m}^3 \text{ (Ans.)}}$$

$$\therefore \quad \rho_2 = \frac{\rho_1}{1.25} = \frac{5.61}{1.25} = \mathbf{4.488 \text{ kg/m}^3 \text{ (Ans.)}}$$

(ii) Velocities, V_1 and V_2 :

According to continuity equation:

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

$$\text{or,} \quad \frac{V_2}{V_1} = \frac{\rho_1 A_1}{\rho_2 A_2} = 1.25 \times \frac{0.01131}{2.827 \times 10^{-3}} = 5.0$$

Applying Bernoulli's equation at sections 1 and 2 for isothermal condition, we get:

$$\frac{p_1}{\rho_1 g} \ln(p_1) + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho_2 g} \ln(p_2) + \frac{V_2^2}{2g} + z_2$$

Assuming $z_1 = z_2$, we have:

$$\frac{p_1}{\rho_1 g} \ln(p_1) + \frac{V_1^2}{2g} = \frac{p_2}{\rho_2 g} \ln(p_2) + \frac{V_2^2}{2g}$$

$$\text{or,} \quad \frac{p_1}{\rho_1 g} \ln(p_1) - \frac{p_2}{\rho_2 g} \ln(p_2) = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

Cancelling 'g' on both the sides, we get:

$$\frac{p_1}{\rho_1} \ln(p_1) - \frac{p_2}{\rho_2} \ln(p_2) = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\text{or,} \quad \frac{p_1}{\rho_1} \ln(p_1) - \frac{p_1}{\rho_1} \ln(p_2) = \frac{V_2^2}{2} - \frac{V_1^2}{2} \quad \left(\because \frac{p_1}{\rho_1} = \frac{p_2}{\rho_2} \right)$$

$$\text{or,} \quad \frac{p_1}{\rho_1} \ln\left(\frac{p_1}{p_2}\right) = \frac{V_2^2}{2} - \frac{V_1^2}{2} = \frac{(5V_1)^2}{2} - \frac{V_1^2}{2} = 12V_1^2 \quad (\because V_2 = 5V_1)$$

$$\text{or,} \quad \frac{480 \times 10^3}{5.61} \ln\left(\frac{480 \times 10^3}{384 \times 10^3}\right) = 12V_1^2$$

$$\text{or,} \quad 85561.5 \times 0.223 = 12V_1^2 \quad \text{or} \quad V_1^2 = 1590 \quad \text{or} \quad V_1 = \mathbf{39.87 \text{ m/s (Ans.)}}$$

$$\therefore \quad V_2 = 5V_1 = 5 \times 39.87 = \mathbf{199.35 \text{ m/s (Ans.)}}$$

(iii) Mass rate of flow through the pipe, m :

$$m = \rho_1 A_1 V_1 (= \rho_2 A_2 V_2) \quad \text{or} \quad m = 5.61 \times 0.01131 \times 39.87 = \mathbf{2.53 \text{ kg/s (Ans.)}}$$

Example 15.4. A gas with a velocity of 300 m/s is flowing through a horizontal pipe at a section where pressure is 78 kN/m² absolute and temperature 40° C. The pipe changes in diameter and at this section, the pressure is 117 kN/m² absolute. Find the velocity of the gas at this section if the flow of the gas is adiabatic. Take $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$. [Delhi University]

Solution.

Section 1: Velocity of the gas, $V = 300 \text{ m/s}$

Pressure, $p_1 = 78 \text{ kN/m}^2$

Temperature, $T_1 = 40 + 273 = 313 \text{ K}$

Section 2: Pressure, $p_2 = 117 \text{ kN/m}^2$

$R = 287 \text{ J/kg K}$, $\gamma = 1.4$

Velocity of gas at section 2, V_2 :

Applying Bernoulli's equations at sections 1 and 2 for *adiabatic process*, we have:

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1 g} + \frac{V_1^2}{2g} + z_1 = \left(\frac{\gamma}{\gamma-1}\right) \frac{p_2}{\rho_2 g} + \frac{V_2^2}{2g} + z_2 \quad [\text{Eqn. (15.26)}]$$

But, $z_1 = z_2$, since the pipe is horizontal.

$$\therefore \left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1 g} + \frac{V_1^2}{2g} = \left(\frac{\gamma}{\gamma-1}\right) \frac{p_2}{\rho_2 g} + \frac{V_2^2}{2g}$$

Cancelling 'g' on both sides, we get:

$$\left(\frac{\gamma}{\gamma-1}\right) \left(\frac{p_1}{\rho_1} - \frac{p_2}{\rho_2}\right) = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\text{or, } \left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left(1 - \frac{p_2}{p_1} \times \frac{\rho_1}{\rho_2}\right) = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\therefore \left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left(1 - \frac{p_2}{p_1} \times \frac{\rho_1}{\rho_2}\right) = \frac{V_2^2}{2} - \frac{V_1^2}{2} \quad \dots(i)$$

$$\text{For an adiabatic flow : } \frac{p_1}{\rho_1^\gamma} = \frac{p_2}{\rho_2^\gamma} \quad \text{or} \quad \frac{p_1}{p_2} = \left(\frac{\rho_1}{\rho_2}\right)^\gamma \quad \text{or} \quad \frac{\rho_1}{\rho_2} = \left(\frac{p_1}{p_2}\right)^{\frac{1}{\gamma}}$$

Substituting the value of $\frac{\rho_1}{\rho_2}$ in eqn. (i), we get:

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left\{1 - \frac{p_2}{p_1} \times \left(\frac{p_1}{p_2}\right)^{\frac{1}{\gamma}}\right\} = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left\{1 - \left(\frac{p_2}{p_1}\right)^{1-\frac{1}{\gamma}}\right\} = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\text{or, } \left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left\{1 - \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}\right\} = \frac{V_2^2 - V_1^2}{2} \quad \dots(ii)$$

$$\text{At section 1 : } \frac{p_1}{\rho_1} = RT_1 = 287 \times 313 = 89831,$$

$$\frac{p_2}{p_1} = \frac{117}{78} = 1.5, \quad \text{and} \quad V_1 = 300 \text{ m/s}$$

Substituting the values in eqn. (ii), we get:

$$\left(\frac{1.4}{1.4-1}\right) \times 89831 \left\{1 - (1.5)^{\frac{1.4-1}{1.4}}\right\} = \frac{V_2^2}{2} - \frac{300^2}{2}$$

$$314408.5 (1 - 1.1228) = \frac{V_2^2}{2} - 45000$$

$$\text{or,} \quad -38609.4 = \frac{V_2^2}{2} - 45000$$

$$\text{or,} \quad V_2^2 = 12781.2 \quad \text{or} \quad V_2 = 113.05 \text{ m/s (Ans.)}$$

Example 15.5. In the case of air flow in a conduit transition, the pressure, velocity and temperature at the upstream section are 35 kN/m², 30 m/s and 150°C respectively. If at the downstream section the velocity is 150 m/s, determine the pressure and the temperature if the process followed is isentropic. Take $\gamma = 1.4$, $R = 290 \text{ J/kg K}$.

Solution.

Section 1 (upstream) : Pressure, $p_1 = 35 \text{ kN/m}^2$,

Velocity, $V_1 = 30 \text{ m/s}$

Temperature, $T_1 = 150 + 273 = 423 \text{ K}$

Velocity, $V_2 = 150 \text{ m/s}$

$R = 290 \text{ J/kg K}$, $\gamma = 1.4$

Section 2 (downstream) :

Pressure, p_2 :

Applying Bernoulli's equation at sections 1 and 2 for *isentropic (reversible adiabatic) process*, we have:

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1 g} + \frac{V_1^2}{2g} + z_1 = \left(\frac{\gamma}{\gamma-1}\right) \frac{p_2}{\rho_2 g} + \frac{V_2^2}{2g} + z_2$$

Assuming $z_1 = z_2$, we have:

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1 g} + \frac{V_1^2}{2g} = \left(\frac{\gamma}{\gamma-1}\right) \frac{p_2}{\rho_2 g} + \frac{V_2^2}{2g}$$

Cancelling 'g' on both the sides, and rearranging we get:

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left(1 - \frac{p_2}{p_1} \times \frac{\rho_1}{\rho_2}\right) = \frac{V_2^2}{2} - \frac{V_1^2}{2} \quad \dots(i)$$

$$\text{For an isentropic flow:} \quad \frac{p_1}{\rho_1^\gamma} = \frac{p_2}{\rho_2^\gamma} \quad \text{or} \quad \frac{p_1}{p_2} = \left(\frac{\rho_1}{\rho_2}\right)^\gamma \quad \text{or} \quad \frac{\rho_1}{\rho_2} = \left(\frac{p_1}{p_2}\right)^{\frac{1}{\gamma}}$$

Substituting the value of $\frac{\rho_1}{\rho_2}$ in eqn. (i), we have:

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left\{1 - \frac{p_2}{p_1} \times \left(\frac{p_1}{p_2}\right)^{\frac{1}{\gamma}}\right\} = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left\{1 - \left(\frac{p_2}{p_1}\right)^{1-\frac{1}{\gamma}}\right\} = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left\{1 - \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}\right\} = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

Substituting the values, we get:

$$\frac{1.4}{1.4-1} \times 122670 \left\{ 1 - \left(\frac{p_2}{p_1} \right)^{\frac{1.4-1}{1.4}} \right\} = \frac{150^2}{2} - \frac{30^2}{2} = 10800$$

$$\left(\because \frac{p_1}{\rho_1} = RT_1 = 290 \times 423 = 122670 \right)$$

$$429345 \left\{ 1 - \left(\frac{p_2}{p_1} \right)^{0.2857} \right\} = 10800$$

$$\text{or,} \quad \left(\frac{p_2}{p_1} \right)^{0.2857} = 1 - \frac{10800}{429345} = 0.9748$$

$$\text{or,} \quad \frac{p_2}{p_1} = (0.9748)^{1/0.2857} = (0.9748)^{3.5} = 0.9145$$

$$\text{or,} \quad p_2 = 35 \times 0.9145 = \mathbf{32 \text{ kN/m}^2} \quad (\text{Ans.})$$

Temperature, T_2 :

For an *isentropic process*, we have:

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} = (0.9145)^{\frac{1.4-1}{1.4}} = (0.9145)^{0.2857} = 0.9748$$

$$\therefore T_2 = 423 \times 0.9748 = 412.3 \text{ K or } t_2 = 412.3 - 273 = \mathbf{139.3^\circ \text{C}} \quad (\text{Ans.})$$

15.5. PROPAGATION OF DISTURBANCES IN FLUID AND VELOCITY OF SOUND

The solids as well as fluids consist of molecules. Whereas the molecules in solids are close together, these are relatively apart in fluids. Consequently whenever there is a minor disturbance, it travels instantaneously in case of solids; but in case of fluid the molecules change in position before the transmission or propagation of the disturbance depends upon its *elastic properties*. The *velocity of disturbance depends upon the changes of pressure and density of the fluid*.

The propagation of disturbance is similar to the propagation of sound through a media. The *speed of propagation of sound in a media* is known as **acoustic or sonic velocity** and depends upon the difference of pressure. Incompressible flow, velocity of sound (sonic velocity) is of paramount importance.

15.5.1. Derivation of Sonic Velocity (velocity of sound)

Consider a one-dimensional flow through long straight rigid pipe of uniform cross-sectional area fitted with a frictionless piston at one end as shown in Fig. 15.3. The tube is filled with a compressible fluid initially at rest. If the piston is moved suddenly to the right with velocity, a *pressure wave* would be propagated through the fluid with a velocity of sound wave.

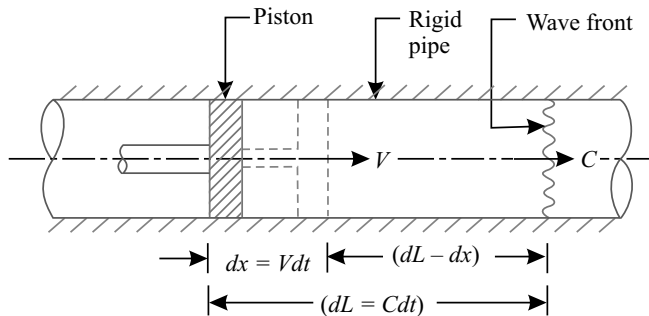


Fig. 15.3. One dimensional pressure wave propagation.

Let,

A = Cross-sectional area of the pipe,

V = Piston velocity,

p = Fluid pressure in the pipe before the piston movement,

ρ = Fluid density before the piston movement,

dt = A small interval of time during which piston moves, and

C = Velocity of pressure wave or sound wave

(travelling in the fluid).

Before the movement of the piston the length dL has an initial density ρ , and its total mass

$$= \rho \times dL \times A$$

When the piston moves through a distance dx , the fluid density within the compressed region of length $(dL - dx)$ will be increased and becomes $(\rho + d\rho)$ and subsequently the total mass of fluid in the compressed region $= (\rho + d\rho) (dL - dx) \times A$

$$\therefore \rho \times dL \times A = (\rho + d\rho) (dL - dx) \times A \quad \dots \text{by principle of continuity.}$$

But, $dL = C dt$ and $dx = V dt$; therefore, the above equation becomes:

$$\rho C dt = (\rho + d\rho) (C - V) dt$$

$$\text{or, } \rho C = (\rho + d\rho) (C - V) \quad \text{or } \rho C = \rho C - \rho V + d\rho.C - d\rho.V$$

$$\text{or, } 0 = -\rho V + d\rho.C - d\rho.V$$

Neglecting the term $d\rho.V$ (V being much smaller than C), we get:

$$d\rho.C = \rho V \quad \text{or } C = \frac{\rho V}{d\rho} \quad \dots(15.27)$$

Further in the region of compressed fluid, the fluid particles have attained a velocity which is apparently equal to V (velocity of the piston), accompanied by an increase in pressure dp due to sudden motion of the piston. Applying impulse-momentum equation for the fluid in the compressed region during dt , we get:

$$dp \times A \times dt = \rho \times dL \times A (V - 0)$$

$$(\text{Force on the fluid}) \quad (\text{Rate of change of momentum})$$

$$\text{or, } dp = \rho \frac{dL}{dt} V = \rho \times \frac{C dt}{dt} \times V = \rho C V \quad (\because dL = C dt)$$

$$\text{or, } C = \frac{dp}{\rho V} \quad \dots(15.28)$$

Multiplying eqns. (15.2) and (15.3), we get:

$$C^2 = \frac{\rho V}{dl} \times \frac{dp}{\rho V} = \frac{dp}{d\rho}$$

$$\therefore C = \sqrt{\frac{dp}{d\rho}} \quad \dots(15.29)$$

15.5.2. Sonic Velocity in terms of Bulk Modulus

The bulk modulus of elasticity of fluid (K) is defined as:

$$K = \frac{dp}{\left(-\frac{dv}{v}\right)} \quad \dots(i)$$

where, dv = decrease in volume, and v = original volume.

(-ve sign indicates that volume decreases with increase in pressure)

Also, volume $v \propto \frac{1}{\rho}$, or $v \rho = \text{constant}$

Differentiating both sides, we get

$$v d\rho + \rho dv = 0 \quad \text{or} \quad -\frac{dv}{v} = \frac{d\rho}{\rho}$$

Substituting the value of $-\frac{dv}{v} \left(= \frac{dp}{K} \right)$ from eqn. (i), we get:

$$\frac{dp}{K} = \frac{dp}{\rho} \quad \text{or} \quad \frac{dp}{d\rho} = \frac{K}{\rho}$$

Substituting this value of $\frac{dp}{d\rho}$ in eqn. (15.29), we get

$$C = \sqrt{\frac{K}{\rho}} \quad \dots(15.30)$$

Eqn. (15.30) is applicable for liquids and gases.

15.5.3. Sonic Velocity for Isothermal Process

For isothermal process : $\frac{p}{\rho} = \text{constant}$

Differentiating both sides, we get:

$$\frac{\rho \cdot dp - p \cdot d\rho}{\rho^2} = 0 \quad \text{or} \quad \frac{dp}{\rho} - \frac{p \cdot d\rho}{\rho^2} = 0$$

$$\text{or,} \quad \frac{dp}{\rho} = \frac{p \cdot d\rho}{\rho^2} \quad \text{or} \quad \frac{dp}{d\rho} = \frac{p}{\rho} = RT \quad \dots(15.31)$$

$$\left(\frac{p}{\rho} = RT \quad \dots \text{equation of state} \right)$$

Substituting the value of $\frac{dp}{d\rho}$ in eqn. (15.29), we get:

$$C = \sqrt{\frac{p}{\rho}} = \sqrt{RT} \quad \dots(15.32)$$

15.5.4 Sonic Velocity for Adiabatic Process

For isentropic (reversible adiabatic) process: $\frac{p}{\rho^\gamma} = \text{constant}$

$$\text{or,} \quad p \cdot \rho^{-\gamma} = \text{Constant}$$

Differentiating both sides, we have $p(-\gamma) \cdot \rho^{-\gamma-1} d\rho + \rho^{-\gamma} dp = 0$

$$\text{Dividing both sides by } \rho^{-\gamma}, \text{ we get:} \quad -p\gamma\rho^{-1}d\rho + dp = 0 \quad \text{or} \quad dp = p\gamma\rho^{-1}d\rho$$

$$\text{or,} \quad \frac{dp}{d\rho} = \frac{p}{\rho}\gamma = \gamma RT \quad \left(\because \frac{p}{\rho} = RT \right)$$

Substituting the value of $\frac{dp}{d\rho}$ in eqn. (15.29), we get:

$$C = \sqrt{\gamma RT} \quad \dots(15.33)$$

The following points are worth noting :

- (i) The process is assumed to be *adiabatic* when minor disturbances are to be propagated through air; due to *very high velocity* of disturbances/pressure waves appreciable heat transfer does not take place.
- (ii) For calculation of velocity of the sound/pressure waves, **isothermal process is considered only when it is mentioned in the numerical problem** (that the process is isothermal). *When no process is mentioned in the problem, calculations are made assuming the process to be adiabatic.*

15.6. MACH NUMBER

The **Mach number** is an important parameter in dealing with the flow of compressible fluids, when elastic forces become important and predominant.

Mach number is defined as the square root of the ratio of the inertia force of a fluid to the elastic force.

$$\therefore \text{Mach number, } M = \sqrt{\frac{\text{Inertia force}}{\text{Elastic force}}} = \sqrt{\frac{\rho A V^2}{K A}} = \sqrt{\frac{V^2}{K / \rho}} = \frac{V}{\sqrt{K / \rho}} = \frac{V}{C} \quad \left[\because \sqrt{K / \rho} = C \right] \quad \text{...eqn. (15.30)}$$

$$\text{i.e.} \quad M = \frac{V}{C} \quad \text{...(15.34)}$$

$$\text{Thus,} \quad M = \frac{\text{Velocity at a point in a fluid}}{\text{Velocity of sound at that point at a given instant of time}}$$

Depending on the value of Mach number, the flow can be classified as follows :

1. *Subsonic flow* : Mach number is less than 1.0 (or $M < 1$); in this case $V < C$.
2. *Sonic flow* : Mach number is equal to 1.0 (or $M = 1$); in this case $V = C$.
3. *Supersonic flow* : Mach number is greater than 1.0 (or $M > 1$); in this case $V > C$.

When the Mach number in flow region is slightly less to slightly greater than 1.0, the flow is termed as **transonic flow**.

The following points are worth noting :

- (i) Mach number is important in those problems in which the *flow velocity is comparable with the sonic velocity* (velocity of sound). It may happen in case of airplanes travelling at very high speed, projectiles, bullets etc.
- (ii) If for any flow system the *Mach number is less than about 0.4 the effects of compressibility may be neglected (for that flow system)*.

Example 15.6. Find the sonic velocity for the following fluids :

(i) Crude oil of specific gravity 0.8 and bulk modulus 1.5 GN/m^2 .

(ii) Mercury having a bulk modulus of 27 GN/m^2 .

(Delhi University)

Solution. Crude oil: Specific gravity = 0.8

$\therefore$ Density of oil, $\rho = 0.8 \times 1000 = 800 \text{ kg/m}^3$

Bulk modulus, $K = 1.5 \text{ GN/m}^2$

Mercury : Bulk modulus, $K = 27 \text{ GN/m}^2$

Density of mercury, $\rho = 13.6 \times 1000 = 13600 \text{ kg/m}^3$

Sonic velocity, C_{oil} , C_{Hg} :

Sonic velocity is given by the relation :

$$C = \sqrt{\frac{K}{\rho}} \quad \text{...[Eqn. (15.30)]}$$

$$\therefore C_{\text{oil}} = \sqrt{\frac{1.5 \times 10^9}{800}} = 1369.3 \text{ m/s (Ans.)}$$

$$C_{\text{Hg}} = \sqrt{\frac{27 \times 10^9}{13600}} = 1409 \text{ m/s (Ans.)}$$

Example 15.7. An aeroplane is flying at a height of 14 km where temperature is -45°C . The speed of the plane is corresponding to $M = 2$. Find the speed of the plane if $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$.

Solution. Temperature (at a height of 14 km), $t = -45^\circ\text{C}$.

$$T = -45 + 273 = 228\text{ K}$$

$$\text{Mach number, } M = 2$$

$$\text{Gas constant, } R = 287\text{ J/kg K}$$

$$\gamma = 1.4$$

Speed of the plane, V :

Sonic velocity, (C) is given by:

$$C = \sqrt{\gamma RT} \text{ (assuming the process to be adiabatic)} \quad \dots[\text{Eqn. (15.33)}]$$

$$= \sqrt{1.4 \times 287 \times 228} = 302.67\text{ m/s}$$

$$\text{Also, } M = \frac{V}{C} \quad \dots[\text{Eqn. (15.34)}]$$

$$\text{or, } 2 = \frac{V}{302.67}$$

$$\text{or, } V = 2 \times 302.67 = 605.34\text{ m/s} = \frac{605.34 \times 3600}{1000} = \mathbf{2179.2\text{ km/h (Ans.)}}$$

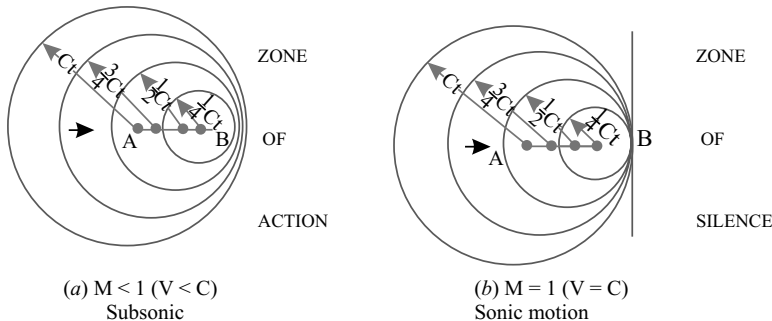
15.7. PROPAGATION OF DISTURBANCE IN COMPRESSIBLE FLUID

When some disturbance is created in a compressible fluid (elastic or pressure waves are also generated), it is propagated in all directions with sonic velocity ($= C$) and its nature of propagation depends upon the Mach number (M). Such disturbance may be created when an object moves in a relatively stationary compressible fluid or when a compressible fluid flows past a stationary object.

Consider a tiny projectile moving in a straight line with velocity V through a compressible fluid which is stationary. Let the projectile is at A when time $t = 0$, then in time t it will move through a distance $AB = Vt$. During this time the disturbance which originated from the projectile when it was at A will grow into the surface of sphere of radius Ct as shown in Fig. 15.4, which also shows the growth of the other disturbances which will originate from the projectile at every $t/4$ interval of time as the projectile moves from A to B .

Let us find nature of propagation of the disturbance for different Mach numbers.

Case I : When $M < 1$ (i.e. $V < C$). In this case since $V < C$ the projectile lags behind the disturbance/pressure wave and hence as shown in Fig. 15.4 (a) the projectile at point B lies inside the sphere of radius Ct and also inside other spheres formed by the disturbances/waves started at intermediate points.



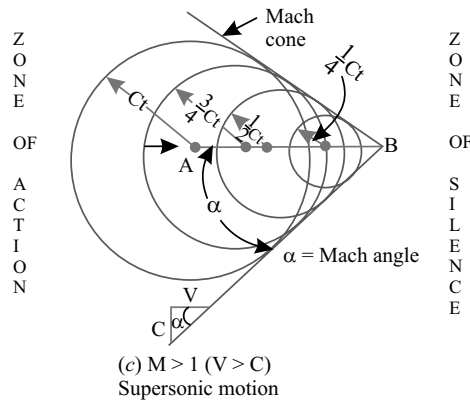


Fig. 15.4. Nature of propagation of disturbances in compressible flow.

Case II: When $M = 1$ (i.e. $V = C$). In this case, the disturbance always travels with the projectile as shown in Fig. 15.4 (b). The circle drawn with centre A will pass through B .

Case III: When $M > 1$ (i.e. $V > C$). In this case the projectile travels faster than the disturbance. Thus the distance AB (which the projectile has travelled) is more than Ct , and hence the projectile at point ' B ' is outside the spheres formed due to formation and growth of disturbance at $t = 0$ and at the intermediate points (Fig. 15.3 (c)). If the tangents are drawn (from the point B) to the circles, the spherical pressure waves form a cone with its vertex at B . It is known as **Mach cone**. The semi-vertex angle α of the cone is known as **Mach angle** which is given by:

$$\sin \alpha = \frac{Ct}{Vt} = \frac{C}{V} = \frac{1}{M} \quad \dots(15.35)$$

In such a case ($M > 1$), the effect of the disturbance is felt only in region inside the Mach cone, this region is called *zone of action*. The region outside the Mach cone is called *zone of silence*.

It has been observed that when an *aeroplane is moving with supersonic speed, its noise is heard only after the plane has already passed over us*.

Example 15.8. Find the velocity of a bullet fired in standard air if its Mach angle is 40° .

Solution. Mach angle, $\alpha = 40^\circ$

$$\gamma = 1.4$$

For standard air: $R = 287 \text{ J/kg K}$, $t = 15^\circ\text{C}$ or $T = 15 + 273 = 288 \text{ K}$

Velocity of the bullet, V :

$$\text{Sonic velocity, } C = \sqrt{\gamma RT} = \sqrt{1.4 \times 287 \times 288} = 340.2 \text{ m/s}$$

$$\text{Now,} \quad \sin \alpha = \frac{C}{V}$$

$$\text{or,} \quad \sin 40^\circ = \frac{340.2}{V} \quad \text{or} \quad V = \frac{340.2}{\sin 40^\circ} = 529.26 \text{ m/s (Ans.)}$$

Example 15.9. A projectile is travelling in air having pressure and temperature as 88.3 kN/m^2 and -2°C . If the Mach angle is 40° , find the velocity of the projectile.

Take $\gamma = 1.4$ and $R = 287 \text{ J/kg K}$.

[M.U.]

Solution. Pressure, $p = 88.3 \text{ kN/m}^2$
 Temperature, $T = -2 + 273 = 271 \text{ K}$
 Mach angle, $M = 40^\circ$
 $\gamma = 1.4, R = 287 \text{ J/kg K}$

Velocity of the projectile, V :

$$\text{Sonic velocity, } C = \sqrt{\gamma RT} = \sqrt{1.4 \times 287 \times 271} \approx 330 \text{ m/s}$$

$$\text{Now,} \quad \sin \alpha = \frac{C}{V} \quad \text{or} \quad \sin 40^\circ = \frac{330}{V}$$

$$\text{or,} \quad V = \frac{330}{\sin 40^\circ} = \mathbf{513.4 \text{ m/s (Ans.)}}$$

Example 15.10. A supersonic aircraft flies at an altitude of 1.8 km where temperature is 4°C . Determine the speed of the aircraft if its sound is heard 4 seconds after its passage over the head of an observer. Take $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$.

Solution. Altitude of the aircraft = 1.8 km = 1800 m

Temperature, $T = 4 + 273 = 277 \text{ K}$

Time, $t = 4 \text{ s}$

Speed of the aircraft, V :

Refer to Fig. 15.5. Let O represent the observer and A the position of the aircraft just vertically over the observer. After 4 seconds, the aircraft reaches the position represented by the point B . Line AB represents the wave front and α the Mach angle.

From Fig. 15.5, we have :

$$\tan \alpha = \frac{1800}{4V} = \frac{450}{V} \quad \dots(i)$$

$$\text{But,} \quad \text{Mach number } M = \frac{V}{C} = \frac{1}{\sin \alpha}$$

$$\text{or,} \quad V = \frac{C}{\sin \alpha} \quad \dots(ii)$$

Substituting the value of V in eqn. (i), we get:

$$\tan \alpha = \frac{450}{(C / \sin \alpha)} = \frac{450 \sin \alpha}{C}$$

$$\text{or,} \quad \frac{\sin \alpha}{\cos \alpha} = \frac{450 \sin \alpha}{C} \quad \text{or} \quad \cos \alpha = \frac{C}{450} \quad \dots(iii)$$

But, $C = \sqrt{\gamma RT}$, where C is the sonic velocity

$$R = 287 \text{ J/kg K and } \gamma = 1.4 \quad \dots(\text{Given})$$

$$\therefore C = \sqrt{1.4 \times 287 \times 277} = 333.6 \text{ m/s}$$

Substituting the value of C in eqn. (ii), we get:

$$\cos \alpha = \frac{333.6}{450} = 0.7413$$

$$\therefore \sin \alpha = \sqrt{1 - \cos^2 \alpha} = \sqrt{1 - 0.7413^2} = 0.6712$$

Substituting the value of $\sin \alpha$ in eqn. (ii), we get:

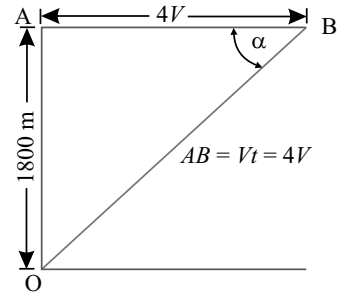


Fig. 15.5

$$V = \frac{C}{\sin \alpha} = \frac{333.6}{0.6712} = 497 \text{ m/s} = \frac{497 \times 3600}{1000} = 1789.2 \text{ km/h (Ans.)}$$

15.8. STAGNATION PROPERTIES

The point on the immersed body where the velocity is zero is called **stagnation point**. At this point velocity head is converted into pressure head. The values of pressure (p_s), temperature (T_s) and density (ρ_s) at stagnation point are called **stagnation properties**.

15.8.1 Expression for Stagnation Pressure (p_s) in Compressible Flow

Consider the flow of compressible fluid past an immersed body where the velocity becomes zero. Consider *frictionless adiabatic (isentropic)* condition. Let us consider two points, O in the free stream and the stagnation point S as shown in the Fig. 15.6.

Let, p_0 = Pressure of compressible fluid at point O ,

V_0 = Velocity of fluid at O ,

ρ_0 = Density of fluid at O ,

T_0 = Temperature of fluid at O ,

and p_s , V_s , ρ_s and T_s corresponding values of pressure, velocity density, and temperature at point S .

Applying Bernoulli's equation for adiabatic (frictionless) flow at points O and S , (given by eqn. 15.26), we get:

$$\left(\frac{\gamma}{\gamma - 1} \right) \frac{p_0}{\rho_0 g} + \frac{V_0^2}{2g} + z_0 = \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_s}{\rho_s g} + \frac{V_s^2}{2g} + z_s$$

But $z_0 = z_s$; the above equation reduces to:

$$\left(\frac{\gamma}{\gamma - 1} \right) \frac{p_0}{\rho_0 g} + \frac{V_0^2}{2g} = \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_s}{\rho_s g} + \frac{V_s^2}{2g}$$

Cancelling 'g' on both the sides, we have:

$$\left(\frac{\gamma}{\gamma - 1} \right) \frac{p_0}{\rho_0} + \frac{V_0^2}{2} = \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_s}{\rho_s} + \frac{V_s^2}{2}$$

At point S the velocity is zero, i.e. $V_s = 0$; the above equation becomes:

$$\left(\frac{\gamma}{\gamma - 1} \right) \left(\frac{p_0}{\rho_0} - \frac{p_s}{\rho_s} \right) = - \frac{V_0^2}{2}$$

$$\text{or, } \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_0}{\rho_0} \left(1 - \frac{p_s}{p_0} \times \frac{\rho_0}{\rho_s} \right) = - \frac{V_0^2}{2}$$

$$\text{or, } \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_0}{\rho_0} \left(1 - \frac{p_s}{p_0} \times \frac{\rho_0}{\rho_s} \right) = - \frac{V_0^2}{2} \quad \dots(i)$$

For adiabatic process:

$$\frac{p_0}{\rho_0^\gamma} = \frac{p_s}{\rho_s^\gamma} \quad \text{or} \quad \frac{p_0}{p_s} = \frac{\rho_0^\gamma}{\rho_s^\gamma}$$

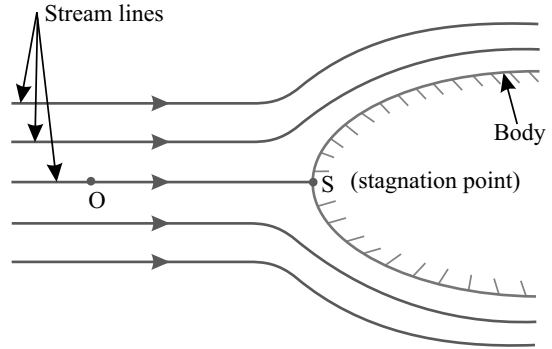


Fig. 15.6 Stagnation properties.

$$\text{or,} \quad \frac{\rho_0}{\rho_s} = \left(\frac{p_0}{p_s} \right)^{\frac{1}{\gamma}} \quad \dots(ii)$$

Substituting the value of $\frac{\rho_0}{\rho_s}$ in eqn. (i), we get:

$$\left(\frac{\gamma}{\gamma-1} \right) \frac{p_0}{\rho_0} \left[1 - \frac{p_s}{p_0} \times \left(\frac{p_0}{p_s} \right)^{\frac{1}{\gamma}} \right] = - \frac{V_0^2}{2}$$

$$\text{or,} \quad \left(\frac{\gamma}{\gamma-1} \right) \frac{p_0}{\rho_0} \left[1 - \left(\frac{p_s}{p_0} \right)^{1-\frac{1}{\gamma}} \right] = - \frac{V_0^2}{2}$$

$$\text{or,} \quad \left[1 - \left(\frac{p_s}{p_0} \right)^{\frac{\gamma-1}{\gamma}} \right] = - \frac{V_0^2}{2} \left(\frac{\gamma-1}{\gamma} \right) \frac{\rho_0}{p_0}$$

$$\text{or,} \quad 1 + \frac{V_0^2}{2} \left(\frac{\gamma-1}{\gamma} \right) \frac{\rho_0}{p_0} = \left(\frac{p_s}{p_0} \right)^{\frac{\gamma-1}{\gamma}} \quad \dots(iii)$$

For adiabatic process, the sonic velocity is given by:

$$C = \sqrt{\gamma RT} = \sqrt{\gamma \frac{p}{\rho}} \quad \left(\because \frac{p}{\rho} = RT \right)$$

$$\text{For point 0,} \quad C_0 = \sqrt{\gamma \frac{p_0}{\rho_0}} \quad \text{or} \quad C_0^2 = \gamma \frac{p_0}{\rho_0}$$

Substituting the value of $\frac{\gamma p_0}{\rho_0} = C_0^2$ in eqn. (iii), we get:

$$1 + \frac{V_0^2}{2} (\gamma-1) \times \frac{1}{C_0^2} = \left(\frac{p_s}{p_0} \right)^{\frac{\gamma-1}{\gamma}}$$

$$\text{or,} \quad 1 + \frac{V_0^2}{2C_0^2} (\gamma-1) = \left(\frac{p_s}{p_0} \right)^{\frac{\gamma-1}{\gamma}}$$

$$1 + \frac{M_0^2}{2} (\gamma-1) = \left(\frac{p_s}{p_0} \right)^{\frac{\gamma-1}{\gamma}} \quad \left(\because \frac{V_0^2}{C_0^2} = M_0^2 \right)$$

$$\text{or,} \quad \left(\frac{p_s}{p_0} \right)^{\frac{\gamma-1}{\gamma}} = \left[1 + \left(\frac{\gamma-1}{2} \right) M_0^2 \right]$$

$$\text{or,} \quad \frac{p_s}{p_0} = \left[1 + \left(\frac{\gamma-1}{2} \right) M_0^2 \right]^{\frac{\gamma}{\gamma-1}} \quad \dots(iv)$$

$$\text{or,} \quad p_s = p_0 \left[1 + \left(\frac{\gamma-1}{2} \right) M_0^2 \right]^{\frac{\gamma}{\gamma-1}} \quad \dots(15.36)$$

Eqn. (15.36) gives the value of **stagnation pressure**.

Compressibility correction factor:

If the right hand side of Eqn. (15.36) is expanded by the binomial theorem, we get:

$$\begin{aligned} p_s &= p_0 \left[1 + \frac{\gamma}{2} M_0^2 + \frac{\gamma}{8} M_0^4 + \frac{\gamma(2-\gamma)}{48} M_0^6 \right] \\ &= p_0 \left[1 + \frac{\gamma M_0^2}{2} \left(1 + \frac{M^2}{4} + \frac{2-\gamma}{24} M_0^4 + \dots \right) \right] \end{aligned}$$

or,

$$p_s = p_0 + \frac{p_0 \gamma M_0^2}{2} \left(1 + \frac{M_0^2}{4} + \frac{2-\gamma}{24} M_0^4 + \dots \right) \quad \dots(15.37)$$

But,

$$M_0^2 = \frac{V_0^2}{C_0^2} = \frac{V_0^2}{\left(\frac{\gamma p_0}{\rho_0} \right)} = \frac{V_0^2 \rho_0}{\gamma p_0} \quad \left(\because C_0^2 = \frac{\gamma p_0}{\rho_0} \right)$$

Substituting the value of M_0^2 in eqn. 15.37, we get:

$$p_s = p_0 + \frac{p_0 \gamma}{2} \times \frac{V_0^2 \rho_0}{\gamma p_0} \left(1 + \frac{M_0^2}{4} + \frac{2-\gamma}{24} M_0^4 + \dots \right)$$

or,

$$p_s = p_0 + \frac{\rho_0 V_0^2}{2} \left(1 + \frac{M_0^2}{4} + \frac{2-\gamma}{24} M_0^4 + \dots \right) \quad \dots(15.38)$$

Also,

$$p_s = p_0 + \frac{\rho_0 V_0^2}{2} \quad (\text{when compressibility effects are neglected}) \quad \dots(15.39)$$

The comparison of eqns. (15.38) and (15.39) shows that the effects of compressibility are isolated in the bracketed quantity and that these effects *depend only* upon the *Mach number*. The bracketed quantity $\left[i.e., \left(1 + \frac{M_0^2}{4} + \frac{2-\gamma}{24} M_0^4 + \dots \right) \right]$ may thus be considered as a **compressibility**

correction factor. It is worth noting that :

- For $M < 0.2$, the compressibility affects the pressure difference ($p_s - p_0$) by *less than 1 per cent* and the simple formula for flow at constant density is then sufficiently accurate.
- For larger value of M , as the terms of binomial expansion become significant, the *compressibility effect must be taken into account*.
- When the Mach number exceeds a value of about 0.3 the *Pitot-static tube used for measuring aircraft speed needs calibration to take into account the compressibility effects*.

15.8.2. Expression for Stagnation Density (ρ_s)

From eqn. (ii), we have:

$$\frac{\rho_0}{\rho_s} = \left(\frac{p_0}{p_s} \right)^{\frac{1}{\gamma}} \quad \text{or} \quad \frac{\rho_s}{\rho_0} = \left(\frac{p_s}{p_0} \right)^{\frac{1}{\gamma}} \quad \text{or} \quad \rho_s = \rho_0 \left(\frac{p_s}{p_0} \right)^{\frac{1}{\gamma}}$$

Substituting the value of $\left(\frac{p_s}{p_0} \right)$ from eqn. (iv), we get:

$$p_s = \rho_0 \left[\left\{ 1 + \left(\frac{\gamma-1}{2} \right) M_0^2 \right\}^{\frac{\gamma}{\gamma-1}} \right]^{\frac{1}{\gamma}}$$

$$\text{or,} \quad \rho_s = \rho_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]^{\frac{1}{\gamma - 1}} \quad \dots(15.40)$$

15.8.3 Expression for stagnation temperature (T_s)

The equation of state is given by : $\frac{p}{\rho} = RT$

For stagnation point, the equation of state may be written as :

$$\frac{p_s}{\rho_s} = RT_s \quad \text{or} \quad T_s = \frac{1}{R} \frac{p_s}{\rho_s}$$

Substituting the values of p_s and ρ_s from eqns. (15.36) and (15.37), we get:

$$\begin{aligned} T_s &= \frac{1}{R} \frac{p_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]^{\frac{\gamma}{\gamma - 1}}}{\rho_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]^{\frac{1}{\gamma - 1}}} \\ &= \frac{1}{R} \frac{p_0}{\rho_0} \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]^{\left(\frac{\gamma}{\gamma - 1} - \frac{1}{\gamma - 1} \right)} \\ &= \frac{1}{R} \frac{p_0}{\rho_0} \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]^{\left(\frac{\gamma - 1}{\gamma - 1} \right)} \\ \text{or,} \quad T_s &= T_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right] \quad \dots(15.41) \end{aligned}$$

$$\left(\because \frac{p_0}{\rho_0} = RT_0 \right)$$

Example 15.11. An aeroplane is flying at 1000 km/h through still air having a pressure of 78.5 kN/m² (abs.) and temperature -8°C . Calculate on the stagnation point on the nose of the plane :

- (i) Stagnation pressure,
- (ii) Stagnation temperature, and
- (iii) Stagnation density.

Take for air : $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$.

Solution. Speed of aeroplane,

$$V = 1000 \text{ km/h} = \frac{1000 \times 1000}{60 \times 60} = 277.77 \text{ m/s}$$

$$\text{Pressure of air, } p_0 = 78.5 \text{ kN/m}^2$$

$$\text{Temperature of air, } T_0 = -8 + 273 = 265 \text{ K}$$

$$\text{For air :} \quad R = 287 \text{ J/kg K, } \gamma = 1.4$$

The sonic velocity for adiabatic flow is given by:

$$C_0 = \sqrt{\gamma R T_0} = \sqrt{1.4 \times 287 \times 265} = 326.31 \text{ m/s}$$

$$\therefore \text{Mach number, } M_0 = \frac{V_0}{C_0} = \frac{277.77}{326.31} = 0.851$$

(i) Stagnation pressure, p_s :

The stagnation pressure (p_s) is given by the relation:

$$p_s = p_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]^{\frac{\gamma}{\gamma - 1}} \quad \dots[\text{Eqn. (15.36)}]$$

or,

$$\begin{aligned} p_s &= 78.5 \left[1 + \left(\frac{1.4 - 1}{2} \right) \times 0.851^2 \right]^{\frac{1.4}{1.4 - 1}} \\ &= 78.5 (1.145)^{3.5} = \mathbf{126.1 \text{ kN/m}^2} \quad (\text{Ans.}) \end{aligned}$$

(ii) Stagnation temperature, T_s :

The stagnation temperature is given by:

$$\begin{aligned} T_s &= T_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right] \quad \dots[\text{Eqn. (15.41)}] \\ &= 265 \left[1 + \frac{1.4 - 1}{2} \times 0.851^2 \right] = 303.4 \text{ K} \quad \text{or} \quad \mathbf{30.4^\circ \text{C}} \quad (\text{Ans.}) \end{aligned}$$

(iii) Stagnation density, ρ_s :

The stagnation density (ρ_s) is given by:

$$\frac{p_s}{\rho_s} = RT_s \quad \text{or} \quad \rho_s = \frac{p_s}{RT_s}$$

or,

$$p_s = \frac{126.1 \times 10^3}{287 \times 303.4} = \mathbf{1.448 \text{ kg/m}^3} \quad (\text{Ans.})$$

Example 15.12. Air has a velocity of 1000 km/h at a pressure of 9.81 kN/m² vacuum and a temperature of 47°C. Compute its stagnation properties and the local Mach number. Take atmospheric pressure = 98.1 kN/m², $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$.

What would be the compressibility correction factor for a pitot-static tube to measure the velocity at a Mach number of 0.8. [PEC]

Solution. Velocity of air, $V_0 = 1000 \text{ km/h} = \frac{1000 \times 1000}{60 \times 60} = 277.78 \text{ m/s}$

Temperature of air, $T_0 = 47 + 273 = 320 \text{ K}$

Atmospheric pressure, $p_{atm} = 98.1 \text{ kN/m}^2$

Pressure of air (static), $p_0 = 98.1 - 9.81 = 88.29 \text{ kN/m}^2$

$R = 287 \text{ J/kg K}$, $\gamma = 1.4$

Sonic velocity, $C_0 = \sqrt{\gamma R T_0} = \sqrt{1.4 \times 287 \times 320} = 358.6 \text{ m/s}$

$\therefore$ Mach number, $M_0 = \frac{V_0}{C_0} = \frac{277.78}{358.6} = 0.7746$

Stagnation pressure, (p_s) :

The stagnation pressure is given by:

$$p_s = p_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]^{\frac{\gamma}{\gamma - 1}} \quad \dots[\text{Eqn. (15.36)}]$$

or,

$$\begin{aligned} p_s &= 88.29 \left[1 + \frac{1.4 - 1}{2} \times 0.7746^2 \right]^{\frac{1.4}{1.4 - 1}} \\ &= 88.29 (1.12)^{3.5} = \mathbf{131.27 \text{ kN/m}^2} \quad (\text{Ans.}) \end{aligned}$$

Stagnation temperature, T_s :

$$T_s = T_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right] \quad \dots[\text{Eqn. (15.41)}]$$

or,

$$T_s = 320 \left[1 + \frac{1.4-1}{2} \times 0.7746^2 \right] = 358.4 \text{ K or } 85.4^\circ\text{C} \quad (\text{Ans.})$$

Stagnation density, ρ_s :

$$p_s = \frac{p_s}{RT_s} = \frac{131.27 \times 10^3}{287 \times 358.4} = 1.276 \text{ kg/m}^3 \quad (\text{Ans.})$$

Compressibility factor at $M = 0.8$:

$$\begin{aligned} \text{Compressibility factor} &= 1 + \frac{M_0^2}{4} + \frac{2-\gamma}{24} M_0^4 + \dots \\ &= 1 + \frac{0.8^2}{4} + \frac{2-1.4}{24} \times 0.8^4 = 1.1702 \quad (\text{Ans.}) \end{aligned}$$

Example 15.13. Air at a pressure of 220 kN/m^2 and temperature 27°C is moving at a velocity of 200 m/s . Calculate the stagnation pressure if

(i) Compressibility is neglected; (ii) Compressibility is accounted for.

For air, take $R = 287 \text{ J/kg K}$, $\gamma = 1.4$

Solution. Pressure of air, $p_0 = 200 \text{ kN/m}^2$

Temperature of air, $T_0 = 27 + 233 = 300 \text{ K}$

Velocity of air, $V_0 = 200 \text{ m/s}$

Stagnation pressure, p_s :

(i) Compressibility is neglected :

$$p_s = p_0 + \frac{\rho_0 V_0^2}{2}$$

where,

$$\rho_0 = \frac{p_0}{RT_0} = \frac{220 \times 10^3}{287 \times 300} = 2.555 \text{ kg/m}^3$$

$$\therefore p_s = 220 + \frac{2.555 \times 200^2}{2} \times 10^{-3} \text{ (kN/m}^2\text{)} = 271.1 \text{ kN/m}^2 \quad (\text{Ans.})$$

(ii) Compressibility is accounted for :

The stagnation pressure, when compressibility is accounted for, is given by:

$$p_s = p_0 + \frac{\rho_0 V_0^2}{2} \left(1 + \frac{M_0^2}{4} + \frac{2-\gamma}{24} M_0^4 + \dots \right) \quad \dots[\text{Eqn. (13.38)}]$$

$$\text{Mach number, } M_0 = \frac{V_0}{C_0} = \frac{200}{\sqrt{\gamma RT_0}} = \frac{200}{\sqrt{1.4 \times 287 \times 300}} = 0.576$$

Whence,

$$p_s = 220 + \frac{2.555 \times 200^2}{2} \times 10^{-3} \left(1 + \frac{0.576^2}{4} + \frac{2-1.4}{24} \times 0.576^4 \right)$$

or,

$$p_s = 220 + 51.1 (1 + 0.0829 + 0.00275) = 275.47 \text{ kN/m}^2 \quad (\text{Ans.})$$

Example 15.14. In aircraft flying at an altitude where the pressure was 35 kPa and temperature -38°C , stagnation pressure measured was 65.4 kPa . Calculate the speed of the aircraft. Take molecular weight of air as 28. (UPSC)

Solution. Pressure of air, $p_0 = 35 \text{ kPa} = 35 \times 10^3 \text{ N/m}^2$

Temperature of air, $T_0 = -38 + 273 = 235 \text{ K}$

Stagnation pressure, $p_s = 65.4 \text{ kPa} = 65.4 \times 10^3 \text{ N/m}^2$

Speed of the aircraft, V_a :

$$p_0 V = mRT_0 = m \times \left(\frac{R_0}{M} \right) T_0$$

$$\text{or, } p_0 = \frac{m}{V} = \frac{p_0 M}{R_0 T_0}$$

where,

R = Characteristic gas constant,

R_0 = Universal gas constant = 8314 Nm/mole K,

M = Molecular weight, for air = 28, and

ρ_0 = Density of air.

Substituting the values, we get:

$$p_0 = \frac{(35 \times 10^3) \times 28}{8314 \times 235} = 0.5 \text{ kg/m}^3$$

$$\text{Now, using the relation: } p_s = p_0 + \frac{\rho_0 V_a^2}{2} \quad \dots \text{ [(Eqn. 15.39)]}$$

$$\begin{aligned} \text{or, } V_a &= \sqrt{\frac{2(p_s - p)}{\rho_0}} = \sqrt{\frac{2(65.4 \times 10^3 - 35 \times 10^3)}{0.5}} \\ &= 348.7 \text{ m/s (Ans.)} \end{aligned}$$

15.9. AREA-VELOCITY RELATIONSHIP AND EFFECT OF VARIATION OF AREA FOR SUBSONIC, SONIC AND SUPERSONIC FLOWS

For an *incompressible flow* the continuity equation may be expressed as :

$AV = \text{Constant}$, which when differentiated gives,

$$AdC + VdA = 0$$

$$\text{or, } \frac{dA}{A} = -\frac{dV}{V} \quad \dots(15.42)$$

But in case of *compressible flow*, the continuity equation is given by:

$\rho AV = \text{Constant}$, which can be differentiated to give

$$\rho d(AV) + AVd\rho = 0$$

$$\text{or, } \rho(AdV + VdA) + AVd\rho = 0$$

$$\text{or, } \rho AdV + \rho VdA + AVd\rho = 0$$

Dividing both sides by ρAV , we get:

$$\frac{dV}{V} + \frac{dA}{A} + \frac{d\rho}{\rho} = 0 \quad \dots(15.93)$$

$$\text{or, } \frac{dA}{A} = -\frac{dV}{V} - \frac{d\rho}{\rho} \quad \dots(15.43 (a))$$

The Euler's equation for compressible fluid is given by:

$$\frac{dp}{\rho} + VdV + gdz = 0$$

Neglecting the z terms the above equation reduces to:

$$\frac{dp}{\rho} + VdV = 0$$

This equation can also be expressed as:

$$\frac{dp}{\rho} \times \frac{d\rho}{d\rho} + VdV = 0$$

$$\text{or, } \frac{dp}{d\rho} \times \frac{d\rho}{\rho} + VdV = 0$$

$$\text{But, } \frac{dp}{d\rho} = C^2 \quad \dots[\text{Eqn. (15.29)}]$$

$$\therefore C^2 \times \frac{d\rho}{\rho} + VdV = 0$$

$$\text{or, } C^2 \frac{d\rho}{\rho} = -VdV \quad \text{or} \quad \frac{d\rho}{\rho} = -\frac{VdV}{C^2}$$

Substituting the value of $\frac{d\rho}{\rho}$ in eqn. (15.43), we get:

$$\frac{dV}{V} + \frac{dA}{A} - \frac{VdV}{C^2} = 0$$

$$\text{or, } \frac{dA}{A} = \frac{VdV}{C^2} - \frac{dV}{V} = \frac{dV}{V} \left(\frac{V^2}{C^2} - 1 \right)$$

$$\therefore \frac{dA}{A} = \frac{dV}{V} (M^2 - 1) \quad \left(\because M = \frac{V}{C} \right) \quad \dots(15.44)$$

*This important equation is due to **Hugoniot**.*

Eqns. (15.42) and (15.44) give variation of $\left(\frac{dA}{A} \right)$ for the flow of incompressible and

compressible fluids respectively. The ratios $\left(\frac{dA}{A} \right)$ and $\left(\frac{dV}{V} \right)$ are respectively fractional variations in the values of area and flow velocity in the flow passage.

Further, in order to study the variation of pressure with the change in flow area, an expression similar to eqn. (15.44), as given below, can be obtained:

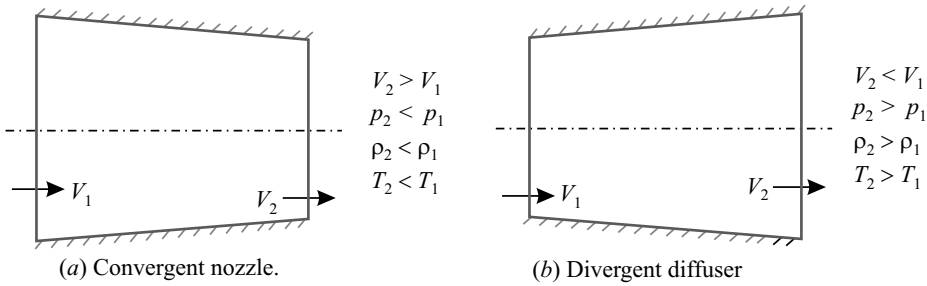
$$dp = \rho V^2 \left(\frac{1}{1 - M^2} \right) \frac{dA}{A} \quad \dots(15.45)$$

From eqns. (15.44) and (15.45), it is possible to formulate the following *conclusions* of practical significance:

(i) **For subsonic flow ($M < 1$) :**

$$\frac{dV}{V} > 0 ; \frac{dA}{A} < 0 ; dp < 0 \text{ (convergent nozzle)}$$

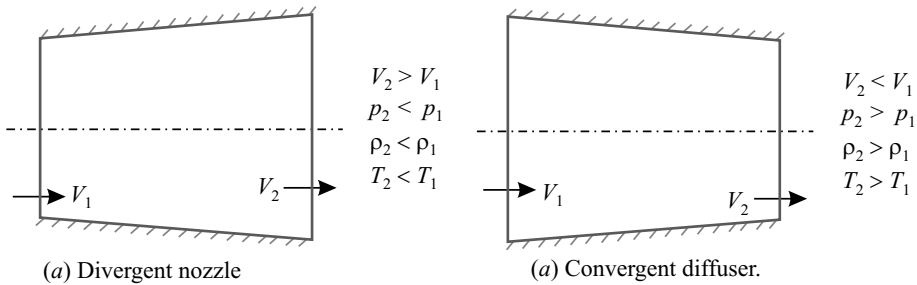
$$\frac{dV}{V} < 0 ; \frac{dA}{A} > 0 ; dp > 0 \text{ (divergent diffuser)}$$

**Fig. 15.7.** Subsonic flow ($M < 1$).

(ii) For supersonic flow ($M > 1$) :

$$\frac{dV}{V} > 0; \frac{dA}{A} > 0; dp < 0 \text{ (divergent nozzle)}$$

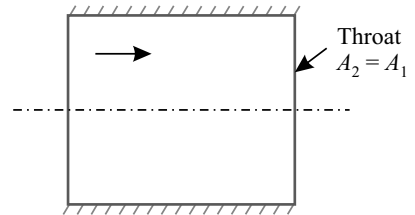
$$\frac{dV}{V} < 0; \frac{dA}{A} < 0; dp > 0 \text{ (convergent diffuser)}$$

**Fig. 15.8.** Supersonic flow ($M > 1$).

(iii) For sonic flow ($M = 1$) :

$$\frac{dA}{A} = 0 \text{ (straight flow passage since } dA \text{ must be zero)}$$

and, $dp = (\text{zero}/\text{zero})$ i.e. indeterminate, but when evaluated, the change of pressure $dp = 0$, since $dA = 0$ and the flow is frictionless.

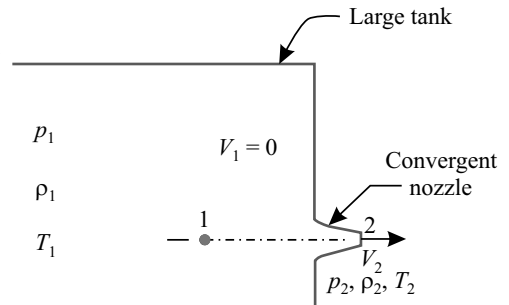
**Fig. 15.9.** Sonic Flow ($M = 1$)

15.10. FLOW OF COMPRESSIBLE FLUID THROUGH A CONVERGENT NOZZLE

Fig. 15.10 shows a large tank/vessel fitted with a short convergent nozzle and containing a compressible fluid. Consider two points 1 and 2 inside the tank and exit of the nozzle respectively.

Let, p_1 = Pressure of fluid at the point 1,
 V_1 = Velocity of fluid in the tank ($= 0$),
 T_1 = Temperature of fluid at point 1,
 ρ_1 = Density of fluid at point 1, and

p_2, V_2, T_2 and ρ_2 = Corresponding values of pressure, velocity, temperature and density at point 2.

**Fig. 15.10.** Flow of fluid through a convergent nozzle.

Assuming the flow to take place *adiabatically*, then by using Bernoulli's equation (for adiabatic flow), we have:

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1 g} + \frac{V_1^2}{2g} + z_1 = \left(\frac{\gamma}{\gamma-1}\right) \frac{p_2}{\rho_2 g} + \frac{V_2^2}{2g} + z_2 \quad [\text{Eqn. (15.26)}]$$

But $z_1 = z_2$ and $V_1 = 0$

$$\therefore \frac{\gamma}{\gamma-1} \frac{p_1}{\rho_1 g} = \left(\frac{\gamma}{\gamma-1}\right) \frac{p_2}{\rho_2 g} + \frac{V_2^2}{2g}$$

$$\text{or, } \left(\frac{\gamma}{\gamma-1}\right) \left[\frac{p_1}{\rho_1 g} - \frac{p_2}{\rho_2 g} \right] = \frac{V_2^2}{2g}$$

$$\text{or, } \frac{\gamma}{\gamma-1} \left(\frac{p_1}{\rho_1} - \frac{p_2}{\rho_2} \right) = \frac{V_2^2}{2}$$

$$\text{or, } V_2 = \sqrt{\frac{2\gamma}{(\gamma-1)} \left(\frac{p_1}{\rho_1} - \frac{p_2}{\rho_2} \right)}$$

$$\text{or, } V_2 = \sqrt{\frac{2\gamma}{(\gamma-1)} \frac{p_1}{\rho_1} \left(1 - \frac{p_2}{p_1} \times \frac{\rho_1}{\rho_2} \right)} \quad \dots(1)$$

For adiabatic flow : $\frac{p_1}{\rho_1^\gamma} = \frac{p_2}{\rho_2^\gamma} \quad \text{or} \quad \frac{p_1}{p_2} = \left(\frac{\rho_1}{\rho_2} \right)^\gamma$

$$\text{or, } \frac{\rho_1}{\rho_2} = \left(\frac{p_1}{p_2} \right)^{\frac{1}{\gamma}} \quad \dots(i)$$

Substituting the value of $\frac{\rho_1}{\rho_2}$ in eqn. (1), we get:

$$V_2 = \sqrt{\frac{2\gamma}{(\gamma-1)} \frac{p_1}{\rho_1} \left[1 - \frac{p_2}{p_1} \times \left(\frac{p_1}{p_2} \right)^{\frac{1}{\gamma}} \right]} = \sqrt{\frac{2\gamma}{(\gamma-1)} \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{1-\frac{1}{\gamma}} \right]}$$

$$\text{or, } V_2 = \sqrt{\frac{2\gamma}{(\gamma-1)} \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right]} \quad \dots(15.46)$$

The mass rate of flow of the compressible fluid,

$$m = \rho_2 A_2 V_2 \quad (A_2 \text{ being the area of the nozzle at the exit})$$

$$= \rho_2 A_2 \sqrt{\frac{2\gamma}{(\gamma-1)} \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right]}, \quad [\text{substituting } V_2 \text{ from eqn. (15.46)}]$$

$$\text{or, } m = A_2 \sqrt{\frac{2\gamma}{(\gamma-1)} \frac{p_1}{\rho_1} \times \rho_2^2 \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right]}$$

From eqn. (i), we have:

$$\rho_2 = \frac{\rho_1}{(p_1/p_2)^{1/\gamma}} = \rho_1 \left(\frac{p_2}{p_1} \right)^{\frac{1}{\gamma}}$$

$$\therefore \rho_2^2 = \rho_1^2 \left(\frac{p_1}{p_2} \right)^{2/\gamma}$$

Substituting this value in the above equation, we get:

$$\begin{aligned} m &= A_2 \sqrt{\frac{2\gamma}{\gamma-1} \frac{p_1}{\rho_1} \times \rho_1^2 \left(\frac{p_2}{p_1} \right)^{2/\gamma} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right]} \\ &= A_2 \sqrt{\frac{2\gamma}{\gamma-1} p_1 \rho_1 \left[\left(\frac{p_2}{p_1} \right)^{2/\gamma} - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma} + \frac{2}{\gamma}} \right]} \\ m &= A_2 \sqrt{\frac{2\gamma}{\gamma-1} p_1 \rho_1 \left[\left(\frac{p_2}{p_1} \right)^{2/\gamma} - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma+1}{\gamma}} \right]} \quad \dots(15.47) \end{aligned}$$

The mass rate of flow (m) depends on the value of $\frac{p_2}{p_1}$ (for the given values of p_1 and ρ_1 at point 1).

Value of $\frac{p_2}{p_1}$ for maximum value of mass rate of flow:

For maximum value of m we have: $\frac{dm}{d\left(\frac{p_2}{p_1}\right)} = 0$

As other quantities except the ratio $\frac{p_2}{p_1}$ are constant, therefore:

$$\begin{aligned} \frac{d}{d\left(\frac{p_2}{p_1}\right)} \left[\left(\frac{p_2}{p_1} \right)^{2/\gamma} - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma+1}{\gamma}} \right] &= 0 \\ \text{or, } \frac{2}{\gamma} \left(\frac{p_2}{p_1} \right)^{\frac{2}{\gamma}-1} - \left(\frac{\gamma+1}{\gamma} \right) \left(\frac{p_2}{p_1} \right)^{\frac{\gamma+1}{\gamma}-1} &= 0 \\ \text{or, } \left(\frac{p_2}{p_1} \right)^{\frac{2}{\gamma}-1} &= \frac{\gamma+1}{2} \left(\frac{p_2}{p_1} \right)^{\frac{1}{\gamma}} \quad \text{or} \quad \left(\frac{p_2}{p_1} \right)^{\frac{2-\gamma}{\gamma}} = \frac{\gamma+1}{2} \left(\frac{p_2}{p_1} \right)^{\frac{1}{\gamma}} \\ \text{or, } \left(\frac{p_2}{p_1} \right)^{2-\gamma} &= \left(\frac{\gamma+1}{2} \right)^\gamma \left(\frac{p_2}{p_1} \right) \\ \text{or, } \left(\frac{p_2}{p_1} \right)^{2-\gamma-1} &= \left(\frac{\gamma+1}{2} \right)^\gamma \quad \text{or} \quad \left(\frac{p_2}{p_1} \right)^{1-\gamma} = \left(\frac{\gamma+1}{2} \right)^\gamma \\ \text{or, } \left(\frac{p_2}{p_1} \right)^{\gamma-1} &= \left(\frac{2}{\gamma+1} \right)^\gamma \quad \text{or} \quad \left(\frac{p_2}{p_1} \right) = \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}} \quad \dots(15.48) \end{aligned}$$

Eqn. (15.48) is the condition for maximum mass flow rate through the nozzle.

- It may be pointed out that a convergent nozzle is employed when the exit pressure is equal to or more than the critical pressure, and a convergent-divergent nozzle is used when the discharge pressure is less than the critical pressure.

For air with $\gamma = 1.4$, the *critical pressure ratio*,

$$\frac{p_2}{p_1} = \left(\frac{2}{1.4 + 1} \right)^{\frac{1.4}{1.4 - 1}} = 0.528 \quad \dots(15.49)$$

Relevant relations for critical density and temperature are :

$$\frac{\rho_2}{\rho_1} = \left(\frac{2}{\gamma + 1} \right)^{\frac{1}{\gamma - 1}} \quad \dots[15.49(a)]$$

$$\frac{T_2}{T_1} = \frac{2}{\gamma + 1} \quad \dots[15.49(b)]$$

Value of V_2 for maximum rate of flow of fluid:

Substituting the value of $\frac{p_2}{p_1}$ from eqn. (15.48) in eqn. (15.46), we get:

$$\begin{aligned} V_2 &= \sqrt{\frac{2\gamma}{\gamma - 1} \frac{p_1}{\rho_1} \left[1 - \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1} \times \frac{\gamma - 1}{\gamma}} \right]} = \sqrt{\frac{2\gamma}{\gamma - 1} \frac{p_1}{\rho_1} \left(1 - \frac{2}{\gamma + 1} \right)} \\ &= \sqrt{\frac{2\gamma}{\gamma - 1} \frac{p_1}{\rho_1} \left(\frac{\gamma + 1 - 2}{\gamma + 1} \right)} = \sqrt{\frac{2\gamma}{\gamma - 1} \frac{p_1}{\rho_1} \left(\frac{\gamma - 1}{\gamma + 1} \right)} \end{aligned}$$

$$\text{or,} \quad V_2 = \sqrt{\frac{2\gamma}{\gamma + 1} \frac{p_1}{\rho_1}} (= C_2) \quad \dots(15.50)$$

Maximum rate of flow of fluid through nozzle, $m_{\max}$:

Substituting the value of $\frac{p_2}{p_1}$ From eqn. (15.49) in eqn. (15.47), we get:

$$\begin{aligned} m_{\max} &= A_2 \sqrt{\left(\frac{2\gamma}{\gamma - 1} \right) p_1 \rho_1 \left[\left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1} \times \frac{2}{\gamma}} - \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1} \times \frac{\gamma + 1}{\gamma}} \right]} \\ &= A_2 \sqrt{\left(\frac{2\gamma}{\gamma - 1} \right) p_1 \rho_1 \left[\left(\frac{2}{\gamma + 1} \right)^{\frac{2}{\gamma - 1}} - \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma + 1}{\gamma - 1}} \right]} \end{aligned}$$

For air, $\gamma = 1.4$,

$$\begin{aligned} \therefore m_{\max} &= A_2 \sqrt{\left(\frac{2 \times 1.4}{1.4 - 1} \right) p_1 \rho_1 \left[\left(\frac{2}{1.4 + 1} \right)^{\frac{2}{1.4 - 1}} - \left(\frac{2}{1.4 + 1} \right)^{\frac{1.4 + 1}{1.4 - 1}} \right]} \\ &= A_2 \sqrt{7 p_1 \rho_1 (0.4018 - 0.3348)} \end{aligned}$$

$$\text{or,} \quad m_{\max} = 0.685 A_2 \sqrt{p_1 \rho_1} \quad \dots(15.51)$$

Variation of mass flow rate of compressible fluid with pressure ratio $\left(\frac{p_2}{p_1} \right)$:

A passage in which the sonic velocity has been reached and thus in which the mass flow rate is maximum is often said to be **choked** or in **choking conditions**. It is evident from eqn. (15.47) that for a fixed value of inlet pressure the mass flow depends on nozzle exit pressure.

Fig. 15.11 depicts the variation of actual and theoretical mass flow rate versus $\frac{p_2}{p_1}$. Following points are *worth noting* :

(i) The flow rate increases with a decrease in the pressure ratio $\frac{p_2}{p_1}$ and attains the maximum value of the critical pressure ratio $\frac{p_2}{p_1} = 0.528$ for air.

(ii) With further decrease in exit pressure below the critical value, the theoretical mass flow rate decreases. This is contrary to the actual results where the mass flow rate remains constant after attaining the maximum value.

This may be explained as follows :

At critical pressure ratio, the velocity V_2 at the throat is equal to the sonic speed (derived below). For an accelerating flow of a compressible fluid in a convergent nozzle the velocity of flow within the nozzle is subsonic with a maximum velocity equal to the sonic velocity at the throat. Thus once the velocity V_2 at the throat has attained the sonic speed at the critical pressure ratio, it remains at the same value for all the values of $\left(\frac{p_2}{p_1}\right)$ less than critical pressure ratio, since the flow in the nozzle is being continuously accelerated with the reduction in the throat pressure below the critical values and hence the velocity cannot reduce. Thus, the mass flow rate for all values of $\left(\frac{p_2}{p_1}\right)$ less than critical pressure ratio remains constant at the maximum value (indicated by the solid horizontal line in Fig. 15.11). This fact has been verified experimentally too.

Velocity at outlet of nozzle for maximum flow rate :

The velocity at outlet of nozzle for *maximum flow rate* is given by:

$$V_2 = \sqrt{\left(\frac{2\gamma}{\gamma+1}\right) \frac{p_1}{\rho_1}} \quad \dots[\text{Eqn. (15.50)}]$$

Now pressure ratio,
$$\frac{p_2}{p_1} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$$

$$\therefore p_1 = \frac{p_2}{\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}} = p_2 \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$$

For adiabatic flow:
$$\frac{p_1}{\rho_1^\gamma} = \frac{p_2}{\rho_2^\gamma} \quad \text{or} \quad \frac{p_1}{p_2} = \left(\frac{\rho_1}{\rho_2}\right)^\gamma$$

or,
$$\frac{\rho_1}{\rho_2} = \left(\frac{p_1}{p_2}\right)^{\frac{1}{\gamma}} = \left(\frac{p_2}{p_1}\right)^{-\frac{1}{\gamma}}$$

$$\therefore \rho_1 = \rho_2 \left(\frac{p_2}{p_1}\right)^{-\frac{1}{\gamma}} \quad \text{or} \quad \rho_2 \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1} \times \frac{-1}{\gamma}} \quad \text{or} \quad \rho_2 \left(\frac{2}{\gamma+1}\right)^{\frac{1}{1-\gamma}}$$

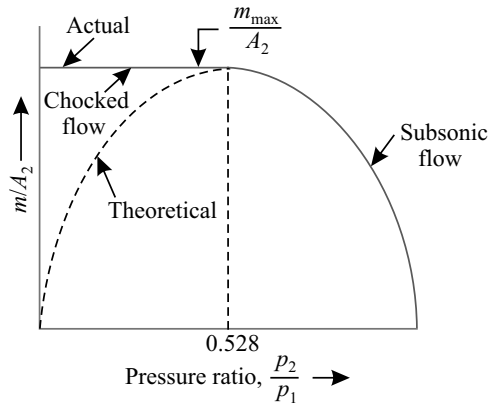


Fig. 15.11. Mass flow rate through a convergent nozzle.

Substituting the values of p_1 and ρ_1 in the above eqn. (15.50), we get:

$$\begin{aligned} V_2 &= \sqrt{\left(\frac{2\gamma}{\gamma+1}\right) \times p_2 \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{1-\gamma}} \times \left\{\frac{1}{\rho_2} \times \left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}}\right\}} \\ &= \sqrt{\left(\frac{2\gamma}{\gamma+1}\right) \times \frac{p_2}{\rho_2} \times \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{1-\gamma} + \frac{1}{\gamma-1}}} = \sqrt{\left(\frac{2}{\gamma+1}\right) \times \frac{p_2}{\rho_2} \left(\frac{2}{\gamma+1}\right)^{-1}} \end{aligned}$$

or,
$$V_2 = \sqrt{\left(\frac{2\gamma}{\gamma+1}\right) \times \frac{p_2}{\rho_2} \left(\frac{\gamma+1}{2}\right)} = \sqrt{\frac{\gamma p_2}{\rho_2}} = C_2$$

i.e.
$$V_2 = C_2 \quad \dots(15.52)$$

Hence, the *velocity* at the outlet of nozzle for *maximum flow rate equals sonic velocity*.

15.11. VARIABLES OF FLOW IN TERMS OF MACH NUMBER

In order to obtain relationship involving change in velocity, pressure, temperature and density in terms of the Mach number use is made of the continuity, perfect gas, isentropic flow and energy equations.

For *continuity equation*, we have:

$$\rho AV = \text{constant}$$

Differentiating the above equation, we get:

$$\rho [AdV + VdA] + AVd\rho = 0$$

Dividing throughout by ρAV , we have:

$$\frac{dV}{V} + \frac{dA}{A} + \frac{d\rho}{\rho} = 0$$

From *isentropic flow*, we have:

$$\frac{p}{\rho^\gamma} = \text{constant} \quad \text{or} \quad \frac{dp}{p} = \gamma \frac{d\rho}{\rho}$$

For *perfect gas*, we have: $p = \rho RT$

or,
$$\frac{dp}{p} = \frac{d\rho}{\rho} + \frac{dT}{T}$$

From *energy equation*, we have: $c_p T + \frac{V^2}{2} = \text{constant}$

Differentiating throughout, we get:

$$c_p dT + VdV = 0$$

or,
$$\left(\frac{\gamma R}{\gamma-1}\right) dT + VdV = 0 \quad \left(\because c_p = \frac{\gamma R}{\gamma-1}\right)$$

or,
$$\frac{\gamma R}{\gamma-1} \frac{dT}{V^2} + \frac{dV}{V} = 0 \quad \dots(i)$$

Also, Sonic velocity, $C = \sqrt{\gamma RT} \quad \therefore \gamma R = \frac{C^2}{T}$

Substituting the value of $\gamma R = \frac{C^2}{T}$ in eqn. (i), we get:

$$\frac{C^2}{(\gamma - 1) T} \times \frac{dT}{V^2} + \frac{dV}{V} = 0$$

or, $\frac{1}{(\gamma - 1) M^2} \frac{dT}{T} + \frac{dV}{V} = 0 \quad \left(\because M = \frac{V}{C} \right) \quad \dots(15.52)$

From the Mach number relationship:

$$M = \frac{V}{\sqrt{\gamma RT}} \quad (\text{where } \sqrt{\gamma RT} = C)$$

$$\frac{dM}{M} = \frac{dV}{V} - \frac{1}{2} \frac{dT}{T} \quad \dots(15.53)$$

Substituting the value of $\frac{dT}{T}$ from eqn. (15.52) in eqn. (15.53), we get:

$$\begin{aligned} \frac{dM}{M} &= \frac{dV}{V} - \frac{1}{2} \left[-\frac{dV}{V} \times (\gamma - 1) M^2 \right] \\ &= \frac{dV}{V} + \frac{1}{2} \frac{dV}{V} \times (\gamma - 1) M^2 \\ \text{or, } \frac{dM}{M} &= \frac{dV}{V} \left[1 + \frac{\gamma - 1}{2} M^2 \right] \\ \text{or, } \frac{dV}{V} &= \frac{1}{\left[1 + \left(\frac{\gamma - 1}{2} \right) M^2 \right]} \frac{dM}{M} \quad \dots(15.54) \end{aligned}$$

Since the quantity within the bracket is *always positive*, the *trend of variation of velocity and Mach number is similar*. For temperature variation, one can write:

$$\frac{dT}{T} = \left[\frac{-(\gamma - 1) M^2}{1 + \left(\frac{\gamma - 1}{2} \right) M^2} \right] \frac{dM}{M} \quad \dots(15.55)$$

Since the right hand side is *negative* the *temperature changes follow an opposite trend to that of Mach number*. Similarly for *pressure and density*, we have:

$$\frac{dp}{p} = \left[\frac{-\gamma M^2}{1 + \left(\frac{\gamma - 1}{2} \right) M^2} \right] \frac{dM}{M} \quad \dots(15.56)$$

$$\text{and, } \frac{d\rho}{\rho} = \left[\frac{-M^2}{1 + \left(\frac{\gamma - 1}{2} \right) M^2} \right] \frac{dM}{M} \quad \dots(15.57)$$

For changes in *area* we have:

$$\frac{dA}{A} = \left[\frac{-(1 - M^2)}{1 + \left(\frac{\gamma - 1}{2} \right) M^2} \right] \frac{dM}{M} \quad \dots(15.58)$$

The quantity within the brackets may be *positive* or *negative* depending upon the *magnitude of Mach number*. By integrating eqn. (15.58), we can obtain a relationship between the critical throat area A_c , where Mach number is *unity* and the area A at any section where $M \geq 1$,

$$\frac{A}{A_c} = \frac{1}{M} \left[\frac{2 + (\gamma - 1) M^2}{\gamma + 1} \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} \quad \dots(15.59)$$

Example 15.15. The pressure leads from Pitot-static tube mounted on an aircraft were connected to a pressure gauge in the cockpit. The dial of the pressure gauge is calibrated to read the aircraft speed in m/s. The calibration is done on the ground by applying a known pressure across the gauge and calculating the equivalent velocity using incompressible Bernoulli's equation and assuming that the density is 1.224 kg/m^3 .

The gauge having been calibrated in this way the aircraft is flown at 9200 m, where the density is 0.454 kg/m^3 and ambient pressure is 30 kN/m^2 . The gauge indicates a velocity of 152 m/s. What is the true speed of the aircraft ? [UPSC Fluid Mechanics.]

Solution. Bernoulli's equation for an *incompressible* flow is given by:

$$p + \frac{\rho V^2}{2} = \text{constant}$$

The stagnation pressure (p_s) created at Pitot-static tube,

$$p_s = p_0 + \frac{\rho_0 V_0^2}{2} \quad (\text{neglecting compressibility effects}) \quad \dots(i)$$

Here, $p_0 = 30 \text{ kN/m}^2$, $V_0 = 152 \text{ m/s}$, $\rho_0 = 1.224 \text{ kg/m}^3$...(Given)

$$\therefore p_s = 30 + \frac{1.224 \times 152^2}{2} \times 10^{-3} = 44.139 \text{ kN/m}^2$$

Neglecting *compressibility effect*, the speed of the aircraft when

$\rho_0 = 0.454 \text{ kg/m}^3$ is given by [using eqn. (i)]:

$$44.139 \times 10^3 = 30 \times 10^3 + \frac{0.454 \times V_0^2}{2}$$

$$\text{or, } V_0^2 = \frac{(44.139 - 30) \times 10^3 \times 2}{0.454} = 62286.34$$

$$\therefore V_0 = 249.57 \text{ m/s}$$

$$\text{Sonic velocity, } C_0 = \sqrt{\gamma R T_0} = \sqrt{\gamma \frac{p_0}{\rho_0}} = \sqrt{1.4 \times \frac{30 \times 10^3}{0.454}} = 304.16 \text{ m/s}$$

$$\text{Mach number, } M_0 = \frac{V_0}{C_0} = \frac{249.57}{304.16} = 0.82$$

Compressibility correction factor = $\left(1 + \frac{M_0^2}{4} \right)$, neglecting the terms containing higher powers of M_0 (from eqn 15.38)

$$= \left(1 + \frac{0.82^2}{4} \right) = 1.168$$

$$\therefore \text{True speed of aircraft} = \frac{249.57}{\sqrt{1.168}} = 230.9 \text{ m/s}$$

Hence, *true speed of aircraft* = **230.9 m/s (Ans.)**

Example 15.16. (a) In case of isentropic flow of a compressible fluid through a variable duct, show that

$$\frac{A}{A_c} = \frac{1}{M} \left[\frac{1 + \frac{1}{2}(\gamma - 1) M^2}{\frac{1}{2}(\gamma + 1)} \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

where γ is the ratio of specific heats, M is the Mach number at a section whose area is A and A_c is the critical area of flow.

(b) A supersonic nozzle is to be designed for air flow with Mach number 3 at the exit section which is 200 mm in diameter. The pressure and temperature of air at the nozzle exit are to be 7.85 kN/m² and 200 K respectively. Determine the reservoir pressure and temperature and the throat area. Take : $\gamma = 1.4$. [UPSC Exams Fluid Mechanics.]

Solution. (a) Refer to Art. 15.11.

(b) Mach number, $M = 3$

$$\text{Area at the exit section, } A = \frac{\pi}{4} \times 0.2^2 = 0.0314 \text{ m}^2$$

$$\text{Pressure of air at the nozzle, } (p)_{\text{nozzle}} = 7.85 \text{ kN/m}^2$$

$$\text{Temperature of air at the nozzle, } (T)_{\text{nozzle}} = 200 \text{ K}$$

Reservoir pressure, $(p)_{\text{res.}}$:

$$\text{From eqn. (15.36), } (p)_{\text{res.}} = (p)_{\text{nozzle}} \left[1 + \left(\frac{\gamma - 1}{2} \right) M^2 \right]^{\left(\frac{\gamma}{\gamma - 1} \right)}$$

$$\text{or, } (p)_{\text{res.}} = 7.85 \left[1 + \left(\frac{1.4 - 1}{2} \right) \times 3^2 \right]^{\left(\frac{1.4}{1.4 - 1} \right)} = 288.35 \text{ kN / m}^2 \text{ (Ans.)}$$

Reservoir temperature, $(T)_{\text{res.}}$:

$$\text{From eqn. (15.41), } (T)_{\text{res.}} = (T)_{\text{nozzle}} \left[1 + \left(\frac{\gamma - 1}{2} \right) M^2 \right]$$

$$\text{or, } (T)_{\text{res.}} = 200 \left[1 + \left(\frac{1.4 - 1}{2} \right) \times 3^2 \right] = 560 \text{ K (Ans.)}$$

Throat area (critical), A_c :

$$\text{From eqn. (15.59), } \frac{A}{A_c} = \frac{1}{M} \left[\frac{2 + (\gamma - 1) M^2}{\gamma + 1} \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

$$\text{or, } \frac{0.0314}{A_c} = \frac{1}{3} \left[\frac{2 + (1.4 - 1) 3^2}{1.4 + 1} \right]^{\frac{1.4 + 1}{2(1.4 - 1)}}$$

$$\text{or, } \frac{0.0314}{A_c} = \frac{1}{3} (2.333)^3 = 4.23$$

$$\text{or, } A_c = \frac{0.0314}{4.23} = 0.00742 \text{ m}^2 \text{ (Ans.)}$$

15.12. FLOW THROUGH LAVAL NOZZLE (CONVERGENT-DIVERGENT NOZZLE)

Laval nozzle is a convergent-divergent nozzle (named after de Laval, the swedish scientist who invented it) in which *subsonic flow prevails in the converging section, critical or transonic conditions in the throat and supersonic flow in the diverging section.*

Let, $p_2 (= p_c)$ = Pressure in the throat when the flow is sonic for given pressure p_1 .

- When the pressure in the receiver, $p_3 = p_1$, there will be no flow through the nozzle, this is shown by line a in Fig. 15.12 (b).

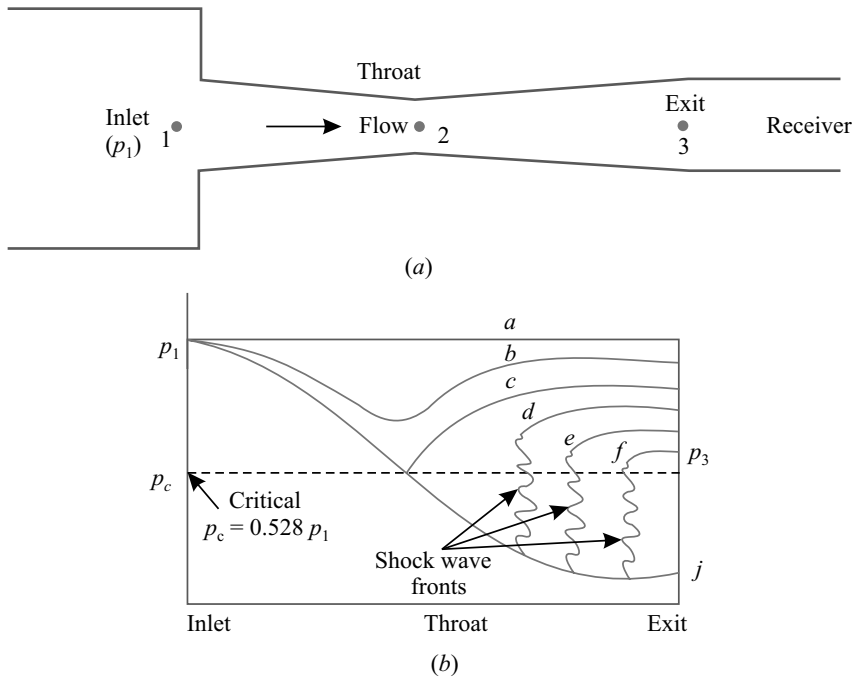


Fig. 15.12. (a) Laval nozzle (convergent-divergent nozzle); (b) Pressure distribution through a convergent-divergent nozzle with flow of compressible fluid.

- When the receiver pressure is reduced, flow will occur through the nozzle. As long as the value of p_3 is such that throat pressure p_2 is greater than the critical pressure $0.528 p_1$, the flow in the converging and diverging sections will be subsonic. This condition is shown by line 'b'.
- With further reduction in p_3 , a stage is reached when p_2 is equal to critical pressure $p_c = 0.528 p_1$, at this line $M = 1$ in the throat. This condition is shown by line 'c'. Flow is subsonic on the upstream as well the downstream of the throat. The flow is also *isentropic*.
- If p_3 is further reduced, it does not effect the flow in convergent section. The flow in throat is sonic, downstream it is supersonic. Somewhere in the diverging section a shock wave occurs and flow changes to subsonic (curve d). The flow across the shock is non-isentropic. Downstream of the shock wave the flow is subsonic and decelerates.
- If the value of p_3 is further reduced, the shock wave forms somewhat downstream (curve e).
- For p_3 equal to p_j , the shock wave will occur just at the exit of divergent section.
- If the value of p_3 lies before p_j and p_j oblique waves are formed at the exit :

Example 15.17. A large tank contains air at 284 kN/m^2 gauge pressure and 24°C temperature. The air flows from the tank to the atmosphere through a convergent nozzle. If the diameter at the outlet of the nozzle is 20 mm , find the maximum flow rate of air.

Take: $R = 287 \text{ J/kg K}$, $\gamma = 1.4$ and atmospheric pressure $= 100 \text{ kN/m}^2$.

[Roorkee University]

Solution. Pressure in the tank, $p_1 = 284 \text{ kN/m}^2$ (gauge) $= 284 + 100 = 384 \text{ kN/m}^2$ (absolute)

Temperature in the tank, $T_1 = 24 + 273 = 297 \text{ K}$

Diameter at the outlet of the nozzle,

$$D = 20 \text{ mm} = 0.02 \text{ m}$$

$$\therefore \text{Area, } A_2 = \frac{\pi}{4} \times 0.02^2 = 0.0003141 \text{ m}^2$$

$$R = 287 \text{ J/kg K}, \gamma = 1.4$$

(Two points are considered. Point 1 lies inside the tank and point 2 lies at the exit of the nozzle)

Maximum flow rate, $m_{\max}$:

Equation of state is given by: $p = \rho RT$ or $\rho = \frac{p}{RT}$

$$\therefore \rho_1 = \frac{p_1}{RT_1} = \frac{384 \times 10^3}{287 \times 297} = 4.5 \text{ kg/m}^3$$

The fluid parameters in the tank correspond to the stagnation values, and maximum flow rate of air is given by:

$$\begin{aligned} m_{\max} &= 0.685 A_2 \sqrt{p_1 \rho_1} \quad \dots [\text{Eqn. (15.51)}] \\ &= 0.685 \times 0.0003141 \sqrt{384 \times 10^3 \times 4.5} = 0.283 \text{ kg/s} \end{aligned}$$

Hence maximum flow rate of air = **0.283 kg/s (Ans.)**

Example 15.18. A large vessel, fitted with a nozzle, contains air at a pressure of 2500 kN/m^2 (abs.) and at a temperature of 20°C . If the pressure at the outlet of the nozzle is 1750 kN/m^2 find the velocity of air flowing at the outlet of the nozzle.

Take : $R = 287 \text{ J/kg K}$, and $\gamma = 1.4$.

Solution. Pressure inside the vessel, $p_1 = 2500 \text{ kN/m}^2$ (abs.)

Temperature inside vessel, $T_1 = 20 + 273 = 293 \text{ K}$

Pressure at the outlet of the nozzle, $p_2 = 1750 \text{ kN/m}^2$ (abs.)

$$R = 287 \text{ J/kg K}, \gamma = 1.4$$

Velocity of air, V_2 :

$$V_2 = \sqrt{\left(\frac{2\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}\right]} \quad \dots [\text{Eqn. (15.46)}]$$

where,

$$\begin{aligned} \rho_1 &= \frac{p_1}{RT_1} \quad (\text{From equation of state : } \frac{p}{\rho} = RT) \\ &= \frac{2500 \times 10^3}{287 \times 293} = 29.73 \text{ kg/m}^3 \end{aligned}$$

Substituting the values in the above equation, we get:

$$\begin{aligned} V_2 &= \sqrt{\left(\frac{2 \times 1.4}{1.4-1}\right) \times \frac{2500 \times 10^3}{29.73} \left[1 - \left(\frac{1750}{2500}\right)^{\frac{1.4-1}{1.4}}\right]} \\ &= \sqrt{7 \times 84090 (1 - 0.903)} = 238.9 \text{ m/s} \end{aligned}$$

i.e.

$$V_2 = \mathbf{238.9 \text{ m/s (Ans.)}}$$

Example 15.19. A tank fitted with a convergent nozzle contains air at a temperature of 20°C . The diameter at the outlet of the nozzle is 25 mm . Assuming adiabatic flow, find the mass rate of flow of air through the nozzle to the atmosphere when the pressure in the tank is :

(i) 140 kN/m^2 (abs.); (ii) 300 kN/m^2 .

Take for air : $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$. Barometric pressure = 100 kN/m^2

Solution. Temperature of air in the tank,

$$T_1 = 20 + 273 = 293 \text{ K}$$

Diameter at the outlet of the nozzle,

$$D_2 = 25 \text{ mm} = 0.025 \text{ m}$$

$$\text{Area, } A_2 = \frac{\pi}{4} \times 0.025^2 = 0.0004908 \text{ m}^2$$

$$R = 287 \text{ J/kg K}, \gamma = 1.4$$

(i) Mass rate of flow of air when pressure in the tank is 140 kN/m^2 (abs.), m :

$$\rho_1 = \frac{p_1}{RT_1} = \frac{140 \times 10^3}{287 \times 293} = 1.665 \text{ kg/m}^3$$

$$p_1 = 140 \text{ kN/m}^2 \text{ (abs.)}$$

$$\text{Pressure at the nozzle, } p_2 = \text{Atmospheric pressure} = 100 \text{ kN/m}^2$$

$$\therefore \text{Pressure ratio, } \frac{p_2}{p_1} = \frac{100}{140} = 0.7143$$

Since the pressure ratio is more than the critical value, flow in the nozzle will be *subsonic*, hence mass rate of flow of air is given by eqn. 15.47, as:

$$\begin{aligned} m &= A_2 \sqrt{\frac{2\gamma}{\gamma-1}} p_1 \rho_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{2}{\gamma}} - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma+1}{\gamma}} \right] \\ &= 0.0004908 \sqrt{\left(\frac{2 \times 1.4}{1.4-1} \right) \times 140 \times 10^3 \times 1.665 \left[(0.7143)^{\frac{2}{1.4}} - (0.7143)^{\frac{1.4+1}{1.4}} \right]} \\ &= 0.0004908 \sqrt{1631700 (0.7143)^{1.4285} - (0.7143)^{1.7142}} \end{aligned}$$

$$\text{or, } m = 0.0004908 \sqrt{1631700 (0.6184 - 0.5617)} = \mathbf{0.1493 \text{ kg/s (Ans.)}}$$

(ii) Mass rate of flow of air when pressure in the tank is 300 kN/m^2 (abs.) :

$$p_1 = 300 \text{ kN/m}^2 \text{ (abs.)}$$

$$p_2 = \text{Pressure at the nozzle} = \text{atmospheric pressure} = 100 \text{ kN/m}^2$$

$$\therefore \text{Pressure ratio, } \frac{p_2}{p_1} = \frac{100}{300} = 0.33$$

The pressure ratio being less than the critical ratio 0.528, the flow in the nozzle will be *sonic*, the flow rate is *maximum* and is given by eqn. (5.51), as:

$$m_{\max} = 0.685 A_2 \sqrt{p_1 \rho_1}$$

$$\text{where, } \rho_1 = \frac{p_1}{RT_1} = \frac{300 \times 10^3}{287 \times 293} = 3.567 \text{ kg/m}^3$$

$$m_{\max} = 0.685 \times 0.0004908 \sqrt{300 \times 10^3 \times 3.567} = \mathbf{0.3477 \text{ kg/s (Ans.)}}$$

Example 15.20. At some section in the convergent-divergent nozzle, in which air is flowing, pressure, velocity, temperature and cross-sectional area are 200 kN/m^2 , 170 m/s , 200°C and 1000 mm^2 respectively. If the flow conditions are isentropic, determine :

(i) Stagnation temperature and stagnation pressure.

(ii) Sonic velocity and Mach number at this section.

(iii) Velocity, Mach number and flow area at outlet section where pressure is 110 kN/m^2 .

(iv) Pressure, temperature, velocity and flow area at throat of the nozzle.

Take for air : $R = 287 \text{ J/kg K}$, $c_p = 1.0 \text{ kJ/kg K}$, and $\gamma = 1.4$.

Solution. Let subscripts 1, 2 and t refer to the conditions at given section, outlet section and throat section of the nozzle respectively.

Pressure in the nozzle, $p_1 = 200 \text{ kN/m}^2$

Velocity of air, $V_1 = 170 \text{ m/s}$

Temperature, $T_1 = 200 + 273 = 473 \text{ K}$

Cross-sectional area, $A_1 = 1000 \text{ mm}^2 = 1000 \times 10^{-6} = 0.001 \text{ m}^2$

For air : $R = 287 \text{ J/kg K}$; $c_p = 1.0 \text{ kJ/kg K}$; $\gamma = 1.4$

(i) Stagnation temperature (T_s) and stagnation pressure (p_s) :

$$\begin{aligned} \text{Stagnation temperature, } T_s &= T_1 + \frac{V_1^2}{2 \times c_p} \\ &= 473 + \frac{170^2}{2 \times (1.0 \times 1000)} = 487.45 \text{ K} \quad (\text{or } 214.45^\circ\text{C}) \quad (\text{Ans.}) \end{aligned}$$

$$\text{Also, } \frac{p_s}{p_1} = \left(\frac{T_s}{T_1} \right)^{\frac{\gamma}{\gamma-1}} = \left(\frac{487.45}{473} \right)^{\frac{1.4}{1.4-1}} = 1.111$$

$\therefore$ Stagnation pressure, $p_s = 200 \times 1.111 = 222.2 \text{ kN/m}^2$ (Ans.)

(ii) Sonic velocity and Mach number at this section :

Sonic velocity, $C_1 = \sqrt{\gamma R T_1} = \sqrt{1.4 \times 287 \times 473} = 435.9 \text{ m/s}$ (Ans.)

Mach number, $M_1 = \frac{V_1}{C_1} = \frac{170}{435.9} = 0.39$ (Ans.)

(iii) Velocity, Mach number and flow area at outlet section where pressure is 110 kN/m^2 :

Pressure at outlet section, $p_2 = 110 \text{ kN/m}^2$... (Given)

$$\begin{aligned} \text{From eqn (15.36), } \frac{p_s}{p_2} &= \left[1 + \left(\frac{\gamma-1}{2} \right) M_2^2 \right]^{\frac{\gamma}{\gamma-1}} \\ \frac{222.2}{110} &= \left[1 + \left(\frac{1.4-1}{2} \right) M_2^2 \right]^{\frac{1.4}{1.4-1}} = (1 + 0.2 M_2^2)^{3.5} \end{aligned}$$

$$\text{or, } (1 + 0.2 M_2^2) = \left(\frac{222.2}{110} \right)^{\frac{1}{3.5}} = 1.222$$

$$\text{or, } M_2 = \left(\frac{1.222 - 1}{0.2} \right)^{1/2} = 1.05 \quad (\text{Ans.})$$

$$\text{Also, } \frac{T_2}{T_s} = \left(\frac{p_2}{p_s} \right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{110}{222.2} \right)^{\frac{1.4-1}{1.4}} = 0.818$$

$$\text{or, } T_2 = 0.818 \times 487.45 = 398.7 \text{ K}$$

Sonic velocity at outlet section, $C_2 = \sqrt{\gamma R T_2} = \sqrt{1.4 \times 287 \times 398.7} = 400.25 \text{ m/s}$

$\therefore$ Velocity at outlet section, $V_2 = M_2 \times C_2 = 1.05 \times 400.25 = 420.26 \text{ m/s}$ (Ans.)

Now, Mass flow at the given section = Mass flow at outlet section (exit)

... Continuity equation

$$i.e. \quad \rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad \text{or} \quad \frac{p_1}{RT_1} A_1 V_1 = \frac{p_2}{RT_2} A_2 V_2$$

∴ Flow area at the outlet section,

$$A_2 = \frac{p_1 A_1 V_1 T_2}{T_1 p_2 V_2} = \frac{200 \times 0.001 \times 170 \times 398.7}{473 \times 110 \times 420.26}$$

$$= 6.199 \times 10^{-4} \text{ m}^2$$

Hence,

$$A_2 = 6.199 \times 10^{-4} \text{ m}^2 \quad \text{or} \quad \mathbf{619.9 \text{ mm}^2 \text{ (Ans.)}}$$

(iv) Pressure (p_t), temperature (T_t), velocity (V_t), and flow area (A_t) at throat of the nozzle :

At throat, *critical conditions prevail*, i.e. the flow velocity becomes *equal to the sonic velocity* and *Mach number attains a unit value*.

$$\text{From eqn. (15.41),} \quad \frac{T_s}{T_t} = \left[1 + \left(\frac{\gamma - 1}{2} \right) M_t^2 \right]$$

$$\text{or,} \quad \frac{487.45}{T_t} \left[1 + \left(\frac{1.4 - 1}{2} \right) \times 1^2 \right] = 1.2 \quad \text{or} \quad T_t = 406.2 \text{ K}$$

Hence,

$$T_t = \mathbf{406.2 \text{ K (or } 133.2^\circ\text{C) (Ans.)}}$$

$$\text{Also,} \quad \frac{p_t}{p_s} = \left(\frac{T_t}{T_s} \right)^{\frac{\gamma}{\gamma - 1}}$$

$$\text{or,} \quad \frac{p_t}{222.2} = \left(\frac{406.2}{487.45} \right)^{\frac{1.4}{1.4 - 1}} = 0.528$$

or,

$$p_t = 222.2 \times 0.528 = \mathbf{117.32 \text{ kN/m}^2 \text{ (Ans.)}}$$

Sonic velocity (corresponding to throat conditions),

$$C_t = \sqrt{\gamma R T_t} = \sqrt{1.4 \times 287 \times 406.2} = 404 \text{ m/s}$$

∴ Flow velocity, $V_t = M_t \times C_t = 1 \times 404 = 404 \text{ m/s}$

By continuity equation, we have :

$$\rho_1 A_1 V_1 = \rho_t A_t V_t$$

$$\text{or,} \quad \frac{p_1}{RT_1} A_1 V_1 = \frac{p_t}{RT_t} A_t V_t$$

$$\therefore \text{ Flow area at throat, } A_t = \frac{p_1 A_1 V_1 T_t}{T_1 p_t V_t} = \frac{200 \times 0.001 \times 170 \times 406.2}{473 \times 117.32 \times 404} = 6.16 \times 10^{-4} \text{ m}^2$$

Hence,

$$A_t = 6.16 \times 10^{-4} \text{ m}^2 \quad \text{or} \quad \mathbf{616 \text{ mm}^2 \text{ (Ans.)}}$$

15.13. SHOCK WAVES

Whenever a supersonic flow (compressible) abruptly changes to subsonic flow, a shock wave (analogous to hydraulic jump in an open channel) is produced, resulting in a sudden rise in pressure, density, temperature and entropy. This occurs due to pressure differentials and when the Mach number of the approaching flow $M_1 > 1$. A shock wave is a pressure wave of finite thickness, of the order of 10^{-2} to 10^{-4} mm in the atmospheric pressure. A shock wave takes place in the diverging section of a nozzle, in a diffuser, throat of a supersonic wind tunnel, in front of sharp-nosed bodies.

Shock waves are of two types :

1. Normal shocks which are almost perpendicular to the flow.
2. Oblique shocks which are inclined to the flow direction.

15.13.1 Normal shock wave

Consider a duct having a compressible sonic flow (See Fig. 15.13)

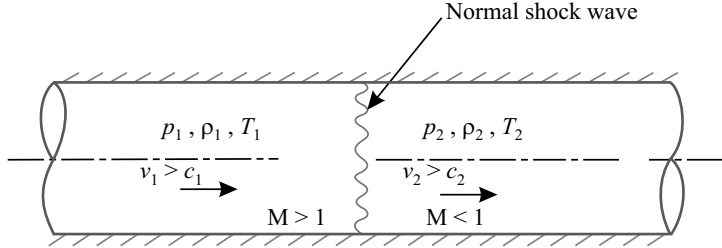


Fig. 15.13. Normal shock wave.

Let p_1, ρ_1, T_1 , and V_1 be the pressure, density, temperature and velocity of the flow ($M_1 > 1$) and p_2, ρ_2, T_2 and V_2 the corresponding values of pressure, density, temperature and velocity after a shock wave takes place ($M_2 < 1$).

For analysing a normal shock wave, use will be made of the *continuity, momentum and energy equations*.

Assume unit area cross-section, $A_1 = A_2 = 1$.

$$\text{Continuity equation : } m = \rho_1 V_1 = \rho_2 V_2 \quad \dots(i)$$

$$\text{Momentum equation : } \Sigma F_x = p_1 A_1 - p_2 A_2 = m (V_2 - V_1) = \rho_2 A_2 V_2^2 - \rho_1 A_1 V_1^2$$

for $A_1 = A_2 = 1$, the pressure drop across the shock wave,

$$p_1 - p_2 = \rho_2 V_2^2 - \rho_1 V_1^2 \quad \dots(ii)$$

$$p_1 + \rho_1 V_1^2 = p_2 + \rho_2 V_2^2$$

Consider the flow across the shock wave as adiabatic.

$$\text{Energy equation : } \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_1}{\rho_1} + \frac{V_1^2}{2} = \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_2}{\rho_2} + \frac{V_2^2}{2} \quad \dots[\text{Eqn. (15.26)}]$$

($z_1 = z_2$, duct being in horizontal position)

$$\text{or, } \frac{\gamma}{\gamma - 1} \left(\frac{p_2}{\rho_2} - \frac{p_1}{\rho_1} \right) = \frac{V_1^2 - V_2^2}{2} \quad \dots(iii)$$

Combining continuity and momentum equations [refer to eqns. (i) and (ii)], we get:

$$p_1 + \frac{(\rho_1 V_1)^2}{\rho_1} = p_2 + \frac{(\rho_2 V_2)^2}{\rho_2} \quad \dots(15.60)$$

This equation is known as **Rankine Line Equation**.

Now combining continuity and energy equations [refer to eqns. (i) and (iii)], we get:

$$\frac{\gamma}{\gamma - 1} \left(\frac{p_1}{\rho_1} \right) + \frac{(\rho_1 V_1)^2}{2\rho_1^2} = \frac{\gamma}{\gamma - 1} \left(\frac{p_2}{\rho_2} \right) + \frac{(\rho_2 V_2)^2}{2\rho_2^2} \quad \dots(15.61)$$

This equation is called **Fanno Line Equation**.

Further combining eqns. (i), (ii) and (iii) and solving for $\frac{p_2}{p_1}$, we get:

$$\frac{p_2}{p_1} = \frac{\left(\frac{\gamma+1}{\gamma-1}\right) \frac{\rho_2}{\rho_1} - 1}{\left(\frac{\gamma+1}{\gamma-1}\right) - \frac{\rho_2}{\rho_1}} \quad \dots(15.62)$$

Solving for density ratio $\frac{\rho_2}{\rho_1}$, the same equations yield:

$$\frac{\rho_2}{\rho_1} = \frac{V_1}{V_2} = \frac{1 + \left(\frac{\gamma+1}{\gamma-1}\right) \frac{p_2}{p_1}}{\left(\frac{\gamma+1}{\gamma-1}\right) + \frac{p_2}{p_1}} \quad \dots(15.63)$$

The eqns. (15.62) and (15.63) are called **Rankine-Hugoniot equations**.

One can also express $\frac{p_2}{p_1}$, $\frac{V_2}{V_1}$, $\frac{\rho_2}{\rho_1}$ and $\frac{T_2}{T_1}$ in terms of Mach number as follows :

$$\frac{p_2}{p_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{(\gamma + 1)} \quad \dots(15.64)$$

$$\frac{V_1}{V_2} = \frac{\rho_2}{\rho_1} = \frac{(\gamma + 1) M_1^2}{(\gamma - 1) M_1^2 + 2} \quad \dots(15.65)$$

$$\frac{T_2}{T_1} = \frac{[(\gamma - 1) M_1^2 + 2][2\gamma M_1^2 - (\gamma - 1)]}{(\gamma + 1)^2 M_1^2} \quad \dots(15.66)$$

By algebraic manipulation the following equation between M_1 and M_2 can be obtained:

$$M_2^2 = \frac{(\gamma - 1) M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)} \quad \dots(15.67)$$

Example 15.21. For a normal shock wave in air Mach number is 2. If the atmospheric pressure and air density are 26.5 kN/m^2 and 0.413 kg/m^3 respectively, determine the flow conditions before and after the shock wave. Take $\gamma = 1.4$.

Solution. Let subscripts 1 and 2 represent the flow conditions before and after the shock wave.

Mach number, $M_1 = 2$

Atmospheric pressure, $p_1 = 26.5 \text{ kN/m}^2$

Air density, $\rho_1 = 0.413 \text{ kg/m}^3$.

Mach number, M_2 :

$$M_2^2 = \frac{(\gamma - 1) M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)} \quad \dots[\text{Eqn. 15.67}]$$

$$= \frac{(1.4 - 1) \times 2^2 + 2}{2 \times 1.4 \times 2^2 - (1.4 - 1)} = \frac{3.6}{11.2 - 0.4} = 0.333$$

$$\therefore M_2 = \mathbf{0.577 \text{ (Ans.)}}$$

Pressure, p_2 :

$$\frac{p_2}{p_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{(\gamma + 1)} \quad \dots[\text{Eqn. (15.64)}]$$

$$= \frac{2 \times 1.4 \times 2^2 - (1.4 - 1)}{(1.4 + 1)} = \frac{11.2 - 0.4}{2.4} = 4.5$$

$$\therefore p_2 = 26.5 \times 4.5 = \mathbf{119.25 \text{ kN/m}^2 \text{ (Ans.)}}$$

Density, ρ_2 :

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1) M_1^2}{(\gamma - 1) M_1^2 + 2} \quad \dots[\text{Eqn. (15.65)}]$$

$$= \frac{(1.4 + 1) 2^2}{(1.4 - 1) 2^2 + 2} = \frac{9.6}{1.6 + 2} = 2.667$$

$$\therefore \rho = 0.413 \times 2.667 = \mathbf{1.101 \text{ kg/m}^3 \text{ (Ans.)}}$$

Temperature, T_1 :

Since,

$$p_1 = \rho_1 R T_1,$$

$\therefore$

$$T_1 = \frac{p_1}{\rho_1 R} = \frac{26.5 \times 10^3}{0.413 \times 287} = 223.6 \text{ K} \quad \text{or} \quad -49.4^\circ\text{C} \text{ (Ans.)}$$

Temperature, T_2 :

$$\begin{aligned} \frac{T_2}{T_1} &= \frac{[(\gamma - 1) M_1^2 + 2] [2\gamma M_1^2 - (\gamma - 1)]}{(\gamma + 1)^2 M_1^2} \\ &= \frac{[(1.4 - 1) 2^2 + 2] [2 \times 1.4 \times 2^2 - (1.4 - 1)]}{(1.4 + 1)^2 \times 2^2} \\ &= \frac{(1.6 + 2) (11.2 - 0.4)}{23.04} = 1.6875 \end{aligned}$$

$\therefore$

$$T_2 = 223.6 \times 1.6875 = 377.3 \text{ K} \quad \text{or} \quad \mathbf{104.3^\circ\text{C} \text{ (Ans.)}}$$

Velocity, V_1 :

$$C_1 = \sqrt{\gamma R T_1} = \sqrt{1.4 \times 287 \times 223.6} = 299.7 \text{ m/s}$$

Since

$$\frac{V_1}{C_1} = M_1 = 2$$

$\therefore$

$$V_1 = 299.7 \times 2 = \mathbf{599.4 \text{ m/s} \text{ (Ans.)}}$$

Velocity, V_2 :

$$C_2 = \sqrt{\gamma R T_2} = \sqrt{1.4 \times 287 \times 377.3} = 389.35 \text{ m/s}$$

Since,

$$\frac{V_2}{C_2} = M_2 = 0.577$$

$\therefore$

$$V_2 = 389.35 \times 0.577 = \mathbf{224.6 \text{ m/s} \text{ (Ans.)}}$$

15.13.2. Oblique Shock Wave

As shown in Fig. 15.14, when a supersonic flow undergoes a sudden turn through a small angle α (positive), an oblique wave is established at the corner. In comparison with normal shock waves, the *oblique shock waves, being weaker, are preferred*.

The shock waves should be avoided or made as weak as possible, since during a shock wave conversion of mechanical energy into heat energy takes place.

15.13.3. Shock Strength

The **strength of shock** is defined as the ratio of pressure rise across the shock to the upstream pressure.

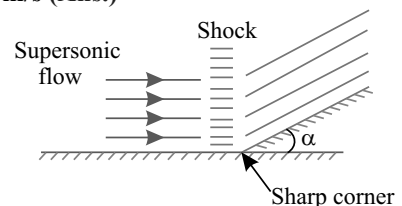


Fig. 15.14. Oblique shock wave.

$$\begin{aligned}
 \text{i.e. Strength of shock} &= \frac{p_2 - p_1}{p_1} = \frac{p_2}{p_1} - 1 \\
 &= \frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1} - 1 = \frac{2\gamma M_1^2 - (\gamma - 1) - (\gamma + 1)}{\gamma + 1} \\
 &= \frac{2\gamma M_1^2 - \gamma + 1 - \gamma - 1}{\gamma + 1} = \frac{2\gamma M_1^2 - 2\gamma}{\gamma + 1} = \frac{2\gamma}{\gamma + 1} (M_1^2 - 1)
 \end{aligned}$$

$$\text{Hence, strength of shock} = \frac{2\gamma}{\gamma + 1} (M_1^2 - 1) \quad \dots(15.68)$$

Example 15.22. In a duct in which air is flowing, a normal shock wave occurs at a Mach number of 1.5. The static pressure and temperature upstream of the shock wave are 170 kN/m² and 23°C respectively. Determine :

- (i) Pressure, temperature and Mach number downstream of the shock, and
 (ii) Strength of shock.

Take $\gamma = 1.4$

Solution. Let subscripts 1 and 2 represent flow conditions upstream and downstream of the shock wave respectively.

$$\text{Mach number, } M_1 = 1.5$$

$$\text{Upstream pressure, } p_1 = 170 \text{ kN/m}^2$$

$$\text{Upstream temperature, } T_1 = 23 + 273 = 296 \text{ K}$$

$$\gamma = 1.4$$

- (i) Pressure, temperature and Mach number downstream of the shock :

$$\frac{p_2}{p_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1} \quad \dots[\text{Eqn. (15.64)}]$$

$$= \frac{2 \times 1.4 \times 1.5^2 - (1.4 - 1)}{1.4 + 1} = \frac{6.3 - 0.4}{2.4} = 2.458$$

$$\therefore p_2 = 170 \times 2.458 = \mathbf{417.86 \text{ kN/m}^2} \text{ (Ans.)}$$

$$\frac{T_2}{T_1} = \frac{[(\gamma - 1) M_1^2 + 2][2\gamma M_1^2 - (\gamma - 1)]}{(\gamma + 1)^2 M_1^2} \quad \dots[\text{Eqn. (15.66)}]$$

$$= \frac{[(1.4 - 1) \times 1.5^2 + 2][2 \times 1.4 \times 1.5^2 - (1.4 - 1)]}{(1.4 + 1)^2 \times 1.5^2} = \frac{2.9 \times 5.9}{12.96} = 1.32$$

$$\therefore T_2 = 296 \times 1.32 = \mathbf{390.72 \text{ K}} \quad \text{or} \quad \mathbf{117.72^\circ\text{C}} \text{ (Ans.)}$$

$$M_2^2 = \frac{(\gamma - 1) M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)} \quad \dots[\text{Eqn. (15.67)}]$$

$$= \frac{(1.4 - 1) \times 1.5^2 + 2}{2 \times 1.4 \times 1.5^2 - (1.4 - 1)} = \frac{2.9}{5.9} = 0.49$$

$$\therefore M_2 = \mathbf{0.7 \text{ (Ans.)}}$$

- (ii) Strength of shock :

$$\text{Strength of shock} = \frac{p_1}{p_2} - 1 = 2.458 - 1 = \mathbf{1.458} \text{ (Ans.)}$$

15.14. MEASUREMENT OF COMPRESSIBLE FLOW

A. Measurement of discharge (flow rate) :

1. Convergent nozzle.
2. Orificemeter.
3. Convergent-divergent nozzle.
4. Venturimeter—If the *pressure drop* between the entrance and the throat is *small*, the flow computation may be made considering it as an *isothermal process*; if *pressure drop is appreciable*, the flow will be *adiabatic with a rapid fall of temperature at the throat* :

B. Measurement of velocity :

1. Pitot tube....*works on the principle of stagnation pressure.*
2. Hot-wire anemometer—*works on the principle that the rate of heat loss varies with the flow velocity;*
 - Particularly used in a *supersonic wind tunnel.*
 - Constant-current hot-wire anemometer, because of its high sensitivity, is especially suitable for flow in which velocity fluctuations are small.
 - Since it responds very rapidly to fluctuations of velocity it is widely used in conjunction with oscilloscopes and similar electronic instruments for measuring the intensity of turbulence.

C. Measurement of flow direction :

The instruments which may be used to determine both the magnitudes as well as the direction of the velocity are :

1. Pitot cylinder...suitable for determination of both magnitude and direction of velocity in a *two-dimensional flow.*
2. Pitot sphere....may be used to determine the magnitude and direction of the velocity in a *three-dimensional flow.*

15.15. FLOW OF COMPRESSIBLE FLUID THROUGH VENTURIMETER

Consider a compressible fluid flowing through a horizontal venturimeter. Let suffices 1 and 2 denote main and throat diameters of venturimeter respectively. Considering the flow to be *adiabatic*, we have:

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1 g} + \frac{V_1^2}{2g} + z_1 = \left(\frac{\gamma}{\gamma-1}\right) \frac{p_2}{\rho_2 g} + \frac{V_2^2}{2g} + z_2 \quad \dots[\text{Eqn. (15.26)}]$$

Taking $z_1 = z_2$ (venturimeter being horizontal) and cancelling 'g' we get:

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} + \frac{V_1^2}{2} = \left(\frac{\gamma}{\gamma-1}\right) \frac{p_2}{\rho_2} + \frac{V_2^2}{2}$$

$$\text{or,} \quad \frac{\gamma}{\gamma-1} \left[\frac{p_1}{\rho_1} - \frac{p_2}{\rho_2} \right] = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\text{or,} \quad \left(\frac{\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left[1 - \frac{p_2}{p_1} \times \frac{\rho_1}{\rho_2} \right] = \frac{V_2^2}{2} - \frac{V_1^2}{2} \quad \dots(i)$$

$$\text{For adiabatic flow :} \quad \frac{p_1}{\rho_1^\gamma} = \frac{p_2}{\rho_2^\gamma}$$

$$\therefore \quad \frac{p_1}{p_2} = \left(\frac{\rho_1}{\rho_2} \right)^\gamma$$

$$\text{or,} \quad \frac{\rho_1}{\rho_2} = \left(\frac{p_1}{p_2} \right)^{\frac{1}{\gamma}} \quad \dots(ii)$$

Substituting this value of $\frac{\rho_1}{\rho_2}$ in eqn. (i), we get:

$$\text{or,} \quad \left(\frac{\gamma}{\gamma-1} \right) \frac{p_1}{\rho_1} \left[1 - \frac{p_2}{p_1} \times \left(\frac{p_2}{p_1} \right)^{-\frac{1}{\gamma}} \right] = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\text{or,} \quad \left(\frac{\gamma}{\gamma-1} \right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{1-\frac{1}{\gamma}} \right] = \frac{V_2^2}{2} - \frac{V_1^2}{2}$$

$$\text{or,} \quad \left(\frac{\gamma}{\gamma-1} \right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right] = \frac{V_2^2}{2} - \frac{V_1^2}{2} \quad \dots(iii)$$

Also, $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$...Continuity equation

$$\therefore \quad V_1 = \frac{\rho_2 A_2 V_2}{\rho_1 A_1}$$

Substituting the value of V_1 in eqn. (iii), we get:

$$\left(\frac{\gamma}{\gamma-1} \right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right] = \frac{V_2^2}{2} - \left(\frac{\rho_2 A_2 V_2}{\rho_1 A_1} \right)^2 \times \frac{1}{2} = \frac{V_2^2}{2} \left[1 - \frac{\rho_2^2 A_2^2}{\rho_1^2 A_1^2} \right] \quad \dots(iv)$$

$$\text{But from eqn. (ii)} \quad \left(\frac{\rho_2}{\rho_1} \right)^2 = \left(\frac{p_2}{p_1} \right)^{2/\gamma}$$

Substituting this value in eqn. (iv), we get:

$$\left(\frac{\gamma}{\gamma-1} \right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right] = \frac{V_2^2}{2} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{2}{\gamma}} \times \frac{A_2^2}{A_1^2} \right] \quad \dots(iv)$$

$$\therefore \quad V_2^2 = \frac{\left(\frac{2\gamma}{\gamma-1} \right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right]}{\left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{2}{\gamma}} \times \frac{A_2^2}{A_1^2} \right]}$$

$$\therefore \quad V_2 = \sqrt{\frac{\left(\frac{2\gamma}{\gamma-1} \right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right]}{1 - \left(\frac{p_2}{p_1} \right)^{\frac{2}{\gamma}} \times \frac{A_2^2}{A_1^2}}}$$

∴ Mass rate of flow through venturimeter,

$$m = \rho_2 A_2 V_2 = \rho_2 A_2 \sqrt{\frac{\left(\frac{2\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}\right]}{1 - \left(\frac{p_2}{p_1}\right)^{\frac{2}{\gamma}} \times \left(\frac{A_2}{A_1}\right)^2}} \quad \dots(15.69)$$

Example 15.23. Find the mass rate of flow of air through a venturimeter having inlet diameter 300 mm and throat diameter 150 mm. The pressure and temperature of air at inlet section of the venturimeter are 137 kN/m² absolute and 15°C respectively, and the pressure at the throat is 127 kN/m² absolute. Take $R = 287 \text{ J/kg K}$ and adiabatic exponent $\gamma = 1.4$. [Delhi University]

Solution. Let suffices 1 and 2 represent the conditions at the inlet and throat sections of venturimeter.

Diameter at inlet, $D_1 = 300 \text{ mm} = 0.3 \text{ m}$

∴ Area, $A_1 = \frac{\pi}{4} \times 0.3^2 = 0.07068 \text{ m}^2$

Diameter of throat, $D_2 = 150 \text{ mm} = 0.15 \text{ m}$

∴ Area, $A_2 = \frac{\pi}{4} \times 0.15^2 = 0.01767 \text{ m}^2$

Pressure, $p_1 = 137 \text{ kN/m}^2 \text{ (abs.)}$

Temperature, $T_1 = 15 + 273 = 288 \text{ K}$

Pressure, $p_2 = 127 \text{ kN/m}^2$

$R = 287 \text{ J/kg K}$, $\gamma = 1.4$

Mass rate of flow, m :

$$\rho_1 = \frac{p_1}{RT_1} = \frac{137 \times 10^3}{287 \times 288} = 1.657 \text{ kg/m}^3$$

$$\frac{\rho_2}{\rho_1} = \left(\frac{p_2}{p_1}\right)^{\frac{1}{\gamma}}$$

or,

$$\rho_2 = \rho_1 \times \left(\frac{p_2}{p_1}\right)^{\frac{1}{\gamma}} = 1.657 \times \left(\frac{127}{137}\right)^{\frac{1}{1.4}} = 1.57 \text{ kg/m}^3$$

Mass flow rate through a venturimeter is given by:

$$m = \rho_2 A_2 \sqrt{\frac{\left(\frac{2\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}\right]}{1 - \left(\frac{p_2}{p_1}\right)^{\frac{2}{\gamma}} \times \left(\frac{A_2}{A_1}\right)^2}} \quad \dots[\text{Eqn. (15.69)}]$$

$$m = 1.57 \times 0.01767 \sqrt{\frac{\left(\frac{2 \times 1.4}{1.4-1}\right) \times \frac{137 \times 10^3}{1.657} \left[1 - \left(\frac{127}{137}\right)^{\frac{1.4-1}{1.4}}\right]}{1 - \left(\frac{127}{137}\right)^{\frac{2}{1.4}} \times \left(\frac{0.01767}{0.07068}\right)^2}}$$

or,

$$m = 0.02774 \sqrt{\frac{7 \times 82.68 \times 10^3 (1 - 0.09786)}{1 - 0.897 \times 0.0625}} = 3.177 \text{ kg/s (Ans.)}$$

HIGHLIGHTS

1. A compressible flow is that flow in which the density of the fluid changes during flow.
2. The characteristic equation of state is given by :

$$\frac{p}{\rho} = RT$$

where, p = Absolute pressure, N/m^2 ,
 ρ = Density of gas, kg/m^3 ,
 R = Characteristic gas constant, J/kg K , and
 T = Absolute temperature ($= t^\circ\text{C} + 273$).

3. The pressure and density of a gas are related as :

For isothermal process $\frac{p}{\rho} = \text{constant}$

For adiabatic process : $\frac{p}{\rho^\gamma} = \text{constant}$

4. The continuity equation for compressible flow is given as :

$$\rho AV = \text{constant}$$

$$\frac{d\rho}{\rho} + \frac{dA}{A} + \frac{dV}{V} = 0 \quad \dots \text{in differential form.}$$

5. For compressible fluids Bernoulli's equation is given as :

$$\frac{p}{\rho g} \ln(p) + \frac{V^2}{2g} + z = \text{constant} \quad \dots \text{for isothermal process}$$

$$\left(\frac{\gamma}{\gamma - 1} \right) \frac{p}{\rho g} + \frac{V^2}{2g} + z = \text{constant} \quad \dots \text{for adiabatic process}$$

6. Sonic velocity is given by :

$$C = \sqrt{\frac{dp}{d\rho}} = \sqrt{\frac{K}{\rho}} \quad \dots \text{in terms of bulk modulus}$$

$$C = \sqrt{\frac{p}{\rho}} = \sqrt{RT} \quad \dots \text{for isothermal process}$$

$$C = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\gamma RT} \quad \dots \text{for adiabatic process.}$$

7. Mach number, $M = \frac{V}{C}$

(i) Subsonic flow : $M < 1$, $V < C$... disturbance always moves ahead of the projectile

(ii) Sonic flow : $M = 1$, $V = C$... disturbance moves along the projectile

(iii) Supersonic flow : $M > 1$, $V > C$... The projectile always moves ahead of the disturbance.

Mach angle is given by: $\sin \alpha = \frac{C}{V} = \frac{1}{M}$

8. The pressure, temperature and density at a point where velocity is zero are called stagnation pressure (p_s), temperature, (T_s) and stagnation density ρ_s . Their values are given as :

$$p_s = p_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]^{\frac{\gamma}{\gamma - 1}}$$

$$\rho_s = \rho_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]^{\frac{1}{\gamma - 1}}$$

$$T_s = T_0 \left[1 + \left(\frac{\gamma - 1}{2} \right) M_0^2 \right]$$

where p_0 , ρ_0 and T_0 are the pressure, density and temperature at any point O in the flow.

9. Area-velocity relationship for compressible fluid is given as :

$$\frac{dA}{A} = \frac{dV}{V} (M^2 - 1)$$

(i) Subsonic flow ($M < 1$) : $\frac{dV}{V} > 0$; $\frac{dA}{A} < 0$; $dp < 0$ (convergent nozzle)

$\frac{dV}{V} < 0$; $\frac{dA}{A} > 0$; $dp > 0$ (divergent diffuser)

(ii) Supersonic flow ($M > 1$) : $\frac{dV}{V} > 0$; $\frac{dA}{A} > 0$; $dp < 0$ (divergent nozzle)

$\frac{dV}{V} < 0$; $\frac{dA}{A} < 0$; $dp > 0$ (convergent diffuser)

(iii) Sonic flow ($M = 1$) : $\frac{dA}{A} = 0$ (straight flow passage since dA must be zero)

$dp = \frac{\text{zero}}{\text{zero}}$ i.e. indeterminate, but when evaluated, the

change of pressure $dp = 0$, since $dA = 0$ and the flow is frictionless.

10. Flow of compressible fluid through a convergent nozzle :

- (i) Velocity through a nozzle or orifice fitted to a large tank :

$$V_2 = \sqrt{\left(\frac{2\gamma}{\gamma - 1} \right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma}{\gamma - 1}} \right]}$$

- (ii) The mass rate of flow is given by :

$$m = A_2 \sqrt{\left(\frac{2\gamma}{\gamma - 1} \right) p_1 \rho_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{2}{\gamma}} - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma + 1}{\gamma}} \right]}$$

- (iii) Value of $\left(\frac{p_2}{p_1} \right)$ for maximum value of mass rate of flow is given by :

$$\left(\frac{p_2}{p_1} \right) = \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}} = 0.528 \quad (\text{when } \gamma = 1.4)$$

- (iv) Value of V_2 for maximum rate of flow of liquid is given as:

$$V_2 = \sqrt{\left(\frac{2\gamma}{\gamma + 1} \right) \frac{p_1}{\rho_1}} (= C_2)$$

(v) Maximum rate of flow of fluid through nozzle,

$$m_{\max} = A_2 \sqrt{\left(\frac{2\gamma}{\gamma-1}\right) p_1 \rho_1 \left[\left(\frac{2}{\gamma+1}\right)^{\frac{2}{\gamma-1}} - \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}} \right]}$$

For air, substituting $\gamma = 1.4$, we get:

$$m_{\max} = 0.685 A_2 \sqrt{p_1 \rho_1}$$

If the pressure ratio is less than 0.528, the mass rate of flow of the fluid is always corresponding to the pressure ratio of 0.528. But if the pressure ratio is more than 0.528, the mass rate of flow of fluid is corresponding to the given pressure ratio.

11. Whenever a supersonic flow (compressible) changes to subsonic flow, a shock wave (analogous to hydraulic jump in an open channel) is produced, resulting in a sudden rise in pressure, density, temperature and entropy.

$$p_1 + \frac{(\rho_1 V_1)^2}{\rho_1} = p_2 + \frac{(\rho_2 V_2)^2}{\rho_2} \quad \dots \text{Ranking Line Equation}$$

$$\frac{\gamma}{\gamma-1} \left(\frac{p_1}{\rho_1} \right) + \frac{(\rho_1 V_1)^2}{2\rho_1^2} = \frac{\gamma}{\gamma-1} \left(\frac{p_2}{\rho_2} \right) + \frac{(\rho_2 V_2)^2}{2\rho_2^2} \quad \dots \text{Fanno Line Equation}$$

$$\left. \begin{aligned} \frac{p_2}{p_1} &= \frac{\left(\frac{\gamma+1}{\gamma-1}\right) \frac{\rho_2}{\rho_1} - 1}{\left(\frac{\gamma+1}{\gamma-1}\right) - \frac{\rho_2}{\rho_1}} \\ \frac{\rho_2}{\rho_1} &= \frac{V_1}{V_2} = \frac{1 + \left(\frac{\gamma+1}{\gamma-1}\right) \frac{p_2}{p_1}}{\left(\frac{\gamma+1}{\gamma-1}\right) + \frac{p_2}{p_1}} \end{aligned} \right\} \quad \dots \text{Ranking-Hugoniot Equations}$$

One can also express $\frac{p_2}{p_1}$, $\frac{V_2}{V_1}$, $\frac{\rho_2}{\rho_1}$, and $\frac{T_2}{T_1}$ in terms of Mach number as follows :

$$\frac{p_2}{p_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1} \quad \dots (i)$$

$$\frac{V_1}{V_2} = \frac{\rho_2}{\rho_1} = \frac{(\gamma + 1) M_1^2}{(\gamma - 1) M_1^2 + 2} \quad \dots (ii)$$

$$\frac{T_2}{T_1} = \frac{[(\gamma - 1) M_1^2 + 2][2\gamma M_1^2 - (\gamma - 1)]}{(\gamma + 1)^2 M_1^2} \quad \dots (iii)$$

Also,
$$M_2^2 = \frac{(\gamma - 1) M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)}$$

12. Mass rate of flow through venturimeter,

$$m = \rho_2 A_2 \sqrt{\frac{\left(\frac{2\gamma}{\gamma-1}\right) \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}} \right]}{1 - \left(\frac{p_2}{p_1}\right)^{\frac{2}{\gamma}} \times \left(\frac{A_2}{A_1}\right)^2}}$$

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer :

- All real fluids are
 - incompressible
 - compressible to some extent
 - compressible to any extent
 - none of the above.
- A change in the state of a system at constant volume is called
 - isobaric process
 - isochoric process
 - isothermal process
 - adiabatic process.
- A process during which no heat is transferred to or from the gas is called an
 - isochoric process
 - isobaric process
 - adiabatic process
 - isothermal process.
- An adiabatic process is one which follows the relation
 - $\frac{P}{\rho} = \text{constant}$
 - $\frac{P}{\rho^\gamma} = \text{constant}$
 - $\frac{P}{\rho^n} = \text{constant} \ (n \neq \gamma)$
 - $v = \text{constant}$.
- An isentropic flow is one which is
 - isothermal
 - adiabatic
 - adiabatic and irreversible
 - adiabatic and reversible.
- Indicate upto what Mach number can a fluid flow be considered incompressible ?
 - 0.1
 - 0.3
 - 0.8
 - 1.0.
- Which of the following is the basic equation of compressible fluid flow ?
 - Continuity equation
 - Momentum equation
 - Energy equation
 - Equation of state
 - All of the above.
- The velocity of disturbance in case of fluids is the velocity of the disturbance in solids.
 - less than
 - equal to
 - more than
 - none of the above.
- Sonic velocity (C) for adiabatic process is given as
 - $C = \sqrt{\gamma RT^3}$
 - $C = \sqrt{\gamma RT}$
 - $C = \sqrt{\gamma^2 RT}$
 - $C = \gamma RT$.

where γ = ratio of specific heats, R = gas constant, T = temperature.
- The flow is said to be subsonic when Mach number is
 - equal to unity
 - less than unity
 - greater than unity
 - none of above.
- The region outside the Mach cone is called
 - zone of action
 - zone of silence
 - control volume
 - none of the above.
- A stagnation point is the point on the immersed body where the magnitude of velocity is
 - small
 - large
 - zero
 - none of the above.
- A convergent-divergent nozzle is used when the discharge pressure is
 - less than the critical pressure
 - equal to the critical pressure
 - more than the critical pressure
 - none of the above.
- At critical pressure ratio, the velocity at the throat of a nozzle is
 - equal to the sonic speed
 - less than the sonic speed
 - more than the sonic speed
 - none of the above.
- Laval nozzle is a
 - convergent nozzle
 - divergent nozzle
 - convergent-divergent nozzle
 - any of the above.
- A shock wave is produced when
 - a subsonic flow changes to sonic flow
 - a sonic flow changes to supersonic flow
 - a supersonic flow changes to subsonic flow
 - none of the above.
- The sonic velocity in a fluid medium is directly proportional to
 - Mach number
 - pressure
 - square root of temperature
 - none of the above.

18. The stagnation pressure (p_s) and temperature (T_s) are
 - (a) less than their ambient counterparts
 - (b) more than their ambient counterparts
 - (c) the same as in ambient flow
 - (d) none of the above.
19. Across a normal shock
 - (a) the entropy remains constant
 - (b) the pressure and the temperature rise
 - (c) the velocity and pressure decrease
 - (d) the density and temperature decrease.
20. A normal shock wave
 - (a) is reversible
 - (b) is irreversible
 - (c) is isentropic
 - (d) occurs when approaching flow is supersonic.
21. The sonic speed in an ideal gas varies
 - (a) inversely as bulk modulus
 - (b) directly as the absolute pressure
 - (c) inversely as the absolute temperature
 - (d) none of the above.
22. In a supersonic flow, a diffuser is a conduit having
 - (a) gradually decreasing area
 - (b) converging-diverging passage
 - (c) constant area throughout its length
 - (d) none of the above.
23. Choking of a nozzle fitted to a pressure tank containing gas implies
 - (a) sonic velocity at the throat
 - (b) increase of the mass flow rate
 - (c) obstruction of flow
 - (d) all of the above.
24. A shock wave which occurs in a supersonic flow represents a region in which
 - (a) a zone of silence exists
 - (b) there is no change in pressure, temperature and density
 - (c) there is sudden change in pressure, temperature and density
 - (d) velocity is zero.
25. Which of the following statements regarding a normal shock is correct ?
 - (a) It occurs when an abrupt change takes place from supersonic into subsonic flow condition
 - (b) It causes a disruption and reversal of flow pattern
 - (c) It may occur in sonic or supersonic flow
 - (d) None of the above.
26. For compressible fluid flow the area-velocity relationship is
 - (a) $\frac{dA}{A} = \frac{dV}{V} (1 - M^2)$
 - (b) $\frac{dA}{A} = \frac{dV}{V} (C^2 - 1)$
 - (c) $\frac{dA}{A} = \frac{dV}{V} (M^2 - 1)$
 - (d) $\frac{dA}{A} = \frac{dV}{V} (1 - V^2)$
27. The sonic velocity is largest in which of the following ?
 - (a) Water
 - (b) Steel
 - (c) Kerosene
 - (d) Air.
28. Which of the following expressions does not represent the speed of sound in a medium ?
 - (a) $\sqrt{\frac{K}{\rho}}$
 - (b) $\sqrt{\gamma RT}$
 - (c) $\sqrt{K \frac{p}{\rho}}$
 - (d) $\sqrt{\frac{dp}{d\rho}}$
29. The differential for energy in isentropic flow is of the form
 - (a) $\frac{dV}{V} + \frac{dp}{\rho} + \frac{dA}{A} = 0$
 - (b) $VdV + \frac{dp}{\rho} = 0$
 - (c) $2VdV + \frac{dp}{\rho} = 0$
 - (d) $dp + d(\rho V^2) = 0.$
30. Which of the following statements is *incorrect* ?
 - (a) A shock wave occurs in divergent section of a nozzle when the compressible flow changes abruptly from supersonic to subsonic state.
 - (b) A plane moving at supersonic state is not heard by the stationary observer on the ground until it passes him because zone of disturbance in Mach cone trails behind the plane
 - (c) A divergent section is added to a convergent nozzle to obtain supersonic velocity at the throat.
 - (d) none of above.

ANSWERS

- | | | | | | |
|---------|---------|---------|---------|---------|----------|
| 1. (b) | 2. (b) | 3. (c) | 4. (b) | 5. (d) | 6. (b) |
| 7. (e) | 8. (a) | 9. (b) | 10. (b) | 11. (b) | 12. (c) |
| 13. (a) | 14. (a) | 15. (c) | 16. (c) | 17. (c) | 18. (b) |
| 19. (b) | 20. (d) | 21. (d) | 22. (a) | 23. (d) | 24. (c) |
| 25. (a) | 26. (c) | 27. (b) | 28. (c) | 29. (b) | 30. (c). |

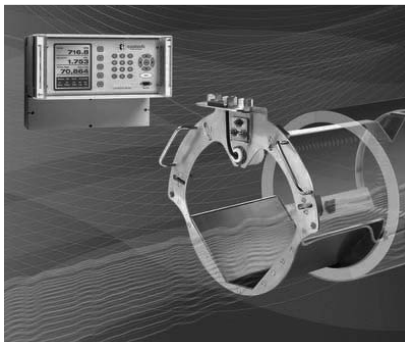
THEORETICAL QUESTIONS

- Differentiate between compressible and incompressible flows.
- Give the examples when liquid is treated as a compressible fluid.
- When is the compressibility of fluid important ?
- What is the difference between isentropic and adiabatic flows ?
- What is the relation between pressure and density of a compressible fluid for (a) isothermal process (b) adiabatic process ?
- Obtain an expression in differential form for continuity equation for one-dimensional compressible flow.
- Derive an expression for Bernoulli's equation when the process is adiabatic.
- How are the disturbances in compressible fluid propagated ?
- What is sonic velocity ? On what factors does it depend ?
- What is Mach number ? Why is this parameter so important for the study of flow of compressible fluids ?
- Prove that velocity of sound wave in a compressible fluid is given by : $C = \sqrt{k/\rho}$ where, k and ρ are the bulk modulus and density of fluid respectively.
- Define the following terms :
(i) Subsonic flow (ii) Sonic flow
(iii) Supersonic flow (iv) Mach cone.
- What is silence zone during the disturbance which propagates when an object moves in still air ?
- What is stagnation point of an object immersed in fluid ?
- What is stagnation pressure ?
- What are static and stagnation temperatures ?
- Derive an expression for mass flow rate of compressible fluid through an orifice or nozzle fitted to a large tank. What is the condition for maximum rate of flow ?
- What is the critical pressure ratio for a compressible flow through a nozzle? On what factors does it depend ?
- Describe compressible flow through a convergent-divergent nozzle. How and where does the shock wave occur in the nozzle ?
- What do you mean by compressibility correction factor ?
- How is a shock wave produced in a compressible fluid ? What do you mean by the term "Shock strength" ?
- Derive an expression for mass rate of flow through a venturimeter.

UNSOLVED EXAMPLES

- A 100 mm diameter pipe reduces to 50 mm diameter through a sudden contraction. When it carries air at 20.16°C under isothermal conditions, the absolute pressures observed in the two pipes just before and after the contraction are 400 kN/m² and 320 kN/m² respectively. Determine the densities and velocities at the two sections. Take $R = 290 \text{ J/kg K}$.
[Ans. 4.7 kg/m³; 3.76 kg/m³; 39.7 m/s; 198.5 m/s]
- A gas with a velocity of 300 m/s is flowing through a horizontal pipe at a section where pressure is 60 kN/m² (abs.) and temperature 40°C. The pipe changes in diameter and at this section the pressure is 90 kN/m². If the flow of gas is adiabatic find the velocity of gas at this section.
Take $R = 287 \text{ J/kgK}$ and $\gamma = 1.4$.
[Ans. 113 m/s]

3. An aeroplane is flying at 21.5 m/s at a low altitude where the velocity of sound is 325 m/s. At a certain point just outside the boundary layer of the wings, the velocity of air relative to the plane is 305 m/s. If the flow is frictionless adiabatic determine the pressure drop on the wing surface near this position.
Assume $\gamma = 1.4$, pressure of ambient air = 102 kN/m². [Ans. 28.46 kN/m²]
4. A jet propelled aircraft is flying at 1100 km/h. at sea level. Calculate the Mach number at a point on the aircraft where air temperature is 20°C. Take : $R = 287$ J/kg K and $\gamma = 1.4$.
[Ans. 0.89]
5. An aeroplane is flying at an height of 20 km where the temperature is -40°C . The speed of the plane is corresponding to $M = 1.8$. Find the speed of the plane.
Take : $R = 287$ J/kg K, $\gamma = 1.4$.
[Ans. 1982.6 km/h]
6. Find the velocity of bullet fired in standard air if its Mach angle is 30° . [Ans. 680.4 m/s]
7. Air, thermodynamic state of which is given by pressure $p = 230$ kN/m² and temperature = 300 K is moving at a velocity $V = 250$ m/s. Calculate the stagnation pressure if (i) compressibility is neglected and (ii) compressibility is accounted for.
Take $\gamma = 1.4$, and $R = 287$ J/kg K.
[Ans. 313 kN/m², 323 kN/m²]
8. A large vessel, fitted with a nozzle, contains air at a pressure of 2943 kN/m² (abs.) and at a temperature of 20°C. If the pressure at the outlet of the nozzle is 2060 kN/m² (abs.) find the velocity of air flowing at the outlet of the nozzle.
Take : $R = 287$ J/kgK, and $\gamma = 1.4$
[Ans. 239.2 m/s]
9. Nitrogen gas ($\gamma = 1.4$) is released through a 10 mm orifice on the side of a large tank in which the gas is at a pressure of 10 bar and temperature 20°C. Determine the mass flow rate if (i) the gas escapes to atmosphere (1 bar); (ii) the gas is released to another tank at (a) 5 bar, (b) 6 bar.
[Ans. (i) 0.183 kg/s; (ii) 0.183 kg/s; 0.167 kg/s]
10. Air is released from one tank to another through a convergent-divergent nozzle at the rate of 12 N/s. The supply tank is at a pressure of 400 kN/m² and temperature 110°C, and the pressure in the receiving tank is 100 kN/m². Determine:
(i) The pressure, temperature, and Mach number in the constriction, (ii) The required diameter of constriction, (iii) The diameter of the nozzle at the exit for full expansion, and the Mach number.
[Ans. (i) 210 kN/m²; 319 K, (ii) 43.5 mm; (iii) 48 mm; 1.56]
11. Oxygen flows in a conduit at an absolute pressure of 170 kN/m². If the absolute pressure and temperature at the nose of small object in the stream are 200 kN/m² and 70.16°C respectively, determine the velocity in the conduit. Take $\gamma = 1.4$ and $R = 281.43$ J/kg K.
[Ans. 175.3 m/s]
12. Air at a velocity of 1400 km/h has a pressure of 10 kN/m² vacuum and temperature of 50.16°C. Calculate local Mach number and stagnation pressure, density and temperature.
Take $\gamma = 1.4$, $R = 281.43$ J/kg K and barometric pressure = 101.325 kN/m²
[Ans. 1.089; 192.358 kN/m²; 1.708 kg/m³; 399.8 K]
13. A normal shock wave occurs in a diverging section when air is flowing at a velocity of 420 m/s, pressure 100 kN/m², and temperature 10°C. Determine : (i) The Mach number before and after the shock, (ii) The pressure rise, and (iii) The velocity and temperature after the shock.
[Ans. (i) 1.25; 0.91; (ii) 66 kN/m², (iii) 292 m/s; 54°C]
14. A normal shock wave occurs in air flowing at a Mach number of 1.5. The static pressure and temperature of the air upstream of the shock wave are 100 kN/m² and 300 K. Determine the Mach number, pressure and temperature downstream of the stockwave. Also estimate the shock strength.
[Ans. 0.7; 246 kN/m²; 396.17 K; 1.46]
15. A 25 mm diameter venturimeter is fixed in a 75 mm diameter pipe to measure the rate of flow of gas. If the absolute pressure at the inlet and the throat of venturimeter are equivalent to 1010 mm and 910 mm of mercury, determine the volumetric flow rate of gas. Assume the flow to be isentropic, $\gamma = 1.4$ and $\rho_1 = 1.6$ kg/m³.
[Ans. 0.0599 m³/s]



FLOW IN OPEN CHANNELS

- 16.1. Introduction
- 16.2. Types of flow in channels
- 16.3. Definitions
- 16.4. Open channel formulae for uniform flow
- 16.5. Most economical section of channel
- 16.6. Open channel section for constant velocity at all depths of flow
- 16.7. Non-uniform flow through open channels
- 16.8. Specific energy and specific energy curve
- 16.9. Hydraulic jump or standing wave
- 16.10. Gradually varied flow
- 16.11. Measurement of flow of irregular channels

Highlights

Answers

Objective Type Questions

Theoretical Questions

Unsolved Examples.

A. UNIFORM FLOW

16.1. INTRODUCTION

16.1.1 Definition of an Open Channel.

An open channel may be defined as a passage in which liquid flows with its upper surface exposed to atmosphere. In open channels the flow is due to gravity, thus the flow conditions are greatly influenced by the slope of the channel.

16.1.2 Comparison between Open Channel and Pipe Flow

The important points of difference between the two types of flows are given below:

S.No.	Aspects	Open channel flow	Pipe flow
1.	<i>Cause of flow</i>	Gravity force (provided by sloping bottom)	The pipe runs full and the flow, in general, takes place at <i>the expense of hydraulic pressure</i> ; the pressure continuously decreases in the direction of flow.
2.	<i>Geometry of cross-section</i>	Open channels may have any shape: triangular, rectangular, trapezoidal, parabolic, circular etc.	Pipes ...generally round in cross-section ... cross-section of flow is fixed, since the flowing liquid entirely fills the pipe section.
3.	<i>Surface roughness</i>	Varies between wide limits, the hydraulic roughness varies with depth of flow.	Roughness co-efficient varies from a low value to a very high value, depending upon the material of the pipe.

4.	<i>Piezometric head</i>	$(z + y)$, where y is the depth of flow. H.G.L. coincides with the water surface.	$\left(z + \frac{p}{w}\right)$, where p is the pressure in the pipe. H.G.L. does not coincide with water surface.
5.	<i>Velocity distribution</i>	The maximum velocity occurs at a little distance below the water surface. The shape of the velocity profile is dependent on the channel roughness.	The velocity distribution is symmetrical about the pipe axis, maximum velocity occurring at the pipe centre and the velocity at the pipe wall reducing to zero.

16.1.3 Types of Channels

The various types of channels are:

1. **Natural channel.** It is the one which has irregular sections of varying shapes, developed in a natural way.

Examples: Rivers, streams etc.

2. **Artificial channel.** It is the one which is built artificially for carrying water for various purposes. They have the cross-sections with regular geometrical shapes (which usually remain same throughout the length of the channel).

Examples: Rectangular channel, trapezoidal channel, parabolic channel etc.

3. **Open channel.** A channel without any cover at the top is known as an *open channel*.

Examples: Irrigation canals, rivers, streams, flumes and water falls.

4. **Covered or closed channels.** The channel having a cover at the top is known as a *covered* or *closed channel*.

Examples: Partly filled conduits carrying public water supply such as sewerage lines, underground drains, tunnels etc. not running full of water.

5. **Prismatic channel.** A channel with constant bed slope and the same cross-section along its length is known as a *prismatic channel*.

The prismatic channels can be further subdivided as:

- (i) **Exponential channel.** It is the one in which area of cross-section of flow is directly proportional to any power of depth of flow in channel.

Examples: Rectangular, triangular and parabolic channels.

- (ii) **Non-exponential channel.** Trapezoidal and circular channels are non-exponential channels.

16.2. TYPES OF FLOW IN CHANNELS

The flow in channels is classified into the following types, *depending upon the change in the depth of flow with respect to space and time*:

1. Steady flow and unsteady flow
2. Uniform flow and non-uniform (or varied) flow
3. Laminar flow and turbulent flow
4. Subcritical flow, critical flow and supercritical flow.

16.2.1 Steady Flow and Unsteady Flow

- When the flow characteristics (such as depth of flow, flow velocity and the flow rate at any cross-section) do not change with respect to time, the flow in a channel is said to be *steady*.

$$\text{Mathematically, } \frac{\partial y}{\partial t} = 0, \frac{\partial V}{\partial t} = 0, \text{ or } \frac{\partial Q}{\partial t} = 0$$

where y , V and Q are depth of flow, velocity and rate of flow respectively.

- The flow is said to be *unsteady flow* when these flow parameters vary with time.

$$\text{Mathematically, } \frac{\partial y}{\partial t} \neq 0; \frac{\partial V}{\partial t} \neq 0 \text{ or } \frac{\partial Q}{\partial t} \neq 0.$$

16.2.2 Uniform and Non-uniform (or varied) Flow

- Flow in a channel is said to be *uniform* if the depth, slope, cross-section and velocity *remain constant* over a given length of the channel.

$$\text{Mathematically, } \frac{\partial y}{\partial l} = 0, \frac{\partial V}{\partial l} = 0$$

Uniform flows are possible only in prismatic channels only. A uniform flow may be either steady or unsteady, depending upon whether or not the discharge varies with time; *unsteady uniform flow is rare in practice*.

- Flow in a channel is said to be *non-uniform (or varied)* when the channel depth *varies* continuously from one section to another.

$$\text{Mathematically, } \frac{\partial y}{\partial l} \neq 0, \frac{\partial V}{\partial l} \neq 0$$

Varied flow may be further classified as:

- Rapidly varied flow (R.V.F.)*. In this type of flow depth of flow *changes abruptly* over a *comparatively small length* of channel.

Examples: Hydraulic jump and the hydraulic drop.

- Gradually varied flow (G.V.F.)*. In this case the change in depth of flow takes place gradually in a long length of the channel.

16.2.3 Laminar Flow and Turbulent Flow

The flow in the open channel may be characterised as laminar or turbulent depending upon the value of Reynolds number, defined as:

$$Re = \frac{\rho VR}{\mu} \quad \dots(16.1)$$

where, V = Average velocity of flow in the channel, and

R = Hydraulic radius (defined as the ratio of area of flow to wetted perimeter)

When $Re < 500$...flow is *laminar*

$Re > 2000$...flow is *turbulent*

$500 < Re < 2000$...flow is *transitional*.

16.2.4 Subcritical flow, Critical Flow and Supercritical Flow

Since gravitational force is a predominant force in the case of channel flow, therefore Froude number, $Fr = \frac{V}{\sqrt{gD}}$ (where V and D are the mean velocity of flow and hydraulic depth of the

channel respectively) is an important parameter for analysing open channel flows. Depending upon Froude number the channel flow may be characterised as:

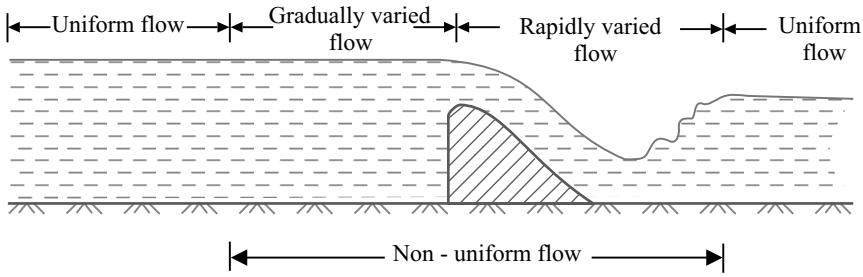


Fig. 16.1. Uniform and non-uniform flow.

(i) When $Fr < 1$ (or $V < \sqrt{gD}$): The flow is described as *subcritical* (or *tranquil* or *streaming*)

(ii) When $Fr = 1$: The flow is said to be in a *critical* state.

(iii) When $Fr > 1$: The flow is said to be *supercritical* (or *rapid* or *shooting* or *torrential*)

Some of the types of channel flow are shown in Fig. 16.1

16.3. DEFINITIONS

1. **Depth of flow (y).** It is the vertical distance of the lowest point of a channel section (bed of the channel) from the free surface.
2. **Depth of flow section.** It is the depth of flow normal to the bed of the channel.

$$d = y \cos \theta \quad \dots(16.2)$$

where, θ = The angle which the channel bed makes with the horizontal.

Since the slopes of the channels are very small,

$$\cos \theta \approx 1 \text{ and } d \approx y.$$

The depth of flow and depth of flow section are assumed equal, unless mentioned otherwise.

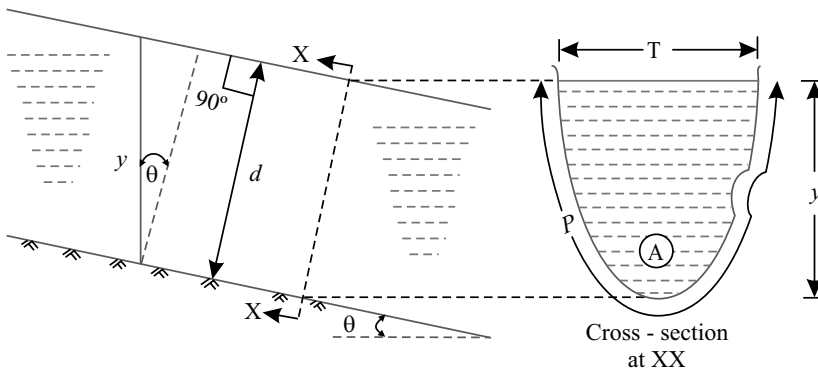
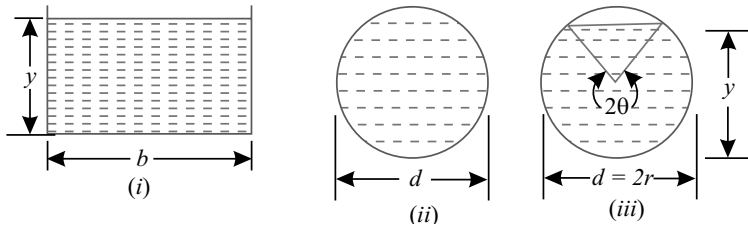


Fig. 16.2. Terms related to flow through open channel.

3. **Top width (T).** It is the width of the channel section at the free surface (*i.e.* the width of the liquid surface exposed to the atmospheric pressure).
4. **Wetted area (A).** It is the cross-sectional area of the flow section of the channel.
5. **Wetted perimeter (P).** It is the length of the channel boundary in contact with the flowing water at any section.
6. **Hydraulic radius (R).** It is ratio of the cross-sectional area of flow to wetted perimeter. It is also called *hydraulic mean depth*.

$$\text{i.e.} \quad R = \frac{A}{P} \quad \dots(16.3)$$

**Fig. 16.3.**

Examples: (i) Rectangular open channel:

$$R = \frac{A}{P} = \frac{b \times y}{b + 2y} \quad \dots(16.4)$$

(ii) Pipe running full:

$$R = \frac{A}{P} = \frac{(\pi/4) \times d^2}{\pi d} = \frac{d}{4} \quad \dots(16.5)$$

(iii) Pipe not running full:

$$R = \frac{A}{P} = \frac{\frac{r^2}{2} (2\theta - \sin 2\theta)}{2r \theta} \quad \dots(16.6)$$

7. **Hydraulic depth (D).** It is the ratio of the wetted area A to the top width T .

$$\text{i.e.} \quad D = \frac{A}{T} \quad \dots(16.7)$$

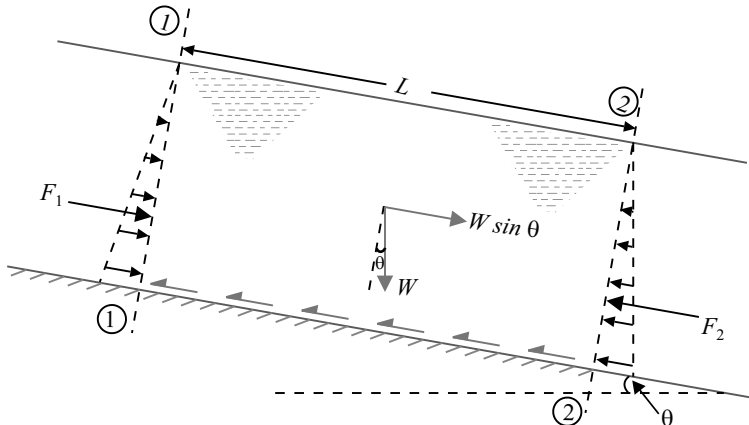
16.4. OPEN CHANNEL FORMULAE FOR UNIFORM FLOW

For uniform flow in open channels, the following formulae will be discussed:

1. Chezy's formula
2. Manning's formula.

16.4.1 Chezy's Formula

Consider a longitudinal section of an open channel in which the flow is steady and uniform, as shown in Fig. 16.4. The forces acting on the free body of water between sections 1-1 and 2-2 in the direction of flow are as follows:

**Fig. 16.4.** Uniform flow in open channel.

(i) Pressure forces F_1 and F_2 acting on the two ends of the body; these forces balance each other since the depth of channel remains constant.

(ii) The component of weight of the water in the direction of flow, which is

$$= W \sin \theta = wAL \sin \theta$$

where, w = Specific weight of water,

A = Wetted cross-sectional area of channel,

L = Length of the channel considered, and

θ = Angle of inclination of channel bottom with the horizontal.

(iii) Frictional resistance offered by the sides of the channel which is $= \tau_0 PL$, where P is the wetted perimeter of the channel and τ_0 is the average shear stress at the channel boundary.

As the flow is steady and uniform, it is neither accelerating nor decelerating; the liquid mass is in equilibrium and the frictional resistance to flow equals the weight of liquid mass acting along the line of fluid motion. Thus

$$wAL \sin \theta = \tau_0 PL$$

Since frictional resistance τ_0 varies with (velocity)², τ_0 may be expressed as fV^2 where f is a *non-dimensional factor* whose value depends upon the *material and nature of flow surface*.

$$\therefore wAL \sin \theta = fV^2 PL$$

$$\text{or, } V_2 = \frac{wAL \sin \theta}{fPL} \quad \text{or} \quad V = \sqrt{\frac{w}{f}} \times \sqrt{\frac{A}{P}} \sin \theta$$

Since $\frac{A}{P}$ is the hydraulic radius (or hydraulic mean depth) and θ is the slope of the channel bed (S), we may write:

$$V = C\sqrt{RS} \quad \dots(16.8)$$

where $C = \sqrt{\frac{w}{f}}$ (a variable which depends on the roughness of the channel surface and the flow Reynolds number).

Eqn. (16.8) is known as **Chezy's formula** (named after the French engineer Antoine Chezy who developed this formula in 1775). The term C is known as *Chezy's constant*.

Discharge through the channel, Q = Area $\times$ velocity

$$= AC\sqrt{RS}$$

which can be written as:

$$Q = K\sqrt{S} \quad \dots(16.9)$$

where,

$$K = AC\sqrt{R}$$

The factor K is called the **conveyance** of the channel section, and is a *measure of the carrying capacity of the channel*. For a channel of constant slope, the conveyance is directly proportional to discharge Q .

Empirical relations for the Chezy's constant C :

Although Chezy's equation is quite simple, the selection of a correct value of C is rather difficult. Some of the important formulae developed for Chezy's constant C are:

(a) Bazin's formula:

A French hydraulician H. Bazin's (1897) proposed the following empirical formula for Chezy's constant:

$$C = \frac{157.6}{181 + \frac{K}{\sqrt{R}}} \quad \dots(16.10)$$

where, R is the hydraulic radius and K is the Bazin's constant whose value depends on surface roughness. Some typical values of K are:

S.No.	Surface of channel	Bazin's constant (K)
1.	Smooth cement plaster or planed wood	0.11
2.	Concrete, brick, or unplanned wood	0.21
3.	Smooth rubble masonry or poor brickwork	0.83
4.	Earth channels in very good condition	1.54
5.	Earth channels in rough condition	3.17
6.	Dredged earth channels, average condition	2.36

(b) Kutter's formula:

Two Swiss engineers Ganguillet and Kutter proposed the following empirical formula (1869) for the determination of Chezy's constant C .

$$C = \frac{23 + \frac{0.00155}{S} + \frac{1}{N}}{1 + \left(23 + \frac{0.00155}{S}\right) \frac{N}{\sqrt{R}}} \quad \dots(16.11)$$

where N is the Kutter's constant whose value depends upon the type of the channel surface. Some typical values of N are given below:

S.No.	Surface of channel	N (Kutter's/Manning's constant)
1.	Smooth cement plaster or planed wood	0.010
2.	Very smooth concrete and planed timber	0.011
3.	Smooth concrete	0.012
4.	Glazed brickwork	0.013
5.	Vitrified clay	0.014
6.	Brick surface lined with cement mortar	0.015
7.	Earth channels in best condition	0.017
8.	Straight unlined earth channels in good condition	0.020
9.	Rivers and earth channels in fair condition	0.025
10.	Canal and river of rough surface with weeds	0.030

(c) Manning's formula:

Robert Manning (an Irish engineer) gave the following empirical relation for determination of Chezy's constant C (1889), which is *simplest* of all used for uniform open channel flow:

$$C = \frac{1}{N} \cdot R^{1/6}$$

where N is the Manning's constant (also known as **rugosity co-efficient**—a term generally used by British engineers) whose value depends on the channel surface.

Example 16.1. Find the rate of flow and conveyance for a rectangular channel 7.5 m wide for uniform flow at a depth of 2.25 m. The channel is having bed slope as 1 in 1000. Take Chezy's constant $C = 55$.

Also state whether the flow is tranquil or rapid.

Solution. Width of the rectangular channel, $b = 7.5$ m

Depth of flow, $y = 2.25$ m

$\therefore$ Area of flow, $A = b \times y = 7.5 \times 2.25 = 16.875 \text{ m}^2$

Bed slope, $S = \frac{1}{1000}$

Chezy's constant, $C = 55$

Wetted perimeter, $P = b + 2y = 7.5 + 2 \times 2.25 = 12.0$ m

$\therefore$ Hydraulic radius (or hydraulic mean depth), $R = \frac{A}{P} = \frac{16.875}{12.0} = 1.406$ m

Rate of flow, Q :

Using Chezy's formula, average velocity,

$$V = C\sqrt{RS} = 55 \sqrt{1.406 \times \frac{1}{1000}} = 2.06 \text{ m/s}$$

$\therefore$ Discharge, $Q = AV = 16.875 \times 2.06 = 34.76 \text{ m}^3/\text{s}$ (Ans.)

Conveyance, K :

$$K = AC\sqrt{R} = 16.875 \times 55 \times \sqrt{1.406} = 1100.5 \text{ (Ans.)}$$

State of flow (tranquil or rapid):

$$\text{Froude number, } Fr = \frac{V}{\sqrt{gy}} = \frac{2.06}{\sqrt{9.81 \times 2.25}} = 0.438$$

Since $Fr < 1.0$, the flow in the channel is **tranquil** in nature. (Ans.)

Example 16.2. A triangular gutter, whose sides include an angle of 60° , conveys water at a uniform depth of 250 mm. If the discharge is $0.04 \text{ m}^3/\text{s}$, determine the gradient of the trough. Use the Chezy's formula assuming that $C = 52$. [Delhi University]

Solution. Depth of flow = 250 mm = 0.25 m.

Discharge through the gutter, $Q = 0.04 \text{ m}^3/\text{s}$

Chezy's constant, $C = 52$

Bed slope, S :

Refer to Fig. 16.5. From ΔACO ,

$$\begin{aligned} \frac{CO}{AO} &= \cos 30^\circ \quad \text{or} \quad AO = \frac{CO}{\cos 30^\circ} \\ &= \frac{0.25}{\cos 30^\circ} = 0.288 \text{ m} \end{aligned}$$

i.e. $AO = BO = 0.288$ m

Further $\frac{AC}{CO} = \tan 30^\circ$

or, $AC = CO \tan 30^\circ = 0.25 \times 0.577 = 0.144$ m

or, $AB = 2AC = 0.288$ m

$\therefore$ Area of flow, $A = \frac{1}{2} \times AB \times CO = \frac{1}{2} \times 0.288 \times 0.25 = 0.036 \text{ m}^2$

Wetted perimeter, $P = AO + BO = 0.288 + 0.288 = 0.576$ m

Hydraulic radius, $R = \frac{A}{P} = \frac{0.036}{0.576} = 0.0625$ m

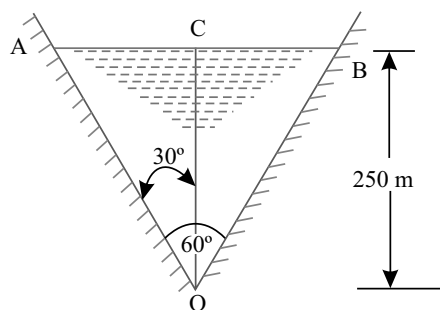


Fig. 16.5

Using Chezy's formula, we have:

$$Q = AV = AC \sqrt{RS} \quad \text{or} \quad 0.04 = 0.036 \times 52 \times \sqrt{0.0625 \times S}$$

$$\text{or,} \quad \sqrt{0.0625 \times S} = \frac{0.04}{0.036 \times 52} = 0.02137$$

$$\text{or,} \quad S = \frac{0.02137^2}{0.0625} = \frac{1}{137} \quad (\text{Squaring both sides})$$

Hence, *gradient of the trough (or bed slope) is 1 in 137 (Ans.)*

Example 16.3. Find the discharge of water through the channel shown in Fig. 16.6. Take the value of Chezy's constant = 60 and slope of the bed as 1 in 950. [UPTU]

Solution. Chezy's constant, $C = 60$

$$\text{Bed slope, } S = \frac{1}{950}$$

Discharge, Q :

Refer to Fig. 16.6.

Area of flow, $A = \text{Area } ABCD + \text{area } DEC$

$$= 1.2 \times 0.6 + \frac{\pi \times 0.6^2}{2} = 1.285 \text{ m}^2$$

$$\begin{aligned} \text{Wetted perimeter, } P &= AD + DEC + CB \\ &= 0.6 + \pi \times 0.6 + 0.6 = 3.085 \text{ m} \end{aligned}$$

$\therefore$ Hydraulic mean radius,

$$R = \frac{A}{P} = \frac{1.285}{3.085} = 0.416 \text{ m}$$

Using Chezy's formula, we have:

$$\begin{aligned} Q &= AV = AC \sqrt{RS} \\ &= 1.285 \times 60 \times \sqrt{0.416 \times \frac{1}{950}} = 1.613 \text{ m}^3/\text{s} \quad (\text{Ans.}) \end{aligned}$$

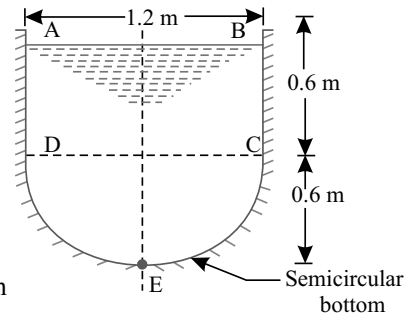


Fig. 16.6

Example 16.4. A canal of trapezoidal section has bed width of 8 m and bed slope of 1 in 4000. If the depth of flow is 2.4 m and side slopes of the channel are 1 horizontal to 3 vertical, determine the average flow velocity and the discharge carried by the channel. Also compute the average shear stress at the channel boundary. Take value of Chezy's constant = 55.

Solution. Width of the channel bed, $b = 8 \text{ m}$

$$\text{Bed slope, } S = \frac{1}{4000}$$

Side slopes = 1 horizontal to 3 vertical

Depth of flow, $y = 2.4 \text{ m}$

Chezy's constant, $C = 55$

Horizontal distance $EA = BF = ny$

where, $n = \text{side slope (1 vertical to } n \text{ horizontal)}$

$$\text{Top width } CD = AB + 2BF = b + 2ny$$

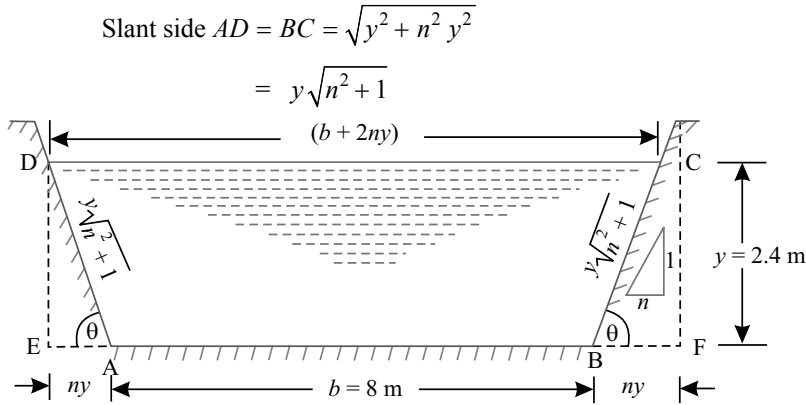


Fig. 16.7

$\therefore$ Wetted perimeter, $P = AB + AD + BC$

$$= 8 + 2y\sqrt{n^2 + 1} = 8 + 2 \times 2.4 \sqrt{\left(\frac{1}{3}\right)^2 + 1} = 13.06 \text{ m } (\because n = 1/3)$$

$$\begin{aligned} \text{Area of flow} &= \left(\frac{\text{Top width} + \text{bottom width}}{2} \right) \times \text{height} = \frac{(b + 2ny) + b}{2} \times y = y(b + ny) \\ &= 2.4 \left(8 + \frac{1}{3} \times 2.4 \right) = 21.12 \text{ m}^2 \end{aligned}$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{21.12}{13.06} = 1.617 \text{ m}$$

Average flow velocity:

$$\begin{aligned} \text{Average flow velocity, } V &= C\sqrt{RS} \\ &= 55 \sqrt{1.617 \times \frac{1}{4000}} = 1.106 \text{ m/s (Ans.)} \end{aligned}$$

Discharge, Q :

$$\text{Discharge through the channel, } Q = AV = 21.12 \times 1.106 = 23.36 \text{ m}^3/\text{s (Ans.)}$$

Shear stress at channel boundary, τ_0 :

Under equilibrium conditions, the frictional resistance to flow equals the weight of liquid mass acting along the line of fluid motion,

$$\text{i.e. } \tau_0 LP = wAL \sin \theta$$

$\therefore$ Shear stress at the channel boundary,

$$\begin{aligned} \tau_0 &= \frac{wAL \sin \theta}{LP} = w \frac{A}{P} \sin \theta = w \times R \times S \quad (\because S = \sin \theta) \\ &= 9810 \times 1.617 \times \frac{1}{4000} = 3.96 \text{ N/m}^2 \text{ (Ans.)} \end{aligned}$$

16.5. MOST ECONOMICAL SECTION OF A CHANNEL

The **most economical section** (also called the *best section* or *most efficient section*) is one which gives the maximum discharge for a given amount of excavation.

From continuity equation it is evident that discharge is maximum when velocity is maximum, the area of cross-section of channel remaining constant. From Chezy's formula and Manning's formula it can be seen that for a given value of slope and surface roughness the velocity of flow will be maximum if hydraulic radius $R = \left(= \frac{A}{P} \right)$ is maximum.

Further the area being constant hydraulic radius is maximum if the *wetted perimeter is minimum*; this condition is used to determine the dimensions of economical sections of different forms of channels. *The best form of channel which complies with this condition is one which has a semi-circular cross-section.*

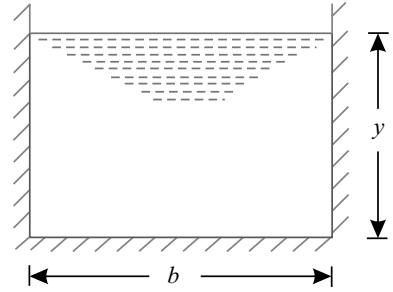


Fig. 16.8 Rectangular channel.

16.5.1 Most Economical Rectangular Channel Section

Fig. 16.8 shows the cross-section of a rectangular channel. Let b and y be the base width and depth of flow respectively.

$$\text{Area of flow, } A = b \times y, \quad \dots(i)$$

$$\text{Wetted perimeter, } P = b + 2y \quad \dots(ii)$$

Substituting the value of $b \left(= \frac{A}{y} \right)$ from eqn. (i) in eqn. (ii), we get:

$$P = \frac{A}{y} + 2y$$

For the section to be most economical/efficient, the wetted perimeter P must be a *minimum*.

$$\text{i.e.} \quad \frac{dP}{dy} = 0 \quad \text{or} \quad \frac{d}{dy} \left[\frac{A}{y} + 2y \right] = 0$$

$$\text{or,} \quad -\frac{A}{y^2} + 2 = 0 \quad \text{or} \quad A = 2y^2 \quad \text{or} \quad b \times y = 2y^2 \quad [\because A = b \times y]$$

$$\text{or,} \quad b = 2y \quad \text{or} \quad y = b/2 \quad \dots(16.12)$$

Hydraulic radius, R :

$$\begin{aligned} \text{Hydraulic radius, } R &= \frac{A}{P} = \frac{b \times y}{b + 2y} \\ &= \frac{2y \times y}{2y + 2y} = \frac{2y^2}{4y} = \frac{y}{2} \quad (\because b = 2y) \end{aligned}$$

$$\text{i.e.} \quad R = \frac{y}{2} \quad \dots(16.13)$$

Thus the rectangular channel section will be most economical when:

(i) The depth of flow is equal to half the base width $\left(y = \frac{b}{2} \right)$, or

(ii) Hydraulic radius is equal to half the depth of flow $\left(R = \frac{y}{2} \right)$.

Example 16.5. A rectangular channel is to be dug in the rocky portion of a soil. Find its most economical cross-section if it is to convey $12 \text{ m}^3/\text{s}$ of water with an average velocity of 3 m/s . Take Chezy's constant $C = 50$.

Solution. Discharge, $Q = 12 \text{ m}^3/\text{s}$
 Average velocity, $V = 3 \text{ m/s}$
 Chezy's constant, $C = 50$

The geometric relations for *optimum discharge* through a rectangular channel are:

$$b = 2y \quad \text{and} \quad R = \frac{y}{2}, \text{ then area } A = b \times y = 2y^2$$

where b , y and R are base width of the channel, depth of flow and hydraulic radius respectively.

Now, $Q = A \times V = 2y^2 \times V$ or $12 = 2y^2 \times 3$ or $y = 1.414 \text{ m}$

i.e. Flow depth, $y = 1.414 \text{ m}$

$\therefore$ Base width of the channel, $b = 2y = 2 \times 1.414 = 2.828 \text{ m}$

Hydraulic radius, $R = \frac{y}{2} = \frac{1.414}{2} = 0.707 \text{ m}$

Also, $V = C\sqrt{RS}$...Chezy's formula

$$\text{or,} \quad S = \frac{V^2}{C^2 R} = \frac{3^2}{50^2 \times 0.707} = \frac{1}{196}$$

(where S = slope bed)

Hence, $b = 2.828 \text{ m}$, $y = 1.414 \text{ m}$, $S = \frac{1}{196}$ (Ans.)

Example 16.6. Determine the most economical section of a rectangular channel carrying water at the rate of $0.5 \text{ m}^3/\text{s}$; the bed slope of the channels being 1 in 2000. Take Chezy's constant $C = 50$.

Solution. Discharge, $Q = 0.5 \text{ m}^3/\text{s}$

$$\text{Bed slope, } S = \frac{1}{2000}$$

Chezy's constant, $C = 50$

Most economical section:

The rectangular channel section will be most economical when:

- (i) Base width, $b = 2y$
 (ii) Hydraulic radius, $R = \frac{y}{2}$ (where, y = depth of flow)

$$\text{Area of flow, } A = b \times y = 2y \times y = 2y^2$$

Now, Discharge $Q = AC\sqrt{RS}$...Chezy's formula

$$\begin{aligned} 0.5 &= 2y^2 \times 50 \sqrt{\frac{y}{2} \times \frac{1}{2000}} \\ &= 100 \sqrt{\frac{1}{4000}} \times y^{5/2} = 1.581 y^{5/2} \end{aligned}$$

$$\therefore y^{5/2} = \frac{0.5}{1.581} = 0.316 \quad \text{or} \quad y = (0.316)^{2/5} = 0.63 \text{ m}$$

$$\text{and,} \quad b = 2y = 2 \times 0.63 = 1.26 \text{ m}$$

Hence, $b = 1.26 \text{ m}$ and $y = 0.63 \text{ m}$ (Ans.)

Example 16.7. A rectangular channel 4 m wide has depth of water 1.5 m. The slope of the bed of the channel is 1 in 1000 and value of Chezy's constant $C = 55$. It is desired to increase the

discharge to a maximum by changing the dimensions of the section for constant area of cross-section, slope of the bed and roughness of the channel. Find the new dimensions of the channel and increase in discharge. [PTU]

Solution. Base width of the channel, $b = 4$ m

Depth of flow, $y = 1.5$ m

$$\text{Bed slope, } S = \frac{1}{1000}$$

Chezy's constant, $C = 55$

New dimensions of the channel and increase in discharge:

$$\text{Area of flow, } A = b \times y = 4 \times 1.5 = 6 \text{ m}^2$$

$$\text{Wetted perimeter, } P = b + 2y = 4 + 2 \times 1.5 = 7.0 \text{ m}$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{6}{7} = 0.857$$

$$\text{Discharge, } Q = AC\sqrt{RS} = 6 \times 55 \sqrt{0.857 \times \frac{1}{1000}} = 9.66 \text{ m}^3/\text{s}$$

For determining maximum discharge, for given area of cross-section, slope of the bed and roughness of the channel, we follow the procedure given below:

Let b' = New base width of the channel, and

y' = New depth of flow,

Then, area $A = b' \times y'$, where $A = 6.0 \text{ m}^2$ (= constant)

$$\therefore b' \times y' = 6$$

Also for maximum discharge, $b' = 2y'$

$$\therefore 2y' \times y' = 6 \quad \text{or} \quad y'^2 = 3 \quad \text{or} \quad y' = \sqrt{3} = 1.732 \text{ m,}$$

$$\text{and, } b' = 2y' = 2 \times 1.732 = 3.464 \text{ m}$$

Hence *new dimensions* of the channel are: $b' = 3.464 \text{ m}$ and $y' = 1.732 \text{ m}$ (Ans.)

$$\text{Wetted perimeter, } P' = b' + 2y' = 3.464 + 2 \times 1.732 = 6.928 \text{ m}$$

$$\therefore \text{Hydraulic radius, } R' = \frac{A}{P'} = \frac{6}{6.928} = 0.866 \text{ m}$$

$$[\text{Alternatively } R' = \frac{y'}{2} = \frac{1.732}{2} = 0.866 \text{ m (maximum discharge conditional)}]$$

$$\text{Maximum discharge, } Q' = AC\sqrt{R'S} = 6 \times 55 \times \sqrt{0.866 \times \frac{1}{1000}} = 9.71 \text{ m}^3/\text{s}$$

$$\therefore \text{Increase in discharge} = Q' - Q = 9.71 - 9.66 = 0.05 \text{ m}^3/\text{s} \text{ (Ans.)}$$

16.5.2 Most Economical Trapezoidal Channel Section

Fig 16.9 shows the cross-section of a trapezoidal channel.

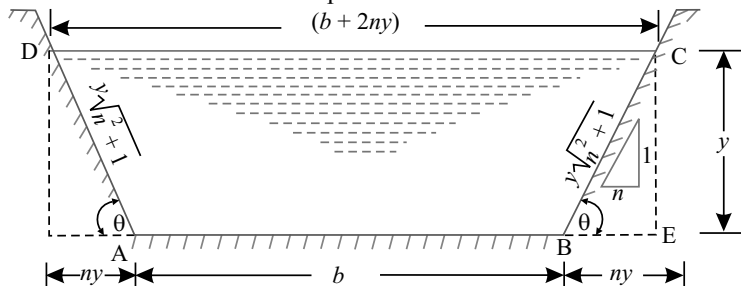


Fig. 16.9. Trapezoidal channel.

Let b = Base width of the channel,
 y = Depth of flow, and
 θ = Angle made by the sides with horizontal.

Side slope = 1 vertical to n horizontal.

$$\text{Area of flow, } A = \left(\frac{AB + CD}{2} \right) \times y = \frac{b + (b + 2ny)}{2} \times y = (b + ny) y \quad \dots(i)$$

$$\therefore \frac{A}{y} = b + ny$$

$$\text{or, } b = \frac{A}{y} - ny \quad \dots(ii)$$

$$\begin{aligned} \text{Wetted perimeter, } P &= AD + AB + BC = AB + 2BC \quad (\because AD = BC) \\ &= b + 2\sqrt{BE^2 + CE^2} \\ &= b + 2\sqrt{n^2 y^2 + y^2} \end{aligned}$$

$$\text{or, } P = b + 2y\sqrt{n^2 + 1} \quad \dots(iii)$$

Substituting the value of b from eqn. (ii) in eqn. (iii), we get:

$$P = \frac{A}{y} - ny + 2y\sqrt{n^2 + 1} \quad \dots(iv)$$

The section of the channel will be *most economical* when its wetted perimeter (P) is *minimum*,

$$\text{i.e. } \frac{dP}{dy} = 0$$

$$\text{or, } \frac{d}{dy} \left[\frac{A}{y} - ny + 2y\sqrt{n^2 + 1} \right] = 0$$

$$\text{or, } -\frac{A}{y^2} - n + 2\sqrt{n^2 + 1} = 0 \quad (\because n \text{ is constant})$$

$$\text{or, } \frac{A}{y^2} + n = 2\sqrt{n^2 + 1}$$

Substituting the value of A from eqn. (i), in the above equation, we get:

$$\frac{(b + ny) y}{y^2} + n = 2\sqrt{n^2 + 1}$$

$$\text{or, } \frac{(b + ny)}{y} + n = 2\sqrt{n^2 + 1}$$

$$\text{or, } \frac{b + ny + ny}{y} + 2\sqrt{n^2 + 1} \quad \text{or} \quad \frac{b + 2ny}{y} = 2\sqrt{n^2 + 1}$$

$$\text{or, } \frac{b + 2ny}{2} = y\sqrt{n^2 + 1} \quad \dots(16.14)$$

[i.e. Half of top width = One of the sloping sides ... Fig 16.9]

Hydraulic radius, R :

$$\text{Hydraulic radius, } R = \frac{A}{P}$$

$$A = (b + ny) \times y \quad [\text{From eqn. (i)}]$$

$$P = b + 2y\sqrt{n^2 + 1} \quad [\text{From eqn. (iii)}]$$

$$\text{But,} \quad 2y\sqrt{n^2 + 1} = b + 2ny \quad [\text{From eqn. (16.14)}]$$

$$\therefore \quad P = b + (b + 2ny) = 2(b + ny)$$

$$\therefore \quad \text{Hydraulic radius, } R = \frac{(b + ny) y}{2(b + ny)} = \frac{y}{2} \quad \dots(16.15)$$

i.e., The hydraulic radius equals half the flow depth.

Fig. 16.10 shows a trapezoidal channel of most economical section.

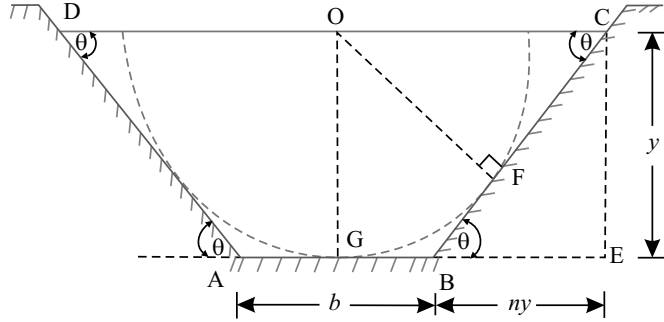


Fig. 16.10. Most economical section of a trapezoidal channel.

Let, θ = Angle made by the sloping side with the horizontal,

O = Centre of the top width DC , and

OF = A perpendicular to the sloping side BC .

The $\triangle OCF$ is then a right angled triangle with $\angle OCF = \theta$

$$\therefore \quad \sin \theta = \frac{OF}{OC} \quad \text{or} \quad OF = OC \sin \theta \quad \dots(v)$$

$$\text{Also, from } \triangle BCE, \quad \sin \theta = \frac{CE}{BC} = \frac{y}{\sqrt{y^2 + n^2 y^2}} = \frac{y}{y\sqrt{n^2 + 1}} = \frac{1}{\sqrt{n^2 + 1}}$$

Substituting the value of $\sin \theta$ in eqn. (iv), we have

$$OF = OC \times \frac{1}{\sqrt{n^2 + 1}} = y\sqrt{n^2 + 1} \times \frac{1}{\sqrt{n^2 + 1}} = y, \text{ depth of flow}$$

$$\left[\because OC = \text{Half of top width} = \frac{b + 2ny}{2} = y\sqrt{n^2 + 1} \dots \text{Eqn. (16.14)} \right]$$

Thus a circle with centre O and radius equal to the depth of flow will be *tangential* to the three sides of a most economical trapezoidal section; this condition stipulates that the most economical section of a trapezoidal channel will be a *half-hexagon*.

Hence *conditions for most economical trapezoidal section are*:

$$1. \quad \frac{b + 2ny}{2} = y\sqrt{n^2 + 1} \quad (\text{i.e. Half of top width} = \text{One of the sloping sides})$$

$$2. \quad \text{Hydraulic radius, } R = \frac{y}{2}$$

3. A semicircle drawn from O with radius equal to depth of flow will touch the three sides of the trapezoidal channel.

Best side slope for most economical trapezoidal section:

Side slope will be the best when the section is most-economical or when the wetted perimeter is minimum. For that $\frac{dP}{dn} = 0$

$$\therefore \frac{d}{dn} \left[\frac{A}{y} - ny + 2y\sqrt{n^2 + 1} \right] = 0$$

$$\left[\because P = \frac{A}{y} - ny + 2y\sqrt{n^2 + 1} \text{ ...From eqn. (iv)} \right]$$

$$\text{or,} \quad -y + 2y \times \frac{1}{2} (n^2 + 1)^{\frac{1}{2}-1} \times 2n = 0$$

$$-y + 2ny \times \frac{1}{\sqrt{n^2 + 1}} = 0$$

Cancelling y and rearranging, we have:

$$2n = \sqrt{n^2 + 1}$$

Squaring both sides, we have:

$$4n^2 = n^2 + 1 \quad \text{or} \quad 3n^2 = 1$$

$$\text{or,} \quad n = \frac{1}{\sqrt{3}} \quad \dots(16.16)$$

If the sloping side makes an angle θ with the horizontal, then

$$\tan \theta = \frac{1}{n} = \sqrt{3} = \tan 60^\circ$$

$$\therefore \theta = 60^\circ \quad \dots(16.17)$$

Hence *best side slope is at 60° to the horizontal.*

For the most economical section,

Half of top width = Length of the sloping side

$$\frac{b + 2ny}{2} = y\sqrt{n^2 + 1}$$

Substituting the value of side slope $n = \frac{1}{\sqrt{3}}$ in the above eqn. we get:

$$\frac{b + 2 \times \frac{1}{\sqrt{3}} y}{2} = y \sqrt{(1/\sqrt{3})^2 + 1} = \frac{2y}{\sqrt{3}}$$

$$\text{or,} \quad \frac{\sqrt{3}b + 2y}{2 \times \sqrt{3}} = \frac{2y}{\sqrt{3}}$$

$$\text{or,} \quad \sqrt{3}b + 2y = 4y \quad \text{or} \quad b = \frac{2y}{\sqrt{3}} \quad \dots(vi)$$

Now, wetted perimeter, $P = b + 2y\sqrt{n^2 + 1}$

$$= \frac{2y}{\sqrt{3}} + 2y \sqrt{\left(\frac{1}{\sqrt{3}}\right)^2 + 1} \quad \left(\because b = \frac{2y}{\sqrt{3}} \text{ and } n = \frac{1}{\sqrt{3}} \right)$$

$$= \frac{2y}{\sqrt{3}} + 2y \times \frac{2}{\sqrt{3}} = \frac{6y}{\sqrt{3}} = 3 \times \frac{2y}{\sqrt{3}} = 3b$$

$$\text{i.e.} \quad P = 3b \quad \left(\because b = \frac{2y}{\sqrt{3}} \right)$$

Thus for a side slope of 60° , the length of sloping side is equal to the base width of the trapezoidal section.

Example 16.8. A trapezoidal channel has side slopes of 3 horizontal to 4 vertical and the slope of its bed is 1 in 2000. Determine the optimum dimensions of the channel if it is to carry water at $0.5 \text{ m}^3/\text{s}$. Take Chezy's constant as 80. [RGPV, Bhopal]

Solution. Side slope, $n = \frac{\text{Horizontal}}{\text{Vertical}} = \frac{3}{4}$

Bed slope, $S = \frac{1}{2000}$

Discharge, $Q = 0.5 \text{ m}^3/\text{s}$

Chezy's constant, $C = 80$

Optimum dimensions of the channel:

For the most economical section, using eqn. (16.14), we have:

$$\frac{b + 2ny}{2} = y\sqrt{n^2 + 1}$$

[where, b = base width of the channel section, and y = depth of flow.]

$$\text{or,} \quad \frac{b \times 2 \times \frac{3}{4}y}{2} = y\sqrt{\left(\frac{3}{4}\right)^2 + 1} = \frac{5}{4}y$$

$$\text{or,} \quad \frac{b + 1.5y}{2} = 1.25y \quad \text{or} \quad b = 2 \times 1.25y - 1.5y = y$$

$$\text{i.e.} \quad b = y \quad \dots(i)$$

Also discharge, $Q = AC\sqrt{RS}$

[where, R = hydraulic radius, and S = bed slope]

$$0.5 = A \times 80 \sqrt{\frac{y}{2} \times \frac{1}{2000}} \quad \left(\because R = \frac{y}{2} \right)$$

But area,

$$\begin{aligned} A &= (b + ny) \times y \\ &= \left(y + \frac{3}{4}y \right) \times y = 1.75y^2 \end{aligned} \quad [\because b = y \dots \text{eqn. (i)}]$$

$$\therefore 0.5 = 1.75y^2 \times 80 \sqrt{\frac{y}{2} \times \frac{1}{2000}} = 2.2136y^{5/2}$$

$$\therefore y = \left(\frac{0.5}{2.2136} \right)^{2/5} = 0.55 \text{ m (Ans.)}$$

$$b = y = 0.55 \text{ m (Ans.)}$$

$\therefore$ Optimum dimensions of the channel are:

$$\text{Width (b) = Depth of flow (y) = 0.55 m (Ans.)}$$

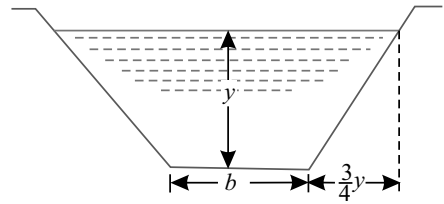


Fig. 16.11

Example 16.9. A trapezoidal channel having the side slope equal to 60° with the horizontal and laid on a slope of 1 in 750, carries a discharge of $10 \text{ m}^3/\text{s}$. Find the width at the base and depth of flow for most economical section. Take the value of Chezy's resistance co-efficient $C = 66$.

[AMIE]

Solution. Bed slope, $S = \frac{1}{750}$

Discharge, $Q = 10 \text{ m}^3/\text{s}$

Chezy's constant, $C = 66$

Side slope with the horizontal = 60°

Dimensions for most economical section, b and y :

For a trapezoidal channel of most economical (optimum) cross-section, the geometric parameters have the following proportions:

$$(i) \frac{b + 2ny}{2} = y\sqrt{n^2 + 1},$$

$$(ii) R = \frac{y}{2}$$

$$(iii) \tan \theta = \tan 60^\circ = \sqrt{3} = \frac{1}{n} \text{ or } n = \frac{1}{\sqrt{3}}$$

Thus from (i) and (iii), we have:

$$\frac{b + 2 \times \frac{1}{\sqrt{3}} y}{2} = y\sqrt{\left(\frac{1}{\sqrt{3}}\right)^2 + 1}; \frac{\sqrt{3}b + 2y}{2\sqrt{3}} = \frac{2y}{\sqrt{3}} \text{ or } \sqrt{3}b + 2y = 4y \text{ or } b = \frac{2}{\sqrt{3}} y$$

$$\text{Area of flow, } A = (b + ny)y = \left(b + \frac{1}{\sqrt{3}} y\right)y = \left(\frac{2}{\sqrt{3}} y + \frac{1}{\sqrt{3}} y\right)y = \sqrt{3}y^2$$

Also, discharge, $Q = AC\sqrt{RS}$

(where, R = hydraulic radius)

$$10 = \sqrt{3}y^2 \times 66 \sqrt{\frac{y}{2} \times \frac{1}{750}} = 2.95 y^{5/2}$$

$$\therefore \text{Depth of flow, } y = \left(\frac{10}{2.95}\right)^{2/5} = 1.63 \text{ m (Ans.)}$$

$$\text{Base width, } b = \frac{2}{\sqrt{3}} y = \frac{2}{\sqrt{3}} \times 1.63 = 1.88 \text{ m (Ans.)}$$

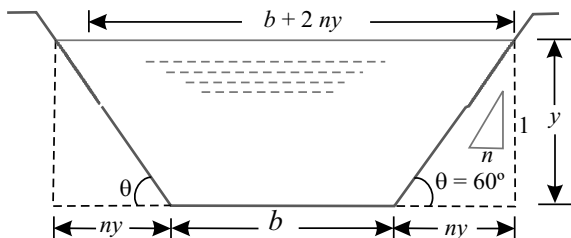


Fig. 16.12

Example 16.10. An open channel of most economical section, having the form of a half hexagon with horizontal bottom is required to give a maximum discharge of $20.2 \text{ m}^3/\text{s}$ of water. The slope of the channel bottom is 1 in 2500. Taking Chezy's constant, $C = 60$ in Chezy's equation, determine the dimensions of the cross-section.

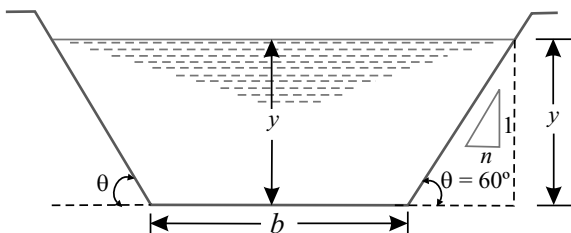


Fig. 16.13

Solution. Maximum Discharge,

$$Q = 20.2 \text{ m}^3/\text{s}$$

$$\text{Bed slope, } S = \frac{1}{2500}$$

$$\text{Chezy's constant } C = 60$$

Dimensions of the cross-section:

Since the channel has a form of a *half hexagon* (Fig. 16.13), therefore, the angle made by the sloping side with the horizontal is 60° .

$$\therefore \tan \theta = \tan 60^\circ = \sqrt{3} = \frac{1}{n} \quad \text{or} \quad n = \frac{1}{\sqrt{3}}$$

For most economical section the following conditions should be satisfied:

$$(i) \frac{b + 2ny}{2} = y\sqrt{n^2 + 1}, \quad (ii) R = \frac{y}{2}, \quad (iii) n = \frac{1}{\sqrt{3}}$$

$$\text{Now,} \quad \frac{b + 2ny}{2} = y\sqrt{n^2 + 1}$$

$$\text{or,} \quad \frac{b + 2 \times \frac{1}{\sqrt{3}} y}{2} = y \sqrt{\left(\frac{1}{\sqrt{3}}\right)^2 + 1} \quad \text{or} \quad \frac{\sqrt{3}b + 2y}{2\sqrt{3}} = \frac{2y}{\sqrt{3}} \quad \text{or} \quad b = \frac{2}{\sqrt{3}} y$$

$$\text{Area of flow, } A = (b + ny)y = \left(b + \frac{1}{\sqrt{3}} y\right)y = \left(\frac{2}{\sqrt{3}} y + \frac{1}{\sqrt{3}} y\right)y = \sqrt{3}y^2$$

Also, discharge, $Q = AC\sqrt{RS}$, where R is hydraulic radius;

$$\therefore 20.2 = \sqrt{3}y^2 \times 60 \sqrt{\frac{y}{2} \times \frac{1}{2500}} = 1.47 y^{5/2}$$

$$\text{or,} \quad y = \left(\frac{20.2}{1.47}\right)^{2/5} = \mathbf{2.852 \text{ (Ans.)}}$$

$$b = \frac{2}{\sqrt{3}} y = \frac{2}{\sqrt{3}} \times 2.852 = \mathbf{3.29 \text{ m (Ans.)}}$$

Example 16.11. A power canal of trapezoidal section has to be excavated through hard clay at least cost. Determine the dimensions of the channel, given, discharge equal to $14 \text{ m}^3/\text{s}$, bed slope 1:2500 and Manning's $N = 0.02$. [M.U.]

Solution. Discharge, $Q = 14 \text{ m}^3/\text{s}$

$$\text{Bed slope, } S = \frac{1}{2500}$$

$$\text{Manning's } N = 0.02.$$

The canal can be excavated at *least cost* if the trapezoidal section is the *most economical*. The value of side slope (which is not given in this case) is given by (for most economical section),

$$n = \frac{1}{\sqrt{3}} \quad \dots[\text{Eqn. (16.16)}]$$

For most economical section:

$$\frac{b + 2ny}{2} = y\sqrt{n^2 + 1} \quad \dots[\text{Eqn. (16.14)}]$$

(where b = base width, and y = depth of flow)

$$\therefore \frac{b + 2 \times \frac{1}{\sqrt{3}} y}{2} = y \sqrt{\left(\frac{1}{\sqrt{3}}\right)^2 + 1} = \frac{2}{\sqrt{3}} y \quad \left(\because n = \frac{1}{\sqrt{3}}\right)$$

$$\text{or,} \quad b = \frac{2}{\sqrt{3}} y \times 2 - \frac{2}{\sqrt{3}} y = \frac{2}{\sqrt{3}} y \quad \dots(i)$$

$$\text{Area of flow, } A = (b + ny) y = \left(\frac{2}{\sqrt{3}} y + \frac{1}{\sqrt{3}} y\right) y = \sqrt{3} y^2$$

Now, discharge, Q is given by:

$$Q = AC\sqrt{RS}, \quad \text{where } C = \frac{1}{N} R^{1/6}$$

$$\text{or,} \quad Q = \sqrt{3} y^2 \times \frac{1}{N} R^{1/6} \sqrt{RS} = \sqrt{3} y^2 \times \frac{1}{N} R^{2/3} S^{1/2}$$

$$\text{or,} \quad 14.0 = \sqrt{3} y^2 \times \frac{1}{0.02} \times \left(\frac{y}{2}\right)^{2/3} \times \sqrt{\frac{1}{2500}}$$

$$\text{or,} \quad 14.0 = 1.732 y^2 \times y^{2/3} \times \frac{1}{(2)^{2/3}} = 1.09 y^{8/3}$$

$$\text{or,} \quad y = \left(\frac{14}{1.09}\right)^{3/8} = 2.6 \text{ m (Ans.)}$$

$$b = \frac{2}{\sqrt{3}} y = \frac{2}{\sqrt{3}} \times 2.6 = 3.0 \text{ m (Ans.)}$$

Example 16.12. Design an earthen trapezoidal channel for water having a velocity of 0.6 m/s. Side slope of the channel is 1:1.5 and quantity of water flowing is 3 m³/s. Assume C in Chezy's formula as 65. [Delhi University]

Solution. Velocity of flow, $V = 0.6$ m/s

$$\text{Side slope, } n = \frac{\text{Horizontal}}{\text{Vertical}} = \frac{1.5}{1} = 1.5$$

$$\text{Discharge, } Q = 3 \text{ m}^3/\text{s}$$

$$\text{Chezy's constant, } C = 65$$

For design, the most economical trapezoidal section is used, for which the following condition may be used:

$$\text{Hydraulic radius, } R = \frac{y}{2}$$

$$\text{Area of flow, } A = \frac{Q}{V} = \frac{3}{0.6} = 5 \text{ m}^2$$

$$\text{Wetted perimeter, } P = b + 2y\sqrt{n^2 + 1}$$

$$\text{Also,} \quad R = \frac{A}{P} = \frac{5}{b + 2y\sqrt{n^2 + 1}} = \frac{y}{2} \quad \left(\because R = \frac{y}{2}\right)$$

$$\text{or,} \quad 10 = y \left[b + 2y\sqrt{1.5^2 + 1} \right] = y(b + 3.6y) \text{ or } by + 3.6y^2$$

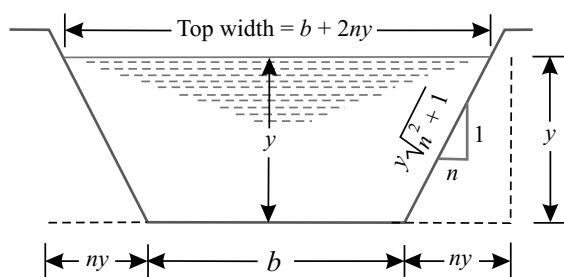


Fig. 16.14

$$\text{i.e.} \quad 10 = by + 3.6y^2 \quad \dots(i)$$

$$\text{Also,} \quad A = (b + ny)y$$

$$\text{or,} \quad S = (b + 1.5y)y \quad \text{or} \quad by + 1.5y^2$$

$$\text{or,} \quad by = 5 - 1.5y^2 \quad \dots(ii)$$

Substituting (ii) in (i), we get:

$$10 = 5 - 1.5y^2 + 3.6y^2 = 5 + 2.1y^2$$

$$\text{or,} \quad y = \left(\frac{5}{2.1}\right)^{1/2} = 1.543 \text{ m}$$

Substituting the value of y ($= 1.543 \text{ m}$) in eqn. (i), we get:

$$10 = b \times 1.543 + 3.6 \times (1.543)^2 = 1.543b + 8.571$$

$$\therefore \quad \text{Bottom width, } b = \frac{10 - 8.571}{1.543} = 0.926 \text{ m}$$

$$\text{Top width, } = b + 2ny = 0.926 + 2 \times 1.5 \times 1.543 = 5.555 \text{ m}$$

$$\text{Velocity, } V = C\sqrt{RS} \quad \dots \text{Chezy's formula}$$

$$0.6 = 65\sqrt{\frac{y}{2}S} \quad \text{or} \quad 0.6 = 65\sqrt{\frac{1.543}{2} \times S} = 57.09\sqrt{S}$$

$$\text{or,} \quad S = \left(\frac{0.6}{57.09}\right)^2 = 1.104 \times 10^{-4} \quad \text{or} \quad \frac{1}{9054}$$

Hence specification of the trapezoidal channel would be:

$$\text{Depth of flow } (y) = 1.543 \text{ m; Slope of the bed } (S) = \frac{1}{9054}$$

$$\text{Bottom width } (b) = 0.926 \text{ m; Top width} = 5.555 \text{ m (Ans.)}$$

Example 16.13. For a trapezoidal channel with bottom width 40 m and side slopes $2H : 1V$, Manning's N is 0.015 and bottom slope is 0.0002. If it carries $60 \text{ m}^3/\text{s}$ discharge, determine the normal depth. [MDU, Haryana]

Solution. Bottom width of the channel, $b = 40 \text{ m}$

Side slopes = $2H : 1V$ i.e. $n = 2$

Manning's constant, $N = 0.015$

Bottom/bed slope, $S = 0.0002$

Discharge, $Q = 60 \text{ m}^3/\text{s}$

Normal depth, y :

$$\text{Now, area } A = (b + ny)y = (40 + 2y) \times y$$

$$\text{and,} \quad \text{Perimeter, } P = b + 2y\sqrt{n^2 + 1} = 40 + 2y\sqrt{2^2 + 1} = 40 + 2\sqrt{5}y = 40 + 4.472y$$

$$\therefore \quad \text{Hydraulic radius, } R = \frac{A}{P} = \frac{(40 + 2y) \times y}{40 + 4.472y}$$

$$\text{Discharge, } Q = A \times V = A \times C \sqrt{RS} \quad \text{where, Chezy's constant, } C = \frac{1}{N} R^{1/6}$$

$$\therefore \quad Q = A \times \frac{1}{N} R^{1/6} \sqrt{RS} = A \times \frac{1}{N} \times R^{2/3} \times S^{1/2}$$

$$\text{or,} \quad 60 = (40 + 2y)y \times \frac{1}{0.015} \times \left[\frac{(40 + 2y) \times y}{40 + 4.472y}\right]^{2/3} \times (0.0002)^{1/2}$$

$$= \frac{[(40 + 2y) \times y]^{5/3}}{0.015 \times (40 + 4.472)^{2/3}} \times 0.01414$$

$$\therefore \frac{60 \times 0.015 \times (40 + 4.472y)^{2/3}}{0.01414} = [(40 + 2y) \times y]^{5/3}$$

$$\text{or, } 63.65 (40 + 4.472 y)^{2/3} = [(40 + 2y) \times y]^{5/3}$$

$$\text{or, } (40y + 2y^2)^{5/3} - 63.65 (40 + 4.472y)^{2/3} = 0$$

By hit and trial method, we get:

$$y = 1.31 \text{ m (Ans.)}$$

Example 16.14. A trapezoidal channel with side slopes of 1:1 has to be designed to convey 10 m³/s at a velocity of 2 m/s, so that the amount of concrete lining for the bed and sides is minimum.

(i) Calculate the area of lining required for one metre length of the canal.

(ii) If the rugosity co-efficient $N = 0.015$, calculate the bed slope of the canal for uniform flow.

[UPSC Exams.]

Solution. Side slope, $n = 1$

Discharge, $Q = 10 \text{ m}^3/\text{s}$

Velocity, $V = 2 \text{ m/s}$

Rugosity/Manning's co-efficient, $N = 0.015$

$$\text{Area of flow, } \frac{Q}{V} = \frac{10}{2} = 5 \text{ m}^2$$

(i) **Lining required for one metre length of the canal:**

For minimum amount of concrete lining, the wetted perimeter must be minimum; for this condition we have (for a trapezoidal channel):

$$(i) \frac{b + 2ny}{2} = y\sqrt{n^2 + 1} \quad \text{and} \quad (ii) \quad R = \frac{y}{2}$$

(where b = base width, y = depth of flow, and R = hydraulic radius)

$$\therefore \frac{b + 2 \times 1 \times y}{2} = y\sqrt{1+1} = \sqrt{2} y$$

$$\text{or, } b + 2y = 2\sqrt{2}y \quad \text{or} \quad b = 2y(\sqrt{2} - 1) = 0.828y$$

$$\text{Area of flow, } A = (b + ny)y$$

$$\text{or, } 5 = (0.828y + y)y = 1.828y^2$$

$$\text{or, } y = \left(\frac{5}{1.828}\right)^{1/2} = 1.65 \text{ m}$$

$$b = 0.828y = 0.828 \times 1.65 = 1.37 \text{ m}$$

Area of lining per metre length

$$\begin{aligned} &= P \times 1 = (b + 2y\sqrt{n^2 + 1}) \times 1 \quad (\because P = b + 2y\sqrt{n^2 + 1}) \\ &= 1.37 + 2 \times 1.65 \sqrt{1+1} = 6.04 \text{ m (Ans.)} \end{aligned}$$

(ii) **Bed slope of the canal, S :**

$$Q = AC\sqrt{RS}, \quad \text{where} \quad C = \frac{1}{N} R^{1/6}$$

$$\text{or, } Q = A \times \frac{1}{N} R^{1/6} \sqrt{RS} = A \times \frac{1}{N} R^{2/3} S^{1/2}$$

$$\text{or, } 10 = 5 \times \frac{1}{0.015} \times \left(\frac{1.65}{2} \right)^{2/3} S^{1/2} = 293.2 S^{1/2}$$

$$\therefore S = \left(\frac{10}{293.2} \right)^2 = 1.163 \times 10^{-3}$$

$$\text{or, } \quad \quad \quad \mathbf{1 \text{ in } 860 \text{ (Ans.)}}$$

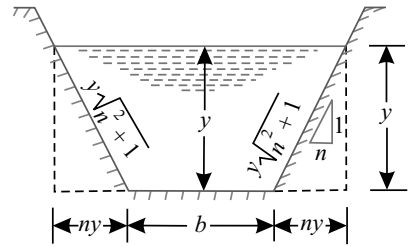


Fig. 16.15

Example 16.15. A trapezoidal channel is required to carry $8 \text{ m}^3/\text{s}$ of water at a velocity of 2 m/s . Find the most economical cross-section if the channel has side slopes 1 horizontal to 2 vertical. For the same discharge what saving in power would result if this trapezoidal section is replaced by a rectangular section 1.5 m deep and 4 m wide. Take Chezy's constant $C = 55$.

Solution. For trapezoidal channel:

$$\text{Discharge, } Q = 8 \text{ m}^3/\text{s}$$

$$\text{Velocity of flow, } V = 2 \text{ m/s}$$

$$\therefore \text{Area of flow, } A = Q/V = 8/2 = 4 \text{ m}^2$$

$$\text{Side slope, } n = \frac{\text{Horizontal}}{\text{Vertical}} = \frac{1}{2}$$

$$\text{Chezy's constant } C = 55.$$

The trapezoidal channel section will be *most economical*, when:

$$\text{Half of top width} = \text{Length of one sloping side}$$

$$\text{or, } \frac{b + 2ny}{2} = y\sqrt{n^2 + 1}$$

(where b = base width, y = depth of flow)

$$\text{or, } \frac{b + 2 \times \frac{1}{2}y}{2} = y\sqrt{\left(\frac{1}{2}\right)^2 + 1} = \frac{\sqrt{5}}{2}y$$

$$\text{or, } b + y = \sqrt{5}y \quad \text{or} \quad b = y(\sqrt{5} - 1) = 1.236y \quad \dots(i)$$

$$\text{Area of flow, } A = (b + ny)y = \left(b + \frac{1}{2}y\right)y$$

$$\text{or, } 4 = (1.236y + 0.5y)y = 1.732y^2$$

$$\text{or, } y = \left(\frac{4}{1.736}\right)^{1/2} = \mathbf{1.52 \text{ m (Ans.)}}$$

$$b = 1.236y = 1.236 \times 1.52 = \mathbf{1.88 \text{ m (Ans.)}}$$

$$\text{Also hydraulic radius, } R \text{ (for most economical section)} = y/2 = \frac{1.52}{2} = 0.76 \text{ m}$$

Now, velocity, $V = C\sqrt{RS}$, where S is the bed slope

$$\therefore 2 = 55\sqrt{0.76 \times S} \quad \text{or} \quad S = \left(\frac{2}{55}\right)^2 \times \frac{1}{0.76} = 1.739 \times 10^{-3} \approx \mathbf{1 \text{ in } 575}$$

For rectangular channel:

$$\text{Base width, } b = 4 \text{ m}$$

$$\text{Depth of flow, } y = 1.5 \text{ m}$$

$$\therefore \text{Area of flow, } A = b \times y = 4 \times 1.5 = 6.0 \text{ m}^2$$

$$\text{Wetted perimeter, } P = b + 2y = 4 + 2 \times 1.5 = 7 \text{ m}$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{6}{7} = 0.857 \text{ m}$$

$$\text{Flow velocity, } V = \frac{Q}{A} = \frac{8}{6} = 1.333 \text{ m/s}$$

$$\text{From Chezy's formula, } V = C\sqrt{RS}$$

$$\text{or, } 1.333 = 55\sqrt{0.857 \times S}$$

$$\text{or, } S = \left(\frac{1.333}{55}\right)^2 \times \frac{1}{0.857} = 6.854 \times 10^{-4} \approx 1 \text{ in } 1459$$

$\therefore$ Saving in head *per kilometre of channel run*

$$h = (1.739 \times 10^{-3} - 6.854 \times 10^{-4}) \times 1000 = 1.054 \text{ m}$$

$$\text{Hence, saving in power} = \frac{wQh}{1000} \text{ kW} = \frac{9810 \times 8 \times 1.054}{1000} = \mathbf{82.7 \text{ kW (Ans.)}}$$

Example 16.16. A hydraulically efficient trapezoidal channel has side slopes of 1:1. It is required to discharge $14 \text{ m}^3/\text{s}$ with a gradient (channel slope) of 1 in 1000. If unlined, the value of Chezy's C is 45. If lined with concrete, the value is 70. If the least cost per m^3 of excavation is three times the cost m^2 of lining, will the lined or the unlined channel be cheaper? [Anna University]

Solution. Side slope, $n = \frac{1}{1} = 1$

$$\text{Discharge, } Q = 14 \text{ m}^3/\text{s}$$

$$\text{Bed slope, } S = \frac{1}{1000}$$

Chezy's constant C : Unlined channel = 45,

Channel lined with concrete = 70.

Hydraulically efficient trapezoidal channel must satisfy the following conditions:

$$(i) \frac{b + 2ny}{2} = y\sqrt{n^2 + 1}, \quad \text{and} \quad (ii) \quad R = \frac{y}{2}$$

where, b = Base width of channel,

y = Depth of flow, and

R = Hydraulic radius

$$\text{or, } \frac{b + 2y}{2} = y\sqrt{1 + 1} = \sqrt{2}y \quad \text{or} \quad b + 2y = 2\sqrt{2}y$$

$$\text{or, } b = 2y(\sqrt{2} - 1) = 0.828y$$

Unlined channel:

$$Q = AC\sqrt{RS} = (b + ny)y \times C\sqrt{\frac{y}{2}S}$$

$$14 = (0.828y + y)y \times 45 \times \sqrt{\frac{y}{2} \times \frac{1}{1000}} = 1.839y^{5/2} \quad (\because n = 1)$$

$$\text{or, } y = \left(\frac{14}{1.839}\right)^{2/5} = 2.25 \text{ m}$$

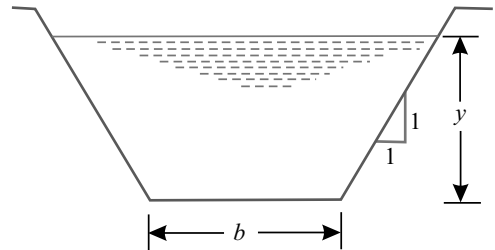


Fig. 16.16

$$b = 0.828y = 0.828 \times 2.25 = 1.863 \text{ m}$$

Let, cost of lining per m^2 of the channel surface = K

Then, cost of excavation per $m^3 = 3K$

...(Given)

Consider one metre length of channel.

$$\text{Amount of excavation} = A \times 1 = (b + ny)y \times 1 = (1.863 + 1 \times 2.25) \times 2.25 \times 1 = 9.254 \text{ m}^3$$

$$\therefore \text{The cost of excavation} = 3K \times 9.254 = 27.76K$$

Lined channel:

$$Q = AC\sqrt{RS} = (b + ny)y \times C\sqrt{\frac{y}{2}S}$$

$$\text{or, } 14 = (0.828y + y)y \times 70 \times \sqrt{\frac{y}{2} \times \frac{1}{1000}} = 2.86y^{5/2}$$

$$\text{or, } y = \left(\frac{14}{2.86}\right)^{2/5} = 1.88 \text{ m}$$

$$b = 0.828y = 0.828 \times 1.88 = 1.55 \text{ m}$$

Considering one metre length of channel, cost of excavation

$$= A \times 1 \times 3K = (b + ny)y \times 3K = (1.55 + 1 \times 1.88) \times 1.88 \times 3K \\ = 19.34K$$

$$\text{Cost of lining} = \text{Perimeter } (P) \times 1 \times K = (b + 2y\sqrt{n^2 + 1}) \times 1 \times K$$

$$= (1.55 + 2 \times 1.88\sqrt{1 + 1})K = 6.86K$$

$$\therefore \text{Total cost of lined channel} = 19.34K + 6.86K = 26.2K$$

Since the cost of excavation of unlined channel (27.76K) is greater than the total cost of lined channel, hence the **lined channel is cheaper. (Ans.)**

Example 16.17. Design a concrete lined channel to carry a discharge of $500 \text{ m}^3/\text{s}$ at a slope of 1 in 4000. The side slopes of channel may be taken as 1:1. The Manning's roughness co-efficient for the lining is 0.014. Assume the permissible velocity in the section as 2.5 m/s . [UPSC Exams.]

Solution.

$$\text{Discharge, } Q = 500 \text{ m}^3/\text{s}$$

$$\text{Bed slope, } S = \frac{1}{4000}$$

$$\text{Side slope, } n = \frac{1}{1} = 1$$

Manning's roughness co-efficient, $N = 0.0014$

Permissible velocity, $V = 2.5 \text{ m/s}$

Base width (b) and depth of flow (y):

$$\text{Area of flow, } A = \frac{Q}{V} = \frac{500}{2.5} = 200 \text{ m}^2$$

$$\text{Also, } A = (b + ny)y = (b + y)y \quad (\because n = 1)$$

$$\text{or, } A = (b + y)y = 200 \quad \dots(i)$$

$$\text{Perimeter, } P = b + 2y\sqrt{n^2 + 1} = b + 2y\sqrt{1 + 1} = b + 2\sqrt{2}y$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{200}{b + 2\sqrt{2}y}$$

$$= b + y \{\sqrt{n^2 + 1} + 1\}$$

$$\text{or,} \quad P = b + y (\sqrt{5} + 1) \quad (\because n = 2)$$

$$\begin{aligned} \text{Hydraulic radius, } R &= \frac{A}{P} = \frac{75}{b + y (\sqrt{5} + 1)} \\ &= \frac{75}{\left(\frac{75}{y} - y\right) + y\sqrt{5} + y} \\ &= \frac{75}{\frac{75}{y} + y\sqrt{5}} \end{aligned}$$

Now, discharge (Q) is given by:

$$Q = AC\sqrt{RS} \quad \text{where, } C = \frac{1}{N} (R)^{\frac{1}{6}}$$

$$\text{or,} \quad Q = A \times \frac{1}{N} (R)^{1/6} \sqrt{RS} = A \times \frac{1}{N} (R)^{\left(\frac{1}{6} + \frac{1}{2}\right)} \sqrt{S} = A \times \frac{1}{N} (R)^{2/3} \sqrt{S}$$

$$\text{or,} \quad \sqrt{S} = \frac{QN}{A(R)^{2/3}} = \frac{45 \times 0.013}{75(R)^{2/3}} \quad \dots(i)$$

For S to be minimum, R has to be maximum.

$$\text{or,} \quad \frac{75}{\frac{75}{y} + y\sqrt{5}} \text{ is to be maximum}$$

$$\text{or,} \quad \frac{75}{y} + y\sqrt{5} \text{ is to be minimum}$$

$$\begin{aligned} \text{or,} \quad \frac{d}{dy} \left[\frac{75}{y} + y\sqrt{5} \right] &= 0 \\ -\frac{75}{y^2} + \sqrt{5} &= 0 \quad \text{or} \quad y^2 = \frac{75}{\sqrt{5}} \end{aligned}$$

$$\text{or,} \quad y = 5.79 \text{ m}$$

Hence, for minimum slope,

$$R = \frac{75}{\frac{75}{y} + y\sqrt{5}} = \frac{75}{\frac{75}{5.79} + 5.79\sqrt{5}} = 2.896 \text{ m}$$

The minimum hydraulic slope is obtained by substituting their value of R in (i).

$$\therefore \sqrt{S_{\min}} = \frac{45 \times 0.013}{75 \times (2.896)^{2/3}} = 3.839 \times 10^{-3}$$

$$\text{or,} \quad S_{\min} = (3.839 \times 10^{-3})^2 = \mathbf{1.474 \times 10^{-5} \text{ (Ans.)}}$$

Example 16.19. A trapezoidal channel having a cross-sectional area A_1 , wetted perimeter P_1 , Manning's co-efficient N , and laid to a slope S carries a discharge Q , at a depth of flow equal to y . To increase the discharge, the base width of the channel is widened by x , keeping all other parameters same. Prove that

$$\left(\frac{Q_2}{Q_1}\right)^3 \times \left(1 + \frac{x}{P_1}\right)^2 = \left(1 + \frac{xy}{A_1}\right)^5$$

where, Q_2 is the new discharge in the channel.

[UPSC Exams.]

Solution. The Chezy's constant (C), using Manning's formula, is given by:

$$C = \frac{1}{N} (R)^{1/6}$$

$$\therefore \text{Velocity of flow, } V = C \sqrt{RS} = \frac{1}{N} R^{1/6} \times (RS)^{1/2} = \frac{R^{2/3} S^{1/2}}{N}$$

where, R = Hydraulic radius (or hydraulic mean depth), and

S = Slope of the channel bed.

$$\begin{aligned} \text{Discharge, } Q &= AV = A \times \frac{R^{2/3} S^{1/2}}{N} \\ &= KAR^{2/3} = KA \left(\frac{A}{P}\right)^{2/3} = K \frac{A^{5/3}}{P^{2/3}} \quad \left(\because R = \frac{A}{P}\right) \quad \dots(i) \end{aligned}$$

$$\text{where, } K \text{ (a constant)} = \frac{S^{1/2}}{N} \quad (S \text{ and } N \text{ kept constant})$$

Area of cross-section of the widened canal $A_2 = (A_1 + xy)$

Wetted perimeter of the original channel, $P_2 = (P_1 + x)$

Where A_1 and P_1 are the area of cross-section and wetted perimeter respectively of the original channel.

Then from expression (i), we have:

$$Q_1 = K \frac{A_1^{5/3}}{P_1^{2/3}} \quad \text{and} \quad Q_2 = K \frac{A_2^{5/3}}{P_2^{2/3}}$$

$$\therefore \frac{Q_2}{Q_1} = \left(\frac{A_2}{A_1}\right)^{5/3} \times \left(\frac{P_1}{P_2}\right)^{2/3}$$

Substituting the values of A_2 and P_2 in the above equation, we have:

$$\begin{aligned} \frac{Q_2}{Q_1} &= \left(\frac{A_1 + xy}{A_1}\right)^{5/3} \times \left(\frac{P_1}{P_1 + x}\right)^{2/3} \\ &= \left(1 + \frac{xy}{A_1}\right)^{5/3} \times \left(\frac{1}{1 + \frac{x}{P_1}}\right)^{2/3} \end{aligned}$$

Taking cube on both sides, we get:

$$\left(\frac{Q_2}{Q_1}\right)^3 = \left(1 + \frac{xy}{A_1}\right)^5 \times \left(\frac{1}{1 + \frac{x}{P_1}}\right)^2$$

$$\text{or, } \left(\frac{Q_2}{Q_1}\right)^3 \times \left(1 + \frac{x}{P_1}\right)^2 = \left(1 + \frac{xy}{A_1}\right)^5$$

...Proved

16.5.3 Most Economical Triangular Channel Section

Fig. 16.18 shows a triangular channel. The side slopes are n (horizontal) to 1 (vertical)

Let, y = Depth of flow, and

θ = Angle made by the sides with the vertical.

From $\triangle ODC$, $\frac{CD}{DO} = \tan \theta$ or $\frac{CD}{y} = \tan \theta$

or, $CD = y \tan \theta$

Also, $\frac{DO}{CO} = \cos \theta$ or $\frac{y}{CO} = \cos \theta$

or, $CO = y \sec \theta$

Area of flow,

$$A = \frac{1}{2} \times BC \times DO = \frac{1}{2} \times 2CD \times DO$$

$$= \frac{1}{2} \times 2y \tan \theta \times y = y^2 \tan \theta$$

i.e. $A = y^2 \tan \theta$... (i)

Perimeter, $P = BO + OC = 2OC = 2y \sec \theta$ ($\because BO = OC$) ... (ii)

Substituting the value of y ($= \sqrt{\frac{A}{\tan \theta}}$) from eqn. (i) in eqn. (ii), we get:

$$P = 2 \sqrt{\frac{A}{\tan \theta}} \sec \theta = 2 \frac{\sqrt{A}}{\sqrt{\tan \theta}} (\sec \theta) \quad \dots (iii)$$

Assuming the area to be constant, eqn. (iii) can be differentiated with respect to θ and equated to zero for obtaining the condition for minimum P .

i.e. $\frac{dP}{d\theta} = \frac{d}{d\theta} \left[2 \frac{\sqrt{A}}{\sqrt{\tan \theta}} (\sec \theta) \right] = 0$

$$= 2\sqrt{A} \left[\frac{\sqrt{\tan \theta} \times \sec \theta \cdot \tan \theta - \sec \theta \times \frac{1}{2} (\tan \theta)^{-1/2} \sec^2 \theta}{\tan \theta} \right] = 0$$

$$= 2\sqrt{A} \left[\frac{\sec \theta \tan \theta}{\sqrt{\tan \theta}} - \frac{\sec^3 \theta}{2 (\tan \theta)^{3/2}} \right] = 0$$

or, $\sec \theta (2 \tan^2 \theta - \sec^2 \theta) = 0$

Since $\sec \theta \neq 0$,

$\therefore 2 \tan^2 \theta - \sec^2 \theta = 0$ or $2 \tan^2 \theta = \sec^2 \theta$

or, $\sqrt{2} \tan \theta = \sec \theta$

$\therefore \theta = 45^\circ$; and $n = 1$... (16.18)

Hence, a triangular channel section will be most economical when each of its sloping sides makes an angle of 45° with the vertical.

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{y^2 \tan \theta}{2y \sec \theta}$$

Substituting the value of θ from eqn. (16.18) in the above eqn., we get:

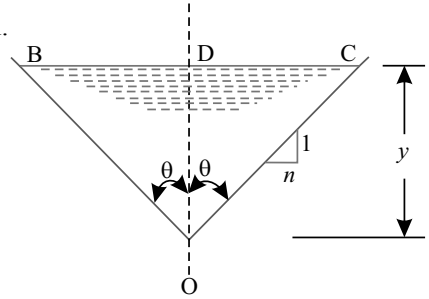


Fig. 16.18. Triangular channel.

$$R = \frac{y^2 \tan 45^\circ}{2y \sec 45^\circ} = \frac{y^2}{2y \times \sqrt{2}} = \frac{y}{2\sqrt{2}} \quad \dots(16.19)$$

Thus it can be seen that the *most economical triangular channel section will be a half square described on a diagonal and having equal sloping sides.*

Example 16.20. Water flows in a channel of the shape of isosceles triangle of bed width 'a' and sides making an angle of 45° with the bed. Determine the relation between depth of flow 'd' and the bed width 'a' for maximum velocity condition and for maximum discharge condition. Use Manning's formula and note that 'd' is less than 0.5 a. [UPSC, CES, Exams.]

Solution. Bed width of the channel = a

Angle of sides with the bed = 45°

Depth of flow = d ($d < 0.5a$)

Velocity, $V = C\sqrt{RS}$

But $C = \frac{1}{N} R^{1/6}$, where N is Manning's constant

$$\therefore V = \frac{1}{N} R^{1/6} \sqrt{RS} = \frac{1}{N} R^{2/3} S^{1/2}$$

where, R = hydraulic radius, S = bed slope.

$$\text{Area of flow, } A = \left(\frac{GH + BC}{2} \right) \times d$$

$$\text{But } GH = BC - 2JC = a - 2d$$

$$\left[\begin{array}{l} \because \frac{HJ}{JC} = \tan 45^\circ = 1 \\ \text{or } \frac{d}{JC} = 1 \text{ or } JC = d \end{array} \right]$$

$$\therefore A = \left[\frac{(a - 2d) + a}{2} \right] \times d = (a - d) d$$

$$\text{Wetted perimeter, } P = BC + BG + HC = BC + 2BG$$

$$(\because BG = HC)$$

$$\text{But } BG = \sqrt{d^2 + d^2} = d\sqrt{2}$$

$$\therefore P = a + 2\sqrt{2} d$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{(a - d) d}{a + 2\sqrt{2} d}$$

$$\therefore V = \frac{1}{N} \left[\frac{(a - d) d}{a + 2\sqrt{2} d} \right]^{2/3} S^{1/2}$$

For maximum velocity, $\frac{dV}{dd} = 0$

$$\frac{d}{dd} \left[\frac{S^{1/2}}{N} \left\{ \frac{(a - d) d}{a + 2\sqrt{2} d} \right\}^{2/3} \right] = 0$$

$$\frac{S^{1/2}}{N} \left[\frac{2}{3} \left\{ \frac{(a - d) d}{a + 2\sqrt{2} d} \right\}^{-1/3} \left\{ \frac{(a + 2\sqrt{2} d)(a - 2d) - (a - d) d \times 2\sqrt{2}}{(a + 2\sqrt{2} d)^2} \right\} \right] = 0$$

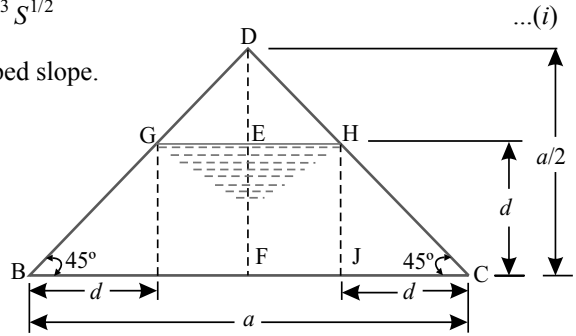


Fig. 16.19

On simplification, we have:

$$a^2 - 2ad - 2\sqrt{2}d^2 = 0$$

$$\text{or, } a = \frac{2d \pm \sqrt{4d^2 + 8\sqrt{2}d^2}}{2} = \frac{2d \pm 3.91d}{2} = 2.955d \quad (\text{neglecting } -ve \text{ value})$$

$$\text{or, } \frac{d}{a} = \frac{1}{2.955} = \mathbf{0.338 \text{ (Ans.)}}$$

For maximum discharge, $\frac{dQ}{dd} = 0$

$$Q = A.V = (a-d)d \times \frac{1}{N} \left\{ \frac{(a-d)d}{a+2\sqrt{2}d} \right\}^{2/3} S^{1/2}$$

$$\therefore \frac{d}{dd} \left[(a-d)d \times \frac{1}{N} \left\{ \frac{(a-d)d}{a+2\sqrt{2}d} \right\}^{2/3} S^{1/2} \right] = 0$$

$$\text{or, } \frac{d}{dd} \left[\frac{S^{1/2}}{N} \times \{(a-d)d\}^{5/3} \times \frac{1}{(a+2\sqrt{2}d)^{2/3}} \right] = 0$$

$$\text{or, } \frac{S^{1/2}}{N} \left[\frac{(a+2\sqrt{2}d)^{2/3} \times 5/3 \{(a-d)d\}^{2/3} \times (a-2d) - \{(a-d)d\}^{5/3}}{(a+2\sqrt{2}d)^{4/3}} \times \left\{ \frac{2}{3} (a+2\sqrt{2}d)^{-1/3} \times 2\sqrt{2} \right\} \right] = 0$$

On simplification, we get:

$$5a^2 - 1.5147ad - 22.6274d^2 = 0$$

$$\begin{aligned} \text{or, } a &= \frac{1.5147d \pm d \sqrt{(1.5147)^2 + 4 \times 5 \times 22.6274}}{10} \\ &= \frac{1.5147d \pm 21.327d}{10} = 2.284d \quad (\text{neglecting } -ve \text{ value}) \end{aligned}$$

$$\text{or, } \frac{d}{a} = \frac{1}{2.284} = \mathbf{0.4378 \text{ (Ans.)}}$$

16.5.4. Most Economical Circular Channel Section

Circular pipes and culverts which are partly filled are treated as channels. In case of conduits the condition of area remaining constant does not hold good since both the wetted perimeter and wetted area vary with depth of flow. The most economical section (optimum section) is designed both for conditions of maximum mean velocity and maximum flow rate.

$$\text{Velocity of flow, } V = C\sqrt{RS} = C\sqrt{\frac{A}{P}S} \quad \dots \text{Chezy's formula}$$

$$\text{Discharge, } Q = AV = AC\sqrt{RS} = C\sqrt{\left(\frac{A^3}{P}\right)S}$$

Thus the *flow velocity* will have a maximum value when hydraulic radius $\left(\frac{A}{P}\right)$ is *maximum*, and a maximum discharge is obtained when $\left(\frac{A^3}{P}\right)$ is *maximum*.

Fig. 16.20 shows a circular channel through which water is flowing.

Let, y = Depth of flow,

r = Radius of the channel, and

2θ = Angle subtended by water surface AB at the centre in radians.

Wetted perimeter, P = Length of arc AD

$$= \frac{2\pi r}{2\pi} \times 2\theta = 2r\theta$$

i.e. $P = 2r\theta$

... (16.20)

Wetted area, A = Area $A DBA$

= Area of sector $O A D B O$ – area of $\Delta O A B$

$$= \frac{\pi r^2}{2\pi} \times 2\theta - \frac{1}{2} AB \times CO$$

$$= r^2\theta - \frac{1}{2} \times 2BC \times CO \quad (\because AB = 2BC)$$

$$= r^2\theta - \frac{1}{2} \times 2 \times r \sin \theta \times r \cos \theta$$

$$(\because BC = r \sin \theta, CO = r \cos \theta)$$

$$= r^2\theta - \frac{1}{2} r^2 \times 2 \sin \theta \cos \theta$$

or, $A = r^2 \left(\theta - \frac{\sin 2\theta}{2} \right)$... (16.21)

$$(\because 2 \sin \theta \cos \theta = \sin 2\theta)$$

(i) **Condition for maximum velocity:**

The velocity will be maximum when:

$$\frac{d}{d\theta} \left(\frac{A}{P} \right) = 0$$

(where, A and P both are functions of θ).

$$\text{or, } \frac{P \frac{dA}{d\theta} - A \cdot \frac{dP}{d\theta}}{P^2} = 0 \quad \text{or} \quad P \frac{dA}{d\theta} - A \cdot \frac{dP}{d\theta} = 0 \quad \dots (i)$$

Now, $A = r^2 \left(\theta - \frac{\sin 2\theta}{2} \right)$ [Eqn. (16.21)]

$$\frac{dA}{d\theta} = r^2 \left(1 - \frac{\cos 2\theta}{2} \times 2 \right) = r^2 (1 - \cos 2\theta)$$

Again, $P = 2r\theta$ [Eqn. (16.20)]

$$\frac{dP}{d\theta} = 2r$$

Substituting the values of A , P , $\frac{dA}{d\theta}$ and $\frac{dP}{d\theta}$ in eqn. (i), we get:

$$2r\theta [r^2 (1 - \cos 2\theta)] - r^2 \left(\theta - \frac{\sin 2\theta}{2} \right) \times 2r = 0$$

$$\text{or, } 2r^3\theta (1 - \cos 2\theta) - 2r^3 \left(\theta - \frac{\sin 2\theta}{2} \right) = 0$$

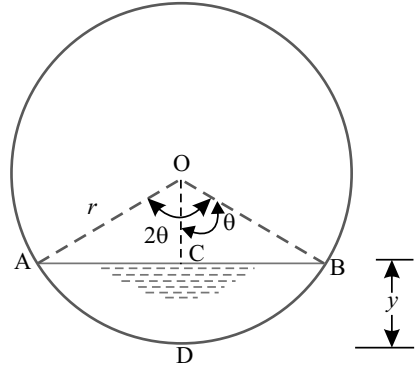


Fig. 16.20. Circular channel.

$$\text{or,} \quad \theta (1 - \cos 2\theta) = \left(\theta - \frac{\sin 2\theta}{2} \right) = 0 \quad (\text{cancelling } 2r^3)$$

$$\text{or, } \theta - \theta \cdot \cos 2\theta - \theta + \frac{\sin 2\theta}{2} = 0$$

$$\text{or,} \quad \theta \cdot \cos 2\theta = \frac{\sin 2\theta}{2}$$

$$\therefore \quad \tan 2\theta = 2\theta$$

Solution gives : $2\theta = 257.5^\circ$ (approximately) ... by hit and trial method, or $\theta = 128.75^\circ$

$$\text{Depth of flow, } y = OD - OC = r - r \cos \theta \quad (\text{Fig. 16.16})$$

$$= r(1 - \cos \theta) = r(1 - \cos 128.75^\circ) \approx 1.62r \approx 0.81d$$

$$\text{i.e.} \quad y \approx 0.81d \quad \dots(16.22)$$

where, d = Diameter of the circular channel.

Thus, *maximum velocity occurs when the depth of flow is 0.81 times the diameter of the circular channel.*

Hydraulic radius (or hydraulic mean depth) for maximum velocity,

$$R = \frac{A}{P} = \frac{r^2 \left(\theta - \frac{\sin 2\theta}{2} \right)}{2r\theta} = \frac{r}{2\theta} \left(\theta - \frac{\sin 2\theta}{2} \right) \quad \dots(16.23)$$

$$\text{where,} \quad \theta = 128.75^\circ = 128.75 \times \frac{\pi}{180} = 2.247 \text{ radians}$$

$$\therefore \quad R = \frac{r}{2 \times 2.247} \left(2.247 - \frac{\sin 257.5^\circ}{2} \right)$$

$$\text{or,} \quad R \approx 0.6086r \approx 0.305d \quad \dots(16.24)$$

Thus, *for maximum mean velocity in a channel of circular section hydraulic radius equals 0.305 times the channel diameter.*

(ii) **Condition for maximum discharge:**

The discharge will be maximum when:

$$\frac{d}{d\theta} \left(\frac{A^3}{P} \right) = 0 \quad \text{or} \quad \frac{P \times 3A^2 \frac{dA}{d\theta} - A^3 \frac{dP}{d\theta}}{P^2} = 0$$

$$\text{or,} \quad 3PA^2 \frac{dA}{d\theta} - A^3 \frac{dP}{d\theta} = 0$$

Dividing, both sides by A^2 , we get:

$$\text{or,} \quad 3P \frac{dA}{d\theta} - A \frac{dP}{d\theta} = 0 \quad \dots(i)$$

$$P = 2r\theta \quad (\text{Eqn. 16.20})$$

$$\therefore \quad \frac{dP}{d\theta} = 2r$$

$$A = r^2 \left(\theta - \frac{\sin 2\theta}{2} \right) \quad (\text{Eqn. 16.21})$$

$$\therefore \quad \frac{dA}{d\theta} = r^2 (1 - \cos 2\theta)$$

Substituting the values of P , A , $\frac{dP}{d\theta}$ and $\frac{dA}{d\theta}$ in eqn. (i), we have:

$$3 \times 2r\theta \times r^2 (1 - \cos 2\theta) - r^2 \left(\theta - \frac{\sin 2\theta}{2} \right) \times 2r = 0$$

$$\text{or,} \quad 6r^3\theta (1 - \cos 2\theta) - 2r^3 \left(\theta - \frac{\sin 2\theta}{2} \right) = 0$$

Dividing by $2r^3$, we get:

$$\text{or,} \quad 3\theta (1 - \cos 2\theta) - \left(\theta - \frac{\sin 2\theta}{2} \right) = 0$$

$$\text{or,} \quad 3\theta - 3\theta \cos 2\theta - \theta + \frac{\sin 2\theta}{2} = 0$$

$$\text{or,} \quad 2\theta - 3\theta \cos 2\theta + \frac{\sin 2\theta}{2} = 0$$

$$\text{or,} \quad 4\theta - 6\theta \cos 2\theta + \sin 2\theta = 0$$

Solution gives: $2\theta = 308^\circ$... by hit and trial method

$$\theta = \frac{308}{2} = 154^\circ$$

Depth of flow for maximum discharge,

$$y = r(1 - \cos \theta) \quad [\text{Fig. 16.16}]$$

$$= r(1 - \cos 154^\circ) \approx 1.9r \approx 0.95d$$

$$\text{i.e.} \quad y \approx 0.95d \quad (16.25)$$

where, d is the diameter of the circular channel.

Thus for maximum discharge through a circular channel, the depth of flow is equal to 0.95 times its diameter.

Hydraulic radius for maximum discharge,

$$R = \frac{A}{P} = \frac{r^2 \left(\theta - \frac{\sin 2\theta}{2} \right)}{2r\theta} = \frac{r}{2\theta} \left(\theta - \frac{\sin 2\theta}{2} \right)$$

$$\text{where,} \quad \theta = 154^\circ = 154 \times \frac{\pi}{180} = 2.687 \text{ radians}$$

$$\therefore R = \frac{r}{2 \times 2.687} \left(2.687 - \frac{\sin 308^\circ}{2} \right)$$

$$\text{or,} \quad R \approx 0.573r \approx 0.29d \quad \dots(16.26)$$

Thus for maximum discharge through a circular channel, the hydraulic radius equals 0.29 times channel diameter.

Example 16.21. A concrete lined circular channel of 3.6 m diameter has a bed slope of 1 in 600. Determine the velocity and flow rate for the conditions of:

(i) Maximum velocity, and

(ii) discharge.

Take Chezy's constant, $C = 50$.

Solution. Diameter of the circular channel, $d = 3.6$ m

$$\text{Bed slope, } S = \frac{1}{600}$$

$$\text{Chezy's constant, } C = 50$$

Let 2θ = Total angle subtended by the water surface at the centre of the channel.

Flow velocity, V ; flow rate, Q :

(i) *Maximum velocity condition:*

For maximum velocity condition,

$$2\theta = 257.5^\circ = 257.5 \times \frac{\pi}{180} = 4.49 \text{ radians}$$

$$\text{Depth of flow, } y = 0.81d = 0.81 \times 3.6 = 2.92 \text{ m}$$

$$\begin{aligned} \text{Area of flow, } A &= \frac{r^2}{2} (2\theta - \sin 2\theta) \\ &= \frac{1.8^2}{2} (4.49 - \sin 257.5^\circ) = 8.85 \text{ m}^2 \end{aligned}$$

$$\text{Wetted perimeter, } P = 2r\theta = r \times 2\theta = 1.8 \times 4.49 = 8.08 \text{ m}$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{8.85}{8.08} = 1.095$$

$$\therefore \text{Flow velocity, } V = C\sqrt{RS} = 50\sqrt{1.095 \times \frac{1}{600}} \approx \mathbf{2.14 \text{ m/s (Ans.)}}$$

$$\text{Flow rate, } Q = AV = 8.85 \times 2.14 = \mathbf{18.94 \text{ m}^3/\text{s (Ans.)}}$$

(ii) *Maximum discharge condition:*

For maximum discharge condition,

$$2\theta = 308^\circ = 308 \times \frac{\pi}{180} = 5.375 \text{ radians}$$

$$\text{Depth of flow, } y = 0.95d = 0.95 \times 3.6 = 3.42 \text{ m}$$

$$\begin{aligned} \text{Area of flow, } A &= \frac{r^2}{2} (2\theta - \sin 2\theta) \\ &= \frac{1.8^2}{2} (5.375 - \sin 308^\circ) = 6.984 \text{ m}^2 \end{aligned}$$

$$\text{Wetted perimeter, } P = 2r\theta = r \times 2\theta = 1.8 \times 5.375 = 9.675 \text{ m}$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{6.984}{9.675} = 1.032 \text{ m}$$

$$\therefore \text{Flow velocity, } V = C\sqrt{RS} = 50\sqrt{1.032 \times \frac{1}{600}} = \mathbf{2.07 \text{ m/s (Ans.)}}$$

$$\text{Flow rate, } Q = AV = 6.984 \times 2.07 = \mathbf{20.66 \text{ m}^3/\text{s (Ans.)}}$$

16.6. OPEN CHANNEL SECTION FOR CONSTANT VELOCITY AT ALL DEPTHS OF FLOW

It has been observed that according to Chezy's or Manning's formulae the hydraulic radius is the sole shape parameter on which the velocity of flow in a channel laid on a constant bottom slope depends. Thus, *if the hydraulic radius is constant for any depth of flow the velocity of flow will be constant.*

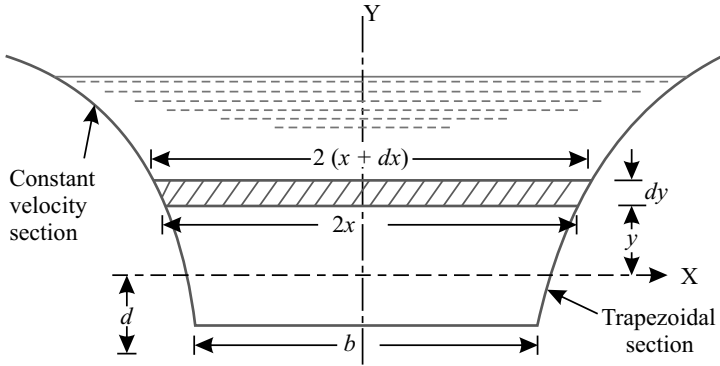


Fig. 16.21. Channel section for constant velocity at all depths.

Consider a profile of channel section (shown in Fig. 16.21) having a constant hydraulic radius R for any depth of flow. For constant velocity V , hydraulic radius has to be constant, which means that $\frac{dA}{dP}$ must remain constant at all depth of flow.

$$\text{i.e.,} \quad \frac{dA}{dP} = R$$

where, cross-sectional area, $dA = 2x dy$

and, wetted perimeter, $dP = 2\sqrt{(dx)^2 + (dy)^2}$

For a small portion of the channel section considered at a depth of y and dy in thickness, as shown in Fig. 16.21.

$$\therefore \quad \frac{2x dy}{2\sqrt{(dx)^2 + (dy)^2}} = R \quad \text{or} \quad x^2 (dy)^2 = R^2 [(dx)^2 + (dy)^2]$$

Dividing both sides by $(dx)^2$, we get:

$$x^2 \left(\frac{dy}{dx} \right)^2 = R^2 \left[1 + \left(\frac{dy}{dx} \right)^2 \right] \quad \text{or} \quad x^2 \left(\frac{dy}{dx} \right)^2 = R^2 + R^2 \left(\frac{dy}{dx} \right)^2$$

$$\text{or,} \quad \left(\frac{dy}{dx} \right)^2 (x^2 - R^2) = R^2 \left(\frac{dy}{dx} \right)^2 = \frac{R^2}{x^2 - R^2}$$

$$\text{or,} \quad \frac{dy}{dx} = \frac{R}{\sqrt{x^2 - R^2}} \quad \text{or} \quad dy = \frac{R}{\sqrt{x^2 - R^2}} \cdot dx$$

Integrating both sides, we get the following two forms:

$$y = R \cos^{-1} \left(\frac{x}{R} \right) + C \quad \dots(16.27)$$

$$\text{or,} \quad y = R \ln (x + \sqrt{x^2 - R^2}) + C_1 \quad \dots(16.27 (a))$$

where, C and C_1 are the constants of integration.

However, we shall consider the first form.

The eqn. (16.27) is the equation of curves forming the sides of the section, the channel is bottomless. The value of C can be obtained if the width of the section of X-axis is known. Let the width be $2R$ at $y = 0$ i.e. $x = R$ at $y = 0$

$$\therefore \quad 0 = R \cos^{-1} \left(\frac{R}{R} \right) + C \quad (\text{From eqn. 16.27})$$

$$\text{or,} \quad C = 0$$

Thus eqn. (16.27) becomes:

$$y = R \cos h^{-1} \left(\frac{x}{R} \right) \quad \dots(16.28)$$

The channel section, below the X-axis may be of any regular shape (e.g. rectangular, trapezoidal, triangular, semicircular etc.). If the section is trapezoidal (as shown in Fig 16.17) for the section below the X-axis, then:

$$\text{Area, } A_1 = (b + 2R) \frac{d}{2}$$

$$\text{Perimeter, } P_1 = b + 2\sqrt{\left(R - \frac{b}{2}\right)^2 + d^2}$$

It has been observed in the case of an open channel that the velocity increases with the increase in depth of flow, thereby causing damage (scouring of the bed and sides) to the channel section. On the contrary if the depth of flow decreases, the velocity decreases which may cause silting of the suspended matter in the liquid. Both these defects are removed by having **constant velocity channels** (where in the large fluctuations in the velocity are avoided).

Channel sections of constant velocity are designed particularly in the case of large sewers in which the discharge ranges from a certain minimum value that flows daily to a very large value during rainy season. In such sewers, the bottom portion (triangular or trapezoidal) is designed for the minimum discharge which flows during lean period, when the discharge increases further, the constant-velocity section becomes effective and discharges the increased flow at the constant velocity.

Example 16.22. It is required to design a channel to give a constant mean velocity of flow of 1.8 m/s at all depths of flow. The lower portion of the channel to carry the minimum discharge is rectangular and has the best proportion, the bottom width being 1.5 m. Determine:

(i) The channel bed slope;

(ii) The depth of flow when the width of water surface is 9 m.

Take Manning's $N = 0.015$.

Solution. Velocity of flow at all depths, $V = 1.8$ m/s

Bottom width, $b = 1.5$ m

Manning's $N = 0.015$

Width of water surface = 9 m

The bottom portion is rectangular and has the best proportion, thus $b = 2d$, where b and d are the base width and depth of flow respectively.

or, $1.5 = 2d; \therefore d = 0.75$ m

Also for the best channel section, $R = \frac{d}{2} = \frac{0.75}{2} = 0.375$ m

(i) **The channel bed slope, S :**

Using Manning's formula, $V = \frac{1}{N} R^{2/3} S^{1/2}$

(where, R = hydraulic radius)

or, $1.8 = \frac{1}{0.015} \times (0.375)^{2/3} S^{1/2}$

or, $S^{1/2} = \frac{1.8 \times 0.015}{(0.375)^{2/3}} = 0.05192$ or $S = 0.0027$ or 1 in 370

Hence, the channel bed slope = **1 in 370 (Ans.)**

(ii) **Depth of flow:**

$$y = R \ln[x + \sqrt{x^2 - R^2}] + C \quad [\text{Ean. 16.27 (a)}]$$

When, $x = \frac{1.5}{2} = 0.75 \text{ m}, y = 0$

$$\therefore C = -R \ln[0.75 + \sqrt{0.75^2 - R^2}]$$

Substituting this value of C in the above eqn., we have:

$$y = R \ln[x + \sqrt{x^2 - R^2}] - R \ln[0.75 + \sqrt{0.75^2 - R^2}]$$

or,
$$y = R \ln \left[\frac{x + \sqrt{x^2 - R^2}}{0.75 + \sqrt{0.75^2 - R^2}} \right]$$

When, $x = \frac{9}{2} = 4.5 \text{ m}$ (given), $R = 0.375 \text{ m}$ (calculated earlier), substituting these values, we have:

$$y = 0.375 \ln \left[\frac{4.5 + \sqrt{4.5^2 - 0.375^2}}{0.75 + \sqrt{0.75^2 - 0.375^2}} \right] = 0.375 \ln \left[\frac{4.5 + 4.484}{0.75 + 0.694} \right] = 0.697 \text{ m}$$

$$\therefore \text{Total depth of flow} = d + y = 0.75 + 0.697 = \mathbf{1.447 \text{ m (Ans.)}}$$

B. NON-UNIFORM FLOW

16.7. NON-UNIFORM FLOW THROUGH OPEN CHANNELS

Whereas in *uniform flow* the gravity force on the flowing liquid just *balances* the frictional force between the flowing liquid and that inside surface of the channel which is in contact with this liquid, the friction force and gravity force are *not in balance* in case of a *steady non-uniform flow*. Non-uniform flow may be *caused by*:

- (i) The change in width, depth, bed slope etc. of a channel;
- (ii) An obstruction, constructed across a channel of uniform width.
- *Waves and surges in open channel produce unsteady non-uniform flow.*

Non-uniform flow is also known as the *flow of varying depth or, the varied flow*. The varied flow may be:

- (i) *Gradually varied flow (G.V.F.)*. In this case of flow the depth of flow increases or decreases *gradually* in the direction of flow; this change from one depth of flow to another occurs gradually in a distance of *appreciable length*.
- (ii) *Rapidly varied flow (R.V.F.)*. In this case a sudden change of depth occurs at a particular point of a channel and the change from one depth to another takes place in a distance of *very short length*.

16.8. SPECIFIC ENERGY AND SPECIFIC ENERGY CURVE

The total energy of flow per *unit weight* of liquid is given by:

$$\text{Total energy} = z + y + \frac{V^2}{2g}$$

where, z = Elevation of the channel bottom above the horizontal bottom,
 y = Depth of flow, and
 V = Average velocity of flow.

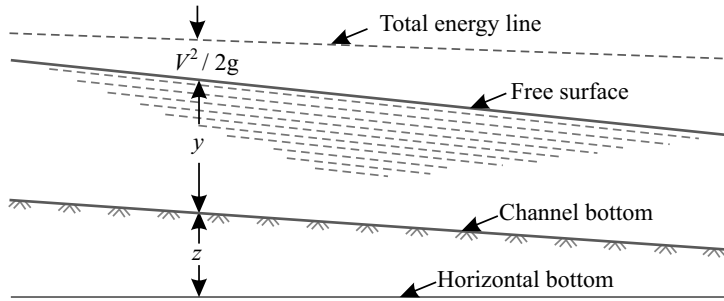


Fig. 16.22. Specific energy.

If the channel bottom itself is taken as the datum (Fig. 16.22), then total energy for unit weight of liquid,

$$E = y + \frac{V^2}{2g} \quad \dots(16.29)$$

The energy E given by eqn. (16.29) is known as *specific energy*. Thus **specific energy** is defined as the energy per unit weight of flowing liquid above the channel bottom. Although the total (or Bernoulli's) energy is reduced by friction, the specific energy can increase or decrease from section to section if the *bed elevation changes*; however, for uniform flow the specific energy remains constant along the flow.

It is evident from eqn. (16.29) that specific energy comprises:

(i) Potential energy of flow (E_p), y , and

(ii) Kinetic energy of flow (E_k), $\frac{V^2}{2g}$.

$$\begin{aligned} \text{i.e.} \quad E &= y + \frac{V^2}{2g} \\ &= E_p + E_k \end{aligned}$$

For the sake of simplicity let us consider a channel of *rectangular section*.

Let, b = Width of channel,
 y = Depth of flow, and
 Q = Discharge through the channel.

$$\text{Now, Velocity of flow, } V = \frac{\text{Discharge}}{\text{Area}} = \frac{Q}{b \times y} = \frac{q}{y} \quad \dots(16.30)$$

(where q = discharge per unit width)

$$\therefore \text{ Specific energy, } E = y + \frac{(q/y)^2}{2g}$$

$$\text{or, } E = y + \frac{q^2}{2gy^2} = E_p + E_k \quad \dots(16.31)$$

For a given channel section and discharge, eqn. (16.31) can be represented graphically as a plot of specific energy E against the depth of flow. Such a plot is called the *specific energy curve/diagram* and it consists of a family of similar curves each representing a given unit discharge.

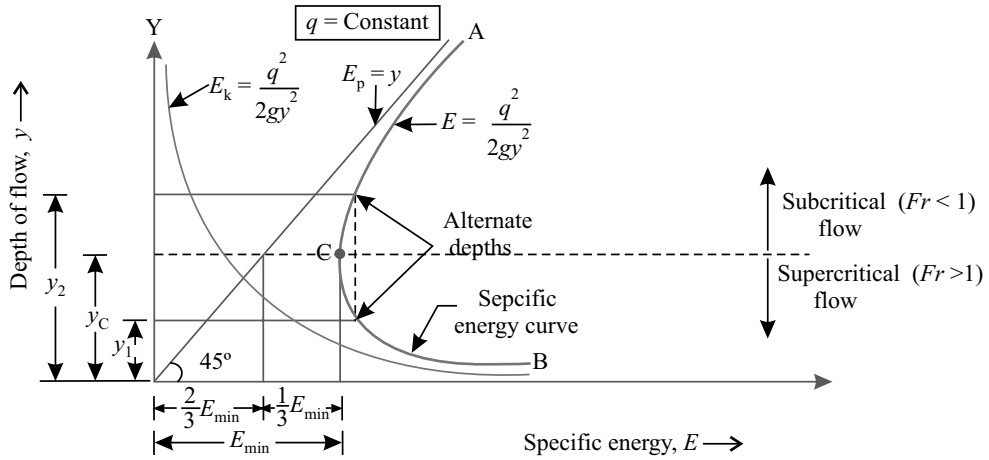


Fig. 16.23. Specific energy curve.

The specific energy plot of Fig. 16.23 entails the following information:

- The curve for potential energy (i.e. $E_p = y$) is a *straight line* passing through the origin, making an angle of 45° with each of the two axes (X and Y),
- The curve for kinetic energy is a *parabola*. Plot for *specific energy* is obtained by adding kinetic energy to potential energy.
- Specific energy is asymptotic to the horizontal axis for small values of y and asymptotic to 45° line for high values of y .
- At a certain depth y_c , called the *critical depth*, the specific energy curve has a point of *minimum specific energy*, the corresponding flow velocity is called the critical velocity V_c .
- For every value of specific energy other than minimum there are two possible depths of flow (y_1 and y_2), one greater and other less than critical depth y_c ; these two depths (for same specific energy) are referred to as *alternate or conjugate depths*.

Critical depth, y_c . It can be seen from the specific energy curve ACB (Fig 16.23) that, there is one point C on the curve which has a *minimum specific energy*, thereby indicating that below this value of specific energy the given discharge cannot occur. *The depth of flow at which the specific energy is minimum is called **critical depth** y_c .*

The mathematical expression for critical depth can be obtained by differentiating the specific energy equation, $E = y + \frac{q^2}{2gy^2}$ with respect to y and equating the derivative to zero. Thus:

$$\frac{dE}{dy} = \frac{d}{dy} \left[y + \frac{q^2}{2gy^2} \right] = 0$$

$$\text{or,} \quad 1 + \frac{q^2}{2g} \left(-\frac{2}{y^3} \right) = 0 \quad \text{or} \quad 1 = \frac{2q^2}{2gy^3}$$

$$\text{or,} \quad y = \left(\frac{q^2}{g} \right)^{1/3}$$

But when the specific energy is minimum the depth of flow is critical, denoted by y_c .

$$\therefore \quad y_c = \left(\frac{q^2}{g} \right)^{1/3} \quad \dots(16.32)$$

Critical velocity, V_c . The velocity of flow at critical depth is known as *critical velocity*; denoted by V_c . Its value is obtained as follows:

$$\text{Velocity} = \frac{\text{Discharge}}{\text{Area}} = \frac{Q}{b \cdot y} = \frac{q}{y}$$

$$\therefore V_c = \frac{q}{y_c} = \frac{q}{\left(\frac{q^2}{g}\right)^{1/3}} = q^{1/3} g^{1/3} \quad \left[\because y_c = \left(\frac{q^2}{g}\right)^{1/3} \right]$$

$$\text{or,} \quad V_c^3 = qg \quad \left(\text{where, } q = \text{discharge per unit width} = \frac{Q}{b} \right)$$

$$\text{Also} \quad q = V_c \times y_c$$

$$\therefore V_c^3 = V_c y_c g \quad \text{or} \quad V_c^2 = g y_c$$

$$\text{or,} \quad V_c = \sqrt{g y_c} \quad \dots(16.33)$$

Minimum specific energy in terms of critical depth. The specific energy is given by:

$$E = y + \frac{q^2}{2gy^2} \quad (\text{Eqn. 16.31})$$

The specific energy is *minimum* when *depth of flow is critical* and hence the above equation becomes:

$$E_{\min} = y_c + \frac{q^2}{2gy_c^2}$$

$$\text{But,} \quad y_c = \left(\frac{q^2}{g}\right)^{1/3} \quad \text{or} \quad y_c^3 = \frac{q^2}{g} \quad (\text{Eqn. 16.32})$$

$$\therefore E_{\min} = y_c + \frac{y_c^3}{2y_c^2} = y_c + \frac{y_c}{2} = \frac{3y_c}{2} \quad (\text{Eqn. 16.34})$$

$$\left(\text{or } y_c = \frac{2}{3} E_{\min} \right)$$

Critical flow. Refer to Fig. 16.23. A critical flow is one in which specific energy is minimum. A flow corresponding to critical depth is also known as **critical flow**.

From eqn. (16.33), we have:

$$V_c = \sqrt{g y_c} \quad \text{or} \quad \frac{V_c}{\sqrt{g y_c}} = 1$$

$$\text{But,} \quad \frac{V_c}{\sqrt{g y_c}} = \text{Froude number (Fr)}$$

$$\therefore Fr = 1 \text{ for critical flow.}$$

Subcritical flow. The flow is *subcritical* (or *streaming* or *tranquil*) when the depth of flow in a channel is *greater than the critical depth* (y_c). In this type of flow, $Fr < 1$.

Supercritical flow. The flow is *supercritical* (or *shooting* or *torrential*) when the depth of flow in a channel is *less than the critical depth* (y_c). In this case $Fr > 1$.

Condition for maximum discharge for a given value of specific energy:

The specific energy (E) at any section of a channel is given by:

$$E = y + \frac{V^2}{2g}, \quad \text{where, } V = \frac{Q}{A} = \frac{Q}{b \times y}$$

$$\therefore E = y + \frac{Q^2}{b^2 y^2} \times \frac{1}{2g} = y + \frac{Q^2}{2gb^2 y^2}$$

$$\text{or, } Q^2 = 2gb^2 y^2 (E - y)$$

$$\text{or, } Q = b\sqrt{2g(Ey^2 - y^3)}$$

For discharge Q to be maximum the expression $(Ey^2 - y^3)$ should be *maximum*.

$$\text{i.e. } \frac{d}{dy} (Ey^2 - y^3) = 0$$

$$\text{or, } 2Ey - 3y^2 = 0 \quad (\because E \text{ is constant})$$

$$\text{or, } 2Ey = 3y^2$$

$$\text{or, } y = \frac{2}{3} E \quad \dots(16.35)$$

$$\text{or, } E = \frac{3y}{2} \quad \dots[16.35 (a)]$$

According to eqn. (16.34) specific energy is *minimum* when it is equal to $\frac{3}{2}$ times the value of depth of critical flow. Here according to eqn. [16.35 (a)] the specific energy is equal to $\frac{3}{2}$ times the depth of flow. Thus eqn. [16.35 (a)] represents minimum specific energy and y is the critical depth. Hence the *condition for maximum discharge for given value of specific energy* is that the *depth of flow should be critical*.

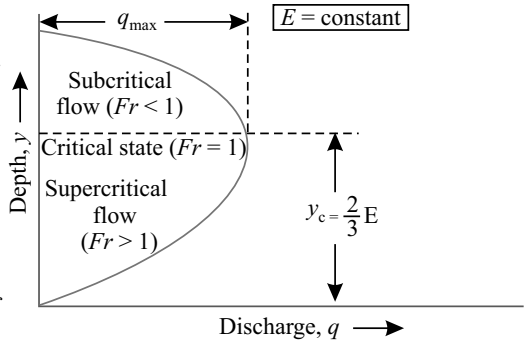


Fig. 16.24 shows the discharge curve.

Fig. 16.24. Discharge curve.

Example 16.23. A 8 m wide channel conveys 15 m³/s of water at a depth of 1.2 m. Calculate:

- (i) Specific energy of the flowing water;
- (ii) Critical depth, critical velocity and minimum specific energy;
- (iii) Froude number and state whether flow is subcritical or supercritical.

Solution. Width of the channel, $b = 8$ m

Discharge, $Q = 15$ m³/s

Depth of flow, $y = 1.2$ m

(i) **Specific energy of the flowing water:**

Average flow velocity,

$$V = \frac{Q}{b \times y} = \frac{15}{8 \times 1.2} = 1.5625 \text{ m/s}$$

Discharge per unit width,

$$q = \frac{Q}{b} = \frac{15}{8} = 1.875 \text{ m}^3/\text{s per m}$$

∴ Specific energy,

$$E = y + \frac{V^2}{2g} = 1.2 + \frac{1.5625^2}{2 \times 9.81} = \mathbf{1.324 \text{ m (Ans.)}}$$

(ii) Critical depth (y_c), critical velocity (V_c) and $E_{\min}$:

$$\text{Critical depth, } y_c = \left(\frac{q^2}{g} \right)^{1/3} = \left(\frac{1.875^2}{9.81} \right)^{1/3} = \mathbf{0.71 \text{ m/s (Ans.)}}$$

$$\text{Critical velocity, } V_c = \sqrt{gy_c} = \sqrt{9.81 \times 0.71} = \mathbf{2.64 \text{ m/s (Ans.)}}$$

$$\text{Minimum specific energy, } E_{\min} = \frac{3}{2} y_c = \frac{3}{2} \times 0.71 = \mathbf{1.065 \text{ m (Ans.)}}$$

$$\left(\text{Alternatively: } E_{\min} = y_c + \frac{V_c^2}{2g} = 0.71 + \frac{2.64^2}{2 \times 9.81} = 1.065 \text{ m} \right)$$

(iii) Froude number and nature of flow:

$$\text{Froude number, } Fr = \frac{V}{\sqrt{gy}} = \frac{1.5625}{\sqrt{9.81 \times 1.2}} = 0.455$$

Since $Fr < 1$, the flow is *subcritical* or *tranquil state*. This is also evident from the fact that $y > y_c$ i.e., $\frac{y}{y_c} > 1$.

Example 16.24. The specific energy for a 3 m wide channel is to be 3 Nm/N. What would be the maximum possible discharge? [PTU]

Solution. Width of channel, $b = 3 \text{ m}$

Specific energy, $E = 3 \text{ Nm/N}$

For the given value of specific energy, the discharge will be maximum, when depth of flow is critical. From eqn. (16.35), for maximum discharge, we have:

$$y_c = y = \frac{2}{3} E = \frac{2}{3} \times 3 = 2 \text{ m}$$

$$\therefore \text{Maximum discharge, } Q_{\max} = \text{Area} \times \text{velocity} \\ = (b \times y_c) \times V_c$$

(∵ At critical depth, y_c , the velocity will be critical.)

$$\text{But, } V_c = \sqrt{gy_c} = \sqrt{9.81 \times 2} = 4.43 \text{ m/s}$$

Substituting the values, we have:

$$Q_{\max} = 3 \times 2 \times 4.43 = \mathbf{26.58 \text{ m}^3/\text{s (Ans.)}}$$

Example 16.25. Water flows at a steady and uniform depth of 2 m in an open channel of rectangular cross-section having base width equal to 5 m and laid at a slope of 1 in 1000. It is desired to obtain critical flow in the channel by providing a hump in the bed. Calculate the height of the hump and sketch the flow profile. Consider the value of Manning's rugosity co-efficient $N = 0.02$ for the channel surface. [UPSC Exams.]

Solution. Depth of flow, $y = 2 \text{ m}$

Base width of channel, $b = 5 \text{ m}$

Bed slope, $S = 1 \text{ in } 1000$

Manning's co-efficient, $N = 0.02$

Height of the hump, h :

For rectangular channel : Area, $A = b \times y$, and

$$\text{Perimeter, } P = b + 2y$$

$$\therefore \text{Hydraulic radius, } R = \frac{A}{P} = \frac{b \times y}{b + 2y} = \frac{5 \times 2}{5 + 2 \times 2} = 1.111 \text{ m}$$

$$\text{Discharge, } Q = A \times V = A \times C \sqrt{RS}$$

$$\text{where, Chezy's constant } C = \frac{1}{N} R^{1/6}$$

$$\begin{aligned} \therefore Q &= A \times \frac{1}{N} R^{1/6} \sqrt{RS} = \frac{A}{N} R^{2/3} S^{1/2} \\ &= \frac{(5 \times 2)}{0.02} \times (1.111)^{2/3} \times \left(\frac{1}{1000}\right)^{1/2} \quad (\text{substituting the values}) \\ &= 16.96 \text{ m}^3/\text{s} \end{aligned}$$

$$\text{Discharge per unit width, } q = \frac{Q}{b} = \frac{16.95}{5} = 3.392 \text{ m}^3/\text{s}$$

$$\text{Critical depth, } y_c = \left(\frac{q^2}{g}\right)^{1/3} = \left(\frac{3.392^2}{9.81}\right)^{1/3} = 1.055 \text{ m}$$

$$\text{Minimum specific energy, } E_{\min} = \frac{3}{2} \times 1.055 = 1.5825 \text{ m}$$

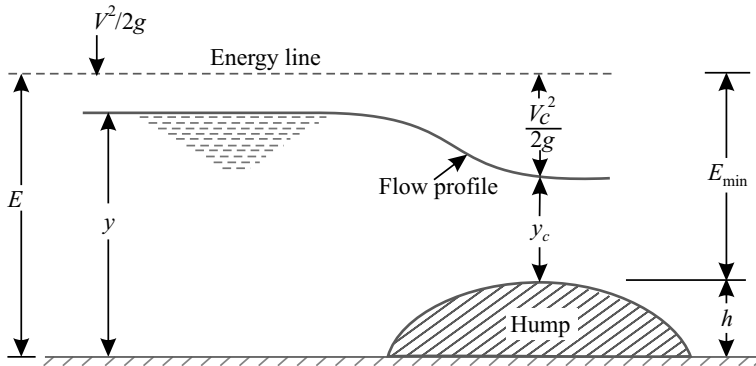


Fig. 16.25.

Specific energy in normal flow,

$$E = y + \frac{V^2}{2g} = 2 + \frac{1.696^2}{2 \times 9.81} = 2.147 \text{ m}$$

$$\text{where, } V = \frac{1}{N} R^{2/3} S^{1/2} = \frac{1}{0.02} \times (1.111)^{2/3} \times \left(\frac{1}{1000}\right)^{1/2} = 1.696 \text{ m/s}$$

$$\text{Height of hump provided, } h = E - E_{\min} = 2.147 - 1.5825 = \mathbf{0.5645 \text{ m (Ans.)}}$$

The flow profile has been shown in Fig. 16.25.

16.9. HYDRAULIC JUMP OR STANDING WAVE

In an open channel when rapidly flowing stream abruptly changes to slowly flowing stream, a distinct rise or jump in the elevation of liquid surface takes place, this phenomenon is known as

hydraulic jump (which is analogous to shock wave in compressible fluids). The hydraulic jump converts *kinetic energy of stream rapidly flowing into potential energy*. Due to this there is a *loss of kinetic energy*. At the place where hydraulic jump occurs *rollers of turbulent water* (eddying turbulences) form, which *cause dissipation of energy*. A hydraulic jump occurs in practice at the toe of spillways or below a sluice gate where the velocity is very high.

The hydraulic jump is also known as a **standing wave** because it is, in essence, a wave which is stationary (i.e., at stand-still) at one place. Such a standing wave is shown in Fig. 16.26.

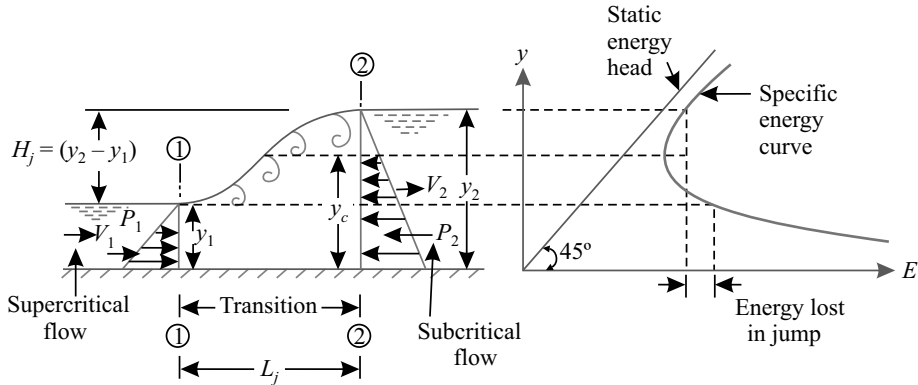


Fig. 16.26. Hydraulic jump.

Analysis of hydraulic jump:

The following *assumptions* are made in the analysis of hydraulic jump:

1. Loss of head due to friction at the walls and channel bed is negligible.
2. The flow is uniform and the pressure distribution is hydrostatic before and after the jump.
3. The channel is horizontal or it has a very small slope. The weight component in the direction of flow is neglected.
4. The momentum correction factor (β) is unity.

Height of hydraulic jump (H_j):

Refer to Fig. 16.26.

$$\text{Discharge per unit width, } q = V_1 y_1 = V_2 y_2 \quad \dots \text{Continuity equation} \quad \dots(i)$$

$$\therefore V_1 = \frac{q}{y_1} \quad \text{and} \quad V_2 = \frac{q}{y_2}$$

In case of hydrostatic pressure distribution, the pressure force at any section,

$$P = wA\bar{y}$$

where, $\bar{y}$ = Vertical depth of centroid of wetted area from the liquid surface.

$$\therefore P_1 = w \times (y_1 \times 1) \times \frac{y_1}{2} = \frac{wy_1^2}{2} \quad \dots \text{Pressure force at section 1-1}$$

$$P_2 = w \times (y_2 \times 1) \times \frac{y_2}{2} = \frac{wy_2^2}{2} \quad \dots \text{Pressure force at section 2-2}$$

Net force acting on mass of water between 1-1 and 2-2

$$= P_2 - P_1 = \frac{wy_2^2}{2} - \frac{wy_1^2}{2} = \frac{w}{2} (y_2^2 - y_1^2) \quad \dots(ii)$$

$$[\because P_2 > P_1 \text{ as } y_2 > y_1]$$

$$\text{Now, change in linear momentum} = \rho q (V_1 - V_2) \quad \dots(iii)$$

But, according to *impulse-momentum equation*:

Net force acting on a mass of fluid = Rate of change of momentum in the same direction

$$\therefore P_2 - P_1 = \rho q (V_1 - V_2) \quad \text{or} \quad \frac{w}{2} (y_2^2 - y_1^2) = \rho q (V_1 - V_2)$$

Substituting, $V_1 = \frac{q}{y_1}$ and $V_2 = \frac{q}{y_2}$, we have:

$$\begin{aligned} \frac{\rho g}{2} (y_2^2 - y_1^2) &= \rho q \left(\frac{q}{y_1} - \frac{q}{y_2} \right) = \rho q^2 \left(\frac{1}{y_1} - \frac{1}{y_2} \right) & (\because w = \rho g) \\ \frac{g}{2} (y_2 + y_1) (y_2 - y_1) &= q^2 \left(\frac{1}{y_1} - \frac{1}{y_2} \right) = q^2 \left(\frac{y_2 - y_1}{y_1 y_2} \right) \end{aligned}$$

Dividing both sides by $(y_2 - y_1)$, we get:

$$\frac{g}{2} (y_2 + y_1) = \frac{q^2}{y_1 y_2} \quad \text{or} \quad (y_2 + y_1) = \frac{2q^2}{gy_1 y_2} \quad \dots(iv)$$

Multiplying both sides by y_2 , we have:

$$y_2^2 + y_1 y_2 = \frac{2q^2}{gy_1} \quad \text{or} \quad y_2^2 + y_1 y_2 - \frac{2q^2}{gy_1} = 0 \quad \dots(v)$$

$$\text{or,} \quad y_2 = \frac{-y_1 \pm \sqrt{y_1^2 + 4 \times 1 \times \frac{2q^2}{gy_1}}}{2} = -\frac{y_1}{2} \pm \sqrt{\frac{y_1^2}{4} + \frac{2q^2}{gy_1}}$$

$$\text{or,} \quad y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2q^2}{gy_1}} \quad \text{or} \quad -\frac{y_1}{2} - \sqrt{\frac{y_1^2}{4} + \frac{2q^2}{gy_1}}$$

Neglecting the second root (being impossible, $-ve$ depth), we have:

$$y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2q^2}{gy_1}} \quad \dots(16.36)$$

$$= -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2 \times (V_1 y_1)^2}{gy_1}} \quad (\because q = V_1 y_1)$$

$$\text{or,} \quad y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2 V_1^2 y_1}{g}} \quad \dots(16.37)$$

Expression of y_2 in terms of Froude number (Fr):

Eqn. (16.37) can be written as:

$$y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} \left(1 + \frac{8V_1^2}{gy_1} \right)}$$

$$\text{or,} \quad y_2 = -\frac{y_1}{2} + \frac{y_1}{2} \sqrt{1 + \frac{8V_1^2}{gy_1}} \quad \dots(vi)$$

$$\text{But,} \quad F_{r_1} = \frac{V_1}{\sqrt{gy_1}} \quad \text{or} \quad (F_{r_1})^2 = \frac{V_1^2}{gy_1}$$

$\therefore$ Substituting this value in expression (vi), we have:

$$y_2 = -\frac{y_1}{2} + \frac{y_1}{2} \sqrt{1 + 8 (F_{r_1})^2}$$

$$\text{or,} \quad y_2 = \frac{y_1}{2} \left(\sqrt{1 + 8 (Fr_1)^2} - 1 \right) \quad \dots(16.38)$$

$$\therefore \text{ Height of hydraulic jump, } H_j = y_2 - y_1 \quad \dots(16.39)$$

Length of hydraulic jump (L_j). Length of hydraulic jump represents *that short distance over which the jump occurs* (Refer Fig. 16.26). For rectangular channels with horizontal floor, length of a jump has been found to vary between 5 to 7 times the height of the jump.

$$\text{i.e.,} \quad L_j = 5 \text{ to } 7 H_j \quad \dots(16.40)$$

Loss of energy due to hydraulic jump:

The loss of energy due to hydraulic jump is equal to the difference of specific energies at the upstream (1-1) and downstream (2-2) sections.

$$\begin{aligned} \text{i.e.} \quad E_2 &= E_1 - E_L \\ &= \left(y_1 + \frac{V_1^2}{2g} \right) - \left(y_2 + \frac{V_2^2}{2g} \right) = \left(\frac{V_1^2}{2g} - \frac{V_2^2}{2g} \right) - (y_2 - y_1) \\ &= \left(\frac{q^2}{2gy_1^2} - \frac{q^2}{2gy_2^2} \right) - (y_2 - y_1) \quad \left(\because V_1 = \frac{q}{y_1}, V_2 = \frac{q}{y_2} \right) \\ \text{or,} \quad &= \frac{q^2}{2g} \left(\frac{1}{y_1^2} - \frac{1}{y_2^2} \right) - (y_2 - y_1) \quad \dots(vii) \end{aligned}$$

$$\text{or,} \quad E_L = \frac{q^2}{2g} \left(\frac{y_2^2 - y_1^2}{y_1^2 y_2^2} \right) - (y_2 - y_1)$$

$$\text{But,} \quad q^2 = gy_1 y_2 \left(\frac{y_2 + y_1}{2} \right) \quad \dots(\text{From expression iv})$$

$$\begin{aligned} \therefore \text{ Loss of energy, } E_L &= gy_1 y_2 \left(\frac{y_2 + y_1}{2} \right) \times \frac{(y_2^2 - y_1^2)}{2 gy_1^2 y_2^2} - (y_2 - y_1) \\ &= \frac{(y_2 + y_1)(y_2^2 - y_1^2)}{4y_1 y_2} - (y_2 - y_1) \\ &= \frac{(y_2 + y_1)(y_2 + y_1)(y_2 - y_1)}{4y_1 y_2} - (y_2 - y_1) \\ &= (y_2 - y_1) \left[\frac{(y_2 + y_1)^2}{4y_1 y_2} - 1 \right] \\ &= (y_2 - y_1) \left[\frac{y_2^2 + y_1^2 + 2y_1 y_2 - 4y_1 y_2}{4y_1 y_2} \right] \\ &= (y_2 - y_1) \left[\frac{(y_1 - y_2)^2}{4y_1 y_2} \right] \\ \text{or,} \quad E_L &= \frac{(y_2 - y_1)^3}{4y_1 y_2} \quad \dots(16.41) \end{aligned}$$

Example 16.26. A sluice gate discharges water into horizontal rectangular channel with a velocity of 10 m/s and depth of flow of 1 m. Determine the depth of flow of water after the jump and consequent loss in total head [NU]

Solution. Velocity of flow before hydraulic jump, $V_1 = 10$ m/s.

Depth of flow before hydraulic jump, $y_1 = 1$ m

Depth of flow after the jump, y_2 :

Discharge per unit width, $q = V_1 \times y_1 = 10 \times 1 = 10$ m³/s per m

The depth of flow after the jump is given by:

$$y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2q^2}{gy_1}} \quad \dots[\text{Eqn. (16.36)}]$$

or,

$$y_2 = -\frac{1}{2} + \sqrt{\frac{1^2}{4} + \frac{2 \times 10^2}{9.81 \times 1}} = \mathbf{4.043 \text{ m (Ans.)}}$$

Loss in total head, E_L :

Loss in total head is given by:

$$E_L = \frac{(y_2 - y_1)^3}{4y_1y_2} \quad \dots[\text{Eqn. (16.31)}]$$

or,

$$E_L = \frac{(4.043 - 1)^3}{4 \times 1 \times 4.043} = \mathbf{1.742 \text{ m (Ans.)}}$$

Example 16.27. A 3.6 m wide rectangular channel conveys 9.0 m³/s of water with a velocity of 6 m/s.

(i) Is there a condition for hydraulic jump to occur? If so, calculate the height, length and strength of the jump.

(ii) What is loss of energy per kg of water?

Solution. Width of channel, $b = 3.6$ m

Discharge, $Q = 9.0$ m³/s

Velocity of flow before jump, $V_1 = 6$ m/s

(i) Is there a condition for hydraulic jump to occur?

$$\text{Depth of water before jump, } y_1 = \frac{Q}{b \times V_1} = \frac{9.0}{3.6 \times 6} = 0.4167 \text{ m}$$

$$\text{Discharge per unit width, } q = \frac{Q}{b} = \frac{9.0}{3.6} = 2.5 \text{ m}^3/\text{s per m}$$

$$\text{Critical depth, } y_c = \left(\frac{q^2}{g}\right)^{1/3} = \left(\frac{2.5^2}{9.81}\right)^{1/3} = 0.86 \text{ m}$$

Since $y_1 < y_c$, a jump would occur. (Ans.)

Froude number ahead of jump,

$$Fr_1 = \frac{V_1}{\sqrt{gy_1}} = \frac{6}{\sqrt{9.81 \times 0.4167}} = 2.967$$

Depth of water downstream the jump,

$$y_2 = \frac{y_1}{2} \left[\sqrt{1 + 8(Fr_1)^2} - 1 \right] \quad \dots(\text{Eqn. 16.38})$$

$$y_2 = \frac{0.4167}{2} \left[\sqrt{1 + 8 \times 2.967^2} - 1 \right] = 1.5525 \text{ m}$$

$\therefore$ **Height of jump, $H_j = y_2 - y_1 = 1.5525 - 0.4167 = 1.1358$ m (Ans.)**

Length of jump, $L_j \approx 6(y_2 - y_1) = 6 \times 1.1358 = 6.8148 \text{ m}$ (Ans.)

$$\text{Strength of jump} = \frac{y_2}{y_1} = \frac{1.5525}{0.4167} = 3.726 \text{ (Ans.)}$$

(ii) **Loss of energy per kg of water, E_L :**

Velocity before jump, $V_1 = 6 \text{ m/s}$... (Given)

$$\text{Velocity after jump, } V_2 = \frac{q}{y_2} = \frac{2.5}{1.5525} = 1.61 \text{ m/s}$$

$$E_1 = y_1 + \frac{V_1^2}{2g} = 0.4167 + \frac{6^2}{2 \times 9.81} = 2.25 \text{ m}$$

$$E_2 = y_2 + \frac{V_2^2}{2g} = 1.5525 + \frac{1.61^2}{2 \times 9.81} = 1.68 \text{ m}$$

$\therefore$ Loss of energy in the jump, $E_L = E_1 - E_2 = 2.25 - 1.68 = 0.57 \text{ m}$ (Ans.)

$$\left[\text{Alternatively, } E_L = \frac{(y_2 - y_1)^3}{4y_1y_2} = \frac{(1.5525 - 0.4167)^3}{4 \times 1.5525 \times 0.4167} = 0.57 \text{ m} \right]$$

Example 16.28. In a rectangular channel of 0.5 m width, a hydraulic jump occurs at a point where depth of water flow is 0.15 m and Froude number is 2.5. Determine:

- (i) The specific energy; (ii) The critical and subsequent depths,
(iii) Loss of head, and; (iv) Energy dissipated.

Solution. Width of the channel, $b = 0.5 \text{ m}$

Depth of flow, $y_1 = 0.15 \text{ m}$

Froude number, $Fr = 2.5$

Now, $Fr = \frac{V_1}{\sqrt{gy_1}}$, where V_1 is the upstream velocity

$$\therefore 2.5 = \frac{V_1}{\sqrt{9.81 \times 0.15}} \quad \text{or} \quad V_1 = 3.03 \text{ m/s}$$

Discharge per unit width, $q = V_1 y_1 = 3.03 \times 0.15 = 0.4545 \text{ m}^3/\text{s per m}$

(i) **Specific energy, E :**

$$E = y_1 + \frac{V_1^2}{2g} = 0.15 + \frac{3.03^2}{2 \times 9.81} = 0.618 \text{ m (Ans.)}$$

(ii) **Critical depth, y_c :**

$$y_c = \left(\frac{q^2}{g} \right)^{1/3} = \left[\frac{0.4545^2}{9.81} \right]^{1/3} = 0.276 \text{ m (Ans.)}$$

$$\text{Subsequent depth, } y_2 = \frac{y_1}{2} \left[\sqrt{1 + 8 (Fr_1)^2} - 1 \right]$$

$$\text{or, } y_2 = \frac{0.15}{2} \left[\sqrt{1 + 8 \times 2.5^2} - 1 \right] = 0.461 \text{ m (Ans.)}$$

(iii) **Loss head, E_L :**

$$E_L = \frac{(y_2 - y_1)^3}{4y_1y_2} = \frac{(0.461 - 0.15)^3}{4 \times 0.15 \times 0.461} = 0.108 \text{ m (Ans.)}$$

(iv) Power dissipated, P :

$$\begin{aligned} P &= wQE_L \\ \text{where } Q &= A_1 V_1 = (b \times y_1) V_1 = (0.5 \times 0.15) \times 3.03 = 0.227 \text{ m}^3/\text{s} \\ \therefore P &= 9810 \times 0.227 \times 0.108 = \mathbf{240.5 \text{ W (Ans.)}} \end{aligned}$$

Example 16.29. Find in terms of specific energy E , an expression for the critical depth in a trapezoidal channel with bottom width b and side slope 1 vertical to n horizontal.

[MDU, Haryana]

Solution. The specific energy (E) of a channel is given as:

$$E = y + \frac{V^2}{2g} = y + \frac{Q^2}{2gA^2} \quad (\because V = Q/A)$$

where,

y = Depth of flow,

V = Average velocity of flow,

Q = Discharge, and

A = Area of cross-section of the channel.

The condition for minimum specific energy ($F_{\min}$) can be obtained by differentiating the specific energy equation with respect to y and equating the derivative to zero. Thus:

$$\begin{aligned} \frac{dE}{dy} &= \frac{d}{dy} \left[y + \frac{Q^2}{2gA^2} \right] = 0 \\ &= 1 + \frac{Q^2}{2g} \times \left(-2 \times A^{-3} \cdot \frac{dA}{dy} \right) = 1 - \frac{Q^2}{gA^3} \times \frac{dA}{dy} = 0 \end{aligned}$$

($\because Q = \text{constant}$)

$$\therefore \frac{dA}{dy} = \frac{gA^3}{Q^2} = \frac{gA_c^3}{Q^2}, \quad \text{for critical condition} \quad \dots(i)$$

In case of a trapezoidal channel,

$$\begin{aligned} A &= (b + ny)y = by + ny^2 \\ \frac{dA}{dy} &= b + 2ny = b + 2ny_c, \quad \text{for critical condition} \quad \dots(ii) \end{aligned}$$

From expressions (i) and (ii), we have:

$$\frac{gA_c^3}{Q^2} = b + 2ny_c \quad \text{or} \quad \frac{Q^2}{g} = \frac{A_c^3}{b + 2ny_c} \quad \dots(iii)$$

The specific energy for critical conditions becomes:

$$E = y_c + \frac{Q^2}{2gA_c^2}$$

Substituting the value of $\frac{Q^2}{g}$ from expression (iii), we get:

$$E = y_c + \frac{A_c^3}{2A_c^2(b + 2ny_c)} = y_c + \frac{A_c}{2(b + 2ny_c)}$$

$$\text{or,} \quad E = y_c + \frac{(b + ny_c)y_c}{2(b + 2ny_c)} \quad \dots(iv)$$

$$\text{or,} \quad E \times 2(b + 2ny_c) = y_c \times 2(b + 2ny_c) + (b + ny_c)y_c$$

$$\text{or,} \quad 2bE + 4nEy_c = 2by_c + 4ny_c^2 + by_c + ny_c^2$$

Rearranging the above equation, we have:

$$5ny_c^2 + (3b - 4nE)y_c - 2bE = 0$$

$$\text{or, } y_c = \frac{-(3b - 4nE) \pm \sqrt{(3b - 4nE)^2 - 4 \times 5n \times (-2bE)}}{2 \times 5n}$$

$$\text{or, } y_c = \frac{(4nE - 3b) \pm \sqrt{(3b - 4nE)^2 + 40nbE}}{10n} \quad (\text{Ans.})$$

Note : When $n = 0$, the expression (iv) becomes:

$$E = y_c + \frac{b \cdot y_c}{2b} = y_c + \frac{y_c}{2} = \frac{3y_c}{2} \quad \text{or } y_c = \frac{2}{3} E$$

which is the condition for maximum discharge for a given value of specific energy in a rectangular channel.

Example 16.30. (Flow in venturiflume) A venturiflume is 1.30 m wide at entrance and 0.65 m in the throat. Neglecting hydraulic losses in the flume, calculate the flow if the depths at the entrance and throat are 0.65 m and 0.60 m respectively. A hump is now installed at the throat, of height 200 mm, so that a standing wave (hydraulic jump) is formed beyond the throat. What is the increase in the upstream depth when the same flow as before passes through the flume?

[Roorkee University]

Solution. Width of venturiflume at entrance, $b_1 = 1.3$ m

Width at throat, $b_2 = 0.65$ m

Depth of flow at section 1, $y_1 = 0.65$ m

Depth of flow at section 2, $y_2 = 0.6$ m

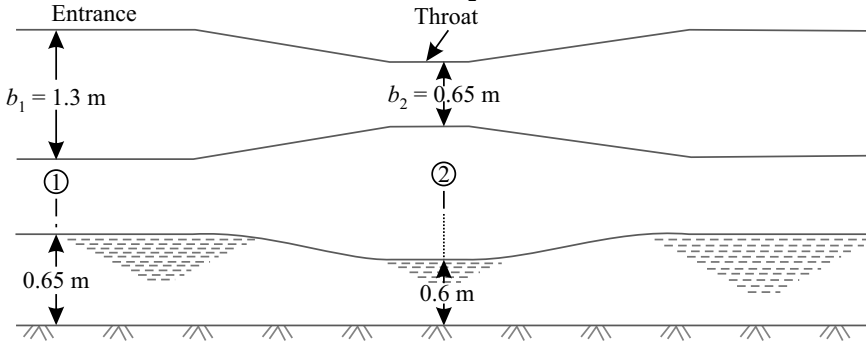


Fig. 16.27. Flow in venturiflume.

Using continuity equation, we have:

$$\text{Discharge, } Q = b_1 y_1 V_1 = b_2 y_2 V_2$$

$$\text{or, } Q = 1.3 \times 0.65 \times V_1 = 0.65 \times 0.6 V_2, \therefore V_2 = \frac{1.3 \times 0.65 V_1}{0.65 \times 0.6} = 2.17 V_1$$

Neglecting losses, Specific energy at (1) = Specific energy at (2)

$$\text{i.e. } y_1 + \frac{V_1^2}{2g} = y_2 + \frac{V_2^2}{2g}$$

$$0.65 + \frac{V_1^2}{2g} = 0.6 + \frac{(2.17 V_1)^2}{2g}$$

$$\text{or } \frac{(2.17 V_1)^2}{2g} - \frac{V_1^2}{2g} = 0.65 - 0.6 = 0.05$$

$$\text{or, } \frac{V_1^2}{2g} (2.17^2 - 1) = 0.05$$

$$\text{or, } V_1^2 = \frac{0.05 \times 2g}{(2.17^2 - 1)} = \frac{0.05 \times 2 \times 9.81}{2.17^2 - 1}$$

$$\text{or, } V_1 = 0.514 \text{ m/s}$$

$$\text{The discharge, } Q = b_1 y_1 V_1 = 1.3 \times 0.65 \times 0.514 = 0.434 \text{ m}^3/\text{s}$$

Critical depth in contracted portion,

$$y_c = \left(\frac{q^2}{g} \right)^{1/3}$$

$$\text{where, } q = \frac{Q}{b_1} = \frac{0.434}{0.65} = 0.67 \text{ m}^3/\text{s per m}$$

$$\therefore y_c = \left(\frac{0.67^2}{9.81} \right)^{1/3} = 0.357 \text{ m}$$

The new specific energy corresponding to critical flow at the throat when hump of height h is installed,

$$y_c = h + y_c + \frac{V_c^2}{2g} = h + \frac{3}{2} y_c = 0.2 + \frac{3}{2} \times 0.357 = 0.735 \text{ m}$$

(where, $h = 200 \text{ mm} = 0.2 \text{ m}$) ... (Given)

The upstream surface will rise till the upstream specific energy equals 0.735 m.

$$0.735 = y_1 + \frac{V_1^2}{2g}$$

$$= y_1 + \frac{[Q / (b_1 \times y_1)]^2}{2g}$$

$$\text{or, } 0.735 = y_1 + \frac{Q^2}{2g \times (1.3y_1)^2} = y_1 + \frac{0.435^2}{2 \times 9.81 \times 1.69y_1^2} = y_1 + \frac{0.0057}{y_1^2}$$

$$\text{i.e. } y_1^3 \cdot 0.735 y_1^2 + 0.0057 = 0$$

Solving by trial and error, $y_1 = 0.72 \text{ m}$

The increase in the upstream depth $= 0.72 - 0.65 = 0.07 \text{ m} = 70 \text{ mm}$ (Ans.)

Example 16.31. A sluice across a channel 7.2 m wide discharges a stream 1.2 m deep. What is the flow rate when the depth upstream of the sluice is 8.4 m? On the downstream side concrete blocks have been placed to create condition for hydraulic jump to occur. Calculate the force on the blocks if the downstream depth is 3.6 m.

Solution. Refer to Fig. 16.28.

Applying continuity equation at sections (1-1), (2-2) and (3-3), we have:

$$\text{Discharge, } Q = (b_1 \times y_1) V_1 = (b_2 \times y_2) V_2 = (b_3 \times y_3) V_3$$

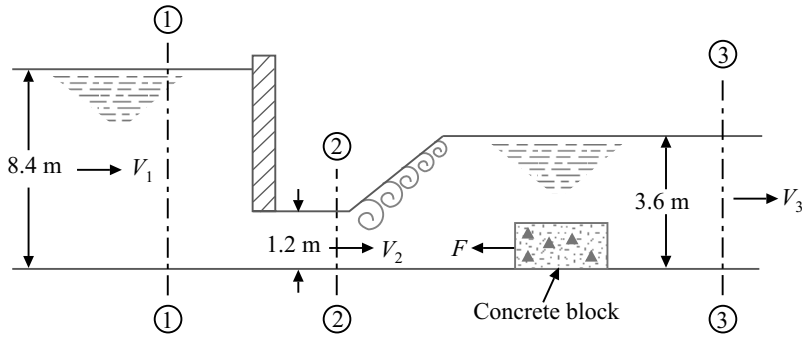


Fig. 16.28

But, $b_1 = b_2 = b_3 = 7.2 \text{ m}$, $y_1 = 8.4 \text{ m}$, $y_2 = 1.2 \text{ m}$, $y_3 = 3.6 \text{ m}$

$\therefore Q = (7.2 \times 8.4) V_1 = (7.2 \times 1.2) V_2 = (7.2 \times 3.6) V_3$

From which, $V_2 = 7V_1$ and $V_3 = 2.333V_1$

Neglecting frictional losses between sections (1-1) and (2-2), the specific energies at (1-1) and (2-2) are equal.

$$\text{or, } y_1 + \frac{V_1^2}{2g} = y_2 + \frac{V_2^2}{2g}$$

$$\text{or, } 8.4 + \frac{V_1^2}{2g} = 1.2 + \frac{(7V_1)^2}{2g} = 1.2 + \frac{49V_1^2}{2g} \quad (\because V_2 = 7V_1)$$

$$\text{or, } \frac{48V_1^2}{2g} = 7.2 \quad \text{or} \quad V_1^2 = \frac{7.2 \times 2 \times 9.81}{48} = 2.943$$

$$\therefore V_1 = 1.715 \text{ m/s}$$

Flow rate, Q :

$$Q = (b_1 y_1) V_1 = 7.2 \times 8.4 \times 1.715 = 103.72 \text{ m}^3/\text{s} \text{ (Ans.)}$$

Force on the blocks, F :

Applying momentum equation to sections (2-2) and (3-3), neglecting the boundary friction, we have:

$$P_2 - F - P_3 = \frac{wQ}{g} (V_3 - V_2)$$

$$w A_2 \bar{y}_2 - F - w A_3 \bar{y}_3 = \frac{wQ}{g} (V_3 - V_2)$$

$$9810 \times (7.2 \times 1.2) \times \frac{1.2}{2} - F - 9810 \times (7.2 \times 3.6) \times \frac{3.6}{2} = \frac{9810 \times 103.72}{9.81} (2.333 V_1 - 7V_1)$$

$$50855 - F - 457695 = 103720 (-4.667 \times 1.715)$$

$$\begin{aligned} \text{or, } F &= 50855 - 457695 + 103720 (4.667 \times 1.715) \\ &= 423325 \text{ N or } 423.325 \text{ kN} \end{aligned}$$

Hence, the force on the concrete blocks = **423.325 kN** which acts in a direction *opposite* to F (Ans.)

Example 16.32. Uniform flow occurs at a depth of 1.5 m in a long rectangular channel 3 m wide and laid to a slope of 0.0009. If Manning's $N = 0.015$ calculate:

(i) Maximum height of hump on the floor to produce critical depth.

(ii) Width of contraction which will produce critical depth without increasing the upstream depth of flow. [IIT Madras]

Solution. Depth of flow, $y = 1.5$ m
 Width of channel, $b = 3$ m
 Bed slope, $S = 0.0009$
 Manning's $N = 0.015$

Height of the hump, h :

Discharge, $Q = A \times V = A \times C \sqrt{RS}$

where, C (Chezy's constant) $= \frac{1}{N} R^{1/6}$

$$\therefore Q = A \times \frac{1}{N} R^{1/6} \sqrt{RS} \quad \dots(i)$$

$$= A \times \frac{1}{N} R^{2/3} S^{1/2}$$

(where, V = average velocity of flow, and

R = hydraulic radius)

Here, area, $A = b \times y = 3 \times 1.5 = 4.5 \text{ m}^2$

Perimeter, $P = b + 2y = 3 + 2 \times 1.5 = 6 \text{ m}$ and $R = \frac{A}{P} = \frac{4.5}{6} = 0.75 \text{ m}$

Substituting the values in expression (i), we get:

$$Q = 4.5 \times \frac{1}{0.015} \times (0.75)^{2/3} \times (0.0009)^{1/2} = 7.43 \text{ m}^3/\text{s}$$

Discharge per unit width, $q = \frac{Q}{b} = \frac{7.43}{3} = 2.477 \text{ m}^3/\text{s per m}$

$$\text{Critical depth, } y_c = \left(\frac{q^2}{g} \right)^{1/3} = \left(\frac{2.477^2}{9.81} \right)^{1/3} = 0.855 \text{ m}$$

Now, equating the specific energies upstream and at the hump, we get:

$$1.5 + \frac{V^2}{2g} = h + y_c + \frac{V_c^2}{2g} \quad \dots(ii)$$

Here, $V = \frac{Q}{A} = \frac{7.43}{4.5} = 1.65 \text{ m/s}$, and

$$V_c = \sqrt{gy_c} \text{ or } V_c^2 gy_c \text{ or } y_c = \frac{V_c^2}{g} \text{ or } \frac{V_c^2}{2g} = \frac{y_c}{2}$$

Substituting the values in expression (ii), we have:

$$1.5 + \frac{1.65^2}{2 \times 9.81} = h + 0.855 + \frac{0.855}{2} \text{ or } 1.6387 = h + 1.2825$$

$$\therefore h = 0.3562 \text{ m (Ans.)}$$

(ii) **Width of contraction:**

Let, b_c = Width at the contracted portion to produce critical depth.

Now, Upstream specific energy = Specific energy at the contracted portion.

$$\begin{aligned} 1.6387 &= y_c + \frac{V_c^2}{2g} = y_c + \frac{y_c}{2} = \frac{3}{2} y_c \\ &= \frac{3}{2} \left[\frac{q^2}{g} \right]^{1/3} = \frac{3}{2} \left[\frac{(Q/b_c)^2}{g} \right]^{1/3} = \frac{3}{2} \left[\frac{(7.43)^2}{b_c^2 \times 9.81} \right]^{1/3} \end{aligned}$$

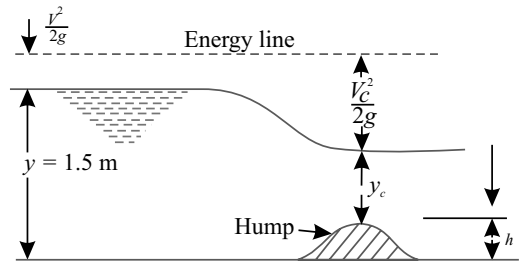


Fig. 16.29

$$\text{or, } \left[\frac{(7.43)^2}{b_c^2 \times 9.81} \right] = 1.6387 \times \frac{2}{3} = 1.0925$$

$$\text{or, } \frac{7.43^2}{b_c^2 \times 9.81} = (1.0925)^3 = 1.304$$

$$\therefore b_c = \left(\frac{7.43^2}{1.304 \times 9.81} \right)^{\frac{1}{2}} = 2.077 \text{ m (Ans.)}$$

Example 16.33. Water flows at a velocity of 1 m/s and a depth of 2 m in an open channel of rectangular cross-section, 3 m wide. At a certain section the width is reduced to 1.8 m and the bed is raised by 0.65 m. Will the upstream depth be affected? If so, to what extent?

[UPSC, CES Exams.]

Solution. Velocity of flow, $V = 1 \text{ m/s}$

Depth of flow, $y = 2 \text{ m}$

Width of channel, $b = 3 \text{ m}$

Width of contracted section, $b_c = 1.8 \text{ m}$

At section (1): Refer to Fig. 16.30.

Specific energy at the section,

$$E_1 = y_1 + \frac{V_1^2}{2g} = 2 + \frac{(1)^2}{2 \times 9.81} = 2.051 \text{ m}$$

$$\text{Discharge, } Q = A.V = (b_1 \times y_1) \times V_1 \\ = (3 \times 2) \times 1 = 6 \text{ m}^3/\text{s}$$

Discharge per unit width,

$$q_1 = \frac{Q}{b_1} = \frac{6}{3} = 2 \text{ m}^3/\text{s per m}$$

Critical depth,

$$(y_c)_1 = \left(\frac{q_1^2}{g} \right)^{1/3} = \left[\frac{2^2}{9.81} \right]^{1/3} = 0.7415 \text{ m}$$

Since $y_1 > y_c$, the flow in the channel is *sub-critical*.

Minimum specific energy at section 1,

$$(E_{\min})_1 = \frac{3}{2} (y_c)_1 = \frac{3}{2} \times 0.7415 = 1.1122 \text{ m}$$

At the section (2) (contracted and humped section):

$$\text{Discharge per unit width, } q_2 = \frac{Q}{b_2} = \frac{6}{1.8} = 3.333 \text{ m}^3/\text{per m}$$

Critical depth,

$$(y_c)_2 = \left(\frac{q_2^2}{g} \right)^{1/3} = \left(\frac{3.333^2}{9.81} \right)^{1/3} = 1.0423 \text{ m}$$

Minimum specific energy,

$$(E_{\min})_2 = \frac{3}{2} (y_c)_2 = \frac{3}{2} \times 1.0423 = 1.5634 \text{ m}$$

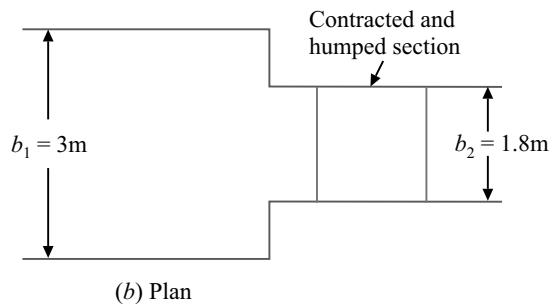
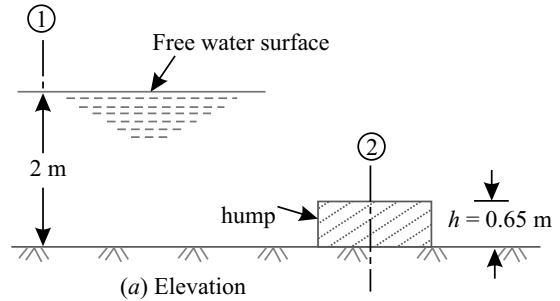


Fig. 16.30

Specific energy w.r.t. channel bed at section (2),

$$E_2 = (E_{\min})_2 + h = 1.5634 + 0.65 = 2.2134 \text{ m}$$

Since $E_2 > E_1$ the upstream depth will be affected. The flow will be possible only when the upstream water level is increased such that:

$$E_1 = E_2 \quad \text{or} \quad y_1 + \frac{V_1^2}{2g} = 2.2134 \quad \dots(i)$$

Also

$$Q = b_1 y_1 \times V_1 \quad \dots \text{Continuity equation}$$

$$6 = 3 \times y_1 \times V_1 \quad \text{or} \quad V_1 y_1 = 2 \quad \dots(ii)$$

From expressions (i) and (ii), we have:

$$y_1 + \frac{(2/y_1)^2}{2g} = 2.2134$$

$$\text{or,} \quad y_1 + \frac{4}{2g \times y_1^2} = 2.2134$$

$$\text{or,} \quad y_1 + \frac{0.204}{y_1^2} = 2.2134 \quad \text{or} \quad y_1^3 = 2.2134 y_1^2 + 0.204 = 0$$

Solving by trial and error, we get $y_1 = 2.17 \text{ m}$

Hence the water level on the upstream side will be headed up by,

$$(2.17 - 2) = 0.17 \text{ m or } 170 \text{ mm (Ans.)}$$

Example 16.34. A hydraulic jump occurs in a V-shaped channel having sides sloping at 45° . Derive an equation relating the two depths and the flow rate.

If the depths before and after the jump in the above channel are 0.50 m and 1.0 m , determine:

(i) The flow rate;

(ii) Froude numbers before and after the jump.

[Roorkee University]

Solution. Let,

y_1 = Depth of flow before hydraulic jump,

V_1 = Velocity of flow before hydraulic jump, and

y_2, V_2 = Depth of flow and velocity of flow respectively after hydraulic jump.

Refer to Fig. 16.31:

According to impulse-momentum equation:

Net force acting on a mass of fluid = Rate of change of momentum in the same direction

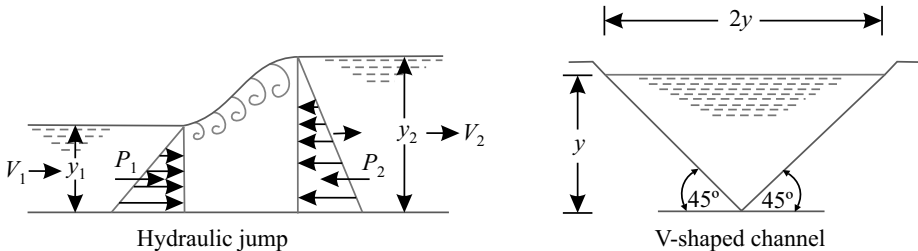


Fig. 16.31. Hydraulic jump – V-shaped channel.

$$P_1 - P_2 = \frac{wQ}{g} (V_1 - V_2)$$

(where, Q = discharge or flow rate)

$$\text{Area, } A = \frac{1}{2} \times 2y \times y = y^2 \quad \text{and} \quad \bar{y} = \frac{1}{3} y$$

$$\therefore P_1 = wA_1\bar{y}_1 = w \times y_1^2 \times \left(\frac{1}{3}y_1\right) = \frac{1}{3}wy_1^3$$

$$\text{and, } P_2 = wA_2\bar{y}_2 = w \times y_2^2 \times \left(\frac{1}{3}y_2\right) = \frac{1}{3}wy_2^3$$

From continuity equation, we have:

$$Q = A_1V_1 = A_2V_2; \quad V_1 = \frac{Q}{A_1} = \frac{Q}{y_1^2} \quad \text{and} \quad V_2 = \frac{Q}{A_2} = \frac{Q}{y_2^2}$$

Substituting these quantities in expression (i), we have:

$$\frac{1}{3}wy_2^3 - \frac{1}{3}wy_1^3 = \frac{wQ}{g} \left(\frac{Q}{y_1^2} - \frac{Q}{y_2^2} \right)$$

$$\text{or, } \frac{1}{3}(y_2^3 - y_1^3) = \frac{Q^2}{g} \left(\frac{1}{y_1^2} - \frac{1}{y_2^2} \right) \quad (\text{cancelling } w \text{ on both sides})$$

$$\text{or, } \frac{1}{3}(y_2^3 - y_1^3) = \frac{Q^2}{g} \left(\frac{y_2^2 - y_1^2}{y_1^2 y_2^2} \right)$$

$$\text{or, } Q^2 = \frac{g}{3} \times y_1^2 y_2^2 \left(\frac{y_2^3 - y_1^3}{y_2^2 - y_1^2} \right)$$

$$\text{or, } Q = y_1 y_2 \sqrt{\frac{g}{3} \left(\frac{y_2^3 - y_1^3}{y_2^2 - y_1^2} \right)}$$

This is the required equation relating the two depths and the flow rate.

Depths: $y_1 = 0.5 \text{ m}, y_2 = 1.0 \text{ m}$

...(Given)

(i) Flow rate, Q:

$$Q = y_1 y_2 \sqrt{\frac{g}{3} \left(\frac{y_2^3 - y_1^3}{y_2^2 - y_1^2} \right)}$$

$$\text{or } Q = 0.5 \times 1.0 \sqrt{\frac{9.81}{3} \left(\frac{1^3 - 0.5^3}{1^2 - 0.5^2} \right)} = 0.5 \sqrt{3.27 \left(\frac{1 - 0.125}{1 - 0.25} \right)} = \mathbf{0.977 \text{ m}^3/\text{s} \text{ (Ans.)}}$$

(ii) Froude number before and after jump, Fr_1, Fr_2 :

$$\text{Froude number, } Fr = \frac{V}{\sqrt{gD}}$$

$$\text{where, } D = \text{Hydraulic depth} = \frac{A}{T}$$

(T = top width of the channel, A = area of cross-section of the channel)

$$V_1 = \frac{Q}{y_1^2} = \frac{0.977}{0.5^2} = 3.91 \text{ m/s}$$

$$V_2 = \frac{Q}{y_2^2} = \frac{0.977}{1^2} = 0.977 \text{ m/s}$$

$$D_1 = \frac{A_1}{T} = \frac{y_1^2}{2y_1} = \frac{y_1}{2} = \frac{0.5}{2} = 0.25 \text{ m}$$

$$D_2 = \frac{A_2}{T} = \frac{y_2^2}{2y_2} = \frac{y_2}{2} = \frac{1.0}{2} = 0.5 \text{ m}$$

$$Fr_1 = \frac{V_1}{\sqrt{gD_1}} = \frac{3.91}{\sqrt{9.81 \times 0.25}} = 2.5 \text{ m/s (Ans.)}$$

$$Fr_1 = \frac{V_2}{\sqrt{gD_2}} = \frac{0.977}{\sqrt{9.81 \times 0.5}} = 0.44 \text{ (Ans.)}$$

Example 16.35. (Surges in open channels). A horizontal rectangular channel of 3 m width and 2 m water depth conveys water at $18 \text{ m}^3/\text{s}$. If the flow rate is suddenly reduced to $\frac{2}{3}$ of its original value, compute the magnitude and speed of the upstream surge.

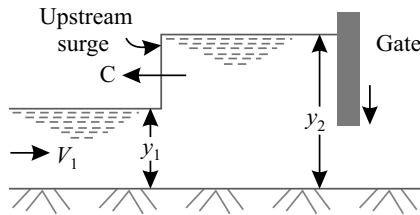
Assume that the front of the surge is rectangular and friction in the channel is neglected.

[UPSC Exams.]

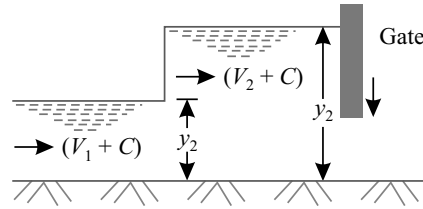
Solution. Width of channel, $b = 3 \text{ m}$

Depth of water, $y_1 = 2 \text{ m}$

The flow rate or discharge, $Q_1 = 18 \text{ m}^3/\text{s}$



(a) Observer stationary.



(b) Observer moving with the surge.

Fig. 16.32

In a channel, when discharge is suddenly reduced by operating a gate, an upstream surge will be developed which will move with a constant velocity C (also known as *celerity* of the wave) as shown in Fig. 16.32 (a). An observer standing on the canal bank will notice the surge moving upstream. This unsteady flow case can be transformed into a steady one by *superimposing* flow with velocity C in opposite direction as shown in Fig. 16.32 (b)

$$\text{Also, } by_1(V_1 + C) = by_2(V_2 + C) \quad \dots \text{Continuity equation}$$

$$\text{or, } y_1(V_1 + C) = y_2(V_2 + C)$$

$$\text{Again, } \frac{wb}{2}(y_2^2 - y_1^2) = \frac{wb}{g}y_1(V_1 + C)(V_1 - V_2) \quad \dots \text{Momentum equation}$$

$$\text{or, } (y_2^2 - y_1^2) = \frac{2y_1}{g}(V_1 + C)(V_1 - V_2) \quad \dots(i)$$

$$\text{Now, } V_1 = \frac{Q_1}{b \times y_1} = \frac{18}{3 \times 2} = 3 \text{ m/s}$$

$$Q_2 = \frac{2}{3} Q_1 \quad \dots(\text{Given})$$

$$\therefore Q_2 = \frac{2}{3} \times 18 = 12 \text{ m}^3/\text{s}$$

$$Q_2 = (b_2 \times y_2) V_2 = b_2 \times V_2 y_2$$

$$\therefore V_2 y_2 = \frac{Q_2}{b_2} = \frac{12}{3} = 4 \text{ m}^2/\text{s per m}$$

Now,

$$V_1 y_1 = V_2 y_2 + C (y_2 - y_1)$$

...Continuity equation

$$3 \times 2 = 4 + C (y_2 - 2)$$

or,

$$C = \frac{2}{y_2 - 2}$$

Substituting the values in expression (i) we have:

$$(y_2^2 - 2^2) = \frac{2 \times 2}{9.81} \left(3 + \frac{2}{y_2 - 2} \right) (3 - V_2)$$

$$(y_2^2 - 4) = 0.41 \left(3 + \frac{2}{y_2 - 2} \right) \left(3 - \frac{4}{y_2} \right) \left[\begin{array}{l} \because V_2 y_2 = 4 \\ \text{or } V_2 = 4 / y_2 \end{array} \right]$$

Solving by trial and error, $y_2 = 2.8$ m and

Height of the surge $= y_2 - y_1 = 2.8 - 2 = 0.8$ m (Ans.)

Velocity of the upstream surge, $C = \frac{2}{y_2 - 2} = \frac{2}{2.8 - 2} = 2.5$ m/s (Ans.)

16.10. GRADUALLY VARIED FLOW

Gradually varied flow (G.V.F.) is one in which the depth changes gradually over a long distance. In a rapidly varied flow, the change in depth takes place in a short distance.

Gradually varied flow may be caused due to one or more of the following factors:

1. The change in the shape and size of the channel cross-section,
2. The change in slope of the channel,
3. The presence of obstruction (e.g., weir etc.), and
4. The change in frictional forces at the boundaries.

16.10.1. Equation of Gradually Varied Flow

The following assumptions are made for analysing gradually varied flow:

1. The channel is a prismatic (a channel with constant section and alignment).
2. The bed slope is small.
3. The flow is steady and hence discharge is constant.
4. The pressure distribution over the channel section is hydrostatic i.e. stream lines are practically straight and parallel.
5. The energy correction factor (α) is unity.
6. The roughness co-efficient is constant for the length of the channel and it does not depend on the depth of flow.
7. The Chezy and Manning correlations are equally applicable to gradually varied flow for determining the slope of energy line.

Consider a rectangular channel having gradually varied flow (Fig. 16.33), the depth of flow gradually decreasing in the direction of flow.

Let,

b = Width of the channel,

Q = Discharge through the channel,

z = Height of bottom of channel above datum,

y = Depth of flow,

V = Mean velocity of flow,

$S_b = \tan i \approx i$ = slope of the channel bed, and

$S_e = \tan j \approx j$ = slope of energy line.

According to Bernoulli's equation, the energy equation at any section is given by:

$$E = z + y + \frac{V^2}{2g} \quad \dots(i)$$

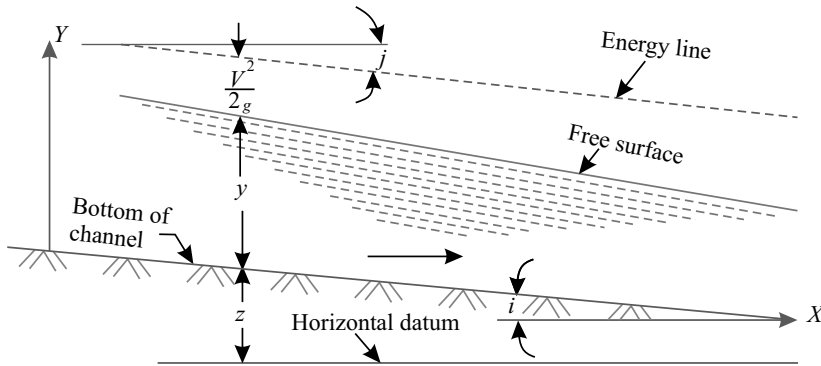


Fig. 16.33. Gradually varied flow in a channel.

Taking the bottom of the channel on the X -axis and the vertically upwards direction measured from the channel bottom, as the Y -axis, differentiation of eqn. (i), with respect to x yields:

$$\frac{dE}{dx} = \frac{dz}{dx} + \frac{dy}{dx} + \frac{d}{dx} \left(\frac{V^2}{2g} \right) \quad \dots(ii)$$

Now,

$$\begin{aligned} \frac{d}{dx} \left(\frac{V^2}{2g} \right) &= \frac{d}{dx} \left[\frac{(Q/A)^2}{2g} \right] = \frac{d}{dx} \left[\frac{\{Q/(b \cdot y)\}^2}{2g} \right] \quad \left[\because V = \frac{Q}{A} \right. \\ &\quad \left. \text{and } A = b \cdot y \right] \\ &= \frac{d}{dx} \left[\frac{Q^2}{b^2 \cdot y^2 \times 2g} \right] = \frac{Q^2}{b^2 \times 2g} \frac{d}{dx} \left(\frac{1}{y^2} \right) \quad \left[\because Q, b \text{ and } y \right. \\ &\quad \left. \text{are constant} \right] \\ &= \frac{Q^2}{b^2 \times 2g} \frac{d}{dy} \left(\frac{1}{y^2} \right) \frac{dy}{dx} \\ &= \frac{Q^2}{b^2 \times 2g} \left(-\frac{2}{y^3} \right) \frac{dy}{dx} = -\frac{2Q^2}{b^2 \times 2gy^3} \frac{dy}{dx} \\ \text{or,} \quad \frac{d}{dx} \left(\frac{V^2}{2g} \right) &= \frac{-Q^2}{b^2 \cdot y^2 \times gy} \frac{dy}{dx} = -\frac{V^2}{gy} \frac{dy}{dx} \quad \left(\because \frac{Q}{b \cdot y} = V \right) \end{aligned}$$

Substituting the value of $\frac{d}{dx} \left(\frac{V^2}{2g} \right)$ in expression (ii), we get:

$$\frac{dE}{dx} = \frac{dz}{dx} + \frac{dy}{dx} - \frac{V^2}{gy} \frac{dy}{dx}$$

or,

$$\frac{dE}{dx} = \frac{dz}{dx} + \frac{dy}{dx} \left(1 - \frac{V^2}{gy} \right) \quad \dots(iii)$$

Now,

$$\frac{dE}{dx} = \text{Slope of energy line} = -S_e$$

and,

$$\frac{dz}{dx} = \text{Slope of bed of the channel} = -S_b$$

– ve signs with S_e and S_b indicate that the values of E and z decrease with the increase of x .

Substituting the values of $\frac{dE}{dx}$ and $\frac{dz}{dx}$ in expression (iii), we get:

$$-S_e = -S_b + \frac{dy}{dx} \left(1 - \frac{V^2}{gy} \right) \quad \text{or} \quad \frac{dy}{dx} = \frac{(S_b - S_e)}{\left(1 - \frac{V^2}{gy} \right)} \quad \dots(16.42)$$

$$\text{or,} \quad \frac{dy}{dx} = \frac{(S_b - S_e)}{[1 - (Fr)^2]} \quad \left(\because \frac{V}{\sqrt{gy}} = Fr \right) \quad \dots(16.43)$$

$\frac{dy}{dx}$ represents the variation of depth along the bottom of the channel and is also called the *slope of the free water surface*.

- (i) When $\frac{dy}{dx} = 0$: y is constant (or depth of water above the bottom of channel is *constant*); it means that *free water surface is parallel to the channel bed*.
- (ii) When $\frac{dy}{dx} > 0$ (or $\frac{dy}{dx}$ is +ve) : It indicates that the depth of water increases in the direction of flow, the profile of water so obtained is called **back water curve**.
- (iii) When $\frac{dy}{dx} < 0$ or $\frac{dy}{dx}$ is -ve : It indicates that the depth of water decreases in the direction of flow. The profile of water so obtained is known as **drop down curve**.

16.10.2. Back Water Curve and Afflux

In an open channel when the flow is uniform, the flow has constant depth at all the sections and the surface of the free water lies parallel to bed of the channel. But when an obstruction like a dam, weir etc. comes across the channel width the water level rises and it has maximum depth from the bed at some section (Fig. 16.34). If y_1 is the depth of water at the point, where the water starts rising up and y_2 is the maximum height of rising water from the bed, then this increase in depth (i.e. $y_2 - y_1$) is known as '**afflux**' and the curved surface of the liquid with its *concavity upwards*, is known as '**back water curve**'.

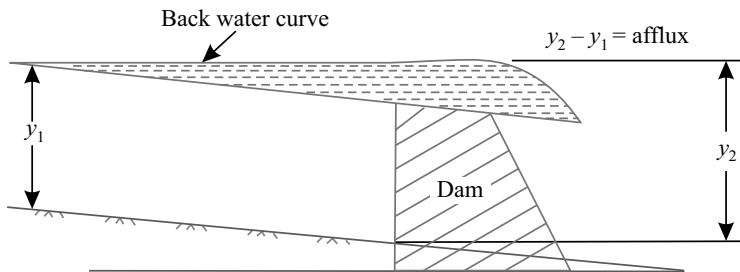


Fig. 16.34. Back water curve and afflux.

Length of back water curve:

The length of back water curve is the distance along the bed of the channel between the section where water starts rising to the section and where water has maximum depth.

Consider a channel in which a back water curve is formed as shown in Fig. 16.35. Let two sections 1-1 and 2-2 are so chosen that distance between them represents the length of backwater curve.

Let,

- y_1 = Depth of flow at section 1-1,
 V_1 = Velocity of flow at section 1-1,
 y_2 = Depth of flow at section 2-2,
 V_2 = Velocity of flow at section 2-2,
 S_b = Bed slope,
 S_e = Energy line slope, and
 l = Length of back water curve.

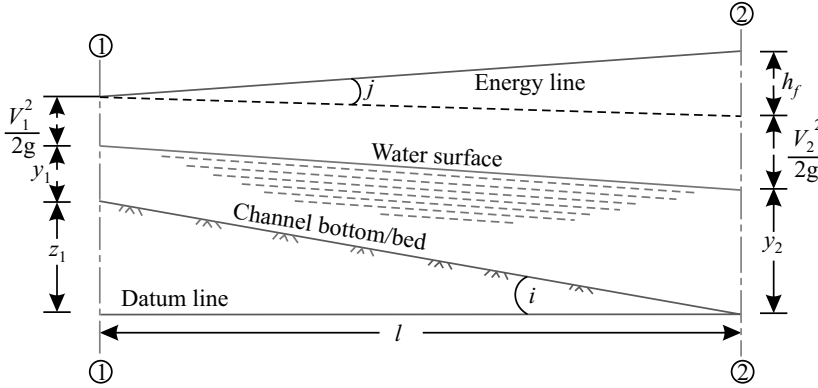


Fig. 16.35. Length of back water curve.

Applying Bernoulli's equation at the two sections with channel bed at section 2-2 as the datum for potential head, we have:

$$z_1 + y_1 + \frac{V_1^2}{2g} = y_2 + \frac{V_2^2}{2g} + h_f \quad (\because z_2 = 0)$$

where, h_f = Loss of head due to friction = $S_e \times l$, and $z_1 = S_b \times l$

$$\therefore S_b \times l + y_1 + \frac{V_1^2}{2g} = y_2 + \frac{V_2^2}{2g} + S_e \times l$$

$$\text{or, } S_b \times l - S_e \times l = \left(y_2 + \frac{V_2^2}{2g} \right) - \left(y_1 + \frac{V_1^2}{2g} \right)$$

$$\text{or, } l (S_b - S_e) = E_2 - E_1 \quad \left(\text{where } E_2 = y_2 + \frac{V_2^2}{2g}, E_1 = y_1 + \frac{V_1^2}{2g} \right)$$

$$\therefore l = \frac{E_2 - E_1}{S_b - S_e} \quad \dots(16.43)$$

where, E_1 and E_2 represent the specific energies at the beginning and end of the backwater curve. The value of S_e (slope of energy line) is determined by Manning's formula or Chezy's formula corresponding to flow conditions at mean/average depth of flow.

Example 16.36. In a rectangular channel 12 m wide and 3.6 m deep water is flowing with a velocity of 1.2 m/s. The bed slope of the channel is 1 in 4000. If flow of water through the channel is regulated in such a way that energy line is having a slope of 0.00004 find the rate of change of depth of water in the channel.

Solution. Width of channel, $b = 12$ m

Depth of the channel, $y = 3.6$ m

Velocity of flow, $V = 1.2$ m/s

Bed slope, $S_b = \frac{1}{4000} = 0.00025$

Slope of the energy line, $S_e = 0.00004$

Rate of change of depth of water, $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{S_b - S_e}{\left(1 - \frac{V^2}{gy}\right)} \quad \dots[\text{Eqn. (16.42)}]$$

Substituting the values, we get:

$$\frac{dy}{dx} = \frac{0.00025 - 0.00004}{\left(1 - \frac{1.2^2}{9.81 \times 3.6}\right)} = \frac{0.00021}{0.9592} = \mathbf{0.0002189 \text{ (Ans.)}}$$

Example 16.37. In a rectangular channel of width 24 m and depth of flow 6 m, the rate of flow of water is $86.4 \text{ m}^3/\text{s}$. If the bed slope of the channel is 1 in 4000 find the slope of the free water surface. Take Chezy's constant $C = 60$.

Solution. Width of the channel, $b = 24 \text{ m}$

Depth of flow, $y = 6 \text{ m}$

Rate of flow or discharge, $Q = 86.4 \text{ m}^3/\text{s}$

$$\text{Bed slope, } S_b = \frac{1}{4000} = 0.00025$$

Chezy's constant, $C = 60$.

Slope of the free water surface, $\frac{dy}{dx}$:

$$\text{Discharge, } Q = A \times V = A \times C \sqrt{RS_b}$$

where, $A = \text{Area of flow} = b \times y = 24 \times 6 = 144 \text{ m}^2$

$$\text{Hydraulic radius} = \frac{A}{P} = \frac{144}{b + 2y} = \frac{144}{24 + 2 \times 6} = 4 \text{ m}$$

The slope of the energy line (S_e) is determined from Chezy's formula.

$$\therefore 86.4 = 144 \times 60 \sqrt{4 \times S_e} = 17280 \sqrt{S_e} \quad [\text{Art. 16.10, point 7}]$$

or
$$S_e = \left(\frac{86.4}{17280}\right)^2 = 0.000025$$

Now,
$$\frac{dy}{dx} = \frac{S_b - S_e}{1 - \frac{V^2}{gy}} = \frac{0.00025 - 0.000025}{1 - \frac{0.6^2}{9.81 \times 6}}$$

$$\left(V = \frac{Q}{b \times y} = \frac{86.4}{24 \times 6} = 0.6 \text{ m/s}\right)$$

Hence, slope of the free water surface = **0.000226 (Ans.)**

Example 16.38. A wide channel laid to a slope of 1 in 1000 carries a discharge of $3.5 \text{ m}^3/\text{s}$ per metre width at a depth of 1.6 m. Find out the value of Chezy's constant C . Consider the flow to be uniform.

If the actual depth varies from 1.5 m at an upstream location to 1.7 m at a location 300 m downstream or in other words the flow is gradually varied, what will be the value of Chezy's coefficient C . [Roorkee University]

Solution. Bed slope of channel, $S_b = \frac{1}{1000}$

Discharge, $q = 3.5 \text{ m}^3/\text{s}$ per metre width

Depth of water, $y = 1.6 \text{ m}$

$$\therefore \text{Velocity of flow} = \frac{q}{y} = \frac{3.5}{1.6} = 2.1875 \text{ m/s}$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{b \times y}{b + 2y}$$

For a wide channel, the width b of the stream is large in comparison with depth of flow y . Therefore,

$$R \approx \frac{b \times y}{b} = y = 1.6 \text{ m}$$

(i) **Uniform flow:**

$$V = C \sqrt{RS_b}$$

...Chezy's formula

$$\therefore 2.1875 = C \sqrt{1.6 \times \frac{1}{1000}} = 0.04 C$$

$$\therefore C = \frac{2.1875}{0.04} = \mathbf{54.68 \text{ (Ans.)}}$$

(ii) **Gradually varied flow:**

$$\text{Slope of the free water surface, } \frac{dy}{dl} = \frac{1.7 - 1.5}{300} = 0.000667$$

$$\text{Average flow depth, } y = \frac{y_1 + y_2}{2} = \frac{1.7 + 1.5}{2} = 1.6 \text{ m}$$

$$\text{Velocity at average flow depth, } V = \frac{q}{y} = \frac{3.5}{1.6} = 2.1875 \text{ m/s}$$

$$\text{Hydraulic radius, } R \approx y = 1.6 \text{ m}$$

The rate of change of depth is given by:

$$\frac{dy}{dl} = \frac{S_b - S_e}{1 - \frac{V^2}{gy}}$$

$$\text{or, } 0.000667 = \frac{0.001 - S_e}{1 - \frac{2.1875^2}{9.81 \times 1.6}} = \frac{0.001 - S_e}{0.695}$$

$$\text{or, } S_e = 0.001 - 0.000667 \times 0.695 = 0.000536$$

$$\text{Now, } V = C \sqrt{RS_e} \quad \dots \text{Chezy's formula}$$

$$2.1875 = C \sqrt{1.6 \times 0.000536} = 0.0293 C$$

$$\text{or, } C = \frac{2.1875}{0.0293} = \mathbf{74.65 \text{ (Ans.)}}$$

Example 16.39. The normal depth of flow of water, in a rectangular channel 1.5 m wide, is one metre. The bed slope of the channel is 0.0006 and Manning's rugosity co-efficient N 0.012. Find the critical depth.

At a certain section of the same channel the depth is 0.92 m while at a second section the depth is 0.86 m. Find the distance between the two sections. Also find whether the section is located downstream or upstream with respect to the first section. [UPSC Exams.]

Solution. Width of the channel, $b = 1.5$ m

Normal depth of water, $y_n = 1$ m

$$\therefore \text{Area of flow, } A = b \times y_n = 1.5 \times 1 = 1.5 \text{ m}^2$$

$$\text{Perimeter, } P = b + 2y_n = 1.5 + 2 \times 1 = 3.5 \text{ m}$$

$$\therefore \text{Hydraulic radius, } R = \frac{A}{P} = \frac{1.5}{3.5} = 0.4286 \text{ m}$$

Manning's co-efficient, $N = 0.012$

Bed slope, $S_b = 0.0006$

Critical depth:

$$\text{Discharge, } Q = A \times V = A \times C \sqrt{RS_b} = A \times \frac{1}{N} R^{1/6} \sqrt{RS_b} = A \times \frac{1}{N} R^{2/3} \sqrt{S_b}$$

(where, Chezy's constant, $C = \frac{1}{N} R^{1/6}$)

$$\text{or, } Q = 1.5 \times \frac{1}{0.012} \times (0.4286)^{2/3} \times (0.0006)^{1/2} = 1.74 \text{ m}^3/\text{s}$$

$$\text{Discharge per unit width, } q = \frac{Q}{b} = \frac{1.74}{1.5} = 1.16 \text{ m}^3/\text{s per m}$$

$$\text{The critical depth, } y_c = \left(\frac{q^2}{g} \right)^{1/3} = \left[\frac{1.16^2}{9.81} \right]^{1/3} = \mathbf{0.516 \text{ m (Ans.)}}$$

Specific energy at 0.92 m depth:

$$E_1 = 0.92 + \frac{V_1^2}{2g}$$

$$\text{where, } V_1 = \frac{Q}{b \times 0.92} = \frac{1.74}{1.5 \times 0.92} = 1.26 \text{ m/s}$$

$$\therefore E_1 = \frac{1.26^2}{2 \times 9.81} = 1.0 \text{ m}$$

Specific energy at 0.86 m depth:

$$E_2 = 0.86 + \frac{V_2^2}{2g}$$

$$\text{where, } V_2 = \frac{Q}{b \times 0.86} = \frac{1.74}{1.5 \times 0.86} = 1.35 \text{ m/s}$$

$$\therefore E_2 = \frac{1.35^2}{2 \times 9.81} = 0.953 \text{ m}$$

Slope of energy line (S_e) at the mean section:

$$y = \frac{y_1 + y_2}{2} = \frac{0.92 + 0.86}{2} = 0.89 \text{ m}$$

Now,

$$Q = A \times \frac{1}{N} R^{2/3} (S_e)^{1/2} \quad \text{or} \quad Q^2 = A^2 \times \frac{R^{4/3}}{N^2} S_e$$

$\therefore$

$$S_e = \frac{Q^2 N^2}{A^2 R^{4/3}} = \frac{1.74^2 \times 0.012^2}{(1.5 \times 0.89)^2 \times (0.407)^{4/3}} = 8.11 \times 10^{-4} = 0.000811$$

$$\left(\because R = \frac{A}{P} = \frac{1.5 \times 0.89}{1.5 + 2 \times 0.89} = 0.407 \right)$$

Distance between the two sections,

$$\Delta x = \frac{E_2 - E_1}{S_b - S_e} = \frac{0.953 - 1.0}{0.0006 - 0.000811} = 222.75 \text{ m (Ans.)}$$

Slope of water surface, $\frac{dy}{dx} = \frac{S_b - S_e}{1 - \frac{V^2}{gy}}$

Average depth of flow = 0.89 m (calculated above)

Velocity at mean section, $V = \frac{Q}{1.5 \times 0.89} = \frac{1.74}{1.5 \times 0.89} = 1.3 \text{ m/s}$

$\therefore$

$$\frac{dy}{dx} = \frac{0.0006 - 0.000811}{1 - \frac{1.3^2}{9.81 \times 0.89}} = -2.616 \times 10^{-4}$$

Since $\frac{dy}{dx}$ is $-ve$, therefore, the second section is **downstream** (Ans.)

Example 16.40. (Length of backwater curve). Draw the specific energy diagram for various constant discharges and show the alternate and critical depths.

A weir is installed across a rectangular open channel thereby raising the flow depth from 1.5 m in a normal flow to 2.5 m at the weir. The width of the channel is 10 m and it is laid to a slope of 1 in 10000. Find an approximate length of the backwater curve considering the average velocity, average depth and average slope midway between the two sections. Take the value of Manning's rugosity co-efficient equal to 0.02. [Delhi University]

Solution. Upstream section 1-1:

Width of the channel, $b_1 = 10 \text{ m}$

Depth of flow, $y_1 = 1.5 \text{ m}$

$\therefore$ Area of flow, $A_1 = b_1 \times y_1 = 10 \times 1.5 = 15 \text{ m}^2$

Wetted perimeter, $P_1 = b_1 + 2y_1 = 10 + 2 \times 1.5 = 13 \text{ m}$

$\therefore$ Hydraulic radius, $R_1 = \frac{A_1}{P_1} = \frac{15}{13} = 1.154 \text{ m}$

Chezy's constant, $C_1 = \frac{1}{N} (R)^{1/6} = \frac{1}{0.02} \times (1.154)^{1/6} = 51.2$
(where, $N = 0.02$... Given)

Velocity of flow, $V_1 = C_1 \sqrt{RS_b} = 51.2 \sqrt{1.154 \times \frac{1}{10000}} = 0.55 \text{ m/s}$

(where, slope of the channel bed, $S_b = \frac{1}{10000}$... Given)

Specific energy, $E_1 = y_1 + \frac{V_1^2}{2g} = 1.5 + \frac{0.55^2}{2 \times 9.81} = 1.515 \text{ m}$

Downstream section 2-2:

Width of the channel, $b_2 = b_1 = 10$ m

Depth of flow, $y_2 = 2.5$ m

Area of flow, $A_2 = b_2 \times y_2 = 10 \times 2.5 = 25$ m²

Wetted perimeter, $P_2 = b_2 + 2y_2 = 10 + 2 \times 2.5 = 15$ m

$$\therefore \text{Hydraulic radius, } R_2 = \frac{A_2}{P_2} = \frac{25}{15} = 1.667 \text{ m}$$

Also,

$$A_1 V_1 = A_2 V_2$$

...Continuity equation

$$\therefore V_2 = \frac{A_1 V_1}{A_2} = \frac{15 \times 0.55}{25} = 0.33 \text{ m/s}$$

$$\text{Specific energy, } E_2 = y_2 + \frac{V_2^2}{2g} = 2.5 + \frac{0.33^2}{2 \times 9.81} = 2.505 \text{ m}$$

The value of S_e (slope of energy line) is calculated by Chezy's formula corresponding to flow conditions at the average depth of flow.

$$\text{Average depth of flow, } y = \frac{y_1 + y_2}{2} = \frac{1.5 + 2.5}{2} = 2 \text{ m}$$

At the average depth of flow:

Area of flow, $A = b \times y = 10 \times 2 = 20$ m²

($\because b_1 = b_2 = b = 10$ m)

Wetted perimeter, $P = b + 2y = 10 + 2 \times 2 = 14$ m

$$\therefore \text{Hydraulic radius, } R = \frac{A}{P} = \frac{20}{14} = 1.428 \text{ m}$$

Again,

$$AV = A_1 V_1$$

$$\therefore \text{Velocity of flow, } V = \frac{A_1 V_1}{A} = \frac{15 \times 0.55}{20} = 0.4125 \text{ m/s}$$

$$\text{Chezy's constant, } C = \frac{1}{N} (R)^{1/6} = \frac{1}{0.02} \times (1.428)^{1/6} = 53.06$$

$$\text{Velocity, } V = C \sqrt{R \times S_e}$$

$$\text{or, } 0.4125 = 53.06 \sqrt{1.428 \times S_e} = 63.4 \sqrt{S_e}$$

$$\text{or, } S_e = \left(\frac{0.4125}{63.4} \right)^2 = 0.0000423$$

Length of back water curve,

$$l = \frac{E_2 - E_1}{S_b - S_e}, \text{ where } S_b \text{ is the slope of channel bed}$$

$$= \frac{2.505 - 1.515}{0.0001 - 0.0000423} = 17157 \text{ or } \mathbf{17.157 \text{ km (Ans.)}}$$

Example 16.41. (Back water curve). A river 45 m wide has a normal depth of flow of 3 m and an average bed slope of 1 in 10000. A weir is built across the river raising the water surface level at the weir site to 5 m above the bottom of the river. Assuming that the back water curve is an arc of circle, calculate the approximate length of the backwater curve. Consider that the river is prismatic. Take the value of N in Manning's formula as 0.025. [UPSC Exams.]

Solution. Width of the bed, $b = 45$ m

Depth of flow (normal), $y_n = 3$ m

$$\text{Average bed slope, } S_b = \frac{1}{10000} = 0.0001$$

$$\text{Depth of flow at weir site, } y = 5 \text{ m}$$

$$\text{Manning's co-efficient, } N = 0.025$$

$$\text{Afflux, } h = y - y_n = 5 - 3 = 2 \text{ m}$$

Length of back water curve, l :

Length of backwater curve, *by circular arc method*, is given as:

$$l = \frac{2h}{dy/dx} \quad \dots(i)$$

$$\text{Area of flow, } A = 45 \times 3 = 135 \text{ m}^2$$

$$\text{Perimeter, } P = 45 + 2 \times 3 = 51 \text{ m}$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{135}{51} = 2.65 \text{ m}$$

$$\text{Discharge, } Q = A \times V = A \times C \sqrt{RS_b} = A \times \frac{1}{N} R^{1/6} \sqrt{RS_b} = A \times \frac{1}{N} R^{2/3} S_b^{1/2}$$

$$(\text{where, Chezy's constant, } C = \frac{1}{N} \times R^{1/6})$$

$$\text{or, } Q = 135 \times \frac{1}{0.025} \times (2.65)^{2/3} \times (0.0001)^{1/2} = 103.4 \text{ m}^3/\text{s}$$

At the weir site:

$$y = 5 \text{ m, } V = \frac{Q}{45 \times 5} = \frac{103.4}{45 \times 5} = 0.46 \text{ m/s}$$

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{45 \times 5}{45 + (2 \times 5)} = 4.09 \text{ m}$$

Slope of water surface at the weir,

$$\frac{dy}{dx} = \frac{S_b - S_e}{1 - \frac{V^2}{gy}}$$

where, S_e is the slope of the total energy line at the weir, V and y are the velocity and depth of flow respectively at the weir.

$$S_e = \frac{Q^2 N^2}{A^2 R^{4/3}} \quad (\text{Refer to example 16.39})$$

$$= \frac{V^2 N^2}{R^{4/3}} = \frac{0.46^2 \times 0.025^2}{(4.09)^{4/3}} = 2.02 \times 10^{-5} = 0.0000202$$

$$\frac{V^2}{gy} = \frac{0.46^2}{9.81 \times 5} = 0.0043$$

$$\therefore \frac{dy}{dx} = \frac{0.0001 - 0.0000202}{1 - 0.0043}$$

Substituting the value of $\frac{dy}{dx}$ in expression (i), we have:

$$l = \frac{2 \times 2}{0.00008} = 50000 \quad \text{or} \quad \mathbf{50 \text{ km (Ans.)}}$$

16.11. MEASUREMENT OF FLOW OF IRREGULAR CHANNELS

The term “irregular channels” includes *large rivers* and *small streams*. In case of a small stream it is possible to obtain the quantity of flow by fitting a notch or weir across the stream; the discharge may then be calculated by measuring the head over the notch. However, this method cannot be employed for large rivers on account of the expense and the obstruction which may be caused to navigation. In order to obtain the discharge through a large river (or irregular channel), we require: (i) Area of flow, and (ii) Mean velocity of flow. By knowing this data discharge is calculated as follows:

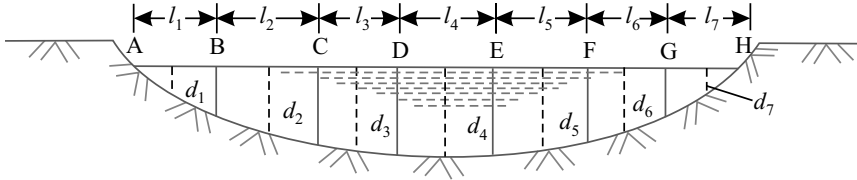


Fig. 16.36. Cross-section of river with unequal segments (Segments method).

16.11.1. Area of Flow

The area of flow may be *calculated* by the several methods but the following are important ones:

1. Simple segments method.
2. Simpson's rule.

1. Simple segments method :

In this method, the cross-section of the river is divided into a number of segments *AB*, *BC*, *CD*, etc. as shown in Fig. 16.36.

Let, $l_1, l_2, l_3 \dots$ = Lengths of the segments *AB*, *BC*, *CD* ... respectively, and
 $d_1, d_2, d_3 \dots$ = Mean depths of the respective segment.

Then, Area of flow, A = Area of segment *AB* + area of segment *BC* + area of segment *CD* + ...
 $= l_1 d_1 + l_2 d_2 + l_3 d_3 + \dots$

2. Simpson's rule:

A greater accuracy in the computation of discharge may be obtained by using *Simpson's rule*. In this method the whole river width is divided into an *even number of equal segments* so that there are *odd number of depths* taken at the end of each segment as shown in Fig. 16.28. Then area of flow,

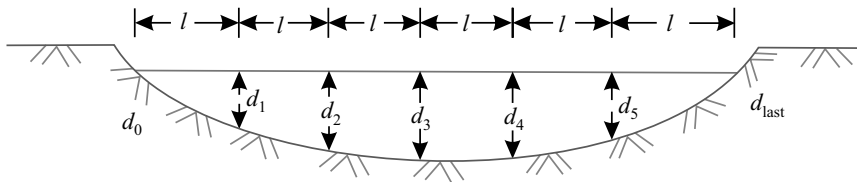


Fig. 16.37. Cross-section of river with equal segments (Simpson's) rule.

$$A = \frac{l}{3} (d_0 + d_{last}) + 2 (d_1 + d_3 + d_5) + 4 (d_2 + d_4 + d_6)$$

where,

l = Length of each segment, and

$d_0, d_1, d_2 \dots$ = Depths taken at the end of segments.

16.11.2. Mean Velocity of Flow

The mean velocity of flow may be *measured* by the following methods :

1. Pitot tube
2. Floats
3. Current meter.

1. **Pitot tube.** A pitot tube is a simple device used for measuring the velocity of flow at the required point in the flowing stream. In its simplest form it consists of a glass tube (large enough for capillary effects to be negligible) bent at *right angle*. The tube is dipped vertically in the flowing stream with its lower open end facing the direction of flow and upper open end projecting above the water surface in the stream as shown in the Fig. 16.38. The water rises up in the tube, due to pressure exerted by the flowing water. By measuring the rise of water in the tube, the velocity of water (V) is calculated by using the following relation:

$$V = \sqrt{2gh}$$

where,

h = Height of water in the tube above the water surface, and

g = Acceleration due to gravity.

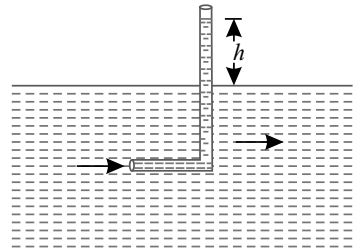


Fig. 16.38. Pitot tube.

2. **Floats.** The velocity of flow can be measured in a simple way by means of *floats*. A float is a small object made of wood or other suitable material which is lighter than water and thus capable of floating on the surface. The *surface velocity at any section may be obtained by using a single float*. The time taken by the float to traverse a known distance is measured; the velocity is then calculated by dividing the distance travelled (by the float) by the time taken to travel that distance. Since the mean velocity of flow is equal to 0.8 to 0.95 times the surface velocity, the approximate value of the mean velocity of flow may then be determined from the known value of the surface velocity.

A better method is to use *double floats*. A double float consists of a *surface float* on to which is attached a *hollow metal sphere*, heavier than water, and suspended from it by a cord of known length (Fig. 16.39). The depth of the lower float may be regulated by the length of the cord. The velocity is then obtained by noting the time taken by the float to traverse a known distance (as explained in the previous case of single float)

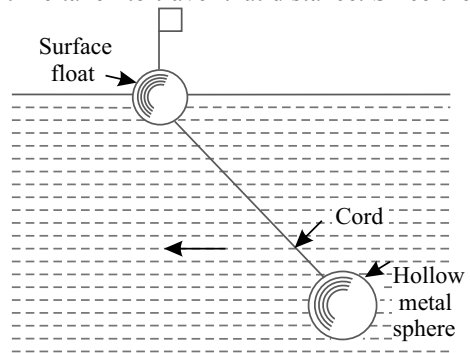


Fig. 16.39. Double float.

The double float method *directly* gives the value of mean velocity of flow.

The *best type of float* is the *rod float* (Fig. 16.40). It consists of a vertical wooden rod which is weighted at the bottom to keep it vertical. The length of the rod is so adjusted that it reaches the bottom of stream (without touching the weeds, sand or mud at the bottom of the river) and its top should be above the water surface. Some types of rod are made telescopic, so that length may be adjusted to suit any depth. *The rod will travel with a velocity equal to the mean velocity of the section.*

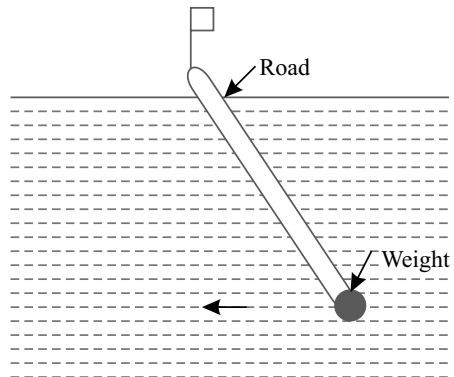


Fig. 16.40. Rod float.

3. **Current meter.** A current meter is an instrument used to measure the velocity of flow at a required point in the flowing stream. In general it consists of a *wheel or revolving element containing blades or cups*, and a *tail on which flat vanes or fins are fixed*. The current meters, according to the shape of the revolving element, may be classified as follows:

(i) Cup type

(ii) Screw type or propeller type.

In a *Cup type current meter* (Fig. 16.41) the wheel or revolving element has the form of a series of conical cups, mounted on a spindle. The spindle is held vertical at right angle to the direction of flow.

In a *screw or propeller type current meter* (Fig. 16.42) the revolving element consists of a shaft, with its axis parallel to the direction of flow, which carries a number of curved vanes (or propeller blades) mounted on the periphery of the shaft. This type of meter is more sensitive than cup type because it gives higher r.p.m. for the same velocity of flow.

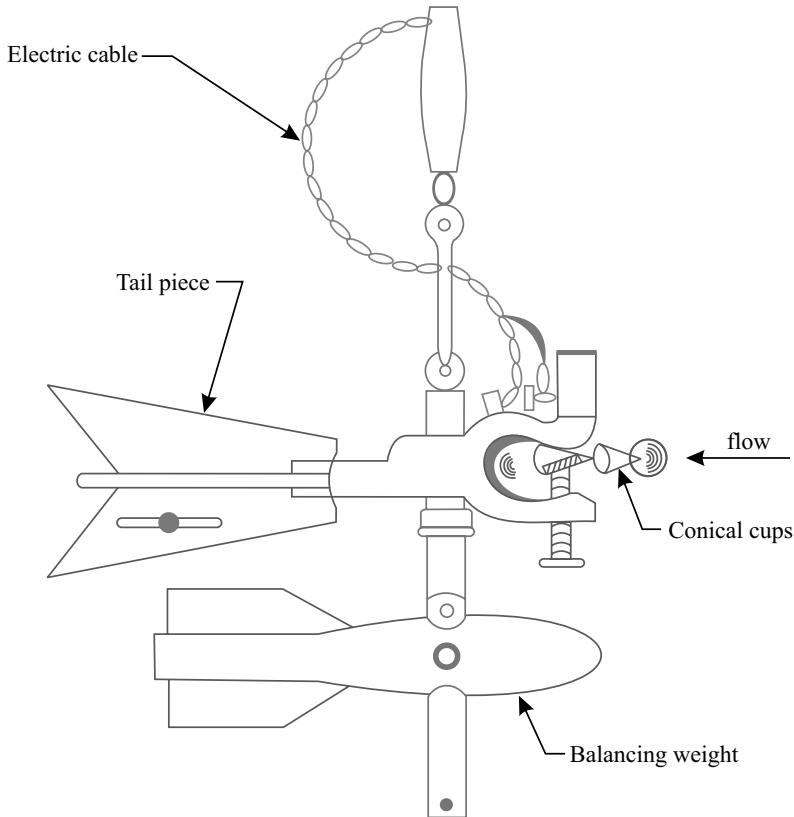


Fig. 16.41. Cup type current meter.

In order to measure the velocity of flow, meter is submerged under water and motion of water in the stream activates it, driving the wheel (or rotatory elements) at a *speed proportional to the velocity of flow*. An electric current is passed from the battery to the wheel by means of wire. The rotation of wheel makes and breaks the electric circuit, which causes an electric bell to ring. Thus by counting the ringing of bell, the rotations of the wheel and hence the velocity of flowing water is obtained.

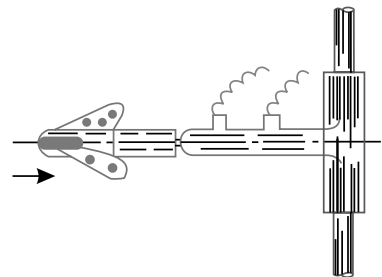


Fig. 16.42. Screw or propeller type current meter.

HIGHLIGHTS

1. An *open channel* may be defined as a passage in which liquid flows with its upper surface exposed to atmosphere.
2. Flow in a channel is said to be *uniform*, if the depth, slope, cross-section and velocity remain constant over a given length of the channel. Flow in a channel is said to be *non-uniform* (or varied) when the channel depth *varies* continuously from one section to another.
3. The flow in the open channel may be characterised as laminar or turbulent depending upon the value of Reynolds number:
When $Re < 500$...flow is laminar;
When $Re > 2000$...flow is turbulent.
When $500 < Re < 2000$...flow is transitional.
4. If Froude number (Fr) is less than 1.0, the flow is subcritical or streaming. If Fr is equal to 1.0, the flow is critical. If Fr is greater than 1.0, the flow is supercritical or shooting.
5. Velocity by Chezy's formula is given by

$$V = C\sqrt{RS}$$

where,

C = Chezy's constant,

R = Hydraulic radius (or hydraulic mean depth)

$$= \frac{A \text{ (area)}}{P \text{ (wetted perimeter)}}, \text{ and}$$

S = Slope of the bed.

6. Empirical relations for the Chezy's constant, C

$$(i) \quad C = \frac{157.6}{1.81 + \frac{K}{\sqrt{R}}} \quad \dots \text{Bazin's formula}$$

where,

K = Bazin's constant,

R = Hydraulic radius (or hydraulic mean depth)

$$(ii) \quad C = \frac{23 + \frac{0.00155}{S} + \frac{1}{N}}{1 \left(23 + \frac{0.00155}{S} \right) \frac{N}{\sqrt{R}}} \quad \dots \text{Kutter's formula}$$

where,

N = Kutter's constant, and S = bed slope.

$$(iii) \quad C = \frac{1}{N} R^{1/6} \quad \dots \text{Manning's formula}$$

where,

N = Manning's constant = Kutter's constant.

7. The *most economical section* (also called the best section or most efficient section) is one which gives the maximum discharge for a given amount of excavation.
8. *Conditions for maximum discharge* through different channel sections:

(a) *Rectangular section:*

$$(i) \quad b = 2y; \quad (ii) \quad R = \frac{y}{2}$$

(b) *Trapezoidal section:*

(i) Half top width = Sloping side

$$\text{or,} \quad \frac{b + 2ny}{2} = y\sqrt{n^2 + 1}$$

$$(ii) \quad R = \frac{y}{2}$$

- (iii) A semicircle drawn from the mid-point of the top width with radius equal to depth of flow will touch the three sides of the channel. *Best side slope* for most economical trapezoidal section is

$$\theta = 60^\circ \quad \text{or} \quad n = \frac{1}{\sqrt{3}} = \frac{1}{\tan \theta}$$

- (c) *Triangular section:*

- (i) Each sloping side makes an angle of 45° with the vertical.

(ii) Hydraulic radius, $R = \frac{y}{2\sqrt{2}}$.

- (d) *Circular section:*

- (i) Condition for *maximum discharge*:

Depth of flow, $y = 0.95$ diameter of circular channel;

Hydraulic radius, $R = 0.29$ times channel diameter.

- (ii) Condition for *maximum velocity*:

Depth of flow, $y = 0.81$ diameter of circular channel;

Hydraulic radius, $R = 0.305$ diameter.

9. For a *circular channel*:

$$\text{Area of flow, } A = r^2 \left(\theta - \frac{\sin 2\theta}{2} \right)$$

$$\text{Wetted perimeter, } P = 2r \theta$$

where,

r = Radius of circular channel, and

θ = Half the angle subtended by the water surface at the centre.

10. Channel sections of constant velocity are designed particularly in the case of large sewers in which the discharge ranges from a certain minimum value that flows daily to a very large value during rainy season.

11. The total energy of flow per unit weight of liquid is given by:

$$\text{Total energy} = z + y + \frac{V^2}{2g}$$

12. Specific energy of a flowing liquid per unit weight,

$$E = y + \frac{V^2}{2g}$$

13. The depth of flow at which specific energy is minimum is called *critical depth*, which is

given by $y_c = \left(\frac{q^2}{g} \right)^{1/3}$, where q = discharge per unit width.

14. The velocity of flow at critical depth is known as *critical velocity*, which is given by:

$$V_c = \sqrt{g \times y_c}$$

15. Minimum specific energy is given by:

$$E_{\min} = \frac{3}{2} y_c, \quad \text{where } y_c = \text{critical depth.}$$

16. (i) A flow corresponding to critical depth (or when Froude number, $Fr = 1$) is known as *critical flow*.

- (ii) When the depth of flow in a channel is greater than critical depth (when $Fr < 1$) the flow is said to be *sub-critical* or *streaming flow*.
- (iii) The flow is *supercritical* (or shooting or torrential) when the depth of flow in a channel is less than the critical depth (when $Fr > 1$).
17. The condition for maximum discharge for given value of specific energy is that the depth of flow should be *critical*.
18. **Hydraulic jump.** In an open channel when rapidly flowing stream abruptly changes to slowly flowing stream, a distinct rise or jump in the elevation of liquid surface takes place, this phenomenon is known as *hydraulic jump*. The hydraulic jump is also known as '*standing wave*'. The depth of flow after the jump is given by:

$$y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2q^2}{2gy_1}} \quad \dots(\text{in terms of } q)$$

$$= -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2V_1^2 y_1}{g}} \quad \dots(\text{in terms of } V_1)$$

$$= \frac{y_1}{2} (\sqrt{1 + 8Fr_1^2} - 1) \quad \dots(\text{in terms of } Fr_1)$$

(where, y_1 = depth of flow of water before the jump)

Height of hydraulic jump, $H_j = y_2 - y_1$

Length of hydraulic jump, $L_j = 5 \text{ to } 7 H_j$

Loss of energy due to hydraulic jump, $E_L = \frac{(y_2 - y_1)^3}{4y_1 y_2}$

19. **Gradually varied flow** (G.V.F.) is one in which the depth changes gradually over a long distance. Equation of gradually varied flow is given by:

$$\frac{dy}{dx} = \frac{S_b - S_e}{\left(1 - \frac{V^2}{gy}\right)} \quad \dots(\text{in terms of } V)$$

$$= \frac{S_b - S_e}{(1 - Fr^2)} \quad \dots(\text{in terms of } Fr)$$

where, $\frac{dy}{dx}$ = Slope of free water surface,

S_b = Slope of the channel bed,

S_e = Slope of the energy line, and

V = Velocity of flow.

20. **Afflux** is the increase in water level due to some obstruction across the flowing liquid; the curved surface of the liquid with its concavity upwards, is known as **back water curve**.

Length of back water curve, $l = \frac{E_2 - E_1}{S_b - S_e}$

where, $E_1 \left(= y_1 + \frac{V_1^2}{2g} \right)$ and $E_2 \left(= y_2 + \frac{V_2^2}{2g} \right)$ represent the specific energies at the beginning and end of back water curve.

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer:

1. Which of the following is the most essential condition for a hydraulic jump to form?
 - (a) The constancy of specific energy
 - (b) The existence of subcritical flow before the jump
 - (c) The existence of supercritical flow before the jump
 - (d) None of the above.
2. In a hydraulic jump the energy loss is expressed as
 - (a) $\frac{(y_2 - y_1)^2}{4y_1y_2}$
 - (b) $\frac{(y_2 - y_1)^3}{4y_1y_2}$
 - (c) $\frac{(y_2 - y_1)}{4y_1y_2}$
 - (d) $\frac{\sqrt{(y_2 - y_1)}}{4y_1y_2}$.
3. The water surface slope $\frac{dy}{dx}$, in case of uniform flow in the channel, is equal to
 - (a) 0
 - (b) 1
 - (c) 1000
 - (d) ∞ .
4. In open channels, gradually varied flow is caused
 - (a) when the channel slope is equal to the normal slope
 - (b) when the pressure forces and the change of momentum are different from each other
 - (c) when the force causing the flow is not equal to the resistance force
 - (d) when there is an equilibrium between the forces causing the flow and those opposing it.
5. In a channel, the alternate depths of flow are the depths
 - (a) which occur at the same specific energy
 - (b) at which total energies are same
 - (c) for the same specific force
 - (d) none of the above.
6. The critical depth is the depth of flow at which
 - (a) the Froude number is less than unity
 - (b) the specific energy is maximum
 - (c) the specific energy is minimum
 - (d) the unit discharge is minimum.
7. In a rectangular channel, the critical depth is given by
 - (a) $\left(\frac{q^2}{g}\right)^{1/2}$
 - (b) $\left(\frac{q^2}{g}\right)^{1/3}$
 - (c) $\left(\frac{q^2}{g}\right)^{1/4}$
 - (d) $\left(\frac{q^3}{g}\right)^{1/3}$.
8. In open channels, the specific energy is
 - (a) the total energy per unit discharge
 - (b) the total energy measured above a horizontal datum
 - (c) the total energy measured with respect to the channel bottom which is taken as datum
 - (d) the kinetic energy plotted above the free-surface.
9. An open channel flow is one in which
 - (a) the solid boundaries confining the flow are open at the top
 - (b) the liquid flowing in a closed conduit has a free-surface
 - (c) a closed conduit is full of flowing liquid
 - (d) none of the above.
10. In case of open channels, uniform flow is characterised by
 - (a) a constant slope of channel bottom
 - (b) a constant depth of flow
 - (c) a changing depth of flow
 - (d) none of the above.
11. For flow in open channels, the Manning's equation is expressed as
 - (a) $C = \frac{1}{N} R^{1/4}$
 - (b) $V = C \sqrt{RS}$
 - (c) $V = \frac{1}{N} R^{2/3} S^{1/2}$
 - (d) $C = 87/(1 + m/\sqrt{R})$.
12. The maximum velocity in open channels occurs
 - (a) near the channel bottom
 - (b) a little below the free-surface
 - (c) at the free surface
 - (d) none of the above.
13. Under which of the following conditions steady non-uniform flow in open channels occurs?
 - (a) When the discharge and depth both vary along the channel length
 - (b) When a constant discharge flows at the constant depth of flow
 - (c) When a constant discharge flows in a channel laid at a fixed slope
 - (d) When for a constant discharge the liquid depth in the channel varies along its length.
14. Hydraulically efficient channel cross-section is one

- (a) which carries maximum discharge under given conditions of slope, roughness and flow area
 (b) which has the minimum hydraulic radius
 (c) which has the maximum wetted perimeter
 (d) none of the above.
15. The equation of gradually varied flow is expressed by
 (a) $\frac{dy}{dx} = \frac{S_b - S_e}{1 - Fr^2}$ (b) $\frac{dy}{dx} = \frac{S_b - S_e}{1 + Fr^2}$
 (c) $\frac{dy}{dx} = \frac{S_b - S_e}{1 - Fr^3}$ (d) $\frac{dy}{dx} = \frac{S_b - S_e}{1 - \sqrt{Fr}}$.
16. A channel without any cover at the top is known as
 (a) natural channel (b) open channel
 (c) artificial channel (d) none of above.
17. A channel with constant bed slope and the same cross-section along its length is known as
 (a) natural channel
 (b) artificial channel
 (c) prismatic channel
 (d) open channel.
18. The flow in the open channel may be characterised as laminar when
 (a) $Re < 500$ (b) $Re > 2000$
 (c) $500 < Re < 2000$ (d) none of the above.
19. The channel flow is subcritical when
 (a) $Fr < 1$ (b) $Fr = 1$
 (c) $Fr > 1$ (d) any of the above.
20. Non-uniform flow may be caused by
 (a) the change in width, depth, bed slope etc. of a channel
 (b) An obstruction, constructed across a channel of uniform width
 (c) both (a) and (b)
 (d) none of the above.
21. Prismatic channels are those which have
 (a) a constant bed slope downstream
 (b) the same cross-section and bed slope throughout
 (c) the shape of a prism
 (d) a uniform cross-section throughout.
22. For the best trapezoidal section
 (a) depth of flow = half the bed width
 (b) side slope is 45°
 (c) the shape is of a half hexagon
 (d) none of the above.
23. For the best rectangular section
 (a) $y = b/3$ (b) $y = b$
 (c) $y = \frac{b}{2}$ (d) $y = \frac{b}{4}$
24. Manning and Chezy formulae are valid for
 (a) steady
 (b) steady uniform flow
 (c) steady non-uniform flow
 (d) unsteady uniform flow.
25. The hydraulic jump occurs in a channel when
 (a) the bed slope is adverse
 (b) the bed slope changes from steep to mild
 (c) flow changes from subcritical to super-critical
 (d) none of the above.
26. In an open channel flow, shooting flow cannot
 (a) occur just after a hydraulic jump
 (b) be gradually varied
 (c) follow tranquil flow
 (d) none of the above.
27. When Froude's number is equal to unity, the flow in an open channel is called
 (a) critical flow (b) tranquil flow
 (c) streaming flow (d) shooting flow.
28. The cross-section of a channel is said to be best, if the
 (a) hydraulic mean depth is maximum
 (b) section has the least perimeter for a given area
 (c) roughness co-efficient is maximum
 (d) section gives maximum area for a given flow.
29. The strength of a jump is governed by the
 (a) upstream velocity
 (b) downstream velocity
 (c) upstream Froude number
 (d) bed slope.
30. For maximum discharge through a circular channel, the depth of flow should be equal to
 (a) 0.6 times the diameter of the channel
 (b) 0.8 times the diameter of the channel
 (c) 0.95 times the diameter of the channel
 (d) 1.2 times the diameter of the channel.

ANSWER

- | | | | | | |
|---------|---------|---------|---------|---------|----------|
| 1. (c) | 2. (b) | 3. (a) | 4. (c) | 5. (a) | 6. (c) |
| 7. (b) | 8. (c) | 9. (b) | 10. (b) | 11. (c) | 12. (b) |
| 13. (d) | 14. (a) | 15. (a) | 16. (b) | 17. (c) | 18. (a) |
| 19. (a) | 20. (c) | 21. (b) | 22. (c) | 23. (c) | 24. (a) |
| 25. (b) | 26. (a) | 27. (a) | 28. (b) | 29. (c) | 30. (c). |

THEORETICAL QUESTIONS

- What is an open channel ?
- What are the different types of channels? Give examples in each case.
- What is the purpose of providing bed slope in open channels?
- Explain briefly the following:
 - Uniform and non-uniform flows,
 - Laminar and turbulent flows,
 - Steady and unsteady flows, and
 - Subcritical and supercritical flows.
- State the conditions under which uniform and non-uniform flows are produced.
- Differentiate between 'Gradually varied flow' and 'Rapidly varied flow'.
- What is Chezy's formula? How is it derived ?
[UPSC]
- How does the roughness of channel affect the Chezy's constant? [BHU]
- What is Bazin's formula and how is it used ?
[UPSC]
- State the following formulae for the values of C:
 - Bazin's formula, (ii) Kutter's formula, and
 - Manning's formula.
- What do you mean by 'Most-economical section' of an open channel? How is it determined?
- What are the conditions for the rectangular channel of best section?
- Show that the hydraulic mean depth of a trapezoidal channel having the best proportion is half of the minimum depth. [IIT Kharagpur]
- Define the following terms :
 - Hydraulic radius, (ii) Wetted perimeter, and
 - Slope of the bed
- For a trapezoidal channel of most economical section, prove that:
 - Half of top width = Length of one of the sloping sides;
 - Hydraulic mean depth = $\frac{1}{2} \times$ depth of flow.
- State and prove the condition under which the trapezoidal section of an open channel will be most economical.
- State and prove the conditions of maximum discharge and maximum velocity for circular channel.
- What is a specific energy curve?
- What do you understand by critical depth of an open channel when the flow in it is not uniform?
[IIT Kharagpur]
- Derive expressions for critical depth and critical velocity?
- What is meant by 'Hydraulic jump' in an open channel?
 - Determine from first principle the conditions required for the formation of such a jump in the case of a rectangular channel of constant width and calculate the loss of head in terms of depth just before and after the jump. [UPSC]
- Derive an expression for loss of energy head for a hydraulic jump.
- Define the terms: (i) Afflux and (ii) Back water curve.
Prove that the length of the backwater curve is given by

$$l = \frac{E_2 - E_1}{S_b - S_e}$$
 where, l = length of backwater curve, E_2 = specific energy at the end of backwater curve, E_1 = specific energy at the section where water starts rising, S_b = slope of bed, and S_e = slope of energy gradient.

UNSOLVED EXAMPLES

1. Find the rate of flow and conveyance for a rectangular channel 5.0 m wide for uniform flow at a depth of 1.5 m. The channel is having bed slope of 1 in 1000. Take Chezy's constant $C = 50$. Also state whether the flow is tranquil or rapid.

[Ans. $11.48 \text{ m}^3/\text{s}$; 363.09 , tranquil]

2. A flow of water of 100 litres per second flows down in a rectangular flume of width 60 cm having adjustable bottom slope. If the Chezy's constant C is 56, find the bottom slope necessary for uniform flow with a depth of flow of 30 cm. Also find the conveyance K of the flume.

[Ans. $\frac{1}{1524}$; $3.9 \text{ m}^3/\text{s}$]

3. A triangular gutter, whose sides include an angle of 60° , conveys water at a uniform depth of 4 m. If the slope of the bed is 1 in 1000 find the rate of flow of water. Take Chezy's constant $C = 55$.

[Ans. $16.066 \text{ m}^3/\text{s}$]

4. Calculate the discharge of water in such a channel having semicircular bottom of 3 m diameter and two sides as vertical when the depth of flow is 2.7 m. Take Chezy's constant equal to 60 and slope of the bed as 1 in 2000. [Ans. $9.585 \text{ m}^3/\text{s}$]

5. Determine the discharge through a trapezoidal channel of width of 8 m and side slopes of 1 horizontal to 3 vertical. The depth of flow of water is 2.4 m and the slope of the bed is 1 in 4000. Take Chezy's constant $C = 60$.

[Ans. $25.47 \text{ m}^3/\text{s}$]

6. Find the most economical cross-section of a rectangular channel which is to be dug in the rocky portion of a soil. The channel is to convey $8 \text{ m}^3/\text{s}$ of water with an average velocity of 2 m/s. Take Chezy's constant $C = 65$.

[Ans. $b = 2.828 \text{ m}$; $y = 1.414 \text{ m}$, $S = \frac{1}{746}$]

7. Determine the most economical section of a rectangular channel carrying water at the rate of $0.4 \text{ m}^3/\text{s}$; The bed slope of the channel being 1 in 2000. Take Chezy's constant $C = 50$.

[Ans. $b = 1.154 \text{ m}$; $y = 0.577 \text{ m}$]

8. A trapezoidal channel carries a discharge of $2.5 \text{ m}^3/\text{s}$. Design the section if the slope is 1 in 1200 and the side slopes are 1 in 1. Use Chezy's formula, $C = 55$. [Ans. $b = 0.9 \text{ m}$; $y = 1.085 \text{ m}$]

9. Determine the dimensions of the most economical trapezoidal earth-lined channel (Manning's $N = 0.020$) to carry $14 \text{ m}^3/\text{s}$ at a slope of 1 in 2500. [Ans. $b = 2.98 \text{ m}$; $y = 2.58 \text{ m}$]

[Hint. Take $n = \frac{1}{\sqrt{3}}$]

10. A trapezoidal channel has side slopes of 1 horizontal to 2 vertical and the slope of the bed is 1 in 1500. The area of the section is 40 m^2 . If Chezy's constant $C = 60$, determine:

(i) The dimensions of the section if it is most economical, and

(ii) Discharge of the most economical section.

[Ans. (i) $b = 5.93 \text{ m}$; $y = 4.8 \text{ m}$ (ii) $96 \text{ m}^3/\text{s}$]

11. Design a trapezoidal channel for carrying $30 \text{ m}^3/\text{s}$ of water. The bed slope of the channel is 1:1800 and side slope of horizontal to 1 vertical. Assume C in Chezy's formula as 50.

[Rajputana University]

[Ans. $b = 1.872 \text{ m}$; $y = 3.12 \text{ m}$]

12. A concrete lined circular channel of 3 m diameter has a bed slope of 1 in 500. Determine the velocity and the flow rate for the condition of (i) maximum velocity, and (ii) maximum discharge. Take Chezy's constant $C = 50$.

[Ans. (i) 2.13 m/s , $13.11 \text{ m}^3/\text{s}$

(ii) 2.073 m/s ; $14.37 \text{ m}^3/\text{s}$]

13. A 3 m wide rectangular channel conveys $12 \text{ m}^3/\text{s}$ of water at a depth of 2 m. Calculate:

(i) Specific energy of flowing fluid;

(ii) Critical depth, critical velocity and the minimum specific energy;

(iii) Froude number and state whether flow is subcritical or supercritical.

[Ans. (i) 2.2038 m ; (ii) 1.177 m ; 3.398 m/s ; 1.765 m ; (iii) 0.453 ; subcritical]

14. Calculate the specific energy of $12 \text{ m}^3/\text{s}$ of water flowing with a velocity of 1.5 m/s in a rectangular channel 7.5 m wide. Find the depth of water in the channel when the specific energy would be minimum. What would be the value of critical velocity as well as minimum specific energy?

[Ans. 1.1825 m , 0.639 m ; 2.5 m/s ; 0.0958 m]

15. A 3.6 m wide rectangular channel carries water to a depth of 1.8 m. In order to measure the discharge, the channel width is reduced to 2.4 m and a hump of 0.30 m height is provided at the bottom. Calculate the discharge if water surface in the contracted section drops by 0.15 m. Assume no losses.

[UPSC, CES Exams.]

[Ans. $6.418 \text{ m}^3/\text{s}$]

16. A control sluice spanning the entry of a 4.0 m wide rectangular channel having a mild slope admits water at the rate of $16.0 \text{ m}^3/\text{s}$ and at a velocity of 3.0 m/s . Find whether a hydraulic jump is expected in the channel downstream from the sluice. [UPSC Exams.]
[Ans. Jump not expected.]
17. Water flows at the rate of $1 \text{ m}^3/\text{s}$ along a channel of rectangular section, 1.75 metres in width. Calculate the critical depth. If a hydraulic jump is formed at a point where the upstream depth is 0.25 m, what would be the rise in water level and power lost in the jump? [IIT Delhi]
[Ans. 0.322 m; 0.158 m; 90.15 W]
18. A hydraulic jump occurs in a rectangular channel and the depths of flow before and after the jump are 0.5 m and 2.0 m respectively. Calculate the critical depth of flow. (Roorkee University)
[Ans. 1.077 m]
19. In a rectangular channel, 10 m wide and 3 m deep, water is flowing with a velocity of 1 m/s . The bed slope of the channel is 1 in 4000. If flow of water through the channel is regulated in such a way that energy line is having a slope of 0.00004 find the rate of change of depth of water in the channel. [Ans. 0.000217]
20. In a rectangular channel of width 20 m and depth of flow 5 m, the rate of flow of water is $50 \text{ m}^3/\text{s}$. If the bed slope of the channel is 1 in 4000 find the slope of the free water surface. Take Chezy's constant $C = 60$.
21. A concrete lined rectangular channel 5.5 m wide carries water at a rate of $10 \text{ m}^3/\text{s}$. Calculate the critical depth, critical velocity and the corresponding minimum specific energy. Would the flow be subcritical or supercritical at a point where the flow depth is 0.5 m. Also find the slope of free water surface at this point if the channel bed is having a slope of 1 in 2000. Take Manning's constant $N = 0.01$.
[Ans. 0.696 m; 2.613 m/s; 1.044 m; supercritical; 0.00216]
22. Find the length of the backwater curve caused by an afflux of 2.0 m, in a rectangular channel of width 40 m and depth 2.5 m. The slope of the bed is given as 1 in 11000. Take Manning's $N = 0.03$.
Hint : Afflux = $y_2 - y_1 = 2.0 \text{ m}$; $y_1 = 2.5 \text{ m}$
[Ans. 33.584 km]

SHORT ANSWER QUESTIONS

Q. 1. *What is 'Continuum?'*

Ans. A continuous and homogeneous medium is called '*Continuum*.' From the continuum view point, the overall properties and behaviour of fluids can be studied without regard for its atomic and molecular structure.

Q. 2. *What is 'viscosity?'*

Ans. **Viscosity** may be defined as the property of a fluid which determines its resistance to shearing stresses. Viscosity of fluids is due to cohesion and interaction between particles. An 'ideal fluid' has no viscosity.

Q. 3. *What is 'hydrostatic law?'*

Ans. The hydrostatic law states : "The rate of increase of pressure in a vertically downward direction must be equal to the specific weight of the fluid at that point."

This law is used to determine the pressure at any point.

Q. 4. *What is 'Standard atmospheric pressure?'*

Ans. The atmospheric pressure at sea level (above absolute zero) is called 'Standard atmospheric pressure.'

It has the following equivalent values:

101.3 kN/m² or 101.3 kPa; 10.3 m of water; 760 mm of mercury; 1013 mb (millibar); $\approx$ 1 bar $\approx$ 100 kPa = 10^5 N/m².

Q. 5. *What is 'gauge pressure?'*

Ans. It is the pressure measured with the help of pressure measuring instrument in which the atmospheric pressure is taken as datum. The atmospheric pressure on the scale is marked as zero.

Q. 6. *What is a 'differential manometer?'*

Ans. It is a device used to measure the difference in pressures between two points in a pipe, or in two different pipes. In its simplest form a differential manometer consists of a U-tube, containing a heavy liquid, whose two ends are connected to the points, whose difference of pressures is required to be found out.

Q. 7. *What is the use of a micromanometer?*

Ans. It is used for measuring small pressure differences. It utilizes two manometer liquids which are immiscible with each other and also with the fluid whose pressure difference is to be measured.

Q. 8. *How does 'total pressure' differ from 'centre of pressure?'*

Ans. **Total pressure** is defined as the force exerted by static fluid on a surface (either plane or curved) when the fluid comes in contact with the surface. This force is always at right angle (or normal) to the surface.

Centre of pressure is defined as the point of application of the total pressure on the surface.

Q. 9. List the possibilities of dam failure?

Ans. The possibilities of dam failure are :

- (i) Failure due to sliding along its base.
- (ii) Failure due to tension or compression.
- (iii) Failure due to shear at the weakest section.
- (iv) Failure due to overturning.

Q. 10. What is 'buoyancy'?

Ans. Whenever a body is immersed wholly or partially in a fluid it is subjected to an upward force which tends to lift (or buoy) it up. This tendency for an immersed body to be lifted up in the fluid, due to an upward force opposite to action of gravity is known as **buoyancy**.

Q. 11. What is 'centre of buoyancy'?

Ans. The point of application of the force of buoyancy on the body is known as the **centre of buoyancy**. It is always the centre of gravity of the volume of fluid displaced.

Q. 12. Define the term 'metacentre'?

Ans. **Metacentre** may be defined as a point of intersection of the axis of the body passing through e.g. (G) and original centre of buoyancy (B) and a vertical line passing through the centre of buoyancy (B_1) of the tilted position of the body (floating). The position of the centre (M) remains practically constant for the small angle of tilt θ .

Q. 13. What do you mean by 'unstable equilibrium'?

Ans. If the body does not return to its original position from the slightly displaced angular position and heels farther away, when given a small angular displacement, such an equilibrium is called an **unstable equilibrium**.

Q. 14. What is the difference between 'steady and unsteady flows'?

Ans. The type of flow in which the fluid characteristics like velocity, pressure, density, etc. at a point *do not change* with time is called **steady flow**.

An **unsteady flow** is that type of flow in which the velocity, pressure or density at a point *change* w.r.t. time.

Q. 15. What is a 'non-uniform flow'? Give two examples.

Ans. A **non-uniform flow** is that type of flow in which the velocity at any given time *changes* with respect to space.

- Examples:* (i) Flow through a non-prismatic conduit;
(ii) Flow around a uniform pipe-bend or canal bend.

Q. 16. Name the type of flow in the following cases :

- (i) Ground water flow.
- (ii) Flow in a converging or diverging pipe or channel.
- (iii) Flow over a drain hole of a stationary tank or wash basin.
- (iv) Subsonic aerodynamics

- Ans.** (i) Laminar flow;
(ii) Three dimensional flow;
(iii) Irrotational flow;
(iv) Incompressible flow

Q. 17. What is the difference between 'path line and 'stream line'?

Ans. A **path line** is the path followed by a fluid particle in motion, whereas, a **stream line** is an imaginary line within the flow so that the tangent at any point on it indicates the velocity at the point.

Q. 18. What is a 'stream tube'? Give examples.

Ans. A **stream tube** is a fluid mass bounded by a group of stream lines. The contents of a stream tube are known as 'current filament.'

Examples: Pipes and nozzles.

Q. 19. What is 'continuity equation'?

Ans. A 'continuity equation' is based on the principle of conservation of mass. It states as follows: "If no fluid is added or removed from the pipe in any length then the mass passing across different sections shall be same."

Q. 20. What do you mean by the terms 'circulation' and 'vorticity'?

Ans. **Circulation** is defined mathematically as the line integral of the tangential velocity about a closed path (contour). Circulation around regular curves can be obtained by integration.

Vorticity is defined as the circulation per unit of enclosed area. If a flow possesses vorticity, it is rotational.

Q. 21. State 'Bernoulli's equation' and list its practical applications?

Ans. **Bernoulli's equation** states: "In an ideal incompressible fluid when the flow is steady and continuous, the sum of pressure energy, kinetic energy and potential (or datum) energy is constant along a stream line".

Practical applications: (i) Venturimeter; (ii) Orificemeter; (iii) Rotameter and elbow meter; (iv) Pilot tube.

Q. 22. What is a 'venturimeter'?

Ans. A **venturimeter** is a device which is inserted into a pipeline to measure incompressible fluid flow rates. It consists of a convergent section which reduces the diameter to between one-half to one-fourth of the pipe diameters. This is followed by a divergent section. The pressure difference between the position just before the venturi and the throat of the venturi is measured by a differential manometer. The working of the venturimeter is based on the Bernoulli's principle, that when the velocity head increases in an accelerated flow, there is a corresponding reduction in the piezometric head.

Q. 23. What are the main points of difference between a venturimeter and orificemeter?

Ans. Following are the main points of difference :

(i) The venturimeter can be used for measuring the flow rates of all incompressible flows, whereas orifice meters are generally used for measuring the flow rates of liquids; (ii) Venturimeter is installed in pipeline only, and the accelerated flow through the apparatus is subsequently decelerated to the original velocity at the outlet of the venturimeter. The flow continues through the pipeline. In the orificemeter the entire potential energy of the fluid is converted to kinetic energy and the jet discharges freely into the open atmosphere; (iii) In the venturimeter, the flow velocity is measured by noting the pressure difference between the inlet and throat of venturimeter, whereas in the orificemeter the discharge velocity is measured by using pitot tube or by trajectory method.

Q. 24. What assumptions are made while deriving Bernoulli's equation?

Ans. The following assumptions are made while deriving Bernoulli's equation :

- (i) The liquid is ideal and incompressible;
- (ii) The flow is steady and continuous;
- (iii) The flow is along stream lines, i.e., one dimensional;
- (iv) The velocity is uniform over the section and is equal to the mean velocity;
- (v) The only forces acting on the fluid are the gravity forces and pressure forces.

Q. 25. *What is a Pitot tube? On what principle does it work?*

Ans. **Pitot tube** is one of the most accurate devices for velocity measurement.

It works on the principle that if the velocity of flow at a point becomes zero, the pressure there is increased due to conversion of kinetic energy into pressure energy.

It consists of a glass tube in the form of 90° bend of short length open at both its ends. It is placed in the flow with its bent leg directed upstream so that a stagnation point is created immediately in the front of the opening. The kinetic energy at this point gets converted into pressure energy causing the liquid to rise in the vertical limb, to a height equal to the stagnation pressure.

Q. 26. *What is 'Impulse-momentum equation?' What are its applications?*

Ans. **Impulse-momentum equation** states: "The momentum of a force F acting on a fluid mass ' m ' in a short interval of time dt is equal to the change of momentum $d(mv)$ in the direction of force".

Applications: This equation is used to the following types of problems:

1. To determine the resultant force acting on the boundary of flow passage by a stream of fluid as the stream changes its direction, magnitude or both. Problems of this type are :

(i) Pipe bends; (ii) Reducers; (iii) Moving valves; (iv) Jet propulsion, etc.

2. To determine the characteristic of flow when there is an abrupt change of flow section. Problems of this type are :

(i) Sudden enlargement in a pipe; (ii) Hydraulic jump in a channel, etc.

Q. 27. *Define kinetic energy and momentum correction factors (Coriolis coefficients).*

Ans. **Kinetic energy correction factor:** It is defined as the ratio of the kinetic energy of flow per second based on actual velocity across a section to the kinetic energy of flow per second based on average velocity across the same section. It is denoted by ' α '.

Momentum correction factor: It is defined as the ratio of momentum of the flow per second based on actual velocity to the momentum of the flow per second based on average velocity across a section. It is denoted by ' β '.

Q. 28. *What is moment of momentum principle?*

Ans. **Moment of momentum principle** states: "The resulting torque acting on a rotating fluid is equal to the rate of change of moment of momentum".

When the moment of momentum of flow leaving a control volume is different from that entering it, the result is a torque acting over the control volume.

Q. 29. *What is 'vortex motion'? How is it characterised?*

Ans. The **vortex motion** is defined as a motion in which the whole fluid mass rotates about an axis.

A vortex motion is characterised by a flow pattern wherein the stream lines are curved. When fluid flows between curved stream lines, centrifugal forces are set up and these are counter-balanced by the pressure force acting in the radial direction.

Q. 30. *What is 'dimensional analysis'?*

Ans. **Dimensional analysis** is a mathematical technique which makes use of the study of the dimensions for solving several engineering problems. It is based on the 'principle of dimensional homogeneity' and uses the dimensions of relevant variables affecting the phenomenon.

It is specially useful in presenting experimental results in a concise form.

Q. 31. *What are the applications of dimensional homogeneity?*

Ans. The applications of dimensional homogeneity are :

(i) It facilitates to determine the dimensions of a physical quantity;

- (ii) It helps to check whether an equation of any physical phenomenon is dimensionally homogeneous or not;
- (iii) It facilitates conversion of units from one form system to another;
- (iv) It provides a step towards dimensional analysis which is fruitfully employed to plan experiments and to present the results meaningfully.

Q. 32. *What is the difference between 'model' and 'prototype'?*

Ans. The **model** is the small scale replica of the actual structure or machine. The actual structure of machine is called **prototype**.

The models are not always smaller than the prototype, in some cases a model may be even larger or of the same size as prototype depending upon the need and purpose (e.g., the working of a wrist watch or a carburettor can be studied in a large scale model).

Q. 33. *Enumerate the forces which influence hydraulic phenomena.*

Ans. (i) Inertia force; (ii) Viscous force; (iii) Gravity force; (iv) Pressure force; (v) Surface tension force; (vi) Elastic force.

Q. 34. *What is Mach number?*

Ans. It is defined as the square root of the ratio of inertia force to the elastic force.

Q. 35. *What is a 'distorted model'?*

Ans. A **distorted model** is one which is not geometrically similar to its prototype. In such a model different scale ratios for linear dimensions are adopted. For example in case of a wide and shallow river it is not possible to obtain the same horizontal and vertical scale ratios, however, if these ratios are taken to be same then because of the small depth of flow the vertical dimensions of the model will become too less in comparison to its horizontal length. Thus in distorted models the plan form is geometrically similar to that of prototype but the cross-section is distorted.

Q. 36. *What do you understand by 'Scale effect in models'?*

Ans. By model testing it is not possible to predict the exact behaviour of the prototype. The behaviour of the prototype as predicted by two models with different scale ratios is generally not the same. Such a discrepancy or difference in the prediction of behaviour of the prototype is termed as **scale effect**. The magnitude of the scale effect is affected by the type of the problem and the scale ratio used for the performance of experiments on models. The scale effect can be positive and negative and when applied to the results accordingly, the corrected results then hold good for prototype.

Scale effect can be known by testing a number of models using different scale ratios, and the exact behaviour of the prototype can then be predicted.

Q. 37. *What are the limitations of model investigation/hydraulic similitude?*

Ans. 'Model investigation', although very important and valuable, may not provide ready solution to all problems. It has the following 'limitations':

- (i) The model results, in general, are qualitative but not quantitative;
- (ii) As compared to the cost of analytical work, models are usually expensive;
- (iii) Transferring results to the prototype requires some judgement (the scale effect should be allowed for);
- (iv) The selection of size of a model is a matter of experience.

Q. 38. *What is the difference between an 'orifice' and a 'mouthpiece'?*

Ans. An **orifice** is an opening in the wall or base of a vessel through which the fluid flows. The top edge of the orifice is always *below* the free surface (If the free surface is below the top edge of the orifice, becomes a weir).

A **mouthpiece** is an attachment in the form of a small tube or pipe fixed to the orifice (the length of pipe extension is usually 2 to 3 times the orifice diameter) and is used to increase the amount of discharge.

Orifices as well as mouthpieces are used to measure the discharge.

Q. 39. *What is 'coefficient of resistance'?*

Ans. The ratio of loss of head (or loss of kinetic energy) in the orifice to the head of water (actual kinetic energy) available at the exit of the orifice is known as **Coefficient of resistance**. It is denoted by C_r .

The loss of head in the orifice takes place, because the walls of the orifice offer some resistance to the liquid, as it comes out. While solving numerical problems C_r is generally neglected.

Q. 40. *What is the difference between a 'notch' and a 'weir'?*

Ans. A **notch** may be defined as an opening provided in the side of a tank or vessel such that the liquid surface in the tank is below the top edge of the opening. It is generally made of metallic plate and is used for measuring the rate of flow of a liquid through a small channel or a tank.

A **weir** may be defined as any regular obstruction in an open stream over which the flow takes place. It is made of masonry or concrete. The conditions of flow, in the case of a weir are practically the same, as those of a rectangular notch. That is why, a notch is sometimes called as a weir or vice versa.

Q. 41. *What is the formula for calculating discharge over a triangular notch?*

Ans. The formula for calculating discharge (Q) over a triangular notch is given by :

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} H^{5/2}$$

where,

H = Head of water above the apex of the notch,

θ = Angle of notch, and

C_d = Coefficient of discharge.

Q. 42. *What is velocity of approach?*

Ans. The velocity with which the water approaches or reaches the weir or notch before it flow over is known as **velocity of approach**. Thus if V_a is the velocity of approach, then an additional head H_a $\left(= \frac{V_a^2}{2g} \right)$ due to the velocity of approach is acting on water flowing over

the notch or weir. Then initial and final heights of water over the notch or weir will be $(H + H_a)$ and H_a respectively.

Q. 43. *Name the type of flow for each of the following Reynolds (Re) numbers :*

(i) $Re < 2000$; (ii) $Re > 4000$; (iii) Re between 2000 and 4000.

Ans. (i) Laminar flow; (ii) Turbulent flow; (iii) Unpredictable flow.

Q. 44. *What are the important applications of Navier-Stokes equations?*

Ans. The important applications of Navier-stokes equations are :

- (i) Laminar flow in circular pipes;
- (ii) Laminar flow between concentric rotating cylinders;
- (iii) Laminar uni-directional flow between stationary parallel plates;
- (iv) Laminar uni-directional flow between parallel plates having relative motion.

Q. 45. How is loss of head due to friction in pipe flow expressed?

Ans. Loss of head (h_f) due to friction in pipe flow is given by :

$$h_f = \frac{4fLV^2}{D \times 2g} = \frac{f_1 LV^2}{D \times 2g}$$

where,

f = Friction coefficient; $f_1 (= 4f)$ = friction factor,

L = Length of the pipe between the two sections considered,

D = Diameter of the pipe, and

V = Average flow velocity.

Q. 46. When does a pipe behave like an open channel?

Ans. A pipe is a closed conduit (generally of circular cross-section) which is used for carrying fluids under pressure. The flow in a pipe is termed pipe flow only when the fluid *completely fills* the cross-section and there is no free surface of liquid. The pipe running *partially full* (in such a case atmospheric pressure exists inside the pipe) behaves like an open channel.

Q. 47. What is the difference between 'Energy' and 'Hydraulic' gradient lines?

Ans. **'Energy gradient line (E.G.L.):** If the total energy at various points along the axis of the pipe is plotted and joined by a line, the line so obtained is called the 'Energy gradient line.'

Hydraulic gradient line: If a line is drawn joining the piezometric levels at various points, the line so obtained is called the 'Hydraulic gradient line.'

Q. 48. What do you mean by water hammer in pipes?

Ans. The phenomenon of sudden rise in pressure in a pipe when water flowing in it is suddenly brought to rest by closing the valve is known as **water hammer** or **hammer blow**.

Q. 49. What is a boundary layer?

Ans. The layer adjacent to the boundary is known as **boundary layer**. Boundary layer is formed whenever there is relative motion between the boundary and the fluid.

According to boundary layer theory the extensive fluid medium around bodies moving in the fluids can be divided into following two regions :

- (i) A thin layer adjoining the boundary is called the 'boundary layer' where the viscous shear takes place.
- (ii) A region outside the boundary layer where the flow behaviour is quite like that of an ideal fluid and the potential flow theory is applicable.

Q. 50. What is boundary layer thickness?

Ans. The velocity within the boundary layer increases from zero at the boundary surface to the velocity of the main stream asymptotically. Therefore the **thickness boundary layer** is arbitrarily defined as that distance from the boundary in which the velocity reaches 99 per cent of the velocity of the free stream ($u = 0.99U$). It is defined by the symbol δ . This definition however gives an approximate value of the boundary layer thickness and hence δ is generally termed as 'nominal thickness' of boundary layer.

The commonly adopted definitions of the boundary layer thickness are : (i) Displacement thickness (δ^*); (ii) Momentum thickness (θ); (iii) Energy thickness (δ_e).

Q. 51. How does 'displacement thickness (δ^*)' differ from momentum thickness (θ)?

Ans. **Displacement thickness** is the distance measured perpendicular to the boundary, by which the main/free stream is displaced on account of formation of boundary layer (δ).

Momentum thickness is the distance through which the total loss of momentum per second be equal to it if it were passing a stationary plate. The momentum thickness is useful in kinetics.

Q. 52. What is 'Von Karman momentum equation?' What is the use of this equation?

Ans. *Von Karman momentum equation* is given by :

$$\frac{\tau_0}{\rho U^2} = \frac{d\theta}{dx}$$

where,
$$\theta = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy, \text{ and}$$

τ_0 = Shear stress at the surface,

ρ = Density of fluid,

u = Velocity at the section considered,

U = Free stream velocity, and

dy = Thickness of the section considered.

This equation is used to find out the frictional drag on smooth flat plate for both laminar and turbulent boundary layers.

Q. 53. On what factors does the flow separation depends?

Ans. The flow separation depends on the following factors :

(i) The curvature of the surface, (ii) The Reynolds number, and (iii) Roughness of surface.

Q. 54. Name the methods used to control separation.

Ans. The following methods are used to control separation:

(i) Motion of boundary layer, (ii) Acceleration of fluid in the boundary layer, (iii) Suction of fluid from the boundary layer, and (iv) Streamlining of body shapes.

Q. 55. Name the forces to which a body wholly immersed in a real fluid may be subjected to?

Ans. A body wholly immersed in a real fluid may be subjected to the following forces:

(i) Drag force (F_D): It is the force exerted by the fluid in the direction of flow (free stream).

(ii) Lift force (F_L): It is the force exerted by fluid at right angles to the direction of flow.

Q. 56. What is the difference between a 'streamlined body' and a 'bluff body'?

Ans. A body whose surface *coincides* with the stream lines when placed in a flow, is called a **streamlined body**. If the surface of the body *does not coincide* with the stream lines the body is called **bluff body**.

Q. 57. What is the 'terminal velocity' in relation to the falling body? What is the formula for calculating it for a sphere falling through a liquid at rest.

Ans. The **terminal velocity** is the maximum velocity attained by a falling body. The terminal velocity (U) of a sphere falling through a liquid at rest is calculated from the following relation:

$$U = \frac{D^2}{18\mu} (w_s - w_f)$$

where, D = Diameter of the sphere, μ = Dynamic viscosity of the fluid, w_s = Specific weight of the material of sphere, and w_f = Specific weight of fluid.

Q. 58. What is D'Alembert's paradox?

Ans. The concept of zero drag on bodies immersed in a steady flow of ideal fluid is called **D'Alembert's paradox**.

Q. 59. What is 'Magnus effect'? What are its uses?

Ans. The generation of lift by spinning cylinder in a fluid stream is called **Magnus effect**.

Uses :

(i) This effect has been successfully employed in the propulsion of ships;

(ii) The Magnus effect may also be used with advantage in the games like table tennis, golf, cricket etc.

Q. 60. How is Kutta-Joukowski equation expressed?

Ans. According to **Kutta-Joukowski equation**, the total lift (F_L) of a cylinder of length L is given by :

$$F_L = \rho L U \Gamma$$

where,

ρ = Density of fluid,

L = Length of the cylinder,

U = Velocity of uniform flow, and

Γ = Circulation.

Q. 61. What do you mean by stagnation point and stagnation properties?

Ans. The point on the immersed body where the velocity is zero is called **stagnation point**. At this point velocity head is converted into pressure head. The values of pressure (p_s), temperature (T_s) and density (ρ_s) at stagnation point are called **stagnation properties**.

Q. 62. What are shock waves in compressible flow?

Ans. Whenever a supersonic flow (compressible) abruptly changes to subsonic flow, a shock wave (analogous to hydraulic jump in an open channel) is produced resulting in a sudden rise in pressure, density, temperature and entropy. This occurs due to pressure differentials and when the Mach number of the approaching flow is greater than one (i.e., $M > 1$).

Q. 63. What is an open channel?

Ans. An **open channel** may be defined as a passage in which liquid flows with its upper surface exposed to atmosphere. In open channels the flow is due to gravity, thus the flow conditions are greatly influenced by the slope of the channel.

Q. 64. What is hydraulic radius (R)?

Ans. **Hydraulic radius** is the ratio of the cross-sectional area of flow to wetted parameter. It is also called 'hydraulic mean depth.'

Q. 65. What do you mean by most economical section of a channel?

Ans. The **most economical section** (also called the best section or most efficient section) is one which gives the maximum discharge for a given amount of excavation.

Q. 66. What is a 'hydraulic jump' or 'standing wave'?

Ans. In an open channel when rapidly flowing stream abruptly changes to slowly flowing stream, a distinct rise or jump in the elevation of fluid surface takes place, this phenomenon is known as **hydraulic jump** (which is analogous to shock wave in compressible fluids). The hydraulic jump converts kinetic energy of stream rapidly flowing into potential energy. Due to this there is a loss of kinetic energy. At the place where hydraulic jump occurs rollers of turbulent water (eddying turbulences) form, which cause dissipation of energy. A hydraulic jump occurs in practice at the toe of spillways or below a sluice gate where the *velocity is very high*.

The hydraulic jump is also known as a **standing wave** because it is, in essence, a *wave which is stationary (i.e., at stand-still) at one place*.

Q. 67. How does 'afflux' differ from 'back water curve'?

Ans. **Afflux** is the increase in water level due to some obstruction across the flowing liquid; the curved surface of the liquid with its concavity upwards, is known as **back water curve**.

GATE and UPSC Examinations' Questions with Answers/Solutions

A. Conventional Questions with Answers/Solutions

B. Multiple-choice Questions with Answers and Explanations

SECTION-A: Conventional Questions with Answers/ Solutions

Q. 1. Define a distorted model. Explain why models of rivers and harbours are made distorted models. Write down the merits and demerits of distorted models. [CSE (Main) 2016 : 10 Marks]

Solution. A **distorted model** is one which is not geometrically similar to its prototype.

In such a model *different scale ratios* for the linear dimensions are adopted. For example, in case of rivers, harborers etc. it is *not possible* to obtain the same horizontal and vertical ratios, however, if these ratios are taken to be same then because of the small depth of flow the vertical dimensions of the model will become two less in comparison to its horizontal length. Thus in distorted models the *plan form is geometrically similar to that of prototype but the cross-section is distorted*.

Merits and Demerits of Distorted Models:

Merits:

1. Due to increase in the depth of fluid or height of waves accurate measurements are made possible.
2. The surface tension can be reduced to minimum.
3. Model size can be sufficiently reduced, thereby *its operation is simplified and also cost is lowered considerably*.
4. Sufficient tractive force can be developed to move the bed material of the model.
5. The Reynolds number of flow in a model can be increased that will yield better results.

Demerits:

1. The pressure and velocity distributions are not truly reproduced.
2. A model wave may differ in type and possibly in action from that of the prototype.
3. Slopes of river beds, earth cuts and dikes cannot be truly reproduced.
4. It is difficult to extrapolate and interpolate results obtained from distorted models.
5. The observer experiences an unfavourable psychological effect.

Note: Although the distorted models entail a number of demerits, yet by incorporating judicious allowances in the interpretation of the results obtained from such models, useful information can be obtained (which is otherwise not possible).

Q. 2. What do you mean by boundary layer separation? Briefly explain the various methods of controlling of boundary layer separation. [CSE (Main) 2016 : 10 Mark]

Solution. When a solid body is immersed in a flowing fluid, a thin layer of fluid called *boundary layer* is formed, adjacent to the solid body. In this thin layer of fluid, the velocity varies from *zero to*

free stream velocity in the direction normal to the solid body. Along the length of the solid body, the thickness of the boundary layer increases. The fluid layer adjacent to the solid surface has to do work against surface friction at the expense of kinetic energy. This loss of kinetic energy is recovered from the immediate fluid layer in contact with the layer adjacent to the solid surface through momentum exchange process. Thus the velocity of the layer goes on decreasing. Along the length of solid body at a certain point a stage may come when the boundary layer may not be able to keep sticking to the solid body, if it cannot provide kinetic energy to overcome the resistance offered by the solid body. In other words, the boundary layer will be separated from the surface. This phenomenon is called the **boundary layer separation**. The point on the body at which the boundary layer is on the verge of separation from the surface is called **point of separation**.

Methods to control separation

1. Motion of solid boundary:

By rotating a circular cylinder lying in a stream of fluid, so that the upper side of cylinder where the fluid as well as the cylinder move in the same direction, the boundary layer does not form. However on the lower side of cylinder where the fluid motion is opposite to that of cylinder separation would occur (Fig. 1).

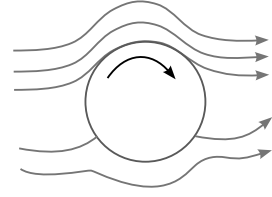


Fig. 1.

2. Acceleration of fluid in the boundary layer:

This method of controlling separation consists of supplying additional energy to particles of fluid which are being retarded in the boundary layer. This may be achieved either injecting the fluid into the region of boundary layer from the interior of the body with the help of some available device as shown in Fig. 2 or by diverting a portion of fluid of the main stream from the region of high pressure to the retarded region of boundary layer through a slot provided in the body (Fig. 3).

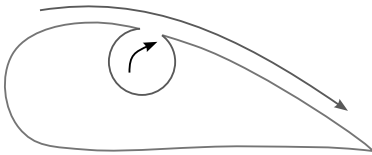


Fig. 2. Injecting fluid into boundary layer

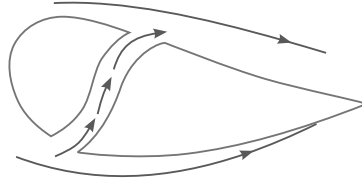


Fig. 3. Slotting wing.

3. Suction of fluid from the boundary layer.

In this method, the slow moving fluid in the boundary layer is removed by suction through slots or through a porous surface as shown in the Fig. 4.

4. Streamlining of body shapes:

By the use of suitably shaped bodies, the point of transition of the boundary layer from laminar to turbulent can be moved downstream which results in the reduction of the skin friction drag. Further more by streamlining of body shapes, the separation may be eliminated.

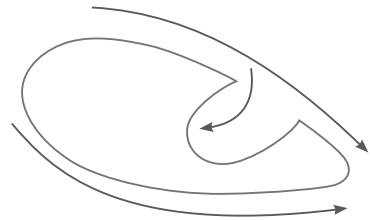


Fig. 4.

Q. 3. Air flows in a circular disc which suddenly contracts in the cross-sectional area. Draw the flow sketch and locate the points on the sketch where turbulent flow will occur and calculate the dynamic loss co-efficient. Take the co-efficient of contraction as 0.62.

[CSE (Main) 2015 : 10 Marks]

Solution. Refer to Fig. 5.

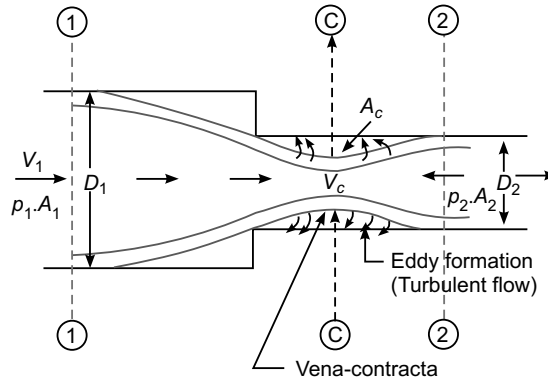


Fig. 5.

Due to sudden contraction, the *stream lines* converge to a minimum cross-section (C–C) called the *Vena-contract* and then expands to fill the downstream pipe. At section C–C a sudden enlargement of the area takes place. The loss of head due to sudden contraction is actually due to sudden enlargement from vena-contract a to smaller pipe.

Let the dynamic loss coefficient be k and co-efficient of contraction be C_c .

We know that
$$k = \left(\frac{1}{C_c} - 1 \right)^2$$

$$\therefore k = \left(\frac{1}{0.62} - 1 \right)^2 = 0.375 \text{ (Ans.)} \quad (\because C_c = 0.62 \text{ given})$$

Q. 4. A ship with a hull length of 140 metres has to travel with 7.6 m/s velocity. Compute the Froude number. For dynamic similarity, at what velocity should a 1:30 scale down model flow through water? [ESE 2016 : 5 Marks]

Solution. Given: $L = 140 \text{ m}$; $V_p = 7.6 \text{ m/s}$, $L_m / L_p = 1 : 30$

In case of ship model, Froude number,

$$Fr = \frac{V}{\sqrt{Lg}} = \frac{7.6}{\sqrt{140 \times 9.81}} = 0.205 \text{ (Ans.)}$$

Now, $(Fr)_{\text{model}} = (Fr)_{\text{prototype}}$

or,
$$\left(\frac{V}{\sqrt{Lg}} \right)_m = \left(\frac{V}{\sqrt{Lg}} \right)_p$$

or,
$$\frac{V_m}{V_p} = \sqrt{\frac{L_m}{L_p}}$$

or,
$$V_m = V_p \sqrt{\frac{1}{30}} = 7.6 \sqrt{\frac{1}{30}} = 1.387 \text{ m/s (Ans.)}$$

Q. 5. A sphere of diameter D and density ρ_s falls at a steady rate through a liquid of density ρ and viscosity μ . If the Reynolds number is less than 1, show that the viscosity can be determined from $\mu = \frac{gD^2(\rho_s - \rho)}{18U}$ where the Reynolds number is evaluated as $\frac{\rho DU}{\mu}$. [ESE 2016 : 7 Marks]

Solution. Refer to Fig. 6

Given: Diameter of sphere = D ;

Density of sphere = ρ_s

Density of liquid = ρ

Viscosity = μ ; $Re < 1$

Now, under steady condition,

$$(F_{\text{net}})_{\text{sphere}} = 0$$

$$F_B + F_D = W \quad \dots(i)$$

Also,

$$F_B = \rho g V$$

$$= \rho g \times \left(\frac{4}{3} \times \pi \times \left(\frac{D}{2} \right)^3 \right)$$

$$= \frac{4 \pi \rho g D^3}{24}$$

and,

$$F_D = \frac{1}{2} C_D \rho A U^2 = \text{Drag force}$$

$$C_D = \frac{24}{Re}, \left(Re = \frac{\rho D U}{\mu} \right)$$

$$A = \text{Area of projection} = \frac{\pi}{4} D^2$$

Weight,

$$W = g \times \frac{4}{3} \pi \rho_s \left(\frac{D}{2} \right)^3 = \frac{4 \pi \rho_s g D^3}{24}$$

Substituting the values in (i), we get:

$$\frac{4 \pi \rho g D^3}{24} + \frac{1}{2} \times \frac{24}{Re} \times \rho \times \frac{\pi}{4} D^2 \times U^2 = \frac{4 \pi \rho_s g D^3}{24}$$

$$\text{or, } \frac{12 \mu \rho \times \pi D^2 U^2}{\rho D U \times 4} = \frac{4 \pi g D^3}{24} (\rho_s - \rho)$$

$$\text{or, } \mu = \frac{g D^2 (\rho_s - \rho)}{18 U} \quad \text{Proved. (Ans.)}$$

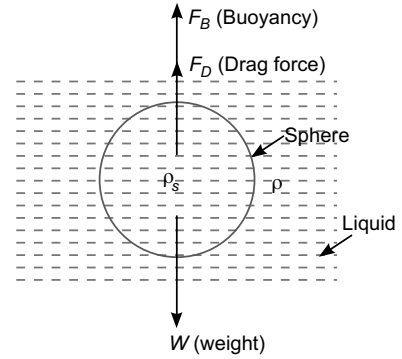


Fig. 6.

Q. 6. In a circular pipe the velocity flow is given as $u = U_m \left[1 - \left(\frac{r}{R} \right)^2 \right]$ where u is the

velocity at any radius r , U_m is the velocity at the pipe axis, R is the radius of the pipe. Find:

(i) Average velocity, (ii) Energy correction factor, and (iii) Momentum correction factor.

[CSE (Mains) 2015 : 20 Marks]

Solution. Refer to Fig. 7. Consider an elementary area dA in the form of a ring at a radius r and thickness dr , then $dA = 2 \pi r \cdot dr$

Flow through the ring = dQ = Elemental area $\times$ local velocity = $2 \pi r \cdot dr \cdot u$

$$\text{Total flow, } Q = \int_0^R 2 \pi r \cdot u \cdot dr$$

$$\begin{aligned}
 &= \int_0^R 2\pi U_m \left(1 - \frac{r^2}{R^2}\right) r \, dr \\
 &= 2\pi U_m \int_0^R \left(r - \frac{r^3}{R^2}\right) dr = 2\pi U_m \left[\frac{r^2}{2} - \frac{r^4}{4R^2}\right]_0^R \\
 &= 2\pi U_m \left(\frac{R^2}{2} - \frac{R^2}{4}\right) = 2\pi U_m \left(\frac{R^2}{4}\right)
 \end{aligned}$$

$$Q = 2\pi U_m \left(\frac{R^2}{4}\right) \quad \dots(i)$$

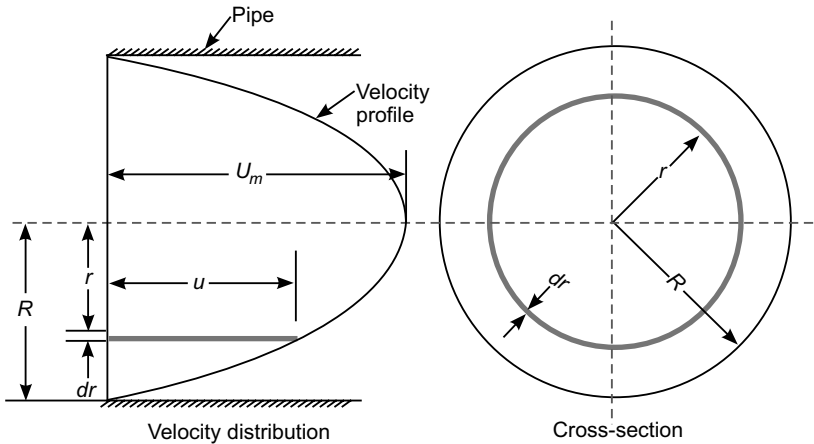


Fig. 7. Velocity distribution and cross-section of a pipe.

(i) **Average velocity, $\bar{u}$:**

If $\bar{u}$ is the average flow velocity, then:

$$Q = A\bar{u} = \pi R^2\bar{u} \quad \dots(ii)$$

From (i) and (ii), we get:

$$\begin{aligned}
 \pi R^2\bar{u} &= 2\pi U_m \left(\frac{R^2}{4}\right) \\
 \bar{u} &= \frac{2\pi U_m \left(\frac{R^2}{4}\right)}{\pi R^2} = \frac{U_m}{2} \quad \text{(Ans.)}
 \end{aligned}$$

Energy correction factor, α :

$$\alpha = \frac{1}{A\bar{u}^3} \int_0^R u^3 dA \quad \dots[\text{Eqn. 6.28}]$$

$$\begin{aligned}
 &= \frac{1}{A\bar{u}^3} \int_0^R U_m^3 \left[1 - \left(\frac{r}{R}\right)^2\right]^3 2\pi r \, dr \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \int_0^R \left(1 - \frac{3r^2}{R^2} + \frac{3r^4}{R^4} - \frac{r^6}{R^6}\right) r \, dr
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \int_0^R \left(r - \frac{3r^3}{R^2} + \frac{3r^5}{R^4} - \frac{r^7}{R^6} \right) dr \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \left[\frac{r^2}{2} - \frac{3r^4}{4R^2} + \frac{3r^6}{6R^4} - \frac{r^8}{8R^6} \right]_0^R \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \left[\frac{R^2}{2} - \frac{3}{4}R^2 + \frac{R^2}{2} - \frac{R^2}{8} \right] \\
 &= \frac{2\pi U_m^3}{A\bar{u}^3} \left(\frac{R^2}{8} \right)
 \end{aligned}$$

Substituting the values $A = \pi R^2$ and $\bar{u} = \frac{U_m}{2}$, we get:

$$\alpha = \frac{2\pi U_m^3}{\pi R^2 \times \left(\frac{U_m}{2} \right)^3} \times \left(\frac{R^2}{8} \right) = 2 \text{ (Ans.)}$$

(iv) **Momentum correction factor, β :**

...[Eqn. 6.30]

$$\begin{aligned}
 \beta &= \frac{1}{A\bar{u}^2} \int u^2 dA \\
 &= \frac{1}{A\bar{u}^2} \int_0^R U_m^2 \left[1 - \left(\frac{r}{R} \right)^2 \right]^2 2\pi r dr \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \int_0^R \left(1 - 2 \times \frac{r^2}{R^2} + \frac{r^4}{R^4} \right) r dr \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \int_0^R \left(r - 2 \times \frac{r^3}{R^2} + \frac{r^5}{R^4} \right) dr \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \left[\frac{r^2}{2} - 2 \times \frac{r^4}{4R^2} + \frac{1}{6} \times \frac{r^6}{R^4} \right]_0^R \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \left[\frac{R^2}{2} - \frac{R^2}{2} + \frac{R^2}{6} \right] \\
 &= \frac{2\pi U_m^2}{A\bar{u}^2} \left(\frac{R^2}{6} \right)
 \end{aligned}$$

Substituting the values $A = \pi R^2$ and $\bar{u} = \frac{U_m}{2}$, we get:

$$\beta = \frac{2\pi U_m^2}{\pi R^2 \times \left(\frac{U_m}{2} \right)^2} \left(\frac{R^2}{6} \right) = 1.33 \text{ (Ans.)}$$

Q. 7. The efficiency η of fan depends on the density ρ , the dynamic viscosity μ of the fluid, the angular velocity ω , diameter D of the rotor and the discharge Q . Express η in terms of dimensionless parameters. [CSE (Mains) 2014 : 10 Marks]

Solution. The efficiency η of a fan is a function of:

- (i) Density ρ , (ii) Viscosity μ ,
 (iii) Angular velocity ω , (iv) Diameter D , and
 (v) Discharge Q .

Mathematically, $\eta = f(\rho, \mu, \omega, D, Q)$... (1)

or, $\eta = C(\rho^a, \mu^b, \omega^c, D^d, Q^e)$

Where C is a non-dimensional constant.

Using $M-L-T$ system, the corresponding equation for dimensions is:

$$M^0 L^0 T^0 = C [(ML^{-3})^a (ML^{-1} T^{-1})^b (T^{-1})^c (L)^d (L^3 T^{-1})^e]$$

For dimensional homogeneity the exponents of each dimension on both sides of the equation must be identical. Thus:

For M : $0 = a + b$

For L : $0 = -3a - b + d + 3e$

For T : $0 = -b - c - e$

There are five variables and we have only three equations. Experience has shown that recognized dimensionless groups appear if the exponents of D , ω and ρ are evaluated in terms of b and e (exponents of viscosity and discharge which are more important)

$$a = -b; c = -(b + e);$$

$$d = 3a + b - 3e = 3(-b) + b - 3e = -2b - 3e = -(2b + 3e)$$

Substituting these values of exponents in eqn. (1), we get:

$$\begin{aligned} \eta &= C(\rho^{-b} \cdot \mu^b \cdot \omega^{-(b+e)} \cdot D^{-(2b+3e)} \cdot Q^e) \\ &= C(\rho^{-b} \cdot \mu^b \cdot \omega^{-b} \cdot \omega^{-e} D^{-2b} \cdot D^{-3e} Q^e) \\ &= C \left[\left(\frac{\mu}{\rho \omega D^2} \right)^b \left(\frac{Q}{\omega D^3} \right)^e \right] = \phi \left[\left(\frac{\mu}{\rho \omega D^2} \right), \left(\frac{Q}{\omega D^3} \right) \right] \quad (\text{Ans.}) \end{aligned}$$

Q. 8. A closed cylindrical vessel 0.2 m in diameter and 1.2 m long is filled with water upto a height of 0.8 m from the bottom. Find the speed of the vessel about its vertical axis, when the axial depth of water is zero. [CSE (Mains) 2012 : 10 Marks]

Solution. Refer to Fig. 8.

Given: $D = 0.2 \text{ m}; z = 1.2 \text{ m}$

Angular speed, ω :

$$z = \frac{\omega^2 r^2}{2g} \quad \dots [\text{Eqn. 6.38}]$$

or, $\omega^2 r^2 = 1.2 \times 2 \times 9.81 = 23.54 \quad \dots (i)$

Volume of air before rotation = Volume of air after paraboloid

$$\pi R^2 (1.2 - 0.8) = \frac{1}{2} \pi r^2 \times z = \frac{\pi r^2}{2} \times 1.2$$

or $\pi \times 0.1^2 \times 0.4 = 0.6 \pi r^2$

or, $r^2 = 6.67 \times 10^{-3} \quad \dots (ii)$

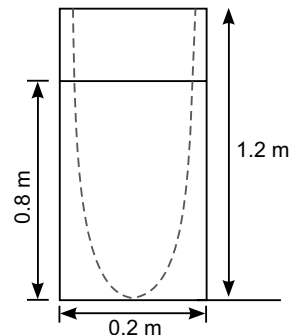


Fig. 8.

From (i) and (ii), we get:

$$\omega^2 \times 6.67 \times 10^{-3} = 23.54$$

$$\omega = 59.41 \text{ rad/s}$$

Also,
$$\omega = \frac{2\pi N}{60} = 59.41$$

or,
$$N = \frac{59.4 \times 60}{2\pi} = \mathbf{567.23 \text{ rpm (Ans.)}}$$

Q. 9. A circular pipe 100 mm in diameter has a 2.5 m length which is porous. In this porous section the velocity of exit is known to be constant (Fig. 9). If the velocities at inlet and outlet of the porous section are 1.8 m/s and 1.1 m/s respectively, calculate :

- The discharge emitted out through walls of the porous pipe, and
- The average velocity of this emitted discharge.

[CSE]

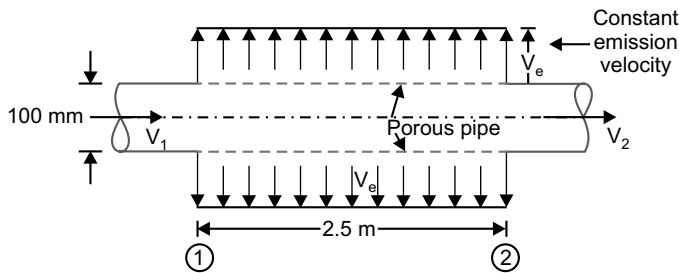


Fig. 9

Solution. Given: Dia. of the pipe, $D = 100 \text{ mm} = 0.1 \text{ m}$; Length, $L = 2.5 \text{ m}$; Velocities at inlet and outlet of the porous pipe, $V_1, V_2 = 1.8 \text{ m/s}$ and 1.1 m/s respectively.

Area of the pipe cross-section, $A = \frac{\pi}{4} \times (0.1)^2 = 7.854 \times 10^{-3} \text{ m}^2$

- (i) The discharge emitted out through walls of the porous pipe, Q_e :

Inlet discharge, $Q_1 = AV_1 = 7.854 \times 10^{-3} \times 1.8 = 0.014137 \text{ m}^3/\text{s}$

Outlet discharge, $Q_2 = AV_2 = 7.854 \times 10^{-3} \times 1.1 = 0.0086394 \text{ m}^3/\text{s}$

$\therefore$ Discharge emitted through walls of the porous pipe,

$$Q_e = Q_1 - Q_2 = 0.014137 - 0.0086394 \\ = \mathbf{0.0054976 \text{ m}^3/\text{s (Ans.)}}$$

- (ii) The average velocity of the emitted discharge V_e :

Surface area of the emission, $A_e = \pi DL$

$$= \pi \times 0.1 \times 2.5 = 0.7854 \text{ m}^2$$

Velocity of emission, $V_e = \frac{Q_e}{A_e} = \frac{0.0054976}{0.7854} = \mathbf{0.007 \text{ m/s (Ans.)}}$

Q. 10. Fig. 10. shows a standard lined triangular section. The section consists of a triangular section of side slope 'n' horizontal : 1 vertical with its bottom being rounded off by a circular curve of radius equal to the full supply depth. For such a channel calculate fully supply depth corresponding to a full supply discharge of $20 \text{ m}^3/\text{s}$. The side slopes are 2 horizontal : 1 vertical, the longitudinal slope is 1 in 2500 and Manning's $N = 0.018$.

[IES]

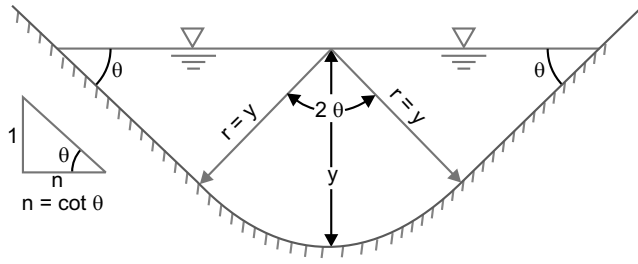


Fig. 10. Standard lined triangular canal section.

Solution. *Given:* Full supply discharge, $Q = 20 \text{ m}^3$; Side slopes = 2 horizontal : 1 vertical; Longitudinal slopes = 1 in 2500; Manning's $N = 0.018$.

Let,

y_0 = Normal (or full supply) depth = r

θ = Inclination of the sides to the horizontal,

$$\cot \theta = n, \quad \text{or,} \quad \theta = \tan^{-1} \frac{1}{n}$$

$$\text{Area, } A = 2 \left(\frac{1}{2} y^2 \cot \theta \right) + \frac{1}{2} y^2 \cdot 2\theta$$

$$= y^2 (\theta + \cot \theta)$$

...(i)

$$= \epsilon y^2, \quad \text{where, } \epsilon = \theta + \cot \theta$$

$$= \left(\tan^{-1} \frac{1}{n} + n \right)$$

$$\text{Wetted perimeter, } P = 2y \cot \theta + 2y \theta$$

$$= 2y (\cot \theta + \theta) = 2\epsilon y$$

...(ii)

$$\text{Hydraulic radius, } R = \frac{A}{P} = \frac{\epsilon y^2}{2\epsilon y} = \frac{y}{2}$$

...(iii)

In the present case, $n = 2$

$$\text{and,} \quad \tan^{-1} \frac{1}{n} = \tan^{-1} \frac{1}{2} = 26.565 \times \frac{\pi}{180} = 0.4636 \text{ rad}$$

$$\therefore \quad \epsilon = \theta + \cot \theta = \tan^{-1} \frac{1}{n} + n$$

$$= 0.4636 + 2 = 2.4636$$

$$\text{and,} \quad A = \epsilon y^2 = 2.4636 y^2$$

[from (i)]

$$R = 0.5 y$$

[from (iii)]

By using Manning's equation, we have :

$$Q = A \times \frac{1}{N} R^{2/3} S^{1/2}$$

$$20 = 2.4636 y^2 \times \frac{1}{0.018} \times (0.5 y)^{2/3} \times \left(\frac{1}{2500} \right)^{1/2}$$

$$20 = 1.7244 y^{8/3}$$

$$\text{or,} \quad y = \left(\frac{20}{1.7244} \right)^{3/8} = \mathbf{2.507 \text{ m (Ans.)}}$$

Q. 11. In Fig. 11 is shown a standard lined trapezoidal section which has a bottom width of 30 m and side slopes of 1.5 horizontal : 1 vertical. The longitudinal slope is 1 in 4500 and the Manning's N can be assumed to be 0.017. Determine the discharge if the full supply depth y is 3.2 m. [CSE]

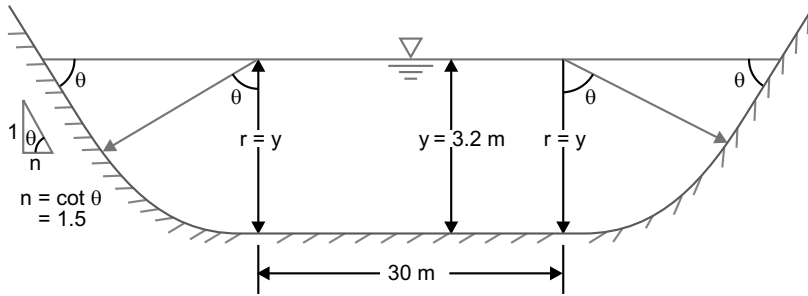


Fig. 11. Standard lined trapezoidal section.

Solution. Given: Bottom width, $b = 30$ m; Side slopes = 1.5 horizontal : 1 vertical ($n = 1.5$); Longitudinal slope; $S = \frac{1}{4500}$; Supply depth, $y = 3.2$ m; Manning's $N = 0.017$.

Discharge with full supply width, Q :

Let the side slopes be n horizontal : 1 vertical. If θ is the inclination of side to the horizontal, then:

$$\cot \theta = n, \text{ and } \theta = \tan^{-1} \frac{1}{n}$$

If, $\epsilon = \cot \theta + \theta = n + \tan^{-1} \frac{1}{n} \left(= 1.5 + \tan^{-1} \frac{1}{1.5} \times \frac{\pi}{80} = 2.088 \right)$, then

$$\text{Area, } A = by + y^2 (\cot \theta + \theta) = (b + \epsilon y) y$$

$$\text{Perimeter, } P = b + 2y (\cot \theta + \theta) = b + 2y\epsilon$$

Inserting the values, we get :

$$A = (b + \epsilon y) y = (32 + 2.088 \times 3.2) \times 3.2 = 123.78 \text{ m}^2$$

$$\text{and, } P = b + 2y\epsilon = 32 + 2 \times 3.2 \times 2.088 = 45.363 \text{ m}$$

$$\text{and, hydraulic radius, } R = \frac{A}{P} = \frac{123.78}{45.363} = 2.729 \text{ m}$$

By using Manning's formula, we get :

$$\begin{aligned} \text{Discharge, } Q &= A \times V = A \times \frac{1}{N} R^{2/3} S^{1/2} \\ &= 123.78 \times \frac{1}{0.017} \times (2.729)^{2/3} \times \left(\frac{1}{4500} \right)^{1/2} = 211.96 \text{ m}^3/\text{s (Ans.)} \end{aligned}$$

Q. 12. (Flow through pipes) Two pipes have length L each. One of them has a diameter of D and other d . If the pipes are arranged in parallel, the head loss is h for a discharge of Q m³/s. When the pipes are connected in series, the head lost is H for same discharge. Find the ratio of H to h , for $D = 2.25 d$. (All dimensions in m) [IES]

Solution. Given: Diameter of pipe 1, $D_1 = D$; Length of pipe; $L_1 = L$; Diameter of pipe 2, $D_2 = d$, Length of pipe 2, $L_2 = L$; Total discharge = Q .

Ratio $\frac{H}{h}$:

Refer to Fig. 12.

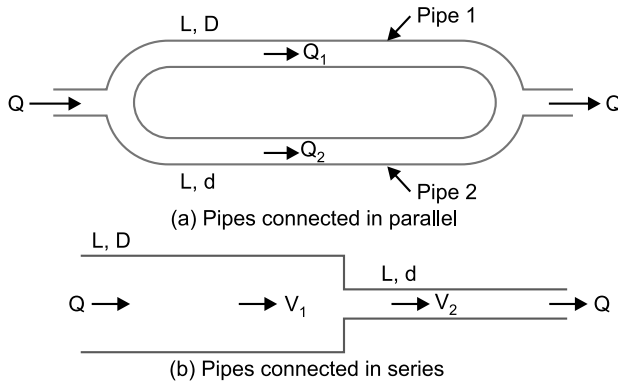


Fig. 12.

Case I When pipes are connected in “parallel.” Refer to Fig. 12 (a).

When pipes are connected in *parallel*,

$$Q = Q_1 + Q_2 \quad \dots(i)$$

Loss of head in each pipe = h

For pipe 1:

$$h = \frac{4fL_1V_1^2}{D \times 2g}$$

where,

$$V_1 = \frac{Q_1}{A_1} = \frac{Q_1}{(\pi/4) \times D^2} = \frac{4Q_1}{\pi D^2}$$

$$\therefore h = \frac{4fL \times \left(\frac{4Q_1}{\pi D^2}\right)^2}{D \times 2g} = \frac{32fLQ_1^2}{\pi^2 D^5 \times g} \quad (\because L_1 = L) \quad \dots(ii)$$

For pipe 2:

$$h = \frac{32fLQ_2^2}{\pi^2 d^5 \times g} \quad \dots(iii)$$

From equal (ii) and (iii), we get:

$$\frac{32fLQ_1^2}{\pi^2 D^5 \times g} = \frac{32fLQ_2^2}{\pi^2 d^5 \times g}$$

or,

$$\frac{Q_1^2}{D^5} = \frac{Q_2^2}{d^5}$$

or,

$$\left(\frac{Q_1}{Q_2}\right)^2 = \left(\frac{D}{d}\right)^5 = \left(\frac{2.25d}{d}\right)^5 = 57.665 \quad (D = 2.25d \dots \text{given})$$

or,

$$\frac{Q_1}{Q_2} = 7.594 \quad \text{or} \quad Q_1 = 7.594 Q_2$$

Substituting the value of Q_1 in eqn. (i), we have:

$$Q = 7.594 Q_2 + Q_2 = 8.594 Q_2$$

or,

$$Q_2 = 0.116 Q \quad \dots(iv)$$

and,

$$Q_1 = 7.594 \times 0.116 Q = 0.881 Q \quad \dots(v)$$

Case II Pipes connected in "series": Refer to Fig. 12 (b).

In this case,

Total loss = Sum of head losses in two pipes

$$\text{or} \quad H = \frac{4fLV_1^2}{D \times 2g} + \frac{4fLV_2^2}{d \times 2g}$$

$$\text{where,} \quad V_1 = \frac{Q}{(\pi/4) \times D^2} = \frac{4Q}{\pi D^2}$$

$$\text{and,} \quad V_2 = \frac{Q}{(\pi/4) \times d^2} = \frac{4Q}{\pi d^2}$$

$$H = \frac{4fL \times \left(\frac{4Q}{\pi D^2}\right)^2}{D \times 2g} + \frac{4fL \times \left(\frac{4Q}{\pi d^2}\right)^2}{d \times 2g}$$

$$\text{or,} \quad H = \frac{32fLQ^2}{\pi^2 D^5 \times g} + \frac{32fLQ^2}{\pi^2 d^5 \times g} \quad \dots (vi)$$

$$\text{From eqn. (ii),} \quad \frac{32fL}{\pi^2 D^5 \times g} = \frac{h}{Q_1^2}$$

$$\text{and, From eqn. (iii)} \quad \frac{32fL}{\pi^2 d^5 \times g} = \frac{h}{Q_2^2}$$

Inserting these values in eqn. (vi), we get:

$$H = Q^2 \times \frac{h}{Q_1^2} + Q^2 \times \frac{h}{Q_2^2} = h \left(\frac{Q^2}{Q_1^2} + \frac{Q^2}{Q_2^2} \right)$$

$$\text{or,} \quad \frac{H}{h} = \frac{Q^2}{Q_1^2} + \frac{Q^2}{Q_2^2}$$

From eqns (iv) and (v), we have:

$$Q_1 = 0.881 Q \quad \text{and} \quad Q_2 = 0.116 Q$$

$$\therefore \quad \frac{H}{h} = \frac{Q^2}{(0.881 Q)^2} + \frac{Q^2}{(0.116 Q)^2} = 1.288 + 74.316 = 75.6 \quad (\text{Ans.})$$

Q. 13. (Triangular weir) Develop (in general terms) the expression for the per cent of error in Q over a triangular weir if there is a small error in the measurement of the vertex angle. Assume there is no error in the weir coefficient. Compute the per cent error in Q , if there is a 1° error in the measurement of total vertex angle of a triangular weir which is having a total vertex angle of 60° . [GATE]

Solution. Given: Error in measurement, $dQ = 1^\circ$; Total vertex angle, $\theta = 60^\circ$

Percentage error in Q :

In case of triangular weir,

$$Q = \frac{8}{15} C_d \sqrt{2g} \cdot \tan \frac{\theta}{2} \cdot H^{5/2}$$

If C_d and H are constant, then:

$$Q = k \cdot \tan \frac{\theta}{2}, \quad \text{where } k \text{ is constant}$$

$$\log Q = \log k + \log \tan \frac{\theta}{2}$$

Differentiating both sides, we get:

$$\frac{dQ}{Q} = 0 + \frac{\sec^2\left(\frac{\theta}{2}\right)}{\tan\left(\frac{\theta}{2}\right)} \cdot \frac{d\theta}{2} \quad \dots(i)$$

When θ is very small or $\theta \rightarrow 0$

$$\sec \theta \rightarrow 1, \quad \tan \theta \rightarrow 0$$

Now eqn. (i) reduces to:

$$\frac{dQ}{Q} = \frac{1}{\left(\frac{\theta}{2}\right)} \cdot \frac{d\theta}{2} = \frac{d\theta}{\theta}$$

For, $d\theta = 1^\circ$ and $\theta = 60^\circ$, we have:

$$\frac{dQ}{Q} = \frac{1}{60} = 0.0167 \quad \text{or} \quad 1.67\% \quad (\text{Ans.})$$

Q. 14. (Fluid kinematics) A flow field is defined by $u = 2y$ and $v = xy$. Derive expressions for the acceleration components. Find the magnitude of the velocity and acceleration at the point (2, 3). Specify units in terms of L and T. [IES]

Solution. Given: $u = 2y$; $v = xy$

Expressions for the acceleration components:

$$\begin{aligned} a_x &= u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \\ &= 2y \times 0 + xy \times 2 = 2xy = 2 \times 2 \times 3 = 12 \\ a_y &= u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \\ &= 2y \times y + xy \times x = xy^2 + x^2y = 2 \times 3^2 + 2^2 \times 3 = 30 \end{aligned}$$

$$\begin{aligned} \text{Magnitude of acceleration} &= \sqrt{a_x^2 + a_y^2} = \sqrt{(12)^2 + (30)^2} \\ &= 32.31 \text{ m/s}^2 \quad \text{or} \quad 32.31 \text{ LT}^{-2} \quad (\text{Ans.}) \end{aligned}$$

$$\begin{aligned} \text{Magnitude of velocity} &= \sqrt{u^2 + v^2} = \sqrt{(2 \times 3)^2 + (2 \times 3)^2} \\ &= 8.48 \text{ m/s} \quad \text{or} \quad 8.48 \text{ LT}^{-1} \quad (\text{Ans.}) \end{aligned}$$

Q. 15. (Boundary layer thickness) For the velocity distribution prescribed by $u/u_0 = (y/\delta)^{1/7}$, show that the ratio of displacement thickness (δ^*) to momentum thickness (θ) is 1.285. [GATE]

Solution. Given: $\frac{u}{u_0} = \left(\frac{y}{\delta}\right)^{1/7}$

$$\frac{\delta^*}{\theta}:$$

Displacement thickness, δ^* :

$$\begin{aligned} \delta^* &= \int_0^\delta \left(1 - \frac{u}{u_0}\right) dy \\ &= \int_0^\delta \left[1 - \left(\frac{y}{\delta}\right)^{1/7}\right] dy = \left[y - \frac{7}{8} \frac{y^{8/7}}{\delta^{1/7}}\right]_0^\delta = \left[\delta - \frac{7}{8} \times \frac{\delta^{8/7}}{\delta^{1/7}}\right] = \frac{\delta}{8} \end{aligned}$$

Momentum thickness, θ :

$$\begin{aligned}\theta &= \int_0^\delta \frac{u}{u_0} \left(1 - \frac{u}{u_0}\right) dy \\&= \int_0^\delta \left(\frac{y}{\delta}\right)^{1/7} \left[1 - \left(\frac{y}{\delta}\right)^{1/7}\right] dy = \int_0^\delta \left[\left(\frac{y}{\delta}\right)^{1/7} - \left(\frac{y}{\delta}\right)^{2/7}\right] dy \\&= \left[\frac{7}{8} \cdot \frac{y^{8/7}}{\delta^{1/7}} - \frac{7}{9} \cdot \frac{y^{9/7}}{\delta^{2/7}}\right]_0^\delta = \left[\frac{7}{8} \cdot \frac{\delta^{8/7}}{\delta^{1/7}} - \frac{7}{9} \cdot \frac{\delta^{9/7}}{\delta^{2/7}}\right] \\&= \left[\frac{7}{8} \delta - \frac{7}{9} \delta\right] = \frac{7}{72} \delta\end{aligned}$$

$$\therefore \frac{\delta^*}{\theta} = \frac{(\delta/8)}{\left(\frac{7}{72} \delta\right)} = 1.285 \dots \text{Proved (Ans.)}$$

Q. 16. (Venturimeter) For a rate of flow exceeding certain value, the coefficient of discharge for a venturimeter used for measuring the discharge of an incompressible fluid is found to be constant. Prove that loss of head in the convergent portion of the venturi can be expressed as KQ^2 under these conditions, where K is a constant (function of C_d , areas of the venturimeter) and Q is flow rate in m^3/s . What will be the value of K , assuming the C_d constant? [IES]

Solution. Given: $C_d = \text{constant}$

Refer to fig. 6.29 (Venturimeter).

Applying Bernoulli's equation at section 1 and 2, we get:

$$\begin{aligned}\frac{p_1}{w} + \frac{V_1^2}{2g} + z_1 &= \frac{p_2}{w} + \frac{V_2^2}{2g} + z_2 \\ \frac{V_2^2 - V_1^2}{2g} &= h_2 - h_1 = \Delta h\end{aligned}$$

where,

$$h = \frac{p}{w} + z = \text{piezometric head}$$

Applying continuity equation at sections 1 and 2, we have:

$$A_1 V_1 = A_2 V_2 \quad \text{or} \quad V_1 = \frac{A_2}{A_1} V_2$$

$$\text{or,} \quad \frac{V_2^2 - \left(\frac{A_2}{A_1} V_2\right)^2}{2g} = \Delta h$$

$$\text{or,} \quad V_2^2 \left[1 - \left(\frac{A_2}{A_1}\right)^2\right] = 2g\Delta h$$

$$\text{or,} \quad V_2 = \frac{\sqrt{2g\Delta h}}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}}$$

$$\text{Now,} \quad Q = \frac{C_d A_2 \sqrt{2g\Delta h}}{\sqrt{1 - \left(\frac{D_2}{D_1}\right)^2}} \quad \text{or} \quad \Delta h = \frac{Q^2}{2gA_2^2 C_d^2} \left\{1 - \left(\frac{D_2}{D_1}\right)^4\right\}$$

Head lost between inlet and thrust is given by:

$$H = \frac{V_2^2}{2g} \left(\frac{1}{C_d^2} - 1 \right) \left\{ 1 - \left(\frac{D_2}{D_1} \right)^4 \right\}$$

$$= (1 - C_d^2) \frac{\left\{ 1 - \left(\frac{D_2}{D_1} \right)^4 \right\} V_2^2}{2g C_d^2}$$

or,

$$H = (1 - C_d^2) \Delta h$$

or,

$$H = \frac{1 - C_d^2}{C_d^2} \cdot \frac{Q^2}{2g A_2^2} \left\{ 1 - \left(\frac{D_2}{D_1} \right)^4 \right\}$$

As C_d , g , A_2 , D_2 , D_1 are constant, therefore,

$$H = KQ^2 \dots \text{Proved} \quad (\text{Ans.})$$

Now,

$$H = (1 - C_d^2) \Delta h = KQ^2$$

$\therefore$

$$K = \frac{1 - C_d^2}{Q^2} \Delta h \quad (\text{Ans.})$$

Q. 17. (Flow through orifices) A cylindrical water tank of 50 cm diameter and 120 cm height is initially filled with water. The top of the tank is open to the atmosphere. A discharge plug is fitted at the bottom of the tank. When the plug near the bottom of the tank is pulled out, a water jet streams out of the tank with a jet diameter of 1 cm. The mean velocity of the jet is given by $V = \sqrt{2gh}$ where h is the height of the water in the tank measured from the centre of the hole. Determine how long will it take for the water level in the tank to drop to 60 cm level from the bottom. Neglect the distance between the bottom of the tank and the centre of the hole compared to the total water height.

[GATE]

Solution. Given: $D = 50 \text{ cm} = 0.5 \text{ m}$; $H_1 = 120 \text{ cm} = 1.2 \text{ m}$; $d = 1 \text{ cm} = 0.01 \text{ m}$; $H_2 = 60 \text{ cm} = 0.6 \text{ m}$.

Refer to Fig. 13.

Time taken for water level to drop to 0.6 m:

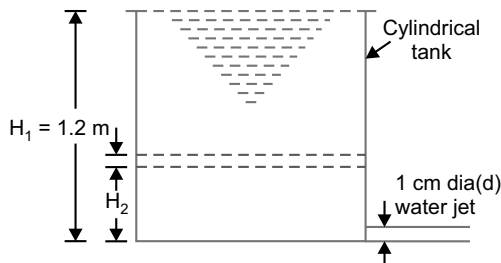


Fig. 13

When the height of water level above the orifice is H , then velocity of jet (V) is given by:

$$V = \sqrt{2gH}$$

If the water level in the tank lowers by dH , in a time t ,

$$\text{then the volume of water} = -\left(\frac{\pi}{4} D^2\right) dH \quad \dots(i)$$

Volume of water flowed out of orifice

$$= V \times a \times dt = V \times \left[\frac{\pi}{4} \times d^2\right] \times dt \quad \dots(ii)$$

Equating both volumes (i.e., (i) and (ii)) we get:

$$-\left(\frac{\pi}{4} D^2\right) dH = V \times \left(\frac{\pi}{4} d^2\right) \times dt = \sqrt{2gH} \times \left(\frac{\pi}{4} d^2\right) \times dt$$

$$\text{or,} \quad \frac{dH}{\sqrt{2gH}} = -\frac{d^2}{D^2} dt$$

Integrating both sides we have:

$$\frac{1}{\sqrt{2g}} \int_{H_1=1.2 \text{ m}}^{H_2=0.6 \text{ m}} \frac{dH}{(H)^{1/2}} = \int_0^T -\frac{0.01^2}{0.5^2} \times dt = -0.0004 \, dt$$

$$\text{or} \quad \int_{H_1}^{H_2} \frac{dH}{(H)^{1/2}} = -\int_0^T \sqrt{2 \times 9.81} \times 0.0004 \, dt = -\int_0^T 0.00177 \, dt$$

$$\text{or} \quad \int_{H_1}^{H_2} \frac{H^{1/2}}{\left(\frac{1}{2}\right)} = -\int_0^T 0.00177 \, dt \quad \text{or} \quad \left[H_2^{1/2} - H_1^{1/2}\right]_{1.2}^{0.6} = -0.00177 \, T$$

$$\text{or,} \quad T = -\left[\frac{2\{(0.6)^{1/2} - (1.2)^{1/2}\}}{0.00177}\right] = 362 \text{ s} = \mathbf{6.04 \text{ minutes (Ans.)}}$$

Q. 18. (Buoyancy and floatation) Determine the position in which a solid cylindrical block of wood of diameter 0.3 m and length 0.4 m will float in water. Take specific gravity of wood as 0.5. [IES]

Solution. Given: $d = 0.3 \text{ m}$; $l = 0.4 \text{ m}$; $S = 0.5$.

Refer to Fig. 14.

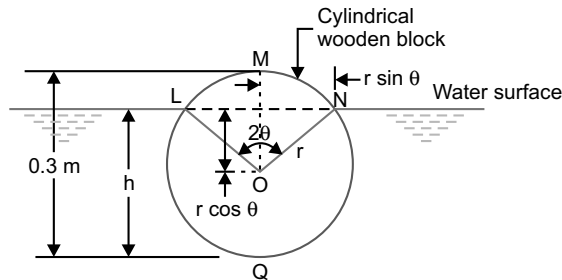


Fig. 14

$$\text{Weight of the wooden block} = (0.5 \times 9.81) \times \frac{\pi}{4} d^2 l$$

$$= (0.5 \times 9.81) \times \frac{\pi}{4} \times 0.3^2 \times 0.4 = 0.1387 \text{ kN}$$

This also represents the weight of water displaced.

$$\therefore \text{Volume of water displaced} = \frac{0.1387}{9.81} = 0.01414 \text{ m}^3$$

Let, h = Depth of immersion

$$\therefore \text{Volume of wooden block inside water} = \text{Volume of water displaced} = 0.01414 \text{ m}^3$$

$$\text{Now, } 0.01414 = \text{Area } LQNL \times 0.4$$

$$\text{or, } \text{Area } LQNL = \frac{0.01414}{0.4} = 0.03535 \text{ m}^2$$

$$\begin{aligned} \text{Also, } \text{Area } LQNL &= \text{Area } LQNL + \text{Area } LON \\ &= \pi r^2 \left[\frac{360 - 2\theta}{360} \right] + \frac{1}{2} \times 2r \sin \theta \times r \cos \theta \\ &= \pi r^2 \left(1 - \frac{\theta}{180} \right) + r^2 \sin \theta \cos \theta \end{aligned}$$

$$\text{or, } 0.03535 = \pi \times 0.15^2 \left(1 - \frac{\theta}{180} \right) + (0.15)^2 \sin \theta \cos \theta$$

$$\text{or, } 0.03535 = 0.0707 - 0.0003927 \theta + 0.000225 \sin \theta \cos \theta$$

$$\text{or, } 0.0003927 \theta - 0.0225 \sin \theta \cos \theta = 0.03535$$

$$\text{or, } \theta - 57.3 \sin \theta \cos \theta = 90.02$$

By trial and error, we get $\theta \approx 90.1^\circ$

Depth of wooden block in water,

$$\begin{aligned} h &= r + r \cos \theta \\ &= 0.15 + 0.15 \times \cos 90.1^\circ = \mathbf{0.1497 \text{ m (Ans.)}} \end{aligned}$$

Q. 19. (Turbulent flow in rough pipe) A rough pipe of diameter 0.1 m carries water at 20°C at the rate of 50 L/s. If the average height of the roughness projections on the pipe surface is 0.15 mm, determine:

- Friction factor.
- Shear velocity.
- Shear stress at the pipe surface.
- Maximum velocity.

Take for water at 20°C

$$\nu = \text{Kinematic viscosity} = 1 \times 10^{-6} \text{ m}^2/\text{s}.$$

$$\rho = \text{Mass density} = 1000 \text{ kg/m}^3.$$

[IES]

Solution. Given: $D = 0.1 \text{ m}$; $Q = 50 \text{ litres/s} = 0.05 \text{ m}^3/\text{s}$; $k = 0.15 \text{ mm} = 0.15 \times 10^{-3} \text{ m}$; $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$; $\rho = 1000 \text{ kg/m}^3$.

(i) Friction factor f_1 :

$$\text{Mean velocity of water, } \bar{U} = \frac{Q}{A} = \frac{0.05}{\frac{\pi}{4} \times 0.1^2} = 6.37 \text{ m/s}$$

$$\text{Reynolds number, } Re = \frac{\bar{U}D}{\nu} = \frac{6.37 \times 0.1}{1 \times 10^{-6}} = 6.37 \times 10^5$$

Since $Re > 10^5$, therefore the flow is *turbulent* in the rough pipe.

We know that for rough turbulent flow,

$$\frac{1}{\sqrt{f_1}} = 2.0 \log_{10} \left(\frac{R}{k} \right) + 1.74 \quad \dots [\text{Eqn. (11.39)}]$$

$$\text{or,} \quad \frac{1}{\sqrt{f_1}} = 2.0 \log_{10} \left(\frac{0.05}{0.15 \times 10^{-3}} \right) + 1.74 = 6.78$$

$$\therefore f_1 = \left(\frac{1}{6.78} \right)^2 = \mathbf{0.0217 \text{ (Ans.)}}$$

(ii) Shear velocity, u_f :

$$\text{Friction coefficient } f = \frac{f_1}{4} = \frac{0.0217}{4} = 0.00542$$

$$\text{Also,} \quad u_f = \bar{U} \sqrt{\frac{f}{2}} = 6.37 \times \sqrt{\frac{0.00542}{2}} = \mathbf{0.332 \text{ m/s (Ans.)}}$$

(iii) Shear stress at pipe surface, τ_0 :

$$\text{We know that,} \quad f = \frac{2\tau_0}{\rho \bar{U}^2} \quad \dots [\text{Eqn. 11.5(b)}]$$

$$\text{or,} \quad \tau_0 = \frac{f \rho \bar{U}^2}{2} = \frac{0.00542 \times 1000 \times (6.37)^2}{2} = \mathbf{109.96 \text{ N/m}^2 \text{ (Ans.)}}$$

(iv) Maximum velocity, $U_{\max}$:

$$\frac{U_{\max}}{u_f} = 5.75 \log_{10} (y/k) + 8.5 \quad \dots [\text{Eqn. (11.25)}]$$

(This equation is known as Karman-Prandtl equation for the velocity distribution near hydrodynamically rough boundaries).

$$\text{or,} \quad \frac{U_{\max}}{0.332} = 5.75 \log_{10} \left(\frac{0.05}{0.15 \times 10^{-3}} \right) + 8.5 \quad (\because y = R = 0.05 \text{ m})$$

$$\text{or,} \quad U_{\max} = 0.332 \times \left[5.75 \log_{10} \left(\frac{0.05}{0.15 \times 10^{-3}} \right) + 8.5 \right] = \mathbf{7.64 \text{ m/s (Ans.)}}$$

Q. 20. (Laminar flow) Show that the friction factor is inversely proportional to the Reynolds number in case of Laminar flow in circular pipes. **[GATE]**

Solution. The loss of head for energy (h_f) in pipes due to friction is calculated from Darcy-Weisbach formula which is given by:

$$h_f = \frac{f_1 L \bar{u}^2}{D \times 2g} \quad (\text{where } f_1 \text{ is the friction factor}) \quad \dots (i)$$

From Hagen-Poiseuille formula, the loss of pressure head due to flow of viscous fluid is given as:

$$h_f = \frac{32 \mu \bar{u} L}{\rho g D^2} \quad \dots (ii)$$

From eqns. (i) and (ii), we get:

$$\frac{f_1 L \bar{u}^2}{D \times 2g} = \frac{32 \mu \bar{u} L}{\rho g D^2}$$

$$\text{or, } f_1 = \frac{(D \times 2g) \times 32\mu\bar{u}L}{L\bar{u}^2 \times \rho g D^2} = \frac{64\mu\bar{u}}{\rho \bar{u}^2 D} = \frac{64}{\left(\frac{\rho\bar{u}D}{\mu}\right)} = \frac{64}{Re}$$

$$\text{i.e., } f_1 \propto \frac{1}{Re} \quad \text{.....Proved (Ans.)}$$

Q. 21. (Compressible flow) An airplane flies at an altitude with a velocity of 800 km/hr. The pressure and temperature at that altitude are $1.206 \times 10^4 \text{ N/m}^2$ and 217 K respectively. Calculate:

- The maximum possible temperature on the airplane skin.
- The maximum possible pressure intensity on the airplane body.
- The critical velocity of air relative to the airplane.
- The maximum possible velocity of the air relative to the airplane.

[IES]

Solution. Given: Velocity of air plane, $V_1 = 800 \text{ km/h} = \frac{800 \times 1000}{3600} = 222.22 \text{ m/s}$;
 $p_1 = 1.206 \times 10^4 \text{ N/m}^2$; $T_1 = 217 \text{ K}$.

- (i) Maximum possible temperature on the airplane skin, T_0 :

T_0 will be stagnation temperature because velocity of air also gets converted to temperature.

$$\begin{aligned} \text{Now, } T_0 &= T_1 + \frac{V_1^2}{2c_p} \\ &= 217 + \frac{(222.22)^2}{2 \times 1.005 \times 10^3} = \mathbf{241.57 \text{ K (Ans.)}} \end{aligned}$$

- (ii) Maximum possible pressure intensity on the airplane body, p_0 :

$$\frac{T_0}{T_1} = \left(\frac{p_0}{p_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\text{or, } \frac{p_0}{p_1} = \left(\frac{T_0}{T_1}\right)^{\frac{\gamma}{\gamma-1}} \quad \text{or } p_0 = p_1 \left(\frac{T_0}{T_1}\right)^{\frac{\gamma}{\gamma-1}}$$

$$\text{or, } p_0 = 1.206 \times 10^4 \left(\frac{241.57}{217}\right)^{\frac{1.4}{0.4}} = \mathbf{1.755 \times 10^4 \text{ N/m}^2 \text{ (Ans.)}}$$

- (iii) Critical velocity of air relative to airplane, C :

$$C = \sqrt{\gamma R T_1} = \sqrt{1.4 \times 287 \times 217} = \mathbf{295.28 \text{ m/s (Ans.)}}$$

- (iv) Maximum possible velocity of air relative to the airplane :

$$\begin{aligned} \text{Maximum possible velocity} &= \sqrt{\gamma R T_0} \\ &= \sqrt{1.4 \times 287 \times 241.57} \\ &= 311.55 \text{ m/s or } \mathbf{1121.58 \text{ km/h (Ans.)}} \end{aligned}$$

Q. 22. (Flow through pipes) Develop an expression for the pressure loss across a conical contraction of diameter D_1 and D_2 in a given length L for a laminar flow. [IES]

Solution. Refer to Fig. 15.

Expression for pressure loss, $(p_1 - p_2)$:

Consider a stretch of length dx at a distance x from D_1 diameter end.

Diameter at the section XX ,

$$D = D_1 - \left(\frac{D_1 - D_2}{L} \right) x$$

The head loss due to friction in pipes of uniform diameter D and length L is given by :

$$h_f = \frac{fLV^2}{D \times 2g} \dots \text{Darcy-Weisbach equation}$$

Where, ' f ' is the friction factor.

Now, loss of head (dh_f) through the small stretch dx ,

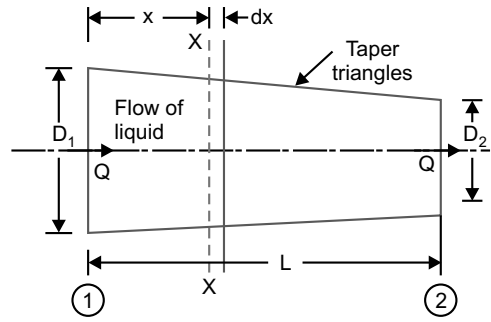


Fig. 15

$$dh_f = \frac{f \cdot (dx) \cdot V^2}{\left[D_1 - \left(\frac{D_1 - D_2}{L} \right) x \right] 2g} \dots (i)$$

We know that, f (friction factor) = $\frac{64}{Re} = \frac{64\mu}{\rho V D}$

Let, $\frac{D_1 - D_2}{L} = k$

Then, $f = \frac{64\mu}{[\rho V (D_1 - kx)]}$

[i.e., diameter at a distance x becomes $(D_1 - kx)$]

Let, $D_1 - kx = D'$

Then, $f = \frac{64\mu}{\rho V D'} = \frac{64\mu}{\rho \left[Q \div \left(\frac{\pi}{4} D'^2 \right) \right] D'} = \frac{16\pi\mu D'}{\rho Q}$

Substituting this value in eqn. (i), we get:

$$dh_f = \frac{16\pi\mu D'}{\rho Q D' \times 2g} \times \left[\frac{Q}{\frac{\pi}{4} D'^2} \right]^2 \times dx$$

$$\left[\because D_1 - \left(\frac{D_1 - D_2}{L} \right) x = D_1 - kx = D' \right]$$

or, $dh_f = \frac{128vQ}{\pi g} \cdot \frac{dx}{(D')^4} \quad \left(\because \frac{\mu}{\rho} = v \right)$

or, $dh_f = \frac{128vQ}{\pi g} \times \frac{dx}{(D_1 - kx)^4}$

Integrating both sides, we get:

$$h_f = \int_0^{h_f} dh_f = \int_0^L \frac{128vQ}{\pi g} \times \frac{dx}{(D_1 - kx)^4}$$

$$= \frac{128vQL}{\pi g \cdot (D_1 - D_2)} \left[\frac{1}{D_2^3} - \frac{1}{D_1^3} \right] \times \frac{1}{3}$$

$$\text{or, } h_f = \frac{p_1 - p_2}{\rho g} = \frac{128\mu QL}{\pi \rho g (D_1 - D_2)} \left[\frac{1}{D_2^3} - \frac{1}{D_1^3} \right] \times \frac{1}{3}$$

$$\text{Hence, } p_1 - p_2 = \frac{128\mu QL}{3\pi (D_1 - D_2)} \left[\frac{1}{D_2^3} - \frac{1}{D_1^3} \right] \quad (\text{Ans.})$$

Q. 23. (Compressible flow-venturimeter) Calculate the mass flow rate of oxygen through a 100 mm diameter horizontal pipe if the inlet pressure is 1.06 m of mercury. The diameter at the throat is 30 mm. The ratio of pressure between the venturimeter and the inlet is 0.9. Take $\gamma = 1.4$ and density of oxygen gas is 1.429 kg/m^3 . [IES]

Solution. Given: Pipe diameter, $D_1 = 100 \text{ mm} = 0.1 \text{ m}$; Throat diameter, $D_2 = 30 \text{ mm} = 0.03 \text{ m}$; Pressure at inlet $p_1 = 1.06 \text{ m of Hg} = (13.6 \times 1000) \times 9.81 \times 1.06 = 1.414 \times 10^5 \text{ N/m}^2$; $\frac{p_2}{p_1} = 0.9$; $\gamma = 1.4$; Density of oxygen gas, $\rho_1 = 1.429 \text{ kg/m}^3$.

Refer to Fig. 16.

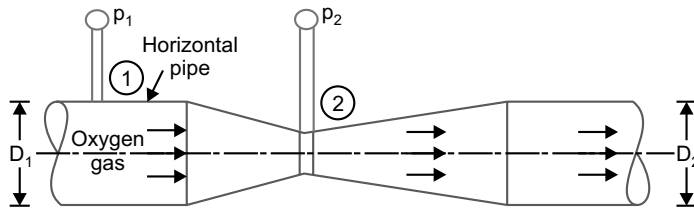


Fig. 16. Venturimeter.

Mass flow rate of Oxygen, m :

Applying Bernoulli's equation at points 1 and 2, we get:

$$\left(\frac{\gamma}{\gamma - 1} \right) \frac{p_1}{\rho_1 g} + \frac{V_1^2}{2g} + z_1 = \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_2}{\rho_2 g} + \frac{V_2^2}{2g} + z_2$$

$$\text{or, } \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_1}{\rho_1} + \frac{V_1^2}{2} = \left(\frac{\gamma}{\gamma - 1} \right) \frac{p_2}{\rho_2} + \frac{V_2^2}{2} \quad (\because z_1 = z_2, \text{ pipe being horizontal})$$

$$\text{or, } \frac{\gamma}{\gamma - 1} \left(\frac{p_1}{\rho_1} - \frac{p_2}{\rho_2} \right) = \frac{V_2^2}{2} - \frac{V_1^2}{2} \quad \dots(i)$$

For compressible fluids,

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \text{ or } V_1 = \frac{\rho_2 A_2 V_2}{\rho_1 A_1}$$

For adiabatic flow,

$$\frac{p_1}{\rho_1^\gamma} = \frac{p_2}{\rho_2^\gamma} \quad \text{or} \quad \frac{\rho_2}{\rho_1} = \left(\frac{p_2}{p_1} \right)^{\frac{1}{\gamma}}$$

Substituting the values in eqn. (i), we have:

$$\frac{\gamma}{\gamma - 1} \left(\frac{p_1}{\rho_1} - \frac{p_2}{\rho_2} \right) = \left[\frac{V_2^2}{2} - \frac{1}{2} \left(\frac{\rho_2 A_2 V_2}{\rho_1 A_1} \right)^2 \right]$$

$$\text{or, } = \frac{V_2^2}{2} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma}{\gamma - 1}} \times \frac{A_2^2}{A_1^2} \right]$$

$$\text{or, } V_2 = \sqrt{\frac{\frac{2\gamma}{\gamma-1} \cdot \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right]}{\left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{2}{\gamma}} \frac{A_2^2}{A_1^2} \right]}}$$

$$\text{Mass flow rate, } m = \rho A_2 V_2$$

$$\text{or, } m = \rho_2 A_2 \sqrt{\frac{\frac{2\gamma}{\gamma-1} \cdot \frac{p_1}{\rho_1} \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \right]}{\left[1 - \frac{A_2^2}{A_1^2} \left(\frac{p_2}{p_1} \right)^{\frac{2}{\gamma}} \right]}} \quad \dots(ii)$$

$$\text{Now, } A_1 = \frac{\pi}{4} D_1^2 = \frac{\pi}{4} \times 0.1^2 = 7.854 \times 10^{-3} \text{ m}^2$$

$$A_2 = \frac{\pi}{4} D_2^2 = \frac{\pi}{4} \times 0.03^2 = 7.068 \times 10^{-4} \text{ m}^2$$

$$\rho_2 = \rho_1 \left(\frac{p_2}{p_1} \right)^{1/\gamma} = 1.429 \times (0.9)^{1/1.4} = 1.325 \text{ kg/m}^3$$

$$\left[\because \frac{p_2}{p_1} = \left(\frac{p_2}{p_1} \right)^{\frac{1}{\gamma}} \right]$$

Inserting the various values in eqn. (ii) we get:

$$\begin{aligned} m &= 1.325 \times (7.068 \times 10^{-4}) \sqrt{\frac{\frac{2 \times 1.4}{(1.4-1)} \times \frac{1.414 \times 10^5}{1.429} \left\{ 1 - (0.9)^{\frac{1.4-1}{1.4}} \right\}}{1 - \left(\frac{7.068 \times 10^{-4}}{7.854 \times 10^{-3}} \right)^2 \times (0.9)^{\frac{2}{1.4}}}} \\ &= 9.365 \times 10^{-4} \sqrt{\frac{(6.926 \times 10^5) \times 0.0296}{0.993}} = \mathbf{0.1346 \text{ kg/s (Ans.)}} \end{aligned}$$

Q. 24. (Model analysis) Show that for a flow governed only by gravity, inertia, and pressure forces, the ratio of rates of flow in two dynamically similar systems equal to the $\frac{5}{2}$ power of the length ratio. [GATE]

Solution. According to Froude model law:

$$(F_r)_m = (F_r)_p$$

$$\text{or, } \frac{V_m}{\sqrt{g_m L_m}} = \frac{V_p}{\sqrt{g_p L_p}} \quad \dots[\text{Eqn. (7.22)}]$$

$$\text{or, } \frac{V_p}{V_m} = \sqrt{\frac{L_p}{L_m}} = \sqrt{L_r} = V_r \quad (\because g_m = g_p)$$

$$\text{Flow rate ratio, } Q_r = \frac{Q_p}{Q_m} = \frac{A_p V_p}{A_m V_m} = L_r^2 V_r = L_r^2 \times L_r = (L_r)^{5/2} \quad (\text{Ans.})$$

Q. 25. (Laminar flow) Oil of viscosity 0.97 poise and 0.9 specific gravity flows through a horizontal circular pipe of 100 mm diameter and 10 m length. Calculate the pressure difference between the ends of the pipe. What is the type of flow?

Solution. Given: $\mu = 0.97 = 0.097 \text{ Ns/m}^2$; $S = 0.9$, $D = 100 \text{ mm} = 0.1 \text{ m}$; $L = 10 \text{ m}$.

$p_1 - p_2$:

We know that :
$$p_1 - p_2 = \frac{32\mu\bar{u}L}{D^2} \quad \dots[\text{Eqn. (10.11)}]$$

Where $\bar{u}$ is an average velocity.

Now,
$$Q = \bar{u} \times A \quad \text{or} \quad \bar{u} = \frac{Q}{A} = \frac{Q}{\frac{\pi}{4} \times 0.1^2} = 127.3 Q$$

$\therefore$
$$p_1 - p_2 = \frac{32 \times 0.097 \times (127.3 Q) \times 10}{0.1^2} = 39.5 \times 10^4 Q \text{ (Ans.)}$$

Type of flow:

Since discharge Q is not given, $\bar{u}$ cannot be found, therefore, type of flow cannot be found,

which depends on the value of Reynolds number Re , where $Re = \frac{\rho\bar{u}D}{\mu}$ (Ans.)

Q. 26. (Flow through pipes) A pipe of 300 mm diameter and 1000 m length connects two reservoirs kept at 15 m level difference. Find the flow through the pipe. If an additional length of 500 m of 300 mm diameter pipe is connected at the later half of the existing pipe, find the increase in discharge. Neglect minor losses. Take $f = (\text{coefficient of friction}) = 0.005$. [GATE]

Solution. Given: $D = 300 \text{ mm} = 0.3 \text{ m}$; $L = 1000 \text{ m}$; $h_1 = h_2 = 500 \text{ m}$; $D_2 = D_1 = 300 \text{ mm} = 0.3 \text{ m}$; $f = 0.005$; $h_f = 15 \text{ m}$.

All losses are negligible except friction.

Increase in discharge:

Refer to Fig. 17.

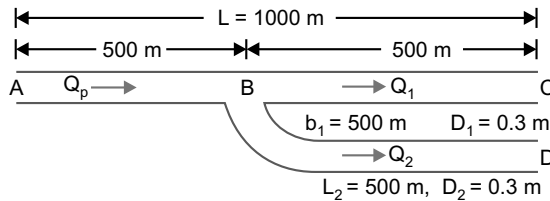


Fig. 17

Case I: Discharge Q for a single pipe of length 1000 m and diameter 0.3 m:

The head lost due to friction in a single pipe is given as:

$$h_f = \frac{4fLV^2}{D \times 2g}, \quad \text{where } V \text{ is the velocity of flow in a single pipe}$$

or,
$$15 = \frac{4 \times 0.005 \times 1000 \times V^2}{0.3 \times 2 \times 9.81}$$

or,
$$V^2 = \left(\frac{15 \times 0.3 \times 2 \times 9.81}{4 \times 0.005 \times 1000} \right)^{1/2} = 2.1 \text{ m/s}$$

$\therefore$ Discharge, $Q = A \times V = \frac{\pi}{4} \times 0.3^2 \times 2.1 = 0.1484 \text{ m}^3/\text{s}$... (i)

Case II: When an additional pipe of length 500 m and diameter 0.3 m is connected in parallel with the last half length of the pipe:

Let, Q_1 = Discharge in first parallel pipe,
 Q_2 = Discharge in second parallel pipe, and
 Q_p = Discharge in the main pipe (when pipes are connected in parallel)

Then, $Q_p = Q_1 + Q_2$

As the pipes in parallel have the same length and diameter, therefore,

$$Q_1 = Q_2 = \frac{Q_p}{2}$$

Consider the flow through ABC or ABD:

Head lost (due to friction) in ABC = Head lost in AB + Head lost in BC ... (ii)

Head lost in ABC = 15 m (given)

Now, Head lost in AB = $\frac{4 \times 0.005 \times 500 \times V_{AB}^2}{0.3 \times (2 \times 9.81)}$

But, $V_{AB} = \frac{Q_p}{\text{Area}} = \frac{Q_p}{\frac{\pi}{4} \times 0.3^2} = 14.15 Q_p$

$\therefore$ Head lost in friction in AB = $\frac{4 \times 0.005 \times 500 \times (14.15 Q_p)^2}{0.3 \times (2 \times 9.81)} = 340.2 Q_p^2$

Head lost due to friction in BC

$$= \frac{4 \times 0.005 \times 500 \times V_{BC}^2}{0.3 \times (2 \times 9.81)}$$

But, $V_{BC} = \frac{(Q_p/2)}{\text{Area}} = \frac{(Q_p/2)}{\frac{\pi}{4} \times 0.3^2} = 7.07 Q_p$

$\therefore$ Head lost due to friction in BC

$$= \frac{4 \times 0.005 \times 500 \times (7.07 Q_p)^2}{0.3 \times (2 \times 9.81)} = 84.9 Q_p^2$$

Substituting the values in eqn. (ii), we get:

$$15 = 340.2 Q_p^2 + 84.9 Q_p^2 = 425.1 Q_p^2$$

or, $Q_p = \left(\frac{15}{425.1} \right)^{1/2}$

$$= 0.1878 \text{ m}^3/\text{s}$$

Hence, increase in discharge = $Q_p - Q = 0.1878 - 0.1484$
 $= 0.0394 \text{ m}^3/\text{s}$ (Ans.)

Q. 27. (Venturimeter) Determine the throat diameter of a venturimeter in a installation of 100 mm diameter pipe carrying water. The maximum range available in water-Hg differential pressure gauge is 50 cm of Hg deflection. Find the maximum throat diameter which will induce full gauge deflection when the flow rate is 20 l/s. Assume $C_d = 0.984$. [IES]

Solution. Given: Diameter of the main pipe, $D = 100 \text{ mm} = 0.1 \text{ m}$; Difference of pressure head, $y = 50 \text{ cm of Hg} = 0.5 \text{ m of Hg}$; Flow rate, $Q = 20 \text{ litres/sec} = 0.02 \text{ m}^3/\text{s}$; $C_d = 0.984$.

Maximum throat diameter, d :

We know that:
$$h = y \left[\frac{S_{Hg}}{S_{water}} - 1 \right] = 0.5 \left[\frac{13.6}{1} - 1 \right] = 6.3 \text{ m}$$

Also,
$$Q = C_d \times \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \times \sqrt{2gh} \quad \dots [\text{Eqn. (6.6)}]$$

$$\begin{aligned} 0.02 &= 0.984 \times \frac{\left(\frac{\pi}{4} \times 0.1^2\right) \times \left(\frac{\pi}{4} \times d^2\right)}{\sqrt{\left(\frac{\pi}{4}\right)^2 [0.1^4 - d^4]}} \times \sqrt{2 \times 9.81 \times 6.3} \\ &= \frac{0.08592 d^2}{\sqrt{0.1^4 - d^4}} \end{aligned}$$

Squaring both sides, we get:

$$(0.02)^2 = \frac{(0.08592)^2 d^4}{(0.1)^4 - d^4} = \frac{0.738 \times 10^{-3} d^4}{1 \times 10^{-4} - d^4}$$

or, $4 \times 10^{-4} (1 \times 10^{-4} - d^4) = 0.738 \times 10^{-3} d^4$

or, $4 \times 10^{-8} = d^4 (4 \times 10^{-4} + 0.738 \times 10^{-3})$

$$d = \left[\frac{4 \times 10^{-8}}{4 \times 10^{-4} + 0.738 \times 10^{-3}} \right]^{1/4} = 0.0476 \text{ m or } 47.6 \text{ mm (Ans.)}$$

Q. 28. (Flow through pipes) A reservoir supplies $45,000 \text{ m}^3$ of water in 8 hours. During this time, the level of water in the reservoir drops from 125 m to 75 m. The head loss in the pipeline is 15 m. Determine the diameter of the supply pipeline using $4f = 0.04$ in Darcy-Weisbach equation. [GATE]

Solution. Given: $Q = 45,000 \text{ m}^3$; Drop in level of water = $125 - 75 = 50 \text{ m}$.

Diameter of supply line d :

Head loss in pipeline is 15 m, so the net driving potential for velocity at any time = $(h - 15) \text{ m}$

Area of cross-section of tank, $A = \frac{45000}{50} = 900 \text{ m}^2$

We know that:
$$\frac{dQ}{dt} = -A \frac{dh}{dt} = C_d \cdot a \cdot \sqrt{2g(h-15)} \quad (\because 'h' \text{ decreases})$$

Assuming, $C_d = 1$, we get:

$$\frac{dh}{\sqrt{h-15}} = -\frac{a}{A} \sqrt{2g} \cdot dt$$

$$\int_{h_1=125}^{h_2=75} \frac{dh}{\sqrt{h-15}} = -\frac{a}{A} \sqrt{2g} \cdot \int_0^t dt = -\frac{a}{A} \cdot \sqrt{2g} \cdot t$$

$$\text{or, } 2[\sqrt{125-15} - \sqrt{75-15}] = \frac{a \times \sqrt{2 \times 9.81}}{900} \times (8 \times 3600)$$

$$\text{or, } 5.484 = a \times 141.74 \quad \text{i.e., } a = 0.0387 \text{ m}^2$$

$$\text{or, } \frac{\pi}{4} d^2 = 0.0387 \quad \text{or } d = 0.222 \text{ m} \quad \text{or } \mathbf{22.2 \text{ cm (Ans.)}}$$

Q. 29. (Hydrostatic forces on surfaces) A ship is sailing in the ocean. The captain of the ship observes an iceberg floating in the sea at a distance and he estimates the visible volume of the iceberg above sea level as 600 m^3 . If the density of the iceberg is 915 kg/m^3 , determine the total volume and weight of the iceberg. The density of the sea water may be taken as 1025 kg/m^3 .

[GATE]

Solution. Given: Volume of iceberg above sea level = 600 m^3 ; Density of iceberg, $\rho_i = 915 \text{ kg/m}^3$; Density of sea water, $\rho_s = 1025 \text{ kg/m}^3$.

Total volume of iceberg, V :

$$\text{Volume of ice in sea water} = (V - 600) \text{ m}^3$$

Now, Weight of iceberg = Weight of water displaced by ice

$$\text{or, } \rho_i \times V \times g = \rho_s (V - 600) \times g$$

$$\text{or, } V = \frac{\rho_s}{\rho_i} (V - 600) = \frac{1025}{915} (V - 600) = 1.1202 (V - 600)$$

$$\text{or, } V = 1.1202 V - 672.12 \quad \text{or } V = \mathbf{5592 \text{ m}^3 \text{ (Ans.)}}$$

Weight of iceberg, W_i :

$$W_i = \rho_i \times V \times g = 915 \times 5592 \times 9.81 = \mathbf{50.194 \times 10^6 \text{ N (Ans.)}}$$

B. MULTIPLE-CHOICE QUESTIONS WITH ANSWERS AND EXPLANATIONS

Q. 1. In a steady laminar flow of a given discharge through a circular pipe of diameter D , the head loss is proportional to

- (a) D^{-1} (b) D^{-2} (c) D^{-3} (d) D^{-4}

[ESE : 2017]

Ans. (d)

Explanation:

$$\text{Head loss, } h_f = 32 = \frac{\mu \bar{u} L}{\rho g D^2} = \frac{32 \times \mu Q L}{\rho g (\pi / 4 D^2) \times D^2} = \frac{128 \mu Q L}{\rho g \pi D^4}$$

or, $h_f \propto D^{-4}$, hence (d) is the correct choice. (Ans.)

Q. 2. A spherical water drop of 1 mm in diameter splits up in air into 64 smaller drops of equal size. The surface tension coefficient of water in air is 0.073 N/m . The work required in splitting up the drop is

- (a) $0.96 \times 10^{-6} \text{ J}$ (b) $0.69 \times 10^{-6} \text{ J}$ (c) $0.32 \times 10^{-6} \text{ J}$ (d) $0.23 \times 10^{-6} \text{ J}$

[ESE : 2017]

Ans. (b)

Explanation: Volume of water drop before splitting
= Volume after splitting

$$\frac{4}{3}\pi R^3 = 64 \times \frac{4}{3}\pi r^3 \quad (\text{where } r \text{ is the radius of each smaller drop})$$

or,
$$r = \left[\frac{(0.5)^3}{(64)} \right]^{1/3} = 0.125 \text{ mm}$$

Work required in splitting up the drop,

$$\begin{aligned} W &= \sigma (\Delta A) \\ &= 0.073 \times (64 \times 4\pi r^2 - 4\pi R^2) \\ &= 0.073 \times 4\pi [64 \times (0.125)^2 - (0.5)^2] \times 10^{-6} \\ &= 0.69 \times 10^{-6} \text{ J} \quad (\text{Ans.}) \end{aligned}$$

Q. 3. An oil flows through a pipe at a velocity of 1.0 m/s. The pipe is 45 m long and has 150 mm diameter. What is the head loss due to friction, if $\rho = 869 \text{ kg/m}^3$ and $\mu = 0.0814 \text{ kg/ms}$?

- (a) 0.61 m (b) 0.51 m (c) 0.41 m (d) 0.31 m

[ESE : 2017]

Ans. (a)

Explanation:

Reynolds number,
$$\text{Re} = \frac{\rho \bar{u} D}{\mu} = \frac{869 \times 1 \times 0.15}{0.0814} = 1601$$

As $\text{Re}, 1601 < 2000$, therefore flow is *laminar*.

Head loss due to friction,
$$h_f = \frac{32 \mu \bar{u} L}{\rho g D^2} = \frac{32 \times 0.0814 \times 1 \times 45}{869 \times 9.81 \times (0.15)^2}$$

= 0.61 m (Ans.)

Q. 4. The centre-line velocity in a pipe flow is 2 m/s. What is the average flow velocity in the pipe if the Reynolds number of the flow is 800?

- (a) 2 m/s (b) 1.5 m/s (c) 1 m/s (d) 0.5 m/s

[ESE : 2017]

Ans. (c)

Explanation:

$$\mu_{\max} = 2 \text{ m/s}, \text{ Re} = 800$$

Now,
$$\mu_{\text{ary}} = \frac{\mu_{\max}}{2} = \frac{2}{1} = 1 \text{ m/s} \quad (\text{Ans.})$$

Q. 5. Consider the following statements regarding Bernoulli's equation:

1. It is assumed that no energy has been supplied.
2. The velocity of a steady stream of fluid flow will depend on the cross-sectional area of the stream.
3. Consider two sections 1 and 2 along a flow stream. In this reach, if q is work done by a pump, w is work absorbed by turbine, ρ is density of water and g is acceleration of gravity, with p , v and z carrying standard meanings, Bernoulli's equation will read

$$\frac{p_1}{\rho} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho} + \frac{v_2^2}{2g} + z_2 + w + q$$

Which of the above statements are correct?

- (a) 1 and 2 only (b) 1 and 3 only (c) 2 and 3 (d) 1, 2 and 3

[ESE : 2017]

Ans. (a)

Explanation:

General energy equation is given as:

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 + h_p = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_T$$

Where, h_p and h_T are pump head and turbine head thus statement 3 is incorrect respectively.

(Ans.)

- Q. 6. Water is coming out from a tap and falls vertically downwards. At the tap opening, the stream diameter is 20 mm with uniform velocity of 2 m/s. Assuming steady inviscid flow, constant pressure atmosphere everywhere, and neglecting curvature and surface tension effects, the diameter of the stream 0.5 m below the tap opening is nearly

- (a) 11.7 mm (b) 14.6 mm (c) 17.5 mm (d) 20.4 mm

[ESE : 2017]

Ans. (b)

Explanation:

$D_1 = 20 \text{ mm}$; $V_1 = 2 \text{ m/s}$, $z_1 = 0$, $z_2 = -0.5 \text{ m}$; $D_2 = ?$

Using energy equation, we have:

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

Now,
$$\frac{p_1}{\rho g} = \frac{p_2}{\rho g} = \frac{p_{\text{atm}}}{\rho g}$$

$$\therefore \frac{(2)^2}{2g} + 0 = \frac{V_2^2}{2g} - 0.5$$

or,
$$\frac{V_2^2}{2g} = \frac{2^2}{2 \times 9.81} + 0.5 = 0.7039$$

or,
$$V_2 = [(2 \times 9.81) \times 0.7039]^{1/2} = 3.716 \text{ m/s}$$

Also,
$$A_1 V_1 = A_2 V_2$$

or,
$$\frac{\pi}{4} D_1^2 \times V_1 = \frac{\pi}{4} D_2^2 \times V_2$$

or,
$$(20)^2 \times 2 = D_2^2 \times 3.716 \text{ or } D_2 = 14.67 \text{ mm (Ans.)}$$

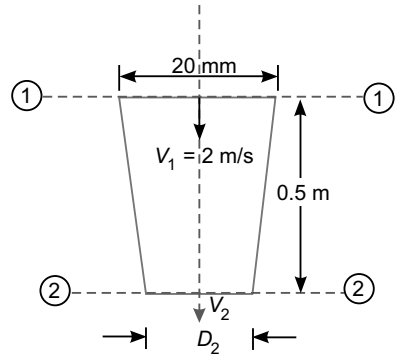


Fig. 1

- Q. 7. A two-dimensional flow field is defined as $\vec{V} = \vec{i}x - \vec{j}y$. The equation of the streamline passing through the point (1, 2) is

- (a) $xy + 2 = 0$ (b) $x^2y + 2 = 0$ (c) $xy - 2 = 0$ (d) $x^2y - 2$

[ESE : 2017]

Ans. (c)

Explanation: $\vec{V} = \vec{i}x - \vec{j}y$

$$u = x; \quad v = -y$$

Now, $\frac{dx}{u} = \frac{dy}{v}$

or, $\int \frac{dx}{x} = \int \frac{dy}{-y}$

or, $\ln(x) = -\ln(y) + \ln(c)$

or, $xy = c$

At point (1, 2) : $1 \times 2 = c$ or $c = 2$

$\therefore xy = 2$

or $xy - 2 = 0$ (Ans.)

- Q. 8.** What is the ratio of displacement thickness to boundary layer thickness for a linear distribution of velocity $\frac{u}{u_\infty} = \frac{y}{\delta}$ in the boundary layer on a flat plate, where ρ is the boundary layer thickness and u_∞ is the free stream velocity ?
- (a) 0.5 (b) 0.67 (c) 0.75 (d) 0.8

[ESE : 2016]

Ans. (a)

Explanation:

The displacement thickness, $\delta = \int_0^\delta \left(1 - \frac{u}{U}\right) dy$

$$= \int_0^\delta \left(1 - \frac{y}{\delta}\right) dy = \left(y - \frac{y^2}{2\delta}\right)_0^\delta$$

$$= \left(\delta - \frac{\delta^2}{2\delta}\right) = \frac{\delta}{2}$$

or, $\frac{\delta}{\delta^*} = \frac{1}{2} = 0.5$ (Ans.)

- Q. 9.** The thickness of the boundary layer for a fluid flowing over a flat plate at a point 20 cm from the leading edge is found to be 4 mm. The Reynolds number at the point (adopting 5 as the relevant constant) is :
- (a) 48400 (b) 57600 (c) 62500 (d) 77600

[ESE : 2016]

Ans. (c)

Explanation: Boundary layer thickness, $\delta = \frac{5x}{\sqrt{\text{Re}_x}} = \frac{5 \times 0.2}{\sqrt{\text{Re}_x}} = 0.004$

or, Reynold number at a point, $\text{Re}_x = \left(\frac{5 \times 0.2}{0.004}\right)^2 = 62500$ (Ans.)

- Q. 10.** A fluid jet is discharging from a 100 mm nozzle and the vena contracta formed has a diameter of 90 mm. If the coefficient of velocity is 0.98, then the coefficient of discharge for the nozzle is :
- (a) 0.673 (b) 0.794 (c) 0.872 (d) 0.971

[ESE : 2016]

Ans. (b)

Explanation:

$$C_d = C_c \times C_v = \frac{a_c}{a} \times C_v$$

$$= \frac{\pi/4 \times 0.09^2}{\pi/4 \times 0.1^2} \times 0.98 = 0.794 \quad (\text{Ans.})$$

Q. 11. For a steady two-dimensional flow, the scalar components of the velocity field are $V_x = -2x$; $V_y = 2y$ and $V_z = 0$. The corresponding components of acceleration a_x and a_y respectively are:

- (a) 0 and $4y$ (b) $4x$ and 0 (c) 0 and 0 (d) $4x$ and $4y$

[ESE : 2016]

Ans. (d)

Explanation:

$$a_x = u \cdot \frac{\partial u}{\partial x} + v \cdot \frac{\partial u}{\partial y}$$

$$= (-2x) \times (-2) + (2y \times 0) = 4x \text{ m/s}^2$$

$$a_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}$$

$$= (-2x) \times 0 + (2y) \times 2 = 4y \text{ m/s}^2$$

Hence (d) is the correct choice. (Ans.)

Q. 12. The velocity of flow from a tap of 12 mm diameter is 8 m/s. What is the diameter of the jet at 1.5 m from the tap when the flow is vertically upward? Assuming that, the jet continues to be circular upto that level.

- (a) 44 mm (b) 34 mm (c) 24 mm (d) 14 mm

[ESE : 2016]

Ans. (d)

Explanation:

$$d_1 = 12 \text{ mm}; s = 1.5 \text{ m}; u = 8 \text{ m/s}; d_2 = ?$$

$$v^2 - u^2 = 2as$$

$$v^2 - 8^2 = -2 \times 9.81 \times 1.5$$

or,

$$v = \sqrt{8^2 - (2 \times 9.81 \times 1.5)} = 5.88 \text{ m/s}$$

Now,

$$a_1 v_1 = a_2 v_2$$

$$\frac{\pi}{4} \times (0.012)^2 \times 8 = \frac{\pi}{4} \times d_2^2 \times 5.88$$

or,

$$d_2 = \left(\frac{(0.012)^2 \times 8}{5.88} \right)^{1/2}$$

$$= 0.014 \text{ m or } 14 \text{ mm} \quad (\text{Ans.})$$

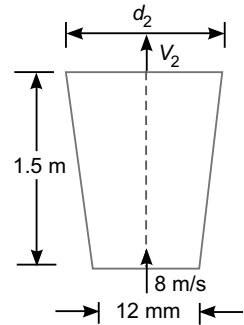


Fig. 2

Q. 13. A 25 cm long prismatic homogeneous solid floats in water with its axis vertical and 10 cm projecting above water surface. If the same solid floats in some oil with its axis vertical and 5 cm projecting above the liquid surface, then the specific gravity of the oil is

- (a) 0.55 (b) 0.65 (c) 0.75 (d) 0.85

[ESE : 2016]

Ans. (c)

Explanation: Refer to Fig. 3. (a & b).

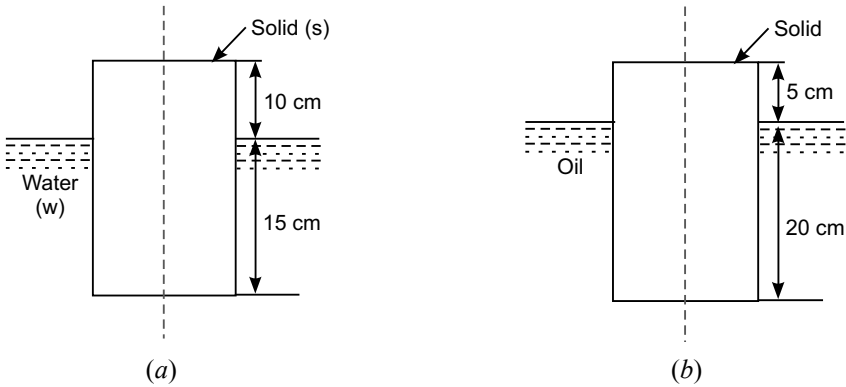


Fig. 3

$$A \times 0.15 \times \rho_w g = A \times (0.25) \times \rho_s g$$

or,
$$\rho_s = \frac{0.15 \times 1000}{0.25} = 600 \text{ kg/m}^3$$

Now,
$$A \times 0.2 \times \rho_{\text{oil}} g = A \times 0.25 \times 600 g$$

$$\therefore \rho_{\text{oil}} = \frac{0.25 \times 600}{0.2} = 750 \text{ kg/m}^3$$

$$\therefore \text{Specific gravity of oil} = \frac{750}{1000} = 0.75 \quad (\text{Ans.})$$

Q. 14. The water level in a dam is 10 m. The total force acting on vertical wall per metre length is:

- (a) 49.05 kN (b) 490.5 kN (c) 981 kN (d) 490.5 N

[ESE : 2016]

Ans. (b)

Explanation: Total force acting on dam,

$$F = [(10 \times 1) \times (10/2)] \times \rho g$$

$$= [50 \times 1000 \times 9.81] \times 10^{-3} \text{ kN} = 490.5 \text{ kN} \quad (\text{Ans.})$$

Q. 15. A solid body of specific gravity 0.5 is 10 m long 3 m wide and 2 m high. When it floats in water with its shortest edge vertical, its metacentric height is:

- (a) 0.75 m (b) 0.45 m
(c) 0.25 m (d) 0.15 m [ESE : 2016]

Ans. (c)

Explanation:

$$\rho_s = 0.5 \times 1000 = 500 \text{ kg/m}^3$$

$$BM = \frac{I_{\min}}{V} = \frac{(10 \times 3^3/12)}{10 \times 3 \times 1} = 0.75 \text{ m}$$

$$BG = OG - OB = 1 - 0.5 = 0.5 \text{ m}$$

$\therefore$ Metacentric height

$$GM = BM - BG = 0.75 - 0.5 = 0.25 \text{ m}$$

(Ans.)

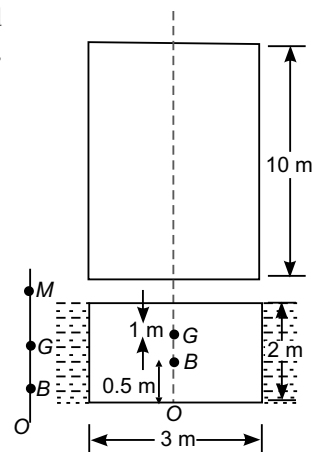


Fig. 4

Q. 16. The pressure at a point in water column is 3.924 N/cm^2 . What is the corresponding height of water?

- (a) 8 m (b) 6 m (c) 4 m (d) 2 m

[ESE : 2016]

Ans. (c)

Explanation:

Pressure, $p = 3.924 \text{ N/cm}^2 = 3.924 \times 10^4 \text{ N/m}^2$ or Pa

Corresponding height, $h = \frac{p}{\rho g} = \frac{3.924 \times 10^4}{1000 \times 9.81} = 4 \text{ m}$ (Ans.)

Q. 17. A vacuum gauge fixed on a steam condenser reads 80 kPa vacuum. The barometer indicates 1.013 bar. The absolute pressure in terms of mercury head is, nearly

- (a) 160 mm of Hg (b) 190 mm of Hg
(c) 380 mm of Hg (d) 760 mm of Hg

[ESE : 2016]

Ans. (a)

Explanation:

$$p_{abc} = 1.013 \times 10^2 \text{ (kPa)} - 80 \text{ (kPa)} = 21.3 \text{ kPa}$$

$$h_{\text{mercury}} = \frac{p_{\text{abs.}}}{\rho_{\text{mercury}} \times g} = \frac{21.3 \times 10^3}{(13.6 \times 1000) \times 9.81}$$

$$= 0.1596 \text{ m or } 160 \text{ mm of Hg (Ans.)}$$

Q. 18. A fluid is flowing over a flat plate. At distance of 8 cm from the leading edge, the Reynolds number is found to be 25600. The thickness of the boundary layer at this point is

- (a) 1.5 mm (b) 2.5 mm (c) 4.0 mm (d) 5.0 mm

[ESE : 2015]

Ans. (b)

Explanation:

$$x = 8 \text{ cm; } Re_x = 25600$$

$$\text{Thickness of boundary layer, } \delta = \frac{5x}{\sqrt{Re_x}} = \frac{5 \times 8}{\sqrt{25600}} = 0.25 \text{ cm or } 2.5 \text{ mm (Ans.)}$$

Q. 19. Consider the following statements pertaining to boundary layer on a flat plate:

1. The thickness of laminar boundary layer at a distance x from the leading edge varies as $x^{1/2}$.
2. The thickness of turbulent boundary layer at a distance x from the leading edge varies as $x^{4/5}$.
3. Boundary layer is laminar when Reynolds number is less than 5×10^5

Which of the above statements are correct?

- (a) 1, 2 and 3 (b) 1 and 2 only (c) 1 and 3 only (d) 2 and 3 only

[ESE : 2015]

Ans. (a)

Explanation:

$$1. \text{ Laminar flow, } \delta = \frac{5x}{\sqrt{Re_x}} = \frac{5x}{\sqrt{\frac{\rho U x}{\mu}}} \text{ i.e. } \delta \propto (x)^{1/2}$$

$$2. \text{ Turbulant flow, } \delta = \frac{0.371 x}{(\text{Re}_x)^{1/5}} = \frac{0.371 x}{\left(\frac{\rho v x}{\mu}\right)^{1/5}} \text{ i.e., } \delta \propto (x)^{4/5}$$

3. Statement is true.

Hence (a) is the correct choice. **(Ans.)**

Q. 20. The laminar flow is characterized by Reynolds number which is

- (a) equal to critical value (b) less than the critical value
(c) more than the critical value (d) zero critical value **[ESE : 2015]**

Ans. (b)

Explanation:

When Reynolds number is *less* than critical value the flow is characterised as *laminar* flow.

(Ans.)

Q. 21. In laminar flow through a circular pipe, the discharge varies

- (a) linearly with fluid density (b) inversely with pressure drop
(c) directly as square of pipe radius (d) inversely with fluid viscosity

[ESE : 2015]

Ans. (d)

Explanation:

According to Hagen – Poiseuille equation,

$$p_1 - p_2 = \frac{32\mu \bar{u} L}{D^2}$$

or,
$$\bar{u} = \left(\frac{p_1 - p_2}{L}\right) \times \frac{D^2}{32\mu}$$

Now,
$$Q = A\bar{u} = \left(\frac{p_1 - p_2}{L}\right) \times \left[\frac{\pi}{4} D^2 \times \left(\frac{D^2}{32\mu}\right)\right]$$

$$= \left(\frac{p_1 - p_2}{L}\right) \times \frac{\pi D^4}{128\mu}$$

$\therefore Q \propto \frac{1}{\mu}$ i.e., Q varies *inversely* with fluid viscosity. **(Ans.)**

Q. 22. Air is flowing over a flat plate with a free stream velocity of 24 m/s, and its kinematic viscosity is $72 \times 10^{-6} \text{ m}^2/\text{s}$. If at a particular point, the Reynolds number is 30000, its location from the leading edge is

- (a) 0.05 m (b) 0.07 m (c) 0.08 m (d) 0.09 m

[ESE : 2015]

Ans. (d)

Explanation:

$$U = 24 \text{ m/s; } \nu = 72 \times 10^{-6} \text{ m}^2/\text{s; } \text{Re}_x = 30,000; x = ?$$

Reynolds number,
$$\text{Re} = \frac{\rho U x}{\mu} = \frac{U x}{\nu}$$

or,
$$x = \frac{\text{Re} \times \nu}{U} = \frac{30,000 \times 72 \times 10^{-6}}{24} = 0.09 \text{ m. (Ans.)}$$

Q. 23. Consider the following statements:

1. If a small upward displacement is given to a floating body, it results in the reduction of the buoyant force acting on the body
2. A slight horizontal displacement does not change either the magnitude or the location of the buoyant force

Which of the above statements is/are correct?

(a) Both 1 and 2

(b) 1 only

(c) 2 only

(d) Neither 1 nor 2

[ESE : 2015]

Ans. (a)

Explanation:

When a small upward displacement is given to a floating body, buoyant force on it *decreases*, since its upward movement will reduce the volume of floating body in liquid on which buoyant force depends.

A slight horizontal displacement does not change either the magnitude or the location of the buoyant force since there is no change in volume of displaced liquid.

Hence (a) is the correct choice. (Ans.)

Q. 24. A tank of length, breadth and height in the ratio of 2 : 1 : 2 is full of water. The ratio of hydrostatic force at the bottom to that at any larger vertical surface is

(a) 1

(b) 4

(c) 2

(d) 3

[ESE : 2015]

Ans. (a)

Explanation:

Bottom area, $A_b = 2a \times a = 2a^2$

Larger surface area, $A_{es} = 2a \times 2a = 4a^2$

$$\frac{F_b}{F_{ls}} = \frac{w A_b \bar{x}}{w A_{ls} \bar{x}} = \frac{w \times 2a^2 \times 2a}{w \times 4a^2 \times (2a/2)} = 1$$

(Ans.)

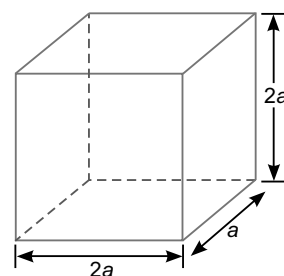


Fig. 5

Q. 25. Which of the following fluids exhibit a certain shear stress at zero shear strain rate followed by a straight line relationship between shear stress and shear strain rate?

(a) Newtonian fluids

(b) Ideal Bingham plastic fluids

(c) Pseudo-plastic fluids

(d) Dilatant fluids

[ESE : 2015]

Ans. (b)

Explanation:

Fig. 6. clearly shows that ideal Bingham plastic fluids exhibit a certain shear stress at zero shear strain rate followed by a straight line relationship between shear stress and shear strain rate. (Ans.)

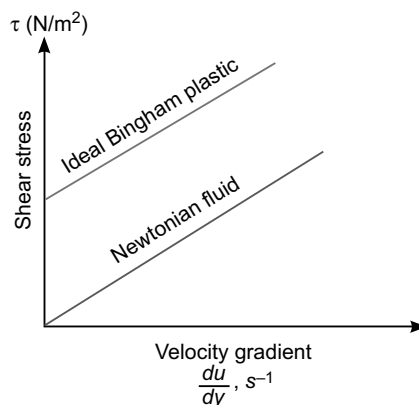


Fig. 6

Q. 26. The boundary layer thickness at a given distance from the leading edge of a flat plate is

- (a) More for lighter fluid (b) More for denser fluid
(c) Less for denser fluid (d) Less for lighter fluid [ESE : 2015]

Ans. (a)

Explanation:

$$\text{Boundary layer thickness, } \delta = \frac{5x}{\sqrt{\text{Re}_x}} = \frac{5x}{\sqrt{\left(\frac{\rho Ux}{\mu}\right)}}$$

$\therefore \delta \propto \frac{1}{\sqrt{\rho}}$, which means that boundary larger thickness is more for light fluid. (Ans.)

Q. 27. The pressure inside a soap bubble of 50 mm diameter is 25 N/m² above the atmospheric pressure. The surface tension in soap film would be

- (a) 0.156 N/m (b) 0.312 N/m (c) 0.624 N/m (d) 0.078 N/m

[ESE : 2014]

Ans. (a)

Explanation:

In a soap bubble, the pressure intensity (p) is given by

$$p = \frac{8\sigma}{d} \text{ or } \sigma = \frac{pd}{8} = \frac{25 \times (50 \times 10^{-3})}{8} = 0.156 \text{ N/m (Ans.)}$$

Q. 28. Unlike the viscosity of liquids, the viscosity of gases increases with increasing temperature. This is due to

- (a) Increased cohesive force between the molecules
(b) Increased momentum transfer in the molecules
(c) Decreased momentum transfer in the molecules
(d) Increases in both cohesive force and momentum transfer

[ESE : 2014]

Ans. (b)

Explanation:

The viscosity of gases increases with increasing temperature due to increase in momentum transfer in the molecules. (Ans.)

Q. 29. A fluid flow is represented by the velocity field $\vec{V} = a\vec{x}\hat{i} + a\vec{y}\hat{i}$, where a is a constant. The equation of streamline passing through a point (1, 2) is

- (a) $x - 2y = 0$ (b) $2x + y = 0$ (c) $2x - y = 0$ (d) $x + 2y = 0$

[ESE : 2014]

Ans. (c)

Explanation:

Given: $\vec{V} = ax\hat{i} + ay\hat{i}$

i.e., $u_x = ax$ and $u_y = ay$

Equation of stream line is given by:

$$\frac{dx}{u_x} = \frac{dy}{u_y}$$

or, $\frac{dx}{ax} = \frac{dy}{ay}$

or, $\frac{dx}{x} = \frac{dy}{y}$

Integrating both sides, we get:

$$\ln(x) = \ln(y) + \ln(c) \quad (\text{where } c \text{ is a constant of integration})$$

$$\therefore x = cy$$

Since the streamline passes through point (1, 2),

Therefore,

$$1 = 2c \quad \text{or} \quad c = \frac{1}{2}$$

Hence, equation of streamline is:

$$x = \frac{1}{2}y$$

or $2x - y = 0$ (Ans.)

- Q. 30.** An incompressible fluid (kinematic viscosity $7.4 \times 10^{-7} \text{ m}^2/\text{s}$, specific gravity, 0.88) is held between two parallel plates. If the top plate is moved with a velocity of 0.5 m/s while the bottom one is held stationary, the fluid attains a linear velocity profile in the gap of 0.5 mm between these plates; the shear stress in Pascals on the surface of top plate is

(a) 0.651×10^{-3}

(b) 0.651

(c) 6.51

(d) 0.651×10^3

Ans. (b)

Explanation: Refer to Fig. 7.

$$\text{Shear stress } \tau = \mu \frac{du}{dy}$$

For linear velocity profile,

$$\tau = \mu \cdot \frac{u}{y}$$

Now, $v = \frac{\mu}{\rho}$

or, $\mu = v \times \rho$

$$= 7.4 \times 10^{-7} \times (0.88 \times 1000)$$

$$= 6.512 \times 10^{-4} \text{ Pa-s}$$

$$\therefore \tau = 6.512 \times 10^{-4} \times \frac{0.5}{0.5 \times 10^{-3}}$$

$$= 0.6512 \text{ N/m}^2 \text{ (Ans.)}$$

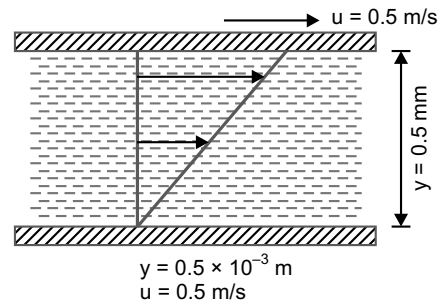


Fig. 7

- Q. 31.** A cylindrical body of cross-sectional area A , height H and density ρ_s , is immersed to a depth h in a liquid of density ρ , and tied to the bottom with a string. The tension in the string is

(a) ρghA

(b) $(\rho_s - \rho) ghA$

(c) $(\rho - \rho_s) ghA$

(d) $(\rho h - \rho_s H)gA$

Ans. (d)

Explanation: Refer to Fig. 8.

$$T + Mg = \text{Buoyancy force}$$

$$T + \rho_s HAg = \rho hAg$$

$$\therefore T = gA(\rho h - \rho_s H) \text{ (Ans.)}$$

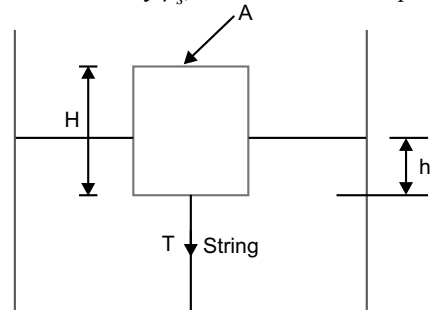


Fig. 8

- Q. 32.** A water container is kept on a weighing balance. Water from a tap is falling vertically into the container with a volume flow rate of Q ; the velocity of the water when it hits the water surface is U . At a particular instant of time the total mass of the container and water is m . The force registered by the weighing balance at this instant of time is

(a) $mg + \rho QU$ (b) $mg + 2\rho QU$ (c) $mg + \rho QU^2/2$ (d) $\rho QU^2/2$

Ans. (a)

Explanation:

Mass of water hitting the surface = ρQ

Force on weighing balance due to water strike

$$= \text{Initial momentum} - \text{final momentum}$$

$$= \rho QU - 0$$

$$= \rho QU$$

Weight of container and water = mg

$\therefore$ Total force on weighing balance

$$= mg + \rho QU \text{ (Ans.)}$$

- Q. 33.** For air flow over a flat plate, velocity (U) and boundary layer thickness (δ) can be expressed respectively as

$$\frac{U}{U_\infty} = \frac{3}{2} \frac{y}{\delta} - \frac{1}{2} \left(\frac{y}{\delta} \right)^3; \delta = \frac{4.64x}{\sqrt{Re_x}}$$

If the free stream velocity is 2 m/s, and air has kinematic viscosity of $1.5 \times 10^{-5} \text{ m}^2/\text{s}$ and density of 1.23 kg/m^3 , the wall shear stress at $x = 1 \text{ m}$, is

(a) $2.36 \times 10^2 \text{ N/m}^2$ (b) $43.6 \times 10^{-3} \text{ N/m}^2$
 (c) $4.36 \times 10^{-3} \text{ N/m}^2$ (d) $2.18 \times 10^{-3} \text{ N/m}^2$

Ans. (c)

Explanation:

Reynold number,

$$Re_x = \frac{ux}{\nu} = \frac{2 \times 1}{1.5 \times 10^{-5}} = 1.33 \times 10^5$$

$$\delta = \frac{4.64x}{\sqrt{Re_x}} = \frac{4.64 \times 1}{\sqrt{1.33 \times 10^5}}$$

$$= 0.0127$$

Now,
$$\frac{du}{dy} = U \left[\frac{3}{2} \cdot \frac{1}{\delta} - \frac{3}{2} \left(\frac{y^2}{\delta^3} \right) \right] = \frac{du}{dy} \Big|_{y=0} = \frac{3U}{2\delta}$$

Now, shear stress,

$$\begin{aligned} \tau_0 &= \mu \left(\frac{du}{dy} \right)_{y=0} = \mu \times \frac{3U}{2\delta} \\ &= \frac{3U \times \nu \times \rho}{2\delta} \quad (\because \mu = \nu\rho) \\ &= \frac{3 \times 2 \times 1.5 \times 10^{-5} \times 1.23}{2 \times 0.0127} \\ &= 4.36 \times 10^{-3} \text{ N/m}^2 \text{ (Ans.)} \end{aligned}$$

- Q. 34.** A centrifugal pump is required to pump water to an open water tank situated 4 km away from the location of the pump through a pipe of diameter 0.2 m having Darcy's friction factor of 0.01. The average speed of water in the pipe is 2 m/s. If it is to maintain a constant head of 5 m in the tank, neglecting other minor losses, the absolute discharge pressure at the pump exit is

(a) 0.449 bar (b) 5.503 bar (c) 44.911 bar (d) 55.203 bar

Ans. (b)

Explanation:

Head loss due to friction

$$h_f = \frac{fLV^2}{2gd} = \frac{0.01 \times 4000 \times (2)^2}{2 \times 9.81 \times 0.2}$$

$$= 40.78 \text{ m}$$

Pressure corresponding to this head

$$= \rho g(h_f + h + h_{atm}) = 1000 \times 9.81 (40.78 + 5 + 10.3)$$

$$= 5.50 \times 10^5 \text{ N/m}^2$$

$$= 5.50 \text{ bar (Ans.)}$$

- Q. 35.** The pressure gauges G_1 and G_2 installed on the system show pressure of $P_{G1} = 5.00$ bar and $P_{G2} = 1.00$ bar. The value of unknown pressure P is

(Atmospheric pressure = 1.01 bar]

(a) 1.01 bar (b) 2.01 bar
(c) 5.00 bar (d) 7.01 bar

Ans. (d)

Explanation: Refer to Fig. 9.

Absolute pressure of $G_2 = 1 + 1.01 = 2.01$ bar

Absolute pressure of $G_1 = \text{Pressure of } P$

$$= 5 + 2.01$$

$$= 7.01 \text{ bar (Ans.)}$$

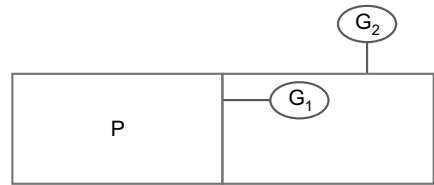


Fig. 9

- Q. 36.** At a hydro electric power plant site, available head and flow rate are 24.5 m and $10.1 \text{ m}^3/\text{s}$ respectively. If the turbine to be installed is required to run at 4.0 revolutions per second (rps) with an overall efficiency of 90%, the suitable type of turbine for this site is

(a) Francis (b) Kaplan (c) Pelton (d) Propeller.

Ans. (a)

Explanation:

Given:

$$H = 24.5 \text{ m,}$$

$$Q = 10.1 \text{ m}^3/\text{s,}$$

$$N = 4 \text{ rev/sec} = 4 \times 60 = 240 \text{ rpm, and}$$

$$\eta_0 = 0.90$$

$$\text{Power generated} = \rho g H \times 0.9 \times Q$$

$$= 1000 \times 9.81 \times 10.1 \times 24.5 \times 0.9 = 2184.74 \text{ kW}$$

$$N_s = \frac{N\sqrt{P}}{H^{5/4}} = \frac{240\sqrt{2184.7}}{(24.5)^{5/4}}$$

$$= 205.80$$

Hence, turbine is *Francis*. (Ans.)

Q. 37. The velocity components in the x and y directions of a two dimensional potential flow are u and v , respectively. Then, $\frac{\partial u}{\partial x}$ is equal to

- (a) $\frac{\partial v}{\partial x}$ (b) $-\frac{\partial v}{\partial x}$ (c) $\frac{\partial v}{\partial y}$ (d) $-\frac{\partial v}{\partial y}$.

Ans. (d)

Explanation:

For two dimensional flow, continuity equation should be satisfied.

i.e.
$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y} \quad (\text{Ans.})$$

Q. 38. In a two-dimensional velocity field with velocities u and v along the x and y directions respectively, the convective acceleration along the x -direction is given by

- (a) $u \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial y}$ (b) $u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial y}$ (c) $u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}$ (d) $v \frac{\partial u}{\partial x} + u \frac{\partial u}{\partial y}$

Ans. (c)

Explanation:

$$a_x = \frac{du}{dt} = u \frac{du}{dx} + v \frac{du}{dy} \quad (\text{Ans.})$$

Q. 39. The velocity profile in fully developed laminar flow in a pipe of diameter D is given by $u = u_0(1 - 4r^2/D^2)$, where r is the radial distance from the center. If the viscosity of the fluid is μ , the pressure drop across a length L of the pipe is

- (a) $\frac{\mu u_0 L}{D^2}$ (b) $\frac{4\mu u_0 L}{D^2}$ (c) $\frac{8\mu u_0 L}{D^2}$ (d) $\frac{16\mu u_0 L}{D^2}$

Ans. (d)

Explanation:

We know that:

Pressure drop across a length L of pipe is,

$$\Delta p = \frac{32\mu u_{av} L}{D^2} \quad \dots(i)$$

and,
$$u_{av} = \frac{u_0}{2} \quad \dots(ii)$$

From equation (i) and (ii)

$$\Delta p = \frac{16\mu u_0 L}{D^2} \quad (\text{Ans.})$$

Q. 40. Consider steady laminar incompressible axi-symmetric fully developed viscous flow through a straight circular pipe of constant cross-sectional area at a Reynolds number of 5. The ratio of inertia force to viscous force on a fluid particle is

- (a) 5 (b) 1/5 (c) 0 (d) ∞ .

Ans. (a)

Explanation:

$$\text{Reynolds number} = \frac{\text{Inertia force}}{\text{viscous force}} = 5 \quad (\text{Ans.})$$

Q. 41. The ratio V_m/U_0 is (Fig. 10)

- (a) $\frac{1}{1 - 2(\delta/H)}$ (b) 1
 (c) $\frac{1}{1 - (\delta/H)}$ (d) $\frac{1}{1 + (\delta/H)}$

Ans. (a)

Explanation:

Assume width of channel is b . Applying continuity equation at A and B , we have:

$$U_0(H \times b) = V_m(H - 2\delta) \times b$$

or,
$$\frac{V_m}{U_0} = \frac{H}{H - 2\delta} = \frac{1}{1 - \frac{2\delta}{H}} \quad (\text{Ans.})$$

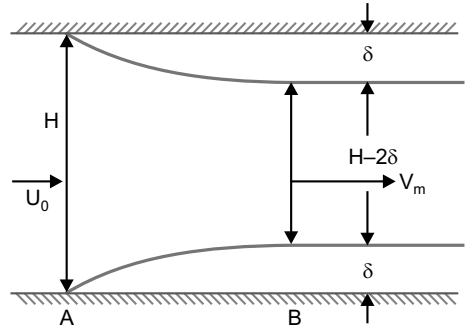


Fig. 10

Q. 42. For the continuity equation given by $\vec{\Delta} \cdot \vec{V} = 0$ to be valid, where $\vec{V}$ is the velocity vector, which one of the following is a necessary condition?

- (a) Steady flow (b) Irrotational flow
 (c) Inviscid flow (d) Incompressible flow.

Ans. (d)

Explanation:

$$\vec{\Delta} \cdot \vec{V} = 0$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

So the above equation represents *incompressible flow* (Ans.)

Q. 43. Water at 25°C is flowing through a 1.0 km long G.I. pipe of 200 mm diameter at the rate of 0.07 m³/s. If value of Darcy friction factor for this pipe is 0.02 and density of water is 1000 kg/m³, the pumping power (in kW) required to maintain the flow is

- (a) 1.8 (b) 17.5 (c) 20.5 (d) 41.0.

Ans. (b)

Explanation:

Given, pipe length, $L = 1000$ m
 Pipe diameter, $D = 200$ mm = 0.2 m
 Flow rate, $Q = 0.07$ m³/s
 Density, $\rho = 1000$ kg/m³
 Friction factor, $f = 0.02$

$$\begin{aligned} \text{Head loss, } h_f &= \frac{fLV^2}{2gD} = \frac{fLQ^2}{12D^5} \\ &= \frac{0.02 \times 1000 \times (0.07)^2}{12 \times (0.2)^5} \\ &= 25.52 \text{ m} \end{aligned}$$

Pumping power required,

$$\begin{aligned}
 P &= \rho g Q h_f \\
 \text{or, } P &= 1000 \times 9.81 \times 0.07 \times 25.52 \\
 &= 17524.5 \text{ W} \\
 &= 17.52 \text{ kW (Ans.)}
 \end{aligned}$$

- Q. 44. The velocity profile of a fully developed laminar flow in a straight circular pipe, as shown in Fig. 11, is given by the expression

$$u(r) = -\frac{R^2}{4\mu} \left(\frac{dp}{dx} \right) \left(1 - \frac{r^2}{R^2} \right)$$

where $\frac{dp}{dx}$ is a constant

The average velocity of fluid in the pipe is

$$(a) -\frac{R^2}{8\mu} \left(\frac{dp}{dx} \right) \quad (b) -\frac{R^2}{4\mu} \left(\frac{dp}{dx} \right) \quad (c) -\frac{R^2}{2\mu} \left(\frac{dp}{dx} \right) \quad (d) -\frac{R^2}{\mu} \left(\frac{dp}{dx} \right)$$

Ans. (a)

Explanation:

The velocity profile of a fully developed laminar flow in a straight circular pipe, as shown in Fig. 11, is given by the expression:

$$\begin{aligned}
 u(r) &= -\frac{R^2}{4\mu} \left(\frac{dp}{dx} \right) \left[1 - \frac{r^2}{R^2} \right] \\
 &= U_{\max} \left[1 - \frac{r^2}{R^2} \right]
 \end{aligned}$$

where, $U_{\max} = -\frac{R^2}{4\mu} \left(\frac{dp}{dx} \right)$

where, $\frac{dp}{dx}$ is a constant

The average velocity of fluid in the pipe is

$$V_{\text{avg}} = \frac{U_{\max}}{2} = -\frac{R^2}{8\mu} \left(\frac{dp}{dx} \right) \text{ (Ans.)}$$

- Q. 45. The maximum velocity of a one-dimensional incompressible fully developed viscous flow, between two fixed parallel plates, is 6 ms^{-1} . The mean velocity (in ms^{-1}) of the flow is
- (a) 2 (b) 3 (c) 4 (d) 5

Ans. (c)

Explanation:

Ratio of maximum velocity to average velocity for one dimensional incompressible flow between two fixed plate is given as:

$$\frac{U_{\max}}{U_{\text{mean}}} = \frac{3}{2}$$

or, $U_{\text{mean}} = \frac{6 \times 2}{3} = 4 \text{ m/s (Ans.)}$

- Q. 46. A hydraulic turbine develops 1000 kW power of a head of 40 m. If the head is reduced to 20 m, the power developed (in kW) is
- (a) 177 (b) 354 (c) 500 (d) 707.

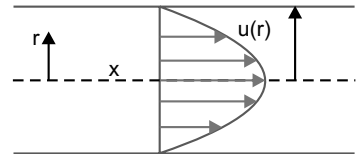


Fig. 11

Ans. (b)

Explanation:

For unit quantities

$$\begin{aligned}\frac{P_1}{H_1^{3/2}} &= \frac{P_2}{H_2^{3/2}} \\ P_2 &= P_1 \left(\frac{H_2}{H_1} \right)^{3/2} \\ &= 1000 \times \left(\frac{20}{40} \right)^{3/2} = 353.6 \approx 354 \text{ kW (Ans.)}\end{aligned}$$

Q. 47. A streamline and an equipotential line in a flow field

- (a) are parallel to each other (b) are perpendicular to each other
(c) intersect at an acute angle (d) are identical.

Ans. (b)

Explanation:

Streamline and equipotential line in flow field are perpendicular to each other. (Ans.)

Q. 48. Calculate the specific weight, specific mass and specific gravity of a liquid having a volume of 6 m³ and weight of 44 kN.

- (a) 5.33 kN/m³, 547.5 kg/m³, 0.447 (b) 6.33 kN/m³, 647.5 kg/m³, 0.547
(c) 7.333 kN/m³, 747.5 kg/m³, 0.747 (d) none of these.

Ans. (c)

Explanation:

Volume of liquid = 6 m³; Weight of the liquid = 44 kN

$$\begin{aligned}\text{Specific weight, } w &= \frac{\text{Weight of liquid}}{\text{Volume of liquid}} = \frac{44}{6} \\ &= 7.333 \text{ kN/m}^3 \text{ (Ans.)}\end{aligned}$$

$$\begin{aligned}\text{Specific mass or mass density, } \rho &= \frac{w}{g} = \frac{7.333 \times 1000}{9.81} \\ &= 747.5 \text{ kg/m}^3 \text{ (Ans.)}\end{aligned}$$

$$\text{Specific gravity, } S = \frac{w_{\text{liquid}}}{w_{\text{water}}} = \frac{7.333}{9.81} = 0.747 \text{ (Ans.)}$$

Q. 49. An isosceles triangular plate of base 3 m and altitude 3 m is immersed vertically in an oil of specific gravity 0.8. The base of the plate coincides with the free surface of oil.

The total pressure on the plate, and centre of pressure are

- (a) 25.3 kN, 1.1 m (b) 35.3 kN, 1.5 m
(c) 45.3 kN, 2.5 m (d) none of the above

Ans. (b)

Explanation: Refer to Fig. 12.

Base of the plate, $b = 3 \text{ m}$

Height of the plate, $h = 3 \text{ m}$

$$\therefore \text{Area, } A = \frac{b \times h}{2} = \frac{3 \times 3}{2} = 4.5 \text{ m}^2$$

Specific gravity of oil, $S = 0.8$

The distance of C.G. from the free surface of oil,

$$\bar{x} = \frac{1}{3} h = \frac{1}{3} \times 3$$

$$= 1 \text{ m}$$

(i) Total pressure on the plate, P :

We know that, $P = wA\bar{x}$

$$P = (0.8 \times 9.81) \times 4.5 \times 1 = 35.3 \text{ kN (Ans.)}$$

(ii) Centre of pressure, h :

Centre of pressure is given by the relation,

$$\bar{h} = \frac{I_G}{A\bar{x}} + \bar{x} = \frac{(bh^3/36)}{A\bar{x}} + \bar{x}$$

$$= \frac{(3 \times 3^3/36)}{4.5 \times 1} + 1$$

$$= 1.5 \text{ m (Ans.)}$$

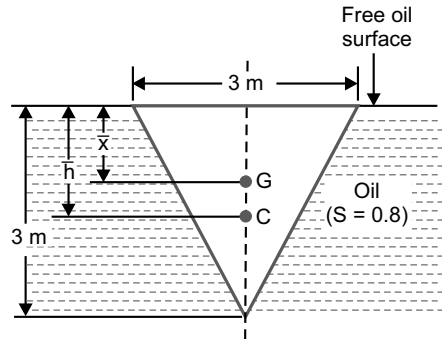


Fig. 12

Q. 50. In a pipe of 90 mm diameter water is flowing with a mean velocity of 2 m/s and a gauge pressure of 350 kN/m². The total head, if pipe is 8 metres above the datum line.

- (a) 23.88 m (b) 33.88 m (c) 43.88 m (d) 53.88 m

Ans. (c)

Explanation:

Diameter of the pipe = 90 mm

Pressure, $p = 350 \text{ kN/m}^2$; velocity of water, $V = 2 \text{ m/s}$

Datum head, $z = 8 \text{ m}$; Specific weight of water, $w = 9.81 \text{ N/m}^3$

Total head of water, H : Using the relation,

$$H = z + \frac{V^2}{2g} + \frac{p}{w}, \quad \text{neglecting friction}$$

$$H = 8 + \frac{2^2}{2 \times 9.81} + \frac{350}{9.81}$$

$$= 43.88 \text{ m (Ans.)}$$

Q. 51. A sub-marine fitted with a pitot tube moves horizontally in sea. Its axis is 12 m below the surface of water. The pitot tube fixed in front of the sub-marine and along its axis is connected to the two limbs of a U-tube containing mercury, the reading for which is found to be 200 mm. The speed of the sub-marine is

(Take the specific-gravity of sea water = 1.025 times fresh water.)

- (a) 24.9 km/h (b) 34.9 km/h
(c) 44.9 km/h (d) none of the above

Ans. (a)

Explanation:

Reading of the manometer, $y = 200 \text{ mm} = 0.2 \text{ m}$ of mercury

Sp. gravity of mercury, $S_h = 13.6$; Sp. gravity of sea water, $S_l = 1.025$

$$h = y \left(\frac{S_h}{S_l} - 1 \right) = 0.2 \left(\frac{13.6}{1.025} - 1 \right) = 2.45 \text{ m}$$

Velocity of the sub-marine, $V = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 2.45}$
 $= 6.93 \text{ m/s or } 24.9 \text{ km/h (Ans.)}$

- Q. 52.** What is the discharge through a rectangular orifice 3.0 m wide and 2.0 deep fitted to a water tank? The water level in the tank is 4.0 m above the top edge of orifice. Take $C_d = 0.62$
- (a) 36.76 m³/s (b) 46.76 m³/s
(c) 56.76 m³/s (d) none of the above

Ans. (a)

Explanation:

Width of the orifice, $b = 3.0$ m

Depth of the orifice = 2.0 m

Height of water above the top of the orifice, $H_1 = 4.0$ m

$\therefore$ Height of the water above the bottom of the orifice, $H_2 = 4 + 2 = 6$ m

Co-efficient of discharge, $C_d = 0.62$

Discharge through the orifice, Q :

Using the relation,

$$Q = \frac{2}{3} C_d b \sqrt{2g} (H_2^{3/2} - H_1^{3/2}) \text{ with usual notations}$$

$$= \frac{2}{3} \times 0.62 \times 3.0 \times \sqrt{2 \times 9.81} (6^{3/2} - 4^{3/2}) = 36.76 \text{ m}^3/\text{s}$$

i.e. $Q = 36.76 \text{ m}^3/\text{s}$ (Ans.)

- Q. 53.** A shaft of 100 mm diameter rotates at 60 r.p.m. in a 200 mm long bearing. Taking that the two surfaces are uniformly separated by a distance of 0.5 mm and taking linear velocity distribution in the lubricating oil having dynamic viscosity 0.04 poise, find the power absorbed in hearing.
- (a) 0.0196 W (b) 0.0296 W (c) 0.0396 W (d) 0.0496 W.

Ans. (d)

Explanation:

Diameter of the shaft, $D = 100 \text{ mm} = 0.1 \text{ m}$

Speed of the shaft, $N = 60$ r.p.m.

Length of the bearing, $L = 200 \text{ mm} = 0.2 \text{ m}$

Thickness of oil film, $t = 0.5 \text{ mm} = 0.005 \text{ m}$

Dynamic viscosity, $\mu = 0.04 \text{ poise} = 0.04 \text{ Ns/m}^2$

Power absorbed, P:

The torque required to overcome viscous resistance in a journal bearing is given by:

$$T = \frac{\mu \pi^2 D^3 N L}{120 t} = \frac{0.004 \times \pi^2 \times 0.1^3 \times 60 \times 0.2}{120 \times 0.0005} = 0.00798 \text{ N}$$

$$\therefore \text{Power absorbed} = \frac{2\pi NT}{60} = \frac{2\pi \times 60 \times 0.00789}{60} = 0.0496 \text{ W (Ans.)}$$

- Q. 54.** In a smooth pipe of diameter 0.5 m and length 1000 m water is flowing at the rate of $0.05 \text{ m}^3/\text{s}$. Assuming the kinematic viscosity of water as 0.02 stokes, find head lost due to friction
- (a) 0.1316 m (b) 0.1416 m (c) 0.1516 m (d) 0.1616 m

Ans. (a)

Explanation:

Diameter of smooth pipe, $D = 0.5$ m

$$\therefore \text{Radius} = \frac{0.5}{2} = 0.25 \text{ m}$$

Length of the pipe, $L = 1000$ m

Discharge through the pipe, $Q = 0.05$ m³/s

Kinematic viscosity of water, $\nu = 0.02 \times 10^{-4}$ m²/s

$$\text{Average velocity, } \bar{u} = \frac{Q}{\text{area}} = \frac{0.05}{\frac{\pi}{4} \times (0.5)^2} = 0.2546 \text{ m/s}$$

$$\therefore \text{Reynolds number, } Re = \frac{V \times D}{\nu} = \frac{\bar{u} \times D}{\nu} = \frac{0.2546 \times 0.5}{0.02 \times 10^{-4}} = 6.365 \times 10^4$$

Since $Re > 4000$, the flow is *turbulent*.

$$\text{We know that, } f = \frac{0.0791}{(Re)^{1/4}} = \frac{0.0791}{(6.365 \times 10^4)^{1/4}} = 0.00498$$

Head lost the to friction, h_f :

$$h_f = \frac{4fLV^2}{D \times 2g} = \frac{4fL\bar{U}^2}{D \times 2g} = \frac{4 \times 0.00498 \times 1000 \times (0.2546)^2}{0.5 \times 2 \times 9.81} = 0.1316 \text{ m (Ans.)}$$

- Q. 55.** The diameter of a horizontal pipe which is 300 mm is suddenly enlarged to 600 mm. If the rate of flow of water through this pipe is 0.4 m³/s determine the loss of head, due to sudden enlargement,

(a) 0.618 m (b) 0.718 m (c) 0.1818 (d) 0.918 m

Ans. (d)

Explanation :

Diameter of the smaller pipe $D_1 = 300$ mm = 0.3 m

$$\text{Area, } A_1 = \frac{\pi}{4} \times 0.3^2 = 0.0707 \text{ m}^2$$

Diameter of longer pipe $D_2 = 600$ mm = 0.6 m

$$\text{Area, } A_2 = \frac{\pi}{4} \times 0.6^2 = 0.2828 \text{ m}^2$$

Rate of flow of water, $Q = 0.4$ m³/s

$$\text{Now, Velocity } V_1 = \frac{Q}{A_1} = \frac{0.4}{0.0707} = 5.66 \text{ m/s}$$

$$\text{Velocity } V_2 = \frac{Q}{A_2} = \frac{0.4}{0.2828} = 1.414 \text{ m/s}$$

$$\text{Loss of head due to sudden enlargement, } h_c = \frac{(V_1 - V_2)^2}{2g} = \frac{(5.66 - 1.414)^2}{2 \times 9.81} = 0.918 \text{ m (Ans.)}$$

- Q. 56.** A truck having a projected area of 6.5 m² travelling at 70 km/h has a total resistance of 2000 N. Of this 20 per cent is due to rolling friction and 10 percent due to surface friction. The rest is due to form drag. The co-efficient of form drag (Take $\rho = 1.22$ kg/m³ for air) is

- (a) 0.734 (b) 0.834
(c) 0.934 (d) none of the above

Ans. (c)

Explanation:

Projected area, $A = 6.5 \text{ m}^2$

Speed of the truck, $U = 70 \text{ km/h} = \frac{70 \times 1000}{60 \times 60} = 19.44 \text{ m/s}$

Density of air, $\rho = 1.22 \text{ kg/m}^3$

Total resistance = 2000 N

Resistance due to rolling friction = $\frac{20}{100} \times 2000 = 400 \text{ N}$

Resistance due to surface friction = $\frac{10}{100} \times 2000 = 200 \text{ N}$

Resistance due to form drag = $2000 - 400 - 200 = 1400 \text{ N}$

Coefficient of form drag, C_D :

Now, Form drag = $C_D \times \frac{\rho U^2}{2} \times A$

$$\therefore 1400 = C_D \times \frac{1.22 \times (19.44)^2}{2} \times 6.5$$

$$\therefore C_D = \frac{1400 \times 2}{1.22 \times (19.44)^2 \times 6.5} = 0.934 \text{ (Ans.)}$$

Q. 57. An aeroplane is flying at an height of 14 km where temperature is -45°C . The speed of the plane is corresponding to $M = 2$. If $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$, the speed of the plane is:

- (a) 2179 km/h (b) 3010 km/h (c) 3250 km/h (d) 3379 km/h

Ans. (a)

Explanation:

Temperature (at a height of 14 km)

$$t = -45^\circ \text{C}$$

$$T = -45 + 273 = 228 \text{ K}$$

Mach number $M = 2$

Gas constant, $R = 287 \text{ J/kg K}$

$$\gamma = 1.4$$

Speed of the plane, V :

Sonic velocity (C) is given by:

$$C = \sqrt{\gamma RT} \quad (\text{assuming the process to be adiabatic})$$

$$\sqrt{1.4 \times 287 \times 228} = 302.67 \text{ m/s}$$

Also, $M = \frac{V}{C}$

or, $M = \frac{V}{302.67}$

or, $V = 2 \times 302.67 = 605.34 \text{ m/s} = \frac{605.34 \times 3600}{1000} = 2179 \text{ km/h (Ans.)}$

Q. 58. Find the velocity of a bullet fired in standard air if its mach angle is 40° .

- (a) 429 m/s (b) 529 m/s (c) 629 m/s (d) 729 m/s

Ans. (b)

Explanation:

Mach angle, $\alpha = 40^\circ$

$$\gamma = 1.4$$

For standard air: $R = 287 \text{ J/kg K}$, $t = 15^\circ \text{ C}$ or $T = 15 + 273 = 288 \text{ K}$

Velocity of the bullet, V :

Sonic velocity, $C = \sqrt{\gamma RT} = \sqrt{1.4 \times 287 \times 288} = 340.2 \text{ m/s}$

Now, $\sin \alpha = \frac{C}{V}$

$$\text{or, } \sin 40^\circ = \frac{340.2}{V} \text{ or } V = \frac{340.2}{\sin 40^\circ} = 529 \text{ m/s (Ans.)}$$

Q. 59. A sluice gate discharges water into horizontal rectangular channel with a velocity of 10 m/s and depth of flow of 1 m. What is the depth of flow of water after the jump and consequent loss in total head?

- (a) 2.2 m, 1.2 m (b) 3.2 m, 1.4 m
(c) 4.043 m, 1.742 m (d) 5.043 m, 2.742 m

Ans. (c)

Explanation:

Velocity of flow before hydraulic jump, $V_1 = 10 \text{ m/s}$

Depth of flow before hydraulic jump, $y_1 = 1 \text{ m}$

Depth of flow after the jump, y_2 :

Discharge per unit width, $q = V_1 \times y_1 = 10 \times 1 = 10 \text{ m}^3/\text{s per m}$

The depth of flow water the jump is given by:

$$y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2q^2}{gy_1}}$$

$$\text{or, } y_2 = -\frac{1}{2} + \sqrt{\frac{1^2}{4} + \frac{2 \times 10^2}{9.81 \times 1}} = 4.042 \text{ m (Ans.)}$$

Loss in total head, E_L :

Loss in total head is given by:

$$E_L = \frac{(y_2 - y_1)^3}{4y_1y_2}$$

$$\text{or, } E_L = \frac{(4.043 - 1)^3}{4 \times 1 \times 4.043} = 1.7432 \text{ m (Ans.)}$$

Q. 60. Flow takes place at Reynolds Number of 1500 in two different pipes with relative roughness of 0.001 and 0.002. The friction factor

- (a) will be higher in the case of pipe with relative roughness of 0.001
(b) will be higher in the case of pipe having relative roughness of 0.002
(c) will be the same in both the pipes
(d) in the two pipes cannot be compared on the basis of data given.

Ans. (c)

Explanation:

The friction factor will be the same for both the pipes because for $Re < 1500$, i.e., laminar flow, it is independent of relative roughness of pipe. Thus (c) is the correct choice.

Q. 61. Decrease in temperature, in general results in

- (a) an increase in viscosities of both gases and liquids
- (b) a decrease in the viscosities of both liquids and gases
- (c) an increase in the viscosity of liquids and a decrease in that of gases
- (d) a decrease in the viscosity of liquids and an increase in that of gases.

Ans. (c)

Explanation:

Viscosity for water, $\mu_1 = [0.001793/(1 + 0.03368 t + 0.000221 t^2)]$, and

Viscosity for air, $\mu_1 = [0.000001702 (1 + 0.00329 t + 0.000007 t^2)]$

From above it is evident that with decrease in temperature, the viscosity of water increases while that of air decreases. Thus, decrease in temperature, in general, results in an increase in the viscosity of liquids and a decrease in that of gases.

Thus the correct choice is (c).

Q. 62. The energy loss between sections (1) and (2) of the pipe shown in Fig. 13 is

- (a) 1.276 kg-m
- (b) 1.00 kg-m
- (c) 0.725 kg-m
- (d) 0.15 kg-m.

Ans. (c)

Explanation:

$$A_1 V_1 = A_2 V_2 \text{ or } \frac{\pi}{4} \times 0.1^2 \times 0.6 = \frac{\pi}{4} \times 0.05^2 \times V_2$$

$$\text{or, } V_2 = 2.4 \text{ m/s}$$

Invoking Bernoulli's equation at sections 1 and 2, we have:

$$\frac{p_1}{\rho} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho} + \frac{V_2^2}{2g} + z_2 + h_c \text{ (energy loss)}$$

$$h_c = \left(\frac{p_1 - p_2}{\rho} \right) + \left(\frac{V_1^2 - V_2^2}{2g} \right) \quad (z_1 = z_2, \text{ the pipe being horizontal})$$

$$= \frac{(3.5 - 3.4) \times 10^4}{1000} + \frac{0.6^2 - 2.4^2}{2 \times 9.81} = 1.0 - 0.275 = 0.725 \text{ kg m (Ans.)}$$

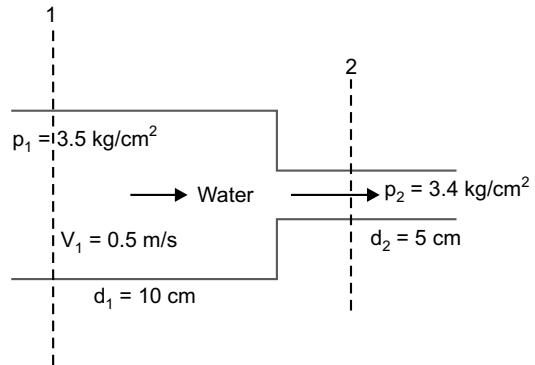


Fig. 13

Q. 63. The reading of the pressure gauge fitted on a vessel is 25 bar. The atmospheric pressure is 1.03 bar and the value of g is 9.81 m/s^2 . The absolute pressure in the vessel is

- (a) 23.97 bar
- (b) 25.00 bar
- (c) 26.03 bar
- (d) 34.84 bar

Ans. (c)

Explanation:

$$p_{abs} = p_{gauge} + p_{atm.} = 25 + 1.03 = 26.03 \text{ bar (Ans.)}$$

Thus (c) is the correct choice.

- Q. 64. In a pipe flow, the head lost due to friction is 6 m. If the power transmitted through the pipe has to be the maximum, then the total head at the inlet of the pipe will have to be maintained at
 (a) 36 m (b) 30 m (c) 24 m (d) 18 m

Ans. (d)

Explanation:

Head lost due to friction, $h_f = 6$ m

For maximum power transmission, the supply head (H) should be equal to $3h_f$

$$\therefore H = 3 \times 6 = 18 \text{ m (Ans.)}$$

Thus (d) is the correct choice.

- Q. 65. The speed of the air emerging from the blades of a running table fan is intended to be measured as a function of time. The point of measurement is very close to the blade and the time period of the speed fluctuation is four times the time taken by the blade to complete one revolution. The appropriate method of measurement would involve the use of
 (a) a Pitot tube (b) a hot wire anemometer
 (c) high speed photography (d) a Schlieren system

Ans. (d)

Explanation:

(d) is the correct choice since other instruments (choices) are used under the following conditions:

Pitot tube for measuring speed in closed duct/pipe

Hot wire anemometer ... for measuring speed over a period of time (does not respond to fact changes as mentioned in the question.)

High speed photography for measuring blade speed (not or air)

- Q. 66. A fluid jet is discharging from a 100 mm nozzle and the vena-contracta formed has a diameter of 90 mm. If the coefficient of velocity is 0.95, then the coefficient of discharge for a nozzle is
 (a) 0.855 (b) 0.81 (c) 0.9025 (d) 0.7695

Ans. (d)

Explanation:

$$\text{Co-efficient of contraction, } C_c = \frac{A_c}{A} = \frac{\frac{\pi}{4} \times 90^2}{\frac{\pi}{4} \times 100^2} = 0.81, \text{ co-efficient of velocity, } C_v = 0.95.$$

$$\therefore \text{Co-efficient of discharge, } C_d = C_c \times C_v = 0.81 \times 0.95 = 0.7695 \text{ (Ans.)}$$

- Q. 67. A circular disc of radius ' r ' is submerged vertically in a static fluid upto a depth ' h ' from the free surface. If $h > r$, then the position of centre pressure will
 (a) be directly proportional to h (b) be inversely proportional to h
 (c) be directly proportional to r (d) not be a function of h or r

Ans. (d)

Explanation:

(d) is the correct choice, since centre of pressure $\left(\frac{I_G}{Ax} + \bar{x} \right)$ is below c.g., and as such it is not a function of h or r alone.

- Q. 68. If a cylindrical wooden pole, 20 cm in diameter, and 1 m in height is placed in a pool of water in a vertical position (the specific gravity of wood is 0.6), then it will
- (a) float in stable equilibrium (b) float in unstable equilibrium
(c) float in neutral equilibrium (d) start moving horizontally

Ans. (b)

Explanation:

Under the given conditions the pole will float with 0.6 m inside water and 0.4 m above water surface; metacentre is below *c.g.* and as such the pole will float in unstable equilibrium.

Thus (b) is the correct choice. (Ans.)

- Q. 69. Drag on cylinders and spheres decreases when the Reynolds number is in the region of 2×10^5 since
- (a) flow separation occurs due to transition to turbulence
(b) flow separation is delayed due to onset of turbulence
(c) flow separation is advanced due to transition to turbulence
(d) flow reattachment occurs.

Ans. (d)

Explanation:

When the Reynolds number is in the region of 2×10^5 the boundary layer on the cylinder and spheres starts becoming unstable and thus boundary layer is said to reattach; as a result of flow reattachment, there is a recovery of pressure over the back side and consequently there is a reduction in drag force.

Thus (d) is the correct choice.

- Q. 70. An open tank contains water to a depth of 2 m and oil over it to a depth of 1 m. If the specific gravity of oil 0.8, then the pressure intensity at the interface of the two fluid layers will be
- (a) 7848 N/m² (b) 8720 N/m² (c) 9347 N/m² (d) 9750 N/m²

Ans. (a)

Explanation:

Pressure intensity at the interface of the two liquids

$$= \rho gh = 1000 \times 9.81 \times (1 \times 0.8) = 7848 \text{ N/m}^2 \text{ (Ans.)}$$

- Q. 71. An inclined manometer, inclined at 30° to the horizontal, measures the pressure differential between two locations of a pipe carrying water. If the manometric liquid is mercury (specific gravity 13.6) and the manometer showed a level difference of 20 cm, then the pressure head difference of water between the two tapping will be
- (a) 1.26 m (b) 1.36 m (c) 2.52 m (d) 2.72 m

Ans. (b)

Explanation:

Vertical level of mercury = $20 \times \sin 30^\circ$ 10 cm or 0.1 m

Pressure head difference of water between the two tapping = $0.1 \times 13.6 = 1.36$ m. Thus (b) is the correct choice.

- Q. 72. The pressure gauge reading in metre of water column shown in Fig. 14 will be

- (a) 3.20 m (b) 2.72 m
(c) 2.52 m (d) 1.52 m

Ans. (d)

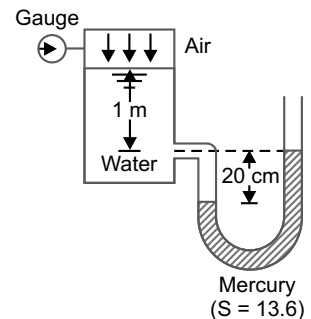


Fig. 14

Explanation:

The pressure gauge reading (p) in *metres of water* may be calculated as follows:

$$p + 1 + \frac{20}{100} = \frac{20}{100} \times 13.6$$

or, $p + 1.2 = 2.72$

or, $p = 1.52 \text{ m (Ans.)}$

- Q. 73.** Two pipe lines at different pressures, p_A and p_B , each carrying the same liquid of specific gravity S_1 , are connected to a U-tube with a liquid of specific gravity S_2 resulting in the level differences h_1 , h_2 and h_3 as shown in the Fig. 15. The difference in pressure head between points A and B in terms of head of water is

- (a) $h_1 S_2 + h_2 S_1 + h_3 S_1$
 (b) $h_1 S_1 + h_2 S_2 - h_3 S_1$
 (c) $h_1 S_1 - h_2 S_2 - h_3 S_1$
 (d) $h_1 S_1 + h_2 S_2 + h_3 S_1$

Ans. (d)

Explanation: Refer to Fig. 15

With reference to datum XX :

New pressure on left side = $h_A - h_1 S_1$ (the pressure due to inverted portion being equal)

Net pressure on right side = $h_B + h_3 S_1 + h_2 S_2$

$$\therefore h_A - h_1 S_1 = h_B + h_3 S_1 + h_2 S_2$$

$$\text{or } h_A - h_B = h_1 S_1 + h_2 S_2 + h_3 S_1$$

Thus (d) is the correct choice.

- Q. 74.** In the situation shown in the Fig. 16, the length BC is 3 m and M is the mid-point of BC . The hydrostatic force on BC measured per unit width (width being perpendicular to the plane of the paper) with 'g' being the acceleration due to gravity, will be

- (a) 16500 g N/m passing through M
 (b) 16500 g N/m passing through a point between M and C
 (c) 14250 g N/m passing through M
 (d) 14250 g N/m passing through a point between M and C .

Ans. (d)

Explanation: Refer to Fig. 17.

$$\text{Hydrostatic force on } BC, F_{BC} = \sqrt{F_V^2 + F_H^2}$$

where, F_V = Vertical component

= weight over area BC

$$= \left(\frac{5.5 + 4.0}{2} \right) \times \frac{3\sqrt{3}}{2} \times 1 \times 1000 \times g = 12340 \text{ gN/m}$$

F_H = Horizontal component = projected area of BC

(i.e., BD) $\times$ depth upto centre of $BD \times (1000 \times g)$

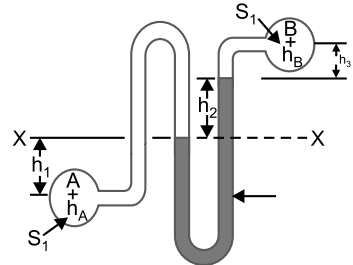


Fig. 15

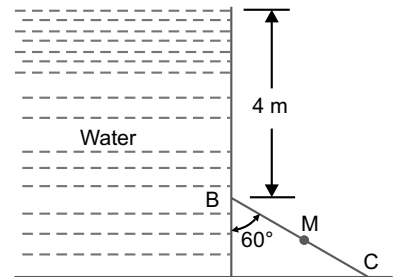


Fig. 16

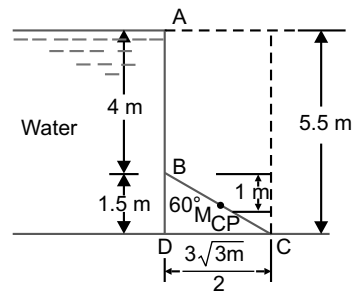


Fig. 17

$$= (1.5 \times 1) \times \left(4 + \frac{1.5}{2}\right) \times (1000 \times g) = 7125 \text{ g N/m}$$

$$\therefore F_{BC} = \sqrt{(12340 \text{ g})^2 + (7125 \text{ g})^2} \approx 14250 \text{ g N/m between } M \text{ and } C \text{ (Ans.)}$$

Q. 75. A simple Pitot tube can be used to measure which of the following quantities ?

1. Static head
2. Datum head
3. Dynamic head
4. Friction head
5. Total head

Select the correct answer using the codes given below

Codes:

- (a) 1, 2 and 4 (b) 1, 3 and 5 (c) 2, 3 and 4 (d) 2, 3 and 5

Ans. (b)

Explanation:

(b) is the correct choice since a simple pitot tube can measure static, dynamic and total heads. (Ans.)

Q. 76. The manometer shown in Fig. 18 connects two pipes, carrying oil and water respectively. From this figure one

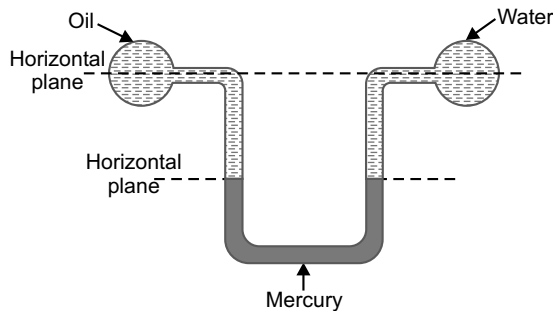


Fig. 18

- (a) can conclude that the pressures in the pipes are equal
- (b) can conclude that the pressure in the oil pipe is higher
- (c) can conclude that the pressure in the water pipe is higher
- (d) cannot compare the pressure in the two pipes for want of sufficient data

Ans. (d)

Explanation:

We cannot compare the pressure in the two pipes since the specific gravity of the fluids is not given. (Ans.)

Q. 77. At the point of boundary layer separation

- (a) shear stress is maximum
- (b) shear stress is zero
- (b) velocity is negative
- (d) density variation is maximum.

Ans. (b)

Explanation:

At the verge of separation, $\left(\frac{\partial u}{\partial y}\right)_{y=0}$ is zero,

$\therefore$ Shear stress, $\tau = \mu \left(\frac{\partial u}{\partial y} \right)_{y=0}$ is also zero. (Ans.)

Q. 78. The turbulent boundary-layer thickness varies as

- (a) $x^{4/5}$ (b) $x^{1/2}$ (c) $x^{1/5}$ (d) $x^{1/7}$.

Ans. (a)

Explanation:

Turbulent boundary layer thickness varies as $\frac{\delta}{x} = \frac{0.477}{x^{0.2}}$ or $\delta \propto x^{4/5}$ (Ans.)

Q. 79. A tank has in its side a very small horizontal cylinder fitted with a frictionless piston. The head of liquid above the piston is 'h' and the piston area 'a', the liquid having a specific weight γ . What is the force that must be exerted on the piston to hold it in position against the hydrostatic pressure?

- (a) $2\gamma ha$ (b) γha (c) $\frac{2\gamma ha}{3}$ (d) $\frac{\gamma ha}{2}$.

Ans. (b)

Explanation:

The pressure of liquid above the piston = γh .

$\therefore$ Force exerted on the piston to hold it in position

$$= \gamma h \times \text{piston area (a)} = \gamma ha \text{ (Ans.)}$$

Q. 80. A jet of oil with relative density 0.7 strikes normally a plate with a velocity of 10 m/s. The jet has an area of 0.03 cm^2 . The force exerted on the plate by the jet is

- (a) 210 N (b) 2.1 kN (c) 20.6 kN (d) 20.6 N.

Ans. (b)

Explanation:

Force exerted by a jet of oil striking normally on a stationary plate,

$$F = \rho a V^2 = (0.7 \times 10^3) \times 0.03 \times 10^2 = 2100 \text{ N or } 2.1 \text{ kN (Ans.)}$$

Q. 81. The parameters which determine the friction for turbulent flow in a rough pipe are:

- (a) Froude number and relative roughness
(b) Froude number and Mach number
(c) Reynolds number and relative roughness
(d) Mach number and relative roughness.

Ans. (c)

Explanation:

Whereas in a 'laminar flow' friction factor depends only on Reynolds number (Re), but in case of '*turbulent flow*', the friction factor depends *both on Re and roughness factor*. (Ans.)

Q. 82. An oil of kinematic viscosity 0.5 stoke is flowing through a pipe of 5 cm diameter. The flow through the pipe becomes critical at velocity of

- (a) 0.2 m/s (b) 2 m/s (c) 2.5 m/s (d) 4 m/s.

Ans. (b)

Explanation:

Given: $\gamma = 0.5 \text{ stoke} = 0.5 \times 10^{-4}$; $D = 5 \text{ cm} = 0.05 \text{ m}$

$$\text{Critical value of } (Re)_{cr} = 2000 = \frac{VD}{\gamma} = \frac{V \times 0.05}{0.5 \times 10^{-4}}$$

$$\therefore V = \frac{2000 \times (0.5 \times 10^{-4})}{0.05} = 2 \text{ m/s (Ans.)}$$

- Q. 83.** The head loss in a sudden expansion from 6 cm diameter pipe to 12 cm diameter pipe in terms of velocity V_1 in the smaller diameter pipe is

(a) $\frac{3}{16} \cdot \frac{V_1^2}{2g}$ (b) $\frac{5}{16} \cdot \frac{V_1^2}{2g}$ (c) $\frac{7}{16} \cdot \frac{V_1^2}{2g}$ (d) $\frac{9}{16} \cdot \frac{V_1^2}{2g}$

Ans. (d)

Explanation:

Refer to Fig. 19.

Losses due to sudden enlargement,

$$h_e = \frac{(V_1 - V_2)^2}{2g}$$

Also, $A_1 V_1 = A_2 V_2$

or, $V_2 = \frac{A_1 V_1}{A_2} = \frac{\frac{\pi}{4} \times 0.06^2}{\frac{\pi}{4} \times 0.12^2} \times V_1 = \frac{V_1}{4}$

$$\therefore h_e = \frac{1}{2g} \left(V_1 - \frac{V_1}{4} \right)^2 = \frac{9}{16} \frac{V_1^2}{2g} \text{ (Ans.)}$$

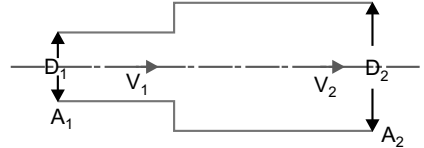


Fig. 19

- Q. 84.** Air (density 1.2 kg/m^3 and kinematic viscosity 15 centistokes) flows over a flat plate at zero angle of incidence, at a velocity of 20 m/s . If Reynolds number at transition is taken as 2.5×10^5 , maximum distance from leading edge up to which the boundary layer remains laminar is

(a) 375 mm (b) 93.75 mm (c) 187.5 mm (d) 250 mm

Ans. (c)

Explanation:

Given: $\rho = 1.2 \text{ kg/m}^3$; $\gamma = 15 \times 10^{-6} \text{ m}^2/\text{s}$; $U_\infty = 20 \text{ m/s}$

As transition taken place at $(Re)_{cr} = 2.5 \times 10^5$, therefore,

$$2.5 \times 10^5 = \frac{20 \times x}{15 \times 10^{-6}} \quad (\text{where } x \text{ is the maximum distance from leading edge upto which boundary layer remains laminar})$$

or, $x = \frac{2.5 \times 10^5 \times 15 \times 10^{-6}}{20} = 0.1875 \text{ m} \quad \text{or } 187.5 \text{ mm (Ans.)}$

- Q. 85.** The discharge, over a rectangular weir of width 100 m when the head of water over the weir is 1.5 m and the head due to velocity of approach is 0.02 m , is given by

(a) $842.5 \text{ m}^3/\text{s}$ (b) $653.4 \text{ m}^3/\text{s}$ (c) $814.5 \text{ m}^3/\text{s}$ (d) $552.5 \text{ m}^3/\text{s}$

Ans. (d)

Explanation:

The discharge over a rectangular weir is given by:

$$Q = \frac{2}{3} C_d \cdot L \cdot \sqrt{2g} \left[(H + H_a)^{3/2} - (H_a)^{3/2} \right] \quad \dots \text{with velocity of approach}$$

$$= \frac{2}{3} \times 1 \times 100 \times \sqrt{2 \times 9.81} \left[(1.5 + 0.02)^{3/2} - (0.02)^{3/2} \right]$$

$$= 295.3 \times 1.871 = 552.5 \text{ (Ans.)}$$

- Q. 86.** The velocity potential function is given by an expression $\phi = -\frac{xy^3}{3} - x^2 + \frac{x^3y}{3} + y^2$.
The velocity component in x direction is

- (a) $xy^2 - \frac{x^3}{3} - 2y$ (b) $\frac{y^3}{3} + 2x - x^2y$
(c) $3x^2y + 2x - xy^2$ (d) $x^3y^2 + 2x^2 - y^2x$

Ans. (b)

Explanation:

Velocity potential function $\phi = -\frac{xy^3}{3} - x^2 + \frac{x^3y}{3} + y^2$...Given

Velocity component in x direction,

$$u = -\frac{\partial \phi}{\partial x} = -\frac{\partial}{\partial x} \left[-\frac{xy^3}{3} - x^2 + \frac{x^3y}{3} + y^2 \right]$$

$$= \frac{y^3}{3} + 2x - x^2y \text{ (Ans.)}$$

- Q. 87.** The stream function in a two-dimensional flow field is given by $\psi = x^2 - y^2$. Then the magnitude of velocity at point (1, 1) is

- (a) 2 (b) $2\sqrt{2}$ (c) 4 (d) 8

Ans. (b)

Explanation:

Stream function, $\psi = x^2 - y^2$...Given

$$u = \frac{\partial \psi}{\partial x} = \frac{\partial}{\partial x} (x^2 - y^2) = -2y$$

$$v = -\frac{\partial \psi}{\partial y} = -\frac{\partial}{\partial y} (x^2 - y^2) = -2x$$

Resultant velocity, $V = \sqrt{u^2 + v^2} = \sqrt{(-2y)^2 + (-2x)^2} = 2\sqrt{(x^2 + y^2)}$

Velocity at (1, 1) = $2\sqrt{1^2 + 1^2} = 2\sqrt{2}$ (Ans.)

- Q. 88.** Consider the following statements:

1. Increasing the metacentric height of a ship causes increase in the period of rolling.
2. The frequency of pitching can be reduced by adding more loads along the length of a ship below its centre of gravity.
3. A ship has different metacentres in rolling and pitching.
4. Stability can be improved by increasing the width and reducing the draft.

Which of these statements are correct?

- (a) 1, 2, 3 and 4 (b) 1, 2 and 4 only (c) 1, 2 and 3 only (d) 2, 3 and 4 only

Ans. (d)

Explanation:

Time period of rolling (*i.e.*, one complete oscillation) ' T ' is given by:

$$T = 2\pi \sqrt{\frac{k^2}{GM \cdot g}} \quad \text{or} \quad T \propto \frac{1}{\sqrt{GM}} \quad \dots(i)$$

where,

k = radius of gyration about G;

GM = metacentric height of the body.

From eqn. (i) it is evident that period of rolling '*decreases*' with the *increase* in the metacentric height. (Ans.)

Q. 89. Consider the following statements related to the stability of floating bodies:

1. The metacentre should be above the centre of gravity of the floating body for stable equilibrium during small oscillations.
2. For a floating body, stability is not determined simply by the relative positions of centre of gravity and centre buoyancy.
3. The position of metacentre of a floating body is fixed irrespective of the axis of oscillations.
4. Large value of metacentric height reduces the period of roll of the vessel.

Which of these statements are correct?

- (a) 1 and 3 only (b) 2 and 4 only (c) 1, 2 and 4 only (d) 1, 2, 3 and 4

Ans. (c)

Explanation:

The position of metacentre of a floating body is *not* fixed; it changes with the change of axis of oscillation. (Ans.)

Q. 90. A circular area of 1.2 m diameter is immersed vertically in a liquid of unit weight 800 N/m^3 with its top edge just on the liquid surface. The depth of centre of pressure on one side, measured below the liquid surface, is

- (a) 0.80 m (b) 0.75 m (c) 0.60 m (d) 0.64 m

Ans. (b)

Explanation:

Refer to Fig. 20.

The position of the centre of pressure,

$$\begin{aligned} \bar{h} &= \bar{x} + \frac{I_G}{A\bar{x}} \\ &= \frac{d}{2} + \frac{\frac{\pi}{64} \times d^4}{\frac{\pi}{4} \times d^2 \times \frac{d}{2}} = \frac{d}{2} + \frac{d}{8} = \frac{5d}{8} \\ &= \frac{5}{8} \times 1.2 = 0.75 \text{ (Ans.)} \end{aligned}$$

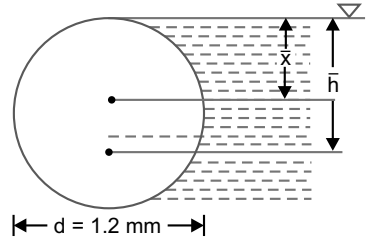


Fig. 20

Q. 91. If angle of contact of a drop of liquid is acute then

- (a) adhesion is more than cohesion
- (b) cohesion is more than adhesion
- (c) cohesion is equal to adhesion
- (d) adhesion and cohesion have no bearing with angle of contact.

Ans. (a)

Explanation:

When *adhesion* > *cohesion* the angle of contact is *acute*.

- For *wetting liquid* (water): $\theta < \frac{\pi}{2}$.

For water: $\theta = 0$ when pure water is in contact with glass. But θ becomes as high as 25° when water is slightly contaminated.

– For *non-wetting liquid* (mercury): $\theta > \frac{\pi}{2}$.

(For mercury: θ varies between 130° to 150°) (Ans.)

Q. 92. Pascal's law states that pressure at a point is equal to all directions in a

- (a) Liquid at rest (b) Fluid at rest
(c) Laminar flow (d) Turbulent flow

Ans. (b)

Explanation:

The Pascal's law states as follows:

"The intensity of pressure at any point in a fluid at rest is the same in all directions." (Ans.)

Q. 93. With increase in pressure the bulk modulus of elasticity

- (a) Decreases (b) Increases
(c) Remains constant (d) None of the above

Ans. (b)

Explanation:

Bulk modulus, $K = \frac{dp \text{ (increase of pressure)}}{-dV/V \text{ (Volumetric strain)}}$

i.e., $K = \frac{dp}{-dV/V}$ i.e., K increases with increase in pressure.

(-ve sign indicates decrease in volume with increase in pressure) (Ans.)

Q. 94. Match List-I with List-II and select the correct answer using the code as given below the Lists:

List-I	List-II
A. Capillarity	1. Cavitation
B. Vapour pressure	2. Density of water
C. Viscosity	3. Shear forces
D. Specific gravity	4. Surface tension

Codes:

	A	B	C	D
(a)	2	3	1	4
(b)	4	3	1	2
(c)	2	1	3	4
(d)	4	1	3	2

Ans. (d)

Explanation:

Capillarity $\rightarrow$ 4; Vapour pressure $\rightarrow$ 1; Viscosity $\rightarrow$ 3; Specific gravity $\rightarrow$ 2,
i.e., d (4, 1, 3, 2) (Ans.)

Q. 95. The stability of a floating body is obtained when its

- (a) Centre of gravity is below the centre of buoyancy
(b) Metacentric height is negative
(c) Metacentric height is positive
(d) Metacentric height is zero.

Ans. (c)

Explanation:

The distance between the centre of gravity (G) of a floating body and the metacentre (M) i.e., distance GM is called metacentric height GM .

For *stable equilibrium*, the position of M remains higher than G , i.e., GM is +ve. (Ans.)

Q. 96. The stream function of a two-dimensional flow is $\psi = 2xy$. The flow is

- (a) Rotational (b) Irrotational
(c) Vortex flow (d) Unsteady pulsating flow.

Ans. (b)

Explanation:

$$\psi = 2xy$$

...(Given)

Now,
$$\frac{\partial \psi}{\partial x} = 2y, \quad \frac{\partial \psi}{\partial y} = 2x$$

and,
$$\frac{\partial^2 \psi}{\partial x^2} = 0, \quad \frac{\partial^2 \psi}{\partial y^2} = 0, \quad \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

Hence, $\psi = 2xy$ represents a possible *irrotational* flow (Ans.)

Q. 97. Consider the following statements pertaining to kinematics and dynamics of fluid flow:

1. For $\psi = x^2 - y^2$, velocity at (1, 1) is $2\sqrt{2}$.
2. For all types of flow, stream and potential functions exist.
3. Bernoulli's equation is not valid for real fluid flow.

Which of these statements are correct?

- (a) 1, 2 and 3 (b) 1 and 2 only (c) 2 and 3 only (d) 1 and 3 only

Ans. (d)

Explanation:

For all types of flow, stream and potential functions *do not exist*. 'Potential function' (ϕ) exists only for *irrotational flow* while 'stream function (ψ)' applies both to rotational or irrotational flows (which are steady and incompressible). (Ans.)

Q. 98. The flow past the cylinder with the vortex motion causes a net transverse motion. This transverse force is known as

- (a) Magnus effect (b) Robin effect
(c) D'Alembert's Pradox (d) Rankine doublet.

Ans. (a)

Explanation:

The generation of lift & transverse force by spinning cylinder in a fluid stream is called 'Magnus effect'. (Ans.)

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A

Afflux, 940

B

Back water curve, 940

length of, 940

Bernoulli's equation, 260

for real fluid, 276

practical applications of, 291

– orificemeter, 303

– pitot tube, 310

– venturimeter, 291

Bluff body, 798

Boundary layer definitions, 726

– boundary layer thickness, 727

– displacement thickness, 727

– energy thickness, 729

– momentum thickness, 728

Boundary layer separation, 774

Buckingham's π -method, 394

Buoyancy, 160

centre of, 160

C

Capillarity, 28

Centre of pressure, 97

Chezy's formula, 639, 884

Circulation and vorticity, 218

Circulation, 804

Compressibility and bulk modulus, 34

Compound pipes, 608

Compressible fluid flow, 825

basic equations of, 829

– continuity equation, 829

– energy equation, 829

– momentum equation, 829

propagation of disturbances in, 837

through a convergent nozzle, 852

through a convergent-divergent nozzle, 860

measurement of, 870

through a venturimeter, 870

Continuity equation, 207

in cartesian co-ordinates, 209

in polar co-ordinates, 211

Coriolis co-efficients, 336

Couette flow, 570

Critical depth, 906

Curved immersed surface, 129

D

Dams, 140

possibility of dam failure, 142

Darcy equation, 606, 639

Density, 3

– mass density, 3

– weight density, 3

Dimensional analysis, 386

introduction to, 386

methods of, 390

Dimensions, 387

Dimensional homogeneity, 389

Dimensionless numbers, 418

Discharge, 207

Displacement thickness, 727

Drag and lift, 785

Drag on a sphere, 798

Drag of a cylinder, 804

E

Efflux viscometers, 594

Elbow meter, 309

Energy thickness, 729

Equivalent pipe, 671

Eulerian method, 193

Euler's equation, 262

Euler's number, 419

F

- Falling sphere method, 594
- Fluid, 2
- Floating bodies, 165
 - oscillation (rolling) of, 187
 - types of equilibrium of, 165
 - neutral equilibrium, 165
 - stable equilibrium, 165
 - unstable equilibrium, 165
- Flownets, 231
 - methods of drawing, 231
 - uses and limitations of, 232
- Flow through nozzle, 706
 - power transmitted, 707
 - condition for maximum power transmission, 707
- Fluid flow, 195
 - compressible and incompressible, 197
 - laminar and turbulent, 197
 - one, two and three-dimensional, 196
 - rotational and irrotational, 197
 - steady and unsteady, 195
 - uniform and non-uniform, 196
- Forced vortex flow, 345
- Free liquid jet, 313
- Free vortex flow, 346
- Froude number, 412

G

- Gradually varied flow, 938
 - equation of, 938

H

- Hazen Poiseuille law, 541
- Hydraulic co-efficients, 458
 - co-efficient of contraction, 458
 - co-efficient of velocity, 459
 - co-efficient of discharge, 459
 - co-efficient of resistance, 459
 - experimental determination of, 460
 - Hydraulic gradient and total energy lines, 657
- Hydraulic jump, 923

I

- Impulse momentum equation, 320
- Inclined immersed surface, 116

K

- Kinetic energy correction factor, 336
- Kutta-Joukowski theorem, 807

L

- Laminar flow, 534
 - between parallel plates, 570
 - through an annulus, 567
 - through porous media, 582
- Laminar boundary layer, 742
- Langrangian method, 192
- Liquids (properties of), 3
- Liquids in relative equilibrium, 364
- Lock gates, 151

M

- Mach number, 420
- Major energy losses, 638
- Magnus effect, 810
- Manning's formula, 873
- Manometers, 54
 - simple manometers, 54
 - differential manometers, 63
 - advantages and limitations of, 81
 - Mechanical gauges, 81
- Metacentre, 165
- Metacentric height, 165
 - determination of, 166
 - analytical method, 166
 - experimental method, 167
- Minor energy losses, 638
 - at the entrance to pipe, 657
 - at the exit of a pipe, 657
 - due to sudden enlargement, 645
 - due to sudden contraction, 652
 - due to obstruction in pipe, 656
 - due to bend in the pipe, 657
 - in various pipe fittings, 657
- Model analysis, 415
- Model laws, 420
- Euler model law, 445

Froude model law, 434
 Mach Model law, 447
 Reynolds model law, 420
 Weber model law, 446
 Models, 449
 distorted, 449
 scale effect in, 450
 undistorted, 449
 Momentum correction factor, 336
 Moment of momentum equation, 343
 Momentum thickness, 728
 Momentum equation for boundary layer, 739
 Most economical section, 889
 – circular channel, 910
 – rectangular channel, 890
 – trapezoidal channel, 892
 – triangular channel, 908
 Mouthpieces, 490

N

Navier-Stokes equations, 537
 Newton's law of viscosity, 5
 Notches, 508
 rectangular, 509
 stepped, 514
 triangular, 511
 trapezoidal, 513

O

Open channels, 880
 formulae for uniform flow in, 884
 – Chezy's formula, 884
 non-uniform flow through, 917
 types of, 881
 types of flow in, 881
 – laminar and turbulent flows, 882
 – steady flow and unsteady flow, 882
 – subcritical, critical and
 supercritical flows, 882
 Orifice, 457
 Classification of, 457
 Orificemeter, 303

P

Pascal's law, 45
 Path line, 198
 Pipes in series, 668
 Pipes in parallel, 674
 Pitot tube, 310
 Potential head or energy, 259
 Power absorbed in bearings, 583
 collar bearing, 586
 foot-step bearing, 585
 journal bearing, 583
 Pressure, 43
 measurement of, 53
 Pressure head (of a liquid), 43

R

Rayleigh's method, 390
 Reynolds number, 418
 Reynolds experiment, 535
 Rotating cylinder method, 591
 Rotameter, 308

S

Shock waves, 865
 Similitude, 416
 Specific volume, 3
 Specific gravity, 3
 Specific energy and specific energy curve, 917
 Steamline, 198
 Steam tube, 198
 Streak line, 199
 Stream function, 228
 Streamlined body, 798
 Stagnation properties, 844
 Surface tension, 25
 Syphon, 699

T

Terminal velocity of a body, 799
 Thermodynamic properties, 23
 Turbulent flow, 605

characteristics of, 608

shear stresses in, 609

– Boussinesq's theory, 609

– Prandtl's mixing length theory, 610

– Reynolds theory, 610

Turbulent boundary layer, 766

V

Vapour pressure, 37

Velocity potential, 227

Velocity head or kinetic energy, 259

Venturimeter, 291

– horizontal, 292

– vertical and inclined, 298

Viscosity, 4

effect of temperature on, 8

effect of pressure on, 8

measurement of, 591

– capillary tube method, 595

– falling sphere method, 594

– rotating cylinder method, 591

Vorticity, 218

Vortex motion, 345

W

Water hammer in pipes, 711

Weber number, 419

Weirs, 508

broad-crested, 522

cipolletti, 521

ogee, 523

narrow-crested, 523

rectangular, 518

submerged or drowned, 523